Circuit Lab Division B

Competition binder · 2026-27 season ("2027" rules) · State and National level

Daniel

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Built from the Obsidian notes in *School/Science Olympiad/Circuit Lab*. Small boxed codes after a name (like A5) tell you which tab that note is in. Page numbers are printed at the outer bottom corner of every page.

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How to use this binder. Start at TAB QR (Quick Reference) for formulas, tables and constants. For a question you can name, go to its tab (A1-A9): each note lists question types from Easy to Nationals, each with a worked example and traps. Scope marks: ✅ IN, ⚠️ borderline, ❌ out for 2027. The rules (TAB R) say answers must be metric with correct significant figures, and math is limited to algebra with at most two equations (no trig).

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Formulas, constants and tables, written for this binder from the checked app data and the notes.

App key card

CircuitLab key card (TI-Nspire CX II)

Home: one key opens a tool

Key	Tool		Key	Tool
o 1	Ohm's law + power		j	Kirchhoff, 2 equations
s 2	Series / parallel		y	n x n equations
d 3	Voltage divider		w	AC peak / rms / period
k	Current divider		m	Mystery resistor
l 4	LED / zener resistor		p	Battery pack (cells)
c 5	Color code (both ways)		z	USB charge time
q 6	Coulomb's law		n	Nearest E6/E12/E24
b 7	Battery capacity		h	Hit a target R from parts
e 8	Energy + cost		t	Truth table / simplify
x 9	Binary convert + add		g	Gates + De Morgan
f	Field + potential		v	Key test
a	Charge Q = It		0	Search the facts
r	Resistivity			
i	Internal resistance			
u	Fuse sizing			

Shift + letter: **N** network solver, **W** Wheatstone, **L** infinite ladder, **P** plates (E = V/d), **R** R(T), **J** energy V^2 t / R, **F** loads per fuse, **B** rated bulbs, **T** tolerance, **Q** point charges, **V** speed test.

Typing rules (all tools)

You want	Type	Example
Pick a box	its letter	`o v12 r4` -> I = 3 A
kilo / milli / micro / nano	`k m u n` after a digit (Shift+M = mega). A space before it is fine	`220k`, `20m`, `4k7`, `4.7 k`
Power of ten	`1.6*10^-19` works everywhere; `1.6x10^-19` too, except in tools with an x box (k, m, i, j, W): there use `*`; EE key; or `e` after a digit (NOT in tools with an e box: a, f, i, e, j, z — use EE there)	`1.6e-19`
Negative	`(-)` or `-` (a `-` at the end also makes it negative)	`a2u-` = -2 uC
Units after a number	allowed: `12V`, `4ohm`, `2000mAh`, `2.5Ah`, `1.2kWh`, `5min` in a time box	`bc2000mAhi.5` -> 4 h
`/` in an ohm box	parallel: `100/2` = 100 || 2 (warning). Divide: `100*1/2`	
SMD resistor code	`c472` -> "If SMD code: 4.7 kohm"; `c4R7` = 4.7 ohm	
Hours, areas, packs	`3h` in a seconds box (moves to box h); `2mm^2`, `3cm^2` in an area box; `p4S2P` = 4 series, 2 parallel. Capital M = mega (warning outside ohm boxes)	`ai2t3h` -> 21.6 kC
Math in a box	`+ - * / ^ ( )`	`v4*1.5` (4 cells), `v24-16`, `p3*60`
Series / parallel in an ohm box	`+` series, `/` parallel; with `*`, `^` or `-` in the box it is plain math (`/` divides)	`b4.7k/6.8k`, `r120^2/1000`

You want	Type	Example
Copy an answer	**enter** copies the selected box; on a box you typed it copies the first answer, enter again = the next one (the message names it)	
Paste it into another tool	`=` in a box, or ctrl+V. A digit right after `=` replaces the paste (`v=12` = 12)	`s100/100` enter, esc, `o i2 r=`
Clear / back / home	ctrl+del / del / esc	

No Enter is needed: the answer updates after every key. Empty boxes are the unknowns.

Network solver (Shift+N) shortcuts

You want	Type
Resistor 100 ohm, nodes 1-2	`1-2 100`
9 V source, + at node 1	`V1-0 9`
Commas are optional	`V1-0 9 1-2 100 2-0 200`
A chain of equal resistors	`1-2-3-4-1 5` = four 5 ohm parts around a loop
Same value as the last resistor	leave the value out: `1-5 2-6 3-7`
All resistors equal R	leave every value out: answer is in units of R
3 equal resistors in parallel	`2-0*3 6` (times key)
Before / after: part removed or shorted	`o2-3 1` (removed after) / `s2-3 1` (shorted after) is its own part: type the bulb that changes with o or s, not twice. Shows each P before -> after, x2.78 brighter
R between nodes 1 and 7	`=1-7`
Voltmeter / ammeter	`V2-3` / `I2-3` (current in the parts between 2 and 3)
Units after a value	allowed: `V1-0 9V`, `I2-0 5mA`, `1-2 100ohm`
Diode	`D2-3` = anode 2, cathode 3, 0.7 V; `D2-3 2` = 2 V LED; `D2-3 0` = ideal (ON/OFF found for you)
Zener	`Z2-3 5.1` = anode 2, cathode 3, Vz 5.1 V (forward 0.7 V); shows ON forward / ON zener / OFF
Dependent source	`V3-0 5*V1-2` (5 Vx), `I3-0 .6*I2-0` (0.6 ix, ix 2->0), `I2-0 2m*V1`; `=a-b` keeps them on
One unknown value + one condition	`2-0 x` and `I2-0=20m` (or `V2=5`, `P=2`, `=1-0 500`)

Cube, corner to corner: `N1-2-3-4-1 5-6-7-8-5 1-5 2-6 3-7 4-8 =1-7` -> 0.8333 R (5R/6).

Other text tools

Tool	Shortcut
Shift+B bulbs	`=120s` only series, `=120p` only parallel
Shift+Q charges	`e` = 1.602E-19 C (`-e` electron, `2e`); commas optional; with commas, a part with only x repeats the charge: `Qe 0, 4, -6`. 2D: `q x y` for each (commas), `=x y`: F with angle, total U, E and V
x binary	`0101^0011` XOR, `1100*1010` AND, `5 or 3`; `h9` hex, `d101` decimal; `101-11` subtract
t truth table	`^` = XOR, `-` or `~` = NOT, `+` = OR, nothing = AND; output column `0110`, with `x` = don't care: `x1x10x0100010001`; minterms `abc m1,3,5 d6`, maxterms `abc M0,2`; every table shows the mux inputs (select = first letters)

Tool	Shortcut
s series/parallel	with `*` or `^` the line is plain math (`/` divides): `s12*2^2` = 48

Search (key 0)

Type 2-5 letters of a title word: `kir`, `gfc`, `rm`. Two words: `co la` (Coulomb's law). Abbreviations work: `kvl kcl led emf rms gfci afci sop pos kmap lsb msb pot mov awg nimh lipo`. One wrong letter is forgiven (words of 4+ letters). Digits stay in the query (`e12`, `7400`); tab or an arrow moves to the list, then 1-9 opens a result. The full code list is in the binder.

Quick Reference

Densest tables first. Formulas use V = volts, I = amps, R = ohms (Ω), P = watts, t = seconds unless a box says hours. Every 2027 answer must be **metric** with **correct significant figures**. Math is algebra only: **no trig, at most 2 equations**. Capacitors, inductors, RC time constants, magnetism, transformers and Thévenin are **OUT for 2027** (a capacitor symbol can still be a wrong answer choice).

Ohm's law and power: every form

Find	From V, I	From I, R	From V, R	From P and one more
V	—	V = I·R	—	V = P/I V = √(P·R)
I	—	—	I = V/R	I = P/V I = √(P/R)
R	R = V/I	—	—	R = V²/P R = P/I²
P	P = V·I	P = I²·R	P = V²/R	—

- **Energy** E = P·t = V·I·t = I²·R·t = V²·t/R (joules if t in s).
- **Charge** Q = I·t. **Current** I = Q/t. **Voltage** V = E/Q (1 V = 1 J/C).
- **kWh** = watts × hours ÷ 1000. **Cost** = kWh × price per kWh.
- **Conductance** G = 1/R, unit siemens (S), old name mho (℧).
- Fixed R: double V → double I → **4× P**. V doubles and R halves → P × 8.
- Resistor rating: I_max = √(P_rated/R), V_max = √(P_rated·R). Common sizes 1/8, 1/4, 1/2 W.

60 W bulb at 120 V: I = 60/120 = 0.5 A, R = 120²/60 = 240 Ω. 100 W for 24 h at $0.15/kWh: 2.4 kWh → $0.36.

Series and parallel

	Series	Parallel
Shared	same **current**	same **voltage**
Adds up	voltages: $V = V_1 + V_2 + \ldots$	currents: $I = I_1 + I_2 + \ldots$
R_eq	$R_1 + R_2 + R_3 + \ldots$	$1/R_{eq} = 1/R_1 + 1/R_2 + \ldots$
Two parts	$R_1 + R_2$	$R_1R_2/(R_1 + R_2)$ (product over sum)
n equal R	n·R	R/n (series ÷ parallel = n^2)
Size	bigger than the biggest	smaller than the smallest
Conductance	$1/G = 1/G_1 + 1/G_2$	$G = G_1 + G_2$
Cells	voltages add, same mAh	mAh adds, same voltage

- **Reduce from the far end** toward the source; then go back out for each branch.
- A wire across a resistor (short) → that resistor carries 0 A: remove it. A dangling resistor carries 0 A.
- R ‖ 0 Ω = 0. R ‖ open = R. Equal parallel pair = R/2.
- **Bridge** balanced when $R_1/R_2 = R_3/R_x$: the middle resistor carries 0 A (remove it).
- **Infinite ladder** (series a first, then shunt b, repeating): R = (a + √(a² + 4ab))/2. a = 1, b = 2 → 2 Ω.

10 Ω ‖ 20 Ω ‖ 20 Ω = 5 Ω. 6 Ω ‖ 3 Ω = 18/9 = 2 Ω. 100, 200, 50, 50 Ω → 200 ‖ (100+50+50) = 100 Ω.

Dividers, internal resistance, bulbs

- **Voltage divider** (series): V_x = V_source · R_x / R_total. Biggest R gets the biggest share.
- **Current divider** (2 branches): $I_1 = I_{total} \cdot R_2/(R_1 + R_2)$. Smaller R takes the bigger current.
- **Real battery** = EMF in series with internal r. V_terminal = EMF − I·r. r = (EMF − V)/I. Short-circuit I = EMF/r.
- **Max power to the load** when R_load = r: P_max = EMF²/(4r) (efficiency then 50%).
- **Bulb brightness** = the power it actually uses. Parallel: each bulb gets full V. Series: the higher-watt bulb (lower R) is **dimmer**.
- **Open** part: I = 0 and the full source voltage appears across the gap. **Shorted** part: V = 0, R ≈ 0.
- **Mystery R** (in series with known R_k): R_x = R_k · V_x / V_k.

15 V across 1 kΩ + 4 kΩ → V(4 kΩ) = 12 V. 0.60 A into 200 Ω ‖ 300 Ω → 0.36 A in 200 Ω, 0.24 A in 300 Ω. EMF 9.0 V, 8.5 V across 10 Ω → I = 0.85 A, r = 0.59 Ω. EMF 12 V, r = 2 Ω → P_max = 18 W at R_load = 2 Ω.

Kirchhoff (KCL, KVL): step by step

KCL (current law, conservation of **charge**): current in = current out at every node. **KVL** (voltage law, conservation of **energy**): the voltage changes around any closed loop add to 0.

Sign rules for KVL: through a battery from – to + = rise (+V). Through a resistor *with* the assumed current = drop (–IR); against it = +IR. A negative answer means the current really flows the other way. Do not change the sign afterward.

Nodal method (best with 1 unknown node)

1. Pick ground (0 V), usually the bottom wire.
2. Name each unknown node voltage (V_x).
3. KCL at each unknown node: sum of (V_x – V_other)/R = 0 for every branch leaving it.
4. Solve, then find each current with I = ΔV/R.

Mesh method (2 loops)

1. Draw a clockwise current I_1, I_2 in each window.
2. KVL around each loop. A shared resistor carries (I_1 – I_2) in loop 1 and (I_2 – I_1) in loop 2.
3. Write as a·x + b·y = c, d·x + e·y = f and solve.

2 × 2 solve (Cramer): D = ae – bd, x = (ce – bf)/D, y = (af – cd)/D.

Example. 10 V source (+ up) → 2 Ω → node X. 5 V source (+ up) → 3 Ω → node X. 4 Ω from X to ground.
Nodal: (V_x–10)/2 + (V_x–5)/3 + V_x/4 = 0 → ×12: 13V_x = 80 → **V_x = 6.15 V**.
I(2 Ω) = 1.92 A into X; I(3 Ω) = 0.385 A out of X into the 5 V source; I(4 Ω) = 1.54 A. Check KCL: 1.92 – 0.385 = 1.54 ✓.
Mesh: $6I_1 – 4I_2 = 10$, $4I_1 – 7I_2 = 5$ → D = –26, I_1 = 1.92 A, I_2 = 0.385 A, shared 4 Ω: $I_1 – I_2$ = 1.54 A ✓.

Only 2 equations at Div B. Thévenin, Norton and delta-Y are OUT.

Resistor color code

Color	Digit	Multiplier	Tolerance	Temp. coeff. (ppm/K)
Black	0	×1	—	250
Brown	1	×10	±1%	100
Red	2	×100	±2%	50
Orange	3	×1 k	—	15
Yellow	4	×10 k	—	25
Green	5	×100 k	±0.5%	20
Blue	6	×1 M	±0.25%	10
Violet	7	×10 M	±0.1%	5
Gray	8	×100 M	±0.05%	1
White	9	×1 G	—	—
Gold	—	×0.1	±5%	—
Silver	—	×0.01	±10%	—
No band	—	—	±20%	—

- **4 bands:** digit, digit, multiplier, tolerance. **5 bands:** 3 digits, multiplier, tolerance. **6 bands:** 5 bands + temp. coeff.
- Tolerance band (gold/silver) goes on the **right**. Gold and silver are never digits. One black band = 0 Ω jumper.
- Memory: **B**ad **B**eer **R**ots **O**ur **Y**oung **G**uts **B**ut **V**odka **G**oes **W**ell.
- Range: R_min = R(1 – t), R_max = R(1 + t). SMD: 103 = 10 kΩ, 471 = 470 Ω, 4R7 = 4.7 Ω.

Red-violet-green-gold = 27 × 10^5 = 2.7 MΩ ±5%. Brown-black-red-gold = 1.0 kΩ ±5% (950 to 1050 Ω). Yellow-violet-black-brown-brown = 470 × 10 = 4.70 kΩ ±1%.

Standard values (E-series)

Series	Tolerance	Values per decade (× 1, 10, 100, 1 k …)
E6	±20%	1.0 1.5 2.2 3.3 4.7 6.8
E12	±10%	1.0 1.2 1.5 1.8 2.2 2.7 3.3 3.9 4.7 5.6 6.8 8.2
E24	±5%	1.0 1.1 1.2 1.3 1.5 1.6 1.8 2.0 2.2 2.4 2.7 3.0 3.3 3.6 3.9 4.3 4.7 5.1 5.6 6.2 6.8 7.5 8.2 9.1

For an LED resistor pick the next value **up** (350 Ω → 390 Ω in E12). Each E12 step is about ×1.21.

SI prefixes and units

Prefix	Sym	Factor	Prefix	Sym	Factor
tera	T	10^{12}	milli	m	10^{-3}
giga	G	10^{9}	micro	μ (u)	10^{-6}
mega	M	10^{6}	nano	n	10^{-9}
kilo	k	10^{3}	pico	p	10^{-12}
centi	c	10^{-2}	femto	f	10^{-15}

m (milli) vs M (mega) differ by 10^9. Kilo is lowercase k. 1 mm² = 10^{-6} m², 1 cm² = 10^{-4} m².

Quantity	Unit	Equals
Current I	ampere A (SI base unit)	C/s
Charge Q	coulomb C	A·s
Voltage V	volt V	J/C = W/A
Resistance R	ohm Ω	V/A
Conductance G	siemens S	A/V = 1/Ω
Power P	watt W	J/s = V·A
Energy E	joule J	W·s = V·C = N·m
Resistivity ρ	Ω·m	
Electric field E	N/C	= V/m
Frequency f	hertz Hz	1/s

A unit named after a person has a capital symbol (A, V, W) but a lowercase name (ampere, volt, watt).

Unit conversions

Unit	Value
1 kWh	3.6×10^6 J (3.6 MJ)
1 Wh	3600 J
1 mAh	3.6 C (1 Ah = 3600 C)
Wh from mAh	Wh = mAh × V ÷ 1000
1 eV	1.602×10^{-19} J
1 calorie	4.184 J (food Calorie = 1000 cal = 4184 J)
1 horsepower	745.7 W
1 hour	3600 s; 1 day = 86 400 s

mAh is charge, not energy. Convert to Wh (multiply by the battery voltage) before comparing batteries or chargers.

5000 mAh × 3.7 V = 18.5 Wh. 18 W charger at 85%: 18.5 / (18 × 0.85) = 1.2 h. 3000 mAh at 1000 mA = 3 h.

Significant figures

- Nonzero digits count. Zeros between them count (2.05 → 3). Leading zeros never count (0.0047 → 2).
- Trailing zeros after a decimal point count (2.50 → 3). In 1200 with no decimal point they are unclear: write 1.20×10^3.
- **× and ÷:** keep the fewest significant figures of the inputs. **+ and –:** keep the fewest decimal places.
- Exact counts and defined numbers (2 resistors, 1000 m/km) do not limit the answer. Round only at the end.

Coulomb's law, field, potential

Constant	Value
Coulomb constant $k = 1/(4\pi\varepsilon_0)$	8.99×10^9 N·m²/C² ($\approx 9 \times 10^9$)
Permittivity of free space ε_0	8.85×10^{-12} C²/(N·m²)
Elementary charge e	1.602×10^{-19} C (exact since 2019)
Electrons in 1 C	6.24×10^{18}
Electron mass	9.11×10^{-31} kg
Proton mass	1.67×10^{-27} kg
Planck constant h	6.626×10^{-34} J·s
Speed of light c	3.00×10^8 m/s
Air breakdown	about 3 kV/mm

- **Force** $F = k \cdot q_1 \cdot q_2 / r^2$. Like charges repel (+), opposite attract (–).
- Double r → F/4. Triple r → F/9. Double one charge → 2F. Double both → 4F.
- **Field** $E = F/q = k \cdot q / r^2$ (N/C = V/m). Points away from +, toward –. Between plates E = V/d.
- **Potential** $V = k \cdot q / r$ (scalar: add with signs). **Potential energy** $U = k \cdot q_1 \cdot q_2 / r$. **Work** $W = q \cdot \Delta V$.
- E = 0 point: between two **like** charges (closer to the smaller); outside two **opposite** charges (on the smaller one's side).
- Charge: rubbing (triboelectric) → opposite signs; contact (conduction) → **same** sign; induction → **opposite** sign. Only electrons move.
- Triboelectric order (+ end to – end): rabbit fur, glass, hair, wool, silk, paper, cotton, rubber, PVC, Teflon.

+6 μC and –7 μC, 0.50 m apart: $F = 8.99 \times 10^9 \times 6 \times 10^{-6} \times 7 \times 10^{-6} / 0.25 = 1.51$ N, attractive (1.512 N with $k = 9 \times 10^9$).

Resistivity and temperature

- **$R = \rho \cdot L / A$**; round wire $A = \pi \cdot d^2/4$. Length ×3 and area ÷2 → R × 6. Double diameter → A × 4 → R ÷ 4.
- **$R(T) = R_0(1 + \alpha \cdot \Delta T)$**. α (/°C): copper 0.0039, aluminum 0.0043, tungsten 0.0045.
- Metals: R rises with temperature. Semiconductors and NTC thermistors: R falls. Superconductor: R = 0 below a critical temperature.
- AWG: a bigger gauge number = a **thinner** wire = more R. 15 A circuit → 14 AWG, 20 A → 12 AWG.

Material	ρ (Ω·m)	Material	ρ (Ω·m)
Silver	1.59×10^{-8}	Aluminum	2.65×10^{-8}
Copper	1.68×10^{-8}	Nichrome	$\approx 1.1 \times 10^{-6}$
Gold	2.44×10^{-8}	Glass	10^{10} to 10^{14}

10 m copper wire, d = 1.0 mm: $R = 1.68 \times 10^{-8} \times 10 / (\pi \times 0.001^2/4) = 0.21$ Ω. Copper 100 Ω at 20 °C → 119.5 Ω at 70 °C.

DC and AC

- **DC**: one direction, fixed polarity. Conventional current + → –; electrons flow – → +.
- **AC**: reverses, usually a sine wave. **V_rms = V_peak/√2 ≈ 0.707·V_peak**. V_peak = 1.414·V_rms. V_pp = 2·V_peak.
- **T = 1/f**. RMS = the DC value that gives the same heating power.
- Grid uses AC because it steps up to high voltage easily: same power, less current, less I²R loss.
- At 50-60 Hz, AC is more dangerous than DC at the same voltage (fibrillation, can't let go).

Mains	V_rms	f	V_peak	V_pp	T
US outlet	120 V	60 Hz	170 V	339 V	16.7 ms
US dryer/range	240 V	60 Hz	339 V	679 V	16.7 ms
Europe	230 V	50 Hz	325 V	651 V	20 ms
Japan	100 V	50/60 Hz	141 V	283 V	20/16.7 ms

US wire colors: hot = black or red; neutral = white or gray; ground = green or bare. Outlet: narrow slot = hot, wide = neutral, round = ground. US house: two 120 V hots, hot to hot = 240 V (split phase).

DC equal to a 100 V-peak AC source: 100/√2 = 70.7 V.

Batteries and USB

Cell	Volts	Cell	Volts
Alkaline, carbon-zinc (AA, AAA, C, D)	1.5	NiMH, NiCd	1.2
Lithium coin CR2032	3	Lead-acid (per cell)	2.0
Silver oxide	1.55	Lithium-ion (full)	3.6-3.7 (4.2)
Zinc-air	1.4	9 V battery	6 × 1.5 V

Size changes capacity, not voltage. 12 V car battery = 6 lead-acid cells. Anode = oxidation = – terminal while discharging ("an ox, red cat").

USB level	Power	USB level	Power
USB 2.0: 5 V 0.5 A	2.5 W	PD: 9 V 3 A	27 W
USB 3.x: 5 V 0.9 A	4.5 W	PD: 15 V 3 A	45 W
USB-C: 5 V 3 A	15 W	PD: 20 V 3.25 A	65 W
BC 1.2: 5 V 1.5 A	7.5 W	PD: 20 V 5 A	100 W

Label "Input 100-240 V ~ 50/60 Hz" = wall side (AC). "Output" = DC levels. Power at each level = V × A.

Diodes

- Current flows when the **anode** (p side, triangle) is higher than the **cathode** (n side, bar) by about V_f.
- n-type: group 15 donors (P, As, Sb), carriers = electrons. p-type: group 13 acceptors (B, Ga, In, Al), carriers = holes. Doped material is still neutral.
- Forward bias: depletion region thinner. Reverse bias: wider; only leakage until breakdown.
- Models: ideal (short/open); **constant drop**: I = (V_source – 0.7)/R for silicon.
- **Zener**: used reverse-biased at V_Z. R = (V_in – V_Z)/I.
- Bridge rectifier: 4 diodes, loses 2 × 0.7 = 1.4 V. Half-wave: 1 diode.
- Diode test on a meter: Si 0.5-0.7 V forward, OL reverse. OL both ways = open; ≈ 0 both ways = shorted.
- Diodes and LEDs do NOT obey Ohm's law (non-linear I-V curve).

Type	V_f (forward drop)	Band gap
Silicon	0.6-0.7 V	1.1 eV
Germanium	about 0.3 V	0.67 eV
Schottky	0.2-0.4 V	(metal-semiconductor)
GaAs (infrared LED)	1.2-1.5 V	1.42 eV

LEDs

LED color	Typical V_f	LED color	Typical V_f
Infrared	1.2-1.5 V	Green (InGaN)	about 3.0 V
Red	1.8-2.1 V	Blue	2.7-3.3 V
Orange / yellow	2.0-2.2 V	White	3.0-3.5 V
Green (old type)	2.0-2.2 V	UV	3.1-4.4 V

- **Series resistor: R = (V_source – n·V_f) / I_LED** (n LEDs in series). Normal I = 10-20 mA.
- Resistor power = I²R. LED power = V_f × I.
- Long leg = anode (+). Short leg and flat rim edge = cathode (–). Reverse breakdown only about 5 V.
- Shorter wavelength → bigger band gap → higher V_f. Photon energy E (eV) ≈ 1240 / λ (nm).
- Light comes from electrons recombining with holes and releasing photons.
- Equal brightness: put identical LEDs in series (same current), or give each parallel LED its own resistor.

9 V, red LED 2.0 V, 20 mA: R = 7.0/0.020 = 350 Ω → use 390 Ω (then I = 18 mA, resistor power 0.13 W). 620 nm red: 1240/620 = 2.0 eV.

Logic gates

Gate	US symbol	Expression	Out for AB = 00 01 10 11	NAND / NOR count	Chip
NOT	triangle + bubble	A′, Ā	A=0→1, A=1→0	1 / 1	7404
AND	D shape	A·B	0 0 0 1	2 / 3	7408
OR	curved shield	A + B	0 1 1 1	3 / 2	7432
NAND	D shape + bubble	(AB)′	1 1 1 0	1 / 4	7400
NOR	shield + bubble	(A + B)′	1 0 0 0	4 / 1	7402
XOR	shield + extra back curve	A⊕B = A′B + AB′	0 1 1 0	4 / 5	7486
XNOR	XOR + bubble	(A⊕B)′ = AB + A′B′	1 0 0 1	5 / 4	74266

- A **bubble** always means NOT. IEC boxes: "&" AND, "≥1" OR, "=1" XOR, "1" NOT/buffer.
- Universal sets: {NAND}, {NOR}, {AND, NOT}, {OR, NOT}. NOT universal: {AND, OR}.
- XOR from 4 NANDs: N1 = NAND(A,B); out = NAND(NAND(A,N1), NAND(B,N1)).
- Logic levels: 1 = HIGH (5 V or 3.3 V), 0 = LOW (0 V). TTL: low < 0.8 V, high > 2.0 V.
- Switches: series = AND, parallel = OR. Diode OR and diode AND work; NOT needs more than diodes.
- 14-pin chips: pin 14 = +5 V (V_cc), pin 7 = ground; count counter-clockwise from the notch. **7402 has a different pinout** (outputs on pins 1, 4, 10, 13).

Boolean algebra

Law	OR form	AND form
Identity	A + 0 = A	A·1 = A
Null	A + 1 = 1	A·0 = 0
Idempotent	A + A = A	A·A = A
Complement	A + A′ = 1	A·A′ = 0
Double NOT	(A′)′ = A	
Commutative	A + B = B + A	AB = BA
Associative	(A+B)+C = A+ (B+C)	(AB)C = A(BC)
Distributive	A + BC = (A+B) (A+C)	A(B+C) = AB + AC
Absorption	A + AB = A	A(A+B) = A
"Disappearing opposite"	A + A′B = A + B	A(A′+B) = AB
Consensus	AB + A′C + BC = AB + A′C	
De Morgan	**(A + B)′ = A′·B′**	**(A·B)′ = A′ + B′**

- De Morgan: "break the bar, change the sign". Works for more inputs: (ABC)′ = A′ + B′ + C′.
- Bubble pushing: NAND = OR with inverted inputs. NOR = AND with inverted inputs. AND-OR → NAND-NAND; OR-AND → NOR-NOR.
- Trap: in Boolean algebra 1 + 1 = 1 (OR). In binary addition 1 + 1 = 10.

Truth tables, binary, adders, MUX

- n inputs → **2^n rows** (2 → 4, 3 → 8, 4 → 16). List rows in binary counting order.
- **Sum of products**: for each row with output 1, AND the inputs (NOT the 0s); OR the terms. Rows 01, 10 → A′B + AB′ (XOR).
- Majority of 3: Y = AB + BC + AC. "Alarm if door open AND armed, OR panic": Y = D·A + P.
- Binary place values: 1, 2, 4, 8, 16, 32, 64, 128. 1011 = 11. 1011 + 0110 = 10001 (11 + 6 = 17).
- Addition: 1+1 = 0 carry 1; 1+1+1 = 1 carry 1. A carry out of the top bit = overflow.
- **Half adder**: S = A⊕B, C = A·B. **Full adder**: S = A⊕B⊕C_in, C_out = AB + C_in(A⊕B) = AB + AC_in + BC_in.
- Full adder = 2 half adders + 1 OR. 4-bit ripple-carry adder = 4 full adders (or 3 full + 1 half).
- **MUX**: 2^n data inputs, **n select lines**, 1 output. 4-to-1 → 2 lines, 8-to-1 → 3, 16-to-1 → 4 (not √N). 2-to-1: Y = $S'D_0 + S \cdot D_1$.
- DEMUX: 1 input to one of 2^n outputs. Decoder: n inputs → 2^n outputs, one on at a time. Encoder = reverse.
- Sequential logic (flip-flops, latches, counters, clocks) is OUT for 2027.

Meters and breadboards

Measure	Connect	Power	Ideal meter R
Voltage	**Across** the part (parallel)	ON	infinite (real DMM ≈ 10 MΩ)
Current	Break the circuit, **in series**	ON	0 Ω
Resistance	Across the part, part removed or one leg lifted	**OFF**	—

- Black lead in **COM**. Red in **VΩ** for volts, ohms, diode test, continuity; red in **mA** or **10 A** for current.
- **Never** put an ammeter straight across a battery: it is a short and blows the meter fuse.
- "OL" = over range or open circuit. DC volts = V with straight line and dashes; AC = V~. Start on a high range.
- Loading: a low-resistance voltmeter across a big resistor reads **low**. Fingers on both probes add body R in parallel.
- Current without an ammeter: measure V across a known R, I = V/R. Power supplied: P = V_source × I.
- Breadboard: each row of 5 holes (a-e, f-j) is connected; the center gap separates halves; chips straddle the gap. Rails run lengthwise (red +, blue –); long boards may split rails in the middle: check with continuity.
- Troubleshoot: check power, battery, switch, fuse, loose wire → expected V at each point → measure V powered → measure R unpowered.

Switches and relays

Switch	Meaning	Terminals
SPST	single pole, single throw: on/off	2
SPDT	single pole, double throw: changeover ("3-way")	3
DPST	two SPSTs moving together	4
DPDT	two SPDTs together (reverse a motor; "4-way" when wired as a crossover)	6

- Pole = number of separate circuits. Throw = positions each pole connects to. NO = normally open, NC = normally closed.
- **3-way**: two SPDT switches joined by two travelers; commons go to supply and load. Light = A XNOR B (or XOR). Each extra location adds one 4-way (DPDT crossover) between them.
- **Relay**: a small coil current moves contacts in a separate circuit. Terminals COM, NO, NC. **Flyback diode** across the coil (cathode to +) absorbs the turn-off spike.

Safety: fuses, breakers, GFCI, shock

- **Fuse**: thin link in **series on the hot wire**; melts above its rating; replace with the **same** rating. Fast-blow vs slow-blow.
- Does it blow? I = V/R_eq or I = P/V, compare with the rating. Exactly equal to the rating: does not blow.
- **Breaker**: resettable. Thermal trip (bimetal strip, slow overloads) + magnetic trip (coil, sudden shorts).
- Fuses and breakers protect **wires from fire**, not people from shock.
- **GFCI** compares hot and neutral currents; trips on a difference (leak, maybe through a person). Class A: trips at 6 mA or more, not below 4 mA; tests use 5 mA; shock-level faults clear in about 25 ms. Needed in bathrooms, kitchens, outdoors, garages, near water. Does not trip on overload or hot-to-neutral contact. Works without a ground wire. TEST monthly.
- **AFCI** detects arcing (sparking) faults.
- Ground wire carries current only during a fault, so the breaker trips and a metal case stays near 0 V. Neutral and ground are bonded only at the main panel.

Body current (60 Hz)	Effect
about 1 mA	felt as a tingle
about 5 mA	highest "harmless" level (GFCI level)
10-20 mA	let-go limit: muscles clamp
50-100 mA	heart may fibrillate
1-4 A and more	heart stops, severe burns

Body R: about 100 kΩ dry, about 1 kΩ wet. 120 V / 1 kΩ = 120 mA (deadly); 120 V / 100 kΩ = 1.2 mA (tingle). Habits: one hand, dry hands, unplug first, keep the ground pin.

History: the seven 2027 scientists

Person	Life, country	Unit	Know for
Ampère	1775-1836, French	ampere A	1820: parallel currents attract, opposite repel (after Oersted); "electrodynamics"; "Newton of electricity"
Coulomb	1736-1806, French	coulomb C	1785: inverse-square law of electric force, measured with his torsion balance
Kirchhoff	1824-1887, German	**none**	1845 (as a student): current and voltage laws; spectroscopy with Bunsen (cesium, rubidium); "black body"
Volta	1745-1827, Italian	volt V	Voltaic pile, first battery (announced 1800); electrophorus 1775; disputed Galvani
Ohm	1789-1854, German	ohm Ω	1827 "Die galvanische Kette": V = IR; used a thermocouple source
Hertz	1857-1894, German	hertz Hz	1886-89: first made and detected radio waves (proved Maxwell); saw photoelectric effect 1887
Boole	1815-1864, **English** (taught in Cork)	**none**	Boolean algebra: "Mathematical Analysis of Logic" 1847, "Laws of Thought" 1854

Others: Franklin (named + and −, kite 1752); Galvani (frog legs, "animal electricity" 1791); Oersted (1820, current moves a compass); De Morgan (laws, 1847); Shannon (1937 thesis: Boolean algebra for relay circuits); Edison (DC) vs Tesla and Westinghouse (AC); Joule (heating I^2R); Watt (steam engine, no electrical work); siemens S = Werner von Siemens. Faraday and Tesla history is OUT for 2027.

TAB A1

History

Units named after people 19
André-Marie Ampère 31
Charles-Augustin de Coulomb 50
Gustav Kirchhoff 66
Alessandro Volta 82
Georg Simon Ohm 100
Heinrich Hertz 116
George Boole 128
Other Historical Figures 142

Each note: key facts, formulas, then question types from Easy to Nationals with worked examples.

Units named after people

Note: Units Named After People

> **INFO: How to use this note**
>
> - Scope icons: ✅ IN = all levels. ✅ States-Nationals = only at State and National tournaments. ⚠️ Borderline = the rules do not settle it (the reason is given). ❌ Out = excluded (the rule is cited).
> - Question types go from Easy to Medium to Hard to Nationals. "Nationals" means the hardest depth, not a different scope.
> - Every number in a worked example was checked in Python.
> - Source tags in square brackets, like [W] or [M], are listed under "Sources".

Scope notes and source tags

Circuit Lab, Division B, 2026-27 season (2027 rules). Built 2026-10-16. Scope rule used here: a question counts only if the 2027 **Div B** rules allow it. Div C tests and older (2026 or before) tests were mined for question shapes. Each item from them was checked against the Div B list in rule 3.c and the exclusions in rule 3.d (see the table at the end of section 3).

Source tags: [R] = 2027 Div B rules, B11-B12 (pdftotext of `/Users/kai/Daniel/Science_Olympiad_Div_B_Rules_2027_for_Web_Secured.pdf`). [W] = Wikipedia page named in section 6. [NIST] = NIST SI writing checklist. [P] = prior files in `research/prior/`. [T:name] = a downloaded test. [Py] = checked in Python (`units_check.py`, scratchpad). [G] = general knowledge, standard SI fact, not checked against a page in this run.

Rule anchors [R]:

- 3.c.i: "Historical perspective of the electricity discoveries made by Ampere, Coulomb, Kirchhoff, Volta, Ohm, Hertz, and Boole".
- 3.c.iv: "Concepts, calculations, and units of current, voltage, resistance, power, and energy".
- 3.b: "answers must be in metric units with appropriate significant figures".
- 3.d excludes capacitance, inductance, AC circuit theory, frequency analysis. Magnetism is in no listed area (the 2026 list said "electricity and magnetism" and named Tesla and Faraday; 2027 drops both) [R][P A_rules.md].

Key facts

1a. The seven 2027 scientists and their units

Scientist (rule 3.c.i)	Unit	Symbol	Quantity	In Div B scope?	Source
André-Marie Ampère (1775-1836, French)	ampere	A	electric current (SI **base** unit)	IN	[W][P]
Charles-Augustin de Coulomb (1736-1806, French)	coulomb	C	electric charge	IN	[W][P]
Alessandro Volta (1745-1827, Italian)	volt	V	electric potential difference, emf	IN	[W][P]
Georg Simon Ohm (1789-1854, German)	ohm	Ω	resistance	IN	[W][P]
Heinrich Rudolf Hertz (1857-1894, German)	hertz	Hz	frequency (1 Hz = 1 s^{-1})	IN (AC frequency is an AC characteristic; frequency analysis is OUT)	[W][P]
Gustav Robert Kirchhoff (1824-1887, German)	**none**	—	—	trap	[W list has no Kirchhoff unit] [P]

Scientist (rule 3.c.i)	Unit	Symbol	Quantity	In Div B scope?	Source
George Boole (1815-1864, English, worked in Cork, Ireland)	**none**	—	— ("Boolean" is a data type and an algebra, not a unit)	trap	[W list has no Boole unit][P]

Kirchhoff and Boole are not in the Wikipedia list of scientists whose names are SI or non-SI units (19 SI rows, 29 non-SI rows) [W]. So "the unit named after Kirchhoff" has no answer. Kirchhoff gives his name to laws (KCL, KVL, thermal radiation law, spectroscopy laws). Boole gives his name to Boolean algebra [P I_bank].

1b. Other SI units named after people (the full set)

The SI has 7 base units and 22 named derived units. **2 base units and 17 derived units are named after people: 19 in total** [W][Py count].

Unit	Symbol	Quantity	Person	Div B scope	Source
ampere	A	current (base)	André-Marie Ampère	IN	[W]
kelvin	K	temperature (base)	William Thomson, Lord Kelvin	OUT (no temperature area)	[W]
newton	N	force	Isaac Newton	IN as a unit (Coulomb's law force); person BORDERLINE	[W]
pascal	Pa	pressure	Blaise Pascal	OUT	[W]
joule	J	energy, work, heat	James Prescott Joule	IN as a unit (3.c.iv energy); person BORDERLINE	[W]
watt	W	power	James Watt	IN as a unit (3.c.iv power); person BORDERLINE	[W]
coulomb	C	charge	Coulomb	IN	[W]
volt	V	potential difference	Volta	IN	[W]
ohm	Ω	resistance	Ohm	IN	[W]
siemens	S	conductance (1/R)	Ernst Werner von Siemens	BORDERLINE (conductance not named, but it is 1/R)	[W]
farad	F	capacitance	Michael Faraday	OUT (capacitance excluded; Faraday removed 2027)	[W][R]
henry	H	inductance	Joseph Henry	OUT (inductance excluded)	[W][R]
weber	Wb	magnetic flux	Wilhelm Eduard Weber	OUT (magnetism)	[W]
tesla	T	magnetic flux density	Nikola Tesla	OUT (magnetism; Tesla removed 2027)	[W]
hertz	Hz	frequency	Hertz	IN	[W]
degree Celsius	°C	temperature	Anders Celsius	OUT	[W]
becquerel	Bq	radioactivity	Henri Becquerel	OUT	[W]
gray	Gy	absorbed dose	Louis Harold Gray	OUT	[W]
sievert	Sv	dose equivalent	Rolf Sievert	OUT	[W]

Named derived units **not** named after people: radian, steradian, lumen, lux, katal [G][Py list]. Base units not named after people: metre, kilogram, second, mole, candela [G].

1c. Non-SI units that a writer can use as distractors or trivia

Unit	Symbol	Quantity	Person / origin	Div B scope	Source
mho	℧ (upside-down Ω)	conductance (old name of siemens)	"ohm" spelled backward; coined by Lord Kelvin, 1883	BORDERLINE	[W Siemens]
electronvolt	eV	energy (1 eV = $1.602176634\times10^{-19}$ J)	built on the volt	BORDERLINE (energy unit, not named in rules)	[G][Py]

Unit	Symbol	Quantity	Person / origin	Div B scope	Source
ampere-hour, milliampere-hour	Ah, mAh	charge (1 mAh = 3.6 C)	built on the ampere	IN (USB charger hands-on, rule Part II c.iv)	[R][Py]
kilowatt-hour	kWh	energy (1 kWh = 3.6×10^6 J)	built on the watt	IN	[Py]
cycles per second	cps	frequency (old name of hertz)	—	BORDERLINE	[W Hertz]
gauss	G	magnetic flux density (1 T = 10^4 G)	Carl Friedrich Gauss	OUT (magnetism)	[W Tesla]
oersted, maxwell, gilbert	Oe, Mx, Gi	magnetic quantities	Ørsted, Maxwell, Gilbert	OUT (magnetism)	[W]
faraday	F	charge (about 96,485 C, a mole of electrons)	Faraday	OUT (Faraday removed; electrochemistry)	[W][G for value]
debye	D	electric dipole moment	Peter Debye	OUT (not in any area)	[W]
franklin (statcoulomb)	Fr	CGS charge	Benjamin Franklin	BORDERLINE (charge, but CGS; very unlikely)	[G]

1d. Adoption history (dates and bodies)

Year	Body / event	What happened to the named units	Source
1861	British Association (BA) meeting	Latimer Clark and Charles Bright proposed "ohma" (and "volt"). Sources conflict on which quantity each name meant (see section 5).	[W Ohm][W Volt]
1864-1867	BA committee	Unit called "B.A. unit, or Ohmad" (1864), then just "ohm" (1867).	[W Ohm]
1867	—	William Henry Preece proposed the symbol Ω, because "ohm" sounds like "omega".	[W Ohm]
1873	British Association	Defined the volt, ohm and farad as practical units (volt ≈ 10^8 CGS units of emf).	[W Volt]
1881	**1st International Electrical Congress, Paris**	Defined the **ohm and volt**; resolutions to define the **ampere, coulomb and farad**.	[W Congress][W Coulomb][W Ampere]
1882	BA presidential address, C. William Siemens	Proposed the names **watt** and **joule**.	[W Watt][W Joule]
1883	Lord Kelvin	Proposed **mho** for conductance.	[W Siemens]
1884	Paris conference	"Legal ohm" = mercury column about 106 cm long. No country made it law.	[W Ohm]
1889	**2nd International Electrical Congress, Paris (31 Aug 1889)**	Adopted the **joule** and **watt**. Joule died on 11 Oct 1889, so he was alive when his unit was named.	[W Joule][W Congress]
1893	Chicago Congress	International ohm (mercury column 106.3 cm, 14.4521 g, 0 °C), international ampere and volt.	[W Ohm][W Congress]
1908	London conference	Formal "international units" (international ohm, ampere, coulomb, watt).	[W Congress][W Coulomb][W Watt]
1930 (proposed) / 1935 (in use)	IEC	Name **hertz** proposed 1930, used from 1935 (Giorgi system). Weber also set by IEC in 1930. Siemens adopted by IEC 1935.	[W Hertz][W IEC via search][W Siemens]
1948	9th CGPM	"Reproducible" international units dropped; absolute (mechanical) definitions for ohm, volt, coulomb, watt, joule.	[W Ohm][W Volt][W Coulomb][W Watt][W Joule]
1960	**11th CGPM** (SI founded)	**Hertz** replaces "cycles per second"; **tesla** announced; watt adopted into SI.	[W Hertz][W Tesla][W Watt]
1971	**14th CGPM**	**Siemens** added as SI derived unit.	[W Siemens]
2019 (20 May)	SI revision	Ampere defined by fixing **e = 1.602176634×10^{-19} C exactly**. Volt and ohm follow.	[W Ampere][W Volt] [W Coulomb]

Old ampere definitions, for "how was it defined before" questions:

- International ampere: current that deposits **1.118 mg of silver per second** from silver nitrate solution (later found = 1.000165 A) [W Ampere].
- 1948 to 2019: current that gives a force of **2×10^{-7} N per metre** between two parallel wires 1 m apart in vacuum [W Ampere]. This definition uses magnetism, so only the number is safe to know; the physics is OUT.

1e. Writing rules (useful for "which is written correctly")

- Unit names are lowercase, even when named after a person: "ampere", "volt", "newton" [W list intro]. Exception: "degree Celsius" [G].
- Symbols from a person's name start with a capital: A, C, V, W, J, N, Hz, Wb, Pa [W list intro]. Ω is a Greek capital omega [W Ohm].
- The unit "ampere" has no accent; the person is "Ampère" [W Ampere][G].
- Symbols are not pluralized: "75 cm", not "75 cms" [NIST #3]. "hertz" and "siemens" are the same in singular and plural [W Hertz][W Siemens]. NIST prefers "henries" [W Henry].
- "amp" is common speech but not the SI symbol; SI deprecates abbreviations such as "sec" [W Ampere][NIST #2].
- A space goes between number and symbol: "25 kg" [NIST #15]. Do not mix names and symbols ("kilogram/m^3" is wrong) [NIST #13].
- Case changes meaning: **S** = siemens, **s** = second; **mA** = milliampere, **MA** = megaampere; **W** = watt, **Wb** = weber; **C** = coulomb (also degree sign °C) [G].

1f. Small facts tied to the named units

- 1 C = 6.241509×10^{18} elementary charges (not a whole number) [W Coulomb][Py].
- e = $1.602176634\times10^{-19}$ C, exact since 2019 [W Ampere].
- Maxwell called Ampère "the Newton of electricity" [P I_bank][T:bullso26 Q6].
- Coulomb and Ampère are among the 72 names on the Eiffel Tower [P I_bank, W].
- US grid 60 Hz, Europe 50 Hz, Japan 50/60 Hz [P I_bank A2][G].
- Volt is near the emf of a Daniell cell, which is why 10^8 CGS units was chosen in 1881 [W Volt].

Formulas

Base dimensions verified in Python [Py]:

Unit	In other SI units	In SI base units
newton N	—	$kg\cdot m\cdot s^{-2}$
joule J	$N\cdot m = W\cdot s = V\cdot C$	$kg\cdot m^2\cdot s^{-2}$
watt W	$J/s = V\cdot A = A^2\cdot\Omega = V^2/\Omega$	$kg\cdot m^2\cdot s^{-3}$
coulomb C	$A\cdot s$	$A\cdot s$
ampere A	C/s	A (base)
volt V	$J/C = W/A = A\cdot\Omega$	$kg\cdot m^2\cdot s^{-3}\cdot A^{-1}$
ohm Ω	$V/A = W/A^2 = V^2/W = J\cdot s/C^2$	$kg\cdot m^2\cdot s^{-3}\cdot A^{-2}$
siemens S	$A/V = 1/\Omega$	$kg^{-1}\cdot m^{-2}\cdot s^3\cdot A^2$
hertz Hz	1/s	s^{-1}
(OUT) farad F	$C/V = C^2/J$	$kg^{-1}\cdot m^{-2}\cdot s^4\cdot A^2$
(OUT) henry H	$\Omega\cdot s = Wb/A$	$kg\cdot m^2\cdot s^{-2}\cdot A^{-2}$
(OUT) weber Wb	$V\cdot s$	$kg\cdot m^2\cdot s^{-2}\cdot A^{-1}$
(OUT) tesla T	Wb/m^2	$kg\cdot s^{-2}\cdot A^{-1}$

Other relations [Py]:

- Electric field: N/C = V/m.
- Resistivity: $\Omega\cdot m = kg\cdot m^3\cdot s^{-3}\cdot A^{-2}$.
- Coulomb constant k = 8.99×10^9 $N\cdot m^2/C^2$ (computed 8.98755×10^9); in base units $kg\cdot m^3\cdot s^{-4}\cdot A^{-2}$.
- 1 kWh = 3,600,000 J = 3.6 MJ. 1 Wh = 3600 J.

- 1 Ah = 3600 C. 1 mAh = 3.6 C.
- 1 eV = $1.602176634 \times 10^{-19}$ J.
- f = 1/T (Hz = 1/s). 60 Hz → T = 16.7 ms.
- Q = I·t, V = W/Q, R = V/I, G = 1/R, P = W/t = V·I, E = P·t.

Question types

This note has **23 question types** (15 ✅ IN, 6 ⚠️ Borderline, 2 ❌ Out). The out-of-scope ones are in "Out of scope (skip)" below.

How the types were found

Tags: IN = all levels. IN-SN = States/Nationals only. BORDERLINE = rules do not settle it (reason given). OUT = excluded (rule cited). Note: no type in this subtopic is IN-SN. Rule 3.c.ix (KVL/KCL) is the only States/Nationals-only area, and unit questions do not need it.

Div C and older tests: Div B check (user request)

Test item	Div	What it asks	Div B 2027 verdict
Rickards 2026 C Q12	C	volt → Volta + meaning	IN (3.c.i, 3.c.iv)
Rickards Dec 2020 C Q30 (= B Q25)	C/B	Coulomb's first name	IN
Rickards Dec 2020 C Q16 (= B Q14)	C/B	conductance in base units	BORDERLINE (conductance)
Bull 2026 C Q1, Q5	C	Coulomb's law, Ohm's law namesake	IN
Bull 2026 C Q4	C	flux density → Tesla	OUT (magnetism)
Bull 2026 C Q6	C	"Newton of electricity" → Ampère	IN
Hawk & Hornet 2026 C Q4 a, b, d	C	V, I, P in C/J/s	IN
Hawk & Hornet 2026 C Q4 c, e	C	capacitance, impedance	OUT (capacitance; AC circuit theory)
Hawk & Hornet 2026 C Q3	C	LC resonant frequency in Hz	OUT (inductance, capacitance, frequency analysis, needs √ and π)
GullSO 2025 Q15, Q16, Q36	C (B/C)	current, energy units; Coulomb's law	IN
GullSO 2025 Q29, Q40	C (B/C)	farad, tesla, gauss	OUT
ATX Regional 2026 C Q34	C	RC time-constant unit	OUT (capacitance)
ATX Regional 2020 C (same as B) Q55	C/B	gauss	OUT (magnetism)

Rule followed: no OUT item above is used as a worked example for an IN type.

History questions in the test-extraction files (8 test groups)

These rows come from `research/tests/` (ATX regionals, UT invitationals, misc-a, misc-b, USC 2026, Florida State 2021). Each row names the question type it belongs to.

Test	Year	Div	Level	Q	Question (short)	Key	Check / scope 2027 B	Type
ATX (UT Austin) Regional	2021	B	Regional	2	The unit of charge bears my name	Coulomb	IN	UN1
ATX (UT Austin) Regional	2026	B	Regional	MC 2	Unit of resistance named after	Ohm	IN	UN1
Cobra Invitational (Michigan)	2019-20 season	B	Invitational	7	Fill-in: Hertz ... unit of frequency	frequency	IN	UN2

Test	Year	Div	Level	Q	Question (short)	Key	Check / scope 2027 B	Type
ATX (UT Austin) Regional	2020	B	Regional	55	Unit before the tesla	c gauss	OUT (magnetism; 3.c areas only)	UN22

Easy

UN1. Unit → namesake

Unit → namesake · ✅ IN · Easy

- **Scope: IN** (3.c.i + 3.c.iv) for ampere, coulomb, volt, ohm, hertz.
- **Where seen:** ATX Regional 2026 B Q2 ("Who has the unit of resistance named after them?", key D = Ohm) [T:atx26reg_Btest/Bkey]; ATX Regional 2021 B Q2 ("The unit of charge bears my name") [T:atx21reg_Btest]; Rickards 2026 C Q12 (volt, Div C test but the content is in Div B 3.c.i and 3.c.iv) [T:rsoi26C_test].
- **How to solve:**
  1. Read the question: unit name or quantity → person.
  2. Map unit → person with table 1a.
- **Worked example:** "The SI unit of frequency honors whom?" → **Heinrich Hertz**.
- **Traps:** Volt vs "Voltaire" (Rickards 2026 C Q12 distractor); "Kirchoff" misspelled in tests; ampere honors Ampère, but current (not magnetism) is its quantity.
- Calculator: look up — type: unit name, symbol, or quantity.

UN2. Namesake → unit (reverse)

Namesake → unit (reverse) · ✅ IN · Easy

- **Scope: IN**.
- **Where seen:** not seen as a stand-alone item; reverse form of UN1 [P I_bank H1].
  - Also seen (test-extraction files): Cobra Invitational (Michigan) 2019-20 season, Div B, Invitational, Q7: "Fill-in: Hertz ... unit of frequency" (key: frequency). Div B 2027: IN.
- **How to solve:**
  1. Read the question: person → unit name, symbol, quantity.
  2. Table 1a.
- **Worked example:** "Which quantity is measured in units named after Coulomb?" → **electric charge (coulomb, C)**.
- **Traps:** Coulomb's *law* is about force, but the coulomb *unit* is charge. Hertz → frequency, not "radio".
- Calculator: look up — type: person name.

UN3. Quantity → SI unit (name and symbol)

Quantity → SI unit (name and symbol) · ✅ IN · Easy

- **Scope: IN** (3.c.iv).
- **Where seen:** GullSO 2025 Q15 (SI unit for current → Ampere) and Q16 (energy → Joule) [T:gull26_test; MC key on page 1 of gull26_key.pdf]; Valparaiso 2013 Q5 (ratio V/I → Ohms) [T:valparaiso 2013].
- **How to solve:**
  1. Recall table in section 2.
- **Worked example:** "SI unit of power?" → **watt (W)**.
- **Traps:** watt (power) vs joule (energy); "kWh" is energy, not power; second appears as a distractor.
- Calculator: look up — type: quantity name.

UN11. Prefixes on named units

Prefixes on named units · ✅ IN · Easy

- **Scope: IN**.
- **Where seen:** national practice deck 2 (12 V across 3 kΩ → 4 mA) [P F_slides UNIT-METRIC-PREFIX].
- **Worked example:** 4.7 kΩ × 2.2 mA = 10.34 → **10. V** (2 s.f.) [Py 10.34].
- **Traps:** k (lowercase) vs K (kelvin); M vs m; μ vs m.

- Calculator: calculate — type: value, prefix, unit.

Medium

UN4. Symbol reading and case traps

Symbol reading and case traps · ✅ IN · Easy to Medium

- **Scope: IN** (3.b metric units, 3.c.iv).
- **Where seen:** not seen as a direct question. Rickards 2026 B Q22 wrongly writes "Weber (W)" [T:rsoi26B_test] (see section 4).
- **How to solve:**
  1. Read the question: a symbol string → unit and quantity, or "which is written correctly".
  2. Check capital vs lowercase and prefix (section 1e).
- **Worked example:** "What is 3 MΩ in mΩ?" 3×10^{6} Ω = 3×10^{9} mΩ → **3×10^{9} mΩ** [Py: 3e6/1e-3 = 3e9].
- **Traps:** S vs s; mA vs MA; W vs Wb; Ω vs ℧.
- Calculator: look up + calculate — type: value, from-prefix, to-prefix. (App note: (symbol table) plus CALCULATE for the prefix step.)

UN5. "No unit" trap: Kirchhoff and Boole

"No unit" trap: Kirchhoff and Boole · ✅ IN · Medium

- **Scope: IN** (3.c.i).
- **Where seen:** not seen as asked; Kirchhoff is a distractor in ATX 2026 B Q2 and GullSO 2025 Q29 [T].
- **How to solve:**
  1. Read the question: list of the seven → which have no SI unit.
  2. Table 1a; only Ampère, Coulomb, Volta, Ohm, Hertz have units.
- **Worked example:** "Of the seven scientists in the rules, which two have no SI unit named after them?" → **Kirchhoff and Boole**.
- **Traps:** "Boolean" sounds like a unit; "Kirchhoff's constant" does not exist as a unit. Answer "none of the above" may be the key.
- Calculator: look up — type: none (fixed list) or a person name.

UN6. Full name / spelling of the namesake

Full name / spelling of the namesake · ✅ IN · Medium

- **Scope: IN** (3.c.i).
- **Where seen:** Rickards Invitational B (Dec 2020) Q25 ("full first name of the person who the SI unit of charge is named after" → key **Charles-Augustin**) [T:rsoiOldB_test/key]; Rickards 2026 C Q12 (full name Alessandro Giuseppe Antonio Anastasio Volta) [T:rsoi26C_test].
- **How to solve:**
  1. Memorize full names: André-Marie Ampère.
  2. Charles-Augustin de Coulomb.
  3. Alessandro Giuseppe Antonio Anastasio Volta.
  4. Georg Simon Ohm.
  5. Gustav Robert Kirchhoff.
  6. Heinrich Rudolf Hertz.
  7. George Boole.
- **Worked example:** "First name of the person the SI unit of current honors?" → **André-Marie**.
- **Traps:** hyphen in "Charles-Augustin" and "André-Marie"; "Georg" not "George" (Ohm) but "George" (Boole).
- Calculator: look up — type: unit or surname.

UN7. Express a unit using other named units

Express a unit using other named units · ✅ IN · Medium

- **Scope: IN** (3.c.iv).

- **Where seen:** Hawk & Hornet 2026 C Q4 ("write the units using only C, J, s": voltage J/C, current C/s, power J/s are IN; capacitance C²/J and impedance J·s/C² are OUT) [T:hh26_test/key]; Pembroke(?) "phi26" B Q33 ("defines the volt": key D J/C) [T:phi26_test/key].
- **How to solve:**
  1. Start from a defining equation: V = W/Q, I = Q/t, P = W/t, R = V/I; substitute.
- **Worked example:** "Write the ohm using only C, J, s." R = V/I = (J/C)/(C/s) = **J·s/C²** [Py: Ohm == J·s/C² True].
- **Traps:** more than one form can be right (V = J/C = W/A).
- Calculator: calculate — type: target unit, allowed units. (App note: (unit algebra).)

UN8. Which combination equals X (and odd one out)

Which combination equals X (and odd one out) · ✅ IN · Medium

- **Scope: IN**.
- **Where seen:** Valparaiso 2013 Q5 [T]; prior bank U2 [P I_bank].
- **How to solve:**
  1. Reduce each choice to base units (section 2) and compare.
- **Worked example:** "Which is NOT a watt? (a) V·A (b) A²·Ω (c) V²/Ω (d) V·C" → **(d)**, V·C is a joule [Py: V·C == J True].
- **Traps:** V·C (energy) vs V·A (power); A·s (charge) vs A/s.
- Calculator: calculate — type: each choice as a unit product.

UN12. Conversions through named units (kWh, Wh, mAh, eV)

Conversions through named units (kWh, Wh, mAh, eV) · ✅ IN · Medium

- **Scope: IN** (3.c.iv, hands-on c.iv USB charger).
- **Where seen:** prior bank U2 (1 kWh = 3.6×10^6 J) [P I_bank]; USB hands-on task [R].
- **Worked example:** a 3000 mAh battery at 5 V: charge 3.0 Ah × 3600 = **10,800 C**; energy 3.0 Ah × 5 V = **15 Wh = 54,000 J**; time at 2 A = **1.5 h** [Py].
- **Traps:** mAh is charge, not energy; multiply by V to get Wh.
- Calculator: calculate — type: capacity (mAh), voltage (V), current (A).

UN13. Electrons per coulomb

Electrons per coulomb · ✅ IN · Medium

- **Scope: IN** (3.c.ii, 3.c.iv).
- **Where seen:** Scioly wiki fact (6.24×10^{18}) [T:cl_wiki]; prior bank [P F_slides STAT-ELECTRON-CHARGE].
- **Worked example:** "How many electrons make 3.0 C?" 3.0 / 1.602×10^{-19} = **1.87×10^{19}** [Py 1.8725e19].
- Calculator: calculate — type: charge in C.

UN14. Units of a constant

Units of a constant · ✅ IN · Medium

- **Scope: IN** (Coulomb's law, 3.c.ii).
- **Where seen:** not seen as asked.
- **Worked example:** "Units of k in $F = kq_1q_2/r^2$?" $k = F\cdot r^2/(q_1q_2)$ → **N·m²/C²** [Py].
- Calculator: look up + calculate — type: constant name. (App note: (common constants) or CALCULATE (rearrange).)

UN15. Hertz in AC facts

Hertz in AC facts · ✅ IN · Easy to Medium

- **Scope: IN** (3.c.iii AC characteristics; f = 1/T is algebra). Frequency analysis (filters, resonance) is **OUT** (3.d).
- **Where seen:** ATX Regional 2026 C Q34 lists Hertz as a distractor for "unit of RC time constant" (question itself OUT: capacitance) [T:atx26reg_Ctest].
- **Worked example:** "Period of 60 Hz mains?" T = 1/60 = **16.7 ms** [Py].
- **Traps:** Hz is cycles per second, not "per minute"; ω (rad/s) is not Hz.
- Calculator: calculate — type: f or T.

UN16. People behind other in-scope units (Watt, Joule, Newton, Siemens)

People behind other in-scope units (Watt, Joule, Newton, Siemens) · ⚠️ Borderline · Medium

- **Scope: BORDERLINE**. The units are IN (3.c.iv), but the people are not in the 3.c.i list. A writer may still ask "the unit of power honors whom?" as a units question.
- **Where seen:** not seen as asked; "Joule's Law" appears in ATX Regional 2020 B [T:atx20reg_Btest].
- **Worked example:** "The unit of energy is named after which brewer-physicist?" → **James Prescott Joule**.
- **Traps:** watt is James Watt (steam engine), not a person named "Watts"; siemens is Werner von Siemens, not Charles William Siemens (who proposed watt/joule).
- Calculator: look up — type: unit.

UN17. Old and non-SI names (mho, cps, eV, Ah)

Old and non-SI names (mho, cps, eV, Ah) · ⚠️ Borderline · Medium

- **Scope: BORDERLINE** for mho/cps/eV (not named; tie to resistance and AC); **IN** for Ah/mAh (USB task).
- **Where seen:** not seen in a test; mho in prior bank [P I_bank].
- **Worked example:** "A 250 Ω resistor has what conductance in mhos?" 1/250 = **0.004 ℧ = 4 mS** [Py].
- **Traps:** mho = siemens, same value; ℧ is an upside-down Ω.
- Calculator: look up + calculate — type: R. (App note: (1/R) plus LOOKUP (old name).)

UN20. Writing rules (name vs symbol, plurals)

Writing rules (name vs symbol, plurals) · ⚠️ Borderline · Medium

- **Scope: BORDERLINE** (3.b asks for metric units; style rules are not named).
- **Where seen:** not seen.
- **Worked example:** "Which is correct SI style? (a) 5 Amps (b) 5 A (c) 5 a (d) 5 As" → **(b)**.
- Calculator: look up — type: the keyword from the question.

Hard

UN9. SI base-unit expression

SI base-unit expression · ✅ IN · Hard

- **Scope: IN** for V, Ω, W, J, C; **BORDERLINE** for S (conductance not named, but = 1/Ω).
- **Where seen:** Rickards Invitational B (Dec 2020) Q14 ("In SI base units, what is the unit of conductance?" key **B: $s^3 \cdot A^2 \cdot m^{-2} \cdot kg^{-1}$**, read from the key PDF image) [T:rsoiOldB_key p.3].
- **How to solve:**
  1. N = kg·m·s^{-2} → J = N·m → W = J/s → V = W/A → Ω = V/A → S = 1/Ω.
- **Worked example:** "Ohm in base units?" → **$kg \cdot m^2 \cdot s^{-3} \cdot A^{-2}$** [Py].
- **Traps:** sign flips for S; distractor s^4 (that is the farad).
- Calculator: calculate — type: unit name. (App note: (exponent vector).)

UN10. Dimensional check of a formula

Dimensional check of a formula · ✅ IN · Hard

- **Scope: IN** (3.c.iv; algebra only, rule 3.e.i).
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Replace each symbol by its unit; simplify.
- **Worked example:** "Show P = V^2/R has units of watts." $V^2/\Omega = V^2/(V/A) = V \cdot A$ = **W** [Py True].
- **Traps:** forgetting R = V/A, not A/V.
- Calculator: calculate — type: formula as product of units.

Nationals

UN18. Adoption timeline (who adopted which unit, when)

Adoption timeline (who adopted which unit, when) · ⚠️ Borderline · Nationals

- **Scope: BORDERLINE**. 3.c.i covers "discoveries made by" the seven; naming of units came later and is not a discovery. Hard writers may still ask.
- **Where seen:** not seen in a test.
- **Worked example:** "Order the adoption: hertz in SI, ohm/volt at Paris, siemens in SI." → **1881 ohm/volt, 1960 hertz, 1971 siemens** [W].
- **Traps:** hertz IEC date 1930 vs 1935 (section 5); joule/watt 1889, not 1881.
- Calculator: look up — type: unit.

UN19. How a unit was defined (historical definitions)

How a unit was defined (historical definitions) · ⚠ Borderline · Nationals

- **Scope: BORDERLINE** (units of current/resistance are IN, but historic definitions are not named). The 1948 wire-force ampere uses magnetism (OUT); know only the number.
- **Where seen:** not seen in a test.
- **Other answers:** international ampere = 1.118 mg Ag/s; international ohm = 106.3 cm mercury column (1893) [W].
- **Worked example:** "Since 2019 the ampere is defined by fixing which constant?" → **the elementary charge, e = 1.602176634×10^{-19} C**.
- Calculator: look up — type: unit, year.

UN21. Counting and trivia about named units

Counting and trivia about named units · ⚠ Borderline · Nationals

- **Scope: BORDERLINE** (outside the seven; general units).
- **Where seen:** not seen in a test.
- **Worked example:** "How many SI units are named after people?" → **19** (2 base + 17 derived) [W][Py]. "Whose unit was named while he was still alive?" → **Joule** (unit 31 Aug 1889, died 11 Oct 1889) [W Joule].
- **Traps:** counting °C; counting non-SI units.
- Calculator: look up — type: the keyword from the question.

Out of scope (skip)

These types appear in tests or books, but the 2027 Div B rules exclude them. Skip them. They stay here so you can recognise them.

UN22. Out-of-scope named units seen on Div C or 2026 tests (farad, henry, tesla, weber, gauss, kelvin)

Out-of-scope named units seen on Div C or 2026 tests (farad, henry, tesla, weber, gauss, kelvin) · ❌ Out · Easy (as asked)

- **Scope: OUT**. Capacitance and inductance are excluded (3.d); magnetism is in no 2027 area; Tesla and Faraday dropped from 3.c.i.
- **Where seen:** GullSO 2025 Q29 (capacitance → Faraday), Q40a-b (tesla, 10^4 gauss) [T:gull26]; BullSO 2026 B Q4 and Bull 2026 C Q4 (magnetic flux density → Tesla) [T:bullso26, bull26C]; Rickards 2026 B Q22 (tesla) [T]; ATX Regional 2020 B Q55 (gauss before tesla) [T]; Hawk & Hornet 2026 C Q4c (capacitance C²/J) [T]; ATX 2026 C Q34 (RC time constant) [T].
  - Also seen (test-extraction files): ATX (UT Austin) Regional 2020, Div B, Regional, Q55: "Unit before the tesla" (key: c gauss). Div B 2027: OUT (magnetism; 3.c areas only).
- **Study advice:** know only the name-symbol pairs so they work as distractors. Do not study the physics.
- Calculator: look up — type: (flag as "not on 2027 Div B").

UN23. Faraday as a unit (the faraday, Faraday constant)

Faraday as a unit (the faraday, Faraday constant) · ❌ Out · Medium

- **Scope: OUT** (Faraday removed from 3.c.i in 2027; electrochemistry and capacitance not listed).
- **Where seen:** not seen in a test.
- Calculator: look up — type: (flag OUT).

Mistakes in tests and keys

1. **"phi26" B test Q33** [T:dlB/tests/phi26_test.txt, key phi26_key.txt line 67]: "Which correctly defines the volt?" Choices include **B) Watt per Ampere** and **D) Joule per Coulomb**. Both are correct (V = W/A = J/C) [Py]. Key gives only D. Fair grading should accept B.
2. **Rickards 2026 B Q22** [T:rsoi26B_test]: choice "a. Weber (W)". The weber symbol is **Wb**; W is the watt [W]. (Item is OUT anyway.)
3. **Prior file I_bank.md, Coulomb entry**: "Wikipedia says named in 1880." The current Wikipedia Coulomb page gives **1881** (International Electrical Congress) and does not mention 1880 [W Coulomb]. Use 1881.
4. **Prior file I_bank.md, Ohm entry**: "The 1881-era legal ohm was a mercury column 106 cm". The legal ohm came from the **1884** Paris conference [W Ohm]. 106.3 cm is the 1893 international ohm. Fix the date.
5. **Prior file I_bank.md, Hertz entry**: "IEC adopted it 1930." Current Wikipedia says the IEC set the name "in 1935". German Wikipedia (citing Ruppert, *IEC History 1906-1956*) says proposed 1930, in use 1935. Write "IEC 1930 (proposed), 1935 (in use)". See section 5.
6. **ATX Regional 2020 B/C Q55** [T]: "Prior to the use of the Tesla ... which unit was used? → Gauss". Gauss is a CGS unit used alongside the tesla, not strictly before it; the key is the intended answer. Item is OUT for 2027.
7. Several tests spell "Kirchoff" (ATX 2026 B, Bull 2026 C). Correct spelling is **Kirchhoff**.
8. Scioly wiki derived-units table labels the newton "Electrostatic Force"; the newton is the unit of any force [T:cl_wiki line 189]. Minor.

Not confirmed

- **1861 Clark and Bright names.** Wikipedia "Ohm" says they proposed "ohma" for a unit of resistance. Wikipedia "Volt" main text says they named the volt; its footnote says "volt" was proposed for resistance. Not resolved. Looked: Wikipedia Ohm, Volt. A primary source (Clark and Bright, 1861, *The Electrician*) was not read.
- **Hertz IEC date (1930 vs 1935).** English Wikipedia: 1935. Older mirrors and German Wikipedia: proposed 1930. No iec.ch primary page found. Looked: Wikipedia Hertz, WebSearch (iec.ch history, encyclopedia mirrors, dewiki).
- **Exact CGPM adoption of named derived units in 1948** (which names were listed). Not read in the BIPM SI Brochure / CGPM resolutions this run. Looked: Wikipedia Volt, Ohm, Coulomb, Watt, Joule (all say 1948 ended the "international" units).
- **Henry adopted at 1893 Chicago congress.** Wikipedia Henry page does not say. Not confirmed.
- **Faraday unit value (96,485 C) and franklin/statcoulomb** are [G], not checked this run (both OUT or BORDERLINE).
- **gull26 MC key letters** are on page 1 of the key PDF as an image; answers above (Q15 A, Q16 B, Q29 C) are the physically correct choices, not read from the key image.
- **phi26 test name**: file is in `dlB/tests/`; I did not confirm the tournament full name ("Pembroke" per task text).
- Kuphaldt *Lessons in Electric Circuits* and OpenStax were not opened for this subtopic; the facts here come from Wikipedia, NIST, the tests and prior files.

Sources

- 2027 Div B rules (local PDF), Circuit Lab B11-B12.
- https://en.wikipedia.org/wiki/List_of_scientists_whose_names_are_used_as_SI_units
- https://en.wikipedia.org/wiki/International_Electrical_Congress
- https://en.wikipedia.org/wiki/Siemens_(unit)
- https://en.wikipedia.org/wiki/Volt
- https://en.wikipedia.org/wiki/Ohm
- https://en.wikipedia.org/wiki/Coulomb
- https://en.wikipedia.org/wiki/Ampere

- https://en.wikipedia.org/wiki/Hertz
- https://en.wikipedia.org/wiki/Watt
- https://en.wikipedia.org/wiki/Joule
- https://en.wikipedia.org/wiki/Tesla_(unit)
- https://en.wikipedia.org/wiki/Henry_(unit)
- https://en.wikipedia.org/wiki/International_Electrotechnical_Commission
- https://dewiki.de/Lexikon/GHz (German Wikipedia mirror, Hertz unit history)
- https://physics.nist.gov/cuu/Units/checklist.html
- https://scioly.org/wiki/index.php/Circuit_Lab (local copy `downloads/dl/cl_wiki.txt`)
- Tests (local, `research/downloads/`): dlG/atx26reg_Btest+Bkey, atx26reg_Ctest, atx21reg_Btest, atx20reg_Btest/Ctest, rsoiOldB_test+key (PDF p.3 image), rsoiOldC_test, rsoi26B_test, rsoi26C_test+key, bull26C_test, gull26_test+key, hh26_test+key, circuit_lab_valparaiso_university_2013_invitational; dlB/tests/bullso26_test, phi26_test+key.
- Prior files: research/prior/I_bank.md, F_slides.md, A_rules.md, G_manifest.md.
- Test-extraction files: `/Users/kai/Daniel/circuitlab/research/tests/` (atx-reg, misc-a, misc-b, nat26, state, ut-a, ut-b, ut-c) and `/Users/kai/Daniel/circuitlab/research/prior/G_tests.md`.
- Raw research file this note was built from: `/Users/kai/Daniel/circuitlab/research/subtopics/units-named.md`.

André-Marie Ampère

Note: Ampere

> **INFO: How to use this note**
> - Scope icons: ✅ IN = all levels. ✅ States-Nationals = only at State and National tournaments. ⚠️ Borderline = the rules do not settle it (the reason is given). ❌ Out = excluded (the rule is cited).
> - Question types go from Easy to Medium to Hard to Nationals. "Nationals" means the hardest depth, not a different scope.
> - Every number in a worked example was checked in Python.
> - Source tags in square brackets, like [W] or [M], are listed under "Sources".

Scope notes and source tags

Scope basis: 2027 Div B rules, manual p. B12, 3.c.i: "Historical perspective of the **electricity** discoveries made by Ampere, Coulomb, Kirchhoff, Volta, Ohm, Hertz, and Boole." The test has "at least one question from each of the following areas only", so at least one history question is certain.

Important scope point. The 2026 wording was "electricity **and magnetism** discoveries". The 2027 wording is "electricity discoveries", and no 2027 area covers magnetism. Rule 3.d does not list magnetism as excluded by name. It is removed because 3.c says "areas only" and no area names it. Ampère's main work (electrodynamics) is about magnetic forces of currents. So this file uses this split:

- **IN**: history facts about Ampère (who, when, where, what he found, what he named, his unit). A test can ask "Ampère showed parallel currents attract" as a history fact.
- **IN**: the unit ampere, through area iv ("units of current"), including Q = I·t and electrons per second.
- **OUT**: any physics or calculation of magnetic fields or magnetic forces (Ampère's force law calculations, Ampère's circuital law, solenoid field, right-hand/swimmer rule).
- **BORDERLINE**: recall of law names or the 1948 force-based unit definition, with no calculation.

Div C note (user request). The Div C 2027 history list is word-for-word the same as Div B (Div C rules text line 878 of `downloads/dl/rulesC2027.txt`). But every Div C test in `downloads/` is from 2026 or earlier, when magnetism, Faraday and Tesla were in scope, and Div C allows trig and full systems of equations. So **every Div C question cited below was checked one by one against the 2027 Div B areas and math limits**. The check result is in each "where seen" line as "Div B 2027 check: ...". Section 3b lists every Ampère-related Div C (and old Div B) item and its result.

Source tags: [W] Wikipedia (Ampère article), [M] MacTutor St Andrews biography, [ML] National MagLab Magnet Academy, [IEEE] ETHW IEEE Milestone "Birth of Electrodynamics 1820-1827", [ITER] ITER Newsline "Lonesome death of Ampère", [PH] Princeton Joseph Henry Project (astatic needles), [CIGRE] CIGRE article, [BIPM]/[NIST] unit history, [SOW] scioly.org wiki (`downloads/dl/cl_wiki.txt`), [P] prior research files (`research/prior/`), [T:...] a test in `research/downloads/`, [U] unverified (see section 5).

Key facts

1.1 Life, places, family

#	Fact	Source
1	Born **20 January 1775, Lyon, France**. French.	[W][M][SOW]. (MagLab says 22 January: error, see section 4.)
2	Grew up at the family country house at **Poleymieux** (Poleymieux-au-Mont-d'Or), about 10 km from Lyon; main home from 1782. The house is now a museum of the history of electricity (Musée Ampère).	[M][IEEE]
3	Mostly **self-taught** in his father's library. Read the *Encyclopédie* and Buffon. Learned Latin to read Euler and Daniel Bernoulli. Taught himself advanced math from about age 12.	[W][M][ML]

#	Fact	Source
4	About age 13: first paper to the Académie de Lyon (constructing a line equal to a circular arc). Not published.	[M]
5	Father **Jean-Jacques Ampère**, merchant, then justice of the peace (juge de paix). **Guillotined in Lyon, 24 November 1793** (Jacobin purges after the siege of Lyon). Some sources say 23 November. Ampère was 18 (Python: 18.8 years).	[W][M]; date conflict noted in search results [U]
6	After the execution he fell into deep depression for about a year (MacTutor: stopped studying Lagrange's *Mécanique analytique* for 18 months). Interest in **botany**, stirred by **Rousseau's letters on botany**, helped bring him back. Austin 2026 C Q12 asks "Who studied botany for a year because his father died?" → Ampère.	[M]; botany link from search snippets of encyclopedia.com and the DSB biography [U: not read in full]
7	Married **Julie Carron** 1799 (met 1796, engaged 1797). Son **Jean-Jacques Ampère** born 1800, later a historian/philologist (Sorbonne chair 1830). Julie died **July 1803**.	[W][M][ML]
8	Second marriage to Jeanne-Françoise ("Jenny") Potot, **1 August 1806**. Daughter Albine born 6 July 1807. Legal separation 1808.	[M] (surname Potot [U])
9	Catholic. Took in the young Frédéric Ozanam (founder of the Society of St Vincent de Paul).	[W]
10	Died **10 June 1836, Marseille**, age 61, of a lung (pulmonary) infection, during an **inspection tour as Inspector General** of the university/education system.	[W][M]; ITER says 9 June (section 4)
11	First buried in Saint-Charles cemetery, Marseille. **Reburied in Montmartre Cemetery, Paris, 1896**, next to his son.	[W][ITER]
12	Tomb text (per ITER): "He loved the human kind; he was simple, good and grand."	[ITER]. "Tandem felix" epitaph: [U]
13	Absent-minded and poor in his last years.	[W]. Specific anecdotes: [U] (section 5)

1.2 Career and honors

#	Fact	Source
14	1799 first job: mathematics teacher (Lyon).	[W]
15	**1802**: professor of physics and chemistry, École Centrale, **Bourg-en-Bresse**.	[W][M]
16	1802/1803: *Considérations sur la théorie mathématique du jeu* (math theory of games, gamblers' probability). Laplace found an error; Ampère fixed it. It earned him notice (Delambre).	[W][M][ML]
17	1803 (MacTutor) or 1804 (MagLab): math post at the Lycée in Lyon, arranged by Delambre.	[M][ML] (date conflict)
18	**1804**: répétiteur (tutor) in analysis, **École Polytechnique**, Paris. **1809**: professor of mathematics there, until 1828.	[W][M]
19	**1808**: named **Inspector General** of the Imperial University (education inspector). Travel for this job wore down his health.	[ITER]
20	**November 1814**: elected to the Institut (Académie des Sciences), mathematics section, beating **Cauchy 28 to 26**. Same year: Chevalier of the Legion of Honour.	[M][W]
21	1819 philosophy, 1820 astronomy at the University of Paris.	[W]
22	**Chair of experimental physics, Collège de France**: 1824 per Wikipedia and IEEE; MacTutor says 1826. Liouville attended his course 1826-27.	[W][IEEE][M] (date conflict)
23	Foreign Member of the **Royal Society 1827**; Royal Swedish Academy 1828; Royal Society of Edinburgh.	[W][M][ML]
24	One of the **72 names on the Eiffel Tower**. Lyon metro station, lunar mountain (Mons Ampère), asteroid 10183 Ampère, NVIDIA "Ampere" GPU.	[W]
25	IEEE Milestone "The Birth of Electrodynamics, 1820-1827", dedicated **3 April 2023**; plaques at the Collège de France (3 rue d'Ulm, Paris) and at Poleymieux.	[IEEE]

1.3 The 1820 discovery chain

#	Fact	Source
26	**Hans Christian Ørsted** (Danish) published, **21 July 1820**, that a current deflects a compass needle.	[W Ørsted, general knowledge]; date [U: not fetched this run]
27	**François Arago** reported Ørsted's result to the Paris Académie on **4 September 1820** and repeated the experiment there **one week later (11 September)**. Sources that give only "11 September" mean the demonstration.	[M]; [CIGRE] says "a week earlier" than 18 Sept
28	**18 September 1820**: Ampère presented his first electrodynamics memoir. Further sessions **25 September** and **2 October 1820**. Python: 7 days after the 11 Sept demonstration, 14 days after the 4 Sept report.	[IEEE][CIGRE][PH]; booksellers' notes in search
29	Before the end of September 1820 he showed that **two parallel wires with currents in the same direction attract, and with opposite currents repel**. This was a new effect: current acting on current, with no magnet.	[W][M][ML]
30	He explained magnets (and the Earth's magnetism) as tiny circulating currents. "**Electrodynamic molecule**": a forerunner of the electron idea. Tested by Einstein and de Haas, 1915.	[W][IEEE]
31	6 November 1820: law of addition of electrodynamic forces. December 1820: symmetry principle. By end of 1820 he had a force formula between current elements.	[M][IEEE]
32	**Ampère's force law**: the mutual action of two lengths of current-carrying wire is proportional to their lengths and to the intensities of their currents. (Princeton 2021 Q4 quotes this sentence.)	[W]; [T:Princeton 2021 Q4]
33	Félix Savary helped with the calculations from 1823.	[M]
34	Biot and Savart reported their own law in October 1820 (rival work).	[M]
35	He thought he found induction in 1822 but later gave full credit to **Faraday** (1831).	[M]

1.4 Instruments, words, ideas

#	Fact	Source
36	**Astatic needle** (astatic pair): two equal magnetized needles, mounted parallel with opposite poles, so the Earth's field has no net effect. Lets the field of a current be studied. Forerunner of the sensitive (astatic) **galvanometer**. Presented in the 18 and 25 Sept 1820 memoirs. Austin 2026 C Q1 answer.	[PH][IEEE]; [T:ATX26 C Q1]
37	He **proposed the name "galvanomètre"** (galvanometer, honoring Galvani) for a current-measuring needle, to separate it from the electrometer, in the 1820 memoirs (Annales de Chimie et de Physique vol. 15). The first working multiplier galvanometer was Schweigger's (1820).	Princeton Henry Project excerpt via search [PH]; prior [P: I_bank "[G]"]. Exact memoir date [U]
38	He was one of the first to **measure** current, not only detect it, with a device of his own.	[ML]
39	**First to clearly separate the ideas of current ("courant") and voltage ("tension")**. He defined current as a flow in a closed circuit.	[IEEE][CIGRE][SOW]
40	He fixed a **direction convention for current** (the flow of positive electricity, from the + pole round the outside circuit).	[U: widely repeated; no primary source read]
41	Coined **"electrodynamics"** (électrodynamique).	[W][ML]; [T:UT20 inv B Q1, MIT20 Q1, ATX20 Q10]
42	Coined **"solenoid"** (from Greek "pipe-shaped").	[W][SOW]; [T:UT20 inv B Q4, Gull 2026 Q28]
43	Coined **"cinématique"** → English "kinematics" (*Essai sur la philosophie des sciences*, 1834).	[W]
44	Gave the "**bonhomme d'Ampère**" rule (Ampère's swimmer / little-man rule) for which way a needle turns near a current.	[IEEE]; physics is OUT
45	Built solenoids and electromagnets, and found the temporary magnetization of soft iron (with Arago, 1820). The practical horseshoe electromagnet is **Sturgeon's (1824)**.	[IEEE]; Sturgeon date [U: general knowledge]

#	Fact	Source
46	**Electromagnetic telegraph** idea, **2 October 1820**, following **Laplace's** suggestion: one pair of wires per letter with a needle at the far end. Never built in a practical form (no coil, weak signal).	Needle telegraph article via search; [IEEE] says "first to propose"

1.5 Publications

#	Fact	Source
47	*Considérations sur la théorie mathématique du jeu*, 1802 (probability).	[W]
48	Memoirs of 18, 25 Sept and 2 Oct 1820, printed in Arago's *Annales de Chimie et de Physique* vol. 15. Offprint Nov 1820: *Mémoires sur l'action mutuelle de deux courants électriques*.	bookseller listings via search [U: secondary]
49	*Recueil d'observations électro-dynamiques*, 1822. *Description d'un appareil électro-dynamique*, 1824.	[W]
50	Magnum opus: ***Théorie (Mémoire sur la théorie) mathématique des phénomènes électro-dynamiques, uniquement déduite de l'expérience***, **1826** (Wikipedia, MacTutor); often cited **1827** (MagLab; the Académie volume printed in 1827). Accept either.	[W][M][ML]
51	*Essai sur la philosophie des sciences*, 1834 (vol. 2, 1843): classification of human knowledge.	[W]

1.6 Other science (distractor material)

#	Fact	Source
52	**Fluorine**: in 1810-1812 (MacTutor: 1811) he suggested that hydrofluoric acid holds an unknown element and proposed the **name "fluorine"** (le fluor), in letters to Humphry Davy. He did **not** isolate it (Moissan, 1886).	[M][ML]; Davy/Moissan [U: general knowledge]
53	1816: a classification of the chemical elements.	[M]
54	Avogadro-type gas law (1814), found independently.	[P: I_bank "[G]"], [U]
55	Optics: wave theory of light supporter (1815-16). Fresnel lodged with him 1822-1827.	[M][ML]
56	Math: classification of partial differential equations (1814); **Monge-Ampère equation**.	[M][W]

1.7 Recognition and the unit

#	Fact	Source
57	**Maxwell**, *A Treatise on Electricity and Magnetism*, 1873, vol. 2, **article 528**: the whole theory and experiment "seems as if it had leaped, full grown and full armed, from the brain of the '**Newton of electricity**'."	Wikiquote and arXiv via search; [T:BullSO 2026 B Q6]
58	Maxwell (1879 edition comments) also wrote that we can scarcely believe Ampère found the law by the experiments he describes (he hid his path to the result).	[M]
59	Unit **ampere (A)** named at the **1881 International Electrical Congress, Paris** (with the volt and ohm). It was set at 1/10 of the CGS electromagnetic unit (1 abampere = 10 A).	[W Ampere][BIPM]
60	1893 Chicago congress: "international" units. 1908 London conference: **international ampere** defined by silver deposited from silver nitrate (0.001118 g/s [U]).	[BIPM][NIST]
61	**1948** (9th CGPM): ampere defined by the **force between two parallel wires 1 m apart in vacuum: 2×10^{-7} N per metre** of length. Python check: $\mu_0/(2\pi) \times 1\ \text{A} \times 1\ \text{A} / 1\ \text{m} = 2.0\times10^{-7}$ N/m.	[BIPM][W Ampere]
62	Ampere is one of the **7 SI base units** (the only SI base unit named after any of the seven history people). SI formally adopted 1960; NIST pages differ on when the ampere joined the base set.	[NIST][P]
63	**2019** redefinition: the elementary charge is fixed exactly, $e = 1.602176634\times10^{-19}$ C, so 1 A = 1 C/s = $1/(1.602176634\times10^{-19}) \approx$ **6.241509×10^{18} elementary charges per second** (Python).	[BIPM][NIST]
64	Unit name is lower-case "ampere" (no accent in SI English), symbol capital **A**, short form "amp".	SI usage [U: SI Brochure not fetched]

Formulas

Item	Value / formula	Units	Scope
Current	$I = Q / t$	A = C/s	IN (area iv)
Charge from current	$Q = I \cdot t$	C = A·s	IN
Battery capacity	1 mAh = 3.6 C; 1 Ah = 3600 C	C	IN
Number of electrons	$N = Q / e$	(count)	IN
Elementary charge	$e = 1.602176634 \times 10^{-19}$ C (exact since 2019)	C	IN
Electrons per second in 1 A	$1/e = 6.241509 \times 10^{18}$ s^{-1}	s^{-1}	IN
Equivalents of 1 A	C/s = V/Ω = W/V = J/(V·s) (Python dimension check: all true; C·s and V·Ω are false)	—	IN
Ampere in base SI	A is itself a base unit. For contrast: V = $kg \cdot m^2 \cdot s^{-3} \cdot A^{-1}$; Ω = $kg \cdot m^2 \cdot s^{-3} \cdot A^{-2}$	—	IN
Prefixes	1 A = 1000 mA = 10^6 µA; 1 kA = 1000 A	—	IN
Force between wires (1948 definition)	$F/L = \mu_0 \cdot I_1 \cdot I_2 / (2\pi \cdot d)$; $\mu_0 = 4\pi \times 10^{-7}$ N/A^2 (pre-2019 exact)	N/m	**OUT** (magnetism) — recall of "2×10^{-7} N/m" is BORDERLINE
Ampère's circuital law	$\oint B \cdot dl = \mu_0 \cdot I$	T·m	**OUT**

Question types

This note has **33 question types** (25 ✅ IN, 3 ⚠️ Borderline, 5 ❌ Out). The out-of-scope ones are in "Out of scope (skip)" below.

How the types were found

Scope tag key: IN = all levels; IN-SN = States/Nationals only; BORDERLINE; OUT.

3b. Every Ampère-related Div C / old-rules item, checked against the 2027 Div B rules

Test (div, season)	Q	What it asks	Key	Div B 2027 result
ATX Regional 2026 C	Q1	Ampère's device to cancel Earth's field	B astatic needle	IN (H-A13)
ATX Regional 2026 C	Q8	"Who wanted to become the Newton of electricity"	A Ampère	IN, wording error (H-A15)
ATX Regional 2026 C	Q12	botany after father's death	A Ampère	IN (H-A08)
ATX Regional 2026 C	Q18, Q26, SA1	solenoid field	—	OUT
BullSO 2026 C (= B text)	Q6	Newton of electricity	A	IN
BullSO 2026 C	Q4	unit of magnetic flux density (Ampère distractor)	D Tesla	OUT (Tesla, magnetism)
GullSO 2026 (C)	Q15, Q17, Q38a	unit of current; 1 A equivalents; mA in 1 A	A; A,D,E (my solve); 1000	IN
GullSO 2026 (C)	Q28	who invented the solenoid	A Ampère	IN (H-A12)
GullSO 2026 (C)	Q20, Q48	solenoid field	—	OUT
Hawk & Hornet 2026 C	Q21	not a Maxwell-equation contributor	a Hertz	BORDERLINE (H-A24)
MIT 2020 C	Q1, Q3	coined electrodynamics; discovered electromagnetism	Ampere; Nobody	IN (H-A12, H-A26)
MIT 2021 C	Q99	law used instead of Biot-Savart	Ampere	BORDERLINE (H-A24)
MIT 2021 C	Q100	wireless power opinion	Faraday, Tesla	OUT

Test (div, season)	Q	What it asks	Key	Div B 2027 result
Northview 2026 C	Q25 + Ampère's-law B question	circuital law	E circle	OUT
Princeton 2026 C	Matching 1	Ampère ↔ currents create magnetic fields	C	IN format; key statement is Ørsted's (section 4)
Princeton 2021 C	Q3, Q4	Faraday cage by Ampère? force-law sentence	False; True (my solve)	IN (H-A19, H-A11)
Princeton 2020 C	Q1	order seven scientists	Ampere (1827)	IN format, list changed for 2027
UT Invitational 2020 C (= B)	Q1, Q4	electrodynamics; solenoid	D; A	IN
ATX Regional 2020 C (= B)	Q10 item, Q32/34, Q33	coined electrodynamics; Ampère's rule; monopoles	Ampere; a; d	Q10 IN; Q32/34 and Q33 OUT
ATX Regional 2021 C (= B)	Q7	wire attracts nearby wires, made a law	Ampere	IN (H-A11)
Rickards 2021 C	Q7	1 A = 1 electron/s?	False	IN (H-A05)
Wichita 2026 C	Q3	unit of current	Ampere	IN (H-A01)
Wichita 2026 C	Q20	solenoid field	—	OUT
PA Central (~2013, C)	—	ampere in base SI	—	IN (H-A04)
cyanophycean314 Test 2 (C)	Q2	SI unit of current	Ampere	IN

History questions in the test-extraction files (8 test groups)

These rows come from `research/tests/` (ATX regionals, UT invitationals, misc-a, misc-b, USC 2026, Florida State 2021). Each row names the question type it belongs to.

Test	Year	Div	Level	Q	Question (short)	Key	Check / scope 2027 B	Type
ATX (UT Austin) Regional	2021	B	Regional	7	Current-carrying wires attract; quantitative law (name bank)	Ampere	IN (asks only the name)	H-A11
ATX (UT Austin) Regional	2020	B	Regional	2	Namesake of the Maxwell equation linking B and current	Ampere	IN (name only; the law itself is magnetism, OUT)	H-A24
ATX (UT Austin) Regional	2020	B	Regional	8	"Oersted's law states the same as this scientist's law"	Faraday (key doubtful; better Ampère)	IN (Ampère)	H-A19
ATX (UT Austin) Regional	2020	B	Regional	10	Coined "electrodynamics"	Ampere	IN	H-A12
Cobra Invitational (Michigan)	2019-20 season	B	Invitational	2	Fill-in paragraph: parallel wires attract/repel; force ∝ lengths; "electrodynamic molecule" → electron	Ampère / lengths / electron	IN	H-A23
UF Northern Regional (Florida)	2020	B	Regional	5	Who invented the solenoid?	E Ampère	IN (person listed; device is magnetic)	H-A12
UT Austin Invitational (2021 season)	2020	B	Invitational	1	Who coined "electrodynamics"?	Kirchhoff (KEY ERROR: Ampère)	IN	H-A12
UT Austin Invitational (2021 season)	2020	B	Invitational	4	Who invented the solenoid?	Ampère	IN	H-A12

Test	Year	Div	Level	Q	Question (short)	Key	Check / scope 2027 B	Type
USC Invitational (not the 2026 Nationals test)	Jan 2026	B	Invitational (top tier)	26	Match 4 facts → physicist (current-field relation → Ampère)	C: Faraday; Ohm; Ampère; Faraday	BORDERLINE (Faraday rows not listed)	H-A26
Florida State Tournament	2021	B (= C)	State	5	Quote: "Écoute les savants, mais ne les écoute que d'une oreille!..."	Ampere	IN	H-A32
UT Austin Invitational, fall	2025	B/C	Invitational	4b	Pick the letters (A, C, K, V, O, T, F) whose law has 1/r dependence	A, V (V dubious)	BORDERLINE (Ampère's law is magnetism)	H-A33
UT Austin Invitational, fall	2025	B/C	Invitational	4d	Pick who did most work in France	A, V, C (KEY ERROR: A, C)	IN	H-A33
ATX (UT Austin) Regional	2026	B	Regional	MC6	Forgot his own name and house	Ampère	IN	H-A09
ATX (UT Austin) Regional	2026	C	Regional	MC1	What Ampère made to measure tiny currents without Earth's field	astatic needle	IN	H-A13
ATX (UT Austin) Regional	2026	C	Regional	MC8	Wanted to be the "Newton of electricity"	Ampère	IN	H-A15
ATX (UT Austin) Regional	2026	C	Regional	MC12	Studied botany for a year after his father died	Ampère	IN	H-A08
F-M Invitational (NY), fall	2019	not stated	Invitational	1	Match 7 discoveries to Ampere, Coulomb, Faraday, Kirchhoff, Ohm, Tesla, Volta	Ohm, Faraday, Tesla, Volta, Coulomb, Ampere, Kirchhoff	IN for 5 rows; Faraday, Tesla rows OUT	H-A26

Easy

H-A01. Unit of current → person (and reverse)

Unit of current → person (and reverse) · ✅ IN · Easy

- **Scope: IN** (area i and iv).
- **Where seen:** Wichita Heights 2026 C Q3 ("The unit of current is" → Ampere) — Div B 2027 check: IN. GullSO 2026 Q15 (SI unit for current) — Div B 2027 check: IN. cyanophycean314 Test 2 Q2 (SI unit of current) — Div B 2027 check: IN. NC Shock Value student test Q1 "Ampere:" (define) (Div B) — IN.
- **How to solve:**
  1. Read the question: "current" (or "A") → Ampère; or Ampère → "current, ampere, A".
  2. Recall.
  3. Of the seven listed people, units: coulomb (charge), volt (potential), ampere (current), ohm (resistance), hertz (frequency)
  4. Kirchhoff and Boole have no SI unit.
- **Worked example:** "The SI unit of electric current is named for which listed scientist?" → **André-Marie Ampère**.
- **Traps:** choosing Volta or Ohm; "Coulomb" is charge, not current; Tesla/Faraday are OUT in 2027 but still appear as distractors.
- Calculator: look up — type: unit name, symbol or quantity (e.g. "A", "current").

H-A02. Symbol, spelling, abbreviation

Symbol, spelling, abbreviation · ✅ IN · Easy

- **Scope: IN** (area iv units).
- **Where seen:** ATX 2021 B fill-in (spell "Ampere" exactly) — Div B. Not seen as its own question otherwise.

- **How to solve:**
  1. Read the question: "symbol for ampere?" → A. "Is 'Amp' an SI symbol?" → no, the symbol is A.
  2. Unit names are lower case (ampere); symbols named for people are capitals (A, V, Ω for ohm is Greek omega, Hz, C).
- **Worked example:** "Write 45 milliamperes with SI symbols." → **45 mA**.
- **Traps:** "ma" or "MA" (MA = megaampere). ATX 2021 grading required exact spelling "Ampere".
- Calculator: look up — type: unit name.

H-A07. Birth, death, nationality, places

Birth, death, nationality, places · ✅ IN · Easy

- **Scope: IN**.
- **Where seen:** not seen as a direct question in the downloaded tests; facts appear in the scioly wiki [SOW] and [P: l_bank H6].
- **How to solve:**
  1. Read the question: person → country/city/years; or "Which listed scientist was French and born in Lyon?" → Ampère (Coulomb is also French: Angoulême).
  2. Recall table (fact 1, 10, 11).
- **Worked example:** "Where did Ampère die?" → **Marseille, 1836** (on an inspection tour).
- **Traps:** Coulomb is the other Frenchman; died in Marseille but buried in Paris (Montmartre); 1775 vs 1785 (1785 is Coulomb's law).
- Calculator: look up — type: name or city.

H-A12. Words he coined

Words he coined · ✅ IN · Easy

- **Scope: IN** (who coined a word is history; the physics of a solenoid is OUT).
- **Where seen:** UT Austin Invitational 2020 (2021 season) B Q1 "coined electrodynamics" → D Ampere and Q4 "invented the solenoid" → A Ampere (C test has the same Q1, Q4) — Div B 2027 check: IN. MIT 2020 C Q1 "Coined electrodynamics" → Ampere — Div B 2027 check: IN. ATX 2020 B/C "Coined the name Electrodynamics" → Ampere — Div B 2027 check: IN. GullSO 2026 Q28 "Which scientist from the rules invented the solenoid?" → A Ampère — Div B 2027 check: IN. MIT 2020 Q2 "black-body" → Kirchhoff (same format).
  - Also seen (test-extraction files): ATX (UT Austin) Regional 2020, Div B, Regional, Q10: "Coined "electrodynamics"" (key: Ampere). Div B 2027: IN.
  - Also seen (test-extraction files): UF Northern Regional (Florida) 2020, Div B, Regional, Q5: "Who invented the solenoid?" (key: E Ampère). Div B 2027: IN (person listed; device is magnetic).
  - Also seen (test-extraction files): UT Austin Invitational (2021 season) 2020, Div B, Invitational, Q1: "Who coined "electrodynamics"?" (key: Kirchhoff (KEY ERROR: Ampère)). Div B 2027: IN.
  - Also seen (test-extraction files): UT Austin Invitational (2021 season) 2020, Div B, Invitational, Q4: "Who invented the solenoid?" (key: Ampère). Div B 2027: IN.
- **How to solve:**
  1. Read the question: word → person.
  2. Ampère: electrodynamics, solenoid, cinématique (kinematics), galvanometer (name proposed)
  3. Kirchhoff: black body.
  4. Faraday (OUT list): electrode, anode, cathode, ion.
- **Worked example:** "Which listed scientist coined 'electrodynamics'?" → **Ampère**.
- **Traps:** "invented the solenoid" (tests) vs "coined the word" (sources); "electrode" is Faraday (BullSO B Q3).
- Calculator: look up — type: the word.

H-A15. "The Newton of electricity"

"The Newton of electricity" · ✅ IN · Easy

- **Scope: IN**.

- **Where seen:** BullSO 2026 B Q6 and BullSO 2026 C Q6 (same test text; key A Ampere) — Div B 2027 check: IN. ATX Regional 2026 C Q8 ("Who wanted to become the 'Newton of Electricity'?" key A Ampere) — Div B 2027 check: IN, but the wording is off (section 4).
- **How to solve:**
  1. Read the question: nickname → person; or "Who gave the nickname?" → Maxwell (Treatise, 1873, art. 528).
  2. Recall.
- **Worked example:** BullSO Q6 → **A, Ampère**.
- **Traps:** Kirchhoff or Volta as distractors; the nickname came from Maxwell, not from Ampère himself.
- Calculator: look up — type: "Newton".

Medium

H-A03. Which units equal 1 ampere

Which units equal 1 ampere · ✅ IN · Easy–Medium

- **Scope: IN** (area iv).
- **Where seen:** GullSO 2026 Q17 (select all: $C \cdot s^{-1}$, $C \cdot s$, $V \cdot \Omega$, $V \cdot \Omega^{-1}$, $J \cdot V^{-1} \cdot s^{-1}$) — Div B 2027 check: IN (only unit algebra). NC Shock Value key Q1 (V/Ω half credit, C/s preferred) — Div B. Philadelphia/Pembroke 2026 B Q33 (volt defined as J/C; "watt per ampere" also correct) — Div B.
- **How to solve:**
  1. Read the question: list of unit combinations → which equal A.
  2. Use V = J/C, Ω = V/A, W = J/s = V·A. Reduce each option.
- **Worked example:** 1 A = C/s ✔; C·s ✘; V·Ω ✘; V/Ω ✔ (Ohm's law I = V/R); J/(V·s) = (J/s)/V = W/V ✔. Answer **A, D, E** (Python dimension check confirms; the GullSO MC key is a page image and was not readable, see section 5).
- **Traps:** "V·Ω"; "amps per second" (Shock Value Q on current units); in Pembroke Q33 both B (W/A) and D (J/C) are correct.
- Calculator: calculate — type: a unit expression (e.g. "J/(V*s)"); app reduces to kg, m, s, A exponents and compares with A. (App note: (unit-dimension checker).)

H-A04. Ampere as an SI base unit / base-unit expressions

Ampere as an SI base unit / base-unit expressions · ✅ IN · Medium

- **Scope: IN** (area iv).
- **Where seen:** PA Central regional (~2013, Div C) "What is an Ampere in BASE SI units?" — Div B 2027 check: IN (units only). Not seen in a 2026-27 Div B test.
- **How to solve:**
  1. Read the question: "Which of these is a base unit: coulomb, ampere, volt, ohm?" → ampere.
  2. The 7 SI base units are s, m, kg, A, K, mol, cd.
  3. The coulomb is derived (A·s).
- **Worked example:** "Express 1 coulomb in base units." → **1 C = 1 A·s**. "Express the volt" → **$kg \cdot m^2 \cdot s^{-3} \cdot A^{-1}$** (Python check).
- **Traps:** thinking the coulomb is the base unit (it was the other way round historically, and still is after 2019: the ampere is base, defined through e).
- Calculator: look up — type: (table of base-unit expressions) plus the H-A03 checker.

H-A05. One ampere = how many electrons per second

One ampere = how many electrons per second · ✅ IN · Medium

- **Scope: IN** (area iv).
- **Where seen:** Rickards 2021-season B Q6 T/F "One Ampere of current is equal to one electron passing a point every one second." (key: False) — Div B. Rickards 2021 C Q7 same — Div B 2027 check: IN.
- **How to solve:**
  1. Read the question: I and t (or N) → N (or I).
  2. N = I·t / e.

- **Worked example:** electrons per second in 1 A = 1 / $1.602176634\times10^{-19}$ = **6.24×10^{18}** (Python 6.241509×10^{18}). One electron per second is only 1.60×10^{-19} A, so the T/F is **False**.
- **Worked example:** 2.0×10^{19} electrons pass in 4.0 s → I = 2.0×10^{19} × 1.602×10^{-19} / 4.0 = **0.80 A** (Python 0.8011).
- **Traps:** dividing by e the wrong way; sign of the electron charge (current is a magnitude here); electron flow vs conventional direction.
- Calculator: calculate — type: any two of I (A or mA), t (s/min/h), N.

H-A06. Charge, current, time with mA and mAh

Charge, current, time with mA and mAh · ✅ IN · Easy–Medium

- **Scope: IN** (area iv; hands-on iv USB charger).
- **Where seen:** GulISO 2026 Q38(a) "How many mA are in a single Ampere?" → 1000 — Div B 2027 check: IN. MIT 2020 Q11 (total charge from a battery charge curve) — Div B 2027 check: BORDERLINE (needs area under a graph; fine if piecewise constant, OUT if it needs integration).
- **How to solve:**
  1. Read the question: I, t → Q; or Q, I → t.
  2. Q = I·t; 1 mAh = 3.6 C.
- **Worked example:** 0.25 A for 120 s → **30 C**. A 2500 mAh battery holds 2.5 A × 3600 s = **9000 C** (Python).
- **Traps:** mixing minutes and seconds; mAh is charge, not energy.
- Calculator: calculate — type: I, t, unit choices.

H-A08. Family tragedy and botany

Family tragedy and botany · ✅ IN · Medium

- **Scope: IN**.
- **Where seen:** Austin (ATX) Regional 2026 C Q12 ("Who studied botany for a year because his father died?" key **A Ampere**) — Div B 2027 check: IN (biography only; Coulomb/Tesla/Volta distractors, Tesla OUT for 2027).
- **How to solve:**
  1. Read the question: anecdote → person.
  2. Father guillotined 24 Nov 1793 in Lyon → depression → Rousseau's botany letters revived him; first wife Julie died 1803.
- **Worked example:** "Which listed scientist's father was executed during the French Revolution?" → **Ampère**. Age at the time: 18 (Python 18.8).
- **Traps:** Lavoisier (chemist) was also guillotined (1794); he is not on the list.
- Calculator: look up — type: keyword ("guillotine", "botany", "father").

H-A09. Absent-mindedness anecdotes

Absent-mindedness anecdotes · ✅ IN · Medium

- **Scope: IN** (but facts weakly sourced).
- **Where seen:** ATX Regional 2026 B Q6 ("Who failed to remember their own name and even their house?" key **C Ampere**) — Div B.
- **How to solve:**
  1. Read the question: anecdote → person.
  2. Recall.
  3. The common stories: he wrote equations on the back of a cab, which then drove off; he forgot appointments (one story: a dinner with Napoleon)
  4. Only "absent-minded in later years" is sourced [W].
- **Worked example:** ATX26 B Q6 → **Ampère**.
- **Traps:** the stories are folklore; the test key is the only "source". Do not use them to rule out other answers.
- Calculator: look up — type: keyword ("absent-minded", "forgot", "cab").

H-A10. What started his work: Ørsted, Arago, the one-week gap

What started his work: Ørsted, Arago, the one-week gap · ✅ IN · Medium

- **Scope: IN**.
- **Where seen:** not seen in the downloaded tests as a direct question. Prior files: [P: I_bank H7] "Ørsted → Ampère".
- **How to solve:**
  1. Read the question: "Whose 1820 discovery did Ampère build on?" → Ørsted. "Who showed it in Paris?" → Arago. "How long after?" → about one week.
  2. Timeline: 21 Jul 1820 Ørsted publishes.
  3. 4 Sep Arago reports.
  4. 11 Sep Arago demonstrates.
  5. 18 Sep Ampère's first memoir.
- **Worked example:** days from the 11 Sep demonstration to the 18 Sep memoir = **7 days** (Python); from the 4 Sep report = 14 days; from Ørsted's 21 Jul paper = 59 days.
- **Traps:** Ørsted is not on the 2027 list (he is a distractor); Ampère did **not** discover that currents make magnetic effects (that is Ørsted).
- Calculator: look up + calculate — type: two dates. (App note: (dates) + CALCULATE (date difference).)

H-A11. The parallel-current discovery (history framing)

The parallel-current discovery (history framing) · ✅ IN · Easy–Medium

- **Scope: IN** (as a history fact only).
- **Where seen:** ATX Regional 2021 B Q7 ("I discovered that a wire with flowing current attracts other nearby wires. I also made a law that described this ... quantitatively." → Ampere) — Div B, 2021 rules; Div B 2027 check: IN as recognition of the discoverer. Princeton 2021 C Q4 T/F (the force-law sentence; answer True) — Div B 2027 check: IN (history statement, no calculation). Princeton 2026 C Matching 1 (key: Ampère ↔ "Demonstrated that electric currents create magnetic fields") — Div B 2027 check: IN as a matching item, but the statement is really Ørsted (section 4).
  - Also seen (test-extraction files): ATX (UT Austin) Regional 2021, Div B, Regional, Q7: "Current-carrying wires attract; quantitative law (name bank)" (key: Ampere). Div B 2027: IN (asks only the name).
- **How to solve:**
  1. Read the question: description → Ampère.
  2. Same direction → attract; opposite → repel; force ∝ lengths × currents.
- **Worked example:** "Who showed in 1820 that two parallel currents in the same direction attract?" → **Ampère**.
- **Traps:** Coulomb's law looks similar (like charges repel) but parallel same-direction currents **attract**.
- Calculator: look up — type: keyword ("parallel wires", "attract").

H-A13. Instruments: astatic needle, galvanometer

Instruments: astatic needle, galvanometer · ✅ IN · Medium

- **Scope: IN**.
- **Where seen:** ATX Regional 2026 C Q1 ("What did Ampere create to accurately measure tiny currents without measuring the noise of Earth's magnetic field?" key **B Astatic Needle**; distractor D "Torsional Galvanometer") — Div B 2027 check: IN (instrument history; no field calculation). Princeton 2020 C Part 2 key lists "Galvanometer" (component use) — IN as a meter.
- **How to solve:**
  1. Read the question: description → instrument, or instrument → person.
  2. Astatic pair = two opposite needles that cancel Earth's field → more sensitive current detector → galvanometer.
  3. Coulomb = torsion balance.
  4. Ohm = torsion galvanometer + thermocouple.
- **Worked example:** "Ampère's device that cancels the Earth's magnetic field so a weak current can be detected" → **astatic needle (astatic galvanometer forerunner)**.
- **Traps:** torsion balance (Coulomb) and torsion galvanometer (Ohm, Coulomb-style) are distractors; Schweigger built the first multiplier galvanometer.

- Calculator: look up — type: instrument name or keyword.

H-A14. Current vs voltage; current as a quantity

Current vs voltage; current as a quantity · ✅ IN · Medium

- **Scope: IN**.
- **Where seen:** not seen in downloaded tests. [SOW] states it.
- **How to solve:**
  1. Read the question: "Who first clearly separated current from voltage (tension)?" → Ampère.
  2. Recall fact 39 (and 40 with the caveat).
- **Worked example:** → **Ampère** (he called them "courant" and "tension").
- **Traps:** Ohm (1827) linked them in a law, but Ampère first separated the two ideas (1820). Volta's "tension" was the older term.
- Calculator: look up — type: the keyword from the question.

H-A24. Name recognition of "Ampère's law" (no physics)

Name recognition of "Ampère's law" (no physics) · ⚠️ Borderline · Medium

- **Scope: BORDERLINE**. Reason: naming which law carries his name is history, but what the law says is magnetism (OUT).
- **Where seen:** MIT 2021 C Q99 ("A certain law of electromagnetism can be used instead of the Biot-Savart law ... After which scientist is it named?" → Ampere) — Div B 2027 check: BORDERLINE (needs Biot-Savart knowledge, which is OUT). ATX 2020 key ("Namesake of the Maxwell equation that relates magnetic fields and electric currents" → Ampere) — Div B 2027 check: BORDERLINE/OUT. Hawk & Hornet 2026 C Q21 ("Which figure did not contribute to Maxwell's equations?" key **a Hertz**; Ampère is a contributor) — Div B 2027 check: BORDERLINE (Maxwell's equations are magnetism; but it uses two listed names).
  - Also seen (test-extraction files): ATX (UT Austin) Regional 2020, Div B, Regional, Q2: "Namesake of the Maxwell equation linking B and current" (key: Ampere). Div B 2027: IN (name only; the law itself is magnetism, OUT).
- **How to solve:**
  1. Read the question: description of the law → Ampère.
- **Worked example:** → **Hertz** (Gauss, Ampère, Faraday each have a Maxwell equation named after them; Hertz tested the theory).
- **Traps:** "Ampère's law" can mean the force law (1820s) or the circuital law (Maxwell's form).
- Calculator: look up — type: the keyword from the question.

H-A33. Multi-select: which listed scientists have property X?

Multi-select: which listed scientists have property X? · ✅ IN · Medium

- **Scope:** IN (3.c.i). BORDERLINE when the property is about a magnetic law (the "1/r" row is Ampère's magnetic law).
- **Where seen:** UT Austin Invitational fall 2025, Div B/C, Invitational, Q4a-d (letters A, C, K, V, O, T, F; pick every letter that fits). Ampère is a right letter in 4b (law with 1/r), 4c (steady state) and 4d (worked in France).
- **How to solve:**
  1. Write the letter code of each listed scientist (A = Ampère, C = Coulomb, K = Kirchhoff, V = Volta, O = Ohm, Hz = Hertz, B = Boole).
  2. Test each scientist against the property, one at a time.
  3. Pick every scientist that passes. A multi-select question can have more than one right letter.
  4. Check each letter again. Keys often add a wrong letter (UT 2025 put Volta under "France").
- **Worked example:** "Which did most of their work in France? A, C, K, V, O" → **A and C** (Ampère in Paris, Coulomb in Paris). The UT key also put V; that is wrong, because Volta worked in Como and Pavia in Italy.
- **Traps:** Volta visited Paris in 1801 to show the pile to Napoleon, but he did not work there. Ohm, Kirchhoff and Hertz are German. Boole is English and worked in Ireland.
- Calculator: look up — type: the property (e.g. "France", "1/r²", "steady state").

Hard

H-A16. Publications

Publications · ✅ IN · Hard

- **Scope: IN**.
- **Where seen:** not seen in downloaded tests. [P: I_bank H4].
- **How to solve:**
  1. Read the question: title → person/year.
  2. 1802 game-theory paper.
  3. 1820 memoirs (Annales vol.
  4. 15.
  5. 1822 *Recueil*
  6. 1824 *Description d'un appareil*
  7. 1826/27 *Théorie mathématique des phénomènes électro-dynamiques, uniquement déduite de l'expérience*
  8. 1834 *Essai sur la philosophie des sciences*.
- **Worked example:** "*Théorie des phénomènes électro-dynamiques, uniquement déduite de l'expérience* was written by?" → **Ampère (1826; often given as 1827)**.
- **Traps:** Ohm's *Die galvanische Kette, mathematisch bearbeitet* is also 1827 and also "mathematical". Tell them apart by language (French vs German) and by topic (electrodynamics vs the galvanic circuit).
- Calculator: look up — type: words from the title.

H-A17. Career posts and honors

Career posts and honors · ✅ IN · Hard

- **Scope: IN**.
- **Where seen:** not seen in downloaded tests.
- **How to solve:**
  1. Read the question: post/honor → person/year.
  2. Facts 15-25.
- **Worked example:** "Which listed scientist taught at the École Polytechnique and held the chair of experimental physics at the Collège de France?" → **Ampère**. "Whom did he beat for the 1814 Académie seat?" → **Cauchy, 28-26**.
- **Traps:** Coulomb was also an Académie member and also French; Collège de France date 1824 vs 1826.
- Calculator: look up — type: the keyword from the question.

H-A18. Non-electric work ("which is NOT his")

Non-electric work ("which is NOT his") · ✅ IN · Medium–Hard

- **Scope: IN** (as distractor material).
- **Where seen:** ATX Regional 2026 C Q4 (methane → Volta, with Ampere as distractor) — Div B 2027 check: IN. Not seen for Ampère's own non-electric work.
- **How to solve:**
  1. Read the question: list → the one that is not Ampère's (or the one that is).
  2. Ampère: probability (1802), PDEs/Monge-Ampère, name "fluorine" (1811-12), element classification (1816), wave theory of light, kinematics word, gas law.
  3. Not Ampère: methane (Volta), soil mechanics/friction (Coulomb), caesium/rubidium (Kirchhoff), photoelectric effect observation (Hertz).
- **Worked example:** "Ampère proposed the name of which element?" → **fluorine** (he did not isolate it).
- **Traps:** "discovered fluorine" (wiki wording) — isolation was Moissan 1886.
- Calculator: look up — type: the keyword from the question.

H-A19. Mis-attribution and true/false traps

Mis-attribution and true/false traps · ✅ IN · Hard

- **Scope: IN**.
- **Where seen:** Princeton 2021 C Q3 T/F "Faraday cages were actually invented by Ampere" (answer **False**) — Div B 2027 check: IN only as a history T/F; Faraday is OUT for 2027 but a T/F about Ampère is still about a listed person. UT20 inv B Q2 (chemical electricity → Volta, Ampere distractor) — Div B. ATX26 B Q1 ("first chemical battery" → Volta; Ampere distractor) — Div B. Berkeley 2026 B Q1-2 (AC → Tesla, cage → Faraday; Ampère distractor) — Div B (2026 rules; Tesla/Faraday OUT 2027).
  - Also seen (test-extraction files): ATX (UT Austin) Regional 2020, Div B, Regional, Q8: ""Oersted's law states the same as this scientist's law"" (key: Faraday (key doubtful; better Ampère)). Div B 2027: IN (Ampère).
- **How to solve:**
  1. Read the question: a claim → true/false, or which pairing is wrong.
  2. Check against the fact table.
  3. Common false claims: Ampère discovered that current makes magnetism (Ørsted); invented the battery (Volta); invented the Faraday cage (Faraday); invented the practical electromagnet (Sturgeon); isolated fluorine (Moissan); stated V = IR (Ohm).
- **Worked example:** "Which is mis-attributed? A Ampère–solenoid; B Ampère–astatic needle; C Ampère–voltaic pile; D Ampère–electrodynamics" → **C** (Volta).
- **Traps:** half-true statements ("Ampère invented the galvanometer": he named it and built a forerunner).
- Calculator: look up — type: (claim checker keyed on keywords).

H-A20. Timeline ordering and age arithmetic

Timeline ordering and age arithmetic · ✅ IN · Medium (ordering) to Hard (overlaps, ages)

- **Scope: IN**.
- **Where seen:** Princeton 2020 C Q1 (order seven scientists by crowning contribution; key gives "Ampere (1827)") — Div B 2027 check: IN for ordering, but the 2027 list differs (Hertz and Boole replace Tesla and Faraday) and 1827 for Ampère is debatable (section 4).
- **How to solve:**
  1. Read the question: events → chronological order; or "who was alive when ...".
  2. Coulomb's law 1785 → Volta's pile 1800 → Ampère's electrodynamics 1820 (treatise 1826/27) → Ohm's law 1827 → Kirchhoff's laws 1845 → Boole 1847/1854 → Hertz's waves 1886-88.
- **Worked example:** order → **Coulomb, Volta, Ampère, Ohm, Kirchhoff, Boole, Hertz** (Python sort).
- **Worked example:** "Which listed scientist was born after Ampère died?" → **Hertz (born 1857)**. All other five were alive during part of Ampère's life (Python).
- **Worked example:** Ampère's age at his 18 Sep 1820 memoir = **45** (Python 45.7). Age at death = **61**. Years from his first memoir to the naming of the unit (1881) = **61**.
- **Traps:** Ampère (1826/27 treatise) and Ohm (1827) tie if the treatise is used; use 1820 for "discovery".
- Calculator: look up + calculate — type: names/events. (App note: (dates) + CALCULATE (differences, sort).)

H-A25. The 1948 definition number (recall, no calculation)

The 1948 definition number (recall, no calculation) · ⚠️ Borderline · Hard

- **Scope: BORDERLINE**. Reason: unit history is IN, but the definition is a magnetic force between wires (OUT physics). A recall of "2×10^{-7} N per metre at 1 m" needs no magnetism math.
- **Where seen:** not seen in a test.
- **Worked example:** "Under the 1948 definition, 1 A in each of two parallel wires 1 m apart gives what force per metre?" → **2×10^{-7} N/m** (Python: $\mu_0/(2\pi) = 2.0\times10^{-7}$).
- Calculator: look up — type: (store the number; do not build the force formula into the Div B app).

H-A26. Matching columns and short essays (multi-person)

Matching columns and short essays (multi-person) · ✅ IN · Medium (matching) to Hard (essay)

- **Scope: IN**.
- **Where seen:** Princeton 2026 C Matching Part 1 (Ampère→C) — Div B 2027 check: IN in format; for 2027 swap Tesla/Faraday for Hertz/Boole. MIT 2020 C Q1-10 (fill in "last name from the rules or Nobody") — Div B 2027 check: IN in format; Q3 "Discovered electromagnetism → Nobody" is a good trap (Ørsted is not on the list). ATX

2021 B fill-in Q2-7 — Div B. MIT 2021 C Q100 (opinion: which two scientists matter most for wireless power; key Faraday & Tesla) — Div B 2027 check: OUT as written (both OUT for 2027).
  - Also seen (test-extraction files): USC Invitational (not the 2026 Nationals test) Jan 2026, Div B, Invitational (top tier), Q26: "Match 4 facts → physicist (current-field relation → Ampère)" (key: C: Faraday; Ohm; Ampère; Faraday). Div B 2027: BORDERLINE (Faraday rows not listed).
  - Also seen (test-extraction files): F-M Invitational (NY), fall 2019, Div not stated, Invitational, Q1: "Match 7 discoveries to Ampere, Coulomb, Faraday, Kirchhoff, Ohm, Tesla, Volta" (key: Ohm, Faraday, Tesla, Volta, Coulomb, Ampere, Kirchhoff). Div B 2027: IN for 5 rows; Faraday, Tesla rows OUT.
- **How to solve:**
  1. Build a person × (unit, invention, word, nationality, date, nickname) grid; eliminate.
- **Worked example:** "Coined 'electrodynamics' / Discovered 'electromagnetism' (2 blanks, name or Nobody)" → **Ampère / Nobody** (MIT 2020 key).
- **Traps:** "Nobody" answers; one-use-only matching with a wrong key item (Princeton 2026, section 4).
- Calculator: look up — type: (keyword search across all seven people).

H-A32. Quote attribution (who said it?)

Quote attribution (who said it?) · ✅ IN · Hard

- **Scope:** IN (3.c.i). A quote is a "historical perspective" fact about a listed scientist.
- **Where seen:** Florida State Tournament 2021, Div B (same as Div C), State, Q2-Q8 (seven quotes, 2 points each, "identify the last name"). Q5 is Ampère: "Écoute les savants, mais ne les écoute que d'une oreille!... Listen to learned men, but do so only with one ear!... Let the other be always ready to receive the sweet accents of the voice of your heavenly Friend." Not seen on another test.
- **How to solve:**
  1. Look for one key word or idea in the quote (an invention, a place, a belief, a language).
  2. Match that idea to one of the seven listed scientists.
  3. If the quote is in French, think of the two French scientists first: Ampère and Coulomb.
  4. If nothing fits the seven, the answer may be a scientist who is not on the 2027 list (Tesla, Faraday). Those items are out of scope in 2027.
  5. Clues for Ampère: French text, religious (Catholic) faith, advice to a friend or son, his journal.
- **Worked example:** "Listen to learned men, but do so only with one ear... the voice of your heavenly Friend." Who said it? → **Ampère** (key of Florida State 2021 Q5).
- **Traps:** Coulomb is also French, but the religious advice fits Ampère. Maxwell's "Newton of electricity" is a quote ABOUT Ampère, not BY him.
- Calculator: look up — type: a key word from the quote (e.g. "oreille", "heavenly").

Nationals

H-A21. History of the unit ampere

History of the unit ampere · ✅ IN · Hard to Nationals

- **Scope: IN** (area i + area iv units).
- **Where seen:** not seen in downloaded tests. [P: I_bank H12].
- **How to solve:**
  1. Read the question: "When/where was the ampere named?" → 1881, International Electrical Congress, Paris. "What fixes the ampere since 2019?" → the exact value of e.
  2. 1881 Paris (1/10 CGS unit) → 1893 Chicago / 1908 London international ampere (silver voltameter) → 1948 force definition → 2019 e fixed.
- **Worked example:** "Since 2019, 1 A corresponds to how many elementary charges per second?" → **6.241509074×10^{18}** (Python, exact definition $1/1.602176634\times10^{-19}$).
- **Traps:** the 1881 congress is the "International Electrical Congress" held during the Paris International Exposition of Electricity; 1948 vs 1960 (SI) vs 2019.
- Calculator: look up + calculate — type: year or keyword. (App note: CALCULATE (for e-based numbers).)

H-A22. The telegraph proposal

The telegraph proposal · ✅ IN · Nationals

- **Scope: IN** (history of an application of current).
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: "Which listed scientist proposed an electromagnetic telegraph in 1820?" → Ampère (on Laplace's suggestion).
  2. Recall fact 46.
- **Worked example:** → **Ampère, 2 October 1820, one pair of wires per letter with needles at the receiver**.
- **Traps:** practical telegraphs came later (Schilling 1832, Gauss-Weber 1833, Cooke-Wheatstone 1837 [U: general knowledge]); Morse is not on the list.
- Calculator: look up — type: the keyword from the question.

H-A23. Electrodynamic molecule (magnets as tiny currents)

Electrodynamic molecule (magnets as tiny currents) · ⚠️ Borderline · Nationals

- **Scope: BORDERLINE**. Reason: it is a history fact about Ampère, but its content is the origin of magnetism, which 2027 removed. A supervisor could call it "electricity discovery" history.
- **Where seen:** not seen in a test. [SOW][W].
  - Also seen (test-extraction files): Cobra Invitational (Michigan) 2019-20 season, Div B, Invitational, Q2: "Fill-in paragraph: parallel wires attract/repel; force ∝ lengths; "electrodynamic molecule" → electron" (key: Ampère / lengths / electron). Div B 2027: IN.
- **How to solve:**
  1. Read the question: "Who proposed that magnetism comes from tiny circulating currents (a forerunner of the electron)?" → Ampère.
- **Worked example:** → **Ampère ("electrodynamic molecule"; tested by Einstein-de Haas 1915)**.
- **Traps:** do not go into magnetic dipole physics.
- Calculator: look up — type: the keyword from the question.

Out of scope (skip)

These types appear in tests or books, but the 2027 Div B rules exclude them. Skip them. They stay here so you can recognise them.

H-A27. Parallel-wire physics without history (attract or repel?)

Parallel-wire physics without history (attract or repel?) · ❌ Out · Easy

- **Scope: OUT**. Reason: magnetic force between currents; no 2027 area covers magnetism (3.c "areas only").
- **Where seen:** not seen as a pure parallel-wire question in the downloaded tests. Nearest: Berkeley 2026 B magnetism items (loop entering a solenoid) — Div B, 2026 rules, OUT for 2027.
- **Worked example:** same-direction currents → attract.
- Calculator: none (skip). (do not include in the Div B app except as the H-A11 history fact).

H-A28. Force between two wires (calculation)

Force between two wires (calculation) · ❌ Out · Medium

- **Scope: OUT**. Reason: magnetism, and μ_0 is not in any 2027 area.
- **Where seen:** not seen in a downloaded Div B test.
- **Worked example:** 3 A and 2 A, 5 cm apart → $F/L = 4\pi\times10^{-7}\times3\times2/(2\pi\times0.05) = 2.4\times10^{-5}$ N/m (Python).
- Calculator: none (skip).

H-A29. Ampère's circuital law, Maxwell's equations, magnetic field of a wire

Ampère's circuital law, Maxwell's equations, magnetic field of a wire · ❌ Out · Medium–Hard

- **Scope: OUT**. Reason: magnetism; also the field formula is not in the math/area list.

- **Where seen:** Northview 2026 C Q25 ("optimal shape for Ampère's circuital law" key E circle) — Div B 2027 check: OUT. Northview 2026 C (Ampère's law B at 0.1 m) — OUT. ATX 2020 B/C Q33 ("monopoles" → Gauss's law, Ampère's Law distractor) — OUT. UT20 inv B Q13 TB (KCL restates? Ampère's Law distractor) — the KCL part is IN-SN; the Ampère's-law option is a distractor only.
- Calculator: none (skip).

H-A30. Ampère's rule / swimmer rule / right-hand rule for direction

Ampère's rule / swimmer rule / right-hand rule for direction · ✗ Out · Medium

- **Scope: OUT**. Reason: right-hand rule and magnetic direction were removed in 2027 [P: A_rules "What changed"].
- **Where seen:** ATX 2020 B Q32 / C Q34 ("used to assign the positive direction of current in Ampere's Law?" a Ampère's Rule / b Right Hand Rule) — Div B 2027 check: OUT.
- Calculator: none (skip). (The name "bonhomme d'Ampère" as a history word could be asked; treat as H-A12 trivia.)

H-A31. Solenoid field physics

Solenoid field physics · ✗ Out · Medium

- **Scope: OUT**. Reason: magnetism.
- **Where seen:** ATX 2026 C Q18, Q26, SA1; GullSO Q20, Q48; Wichita 2026 C Q20; Princeton 2021 C Q45; MIT 2020 C Q53; Rickards 2026 B Q24; ATX 2019 B/C Q9 — all Div B 2027 check: OUT. (Only "who coined/invented the solenoid" stays IN, H-A12.)
- Calculator: none (skip).

Mistakes in tests and keys

1. **MagLab Magnet Academy** gives Ampère's birth date as **22 January 1775**. Wikipedia, MacTutor and the scioly wiki give **20 January 1775**. Use 20 January.
2. **ITER Newsline** says he died "in the early hours of **9 June** 1836". Wikipedia, MacTutor and the scioly wiki say **10 June**. Use 10 June; accept 9 June only if a key uses it.
3. **ATX Regional 2026 C Q8** asks "Who **wanted to become** the 'Newton of Electricity'?" (key A Ampère). The nickname was given by **Maxwell** in 1873, 37 years after Ampère died. No source read says Ampère sought the title. The answer letter is fine; the stem is wrong.
4. **Princeton 2026 C Matching Part 1**: the key pairs Ampère with "Demonstrated that electric currents create magnetic fields". That was **Ørsted's** 1820 discovery. Ampère's distinct result was that currents exert forces **on each other** (parallel wires). The key works only by elimination.
5. **ATX Regional 2020 key**: "Oersted's law states the same findings as this scientist's law" → key **Faraday**. "Ørsted's law" is usually a name for the magnetic field around a current, which matches **Ampère's** (circuital) law, not Faraday's induction law. Likely key error. (OUT for 2027 anyway.)
6. **Princeton 2020 C Q1 key** dates Ampère's "crowning contribution" as **1827**. His discovery was 1820; 1826/1827 is the treatise. Not wrong, but a 2027 ordering question should accept 1820.
7. **Scioly wiki** (`cl_wiki.txt`, Ampère section): "He discovered and named the chemical element fluorine." He **proposed the name** (1810-1812, MacTutor: 1811); he did **not** isolate it (Henri Moissan, 1886 [U: general knowledge]). Also "This led to the invention of the electromagnet": the practical electromagnet was William Sturgeon's (1824); Ampère and Arago magnetized iron/steel in a coil in 1820. Overstated.
8. **Source date conflicts (not errors, but traps):** Arago's report 4 Sept vs demonstration 11 Sept 1820; Lyon lycée 1803 (MacTutor) vs 1804 (MagLab); Collège de France 1824 (Wikipedia, IEEE) vs 1826 (MacTutor); treatise 1826 vs 1827; father's execution 23 vs 24 November 1793.
9. **GullSO 2026 Q28** says Ampère "invented the solenoid". He **coined the word** and used coils; "invented" is loose but is the intended answer.
10. Prior file [P: G_tests.md line 155] lists "MIT 2020 Q1-10 (Ampère 'electrodynamics'; Ohm's false acoustic law)" — confirmed correct against `mit20_key.txt`.

- **UT Austin Invitational 2020, Div B, Q1:** the key says Kirchhoff coined "electrodynamics". Ampère coined *électrodynamique* in the 1820s. Key error.
- **ATX Regional 2020, Div B, Q8:** "Oersted's law states the same as this scientist's law", key Faraday. Ørsted's finding (a current makes a magnetic field) matches Ampère's law, not Faraday's. Key doubtful.
- **UT Austin Invitational fall 2025, Q4d:** "did most work in France", key A, V, C. Volta worked in Italy. The right answer is A, C.

Not confirmed

- **Absent-minded anecdotes** (cab-back equations, forgetting his name/house, missed dinner with Napoleon). Looked: Wikipedia (only "absent-minded" in later years), MacTutor, MagLab, ITER. Not found. ATX 2026 B Q6 key is the only support.
- **Rousseau's botany letters as the cure for his depression.** Seen only in search snippets (encyclopedia.com, a DSB-based biography PDF). MacTutor and Wikipedia do not mention it. Plausible; not read in full.
- **"Tandem felix" epitaph.** Searched; not found. ITER gives a different inscription.
- **Second wife's surname "Potot".** MacTutor gives "Jenny" only. Surname from memory.
- **Current direction convention (fact 40).** Widely repeated; no primary source read this run.
- **Galvanometer naming** date and memoir. Princeton Joseph Henry Project excerpt via a search summary only; I did not read the Annales page.
- **Ørsted's 21 July 1820 date, Sturgeon 1824, Moissan 1886, later telegraphs (Schilling 1832, Gauss-Weber 1833, Cooke-Wheatstone 1837).** General knowledge; not fetched this run.
- **0.001118 g/s silver international ampere.** Search did not show the figure.
- **When the ampere became a base unit** (NIST pages differ: 1921 vs 1960; 1954 CGPM is also cited elsewhere).
- **GullSO 2026 MC key (Q15, Q17, Q28).** The key PDF stores MC answers as an image on page 1; pdftotext returns nothing. Answers given above are my own (Python dimension check for Q17).
- **Princeton 2021 Q3/Q4 and UT 2020 Q1/Q4 answers.** The key text files do not mark the chosen option (highlighting lost). Answers are my own and agree with the facts.
- **Avogadro-type gas law (1814)** — from prior file only ([G] tag in I_bank), not re-checked.
- SI Brochure (spelling/symbol rules) not fetched.

- Florida State 2021 Q5 quote ("Écoute les savants..."): the key says Ampère. I did not find the primary source. Searched: `research/downloads/` (only the test itself). Not searched online in this run.

Sources

- 2027 Div B rules: `/Users/kai/Daniel/Science_Olympiad_Div_B_Rules_2027_for_Web_Secured.pdf` (Circuit Lab B12-B13); 2027 Div C rules: `downloads/dl/rulesC2027.txt`.
- Tests (all under `/Users/kai/Daniel/circuitlab/research/downloads/`): dlG/atx26reg_Btest, atx26reg_Bkey, atx26reg_Ctest, atx26reg_Ckey; dlG/atx21reg_Bkey; dlG/atx20reg_Btest, atx20reg_key; dlG/ut20inv_Btest/Bkey/Ctest; dlG/bull26C_test; dlB/tests/bullso26_test/key; dlB/tests/berk26_key; dlB/tests/phi26_test; dlG/gull26_test/key; dlG/hh26_test/key; dlG/mit20_key; dlG/mit21_test/key; dlG/nv26C_test/key; dlG/puso26_test/key; dlG/pz/p20, pz/p21 Princeton; dlG/rsoiOldB_key, rsoiOldC_key; dlG/wich26C_test/key; dlG/wiki/central.txt, Circuit_Lab_2_cyanophycean314*, Shock_value_test*; dl/cl_wiki.txt.
- Prior files: `research/prior/A_rules.md`, `B_topics.md`, `F_slides.md`, `G_tests.md`, `G_manifest.md`, `I_bank.md`.
- https://en.wikipedia.org/wiki/André-Marie_Ampère
- https://mathshistory.st-andrews.ac.uk/Biographies/Ampere/
- https://nationalmaglab.org/magnet-academy/history-of-electricity-magnetism/pioneers/andre-marie-ampere/
- https://ethw.org/Milestones:The_Birth_of_Electrodynamics,_1820-1827
- https://www.iter.org/node/24467
- https://cigre.org/article/andre-marie-ampere-and-the-two-hundred-years-of-electrodynamics
- https://commons.princeton.edu/josephhenry/astatic-needles

- https://en.wikipedia.org/wiki/Needle_telegraph
- https://en.wikiquote.org/wiki/André-Marie_Ampère (Maxwell art. 528 quote)
- https://www.bipm.org/en/history-si/ampere
- https://www.nist.gov/si-redefinition/ampere/ampere-history
- https://en.wikipedia.org/wiki/Ampere
- Python checks: `/private/tmp/claude-502/-Users-kai-Daniel/893762dd-d9b3-4158-937c-ea93eb65d1e6/scratchpad/amp.py` (electrons per second, ages, date gaps, unit dimensions, Q = I·t, 2×10^{-7} N/m, timeline sort, lifespan overlaps).
- Test-extraction files: `/Users/kai/Daniel/circuitlab/research/tests/` (atx-reg, misc-a, misc-b, nat26, state, ut-a, ut-b, ut-c) and `/Users/kai/Daniel/circuitlab/research/prior/G_tests.md`.
- Raw research file this note was built from: `/Users/kai/Daniel/circuitlab/research/subtopics/ampere.md`.

Charles-Augustin de Coulomb

Note: Coulomb

> **INFO: How to use this note**
>
> - Scope icons: ✅ IN = all levels. ✅ States-Nationals = only at State and National tournaments. ⚠️ Borderline = the rules do not settle it (the reason is given). ❌ Out = excluded (the rule is cited).
> - Question types go from Easy to Medium to Hard to Nationals. "Nationals" means the hardest depth, not a different scope.
> - Every number in a worked example was checked in Python.
> - Source tags in square brackets, like [W] or [M], are listed under "Sources".

Scope notes and source tags

Slug: `coulomb`. Area: rule 3.c.i (history) with links to rule 3.c.ii (Coulomb's law). Built 2026-10-16 from the prior files, the downloaded tests and web sources. Every number below was checked with a Python calculation.

Scope rule that governs this page. The 2027 Div B text (manual p. B11, rule 3.c.i) says: "Historical perspective of the electricity discoveries made by Ampere, Coulomb, Kirchhoff, Volta, Ohm, Hertz, and Boole". The 2026 text said "electricity **and magnetism** discoveries". All magnetism is gone in 2027. So Coulomb's electrical work is IN. His magnetism, friction, soil and engineering work is not an "electricity discovery". It is tagged BORDERLINE.

Div C source check (user request). Many sources below are Div C tests. For each type that came from a Div C test, the block has a "Div C check" line. That line says if the question fits the 2027 Div B rules (topic list 3.c, exclusions 3.d, math limits 3.e). A Div C question that fails the check is tagged BORDERLINE or OUT, not IN.

Source tags: **[R]** 2027 Div B rules PDF. **[W]** Wikipedia (fetched 2026-10-16). **[DSB]** Gillmor, Dictionary of Scientific Biography (via encyclopedia.com / MacTutor PDF). **[MT]** MacTutor biography. **[ML]** National MagLab biography. **[SK]** skullsinthestars.com 2026 article on the 1785 memoir. **[T]** a downloaded test (named). **[P]** a prior research file.

Key facts

1.1 Life

#	Fact	Source
F1	Full name Charles-Augustin de Coulomb. French.	[W] [DSB]
F2	Born 14 June 1736, Angoulême, France. Died 23 August 1806, Paris, age 70.	[W] [DSB] [MT]
F3	Parents: Henry Coulomb (from Montpellier) and Catherine Bajet.	[W] [DSB]
F4	Studied at the Collège Mazarin in Paris. The family lost money. He followed his father to Montpellier after an argument with his mother about his career.	[DSB] [MT]
F5	Joined the Société des Sciences de Montpellier, March 1757. Read papers on astronomy and mathematics.	[DSB] [MT]
F6	Entered the École royale du génie (military engineering school) at Mézières in February 1760. Graduated November 1761 as lieutenant (lieutenant en premier) in the Corps du Génie. Friends: Borda and his teacher Bossut.	[DSB] [MT]
F7	First post Brest. Sent to **Martinique** (West Indies) February 1764 to June 1772 to build **Fort Bourbon** (now Fort Desaix). Cost six million livres. He was ill there and his health stayed poor for life.	[W] [DSB]
F8	Later posts: Bouchain, Cherbourg, Rochefort (1779, with Montalembert, a wooden fort on Île d'Aix).	[W] [DSB]
F9	One week in prison, November 1783, after a critical report on Brittany canals.	[DSB] [MT]
F10	July 1784: intendant of the royal waters and fountains (part of the Paris water supply).	[DSB] [MT]

#	Fact	Source
F11	1787: went to England with Tenon to study hospitals.	[W] [DSB]
F12	Resigned from the Corps du Génie April 1791, rank lieutenant-colonel, holder of the Croix de St. Louis.	[DSB]
F13	Academy abolished 8 August 1793. He worked on the new weights and measures committee until it was purged in December 1793. He and Borda then retired to his property near **Blois**.	[DSB] [W]
F14	December 1795: elected to the new Institut de France (experimental physics). Returned to Paris.	[DSB] [MT]
F15	Sons born 26 Feb 1790 and 30 Jul 1797. Married their mother Louise Françoise LeProust Desormeaux in 1802.	[DSB] [MT]
F16	1802-1806: inspector general of public instruction. Helped set up the French lycée (high school) system.	[DSB] [MT]
F17	He never signed as "de Coulomb". No evidence his family was noble.	[DSB]
F18	Left more than 40,000 francs. The story that he died poor is false.	[DSB]
F19	One of the 72 names on the Eiffel Tower.	[W]
F20	Biot: "It is to Borda and to Coulomb that one owes the renaissance of true physics in France."	[DSB] [MT]

1.2 Engineering and mechanics (not electricity: BORDERLINE)

#	Fact	Source
F21	1773: first memoir to the Paris Academy of Sciences, on maxima and minima in statics for architecture. It held his sliding-wedge theory of **earth pressure** on retaining walls. He is a founder of soil mechanics (Mohr-Coulomb theory carries his name). Wikipedia dates the earth-pressure essay 1776 (publication).	[DSB] [W]
F22	First correct location of the neutral surface of a bent beam (1773 memoir).	[DSB]
F23	Ergonomics: studied how much work people can do per day. First memoir 1778, published 1799. Found 7-8 hours best for heavy work.	[DSB]
F24	**Friction:** "Théorie des machines simples", friction experiments in the Rochefort shipyards. Won the Academy prize in **1781**. Friction is about proportional to the normal force (Amontons-Coulomb laws). Static and sliding friction. "Coulomb friction" carries his name.	[DSB] [W] [MT]
F25	Kragelsky and Schedrov on his friction work: "he created this science."	[DSB] [MT]

1.3 Prizes and the Academy

#	Fact	Source
F26	6 July 1774: named correspondent of Bossut at the Academy.	[DSB] [MT]
F27	**1777:** shared the Academy prize for a memoir on **magnetic compass needles** ("Recherches sur la meilleure manière de fabriquer les aiguilles aimantées"). It held his first torsion work: a thread twist can measure very small forces. Co-winner: Van Swinden (search results and [P] I_bank; MacTutor and DSB do not name him).	[DSB] [MT] [P]
F28	The Paris Observatory adopted his torsion-suspension compass in 1780.	[DSB]
F29	**1781:** prize for the friction memoir. DSB calls it the "double first prize". MagLab says he was the "sole recipient". It got him elected to the Academy (mechanics section, adjoint mécanicien) and a permanent Paris post.	[DSB] [ML] [MT]

#	Fact	Source
F30	He read 25 memoirs at the Academy/Institut (1781-1806) and took part in 310 committee reports.	[DSB] [MT]

1.4 Torsion and the torsion balance (IN as the instrument of his electrical discovery)

#	Fact	Source
F31	Torsion balance: a light horizontal needle hangs from a thin wire. A force on the end of the needle twists the wire. The twist angle measures the force.	[DSB] [SK]
F32	9 September 1784 memoir on torsion of metal wires: torque $M \propto$ (twist angle) $\times D^4 / L$. This corrected his 1777 guess of D^3.	[DSB] [W]
F33	1785 electrical balance: glass cylinder about 12 in wide and 12 in high. Fine **silver** wire about 28 in long. Needle of silk or straw coated with sealing wax. A small **elder-pith ball** (2-3 lignes) on the needle, a second fixed pith ball. A paper vane for balance and damping. A micrometer twist head at the top.	[SK]
F34	Charging: a large-head pin on a wax handle touched the fixed ball, which touched the moving ball. Both got the same charge and repelled.	[SK]
F35	Who invented it: Coulomb said he knew of no earlier work. John Michell had the idea privately. Cavendish used a Michell torsion balance in 1798 to weigh the Earth (gravity, not electricity). At Coulomb's funeral Lalande said "An Englishman seized his idea ... but Coulomb never bothered to complain."	[DSB]
F36	Smallest force he measured: below 9×10^{-4} dyne (9×10^{-9} N).	[DSB]

1.5 Electricity discoveries (IN)

#	Fact	Source
F37	Seven memoirs on electricity and magnetism. The first six are on electricity. The seventh is on magnetism. Dates differ: Wikipedia gives 1785-1789 (Academy volume years). DSB gives reading dates 1785-1791.	[W][DSB]
F38	**First memoir, 1785:** torsion balance. **Like charges repel with a force that varies as the inverse square of the distance.** Measured statically (twist angle at rest).	[W][DSB][SK]
F39	1785 data (three trials): separation 36°, 18°, 8.5° with micrometer twist 0°, 126°, 567°. Total torsion = twist + separation = 36°, 144°, 575.5°. Halve the distance, force × 4.	[SK] + search results; ratios checked in Python
F40	**Second memoir:** unlike charges **attract** with the same inverse-square law. Attraction is unstable in a torsion balance (the balls snap together). So he mostly used an **oscillation method**: time the period of the swinging needle at different distances.	[DSB][W]
F41	He showed $F \propto 1/r^2$ by experiment. He only stated (did not directly prove) $F \propto q_1 q_2$. He never defined a unit of charge.	[DSB]
F42	Third memoir: an isolated charged body loses charge over time. The loss rate is proportional to the charge. It leaks through humid air and the supports.	[W][DSB]
F43	Fourth memoir (1786 per [W]): charge divides between touching conductors and **stays on the outer surface** of a conductor, not inside.	[W][DSB]
F44	Fifth and sixth memoirs: charge distribution and surface density on conductors of different shapes. He used a **proof plane** (small disc on an insulating handle) to sample the charge. Poisson built his 1811 electrostatics theory on this data.	[DSB][W]
F45	Two classes of matter: perfect conductors and dielectrics (insulators). He thought no perfect insulator exists.	[DSB][MT]

#	Fact	Source
F46	He used a fluid theory (he preferred two fluids) and treated electricity like Newton's gravity: action at a distance along the line joining the bodies.	[DSB][W][ML]
F47	Like gravity: inverse square, along the line between centers. Unlike gravity: depends on charge not mass, and it can repel.	[ML]
F48	He found no link between electricity and magnetism (that came with Ørsted, 1820).	[W]

1.6 Magnetism (BORDERLINE: removed in 2027)

#	Fact	Source
F49	Seventh memoir (1789 per [W], 1791 per [DSB]): magnetic poles also follow an inverse-square law. Molecular theory of magnetism (each particle is a tiny magnet). His work led to Poisson's theory of magnetism.	[W] [DSB] [ML]

1.7 Precursors of the inverse-square law (historical perspective)

#	Fact	Source
F50	1767 Priestley guessed an inverse-square law. 1769 Robison measured an exponent of 2.06. Cavendish found it in the early 1770s but did not publish. Coulomb published first with direct measurements (1785) and covered both like and unlike charges.	[W] Coulomb's law page

1.8 The unit

#	Fact	Source
F51	The coulomb (C) is the SI unit of charge. 1 C = 1 A × 1 s.	[W][ML]
F52	Named for him in 1881 by the International Electrical Congress in Paris (Wikipedia unit page). The Wikipedia biography says 1880. This is after his death (1806), so the name is posthumous.	[W] (two pages disagree)
F53	"International coulomb" 1908. Modern coulomb from 1948. Since 2019, e = 1.602176634 × 10^{-19} C exactly. So 1 C ≈ 6.2415 × 10^{18} elementary charges.	[W]

Formulas

Item	Formula / value	Units
Coulomb's law (magnitude)	$F = k\cdot$	$q_1 \cdot q_2$
Coulomb constant	$k = 8.9875517862 \times 10^9 \approx 8.99 \times 10^9 \approx 9 \times 10^9$	$N \cdot m^2/C^2$
From ε_0	$k = 1/(4\pi\,\varepsilon_0)$, $\varepsilon_0 = 8.8541878188 \times 10^{-12}$ F/m (Python: 8.98755×10^9)	
Elementary charge	$e = 1.602176634 \times 10^{-19}$ C (exact)	C
Charges in 1 C	$1/e = 6.241509 \times 10^{18}$	count
Charge from current	$Q = I\cdot t$	C = A·s
Battery capacity	1 mAh = 3.6 C; 1 Ah = 3600 C	
Volt	1 V = 1 J/C	
Torsion law (1784)	$M \propto \theta \cdot D^4/L$	torque vs twist angle θ, diameter D, length L
Torsion balance reading	F ∝ total torsion angle = micrometer twist + separation angle	degrees
Scaling	$F_2/F_1 = (q_1'q_2'/q_1q_2)\cdot(r_1/r_2)^2$	ratio

Math-limit note [R 3.e]: algebra and square roots are allowed. Trig is not. In a torsion balance question the writer must use "separation angle ∝ distance" (small-angle idea) or give distances directly. The oscillation method (period ∝ distance) uses only ratios, so it is legal. The physics behind it (simple harmonic motion) is not needed.

Question types

This note has **34 question types** (25 ✅ IN, 6 ⚠️ Borderline, 3 ❌ Out). The out-of-scope ones are in "Out of scope (skip)" below.

How the types were found

Order: easiest first inside each group. Group A = history (IN). Group B = history BORDERLINE. Group C = Coulomb's-law links (IN). Group D = OUT. Note: full Coulomb's-law calculation types (2D vectors, superposition, field zero points) belong to the area-ii subtopic file. Group C has only the types that tests ask beside the history.

- Group A: History, IN
- Group B: History, BORDERLINE
- Group C: Coulomb's-law links, IN (rule 3.c.ii)
- Group D: OUT

History questions in the test-extraction files (8 test groups)

These rows come from `research/tests/` (ATX regionals, UT invitationals, misc-a, misc-b, USC 2026, Florida State 2021). Each row names the question type it belongs to.

Test	Year	Div	Level	Q	Question (short)	Key	Check / scope 2027 B	Type
ATX (UT Austin) Regional	2021	B	Regional	2	The unit of charge bears my name	Coulomb	IN	CL-H03
ATX (UT Austin) Regional	2021	B	Regional	8	Torsion-balance scientist	Coulomb	IN	CL-H04
ATX (UT Austin) Regional	2021	B	Regional	9	Explain how a torsion balance measures force	twist measured → force	IN	CL-H07
ATX (UT Austin) Regional	2020	B	Regional	4	Law that derives Gauss's law and vice versa	Coulomb	Test file says IN (name only); this note keeps the Gauss/Maxwell content OUT (CL-H24)	CL-H24
UT Austin Invitational (2021 season)	2020	B	Invitational	3	Who set the inverse-square law of attraction/repulsion?	Coulomb	IN	CL-H01
UT Austin Invitational (2020 season)	2019	B	Invitational	4	Epithet: fled Paris around 1800	Coulomb	IN (he left Paris in the Revolution, 1790s, for Blois)	CL-H21
UT Austin Invitational, fall (2020 season)	2019	C	Invitational	4	Epithet: fled Paris during the French Revolution	Coulomb	IN	CL-H21
UT Austin Invitational (2020 season)	2019	B	Invitational	9	Epithet: published "On Magnetism"	Coulomb (doubtful: De Magnete is Gilbert)	BORDERLINE (key doubtful)	CL-H22
UT Austin Invitational, fall (2020 season)	2019	C	Invitational	9	Epithet: published "On Magnetism"	Coulomb (doubtful)	BORDERLINE (key doubtful)	CL-H22

Test	Year	Div	Level	Q	Question (short)	Key	Check / scope 2027 B	Type
ATX (UT Austin) Regional	2026	B	Regional	MC10	Built fortifications in Martinique	Coulomb (Fort Bourbon)	IN	CL-H19
ATX (UT Austin) Regional	2026	C	Regional	MC6	Proved the inverse-square law for magnetic forces	Coulomb	IN as history; the physics is magnetism	CL-H17
Florida State Tournament	2021	B (= C)	State	6	Quote: "On graduating from the school, a studious young man who would withstand the tedium..."	Coulomb	IN (attribution not confirmed)	CL-H25
UT Austin Invitational, fall	2025	B/C	Invitational	4a	Pick the letter whose law has $1/r^2$	C	IN	CL-H26
UT Austin Invitational, fall	2025	B/C	Invitational	4d	Pick who did most work in France	A, V, C (KEY ERROR: A, C)	IN	CL-H26
F-M Invitational (NY), fall	2019	not stated	Invitational	1	Match 7 discoveries to 7 scientists	Coulomb row correct	IN for the Coulomb row	CL-H02

Easy

CL-H01. Law or statement → person

Law or statement → person · ✅ IN · Easy

- **Scope:** IN (3.c.i).
- **Where seen:** BullSO 2026 B Q1; UT Austin 2020 Inv B/C Q3 ("established the inverse square law of attraction and repulsion"); Rickards 2020 B Q24 / C Q28; GullSO 2026 Q36; ATX 2021 Reg B Q8.
- **Div B 2027 check:** GullSO Q36 and Rickards C Q28 ask only who made the electrostatic force law. Fits Div B 3.c.i. Pass.
- **How to solve:**
  1. Read the question: a description ("force $\propto q_1q_2/r^2$", "inverse-square law of charges") → "Coulomb".
  2. Inverse square + charges = Coulomb.
  3. Inverse square + magnetic poles = also Coulomb (see H17)
  4. Force between currents = Ampère.
- **Worked example:** "Who discovered the inverse relation between the force on charges and distance squared?" → **Coulomb**.
- **Traps:** Newton made the inverse-square law for gravity, not charge. Priestley guessed it first but did not measure it.
- Calculator: look up — type: keyword ("inverse square", "torsion", "charge").

CL-H02. Person → contribution (matching)

Person → contribution (matching) · ✅ IN · Easy

- **Scope:** IN.
- **Where seen:** PUSO (Princeton) 2026 C Matching Part 1 (2 → E "Formulated the law of electrostatic force between charges"); ATX 2020 Reg B/C matching; ATX 2021 Reg B Q2-8 (name bank).
  - Also seen (test-extraction files): F-M Invitational (NY), fall 2019, Div not stated, Invitational, Q1: "Match 7 discoveries to 7 scientists" (key: Coulomb row correct). Div B 2027: IN for the Coulomb row.

- **Div B 2027 check:** PUSO 2026 is Div C. The Coulomb row fits Div B. Its list holds Tesla and Faraday, who are not in the 2027 list. A 2027 Div B version would use Hertz and Boole. Pass for the Coulomb row.
- **How to solve:**
  1. Read the question: a list of 7 names and 7 statements → pairs.
  2. Tie Coulomb to "electrostatic force", "torsion balance", "unit of charge".
- **Worked example:** PUSO 2026: Coulomb ↔ "Formulated the law of electrostatic force between charges" → **E** (key confirms 2 → E).
- **Traps:** "first chemical battery" is Volta. "Junction and loop laws" is Kirchhoff.
- Calculator: look up — type: scientist name → fact card.

CL-H03. Unit of charge ↔ person, unit definition

Unit of charge ↔ person, unit definition · ✅ IN · Easy

- **Scope:** IN (3.c.i and 3.c.iv).
- **Where seen:** ATX 2021 Reg B Q2 ("The unit of charge bears my name" → Coulomb); Shock Value practice key Q61; CircuitLabRegTest 2013 (VA, C) Q29 ("Coulomb per second" → ampere); Coulomb is a distractor in ATX 2026 Reg B Q2 (unit of resistance → Ohm) and Rickards 2026 B Q22.
- **Div B 2027 check:** VA 2013 C Q29 (C/s = ampere) fits Div B 3.c.iv. Pass.
- **How to solve:**
  1. Read the question: "unit of charge" → coulomb (C); or "1 C = ?" → 1 A·s.
  2. C = A·s; A = C/s; V = J/C.
- **Worked example:** "What is a coulomb per second?" → **ampere**.
- **Traps:** farad (capacitance) is named after Faraday, not Coulomb. Capacitance is OUT in 2027.
- Calculator: look up — type: unit name. (App note: (unit table).)

CL-H04. Instrument → person (torsion balance)

Instrument → person (torsion balance) · ✅ IN · Easy

- **Scope:** IN.
- **Where seen:** ATX 2021 Reg B Q8 (torsion balance setup → Coulomb).
- **How to solve:**
  1. Read the question: "bar on a thin fibre with a charge on each end and a fixed charge" → Coulomb.
  2. Torsion balance → Coulomb (electricity, 1785)
  3. Same instrument for gravity → Cavendish (1798).
- **Worked example:** ATX 2021 B Q8 → **Coulomb**.
- **Traps:** Cavendish also used a torsion balance (for gravity). The galvanometer and astatic needle are Ampère-era tools.
- Calculator: look up — type: instrument name.

CL-H05. Name, nationality, life dates, profession

Name, nationality, life dates, profession · ✅ IN · Easy

- **Scope:** IN (historical perspective).
- **Where seen:** Rickards 2020 B Q25 / C Q30 ("full first name of the person the SI unit of charge is named after" → Charles-Augustin).
- **Div B 2027 check:** Rickards C Q30 is the same as B Q25. Pass.
- **How to solve:**
  1. Read the question: clue → "Charles-Augustin", "French", "1736-1806", "military engineer".
  2. Recall F1-F2, F6.
- **Worked example:** "How old was Coulomb when he died?" 1806 – 1736 = **70** (his birthday was in June, he died in August).
- **Traps:** "Charles" alone may lose credit when the key wants the hyphenated full first name. Ampère is the other French scientist in the list.
- Calculator: look up — type: name.

CL-L01. State or write Coulomb's law; name each symbol

State or write Coulomb's law; name each symbol · ✅ IN · Easy

- **Scope:** IN.
- **Where seen:** Wichita Heights 2026 C Q9; ATX 2021 Reg B Q10; cyanophycean 2 (practice C) Q8.
- **Div B 2027 check:** fits Div B 3.c.ii. Pass.
- **Worked example: $F = k \cdot q_1 \cdot q_2 / r^2$**; F in N, q in C, r in m, $k = 8.99 \times 10^9$ N·m²/C².
- **Traps:** r^2, not r. r is center-to-center distance.
- Calculator: look up — type: the keyword from the question.

CL-L02. Proportional reasoning (scaling)

Proportional reasoning (scaling) · ✅ IN · Easy

- **Scope:** IN.
- **Where seen:** Pembroke Hill 2026 B Q9 (distance tripled → force ÷ 9, key confirms); Berkeley 2026 B §1(d) (charge tripled → force tripled).
- **Worked example:** distance ×3 → **1/9**. Both charges ×2 and distance ×2 → **×1**. One charge ×3 and distance ×½ → **×12**.
- **Traps:** squaring the charge factor; forgetting to square the distance factor.
- Calculator: calculate — type: q_1 factor, q_2 factor, r factor.

CL-L06. Value of k; compare magnitudes

Value of k; compare magnitudes · ✅ IN · Easy

- **Scope:** IN.
- **Where seen:** Shock Retryout (B practice) Q19 (value of the Coulomb constant); Princeton 2021 C Q29 (greatest magnitude: e, k, Earth's field, proton mass → k).
- **Div B 2027 check:** Princeton Q29 has an Earth-magnetic-field choice (magnetism, OUT for 2027). The k part passes. A 2027 Div B version would drop that choice.
- **Worked example: $k \approx 8.99 \times 10^9$ N·m²/C²**, much larger than e (1.6×10^{-19} C) or the proton mass (1.67×10^{-27} kg).
- Calculator: look up — type: (constants table).

Medium

CL-H06. Year of the law; timeline ordering

Year of the law; timeline ordering · ✅ IN · Medium

- **Scope:** IN.
- **Where seen:** Princeton 2020 C Q1 (order the seven scientists, write years; key gives "Coulomb (1786)").
- **Div B 2027 check:** the task fits Div B 3.c.i. The 2020 list had Tesla and Faraday. The 2027 Div B list is Ampère, Coulomb, Kirchhoff, Volta, Ohm, Hertz, Boole. Pass with the new list.
- **How to solve:**
  1. Read the question: names → order and years.
  2. Coulomb 1785 → Volta 1800 → Ampère 1820-1826 → Ohm 1827 → Kirchhoff 1845 → Boole 1847/1854 → Hertz 1886-1888.
- **Worked example:** "How many years after Coulomb's law was Ohm's law published?" 1827 − 1785 = **42**.
- **Traps:** Coulomb is always first of the seven. Some keys write 1786 or 1787 (later memoirs). Write 1785 and add "1785-1791 memoirs" if the box allows.
- Calculator: look up + calculate — type: names. CALCULATE only for year differences. (App note: (timeline table).)

CL-H07. Explain how the torsion balance measures force

Explain how the torsion balance measures force · ✅ IN · Medium

- **Scope:** IN.
- **Where seen:** ATX 2021 Reg B Q9 (2 pts), Q10 (write the law: $F = kQq/r^2$).

- **How to solve:**
  1. Read the question: a diagram or description → explanation.
  2. (1) the force between the balls pushes the needle.
  3. (2) the wire twists.
  4. (3) twist angle ∝ force (calibrated)
  5. (4) turn the top micrometer to bring the balls to a set separation; total twist = force reading.
  6. (5) change the distance and repeat.
- **Worked example:** "Why does a bigger twist mean a bigger force?" → **The wire's restoring torque is proportional to the twist angle, so at rest the twist angle is proportional to the electric force.**
- **Traps:** the ATX key wants "oscillation". For repulsion Coulomb read the static twist. He used oscillation mainly for attraction (see section 4). Mention both.
- Calculator: look up — type: none. (App note: (stored explanation).)

CL-H15. Compare Coulomb's law with Newton's gravity

Compare Coulomb's law with Newton's gravity · ✅ IN · Medium

- **Scope:** IN.
- **Where seen:** not seen in a test (MagLab biography text).
- **How to solve:**
  1. Read the question: "Name one likeness and one difference." → both inverse square along the center line; charge vs mass; electric can repel.
  2. Recall F46-F47.
- **Worked example:** double the distance in both laws → both forces become ¼ (Python: $1/2^2 = 0.25$).
- **Traps:** electric force is far stronger than gravity for particles.
- Calculator: look up — type: the keyword from the question.

CL-H16. Was the unit named in his lifetime? (T/F)

Was the unit named in his lifetime? (T/F) · ✅ IN · Medium

- **Scope:** IN.
- **Where seen:** Princeton 2021 C Q7 ("The unit coulomb was one of the only units named after a scientist not named posthumously" → False).
- **Div B 2027 check:** unit and naming history fit Div B 3.c.i. Pass.
- **How to solve:**
  1. Died 1806; unit named 1881.
  2. 1881 − 1806 = **75** years after death (Python).
- **Worked example:** Princeton 2021 Q7 → **False**.
- **Traps:** some sources say 1880. Either way it was after his death.
- Calculator: look up — type: unit name → year named, death year.

CL-H22. "Which is NOT Coulomb's" and distractor elimination

"Which is NOT Coulomb's" and distractor elimination · ✅ IN · Medium

- **Scope:** IN.
- **Where seen:** Coulomb is a distractor in NV 2026 C Q49 (methane → Volta), ATX 2026 Reg C Q4 (methane), Q5 (stationary waves in the Earth → Tesla), Q9, Q12 (botany after father's death → Ampère); ATX 2026 Reg B Q6 (key C: Ampère); Princeton 2021 C Q29.
  - Also seen (test-extraction files): UT Austin Invitational (2020 season) 2019, Div B, Invitational, Q9: "Epithet: published "On Magnetism"" (key: Coulomb (doubtful: De Magnete is Gilbert)). Div B 2027: BORDERLINE (key doubtful).
  - Also seen (test-extraction files): UT Austin Invitational, fall (2020 season) 2019, Div C, Invitational, Q9: "Epithet: published "On Magnetism"" (key: Coulomb (doubtful)). Div B 2027: BORDERLINE (key doubtful).
- **Div B 2027 check:** Coulomb-distractor items whose answer is Tesla or Faraday fail Div B 2027 (those people are gone). Items with answer Volta, Ampère, Kirchhoff, Ohm pass.

- **How to solve:**
  1. Build a "not Coulomb" list: battery, methane, solenoid, electrodynamics, current law, Ohm's law, radio waves, Boolean algebra.
- **Worked example:** NV 2026 Q49 "First to discover and isolate methane gas" → **Volta**, not Coulomb (the NV key says B, which looks wrong; see section 4).
- **Traps:** Ampère's father was executed in the Revolution; Coulomb's was not. Both men are French.
- Calculator: look up — type: (fact → person index).

CL-H23. Essay: argue Coulomb's importance

Essay: argue Coulomb's importance · ✅ IN · Medium

- **Scope:** IN.
- **Where seen:** Princeton 2020 C Q3 ("who most deserved a Nobel Prize", 6 pts: claim, evidence, analysis).
- **Div B 2027 check:** the essay format fits Div B 3.c.i if limited to the 2027 list. Pass.
- **How to solve:**
  1. Claim; evidence (first measured force law of electricity, 1785; torsion balance; surface charge; the base of Poisson's theory); analysis (made electricity quantitative, like Newton for gravity).
- **Worked example (model answer): "Coulomb: he turned electricity into a measured science. His torsion balance gave the first direct force law (1785), on which all later field theory builds."**
- **Traps:** no evidence, or facts from other scientists.
- Calculator: look up — type: (outline template).

CL-H17. Magnetism: inverse-square law for poles

Magnetism: inverse-square law for poles · ⚠️ Borderline · Medium

- **Scope:** BORDERLINE (lean OUT). Rule 3.c.i now says "electricity discoveries" and all magnetism was removed in 2027.
- **Where seen:** ATX 2026 Reg C Q6 ("Who proved the inverse square law applied to magnetic fields?" → Coulomb).
- **Div B 2027 check: fails** the 2027 Div B rules (magnetism removed; 2026 Div B also had magnetism, 2027 does not). Keep as a one-line fact only.
- **Worked example:** ATX 2026 C Q6 → **Coulomb**.
- Calculator: look up — type: the keyword from the question.

CL-H18. Friction (Amontons-Coulomb laws)

Friction (Amontons-Coulomb laws) · ⚠️ Borderline · Medium

- **Scope:** BORDERLINE. Not electricity. Likely only as "Coulomb is also known for" or a distractor.
- **Where seen:** not seen in a test as a Coulomb item.
- **Worked example:** "Coulomb's 1781 prize memoir was about?" → **friction (Théorie des machines simples)**.
- **Traps:** Amontons (1699) came first; Coulomb extended the work.
- Calculator: look up — type: the keyword from the question.

CL-H19. Military engineering, Martinique, soil mechanics, ergonomics

Military engineering, Martinique, soil mechanics, ergonomics · ⚠️ Borderline · Medium

- **Scope:** BORDERLINE (biography, not an electricity discovery). A Div B test did ask it.
- **Where seen:** ATX 2026 Reg **B** Q10 ("Who built structural fortifications in Martinique?" → Coulomb; key A).
- **Worked example:** ATX 2026 B Q10 → **Coulomb** (Fort Bourbon, 1764-1772). Years there: 1772 – 1764 = **8** (Python).
- **Traps:** Wikipedia says health problems came from "three years" there; DSB says he stayed 1764-1772.
- Calculator: look up — type: the keyword from the question.

CL-L03. Force between two point charges, with attract/repel

Force between two point charges, with attract/repel · ✅ IN · Medium

- **Scope:** IN.

- **Where seen:** BullSO 2026 B Q11 (+6 μC, −7 μC, 0.5 m, k = 9 × 10^9 → 1.512 N attractive; key A); Rickards 2026 B Q33-34 (+1.5 nC, −2.0 nC, 1.5 cm → 1.2 × 10^{-4} N attractive); NV 2026 C Q3 (0.41 C, 0.67 C, 6067 m → C, 67 N); Wichita 2026 C Q12.
- **Div B 2027 check:** NV Q3 and Wichita Q12 are one-step algebra. Pass.
- **How to solve:**
  1. Convert prefixes (μ = 10^{-6}, n = 10^{-9}, cm → m)
  2. F = k|q_1q_2|/r^2; sign: unlike → attract.
- **Worked example:** BullSO: 9e9 × 6e-6 × 7e-6 / 0.5^2 = **1.512 N, attractive** (with k = 8.99 × 10^9: 1.510 N). NV: 8.99e9 × 0.41 × 0.67 / 6067^2 = **67.1 N** → C.
- **Traps:** not squaring r; cm not converted; reporting a negative "magnitude".
- Calculator: calculate — type: q_1 (with prefix), q_2, r (with unit), k choice (9e9 or 8.99e9). Output: F and attract/repel.

CL-L04. Sense of size: "Can 1 C at 1 m give 1 N?"

Sense of size: "Can 1 C at 1 m give 1 N?" · ✅ IN · Medium

- **Scope:** IN.
- **Where seen:** Princeton 2019 C Part 2(d) (answer no; F = 9 × 10^9 N).
- **Div B 2027 check:** one-step algebra. Pass.
- **Worked example:** 9e9 × 1 × 1 / 1^2 = **9 × 10^9 N**. So no. A coulomb is a huge static charge.
- Calculator: calculate — type: (same as L03).

CL-L05. Units and charge counting

Units and charge counting · ✅ IN · Easy-Medium

- **Scope:** IN (3.c.iv).
- **Where seen:** HH 2026 C Q4 (write units with C, J, s: voltage J/C, current C/s, power J/s, resistance J·s/C^2; part c capacitance C^2/J); Berkeley 2026 B Q5 (e to nearest power of ten → 10^{-19}); Shock value test Q27-28 (charge of proton/electron); Rickards 2020 C Q7 (1 A = 1 electron per second? → False).
- **Div B 2027 check:** HH parts a, b, d, e pass. **Part c (capacitance) fails**: capacitance is excluded in 2027 Div B (3.d).
- **Worked example:** 2000 mAh = 2 × 3600 = **7200 C**. Electrons in 1 C = **6.24 × 10^{18}**. Q for 2 A over 5 s = **10 C**.
- **Traps:** 1 A is 6.24 × 10^{18} electrons per second, not 1.
- Calculator: calculate — type: mAh or (I, t) or charge → C, electron count.

CL-H26. Multi-select: which listed scientists have property X?

Multi-select: which listed scientists have property X? · ✅ IN · Medium

- **Scope:** IN (3.c.i).
- **Where seen:** UT Austin Invitational fall 2025, Div B/C, Invitational, Q4a-d (letters A, C, K, V, O, T, F; pick every letter that fits). Coulomb is the right letter in 4a (law with 1/r^2), and a right letter in 4c (steady state) and 4d (France).
- **How to solve:**
  1. Write the letter code of each listed scientist (A = Ampère, C = Coulomb, K = Kirchhoff, V = Volta, O = Ohm, Hz = Hertz, B = Boole).
  2. Test each scientist against the property, one at a time.
  3. Pick every scientist that passes. A multi-select question can have more than one right letter.
  4. Check each letter again. Keys often add a wrong letter (UT 2025 put Volta under "France").
- **Worked example:** "Whose law has a 1/r^2 dependence? A, C, K, V, O" → **C (Coulomb)**: F = k·q_1·q_2/r^2.
- **Traps:** Ampère's force between wires goes as 1/r, not 1/r^2. Newton's gravity is also 1/r^2, but Newton is not a choice.
- Calculator: look up — type: the property (e.g. "1/r^2", "France").

Hard

CL-H09. Analyse the 1785 data: show the inverse-square law

Analyse the 1785 data: show the inverse-square law · ✅ IN · Hard

- **Scope:** IN (algebra, ratios).
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: table of separation and total torsion → exponent or "inverse square".
  2. Force ∝ total torsion.
  3. Distance ∝ separation angle.
  4. Compare ratios: if distance ÷2 gives force ×4, then $F \propto 1/r^2$.
- **Worked example:** separations 36°, 18°, 8.5°; total torsion 36°, 144°, 575.5°. Python: 144/36 = **4.00** for distance ×½. 575.5/144 = **3.997** for distance 18 → 8.5 (≈ ×½). So force ∝ $1/r^2$.
- **Traps:** 126° and 567° are the micrometer twists, not total torsion. Add the separation angle. Strictly, 18/8.5 = 2.12 and $2.12^2 = 4.48$, so a perfect law would need about 646° total. Coulomb's 8.5° is a little small. He blamed charge leakage.
- Calculator: calculate — type: d_1, F_1, d_2, F_2 → ratio F_2/F_1, ratio $(d_1/d_2)^2$, exponent $n = \log(F_2/F_1)/\log(d_1/d_2)$. (The app can use log internally; a student answer needs only ratios.).

CL-H13. His other electricity discoveries

His other electricity discoveries · ✅ IN · Medium-Hard

- **Scope:** IN (also ties to 3.c.ii static electricity).
- **Where seen:** not seen in a test as a Coulomb item. Princeton 2019 C Part 2 asks related charge properties.
- **How to solve:**
  1. Read the question: clue → fact. "Where does charge sit on a conductor?" → outer surface. "How fast is charge lost?" → in proportion to the charge; more in humid air.
  2. Recall F42-F45.
- **Worked example:** "Which tool did Coulomb use to sample surface charge?" → **proof plane**.
- **Traps:** Faraday's ice-pail and cage experiments (1830s) also show surface charge; Faraday is not in the 2027 list. Humidity link: static is worse in dry air.
- Calculator: look up — type: the keyword from the question.

CL-H14. Priority: who came before Coulomb

Priority: who came before Coulomb · ✅ IN · Hard

- **Scope:** IN (historical perspective of his discovery). The rules do not name the precursors, so this is rare.
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: "Who first guessed the inverse-square law for charge?" → Priestley (1767).
  2. Priestley 1767 guess.
  3. Robison 1769 exponent 2.06.
  4. Cavendish early 1770s, unpublished.
  5. Michell had the torsion idea.
  6. Coulomb 1785 published measurements.
- **Worked example:** "Why is the law named for Coulomb if Cavendish found it first?" → **Coulomb published direct measurements; Cavendish did not publish.**
- **Traps:** Cavendish's 1798 torsion experiment measured gravity.
- Calculator: look up — type: the keyword from the question.

CL-H08. The torsion law of wires

The torsion law of wires · ⚠️ Borderline · Hard

- **Scope:** BORDERLINE. It is the physics of his instrument, but it is mechanics, not an "electricity discovery".

- **Where seen:** not seen in a test.
- **How to solve:**
  1. Torque ∝ θ·D⁴/L.
- **Worked example:** double the wire diameter → torque per degree × 2^4 = **16**. Double the length → ×½ (Python).
- **Traps:** his 1777 memoir used D^3; the 1784 memoir corrected it to D^4.
- Calculator: calculate — type: D ratio, L ratio → stiffness ratio.

CL-H20. Prizes and Academy election

Prizes and Academy election · ⚠️ Borderline · Hard

- **Scope:** BORDERLINE (the 1777 prize memoir introduced torsion suspension, which ties to the electrical work).
- **Where seen:** not seen in a test.
- **Worked example:** "Coulomb shared the 1777 Academy prize for a memoir on?" → **magnetic compass needles**. "What got him elected to the Academy in 1781?" → **the friction memoir**.
- **Traps:** 1781 "sole" vs "double" prize: sources disagree (section 5).
- Calculator: look up — type: the keyword from the question.

CL-H21. Life anecdotes

Life anecdotes · ⚠️ Borderline · Hard

- **Scope:** BORDERLINE (trivia).
- **Where seen:** not seen in a test.
  - Also seen (test-extraction files): UT Austin Invitational (2020 season) 2019, Div B, Invitational, Q4: "Epithet: fled Paris around 1800" (key: Coulomb). Div B 2027: IN (he left Paris in the Revolution, 1790s, for Blois).
  - Also seen (test-extraction files): UT Austin Invitational, fall (2020 season) 2019, Div C, Invitational, Q4: "Epithet: fled Paris during the French Revolution" (key: Coulomb). Div B 2027: IN.
- **Facts:** one week in prison (1783); retired to Blois with Borda; weights and measures committee; Institut 1795; inspector general of public instruction (lycées) 1802; never called himself "de Coulomb"; name on the Eiffel Tower; did not die poor.
- **Worked example:** "Which scientist helped set up France's lycée system?" → **Coulomb**.
- Calculator: look up — type: the keyword from the question.

CL-H25. Quote attribution (who said it?)

Quote attribution (who said it?) · ✅ IN · Hard

- **Scope:** IN (3.c.i).
- **Where seen:** Florida State Tournament 2021, Div B (same as Div C), State, Q2-Q8 (seven quotes, 2 points each, "identify the last name"). Q6 is keyed Coulomb: "On graduating from the school, a studious young man who would withstand the tedium and monotony of his duties has no choice but to lose himself in some branch of science or literature completely irrelevant to his assignment."
- **How to solve:**
  1. Look for one key word or idea in the quote (an invention, a place, a belief, a language).
  2. Match that idea to one of the seven listed scientists.
  3. If the quote is in French, think of the two French scientists first: Ampère and Coulomb.
  4. If nothing fits the seven, the answer may be a scientist who is not on the 2027 list (Tesla, Faraday). Those items are out of scope in 2027.
  5. Clues for Coulomb: "the school" = the military engineering school at Mézières; dull army duties; a young officer turning to science.
- **Worked example:** "On graduating from the school, a studious young man ... has no choice but to lose himself in some branch of science." → **Coulomb** (Florida 2021 key). He trained at Mézières and served as an army engineer.
- **Traps:** Ampère is the other French scientist, but he was never an army engineer. The attribution is not confirmed (see "Not confirmed").
- Calculator: look up — type: a key word from the quote (e.g. "graduating", "duties").

Nationals

CL-H10. Predict the twist for a new separation

Predict the twist for a new separation · ✅ IN · Nationals

- **Scope:** IN.
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: one reading (separation, total torsion), new separation → micrometer twist needed.
  2. $\text{Total}_2 = \text{total}_1 \times (d_1/d_2)^2$. Micrometer twist = $\text{total}_2 - d_2$.
- **Worked example:** balls rest at 36° with total torsion 36°. For 12°: total = $36 \times (36/12)^2$ = **324°**. Micrometer twist = 324 − 12 = **312°** (Python).
- **Traps:** forgetting to subtract the separation angle; using $1/r$ instead of $1/r^2$.
- Calculator: calculate — type: d_1, total_1, d_2.

CL-H11. Why the oscillation method for attraction?

Why the oscillation method for attraction? · ✅ IN · Nationals

- **Scope:** IN.
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: "Why was attraction hard to measure with the static balance?" → explanation.
  2. Electric force grows as $1/r^2$ when balls come closer.
  3. Wire torque grows only linearly with twist.
  4. So attraction gives unstable balance: the balls snap together and lose their charge.
  5. He timed oscillations instead: the period is proportional to distance when the force is inverse square.
- **Worked example:** "If the period of oscillation doubles, the distance has?" → **doubled** (period $\propto$ distance).
- **Traps:** saying "he could not measure attraction at all". He did, mostly by oscillation.
- Calculator: look up + calculate — type: T_1, T_2, d_1. (App note: Ratio form: CALCULATE (period ratio = distance ratio).)

CL-H12. What Coulomb proved versus what he assumed

What Coulomb proved versus what he assumed · ✅ IN · Nationals

- **Scope:** IN.
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: "Did Coulomb measure $F \propto q_1 q_2$?" → No; he measured $1/r^2$ and stated the charge product.
  2. Recall F41.
- **Worked example:** T/F "Coulomb defined a unit of charge." → **False** (he used only relative "electric mass").
- **Traps:** the unit named after him does not mean he defined it.
- Calculator: look up — type: the keyword from the question.

Out of scope (skip)

These types appear in tests or books, but the 2027 Div B rules exclude them. Skip them. They stay here so you can recognise them.

CL-H24. Maxwell's equations / Gauss's law link

Maxwell's equations / Gauss's law link · ❌ Out · Medium

- **Scope:** OUT. Needs Maxwell's equations (Faraday induction, Gauss's law for magnetism). Magnetism was removed in 2027 and Gauss's law is not in 3.c.
- **Where seen:** Rickards 2020 B Q10 / C Q12 ("Which is NOT one of Maxwell's equations?" → Coulomb's law); ATX 2020 Reg B/C matching ("law can be used to derive Gauss's law" → Coulomb).
- **Div B 2027 check:** Rickards C Q12 **fails** for 2027 Div B.

- Calculator: look up — type: only if the app keeps a "past-test trivia" page.

CL-O01. Coulomb friction, Mohr-Coulomb, Coulomb damping calculations

Coulomb friction, Mohr-Coulomb, Coulomb damping calculations · ✗ Out · difficulty not given

- **Scope:** OUT. Mechanics, not electricity (3.c lists electricity areas only).
- **Where seen:** not seen in a test.
- Calculator: none (skip).

CL-O02. Charge on a capacitor (Q = CV) in coulombs

Charge on a capacitor (Q = CV) in coulombs · ✗ Out · difficulty not given

- **Scope:** OUT. Capacitance excluded (3.d).
- **Where seen:** BullSO 2026 B/C Q31 (804 C at 12 V → 67 F); ATX 2019 Reg capacitor charge; UT 2020 Inv B/C μC capacitor items.
- Calculator: none (skip).

Mistakes in tests and keys

1. **Princeton 2020 C key, Q1**: writes "Coulomb (1786)". The first memoir with the inverse-square law is 1785 [W] [DSB]. 1786 is the fourth memoir (surface charge). The key also gives Volta 1799 (pile built late 1799, announced 1800).
2. **Prior file G_tests.md line 156** says Princeton 2020 Q1 gives "Coulomb 1785". The key actually says 1786.
3. **ATX 2021 Reg B/C key Q9** gives half credit without "oscillation". For repulsion (1785 first memoir), Coulomb read the static twist angle. He used oscillation mainly for attraction [DSB]. A static-twist answer is correct.
4. **DSB (Gillmor) text**: says the first memoir showed the law "for the case of two bodies of opposite electrical charge", then says he limited it to repulsive forces. The first memoir was like charges (repulsion) [W][SK]. The "opposite" is a slip.
5. **Wikipedia biography** says the unit was named in 1880. The Wikipedia "Coulomb" unit page says 1881 (International Electrical Congress). The two pages disagree.
6. **BullSO 2026 B Q11** choices give a negative magnitude ("−1.512") and no unit (N).
7. **Shock Retryout test Q19** says the coulomb is named after Coulomb "who discovered all of this" (the ampere and the elementary charge). He did not. The electron was found in 1897 and he never defined a unit of charge [DSB].
8. **Prior file F_slides.md line 168** says "2027 adds Coulomb". Coulomb was in the 2026 list too (A_rules.md line 157). 2027 adds Hertz and Boole.
9. **Prior file I_bank.md** gives "Seven memoirs in total, 1785-1789". This uses Academy volume years. DSB gives reading dates 1785-1791. Both appear in sources.
10. **NV 2026 C key Q49** (methane): key B (Ohm). Volta found methane (1776). Already flagged in G_tests.md; still stands.
11. **MagLab** spells Blois as "Blous" and Montpellier as "Montpelier".
12. **skullsinthestars** writes 567° once and 576° once for trial 3. 567° is the micrometer twist; 575.5° is the total. Its 1/340 vs 1/3400 grain figures also disagree.

- **UT Austin Invitational 2019, Div B, Q9 and fall 2019 Div C, Q9:** "Published On Magnetism", key Coulomb. The famous book *De Magnete* ("On the Magnet", 1600) is by William Gilbert. Coulomb wrote memoirs on electricity and magnetism. Key doubtful.
- **UT Austin Invitational fall 2025, Q4d:** key A, V, C for "France". Right answer A, C.

Not confirmed

- **1781 prize, sole or shared**: DSB says "double first prize"; MagLab says "sole recipient"; MacTutor says he won it. Not settled. Looked: DSB PDF, MagLab, MacTutor, one web search.
- **1777 co-winner Van Swinden**: in a search snippet and prior I_bank.md only. MacTutor and DSB do not name him. Not checked in a primary source.

- **Year the unit was named, 1880 vs 1881**: two Wikipedia pages disagree. The arXiv Daniell-cell paper and other search results point to the 1881 Paris congress. I did not read the 1881 congress minutes.
- **Memoir dates (1785-1789 vs 1785-1791)**: depends on reading vs publication date. Not checked against the Academy volumes.
- **Exact 1785 data**: 36/18/8.5 and 126/567 come from skullsinthestars and a Spanish paper (search results). The Martinez replication paper (2006) says the results are hard to reproduce. I did not read the 1785 memoir itself.
- **Martinique "three years" of illness (Wikipedia) vs eight-year stay (DSB)**: both may be true (illness during part of the stay). Not settled.
- **ATX 2026 Reg B Q6 key (Ampère forgot his own name and house)**: I did not verify this anecdote. It is not a Coulomb fact.
- **Eiffel Tower side for his name**: not checked.
- **Legion of Honour**: no source says he got it. DSB says Croix de St. Louis. "Couldn't find it."
- **2026 Nationals Div B test**: paid packet, not seen. Coulomb questions there are unknown.
- **No free state or national Circuit Lab test** was found (see G_manifest.md). Difficulty tags for "Nationals" are my estimate.

- Florida State 2021 Q6 quote ("On graduating from the school..."): keyed Coulomb. No source in `research/downloads/`. Not searched online in this run.

Sources

- 2027 Div B rules: /Users/kai/Daniel/Science_Olympiad_Div_B_Rules_2027_for_Web_Secured.pdf (pp. B11-B12)
- Wikipedia, Charles-Augustin de Coulomb: https://en.wikipedia.org/wiki/Charles-Augustin_de_Coulomb
- Wikipedia, Coulomb (unit): https://en.wikipedia.org/wiki/Coulomb
- Wikipedia, Coulomb's law: https://en.wikipedia.org/wiki/Coulomb%27s_law
- MacTutor biography: https://mathshistory.st-andrews.ac.uk/Biographies/Coulomb/
- Gillmor, Dictionary of Scientific Biography (PDF via MacTutor): https://mathshistory.st-andrews.ac.uk/DSB/Coulomb.pdf
- National MagLab, Coulomb: https://nationalmaglab.org/magnet-academy/history-of-electricity-magnetism/pioneers/charles-augustin-de-coulomb/
- National MagLab, torsion balance 1785: https://nationalmaglab.org/magnet-academy/history-of-electricity-magnetism/museum/torsion-balance-1785/
- Coulomb's 1785 experiment: https://skullsinthestars.com/2026/03/19/coulombs-remarkable-experiment-in-electricity-1785/
- Martinez, replication of Coulomb's torsion balance: https://martinezwritings.com/m/Coulomb.html
- Daniell cell and the SI units (unit history): https://arxiv.org/pdf/1512.07306
- Tests (local, /Users/kai/Daniel/circuitlab/research/downloads/): dlB/tests/bullso26_, *berk26_*, phi26_*; dlG/atx21reg_B*, atx21reg_C*, atx26reg_B*, atx26reg_C*, atx20reg_key.txt, ut20inv_B*, rsoiOldB_, *rsoiOldC_*, rsoi26B_, *puso26_*, gull26_test.txt, wich26C_test.txt, nv26C_, *hh26_*, mit20_key.txt, pz/p19, pz/p20, pz/p21, wiki/Shock_, *wiki/CircuitLabRegTest2013.txt, wiki/Circuit_Lab_2_cyanophycean314*.
- Prior files: /Users/kai/Daniel/circuitlab/research/prior/ (A_rules.md, B_topics.md, F_slides.md, G_tests.md, G_manifest.md, I_bank.md).
- Test-extraction files: `/Users/kai/Daniel/circuitlab/research/tests/` (atx-reg, misc-a, misc-b, nat26, state, ut-a, ut-b, ut-c) and `/Users/kai/Daniel/circuitlab/research/prior/G_tests.md`.
- Raw research file this note was built from: `/Users/kai/Daniel/circuitlab/research/subtopics/coulomb.md`.

Gustav Kirchhoff

Note: Kirchhoff

> **INFO: How to use this note**
> - Scope icons: ✅ IN = all levels. ✅ States-Nationals = only at State and National tournaments. ⚠️ Borderline = the rules do not settle it (the reason is given). ❌ Out = excluded (the rule is cited).
> - Question types go from Easy to Medium to Hard to Nationals. "Nationals" means the hardest depth, not a different scope.
> - Every number in a worked example was checked in Python.
> - Source tags in square brackets, like [W] or [M], are listed under "Sources".

Scope notes and source tags

Slug: `kirchhoff`. Area: History (rule 3.c.i). The circuit-analysis use of his laws is rule 3.c.ix (States/Nationals only).

Rule text (2027 Div B, manual pages B12-B13 in `Science_Olympiad_Div_B_Rules_2027_for_Web_Secured.pdf`, checked with pdftotext):

> 3.c.i. Historical perspective of the electricity discoveries made by Ampere, Coulomb, **Kirchhoff**, Volta, Ohm, Hertz, and Boole 3.c.ix. States/Nationals only: Circuit analysis using Kirchhoff's Voltage & Current Laws 3.e.i. Functions, algebraic manipulations, and systems of no more than two equations

How to read the scope tags in this file.

- The rule says "**electricity** discoveries". So his circuit laws and his 1857 signal-speed result are clearly IN.
- His spectroscopy, cesium/rubidium, black-body and thermal-radiation work are **not** electricity discoveries. The rules do not settle them. They are tagged BORDERLINE. But test writers ask them often, even at a Div B **regional** (ATX 2026 B regional Q8, black-body). Learn them anyway.
- Naming or stating a Kirchhoff law is history (IN at all levels). **Using** the laws to solve a circuit is IN-SN (States/Nationals only), and limited to two simultaneous equations.

Div C check (user request). Many questions below were first seen on Div C tests. Each one was checked against the Div B 2027 rule text above, not the Div C text. Div C 2027 differs in three ways that matter here: KVL/KCL is at **all** levels in C (C 3.c.vii) but States/Nationals only in B; C allows unlimited systems of equations, B allows at most two; C adds op-amps, BJTs and flip-flops, which are excluded in B. Div C's *Thermodynamics* event (not Circuit Lab) has black-body radiation and Stefan-Boltzmann as States/Nationals topics; that does not carry over to Circuit Lab B. Table 3.0 lists each Div C item and its Div B verdict.

Key facts

Source tags: [W] en.wikipedia.org Gustav_Kirchhoff; [M] MacTutor (St Andrews); [APS] APS News "This Month in Physics History", May 2018; [P] Purdue chemed history page; [Cs] Wikipedia Caesium; [Rb] Wikipedia Rubidium; [Bu] Wikipedia Robert_Bunsen; [FL] Wikipedia Fraunhofer_lines; [TR] Wikipedia Kirchhoff's law of thermal radiation; [CL] Wikipedia Kirchhoff's circuit laws; [SW] scioly.org Circuit Lab wiki (`downloads/dl/cl_wiki.txt` line 900-918); [T] a past test (named); [IB] prior `I_bank.md`; [S] other web source named in section 6. "V" = verified in two or more independent sources.

1a. Life, dates, places

Fact	Source
Full name **Gustav Robert Kirchhoff**. Spelling: **two h's, two f's** ("Kirchhoff"). Many tests misspell it "Kirchoff".	[W][M][SW] V
Born **12 March 1824**, **Königsberg, Prussia** (now **Kaliningrad, Russia**).	[W][M][SW] V
Father Friedrich Kirchhoff, a lawyer / law councillor. Mother Johanna Henriette Wittke.	[W][M] V

Fact	Source
Studied at the **University of Königsberg** (Albertus University). Attended the **Neumann–Jacobi seminar**, 1843-1846. Teacher/advisor **Franz Ernst Neumann**. Math teacher **Friedrich Richelot**.	[W][M] V
Graduated (doctorate) **1847**, Königsberg. Dissertation title = his 1845 paper title.	[M] (graduated 1847); Mathematics Genealogy Project [S]
Died **17 October 1887**, **Berlin**, age **63** (Python check). Buried Alter St.-Matthäus-Kirchhof, Schöneberg, Berlin.	[W][M][P] V
Nationality: **German (Prussian)**.	[W][M] V
Posts: **Berlin** Privatdozent 1847/48-1850 (unpaid) → **Breslau** extraordinary professor **1850-1854** (age 26) → **Heidelberg** professor of physics **1854-1875** → **Berlin** chair of mathematical (theoretical) physics **1875-1887**.	[W][M][P] V
Met **Robert Bunsen** in **Breslau** (Bunsen there 1851-52). Bunsen persuaded him to move to **Heidelberg** in **1854**.	[M][APS][P] V
Married **Clara Richelot** (daughter of his math professor) in **1857**. Five children (3 sons, 2 daughters). Clara died 1869. Married **Luise Brömmel** 1872.	[W][M] V
Had a disability. He spent much of his life on **crutches or in a wheelchair**. This pushed him toward theoretical work.	[M][P] V. Cause (an "accident") NOT confirmed — see section 5.
Wrote *Vorlesungen über mathematische Physik* (Lectures on Mathematical Physics), 4 volumes, 1876-1894.	[M]

1b. Electricity discoveries (clearly IN)

Fact	Source
1845: Kirchhoff's circuit laws, stated **while still a student** (at age 20-21) in Neumann's seminar. They **generalized Ohm's law** (1827) to networks with many loops. They came **before Maxwell's** equations.	[W][M][CL] V
First paper: "Ueber den Durchgang eines elektrischen Stromes durch eine Ebene, insbesondere durch eine kreisförmige" (On the passage of an electric current through a plane, especially a circular one), *Annalen der Physik und Chemie*, 1845, pp. 497-514. Volume is cited as **64** (Poggendorff series) or **140** (continuous numbering). These are the same volume (offset 76).	[W][S Heidelberg KIP sheet][S IEEE CAS Mag. 2023]
Current law (KCL) = Kirchhoff's **first law** = **junction rule** = **node rule** = point rule. "The algebraic sum of currents in a network of conductors meeting at a point is zero." Sum of currents in = sum of currents out. Basis: **conservation of charge**.	[CL][T ATX 2021 B Q5][T Berkeley 2026 B Q17] V
Voltage law (KVL) = Kirchhoff's **second law** = **loop rule** = **mesh rule**. "The directed sum of the potential differences (voltages) around any closed loop is zero." Basis: **conservation of energy** (the textbook statement; Wikipedia instead derives it from the Maxwell-Faraday equation).	[CL][T Berkeley 2026 B Q18] V
Nodal analysis is built on KCL. **Mesh analysis** is built on KVL.	[SW][T UT 2020 B Q12]
1847: paper on solving the circuit equations of a network ("Über die Auflösung der Gleichungen ... galvanischer Ströme", *Ann. Phys. Chem.* 72, 497-508). Source of the **matrix-tree theorem** (spanning trees in graph theory). A 2016 review says he did not explicitly count spanning trees.	[W][S Croatica Chemica Acta 2016]
1848-1850 (Berlin): corrected the understanding of current in conductors. He showed that Ohm's "electroscopic force" (tension) is the same as electrostatic potential.	[M] (wording "corrected the accepted understanding"); exact claim from [SW] "redefined electrical potential differences" — partly unverified, see 5
1857: calculated that an electric **signal** on a **resistanceless (ideal) wire** travels at the **speed of light**. Paper: "On the motion of electricity in wires", *Phil. Mag.* 13 (1857). He and Weber treated it as a coincidence; Maxwell made the light-is-electromagnetic link ~5 years later.	[W][M][SW][T GullSO 2026 Q31][T Princeton 2021 Q6] V
No SI unit is named after Kirchhoff.	[IB]; checked against SI derived-unit list (none)

1c. Spectroscopy, chemistry, radiation (BORDERLINE for this event)

Fact	Source
1859, Heidelberg: with **Bunsen**, built the **Bunsen-Kirchhoff spectroscope** (prism spectroscope) and founded **spectrum analysis** (spectrochemical analysis). Bunsen's **Bunsen burner** (perfected **1855** with assistant **Peter Desaga**) gave a nearly colorless flame.	[APS][P][Bu][Cs] V
Caution: the spectroscope as an instrument goes back to **Fraunhofer (1814)**. Saying "Kirchhoff and Bunsen invented the spectroscope" is common (Wikipedia Caesium says so) but contested.	[W][S Spectroscopy Online]
Each element has its own unique spectrum (a "fingerprint").	[M][APS]
1859: matched the yellow **sodium** flame lines to the dark **D lines** in sunlight. Concluded the **Sun contains sodium**. Paper "Über die Fraunhofer'schen Linien", *Monatsberichte* Berlin Academy, 1859, pp. 662-665.	[W][FL][P][S] V
Explained the dark **Fraunhofer lines** as **absorption** by elements in the **Sun's atmosphere** (cooler gas in front of a hotter source). He identified about **30** elements in the Sun.	[M][FL][APS][P]
Dark solar lines: first recorded by **Wollaston (1802)**; mapped (570+ lines, letters A-K) by **Fraunhofer (1814)**. Sodium D2 = 588.995 nm, D1 = 589.592 nm.	[FL]
Kirchhoff's three laws of spectroscopy: (1) a hot dense solid/liquid/gas gives a **continuous** spectrum; (2) a hot thin gas gives a **bright-line (emission)** spectrum; (3) a cooler thin gas in front of a continuous source gives a **dark-line (absorption)** spectrum at the same wavelengths it would emit.	[W] (named only); statements standard astronomy wording [S]
Caesium (cesium), 1860 (Bunsen's notebook date often given as **10 May 1860** [APS]). Found in **Dürkheim** mineral water (Bad Dürkheim). **First element discovered with a spectroscope.** Two **bright blue** lines. Name from Latin ***caesius*** = bluish-grey / sky blue. About **44,000 L** of water evaporated → **7.3 g** CsCl ([Cs]); APS says **12,000 gallons** (= 45,425 L, Python).	[Cs][APS][Bu] V. Kirchhoff's own Wikipedia page says 1861 (error, see 4).
Rubidium, 1861, from the mineral **lepidolite**. **Deep red** lines. Latin ***rubidus*** = deep red. Bunsen isolated rubidium metal; caesium metal was first isolated by **Carl Setterberg, 1882**.	[Rb][APS][Bu][Cs] V
Law of thermal radiation, 1859-1860: at thermal equilibrium, a body's **emissivity equals its absorptivity** at each wavelength and temperature. Short form: **a good absorber is a good emitter**.	[W][TR][SW] V
Coined **"black body"** (perfect absorber of all radiation). Year: **1860** [W, citing Annalen 185 (1860)] or **1862** [M][P][APS-era sources]. Both appear; accept either.	conflict — see 4
The black-body problem he posed led **Max Planck (1900)** to the quantum.	[W][TR]
Kirchhoff's law of thermochemistry (1858): how reaction heat changes with temperature (difference in heat capacities).	[W]
Also: Kirchhoff's **diffraction** formula (optics, Huygens' principle); **Kirchhoff-Love plate theory** (1850, elastic plates, Breslau).	[W][M]

1d. People links, honors, anecdotes

Fact	Source
Students/attendees: **Heinrich Hertz** (studied in Berlin under Kirchhoff and Helmholtz), **Max Planck**, Gabriel Lippmann, Loránd Eötvös, Arthur Schuster, Max Noether, Kamerlingh Onnes, Sofia Kovalevskaya.	[W][IB]
Kirchhoff died (Oct 1887) during Hertz's radio-wave work (1886-1888).	Python date compare; [W]
Rumford Medal 1862; Pour le Mérite 1874; Cothenius 1876; Matteucci 1877; **Davy Medal 1877** (with Bunsen, the first Davy Medal, "for their researches & discoveries in spectrum analysis"). FRS 1875; FRSE 1868.	[W][APS][M]
Anecdote: a banker asked "of what use is gold in the Sun if it cannot be brought to Earth?" Kirchhoff later deposited his prize money (a medal/prize in gold sovereigns) with him: "**Here is gold from the Sun.**"	[W]
Lunar crater **Kirchhoff**; Bunsen-Kirchhoff Award.	[M][W]
Bunsen: born 1811 Göttingen, died 1899 Heidelberg; lost sight of right eye (cacodyl explosion); never married.	[Bu][APS]

Formulas

Item	Form	Units	Scope
KCL	Σ I_in = Σ I_out (or Σ I = 0 at a node, signed)	A	IN-SN (state it: IN)
KVL	Σ ΔV = 0 around any closed loop; rises = drops	V	IN-SN (state it: IN)

Item	Form	Units	Scope
Sign rule	Battery – → +: +V (rise). Resistor in current direction: –IR (drop).	V	IN-SN
Ohm's law (used with KVL)	V = I R	V, A, Ω	IN
Speed of light	$c = 2.998 \times 10^8$ m/s ≈ 3.00×10^8 m/s	m/s	BORDERLINE use (K13)
Travel time of a signal (ideal wire)	t = d / c	s	BORDERLINE
Kirchhoff radiation law	ε(λ,T) = α(λ,T) (emissivity = absorptivity)	unitless	BORDERLINE (statement only)
Age	age = event year – 1824 (–1 if before 12 March)	years	IN
Stefan-Boltzmann $P = \varepsilon\sigma AT^4$, Wien $\lambda_{max} T = b$	—	—	**OUT** (no rule area; Div C Thermodynamics only)

Question types

This note has **31 question types** (17 ✅ IN, 8 ⚠️ Borderline, 3 ✅ States-Nationals, 3 ❌ Out). The out-of-scope ones are in "Out of scope (skip)" below.

How the types were found

3.0 Div C (and other-rules) source items checked against Div B 2027

Item seen	Test (Div)	Div B 2027 verdict
Matching "Kirchhoff → junction law and loop law"	Princeton 2026 Matching Part 1 #3 (C)	IN (history, 3.c.i)
"Created laws generalizing the work of Ohm"	ATX 2020 regional matching (B/C shared key)	IN
"Signals in an ideal wire travel at c → Kirchhoff"	GullSO 2026 Q31 (C)	IN (electricity discovery)
T/F "in a wire with no resistance, electricity travels at the speed of light" (key True)	Princeton 2021 Q6 (C)	IN
"Coined black-body radiation"	MIT 2020 Q2 (C); Rickards 2020 C Q26	BORDERLINE (not an electricity discovery); also asked on Div B tests
T/F "Kirchhoff proved the moon contained sodium" (False)	Rickards 2020 C Q1 (also B Q1)	BORDERLINE
Order 7 scientists by date (Kirchhoff 1845)	Princeton 2020 Q1 (C)	IN, but Tesla and Faraday are **OUT in 2027** (removed). Re-do with Hertz and Boole.
"State Kirchhoff's laws"	MIT 2021 Q98 (C)	IN (stating = history)
KVL/KCL statements, fill-in 0 V	Northview 2026 Q21-22 (C)	IN (statement); IN-SN if a number must be computed from a circuit
"Using only Kirchhoff's laws" find a drop	UT 2020 C Q74 (same as B Q68)	IN-SN, only if ≤ 2 unknowns (B math limit)
KVL/KCL with an op-amp box	UT 2025 Inv. FRQ (B/C)	**OUT** (op-amps excluded in B)
"Kirchhoff's laws are not violated by op-amps (power pins)"	ATX 2021 C key	**OUT**
Kirchhoff as distractor for methane, Sandemanians, stationary waves	ATX 2026 C Q4, Q5, Q14	the Kirchhoff part is IN; the asked person (Volta) IN, (Faraday, Tesla) **OUT 2027**
Kirchhoff "Junction Rule" in a static-shock question	ATX 2020 B/C Q49/Q53	IN (as a distractor)

History questions in the test-extraction files (8 test groups)

These rows come from `research/tests/` (ATX regionals, UT invitationals, misc-a, misc-b, USC 2026, Florida State 2021). Each row names the question type it belongs to.

Test	Year	Div	Level	Q	Question (short)	Key	Check / scope 2027 B	Type
ATX (UT Austin) Regional	2021	B	Regional	5	"Algebraic sum of currents at a point is zero" (name bank)	Kirchhoff	IN	K 2
ATX (UT Austin) Regional	2020	B	Regional	6	Generalized Ohm's work	Kirchhoff	IN	K 3
UF Northern Regional (Florida)	2020	B	Regional	7	Who formulated KCL?	C Kirchhoff	IN	K 2
UF Northern Regional (Florida)	2020	C	Regional	1	___ says current in = current out at a node	Kirchhoff's law	IN	K 5
Florida State Tournament	2021	B (= C)	State	4	Quote: "Look here, I have succeeded at last in fetching some gold from the sun... All is done and understood"	Kirchhoff (half-wrong: second part is usually credited to Philipp von Jolly)	IN	K 30
UT Austin Invitational, fall	2025	B/C	Invitational	4c	Pick the letters whose laws are useful in steady state	A, C, K, V, O	IN	K 31
ATX (UT Austin) Regional	2026	B	Regional	MC 8	Breakthrough in blackbody radiation	Kirchhoff	BORDERLINE in this note (not an electricity discovery), but asked at a Div B regional	K 20
UT Austin Invitational (2020 season)	2019	B	Invitational	7	Epithet: showed the Sun has sodium	Kirchhoff	BORDERLINE (spectroscopy), asked on Div B	K 18
UT Austin Invitational, fall (2020 season)	2019	C	Invitational	7	Epithet: showed the sun has sodium	Kirchhoff	BORDERLINE	K 18
F-M Invitational (NY), fall	2019	not stated	Invitational	1	Match 7 discoveries to 7 scientists	Kirchhoff row correct	IN	K 3

Easy

K1. Spell or write the name

Spell or write the name · ✅ IN · Easy

- **Scope:** IN (3.c.i).
- **Where seen:** ATX 2021 regional B (and C), name bank "spelled and capitalized exactly as written"; Q5 answer "Kirchhoff". Rickards 2020 B Q22 "(Last name only)".
- **How to solve:**
  1. Read the question: a description → the last name, spelled right.
  2. Write K-i-r-c-h-h-o-f-f. Two h, two f.
- **Worked example:** "I found that the algebraic sum of currents in a network of conductors meeting at a point is zero." → **Kirchhoff**.
- **Traps:** "Kirchoff" and "Kirchof" are wrong. Many tests themselves misspell it (see section 4), so do not copy the test's spelling.

- Calculator: look up — type: keyword ("spell", "junction").

K2. Law statement or equation → person

Law statement or equation → person · ✅ IN · Easy

- **Scope:** IN.
- **Where seen:** ATX 2021 regional B Q5; Princeton 2026 Matching Part 1 (C); MIT 2021 Q98 (C).
  - Also seen (test-extraction files): UF Northern Regional (Florida) 2020, Div B, Regional, Q7: "Who formulated KCL?" (key: C Kirchhoff). Div B 2027: IN.
- **How to solve:**
  1. Read the question: "Σ I_in = Σ I_out", "sum of voltages around a loop is zero", "junction rule" → Kirchhoff.
  2. Junction/node/loop/mesh words = Kirchhoff.
  3. V = IR = Ohm.
  4. F = kq_1q_2/r^2 = Coulomb.
- **Worked example:** "Whose law says the voltages around any closed loop add to zero?" → **Kirchhoff (KVL, 1845)**.
- **Traps:** "Kirchhoff's law" alone can also mean his **radiation** law or **thermochemistry** law. Read the context.
- Calculator: look up — type: keywords of the law.

K3. Person → contribution (matching column)

Person → contribution (matching column) · ✅ IN · Easy

- **Scope:** IN.
- **Where seen:** Princeton 2026 Matching Part 1 #3 (C; key 3 → A "Developed the junction law and loop law used in circuit analysis"). ATX 2020 regional matching (key: "Created laws generalizing the work of Ohm (Kirchoff)").
  - Also seen (test-extraction files): F-M Invitational (NY), fall 2019, Div not stated, Invitational, Q1: "Match 7 discoveries to 7 scientists" (key: Kirchhoff row correct). Div B 2027: IN.
- **How to solve:**
  1. Read the question: a list of 7 names and 7 statements → pairs.
  2. Match the unique keyword first (junction/loop, generalized Ohm, ideal-wire signal speed)
  3. Then eliminate.
- **Worked example:** Statements: A "junction law and loop law", B "first chemical battery", D "law relating V, I, R". Kirchhoff → **A**; Volta → B; Ohm → D.
- **Traps:** "Generalized Ohm's work" is Kirchhoff, not Ohm.
- Calculator: look up — type: scientist name → list of contributions.

K4. Kirchhoff as a wrong option (eliminate the distractor)

Kirchhoff as a wrong option (eliminate the distractor) · ✅ IN · Easy

- **Scope:** IN.
- **Where seen:** BullSO 2026 B Q2 (battery → Volta, Kirchoff option D), Q6 ("Newton of electricity" → Ampère); Berkeley 2026 B Q6 (V = IR → Ohm, Kirchhoff option A); ATX 2026 regional B Q1 (battery), Q2 (unit of resistance → Ohm), Q6 (forgot own name → Ampère, key C); GullSO 2026 Q29 (capacitance unit → Faraday), Q30, Q34, Q36 (C); UT 2020 B Q1-5.
- **How to solve:**
  1. Read the question: a fact about another scientist → the right person.
  2. Know what Kirchhoff did **not** do: no battery, no unit, no V = IR, no electromagnetic induction, no solenoid, no methane, no torsion balance.
- **Worked example:** "Who has the unit of resistance named after them? a) Kirchoff b) Coulomb c) Tesla d) Ohm" → **d) Ohm** (ATX 2026 B Q2, key D).
- **Traps:** Kirchhoff "sounds right" for any circuit law. Ohm's law is not his.
- Calculator: look up — type: the fact keyword → person.

K5. Name the two laws and their other names

Name the two laws and their other names · ✅ IN · Easy

- **Scope:** IN (naming is history; no circuit solving).
- **Where seen:** ATX 2021 regional B Q46 (key "Kirchhoff Loop Rule / KVL / Kirchhoff Second Law"); cyanophycean314 practice C Q13 "What is Kirchhoff's First Law?"; Whiting 2013 B matching (d, f); Virginia 2013 regional C Q21-22 matching; UT 2020 B Q12 (nodal analysis → KCL, key A); BullSO 2026 B FRQ 5a (law to find I1 → KCL).
  - Also seen (test-extraction files): UF Northern Regional (Florida) 2020, Div C, Regional, Q1: "___ says current in = current out at a node" (key: Kirchhoff's law). Div B 2027: IN.
- **How to solve:**
  1. Read the question: a name ("first law", "loop rule", "node law") → which law (KCL or KVL), or the reverse.
  2. First = current = junction = node = KCL. Second = voltage = loop = mesh = KVL. Nodal analysis uses KCL. Mesh analysis uses KVL.
- **Worked example:** "Nodal analysis is based on which law? A) Kirchhoff's Junction Law B) Kirchhoff's Voltage Law C) Ohm's Law D) Ampère's Law" → **A** (UT 2020 B Q12, key A).
- **Traps:** "Kirchhoff's current law" is not "a law about current in one wire". ATX 2021 Q46 asks "What **current** law" but wants KVL (wording error, section 4).
- Calculator: look up — type: alias → KCL/KVL.

K7. Year and setting of the circuit laws

Year and setting of the circuit laws · ✅ IN · Easy

- **Scope:** IN.
- **Where seen:** Princeton 2020 Q1 (C; "Kirchhoff (1845)").
- **How to solve:**
  1. Read the question: "When did Kirchhoff state his laws?" → **1845**, as a student in Königsberg (Neumann's seminar); they extend Ohm (1827).
  2. Recall 1845. Link: student → Königsberg → teacher Neumann.
- **Worked example:** "Kirchhoff formulated his circuit laws in A 1827 B 1845 C 1859 D 1887" → **B 1845**. (1827 = Ohm, 1859 = spectroscopy, 1887 = his death and Hertz's work.)
- **Traps:** 1847 is his graduation and the network-equations paper. 1845 is the laws.
- Calculator: look up — type: "Kirchhoff laws year".

K14. Unit trick (no unit for Kirchhoff)

Unit trick (no unit for Kirchhoff) · ✅ IN · Easy

- **Scope:** IN.
- **Where seen:** GullSO 2026 Q29 (C; capacitance unit, Kirchhoff option D; key C Faraday); ATX 2026 B Q2.
- **How to solve:**
  1. Read the question: "Which unit is named after Kirchhoff?" → **none**.
  2. Of the 2027 seven, units exist for Ampère (A), Coulomb (C), Volta (V), Ohm (Ω), Hertz (Hz)
  3. None for Kirchhoff or Boole.
- **Worked example:** "Which pair of scientists on the 2027 list has no SI unit?" → **Kirchhoff and Boole**.
- **Traps:** "mho" is Ohm backwards (conductance; now siemens). Not Kirchhoff.
- Calculator: look up — type: scientist → unit or "none".

K24. Apply KCL at one node (missing current)

Apply KCL at one node (missing current) · ✅ States-Nationals · Easy (at States)

- **Scope:** IN-SN (3.c.ix).
- **Where seen:** Berkeley 2026 B Q19 (key A, 3 A); Northview 2026 Q22 (C, statement); BullSO 2026 B FRQ 5 (I1 by KCL, key 2.625 A, verified).
- **How to solve:**
  1. Read the question: known currents in/out at a node → the unknown current and its direction.
  2. Σ in = Σ out.
  3. Solve the one unknown.

4. A negative answer means the direction is opposite to the guess.

- **Worked example:** In: 4 A and x. Out: 7 A. x = 7 − 4 = **3 A** in. Example 2: in 2.5 A and 0.8 A, out 1.1 A and y → y = **2.2 A** out (Python). BullSO FRQ 5: 10 V across 16 Ω → I2 = **0.625 A** (Ohm); I1 = 2 A + 0.625 A = **2.625 A** (KCL).
- **Traps:** Signs. Counting a current twice.
- Calculator: calculate — type: list of "in" currents, list of "out" currents (one blank) → missing value with sign.

Medium

K6. Which conservation law is behind each Kirchhoff law

Which conservation law is behind each Kirchhoff law · ✅ IN · Easy to Medium

- **Scope:** IN (concept of his discovery; no computing). Note: it sits near 3.c.ix, but it asks no circuit analysis, and Div B regionals have asked it (ATX 2021 regional B, Pembroke).
- **Where seen:** Berkeley 2026 B Q17 (node law → charge, key D) and Q18 (voltage law → energy, key B); Pembroke Hill 2026 B Q31 (KCL → charge, key C); UT 2020 B and C Q13 (TIEBREAKER #6; key marks **energy**, which is wrong — section 4).
- **How to solve:**
  1. Read the question: KCL or KVL → conservation of charge / energy.
  2. Charge cannot pile up at a node → KCL = **charge**
  3. A charge that goes round a loop comes back to the same potential, so it gains and loses the same energy → KVL = **energy**.
- **Worked example:** "Kirchhoff's Voltage Law is fundamentally a statement of: A mass B energy C momentum D charge" → **B, conservation of energy**.
- **Traps:** Swapping them. "Momentum" and "mass" are fillers. A key may be wrong; trust the physics and appeal.
- Calculator: look up — type: "KCL" or "KVL".

K8. Timeline ordering among the seven scientists

Timeline ordering among the seven scientists · ✅ IN · Medium

- **Scope:** IN.
- **Where seen:** Princeton 2020 Q1 (C) — seven 2020-list scientists by "crowning contribution". Kirchhoff (1845). 2027 list must swap Tesla/Faraday for Hertz/Boole.
- **How to solve:**
  1. Read the question: 4-7 discoveries → chronological order or "which came just before/after Kirchhoff".
  2. Coulomb 1785 → Volta 1800 → Ampère 1820-26 → Ohm 1827 → **Kirchhoff 1845** → Boole 1847/1854 → Hertz 1886-88.
- **Worked example:** Order: Hertz's radio waves, Kirchhoff's laws, Boole's *Mathematical Analysis of Logic*, Ohm's law. → **Ohm 1827, Kirchhoff 1845, Boole 1847, Hertz ~1887** (Python sort check).
- **Traps:** Kirchhoff (1845) and Boole (1847) are only 2 years apart. If the test uses *Laws of Thought* (1854) the order stays the same.
- Calculator: look up — type: pick names → app sorts. (App note: (a sorted table of the seven with years).)

K9. Age and date arithmetic

Age and date arithmetic · ✅ IN · Medium

- **Scope:** IN (algebra only).
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: two dates → age or gap in years.
  2. Subtract years.
  3. Subtract 1 more if the event is before 12 March.
- **Worked example:** "How old was Kirchhoff when he died?" 17 Oct 1887 − 12 Mar 1824 → **63 years** (Python). "How many years after Ohm's law?" 1845 − 1827 = **18 years**. "Age when he stated his laws (1845)?" → **20 or 21** (21 if after 12 March); most sources say "21" or "as a student".
- **Traps:** Month of birth. The 1845 month is not given in the sources, so "about 21".

- Calculator: calculate — type: birth date (preset 1824-03-12), event date → age; or two years → gap.

K10. Places and posts

Places and posts · ✅ IN · Easy to Medium

- **Scope:** IN.
- **Where seen:** not seen as a direct question (prior I_bank H6 type).
- **How to solve:**
  1. Read the question: city/university → event, or the reverse.
  2. Königsberg (born, studied, 1845 laws) → Berlin (1847-50) → Breslau (1850-54, met Bunsen) → Heidelberg (1854-75, spectroscopy, Cs, Rb) → Berlin (1875-87, died).
- **Worked example:** "In which city did Kirchhoff and Bunsen do their spectroscopy and discover cesium?" → **Heidelberg**.
- **Traps:** They **met** in Breslau, but **worked together** in Heidelberg. Königsberg is now Kaliningrad (Russia), not in Germany today.
- Calculator: look up — type: city or event.

K11. Teachers, students, collaborators

Teachers, students, collaborators · ✅ IN · Medium

- **Scope:** IN.
- **Where seen:** ATX 2020 regional key ("generalizing the work of Ohm"); not seen otherwise.
- **How to solve:**
  1. Read the question: a person → relation to Kirchhoff.
  2. Neumann = teacher.
  3. Bunsen = collaborator (chemist)
  4. Hertz, Planck = students in Berlin.
  5. Ohm = the law he generalized.
  6. Maxwell = came later and linked c to light.
- **Worked example:** "Which scientist on the 2027 list studied under Kirchhoff?" → **Hertz**.
- **Traps:** Helmholtz also taught Hertz; the 2027 list has only Hertz.
- Calculator: look up — type: person name.

K12. 1857 signal-speed result

1857 signal-speed result · ✅ IN · Medium

- **Scope:** IN (electricity discovery).
- **Where seen:** GullSO 2026 Q31 (C; 2 pts; key C = Kirchhoff, verified on the ZipGrade key page); Princeton 2021 Q6 (C; T/F, key **True**).
- **How to solve:**
  1. Read the question: "Who found that signals on an ideal wire travel at the speed of light?" → Kirchhoff (1857).
  2. Ideal (resistanceless) wire + speed of light + 1857 = Kirchhoff.
- **Worked example:** GullSO Q31: A Ampère B Faraday C Kirchhoff D Ohm E Volta → **C**.
- **Traps:** It is the **signal** (the wave of field), not the electrons. Electron drift is very slow (mm/s range). A T/F "electrons in a wire move at the speed of light" is **False**. He did not make the light = EM wave claim; Maxwell did.
- Calculator: look up — type: "speed of light wire".

K13. Signal travel time on an ideal wire (calc)

Signal travel time on an ideal wire (calc) · ⚠️ Borderline · Medium

- **Scope:** BORDERLINE. It uses only algebra (allowed), and it is tied to an electricity discovery, but no rule area lists signal speed as a calculation topic.
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: distance d → time t = d/c.

2. Convert km to m. Divide by 3.00×10^8 m/s.

- **Worked example:** A 3000 km ideal telegraph line. t = 3.0×10^6 m / 3.00×10^8 m/s = **0.010 s = 10 ms** (Python: 10.0 ms with c = 3.00e8; 10.007 ms with 2.998e8).
- **Traps:** km → m. Real cables are slower than c.
- Calculator: calculate — type: distance and unit (m/km); option c = 3.00e8 or 2.998e8.

K15. Significance / short essay

Significance / short essay · ✅ IN · Medium

- **Scope:** IN.
- **Where seen:** MIT 2021 Q98 (C; 5 pts, "State Kirchhoff's laws"); MIT 2021 Q100 / Princeton 2020 Q2 style essays (other scientists).
- **How to solve:**
  1. Read the question: "Explain why Kirchhoff's laws matter" → 3-5 points.
  2. Give: (1) both statements.
  3. (2) basis (charge, energy)
  4. (3) they extend Ohm's law from one wire to networks.
  5. (4) basis of nodal and mesh analysis, used for every circuit since.
  6. (5) valid for DC and low-frequency circuits.
- **Worked example:** "KCL: the sum of currents into a node equals the sum out (charge conservation). KVL: the sum of voltage changes around any closed loop is zero (energy conservation). Stated in 1845 as a student, they let engineers solve circuits with many loops, which Ohm's law alone cannot." (MIT 2021 key wording: "the (signed) sum of voltage drops around a closed loop is zero... the (signed) sum of currents entering and exiting a node is zero.")
- **Traps:** Leaving out "closed loop" or "node". Writing "conservation of current".
- Calculator: look up — type: "essay Kirchhoff". (App note: (show a model answer).)

K16. True/False or "which is NOT Kirchhoff's" (mixed)

True/False or "which is NOT Kirchhoff's" (mixed) · ✅ IN · Medium

- **Scope:** IN (each statement's own scope may be BORDERLINE, see K17-K23).
- **Where seen:** Rickards 2020 B Q1 (and C Q1); Princeton 2021 Q6 (C).
- **How to solve:**
  1. Read the question: a statement → True/False, or 4 statements → the misattributed one.
  2. Check each word: Sun vs Moon, signal vs electrons, 1845 vs 1827, cesium vs sodium (he found sodium in the Sun but did not discover sodium).
- **Worked example:** "Which is NOT a Kirchhoff achievement? A KCL and KVL B co-discovery of cesium C invention of the voltaic pile D the term 'black body'" → **C** (Volta, 1800).
- **Traps:** "Discovered sodium" is false (Davy isolated it, 1807). "Discovered helium" is false (Janssen/Lockyer, 1868).
- Calculator: look up — type: keyword of the statement → true/false + reason.

K17. Spectroscopy with Bunsen

Spectroscopy with Bunsen · ⚠️ Borderline · Medium

- **Scope:** BORDERLINE (not an electricity discovery; rule 3.c.i says "electricity discoveries").
- **Where seen:** not seen as its own question (prior I_bank H3/H8 types).
- **How to solve:**
  1. Read the question: "spectroscope / spectrum analysis / Bunsen" → Kirchhoff (1859, Heidelberg).
  2. 1859 + Heidelberg + Bunsen + prism + flame = spectrum analysis.
- **Worked example:** "Kirchhoff worked with which chemist to develop spectrum analysis?" → **Robert Bunsen** (Bunsen burner 1855).
- **Traps:** The spectroscope instrument came from Fraunhofer (1814). If a choice says "Fraunhofer invented the first spectroscope", that is also true.

- Calculator: look up — type: the keyword from the question.

K18. Sodium in the Sun, Fraunhofer lines (and the Moon trap)

Sodium in the Sun, Fraunhofer lines (and the Moon trap) · ⚠️ Borderline · Medium

- **Scope:** BORDERLINE.
- **Where seen:** Rickards 2020 B Q1 and C Q1 (T/F "Gustav Kirchhoff proved the moon contained sodium", key **False**, verified on the key PDF).
  - Also seen (test-extraction files): UT Austin Invitational (2020 season) 2019, Div B, Invitational, Q7: "Epithet: showed the Sun has sodium" (key: Kirchhoff). Div B 2027: BORDERLINE (spectroscopy), asked on Div B.
  - Also seen (test-extraction files): UT Austin Invitational, fall (2020 season) 2019, Div C, Invitational, Q7: "Epithet: showed the sun has sodium" (key: Kirchhoff). Div B 2027: BORDERLINE.
- **How to solve:**
  1. Read the question: a claim about what he found in the Sun → T/F or element name.
  2. The yellow sodium flame lines line up with the dark solar **D lines** → sodium is in the **Sun** (1859)
  3. Dark lines = absorption by cooler gas.
- **Worked example:** "The dark lines in the solar spectrum are caused by:" → **absorption by elements in the Sun's (cooler) outer atmosphere** (Kirchhoff, 1859).
- **Traps:** Moon (reflects sunlight, so it shows the same lines, but he did not "prove" this). Dark lines are not "missing elements".
- Calculator: look up — type: the keyword from the question.

K19. Cesium and rubidium

Cesium and rubidium · ⚠️ Borderline · Medium

- **Scope:** BORDERLINE.
- **Where seen:** not seen in a downloaded test (prior I_bank H8 example).
- **How to solve:**
  1. Read the question: "Which elements did Kirchhoff and Bunsen discover?" → cesium (1860, blue, Dürkheim mineral water) and rubidium (1861, red, lepidolite).
  2. Blue → caesius → cesium.
  3. Red → rubidus → rubidium.
  4. Cesium first (first element ever found with a spectroscope).
- **Worked example:** "The first element discovered by spectrum analysis was named for its color. Which element and which color?" → **Cesium, sky-blue (bluish-grey)**.
- **Traps:** Helium (1868) and thallium (Crookes 1861) are other spectroscope finds, not Kirchhoff's. Some sources say 1861 for both (section 4).
- Calculator: look up — type: the keyword from the question.

K20. "Black body" term

"Black body" term · ⚠️ Borderline · Medium

- **Scope:** BORDERLINE (not electricity). Seen at a Div B **regional**, so expect it.
- **Where seen:** ATX 2026 regional **B** Q8 ("Who made a breakthrough in Blackbody Radiation?" key B Kirchoff); Rickards 2020 B Q22 (3 pts, "Who coined the term black-body radiation?" → Kirchhoff); Rickards 2020 C Q26; MIT 2020 Q2 (C).
- **How to solve:**
  1. Read the question: "coined black body" → Kirchhoff (1860/1862).
  2. Black body = perfect absorber; named by Kirchhoff; solved by Planck (1900).
- **Worked example:** ATX 2026 B Q8: a) Ampère b) Kirchoff c) Ohm d) Faraday → **b**.
- **Traps:** Planck solved the black-body spectrum (1900); he did not coin the term. Year 1860 vs 1862 (accept both).
- Calculator: look up — type: the keyword from the question.

K25. Apply KVL around one loop

Apply KVL around one loop · ✅ States-Nationals · Easy to Medium

- **Scope:** IN-SN.
- **Where seen:** Northview 2026 Q21 (C; sum = 0 V); ATX 2021 regional B Q44-46; BullSO 2026 B FRQ 6 (two sources in one loop, key V1 = 14.4 V, V2 = 17.6 V, verified).
- **How to solve:**
  1. Read the question: sources and resistors in one loop → current and each drop.
  2. Walk the loop.
  3. Add rises, subtract drops, set = 0.
  4. Sources aiding → add; opposing → subtract.
  5. I = (net EMF)/(ΣR)
  6. Each drop = I R.
- **Worked example:** BullSO FRQ 6: 20 V and 12 V aiding, 18 Ω and 22 Ω. I = 32/40 = **0.8 A**; V1 = 0.8 × 18 = **14.4 V**; V2 = 0.8 × 22 = **17.6 V** (Python). Example 2: 12 V source, drops 5 V and 3 V → third drop = **4 V**.
- **Traps:** Polarity of the second source (read + and − on the figure). If they opposed, I = 8/40 = 0.2 A.
- Calculator: calculate — type: each source value with +/− sign, each resistor value → I, each V drop, check Σ = 0.

K31. Multi-select: which listed scientists have property X?

Multi-select: which listed scientists have property X? · ✅ IN · Medium

- **Scope:** IN (3.c.i).
- **Where seen:** UT Austin Invitational fall 2025, Div B/C, Invitational, Q4a-d (letters A, C, K, V, O, T, F; pick every letter that fits). Kirchhoff is a right letter in 4c (laws useful in steady state).
- **How to solve:**
  1. Write the letter code of each listed scientist (A = Ampère, C = Coulomb, K = Kirchhoff, V = Volta, O = Ohm, Hz = Hertz, B = Boole).
  2. Test each scientist against the property, one at a time.
  3. Pick every scientist that passes. A multi-select question can have more than one right letter.
  4. Check each letter again. Keys often add a wrong letter (UT 2025 put Volta under "France").
- **Worked example:** "Whose laws are useful in steady state? A, C, K, V, O, T, F" → Kirchhoff is one of them: **K** (with A, C, O).
- **Traps:** Kirchhoff's laws work for any lumped circuit, but the question only asks about steady state.
- Calculator: look up — type: the property.

Hard

K21. Kirchhoff's law of thermal radiation (statement)

Kirchhoff's law of thermal radiation (statement) · ⚠️ Borderline · Hard

- **Scope:** BORDERLINE (statement only; any calculation is OUT, see K27).
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: "State Kirchhoff's law of thermal radiation" / pick the correct statement.
  2. At thermal equilibrium, emissivity = absorptivity, at each wavelength and temperature.
  3. Good absorber = good emitter.
  4. A perfect absorber (black body) is the best emitter.
- **Worked example:** "A surface absorbs 30% of infrared at one wavelength, in thermal equilibrium. Its emissivity at that wavelength?" → **0.30** (equal, by Kirchhoff's law).
- **Traps:** "Good reflector = good emitter" is wrong (good reflector = poor emitter).
- Calculator: look up — type: (statement). The 0.30 case is a copy, so LOOKUP is enough.

K22. Which "Kirchhoff's law"? (disambiguation)

Which "Kirchhoff's law"? (disambiguation) · ⚠️ Borderline · Hard

- **Scope:** BORDERLINE.
- **Where seen:** not seen in a test.

- **How to solve:**
  1. Read the question: a description → circuit laws / spectroscopy laws / radiation law / thermochemistry law / diffraction formula / plate theory / matrix-tree theorem.
  2. Currents and voltages → circuit (1845)
  3. Continuous/emission/absorption spectra → spectroscopy (1859)
  4. Emissivity = absorptivity → radiation (1859-60)
  5. Heat of reaction vs temperature → thermochemistry (1858)
  6. Counting spanning trees → matrix-tree (1847).
- **Worked example:** "Kirchhoff's law relating how a reaction's heat changes with temperature belongs to which field?" → **thermochemistry (1858)**.
- **Traps:** Thinking only one "Kirchhoff's law" exists.
- Calculator: look up — type: keyword.

K23. Anecdotes, honors, family, disability

Anecdotes, honors, family, disability · ⚠️ Borderline · Medium to Hard

- **Scope:** BORDERLINE (personal trivia; 2026 ATX tests asked this style for other scientists).
- **Where seen:** not seen for Kirchhoff (ATX 2026 B Q6 / C Q9-13 ask this style about Ampère, Volta, Faraday).
- **How to solve:**
  1. Read the question: an anecdote → Kirchhoff, or Kirchhoff → the medal/year.
  2. "Gold from the Sun" banker story.
  3. Crutches/wheelchair.
  4. Married his professor's daughter (Clara Richelot, 1857)
  5. Rumford 1862; first Davy Medal 1877 with Bunsen.
- **Worked example:** "Which scientist, mocked by his banker about gold in the Sun, later handed him gold from a prize, saying 'Here is gold from the Sun'?" → **Kirchhoff**.
- **Traps:** Davy Medal ≠ Copley Medal (Bunsen got the Copley in 1860; Ohm 1841).
- Calculator: look up — type: the keyword from the question.

K30. Quote attribution (who said it?)

Quote attribution (who said it?) · ✅ IN · Hard

- **Scope:** IN (3.c.i).
- **Where seen:** Florida State Tournament 2021, Div B (same as Div C), State, Q2-Q8 (seven quotes, 2 points each, "identify the last name"). Q4 is keyed Kirchhoff: "Look here, I have succeeded at last in fetching some gold from the sun... Why do you want to come into physics? All is done and understood."
- **How to solve:**
  1. Look for one key word or idea in the quote (an invention, a place, a belief, a language).
  2. Match that idea to one of the seven listed scientists.
  3. If the quote is in French, think of the two French scientists first: Ampère and Coulomb.
  4. If nothing fits the seven, the answer may be a scientist who is not on the 2027 list (Tesla, Faraday). Those items are out of scope in 2027.
  5. Clues for Kirchhoff: "gold from the Sun" (spectroscopy, his banker story).
- **Worked example:** "Look here, I have succeeded at last in fetching some gold from the sun." → **Kirchhoff**.
- **Traps:** The test joined two quotes. "All is done and understood" is usually credited to Philipp von Jolly, who said it to the young Max Planck. Answer Kirchhoff for the "gold" part.
- Calculator: look up — type: "gold from the sun".

Nationals

K26. Two-unknown network using Kirchhoff's laws (Nationals)

Two-unknown network using Kirchhoff's laws (Nationals) · ✅ States-Nationals · Hard / Nationals

- **Scope:** IN-SN, only with ≤ 2 unknowns (math limit 3.e.i).

- **Where seen:** UT 2020 B Q68 / C Q74 ("using only Kirchhoff's Laws", key 8.94 V; figure not checked); Princeton 2019 "Kirchhoff's Word is Law" (C; key V = 70, V′ = 50, currents 15, 8, 6, 3 A; figure not checked); BullSO 2026 B FRQ 7 (two node voltages, key 28 V and 30 V; figure not checked).
- **How to solve:**
  1. Read the question: a 2-loop circuit (two sources, three resistors) → branch currents and node voltage.
  2. Pick one node voltage Vn (one KCL equation), or two mesh currents (two KVL equations)
  3. Solve.
  4. Check KCL at the node.
- **Worked example:** Left: 12 V source through R1 = 2 Ω to node N. Right: 8 V source through R2 = 2 Ω to node N. Middle: R3 = 1 Ω from N to ground. KCL: (12 – Vn)/2 + (8 – Vn)/2 = Vn/1 → **Vn = 5 V**; I1 = **3.5 A**, I2 = **1.5 A**, I3 = **5 A**; check 3.5 + 1.5 = 5 (Python; mesh solve gives the same; power in 54 W = power out 54 W).
- **Traps:** A negative current means the guessed direction was wrong (e.g. with R1 = 4, R2 = 2, R3 = 6, 12 V and 6 V, I2 = –0.273 A). Do not set up 3 equations when 1 node equation will do.
- Calculator: calculate — type: V1, R1, V2, R2, R3 (preset topology) → Vn, I1, I2, I3; general mode: 2×2 coefficient entry solved by substitution (no matrices shown to the student).

Out of scope (skip)

These types appear in tests or books, but the 2027 Div B rules exclude them. Skip them. They stay here so you can recognise them.

K27. Black-body / spectral-line calculations

Black-body / spectral-line calculations · ❌ Out · Hard

- **Scope:** OUT. No 2027 Div B Circuit Lab area covers radiation physics; "frequency analysis" is excluded; Stefan-Boltzmann is a Div C *Thermodynamics* States/Nationals topic, not Circuit Lab.
- **Where seen:** not seen in a Circuit Lab test.
- **Worked example:** $P = \varepsilon\sigma AT^4$; Wien's law; D-line wavelengths (588.995 / 589.592 nm).
- Calculator: none (skip).

K28. KVL/KCL with op-amps, transistors, or 3+ equations

KVL/KCL with op-amps, transistors, or 3+ equations · ❌ Out · Nationals (Div C)

- **Scope:** OUT (op-amps/BJTs = "semiconductors beyond those listed"; 3+ simultaneous equations break 3.e.i). A dependent-source circuit solvable with ≤ 2 equations is BORDERLINE (not in the exclusion list) and belongs to the KVL/KCL subtopic.
- **Where seen:** UT 2025 Inv. FRQ (op-amp gain box with KVL/KCL); ATX 2021 C key (op-amp does not violate Kirchhoff's laws).
- Calculator: none (skip).

K29. Limits of Kirchhoff's laws

Limits of Kirchhoff's laws · ❌ Out · Hard

- **Scope:** OUT (needs AC circuit theory / high frequency / changing magnetic flux; AC theory, frequency analysis and magnetism are excluded).
- **Where seen:** not seen in a test. Fact: the laws hold for the lumped-element model; they fail when wavelength is not much larger than the circuit, or with leaking changing magnetic fields [CL].
- Calculator: none (skip).

Mistakes in tests and keys

1. **UT Austin 2020 Invitational, B and C, Q13 (TIEBREAKER #6):** key marks **C "conservation of energy"** for "Kirchhoff's Junction Rule is most directly a restatement of". Correct: **B, conservation of charge**. Checked on the rendered key pages (`dlG/ut20inv_Bkey.pdf` p.3, `ut20inv_Ckey.pdf` p.3).
2. **ATX Regional 2021 B and C, Q46:** asks "What **current** law helps us understand this circuit?" but the key answer is "Kirchhoff Loop Rule / KVL / Second Law". The wording mixes up the two laws.

3. **Spelling "Kirchoff"** (wrong) in ATX 2020 B/C, ATX 2026 B/C, UT 2020 B/C, BullSO 2026 B and C, Northview 2026 C, Whiting 2013 B, cyanophycean314 practice, supervisor deck 10.
4. **Wikipedia, Gustav Kirchhoff page:** says caesium and rubidium were both discovered in **1861**. Wikipedia's own Caesium page, the Bunsen page and APS say caesium **1860** (rubidium 1861).
5. **Black-body coinage year conflict:** Wikipedia says **1860**; MacTutor and Purdue say **1862**. Not an error that can be resolved here; accept either.
6. **Spectroscope "invented" by Kirchhoff and Bunsen** (Wikipedia Caesium; prior I_bank "invented spectroscopy / the spectroscope"; scioly wiki "he invented the spectroscope with Robert Bunsen"): overstated. Fraunhofer built a spectroscope in 1814. Safer: "developed spectrum analysis / built the Bunsen-Kirchhoff spectroscope (1859)".
7. **Purdue chemed page:** "moved to Berlin after marrying the daughter of his mathematics professor". He moved to Berlin in 1847 and married in 1857. Also "chair of mathematics at Berlin, 1875": it was the chair of mathematical (theoretical) physics [M].
8. **Prior I_bank.md:** "He used crutches or a wheelchair **after an accident**" — the disability is confirmed [M][P], the accident cause is not (neither MacTutor nor Purdue gives a cause). "**40 tons** of water" — Wikipedia says 44,000 L (≈ 44 t), APS says 12,000 US gallons (= 45,425 L, Python); "40 tons" is a rough low figure.
9. **Task brief / prior notes:** say Circuit Lab is on manual pages **B11-B12**. In this PDF it is on **B12-B13** (page footers "©2027 - B12" and "B13"; B11 is the event before).
10. **Wikisource vs IEEE volume number for the 1845 paper (140 vs 64):** not an error. Annalen has two numbering schemes; 140 − 64 = 76, the same offset as 185 − 109 for the 1860 paper (Python).
11. **Supervisor deck 10** (prior F_slides): KVL slide titled "Kirchhoff's Voltage Law (KCL)", and both laws marked "Division C only" (in 2027 they are Div B States/Nationals).
12. **Prior G_tests.md:** says NV Q49 methane key "looks wrong"; not re-checked here (not a Kirchhoff item). Prior G_tests lists "Princeton 2020 Q1: ... Kirchhoff 1845" — confirmed.

- **Florida State 2021, Q4:** the quote joins two quotes. "Gold from the sun" is Kirchhoff. "All is done and understood" is usually credited to Philipp von Jolly (advice to Max Planck). Half-wrong key. Not verified online.

Not confirmed

- **Cause of his disability** (accident? which year?): looked in Wikipedia (not mentioned), MacTutor (disability only, no cause), Purdue (no cause). Not confirmed.
- **Exact month of the 1845 laws** (so age 20 vs 21): Wikipedia, MacTutor, Heidelberg KIP sheet (search snippet only). Not found.
- **"Redefined electrical potential / rejected static currents in a conductor at equilibrium"** (scioly wiki): MacTutor only says he "corrected the accepted understanding of electric currents and electrostatics" (1848-50). The exact claim is not confirmed.
- **The three laws of spectroscopy were stated by Kirchhoff in this exact form:** Wikipedia names them; the wording above is standard textbook wording, not checked against his 1859-60 papers.
- **Cesium date "10 May 1860"**: only APS. Not cross-checked.
- **"Identified about 30 elements in the Sun"**: APS and Purdue. Not cross-checked with a primary source.
- **"Here is gold from the Sun" details** (medal vs prize money in sovereigns): Wikipedia only; many popular retellings differ.
- **Figure-based keys** UT 2020 B Q68 (8.94 V), Princeton 2019 "Kirchhoff's Word is Law", BullSO 2026 B FRQ 7: figures not viewed, so the key numbers are not re-computed. BullSO FRQ 5 and 6 figures were viewed and the keys check out.
- **Britannica** biography: fetch returned HTTP 403; not used.
- **IEEE / ETHW biography of Kirchhoff**: not fetched (no time); the IEEE CAS Magazine 2023 citation came from a search snippet only.
- No State or National Circuit Lab test was available (prior G_manifest: paid packets), so no Nationals-level Kirchhoff history question was seen.

Sources

- 2027 Div B rules PDF: `/Users/kai/Daniel/Science_Olympiad_Div_B_Rules_2027_for_Web_Secured.pdf` (pp. B12-B13); 2027 Div C rules text: `circuitlab/research/downloads/dl/rulesC2027.txt` lines 878-905; 2026 Div B rules `dl/rulesB2026.txt` line 699.
- Past tests (local, `circuitlab/research/downloads/`): `dlB/tests/berk26_*`, `bullso26_*`, `phi26_*`; `dlG/atx20reg_*`, `atx21reg_*`, `atx26reg_*`, `ut20inv_*`, `ut25inv_test`, `rsoiOld{B,C}_*`, `gull26_*`, `puso26_*`, `nv26C_test`, `mit20_*`, `mit21_*`, `pz/p19`, `pz/p20`, `pz/p21`, `whiting_circuit_lab_test`, `wiki/CircuitLabRegTest2013`, `wiki/Circuit_Lab_cyanophycean314`.
- Prior files: `circuitlab/research/prior/I_bank.md` (§1.0-1.1), `B_topics.md` (§5e, history table), `G_tests.md`, `G_manifest.md`, `F_slides.md`, `A_rules.md`.
- Scioly.org wiki Circuit Lab (local copy `downloads/dl/cl_wiki.txt`): https://scioly.org/wiki/index.php/Circuit_Lab
- Wikipedia, Gustav Kirchhoff: https://en.wikipedia.org/wiki/Gustav_Kirchhoff
- MacTutor, Kirchhoff biography: https://mathshistory.st-andrews.ac.uk/Biographies/Kirchhoff/
- APS News, This Month in Physics History (May 2018, cesium): https://www.aps.org/apsnews/2018/05/this-month-in-physics-history
- Purdue chemed history: https://chemed.chem.purdue.edu/genchem/history/kirchhoff.html
- Wikipedia, Kirchhoff's circuit laws: https://en.wikipedia.org/wiki/Kirchhoff%27s_circuit_laws
- Wikipedia, Kirchhoff's law of thermal radiation: https://en.wikipedia.org/wiki/Kirchhoff%27s_law_of_thermal_radiation
- Wikipedia, Fraunhofer lines: https://en.wikipedia.org/wiki/Fraunhofer_lines
- Wikipedia, Caesium: https://en.wikipedia.org/wiki/Caesium ; Rubidium: https://en.wikipedia.org/wiki/Rubidium ; Robert Bunsen: https://en.wikipedia.org/wiki/Robert_Bunsen
- Wikipedia, Kirchhoff's theorem (matrix tree): https://en.wikipedia.org/wiki/Kirchhoff%27s_theorem
- Croatica Chemica Acta 2016 review (Kirby, Mallion, Pollak, Skrzyński): https://hrcak.srce.hr/172895
- Heidelberg KIP Kirchhoff leaflet: https://www.kip.uni-heidelberg.de/image/f/oeffwiss/kirchhoff/Faltblatt-Kirchhoff.pdf
- Mathematics Genealogy Project: https://mathgenealogy.org/id.php?id=46968
- German Wikisource bibliography: https://de.wikisource.org/wiki/Gustav_Robert_Kirchhoff
- Spectroscopy Online, "Happy sesquicentennial, spectroscopy": https://www.spectroscopyonline.com/view/happy-sesquicentennial-spectroscopy
- IEEE Circuits and Systems Magazine Q3 2023 (citation only): https://nxtbook.com/nxtbooks/ieee/circuitsandsystems_2023Q3/index.php?startid=69
- Python checks: dates/ages, year gaps, timeline sort, d/c times, gallon→litre, Annalen offset, every KCL/KVL example (script `/private/tmp/claude-502/-Users-kai-Daniel/893762dd-d9b3-4158-937c-ea93eb65d1e6/scratchpad/k/v.py`).
- Test-extraction files: `/Users/kai/Daniel/circuitlab/research/tests/` (atx-reg, misc-a, misc-b, nat26, state, ut-a, ut-b, ut-c) and `/Users/kai/Daniel/circuitlab/research/prior/G_tests.md`.
- Raw research file this note was built from: `/Users/kai/Daniel/circuitlab/research/subtopics/kirchhoff.md`.

Alessandro Volta

Note: Volta

> **INFO: How to use this note**
> - Scope icons: ✅ IN = all levels. ✅ States-Nationals = only at State and National tournaments. ⚠️ Borderline = the rules do not settle it (the reason is given). ❌ Out = excluded (the rule is cited).
> - Question types go from Easy to Medium to Hard to Nationals. "Nationals" means the hardest depth, not a different scope.
> - Every number in a worked example was checked in Python.
> - Source tags in square brackets, like [W] or [M], are listed under "Sources".

Scope notes and source tags

Circuit Lab Division B, 2027 rules. Rule anchor: Part I 3.c.i, "Historical perspective of the electricity discoveries made by Ampere, Coulomb, Kirchhoff, Volta, Ohm, Hertz, and Boole" (manual page B11, quoted from `pdftotext` of `/Users/kai/Daniel/Science_Olympiad_Div_B_Rules_2027_for_Web_Secured.pdf`). The battery calculations also fall under 3.c.iii (sources of DC), 3.c.iv (units of voltage, energy) and 3.c.v (DC circuits). The KVL/KCL two-pile type falls under 3.c.ix (States/Nationals only).

Div C check (per the user's request). Many Volta questions below were seen first in Division C tests. For each one, the "Scope" line says whether the Div B 2027 rules still cover it. A Div C question counts as IN only when it fits a Div B 3.c area and needs no excluded topic (3.d) and no math beyond 3.e. Div C questions that need capacitance, electrochemistry detail or magnetism are tagged BORDERLINE or OUT, with the reason.

Source tags: [W] Wikipedia, [MG] Museo Galileo, [NE] napoleon-empire.org, [EC] Electrochemical Society encyclopedia, [RS] Royal Society archive / Wellcome Collection catalogue, [KY] Kenyon early apparatus, [ETHW] IEEE ETHW, [LR] Larousse, [AV] alessandrovolta.it, [PR] prior file in `research/prior/`, [T] seen in a past test, [R] 2027 rules, [G] general physics knowledge (standard textbook, e.g. OpenStax College Physics 2e ch. 19-21).

Key facts

1.1 Life and posts

- Full name: **Alessandro Giuseppe Antonio Anastasio Volta**. [W] (A Rickards 2026 C answer choice uses the full name. [T])
- Born **18 February 1745, Como**, Duchy of Milan (Austrian rule at the time). [W][NE]
- Died **5 March 1827, Como** (estate at Camnago, now called Camnago Volta, a part of Como). Age 82. [W][MG] [NE]
- Nationality: **Italian** (Lombard). Father Filippo Volta (noble family), mother Maddalena Inzaghi. [W]
- Catholic all his life; wrote a declaration of faith. He was not ordained, although his family expected it. [W]
- Married Teresa Peregrini, 1794. Three sons: Zanino, Flaminio, Luigi. [W]
- **1769: first scientific memoir, *De vi attractiva ignis electrici*, addressed to Giovanni Battista Beccaria.** [AV] [LR] (Princeton 2019 statement (c) uses this fact. [T])
- **1774:** physics professor (and director of public schools) at the Royal School in Como. [W][MG][NE]
- **Pavia:** chair of experimental physics at the University of Pavia, **1779** [W][NE] (Museo Galileo says elected **1778** [MG]). Held it about 40 years, to **1819**. [W] Rector of Pavia in 1785. [MG]
- 1780-1783: travels in Switzerland, Germany, Holland, England, France; met Lavoisier and Laplace. [MG][NE]
- **1819:** retired to Como. [W][MG][NE]

1.2 Electrophorus (static electricity)

- **1775: Volta improved and popularized the electrophorus.** He is often (wrongly) called its inventor. **Johan Carl Wilcke** described a similar device in **1762**. [W]

- Volta **coined the name** "electrophorus" (Greek, "electricity bearer"). He called it the *elettroforo perpetuo* ("perpetual"). [W]
- Parts: an insulating (dielectric) plate (resin, pitch, wax; now plastic) and a **metal plate with an insulating handle**. [W]
- Method (charging by **induction**): (1) rub the dielectric (fur or cloth) so it gets a charge by the triboelectric effect; (2) set the metal plate on it; charges in the plate separate; (3) touch (ground) the top of the plate briefly; (4) lift the plate. The plate now has a **charge opposite in sign to the dielectric**. [W]
- The dielectric keeps its charge, so the plate can be charged **many times** from one rubbing. (In practice the charge leaks away in a few days.) [W]
- Energy source: the **work done lifting the plate** against the attraction. [W]
- Lichtenberg built a large one in 1777 (Lichtenberg figures). [W]

1.3 Methane, eudiometer, "Volta pistol"

- **November 1776:** Volta found "inflammable air" (**methane**) in the marshes at **Angera, Lake Maggiore**. He read a Benjamin Franklin paper on flammable air first. **By 1778 he had isolated it.** [W] Letters on marsh gas published in Milan, 1777. [AV]
- He **ignited methane with an electric spark** in a closed vessel. [W]
- **Eudiometer** (gas-volume tube with spark ignition, used to test the "goodness" of air, i.e. its oxygen content): Wikipedia says 1777; Kenyon says letters of 1776-77. [W][KY]
- **Voltaic (electric) pistol:** a spark from a **Leyden jar** explodes an inflammable gas (methane or hydrogen with air) and blows out a cork. Museo Galileo's specimen is dated 1778-1793 and used hydrogen. [MG]
- Princeton 2019 statement (k) says the spark ignition of methane was "the basis of the internal combustion engine". That is a test writer's claim (see section 4). [T]

1.4 Other electrical work before the pile

- **Condensing electroscope / condenser** (makes small charges detectable), about **1782**. [LR][AV] Straw electrometer (not dated in any source I read).
- **Volta's law of capacitance:** for a given object, the charge and the electric potential are **proportional** ($Q \propto V$). [W] Rickards 2020 B Q20 and C Q24 asked it ("Who discovered that electric potential and charge are proportional?" key: Volta). [T] Capacitance is excluded in 2027 (see Q-V24).
- He introduced the idea of electric **"tension"** (potential) during his Pavia years. [EC]

1.5 Galvani dispute and contact electricity

- **Luigi Galvani** (1737-1798, Bologna): frog-leg experiments from **1780** [W]; the famous muscle contraction with metals is dated **1786** by [EC]; book ***De viribus electricitatis in motu musculari commentarius*, 1791**. Theory: **"animal electricity"** made by the animal. [W]
- **1792:** Volta repeated the frog experiments and rejected Galvani's conclusion. [NE]
- Volta's view: the current comes from the **contact of two different (dissimilar) metals**. The frog leg is only a **conductor and detector**. He replaced the frog with **brine-soaked paper**. [W]
- Volta coined the word **"galvanism"**. Galvani's nephew **Giovanni Aldini** defended animal electricity. The two men disagreed respectfully. [W]
- Modern view: both were partly right. Nerves do carry electrical signals (Galvani), and two metals in an electrolyte make a current (Volta). [G] (Wikipedia does not phrase it this way; see section 5.)
- Volta ranked metals into the **electrochemical (contact) series**: "Volta's law" of the series. [W] NY Lake Erie 2019 Q1: "Volta's discovery of the electrochemical series led to the creation of the battery." [T]
- Galvani refused an oath to the Cisalpine Republic in 1798, lost his posts, and died 4 December 1798. [W] So Galvani never saw the voltaic pile. [G, from the dates]
- UT 2020 Q2 (B and C): "Which scientist proved that electricity could be generated chemically and not solely by living beings?" key: Volta. [T]

1.6 The voltaic pile (1799-1800)

- Built **late 1799** [EC]; announced in **1800**. Tests accept "1800" (and Princeton 2020 key used 1799). [T]

- The **first electric battery** that gave a **continuous (steady) current**. [W][ETHW] Before it, only instant discharges (Leyden jar, electrophorus) existed. [MG: "the first generator of non-instantaneous electric current"]
- Construction: alternating **zinc** and **copper (or silver)** discs, each pair separated by **cardboard or cloth (or leather) soaked in brine (salt water)**. [W][EC] Zinc + copper was his best pair. [W] Up to about **30 discs** in the first piles. [EC search result; see section 5]
- Volta's originals had an **extra copper disc on top and an extra zinc disc at the bottom**; later shown unnecessary. [W]
- Volta called it an **"artificial electric organ"**, built in imitation of the electric organ of the **torpedo fish** (electric ray). [EC] In French he called it an *appareil à colonne*; "pile" (heap) came from that. [EC]
- **Crown of cups** (*corona di tazze*): a ring of cups/glasses of salt or acidulated water, joined by metal strips (zinc on one end, copper on the other). Described in the same 1800 letter. [KY] Princeton 2021 Q8 key: "Volta originally used wine glasses filled with brine instead of cardboard soaked in brine": **True**. [T][W] (MIT 2020 Q10 key: "Enjoyed testing currents in wine goblets" → Volta. [T])
- EMF per Zn/brine/Cu cell: about **0.76 V** [W], or about **0.75 V** for the prototype [EC]. Cells in series **add**. [W]
- Volta thought the EMF came from the **metal contact** and that the wet layer did not matter. That was **wrong**: the source is a **chemical reaction**. Zinc is oxidized ($Zn \rightarrow Zn^{2+} + 2e^-$) at the zinc; hydrogen ions take electrons at the copper and make **H_2 bubbles**; the copper does not react. **Humphry Davy** showed the chemical origin. [W] Nernst gave the first full explanation around 1890. [EC]
- Credits Volta gave: William Nicholson, Tiberius Cavallo, Abraham Bennet. [W]

1.7 The letter to Banks (1800)

- **Dated Como, 20 March 1800**, to **Sir Joseph Banks, President of the Royal Society of London**. Written in **French**. [RS][W][EC]
- A **second part** followed: dated **1 April** [EC] or **2 April 1800** [RS Wellcome catalogue]. Sources differ.
- **Read to the Royal Society on 26 June 1800.** [RS]
- Published in ***Philosophical Transactions*** **vol. 90 (1800), pp. 403-431**, titled **"On the Electricity excited by the mere Contact of conducting Substances of different kinds."** [RS][W] (21 pages French + English translation. [RS]) The sketch appeared in the September 1800 issue. [EC]
- Note: the title states Volta's **contact theory** ("mere contact"). [RS]

1.8 Right after the pile

- **Nicholson and Carlisle**, London, **1800** (weeks after the letter reached London): first **electrolysis of water** (hydrogen and oxygen) with a pile. [W][NE] ([EC] says 1801; see section 4.)
- **Davy** (1807-1808): with very large piles (2000 pairs at the Royal Institution, 1808), isolated potassium, sodium, and barium, calcium, boron, strontium, magnesium. [W][EC]
- Petrov used a pile for an electric arc, 1802. Ritter's dry pile, 1802. [W]
- Daniell and Grove cells descended from the pile. [W]

1.9 Honors

- **1794: Copley Medal** of the Royal Society (gold). [W][MG] Fellow of the Royal Society (FRS). [W]
- **1801 (November): demonstrated the pile in Paris at the Institut de France**, with First Consul **Napoleon** present. Gold medal and a pension. [NE][MG]
- **1805:** Knight of the **Legion of Honour** and an annuity. [MG] (1806 Iron Crown, one source only.)
- **1809:** Senator of the Kingdom of Italy [MG] ([NE] says 1810, "Kingdom of Lombardy").
- **1810: made a Count** by Napoleon. [W][MG][NE] ATX 2026 C Q13: "What rank was Volta handed by Napoleon Bonaparte?" key **b) Count**. [T] Shock Value Network Analysis tiebreaker 2: "Count Alessandro Volta". [T]
- **Italian 10,000-lire banknote** (1990-1997), with a sketch of the pile. [W]
- 1927: Como centenary physics conference; Italian stamp. [W][NE] **Tempio Voltiano** museum, Como (opened 1928). [W][NE]
- Eponyms: unit **volt (V)**; lunar crater Volta; Volta Prize; Nvidia "Volta" GPU (2017); electric eel ***Electrophorus voltai*** (2019). [W]

- **Unit volt** adopted **1881** (International Electrical Congress, Paris) [NE][PR]; [EC] says 1880 (BAAS + congress). See section 5.

1.10 What is NOT Volta (distractor list from tests)

- "Electrode", "electrolyte", "anode", "cathode" were coined by **Faraday**, not Volta. [W] (BullSO 2026 C Q3: "Who coined the term electrode?" key C Faraday; Volta is a distractor. [T])
- "Electrodynamics" and "solenoid": **Ampère**. [T UT 2020 Q1, Q4]
- Torsion balance, inverse-square law: **Coulomb**. [T]
- Leyden jar (1745-46): von Kleist / van Musschenbroek, **not** Volta (Volta was born in 1745). [PR][G]
- "Who experimented with an early Ohm's law using Leyden jars and glass tubes?" (ATX 2019 Q2): **Henry Cavendish** is the standard answer [G; key not in the text file], Volta is a distractor.
- Electromagnetic induction (Faraday), AC (Tesla), V = IR (Ohm). [T]

Formulas

Item	Formula	Units	Note
Volt definition	1 V = 1 J/C = 1 W/A	V	[G]; asked as "volt = joule per ___"
Energy given to charge	W = QV	J = C·V	[G]
Pile EMF, n identical cells in series	E = n·e_cell	V	[W] cells add
Zn/brine/Cu cell EMF	e_cell ≈ 0.76 V (≈ 0.75 V [EC])	V	Use the value the test gives
Cells needed	n = ceil(V_target / e_cell)	cells	always round **up**
One cell reversed in a series pile of n	E = (n − 2)·e_cell	V	reversed cell subtracts its own EMF and cancels one more
Internal resistance, n cells in series	r_total = n·r	Ω	[G]
Current from pile into load R	I = n·e / (R + n·r)	A	Ohm's law, one equation
Terminal voltage	V_t = E − I·r_total = I·R	V	
Power in load	P = I²R = V_t²/R	W	
Identical piles in parallel	E unchanged, r_total = r/m	V, Ω	[G]
Unequal piles in parallel (S/N)	(E1−V)/r1 + (E2−V)/r2 = V/R (KCL at top node)	V	IN-SN
Electrochemical series (BORDERLINE)	E_cell = E°(cathode) − E°(anode); Volta's law: V(A→C) = V(A→B) + V(B→C)	V	E°: Zn −0.76, Cu +0.34, Ag +0.80 V [G]
Volta's law of capacitance (OUT for calc)	Q = C·V	C, F, V	Capacitance excluded in 2027

Key dates (for the app's lookup table): 1745 birth; 1769 Beccaria memoir; 1774 Como; 1775 electrophorus; 1776 methane (Nov); 1777 eudiometer; 1778/1779 Pavia; ~1782 condenser; 1791 Galvani's book; 1792 Volta's rebuttal; 1794 Copley; 1799 pile built; 20 Mar 1800 letter; 26 Jun 1800 read; 1801 Paris/Napoleon; 1805 Legion of Honour; 1809 senator; 1810 count; 1819 retired; 5 Mar 1827 death; 1881 volt named.

Question types

This note has **34 question types** (27 ✅ IN, 4 ⚠️ Borderline, 3 ❌ Out). The out-of-scope ones are in "Out of scope (skip)" below.

How the types were found

Order: easiest first. Each Python-verified example was run in `/private/tmp/claude-502/-Users-kai-Daniel/893762dd-d9b3-4158-937c-ea93eb65d1e6/scratchpad/volta_chk.py` (output quoted in each example).

History questions in the test-extraction files (8 test groups)

These rows come from `research/tests/` (ATX regionals, UT invitationals, misc-a, misc-b, USC 2026, Florida State 2021). Each row names the question type it belongs to.

Test	Year	Div	Level	Q	Question (short)	Key	Check / scope 2027 B	Type
ATX (UT Austin) Regional	2021	B	Regional	11	Picture of a stacked-disc device: what is it?	voltaic pile / first battery	IN	Q-V04
ATX (UT Austin) Regional	2021	B	Regional	12	Who made it?	Volta	IN	Q-V01
ATX (UT Austin) Regional	2021	B	Regional	13	Label the 3 layers (blue/red/green)	zinc, copper, brine-soaked cardboard	IN	Q-V04
ATX (UT Austin) Regional	2020	B	Regional	7	His invention was dangerous (sulfuric acid)	Volta (pile used brine or dilute acid)	IN	Q-V22
Cobra Invitational (Michigan)	2019-20 season	B	Invitational	1	Fill-in paragraph: Volta, voltaic pile 1799, electricity made "chemically"	Volta / Voltaic / chemically	IN	Q-V01
North Pocono Invitational (PA)	2020	C	Invitational	16	Describe or draw a voltaic pile	Zn/Cu discs, separator, electrolyte, order, end wires	IN	Q-V04
MIT Invitational	2019	C	Invitational (top tier)	40	Electrodes of Volta's first pile	zinc, copper (silver also used)	IN	Q-V04
UF Northern Regional (Florida)	2019	C	Regional	1-8	Match Volta/Ohm/Tesla/Hertz/Faraday to achievements	A, C, C, A, E, D, D, A	IN for Volta/Hertz rows (1, 4, 6, 7, 8); Tesla, Faraday OUT	Q-V14
Univ. of Michigan Invitational (UMSO)	2019 (file label)	not stated (likely C)	Invitational	1-5	Match Faraday, Tesla, Hertz, Volta, Ohm to discoveries	A, C, E, B, D	Hertz, Volta, Ohm IN; Faraday, Tesla OUT	Q-V14
UF Northern Regional (Florida)	2020	B	Regional	2	Inventor of the first battery	D Volta	IN	Q-V01
NY Lake Erie/Niagara Invitational	Dec 2018	C	Invitational	1	Volta's electrochemical series led to	B battery	IN	Q-V15
Florida State Tournament	2021	B (= C)	State	8	Quote: "The language of experiment is more authoritative than any reasoning"	Volta	IN (common attribution)	Q-V33
UT Austin Invitational, fall	2025	B/C	Invitational	4d	Pick who did most work in France	A, V, C (KEY ERROR: Volta worked in Como and Pavia, Italy)	IN	Q-V34
ATX (UT Austin) Regional	2026	B	Regional	MC1	First chemical battery	Volta	IN	Q-V01
ATX (UT Austin) Regional	2026	C	Regional	MC4	Who isolated methane?	Volta (1776-78)	IN	Q-V09

Test	Year	Div	Level	Q	Question (short)	Key	Check / scope 2027 B	Type
ATX (UT Austin) Regional	2026	C	Regional	MC13	Rank Napoleon gave Volta	Count	IN	Q-V12
UT Austin Invitational (2021 season)	2020	B	Invitational	2	Who showed electricity can be made chemically, not only by living things?	Volta	IN	Q-V06
UT Austin Invitational (2020 season)	2019	B	Invitational	2	Epithet: contested with a man over frogs	Volta (vs Galvani)	IN	Q-V06
UT Austin Invitational, fall (2020 season)	2019	C	Invitational	2	Epithet: contested with a man over frogs	Volta	IN	Q-V06
UT Austin Invitational, fall (no key)	2018	B/C	Invitational	1	Who invented the battery?	Volta (no key)	IN	Q-V01
ATX (UT Austin) Regional	2019	B	Regional	11	Who invented the first true battery?	Volta	IN	Q-V01
F-M Invitational (NY), fall	2019	not stated	Invitational	1	Match 7 discoveries to 7 scientists	Volta row correct	IN for the Volta row	Q-V14

Easy

Q-V01. Invention → person ("Who invented the first battery?")

Invention → person ("Who invented the first battery?") · ✅ IN · Easy

- **Scope:** IN (3.c.i).
- **Where seen:** BullSO 2026 C Q2 and BullSO 2026 B Q2 (`dlB/tests/bullso26_test.txt`); ATX 2019 Reg B/C Q11 ("first true battery"); ATX 2026 Reg B Q1 ("first chemical battery", key B Volta); Hawk & Hornet 2026 C Q4 (key D + bonus "Voltaic Pile"); PUSO 2026 matching 4 → B "Invented the first chemical battery"; Gull 2026 Q30 ("During a feud with Luigi Galvani, which scientist invented the voltaic pile?" key E Volta); Shock Value Network Analysis tiebreaker 2; MIT 2020 Q7 ("Breakthrough invention in 1800" → Volta). Div C items: the question needs only history, so it is IN for Div B.
  - Also seen (test-extraction files): Cobra Invitational (Michigan) 2019-20 season, Div B, Invitational, Q1: "Fill-in paragraph: Volta, voltaic pile 1799, electricity made "chemically"" (key: Volta / Voltaic / chemically). Div B 2027: IN.
  - Also seen (test-extraction files): UF Northern Regional (Florida) 2020, Div B, Regional, Q2: "Inventor of the first battery" (key: D Volta). Div B 2027: IN.
  - Also seen (test-extraction files): UT Austin Invitational, fall (no key) 2018, Div B/C, Invitational, Q1: "Who invented the battery?" (key: Volta (no key)). Div B 2027: IN.
- **How to solve:**
  1. Read the question: invention or description → scientist.
  2. "battery / pile / steady current / chemical electricity / 1800" → Volta.
- **Worked example:** "Who invented the first electric battery?" → **Alessandro Volta** (voltaic pile, 1800).
- **Traps:** Galvani (frog, no battery), Faraday (electrochemistry words), Franklin (he used the word "battery" for a row of Leyden jars [G]). "Baghdad battery" is a myth distractor [G].
- Calculator: look up — type: keyword ("battery", "pile").

Q-V02. Unit "volt" ↔ person, and what the volt measures

Unit "volt" ↔ person, and what the volt measures · ✅ IN · Easy

- **Scope:** IN (3.c.i + 3.c.iv units).

- **Where seen:** Rickards 2026 C Q12 (`rsoi26C_test.txt`: "Who is the SI Unit of Voltage named after, and what does Voltage represent?" key D, Volta, potential difference) - Div C, but units of voltage are Div B area iv, so IN. Princeton 2019 statement (a) ("SI unit of electric potential is named after him" → V).
- **How to solve:**
  1. Read the question: unit → person, or person → unit/quantity.
  2. Volt = unit of potential difference (voltage, EMF)
  3. 1 V = 1 J/C.
- **Worked example:** "1 volt equals one joule per ___?" → **coulomb**. (1 V = 1 J/C.)
- **Traps:** "voltage represents electron flow" (that is current). Joke names (Voltaire, "John Voltage"). Do not confuse V (volt) with the variable V.
- Calculator: look up — type: unit name or symbol.

Q-V03. True/False on Volta and the pile

True/False on Volta and the pile · ✅ IN · Easy

- **Scope:** IN.
- **Where seen:** Rickards 2026 B Q9 ("Alessandro Volta invented the first electric battery called the voltaic pile." True); Princeton 2021 Q8 (wine glasses with brine: True, key verified in the PDF), Q5 (living beings only: False), Q9 ("Ohm's main rival was Galvani": False; Galvani's rival was Volta).
- **How to solve:**
  1. Read the question: statement → True/False.
  2. Check each noun (person, date, material) against section 1.
- **Worked example:** "Volta's pile used zinc and copper discs separated by brine-soaked cardboard." → **True** [W].
- **Traps:** swapped names (Ohm vs Volta as Galvani's rival); "acid" vs "brine" (both appear in sources; brine is the textbook answer).
- Calculator: look up — type: keyword.

Medium

Q-V04. Name the device / label the parts (picture)

Name the device / label the parts (picture) · ✅ IN · Easy to Medium

- **Scope:** IN.
- **Where seen:** ATX 2021 Reg B and C Q11-13 (`atx21reg_Bkey.txt`: picture → "Voltaic Pile/First Battery", inventor Volta, labels **Zinc, Copper, Cardboard soaked in weak** [acid/brine; text cut off]). Northview 2026 C Q30 ("name of Volta's early electrochemical cell invention" → B Voltaic Pile; E Electrophorus is a distractor) - Div C, IN for Div B (history + DC source).
  - Also seen (test-extraction files): North Pocono Invitational (PA) 2020, Div C, Invitational, Q16: "Describe or draw a voltaic pile" (key: Zn/Cu discs, separator, electrolyte, order, end wires). Div B 2027: IN.
  - Also seen (test-extraction files): MIT Invitational 2019, Div C, Invitational (top tier), Q40: "Electrodes of Volta's first pile" (key: zinc, copper (silver also used)). Div B 2027: IN.
- **How to solve:**
  1. Read the question: picture of a stack (or ring of cups) → name, inventor, layer materials.
  2. Stack of alternating discs with wet layers = voltaic pile; ring of cups with strips = crown of cups; plate with insulating handle on a resin cake = electrophorus.
- **Worked example:** A drawing shows a column: grey disc, orange disc, wet pad, repeated. Name and parts? → **Voltaic pile: zinc, copper, brine-soaked cardboard.**
- **Traps:** Leyden jar picture (a jar with foil, not Volta's); Daniell cell (two solutions, 1836); dry cell.
- Calculator: look up — type: device keyword (stack / cups / plate).

Q-V05. Biography: birthplace, nationality, full name, university

Biography: birthplace, nationality, full name, university · ✅ IN · Easy to Medium

- **Scope:** IN.
- **Where seen:** Rickards 2026 C Q12 (full name in the answer); prior I_bank H6 type. Not otherwise seen.

- **How to solve:**
  1. Read the question: clue → Volta or a place/date.
  2. Como (born/died), Pavia (professor), Italian, 1745-1827.
- **Worked example:** "Which of the seven listed scientists was Italian?" → **Volta** (the other six are French, German or English).
- **Traps:** Galvani was also Italian (Bologna) but is not on the list. Pavia 1778 vs 1779 (accept either if asked; see section 5).
- Calculator: look up — type: person or place.

Q-V06. Galvani dispute: who, what, outcome

Galvani dispute: who, what, outcome · ✅ IN · Medium

- **Scope:** IN (history of Volta's discovery). Galvani himself is not on the list, so a question only about Galvani is BORDERLINE.
- **Where seen:** Gull 2026 Q30; UT Austin 2020 Inv B/C Q2 (`ut20inv_Bkey.txt`, key B Volta); Princeton 2021 Q5, Q9; Princeton 2019 (d) "contact electricity" → V.
  - Also seen (test-extraction files): UT Austin Invitational (2020 season) 2019, Div B, Invitational, Q2: "Epithet: contested with a man over frogs" (key: Volta (vs Galvani)). Div B 2027: IN.
  - Also seen (test-extraction files): UT Austin Invitational, fall (2020 season) 2019, Div C, Invitational, Q2: "Epithet: contested with a man over frogs" (key: Volta). Div B 2027: IN.
- **How to solve:**
  1. Read the question: description of the dispute → who said what.
  2. Galvani = "animal electricity" (from the frog)
  3. Volta = "contact of two dissimilar metals"; frog only a detector.
  4. Result = the pile.
- **Worked example:** "Galvani saw frog legs twitch when touched by two different metals. What did Volta say caused it?" → **The two dissimilar metals (with a moist conductor) made the current; the frog only detected it.**
- **Traps:** answering "the frog made it" (Galvani's view). Saying Volta was totally right (nerves are electrical, so Galvani was partly right).
- Calculator: look up — type: "Galvani".

Q-V07. Electrophorus: history

Electrophorus: history · ✅ IN · Medium

- **Scope:** IN.
- **Where seen:** Northview 2026 C Q30 (Electrophorus as a distractor). Not seen as its own question.
- **How to solve:**
  1. Read the question: "who improved/popularized/named the electrophorus, when" → Volta, 1775 (Wilcke 1762 first).
  2. Lookup.
- **Worked example:** "Volta is often called the inventor of the electrophorus. Who described a similar device first, and when?" → **Johan Carl Wilcke, 1762.**
- **Traps:** "Volta invented it" (the key may accept it on easy tests; "improved/popularized" is correct). Confusing it with the electric eel genus *Electrophorus* (named for the same Greek word; *E. voltai* honors Volta).
- Calculator: look up — type: "electrophorus".

Q-V08. Electrophorus: physics (charging by induction)

Electrophorus: physics (charging by induction) · ✅ IN · Medium

- **Scope:** IN (3.c.ii properties of charges, sources of static electricity).
- **Where seen:** not seen in a test.
- **How to solve:**

1. Read the question: sign of the dielectric's charge + steps → sign on the metal plate; why it can repeat; where the energy comes from.
2. (1) dielectric charged by rubbing.
3. (2) plate set on top: induction separates charge.
4. (3) grounding removes the like charge.
5. (4) lift: plate keeps the opposite charge.
6. Energy = work to lift the plate.

- **Worked example:** A resin plate rubbed with fur becomes negative. The metal plate is set on it, touched, then lifted. Plate charge? → **Positive** (opposite to the dielectric).
- **Traps:** "same sign as the dielectric" (that is charging by conduction); thinking the dielectric loses charge each cycle; forgetting the ground step (without it the lifted plate is neutral).
- Calculator: look up — type: dielectric sign. (App note: (rule: plate sign = opposite of dielectric sign).)

Q-V09. Methane discovery

Methane discovery · ✅ IN · Medium

- **Scope:** IN (a discovery made by Volta). The gas chemistry itself is OUT (Q-V31).
- **Where seen:** Northview 2026 C Q49 ("First to discover and isolate methane gas" → correct A Volta; the key says B, an error, section 4); ATX 2026 Reg C Q4 ("Who isolated methane gas?" key D Volta); Princeton 2019 (k). Div C items: plain history of a listed scientist, so IN for Div B.
- **How to solve:**
  1. Read the question: methane / marsh gas / Lake Maggiore / 1776 → Volta.
  2. Lookup.
- **Worked example:** "In 1776, which scientist on the list found 'inflammable air' in a marsh by Lake Maggiore?" → **Volta** (methane, Angera, November 1776).
- **Traps:** Ohm, Coulomb, Kirchhoff as distractors (Kirchhoff discovered elements caesium/rubidium, not a gas).
- Calculator: look up — type: "methane".

Q-V12. Honors: count, medals, banknote

Honors: count, medals, banknote · ✅ IN · Medium

- **Scope:** IN.
- **Where seen:** ATX 2026 Reg C Q13 (rank from Napoleon, key b Count) - Div C, plain history, IN for Div B; Shock Value Network Analysis tiebreaker 2 ("include titles": "Count Alessandro Volta").
- **How to solve:**
  1. Read the question: honor → person, or person → honor.
  2. Copley 1794.
  3. Napoleon gold medal 1801.
  4. Legion of Honour 1805; senator 1809.
  5. **count 1810**
  6. 10,000-lire note.
- **Worked example:** "What title did Napoleon give Volta, and in what year?" → **Count, 1810.**
- **Traps:** Duke / Baron / Marquis distractors; ETHW wrongly gives 1801 for the count (section 4); "Sir" (that is Banks or Davy, not Volta).
- Calculator: look up — type: honor or year.

Q-V14. Matching column / letter-code statements

Matching column / letter-code statements · ✅ IN · Medium

- **Scope:** IN.
- **Where seen:** PUSO 2026 matching part 1 (4 → B); Princeton 2019 Part 1 (a, c, d, f, k, q → V); MIT 2020 Q1-10 fill-in (Q7, Q10 → Volta); ATX 2021 Q2-7 name bank; ATX 2020 Reg "this scientist" list (sulfuric acid → Volta). All Div B/C invitationals; the Volta lines are pure history, IN. (Lines about Faraday/Tesla in those same sets are OUT for 2027: names removed from the list.)

  - Also seen (test-extraction files): UF Northern Regional (Florida) 2019, Div C, Regional, Q1-8: "Match Volta/Ohm/Tesla/Hertz/Faraday to achievements" (key: A, C, C, A, E, D, D, A). Div B 2027: IN for Volta/Hertz rows (1, 4, 6, 7, 8); Tesla, Faraday OUT.
  - Also seen (test-extraction files): Univ. of Michigan Invitational (UMSO) 2019 (file label), Div not stated (likely C), Invitational, Q1-5: "Match Faraday, Tesla, Hertz, Volta, Ohm to discoveries" (key: A, C, E, B, D). Div B 2027: Hertz, Volta, Ohm IN; Faraday, Tesla OUT.
  - Also seen (test-extraction files): F-M Invitational (NY), fall 2019, Div not stated, Invitational, Q1: "Match 7 discoveries to 7 scientists" (key: Volta row correct). Div B 2027: IN for the Volta row.
- **How to solve:**
  1. Read the question: 5-10 statements × 5-7 names → one letter each.
  2. Tick the easy ones first, eliminate, then decide the obscure ones.
- **Worked example:** Statements: (1) unit of potential; (2) contact electricity of two metals; (3) Q ∝ V for an object; (4) first battery. Answer for all four → **Volta**. (Princeton 2019 key a, d, q, f = V.)
- **Traps:** "used only once" vs "may repeat" instructions; Faraday's electrochemistry words.
- Calculator: look up — type: keyword per statement.

Q-V15. Contact theory and the electrochemical series (history level)

Contact theory and the electrochemical series (history level) · ✅ IN · Medium

- **Scope:** IN (history). Numerical electrode potentials are Q-V23 (BORDERLINE).
- **Where seen:** NY Lake Erie/Niagara 2019 Inv Q1 (`wiki/CircuitLabTest2019I.txt`: series led to the B battery); Princeton 2019 (d).
  - Also seen (test-extraction files): NY Lake Erie/Niagara Invitational Dec 2018, Div C, Invitational, Q1: "Volta's electrochemical series led to" (key: B battery). Div B 2027: IN.
- **How to solve:**
  1. Read the question: "what did Volta's series lead to / what was his theory / was it right".
  2. Volta ranked metals; the farther apart two metals are, the larger the voltage; he thought contact itself caused the EMF; Davy showed it is chemical.
- **Worked example:** "Volta's contact theory said the EMF arises where? What did later chemists find?" → **At the contact between the two metals; actually from the chemical reaction at the metal-electrolyte surfaces (zinc oxidizes).**
- **Traps:** Choice "capacitor" or "PN junction" in the 2019 item; saying the copper is consumed.
- Calculator: look up — type: "contact theory".

Q-V16. Pile voltage from number of cells

Pile voltage from number of cells · ✅ IN · Medium

- **Scope:** IN (3.c.iii DC sources, 3.c.iv voltage, 3.c.v series). Math: multiplication only.
- **Where seen:** not seen in a test (Wikipedia's 6-cell example).
- **How to solve:**
  1. Read the question: n cells, EMF per cell → total EMF.
  2. E = n × e_cell (series cells add)
  3. Count cells = number of electrolyte layers (each Zn-wet-Cu set).
- **Worked example:** 6 Zn/brine/Cu cells at 0.76 V → **4.56 V** (Python 6×0.76 = 4.56).
- **Traps:** counting discs instead of cells (30 metal discs, alternating Zn-Cu = 15 cells → 11.4 V; Python 15×0.76 = 11.4); Volta's extra end discs add no cell; using 1.5 V (alkaline) or 1.10 V (Daniell) instead of the given value.
- Calculator: calculate — type: n (or number of wet layers), e_cell.

Q-V17. Number of cells for a target voltage

Number of cells for a target voltage · ✅ IN · Medium

- **Scope:** IN.
- **Where seen:** not seen in a test.
- **How to solve:**

1. Read the question: target V, e_cell → minimum cells.
2. N = V_target / e_cell, then **round up**.

- **Worked example:** Minimum Zn-Cu cells (0.76 V each) to reach at least 9 V? 9/0.76 = 11.84 → **12 cells, 9.12 V** (Python ceil = 12, 12×0.76 = 9.12).
- **Traps:** rounding down (11 cells give only 8.36 V); crown-of-cups version: 12 cups, each with one Zn and one Cu strip, also → 9.12 V.
- Calculator: calculate — type: V_target, e_cell.

Q-V20. Energy and charge with the volt (W = QV)

Energy and charge with the volt (W = QV) · ✅ IN · Medium

- **Scope:** IN (3.c.iv energy, units).
- **Where seen:** ATX 2019 Q15 asks a 2 C / 10 J charge-energy question (not Volta-themed). Not seen with a pile.
- **How to solve:**
  1. Read the question: two of {Q, V, W} → the third.
  2. W = Q·V (V = J/C).
- **Worked example:** 3.0 C passes through a 6-cell pile (4.56 V). Energy given? → 3 × 4.56 = **13.7 J** (Python 13.68). Reverse: 9.0 J at 4.56 V → Q = **1.97 C** (Python 1.9737).
- **Traps:** using P = VI without a time; mixing W (watt) and W (work).
- Calculator: calculate — type: Q, V or W.

Q-V24. Volta's law of capacitance (Q ∝ V)

Volta's law of capacitance (Q ∝ V) · ⚠️ Borderline · Medium (name only)

- **Scope:** BORDERLINE. The name/attribution is history of Volta (3.c.i), but 3.d excludes "capacitance". A one-line "who found Q ∝ V" is possible; any capacitance calculation is OUT (Q-V30).
- **Where seen:** Rickards 2020 B Q20 and C Q24 (key: Volta); Princeton 2019 (q) → V.
- **How to solve:**
  1. Read the question: "who found charge ∝ potential" → Volta.
  2. Lookup.
- **Worked example:** "Who discovered that electric potential and charge are proportional?" → **Volta.**
- **Traps:** answering Coulomb (coulomb is the unit of charge) or Faraday (farad is the unit of capacitance).
- Calculator: look up — type: "Q ∝ V".

Q-V25. Legacy and applications

Legacy and applications · ✅ IN · Medium

- **Scope:** IN (the pile's uses). Davy, Nicholson and Carlisle are not on the list, so a question about them alone is BORDERLINE.
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: "what did the pile make possible first".
  2. Electrolysis of water (Nicholson & Carlisle, 1800); new elements (Davy, 1807-1808); electric arc (1802); steady DC for Ohm's experiments (Ohm first used a voltaic pile, then a thermocouple [PR]); modern batteries.
- **Worked example:** "Which experiment was first done with a voltaic pile in 1800?" → **Electrolysis of water into hydrogen and oxygen (Nicholson and Carlisle).**
- **Traps:** "electromagnetic induction" (Faraday 1831, OUT topic); Ohm's law (Ohm switched away from the pile because it was unsteady).
- Calculator: look up — type: keyword.

Q-V27. Volta-cell hands-on (coin/lemon battery)

Volta-cell hands-on (coin/lemon battery) · ⚠️ Borderline · Medium

- **Scope:** BORDERLINE. The 2027 hands-on list (Part II 3.c) does not name a home-made cell, but says "may include but are not limited to". It is low voltage and safe (3.d allows it). Measuring its voltage with a multimeter fits 3.b.

- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: measured voltage of one cell, number of cells → predicted stack voltage; or whether it can light an LED.
  2. E = n·e; an LED needs about its forward voltage (red ≈ 1.8-2 V [G]) and enough current.
- **Worked example:** one penny-zinc washer-vinegar cell reads 0.76 V (assume, same as the pile value). Cells for at least 1.5 V? 1.5/0.75 = 2.0 → with 0.76 V cells, **2 cells (1.52 V)**. (Python 1.5/0.75 = 2.0; 2×0.76 = 1.52.)
- **Traps:** expecting the LED to light from 1.5 V (red LED needs ~1.8 V) [G]; reversed cell in the stack.
- Calculator: calculate — type: e_cell measured, target V.

Q-V28. Static electricity link: why the pile was new

Static electricity link: why the pile was new · ✅ IN · Medium

- **Scope:** IN (3.c.ii vs 3.c.iii: static vs current electricity).
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: compare the electrophorus/Leyden jar with the pile.
  2. Static devices give one short discharge (high voltage, tiny charge); the pile gives a steady low-voltage current (DC).
- **Worked example:** "Why was the voltaic pile more useful than a Leyden jar for experiments on circuits?" → **It gave a continuous, steady current (DC) instead of a single spark.**
- **Traps:** "higher voltage" (a Leyden jar has much higher voltage).
- Calculator: look up — type: the keyword from the question.

Q-V33. Quote attribution (who said it?)

Quote attribution (who said it?) · ✅ IN · Medium

- **Scope:** IN (3.c.i).
- **Where seen:** Florida State Tournament 2021, Div B (same as Div C), State, Q2-Q8 (seven quotes, 2 points each, "identify the last name"). Q8 is Volta: "The language of experiment is more authoritative than any reasoning."
- **How to solve:**
  1. Look for one key word or idea in the quote (an invention, a place, a belief, a language).
  2. Match that idea to one of the seven listed scientists.
  3. If the quote is in French, think of the two French scientists first: Ampère and Coulomb.
  4. If nothing fits the seven, the answer may be a scientist who is not on the 2027 list (Tesla, Faraday). Those items are out of scope in 2027.
  5. Clues for Volta: experiment over theory (his fight with Galvani), Italian, Napoleon, the pile.
- **Worked example:** "The language of experiment is more authoritative than any reasoning..." → **Volta** (Florida 2021 key; a common attribution).
- **Traps:** Ohm's quote in the same test is about "principles obtained chiefly by experiment". Read the whole quote: Ohm's is about a "Memoir" and "contact of two or more bodies".
- Calculator: look up — type: a key word from the quote (e.g. "language of experiment").

Q-V34. Multi-select: which listed scientists have property X? (France trap)

Multi-select: which listed scientists have property X? (France trap) · ✅ IN · Medium

- **Scope:** IN (3.c.i).
- **Where seen:** UT Austin Invitational fall 2025, Div B/C, Invitational, Q4a-d (letters A, C, K, V, O, T, F; pick every letter that fits). In 4d ("did most work in France") the key gave A, V, C. Volta is wrong there.
- **How to solve:**
  1. Write the letter code of each listed scientist (A = Ampère, C = Coulomb, K = Kirchhoff, V = Volta, O = Ohm, Hz = Hertz, B = Boole).
  2. Test each scientist against the property, one at a time.
  3. Pick every scientist that passes. A multi-select question can have more than one right letter.

4. Check each letter again. Keys often add a wrong letter (UT 2025 put Volta under "France").

- **Worked example:** "Which did most of their work in France? A, C, K, V, O" → **A and C**. Volta worked at Como and the University of Pavia (Italy).
- **Traps:** Volta went to Paris in 1801 and Napoleon honoured him. That visit does not make France his place of work.
- Calculator: look up — type: the property (e.g. "France", "Italy").

Hard

Q-V10. Eudiometer and Volta pistol

Eudiometer and Volta pistol · ✅ IN · Hard

- **Scope:** IN (Volta's discovery; the device uses an electric spark).
- **Where seen:** Princeton 2019 (k) (spark ignition of methane-air). Not seen by name in a test.
- **How to solve:**
  1. Read the question: device description → name/inventor/purpose.
  2. Eudiometer = closed glass tube, spark ignites a gas mixture, volume change measures oxygen ("goodness of air")
  3. Pistol = spark from a Leyden jar (or electrophorus) explodes gas and fires a cork.
- **Worked example:** "Volta's electric pistol used what to ignite the gas?" → **An electric spark (from a Leyden jar).** [MG]
- **Traps:** "a match/flame"; "a battery" (the pistol is older than the pile).
- Calculator: look up — type: "pistol" / "eudiometer".

Q-V13. Timeline ordering and date arithmetic

Timeline ordering and date arithmetic · ✅ IN · Medium (order) to Hard (arithmetic, exact dates)

- **Scope:** IN.
- **Where seen:** Princeton 2020 C Q1 (order the scientists by crowning contribution; key gives Volta 1799, Coulomb 1786). Div C, but ordering is pure history: IN for Div B, using the 2027 list (Hertz, Boole instead of Tesla, Faraday).
- **How to solve:**
  1. Read the question: list of events → order; or two dates → years between / age.
  2. Map each to its year, sort.
  3. Ages: subtract years; subtract 1 if the birthday (18 Feb) has not yet passed.
- **Worked example:** Order: Ohm's law, Galvani's book, voltaic pile, Coulomb's law, Ampère's force law, Boole's *Laws of Thought*, Kirchhoff's laws, Hertz's waves. → Python sort: **Coulomb 1785, Galvani 1791, pile 1800, Ampère 1820, Ohm 1827, Kirchhoff 1845, Boole 1854, Hertz 1887.**
- **Worked example:** Volta's age when he wrote to Banks (20 Mar 1800)? → **55**. Age at death? → **82**. Years from Galvani's 1791 book to the pile letter? → **9**. Years from the pile to the volt's naming (1881)? → **81**. (All Python-verified.)
- **Traps:** 1799 vs 1800 for the pile; Volta died in 1827, the same year Ohm published; Galvani died 1798, before the pile.
- Calculator: look up + calculate — type: two events (pick from list) or two dates (y, m, d). (App note: (date difference, age) on top of a LOOKUP date table.)

Q-V18. One cell put in backward

One cell put in backward · ✅ IN · Hard

- **Scope:** IN (series EMF addition with sign; 3.c.v).
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: n cells, k reversed, e_cell → total EMF.
  2. E = (n − 2k)·e_cell (a reversed cell loses its +e and adds −e).
- **Worked example:** 10 cells, 1 reversed, 0.76 V each → (10 − 2)×0.76 = **6.08 V** (Python).
- **Traps:** answering (n − 1)·e = 6.84 V.

- Calculator: calculate — type: n, k, e_cell.

Q-V19. Pile with internal resistance driving a load

Pile with internal resistance driving a load · ✅ IN · Hard

- **Scope:** IN (Ohm's law, series circuit, one equation; 3.c.v). Real piles had high internal resistance [G].
- **Where seen:** not seen in a test with a pile (internal resistance appears in many DC tests).
- **How to solve:**
  1. Read the question: n, e_cell, r per cell, load R → I, terminal V, load power.
  2. E = n·e; r_tot = n·r.
  3. I = E/(R + r_tot)
  4. V_t = I·R; P = I²R.
- **Worked example:** 10 cells, 0.76 V and 2.0 Ω each, load 10 Ω. E = 7.6 V, r_tot = 20 Ω, I = 7.6/30 = **0.253 A**, V_t = **2.53 V**, P = **0.642 W** (Python 0.25333, 2.5333, 0.64178).
- **Traps:** forgetting r_tot is n·r; using E across R (gives 0.76 A, wrong).
- Calculator: calculate — type: n, e_cell, r_cell, R.

Q-V22. How the pile works (chemistry, hazards)

How the pile works (chemistry, hazards) · ⚠️ Borderline · Medium to Hard

- **Scope:** BORDERLINE. Batteries are a DC "source" (3.c.iii), so "zinc is used up, hydrogen bubbles form, the pile weakens" fits. Redox half-reactions are chemistry beyond the listed areas, so deep chemistry is likely not tested.
- **Where seen:** ATX 2020 Reg B/C history list ("invention was extremely dangerous to handle, due to it producing sulfuric acid" → Volta; the wording is wrong, section 4); Philadelphia-area 2026 B Q18 (`dlB/tests/phi26_test.txt`: "In a standard alkaline battery or voltaic cell, where does oxidation occur?" → A Anode).
- **How to solve:**
  1. Read the question: "which metal is used up / what gas / why did it stop working".
  2. Zinc oxidizes (anode, negative terminal)
  3. H_2 forms at the copper; electrolyte dries/squeezes out; bubbles block the copper (polarization).
- **Worked example:** "In a Zn-Cu voltaic pile, which metal is consumed?" → **Zinc.**
- **Traps:** copper is consumed (it is not); "anode = positive" (in a battery the anode is the negative terminal).
- Calculator: look up — type: keyword.

Q-V23. Electrode potentials and Volta's series (numeric)

Electrode potentials and Volta's series (numeric) · ⚠️ Borderline · Hard

- **Scope:** BORDERLINE. Volta's law of the series is a Volta discovery (3.c.i) and the math is subtraction; but standard electrode potentials are chemistry, not in the 3.c list. Seen only in Div C tests. A Div B writer might give the potentials in the question.
- **Where seen:** MIT 2020 C Q42 (Cu/Ag cell 0.46 V, "electrolytic or voltaic"; per `prior/G_tests.md`); Rickards 2026 C Q30 (voltaic vs electrolytic electron direction).
- **How to solve:**
  1. Read the question: two E° values → cell EMF; or two pair voltages → third (Volta's additive law).
  2. E_cell = E°(higher) − E°(lower)
  3. Additive: V(Zn,Ag) = V(Zn,Cu) + V(Cu,Ag).
- **Worked example:** E°: Zn −0.76, Cu +0.34, Ag +0.80 V. Zn-Cu = **1.10 V**, Cu-Ag = **0.46 V**, Zn-Ag = **1.56 V** = 1.10 + 0.46 (Python).
- **Traps:** 1.10 V (Daniell standard) vs 0.76 V (brine pile, hydrogen at copper); sign errors.
- Calculator: look up + calculate — type: E°1, E°2 (with a LOOKUP table of common metals).

Q-V26. "Which is NOT Volta's?" / mis-attribution

"Which is NOT Volta's?" / mis-attribution · ✅ IN · Medium to Hard

- **Scope:** IN.

- **Where seen:** BullSO 2026 C Q3 (electrode → Faraday, Volta a distractor); UT 2020 Q1, Q3-Q5 (Volta as a distractor); Northview 2026 C Q46 (Volta as a distractor for "Nobel prize for AC": key E Nobody); ATX 2019 Q2 (Leyden jar/Ohm's law: Volta a distractor).
- **How to solve:**
  1. Read the question: 4-5 claims → the false one.
  2. Check each claim against section 1.
  3. Volta never worked on magnetism or AC.
- **Worked example:** "Which is NOT a Volta achievement? A. electrophorus improvement B. methane C. the word 'electrode' D. voltaic pile" → **C** (Faraday coined "electrode").
- **Traps:** "invented the electrophorus" (popularized, not invented; the expected answer is usually still "Volta's").
- Calculator: look up — type: claim keyword.

Q-V29. Volta in a mixed "this scientist" history list (essay/short answer, tiebreaker)

Volta in a mixed "this scientist" history list (essay/short answer, tiebreaker) · ✅ IN · Hard

- **Scope:** IN.
- **Where seen:** Princeton 2020 C Q2 style short response (War of Currents; not Volta). Shock Value tiebreaker 2 ("include titles"). Not seen as a Volta essay.
- **How to solve:**
  1. Read the question: "Describe the Galvani-Volta dispute and its result in 3-4 sentences.".
  2. Claim (who), evidence (frog, metals, brine paper), result (pile, letter 1800), significance (first steady current).
- **Worked example:** answer outline → **Galvani (1791) said frog muscle made "animal electricity". Volta (1792 on) showed two dissimilar metals with a wet conductor make the current. To prove it, he built the voltaic pile (letter to Banks, 20 March 1800), the first steady current source.**
- **Traps:** missing the date or the metals; "include titles" → Count.
- Calculator: look up — type: (stored outline).

Nationals

Q-V11. The 1800 letter to Banks

The 1800 letter to Banks · ✅ IN · Hard (States/Nationals history depth)

- **Scope:** IN.
- **Where seen:** not seen in a test (prior I_bank lists the fact).
- **How to solve:**
  1. Read the question: "To whom / when / which society / what title / what language" → answer.
  2. Lookup section 1.7.
- **Worked example:** "Volta announced the pile in a letter dated 20 March 1800. To whom?" → **Sir Joseph Banks, President of the Royal Society of London.** Days from the letter date to its reading (26 June 1800): Python `date(1800,6,26)-date(1800,3,20)` = **98 days**.
- **Traps:** "Napoleon" (1801 demo was later); "French Academy"; publication date ≠ letter date (Wikipedia's wording blurs this).
- Calculator: look up + calculate — type: two dates. (App note: (dates). CALCULATE for day/year gaps.)

Q-V21. Piles in series vs parallel

Piles in series vs parallel · ✅ IN · Medium (identical piles) / Nationals (unequal piles, Q-V21b)

- **Scope:** identical piles IN; unequal piles in parallel **IN-SN** (needs KVL/KCL, 3.c.ix; a system of two equations, allowed by 3.e.i).
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: EMFs, internal resistances, arrangement, load → total EMF, currents.
- **Method (identical):** series → EMFs add; parallel → EMF same, more current capacity, r/m. Method (unequal, S/N): KCL at the top node: (E1 – V)/r1 + (E2 – V)/r2 = V/R, solve for V; then each current.
- **Worked example:** two 4.56 V piles in series → **9.12 V**; in parallel → **4.56 V**.

- **Worked example:** Pile 1: 6 cells (4.56 V), r1 = 1.2 Ω. Pile 2: 5 cells (3.80 V), r2 = 1.0 Ω. Both in parallel across R = 4.0 Ω. Python: V = **3.648 V**, I1 = **0.760 A** (out of pile 1), I2 = **0.152 A** (out of pile 2), I_R = **0.912 A** (check 0.760 + 0.152 = 0.912).
- **Traps:** assuming parallel EMFs average; the weaker pile can be charged backward if its EMF is below V (here 3.80 > 3.648, so it still supplies current).
- Calculator: calculate — type: E1, r1, E2, r2, R (and "series/parallel").

Out of scope (skip)

These types appear in tests or books, but the 2027 Div B rules exclude them. Skip them. They stay here so you can recognise them.

Q-V30. Capacitance calculations from Volta's law

Capacitance calculations from Volta's law · ✗ Out · n/a

- **Scope: OUT** (3.d "capacitance").
- **Where seen:** ATX 2020 Reg Q3-4 (parallel-plate C, Div B/C 2020), not Volta-named.
- Calculator: none (skip).

Q-V31. Methane chemistry (combustion equations, gas laws)

Methane chemistry (combustion equations, gas laws) · ✗ Out · n/a

- **Scope: OUT** (not in any 3.c area; chemistry).
- **Where seen:** not seen.
- Calculator: none (skip).

Q-V32. Voltaic vs electrolytic cell classification, electron flow anode→cathode

Voltaic vs electrolytic cell classification, electron flow anode→cathode · ✗ Out · Hard

- **Scope: OUT** for Div B 2027 as a stand-alone electrochemistry item (Rickards 2026 C Q30, MIT 2020 C Q42 are Div C chemistry). Only the plain "battery = voltaic cell, makes DC" idea is BORDERLINE (Q-V22).
- **Where seen:** not seen in a test.
- Calculator: none (skip). (or LOOKUP of the one-line definition).

Mistakes in tests and keys

1. **Northview Sylvania 2026 C key, Q49** ("First to discover and isolate methane gas": A Volta, B Ohm ...). Key says **B** (checked on the key PDF image, page 3). Correct: **A, Volta** [W]. (prior G_tests.md also flagged it.)
2. **ATX (UT) Regional 2020 B/C**, history list: "This scientist's invention was extremely dangerous to handle, due to it producing sulfuric acid (Volta)." The pile does **not produce** sulfuric acid. Volta used brine; some cups used "acidulated water" [KY]. Hazards were mostly the acid electrolyte some users added, not a product. The intended answer (Volta) is fine, the reason is wrong.
3. **Princeton 2019 statement (k):** spark ignition of methane formed "the basis of the internal combustion engine". Overstated; no source I read makes that link. The answer V is still right.
4. **Princeton 2020 C key Q1:** "Coulomb (1786)". Coulomb's first three electricity memoirs and the torsion-balance law are 1785 [PR][W]. "Volta (1799)" is defensible (built late 1799 [EC]), but most sources say 1800.
5. **IEEE ETHW Volta page:** "1801: Napoleon made him a count." Wikipedia, Museo Galileo and napoleon-empire.org all say **1810**; 1801 was the gold medal.
6. **Wikipedia (Volta article):** says the *Philosophical Transactions* article "appeared ... on 20 March 1800". 20 March is the letter's date; it was read 26 June 1800 [RS] and printed later in 1800 (sketch September 1800 [EC]).
7. **Electrochemical Society encyclopedia:** puts Nicholson and Carlisle's electrolysis of water in **1801**. Wikipedia and napoleon-empire.org put it in **1800** (weeks after the letter). 1800 is the standard date.
8. **Wikipedia (Volta) lead vs body:** lead gives 1799 for the pile, body gives 1800. Not an error, but tests can disagree; accept 1799/1800.
9. **MIT 2020 key Q4** says Ohm's acoustic law was "proven false". Not a Volta item; noted only because it sits in the same fill-in set (Ohm's acoustic law is still taught as partly valid [W]).

- **UT Austin Invitational fall 2025, Q4d:** key A, V, C for "did most work in France". Volta worked at Como and Pavia (Italy). Right answer A, C.
- **UT Austin Invitational fall 2025, Q4b-c:** the key lists V for "law with 1/r" and "laws useful in steady state". There is no "Volta's law" with 1/r. Doubtful.
- **ATX Regional 2020, Div B, Q7:** "invention dangerous (sulfuric acid)" → Volta. The first pile used brine (salt water) or dilute acid on cloth. The acid detail is loose but the answer is fine.

Not confirmed

- **Pavia start, 1778 or 1779.** Museo Galileo: elected 1778. Wikipedia and napoleon-empire.org: 1779. Not settled.
- **Second part of the letter: 1 April or 2 April 1800.** [EC] says 1 April; Wellcome catalogue item M0018845 says 2 April. Not settled.
- **Volt adopted 1880 or 1881.** [EC] 1880 (BAAS + congress); napoleon-empire.org and prior file: 1881 (Paris International Electrical Congress). Britannica returned HTTP 403. Most textbooks say 1881.
- **Senator 1809 (Kingdom of Italy) vs 1810 (Kingdom of Lombardy).** Museo Galileo vs napoleon-empire.org.
- **Legion of Honour year 1805** comes from Museo Galileo only; one search result tied it loosely to 1801.
- **"Up to about 30 discs"** in the first pile comes from a search-result summary of the ECS/JSTOR Daily pages; I did not read the primary text.
- **EMF per cell, 0.76 V vs 0.75 V.** Both are textbook-level estimates ([W], [EC]); real piles varied with electrolyte. Use the test's value.
- **Galvani's first frog experiments: 1780 [W] vs the key metal-contact experiment 1786 [EC].** Both appear; a test could use either.
- **"Volta introduced the term tension"**: only [EC] says he introduced the concept; I did not find the Italian primary use date.
- **Copley Medal reason** (work on animal electricity / contact): not confirmed; I only confirmed the year 1794 [W] [MG].
- **ATX 2019 Q2 answer** (Leyden jars and glass tubes → Cavendish): the key text file has no marked answer; Cavendish is from general history knowledge, not verified in the key.
- **ATX 2021 Q13 third label** text is cut off ("Cardboard soaked in weak ..."); I did not view the PDF image to read the last word.
- **Wikipedia's "Volta coined galvanism"**: seen only in the Galvani article; not cross-checked.
- **Div B Nationals/States Volta questions:** no free State or National Circuit Lab test exists (prior G_manifest: Nationals packets are paid). So the IN-SN depth for Volta is my estimate from the rules, not from a seen test.
- Britannica (403) and the 1800 *Phil. Trans.* full text (not fetched) were not read.

- Florida State 2021 Q8 quote ("The language of experiment is more authoritative than any reasoning"): a common attribution to Volta. Primary source not found. Searched `research/downloads/` only.

Sources

- Official 2027 Div B rules: `/Users/kai/Daniel/Science_Olympiad_Div_B_Rules_2027_for_Web_Secured.pdf` (Circuit Lab B, pp. B11-B12).
- https://en.wikipedia.org/wiki/Alessandro_Volta
- https://en.wikipedia.org/wiki/Voltaic_pile
- https://en.wikipedia.org/wiki/Electrophorus
- https://en.wikipedia.org/wiki/Luigi_Galvani
- https://en.wikipedia.org/wiki/Eudiometer
- https://catalogue.museogalileo.it/biography/AlessandroVolta.html
- https://catalogue.museogalileo.it/object/VoltaicPistol.html
- https://www.napoleon-empire.org/en/personalities/volta.php
- https://knowledge.electrochem.org/encycl/art-v01-volta.htm

- https://ethw.org/Alessandro_Volta
- https://makingscience.royalsociety.org/items/l-and-p_11_137 (Royal Society archive: letter 20 March 1800, read 26 June 1800)
- https://wellcomecollection.org/works/bad5xmq4 and https://content.www.wellcomecollection.org/works/w7du7fha (letters to Banks, 20 March and 2 April 1800)
- https://physics.kenyon.edu/EarlyApparatus/Electricity/Crown_of_Cups/Crown_of_Cups.html
- https://physics.kenyon.edu/EarlyApparatus/Thermodynamics/Eudiometer/Eudiometer.html
- https://www.larousse.fr/encyclopedie/personnage/Alessandro_comte_Volta/149269
- https://alessandrovolta.it/en/?p=1826 , https://alessandrovolta.it/en/?p=1842
- https://daily.jstor.org/electric-fish-and-the-first-battery/
- OpenStax College Physics 2e, ch. 18-21 (charge, EMF, internal resistance) - general formulas only.
- Past tests (local, `/Users/kai/Daniel/circuitlab/research/downloads/`): dlG/atx19reg_{B,C,key}.txt; dlG/atx20reg_key.txt; dlG/atx21reg_{B,C}key.txt; dlG/ut20inv_{B,C}key.txt; dlG/atx26reg_{B,C}{test,key}.txt; dlG/bull26C_{test,key}.txt; dlB/tests/bullso26_test.txt; dlB/tests/berk26_key.txt; dlB/tests/phi26_test.txt; dlG/gull26_test.txt; dlG/nv26C_{test,key}.txt (+ key PDF image); dlG/hh26_{test,key}.txt; dlG/rsoi26{B,C}_key.txt; dlG/rsoiOld{B,C}_key.txt; dlG/puso26_test.txt; dlG/mit20_key.txt; dlG/wiki/CircuitLabTest2019I.txt; dlG/wiki/Shock_Value_Network_Analysis_Test{,_Answer_Key}.txt; dlG/pz/p19, p20, p21 Princeton keys (p21 key PDF image checked).
- Prior files: `/Users/kai/Daniel/circuitlab/research/prior/I_bank.md` (Volta fact sheet, H1-H16), `B_topics.md`, `G_tests.md`, `G_manifest.md`, `F_slides.md`, `A_rules.md`.
- Test-extraction files: `/Users/kai/Daniel/circuitlab/research/tests/` (atx-reg, misc-a, misc-b, nat26, state, ut-a, ut-b, ut-c) and `/Users/kai/Daniel/circuitlab/research/prior/G_tests.md`.
- Raw research file this note was built from: `/Users/kai/Daniel/circuitlab/research/subtopics/volta.md`.

Georg Simon Ohm

Note: Ohm

> **INFO: How to use this note**
> - Scope icons: ✅ IN = all levels. ✅ States-Nationals = only at State and National tournaments. ⚠️ Borderline = the rules do not settle it (the reason is given). ❌ Out = excluded (the rule is cited).
> - Question types go from Easy to Medium to Hard to Nationals. "Nationals" means the hardest depth, not a different scope.
> - Every number in a worked example was checked in Python.
> - Source tags in square brackets, like [W] or [M], are listed under "Sources".

Scope notes and source tags

Event: Circuit Lab, Division B, 2026-27 season ("2027" rules). Levels: invitational, regional, state, national. Rule anchor: 3.c.i "Historical perspective of the electricity discoveries made by Ampere, Coulomb, Kirchhoff, Volta, **Ohm**, Hertz, and Boole". Related rule items: 3.c.iv (units of resistance) and 3.c.v (Ohm's Law). Math limits: rule 3.e (algebra, at most two simultaneous equations, no trig or calculus). Rule location check: in `Science_Olympiad_Div_B_Rules_2027_for_Web_Secured.pdf` (pdftotext), the Circuit Lab B text has page footers "©2027 - B12" and "©2027 - B13". The task brief said B11-B12. The footers say B12-B13.

Div C test rule (user request). Many sources below are Div C tests. I use a Div C question only when its content is inside the 2027 Div B rule. Each Div C citation gets its Div B scope tag. Div C items about Faraday, Tesla, magnetism or other Div-C-only content are tagged OUT and are not used as models.

Source tags used in section 1:

- [WP-O] Wikipedia "Georg Ohm"; [WP-L] Wikipedia "Ohm's law"; [WP-U] Wikipedia "Ohm" (unit); [WP-S] Wikipedia "Siemens (unit)". All fetched 2026-10-16.
- [MT] MacTutor (St Andrews) biography of Ohm, fetched 2026-10-16.
- [ETHW] ETHW pages "Georg Simon Ohm" and "Ohm's Law" (IEEE history wiki). Both contain errors (see section 4).
- [LH] Linda Hall Library, "Scientist of the Day: Georg Ohm".
- [WS] web search result snippets (secondary; named in section 6).
- [T: test] seen in a past test (named).
- [R] 2027 Div B rules.
- [G] general knowledge, not confirmed this run (also listed in section 5).

Key facts

1.1 Life

- Born 16 March 1789, Erlangen (then Brandenburg-Bayreuth, now Bavaria, Germany). [WP-O][MT][LH]
- Died 6 July 1854, Munich, age 65. Buried in the Alter Südfriedhof, Munich. [WP-O][MT]
- Nationality: German. [WP-O]
- Father Johann Wolfgang Ohm was a self-taught **locksmith**. He taught his sons mathematics, physics, chemistry and philosophy at home. [WP-O][MT]
- Mother Maria Elizabeth Beck, a tailor's daughter. She died when Georg was ten. [WP-O]
- Seven children, three survived. Brother **Martin Ohm** became a well-known **mathematician**. Sister Elizabeth Barbara. [WP-O][MT]

1.2 Education and career (chronological)

- About 1800-1804: Erlangen Gymnasium (weak science teaching). [MT]
- 1805: entered the University of Erlangen. He neglected his studies. His father made him leave after three semesters. [MT][ETHW]

- September 1806: mathematics teacher at a school in **Gottstadt bei Nidau, Switzerland**. [WP-O][MT]
- March 1809: private tutor in **Neuchâtel** for two years. Langsdorf advised him to study mathematics on his own (Euler, Laplace, Lacroix). [MT]
- April 1811: back to Erlangen. **Doctorate 25 October 1811** (advisor Karl Christian von Langsdorf). Lecturer for three semesters. [WP-O][MT]
- January 1813: teacher of mathematics and physics at a poor school in **Bamberg** (school closed February 1816). [WP-O][MT]
- 1817: geometry textbook *Grundlinien zu einer zweckmäßigen Behandlung der Geometrie*. [WP-O]
- 11 September 1817: offer from the **Jesuit Gymnasium in Cologne**. It had a well-equipped physics laboratory. [WP-O][MT]
- 1820: began electrical experiments after he learned of **Ørsted's** discovery (current deflects a compass). [MT]
- August 1826: one year's leave on half pay, spent in **Berlin**. [MT]
- March 1828: formally resigned at Cologne, hurt by the poor reception of his book. He taught mathematics at schools in Berlin. [MT]
- 1833: **Polytechnic School of Nuremberg** (title of professor, not a university chair). Stayed to 1852. [WP-O][MT]
- 1849: Munich, curator of the Bavarian Academy's physical cabinet; lectured at the University of Munich. [MT] [ETHW]
- 1852: **chair of (experimental) physics, University of Munich** (Ludwig-Maximilians-Universität), his lifelong ambition. [WP-O][MT]

1.3 The discovery

- 1825: first paper, on how the electromagnetic force of a wire decreases as the wire gets longer. MacTutor says this paper does not yet contain Ohm's law. [WP-O][MT]
- After his 1825 report, he learned of similar experiments by Peter Barlow and Henri Becquerel with different results. [WS]
- He first used a **voltaic pile** as the source. Its output was **not steady**. [WP-L][WS]
- **Poggendorff** (editor of *Annalen der Physik und Chemie*) suggested in a footnote that Ohm use a **thermoelectric circuit** (thermocouple) because it is steadier. [WS: n4trb.com extract]
- The thermocouple effect was found by **Thomas Johann Seebeck in 1821** (the Seebeck effect). [WS]
- His thermocouple: **copper and bismuth**. One junction in ice (0 °C), one in boiling water (100 °C). The voltage stayed steady for about 30 minutes. [WS] [T: ATX Regional 2026 C Q7, answer A,B = copper, bismuth]
- Current meter: a **torsion galvanometer** (a magnetic needle hung on a thin wire; the twist needed to bring it back measures the current). [WS][Uni Flensburg histolab]
- Test wires: different **lengths, diameters (thickness) and materials**. [WP-L][LH]
- Key data date: 8 January 1826 (per ETHW). [ETHW]
- 1826: two papers modelling electrical conduction on **Joseph Fourier's theory of heat conduction**. [MT][WP-O] [LH] [T: ATX Regional 2026 C Q15, answer C = Fourier]
- Ohm's data fit **x = a / (b + ℓ)**: x = galvanometer reading (current), ℓ = length of the test wire, a = a constant set by the thermocouple temperatures (the EMF), b = a constant of the rest of the circuit (in modern terms, the internal and galvanometer resistance). [WP-L]
- One 1826 statement: "the force of the current is as the sum of all the tensions, and inversely as the entire length of the current" ("reduced length" = reduced resistance). [WS]
- **1827 (May): *Die galvanische Kette, mathematisch bearbeitet***, Berlin. English: "The Galvanic Circuit Investigated Mathematically". [WP-O][MT][ETHW] Linda Hall translates it as "The electrical circuit, mathematically determined". [LH]
- In the book: the EMF across any part of a circuit is "the product of the strength of the current, and the resistance of that part of the circuit". [WP-O] Modern form: V = IR, or I = V/R.
- He presented it as a theory of contiguous action (action between neighbouring parts), not action at a distance. [WP-O]
- Wikipedia calls the book the early beginning of circuit theory. [WP-O]

1.4 Priority (who was first)

- **Henry Cavendish** found the same relation in **January 1781**, using Leyden jars and salt-water tubes. He judged current by the strength of the shock he felt. He did not publish. **James Clerk Maxwell** published Cavendish's results in **1879**. [WP-L][WP-O]
- So the T/F "Ohm was not the first to discover Ohm's law" is **True**. [T: Princeton 2021 C Q10, key True]

1.5 Reception

- The book "was received with little enthusiasm". [MT] ETHW: "muted reception". [ETHW]
- Critics called it a "web of naked fancies". [WP-L]
- The (Prussian/German) Minister of Education is quoted: "a professor who preached such heresies was unworthy to teach science". [WP-L][WS] (Primary source not found; see section 5.)
- Reason given: the German physics of the time (influenced by the Hegelian "nature philosophy" school) held that experiments were not needed and truth could be found by reasoning alone. His use of mathematics put off the older physicists. [WP-L][MT] Hegel is not named in WP-L; the "Hegelian" label is [G].
- He did not win the favour of **Johannes Schulze** (Berlin education ministry) or physics professor **Georg Friedrich Pohl**. [MT]
- So the "1827 controversy" was that Ohm **used abstract mathematics to describe physics**. [T: ATX Regional 2026 C Q2, key B]
- Acceptance: not wide until the 1840s; widely known and taken as proven by the 1850s. [WP-L]
- Early users/supporters: Fechner, Lenz, Jacobi, Poggendorff, Pouillet. [WS] Pouillet's 1830s experiments were long credited as the first experimental proof. [WS] Charles Wheatstone drew attention to Ohm's definitions in Britain. [WP-O]

1.6 Honours

- **1841: Copley Medal** of the Royal Society (its highest award). [WP-O][MT][ETHW]
- **1842: foreign member of the Royal Society.** [WP-O][MT]
- Corresponding member of the Berlin and Turin academies (dates not given). [MT]
- **1845: full member of the Bavarian Academy** of Sciences. [WP-O][MT]

1.7 Other work (non-circuit)

- **1843: Ohm's acoustic law**: the ear hears a musical sound as a set of pure harmonic (sine) tones. [WP-O] It led to the **Ohm-Seebeck dispute** with **August Seebeck** (son of T. J. Seebeck [G]). Ohm admitted an error in his derivation. [MT] Wikipedia: "It is well known to be not quite true." [WP-O]
- 1849: *Beiträge zur Molecular-Physik*. Quote: "Man proposes, and God disposes." [WP-O]
- 1852-1853: work on interference in uniaxial crystal plates (optics). 1854: *Grundzüge der Physik* (2 volumes). [WP-O]

1.8 The unit ohm, the mho and the siemens

- Ohm (Ω) = SI unit of resistance. 1 Ω = 1 V/A = $kg \cdot m^2 \cdot s^{-3} \cdot A^{-2}$. [WP-U] (base units checked in Python)
- **1861:** Latimer Clark and Sir Charles Bright proposed names including "**Ohma**". [WP-U]
- **1864:** British Association report: "B.A. unit, or Ohmad". By 1867 just "ohm". The B.A. ohm was about 1.3 % too small. [WP-U]
- **1867: William Henry Preece** suggested the symbol **Ω** (ohm sounds like omega). [WP-U]
- **1881:** International Electrical Congress (Paris) defined a practical ohm (about 104.9 cm of 1 mm^2 mercury at 0 °C). It replaced the older Siemens mercury unit. [WP-U][WP-S]
- **1884:** Paris conference "legal ohm" = 106 cm mercury column. [WP-U]
- **1893:** Chicago congress "international ohm" = **106.3 cm mercury column, 14.4521 g, 0 °C** (cross-section about 1 mm^2). **1908** London conference adopted it. [WP-U]
- **1948:** CGPM redefined the ohm in absolute terms. **2019:** SI revision, defined from fixed constants. [WP-U]
- ETHW gives other dates ("1860s", "1872"). See section 4.
- **Conductance** G = 1/R. Old unit **mho** = "ohm" spelled backward, symbol **℧** (upside-down omega). Suggested by **Lord Kelvin (William Thomson) in 1883**. [WP-S]

- **Siemens (S)** = SI unit of conductance. 1 S = 1/Ω = A/V = $kg^{-1}·m^{-2}·s^3·A^2$. IEC adopted 1935; **14th CGPM approved 1971**. Named for **Ernst Werner von Siemens**. [WP-S][WP-L]
- Writing rules: "siemens" is singular and plural ("1 siemens", not "1 siemen"). Lower-case name, capital S symbol (lower-case s = second). [WP-S]
- Conductivity σ = 1/ρ, unit S/m. Kirchhoff's later form of Ohm's law J = σE. [WP-L]

1.9 Legacy (trivia, low priority)

- Ohm's statue in Munich shows Ω on the plinth. [LH]
- Germany issued a 1994 stamp series on German discoveries; the Ohm's-law stamp shows a resistor. [LH]
- "Ohmic" = a device that obeys V = IR with constant R (straight-line I-V graph through the origin). A diode/LED is non-ohmic. [WP-L][G]
- The Technische Hochschule Nürnberg is named "Georg Simon Ohm". [G]

1.10 Common distractors (facts about other people)

- Voltaic pile 1800 → Volta. Torsion balance 1785 → Coulomb. Astatic needle → Ampère. Galvani → Volta's rival (not Ohm's). Diamagnetism term → Faraday (not on the 2027 B list). Thermoelectric effect → Seebeck. Heat conduction theory → Fourier. First (unpublished) V ∝ I → Cavendish.

Formulas

Item	Formula	Units
Ohm's law	V = I·R, I = V/R, R = V/I	V, A, Ω
Conductance	G = 1/R = I/V	S (siemens) = Ʊ (mho) = A/V
Ohm's law with conductance	I = G·V	A = S·V
Parallel conductances	$G_{total} = G_1 + G_2 + ...$	S
Series conductances	$1/G_{total} = 1/G_1 + 1/G_2 + ...$	S
Wire (resistivity)	R = ρL/A, G = σA/L, σ = 1/ρ	Ω·m, S/m
Ohm's 1826 form	x = a/(b + ℓ)	x in galvanometer units, ℓ in wire length units
Modern twin of Ohm's form	I = ε/(r + R)	V, Ω
Base units	Ω = $kg·m^2·s^{-3}·A^{-2}$; S = $kg^{-1}·m^{-2}·s^3·A^2$	SI
Other forms	Ω = V/A = W/A^2 = V^2/W = $J·s/C^2$	—
International ohm (1893)	106.3 cm Hg, 14.4521 g, 0 °C	→ $\rho_{Hg} \approx 9.407\times10^{-7}$ Ω·m (Python: $10^{-6}/1.063$)
Prefixes	1 mS = 10^{-3} S, 1 µS = 10^{-6} S, 1 kΩ = 10^3 Ω, 1 MΩ = 10^6 Ω	—

Question types

This note has **31 question types** (27 ✅ IN, 3 ⚠️ Borderline, 1 ❌ Out). The out-of-scope ones are in "Out of scope (skip)" below.

How the types were found

Order: easiest first. Difficulty: Easy / Medium / Hard / Nationals. "Where seen" uses the local files in `/Users/kai/Daniel/circuitlab/research/downloads/` (dlG, dlB). For each Div C test, the Div B scope is given. Every worked number was checked in Python (python3 -I) on 2026-10-16.

History questions in the test-extraction files (8 test groups)

These rows come from `research/tests/` (ATX regionals, UT invitationals, misc-a, misc-b, USC 2026, Florida State 2021). Each row names the question type it belongs to.

Test	Year	Div	Level	Q	Question (short)	Key	Check / scope 2027 B	Type
ATX (UT Austin) Regional	2021	B	Regional	4	Law relating V, I, R (name bank)	Ohm	IN	Q1
ATX (UT Austin) Regional	2020	B	Regional	3	Law relating current density to E field	Ohm (J = σE)	IN (name only)	Q1
ATX (UT Austin) Regional	2020	B	Regional	9	Thevenin/Norton valid because of his law	Ohm (linearity)	IN as a name; Thevenin/Norton themselves are OUT (3.d)	Q1
Cobra Invitational (Michigan)	2019-20 season	B	Invitational	4	Fill-in paragraph: Ohm 1827, V ∝ I	Georg Ohm / electric current	IN	Q9
North Pocono Invitational (PA)	2020	C	Invitational	15	First to describe electricity as action between contiguous particles	d Ohm	IN	Q29
UF Northern Regional (Florida)	2020	B	Regional	1	Who found I ∝ V/R?	B Ohm	IN	Q1
USC Invitational (not the 2026 Nationals test)	Jan 2026	B	Invitational (top tier)	26	Match 4 facts → physicist (V = IR → Ohm)	C	BORDERLINE as a whole (Faraday rows)	Q1
Florida State Tournament	2021	B (= C)	State	3	Quote: "The design of this Memoir is to deduce strictly from a few principles..."	Ohm	IN (preface of Die galvanische Kette, 1827)	Q30
UT Austin Invitational, fall	2025	B/C	Invitational	4c	Pick the letters whose laws are useful in steady state	A, C, K, V, O (V questionable)	IN	Q31
ATX (UT Austin) Regional	2026	B	Regional	MC2	Unit of resistance named after	Ohm	IN	Q2
ATX (UT Austin) Regional	2026	C	Regional	MC2	The "Ohmic controversy" of 1827	critics rejected his abstract maths	IN	Q15
ATX (UT Austin) Regional	2026	C	Regional	MC7	Two elements Ohm used as a substitute "ammeter"	copper, bismuth (wording error: it was his voltage source)	IN	Q26
ATX (UT Austin) Regional	2026	C	Regional	MC15	Whose heat-conduction work inspired Ohm	Fourier	IN	Q14
UT Austin Invitational (2020 season)	2019	B	Invitational	5	Epithet: theorized sound is heard as a set of harmonic tones	Ohm (acoustic law)	IN	Q19
UT Austin Invitational, fall (2020 season)	2019	C	Invitational	5	Epithet: said sound is heard as harmonic tones	Ohm	IN	Q19
UT Austin Invitational (2020 season)	2019	B	Invitational	6	Epithet: believed in "contiguous particles"	Ohm (Faraday also used the phrase)	IN (key doubtful)	Q29
UT Austin Invitational, fall (2020 season)	2019	C	Invitational	6	Epithet: believed in "contiguous particles"	Ohm (doubtful)	IN	Q29
ATX (UT Austin) Regional	2019	B	Regional	2	Who tested an early Ohm's law with Leyden jars and glass tubes?	Cavendish	OUT as a person (not listed); the priority fact supports Q16	Q16

Test	Year	Div	Level	Q	Question (short)	Key	Check / scope 2027 B	Type
UF Northern Regional (Florida)	2019	C	Regional	1-8	Match Volta/Ohm/Tesla/Hertz/Faraday to achievements	see Volta note	IN for the Ohm row	Q17
Univ. of Michigan Invitational (UMSO)	2019 (file label)	not stated (likely C)	Invitational	1-5	Match Faraday, Tesla, Hertz, Volta, Ohm to discoveries	A, C, E, B, D	Ohm row IN	Q17
F-M Invitational (NY), fall	2019	not stated	Invitational	1	Match 7 discoveries to 7 scientists	Ohm row correct	IN for the Ohm row	Q17

Easy

Q1. Law → person ("Who formulated V = IR?")

Law → person ("Who formulated V = IR?") · ✅ IN · Easy

- **Scope: IN** (3.c.i, 3.c.v).
- **Where seen:** BullSO 2026 B Q5 (key B = Ohm); Berkeley 2026 B Q6 (Georg Simon Ohm); national supervisor deck 1 (F_slides: "Who first described the relation between voltage, current and resistance?" → Ohm).
  - Also seen (test-extraction files): UF Northern Regional (Florida) 2020, Div B, Regional, Q1: "Who found $I \propto V/R$?" (key: B Ohm). Div B 2027: IN.
  - Also seen (test-extraction files): USC Invitational (not the 2026 Nationals test) Jan 2026, Div B, Invitational (top tier), Q26: "Match 4 facts → physicist (V = IR → Ohm)" (key: C). Div B 2027: BORDERLINE as a whole (Faraday rows).
- **How to solve:**
  1. Read the question: a law statement or equation → the scientist.
  2. V = IR, "current proportional to voltage" → Ohm.
  3. Reject Kirchhoff (current/voltage laws), Volta, Ampère, Lenz.
- **Worked example:** "The law that says current through a conductor is proportional to the voltage across it is named for?" → **Georg Simon Ohm**.
- **Traps:** "Who first discovered it?" → Cavendish (1781, unpublished), not Ohm. Read the verb ("named after" vs "first discovered").
- Calculator: look up — type: keyword ("V=IR", "proportional").

Q2. Unit → person, person → unit

Unit → person, person → unit · ✅ IN · Easy

- **Scope: IN**.
- **Where seen:** ATX Regional 2026 B Q2 ("Who has the unit of resistance named after them?" key D = Ohm).
- **How to solve:**
  1. Read the question: "unit of resistance" → Ohm; or "Ohm's unit measures?" → resistance.
  2. Recall Ω = resistance.
- **Worked example:** "The ohm is the unit of which quantity?" → **electrical resistance**.
- **Traps:** the mho/siemens measures conductance, not resistance. Ohm's name is not on the unit of current (ampere).
- Calculator: look up — type: unit name or quantity.

Q4. What is a mho? What does the siemens measure?

What is a mho? What does the siemens measure? · ✅ IN · Easy

- **Scope: IN** (units of resistance; conductance = 1/R).
- **Where seen:** PA Central Regional ~2013 C ("What is a mho?", `dlG/wiki/central.txt` line 40; Div B scope IN). "Circuit Lab Test (Key)" ~2013 C Q17 ("Siemens is a unit measuring what?" → conductance; Div B scope IN). Shock Value student test ~2011 B part IIa item 2 ("Conductance:" unit equivalent) and fill-in Q20 ("___ is equal to

current divided by voltage" → conductance). NY Lake Erie 2018 C Q6 (option D "Conductance" as a distractor for "opposition to current" → resistance).

- **How to solve:**
  1. Read the question: "mho" → unit of conductance = 1/Ω = siemens.
  2. Mho = ohm spelled backward → reciprocal of ohm.
- **Worked example:** "What is a mho?" → **the old unit of conductance, 1 mho = 1 Ω^{-1} = 1 siemens**.
- **Traps:** the plural is "siemens" (no "siemen"). Symbol S (capital), not s (second). Conductance is not conductivity (S/m).
- Calculator: look up — type: the keyword from the question.

Q5. Conductance ↔ resistance conversion

Conductance ↔ resistance conversion · ✅ IN · Easy

- **Scope: IN** (3.c.iv calculation).
- **Where seen:** not seen as a pure conversion; the idea is in Shock Value ~2011 B Q20 (G = I/V).
- **How to solve:**
  1. Read the question: R → G, or G → R, with prefixes.
  2. G = 1/R; watch the prefix (1 kΩ ↔ 1 mS).
- **Worked example:** R = 250 Ω → G = **0.004 S = 4.0 mS**. G = 40 mS → R = **25 Ω**. 0.25 mho → **4.0 Ω**. 4.7 kΩ → **213 µS** (212.77). (All Python-checked.)
- **Traps:** kΩ maps to mS, MΩ to µS (prefixes flip). Significant figures (rule 3.b).
- Calculator: calculate — type: value, unit (Ω/kΩ/MΩ or S/mS/µS/mho).

Medium

Q3. Symbol Ω and why

Symbol Ω and why · ✅ IN · Easy (symbol) / Medium (why, who)

- **Scope: IN** (units, 3.c.iv).
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: "symbol of the ohm" → Ω; "who proposed Ω and why" → Preece, 1867, "ohm" sounds like "omega".
- **Worked example:** "Which Greek letter is the symbol for the ohm?" → **omega (Ω)**.
- **Traps:** ℧ (upside-down omega) is the mho. Ω (capital) is not ω (lower-case, used for angular frequency, which is OUT for 2027).
- Calculator: look up — type: the keyword from the question.

Q6. Conductances in parallel and in series

Conductances in parallel and in series · ✅ IN · Medium

- **Scope: IN** (3.c.v series/parallel).
- **Where seen:** not seen in a test with conductance wording. Same math as parallel-resistor questions everywhere (e.g. GullSO 2026 Q37g iv, n equal resistors).
- **How to solve:**
  1. Read the question: list of conductances (or resistances) and connection type → total G or R.
  2. Parallel: G adds.
  3. Series: reciprocals add.
  4. Or convert to R and back.
- **Worked example:** 20 Ω, 30 Ω, 60 Ω in parallel. G = 0.05 + 0.0333 + 0.0167 = **0.100 S**, so R_eq = **10 Ω**.
- **Worked example:** 2 mS, 3 mS, 5 mS in parallel → **10 mS** → **100 Ω**. 0.1 S and 0.05 S in series → **0.0333 S (30 Ω)**.
- **Traps:** adding conductances in series (wrong). Mixing S and mS.
- Calculator: calculate — type: n values, unit, "series" or "parallel".

Q7. Conductance of a wire vs length and thickness (Ohm's own variables)

Conductance of a wire vs length and thickness (Ohm's own variables) · ✅ IN · Medium

- **Scope: IN** (resistance concepts; $R \propto L/A$). Resistivity is not named in the rule but is used in many tests; prior file G_tests tags it IN.
- **Where seen:** "Circuit Lab Test (Key)" ~2013 C Q18 (Div B scope IN): "A wire has a measure of .6 Siemens per meter. How many Siemens does 2 meters of this wire have?" key **0.3 S** (wording error, see section 4).
- **How to solve:**
  1. Read the question: G (or R) of one wire, and the change in length/diameter → new G (or R).
  2. $G \propto A/L$, $A \propto d^2$. So $G_{new} = G \cdot (d_{new}/d)^2 \cdot (L/L_{new})$.
- **Worked example:** 1 m of a wire has G = 0.6 S. 2 m of the same wire → **0.3 S**. Same 1 m, double diameter → **2.4 S**. Triple length and double diameter → **0.8 S**.
- **Traps:** conductance goes DOWN with length (opposite of resistance). Diameter is squared.
- Calculator: calculate — type: G or R, length factor, diameter (or area) factor.

Q8. SI base units of the ohm or the siemens

SI base units of the ohm or the siemens · ✅ IN · Medium

- **Scope: IN** (3.c.iv units).
- **Where seen:** Rickards Dec 2020 (2021 season) **B** Q14 "In SI base units, what is the unit of conductance?" (correct choice B = $s^3 \cdot A^2 \cdot m^{-2} \cdot kg^{-1}$; MC key is highlighted in the PDF, not in the text; answer from prior G_tests and my own derivation). Same question as Rickards 2021 C Q16 (Div B scope IN). Hawk & Hornet 2026 C FR4 asked impedance in $J \cdot s/C^2$ (impedance is AC theory → OUT for B; the same answer for resistance $\Omega = J \cdot s/C^2$ is IN).
- **How to solve:**
  1. Read the question: a unit → its base-unit form.
  2. $V = W/A = (kg \cdot m^2 \cdot s^{-3})/A$. $\Omega = V/A = kg \cdot m^2 \cdot s^{-3} \cdot A^{-2}$. S = inverse: $kg^{-1} \cdot m^{-2} \cdot s^3 \cdot A^2$.
- **Worked example:** "Express the siemens in SI base units." → **$kg^{-1} \cdot m^{-2} \cdot s^3 \cdot A^2$**. (Exponents checked in Python.)
- **Traps:** forgetting that A is a base unit (it is; C is not). Sign of every exponent flips for S.
- Calculator: look up + calculate — type: unit name. (App note: (fixed table) with an optional CALCULATE mode (exponent vectors for kg, m, s, A).)

Q9. Book title, year and meaning

Book title, year and meaning · ✅ IN · Easy-Medium

- **Scope: IN**.
- **Where seen:** ATX Regional 2026 C Q2 names the "1827" controversy (Div B scope IN). Title itself not seen in a test.
  - Also seen (test-extraction files): Cobra Invitational (Michigan) 2019-20 season, Div B, Invitational, Q4: "Fill-in paragraph: Ohm 1827, $V \propto I$" (key: Georg Ohm / electric current). Div B 2027: IN.
- **How to solve:**
  1. Read the question: title → author/year, or author → title; or "translate the title".
- **Worked example:** "*Die galvanische Kette, mathematisch bearbeitet* (1827) was written by?" → **Georg Simon Ohm**; meaning **"The Galvanic Circuit Investigated Mathematically"**.
- **Traps:** Ampère's *Théorie des phénomènes électro-dynamiques* is also 1826/1827. "Galvanische" refers to Galvani's name for current electricity, not to Galvani as author.
- Calculator: look up — type: title words or author.

Q10. Timeline: place Ohm among the seven

Timeline: place Ohm among the seven · ✅ IN · Medium

- **Scope: IN**.
- **Where seen:** Princeton 2020 C Q1 (order the seven rule scientists; Div C list had Tesla and Faraday, **OUT** for 2027 B; I use only the 2027 B list). Prior I_bank H5.
- **How to solve:**
  1. Read the question: names or events → chronological order or "which came first".

2. Coulomb's law 1785 → voltaic pile 1800 → Ampère's *Théorie* 1826 → Ohm's book 1827 → Kirchhoff's laws 1845 → Boole *Mathematical Analysis of Logic* 1847 → Hertz radio waves 1886-1888 (1887 commonly used).

- **Worked example:** order Kirchhoff's laws, Ohm's law, voltaic pile, Hertz's waves → **voltaic pile (1800), Ohm (1827), Kirchhoff (1845), Hertz (1887)**. (Sort checked in Python.)
- **Worked example:** years from Cavendish's unpublished work (1781) to Ohm's book (1827) = **46**; from Ohm's book to Maxwell publishing Cavendish (1879) = **52**.
- **Traps:** Ohm and Ampère are both "1827" in some keys (Princeton 2020 key writes Ampère (1827), Ohm (1827)); ties need a tie-break rule. Ohm's 1826 papers vs 1827 book.
- Calculator: look up — type: the list of names/events. (App note: (year table) + sort.)

Q11. Biography: birth, death, family

Biography: birth, death, family · ✅ IN · Easy-Medium

- **Scope: IN** (historical perspective).
- **Where seen:** not seen in a test.
- **Worked example:** "Ohm was born in which city?" → **Erlangen** (1789). "Died where?" → **Munich** (1854). "His father's trade?" → **locksmith**. "His brother Martin was a?" → **mathematician**. "Age when *Die galvanische Kette* came out?" → **38** (1827 – 1789; birthday in March, book in May). "Age at death" → **65**.
- **Traps:** ETHW says born 1787 (wrong). Do not confuse Erlangen with Ohm's later cities (Cologne, Nuremberg, Munich).
- Calculator: look up — type: the keyword from the question.

Q12. Career sequence and places

Career sequence and places · ✅ IN · Medium

- **Scope: IN**.
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: an event → place/year; or order the posts.
- **Order:** Gottstadt, Switzerland (1806) → Neuchâtel tutor (1809) → Erlangen doctorate (1811) → Bamberg (1813) → **Jesuit Gymnasium, Cologne (1817)** → Berlin leave (1826-27) → resigned (1828) → **Nuremberg Polytechnic (1833)** → Munich curator (1849) → **University of Munich chair (1852)**.
- **Worked example:** "Where did Ohm teach when he did the experiments behind Ohm's law?" → **the Jesuit Gymnasium in Cologne**.
- **Traps:** he was a school teacher, not a university professor, in 1827. The Munich chair came only two years before his death.
- Calculator: look up — type: the keyword from the question.

Q13. Apparatus: source, meter, test wires

Apparatus: source, meter, test wires · ✅ IN · Medium

- **Scope: IN** (history of the discovery; thermocouple = a DC source, 3.c.iii).
- **Where seen:** ATX Regional 2026 C Q7 (pick 2 elements → **copper, bismuth**; Div B scope IN; wording error, see section 4). ATX Regional 2026 C Q1 lists "Torsional Galvanometer" as an option (the question is about Ampère).
- **How to solve:**
  1. Read the question: "what source / meter / variables did Ohm use?".
- **Answers:** source first a voltaic pile (unsteady), then a **copper-bismuth thermocouple** (one junction in ice, one in boiling water); meter = **torsion galvanometer**; variables = wire **length, thickness, material**.
- **Worked example:** "Why did Ohm replace the voltaic pile with a thermocouple?" → **the pile's voltage drifted; the thermocouple gave a steady voltage**.
- **Worked example:** "Who suggested the thermocouple?" → **Poggendorff** (the effect is Seebeck's, 1821).
- **Traps:** the thermocouple is the source, not the meter. Seebeck found the effect but did not (per my sources) suggest it to Ohm. How the galvanometer works uses magnetism → that part is OUT (magnetism removed in 2027); "Ohm used a galvanometer" as a fact is IN.
- Calculator: look up — type: the keyword from the question.

Q14. Influence: Fourier and Ørsted

Influence: Fourier and Ørsted · ✅ IN · Medium

- **Scope: IN**.
- **Where seen:** ATX Regional 2026 C Q15 "Who inspired Ohm by their work on heat conduction?" key **C = Fourier** (Div B scope IN).
- **Method/fact:** Ohm treated current like heat flow (Fourier, 1822 [G for the year]); voltage difference like temperature difference. Ørsted's 1820 discovery started his electrical work.
- **Worked example:** "Ohm modelled current on which earlier theory?" → **Fourier's theory of heat conduction**.
- **Traps:** Newton, Gauss and Kelvin are the distractors.
- Calculator: look up — type: the keyword from the question.

Q15. Reception and controversy

Reception and controversy · ✅ IN · Medium

- **Scope: IN**.
- **Where seen:** ATX Regional 2026 C Q2 "What was the 'Ohmic' Controversy of 1827?" key **B = Ohm used abstract math to describe physics** (Div B scope IN). Prior I_bank H9 (resigned his post).
- **Facts:** poor reception; "web of naked fancies"; minister: "unworthy to teach science"; Schulze and Pohl against him; resigned Cologne 1828; accepted in the 1840s (Copley 1841).
- **Worked example:** "What did Ohm do after his 1827 book was poorly received?" → **he resigned his Cologne post (1828) and taught in Berlin, then went to Nuremberg (1833)**.
- **Traps:** the critics did not say his data were faked; the complaint was the mathematical, theory-first style.
- Calculator: look up — type: the keyword from the question.

Q16. Priority: Cavendish found it first

Priority: Cavendish found it first · ✅ IN · Medium

- **Scope: IN** (history of Ohm's law; Cavendish appears only as the answer).
- **Where seen:** Princeton 2021 C Q10 T/F "Ohm was not the first one to discover what is now known as Ohm's Law" key **True** (Div B scope IN; key read from the red highlight in the PDF).
  - Also seen (test-extraction files): ATX (UT Austin) Regional 2019, Div B, Regional, Q2: "Who tested an early Ohm's law with Leyden jars and glass tubes?" (key: Cavendish). Div B 2027: OUT as a person (not listed); the priority fact supports Q16.
- **Worked example:** "Who found $V \propto I$ in 1781 but never published?" → **Henry Cavendish** (published by **Maxwell in 1879**). How did Cavendish judge current? → **by the strength of the shock he felt**.
- Calculator: look up — type: the keyword from the question.

Q17. True/false and rival distractors

True/false and rival distractors · ✅ IN · Easy-Medium

- **Scope: IN** (statements about Ohm). A statement that only names a Div-C topic (for example diamagnetism) is **BORDERLINE**: magnetism is removed in 2027, but "Ohm did not do X" is still a fact about Ohm.
- **Where seen:** Princeton 2021 C Q9 "Ohm's main rival was Galvani" key **False** (Div B IN; Galvani was Volta's rival). Princeton 2021 C Q2 "Ohm termed the phrase diamagnetism" key **False** (Faraday; Div B BORDERLINE).
  - Also seen (test-extraction files): UF Northern Regional (Florida) 2019, Div C, Regional, Q1-8: "Match Volta/Ohm/Tesla/Hertz/Faraday to achievements" (key: see Volta note). Div B 2027: IN for the Ohm row.
  - Also seen (test-extraction files): Univ. of Michigan Invitational (UMSO) 2019 (file label), Div not stated (likely C), Invitational, Q1-5: "Match Faraday, Tesla, Hertz, Volta, Ohm to discoveries" (key: A, C, E, B, D). Div B 2027: Ohm row IN.
  - Also seen (test-extraction files): F-M Invitational (NY), fall 2019, Div not stated, Invitational, Q1: "Match 7 discoveries to 7 scientists" (key: Ohm row correct). Div B 2027: IN for the Ohm row.
- **Worked example:** "T/F: Ohm won the Copley Medal before he published Ohm's law." → **False** (book 1827, medal 1841, 14 years later).
- Calculator: look up — type: (fact table with T/F flags).

Q18. Honours

Honours · ✅ IN · Medium

- **Scope: IN**.
- **Where seen:** not seen for Ohm (prior I_bank H10 lists Copley medals).
- **Worked example:** "Which Royal Society medal did Ohm receive in 1841?" → **the Copley Medal**. "In 1842 he became?" → **a foreign member of the Royal Society**.
- **Traps:** Volta also won the Copley Medal (1794). Copley ≠ Rumford (Kirchhoff, Hertz).
- Calculator: look up — type: the keyword from the question.

Q20. "Which is NOT Ohm's?" / mis-attribution

"Which is NOT Ohm's?" / mis-attribution · ✅ IN · Medium

- **Scope: IN**.
- **Where seen:** prior I_bank H14 style (not seen with Ohm as the key in a test).
- **Worked example:** "A. Die galvanische Kette; B. torsion balance; C. acoustic law; D. Copley Medal 1841 — which is NOT linked to Ohm?" → **B** (Coulomb's torsion balance; Ohm used a torsion **galvanometer**).
- **Traps:** torsion balance vs torsion galvanometer (similar words, different people and devices).
- Calculator: look up — type: the keyword from the question.

Q25. "Ohmic" vs non-ohmic (link to diodes/LEDs)

"Ohmic" vs non-ohmic (link to diodes/LEDs) · ✅ IN · Medium

- **Scope: IN** (Ohm's law 3.c.v; diodes/LEDs 3.c.vii).
- **Where seen:** not seen with the word "ohmic" in a B test. ATX 2026 C Q2 uses "Ohmic" only in the controversy sense.
- **Worked example:** "Which has a straight-line I-V graph through the origin: a resistor or an LED?" → **the resistor (ohmic)**; the LED is non-ohmic.
- **Traps:** a resistor whose R changes with temperature (filament bulb) is also non-ohmic. Filament I-V detail is BORDERLINE ("non-linear devices beyond those listed" is excluded), so keep to resistor vs diode.
- Calculator: look up — type: the keyword from the question.

Q26. Seebeck (thermoelectric) effect as Ohm's DC source

Seebeck (thermoelectric) effect as Ohm's DC source · ⚠️ Borderline · Medium

- **Scope: BORDERLINE**. The rule names sources of DC (3.c.iii) but not the thermoelectric effect. As history of Ohm's apparatus it is IN (Q13); a question on how a thermocouple works goes beyond the rule.
- **Where seen:** Rickards Dec 2020 B Q15 and Rickards Dec 2020 C Q18 (Seebeck effect as a distractor for Joule heating; Div B IN as a distractor).
- **Worked example:** "Ohm's copper-bismuth source makes a voltage because of which effect?" → **the Seebeck (thermoelectric) effect**.
- **Traps:** Peltier effect is the reverse (current → heating/cooling). Joule heating is I^2R.
- Calculator: look up — type: the keyword from the question.

Q31. Multi-select: which listed scientists have property X?

Multi-select: which listed scientists have property X? · ✅ IN · Medium

- **Scope:** IN (3.c.i).
- **Where seen:** UT Austin Invitational fall 2025, Div B/C, Invitational, Q4a-d (letters A, C, K, V, O, T, F; pick every letter that fits). Ohm is a right letter in 4c (laws useful in steady state).
- **How to solve:**
  1. Write the letter code of each listed scientist (A = Ampère, C = Coulomb, K = Kirchhoff, V = Volta, O = Ohm, Hz = Hertz, B = Boole).
  2. Test each scientist against the property, one at a time.
  3. Pick every scientist that passes. A multi-select question can have more than one right letter.
  4. Check each letter again. Keys often add a wrong letter (UT 2025 put Volta under "France").

- **Worked example:** "Whose laws are useful in steady-state (DC) circuits? A, C, K, V, O, T, F" → **A, C, K, O** (the key also gave V, which is doubtful because there is no "Volta's law" for circuits).
- **Traps:** Faraday's law needs changing fields, so it is not steady state. Tesla has no law.
- Calculator: look up — type: the property (e.g. "steady state").

Hard

Q19. Non-electric work (acoustic law, optics, geometry)

Non-electric work (acoustic law, optics, geometry) · ⚠️ Borderline · Medium-Hard

- **Scope: BORDERLINE**. Rule 3.c.i says "electricity discoveries", so acoustics is outside the stated topic. But past tests ask it as "which scientist" trivia.
- **Where seen:** MIT 2020 C Q4 "Published an acoustic law (proven false)" key **Ohm** (Div B BORDERLINE; the key's "proven false" is overstated, see section 4).
  - Also seen (test-extraction files): UT Austin Invitational (2020 season) 2019, Div B, Invitational, Q5: "Epithet: theorized sound is heard as a set of harmonic tones" (key: Ohm (acoustic law)). Div B 2027: IN.
  - Also seen (test-extraction files): UT Austin Invitational, fall (2020 season) 2019, Div C, Invitational, Q5: "Epithet: said sound is heard as harmonic tones" (key: Ohm). Div B 2027: IN.
- **Worked example:** "Ohm's acoustic law (1843) says the ear hears a complex sound as?" → **a set of pure harmonic tones**. Who disputed it? → **August Seebeck**.
- Calculator: look up — type: the keyword from the question.

Q21. History of the unit (ohm, mho, siemens)

History of the unit (ohm, mho, siemens) · ✅ IN · Hard

- **Scope: IN** (units, 3.c.iv; history 3.c.i). Rare.
- **Where seen:** not seen in a test. ATX 2020 Reg B Q55 asks the old unit before the tesla (magnetism, **OUT** for 2027), same style.
- **Facts:** "Ohma" 1861 (Clark and Bright); B.A. unit 1864; Ω symbol 1867 (Preece); practical ohm 1881 (Paris); legal ohm 1884 (106 cm Hg); international ohm 1893 (106.3 cm Hg, 14.4521 g); 1908 London; 1948 absolute; 2019 constants; mho 1883 (Kelvin); siemens IEC 1935, CGPM 1971.
- **Worked example:** "Who suggested the name 'mho', and when?" → **Lord Kelvin, 1883**. "Which unit did the siemens replace?" → **the mho**.
- **Traps:** the old "Siemens mercury unit" was a unit of **resistance**; the modern siemens is **conductance**. Dates differ across sources (see section 4); use Wikipedia's list.
- Calculator: look up — type: the keyword from the question.

Q22. Mercury-column ohm (resistivity from the 1893 definition)

Mercury-column ohm (resistivity from the 1893 definition) · ✅ IN · Hard

- **Scope: IN** ($R = \rho L/A$ is resistance calculation; resistivity not named in the rule, tests use it).
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: the column length and area that give 1 Ω → ρ of mercury; or ρ and a new column → R.
  2. $P = R \cdot A/L$. Then $R = \rho L/A$ for the new case.
- **Worked example:** 1 Ω = 106.3 cm of 1 mm² mercury. $\rho = (1\ \Omega)(1\times10^{-6}\ m^2)/(1.063\ m)$ = **9.41×10^{-7} Ω·m**. A column 2.126 m long, 0.5 mm² → R = **4.0 Ω**. (Python: 9.4073×10^{-7}; 4.000.)
- **Traps:** mm² → m² is 10^{-6}, not 10^{-3}. Doubling length and halving area multiplies R by 4.
- Calculator: calculate — type: R, L, A (with units) → ρ; or ρ, L, A → R.

Q23. Ohm's original law x = a/(b + ℓ): meaning and proportional reasoning

Ohm's original law x = a/(b + ℓ): meaning and proportional reasoning · ✅ IN · Hard

- **Scope: IN** (algebraic function, rule 3.e.i; history of the law).
- **Where seen:** not seen in a test.
- **How to solve:**

1. Read the question: a, b, ℓ → reading x; or "what do a and b mean?"; or the effect of changing ℓ.
2. A ↔ EMF of the thermocouple; b ↔ resistance of the rest of the circuit (source + galvanometer + leads) in "wire-length" units; ℓ ↔ test-wire resistance.
3. Modern form $I = \varepsilon/(r + R)$.

- **Worked example:** b = 20, ℓ = 10 (same length units). Ohm doubles the wire to ℓ = 20. New reading / old = (20 + 10)/(20 + 20) = **0.75** (not 0.5, because b does not change). (Python-checked.)
- **Traps:** assuming current halves when the wire doubles; that is true only when b = 0.
- Calculator: calculate — type: a, b, ℓ (or ℓ_1, ℓ_2 for a ratio).

Q28. Legacy trivia (stamp, statue, university name)

Legacy trivia (stamp, statue, university name) · ⚠️ Borderline · Hard (obscure)

- **Scope: BORDERLINE** (historical perspective, but not an electricity discovery).
- **Where seen:** not seen in a test.
- **Worked example:** "Which symbol is on the plinth of Ohm's Munich statue?" → **Ω**.
- Calculator: look up — type: the keyword from the question.

Q29. "Contiguous particles" / contiguous action

"Contiguous particles" / contiguous action · ✅ IN · Hard

- **Scope:** IN (3.c.i). It is a fact about how Ohm explained his law.
- **Where seen:** North Pocono Invitational 2020, Div C, Invitational, Q15 ("first to describe electricity as action between contiguous particles" → d Ohm). UT Austin Invitational 2019, Div B, Invitational, Q6 and UT Austin Invitational fall 2019, Div C, Q6 (epithet "believed in contiguous particles" → Ohm).
- **How to solve:**
  1. Know that Ohm described conduction as action between neighbouring ("contiguous") parts of the wire, not action at a distance.
  2. If the question asks about the circuit-law scientist, answer Ohm.
  3. If the question is about induction and the 1838 theory of "contiguous particles", it means Faraday (not on the 2027 list).
- **Worked example:** "Which listed scientist described electricity as action between contiguous particles?" → **Ohm** (North Pocono 2020 Q15 key).
- **Traps:** Faraday used the same phrase in 1838 for induction and dielectrics. Faraday is not on the 2027 list, so on a 2027 Div B test pick Ohm.
- Calculator: look up — type: "contiguous".

Q30. Quote attribution (who said it?)

Quote attribution (who said it?) · ✅ IN · Hard

- **Scope:** IN (3.c.i).
- **Where seen:** Florida State Tournament 2021, Div B (same as Div C), State, Q2-Q8 (seven quotes, 2 points each, "identify the last name"). Q3 is Ohm: "The design of this Memoir is to deduce strictly from a few principles, obtained chiefly by experiment, the rationale of those electrical phenomena which are produced by the mutual contact of two or more bodies."
- **How to solve:**
  1. Look for one key word or idea in the quote (an invention, a place, a belief, a language).
  2. Match that idea to one of the seven listed scientists.
  3. If the quote is in French, think of the two French scientists first: Ampère and Coulomb.
  4. If nothing fits the seven, the answer may be a scientist who is not on the 2027 list (Tesla, Faraday). Those items are out of scope in 2027.
  5. Clues for Ohm: "Memoir" (his 1827 book *Die galvanische Kette*), "few principles", "contact of two or more bodies" (the galvanic circuit).
- **Worked example:** "The design of this Memoir is to deduce strictly from a few principles ... phenomena produced by the mutual contact of two or more bodies." → **Ohm** (opening of *The Galvanic Circuit Investigated Mathematically*, 1827).

- **Traps:** "Contact of two bodies" also sounds like Volta's contact theory. The word "Memoir" and "deduce strictly" (maths) point to Ohm. Boole's *Laws of Thought* (1854) also starts "The design of the following treatise...". "Treatise" and "operations of the mind" → Boole (see Boole A1 T27).
- Calculator: look up — type: a key word (e.g. "Memoir", "deduce").

Nationals

Q24. Find a and b from two of Ohm's readings (two-equation system)

Find a and b from two of Ohm's readings (two-equation system) · ✅ IN · **Nationals**

- **Scope: IN** (system of exactly two equations is allowed, rule 3.e.i; not KVL, so not States-only).
- **Where seen:** not seen in a test. Same algebra as the "EMF and internal resistance from two loads" problem.
- **How to solve:**
  1. Read the question: (ℓ_1, x_1), (ℓ_2, x_2) → a, b; then predict x for a new ℓ.
  2. $X_1(b + \ell_1) = a = x_2(b + \ell_2)$
  3. So $b = (x_2\ell_2 - x_1\ell_1)/(x_1 - x_2)$, then $a = x_1(b + \ell_1)$.
- **Modern twin:** a cell gives 1.2 A into 2 Ω and 0.6 A into 5 Ω → **r = 1.0 Ω, ε = 3.6 V**. (Python-checked.)
- **Worked example:** ℓ = 4 in gives x = 300; ℓ = 40 in gives x = 120. Then **b = 20 in, a = 7200**. Prediction at ℓ = 100 in: x = 7200/120 = **60**. (Python-checked.)
- **Traps:** subtracting the wrong products; forgetting the b found is in length units, not ohms.
- Calculator: calculate — type: ℓ_1, x_1, ℓ_2, x_2 (optional ℓ_3) or R_1, I_1, R_2, I_2.

Out of scope (skip)

These types appear in tests or books, but the 2027 Div B rules exclude them. Skip them. They stay here so you can recognise them.

Q27. How Ohm's galvanometer detected current (magnetic needle)

How Ohm's galvanometer detected current (magnetic needle) · ❌ Out · Medium

- **Scope: OUT** for the physics. "All magnetism removed in 2027" (task summary; the 2027 B rule list has no magnetism item, and the 2026 B rule had "electricity and magnetism" in the history line). Naming the device is IN (Q13).
- **Where seen:** ATX Regional 2026 C Q1 (Ampère's astatic needle; Div B OUT as magnetism physics).
- **Worked example:** "A current in a wire deflects a nearby compass needle (Ørsted, 1820)" → **this is why the galvanometer worked**. Do not study the torque math.
- Calculator: none (skip).

Mistakes in tests and keys

1. **Prior file `prior/I_bank.md` (Ohm block):** says Ohm switched to a thermocouple "after a hint from **Seebeck**". My sources say **Poggendorff** suggested it in a footnote; Seebeck discovered the effect in 1821. Use: effect = Seebeck, suggestion = Poggendorff. The I_bank H7 line "Seebeck → Ohm (thermocouple)" is only right in the "effect" sense.
2. **Prior file `prior/I_bank.md`:** "Unit ohm (Ω), adopted 1881, Paris" and "the 1881-era 'legal ohm' was a mercury column 106 cm" mixes two events. Wikipedia "Ohm": 1881 congress = practical ohm (about 104.9 cm); **legal ohm 106 cm = 1884**; international ohm 106.3 cm = 1893.
3. **Prior file `prior/I_bank.md`:** "Henry (USA), Lenz (Russia) and Wheatstone (England) later made the law popular." I could not confirm Henry. Sources name Fechner, Lenz, Jacobi, Poggendorff, Pouillet; Wikipedia credits Wheatstone with drawing attention to Ohm's definitions.
4. **ETHW "Ohm's Law":** gives Ohm's birth as **1787** (correct: 16 March 1789) and says "ohm" was adopted as the unit in **1872** (Wikipedia: name in use by 1867 via the B.A.; international congress 1881).
5. **ETHW "Georg Simon Ohm":** prints "1933" for the Nuremberg post (should be **1833**) and says the unit was formalized "in the 1860s" (only the B.A. unit; international adoption was 1881).

6. **ATX Regional 2026 C Q7:** asks for the two elements Ohm used "as a substitute 'Ammeter'". Copper and bismuth (key A,B) formed his **thermocouple voltage source**, not his current meter (that was the torsion galvanometer). The key's elements are right; the question wording is wrong.
7. **MIT 2020 C Q4:** "Published an acoustic law (proven false)" → Ohm. "Proven false" is too strong. MacTutor: Seebeck showed the derivation's assumptions were not fully justified and Ohm admitted the error; Wikipedia: "not quite true". The law is still taught as an approximation (Helmholtz extended it [G]).
8. **"Circuit Lab Test (Key)" ~2013 C Q18:** "0.6 Siemens per meter" uses the unit pattern of conductivity (S/m, which needs an area). The key (0.3 S for 2 m) reads it as "1 m of this wire has 0.6 S". The phrasing is ambiguous; a student who multiplies (1.2 S) is misled by the wording.
9. **Princeton 2020 C key Q1:** writes Coulomb (1786) and Volta (1799). Coulomb's first memoir is usually dated 1785 and the voltaic pile 1800 (built late 1799). Minor, but a timeline grader may disagree.
10. **Task brief vs rules PDF:** brief says Circuit Lab is on pages B11-B12; the pdftotext footers show **B12-B13**.
11. **Spelling in tests:** ATX 2026 B/C and BullSO 2026 B spell "Kirchoff" (correct: Kirchhoff). Not an Ohm error, but it is in the same history questions.

- **ATX Regional 2026, Div C, MC7:** "two elements Ohm used as a substitute ammeter" → copper and bismuth. The elements are right, but the Bi-Cu thermocouple was his steady voltage source. His current meter was a torsion galvanometer. Wording error.
- **UT Austin Invitational 2019, Q6:** "believed in contiguous particles" → Ohm. Faraday used this phrase in 1838. Ohm described contiguous action, so the key is defensible.

Not confirmed

- **Minister's "unworthy to teach science" quote**: only secondary sources (Wikipedia "Ohm's law", aetherczar substack, edquest.ca, search snippets). No primary letter found. Deutsches Museum archive items NL 267/089, 090, 106 concern Ohm's 1826 leave and a paper review, not the quote.
- **"Web of naked fancies"**: Wikipedia "Ohm's law" only; the critic is not named. Not found in MacTutor.
- **"Hegelian" label for the critics**: Wikipedia describes the attitude but does not name Hegel. Marked [G].
- **Poggendorff footnote wording**: one secondary extract (n4trb.com, Nuts & Volts-style article). Not checked against *Annalen der Physik* 1826.
- **Ice / boiling water and "steady for 30 minutes"**: search snippets (model-engineer.co.uk, n4trb extract). Not in Wikipedia or MacTutor.
- **Ohm's real data constants** (often quoted as a ≈ 7285, b ≈ 20.25, wire lengths in inches): not found. My Q24 numbers are illustrative, not Ohm's data.
- **Date 8 January 1826** for the key data: ETHW only (and ETHW has other errors).
- **August Seebeck as son of T. J. Seebeck**: general knowledge, not checked.
- **Fourier's heat theory year (1822)**: general knowledge, not checked.
- **What Ohm taught in Berlin 1828-1833** (some books say the military school): MacTutor says only "schools in Berlin".
- **1994 stamp**: Linda Hall only. I also recall a 1989 stamp for his 200th birthday; not checked.
- **TH Nürnberg "Georg Simon Ohm" name, lunar crater "Ohm"**: general knowledge, not checked.
- **Britannica**: not tried this run (the prior run got HTTP 403).
- **Official SO resources**: soinc.org Circuit Lab page and the paid 2026 Nationals test packet not opened; no free state/national test found (prior G_manifest). So no State/Nationals history item on Ohm was seen.
- **Rickards Dec 2020 B Q14 answer letter**: the key's highlight is not in the text file. I derived B from the units and the prior G_tests note; I did not view the PDF page.

- Florida State 2021 Q3 quote: matches the opening of the English translation of Ohm's 1827 book (Taylor's *Scientific Memoirs*, 1841). I did not read the translation itself in this run.

Counts

- Question types: 31 (Q1-Q31). Q29-Q31 were added from the test-extraction files.

- IN: 27 (Q1-Q18, Q20-Q25, Q29-Q31). IN-SN: 0 (no KVL/KCL content in this subtopic). BORDERLINE: 3 (Q19, Q26, Q28; the diamagnetism T/F variant inside Q17 is noted, not counted). OUT: 1 (Q27).

Sources

- 2027 Div B rules: `/Users/kai/Daniel/Science_Olympiad_Div_B_Rules_2027_for_Web_Secured.pdf` (Circuit Lab B, footers B12-B13).
- Past tests (local, `/Users/kai/Daniel/circuitlab/research/downloads/`): `dlG/atx26reg_Btest.txt` + `atx26reg_Bkey.txt` (ATX Regional 2026 B Q2); `dlG/atx26reg_Ctest.txt` + `atx26reg_Ckey.txt` (ATX Regional 2026 C Q1, Q2, Q7, Q15); `dlB/tests/bullso26_test.txt` + key (BullSO 2026 B Q5); `dlB/tests/berk26_test.txt` + key (Berkeley 2026 B Q6); `dlG/pz/p21/Circuit Lab/2021 Princeton Circuit Lab Key.pdf` (Q2, Q9, Q10, page 2 viewed); `dlG/pz/p20/Circuit Lab/2020 Princeton Circuit Lab Answer Key.txt` (Q1); `dlG/mit20_key.txt` (Q4); `dlG/rsoiOldB_test.txt` (Q14, Q15); `dlG/rsoiOldC_test.txt` (Q16); `dlG/wiki/central.txt` (mho); `dlG/wiki/Circuit_Lab_test_key_fix.txt` (Q17-19); `dlG/wiki/Shock_value_test_.txt` (conductance items); `dlG/wiki/CircuitLabTest2019I.txt` (Q6); `dlG/atx20reg_Btest.txt` (Q55 style).
- Prior research: `/Users/kai/Daniel/circuitlab/research/prior/I_bank.md`, `B_topics.md`, `G_tests.md`, `G_manifest.md`, `F_slides.md`, `A_rules.md`.
- https://en.wikipedia.org/wiki/Georg_Ohm
- https://en.wikipedia.org/wiki/Ohm%27s_law
- https://en.wikipedia.org/wiki/Ohm
- https://en.wikipedia.org/wiki/Siemens_(unit)
- https://mathshistory.st-andrews.ac.uk/Biographies/Ohm/
- https://ethw.org/Georg_Simon_Ohm
- https://ethw.org/Ohm%27s_Law
- https://www.lindahall.org/about/news/scientist-of-the-day/georg-ohm/
- https://www.uni-flensburg.de/en/working-group-physics/histolab/thematical-subsections/electricity/ohms-torsion-balance
- https://n4trb.com/vol_43_Issue-1-EXTRACT.pdf (Poggendorff footnote, thermocouple details; secondary)
- https://www.model-engineer.co.uk/?p=4355 (thermocouple apparatus model; secondary)
- https://resolve.cambridge.org/core/journals/british-journal-for-the-history-of-science/article/abs/ohmseebeck-dispute-hermann-von-helmholtz-and-the-origins-of-physiological-acoustics/86FAEB166982C6C86C9C98DB5BE701BE (Ohm-Seebeck acoustic dispute)
- https://doi.org/10.17863/CAM.96514 (Cambridge thesis, "A history of Ohm's Law"; not read in full)
- https://www.ee.cityu.edu.hk/~gchen/pdf/Ohm.pdf (early supporters list; snippet only)
- https://library.si.edu/digital-library/book/diegalvanischeke00ohmg (scan of *Die galvanische Kette*; not read)
- https://aetherczar.substack.com/p/341-ohms-law and https://edquest.ca/component/content/article/232 (minister quote; secondary)
- https://digital-dev.deutsches-museum.de/en/digital-catalogue/archive-item/NL%20267/090/ (Ohm papers archive)

- Test-extraction files: `/Users/kai/Daniel/circuitlab/research/tests/` (atx-reg, misc-a, misc-b, nat26, state, ut-a, ut-b, ut-c) and `/Users/kai/Daniel/circuitlab/research/prior/G_tests.md`.
- Raw research file this note was built from: `/Users/kai/Daniel/circuitlab/research/subtopics/ohm.md`.

Heinrich Hertz

Note: Hertz

> **INFO: How to use this note**
> - Scope icons: ✅ IN = all levels. ✅ States-Nationals = only at State and National tournaments. ⚠️ Borderline = the rules do not settle it (the reason is given). ❌ Out = excluded (the rule is cited).
> - Question types go from Easy to Medium to Hard to Nationals. "Nationals" means the hardest depth, not a different scope.
> - Every number in a worked example was checked in Python.
> - Source tags in square brackets, like [W] or [M], are listed under "Sources".

Scope notes and source tags

Slug: `hertz`. Area: rule 3.c.i (history). Built on `prior/I_bank.md` (Hertz fact sheet), `prior/B_topics.md`, `prior/G_tests.md`. Written 2026-10-16.

Rule basis (quoted, 2027 Div B manual p. B12, `Science_Olympiad_Div_B_Rules_2027_for_Web_Secured.pdf`, pdftotext):

> c. The test will consist of at least one question from each of the following areas **only**: i. Historical perspective of the electricity discoveries made by Ampere, Coulomb, Kirchhoff, Volta, Ohm, **Hertz**, and Boole iii. Characteristics, sources, uses, and hazards of direct current (DC) and alternating current (AC) d. Topics not included ... AC circuit theory, **frequency analysis**, capacitance, inductance ... e. ... v. No questions may require the use of trigonometry, complex numbers, matrix algebra, or calculus

(The task brief says B11-B12. In this PDF the Circuit Lab text runs over pages **B12-B13**.)

What this means for Hertz:

- Facts about **his electricity discoveries** are IN at every level: EM (radio) waves, spark gap, confirming Maxwell, speed/reflection/refraction/polarization, the 1887 UV-spark (photoelectric) observation, cathode rays through foil, and the unit hertz.
- Simple frequency facts (Hz = 1/s, T = 1/f, 60 Hz grid) are IN under area iii (AC characteristics). "Frequency analysis" (filters, Bode, cutoff frequency) is OUT.
- Wave calculations (v = f·λ) are **BORDERLINE**. No Circuit Lab area lists waves. The math is allowed (algebra). Div B students do meet `c = λν` in Remote Sensing B (manual p. B39), so a writer could use it.
- No Hertz item is States/Nationals-only (only KVL/KCL is). Harder history items appear at any level, so Daniel should know them all for every level.

Div C test check (per the user's request). Many sources below are Div C tests. Each question type from a Div C test was checked against the Div B 2027 list (areas i-ix, exclusions in 3.d, math limits in 3.e). The result is in each block under "Div B check". Div C items that fail the check are tagged OUT, not dropped, so Daniel knows to skip them.

Source tags: [W] Wikipedia "Heinrich Hertz"; [IEEE] ETHW Milestone page; [OS] OpenStax College Physics 2e §24.1; [ML] National MagLab bio; [WQ] Wikiquote; [IOP] IOP Spark; [NP] nobelprize.org; [AdP] Annalen der Physik bibliographic record; [R] 2027 Div B rules; [T:...] a past test; [P] prior research file. [S] = several secondary sources agree.

Key facts

Life

Fact	Source
Heinrich Rudolf Hertz, born **22 Feb 1857, Hamburg** (German Confederation). Eldest of five children.	[W][ML]

Fact	Source
Died **1 Jan 1894, Bonn**, age **36** (Python: 36.86 years). Complications after surgery; Fölsing (1997) suggests granulomatosis with polyangiitis (a vasculitis). Buried Ohlsdorf Cemetery, Hamburg.	[W]; [IOP] says vasculitis
Studied in Dresden, Munich, Berlin. In Berlin under **Hermann von Helmholtz and Gustav Kirchhoff** (link to the Kirchhoff subtopic).	[W][ML]
1879: won a Berlin prize problem showing that electric current has **no measurable inertia** ("if current had any mass, it must be incredibly small").	[ML]; asked in [T:Princeton 2019 C Part 1 (b)]
PhD Berlin **1880** (magna cum laude), thesis on induction in rotating spheres. Then Helmholtz's assistant about 3 years.	[W][ML]
Kiel lecturer **1883-1885**. **Karlsruhe** professor **1885-1889** (radio-wave work done here). **Bonn** from **3 Apr 1889** to death.	[W]
Married Elisabeth Doll 1886. Daughters Johanna (1887) and Mathilde (1891, biologist).	[W]
Nephew **Gustav Hertz**: Nobel Prize in Physics **1925** with James Franck (Franck-Hertz experiment; awarded in 1926). Grand-nephew Carl Hellmuth Hertz: medical ultrasound.	[NP][W]
Family had converted from Judaism to Lutheranism; Nazi regime removed his name from streets/institutions; a proposal existed to rename the unit after Helmholtz but keep "Hz".	[W]
Honors: Matteucci Medal 1888, Rumford Medal 1890; IEEE Heinrich Hertz Medal (from 1987); lunar crater Hertz; Heinrich-Hertz Institute, Berlin (1928).	[W]

Electromagnetic (radio) waves, 1886-1889

Fact	Source
First to produce and detect electromagnetic waves (radio waves) on purpose, Karlsruhe, from **autumn 1886**; papers Nov 1887 and 1888. IEEE plaque dates: **1886-1888**. OpenStax says "began his experiments in 1887" (see §4).	[W][IEEE][OS]
Trigger: playing with a pair of **Riess (Ries) spirals**; discharging a Leyden jar through one coil made a spark in the other.	[W][IEEE]
Transmitter: a dipole: two collinear ~1 m copper wires, **7.5 mm spark gap**, ending in **30 cm zinc spheres**; driven by **~30 kV** pulses from a **Ruhmkorff induction coil**. Sparks across the gap make the oscillation.	[W][IEEE]
Receiver (detector): a single **wire loop with a tiny adjustable (micrometer) spark gap**. A small spark at the receiver gap = the wave arrived.	[W][IEEE][OS]
First oscillator frequency about **50 MHz** ($\lambda \approx 6$ m). Wikipedia also says waves "about 4 meters long" [citation needed]; the two numbers disagree (§4).	[W]
Later: half-wave dipole of two 13 cm brass rods at the focus of a 1.2 m × 2 m **parabolic sheet-metal reflector**: **~450 MHz**, **66 cm** waves (Python: c/450 MHz = 0.666 m). IEEE: "His 450 MHz transmitter and receiver demonstrated the fundamentals of high-frequency technology."	[W][IEEE]
Detection range up to about **18 m**.	[IEEE]
Standing waves: oscillator ~12 m from a zinc reflecting sheet; incident + reflected waves made nodes and antinodes; node spacing gave the **wavelength**. With the frequency (computed from his circuit), $\mathbf{v = f \cdot \lambda}$ gave the speed.	[W][OS][IEEE]
Result: the waves travel at the **speed of light** → confirmed **Maxwell's** 1860s prediction that light is an EM wave.	[W][OS][IEEE]
He showed the waves **reflect** (metal sheet), **refract** (a large **pitch prism**), and are **polarized** (receiver responds only when parallel to the transmitter dipole; a wire grid passes/blocks them). Also travel in straight lines, can be focused, interfere.	[W][IEEE][ML] [OS]
IEEE plaque: "In this building, Heinrich Hertz first verified Maxwell's equations and prediction of electromagnetic waves in 1886-1888. He observed the reflection, refraction and polarization of the waves and, moreover, the equality of their velocity" [with light]. Dedicated 5 Dec 2014, Karlsruhe (KIT, Kaiserstrasse 12). Original apparatus: **Deutsches Museum, Munich**.	[IEEE]
Waves called **"Hertzian waves"** until about 1910, then "radio waves". **Marconi** began radio telegraphy within about 6 years.	[W][ML]
His 1888 wire experiments gave a speed in wires of about **2/3 of the speed in air** (~200,000 km/s). Theory says equal; Sarasin & de la Rive (Geneva, 1890-92) and J. J. Thomson found equal speeds. Hertz's wire result was an apparatus error.	Nature history piece, O'Hara 2007 (secondary)

Fact	Source
Hertz recast Maxwell's equations into a compact four-equation form at the same time as Oliver **Heaviside** ("Hertz-Heaviside" form).	arXiv 1910.07082; Physics Today (secondary)

Photoelectric effect, 1887

Fact	Source
1887: noticed his receiver spark jumped more easily when **ultraviolet** light (from the transmitter spark) fell on the gap. In a dark box the maximum spark length got shorter.	[W][ML]
Glass between source and gap **shortened** the spark (glass absorbs UV). **Quartz** (passes UV) did not.	[W]
Paper: "Ueber einen Einfluss des ultravioletten Lichtes auf die electrische Entladung", *Annalen der Physik (und Chemie)* 1887, vol. 267 (N.F. 31), pp. 983-1000.	[AdP]
He **did not explain** it. Hallwachs followed up (1888); Lenard studied it; **Einstein** explained it (1905, photons), Nobel 1921.	[W]; Einstein dates [S]

Other work

Fact	Source
1883: tried to show cathode rays are **neutral** (no deflection in an electric field). **Wrong**: J. J. Thomson (1897) showed the field was screened in his tube.	[W]
1892: cathode rays pass through **thin metal foil** (aluminium); student Philipp Lenard continued.	[W]; asked in [T:Princeton 2019 C Part 1 (n)]
1881-82: founded **contact mechanics** (Hertzian contact stress). Not an electricity discovery.	[W]
Die Prinzipien der Mechanik (Principles of Mechanics), published 1894 after his death.	[W][ML]

Unit

Fact	Source
hertz (Hz) = SI unit of frequency = **1 cycle per second = 1 s^{-1}**.	[OS][W]
Name set by the **IEC in 1930**; adopted by the **CGPM in 1960** (11th CGPM, with the SI), replacing **"cycles per second" (cps)**, kc/s, Mc/s. Common use by about 1970. Some mirrors say IEC 1935 (§4).	[W][S]
Grid: **60 Hz** in the US, **50 Hz** in most of Europe/Asia.	[T:Rickards 2026 B Q28c][T:ATX 2021 B Q23]

Quotes

Quote	Attribution quality
"It's of no use whatsoever. This is just an experiment that proves Maestro Maxwell was right — we just have these mysterious electromagnetic waves that we cannot see with the naked eye. But they are there." Asked about applications: "Nothing, I guess."	Secondary only: Norton, *Dynamic Fields and Waves* (2000) p. 83 [WQ]; [IOP]. Student context = folklore.
"I do not think that the wireless waves I have discovered will have any practical application."	[IOP] only; no primary source found
"To the question, 'What is Maxwell's theory?' I know of no shorter or more definite answer than the following: Maxwell's theory is Maxwell's system of equations."	*Electric Waves* (1893) introduction, quoted in arXiv 1910.07082
"One cannot escape the feeling that these mathematical formulas have an independent existence and an intelligence of their own, that they are wiser than we are..."	Kline (1980) p. 338; Bell (1937) [WQ]

Formulas

Item	Formula / value	Units	Scope
Frequency definition	1 Hz = 1 cycle/s = 1 s^{-1}	Hz	IN
Period	T = 1/f	s (f in Hz)	IN (AC characteristic)
Cycles in a time	N = f·t	cycles	IN
Prefixes	kHz = 10^3, MHz = 10^6, GHz = 10^9 Hz		IN

Item	Formula / value	Units	Scope
Wave speed	v = f·λ; for radio/light in air v ≈ c	m/s	BORDERLINE
Speed of light	c = 299,792,458 m/s (exact); use 3.00×10^8 m/s	m/s	BORDERLINE
Standing waves	node-to-node spacing = λ/2, so v = 2·d·f	m	BORDERLINE
Travel time	t = d/c	s	BORDERLINE
Shielding rule of thumb	hole size ≤ λ/10	m	BORDERLINE→OUT (Div C test only)
Photon energy	E = h·f, KE = h·f − ϕ	J, eV	OUT

Question types

This note has **27 question types** (17 ✅ IN, 5 ⚠️ Borderline, 5 ❌ Out). The out-of-scope ones are in "Out of scope (skip)" below.

How the types were found

Format of each block: difficulty · scope · where seen · inputs → output · method · worked example (Python-checked) · traps · app.

History questions in the test-extraction files (8 test groups)

These rows come from `research/tests/` (ATX regionals, UT invitationals, misc-a, misc-b, USC 2026, Florida State 2021). Each row names the question type it belongs to.

Test	Year	Div	Level	Q	Question (short)	Key	Check / scope 2027 B	Type
Cobra Invitational (Michigan)	2019-20 season	B	Invitational	7	Fill-in paragraph: Hertz proved EM waves; unit of frequency	Hertz / frequency	IN	HZ-01
UF Northern Regional (Florida)	2019	B	Regional	4	Who found EM waves experimentally?	C Hertz	IN	HZ-01
UF Northern Regional (Florida)	2019	C	Regional	1-8	Match Volta/Ohm/Tesla/Hertz/Faraday to achievements	A, C, C, A, E, D, D, A	IN for the Hertz rows	HZ-02
Univ. of Michigan Invitational (UMSO)	2019 (file label)	not stated (likely C)	Invitational	1-5	Match Faraday, Tesla, Hertz, Volta, Ohm to discoveries	A, C, E, B, D	Hertz row IN	HZ-02
UT Austin Invitational (2020 season)	2019	B	Invitational	10	Epithet: winner of the Berlin Prize	Nobody (doubtful: Hertz won a Berlin University prize, 1879)	IN; for 2027 answer Hertz	HZ-13
UT Austin Invitational, fall (2020 season)	2019	C	Invitational	10	Epithet: winner of the Berlin Prize	Nobody (doubtful)	IN	HZ-13

Easy

HZ-01. Who first produced and detected electromagnetic (radio) waves?

Who first produced and detected electromagnetic (radio) waves? · ✅ IN · Easy

- **Scope:** IN.
- **Where seen:** Rickards Invitational **Div B** Dec 2020 Q21 (`dlG/rsoiOldB_key.txt`, key "Hertz"); Rickards Div C 2020 Q25 (`rsoiOldC_key.txt`); Princeton 2019 C Part 1 (l). **Div B check:** pure area i history. Passes.
  - Also seen (test-extraction files): Cobra Invitational (Michigan) 2019-20 season, Div B, Invitational, Q7: "Fill-in paragraph: Hertz proved EM waves; unit of frequency" (key: Hertz / frequency). Div B 2027: IN.
  - Also seen (test-extraction files): UF Northern Regional (Florida) 2019, Div B, Regional, Q4: "Who found EM waves experimentally?" (key: C Hertz). Div B 2027: IN.
- **How to solve:**
  1. Read the question: a description → a name.

2. "EM waves / radio waves / Hertzian waves / spark gap / confirmed Maxwell" → Hertz.
3. "Predicted" → Maxwell.
4. "Commercial radio" → Marconi.

- **Worked example:** "Who was the first scientist to prove the existence of electromagnetic waves?" → **Hertz**.
- **Traps:** Maxwell *predicted* (1860s); Marconi *used* them (1890s); Tesla and Faraday are off the 2027 list.
- Calculator: look up — type: keyword (e.g. "radio", "EM wave").

HZ-03. What is the hertz / what does Hz measure?

What is the hertz / what does Hz measure? · ✅ IN · Easy

- **Scope:** IN.
- **Where seen:** Shock Value student test (Div B, ~2011) Q7 "60 Hertz is also known as ____" (key: US AC frequency) (`dlG/wiki/Shock_value_test_answer_key.txt`); ATX 2026 C Q34 (Hz as a distractor; RC time constant itself is OUT). **Div B check:** unit fact passes.
- **How to solve:**
  1. Read the question: unit name → quantity, or equivalence.
  2. Hz = frequency = cycles per second = s^{-1}.
- **Worked example:** "1 Hz in SI base units?" → **s^{-1}**.
- **Traps:** Hz is not "per minute" (rpm); Hz is not a unit of time. Symbol is "Hz" (capital H, lower z); unit name is lowercase "hertz".
- Calculator: look up — type: unit name or symbol.

HZ-08. Predicted vs confirmed: Maxwell and Hertz

Predicted vs confirmed: Maxwell and Hertz · ✅ IN · Easy

- **Scope:** IN (fact). Questions on the *content* of Maxwell's equations → OUT (HZ-24).
- **Where seen:** Princeton 2019 C Part 1 (j) "credited with verifying Maxwell's equations" → H. Hawk & Hornet 2026 **C** Q21 "Which did not contribute to Maxwell's equations?" key A = Hertz (`dlG/hh26_key.txt`). **Div B check:** (j) passes (Hertz fact). H&H Q21 is BORDERLINE for B: Gauss is not on the list and Faraday was removed; it also has a key issue (§4).
- **How to solve:**
  1. Read the question: → name.
  2. Maxwell predicted EM waves (1860s)
  3. Hertz confirmed them (1886-88).
- **Worked example:** "Hertz's experiments confirmed whose theory?" → **Maxwell's**.
- **Traps:** H&H wording: Hertz did not contribute to the *original* equations, but he did help put them in the modern four-equation form.
- Calculator: look up — type: the keyword from the question.

HZ-15. Frequency ↔ period (grid frequency)

Frequency ↔ period (grid frequency) · ✅ IN · Easy

- **Scope:** IN (area iii, AC characteristics). Not "frequency analysis".
- **Where seen:** ATX 2021 **B** Q23 (US socket: 110 V, 60 Hz) (`dlG/atx21reg_Bkey.txt`); Rickards 2026 **B** Q28c (US uses 60 Hz) (`rsoi26B_key.txt`); UT Austin Fall 2025 Inv B/C 1(g)vi δ (period needed for US-grid frequency) (`ut25inv_test.txt`). **Div B check:** the T = 1/f part passes. The rest of the UT item uses a variable-resistance waveform; fine as algebra, but its motor part (vii) is OUT (magnetism).
- **How to solve:**
  1. Read the question: f → T, or T → f.
  2. T = 1/f.
- **Worked example:** 60 Hz → T = **16.7 ms**; 50 Hz → **20.0 ms** (Python: 16.667 ms, 20.0 ms).
- **Traps:** ms vs s; answer with sig figs (rule 3.b).
- Calculator: calculate — type: f (Hz, with prefix) or T (s).

HZ-16. Unit-prefix conversions with Hz

Unit-prefix conversions with Hz · ✅ IN · Easy

- **Scope:** IN.
- **Where seen:** not seen as its own question.
- **Worked example:** 1500 kHz = **1.5 MHz** (Python). 2.4 GHz = 2.4×10^{9} Hz.
- Calculator: calculate — type: value, from-prefix, to-prefix.

HZ-17. Count cycles in a time

Count cycles in a time · ✅ IN · Easy

- **Scope:** IN.
- **Where seen:** not seen.
- **How to solve:**
  1. N = f·t.
- **Worked example:** 60 Hz for 1 min → 60 × 60 = **3600 cycles** (Python).
- **Traps:** each cycle has 2 direction reversals (120 per second at 60 Hz).
- Calculator: calculate — type: f, t.

Medium

HZ-02. Match a statement to the scientist (Hertz among the 7)

Match a statement to the scientist (Hertz among the 7) · ✅ IN · Easy-Medium

- **Scope:** IN.
- **Where seen:** Princeton 2019 C Part 1 (`dlG/pz/p19/...Test.txt`): (b) current mass tiny → H, (j) verifying Maxwell's equations → H, (l) oscillating charges produce EM waves → H, (n) cathode rays through thin foil → H. Princeton 2026 Matching Part 1 (other scientists) [P:G_tests]. **Div B check:** Hertz statements pass. That test's Tesla/Faraday statements (e, h, m, o, p, r, t, u, v, w) are OUT-2027 for B.
  - Also seen (test-extraction files): UF Northern Regional (Florida) 2019, Div C, Regional, Q1-8: "Match Volta/Ohm/Tesla/Hertz/Faraday to achievements" (key: A, C, C, A, E, D, D, A). Div B 2027: IN for the Hertz rows.
  - Also seen (test-extraction files): Univ. of Michigan Invitational (UMSO) 2019 (file label), Div not stated (likely C), Invitational, Q1-5: "Match Faraday, Tesla, Hertz, Volta, Ohm to discoveries" (key: A, C, E, B, D). Div B 2027: Hertz row IN.
- **How to solve:**
  1. Read the question: list of statements + letter list → letter per statement.
  2. Hertz keywords: radio/EM waves, spark gap, Maxwell confirmed, speed of light, polarization, UV/photoelectric 1887, cathode rays through foil, no inertia of current, Karlsruhe, unit Hz.
- **Worked example:** "Demonstrated cathode rays could penetrate very thin metal foil" → **Hertz** (1892).
- **Traps:** "Discovered the electron" is J. J. Thomson. Cathode-ray *neutrality* claim was Hertz's (and wrong).
- Calculator: look up — type: keyword search over the fact table.

HZ-04. When was the hertz adopted, and what did it replace?

When was the hertz adopted, and what did it replace? · ✅ IN · Medium

- **Scope:** IN.
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: → year/body/old name.
  2. IEC 1930 → CGPM 1960 → replaced "cycles per second".
- **Worked example:** "The unit hertz replaced which earlier unit name?" → **cycles per second (cps)**. Years after his death that the IEC named it: 1930 − 1894 = **36** (Python).
- **Traps:** IEC (1930) vs CGPM/SI (1960). Some pages say 1935.
- Calculator: look up — type: the keyword from the question.

HZ-05. Biography facts (dates, places, teachers, age)

Biography facts (dates, places, teachers, age) · ✅ IN · Medium

- **Scope:** IN.
- **Where seen:** not seen for Hertz. Same style for others: MIT 2020 C Q1-10, Rickards 2020 B Q20-23.
- **How to solve:**
  1. Read the question: question → date/place/person/number.
  2. Recall table §1 "Life".
  3. Age = death year − birth year, minus 1 if the birthday had not come yet.
- **Worked example:** "How old was Hertz when he died?" Born 22 Feb 1857, died 1 Jan 1894 → **36** (Python: 36.86 y).
- **Traps:** 1894 − 1857 = 37, but he died before his birthday → 36 (MagLab says 37, §4). Taught by **Kirchhoff** and Helmholtz: a cross-link question. Radio work at **Karlsruhe**, not Bonn.
- Calculator: look up + calculate — type: (+ CALCULATE age from two dates).

HZ-06. Hertz's apparatus: name and explain the parts

Hertz's apparatus: name and explain the parts · ✅ IN · Medium

- **Scope:** IN.
- **Where seen:** not seen in a test. OpenStax Fig. 24.4 is the standard diagram.
- **How to solve:**
  1. Read the question: labelled diagram or description → part names / function.
  2. Induction (Ruhmkorff) coil → high voltage → spark across gap of dipole with metal spheres → waves → receiver loop with small gap → small spark = detection.
- **Worked example:** "What showed Hertz that a wave reached the receiver?" → **a small spark across the gap of the receiver loop**.
- **Traps:** The receiver had no battery or meter. The *transmitter* has the big spark. Circuit details (LC resonance) are OUT (capacitance/inductance).
- Calculator: look up — type: the keyword from the question.

HZ-07. Which wave properties did Hertz demonstrate? (select all)

Which wave properties did Hertz demonstrate? (select all) · ✅ IN · Medium

- **Scope:** IN.
- **Where seen:** not seen in a test; IEEE plaque text, Wikipedia.
- **How to solve:**
  1. Read the question: list of properties → which he showed.
  2. Speed = c; reflection (metal sheet → standing waves); refraction (pitch prism); polarization; straight-line travel; focusing (parabolic mirror); interference.
- **Worked example:** "Which did Hertz NOT show: reflection, refraction, polarization, quantization of photons?" → **quantization of photons** (Planck/Einstein).
- **Traps:** He did not send messages (Marconi). He did not explain the photoelectric effect.
- Calculator: look up — type: the keyword from the question.

HZ-09. Photoelectric effect: the history facts

Photoelectric effect: the history facts · ✅ IN · Medium

- **Scope:** IN (it is Hertz's 1887 discovery).
- **Where seen:** ATX Regional 2019 **B** Q1 and C Q1 (`dlG/atx19reg_B.txt`): no emission with green light → **use UV light**. **Div B check:** the question was in a Div B test; under 2027 rules a qualitative "UV, not brighter light" answer fits Hertz's discovery. Physics beyond that is OUT (HZ-23).
- **How to solve:**
  1. Read the question: → year / light type / who explained.
  2. 1887, UV light helps sparks jump; glass blocks the effect, quartz does not.
  3. Einstein explained it 1905.

- **Worked example:** "Placing a glass plate between the spark and receiver made the receiver spark shorter. Why?" → **glass absorbs ultraviolet light**.
- **Traps:** Brighter red/green light does not work; frequency (colour) matters, not brightness. Hertz observed, Einstein explained.
- Calculator: look up — type: the keyword from the question.

HZ-11. Order scientists by date of main discovery

Order scientists by date of main discovery · ✅ IN · Medium

- **Scope:** IN.
- **Where seen:** Princeton 2020 C Q1 (ordering type, included Faraday and Tesla) [P:G_tests]. **Div B check:** the type passes; drop Faraday/Tesla for 2027.
- **How to solve:**
  1. Read the question: set of names → ordered list.
  2. Coulomb 1785 → Volta 1800 → Ampère 1820-26 → Ohm 1827 → Kirchhoff 1845 → Boole 1847/1854 → **Hertz 1886-88**.
- **Worked example:** order {Hertz, Ohm, Volta} → **Volta (1800), Ohm (1827), Hertz (1887)**.
- **Traps:** Hertz is the latest of the seven. Boole's logic predates Hertz's waves.
- Calculator: look up — type: (stored years) + sort.

HZ-14. Why it mattered / applications (short answer or essay)

Why it mattered / applications (short answer or essay) · ✅ IN · Medium

- **Scope:** IN.
- **Where seen:** essay style in MIT 2020 C Q21-22, Princeton 2020 Q2-3 [P:G_tests]; not on Hertz.
- **How to solve:**
  1. Claim + evidence: proved Maxwell → light is EM; led to radio (Marconi), TV, radar, Wi-Fi, mobile phones; unit Hz on every AC device.
- **Worked example (model answer):** "Hertz's 1886-88 spark-gap experiments proved EM waves exist and move at light speed. This confirmed Maxwell and made radio possible within about 6 years (Marconi)."
- Calculator: look up — type: (stored outline).

HZ-18. Wavelength or frequency of a radio wave (v = f·λ)

Wavelength or frequency of a radio wave (v = f·λ) · ⚠️ Borderline · Medium

- **Scope:** BORDERLINE. Reason: no Circuit Lab area lists waves; area i covers Hertz's discovery and the algebra is allowed (3.e.i); Div B Remote Sensing tests `c = λν` (p. B39), so B students know it.
- **Where seen:** not seen in a Circuit Lab test (ATX 2026 C Q35 uses it inside HZ-21).
- **How to solve:**
  1. Read the question: f → λ, or λ → f.
  2. $\Lambda = c/f$, $c = 3.00\times10^{8}$ m/s.
- **Worked example:** Hertz's 450 MHz transmitter → $\lambda = 3.00\times10^{8} / 4.50\times10^{8}$ = **0.667 m (66.7 cm)** (Python: 0.6667; with exact c 0.6662). First oscillator 50 MHz → **6.00 m**. A 4 m wave → **75 MHz**.
- **Traps:** MHz → $\times10^{6}$; answer in m.
- Calculator: calculate — type: f (with prefix) or λ.

HZ-20. Travel time of a radio wave

Travel time of a radio wave · ⚠️ Borderline · Medium

- **Scope:** BORDERLINE.
- **Where seen:** not seen.
- **How to solve:**
  1. T = d/c.
- **Worked example:** across Hertz's 18 m range → $18 / 3.00\times10^{8}$ = **6.0×10^{-8} s = 60 ns** (Python).
- Calculator: calculate — type: d.

HZ-21. Faraday-cage hole size from wavelength

Faraday-cage hole size from wavelength · ⚠️ Borderline · Medium

- **Scope:** BORDERLINE leaning OUT. Reason: shielding links to area ii (fields), but the λ/10 rule is EM-wave engineering, not listed; Faraday removed from the 2027 history list.
- **Where seen:** ATX Regional 2026 **C** Q35 (5G radio, key D = 5 mm) and Section 2 Q2 (key: hole ≤ λ/10) (`dlG/atx26reg_Ctest.txt`, `atx26reg_Ckey.txt`). **Div B check:** not covered by any B area; key also underdetermined (§4).
- **How to solve:**
  1. Λ = c/f; hole ≤ λ/10.
- **Worked example:** 6 GHz → λ = 5.0 cm → hole ≤ **5.0 mm** (Python). 3.5 GHz → **8.6 mm**.
- Calculator: calculate — type: f.

Hard

HZ-10. Quotes: who said it / what did he mean

Quotes: who said it / what did he mean · ✅ IN · Medium-Hard

- **Scope:** IN.
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: quote → speaker, or speaker → meaning.
  2. Match keywords in §1 Quotes.
- **Worked example:** "'It's of no use whatsoever... proves Maestro Maxwell was right.' Who?" → **Hertz**.
- **Traps:** Wording varies between sources; these are secondary attributions.
- Calculator: look up — type: a keyword from the quote.

HZ-12. True/False and obscure facts

True/False and obscure facts · ✅ IN · Hard

- **Scope:** IN (facts about his electricity work); BORDERLINE for family trivia.
- **Where seen:** Princeton 2021 C Q1-10 T/F format (for others); Rickards 2020 Q20-25 [P:G_tests]. Not seen for Hertz.
- **How to solve:**
  1. Read the question: statement → T/F.
  2. Check against §1.
- **Worked example:** "Hertz showed that cathode rays are deflected by electric fields." → **False** (he saw no deflection and wrongly concluded they were neutral). "Hertz's nephew won a Nobel Prize." → **True** (Gustav Hertz, 1925).
- **Traps:** Gustav Hertz (nephew) ≠ Heinrich; Hertzsprung (astronomer) is unrelated. Wire speed 2/3 c was his error, not a true result.
- Calculator: look up — type: the keyword from the question.

HZ-13. Hertz's 1879 prize: inertia (mass) of electric current

Hertz's 1879 prize: inertia (mass) of electric current · ✅ IN · Hard

- **Scope:** IN.
- **Where seen:** Princeton 2019 C Part 1 (b) → H. **Div B check:** history fact, passes.
  - Also seen (test-extraction files): UT Austin Invitational (2020 season) 2019, Div B, Invitational, Q10: "Epithet: winner of the Berlin Prize" (key: Nobody (doubtful: Hertz won a Berlin University prize, 1879)). Div B 2027: IN; for 2027 answer Hertz.
  - Also seen (test-extraction files): UT Austin Invitational, fall (2020 season) 2019, Div C, Invitational, Q10: "Epithet: winner of the Berlin Prize" (key: Nobody (doubtful)). Div B 2027: IN.
- **Worked example:** "Showed that if electric current had any mass at all, it must be incredibly small" → **Hertz**.
- Calculator: look up — type: the keyword from the question.

HZ-22. Contact mechanics ("Hertzian contact")

Contact mechanics ("Hertzian contact") · ⚠ Borderline · Hard

- **Scope:** BORDERLINE. Reason: a real Hertz discovery but not an *electricity* discovery (rule 3.c.i says "electricity discoveries"). Possible only as a distractor.
- **Where seen:** not seen.
- **Worked example:** "Hertz founded which field in 1881-82?" → **contact mechanics**.
- Calculator: look up — type: the keyword from the question.

Nationals

HZ-19. Hertz's method: speed from standing-wave nodes

Hertz's method: speed from standing-wave nodes · ⚠ Borderline · Hard / Nationals

- **Scope:** BORDERLINE (same reason as HZ-18).
- **Where seen:** not seen in a test; OpenStax §24.1 describes the method.
- **How to solve:**
  1. Read the question: node spacing d, frequency f → speed v.
  2. $\Lambda = 2d$; $v = f \cdot \lambda = 2 \cdot d \cdot f$.
- **Worked example:** nodes 0.333 m apart, f = 450 MHz → $v = 2 \times 0.333 \times 4.50 \times 10^8 = \mathbf{3.00 \times 10^8}$ **m/s** (Python: 2.997×10^8) → matches light → confirms Maxwell.
- **Traps:** node-to-node = λ/2, not λ. Node-to-antinode = λ/4.
- Calculator: calculate — type: d, f (or λ and f).

Out of scope (skip)

These types appear in tests or books, but the 2027 Div B rules exclude them. Skip them. They stay here so you can recognise them.

HZ-23. Photoelectric calculations (photon energy, work function, electron speed)

Photoelectric calculations (photon energy, work function, electron speed) · ✗ Out · Hard

- **Scope:** OUT. Reason: photon/quantum physics is not in any of areas i-ix ("areas only", 3.c); area i covers the historical discovery, not the physics.
- **Where seen:** UT Austin 2019 Inv **B** Q22 (300 eV photon, ϕ = 108.57 eV → electron speed) (`dlG/ut19inv_Bkey.txt`). Python: $v = \sqrt{(2 \cdot 191.43 \text{ eV}/m_e)} = 8.21 \times 10^6$ m/s (choice A). Valid for 2020 rules, OUT for 2027.
- Calculator: none (skip).

HZ-24. Maxwell's equations content

Maxwell's equations content · ✗ Out · Medium-Hard

- **Scope:** OUT. Reason: magnetism was removed in 2027 (not in any area); differential/integral forms need calculus (3.e.v).
- **Where seen:** Rickards 2020 **B** Q10 / C Q12 ("NOT one of Maxwell's equations" → Coulomb's law) (`rsoiOldB_key.txt`); Princeton 2021 C Q18 (equations are linear); Rickards 2026 C Q7 (Maxwell-Faraday law, Heaviside). **Div B check:** fails.
- Calculator: none (skip).

HZ-25. EM wave vector directions (E × B, Poynting vector)

EM wave vector directions (E × B, Poynting vector) · ✗ Out · Hard

- **Scope:** OUT. Reason: magnetism not listed; cross product beyond the 2D-geometry limit (3.e.ii).
- **Where seen:** ATX Regional 2020 B and C (Poynting vector) (`atx20reg_Btest.txt`); MIT 2021 C Q17. **Div B check:** fails.
- Calculator: none (skip).

HZ-26. Why 50 Hz vs 60 Hz (eddy currents, transformer size)

Why 50 Hz vs 60 Hz (eddy currents, transformer size) · ❌ Out · Medium

- **Scope:** OUT. Reason: transformers and eddy currents are magnetism, removed in 2027. A plain "US 60 Hz / Europe 50 Hz" fact stays IN (HZ-15).
- **Where seen:** ATX Regional 2019 B Q4 (`atx19reg_B.txt`).
- Calculator: none (skip).

HZ-27. Generator speed → frequency (rpm ↔ Hz)

Generator speed → frequency (rpm ↔ Hz) · ❌ Out · Medium

- **Scope:** OUT. Reason: generators/motors are magnetism (removed in 2027).
- **Where seen:** MIT 2020 C (loop rotating at 42.0 Hz, turns for peak EMF) (`mit20_key.txt` line ~642). **Div B check:** fails. **Counts:** 27 types. IN 17 (HZ-01 to 17, with HZ-12 partly BORDERLINE), IN-SN 0, BORDERLINE 5 (HZ-18 to 22), OUT 5 (HZ-23 to 27).
- Calculator: none (skip).

Mistakes in tests and keys

1. **Hawk & Hornet 2026 C Q21** (`hh26_test.txt`, key A): "Which did not contribute to Maxwell's equations? Hertz." Defensible for the original 1860s set, but Hertz (with Heaviside) produced the modern compact four-equation form. Ambiguous wording.
2. **ATX Regional 2026 C Q35** (key D = 5 mm): "5G" is not one frequency. λ/10 gives 8.6 mm at 3.5 GHz, 5.0 mm at 6 GHz, 1.07 mm at 28 GHz (Python). The key fits only ~6 GHz. Also λ/10 is a rule of thumb (some use λ/20).
3. **Rickards 2020 B Q11 / C Q13:** "Which is NOT an effect that causes charge separation? Piezo-, photo-, pyro-, triboelectric." The photoelectric effect does separate charge (it ejects electrons; Hallwachs's charged plates). The key's answer is not visible in the text file. Unverified (see §5).
4. **Wikipedia:** first oscillator "about 50 MHz" and waves "about 4 meters long" [citation needed] disagree: 50 MHz → 6.0 m; 4 m → 75 MHz (Python).
5. **National MagLab bio:** "dies ... at age 37" — he was **36** (born 22 Feb 1857, died 1 Jan 1894). It says "blood poisoning"; Wikipedia says surgical complications, likely vasculitis. It also says he used a rotating mirror to find the frequency; I found no other source for that (Hertz computed frequency from his circuit).
6. **OpenStax 2e §24.1:** "began his experiments in 1887". Wikipedia and IEEE: autumn 1886 (published 1887-88).
7. **Hertz's own 1888 result:** wave speed in wires ≈ 2/3 of that in air. This was an error (later shown equal). A test could quote it as if true.
8. **IOP Spark:** "In 1889 Hertz had all his teeth removed". Not confirmed elsewhere.

- **UT Austin Invitational 2019, Div B, Q10 and fall 2019 Div C, Q10:** "Winner of the Berlin Prize", key "Nobody". Hertz won the University of Berlin prize in 1879 (inertia of electric current). For 2027, the answer is probably Hertz.

Not confirmed

- IEC 1930 date: from Wikipedia and mirrors only; iec.ch/history returned 403. Some mirrors say 1935.
- CGPM 1960 = 11th CGPM: secondary sources only; BIPM SI Brochure not fetched.
- "No use whatsoever" quote: only secondary (Norton 2000 via Wikiquote; IOP Spark). The student exchange is not in a primary source. "I do not think the wireless waves... practical application": IOP Spark only.
- "Mathematical formulas ... independent existence" quote: Kline 1980 / Bell 1937 via Wikiquote and MAA; no Hertz primary page.
- 30 kV coil, 7.5 mm gap, 30 cm spheres, 12 m reflector: Wikipedia only (IEEE confirms the 30 cm spheres).
- Hertz-Heaviside form: arXiv 1910.07082 and a Physics Today PDF (search snippet only).
- Wire-speed 2/3 c: Nature history piece and O'Hara (2007) via search snippets; full texts paywalled.
- Rickards 2020 Q11 key answer: the text file has no marked answer; I did not view the PDF highlights.

- Britannica (403). No 2027-season Circuit Lab test exists yet (per [P:B_topics]), so frequency of Hertz questions is unobserved.
- Searched every `.txt` in `downloads/` for "hertz", "Hz", "radio wave", "electromagnetic wave", "photoelectric", "spark", "Maxwell". All hits are cited above.

Sources

- 2027 Div B rules: `/Users/kai/Daniel/Science_Olympiad_Div_B_Rules_2027_for_Web_Secured.pdf` (Circuit Lab B pp. B12-B13; Remote Sensing B p. B39)
- https://en.wikipedia.org/wiki/Heinrich_Hertz
- https://ethw.org/Milestones:First_Generation_and_Experimental_Proof_of_Electromagnetic_Waves,_1886-1888
- https://openstax.org/books/college-physics-2e/pages/24-1-maxwells-equations-electromagnetic-waves-predicted-and-observed
- https://nationalmaglab.org/magnet-academy/history-of-electricity-magnetism/pioneers/heinrich-hertz/
- https://en.wikiquote.org/wiki/Heinrich_Hertz
- https://spark.iop.org/hertzs-useless-discovery
- https://www.nobelprize.org/prizes/physics/1925/summary/
- https://zenodo.org/record/1423827 (Annalen der Physik 1887 UV paper, DOI 10.1002/andp.18872670827)
- https://arxiv.org/pdf/1910.07082 (Hertz on Maxwell's theory; Hertz-Heaviside form)
- https://www.nature.com/articles/44468.pdf ; https://ideas.repec.org/a/cup/eurrev/v15y2007i04p545-559_00.html (wire-speed controversy)
- Past tests (local, `/Users/kai/Daniel/circuitlab/research/downloads/dlG/`): rsoiOldB_key.txt, rsoiOldC_key.txt, pz/p19 Princeton 2019, pz/p21 Princeton 2021, hh26_test/key, atx26reg_Ctest/Ckey, atx19reg_B, ut19inv_Bkey, ut25inv_test, atx21reg_Bkey, rsoi26B_key, rsoi26C_key, atx20reg_Btest, mit20_key, mit21, wiki/Shock_value_test_answer_key.txt
- Prior files: `/Users/kai/Daniel/circuitlab/research/prior/I_bank.md`, `B_topics.md`, `G_tests.md`, `F_slides.md`, `A_rules.md`
- Test-extraction files: `/Users/kai/Daniel/circuitlab/research/tests/` (atx-reg, misc-a, misc-b, nat26, state, ut-a, ut-b, ut-c) and `/Users/kai/Daniel/circuitlab/research/prior/G_tests.md`.
- Raw research file this note was built from: `/Users/kai/Daniel/circuitlab/research/subtopics/hertz.md`.

George Boole

Note: Boole

> **INFO: How to use this note**
>
> - Scope icons: ✅ IN = all levels. ✅ States-Nationals = only at State and National tournaments. ⚠️ Borderline = the rules do not settle it (the reason is given). ❌ Out = excluded (the rule is cited).
> - Question types go from Easy to Medium to Hard to Nationals. "Nationals" means the hardest depth, not a different scope.
> - Every number in a worked example was checked in Python.
> - Source tags in square brackets, like [W] or [M], are listed under "Sources".

Scope notes and source tags

Circuit Lab B, 2026-27 season (2027 rules). The level is all levels: Invitational, Regional, State, National.

Rule basis. Div B 2027 rule 3.c.i: "Historical perspective of the electricity discoveries made by Ampere, Coulomb, Kirchhoff, Volta, Ohm, Hertz, and Boole" (manual page B12; pdftotext of `/Users/kai/Daniel/Science_Olympiad_Div_B_Rules_2027_for_Web_Secured.pdf`). Boole is **new for 2027**. Faraday and Tesla were removed. The related rules are 3.c.viii ("Digital logic gates, truth tables, universal gate sets, boolean expressions, De Morgan's Law, and combinational logic circuits") and 3.e.iii-iv (binary whole numbers and binary addition only; "Logical operations as boolean expressions, including De Morgan's Law and truth tables").

What "historical perspective" means for Boole. Boole made no discovery in electricity. He lived 1815-1864. His algebra of logic became the theory of switching circuits 73 years after his death, through Shannon (1937). A Boole history question therefore asks about the man, his two logic books, his algebra, or the link to digital circuits. Pure logic calculations (gates, De Morgan, simplification) belong to the digital-logic subtopics. This file covers them only where a test ties them to Boole's own laws.

Div C check (user request). Many Boolean questions cited below come from **Div C** tests. Each one has a "Div B check" line that compares it with the Div B 2027 rules. A Div C item counts as practice only when that line says IN. Most Div C tests are from the 2020-2021 seasons. Those rules (C) had their own logic section, which was not the same as the 2027 Div B text.

Tag legend for sources: [R] Div B 2027 rules. [LT] Boole, *Laws of Thought* 1854, Project Gutenberg text (primary source, checked by grep). [MT] MacTutor biography. [SEP] Stanford Encyclopedia of Philosophy. [WP] Wikipedia "George Boole". [WPS] Wikipedia "A Symbolic Analysis of Relay and Switching Circuits". [AMS] AMS Notices Shannon obituary (2002). [UCC] University College Cork Boole papers site. [NLI] National Library of Ireland catalogue. [P] prior research files (`/Users/kai/Daniel/circuitlab/research/prior/`). [Py] checked in Python (`boole_chk.py` in the scratchpad; all outputs agree with this file).

Key facts

1a. Life

#	Fact	Source
1	Born **2 November 1815, Lincoln, Lincolnshire, England**.	[MT] [WP]
2	Died **8 December 1864, Ballintemple, County Cork, Ireland**, age **49** [Py].	[MT] [WP]
3	Cause of death: he walked about two miles to a lecture in heavy rain and lectured in wet clothes. He got a feverish cold, then pneumonia. Wikipedia says "fever-induced pleural effusion". MacTutor repeats a story that his wife threw buckets of water over his bed. It treats this as an anecdote, not a confirmed fact.	[MT] [WP]
4	Buried at St Michael's Church of Ireland, Blackrock, Cork.	[WP]
5	**Nationality: English.** He worked in Ireland from 1849. Trap: he was not Irish by birth.	[MT] [WP]

#	Fact	Source
6	Father **John Boole, a shoemaker (cobbler)** with a shop at 34 Silver Street, Lincoln. Mother Mary Ann Joyce, a lady's maid.	[MT]
7	**Largely self-taught. He had no university degree.** A bookseller's friend taught him Latin. He taught himself Greek, French and German. At 14 he translated a poem by Meleager. His father had it published, and a local schoolmaster doubted that a 14-year-old wrote it.	[MT] [WP]
8	Bainbridge's Commercial Academy, Lincoln, from 10 Sept 1828.	[MT]
9	At **16**: assistant teacher at Heigham's School, Doncaster, to support the family. Then Liverpool (6 months, 1833) and Hall's Academy, Waddington.	[MT]
10	**1834: opened his own school in Lincoln.** MacTutor says "aged 19". 1834 minus 1815 gives 18 before 2 November 1834 and 19 after it [Py]. Accept "19" as the textbook answer.	[MT] [Py]
11	1838: took over Hall's Academy, Waddington. 1840: opened a boarding school in Lincoln.	[MT]
12	He learned calculus from Lacroix's book. Duncan Gregory, editor of the *Cambridge Mathematical Journal*, encouraged him, and Boole published his first papers there.	[MT]
13	Married **Mary Everest, 11 September 1855** (Wickwar, Gloucestershire). She was the niece of **Sir George Everest**, after whom Mount Everest is named.	[MT] [WP]
14	**Five daughters**: Mary Ellen (1856, married Charles Howard Hinton), Margaret (1858, mother of G. I. Taylor FRS), **Alicia** (1860, four-dimensional geometry, "Alicia Boole Stott"), **Lucy Everest** (1862, chemist), **Ethel Lilian** (1864, Ethel Voynich, author of *The Gadfly*).	[MT] [WP]

1b. Career and honors

#	Fact	Source
15	1841: an influential early paper on invariant theory.	[WP]
16	**1844: Royal Medal of the Royal Society**, the first gold medal the Society gave for mathematics. It was for "**On a General Method in Analysis**" (Philosophical Transactions). Age 29 [Py].	[MT] [WP] [UCC search]
17	**1847: *The Mathematical Analysis of Logic, Being an Essay Towards a Calculus of Deductive Reasoning*** (Cambridge and London). It is a short monograph of about 82 pages. In the 1854 preface, Boole says it was "written within a few weeks after its idea had been conceived".	[SEP] [LT preface]
18	What prompted the 1847 book: a public dispute in early 1847 between **Augustus De Morgan** and **Sir William Hamilton** (of Edinburgh, the philosopher) about who first had the "quantification of the predicate". Some sources say the book came out in the same month as De Morgan's *Formal Logic* (1847).	[SEP] [WP]
19	**1849: first Professor of Mathematics, Queen's College, Cork** (now **University College Cork**, UCC). He was named in August 1849 and started in November 1849, at age 33 [Py]. He held the post until he died (15 years). He was a mathematics professor without a degree.	[MT] [WP] [UCC]
20	May 1851: Dean of Science at Cork.	[MT]
21	**1854: *An Investigation of the Laws of Thought, on Which Are Founded the Mathematical Theories of Logic and Probabilities*.** The title page says "By George Boole, LL.D., Professor of Mathematics in Queen's College, Cork". It is dedicated to John Ryall, Vice-President of Queen's College. This book is the full statement of what is now called Boolean algebra. It is **7 years** after *MAL* [Py].	[LT title page] [SEP] [MT]
22	The 1854 preface: *LT* "is not a republication" of *MAL*. It "begins by establishing the same system of fundamental laws, but its methods are more general, and its range of applications far wider".	[LT]
23	Keith Medal, Royal Society of Edinburgh: the Wikipedia text says 1855 and the infobox gives 1855-1857.	[WP]
24	**1857: Fellow of the Royal Society (FRS)**, at age 41-42.	[MT] [WP]
25	Honorary degrees: LL.D. from Dublin and from Oxford. The years were not confirmed.	[MT] [WP]
26	Other books: *Treatise on Differential Equations* (1859). *Treatise on the Calculus of Finite Differences* (1860). Probability (the second half of *LT*). About 50 papers. Eponyms: Boole's inequality, Boole's rule, Boole's expansion theorem.	[MT] [WP]

#	Fact	Source
27	Boole on his own logic (1851 letter to William Thomson): "the most valuable if not the only valuable contribution" he had made to science.	[MT]
28	De Morgan on Boole: "Boole's system of logic is but one of many proofs of genius and patience combined."	[MT]
29	Honors after his death: lunar crater **Boole**. **2015 bicentenary** (200 years [Py]) at UCC. **Google Doodle, 2 November 2015**.	[MT] [WP]
30	**No SI unit is named after Boole.** Kirchhoff has none either. The other five on the list have one (A, C, V, Ω, Hz).	[PI_bank] [SI]

1c. Boole's algebra (his own notation, from *Laws of Thought*)

#	Fact	Source
31	Letters x, y, z stand for **classes** (Boole's "elective symbols"). **xy** is the class of things that are both x and y. This is modern AND (intersection).	[LT ch. II] [SEP]
32	**$x^2 = x$** (from xx = x), "the fundamental law of thought". In *MAL* he wrote it as the "index law" $x^n = x$. Benjamin Peirce later named it the **idempotent** law. Modern form: A·A = A.	[LT ch. II §9, ch. III][SEP]
33	"Of the symbols of Number there are but two, viz. 0 and 1" that obey $x^2 = x$. So Boole's logic is an **algebra of the values 0 and 1**.	[LT ch. II §15][Py]
34	**1 = the Universe. 0 = Nothing.** ("Nothing and Universe are the two limits of class extension.") He also used the term "universe of discourse".	[LT ch. III Prop. II] [SEP]
35	**1 – x = the contrary class** ("not-men"). This is modern NOT.	[LT ch. III Prop. III]
36	**$x(1 - x) = 0$ is the principle of contradiction** (nothing can both have and lack a quality). It comes from $x - x^2 = 0$. Modern form: $A \cdot A' = 0$ (complement law).	[LT ch. III Prop. IV]
37	Boole's **+** was defined only for **disjoint** classes. His "x or y" (inclusive) was x + (1–x)y, or x + y – xy. He wrote the exclusive or as x + y – 2xy. All three were checked on {0,1} [Py].	[SEP][Py]
38	**Jevons** (*Pure Logic*, 1864) made + the inclusive union, so that x + x = x. Boole rejected this. In Boole's numeric algebra x + x = 2x, and 2x = x forces x = 0 [Py]. The modern rule "1 + 1 = 1" therefore comes from Jevons, not from Boole.	[SEP][Py]
39	The name: C. S. Peirce coined "Boolian algebra". Josiah Royce's circle (Sheffer, C. I. Lewis, Wiener) spread the spelling "Boolean" from about 1900.	[SEP]
40	Modern legacy: the Boolean (`bool`) data type, Boolean search (AND/OR/NOT), and digital logic.	[WP]

1d. The link to electricity: Claude Shannon

#	Fact	Source
41	**Claude E. Shannon** (1916-2001) met Boole's logic in a course at the **University of Michigan** (he entered in 1932).	[search: Bentley Library, MacTutor Shannon]
42	**MIT master's thesis "A Symbolic Analysis of Relay and Switching Circuits"**, submitted 10 August **1937**. It was **published 1938** in *Transactions of the AIEE*.	[WPS][search]
43	Main idea: Boolean algebra can describe and simplify relay and switch networks, and relay circuits can solve Boolean problems. Modern mapping: **series contacts = AND, parallel contacts = OR, normally-closed contact = NOT**. Shannon's own paper used "hindrance", with the opposite convention (series = +). That detail is BORDERLINE trivia.	[WPS][PH_logic_diodes A20]
44	Howard Gardner (1985): "possibly the most important, and also the most famous, master's thesis of the century". Goldstine: it changed digital circuit design "from an art to a science".	[WPS]
45	Alfred Noble Prize (the US engineering societies' award). **AMS Notices says 1940. Wikipedia's thesis article says 1939.** See section 4.	[AMS][WPS]
46	Victor Shestakov (USSR) proposed a similar theory independently.	[WP]

#	Fact	Source
47	Time gaps [Py]: *LT* to the thesis = **83 years**. *MAL* to the thesis = 90 years. Boole's death to the thesis = 73 years.	[Py]

1e. De Morgan

#	Fact	Source
48	**Augustus De Morgan** (born 27 June 1806, Madurai, India; died 18 March 1871, London). He was the first professor of mathematics at London University (UCL), from 1828, and the first president of the London Mathematical Society (1865). He coined "mathematical induction" (1838).	[MT De Morgan]
49	Boole and De Morgan wrote to each other **from 1842 to 1864**. The letters are published as *The Boole-De Morgan Correspondence 1842-1864*, ed. G. C. Smith (Oxford, Clarendon Press, 1982).	[MT][NLI]
50	De Morgan wrote a testimonial for Boole's 1846 application for a Queen's College chair (with Cayley, Kelland and Thomson). Boole asked him in 1850 to watch for jobs in England.	[MT]
51	**De Morgan's laws are named after De Morgan, not Boole.** (A·B)′ = A′ + B′ and (A + B)′ = A′·B′ [Py]. They are in the rules (3.c.viii and 3.e.iv).	[R][MT De Morgan] [Py]

1f. Timeline with the other six (for ordering questions)

Coulomb's law 1785. Volta's pile 1800. Ampère 1820-1826. Ohm 1827. Kirchhoff 1845. **Boole *MAL* 1847, *LT* 1854.** Hertz radio waves 1886-1888. (Shannon 1937.) Sorted in Python [Py]. **Boole is the only one of the seven who was not a physicist.**

Formulas

Boole's laws have no units or physical constants. The formulas below are written in Boole's form and in the modern form. Each was checked over every 0/1 input [Py].

Boole (1854)	Modern	Name
xy	A·B (AND)	intersection / conjunction
1 − x	A′ (NOT)	contrary class / complement
1	1 (TRUE)	Universe
0	0 (FALSE)	Nothing
$x^2 = x$	A·A = A	index law / idempotent law
x(1 − x) = 0	A·A′ = 0	principle of contradiction
x + (1 − x)y or x + y − xy	A + B (OR)	inclusive or
x + y − 2xy	A ⊕ B (XOR)	exclusive or
(Jevons 1864) x + x = x	A + A = A	idempotent OR
—	(AB)′ = A′+B′ ; (A+B)′ = A′B′	De Morgan's laws
Shannon: series contacts	A·B	AND
Shannon: parallel contacts	A + B	OR
Count: n inputs	2^n truth-table rows; 2^(2^n) possible functions (n=2: 4 rows, 16 functions)	[Py]

Date arithmetic: age = event year − 1815, minus 1 if the event is before 2 November.

Question types

This note has **26 question types** (20 ✅ IN, 5 ⚠️ Borderline, 1 ❌ Out). The out-of-scope ones are in "Out of scope (skip)" below.

How the types were found

Counts: **26 types**. IN = 20, IN-SN = 0, BORDERLINE = 5, OUT = 1 (T26 is out of this subtopic, not out of Div B). IN-SN: none. No Boole fact is limited to States or Nationals, because only KVL/KCL (3.c.ix) is. The "Nationals" difficulty label marks depth, not scope.

Where seen. No downloaded test (Div B or C, 2010-2026) asks a Boole **history** question. Boole is new to the Div B list in 2027. I searched every .txt under `research/downloads` for "Boole", "Laws of Thought", "Shannon", "De Morgan" and "Cork". All hits were logic questions or rules text. So history types say "not seen in a test" and give the closest format seen. The Div B 2026 history format (BullSO 2026 B Q1-6, Berkeley 2026 B Q1/Q6, Pembroke 2026 B Q15) is 1-point "who / named after whom" multiple choice. Expect Boole to appear in that format first.

Easy

T1. Concept → person ("Boolean algebra is named after?")

Concept → person ("Boolean algebra is named after?") · ✅ IN · Easy

- **Scope:** IN (3.c.i).
- **Where seen:** not seen in a test. Same format as BullSO 2026 B Q1 and Q5 ("law ... named after who?").
- **How to solve:**
  1. Read the question: a term (Boolean algebra, Boolean logic, `bool`, "algebra of logic") → Boole.
  2. "Boolean" = Boole.
  3. Of the seven, he is the only mathematician/logician.
- **Worked example:** "The algebra of AND, OR, NOT on the values 0 and 1 is named after which scientist? A. Ohm B. Boole C. Kirchhoff D. Hertz" → **B. Boole**.
- **Traps:** De Morgan (laws, not the algebra). Shannon (applied it to circuits, did not invent it). Leibniz (binary numbers, not Boolean algebra; not verified here, see section 5).
- Calculator: look up — type: keyword ("boolean").

T2. Book title → author

Book title → author · ✅ IN · Easy

- **Scope:** IN.
- **Where seen:** not seen in a test. Predicted in prior I_bank H4.
- **How to solve:**
  1. Read the question: title (*Laws of Thought*, *Mathematical Analysis of Logic*) → Boole.
  2. Both logic books are Boole's.
  3. Distractors: De Morgan *Formal Logic* (1847)
  4. Ohm *Die galvanische Kette* (1827)
  5. Shannon "A Symbolic Analysis of Relay and Switching Circuits" (1937).
- **Worked example:** "*An Investigation of the Laws of Thought* (1854) was written by?" → **George Boole**.
- **Traps:** *Formal Logic* (De Morgan) came out the same year as *MAL*. Thomson's *Outlines of the Laws of Thought* is a different book by a different author; Boole cites it in his preface [LT].
- Calculator: look up — type: title words.

T10. No SI unit

No SI unit · ✅ IN · Easy

- **Scope:** IN.
- **Where seen:** not seen with Boole. Unit → person is the most common history format (BullSO 2026 B Q4 style, Rickards 2021 C Q30 "full first name of the person the SI unit of charge is named after").
- **How to solve:**
  1. Read the question: "Which of the seven has no SI unit named after them?" → **Kirchhoff and Boole**.
  2. A Ampère, C Coulomb, V Volta, Ω Ohm, Hz Hertz.
- **Worked example:** as above → **Boole (and Kirchhoff)**.

- **Traps:** the "boolean" is a data type, not a unit. Do not mix up the "bit" (Shannon/Tukey) with anything named after Boole.
- Calculator: look up — type: none (a fixed table).

Medium

T3. Person → work and year

Person → work and year · ✅ IN · Easy-Medium

- **Scope:** IN.
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: "Boole" + "what/when" → *MAL* 1847, *LT* 1854.
  2. Memory: "47 analysis, 54 laws".
  3. They are 7 years apart [Py].
- **Worked example:** "In what year did Boole publish *The Mathematical Analysis of Logic*?" → **1847**.
- **Traps:** 1854 is *LT*. 1844 is the Royal Medal. 1849 is Cork. 1937 is Shannon.
- Calculator: look up — type: person + work.

T4. Places and nationality

Places and nationality · ✅ IN · Easy-Medium

- **Scope:** IN.
- **Where seen:** not seen in a test. Format of prior I_bank H6.
- **How to solve:**
  1. Read the question: "which scientist worked in Ireland / was born in Lincoln / taught at Cork" → Boole. Or the reverse: Boole → English; Lincoln; Queen's College Cork (now UCC).
  2. The other six were French (Coulomb, Ampère), Italian (Volta) or German (Ohm, Kirchhoff, Hertz)
  3. Boole is the only one from the British Isles.
- **Worked example:** "Boole was the first professor of mathematics at which college?" → **Queen's College, Cork (now University College Cork), Ireland**.
- **Traps:** "Irish mathematician" is wrong (he was English, worked in Ireland). Trinity College Dublin is wrong (it gave him an honorary degree only). Cambridge is wrong (he published in the *Cambridge Mathematical Journal* but never studied there).
- Calculator: look up — type: place or person.

T5. Early life and self-education

Early life and self-education · ✅ IN · Medium

- **Scope:** IN (biography is part of the "historical perspective"; the low-level details are trivia).
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: clue (shoemaker's son, no degree, taught at 16, own school at 19, taught himself languages) → Boole. Or a T/F question on one fact.
  2. Boole is the clearest "self-taught, no degree" case of the seven.
  3. Ohm also studied partly on his own, but he later took a doctorate; check the Ohm subtopic before using him as a distractor.
- **Worked example:** "True or False: George Boole earned a mathematics degree from Cambridge before Cork hired him." → **False** (no degree; largely self-taught).
- **Traps:** age 19 for the school vs 18 by plain subtraction (fact 10). Accept 19.
- Calculator: look up — type: clue keyword.

T6. Honors and medals

Honors and medals · ✅ IN · Medium

- **Scope:** IN.

- **Where seen:** not seen in a test. Format of prior I_bank H10.
- **How to solve:**
  1. Read the question: honor → person/year. Royal Medal 1844 ("On a General Method in Analysis", the first gold medal for mathematics), Keith Medal (RSE), FRS 1857, honorary degrees from Dublin and Oxford, lunar crater, 2015 Google Doodle.
  2. Lookup table.
- **Worked example:** "For which paper did Boole receive the Royal Society's Royal Medal in 1844?" → **"On a General Method in Analysis"**.
- **Traps:** the Copley Medal went to Volta (1794) and Ohm (1841), not Boole. Boole had no Nobel (it began in 1901, after his death).
- Calculator: look up — type: honor name or year.

T7. Death anecdote

Death anecdote · ✅ IN · Medium

- **Scope:** IN (a favorite trivia item, like Ampère's father or Hertz's early death).
- **Where seen:** not seen in a test. Format of prior I_bank H9.
- **How to solve:**
  1. Read the question: "which scientist died after lecturing in wet clothes" → Boole. Or the year/age: 1864, 49.
  2. Age = 1864 − 1815 = 49 (the death on 8 Dec is after his birthday on 2 Nov) [Py].
- **Worked example:** "How old was Boole when he died?" → **49**.
- **Traps:** 50 (simple subtraction rounded up). Hertz died at 36, which is the other early death.
- Calculator: look up + calculate — type: person. (App note: (CALCULATE age if the app has the date tool in T9).)

T8. Timeline / ordering of the seven

Timeline / ordering of the seven · ✅ IN · Medium

- **Scope:** IN.
- **Where seen:** Princeton 2020 **Div C** Q1 (order scientists by main contribution; prior G_tests). **Div B check:** the format is IN for Div B. Its list includes Faraday 1831 and Tesla 1887, who are OUT for 2027. Replace them with Boole and Hertz before practicing.
- **How to solve:**
  1. Read the question: 4-7 events → chronological order.
  2. Coulomb 1785 → Volta 1800 → Ampère 1820s → Ohm 1827 → Kirchhoff 1845 → **Boole 1847/1854** → Hertz 1886-88 (→ Shannon 1937).
- **Worked example:** order "Kirchhoff's laws, Boole's *Laws of Thought*, Ohm's law, Hertz's radio waves". → **Ohm 1827, Kirchhoff 1845, Boole 1854, Hertz 1887** [Py].
- **Traps:** Kirchhoff (1845) comes *before* Boole's *MAL* (1847). Boole died (1864) before Hertz's experiments.
- Calculator: calculate — type: pick events from a menu; the app sorts by stored year. (App note: (sort).)

T9. Date arithmetic (ages and gaps)

Date arithmetic (ages and gaps) · ✅ IN · Medium

- **Scope:** IN (simple arithmetic, rule 3.e.i).
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: two dates → age or gap.
  2. Gap = year2 − year1.
  3. Age = event year − 1815, minus 1 if the event comes before 2 Nov.
- **Worked example:** "How many years passed between *Laws of Thought* and Shannon's thesis?" → **1937 − 1854 = 83 years** [Py]. "How old was Boole when Cork appointed him (Aug 1849)?" → **33** [Py].
- **Traps:** the age before or after the birthday (*LT* 1854: 38 or 39, because the publication month was not confirmed). Shannon's thesis was 1937; the published paper was 1938.
- Calculator: calculate — type: event A (menu or year, plus month/day if known), event B.

T11. Mis-attribution ("Which was NOT Boole's?")

Mis-attribution ("Which was NOT Boole's?") · ✅ IN · Medium

- **Scope:** IN.
- **Where seen:** not seen with Boole. Format of prior I_bank H14.
- **How to solve:**
  1. Read the question: 4 claims → the false one.
  2. Boole did: *MAL*, *LT*, the Royal Medal paper, differential equations (1859), finite differences (1860), probability, $x^2 = x$.
  3. He did NOT do: De Morgan's laws, the switching-circuit application (Shannon), electronic logic gates (none existed then), a computer (Babbage was his contemporary; BORDERLINE trivia), binary numbers (Leibniz; not verified here).
- **Worked example:** "Which is NOT a contribution of Boole? A. *Laws of Thought* B. Treatise on differential equations C. Applying logic algebra to relay circuits D. The algebraic law $x^2 = x$" → **C** (Shannon, 1937).
- **Traps:** "Boole built logic gates." He did not; his work was on paper, about classes and propositions.
- Calculator: look up — type: claim keyword.

T12. De Morgan connection and De Morgan's laws

De Morgan connection and De Morgan's laws · ✅ IN · Medium

- **Scope:** IN for "De Morgan's Law" (named in 3.c.viii and 3.e.iv) and for De Morgan's role in Boole's story. BORDERLINE for De Morgan's own biography (he is not on the list).
- **Where seen:** Princeton 2021 **Div C** Q19 ("Which is one of DeMorgan's Laws?" Key **A: (A + B)′ = A′·B′**, read from the red highlight on key page 4). Princeton 2020 **Div C** Q37 ("State DeMorgan's Theorem", then draw the gates; key: $\bar{A} + \bar{B}$ and $\bar{A}\bar{B}$). **Div B check:** both are IN (De Morgan's Law is named in the Div B rules). Not seen with history wording.
- **How to solve:**
  1. Read the question: (a) "Boole's friend and correspondent, the logician of the 1847 *Formal Logic*?" → De Morgan. (b) an identity → is it De Morgan?.
  2. "break the bar, change the sign".
  3. Check an identity with a 4-row truth table.
- **Worked example:** "Which scientist exchanged letters with Boole from 1842 to 1864 and has two logic laws named after him?" → **Augustus De Morgan**. Princeton 2021 Q19 → **A** [Py: identity holds for all 4 rows; option D's sides differ].
- **Traps:** "Boole's laws" ≠ De Morgan's laws. (AB)′ ≠ A′B′.
- Calculator: look up + calculate — type: an expression for each side, which the app tests on every row. (App note: for the person. CALCULATE for the identity check.)

T13. Shannon: who linked Boole to circuits

Shannon: who linked Boole to circuits · ⚠️ Borderline · Medium

- **Scope:** BORDERLINE. Shannon is not on the list. However, the rule asks for the "historical perspective of the electricity discoveries made by ... Boole", and the only electrical meaning of Boole's work is Shannon's thesis. A writer is likely to ask it. The fact "Boolean algebra is the basis of digital/switching circuits" is IN.
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: "Who / when / where / title" → Claude Shannon, 1937 (published 1938), MIT, "A Symbolic Analysis of Relay and Switching Circuits".
  2. Recall.
- **Worked example:** "In 1937, an MIT student showed that Boole's algebra could design relay switching circuits. Who?" → **Claude Shannon**.
- **Traps:** 1948 is Shannon's information-theory paper, not the switching thesis. Vannevar Bush was his advisor and the context, not the author (Bush is not in the downloaded sources; not verified). Alan Turing (1936) is a different person and topic.

- Calculator: look up — type: keyword.

T18. Boolean 1 + 1 vs binary 1 + 1

Boolean 1 + 1 vs binary 1 + 1 · ✅ IN · Medium

- **Scope:** IN (3.e.iii binary addition and 3.e.iv logic operations are both allowed; this question tests that they differ).
- **Where seen:** not seen in a test in this form.
- **How to solve:**
  1. Read the question: "1 + 1 in Boolean algebra" → 1. "1 + 1 in binary" → 10_2 (= 2).
  2. In Boolean algebra + means OR. In binary arithmetic + carries.
- **Worked example:** → **Boolean: 1. Binary: 10** [Py].
- **Traps:** answering 2 or 10 for Boolean. Historical twist: in Boole's own 1854 system x + x = 2x was legal; 1 + 1 = 1 came from Jevons (T19).
- Calculator: calculate — type: two bits + mode (OR / binary add).

T24. Matching / multi-fact column across the seven

Matching / multi-fact column across the seven · ✅ IN · Easy-Medium

- **Scope:** IN.
- **Where seen:** Princeton 2026 **Div C** matching Part 1; Princeton 2019 **Div C** Part 1 (prior G_tests). **Div B check:** the format is IN. The old lists include Faraday/Tesla (OUT-2027); swap in Boole and Hertz.
- **How to solve:**
  1. Read the question: a column of people ↔ a column of facts (unit, book, country, century).
- **Worked example:** match "Boole ↔ ?" with {*Laws of Thought*, torsion balance, voltaic pile, spark gap} → ***Laws of Thought*** (torsion balance = Coulomb; pile = Volta; spark gap = Hertz).
- **Traps:** Boole is the only 1854 book and the only Irish post.
- Calculator: look up — type: (per-person fact cards).

Hard

T14. Shannon's mapping: switch network ↔ Boolean expression

Shannon's mapping: switch network ↔ Boolean expression · ✅ IN · Medium-Hard

- **Scope:** IN (switches and relays are 3.c.vi; Boolean expressions are 3.c.viii; the hands-on includes "3-way switch with SPDT", 3.II.c.vii).
- **Where seen:** MIT 2020 **Div C** Q49 and Princeton 2020 **Div C** Q40 (control diagrams with SPDT/DPDT switches from a truth table). **Div B check:** IN for SPDT 3-way switch logic. The 4-way (DPDT) switch is not named in the Div B rules, but "switches" in 3.c.vi covers it. Treat the DPDT part as BORDERLINE. Prior H_logic_diodes A20.
- **How to solve:**
  1. Read the question: switch network → expression, or expression → network.
  2. Series = AND. Parallel = OR. NC contact = complement.
  3. Two 3-way switches = XOR (lamp toggles from either switch).
- **Worked example:** A and B in parallel, then that pair in series with C → **lamp = (A + B)·C**. It is ON only for (A,B,C) = 011, 101, 111 [Py].
- **Traps:** mixing up series and parallel. Shannon's original paper used the opposite "hindrance" convention.
- Calculator: calculate — type: an expression string with + · ′, giving a truth table; or a series/parallel nesting typed as S(A,P(B,C)).

T15. Boole's own symbols: 1 = Universe, 0 = Nothing, 1 – x = NOT

Boole's own symbols: 1 = Universe, 0 = Nothing, 1 – x = NOT · ✅ IN · Hard

- **Scope:** IN (history of Boole's algebra, plus Boolean expressions).
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: a Boole symbol → its meaning or its modern gate.
  2. Xy = AND. 1 – x = NOT. 1 = everything (TRUE)

3. 0 = nothing (FALSE).

- **Worked example:** "In *Laws of Thought*, what did Boole's symbol 1 represent?" → **the Universe (of discourse)** [LT ch. III Prop. II].
- **Traps:** in Boole's own system, + was NOT a plain OR (T17).
- Calculator: look up — type: symbol.

T16. x² = x ("the fundamental law of thought") and why only 0 and 1

x² = x ("the fundamental law of thought") and why only 0 and 1 · ✅ IN · Hard

- **Scope:** IN (algebra within 3.e.i; Boolean expressions).
- **Where seen:** not seen in a test. NY Lake Erie 2019 **Div C** Q28 ("Which is a rule of Boolean algebra? A. A+1=A B. A+0=1 C. A+1=1 D. A+$\bar{A}$=0" → **C**) is the nearest identity item. **Div B check:** IN (Boolean expressions, 3.e.iv). The test has no key (manifest); C was checked in Python.
- **How to solve:**
  1. Read the question: "Which numbers satisfy x² = x?" → 0 and 1. Or "x² = x is the modern ___ law" → idempotent (A·A = A).
  2. X² – x = 0 → x(x – 1) = 0 → x = 0 or 1 (algebra in 3.e.i).
- **Worked example:** "Boole noted that his law x² = x holds for only two numbers. Which?" → **0 and 1** [LT ch. II §15][Py].
- **Traps:** idempotent (A·A = A) ≠ identity (A·1 = A). The name "idempotent" came from Benjamin Peirce, not Boole.
- Calculator: look up + calculate — type: (fixed answer). Optional CALCULATE: a truth-table check of any identity typed in.

T17. x(1 – x) = 0, the principle of contradiction

x(1 – x) = 0, the principle of contradiction · ✅ IN · Hard

- **Scope:** IN.
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: Boole's equation → its name and modern form (A·A′ = 0).
  2. X – x² = 0 → x(1 – x) = 0. "x and not-x" is empty.
- **Worked example:** "Boole derived x(1 – x) = 0. Which modern law is it, and what did he call it?" → **the complement law A·A′ = 0; the principle of contradiction** [LT ch. III Prop. IV][Py].
- **Traps:** A + A′ = 1 is the other complement law (excluded middle), not this one.
- Calculator: look up — type: the keyword from the question.

T20. Bitwise logic on multi-bit numbers ("20 · 13 in Boolean algebra")

Bitwise logic on multi-bit numbers ("20 · 13 in Boolean algebra") · ⚠️ Borderline · Medium-Hard

- **Scope:** BORDERLINE. Rule 3.e.iii allows binary whole numbers and **addition only** ("not including ... any arithmetic operations other than addition"). Rule 3.e.iv allows logical operations as Boolean expressions. A bitwise AND/XOR of two multi-bit numbers mixes the two, and the rules do not say whether it is allowed.
- **Where seen:** Princeton 2021 **Div C** Q25 ("What is the value of 20 * 13 in boolean algebra?" Key **D: 4**, red highlight on key page 5). Rickards (RSOI) 2021 **Div C** Q29 ("What is 108 ⊕ 57?" Key **85**). **Div B check:** BORDERLINE for both, for the reason above. They are useful practice only after the IN types are learned.
- **How to solve:**
  1. Read the question: two whole numbers + an operator → the bitwise result.
  2. Convert to binary, line up the bits, apply the gate to each column, convert back.
- **Worked example:** 20 = 10100_2, 13 = 01101_2. AND per bit = 00100_2 = **4** [Py]. 108 ⊕ 57 = **85** [Py].
- **Traps:** 260 (ordinary multiplication) and 17/30 (other wrong options). The question wording is loose (see section 4).
- Calculator: calculate — type: number 1, number 2, operator (AND/OR/XOR/NAND/NOR).

T22. Family trivia (Everest, daughters)

Family trivia (Everest, daughters) · ⚠️ Borderline · Hard

- **Scope:** BORDERLINE (family facts, not "electricity discoveries").
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: "Boole's wife was the niece of the man Mount Everest is named after. Her name?" → Mary Everest Boole. Daughters: Alicia Boole Stott (4-D geometry), Ethel Voynich (*The Gadfly*), Lucy Everest Boole (chemist).
- **Worked example:** "How many daughters did Boole have?" → **5** [MT].
- **Traps:** none worth memorizing beyond Mary Everest.
- Calculator: look up — type: the keyword from the question.

T23. Other mathematics (distractor bank)

Other mathematics (distractor bank) · ✅ IN · Medium-Hard

- **Scope:** IN when asked as "which is Boole's"; BORDERLINE as a stand-alone question.
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: topic → Boole or not. Boole: differential equations (1859), finite differences (1860), probability (*LT* part 2), invariant theory (1841), Boole's inequality.
- **Worked example:** "Besides logic, Boole wrote treatises on which two subjects (1859, 1860)?" → **differential equations; calculus of finite differences**.
- **Traps:** the 1844 Royal Medal paper was about analysis (differential equations), not logic.
- Calculator: look up — type: the keyword from the question.

Nationals

T19. Boole's + vs Jevons's union (the history of "OR")

Boole's + vs Jevons's union (the history of "OR") · ⚠️ Borderline · Nationals

- **Scope:** BORDERLINE (deep history-of-logic detail; it fits "historical perspective" but goes beyond what a Div B writer usually asks).
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: "Who changed Boole's + to the inclusive OR with x + x = x?" → Jevons (*Pure Logic*, 1864). Or "Boole's expression for 'x or y (or both)'" → x + (1 – x)y or x + y – xy.
  2. Check on 0/1: x + y – xy equals OR; x + y – 2xy equals XOR [Py].
- **Worked example:** "Using Boole's 1854 algebra, write 'x or y, or both'." → **x + y – xy** (equivalently x + (1 – x)y) [SEP][Py].
- **Traps:** saying Boole wrote x + y for inclusive OR.
- Calculator: calculate — type: x, y as 0/1 → shows Boole's form and the modern gate output side by side.

T21. Naming and spelling of "Boolean algebra"

Naming and spelling of "Boolean algebra" · ⚠️ Borderline · Nationals

- **Scope:** BORDERLINE (trivia).
- **Where seen:** not seen in a test.
- **How to solve:**
  1. Read the question: "Who coined 'Boolian algebra'?" → C. S. Peirce. The spelling "Boolean" spread through Royce's circle (Sheffer, Lewis, Wiener) around 1900.
- **Worked example:** → **Charles Sanders Peirce** [SEP].
- **Traps:** Sheffer is often named; SEP gives Peirce for the term and credits Sheffer only with spreading it.
- Calculator: look up — type: the keyword from the question.

T25. Short answer / essay: "Why is Boole in an electricity event?"

Short answer / essay: "Why is Boole in an electricity event?" · ✅ IN · Medium (Nationals for a graded essay)

- **Scope:** IN.

- **Where seen:** essay format in MIT 2020 **Div C** Q21-22 and Princeton 2020 **Div C** Q2-3 (history essays, prior G_tests). **Div B check:** the format is IN; the topics in those tests were other scientists.
- **How to solve:**
  1. Read the question: a prompt → 2-4 sentences.
- **Method (model answer):** Boole (*MAL* 1847, *LT* 1854) made logic an algebra on 0 and 1, with AND (xy), NOT (1 – x) and the law $x^2 = x$. In 1937 Shannon showed that relay and switch circuits obey this algebra (series = AND, parallel = OR). Every digital logic gate, adder and multiplexer in the rules is designed with Boolean algebra.
- **Traps:** saying Boole designed circuits himself.
- Calculator: look up — type: (show the model answer).

T27. Quote attribution (who said it?)

Quote attribution (who said it?) · ✅ IN · Hard

- **Scope:** IN (3.c.i names Boole).
- **Where seen:** not seen with Boole. Florida State 2021, Div B (= C), State, Q2-Q8 gave seven quotes. Ohm, Kirchhoff, Ampère, Coulomb and Volta were in that set. A 2027 State test can add a Boole quote in the same format.
- **How to solve:**
  1. Look for Boole key words: "laws of thought", "operations of the mind", "symbolical language of a Calculus", "Logic", "$x^2 = x$".
  2. Words about currents, wires or galvanic circuits point to Ohm, not Boole.
- **Worked example:** "The design of the following treatise is to investigate the fundamental laws of those operations of the mind by which reasoning is performed..." Who? → **Boole** (*Laws of Thought*, 1854, Chapter I, first sentence; checked in the Gutenberg text).
- **Traps:** Ohm's 1827 preface starts "The design of this Memoir is to deduce strictly from a few principles...". Both quotes start "The design of". "Treatise" + "mind" + "reasoning" → Boole. "Memoir" + "principles" → Ohm.
- Calculator: look up — type: a key word from the quote (e.g. "treatise", "mind").

Out of scope (skip)

These types appear in tests or books, but the 2027 Div B rules exclude them. Skip them. They stay here so you can recognise them.

T26. Logic computations credited to Boole that belong to the logic subtopics

Logic computations credited to Boole that belong to the logic subtopics · ❌ Out · Medium-Nationals

- **Scope: OUT of this subtopic** (they are IN for Div B under 3.c.viii, but they are not history; see the logic files). They are listed so the app routes them correctly.
- **Where seen:** not seen in a test.
- **Where seen (Div C, with Div B check):**
  - Rickards 2021 C Q33: simplify (A′ + B)B + (CB′)(A + C) → key **B + C** [Py: true on all 8 rows]. Div B check: IN (Boolean expressions).
  - MIT 2020 C Q47: minimum NAND gates for A + AB′ + AB′C → key **0** (it simplifies to A) [Py]. Div B check: IN (universal gate sets and expressions).
  - UT Austin "2020 Inv." C Q38: implication p ⇒ q → key **D (none of the above)**. p ⇒ q = NOT p OR q, and options A-C all fail [Py]. Div B check: BORDERLINE (implication is not one of the named gates; truth table → expression is IN).
  - UT Austin "2020 Inv." C Q63: equal to NOT[(AB)·NOT(A + B)] → key **B (1) and E (B′ + B)**. The inner product is always 0 [Py]. Div B check: IN.
  - NY Lake Erie 2019 C Q28 (see T16). Div B check: IN.
- Calculator: calculate — type: expression string(s). (App note: in the logic module (truth-table engine).)

Mistakes in tests and keys

1. **Shannon's Alfred Noble Prize year.** Wikipedia's "A Symbolic Analysis of Relay and Switching Circuits" gives **1939**. AMS Notices (2002 obituary) gives **1940**, which matches most sources. Use 1940 and accept either.
2. **Keith Medal year.** The Wikipedia Boole article says 1855 in the text and 1855-1857 in the infobox. Prior research did not list it. Not settled. Avoid exact-year questions on it.
3. **"Opened his own school at 19" (MacTutor, prior l_bank).** 1834 – 1815 = 18 for any date before 2 Nov 1834 [Py]. The month is not given. The fact is not wrong, but "18" may also be correct.
4. **Princeton 2021 C Q25** ("value of 20 * 13 in boolean algebra"). Boolean algebra has only the values 0 and 1. The intended meaning is a **bitwise** AND of the binary numbers (answer 4). The question should say "bitwise AND". A Div B writer should not copy it as written (also BORDERLINE under 3.e.iii).
5. **Princeton 2020 C Q40.** The text says "three 3-way (SPDT) and one 4-way (DPDT) switches that controls two lights", but then labels only S1-S3 and one light, L1. The rubric awards points for 3 switches and one light. The text was copied from MIT 2020 Q49 (4 switches, 2 lights) and not fully edited.
6. **scioly.org wiki (local copy `downloads/dl/cl_wiki.txt`).** The History section has Coulomb, Volta, Ampère, Ohm, Faraday, Kirchhoff and Tesla. It has **no Boole or Hertz section**, although the page marks Faraday/Tesla as removed in 2027. The live page was blocked (bot check) on 2026-10-16. Its Boolean algebra section also says booleans are "similar to transistors", which is a loose analogy, not a fact.
7. **Prior F_slides.md** notes that the national supervisor decks give no Boole biography. That is still true; it is a gap, not an error.
8. **Common internet claim: "Boole invented 1 + 1 = 1 / the OR gate."** In Boole's 1854 system, + was defined only for disjoint classes and x + x = 2x. The inclusive OR with x + x = x is Jevons's (1864) [SEP]. Tests that say "Boole's law A + A = A" are loose.

Not confirmed

Item	Status	Where I looked
Exact month of *Laws of Thought* publication (so Boole was 38 or 39)	not found	Gutenberg *LT* text (title page has no date or publisher), MacTutor, SEP
LT publisher (Walton and Maberly, London)	not confirmed	Gutenberg title page omits it
Years of the honorary degrees (LL.D. Dublin 1851?, Oxford 1859?)	not confirmed	Wikipedia (no years), MacTutor (no years)
MAL page count "82 pages"	one source only (SEP)	SEP; MacTutor does not give it
MAL and *Formal Logic* in the same month	SEP says "some sources say"	SEP, MacTutor (silent), MacTutor De Morgan (silent)
De Morgan's laws first stated in *Formal Logic* (1847)	not confirmed	MacTutor De Morgan says "introduced De Morgan's laws" with no place
Vannevar Bush as Shannon's advisor/context	not checked	Wikipedia thesis article did not mention him
Shannon's course at Michigan (which department)	Bentley Library calls it an elective; no department named	web search (Bentley, MacTutor Shannon)
Leibniz invented binary numbers (1703)	not checked this run (general knowledge)	none; use only as a distractor
Babbage as Boole's contemporary	not checked (general knowledge)	none
Live scioly.org wiki Boole section	page blocked by the bot check	scioly.org/wiki/index.php/Circuit_Lab, 2026-10-16
Britannica entry	403 Forbidden	britannica.com/biography/George-Boole

Item	Status	Where I looked
UCC Boole page	404 at www.ucc.ie/en/george-boole/	used georgeboole.ucc.ie (search snippet) instead
Any Boole **history** question in a past test	none exists	grep of every .txt in `research/downloads` (dI, dlB, dlF, dlG, including pz/ and wiki/) for Boole, Laws of Thought, Shannon, De Morgan, Cork
NY Lake Erie 2019 C Q28 key	no key exists; answer C checked by truth table	G_manifest (no key)

- The 8 test-extraction groups in `research/tests/` have no Boole history question (grep for "Boole", "Laws of Thought", "Cork", "Shannon"). This agrees with the raw file: Boole is new to the 2027 list.

Sources

- Div B 2027 rules: `/Users/kai/Daniel/Science_Olympiad_Div_B_Rules_2027_for_Web_Secured.pdf` (Circuit Lab, manual pp. B12-B13 in the pdftotext output)
- Boole, *An Investigation of the Laws of Thought* (1854), Project Gutenberg eBook #15114: https://www.gutenberg.org/files/15114/15114-pdf.pdf
- MacTutor, George Boole: https://mathshistory.st-andrews.ac.uk/Biographies/Boole/
- MacTutor, Augustus De Morgan: https://mathshistory.st-andrews.ac.uk/Biographies/De_Morgan/
- MacTutor, Claude Shannon: https://mathshistory.st-andrews.ac.uk/Biographies/Shannon/
- Stanford Encyclopedia of Philosophy, "George Boole": https://plato.stanford.edu/entries/boole/
- Wikipedia, George Boole: https://en.wikipedia.org/wiki/George_Boole
- Wikipedia, A Symbolic Analysis of Relay and Switching Circuits: https://en.wikipedia.org/wiki/A_Symbolic_Analysis_of_Relay_and_Switching_Circuits
- AMS Notices, Shannon obituary (2002): https://community.ams.org/journals/notices/200201/fea-shannon.pdf
- Bentley Historical Library (Shannon at Michigan): https://www.bentley.umich.edu/?p=6953
- UCC, The Papers of George Boole: https://georgeboole.ucc.ie/
- National Library of Ireland, *The Boole-De Morgan Correspondence 1842-1864*: https://catalogue.nli.ie/Record/vtls000062743
- Past tests (local): `research/downloads/dlG/pz/p21/` (Princeton 2021 C, Q19, Q25), `dlG/pz/p20/` (Princeton 2020 C, Q37, Q40), `dlG/mit20_key.txt` (MIT 2020 C, Q47, Q49), `dlG/rsoiOldC_key.txt` (Rickards 2021 C, Q29, Q33), `dlG/ut20inv_Ckey.pdf` (UT 2020 Inv. C, Q38, Q63), `dlG/wiki/CircuitLabTest2019I.txt` (NY 2019 C, Q28), `dlB/tests/` (BullSO, Berkeley, Pembroke 2026 B history format)
- Prior research: `research/prior/I_bank.md` (history fact sheet, H1-H16), `B_topics.md`, `H_logic_diodes.md` (A20, C1), `G_tests.md`, `F_slides.md`, `A_rules.md`
- Test-extraction files: `/Users/kai/Daniel/circuitlab/research/tests/` (atx-reg, misc-a, misc-b, nat26, state, ut-a, ut-b, ut-c) and `/Users/kai/Daniel/circuitlab/research/prior/G_tests.md`.
- Raw research file this note was built from: `/Users/kai/Daniel/circuitlab/research/subtopics/boole.md`.

Other Historical Figures

INFO: How to use this note

- Scope icons: ✅ IN = all levels. ✅ States-Nationals = only at State and National tournaments. ⚠️ Borderline = the rules do not settle it (the reason is given). ❌ Out = excluded (the rule is cited).
- Question types go from Easy to Medium to Hard to Nationals. "Nationals" means the hardest depth, not a different scope.
- Every number in a worked example was checked in Python.
- Source tags in square brackets, like [W] or [CHM], are listed under "Sources".

Scope notes and source tags

Rule anchor. Part I 3.c.i of the 2027 Div B rules says: "Historical perspective of the electricity discoveries made by Ampere, Coulomb, Kirchhoff, Volta, Ohm, Hertz, and Boole" (manual page B12, quoted from `pdftotext` of `/Users/kai/Daniel/Science_Olympiad_Div_B_Rules_2027_for_Web_Secured.pdf`). The list has seven names. No one else is named.

What this means for the people in this note. The rules do not promise a history question about anyone outside the seven. So a pure history question about Franklin, Edison or Tesla is not guaranteed. But test writers still use these people in three ways:

1. **As distractors.** A question about Volta lists Galvani as a wrong choice. You must know enough to reject him. This use is ✅ IN, because the question is about a listed scientist.
2. **As links to a listed scientist.** Ørsted started Ampère's work. Galvani started Volta's work. Maxwell's theory was confirmed by Hertz. De Morgan was Boole's friend, and Shannon used Boole's algebra. These links are ✅ IN as part of the listed person's history.
3. **As the subject of a topic area that is IN.** Edison and the "war of the currents" fit area 3.c.iii (DC and AC hazards and uses). Joule and Watt fit area 3.c.iv (power, energy, units). Ohl, Holonyak and the blue-LED team fit area 3.c.vii (PN junctions, LEDs). Franklin fits area 3.c.ii (charges, static electricity, lightning hazards). These are ⚠️ Borderline as history questions. The physics content is IN. The biography is not on the list.

How I tag each type here.

- ✅ IN: the question is really about a listed scientist, or about IN physics (for example a Joule-heating calculation).
- ⚠️ Borderline: an unlisted person, but the fact belongs to an IN topic area.
- ❌ Out: an unlisted person, and the fact belongs to an excluded or missing topic. Examples: magnetism and induction (not one of the 3.c areas, and 3.c says "the following areas only"), capacitance (3.d), inductance (3.d), 3-phase power (3.d), semiconductors beyond PN junctions (3.d).

Note on magnetism. The 3.d exclusion list does not use the word "magnetism". Magnetism is out because 3.c says "the following areas only", and no area covers magnetism. The test-extraction file `research/tests/misc-a.md` makes the same point.

Source tags: [W] Wikipedia, [CHM] Computer History Museum, [BIPM] Bureau International des Poids et Mesures, [NP] Nobel Prize press material (as reported by Physics World and SPIE), [T] seen in a past test, [PR] prior file in `research/prior/`, [R] 2027 rules, [G] general physics knowledge (standard textbook, for example OpenStax College Physics 2e ch. 18-20, Kuphaldt *Lessons in Electric Circuits* vol. 1 and 3). [G] facts were not re-checked online in this run unless a [W] tag is next to them.

Key facts

How likely is each person on a 2027 Div B test?

This ranking uses two things: how often the person appeared in the 33+ past tests I searched, and how close the person is to a listed scientist or an IN topic.

Person	Likely?	Why
Galvani	**High**	He is the classic Volta distractor and rival. Seen in UT 2019 B/C, ATX 2019 B/C, GullSO 2026, Princeton 2021.
Franklin	**High**	Positive/negative names and conventional current (area ii, area v). Seen in Shock Value, Florida State 2021, UF 2020.
Ørsted	**High (as a link)**	His 1820 discovery started Ampère's work. Seen in ATX 2019, ATX 2020, UF 2020, Shock Value.
Edison	**High**	War of the currents (area iii). Seen in ATX 2020, Pembroke 2026 B, Princeton 2020, Shock Value, Northview 2026.
Westinghouse	Medium	War of the currents. Seen in ATX 2020, GullSO 2026, Northview 2026.
Tesla	Medium as a distractor, Low as a subject	Most-used unlisted name in old tests. Removed from the list; 2026-27 tests that follow the list should not ask about him directly.
Joule	Medium	Joule heating (area iv); unit joule. Seen in ATX 2020 B, Shock Value.
Watt	Medium	Unit of power (area iv). Mostly as a unit, not a person.
Siemens	Medium	Unit of conductance; "mho". See Ohm A1.
Maxwell	Medium (as a link)	Hertz confirmed his theory. Seen in ATX 2019, HH 2026, Princeton 2019, Florida 2021.
De Morgan	Medium	De Morgan's Law is named in 3.c.viii. Boole's friend.
Shannon	Medium (as a link)	He applied Boole's algebra to switching circuits (1937).
Gilbert	Low	Coined "electricus" (1600). Mostly magnetism. Seen as a distractor in ATX 2019 and as a doubtful key in UT 2019.
Thales	Low	Rubbed amber (c. 600 BC). Not seen in any test.
Ohl	Low	Found the PN junction (1940). Area vii. Not seen in any test.
Holonyak	Low	First visible (red) LED, 1962. Area vii. Not seen in any test.
Akasaki, Amano, Nakamura	Low	Blue LED, Nobel 2014. Area vii. Not seen in any test.
Zener	Low	Zener diode named after him. Not seen as history; Zener diodes appear in PUSO 2026 and Rickards 2026 C as devices.
Cavendish	Low	Found an early form of Ohm's law; also measured the inverse-square law. Seen in ATX 2019 B/C Q2.
Wheatstone	Low	Wheatstone bridge (a DC circuit, IN). Seen in Shock Value tiebreaker 5.
Faraday	Low as a subject (✗)	Most common unlisted name in tests. His work (induction, capacitance) is excluded. Know him only to reject him.

1. Thales of Miletus (c. 624 – c. 546 BC)

- Greek philosopher from Miletus (now in Turkey). [W]
- Tradition says he noticed that **rubbed amber attracts light objects**, like straw or feathers. This is static electricity. [W][G]
- Aristotle reports that Thales said a lodestone (a natural magnet) has a "soul" because it moves iron. The amber part comes from later writers. [W]
- The Greek word for amber is **ēlektron**. "Electricity" and "electron" come from this word. [G]
- About 2,200 years separate Thales (c. 600 BC) and Gilbert (1600 AD). The exact count is 2,199 years, because there is no year 0. (Python-checked.)

2. William Gilbert (1544 – 1603)

- English physician. He was physician to Queen Elizabeth I from 1601 to 1603. [W]
- **1600: *De Magnete*** (full title *De Magnete, Magneticisque Corporibus, et de Magno Magnete Tellure*, "On the Magnet and Magnetic Bodies, and on the Great Magnet the Earth"). [W]
- He said **the Earth is a giant magnet**. This is why a compass points north. [W]
- He coined the Latin word **"electricus"**, meaning "like amber". He used it for materials that attract light objects when rubbed. [W]
- The English word **"electricity"** was first used by **Sir Thomas Browne in 1646**, not by Gilbert. [W]

- He built the **versorium**, a pivoted needle that detects static charge. It is an early electroscope. [W]
- He separated electric attraction (amber) from magnetic attraction (lodestone). [G]
- The CGS unit **gilbert** (magnetomotive force) is named after him. It is now replaced by the ampere-turn. [W]

Magnetism: ✗ Out.

3. Benjamin Franklin (1706 – 1790)

- American printer, writer, scientist and statesman. Born 17 January 1706 in Boston. Died 17 April 1790 in Philadelphia. [W]
- **One-fluid theory.** He said electricity is one "fluid". A body with extra fluid is **positive (+)**. A body with too little is **negative (–)**. [G]
- **He chose the names "positive" and "negative".** The choice was arbitrary. Later, the electron turned out to carry the "negative" charge. [G]
- **Conventional current.** Because of Franklin's choice, conventional current flows from + to −. Real electrons in a wire move the other way, from − to +. [G]
- **Conservation of charge.** He argued that charge is not created or destroyed; it only moves. [G]
- He used the word **"battery"** for a row of linked Leyden jars (about 1749). Volta's chemical "battery" came 51 years later (1800). [G] Not confirmed online in this run.
- **Lightning is electricity.** He proposed a test in 1750-1752. **Dalibard did the test first, at Marly-la-Ville, France, in May 1752**, using Franklin's plan. [W]
- **Kite experiment: June 1752**, Philadelphia, with his son William. The date comes from Priestley's book (1767). Franklin's own letter (*Pennsylvania Gazette*, 19 October 1752) did not give the date or place. [W]
- **The kite was not struck by lightning.** The storm cloud induced charge on it. Loose string fibres stood up and repelled each other. He drew a spark from a key and charged a Leyden jar. [W]
- **Georg Richmann was killed in August 1753** in Saint Petersburg while repeating a lightning experiment. [W] This shows the hazard.
- **Lightning rod.** A pointed metal rod on a roof, connected to the ground by a wire. It gives lightning a safe path to the ground. [W][G]
- Other inventions: bifocals, the Franklin stove, the glass harmonica. [W]
- The CGS unit of charge **"franklin" (Fr)**, also called the statcoulomb, is named after him. [G] 1 C ≈ 3×10^9 Fr. [G]

4. Luigi Galvani (1737 – 1798)

- Italian physician and anatomist at the University of Bologna. [W, via Volta A1 note]
- From **1780**, he studied **frog legs** that twitched when touched with metal. [W]
- **1791 book:** *De viribus electricitatis in motu musculari commentarius* ("Commentary on the Effect of Electricity on Muscular Motion"). [W]
- His theory: **"animal electricity"**. He said the frog's body made the electricity. [W]
- **Volta disagreed (from 1792).** Volta said the current came from **two different metals** touching through the wet frog. This led Volta to the voltaic pile (1800). 1800 − 1791 = 9 years. (Python-checked.) [W]
- Words from his name: **galvanism** (Volta coined it), **galvanometer** (a current meter), **galvanize** (to coat iron with zinc; also "to shock into action"), **galvanic cell** (a chemical cell). [W][G]
- Modern view: both men were partly right. Nerves do use electrical signals. Two metals in an electrolyte do make a current. [G]
- Full details: see Volta A1, section "Galvani dispute and contact electricity".

5. Hans Christian Ørsted (1777 – 1851)

- Danish physicist and chemist. Born 14 August 1777 in Rudkøbing. Died 9 March 1851 in Copenhagen. [W]
- **1820: an electric current moves a compass needle.** This was the first link between electricity and magnetism. [W]
- He wrote a short Latin paper in July 1820. An English version is "Experiments on the Effect of a Current of Electricity on the Magnetic Needle" (*Annals of Philosophy*, vol. 16). [W]
- The story that he found it **by accident during a lecture is a myth**, according to Wikipedia. He had been looking for the link since 1818. [W]

- **September 1820:** Arago showed Ørsted's result in Paris. **Ampère** then worked out the forces between currents within weeks. [PR I_bank][W]
- He first made (impure) **aluminium** in 1824-1825. He first extracted piperine (from pepper) in 1819. [W]
- The CGS unit **oersted** (magnetic field strength H) is named after him. [W] Magnetism: ✗ Out.

6. James Watt (1736 – 1819)

- Scottish engineer. He **improved the steam engine** (the separate condenser, patented 1769). [G]
- **He did no work in electricity.** This is a trap. The unit is named after him because power is a general idea. [G]
- He defined the **horsepower**: 33,000 foot-pounds per minute. **1 hp ≈ 745.7 W** (Python-checked: 33,000 × 0.3048 m × 4.44822 N ÷ 60 s = 745.70 W). [G]
- The **watt** was proposed as a unit by (Carl) William Siemens at the British Association in 1882. [W] The **2nd International Electrical Congress (31 August 1889)** adopted the watt and the joule. [W]
- 1 W = 1 J/s = 1 V·A. [G]

7. James Prescott Joule (1818 – 1889)

- English physicist and brewer from Salford, near Manchester. [G]
- **Joule's law of heating (1840-1841):** the heat made by a current in a wire is proportional to **$I^2 \times R \times t$**. In modern form, heat energy **$Q = I^2Rt$**, and power **$P = I^2R$**. [G]
- **Mechanical equivalent of heat (1843-1850):** his paddle-wheel experiment showed that work turns into heat at a fixed rate. Modern value: **1 calorie = 4.186 J**. [G]
- This supported the law of **conservation of energy**. [G]
- The unit **joule (J)** = 1 N·m = 1 W·s. It was adopted in 1889 (2nd International Electrical Congress). [W]
- 1 kWh = 3,600,000 J. [G]

8. Werner von Siemens (1816 – 1892)

- German inventor and industrialist. In 1847 he founded the company **Siemens & Halske** (now Siemens AG). [G]
- He built a **pointer telegraph** (1847). [G]
- **1866-1867: the dynamo-electric principle.** A generator can use its own current to make its magnetic field. No permanent magnet is needed. [G] Magnetism: ✗ Out as physics.
- **1860: the "Siemens mercury unit" of resistance.** It was a column of mercury 1 m long with a 1 mm² cross-section at 0 °C. It came before the ohm. [G]
- First electric passenger railway shown in Berlin, 1879. [G]
- The SI unit of **conductance, the siemens (S)**, was adopted by the **14th CGPM in 1971**. The IEC had adopted it in 1935. [W][BIPM]
- **1 S = 1/Ω = 1 A/V.** The older name was the **mho (℧)**, "ohm" spelled backward. [W] See Ohm A1 for full siemens/mho facts.
- His brother **(Carl) William Siemens** worked in England. William proposed the watt as a unit in 1882. [W]

9. James Clerk Maxwell (1831 – 1879)

- Scottish physicist. Born 13 June 1831 in Edinburgh. Died 5 November 1879 in Cambridge, aged 48. [G]
- **Maxwell's equations** (1861-1865; *A Dynamical Theory of the Electromagnetic Field*, 1865; *Treatise on Electricity and Magnetism*, 1873). They unite electricity, magnetism and light. [G]
- He **predicted electromagnetic waves** that move at the speed of light. He said light is such a wave. [G]
- **He put Faraday's field ideas into mathematics.** (ATX 2019 B/C Q3.) [T]
- **Hertz confirmed Maxwell's waves (1886-1888).** Maxwell had died 8 years before 1887, so he never saw the proof. (Python-checked: 1887 − 1879 = 8.) [G] See Hertz A1.
- He called Ampère **"the Newton of electricity"**. [PR I_bank] See Ampere A1.
- He edited and published **Cavendish's** unpublished electrical papers (1879). These papers showed that Cavendish found an early Ohm's law and the inverse-square law first. [G] See Ohm A1 and Coulomb A1.
- First Cavendish Professor of Physics at Cambridge (1871). He made the first colour photograph (1861). [G]
- The CGS unit **maxwell** (magnetic flux) is named after him. [G] Magnetism and equations with calculus: ✗ Out.

10. The war of the currents (1880s – 1892)

- **Thomas Edison (1847 – 1931)** backed **direct current (DC)**. [W][G]
  - **Pearl Street Station**, New York City, opened 4 September 1882. It supplied **110 V DC**. [W][G]
  - DC could not be stepped up or down easily in the 1880s. So it lost a lot of power in long wires. Power stations had to be close (about 1-2 km) to customers. [G]
  - **Edison's main argument against AC: it is dangerous** (high voltage kills). (ATX 2020 B Q20 / C Q22.) [T][W]
  - Edison made a practical **incandescent light bulb (1879)**. Joseph Swan made one in England at the same time. [G]
  - Edison never won a Nobel Prize. [G]
- **George Westinghouse (1846 – 1914)** backed **alternating current (AC)**. [W]
  - **July 1888:** Westinghouse licensed **Nikola Tesla's** patents for the polyphase AC induction motor. [W]
  - AC voltage is easy to change with a **transformer**. High voltage means low current, and low current means low I^2R loss in the wires. [G]
  - **1893:** Westinghouse lit the **World's Columbian Exposition** (Chicago World's Fair) with AC. He won the contract in May 1892 by under-bidding General Electric. [W]
  - **Niagara Falls:** Westinghouse won the contract to build the AC power station at the end of 1892. [W] Power reached Buffalo in 1896. [G]
- **Nikola Tesla (1856 – 1943):** inventor of the AC induction motor and polyphase system. He is not on the 2027 list. [W][R]
- **Harold P. Brown:** an engineer who attacked AC in 1888. He killed animals with AC in public. In 1889 stolen letters showed he worked with Edison's company. [W]
- **Electric chair:** the first execution was **William Kemmler, 6 August 1890**, Auburn, New York. It used AC. It went badly. [W]
- **End of the war:** In **April 1892**, Edison General Electric merged with Thomson-Houston to form **General Electric**. Edison had lost control by then. [W]
- In 1908 Edison reportedly said "Tell your father I was wrong" to the son of transformer pioneer William Stanley. [W]
- **Topsy the elephant (4 January 1903, Coney Island): Edison had no part in it.** Edison's film company filmed it. The war had ended 11 years earlier. Many websites wrongly blame Edison. [W] (Python-checked: 1903 − 1892 = 11.)
- **Who won?** AC won for power grids. But DC is back today in batteries, electronics, solar panels and long high-voltage DC (HVDC) lines. [G]

11. Augustus De Morgan (1806 – 1871)

- British mathematician, born in Madurai, India. [G]
- First professor of mathematics at London University (later University College London), from 1828. [G]
- **1847: *Formal Logic*.** The same year, Boole published *The Mathematical Analysis of Logic*. [G][PR I_bank] See Boole A1.
- **De Morgan's laws:** NOT(A AND B) = (NOT A) OR (NOT B). NOT(A OR B) = (NOT A) AND (NOT B). Python checked all four input rows. [G]
- He was Boole's friend and supporter. He backed Boole's work in letters. [PR I_bank][G]
- He named and defined **"mathematical induction"** (1838). [G]
- He taught Ada Lovelace by letter. [G]

12. Claude Shannon (1916 – 2001)

- American mathematician and engineer. [G]
- **1937 MIT master's thesis: *A Symbolic Analysis of Relay and Switching Circuits*** (published 1938). [PR I_bank] [G]
- He showed that **Boole's algebra describes switch and relay circuits**. **Switches in series = AND. Switches in parallel = OR.** Python checked both truth tables. [G]

- This is why Boolean algebra runs every digital computer. 1937 − 1854 = 83 years after Boole's *Laws of Thought*. (Python-checked.) [G]
- **1948: "A Mathematical Theory of Communication"**: the start of information theory. It used the word **"bit"** (binary digit). John Tukey coined the word. [G]
- He is called "the father of information theory". [G]

13. Russell Ohl (1898 – 1987)

- American engineer at **Bell Labs** (Holmdel, New Jersey). Born 30 January 1898. Died 20 March 1987. [W]
- He studied **silicon crystals** as radio and radar detectors in the 1930s. [W]
- **23 February 1940: he found the PN junction** in a silicon rod with a crack. When light hit it, a current flowed. [CHM] Wikipedia gives "1939" instead. (See "Not confirmed".)
- With **Jack Scaff**, he saw that the crack separated two regions with different impurities. [CHM]
- The side with extra electrons (from phosphorus) was called **n-type (negative)**. The side with too few electrons (from boron) was called **p-type (positive)**. The boundary was the **p-n junction**. [CHM]
- The light effect led to the **silicon solar cell** (his patent: US 2,402,662, "Light sensitive device"). [W] The first practical silicon solar cell came from Bell Labs in 1954. [CHM]
- Shockley's 1948 junction transistor idea grew from Ohl's PN junction. [CHM] (Transistors: ❌ Out, 3.d.)

14. Clarence Zener (1905 – 1993)

- American physicist. Born 1 December 1905 in Indianapolis. Died 2 July 1993 in Pittsburgh. [W]
- **1934:** he was first to explain the **electrical breakdown of insulators** (the "Zener effect"). [W]
- **He did not build the Zener diode.** Bell Labs used his findings to make it, and named it after him. [W]
- A **Zener diode** is a diode made to conduct in **reverse** at a set voltage (the Zener voltage). It is used to **hold a voltage steady** (a voltage regulator). [G][T PUSO 2026]
- In real Zener diodes above about 5-6 V, the breakdown is mostly **avalanche** breakdown, not the true Zener effect. [G] Not confirmed online in this run.

15. Nick Holonyak Jr. (1928 – 2022)

- American engineer. Born 3 November 1928 in Zeigler, Illinois. Died 18 September 2022. [W]
- **9 October 1962:** at **General Electric**, near Syracuse, New York, he made the **first visible-light LED** (and laser diode). It glowed **red**. [W]
- Material: **gallium arsenide phosphide (GaAsP)**, not silicon. [W]
- He is often called **"the father of the LED"**. [G]
- From 1963 he was a professor at the University of Illinois. [W]
- Earlier: **infrared** LEDs (GaAs) were made in 1961-1962 by Biard and Pittman at Texas Instruments. Infrared is not visible. [G]

16. Isamu Akasaki, Hiroshi Amano, Shuji Nakamura (blue LED)

- **Nobel Prize in Physics 2014**, "for the invention of efficient blue light-emitting diodes which has enabled bright and energy-saving white light sources". [NP]
- Akasaki and Amano worked at **Nagoya University**. Nakamura worked at the **Nichia Corporation**. [NP]
- Material: **gallium nitride (GaN)**. Its band gap fits blue light. Growing good GaN crystals was the hard part. [NP]
- **Why blue mattered:** red and green LEDs already existed. **White light needs blue.** A white LED is usually a blue LED with a yellow phosphor coating. [NP][G]
- The Nobel material says about one fourth of world electricity goes to lighting. LEDs cut that use. [NP] Other sources say one fifth. [NP]
- 2014 − 1962 = 52 years after Holonyak's red LED. (Python-checked.)
- Akasaki died in 2021. [G]

17. Others seen in tests (short)

- **Henry Cavendish (1731 – 1810):** found an early Ohm's law with Leyden jars and glass tubes of salt water. He used his own body as the current meter. His papers were published in 1879 (by Maxwell). (ATX 2019 B/C Q2.) [T]

[G] See Ohm A1.

- **Charles Wheatstone (1802 – 1875):** made the **Wheatstone bridge** popular (1843). **Samuel Christie** invented it (1833). [PR I_bank][G] The bridge is a DC resistor circuit (✅ IN, area v).
- **Michael Faraday (1791 – 1867):** electromagnetic induction (1831), the electric motor (1821), the Faraday cage, electrolysis laws. The farad (capacitance) is named after him. **Not on the 2027 list.** Induction and capacitance are excluded. [T][G]
- **Pieter van Musschenbroek and Ewald von Kleist (1745-1746):** the **Leyden jar**, the first capacitor. Capacitance is ❌ Out (3.d). [G]
- **Stephen Gray (1729):** found conductors and insulators. [G] Area ii/v: ⚠️ .
- **Charles du Fay (1733):** found two kinds of electricity: "vitreous" (glass) and "resinous" (amber). Franklin later renamed them positive and negative. [G]
- **Otto von Guericke (c. 1663):** an early friction machine with a sulfur ball. [G]
- **Thomas Seebeck (1821):** the thermoelectric effect. It gave Ohm a steady source. See Ohm A1.
- **Bardeen, Brattain and Shockley (1947):** the transistor. ❌ Out (3.d). (ATX 2020 C Q11.) [T]

18. Links between unlisted and listed people (the most useful facts)

Unlisted person	Listed person	Link
Galvani	Volta	Frog-leg dispute → voltaic pile (1791 → 1800)
Ørsted	Ampère	Compass deflection (1820) → Ampère's force law (1820)
Franklin	Coulomb	Franklin named + and –; Coulomb measured the force (1785)
Priestley, Cavendish	Coulomb	Both guessed or measured the inverse-square law first
Cavendish	Ohm	Found an early Ohm's law first; unpublished until 1879
Seebeck	Ohm	Thermocouple gave Ohm a steady current
Maxwell	Hertz	Hertz proved Maxwell's waves (1887-1888)
De Morgan	Boole	Friends; both published logic books in 1847
Shannon	Boole	Applied Boolean algebra to relay circuits (1937)
Siemens	Ohm	Siemens (S) = 1/ohm; old name "mho"
Tesla, Westinghouse, Edison	(none)	War of the currents; area iii, not area i

Formulas

These are the formulas that appear in history-style questions about these people. All are ✅ IN as physics (areas iv and v).

Formula	Meaning	Person
$P = I^2R$	Power turned to heat in a resistor	Joule
$Q = I^2Rt$ (energy in joules)	Heat made in time t	Joule
1 cal = 4.186 J	Mechanical equivalent of heat	Joule
P = IV, 1 W = 1 J/s = 1 V·A	Power	Watt (unit only)
1 hp ≈ 745.7 W	Horsepower	Watt
1 kWh = 3.6×10^6 J	Energy unit on electric bills	—
G = 1/R, 1 S = 1/Ω	Conductance	Siemens
Line loss = I^2R_line, with I = P/V	Why high-voltage AC won	War of currents
Series switches = A·B (AND); parallel switches = A + B (OR)	Switch logic	Shannon
NOT(A·B) = A' + B'; NOT(A + B) = A'·B'	De Morgan's laws	De Morgan
I_R = (V_supply – V_Z)/R	Current in a Zener regulator's resistor	Zener (⚠️)
LED: R = (V_supply – V_f)/I	LED series resistor	Holonyak (device only)

Question types

This note has **41 question types** (16 ✅ IN, 17 ⚠️ Borderline, 8 ❌ Out). Numbers run OH-01 to OH-41; the Nationals types are OH-37 to OH-39, and the Out types (OH-31 to OH-36, OH-40, OH-41) are in "Out of scope (skip)" below. The out-of-scope ones are in "Out of scope (skip)" below.

How the types were found

I grepped all 8 test-extraction files in `/Users/kai/Daniel/circuitlab/research/tests/`, `prior/G_tests.md`, `prior/I_bank.md` and every `.txt` file in `research/downloads/` (dlG, dlB, dlFind) for each name in this note. Types not seen in a test come from textbooks (OpenStax College Physics 2e, Kuphaldt *Lessons in Electric Circuits*) and from the rule areas.

History questions in the test-extraction files and downloads (unlisted people)

Test	Year	Div	Level	Q	Question (short)	Key	Check / scope 2027 B	Type
ATX (UT Austin) Regional	2019	B and C	Regional	2	Who tested an early Ohm's law with Leyden jars and glass tubes? (Ronalds, Faraday, Galvani, Cavendish, Volta)	Cavendish	Correct. Person OUT of list	OH-17
ATX (UT Austin) Regional	2019	B and C	Regional	3	Who put Faraday's theories into maths? (Maxwell, Gilbert, Galvani, J.J. Thomson, Ørsted)	Maxwell	Correct. ❌ Out (magnetism)	OH-33
ATX (UT Austin) Regional	2019	B and C	Regional	11	First true battery (Galvani, Tesla, Volta, Faraday, Ørsted)	Volta	✅ IN (distractors)	OH-01
ATX (UT Austin) Regional	2020	B and C	Regional	5	Westinghouse bought this scientist's patents	Tesla	❌ Out (Tesla not listed)	OH-31
ATX (UT Austin) Regional	2020	B and C	Regional	8	Oersted's law states the same as this scientist's law	Faraday (key)	**Key doubtful**: should be Ampère	OH-06
ATX (UT Austin) Regional	2020	B	Regional	11	Earliest capacitors	Leyden jars	❌ Out (capacitance, 3.d)	OH-35
ATX (UT Austin) Regional	2020	B	Regional	17	Law for electrical heat	Joule	✅ IN (area iv)	OH-08
ATX (UT Austin) Regional	2020	B (Q20), C (Q22)	Regional	20 / 22	Edison's main argument against AC	(c) not safe	✅ IN (AC hazards)	OH-10
ATX (UT Austin) Regional	2020	C	Regional	11	Bardeen, Brattain, Shockley invented the...	transistor	❌ Out (3.d)	OH-36
UF Northern Regional (Florida)	2020	B	Regional	2	First battery (Ohm, Faraday, Edison, Volta, Swan)	Volta	✅ IN (Edison/Swan distractors)	OH-01
UF Northern Regional (Florida)	2020	B	Regional	3	Who found that a conductive container blocks fields?	Faraday	⚠️ (Faraday not listed; shielding is a field idea)	OH-24
UF Northern Regional (Florida)	2020	B	Regional	5	Invented the solenoid (Kirchhoff, Ohm, Franklin, Ørsted, Ampère)	Ampère	✅ IN (Franklin/ Ørsted distractors)	OH-01
UF Northern Regional (Florida)	2020	B	Regional	6	Pioneer of AC	Tesla	⚠️ (Tesla not listed)	OH-12
UF Northern Regional (Florida)	2020	C	Regional	2	Who invented AC electricity?	Tesla	⚠️	OH-12
UT Invitational	2019 (2020 season)	B and C	Invitational	2	Contested with a man over frogs	Volta	✅ IN	OH-04

Test	Year	Div	Level	Q	Question (short)	Key	Check / scope 2027 B	Type
UT Invitational	2019 (2020 season)	B	Invitational	8	Won the Nobel Prize for helping invent AC	Nobody	⚠️ trick	OH-13
UT Invitational	2019 (2020 season)	B and C	Invitational	9	Published "On Magnetism"	Coulomb (key)	**Doubtful**: *De Magnete* is Gilbert's	OH-23
UT Invitational	2019 (2020 season)	B and C	Invitational	3	Slept two hours a night	Tesla	❌ Out	OH-31
UT Invitational	fall 2018	B/C	Invitational	5	Who found that current makes a magnetic field?	Oersted (no key)	❌ Out as a subject (magnetism); ✅ as Ampère's trigger	OH-06
Northview Sylvania Invitational	Dec 2025 (2026 season)	C	Invitational	46	Nobel prize for inventing AC (Edison, Volta, Tesla, Westinghouse, Nobody)	E Nobody	⚠️ trick; Div B scope: borderline	OH-13
GullSO Invitational	Dec 2025 (2026 season)	C (B/C)	Invitational	30	During a feud with Luigi Galvani, who invented the voltaic pile?	E Volta	✅ IN	OH-04
GullSO Invitational	Dec 2025 (2026 season)	C (B/C)	Invitational	32	Whose inventions were licensed by Westinghouse?	D Tesla	❌ Out	OH-31
GullSO Invitational	Dec 2025 (2026 season)	C (B/C)	Invitational	34	First remote-control boat	E Tesla	❌ Out	OH-31
GullSO Invitational	Dec 2025 (2026 season)	C (B/C)	Invitational	29	SI unit of capacitance named after	Faraday	❌ Out (3.d)	OH-34
Pembroke Hill Invitational	2025-26	**B**	Invitational	15	Who championed DC in the war of the currents?	Thomas Edison	⚠️ (area iii; Edison not listed)	OH-09
Princeton Invitational	2020	C	Invitational (top tier)	Short response 2	Who won the war of the currents, Edison or Tesla? (claim + evidence, 4 pts)	rubric	⚠️	OH-26
Princeton Invitational	2020	C	Invitational (top tier)	1	Timeline: Coulomb, Volta, Ampère/Ohm, Faraday, Kirchhoff, Tesla	see key	✅ IN for listed names	OH-27
Princeton Invitational	2021	C	Invitational (top tier)	9	T/F "Ohm's main rival was Galvani"	False	✅ IN	OH-04

Test	Year	Div	Level	Q	Question (short)	Key	Check / scope 2027 B	Type
Princeton Invitational	2019	C	Invitational (top tier)	Part 1 (a)-(w)	Identify V, O, T (Tesla), H, F (Faraday) for 23 statements	see key	Mixed: V/O/H items ✅ ; T/F items ❌	OH-28
Hawk & Hornet Invitational	2026	C	Invitational	4	First battery (Edison, Faraday, Tesla, Volta) + name it	d Volta, voltaic pile	✅ IN	OH-01
Hawk & Hornet Invitational	2026	C	Invitational	21	Who is NOT a Maxwell-equation contributor?	Hertz	✅ IN (Hertz)	OH-07
Florida State Tournament	2021	B (= C)	**State**	11	"A rose by any other name" applies to...?	positive vs negative charge names	✅/⚠️ (Franklin's arbitrary naming)	OH-02
Florida State Tournament	2021	B (= C)	**State**	2	Quote "scientists of today think deeply instead of clearly"	Tesla	❌ Out	OH-31
Florida State Tournament	2021	B (= C)	**State**	7	Quote "Nothing is too wonderful to be true"	Faraday	❌ Out	OH-31
USC Invitational (not the Nationals test; see `tests/nat26.md`)	Jan 2026	B	Invitational (top tier)	25	Which Tesla statement is false?	A (scantron says C)	❌ Out; key conflict	OH-31
USC Invitational	Jan 2026	B	Invitational (top tier)	26	Match 4 facts → physicist (Faraday, Ohm, Ampère, Faraday)	C	⚠️ as a whole	OH-28
USC Invitational	Jan 2026	B	Invitational (top tier)	29	Maxwell displacement current is like whose induced-E discovery?	C Faraday	❌ Out	OH-33
F-M Invitational (NY)	2018	—	Invitational	1	Who built the electromagnetic rotation device (picture)?	Faraday	❌ Out	OH-32
Fall F-M Invitational (NY)	2019	—	Invitational	1	Match 7 discoveries to Ampère, Coulomb, Faraday, Kirchhoff, Ohm, Tesla, Volta	see key	✅ for 5 rows; Faraday/Tesla rows ❌	OH-28
UMSO (U. Michigan) Invitational	2019	—	Invitational	1-5	Match Faraday, Tesla, Hertz, Volta, Ohm to discoveries	A, C, E, B, D	Mixed	OH-28
Cobra Invitational (Michigan)	2019-20	B	Invitational	3, 5, 6	Fill-in: Faraday; Tesla (and Ferraris); Maxwell	as stated	❌ Out	OH-31, OH-33
UF Northern Regional (Florida)	2019	B	Regional	1-8	Match Volta/Ohm/Tesla/Hertz/Faraday	A, C, C, A, E, D, D, A	Mixed	OH-28
JC Booth Invitational, "Shock Value"	2011	B (Shock Value era)	Invitational	53	Electricity from magnetism discovered by (Oersted, Einstein, Franklin, Faraday)	d Faraday	❌ Out	OH-32
Shock Value Network Analysis test	~2011	B (Shock Value era)	Practice	Tiebreaker 1	Two most important scientists in the war of currents	Edison and Tesla	⚠️	OH-09
Shock Value Network Analysis test	~2011	B	Practice	Tiebreaker 2	Who invented the battery (with title)	Count Alessandro Volta	✅ IN	OH-01

Test	Year	Div	Level	Q	Question (short)	Key	Check / scope 2027 B	Type
Shock Value Network Analysis test	~2011	B	Practice	Tiebreaker 3	Who created the "predecessor to Ohm's Law" (first, middle, last name)	James Prescott Joule	**Key doubtful** (see Mistakes)	OH-17
Shock Value Network Analysis test	~2011	B	Practice	Tiebreaker 4	Who created conventional flow notation?	Benjamin Franklin	⚠️ / ✅ concept	OH-02
Shock Value Network Analysis test	~2011	B	Practice	Tiebreaker 5	Scientist the "first circuit" is named after	Sir Charles Wheatstone	**Wording doubtful**	OH-18
ATX (UT Austin) Regional	2026	B	Regional	MC7	Another name for Tesla's "death ray"	Teleforce	❌ Out	OH-31
UT/ATX (ut-a group)	2025-26	B/C	Inv./Reg.	MC3-MC14	Faraday induction, Wardenclyffe, mercury, Sandemanians, knighthood, Tesla quote	various	❌ Out	OH-31, OH-32

Easy

OH-01. Reject the unlisted distractor ("Who invented the first battery?")

Reject the unlisted distractor ("Who invented the first battery?") · ✅ IN · Easy

- **Scope:** IN (3.c.i). The question is about a listed scientist. The unlisted names are only wrong choices.
- **Where seen:** ATX Regional 2019 B/C Q11 (Galvani, Tesla, Volta, Faraday, Ørsted → Volta); UF Northern Regional 2020 B Q2 (Ohm, Faraday, Edison, Volta, Swan → Volta); UF Northern Regional 2020 B Q5 (solenoid: Kirchhoff, Ohm, Franklin, Ørsted, Ampère → Ampère); Hawk & Hornet 2026 C Q4 (Edison, Faraday, Tesla, Volta → Volta); Shock Value Network Analysis tiebreaker 2 (~2011 B practice); GullSO 2026 C Q28 (solenoid: Ampère, Faraday, Ohm, Tesla, Volta → Ampère).
- **How to solve:**
  1. Find the key word: battery, pile, solenoid, force law, node law.
  2. Match it to the listed scientist first.
  3. Cross out each unlisted name with its own famous fact. Galvani = frogs. Edison = light bulb, DC. Swan = light bulb. Franklin = kite, + and –. Ørsted = compass (1820). Faraday = induction.
- **Worked example:** "Which scientist invented the first battery? A. Ohm B. Faraday C. Edison D. Volta E. Swan" → **D. Volta** (voltaic pile, 1800). Edison and Swan made light bulbs (1879), 79 years later. (Python-checked: 1879 – 1800 = 79.)
- **Traps:** Galvani studied the "electricity" in frogs but made no battery. Franklin used the word "battery" for a row of Leyden jars; that is not a chemical battery. Faraday's name sounds like "electrochemistry" but he came later (1830s).
- Calculator: look up — type: key word (for example "battery", "solenoid") → listed scientist + one-line reason each distractor is wrong.

OH-02. Franklin named positive and negative (conventional current)

Franklin named positive and negative (conventional current) · ⚠️ Borderline · Easy

- **Scope:** BORDERLINE. Franklin is not in 3.c.i. But charge signs (area ii) and conventional current direction (area v) are IN. So the physics part is IN; the "who" part is borderline.
- **Where seen:** Shock Value Network Analysis test (~2011, B, practice), tiebreaker 4: "Who created conventional flow notation?" → Benjamin Franklin. Florida State 2021 B (= C), State, Q11: "A rose by any other name" applies to → positive vs negative charge names (Franklin's arbitrary choice). Shock Value 2011 test Q52: "Conventional current flows from plus to minus."
- **How to solve:**
  1. "Named + and –", "one-fluid theory", "conventional current" → Franklin.
  2. Remember: conventional current goes + → – outside the source. Electrons go – → +.

- **Worked example:** "Electrons in a wire move from point A to point B. Which way is the conventional current?" → **From B to A.** Conventional current is opposite to electron flow because Franklin called the electron's charge "negative".
- **Traps:** Du Fay (1733) found two kinds of charge first, but called them "vitreous" and "resinous". Franklin gave the + and – names. Do not answer "Coulomb"; he measured the force, he did not name the signs.
- Calculator: look up — type: "named positive/negative" or "conventional current" → Franklin; direction: opposite to electron flow.

OH-03. Franklin's kite and the lightning rod (lightning as static electricity)

Franklin's kite and the lightning rod (lightning as static electricity) · ⚠️ Borderline · Easy

- **Scope:** BORDERLINE. Person not listed. The content (lightning is a static discharge; lightning hazards; grounding) is area ii, "sources/hazards of static electricity".
- **Where seen:** not seen as a question in a test. Franklin appears as a distractor in the Shock Value 2011 Q53 and UF 2020 B Q5. [PR I_bank] lists it as a likely fact.
- **How to solve:**
  1. Kite + key + storm (June 1752) → Franklin. It showed that lightning is electricity.
  2. Lightning rod → gives lightning a safe, low-resistance path to the ground.
- **Worked example:** "Why does a lightning rod protect a house?" → **It is a good conductor connected to the ground. Lightning follows it to the ground instead of passing through the wood of the house.**
- **Traps:** The kite was **not** hit by lightning. If it had been, Franklin would likely have died, like Richmann (1753). Dalibard (France, May 1752) did the lightning test first, using Franklin's plan. "Lightning rods attract lightning from far away" is false; they only give a path.
- Calculator: look up — type: "kite", "lightning rod" → Franklin, 1752; safe path to ground.

OH-04. Galvani vs Volta (the frog dispute)

Galvani vs Volta (the frog dispute) · ✅ IN · Easy

- **Scope:** IN (3.c.i, as Volta's history).
- **Where seen:** UT Invitational 2019 B and C Q2: "Contested with a man over frogs" → Volta. GullSO 2026 C Q30: "During a feud with Luigi Galvani, which scientist invented the voltaic pile?" → E Volta. Princeton 2021 C Q9: T/F "Ohm's main rival was Galvani" → False. ATX 2019 B/C Q2, Q3, Q11: Galvani as a distractor.
- **How to solve:**
  1. "Frog", "animal electricity", "muscle twitch", "1791" → Galvani.
  2. "Disagreed with Galvani", "two metals", "contact electricity", "pile" → Volta.
- **Worked example:** "True or false: Ohm's main rival was Galvani." → **False.** Galvani's rival was Volta. Galvani died in 1798; Ohm's law came in 1827, 29 years later. (Python-checked: 1827 – 1798 = 29.)
- **Traps:** Galvani was a doctor (anatomist), not a physicist. Volta coined the word "galvanism", not Galvani. The two men argued politely; it was a scientific dispute, not a personal war.
- Calculator: look up — type: "frog" → Galvani (rival: Volta); "pile" → Volta.

OH-05. Word origins: amber, "electricus", "electricity"

Word origins: amber, "electricus", "electricity" · ⚠️ Borderline · Easy

- **Scope:** BORDERLINE. Thales and Gilbert are not listed. The content (rubbing makes static charge) is area ii.
- **Where seen:** not seen in a test. Gilbert is a distractor in ATX 2019 B/C Q3. Textbook fact (OpenStax College Physics 2e, ch. 18.1).
- **How to solve:**
  1. "Amber", "ancient Greek", "c. 600 BC" → Thales.
  2. "electricus", "1600", "*De Magnete*", "versorium" → Gilbert.
  3. First English use of "electricity" → Sir Thomas Browne, 1646.
- **Worked example:** "The word 'electron' comes from the Greek word for which material?" → **Amber (ēlektron).**
- **Traps:** Gilbert coined the Latin word, not the English "electricity". Thales is a tradition from later writers; no writing by Thales survives. Amber is fossil tree resin, not a metal.
- Calculator: look up — type: "amber", "electricus" → Thales / Gilbert / Browne.

OH-06. Ørsted's 1820 discovery as the start of Ampère's work

Ørsted's 1820 discovery as the start of Ampère's work · ✅ IN · Easy

- **Scope:** IN as Ampère history (3.c.i). A question only about Ørsted and magnetism is ❌ Out (no magnetism area in 3.c).
- **Where seen:** ATX Regional 2020 B and C Q8: "Oersted's law states the same findings as this scientist's law" → key Faraday (doubtful; Ampère fits). UT Invitational fall 2018 Q5: "Who found that current makes a magnetic field?" → Oersted. Cobra 2019 B: "Furthering Ørsted's experimental work..." → Ampère. ATX 2019 B/C Q3, Q11 distractor.
- **How to solve:**
  1. "Compass needle moved by a current", "1820", "Danish" → Ørsted.
  2. "Built on Ørsted", "within weeks", "force between two currents" → Ampère.
- **Worked example:** "Arago showed Ørsted's result in Paris in September 1820. Which listed scientist then found the force between two currents?" → **Ampère** (also 1820, the same year: 1820 – 1820 = 0 years).
- **Traps:** Faraday's induction (1831) is the reverse effect: magnetism makes current. Do not mix them. The "accident during a lecture" story is a myth, per Wikipedia.
- Calculator: look up — type: "Ørsted" → 1820, compass; follower: Ampère.

OH-07. Maxwell as the link to Hertz

Maxwell as the link to Hertz · ✅ IN · Easy

- **Scope:** IN as Hertz history (3.c.i). Questions about Maxwell's equations themselves are ❌ Out (magnetism, calculus 3.e.v).
- **Where seen:** Hawk & Hornet 2026 C Q21: "Who is NOT a Maxwell-equation contributor?" → Hertz. Princeton 2019 C Part 1 (j), (l): "verified Maxwell's equations", "showed oscillating charges make EM waves as Maxwell predicted" → Hertz. [PR I_bank]: "Hertz's experiments confirmed whose theory?" → Maxwell.
- **How to solve:**
  1. "Predicted electromagnetic waves", "light is an EM wave", "equations" → Maxwell.
  2. "Proved / detected the waves", "spark gap", "1887" → Hertz.
- **Worked example:** "Hertz's experiments of 1887-1888 confirmed whose theory?" → **Maxwell's** electromagnetic theory. Maxwell had died 8 years before 1887.
- **Traps:** Maxwell never saw his waves detected. Hertz did not write Maxwell's equations (Gauss, Faraday, Ampère and Maxwell are the names on them). Marconi, not Hertz, built practical radio.
- Calculator: look up — type: "Maxwell" → predicted EM waves (1865); proved by Hertz (1887).

OH-08. Joule's law of heating (name the law)

Joule's law of heating (name the law) · ✅ IN · Easy

- **Scope:** IN (area iv: power and energy). The name is a standard part of the power topic.
- **Where seen:** ATX Regional 2020 B Q17: "Law for electrical heat" → b Joule. `dl/cl_wiki.txt`: "James Prescott Joule discovered P = I²R (Joule's Law)".
- **How to solve:**
  1. "Heat made by a current", "P = I²R", "resistive heating" → Joule.
  2. Unit of energy → joule (J).
- **Worked example:** "Whose law says the heat made in a resistor is proportional to I²R?" → **Joule.**
- **Traps:** Do not answer "Ohm" (V = IR). Do not answer "Watt": the watt is a unit, and James Watt did no electrical work.
- Calculator: look up — type: "heat", "I²R" → Joule.

OH-09. Who backed DC / who backed AC (war of the currents)

Who backed DC / who backed AC (war of the currents) · ⚠️ Borderline · Easy

- **Scope:** BORDERLINE. Edison, Westinghouse and Tesla are not in 3.c.i. But area iii covers "characteristics, sources, uses, and hazards" of DC and AC. A 2026 Div B test (Pembroke) asked it.

- **Where seen:** Pembroke Hill Invitational 2025-26, **Div B**, Q15: "Who is historically famous for championing the DC standard during the 'War of the Currents'?" → Thomas Edison. Shock Value Network Analysis (~2011 B) tiebreaker 1: "two most important scientists" → Edison and Tesla. Northview 2026 C Q46 (distractor list: Edison, Volta, Tesla, Westinghouse).
- **How to solve:**
  1. DC → **Edison** (Pearl Street, 1882).
  2. AC → **Westinghouse** (businessman) and **Tesla** (inventor of the AC motor).
- **Worked example:** "Who championed DC in the war of the currents?" → **Thomas Edison.**
- **Traps:** Westinghouse was AC, not DC. Tesla once worked for Edison, then left; he is AC. "War of the currents" is not a real war; it is a business fight (about 1886-1892).
- Calculator: look up — type: "DC champion" → Edison; "AC" → Westinghouse + Tesla.

OH-10. Edison's argument against AC (AC hazards)

Edison's argument against AC (AC hazards) · ✅ IN · Easy

- **Scope:** IN (area iii: hazards of AC). The person is only context.
- **Where seen:** ATX Regional 2020 B Q20 and C Q22: "During the war of the currents, what was Thomas Edison's main argument against the use of AC power?" → c (not safe).
- **How to solve:**
  1. Edison's argument was **safety**: high-voltage AC can kill.
  2. He backed it with public animal killings (by Harold Brown) and the AC electric chair (1890).
- **Worked example:** "What was Edison's main argument against AC?" → **AC at high voltage is dangerous (it can kill people).**
- **Traps:** "AC is less efficient" is wrong; AC was better for long lines. "AC cannot light bulbs" is wrong. Remember that AC at 50-60 Hz is in fact very dangerous to the heart, so Edison's point was not silly; the answer is about his argument, not about who was right.
- Calculator: look up — type: "Edison against AC" → safety.

OH-11. Units named after unlisted people (watt, joule, siemens, and the out ones)

Units named after unlisted people (watt, joule, siemens, and the out ones) · ✅ IN · Easy

- **Scope:** IN for watt, joule, siemens (area iv: units of power, energy, resistance/conductance). The tesla, farad, henry, weber, gauss and oersted are ❌ Out (magnetism; 3.d capacitance and inductance). Full detail: Units Named After People [A1].
- **Where seen:** GullSO 2026 C Q29 (capacitance unit → Faraday, ❌); BullSO 2026 B Q4 and C Q4 (flux density unit → Tesla, ❌); ATX 2020 B Q55 (unit before tesla → gauss, ❌); Shock Value 2011 IIa (Watt, Joule as units); UF 2019 C Q14 (units of I, V, R, P, B → watts, tesla).
- **How to solve:**
  1. Power → watt (W) → James Watt.
  2. Energy → joule (J) → James Prescott Joule.
  3. Conductance → siemens (S) → Werner von Siemens. Old name: mho.
  4. If the unit is for capacitance (farad), inductance (henry) or magnetic field (tesla, gauss, weber, oersted), it is out of scope.
- **Worked example:** "A device has a resistance of 50 Ω. What is its conductance, and whose name is on the unit?" → **G = 1/50 = 0.02 S (siemens), after Werner von Siemens.** (Python-checked.)
- **Traps:** Unit names are lower case (watt, joule, siemens). Symbols of person-named units are upper case (W, J, S). "Siemens" ends in s in the singular too: "1 siemens". The S symbol is not "seconds" (s).
- Calculator: look up — type: quantity → unit, symbol, person; calculate — type: R → G = 1/R.

Medium

OH-12. "Who invented AC?" (a loaded question)

"Who invented AC?" (a loaded question) · ⚠️ Borderline · Medium

- **Scope:** BORDERLINE. Tesla is not listed; AC is area iii.

- **Where seen:** UF Northern Regional 2020 B Q6: "pioneer of AC" → Tesla. UF Northern Regional 2020 C Q2: "Who invented AC electricity?" → Tesla.
- **How to solve:**
  1. If the question says "pioneer of AC power systems" or "AC induction motor", answer **Tesla**.
  2. If choices include "Nobody" or the question asks about a Nobel Prize, read OH-13.
- **Worked example:** "Which scientist was a pioneer of AC power used in homes today?" → **Nikola Tesla** (with Westinghouse as the businessman).
- **Traps:** Nobody "invented" AC itself. AC generators existed before Tesla (Faraday's 1831 disk and Pixii's 1832 machine made changing currents). Tesla invented a practical AC motor and the polyphase system. 3-phase power itself is ❌ Out (3.d).
- Calculator: look up — type: "AC pioneer" → Tesla (inventor), Westinghouse (business).

OH-13. Trick question: "Who won a Nobel Prize for AC?" → nobody

Trick question: "Who won a Nobel Prize for AC?" → nobody · ⚠️ Borderline · Medium

- **Scope:** BORDERLINE (unlisted people; area iii context).
- **Where seen:** Northview Sylvania 2026 C Q46: "This person won a Nobel prize for inventing alternating current: A Edison, B Volta, C Tesla, D Westinghouse, E Nobody" → E Nobody. UT Invitational 2019 B Q8: "Won the Nobel Prize for helping invent AC" → Nobody.
- **How to solve:**
  1. The Nobel Prize started in 1901. Check if the person could have won it.
  2. Edison and Tesla never won one. Volta died in 1827, long before 1901.
- **Worked example:** Same as the Northview question → **E. Nobody.** Volta died 74 years before the first Nobel Prize. (Python-checked: 1901 − 1827 = 74.)
- **Traps:** A 1915 newspaper rumour said Edison and Tesla would share a Nobel Prize. It did not happen. [G] Not confirmed online in this run.
- Calculator: look up — type: "Nobel" + person → year or "none".

OH-14. Joule heating calculation in a history frame

Joule heating calculation in a history frame · ✅ IN · Medium

- **Scope:** IN (area iv: power and energy calculations).
- **Where seen:** not seen as a history-framed calculation. Joule's law name: ATX 2020 B Q17. Plain $P = I^2R$ items appear in almost every test (see the Power notes).
- **How to solve:**
  1. Power $P = I^2R$ (watts).
  2. Energy $Q = P \times t$ (joules), with t in seconds.
  3. To get calories, divide joules by 4.186.
- **Worked example:** "Joule passed 2 A through a 10 Ω coil in water for 60 s. How much heat did the coil make?" → $P = 2^2 \times 10 = 40$ W. $Q = 40 \times 60 =$ **2,400 J**, about **573 cal**. (Python-checked: 2400/4.186 = 573.3.)
- **Traps:** Time must be in seconds. Doubling the current makes 4 times the heat (I is squared). Do not use $V \times I$ with the wrong voltage.
- Calculator: calculate — type: I (A), R (Ω), t (s) → P, Q in J and cal.

OH-15. Horsepower and the watt

Horsepower and the watt · ✅ IN · Medium

- **Scope:** IN (area iv: units of power).
- **Where seen:** not seen in a test. Textbook: OpenStax College Physics 2e ch. 7.6 (horsepower); unit conversions appear in Shock Value 2011 IIb.
- **How to solve:**
  1. 1 hp ≈ 746 W (more exactly 745.7 W).
  2. hp = W ÷ 745.7. W = hp × 745.7.

- **Worked example:** "A 1,500 W space heater uses how many horsepower?" → 1500 ÷ 745.7 = **2.01 hp**. (Python-checked.) And 1 kW = 1.34 hp.
- **Traps:** James Watt worked on steam engines, not electricity. The watt is a rate (J/s), not an amount of energy. Do not mix up "kilowatt" (power) and "kilowatt-hour" (energy).
- Calculator: calculate — type: W or hp → the other.

OH-16. Siemens and conductance (the mho)

Siemens and conductance (the mho) · ✅ IN · Medium

- **Scope:** IN (area iv: units; conductance is 1/resistance).
- **Where seen:** `dlG/wiki/central.txt` (mho); Shock Value 2011 IIa ("Conductance:" unit question); `dlG/wiki/Shock_value_test_.txt` item 2 (conductance unit). Full list in Ohm A1 Q4-Q6.
- **How to solve:**
  1. G = 1/R. Unit: siemens (S) = 1/Ω = A/V.
  2. Old name: mho (℧), "ohm" backwards.
  3. Year the SI name was adopted: 1971 (14th CGPM).
- **Worked example:** "A wire has a conductance of 0.25 S. What is its resistance?" → R = 1/0.25 = **4 Ω**. (Python-checked.)
- **Traps:** Werner von Siemens is the usual namesake. His brother William Siemens proposed the watt. The siemens mercury unit (1860) was a resistance unit, not a conductance unit.
- Calculator: calculate — type: G or R → the other.

OH-17. Who found Ohm's law first? (Cavendish; the Joule key error)

Who found Ohm's law first? (Cavendish; the Joule key error) · ⚠️ Borderline · Medium

- **Scope:** BORDERLINE. Cavendish is not listed. The question is about the priority of Ohm's law, which is Ohm's history (IN). See Ohm A1 Q16.
- **Where seen:** ATX Regional 2019 B and C Q2: "Who experimented with an early concept of what eventually became Ohm's law using Leyden jars and glass tubes?" (Ronalds, Faraday, Galvani, Cavendish, Volta) → Cavendish. Princeton 2021 C Q10: T/F "Ohm was not the first one to discover what is now known as Ohm's Law" → True. Shock Value Network Analysis tiebreaker 3: "predecessor to Ohm's Law" → key "James Prescott Joule" (doubtful; see Mistakes).
- **How to solve:**
  1. "Leyden jars", "glass tubes of salt water", "his own body as the meter", "1781" → Cavendish.
  2. His notes were published only in 1879 (by Maxwell), so Ohm (1827) got the credit.
- **Worked example:** "True or false: Ohm was not the first person to find what we call Ohm's law." → **True** (Cavendish, about 1781, unpublished).
- **Traps:** Joule's law (1840-1841) came after Ohm's law (1827), 13 years later. So Joule cannot be a "predecessor". (Python-checked: 1840 – 1827 = 13.)
- Calculator: look up — type: "before Ohm" → Cavendish (c. 1781, published 1879).

OH-18. Wheatstone and the Wheatstone bridge

Wheatstone and the Wheatstone bridge · ⚠️ Borderline · Medium

- **Scope:** BORDERLINE as history (Wheatstone not listed). The bridge circuit is ✅ IN (area v; balanced bridge needs only ratios).
- **Where seen:** Shock Value Network Analysis tiebreaker 5: "the scientist that the first circuit is named after" → Sir Charles Wheatstone (wording doubtful). UT Invitational 2020 B Q16: "Wheatstone bridge is used to measure...?" → unknown resistance. Bridge calculations: Rickards 2020 Q32, Princeton 2026 Comp 1c-1d, ATX 2020 B Q56 [PR G_tests].
- **How to solve:**
  1. Bridge purpose → measure an unknown resistance.
  2. Balanced bridge: R1/R2 = R3/Rx, so Rx = R2 × R3 / R1.
  3. History: invented by Christie (1833); made popular by Wheatstone (1843).

- **Worked example:** "R1 = 100 Ω, R2 = 200 Ω, R3 = 150 Ω, and the bridge is balanced. Find Rx." → Rx = 200 × 150 / 100 = **300 Ω**. (Python-checked; from [PR I_bank].)
- **Traps:** Wheatstone did not invent the bridge. "First circuit" is not a real title; the earliest circuits were Volta's (1800).
- Calculator: calculate — type: R1, R2, R3 → Rx (balanced).

OH-19. De Morgan the person (and De Morgan's laws)

De Morgan the person (and De Morgan's laws) · ✅ IN · Medium

- **Scope:** IN for "whose name is on De Morgan's law" and the link to Boole (3.c.i Boole; 3.c.viii names "De Morgan's Law"). A deep De Morgan biography question would be ⚠️.
- **Where seen:** not seen as a history question in a test. De Morgan's law statements: Princeton 2021 C Q19 ((A+B)' = A'·B'), Princeton 2020 C Q37 (state it and draw gates), MIT 2019 Q45 [PR G_tests]. [PR I_bank]: "Contemporary and friend of Augustus De Morgan".
- **How to solve:**
  1. "Break the bar, change the sign": NOT over AND becomes OR of NOTs; NOT over OR becomes AND of NOTs.
  2. History: Augustus De Morgan, London, *Formal Logic* 1847, Boole's friend.
- **Worked example:** "Use De Morgan's law to rewrite NOT(A OR B)." → **(NOT A) AND (NOT B).** Python checked all four rows of the truth table.
- **Traps:** De Morgan did not invent Boolean algebra (Boole did). Both men published logic books in **1847**. A NAND gate is NOT-AND, which equals OR of inverted inputs.
- Calculator: calculate — type: Boolean expression → De Morgan form + truth table.

OH-20. Shannon: Boole's algebra in switch circuits

Shannon: Boole's algebra in switch circuits · ⚠️ Borderline · Medium

- **Scope:** BORDERLINE as a person (not listed). As Boole's legacy it supports 3.c.i Boole; the switch logic is ✅ IN (3.c.viii and 3.c.vi switches).
- **Where seen:** not seen in a test. [PR I_bank]: "Claude Shannon, 1937 (MIT master's thesis) applied Boolean algebra to relay/switching circuits." See Boole A1.
- **How to solve:**
  1. "Applied Boolean algebra to relays / switching circuits", "1937", "MIT thesis" → Shannon.
  2. Series switches → AND. Parallel switches → OR.
- **Worked example:** "Two switches A and B are in series with a lamp. Write the Boolean expression for 'lamp on'." → **Lamp = A · B (AND).** In parallel → **A + B (OR).** (Truth tables Python-checked.)
- **Traps:** Shannon did not invent Boolean algebra; he applied it 83 years after *The Laws of Thought* (1854). "Father of information theory" = Shannon, not Boole. The word "bit" was coined by John Tukey, first printed by Shannon (1948).
- Calculator: look up — type: "relay circuits 1937" → Shannon; calculate — type: series/parallel switch list → Boolean expression.

OH-21. Ohl and the PN junction

Ohl and the PN junction · ⚠️ Borderline · Medium

- **Scope:** BORDERLINE. Ohl is not listed. The PN junction is area vii (✅ IN as physics).
- **Where seen:** not seen in a test. Source: Computer History Museum (1940 timeline); Wikipedia.
- **How to solve:**
  1. "Discovered the PN junction", "Bell Labs", "1940", "cracked silicon rod" → Russell Ohl.
  2. P = positive side (holes; boron), N = negative side (extra electrons; phosphorus).
- **Worked example:** "In Ohl's silicon rod, one side had extra electrons from phosphorus. What did he call that side?" → **n-type (n for negative).**
- **Traps:** The P side is not positively charged overall; the name means it has "holes" that act like positive carriers. Silicon is the material in the 2027 rules ("silicon PN junctions"). Ohl's find also led to the solar cell. The transistor (1947) is ❌ Out.
- Calculator: look up — type: "PN junction discovery" → Ohl, Bell Labs, 23 Feb 1940 (Wikipedia: 1939).

OH-22. Holonyak and the first visible LED

Holonyak and the first visible LED · ⚠️ Borderline · Medium

- **Scope:** BORDERLINE. Holonyak is not listed. LEDs are area vii (✅ IN as physics).
- **Where seen:** not seen in a test. LED color question: ATX 2019 B/C Q5: "The semiconductor compound in an LED's PN junction is best known for determining its: ... Color" (✅ IN).
- **How to solve:**
  1. "First visible LED", "1962", "General Electric", "red", "GaAsP" → Nick Holonyak Jr.
  2. LED color depends on the semiconductor material (its band gap).
- **Worked example:** "What color was the first visible LED (1962)?" → **Red** (gallium arsenide phosphide).
- **Traps:** The first visible LED was not made of silicon. Silicon diodes give no useful visible light. Infrared LEDs came a little earlier (Texas Instruments, 1961-62). Edison did not invent the LED.
- Calculator: look up — type: "first visible LED" → Holonyak, GE, 9 Oct 1962, red, GaAsP.

OH-23. Gilbert and *De Magnete* (the UT 2019 "On Magnetism" question)

Gilbert and *De Magnete* (the UT 2019 "On Magnetism" question) · ⚠️ Borderline · Medium

- **Scope:** BORDERLINE. Gilbert is not listed. His electricity facts (electricus, versorium) are area ii. His magnetism facts are ❌ Out.
- **Where seen:** UT Invitational 2019 B and C Q9: "Published 'On Magnetism'" → key Coulomb (doubtful). ATX 2019 B/C Q3: Gilbert as a distractor.
- **How to solve:**
  1. "*De Magnete* (On the Magnet)", "1600", "Earth is a magnet", "electricus" → Gilbert.
  2. If only the seven listed names are offered, Coulomb is the closest: he wrote memoirs on electricity and magnetism (1785-1789). See Coulomb A1.
- **Worked example:** "Who wrote *De Magnete* (1600)?" → **William Gilbert.**
- **Traps:** The book title is "On the Magnet", not "On Magnetism". In a 7-name bank, the setter likely meant Coulomb's memoirs. Answer the word bank you are given.
- Calculator: look up — type: "De Magnete" → Gilbert 1600.

OH-24. Faraday cage: who, and why it works

Faraday cage: who, and why it works · ⚠️ Borderline · Medium

- **Scope:** BORDERLINE. Faraday is not listed. The idea (charges sit on the outside of a conductor; the field inside is zero) is area ii, electric fields. [PR I_bank] marks the concept IN, the person OUT.
- **Where seen:** UF Northern Regional 2020 B Q3: "Write the name of the scientist who discovered that a container made of conductive material can block electromagnetic fields" → Faraday. Princeton 2020 Q25 and Rickards 2026 C Q5 (Faraday cage behaviour) [PR G_tests]. UT/ATX MC35 (5G hole size: ❌ Out, frequency).
- **How to solve:**
  1. Name: Faraday (1836).
  2. Reason: free charges in the metal move to cancel the field inside.
- **Worked example:** "Why is a person inside a car usually safe from lightning?" → **The metal body acts like a Faraday cage. The current flows around the outside of the car to the ground.** The tyres are not the reason.
- **Traps:** "The rubber tyres protect you" is false. A Faraday cage does not need to be solid; a mesh works if the holes are small.
- Calculator: look up — type: "conductive box blocks fields" → Faraday cage; reason: charges move to outer surface.

OH-25. Blue LED Nobel Prize and white light

Blue LED Nobel Prize and white light · ⚠️ Borderline · Medium

- **Scope:** BORDERLINE. The laureates are not listed. LED use and energy savings are area vii and area iv.
- **Where seen:** not seen in a test.
- **How to solve:**
  1. "Blue LED", "gallium nitride", "Nobel 2014" → Akasaki, Amano, Nakamura.

2. White LED = blue LED + yellow phosphor (or red + green + blue LEDs).

- **Worked example:** "A 60 W bulb is replaced by a 9 W LED lamp. It runs 5 hours a day for 365 days. How much energy is saved?" → (60 – 9) × 5 × 365 = 93,075 Wh = **93.1 kWh**. (Python-checked.)
- **Traps:** The prize was for **blue**, not for LEDs in general. Red and green came first. The prize was in physics, not chemistry.
- Calculator: calculate — type: old W, new W, hours/day, days → kWh saved; look up — type: "blue LED" → 3 laureates, 2014.

OH-26. Why AC won (transmission-loss reasoning)

Why AC won (transmission-loss reasoning) · ✅ IN · Medium

- **Scope:** IN (area iii: uses of AC and DC; area iv: power calculations). Transformers' inner workings (induction) are ❌ Out. Use them only as "AC voltage is easy to change".
- **Where seen:** Princeton 2020 C Short Response 2: "Who won the war of the currents: Edison or Tesla? Make a strong claim, support your idea with verifiable evidence" (4 pts: claim, evidence, link, clear writing). Northview 2026 C Q45: line loss 2000 A, 2.3 Ω → 9.2 MW. ATX 2020 [PR]: "AC advantage (transmission)". [PR I_bank]: "AC won because of easy voltage change."
- **How to solve:**
  1. Current in the line: I = P / V.
  2. Loss: $P_{loss} = I^2 \times R_{line}$.
  3. Higher voltage → lower current → much lower loss (loss falls with the square).
- **Worked example:** "A plant sends 100 kW through lines with 1 Ω resistance. Find the loss at 1,000 V and at 10,000 V." → At 1,000 V: I = 100 A, loss = $100^2 \times 1$ = **10,000 W**. At 10,000 V: I = 10 A, loss = **100 W**. Ten times the voltage gives 1/100 of the loss. (Python-checked.)
- **Traps:** The loss uses the line resistance, not the load. "AC is always safer" is false. For an essay (Princeton style), give one clear claim, one checkable fact (for example the 1893 World's Fair or Niagara Falls), and link it to the claim.
- Calculator: calculate — type: P (W), V (V), R_line (Ω) → I, P_loss, % lost.

OH-27. Timeline with unlisted people mixed in

Timeline with unlisted people mixed in · ✅ IN · Medium

- **Scope:** IN for the listed names (3.c.i). Unlisted names are anchors.
- **Where seen:** Princeton 2020 C Q1: order Coulomb (1785-86), Volta (1799-1800), Ampère and Ohm (1827), Faraday (1831/1839), Kirchhoff (1845), Tesla (1887). [PR G_tests] "Order scientists by date".
- **How to solve:**
  1. Learn one anchor year per person (table below).
  2. Sort by year. Ties: give both.
- **Worked example:** "Put in order: Ohl's PN junction, Franklin's kite, Shannon's thesis, Volta's pile, Holonyak's LED, Ørsted's compass." → **Franklin 1752, Volta 1800, Ørsted 1820, Shannon 1937, Ohl 1940, Holonyak 1962.** (Python-sorted.)
- Anchor years: Thales c. 600 BC; Gilbert 1600; Franklin kite 1752; Coulomb 1785; Galvani book 1791; Volta pile 1800; Ørsted 1820; Ampère 1820-1827; Ohm 1827; Faraday induction 1831; Joule heating 1840-41; Kirchhoff 1845; De Morgan and Boole 1847; Boole *Laws of Thought* 1854; Maxwell 1865; Siemens dynamo 1866-67; Edison bulb 1879; Pearl Street 1882; Hertz 1887; Westinghouse-Tesla 1888; Zener 1934; Shannon 1937; Ohl 1940; Holonyak 1962; blue-LED Nobel 2014.
- **Traps:** Galvani (1791) comes before Volta (1800). Ørsted (1820) comes before Ampère. Ampère's big book and Ohm's book are both 1827.
- Calculator: look up — type: list of names → sorted years.

Hard

OH-28. Mixed matching (listed and unlisted names in one bank)

Mixed matching (listed and unlisted names in one bank) · ⚠️ Borderline · Hard

- **Scope:** BORDERLINE as a whole. Rows about listed people are ✅ IN; rows about Faraday or Tesla are ❌ Out. A 2027 test that follows the rules should use only the seven names, but old-style banks still appear.
- **Where seen:** USC Invitational Jan 2026 B Q26 (4 facts → Faraday, Ohm, Ampère, Faraday; key C). Fall F-M Invitational 2019 Q1 (7 discoveries ↔ Ampère, Coulomb, Faraday, Kirchhoff, Ohm, Tesla, Volta). UMSO 2019 Q1-5 (Faraday, Tesla, Hertz, Volta, Ohm). UF Northern Regional 2019 B Q1-8 (Volta/Ohm/Tesla/Hertz/Faraday). Princeton 2019 C Part 1 (a)-(w): 23 statements, letters V, O, T, H, F.
- **How to solve:**
  1. Do the sure rows first (unit names, the battery, V = IR).
  2. Assign each unlisted name its "signature": Faraday = induction, motor, dynamo, diamagnetism, farad. Tesla = AC system, induction motor, Niagara, Tesla coil, tesla unit.
  3. Use elimination. Letters can repeat.
- **Worked example (Princeton 2019 style):** "(r) Discovered electromagnetic induction. (s) The SI unit for resistance is named after him. (p) Designed one of the first AC hydroelectric plants at Niagara Falls." → **(r) F (Faraday), (s) O (Ohm), (p) T (Tesla).**
- **Traps:** "Constructed the electric dynamo" → Faraday (1831 disk), not Siemens, in a V/O/T/H/F bank. "Received with little enthusiasm" → Ohm. "Cathode rays through thin foil" → Hertz.
- Calculator: look up — type: statement key word → person letter.

OH-29. Zener: the person vs the diode

Zener: the person vs the diode · ⚠️ Borderline · Hard

- **Scope:** BORDERLINE. Zener is not listed. A Zener diode is a single-junction device (area vii says "e.g., diodes, LEDs"), but 3.d excludes "non-linear devices beyond those listed". Most writers treat Zener diodes as diodes (PUSO 2026 asked one).
- **Where seen:** Zener device: PUSO 2026 (Princeton) short answer 7c: "Explain the behavior of a Zener diode. How are they useful?" (key: works in reverse bias, conducts at the Zener voltage, used as a regulator). Rickards 2026 C (choice "Full Zener Diode", distractor). History: not seen in a test.
- **How to solve:**
  1. Person: Clarence Zener explained insulator breakdown (1934). He did not build the diode; Bell Labs named it after him.
  2. Device: in reverse bias it holds a nearly fixed voltage V_Z. Resistor current $I = (V_{supply} - V_Z)/R$.
- **Worked example:** "A 5.1 V Zener diode and a 470 Ω resistor are in series with a 12 V supply. Find the resistor current (no load)." → $I = (12 - 5.1)/470 =$ **14.7 mA**. (Python-checked.)
- **Traps:** A Zener diode is connected in reverse (cathode to +). Forward-biased, it acts like a normal diode (about 0.7 V). Above about 5-6 V the breakdown is mostly avalanche, but it is still called a Zener diode.
- Calculator: calculate — type: V_{supply}, V_Z, R → I_R.

OH-30. Myth-busting true/false (Edison and Topsy; Franklin's kite; Ørsted's lecture)

Myth-busting true/false (Edison and Topsy; Franklin's kite; Ørsted's lecture) · ⚠️ Borderline · Hard

- **Scope:** BORDERLINE. All unlisted people. These are common online myths that a test writer could copy as "facts".
- **Where seen:** not seen in a test as a myth question. The Topsy story is linked to the war of the currents on many websites.
- **How to solve:**
  1. Ask: is the claim from a reliable source, or a popular story?
  2. Learn the four classic myths below.
- **Worked example:** "True or false: Thomas Edison electrocuted the elephant Topsy to show that AC was dangerous." → **False.** Topsy was killed on 4 January 1903 at Coney Island. Edison had no part in it. His film company only filmed it. The war of the currents had ended in 1892.
- Other myths: "Lightning struck Franklin's kite" → false (it was induced charge). "Ørsted found his effect by accident in a lecture" → Wikipedia calls this a myth. "Edison invented the light bulb alone" → false (Swan, and earlier arc and filament lamps).
- **Traps:** If a test key uses the myth, answer the key's way during the test, then file an appeal with a source.

- Calculator: look up — type: claim → true/false + one-line reason.

Nationals

OH-37. Chain of influence: unlisted → listed (essay or multi-part)

Chain of influence: unlisted → listed (essay or multi-part) · ✅ IN · Nationals

- **Scope:** IN (3.c.i: "historical perspective" of the seven). This is how unlisted people can show up at Nationals depth without breaking the rules.
- **Where seen:** not seen as one question. Parts seen: ATX 2020 Q8 (Ørsted → Ampère), UT 2019 Q2 (Galvani → Volta), HH 2026 Q21 (Maxwell → Hertz), ATX 2019 Q2 (Cavendish → Ohm). [PR I_bank] lists the chains.
- **How to solve:**
  1. Name the listed scientist.
  2. Name who came before (what idea they gave).
  3. Name who came after (what they built on it).
  4. Give one year for each step.
- **Worked example:** "Trace the path from Ørsted to a modern idea through a listed scientist." → **Ørsted (1820: current moves a compass) → Ampère (1820-1827: force between currents; unit of current) → the ampere is now an SI base unit (defined by the elementary charge since 2019).** Another: **Galvani (1791, frogs) → Volta (1800, pile) → the volt; batteries.** Another: **De Morgan and Boole (1847) → Boole (1854) → Shannon (1937, switch circuits) → digital logic gates.**
- **Traps:** Keep the direction right: Maxwell (theory, 1865) came before Hertz (proof, 1887). Seebeck (1821) helped Ohm (1827), not the other way.
- Calculator: look up — type: listed name → predecessor, successor, years.

OH-38. "Which of these people is on the rules list?" / scope reasoning

"Which of these people is on the rules list?" / scope reasoning · ✅ IN · Nationals

- **Scope:** IN. This is not a test question you will be asked word for word. It is the skill of knowing which facts carry the most weight. It matters most when a test mixes in Tesla and Faraday.
- **Where seen:** every mixed bank above. Rules 3.c.i [R].
- **How to solve:**
  1. The seven: **A**mpère, **C**oulomb, **K**irchhoff, **V**olta, **O**hm, **H**ertz, **B**oole.
  2. Memory aid: "**A Cool Kid Visits Old Hidden Books**".
  3. If a bank mixes in other names, the answer is still most often a listed one.
- **Worked example:** "From Faraday, Tesla, Kirchhoff, Franklin, Edison: who is on the 2027 Circuit Lab history list?" → **Kirchhoff.**
- **Traps:** Faraday was on older lists. Tesla was on older lists too. Do not study them as main topics for 2027.
- Calculator: look up — type: name → on list (yes/no) + main fact.

OH-39. War-of-currents numbers (Edison's 110 V DC lamp; Westinghouse line voltage)

War-of-currents numbers (Edison's 110 V DC lamp; Westinghouse line voltage) · ✅ IN · Nationals

- **Scope:** IN (area iii and iv: DC/AC uses, Ohm's law, power).
- **Where seen:** not seen as a history-framed calculation. Related: Northview 2026 C Q45 (line loss 2000 A, 2.3 Ω → 9.2 MW); UT 2020 B Q63 (EV charge at 120 V).
- **How to solve:**
  1. Lamp current $I = P / V$. Lamp resistance $R = V^2 / P$.
  2. Line drop $V_{drop} = I \times R_{line}$. Voltage left at the lamp $= V - V_{drop}$.
  3. Compare two supply voltages with the same power.
- **Worked example:** "Edison's 100 W lamp runs on 110 V DC. Find its current and its hot resistance." → $I = 100/110 =$ **0.909 A**. $R = 110^2/100 =$ **121 Ω**. (Python-checked.) If 100 such lamps share a feeder with 0.1 Ω resistance, the feeder current is 90.9 A, the drop is 9.09 V, and the loss is 826 W.
- **Traps:** Hot filament resistance is much higher than cold resistance (do not measure it with a meter and expect 121 Ω). Keep line resistance separate from load resistance.

- Calculator: calculate — type: P, V, number of lamps, R_line → I, R, V_drop, P_loss.

Out of scope (skip)

These types appear in tests, but the 2027 Div B rules exclude them. Skip them. They stay here so you can recognise them.

OH-31. Tesla trivia (quotes, sleep, death ray, boat, Westinghouse patents, Wardenclyffe)

Tesla trivia (quotes, sleep, death ray, boat, Westinghouse patents, Wardenclyffe) · ❌ Out · n/a

- **Scope: OUT** (3.c.i: Tesla is not one of the seven; his main work, the AC induction motor and polyphase power, is magnetism and 3-phase power, 3.d).
- **Where seen:** ATX Regional 2020 B/C Q5 (Westinghouse bought his patents); GullSO 2026 C Q32 (licensed by Westinghouse), Q34 (first remote-control boat); UT Invitational 2019 B/C Q3 (slept two hours); ATX 2026 B MC7 (death ray = "Teleforce"); UT/ATX MC4 (Wardenclyffe tower), MC9 (quote "As long as I had her..."), MC5 (stationary waves in the Earth, 1899); Florida State 2021 Q2 (quote "think deeply instead of clearly"); USC Jan 2026 B Q25 (which Tesla statement is false; key conflict).
- Know only: Tesla = AC induction motor, polyphase AC, Westinghouse licensed his patents (1888), Tesla coil, unit tesla (magnetic field). That is enough to reject him as a distractor.
- Calculator: look up — type: "Tesla" → AC motor; not on 2027 list.

OH-32. Faraday trivia and induction (knighthood, Sandemanians, mercury, motor, induction)

Faraday trivia and induction (knighthood, Sandemanians, mercury, motor, induction) · ❌ Out · n/a

- **Scope: OUT** (3.c.i: Faraday is not listed; induction and motors are magnetism, which is not a 3.c area; the farad is capacitance, 3.d).
- **Where seen:** ATX 2021 B Q3 (induced current); ATX 2020 B Q1 (eddy currents law); F-M 2018 Q1 (rotation device picture); Shock Value 2011 Q53; UF 2019 C Q3, Q53; UT/ATX MC3, MC5 (mercury), MC9/MC14 (Sandemanians), MC3 (knighthood), MC11 (blood); UT 2019 C Q1 (rejected a knighthood); Florida State 2021 Q7 (quote); GullSO 2026 Q35 (induction law).
- Know only: Faraday = induction (1831), motor (1821), Faraday cage, farad. Enough to reject him.
- Calculator: look up — type: "Faraday" → induction; not on 2027 list.

OH-33. Maxwell's equations and Maxwell trivia

Maxwell's equations and Maxwell trivia · ❌ Out · n/a

- **Scope: OUT** (magnetism is not a 3.c area; the equations need calculus, 3.e.v; Maxwell is not listed). The Maxwell → Hertz link is IN (OH-07).
- **Where seen:** ATX 2019 B/C Q3 (put Faraday into maths); USC Jan 2026 B Q29 (displacement current); Florida State 2021 Q14-16 (div E, div B, Stokes); Cobra 2019 B Q6 (Maxwell unified electricity, magnetism, light); ATX 2020 B Q33 (no monopoles → Gauss).
- Calculator: look up — type: "Maxwell" → link to Hertz only.

OH-34. Units for capacitance, inductance and magnetic field (farad, henry, tesla, weber, gauss, oersted, gilbert, maxwell)

Units for capacitance, inductance and magnetic field (farad, henry, tesla, weber, gauss, oersted, gilbert, maxwell) · ❌ Out · n/a

- **Scope: OUT** (3.d capacitance, inductance; magnetism not a 3.c area).
- **Where seen:** GullSO 2026 C Q29 (farad → Faraday); BullSO 2026 B Q4 (tesla); ATX 2020 B Q55 (gauss before tesla); UF 2019 C Q25b-26 (farad, henry); Princeton 2019 C Part 1 (m), (w).
- See Units Named After People A1 for the full list.
- Calculator: look up — type: unit → quantity; flag "out of scope".

OH-35. Leyden jar and early capacitors (Musschenbroek, Kleist)

Leyden jar and early capacitors (Musschenbroek, Kleist) · ❌ Out · n/a

- **Scope: OUT** (3.d capacitance). Static charge storage as a hazard is area ii, so a simple "a Leyden jar stores static charge" fact is borderline.
- **Where seen:** ATX Regional 2020 B Q11: "Earliest capacitors" → d Leyden jars.
- Know only: Leyden jar = first capacitor, 1745-46, Leiden (Netherlands). Franklin linked several into a "battery".
- Calculator: look up — type: "Leyden jar" → first capacitor (out).

OH-36. Transistor inventors (Bardeen, Brattain, Shockley)

Transistor inventors (Bardeen, Brattain, Shockley) · ✗ Out · n/a

- **Scope: OUT** (3.d "semiconductors (beyond those listed above)"; a transistor has two junctions, so it is not a single-junction device).
- **Where seen:** ATX Regional 2020 C Q11: "Bardeen, Brattain, Shockley invented the" → b transistor.
- Calculator: look up — type: "transistor" → Bell Labs 1947 (out).

OH-40. Siemens's dynamo-electric principle, Ørsted and Gilbert magnetism facts

Siemens's dynamo-electric principle, Ørsted and Gilbert magnetism facts · ✗ Out · n/a

- **Scope: OUT** (magnetism is not one of the 3.c areas). The unit siemens is IN (OH-16); the electricus/versorium facts are borderline (OH-05, OH-23).
- **Where seen:** UT fall 2018 Q5 (Ørsted, current makes a magnetic field). Siemens dynamo: not seen in a test.
- Calculator: look up — type: person → magnetism fact (out).

OH-41. Electric chair and Harold Brown's animal killings (detail)

Electric chair and Harold Brown's animal killings (detail) · ✗ Out · n/a

- **Scope: OUT** as detail. No rule area covers execution history or this level of war-of-currents trivia. The general point "AC was sold as dangerous" is IN under OH-10.
- **Where seen:** not seen in a test.
- Know only: first electric-chair execution, William Kemmler, 6 August 1890, used AC.
- Calculator: look up — type: "electric chair" → 1890, AC (out).

Mistakes in tests and keys

1. **ATX Regional 2020 B and C Q8** ("Oersted's law states the same findings as this scientist's law"): key **Faraday**. Ørsted's finding (a current makes a magnetic field) is described by **Ampère's** law (and Biot-Savart). Faraday's law is the reverse effect. Source: `tests/atx-reg.md` key-error 10.
2. **Shock Value Network Analysis test (~2011 B), tiebreaker 3** ("the scientist who created the predecessor to Ohm's Law", first, middle and last name): key **James Prescott Joule**. Joule's heating law (1840-41) came 13 years after Ohm's law (1827), so it is not a predecessor. **Henry Cavendish** (c. 1781) found an early Ohm's law (ATX 2019 Q2 uses this). The middle-name hint suggests the writer had Joule in mind, so the key is wrong on the history. Source: `downloads/dlG/wiki/Shock_Value_Network_Analysis_Test.txt` lines 124-133 and `..._Answer_Key.txt` lines 56-61.
3. **Shock Value Network Analysis tiebreaker 5** ("the scientist that the first circuit is named after"): key **Sir Charles Wheatstone**. No circuit is the "first circuit". The Wheatstone bridge was invented by Samuel Christie (1833); Wheatstone made it popular (1843). The wording is unclear.
4. **UT Invitational 2019 B and C Q9** ("Published 'On Magnetism'"): key **Coulomb**. The famous book *De Magnete* ("On the Magnet", 1600) is by **Gilbert**. Coulomb wrote memoirs on electricity and magnetism. Source: `tests/ut-b.md` line 133, `tests/ut-c.md` line 264.
5. **USC Invitational Jan 2026 B Q25** (Tesla false statement): the B scantron says C; the B solution PDF and the C scantron say A. Source: `tests/nat26.md` key-error 1. Out of scope for 2027 anyway.
6. **UF Northern Regional 2020 B Q3** asks who found that a conductive container blocks "electromagnetic fields". Faraday showed it for electrostatic fields (1836). Blocking changing EM waves depends on hole size and frequency (3.d frequency analysis is out). The key (Faraday) is fine; the wording mixes two ideas.
7. **ATX Regional 2019 B/C Q3**: the choice "J.J Thompson" is misspelled (Thomson). Not a key error.

Not confirmed

- **Ohl's PN junction date:** the Computer History Museum says **23 February 1940**. Wikipedia's Russell Ohl page says **1939**. I used 1940 (CHM, more specific). Looked at: computerhistory.org timeline, en.wikipedia.org/wiki/Russell_Ohl.
- **Thales and amber:** Wikipedia's Thales page reports Aristotle on the lodestone and amber having "souls". No writing by Thales survives. The amber story comes from writers centuries later. Treat it as tradition.
- **Franklin coined "battery" (1749):** standard textbook fact; not re-checked online in this run. The Wikipedia kite-experiment page did not mention it.
- **Franklin coined "positive", "negative", "charge":** standard fact [G]; not re-checked online in this run.
- **Joule details** (1840-41 heating paper, 1843-1850 paddle wheel, brewer from Salford): [G], not re-checked online. The 1889 adoption of the joule unit is confirmed [W].
- **Watt details** (separate condenser patent 1769, horsepower definition): [G], not re-checked. The horsepower number (745.7 W) was computed in Python from 33,000 ft·lbf/min.
- **Werner von Siemens details** (1847 company, 1860 mercury unit, 1866-67 dynamo, 1879 railway): [G], not re-checked online. The 1971 CGPM adoption of the siemens is confirmed [W][BIPM].
- **Which Siemens the unit honours:** most sources say Werner von Siemens. The CGPM resolution text did not appear in my search snippet.
- **Who proposed the watt in 1882:** one source says William Siemens proposed the watt; another says he proposed the joule (as a unit of heat). Both say the 1889 congress adopted both.
- **Maxwell dates and facts** (born 13 June 1831, died 5 November 1879, 1871 Cavendish chair, 1861 colour photo, editing Cavendish 1879): [G], not re-checked online.
- **Niagara Falls power reaching Buffalo in 1896** and opening in 1895: [G] / [PR I_bank]. Wikipedia's war-of-the-currents page only gave the 1892 contract.
- **Edison/Tesla 1915 Nobel rumour:** [G], not checked.
- **De Morgan details** (born in Madurai, 1828 chair, coined "mathematical induction" 1838, taught Ada Lovelace): [G], not re-checked online.
- **Shannon details** (thesis title, 1948 paper, Tukey coined "bit"): [G] and [PR I_bank]; not re-checked online.
- **Zener diodes above 5-6 V use avalanche breakdown:** [G]; not checked online. Wikipedia's Clarence Zener page does not discuss it.
- **Holonyak "father of the LED" nickname:** [G]; the Wikipedia summary I read did not include it.
- **Akasaki died 2021:** [G]; not re-checked.
- **Biard and Pittman infrared LED (1961-62):** [G]; not re-checked.
- I did not find any past test question about Thales, Gilbert (as the answer), Watt (as a person), Siemens (as a person), Shannon, De Morgan (as a person), Ohl, Holonyak, the blue-LED team or Zener (as a person). Searched: all `.txt` files in `research/downloads/` (dlG, dlB, dlFind, dl), all 8 files in `research/tests/`, `prior/G_tests.md`, `prior/I_bank.md`.

Counts

- Question types: **41** (OH-01 to OH-41).
- ✅ IN: **16** (OH-01, 04, 06, 07, 08, 10, 11, 14, 15, 16, 19, 26, 27, 37, 38, 39).
- ⚠️ Borderline: **17** (OH-02, 03, 05, 09, 12, 13, 17, 18, 20, 21, 22, 23, 24, 25, 28, 29, 30).
- ❌ Out: **8** (OH-31, 32, 33, 34, 35, 36, 40, 41).
- ✅ States-Nationals: 0 (no KVL/KCL content in this note).

Sources

- 2027 Div B rules: `/Users/kai/Daniel/Science_Olympiad_Div_B_Rules_2027_for_Web_Secured.pdf` (Circuit Lab B, Part I 3.c-3.e, manual page B12; text from `research/downloads/dlB/r2027.txt` lines 671-704).
- Test-extraction files: `/Users/kai/Daniel/circuitlab/research/tests/` (atx-reg, misc-a, misc-b, nat26, state, ut-a, ut-b, ut-c) and `/Users/kai/Daniel/circuitlab/research/prior/G_tests.md`.

- Past tests (local, `/Users/kai/Daniel/circuitlab/research/downloads/`): `dlG/atx19reg_B.txt`, `atx19reg_C.txt`, `atx19reg_key.txt` (Q2, Q3, Q11); `dlG/atx20reg_Btest.txt`, `atx20reg_Ctest.txt`, `atx20reg_key.txt` (Q5, Q8, Q11, Q17, Q20/22); `dlG/atx26reg_Btest.txt` (MC2, MC7); `dlFind/nfl20reg_Btest.txt`, `nfl20reg_Bkey.txt` (Q1-Q7); `dlG/gull26_test.txt` (Q28-Q36); `dlG/nv26C_test.txt` (Q45, Q46); `dlG/hh26_test.txt` (Q4); `dlB/tests/phi26_test.txt`, `phi26_key.txt` (Q15); `dlG/pz/p19/Circuit Lab/2019 Princeton Circuit Lab Test.txt` (Part 1); `dlG/pz/p20/Circuit Lab/2020 Princeton Circuit Lab Answer Key.txt` (Q1, SR2); `dlG/pz/p21/Circuit Lab/2021 Princeton Circuit Lab Key.txt` (Q9, Q10); `dlG/ut19inv_Btest.txt`, `ut19inv_Bkey.txt`, `ut19inv_Ctest.txt`, `ut19inv_Ckey.txt` (Q2, Q3, Q8, Q9); `dlG/wiki/Shock_Value_Network_Analysis_Test.txt` and `..._Answer_Key.txt` (tiebreakers 1-5); `dlG/wiki/Shock_value_test.txt` (Q52, Q53); `dlG/wiki/Shock_value_test_.txt` (IIa units); `dlG/puso26_test.txt`, `puso26_key.txt` (Zener 7c); `dlFind/cobra19_Btheory_key.txt`; `dl/cl_wiki.txt` (Franklin, Joule, Galvani lines).
- Prior research: `/Users/kai/Daniel/circuitlab/research/prior/I_bank.md` (lines 69, 84, 92, 156-157, 199-234, 280, 389, 490, 657), `G_tests.md`, `G_manifest.md`, `A_rules.md`.
- Other vault notes: Ampere A1, Boole A1, Coulomb A1, Hertz A1, Kirchhoff A1, Ohm A1, Volta A1, Units Named After People A1.
- https://en.wikipedia.org/wiki/Russell_Ohl
- https://www.computerhistory.org/siliconengine/discovery-of-the-p-n-junction/ (23 Feb 1940; Scaff; p and n names)
- https://en.wikipedia.org/wiki/Nick_Holonyak
- https://en.wikipedia.org/wiki/Clarence_Zener
- https://physicsworld.com/a/isamu-akasaki-hiroshi-amano-and-shuji-nakamura-win-2014-nobel-prize-for-physics/ and https://amp.spie.org/news/1007-nobel-physics (Nobel 2014 motivation; nobelprize.org returned HTTP 403)
- https://en.wikipedia.org/wiki/Siemens_(unit) and https://www.bipm.org/en/-/resolution-cgpm-14- (siemens, 1971)
- https://arxiv.org/pdf/1512.07306 (1882 British Association; 1889 congress adopted joule and watt)
- https://en.wikipedia.org/wiki/Hans_Christian_%C3%98rsted
- https://en.wikipedia.org/wiki/Benjamin_Franklin
- https://en.wikipedia.org/wiki/Kite_experiment
- https://en.wikipedia.org/wiki/War_of_the_currents
- https://en.wikipedia.org/wiki/Topsy_(elephant)
- https://en.wikipedia.org/wiki/William_Gilbert_(physicist)
- https://en.wikipedia.org/wiki/Thales_of_Miletus
- Textbooks for "not seen in a test" types: OpenStax *College Physics 2e* (ch. 7.6 power and horsepower; ch. 18.1 static electricity history; ch. 20 current, resistance, Ohm's law; ch. 20.6 electric hazards); Kuphaldt *Lessons in Electric Circuits* vol. 1 (ch. 1 static electricity, conventional vs electron flow; ch. 5 Wheatstone bridge context) and vol. 3 (diodes, Zener diodes, LEDs); vol. 4 (Boolean algebra, Shannon).
- Python checks: `/private/tmp/claude-502/-Users-kai-Daniel/893762dd-d9b3-4158-937c-ea93eb65d1e6/scratchpad/other/check.py` (horsepower, Joule heat and calories, line loss, Edison lamp, conductance, Zener current, year gaps, timeline sort, series/parallel and De Morgan truth tables, LED kWh savings).

TAB A2

Static Electricity and Coulomb

Charge Basics 169

Charging Methods, Grounding and Electroscopes 177

Coulomb's Law 185

Electric Field and Potential 194

Static Hazards 207

Each note: key facts, formulas, then question types from Easy to Nationals with worked examples.

Charge Basics

This note covers what electric charge is, how to count it, and the triboelectric series. These questions are short, but almost every test has two or three of them.

Everything here is ✅ **IN at every level**. It is rule 3.c.ii: "Properties of electric charges/fields, sources/hazards of static electricity, and Coulomb's Law". A few trivia items are ⚠️ borderline. Each one says why.

Key facts

- **Electric charge** is a property of particles. It makes them push or pull on each other. Its symbol is q or Q.
- There are two kinds of charge: **positive (+)** and **negative (–)**.
- **Like charges repel.** (+ and +, or – and –, push apart.)
- **Unlike charges attract.** (+ and – pull together.)
- The unit of charge is the **coulomb (C)**. 1 C = 1 ampere × 1 second (1 A·s).
- A coulomb is a very large charge. Static charges are usually **nC** (nanocoulomb, 10^{-9} C) or **µC** (microcoulomb, 10^{-6} C).
- A **proton** has charge +e. An **electron** has charge –e. A **neutron** has no charge.
- The **elementary charge** e = 1.602 × 10^{-19} C. It is the smallest free charge. Its order of magnitude is 10^{-19} C.
- **Charge is quantized.** This means every charge is a whole number of e: q = n·e. You can have 3e, but not 1.5e.
- **Charge is conserved.** Charge is never made or destroyed. It only moves from one object to another.
- In a solid, only **electrons** move. Protons stay locked in the nucleus.
- A **positive** object has **lost** electrons. It did not gain protons.
- A **negative** object has **gained** extra electrons.
- A **neutral** object has equal numbers of protons and electrons. Its net charge is 0 C. Most everyday objects are neutral.
- 1 C equals the charge of 6.24 × 10^{18} electrons.
- **Conductors** let electrons move freely. Examples: metals (copper, aluminum, silver, gold, titanium), salt water, the human body, the ground.
- **Insulators** hold electrons in place. Examples: glass, rubber, plastic, wood, dry air, silk, Teflon.
- The names "positive" and "negative" were chosen by Benjamin Franklin. The choice is arbitrary. It could have been the other way round.
- The **triboelectric series** is a list of materials. It is ranked by how easily each material gives away electrons when rubbed.
- The **triboelectric effect** is charging by rubbing or contact. Two different materials touch, then separate. Electrons move from one to the other.

Triboelectric series (common textbook order)

Positive end (loses electrons)	→	→	Negative end (gains electrons)
human skin, rabbit fur, glass, human hair, nylon, wool	lead, silk, aluminum, paper	cotton (near neutral), steel, wood, amber	hard rubber, nickel/copper, polyester, styrofoam, PVC, silicone, Teflon

- The material **higher** in the list (nearer the + end) **loses** electrons. It becomes **positive**.
- The material **lower** in the list **gains** electrons. It becomes **negative**.
- The farther apart two materials are, the more charge usually moves.
- Different sources put the middle of the list in a different order. The two ends are almost always the same.
- Classic pairs: **glass rubbed with silk → glass +, silk –. Rubber (or plastic) rubbed with fur or wool → rubber –, fur +. Balloon rubbed on hair → balloon –, hair +.**

Other ways to separate charge (names a test may ask for)

Effect	What makes the charge	Example
Triboelectric	rubbing or contact between two materials	balloon on hair, shoes on carpet

Effect	What makes the charge	Example
Piezoelectric	squeezing or bending a crystal	quartz watch, gas-grill igniter, crystal microphone
Pyroelectric	heating or cooling a crystal	some motion sensors
Thermoelectric	a temperature difference across two joined metals	thermocouple
Photoelectric	light knocks electrons out of a metal	solar cell idea, light sensors
Induction (charge-induced)	a nearby charge pushes charges around	see Charging Methods and Electroscopes A2

Formulas

What	Formula	Notes
Charge from a count	$q = n \cdot e$	n = number of electrons or protons; sign – for electrons
Count from a charge	$n =$	q
Electrons in 1 C	$1/e = 6.24 \times 10^{18}$	
Conservation	q_total before = q_total after	add charges with their signs
Charge from current	$q = I \cdot t$	t in seconds; see CL 4 - Current, Voltage, Resistance, Power, Energy T
Mass of added electrons	$m = n \cdot m_e$	$m_e = 9.11 \times 10^{-31}$ kg

Constants: $e = 1.602 \times 10^{-19}$ C. $m_e = 9.11 \times 10^{-31}$ kg. $m_p = 1.67 \times 10^{-27}$ kg.

Question types

Easy

1. Attract or repel from the signs

- Scope: ✅ IN.
- Where seen: GullSO Dec 2025 C/B (INV) Q1 ("which pairs attract? select all": –/+ and +/–). Deck 6 practice quiz (STAT-CHARGE-SIGN: proton and electron → attract). North Pocono Jan 2020 C (INV) Q8 (proton and electron → equal, opposite, attractive forces). Berkeley Feb 2026 B (INV) Q8 (a + charge near a – charge moves toward it).
- How to solve:
  1. Look only at the signs.
  2. Same signs → repel. Opposite signs → attract.
  3. The size of the charges does not change the answer.
- Worked example: Which pairs attract: (A) –/–, (B) –/+, (C) +/–, (D) +/+? Answer: **B and C**.
- Traps: "Select all" questions often have two right answers. A neutral object is attracted to both signs (see polarization in Charging Methods and Electroscopes A2).
- Calculator: look up — type: attract repel signs.

2. Value or order of magnitude of the elementary charge

- Scope: ✅ IN.
- Where seen: Berkeley Feb 2026 B (INV) Q5 (nearest order of magnitude → 10^{-19} C). NY Lake Erie/Niagara Dec 2018 C (INV) Q27-28 (charge of a proton and an electron, $\pm 1.602 \times 10^{-19}$ C). "Shock Value" team tryout B (about 2012, practice) I-13 (which particle has the same size charge as an electron → proton). F_slides deck STAT-ELECTRON-CHARGE.
- How to solve:
  1. Memorize $e = 1.602 \times 10^{-19}$ C.
  2. Proton = +e. Electron = –e. Neutron = 0.
- Worked example: To the nearest order of magnitude, what is e? Answer: **10^{-19} C**.
- Traps: The electron's charge is negative. Some keys want "-1.602×10^{-19} C" with the sign. The proton and electron have equal size charges but very different masses.
- Calculator: look up — type: elementary charge.

3. Charge from a number of electrons

- Scope: ✅ IN.
- Where seen: UF Northern Regional Jan 2019 B (REG) Q12 (1.25×10^{19} electrons → –2 C). I_bank Q1.
- How to solve:
  1. q = n × e.
  2. Electrons give a negative charge. Protons (or "electrons removed") give a positive charge.
- Worked example: An object gains 1.25×10^{19} electrons. Its charge = $1.25 \times 10^{19} \times 1.602 \times 10^{-19}$ = **–2.00 C**. (Python: –2.0025 C.)
- Traps: Do not forget the minus sign. "Removed" electrons make the object positive.
- Calculator: calculate — type: electrons count.

4. Number of electrons from a charge

- Scope: ✅ IN.
- Where seen: MIT Jan 2019 C (INV) Q16 (electrons in 1 C → 6.24×10^{18}). I_bank Q2.
- How to solve:
  1. n = |q| ÷ e.
  2. Say "gained" if q is negative and "lost" if q is positive.
- Worked example 1: How many electrons make 1 C? n = $1/1.602 \times 10^{-19}$ = **6.24×10^{18}**.
- Worked example 2: A rod has –3.2 nC. n = $3.2 \times 10^{-9} / 1.602 \times 10^{-19}$ = **2.0×10^{10} electrons gained**.
- Traps: Change nC to C first ($\times 10^{-9}$). A non-whole answer means a rounding slip or an impossible charge.
- Calculator: calculate — type: charge to electrons.

5. Which particle moves: "lost electrons" or "gained protons"?

- Scope: ✅ IN.
- Where seen: Pembroke Hill Dec 2025 B (INV) Q19 (glass rubbed with silk becomes positive: gained protons or lost electrons? → lost electrons). Berkeley Feb 2026 B (INV) Q3 (electrons go from yarn to plastic → yarn +, plastic –).
- How to solve:
  1. Only electrons move in solids.
  2. The object that loses electrons becomes positive.
  3. The object that gains electrons becomes negative.
- Worked example: Electrons transfer from yarn to plastic. Signs of yarn and plastic? Answer: **yarn positive, plastic negative**.
- Traps: Both objects become charged. The two charges are equal in size and opposite in sign (charge is conserved). An answer with "zero" for one object is wrong.
- Calculator: look up — type: which particle moves.

6. Charge conservation (split or combine)

- Scope: ✅ IN.
- Where seen: UF Northern Regional Jan 2019 B (REG) Q39 (a –6 C object breaks; one piece is +2 C; the other → –8 C).
- How to solve:
  1. Total before = total after.
  2. Add with signs.
- Worked example: A –6 C object breaks in two. One piece is +2 C. The other = –6 – (+2) = **–8 C**.
- Traps: Students often write –4 C. Write the equation: –6 = +2 + x.
- Calculator: calculate — type: conservation charge split.

7. Unit of charge and its definition

- Scope: ✅ IN.
- Where seen: UF Northern Regional Jan 2020 B (REG) Q4 (SI unit of charge → coulomb). ATX Regional 2021 B (REG) Q2 ("the unit of charge bears my name" → Coulomb, a history crossover). Rickards Oct 2025 B (INV) Q22 lists "Coulomb (C)" as a wrong choice for the unit of magnetic field (that question itself is ❌ out: magnetism).

- How to solve:
  1. Charge → coulomb (C).
  2. 1 C = 1 A·s. So 1 mAh = 3.6 C.
- Worked example: Express 1 C in base units. Answer: **1 A·s** (ampere-second).
- Traps: "Coulomb" is the unit and the scientist. Do not mix up C (coulomb) with °C or with capacitance's farad (F).
- Calculator: look up — type: unit charge.

8. Net charge of everyday objects

- Scope: ✅ IN.
- Where seen: Florida State 2021 B (STATE) Q24 (net charge of a graham cracker → 0 C).
- How to solve:
  1. Ordinary objects are neutral.
  2. Neutral means net charge 0 C, even though it holds huge numbers of protons and electrons.
- Worked example: Net charge of a graham cracker? Answer: **0 C**.
- Traps: Do not count the protons. The question asks for the net charge.
- Calculator: look up — type: neutral net charge.

Medium

9. Charge quantization: which charge is possible?

- Scope: ✅ IN.
- Where seen: Florida State 2021 B (STATE) Q9 ("why is charge quantized?" → none of the listed reasons; the key is D). The "which value can exist" form is not seen in a test (OpenStax College Physics 2e §18.1).
- How to solve:
  1. Divide the charge by e.
  2. A whole number → possible. Not whole → impossible.
- Worked example: Which charge can an oil drop have: (A) 2.4×10^{-19} C or (B) 4.806×10^{-19} C? A ÷ e = 1.50 (no). B ÷ e = 3.00 (yes). Answer: **B (3e)**.
- Traps: Quarks have charges of ⅓e and ⅔e, but they are never found alone. So free charges are still whole multiples of e.
- Calculator: calculate — type: quantization check.

10. Positive and negative are just names (Franklin's convention)

- Scope: ✅ IN.
- Where seen: Florida State 2021 B (STATE) Q11 (Shakespeare's "a rose by any other name" applies to which idea? → the names positive and negative).
- How to solve:
  1. Know that Benjamin Franklin picked the names around 1747.
  2. The physics does not change if you swap the names.
  3. His choice is why conventional current flows opposite to electron flow.
- Worked example: Why is an electron "negative"? Answer: **by Franklin's arbitrary convention**, not because of anything inside the electron.
- Traps: Franklin is not on the 2027 history list. This question is about charge, not history.
- Calculator: look up — type: Franklin sign convention.

11. Conductor or insulator?

- Scope: ✅ IN.
- Where seen: Rickards Oct 2025 B (INV) Q12 (pick all good conductors: aluminum, titanium; not glass, plastic). Princeton Feb 2020 C (INV) Q24 (rank conductivity: fused quartz < germanium < lead < gold).
- How to solve:
  1. Metals are conductors.
  2. Glass, rubber, plastic, wood, dry air, quartz are insulators.
  3. Silicon and germanium are semiconductors (in between).

- Worked example: Rank from least to most conductive: lead, gold, germanium, fused quartz. Answer: **fused quartz < germanium < lead < gold**.
- Traps: Pure water is a poor conductor. Tap water and salt water conduct. The human body conducts.
- Calculator: look up — type: conductor insulator list.

12. Triboelectric series: which object becomes positive?

- Scope: ✅ IN.
- Where seen: Pembroke Hill Dec 2025 B (INV) Q19. Berkeley Feb 2026 B (INV) Q3. Princeton Feb 2026 C (INV) MC4 ("highest tendency to make static by friction": copper, water, glass, rubber → key: rubber). UF Northern Regional Jan 2019 C (REG) Q9 (a cat covered in packing peanuts → triboelectric effect). F_slides STAT-TRIBO-CHARGING.
- How to solve:
  1. Find both materials in the series.
  2. The one nearer the + end loses electrons → positive.
  3. The other gains electrons → negative.
- Worked example: Wool is rubbed on a PVC pipe. Wool is near the + end. PVC is near the – end. Answer: **wool +, PVC –**.
- Traps: The middle of the list changes between sources. If a test gives its own list, use the test's list. Metals like copper are poor at holding static charge when held in a hand, because the charge leaks away through the body.
- Calculator: look up — type: tribo series pair.

13. Triboelectric series: which pair makes the most charge?

- Scope: ✅ IN.
- Where seen: not seen in a test. Based on the series in OpenStax College Physics 2e §18.1 and the triboelectric tables used by Science Olympiad decks.
- How to solve:
  1. Find each material's place in the list.
  2. The pair farthest apart usually transfers the most charge.
- Worked example: Which pair gives the most charge: glass and silk, glass and Teflon, or wool and paper? Glass is near the + end. Teflon is at the – end. Answer: **glass and Teflon**.
- Traps: Two pieces of the same material give almost no charge.
- Calculator: look up — type: tribo farthest pair.

14. Name the charge-separation effect

- Scope: ✅ IN.
- Where seen: Berkeley Feb 2026 B (INV) Q4 (rubbing → triboelectric effect). Rickards 2021 season B (INV) Q11 (which is NOT a charge-separation effect → photoelectric, in that key's list). UF Northern Regional Jan 2019 C (REG) Q10 (pick the four types: contact-, charge-, heat-, pressure-induced). "Shock Value" tryout B (about 2012, practice) I-3 (electricity from crystals → piezoelectricity). Thunderstorm version: UT Invitational fall 2018 B/C (INV) Q19.
- How to solve:
  1. Rubbing or contact → triboelectric.
  2. Pressure on a crystal → piezoelectric.
  3. Heat on a crystal → pyroelectric.
  4. A nearby charge → induction.
- Worked example: A gas-grill lighter makes a spark when you press a button. A spring hammer hits a crystal. Answer: **piezoelectric effect**.
- Traps: "Thermoelectric" (two metals at different temperatures) and "pyroelectric" (a crystal that is heated) are different effects.
- Calculator: look up — type: charge separation effects.

15. Charge carried by a current

- Scope: ✅ IN (rule 3.c.iv also).

- Where seen: Berkeley Feb 2026 B (INV) Q20 (30 mA for 2 minutes → 3.6 C). MIT Jan 2020 C (INV) Q11 (charge from a current-time graph → 110 kC).
- How to solve:
  1. Change time to seconds and current to amperes.
  2. $q = I \times t$.
- Worked example: 30 mA for 2 min. $q = 0.030 \times 120 =$ **3.6 C**.
- Traps: Minutes must become seconds. mA must become A.
- Calculator: calculate — type: current time charge.

Hard

16. Mass gained by charging

- Scope: ✅ IN.
- Where seen: not seen in a test. OpenStax College Physics 2e §18.1 style.
- How to solve:
  1. $n = |q|/e$.
  2. Added mass = $n \times m_e$.
- Worked example: A sphere gets –8.0 nC. $n = 8.0 \times 10^{-9} / 1.602 \times 10^{-19} = 4.99 \times 10^{10}$ electrons. Mass gained = $4.99 \times 10^{10} \times 9.11 \times 10^{-31} =$ **4.5×10^{-20} kg**. (Python: 4.55×10^{-20} kg.)
- Traps: A positive object loses this much mass. The change is far too small to weigh.
- Calculator: calculate — type: charge to mass.

17. Net charge of the Earth (trivia)

- Scope: ⚠️ Borderline. It is about charge, but it is fact trivia that no textbook chapter teaches.
- Where seen: Florida State 2021 B (STATE) Q23 (key: –500,000 C).
- How to solve:
  1. The ground surface holds about -5×10^5 C.
  2. The atmosphere holds about the same amount of positive charge.
  3. So the Earth plus its air is about neutral.
- Worked example: "Net charge of the Earth, including the atmosphere"? Best physics answer: **about 0 C**. The surface alone is about -5×10^5 C.
- Traps: Read whether the question includes the atmosphere. The key picked –500,000 C anyway (see "Mistakes in tests and keys").
- Calculator: look up — type: earth charge.

18. Quark charges (trivia)

- Scope: ⚠️ Borderline. Charge is IN, but quarks are particle physics. The rules do not list them.
- Where seen: Florida State 2021 B (STATE) Q10 (charge of the "up-dpwn" quark; the key is +⅔, the wording is garbled).
- How to solve:
  1. Up quark = +⅔e. Down quark = –⅓e.
  2. Proton = up + up + down = +1e. Neutron = up + down + down = 0.
- Worked example: Check the proton: ⅔ + ⅔ – ⅓ = **+1** (in units of e).
- Traps: Quarks do not break the "whole number of e" rule for free particles.
- Calculator: look up — type: quark charge.

19. Electric force vs. the particle picture: "electricity is"

- Scope: ✅ IN.
- Where seen: JC Booth 2011 B (INV) Q33 ("electricity is" → charged particles). Cobra 2019 B (INV) Q8 (a charge makes an electric field; a moving charge also makes a magnetic field).
- How to solve:
  1. Electricity is the behavior of charged particles: at rest (static) or moving (current).

- Worked example: A still charge makes a ___ field. Answer: **electric**. (The magnetic part of the Cobra question is ❌ out, since magnetism is not a listed area, rule 3.c.)
- Traps: Do not answer "electrons only". Ions also carry charge in liquids and gases.
- Calculator: look up — type: what is electricity.

Nationals

20. Fraction of electrons removed from a coin

- Scope: ✅ IN (algebra only; uses the Avogadro number as a given constant).
- Where seen: not seen in a test. OpenStax College Physics 2e §18.1 problems use this idea.
- How to solve:
  1. Moles = mass ÷ molar mass.
  2. Atoms = moles × 6.022×10^{23}.
  3. Electrons in the coin = atoms × electrons per atom (29 for copper).
  4. Electrons removed = q/e.
  5. Fraction = removed ÷ total.
- Worked example: A 3.0 g copper coin (63.5 g/mol, 29 electrons per atom) gets +2.0 μC. Total electrons = $3.0/63.5 \times 6.022 \times 10^{23} \times 29 = 8.25 \times 10^{23}$. Removed = $2.0 \times 10^{-6}/1.602 \times 10^{-19} = 1.25 \times 10^{13}$. Fraction = **$1.5 \times 10^{-11}$**.
- Traps: The fraction is tiny. Static charge changes almost nothing in the number of electrons.
- Calculator: calculate — type: fraction electrons removed.

21. Millikan oil drop: find n from a hovering drop

- Scope: ⚠️ Borderline. Millikan is not named in the rules, but the experiment proves charge quantization (3.c.ii). The full worked method is in Electric Field and Potential A2 (type 34).
- Where seen: not seen in a test. I_bank Q31.
- How to solve: q = mg/E, then n = q/e.
- Worked example: see Electric Field and Potential A2. Answer there: **3e**.
- Traps: Millikan is a scientist off the 2027 history list. A history question about him would be out.
- Calculator: calculate — type: millikan.

Out of scope (skip)

- ❌ Magnetic force on a moving charge (qvB), right-hand rules, mass spectrometers. Magnetism is not one of the listed areas (rule 3.c, "areas only"). Seen: ATX Regional 2021 B (REG) Q31, Q50-53.
- ❌ Gauss's law with calculus, Maxwell's equations (div E = ρ/ε_0). Rule 3.e.v: no calculus. Seen: Florida State 2021 B (STATE) Q14-16.
- ❌ Capacitor charge Q = CV. Capacitance is excluded (rule 3.d). Seen: Pembroke Hill Dec 2025 B (INV) Q5; Rickards Oct 2025 B (INV) Q13.

Mistakes in tests and keys

- **Florida State 2021 B Q23**: The key says the Earth's net charge, "including the atmosphere", is −500,000 C. The surface is about -5×10^5 C, but the air above holds about $+5 \times 10^5$ C. So the whole system is about 0 C. The key is likely wrong (from tests/state.md).
- **Florida State 2021 B Q10**: The question says "up-dpwn quark". It is garbled. +⅔ fits an up quark.
- **Princeton Feb 2026 C MC4**: The key says rubber has the "highest tendency to generate static by friction". Glass is also near the end of the series. Rubber is the better answer only because it holds charge well and sits at the far − end. The question is weak but the key is defensible.
- **Berkeley Feb 2026 B Q3**: The choices are fine. Note that "zero" for one object is always wrong in a rubbing question.

Not confirmed

- The Earth surface charge (about -5×10^5 C) and the atmosphere's balancing charge come from the prior research note tests/state.md. I did not check them against a primary source in this run.
- The triboelectric order in the middle of the list differs between sources. I used the order in prior/I_bank.md and prior/B_topics.md. I did not re-check it online in this run.
- The Rickards Oct 2025 B Q12 key answer is highlighted, not text. I did not render the key page, so "aluminum, titanium" is my own answer (looked in rsoi26B_key.txt).

Sources

- Rules: Science Olympiad 2027 Div B rules, Circuit Lab, rule 3.c.ii, 3.d and 3.e (pdftotext of the rules PDF).
- Tests (extractions in /Users/kai/Daniel/circuitlab/research/tests/ and prior/G_tests.md): Berkeley Feb 2026 B; Pembroke Hill Dec 2025 B; GullSO Dec 2025; MIT Jan 2019 C; UF Northern Regional 2019 B/C and 2020 B; Florida State 2021 B; JC Booth 2011 B; NY Lake Erie/Niagara 2018 C; Rickards Oct 2025 B and 2021 season B; Princeton 2020 C and Feb 2026 C; Cobra 2019 B; ATX Regional 2021 B; UT Invitational fall 2018.
- Raw texts read: downloads/dlB/tests/berk26_test.txt, berk26_key.pdf (pages 2-3 rendered), phi26_test.txt, phi26_key.txt; downloads/dlG/puso26_test.txt, puso26_key.txt.
- Textbooks: OpenStax College Physics 2e, chapter 18 (Electric Charge and Electric Field), §18.1-18.2. Kuphaldt, Lessons in Electric Circuits Vol. I, ch. 1 (static electricity).
- Numbers checked in Python (scratchpad a2/chk.py).

Charging Methods, Grounding and Electroscopes

Note: Charging Methods and Electroscopes

This note covers how objects get charged, where the charge goes on a conductor, and how an electroscope shows charge.

Everything here is ✅ **IN at every level** (rule 3.c.ii, "properties of electric charges"). The step-by-step sequence puzzles are the hardest type. Top invitationals use them as tiebreakers.

Key facts

- **Charging by friction (triboelectric charging):** rub two different insulators. Electrons move from one to the other. The two objects get **equal and opposite** charges. See Charge Basics [A2] for the series.
- **Charging by conduction (by contact):** touch a charged object to another object. Some charge flows across. The second object gets the **same sign** as the first.
- **Charging by induction:** bring a charged object **near** a conductor without touching. Ground the conductor, then break the ground, then take the rod away. The conductor gets the **opposite sign** to the rod.
- **Polarization:** a charged object makes the charges inside a nearby neutral object shift. The near side gets the opposite sign. The far side gets the same sign. The object stays neutral overall.
- In a **conductor**, electrons move through the whole object. In an **insulator**, each molecule only stretches a little (its electrons shift toward one side of each atom).
- A neutral object is always **attracted** to a charged object. The near opposite charge is closer, so its pull is stronger than the push on the far side (force falls as $1/r^2$).
- So **attraction does not prove** that an object is charged. Only **repulsion** proves that two objects have the same sign.
- **Grounding** means connecting an object to the Earth with a conductor. The Earth is so big that it can give or take any number of electrons without changing.
- A grounded conductor near a + rod pulls electrons **up from** the ground. A grounded conductor near a – rod pushes electrons **down into** the ground.
- Touching a charged metal object with your hand grounds it through your body. The charge leaves.
- **Bonding** means connecting two objects together with a wire, so they have the same potential. Grounding connects them to Earth.
- On a **conductor**, excess charge goes to the **outer surface**. The field inside the metal is zero.
- Charge **piles up at sharp points** and edges. The field is strongest there. This is why lightning rods are pointed.
- Two **identical** metal spheres that touch share their total charge **equally**.
- Two **different-size** spheres joined by a wire end at the **same potential** (voltage). The bigger sphere gets more charge: Q is proportional to its radius r.
- An **electroscope** is a device that shows charge. The common kind has two thin gold or aluminum **leaves** on a metal rod, with a metal **knob** on top, inside a glass jar.
- When the electroscope is charged, both leaves get the same sign. They **repel and spread apart**.
- A **pith ball** is a light ball hung on a silk thread. It is attracted to any charged object. After it touches the object, it gets the same sign and is **repelled**.
- A **Faraday cage** is a closed metal box or mesh. Charges on it move so that the field inside is zero. A charge inside an ungrounded cage still pushes charge to the outside surface.

Charging methods at a glance

Method	Touch ?	Final sign	Needs a ground?
Friction	rub	opposite on the two objects	no
Conduction	yes	same as the charging object	no
Induction (one object)	no	opposite to the rod	yes, then break it before removing the rod

Method	Touch ?	Final sign	Needs a ground?
Induction (two touching spheres)	no	near sphere opposite to rod, far sphere same as rod	no (separate the spheres while the rod is near)
Polarization	no	stays neutral	no

Electroscope rules

Electroscope state	What you bring near the knob (not touching)	Leaves
Neutral	+ or – rod	spread (charges shift by induction)
Charged +	+ rod	spread **more**
Charged +	– rod	close a little (strong rod: close fully, then spread again)
Charged +	neutral conductor (your hand)	close a little
Charged (any)	touch with a finger	**close fully** (grounded)
Charged –	– rod	spread more
Charged –	+ rod	close a little

Formulas

What	Formula	Notes
Identical spheres touch	each = $(q_1 + q_2)/2$	three spheres touching at once: $(q_1 + q_2 + q_3)/3$
Force after sharing	F_new/F_old = $(q_1' q_2')/(q_1 q_2)$ at the same distance	from Coulombs Law A2
Joined spheres (far apart)	$Q_1/Q_2 = r_1/r_2$	same potential: $kQ_1/r_1 = kQ_2/r_2$
Surface charge density	$\sigma = Q/(4\pi r^2)$	$\sigma_1/\sigma_2 = r_2/r_1$ for joined spheres
Sphere in a metal shell	inner surface of shell = –(inner charge); outer surface = shell net + inner charge	from "E = 0 inside the metal"
Potential of a charged sphere	$V = kQ/r$	outside and on the surface

Question types

Easy

1. Name the charging method

- Scope: ✅ IN.
- Where seen: Berkeley Feb 2026 B (INV) Q4 (rubbing → triboelectric). F_slides deck practice ("charging by friction, conduction, induction"). I_bank Q5.
- How to solve:
  1. Rubbing → friction (triboelectric).
  2. Touching a charged object → conduction.
  3. Near, but not touching, with a ground step → induction.
- Worked example: A charged rod touches a neutral metal ball. The ball becomes charged. Answer: **conduction (contact)**.
- Traps: "Induction" in this event means charge induction, not electromagnetic induction (that is magnetism and ❌ out).
- Calculator: look up — type: charging method.

2. Polarization: why does a charged object attract a neutral one?

- Scope: ✅ IN.
- Where seen: UT Invitational 2020 B (INV) Q7 (balloon rubbed on wool, brought to a wall → it sticks). UF Northern Regional Jan 2019 B (REG) Q14 (charged insulator near a neutral conductor: the conductor stays uncharged, but a force exists). F_slides STAT-BALLOON-INDUCTION.

- How to solve:
  1. The charged object shifts the charges in the neutral one.
  2. The opposite charge is nearer, so its pull is stronger.
  3. The net force is attraction. The neutral object stays neutral.
- Worked example: A balloon is rubbed on hair and held to a wall. Why does it stick? Answer: **The balloon is negative. It polarizes the wall. The near side of the wall becomes slightly positive. The attraction to the near side beats the repulsion from the far side.**
- Traps: The wall does not become charged. It only polarizes. A neutral object is attracted to a positive object too.
- Calculator: look up — type: polarization balloon.

3. Final sign after conduction or induction

- Scope: ✅ IN.
- Where seen: not seen as a stand-alone test question; it is step 1 of types 5-8. I_bank Q5.
- How to solve:
  1. Conduction → same sign as the source.
  2. Induction → opposite sign to the source.
- Worked example: A negative rod charges a metal sphere by induction. Answer: **the sphere becomes positive**.
- Traps: In conduction the rod loses some charge. In induction the rod keeps all its charge.
- Calculator: look up — type: conduction induction sign.

4. Hands-on: charge a pith ball so it is repelled, then attracted

- Scope: ✅ IN (hands-on).
- Where seen: North Pocono Jan 2020 C (INV) hands-on Q36 (rods and furs given; make the pith ball (a) repelled and (b) attracted).
- How to solve:
  1. Rub a rod with fur. The rod is now charged.
  2. Touch the ball with the rod. The ball gets the same sign. Bring the rod back → **repelled**.
  3. For attraction, use a rod with the opposite charge. Or ground the ball to make it neutral; a neutral ball is attracted.
- Worked example: A rubber rod rubbed with fur is –. The ball touches it and becomes –. The same rod now **repels** it. A glass rod rubbed with silk is +. It **attracts** the – ball.
- Traps: The first contact always shows attraction (a neutral ball). Only after contact does it repel. Humid air lets the charge leak away; work quickly.
- Calculator: look up — type: pith ball.

5. Pith ball or electroscope: what proves a charge?

- Scope: ✅ IN.
- Where seen: not seen in a test. OpenStax College Physics 2e §18.2 (conceptual questions).
- How to solve:
  1. Attraction happens for a charged or a neutral object. It proves nothing.
  2. Repulsion happens only between like charges. It proves both are charged.
- Worked example: A rod attracts a pith ball. Is the ball charged? Answer: **cannot tell; it may be neutral**. A rod repels the ball. Answer: **the ball is charged with the rod's sign**.
- Traps: Test writers like this "attraction is not proof" idea.
- Calculator: look up — type: repulsion proof.

Medium

6. Two touching spheres charged by induction

- Scope: ✅ IN.
- Where seen: UT Invitational 2019 B (INV) Q21 and UT Invitational 2019 C (INV) Q21 (neutral spheres A and B touch; a – rod is near A; separate them → B is negative). I_bank Q7.
- How to solve:

1. The rod pushes or pulls electrons through both spheres.
2. The near sphere gets the opposite sign. The far sphere gets the same sign as the rod.
3. Separate the spheres **before** removing the rod. Their charges are now locked in.
4. The two charges are equal in size and opposite in sign.

- Worked example: Spheres A (near) and B (far) touch. A – rod is near A. You separate them, then remove the rod. Answer: **A positive, B negative, equal amounts**.
- Traps: If you remove the rod first, the charges flow back. Both end neutral.
- Calculator: look up — type: two sphere induction.

7. Induction with a ground: the order of the steps

- Scope: ✅ IN.
- Where seen: not seen in a test. I_bank Q6. OpenStax College Physics 2e §18.2 (Figure 18.12 sequence).
- How to solve:
  1. Bring the rod near (do not touch).
  2. Ground the far side (touch with a finger or a wire).
  3. Remove the ground.
  4. Remove the rod.
  5. The object ends with the opposite sign to the rod.
- Worked example: A + rod is near a metal sphere. You touch the sphere, remove your finger, then remove the rod. Electrons came up from the ground. Answer: **the sphere is negative**.
- Traps: Remove the rod before the ground → the extra charge flows back → **neutral**. The question may ask "which step order fails".
- Calculator: look up — type: induction steps ground.

8. Identical spheres touch: share the charge, then compare forces

- Scope: ✅ IN.
- Where seen: UF Northern Regional Jan 2019 B (REG) Q13 (equal and opposite charges; half of one is moved to the other → force becomes ¼). I_bank Q9. Berkeley Feb 2026 B (INV) §3 1d uses a related "charge tripled → force triples" step.
- How to solve:
  1. Add the charges with signs.
  2. Divide by the number of identical spheres that touch.
  3. Use $F \propto q_1 q_2$ at the same distance for the force ratio.
- Worked example 1: +8 μC and –2 μC identical spheres touch, then go back to the same spacing. Each = (8 – 2)/2 = **+3 μC**. F_new/F_old = (3 × 3)/(8 × 2) = **9/16 = 0.5625**. The force changes from **attraction to repulsion**.
- Worked example 2 (NFL 2019 B Q13): +Q and –Q. Half of +Q (that is, +Q/2) moves to the other. Now +Q/2 and –Q/2. Force ratio = (½ × ½) = **¼**.
- Traps: Only identical spheres share equally. Keep track of the sign change (attract → repel).
- Calculator: calculate — type: touch share force ratio.

9. A chain of touches (A touches B, then B touches C)

- Scope: ✅ IN.
- Where seen: UF Northern Regional Jan 2020 C (REG) Q8 (sphere C touches B then A; the force doubles; find Q_C/Q_B → key 7; I could not verify it, see "Not confirmed"). Not seen in a Div B test in a clean form.
- How to solve:
  1. Make a table with one column per sphere.
  2. For each touch, replace both touching spheres by their average.
  3. Spheres not touching keep their charge.
- Worked example: Identical spheres A = +12 μC, B = 0, C = –4 μC. A touches B: A = B = **+6**. Then B touches C: B = C = (6 – 4)/2 = **+1**. Final: **A = +6 μC, B = +1 μC, C = +1 μC**. Check: total = 8 μC before and after.
- Traps: Use the latest charge after each step. Check that the total charge stays the same.
- Calculator: calculate — type: touch sequence.

10. Electroscope: what do the leaves do?

- Scope: ✅ IN.
- Where seen: not seen in any extracted test (I grepped all downloads for "electroscope"; zero hits). Taught in OpenStax College Physics 2e §18.2 and Kuphaldt Lessons in Electric Circuits Vol. I ch. 1. B_topics lists "the electroscope" as a 2027 topic.
- How to solve:
  1. Leaves spread → the leaves have extra charge of one sign.
  2. A rod of the **same** sign near the knob pushes more charge down → leaves spread **more**.
  3. A rod of the **opposite** sign pulls charge up → leaves **close** (partly).
  4. Touch with a finger → grounded → leaves **fall**.
- Worked example: A negatively charged electroscope's leaves close a little as a rod comes near. What is the rod? Answer: **positive** (a neutral conductor also makes a small drop, but "positive" is the best answer).
- Traps: Spreading **more** is the only sure test of sign. Closing can come from an opposite rod or from a neutral object.
- Calculator: look up — type: electroscope leaves.

11. Charging an electroscope by conduction or by induction

- Scope: ✅ IN.
- Where seen: not seen in a test. OpenStax College Physics 2e §18.2. Kuphaldt Vol. I ch. 1.
- How to solve:
  1. By conduction: touch the knob with a charged rod. The electroscope gets the rod's sign.
  2. By induction: hold the rod near, touch the knob with a finger, remove the finger, then remove the rod. The electroscope gets the opposite sign.
- Worked example: You charge an electroscope by induction with a glass rod rubbed with silk. Glass is +. Answer: **the electroscope is negative**. A – rod brought near later makes the leaves spread more.
- Traps: During the induction steps, the leaves collapse while your finger is on the knob. They spread again only after you remove the rod.
- Calculator: look up — type: electroscope charging.

12. Grounding: what happens when you touch a charged object?

- Scope: ✅ IN.
- Where seen: not seen as a stand-alone test question. It appears inside induction questions (UT Invitational 2020 B Q24) and ESD questions (Static Hazards A2).
- How to solve:
  1. Your body and the Earth are conductors.
  2. The charge flows through you to the ground until the object is neutral.
- Worked example: A + metal sphere on an insulating stand is touched by a grounded wire. Which way do electrons flow, and what is the final charge? Answer: **electrons flow from the ground up to the sphere; the sphere ends neutral**.
- Traps: For a + object, electrons move **toward** it. Positive charge does not "flow out" in a metal.
- Calculator: look up — type: grounding flow.

13. Faraday cage: a charge inside, a charge outside

- Scope: ✅ IN (field properties). The scientist Faraday is ❌ out for history (3.c.i lists only Ampere, Coulomb, Kirchhoff, Volta, Ohm, Hertz, Boole).
- Where seen: Princeton Feb 2020 C (INV) Q25 (charge Q inside an ungrounded Faraday cage; another Q outside → the outside object is **repelled** from the cage). Rickards Oct 2025 C (INV) Q5 (true facts about Faraday cages; that question also mixes in magnetic fields). Berkeley Feb 2026 B (INV) Q2 (who invented a cage that blocks EM radiation → Faraday; history ❌ out).
- How to solve:
  1. The +Q inside pulls –Q onto the inner wall of the cage.
  2. The cage is neutral and ungrounded, so +Q goes to the outer wall.

3. Outside the cage, it acts like a +Q charge.
4. If the cage were grounded, the outer +Q would leave; then no force outside.

- Worked example: Princeton 2020 Q25. Answer: **the outside object is repelled** (both act as +Q).
- Traps: A cage blocks outside fields from getting **in**. It does not stop an inside charge's effect from getting **out**, unless the cage is grounded.
- Calculator: look up — type: faraday cage inside outside.

Hard

14. Where charge sits on a conductor: outer surface and sharp points

- Scope: ✅ IN.
- Where seen: ATX Regional 2026 C (REG) SA4 (teardrop conductor with net charge → densest at the pointed end). B_topics §2. I_bank Q10.
- How to solve:
  1. Excess charge spreads to the **outer surface**.
  2. It is densest where the surface curves most sharply (smallest radius).
  3. The field just outside is strongest there.
- Worked example: A teardrop-shaped conductor has charge −5 nC. Where is the charge density largest? Answer: **at the pointed end**.
- Traps: The charge is not spread evenly. There is **zero** excess charge inside the metal.
- Calculator: look up — type: sharp point charge.

15. Two spheres joined by a wire: charge and density ratios

- Scope: ✅ IN (charge/field properties; algebra only).
- Where seen: USC Invitational Jan 2026 B (INV) Q24 (two far-apart conducting spheres joined by a wire: $\sigma_1/\sigma_2 \rightarrow r_2/r_1$).
- How to solve:
  1. Joined conductors have the same potential: $kQ_1/r_1 = kQ_2/r_2$.
  2. So $Q_1/Q_2 = r_1/r_2$. Split the total in that ratio.
  3. Density $\sigma = Q/(4\pi r^2)$. So $\sigma_1/\sigma_2 = r_2/r_1$.
- Worked example: Spheres of radius 2.0 cm and 6.0 cm, far apart, joined by a thin wire. Total charge +8.0 μC. $Q_1 = 8 \times 2/(2 + 6) =$ **2.0 μC**. $Q_2 =$ **6.0 μC**. Potential $= 8.99 \times 10^9 \times 2 \times 10^{-6}/0.02 =$ **8.99×10^5 V** (same for both). $\sigma_1/\sigma_2 = 6/2 =$ **3** (the small sphere has 3 times the density).
- Traps: The bigger sphere has more **charge**. The smaller sphere has more **density** and a stronger surface field. This is the "sharp points" rule in number form.
- Calculator: calculate — type: joined spheres.

16. Sphere inside a metal shell: charge on each surface

- Scope: ✅ IN for the charge bookkeeping. ⚠️ Borderline if the question says "Gauss's law" or "enclosed charge", since Gauss's law is not named in the rules. No calculus is needed.
- Where seen: ATX Regional 2026 B (REG) FR1a-c (metal sphere r = 0.5 m with +10 C inside a metal shell R = 2 m with −10 C: charge inside the sphere metal → 0; enclosed charge inside the shell metal → 0; enclosed charge outside everything → 0).
- How to solve:
  1. Inside any metal, E = 0. So a surface drawn inside the shell's metal encloses zero net charge.
  2. The inner wall of the shell gets −(charge on the inner sphere).
  3. The outer wall gets (shell's net charge) − (inner wall charge).
- Worked example: Inner sphere +3 μC. Shell net +4 μC. Inner wall = **−3 μC**. Outer wall = 4 − (−3) = **+7 μC**. Outside, the pair looks like a **+7 μC** point charge.
- Traps: The inner wall does not depend on the shell's own net charge. Only the outer wall does.
- Calculator: calculate — type: sphere in shell.

17. Contact between different-size spheres (unequal sharing)

- Scope: ✅ IN (concept only).
- Where seen: not seen in a test. Follows from type 15.
- How to solve:
  1. Touching conductors reach the same potential.
  2. The bigger one keeps more charge.
- Worked example: A small charged metal ball touches a large neutral metal ball. Which ends with more charge? Answer: **the large ball**. That is why touching the Earth (huge) takes nearly all the charge.
- Traps: Do not split 50/50 unless the spheres are identical.
- Calculator: look up — type: unequal sphere contact.

Nationals

18. Three-sphere induction + contact puzzle (fixed version of a flawed tiebreaker)

- Scope: ✅ IN.
- Where seen: UT Invitational 2020 B (INV) Q24, tiebreaker 5 (rod + three spheres; induction, separate, touch 2-3, touch 2-1; charge on sphere 1? → key 0 C). The test does not give how much charge the rod moves, so the question cannot be solved as printed.
- How to solve:
  1. After induction, write each sphere's charge in a table.
  2. Apply each touch as an average (type 9).
  3. Check that the total stays the same.
- Worked example (fixed numbers): After induction and separation, sphere 1 = –3 C, sphere 2 = +5 C, sphere 3 = +1 C (separate, uncharged by the rod). Touch 2-3: both = (5 + 1)/2 = **+3 C**. Touch 2-1: both = (3 – 3)/2 = **0 C**. Answer: **sphere 1 = 0 C** (sphere 2 = 0 C, sphere 3 = +3 C). Total = 3 C before and after.
- Traps: The rod only moves charge; it does not add any. Sphere 1 and sphere 2 must be equal and opposite right after induction, plus whatever they held before.
- Calculator: calculate — type: three sphere sequence.

19. Why is a neutral object attracted? Prove it with $1/r^2$

- Scope: ✅ IN (uses Coulomb's law ratios).
- Where seen: not seen in a test. Concept from OpenStax College Physics 2e §18.2.
- How to solve:
  1. Model the polarized object as –q at distance d and +q at distance d + s from a + charge.
  2. Pull $\propto 1/d^2$. Push $\propto 1/(d + s)^2$.
  3. Since d < d + s, the pull wins.
- Worked example: A + rod is 2.0 cm from the near side of a paper bit. The induced charges sit 2.0 cm and 2.1 cm away. Pull/push = $(2.1/2.0)^2$ = **1.10**. So the net force is attraction, about 10% of the pull.
- Traps: The net force is much weaker than the force on a charged object. That is why only light objects (paper, hair) jump.
- Calculator: calculate — type: polarization ratio.

Out of scope (skip)

- ❌ Electromagnetic induction (moving magnets making current). Magnetism is not a listed area (rule 3.c). Seen: USC Invitational Jan 2026 B (INV) Q5 (magnet falling through tubes).
- ❌ Capacitors and "charge stored on a plate". Capacitance is excluded (rule 3.d).
- ❌ Gauss's law with integrals or cylinders. No calculus (rule 3.e.v). Seen: Florida State 2021 B (STATE) Q38 (coax cable).

Mistakes in tests and keys

- **UT Invitational 2020 B Q24 (tiebreaker 5)**: Key = 0 C. From the figure, sphere 1 = +2, sphere 2 = 0, sphere 3 = +1 before the rod acts. The amount the rod moves is not given. The key's 0 C needs sphere 1 = –3 C and sphere 2 = +5 C after separation. The question cannot be solved as printed (from tests/ut-b.md).

- **UF Northern Regional Jan 2020 C Q8**: The text is garbled ("before it is completely discharged"). The key 7 could not be checked.
- **Princeton Feb 2020 C Q25**: The key "repelled" is right only because the cage is **ungrounded**. A grounded cage would show no force.

Not confirmed

- No electroscope question appears in any of the extracted tests (grep of every .txt in downloads/ for "electroscope" found nothing). Types 10-11 come from textbooks only.
- UF Northern Regional Jan 2020 C Q8 (key 7) not verified: the text is garbled and the charge on sphere A is not given.

Sources

- Rules: 2027 Div B Circuit Lab rule 3.c.ii, 3.c (areas only), 3.d, 3.e.
- Tests (from /Users/kai/Daniel/circuitlab/research/tests/ and prior/G_tests.md): Berkeley Feb 2026 B; UT Invitational 2019 B/C and 2020 B; UF Northern Regional 2019 B and 2020 C; North Pocono Jan 2020 C; Princeton Feb 2020 C; Rickards Oct 2025 C; ATX Regional 2026 B and C; USC Invitational Jan 2026 B.
- Raw texts read: downloads/dlG/ut20inv_Btest.txt (Q24), downloads/dlG/pz/p20 Princeton 2020 key (Q25), downloads/dlG/rsoi26C_key.txt (Q5), downloads/dlFind/np20inv_test.txt (Q36).
- Textbooks: OpenStax College Physics 2e §18.2 (Conductors and Insulators; charging by induction; electroscope). Kuphaldt, Lessons in Electric Circuits Vol. I, ch. 1.
- Numbers checked in Python (scratchpad a2/chk.py).

Coulomb's Law

Note: Coulombs Law

Coulomb's law gives the force between two charges. This is the biggest point block in this area. Berkeley Feb 2026 B gave its electrostatics section 32 of 160 points.

All plain Coulomb's law work is ✅ **IN at every level** (rule 3.c.ii names "Coulomb's Law"). The math rules decide how hard it can get:

- Rule 3.e.ii allows "2D geometry for vector directionality: points, parallel and perpendicular lines, and angles in degrees".
- Rule 3.e.v bans trigonometry. So sin, cos and tan are out. Forces at **right angles** (Pythagoras), **along one line**, or **cancelling by symmetry** are in.
- Rule 3.e.i allows systems of at most **two** equations. Quadratics are ordinary algebra.

Key facts

- **Coulomb's law:** the force between two point charges is proportional to the product of the charges. It is inversely proportional to the square of the distance between them.
- A **point charge** is a charge so small that its size does not matter.
- The force acts **along the line** joining the two charges.
- Like charges **repel**. Unlike charges **attract**.
- **Newton's third law:** the two charges feel forces of **equal size** in **opposite directions**. This is true even if one charge is much bigger.
- **k** (Coulomb's constant) = 8.99×10^9 N·m²/C². Many tests let you use 9.0×10^9.
- $k = 1/(4\pi\varepsilon_0)$. $\boldsymbol{\varepsilon_0}$ (permittivity of free space) = 8.85×10^{-12} C²/(N·m²).
- The force is an **inverse-square law**: double the distance → ¼ of the force. Triple → 1/9.
- **Superposition:** with several charges, find each force alone, then add them as vectors (arrows).
- The electric force is far stronger than gravity. For a proton and an electron, it is about 2.3×10^{39} times stronger.
- Charles-Augustin de Coulomb measured the law in 1785 with a **torsion balance** (a twisting wire that measures tiny forces).

Formulas

What	Formula	Notes
Coulomb's law (size)	$F = k\cdot$	q_1
With signs	$F = k\cdot q_1\cdot q_2 / r^2$	+ → repel, − → attract (a common test convention)
Solve for distance	$r = \sqrt{(k\cdot}$	q_1q_2
Solve for one charge	$q_2 = F\cdot r^2 / (k\cdot$	q_1
Two equal charges	$q = r\cdot\sqrt{(F/k)}$	
Scaling	$F_2/F_1 = (q_1'q_2'/q_1q_2) \times (r_1/r_2)^2$	
Gravity (to compare)	$F_g = G\cdot m_1\cdot m_2 / r^2$	$G = 6.67 \times 10^{-11}$ N·m²/kg²
Net force on a line	F_net = sum of signed forces (right = +)	
Two forces at right angles		F
Zero-force point, like charges	$\sqrt{q_1} / x = \sqrt{q_2} / (d - x)$	between them, nearer the smaller
Zero-force point, unlike charges	$\sqrt{}$	q_1
Hover against gravity	$k\cdot q_1\cdot q_2 / r^2 = m\cdot g$	$g = 9.8$ m/s²
Acceleration	$a = F/m$	

Unit prefixes: mC = 10^{-3} C, μC = 10^{-6} C, nC = 10^{-9} C, pC = 10^{-12} C. cm = 10^{-2} m, mm = 10^{-3} m.

Question types

Easy

1. State the law or name its discoverer

- Scope: ✅ IN.
- Where seen: ATX Regional 2021 B (REG) Q10 (equation of the law from the torsion experiment → $F = kQq/r^2$). ATX Regional 2021 B (REG) Q8 (torsion-balance scientist → Coulomb). BullSO Jan 2026 B (INV) Q1 (force ∝ product of charges / distance² → Coulomb). GullSO Dec 2025 (INV) Q36. UT Invitational 2020 B (INV) Q3. ATX Regional 2020 B (REG) Q4 (the law that Gauss's law comes from → Coulomb).
- How to solve:
  1. Write $F = k \cdot q_1 \cdot q_2 / r^2$.
  2. Say it in words: "force is proportional to the product of the charges and inversely proportional to the square of the distance".
- Worked example: Which scientist? "The force is directly proportional to the product of the charges and inversely proportional to the square of the distance." Answer: **Coulomb**.
- Traps: Ampère is the force between currents (magnetism, out). Gauss's law is a different form of the same physics.
- Calculator: look up — type: coulomb law statement.

2. Force between two charges (size and attract/repel)

- Scope: ✅ IN.
- Where seen: BullSO Jan 2026 B (INV) Q11 (+6 μC and −7 μC at 0.5 m, $k = 9 \times 10^9$ → −1.512 N, attractive). Rickards Oct 2025 B (INV) Q33-34 (+1.5 nC and −2.0 nC at 1.5 cm → attract, 1.2×10^{-4} N). Rickards Oct 2025 B (INV) Q11 (3.5 μC and 5.4 μC at 20 cm → 4.25 N). UT Invitational 2020 B (INV) Q17 (1.4 μC and −0.96 μC at 8 cm → 1.89 N). MIT Jan 2019 C (INV) Q1 (+5.0 μC and −2.0 mC at 25 cm → −1438 N). NY Lake Erie/Niagara Dec 2018 C (INV) Q2 (+8 μC and −5 μC at 10 cm → −36 N). UMSO 2019 (INV) Q18 (two +5 C at 0.5 mm → 8.99×10^{17} N) and Q24A (+3 nC and +8 nC at 1 mm → 0.216 N). JC Booth 2011 B (INV) Q49 (two elementary charges 0.5 mm apart → 9.2×10^{-22} N). F_slides deck 6 (two electrons 1 mm apart → 2.31×10^{-22} N).
- How to solve:
  1. Change units: μC → $\times 10^{-6}$, nC → $\times 10^{-9}$, cm → $\times 10^{-2}$.
  2. Square the distance.
  3. $F = k \times |q_1| \times |q_2| \div r^2$.
  4. Like signs → repel. Unlike → attract.
- Worked example (BullSO 2026 B Q11): $F = 9 \times 10^9 \times 6 \times 10^{-6} \times 7 \times 10^{-6} / 0.5^2$ = **1.512 N, attractive**. With signs, F = **−1.512 N** (choice A).
- Worked example 2 (Rickards 2025 B Q34): $F = 8.99 \times 10^9 \times 1.5 \times 10^{-9} \times 2.0 \times 10^{-9} / 0.015^2$ = **1.2×10^{-4} N**, attractive.
- Traps: Forgetting to square r is the most common error. Forgetting cm → m gives an answer 10^4 times too big. A minus sign means "attractive", not "smaller".
- Calculator: calculate — type: coulomb q1 q2 r.

3. Newton's third law pair

- Scope: ✅ IN.
- Where seen: UT Invitational 2020 B (INV) Q18 (force of Q2 on Q1 = force of Q1 on Q2 = 1.89 N). North Pocono Jan 2020 C (INV) Q8 (proton and electron → equal, opposite, attract). I_bank Q15.
- How to solve:
  1. Both charges feel the **same size** force.
  2. The directions are opposite.
- Worked example: A 1 μC charge and a 100 μC charge are 1 m apart. Which feels the bigger force? Answer: **neither; the forces are equal and opposite** (0.899 N each).
- Traps: The small charge (or light electron) **accelerates** more, because a = F/m. The force is still equal.
- Calculator: look up — type: newton third coulomb.

4. Ratio scaling: change the distance or the charges

- Scope: ✅ IN.
- Where seen: Pembroke Hill Dec 2025 B (INV) Q9 (distance tripled → decreases by a factor of 9). UF Northern Regional Jan 2019 B (REG) Q5 (F = 100 N, distance tripled → 11 N). UT Austin Regional 2019 B (REG) Q6 (distance from 6 cm to 1 cm → 36 times). Cobra 2019 B (INV) Q10 (five parts: 2q_A → 2F; both halved → F/4; 3d → F/9; d/2 → 4F; 3q_A and 2d → 3F/4). Berkeley Feb 2026 B (INV) §3 1c (a + charge at 3 m vs 1 m from a –3 C charge → 9 times larger) and 1d (charge tripled → force triples). GullSO Dec 2025 (INV) Q2 (field $\propto 1/r^2$).
- How to solve:
  1. Write F_new/F_old = (charge factors multiplied) ÷ (distance factor)².
  2. Multiply the old force by that ratio.
- Worked example (Cobra Q10e): q_A is tripled and the distance is doubled. Ratio = 3 ÷ 2² = **¾**. So F_new = **3F/4**.
- Worked example 2: F = 100 N, distance tripled. F_new = 100/9 = **11.1 N**.
- Traps: Square the distance factor, not the charge factor. "Halving the distance" gives 4 times the force, not 2.
- Calculator: calculate — type: coulomb scaling.

5. Value and units of k

- Scope: ✅ IN.
- Where seen: JC Booth-era "Shock Value" tryout B (about 2012) VI-19 (value of Coulomb's constant → 8.99×10^9 N·m²/C²). ATX Regional 2020 B (REG) Q45 (dielectric constant of vacuum → 1; ⚠️ borderline, since dielectrics belong to capacitance, rule 3.d).
- How to solve:
  1. $k = 8.99 \times 10^9$ N·m²/C².
  2. $k = 1/(4\pi\varepsilon_0)$ with $\varepsilon_0 = 8.85 \times 10^{-12}$.
- Worked example: Check: $1/(4\pi \times 8.85 \times 10^{-12})$ = **8.99×10^9 N·m²/C²**.
- Traps: The unit is N·m²/C², not N/C.
- Calculator: look up — type: coulomb constant.

Medium

6. Solve for distance or for a charge

- Scope: ✅ IN.
- Where seen: not seen in a test as the main question. I_bank Q14. OpenStax College Physics 2e §18.3 problems.
- How to solve:
  1. Rearrange $F = kq_1q_2/r^2$.
  2. $r = \sqrt{(kq_1q_2/F)}$. Or $q = r\sqrt{(F/k)}$ for equal charges.
- Worked example 1: Two equal charges 0.20 m apart repel with 0.90 N. $q = 0.20 \times \sqrt{(0.90/8.99 \times 10^9)}$ = **2.0×10^{-6} C (2.0 µC)**.
- Worked example 2: Two 3.0 µC charges repel with 0.81 N ($k = 9 \times 10^9$). $r = \sqrt{(9 \times 10^9 \times 9 \times 10^{-12}/0.81)}$ = **0.316 m**.
- Traps: Take the square root at the end, not before.
- Calculator: calculate — type: coulomb solve r or q.

7. Force from particle counts

- Scope: ✅ IN.
- Where seen: UF Northern Regional Jan 2019 C (REG) Q11 (10^{20} electrons and 10^{17} protons, 24 cm apart → 4.0×10^{10} N).
- How to solve:
  1. q = n × e for each group.
  2. Then use Coulomb's law.
- Worked example: $q_1 = 10^{20} \times 1.602 \times 10^{-19} = 16.0$ C. $q_2 = 10^{17} \times 1.602 \times 10^{-19} = 0.0160$ C. $F = 8.99 \times 10^9 \times 16.0 \times 0.0160/0.24^2$ = **4.0×10^{10} N, attractive**.
- Traps: Electrons and protons attract.
- Calculator: calculate — type: coulomb from counts.

8. Electric force vs. gravity

- Scope: ✅ IN.
- Where seen: UT Invitational 2020 B (INV) Q19 (which is stronger between the two charges → electric). I_bank Q16.
- How to solve:
  1. $F_e = ke^2/r^2$. $F_g = Gm_e m_p/r^2$.
  2. The ratio does not depend on r.
- Worked example: For an electron and a proton, $F_e/F_g = (8.99 \times 10^9 \times (1.602 \times 10^{-19})^2)/(6.67 \times 10^{-11} \times 9.11 \times 10^{-31} \times 1.67 \times 10^{-27}) =$ **2.3×10^{39}**.
- Traps: Both forces are inverse-square. So "which is stronger at large distance" has the same answer at every distance.
- Calculator: calculate — type: electric vs gravity.

9. Net force on one charge from charges on a line (1D superposition)

- Scope: ✅ IN.
- Where seen: UT Invitational 2020 B (INV) Q22 (+q, +q (A), +2q in a line, 9 cm spacing, q = 1 μC → 1.11 N left). UT Austin Regional 2019 B (REG) Q26 (three protons on y = 4 at x = 0, 4, –6 m; key 3.12×10^{-30} N is wrong; correct 8.01×10^{-30} N toward –x). UF Northern Regional Jan 2019 B (REG) Q37 (test charge 4 μC; 2 μC 3 cm left; 3 μC 2 cm right → 190 N left). Berkeley Feb 2026 B (INV) §3 4a (+2 C at x = 2 between –4 C at 0 and +1 C at 3 → 4k left). I_bank Q17.
- How to solve:
  1. Draw the line. Choose right = +.
  2. For each other charge, find the size of the force.
  3. Decide the direction from attract/repel. Give it a sign.
  4. Add the signed forces.
- Worked example (UF 2019 B Q37): From the 2 μC (3 cm left, repels): $8.99 \times 10^9 \times 4 \times 10^{-6} \times 2 \times 10^{-6}/0.03^2 = 79.9$ N to the **right**. From the 3 μC (2 cm right, repels): 269.7 N to the **left**. Net = **189.8 N to the left** (key 190 N left).
- Worked example 2 (Berkeley 2026 §3 4a, figure numbers): Force from –4 C at 2 m = $k \times 2 \times 4/2^2 = 2k$, left (attract). Force from +1 C at 1 m = $k \times 2 \times 1/1^2 = 2k$, left (repel). Net = **$4k = 3.6 \times 10^{10}$ N to the left**.
- Traps: Each force uses the distance from **that** charge. Mixed signs need care: draw arrows first.
- Calculator: calculate — type: 1d net force.

10. Find the mass from the acceleration (Coulomb + F = ma)

- Scope: ✅ IN.
- Where seen: UF Northern Regional Jan 2020 C (REG) Q4 (30 nC at 0, 40 nC at 72 cm, 1.2 μC at 28 cm, a = 100 km/s² → 1.9×10^{-8} kg). MIT Jan 2021 C (INV) Scenario A Q14 (acceleration of the central charge → 1.2×10^{13} m/s²).
- How to solve:
  1. Find the net force (type 9).
  2. m = F/a, or a = F/m.
- Worked example (UF 2020 C Q4): From 30 nC at 0.28 m: 4.13×10^{-3} N to the right. From 40 nC at 0.44 m: 2.23×10^{-3} N to the left. Net = 1.90×10^{-3} N. m = $1.90 \times 10^{-3}/1.0 \times 10^5 =$ **1.9×10^{-8} kg**.
- Traps: 100 km/s² = 1.0×10^5 m/s².
- Calculator: calculate — type: coulomb mass acceleration.

11. Direction of the net force by symmetry (no numbers)

- Scope: ✅ IN (3.e.ii: vector direction).
- Where seen: UF Northern Regional Jan 2019 B (REG) Q18 (equilateral triangle: top –q, bottom-left +q, bottom-right –q; direction of the force on the top charge → key "up" is wrong; correct "left"). Berkeley Feb 2026 B (INV) §3 2f and 3c (a + charge 1 m above the midpoint: straight up for +2/+2; straight left for –2/+2).
- How to solve:
  1. Draw an arrow for each force: toward an attracting charge, away from a repelling one.

2. If the two arrows have the same length, look for parts that cancel.
3. The leftover direction is the answer.

- Worked example (UF 2019 B Q18): The pull toward bottom-left +q points down-left. The push from bottom-right −q points up-left. The up and down parts cancel. Answer: **straight left**.
- Traps: Do not guess from one charge alone. Equal distances and equal charges mean equal arrow lengths.
- Calculator: look up — type: symmetry direction.

12. Force after two spheres touch

- Scope: ✅ IN.
- Where seen: UF Northern Regional Jan 2019 B (REG) Q13 (→ ¼). I_bank Q9 (9/16). Full method in Charging Methods and Electroscopes A2 type 8.
- How to solve: Share the charge, then use the ratio $(q_1'q_2')/(q_1q_2)$.
- Worked example: +8 μC and −2 μC touch → +3 and +3. Ratio **9/16**, now repulsive.
- Traps: The sign of the force can flip.
- Calculator: calculate — type: touch share force ratio.

Hard

13. Two forces at right angles (Pythagoras, no trig)

- Scope: ✅ IN. Size by Pythagoras is allowed. The direction may only be asked as a quadrant, an axis, or 45° (when both parts are equal). An angle like 36.9° needs arctan, so it is ❌ out (rule 3.e.v).
- Where seen: not seen in a test with right angles. The out-of-scope version (a 60° angle) is UT Invitational 2020 B (INV) Q23. I_bank Q18.
- How to solve:
  1. Put the charge at the origin. Find the force from the charge on the x-axis (it lies along x).
  2. Find the force from the charge on the y-axis (it lies along y).
  3. $|F| = \sqrt{(F_x^2 + F_y^2)}$.
  4. Give the quadrant from the signs.
- Worked example: +1.0 μC at the origin. +3.0 μC at (0.30 m, 0). −4.0 μC at (0, 0.40 m). $F_x = 8.99 \times 10^9 \times 3 \times 10^{-12}/0.09 = 0.300$ N toward −x (repel). $F_y = 8.99 \times 10^9 \times 4 \times 10^{-12}/0.16 = 0.225$ N toward +y (attract). $|F| = \sqrt{(0.300^2 + 0.225^2)}$ = **0.375 N**, pointing **up and to the left** (second quadrant).
- Worked example 2 (45°): +1.0 μC at the origin. +2.0 μC at (0.30, 0) and +2.0 μC at (0, 0.30). Each force = 0.200 N. $|F| = 0.200 \times \sqrt{2}$ = **0.283 N**, at **45° between −x and −y** (225° from +x).
- Traps: Never add the sizes directly (0.300 + 0.225 is wrong). Get each direction from attract/repel before you combine.
- Calculator: calculate — type: 2d perpendicular force.

14. Zero-force point between two like charges

- Scope: ✅ IN.
- Where seen: Berkeley Feb 2026 B (INV) §3 2c and 2d (two +2 C at ±2 m: a −2 C at x = 0 does not move). I_bank Q20.
- How to solve:
  1. For like charges, the point is **between** them, nearer the smaller one.
  2. Set $k\,q_1/x^2 = k\,q_2/(d - x)^2$.
  3. Take square roots: $\sqrt{q_1}\,(d - x) = \sqrt{q_2}\,x$.
- Worked example: +4q at x = 0 and +q at x = 0.30 m. $\sqrt{4}\,(0.30 - x) = \sqrt{1}\,x \rightarrow 0.60 - 2x = x \rightarrow$ **x = 0.20 m**.
- Traps: The test charge's size and sign do not matter. The answer is the same for any test charge.
- Calculator: calculate — type: zero force like.

15. Zero-force point outside two unlike charges

- Scope: ✅ IN.
- Where seen: Berkeley Feb 2026 B (INV) §3 4b (written as "E = 0", same math; see Electric Field and Potential A2 type 22).

- How to solve:
  1. For unlike charges, the point is **outside**, beyond the **smaller** charge.
  2. Set up $|q_1|/(\text{far distance})^2 = |q_2|/(\text{near distance})^2$.
  3. Take square roots and solve the line equation.
- Worked example (Berkeley text numbers): −3 C at x = −2 m and +1 C at x = +2 m. For x > 2: $3/(x + 2)^2 = 1/(x - 2)^2 \rightarrow \sqrt{3}\,(x - 2) = x + 2 \rightarrow x = (1 + \sqrt{3})^2 = \mathbf{4 + 2\sqrt{3} = 7.46\ m}$.
- Traps: Between unlike charges, both forces point the same way, so there is no zero point there. Check that your answer is in the region you assumed.
- Calculator: calculate — type: zero force unlike.

16. Symmetric polygon with one charge missing

- Scope: ✅ IN. The trick needs no trig. A direct sum of 11 vectors would need trig (❌), so the symmetry is the only legal path.
- Where seen: MIT Jan 2021 C (INV) Scenario A Q12 (11 charges q = 18 μC on 11 of 12 corners of a regular dodecagon, r = 82 cm, centre charge Q = 2.2 μC → net force 0.53 N in x, 0 in y).
- How to solve:
  1. A full ring of equal charges gives zero net force at the centre.
  2. So 11 charges = (full ring) − (the missing one).
  3. The net force equals the force of the missing charge, but pointing the opposite way. It points **toward** the empty corner for like charges.
- Worked example: $F = 8.99 \times 10^9 \times 18 \times 10^{-6} \times 2.2 \times 10^{-6}/0.82^2 = \mathbf{0.53\ N}$, toward the empty corner.
- Traps: The size equals one charge's force, not 11 times it.
- Calculator: calculate — type: missing charge ring.

17. A charge hovering above another (balance with gravity)

- Scope: ✅ IN for the vertical case. The "hanging pith balls at an angle" case needs tan θ, so it is ❌ out (rule 3.e.v).
- Where seen: not seen in a test. I_bank Q23. OpenStax College Physics 2e §18.3 problems.
- How to solve:
  1. Upward electric force = weight.
  2. $kq_1q_2/r^2 = mg$.
- Worked example: Two equal charges +q, one fixed, one hovering 0.10 m above with m = 2.0 g. $q = 0.10 \times \sqrt{(0.0020 \times 9.8/8.99 \times 10^9)} = \mathbf{1.5 \times 10^{-7}\ C}$.
- Traps: Grams → kilograms (2.0 g = 0.0020 kg).
- Calculator: calculate — type: coulomb hover.

18. Force from charges at a right-angle corner (square with 45°)

- Scope: ⚠️ Borderline. Four corners cancel to zero (✅ IN). A diagonal charge needs "split into √2/2 parts", which is close to trig.
- Where seen: GullSO Dec 2025 (INV) Q5 (charges at the corners of a square: field direction and potential at the centre). I_bank Q19.
- How to solve:
  1. Four equal charges on a square → **0** at the centre by symmetry.
  2. Pairs of equal charges on opposite corners cancel.
  3. Whatever is left points along a diagonal or a side.
- Worked example: +q, +q, +q, +q at the corners of a square. Net force on a charge at the centre: **0**.
- Traps: The **potential** at the centre of +q, +q, −q, −q is 0, but the **field** is not 0 (see Electric Field and Potential A2).
- Calculator: look up — type: square symmetry.

Nationals

19. Third charge that puts all three in equilibrium

- Scope: ✅ IN (one line; square roots and two equations).
- Where seen: UMSO 2019 (INV) Q24B (+3 nC at 0, +8 nC at 1 mm; a third charge on the axis makes all three balance → 0.380 mm, −1.155 nC).
- How to solve:
  1. Find where the third charge feels no force (type 14). This gives its **position**.
  2. Write the balance for one of the original charges. This gives the third charge's **size**.
  3. The third charge must have the **opposite** sign.
- Worked example: Position: $\sqrt{3}\,(1 - x) = \sqrt{8}\,x \rightarrow x = 1/(1 + \sqrt{(8/3)}) =$ **0.380 mm**. Size: for the 3 nC charge, $k\cdot 3\cdot 8/1^2 = k\cdot 3\cdot|q|/0.380^2 \rightarrow |q| = 8 \times 0.380^2 =$ **1.15 nC**. Sign: **negative**. Answer: **−1.15 nC at 0.380 mm**.
- Traps: Like charges cannot all balance unless the middle one is opposite. Keep distances in mm throughout; the units cancel in the ratio.
- Calculator: calculate — type: three charge equilibrium.

20. Charge on the perpendicular bisector of two equal charges

- Scope: ✅ IN for equilibrium and the net force using similar triangles (no trig: the vertical part is F × h/R). ⚠️ Borderline for $R = \sqrt{(h^2 + D^2)}$, because 3.e.ii names points, lines and angles, not Pythagoras by name. The small-oscillation (SHM) ending is ❌ out (mechanics, not a listed area).
- Where seen: ATX Regional 2021 B (REG) FRQ6, Q57-61 (+Q at x = ±D, −q on the y-axis: equilibrium at the origin; $R = \sqrt{(h^2 + D^2)}$; net force $-2kQqh/R^3$; small-h approximation; show SHM).
- How to solve:
  1. Distance to each charge: $R = \sqrt{(h^2 + D^2)}$.
  2. Size of each force: $F_1 = kQq/R^2$.
  3. The sideways parts cancel by symmetry.
  4. Each vertical part = $F_1 \times (h/R)$ (similar triangles).
  5. Net = $2kQqh/R^3$, pointing toward the line of charges (attraction).
- Worked example: Q = 4.0 μC, q = 1.0 μC (negative), D = 0.30 m, h = 0.40 m. R = 0.50 m. F_net = $2 \times 8.99 \times 10^9 \times 4 \times 10^{-6} \times 1 \times 10^{-6} \times 0.40/0.50^3 =$ **0.230 N, straight down toward the origin**.
- Traps: Do not add the two 0.144 N forces directly. Only the vertical parts add.
- Calculator: calculate — type: bisector force.

21. Find two charges from their sum and their force

- Scope: ✅ IN (system of two equations, rule 3.e.i).
- Where seen: not seen in a test. Classic textbook problem (OpenStax College Physics 2e §18.3 style).
- How to solve:
  1. $q_1 + q_2 =$ total.
  2. $q_1 \times q_2 = F\,r^2/k$.
  3. Solve the quadratic: $q^2 -$ (total)$q +$ (product) $= 0$.
- Worked example: Two charges total +10 μC. At 0.50 m they repel with 0.864 N ($k = 9 \times 10^9$). Product = $0.864 \times 0.25/9 \times 10^9 = 2.4 \times 10^{-11}$ C². $q^2 - 10 \times 10^{-6}\,q + 2.4 \times 10^{-11} = 0 \rightarrow$ **q_1 = 6.0 μC, q_2 = 4.0 μC**.
- Traps: If they attract, the product is negative: one charge is negative.
- Calculator: calculate — type: two charges sum product.

22. Energy-to-speed: release a charge and find its final speed

- Scope: ✅ IN (energy, algebra, square roots). Full method in Electric Field and Potential A2 type 29.
- Where seen: MIT Jan 2021 C (INV) Scenario A Q15 (the centre charge flies far away → 1.4×10^7 m/s).
- How to solve: Starting potential energy → kinetic energy: $\frac{1}{2}mv^2 = U$.
- Worked example: see Electric Field and Potential A2. Answer: **1.4×10^7 m/s**.
- Traps: Use energy, not force. The force changes as the charge moves.
- Calculator: calculate — type: release speed.

23. Explain the torsion balance

- Scope: ✅ IN (Coulomb history plus Coulomb's law).
- Where seen: ATX Regional 2021 B (REG) Q9 (explain how Coulomb's torsion balance measures force).

- How to solve:
  1. A light rod hangs from a thin wire. A charged ball is on one end.
  2. A second charged ball is brought near. The force twists the wire.
  3. The twist angle is proportional to the force.
  4. Coulomb changed the distance and the charges (by halving with an identical ball) and read the twist.
- Worked example: How did Coulomb get half the charge on a ball? Answer: **touch it to an identical neutral ball; they share equally**.
- Traps: The same device also measured magnetic forces (that physics is out, but the history question is in).
- Calculator: look up — type: torsion balance.

Out of scope (skip)

- ❌ Forces at 60° or other angles that need sin, cos or arctan. Rule 3.e.v (no trigonometry). Seen: UT Invitational 2020 B (INV) Q23 (–2q at 60° below the line; its key is also wrong).
- ❌ Hanging pith balls at an angle (tan θ). Rule 3.e.v.
- ❌ Magnetic force qvB, Lorentz force, right-hand rules. Magnetism is not a listed area (rule 3.c). Seen: ATX Regional 2021 B (REG) Q31, Q50-53; ATX Regional 2020 B (REG) Q22; BullSO Jan 2026 B (INV) Q8, Q25-27.
- ❌ Simple harmonic motion of a charge (ATX Regional 2021 B Q60-61). Mechanics is not a listed area.
- ❌ Cross products (ATX Regional 2020 B Q24, Q31). Beyond the 2D direction allowed by 3.e.ii.
- ❌ Dipole torque between two dipoles (Florida State 2021 B Q40). Trig and vector calculus.

Mistakes in tests and keys

- **UF Northern Regional Jan 2019 B Q18**: key "up". Correct: **left**. The pull to the lower-left + and the push from the lower-right – have equal sizes. Their vertical parts cancel (checked by drawing; from tests/misc-b.md).
- **UT Austin Regional 2019 B Q26**: key 3.12 × 10^{-30} N. Correct with distances in metres: k $e^2(1/4^2 - 1/6^2)$ = **8.01 × 10^{-30} N toward –x** (Python).
- **UT Invitational 2020 B Q23** (out of scope anyway): key 2.26 N, 34.8° N of E. Correct: **2.26 N, 34.8° S of W** (from tests/ut-b.md).
- **UF Northern Regional Jan 2020 B Q36**: the test does not give the charges. The key "36 MN" assumes 1 mC, but 1 mC at 0.5 mm gives **3.6 × 10^{10} N**, not 3.6 × 10^{7} N. The question was copied from JC Booth 2011 Q49, which used proton charges (→ 9.23 × 10^{-22} N).
- **Rickards Oct 2025 B Q11**: it asks for the "force of attraction" between +3.5 μC and +5.4 μC. Two positive charges **repel**. The size 4.25 N (choice d) is right.
- **Berkeley Feb 2026 B §3 Q4**: the text says "–3 C at x = –2 and +1 C at x = 2". The figure and the key use **–4 C at x = 0 and +1 C at x = 3**. With the text numbers, the +2 C charge in 4a would sit on top of the +1 C charge, so 4a only works with the figure. Use the figure.

Not confirmed

- ATX Regional 2021 B FRQ6 diagram was not rendered by me. I used the extraction in tests/atx-reg.md.
- The MIT 2021 dodecagon direction (toward or away from the empty corner) depends on the figure's orientation. The key only gives "x component 0.53 N". I did not render the figure.

Sources

- Rules: 2027 Div B Circuit Lab rule 3.c.ii, 3.c (areas only), 3.e.i, 3.e.ii, 3.e.v.
- Tests (extractions in /Users/kai/Daniel/circuitlab/research/tests/ and prior/G_tests.md): BullSO Jan 2026 B; Berkeley Feb 2026 B (section 3 rendered from berk26_test.pdf pages 7-8; key read from berk26_key.txt); Pembroke Hill Dec 2025 B; Rickards Oct 2025 B (rsoi26B_test.txt, rsoi26B_key.txt); UT Invitational 2019 and 2020 B; UT Austin Regional 2019 B; UF Northern Regional 2019 B/C and 2020 B/C; MIT 2019 C and 2021 C (mit21_test.txt, mit21_key.txt); NY Lake Erie/Niagara 2018 C; UMSO 2019 (umso19_test.txt); Cobra 2019 B; North Pocono 2020 C; GullSO Dec 2025; ATX Regional 2020 B and 2021 B; JC Booth 2011 B.

- Textbooks: OpenStax College Physics 2e §18.3 (Coulomb's Law). Kuphaldt, Lessons in Electric Circuits Vol. I ch. 1.
- Numbers checked in Python (scratchpad a2/chk.py).

Electric Field and Potential

This note covers the electric field, field lines, electric potential, work and the electron-volt. It also covers conductors and shielding. Berkeley Feb 2026 B spent most of its 32-point electrostatics section on these ideas.

The 2027 rules say "Properties of electric charges/fields" (rule 3.c.ii). So the field and its properties are ✅ **IN at every level**. Potential and potential energy are not named. They are still ✅ IN in practice, because they are field properties and "voltage" is in rule 3.c.iv. Recent B tests ask them often (Berkeley 2026, GullSO 2025, BullSO 2026).

The math rules set the limits:

- Rule 3.e.ii allows 2D vector direction with parallel and perpendicular lines and angles in degrees.
- Rule 3.e.v bans trigonometry and calculus. So fields at right angles, along a line, or cancelling by symmetry are in. Gauss's law with integrals is out.
- Rule 3.e.i allows at most two equations at once. Quadratics are ordinary algebra.
- Capacitance is excluded (rule 3.d). Parallel plates are fine only as "a uniform field", not as a capacitor.

Key facts

- **Electric field (E):** the force per unit of positive charge at a point. A charge makes a field in the space around it.
- **Test charge:** a small positive charge we imagine at a point to measure the field. It is too small to move the other charges.
- E is a **vector**. It has a size and a direction.
- The field points **away from** a positive charge. It points **toward** a negative charge.
- A **positive** charge feels a force **along** E. A **negative** charge (an electron) feels a force **opposite** to E.
- The unit of E is the **newton per coulomb (N/C)**. It is the same as the **volt per metre (V/m)**.
- A point-charge field is an **inverse-square** field: double the distance gives ¼ of the field.
- **Superposition:** with several charges, find each field alone, then add the arrows.
- **Field lines:** lines that show the direction of E. They start on + charges and end on − charges (or at infinity). They never cross. Where they are closer together, the field is stronger.
- The number of lines leaving or entering a charge is proportional to the size of the charge.
- **Electric potential (V):** the potential energy per unit of charge at a point. Its unit is the **volt (V) = joule per coulomb (J/C)**.
- Potential is a **scalar** (a number with a sign, no direction). You add potentials as plain numbers.
- **Potential energy (U):** the stored energy of a charge in a field. U = qV.
- **Potential difference (voltage):** the change in potential between two points.
- Positive charges "fall" from high V to low V. Electrons "fall" from low V to high V.
- **Equipotential:** a line or surface where V has one value. Field lines cross equipotentials at **right angles**.
- Moving a charge along an equipotential takes **zero work**.
- **Electron-volt (eV):** the energy one electron gains crossing 1 V. 1 eV = 1.602×10^{-19} J.
- **Conductor in equilibrium:** a metal where the charges have stopped moving. Inside the metal, E = 0. The whole conductor has one potential (it is an equipotential). Excess charge sits on the outer surface. Just outside, E is perpendicular to the surface.
- **Shielding (Faraday cage):** a closed metal shell keeps outside fields out of the hollow inside. A car body protects you from lightning this way.
- By convention, the potential of the Earth (ground) is 0 V. Far away from all charges, V = 0 too.

Formulas

What	Formula	Notes
Field from force	E = F / q	N/C
Force from field	F = qE	sign of q gives the direction
Point-charge field	E = k·	Q
Uniform field between plates	E = V / d	d = gap in metres

What	Formula	Notes
Acceleration in a field	$a = qE / m$	constant in a uniform field
Point-charge potential	$V = k \cdot Q / r$	keep the sign of Q
Several charges	$V = k(Q_1/r_1 + Q_2/r_2 + \ldots)$	plain sum, no vectors
Potential energy of a pair	$U = k \cdot q_1 \cdot q_2 / r$	sum over every pair for 3+ charges
Potential energy at a point	$U = qV$	
Work by the field	$W = q\Delta V$ (size)	ΔV = voltage crossed
Energy to speed	$\frac{1}{2}mv^2 = q\Delta V$	$v = \sqrt{(2q\Delta V/m)}$
Electron-volt	$1\ eV = 1.602 \times 10^{-19}\ J$	J → eV: divide by 1.602×10^{-19}
Zero-field point, like charges	$\sqrt{Q_1} / x = \sqrt{Q_2} / (d - x)$	between, nearer the smaller
Zero-field point, unlike charges	$\sqrt{}$	Q_1
Zero-potential point	$Q_1/r_1 + Q_2/r_2 = 0$	only for unlike charges
Two fields at right angles		E
Electric flux (⚠ borderline)	$\Phi = Q / \varepsilon_0$	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/(N \cdot m^2)$
Millikan balance (⚠ borderline)	$qE = mg$	$g = 9.8\ m/s^2$

Unit prefixes: mC = 10^{-3}, μC = 10^{-6}, nC = 10^{-9}. kN/C = 10^3 N/C. MV/m = 10^6 V/m. cm = 10^{-2} m, mm = 10^{-3} m.

Question types

Easy

1. Define the electric field and give its units

- Scope: ✅ IN.
- Where seen: UF Northern Regional Jan 2019 B (REG) Q7 (true statement about E: "it is a vector field"). Cobra 2019 B (INV) Q8 (a charge makes an electric field; a moving charge also makes a magnetic field). BullSO Jan 2026 B (INV) Q15 (field lines show the direction of force on a positive charge). Wichita Heights 2026 C (INV) Q1 (field strength is inversely proportional to the square of the distance). I_bank Q26 (N/C = V/m).
- How to solve:
  1. E is the force on each coulomb of positive test charge.
  2. E is a vector. Its unit is N/C or V/m.
- Worked example: Which unit equals N/C? (a) N·C (b) V/m (c) J/C. Answer: **(b) V/m**. Check: V/m = (J/C)/m = (N·m/C)/m = N/C.
- Traps: N·C is a wrong unit (UF 2019 B Q7 choice c). J/C is the volt, the unit of potential, not of field.
- Calculator: look up — type: field definition units.

2. Field direction near a + or – charge

- Scope: ✅ IN.
- Where seen: Berkeley Feb 2026 B (INV) Q7 (negative charge at the origin; field lines in the second quadrant → **right and down**). ATX Regional 2020 B (REG) Q21 (field of + and – charges points outward, inward). USC Invitational Jan 2026 B (INV) Q7 (field lines of +Q and –Q go from + to –).
- How to solve:
  1. Draw a line from the charge to the point.
  2. For +, the arrow points away from the charge. For –, it points toward the charge.
- Worked example (Berkeley 2026 Q7): A point in the second quadrant is up and to the left of the origin. Toward the origin is down and to the right. Answer: **D, right and down**.
- Traps: Do not answer "left and up" just because the point is there. That is the position, not the field direction.
- Calculator: look up — type: field direction.

3. Which way does a charge move in a field?

- Scope: ✅ IN.
- Where seen: UT Invitational 2020 B (INV) Q21 (negative charge released where the field points right → **left**). Berkeley Feb 2026 B (INV) Q8 (positive charge on the +x axis next to a negative charge at the origin → **left**).

Berkeley Feb 2026 B (INV) §3 3a (–2 C at x = 0 where E = k to the left → moves **right**). BullSO Jan 2026 B (INV) Q15.

- How to solve:
  1. Find the direction of E at the point.
  2. A positive charge moves along E. A negative charge moves against E.
- Worked example (Berkeley 2026 §3 3a): –2 C at x = –2 m and +2 C at x = +2 m. At x = 0, both fields point left. E = k·2/2² + k·2/2² = k to the left. The test charge is negative. Answer: **it moves to the right**.
- Traps: Field lines are not paths. A charge starting at rest moves along the line only at the first instant (BullSO 2026 B Q15 choices C and D are wrong).
- Calculator: look up — type: force direction in field.

4. Field from the force on a test charge (E = F/q)

- Scope: ✅ IN.
- Where seen: Wichita Heights 2026 C (INV) Q13 (–2.48 × 10^{-16} C feels 2.04 N → key –8.25 × 10^{15} N/C). UF Northern Regional Jan 2019 B (REG) Q37 part 2 (force 190 N on a 4 μC test charge → 4.75 × 10^{7} N/C left). I_bank Q24.
- How to solve:
  1. E = F ÷ q.
  2. If q is negative, E points opposite to the force.
- Worked example: A 2.0 μC charge feels 0.050 N east. E = 0.050/2.0 × 10^{-6} = **2.5 × 10^{4} N/C east**.
- Worked example 2 (Wichita Q13): E = 2.04/2.48 × 10^{-16} = **8.23 × 10^{15} N/C**, opposite to the force (key –8.25 × 10^{15} is the same within rounding).
- Traps: The field does not depend on the test charge. Double the test charge and the force doubles, but E stays the same.
- Calculator: calculate — type: E from F and q.

5. Force on a charge in a known field (F = qE)

- Scope: ✅ IN.
- Where seen: MIT Jan 2019 C (INV) Q4 (electron in 12 MV/m → 1.92 × 10^{-12} N). Wichita Heights 2026 C (INV) Q25 (electron in 30 N/C → 4.8 × 10^{-18} N).
- How to solve:
  1. F = |q| × E.
  2. Direction: along E for +, against E for –.
- Worked example (MIT 2019 Q4): F = 1.602 × 10^{-19} × 12 × 10^{6} = **1.92 × 10^{-12} N**, opposite to the field.
- Traps: "M" in MV/m means 10^{6}. Use e = 1.602 × 10^{-19} C for one electron.
- Calculator: calculate — type: F from q and E.

6. Inverse-square scaling of the field

- Scope: ✅ IN.
- Where seen: GullSO Dec 2025 (INV) Q2 (point charge: field decreases as 1/r² → D). Fall 2019 F-M Invitational (INV) Q2b (E ∝ 1/r²). Wichita Heights 2026 C (INV) Q1 ("inversely proportional to the square of the distance").
- How to solve:
  1. E_new/E_old = (charge factor) ÷ (distance factor)².
- Worked example: A point 2 m from a charge has E = 900 N/C. Find E at 6 m. Factor 3 in distance → E = 900/9 = **100 N/C**.
- Traps: Potential is 1/r, not 1/r². At 6 m the potential is ⅓ of its value at 2 m.
- Calculator: calculate — type: field scaling.

7. Field-line rules (true or false)

- Scope: ✅ IN.
- Where seen: UF Northern Regional Jan 2019 B (REG) Q32 (field lines never cross → True) and Q31 (temperature is not a field → False; ⚠️ general field idea). USC Invitational Jan 2026 B (INV) Q7 (lines go from + to –). BullSO Jan 2026 B (INV) Q15. I_bank Q27.

- How to solve: Use the four rules.
  1. Lines start on + and end on −.
  2. Lines never cross. A crossing would mean two directions at one point.
  3. Close lines = strong field.
  4. More charge = more lines.
- Worked example: True or false: "Field lines can cross where two charges are close." Answer: **False**. The field has only one direction at each point.
- Traps: A field line is not the path a charge takes (see type 3).
- Calculator: look up — type: field line rules.

8. Draw or pick the field-line picture

- Scope: ✅ IN.
- Where seen: Fall 2019 F-M Invitational (INV) Q2a (sketch the field of a negative charge → radial, inward). UT Invitational 2019 B (INV) Q29 (draw lines for ++, +−, −−). UF Northern Regional Jan 2020 C (REG) Q5 (two protons). Wichita Heights 2026 C (INV) Q11 (two like charges; graders told "don't do 11"). North Pocono Jan 2020 C (INV) Q10 (proton-electron picture). GullSO Dec 2025 (INV) Q6a (+1 nC: straight lines outward). F_slides deck 6 quiz (STAT-FIELD-LINES-DRAW).
- How to solve:
  1. Single charge: straight radial lines. Out for +, in for −.
  2. Unlike pair (dipole): lines curve from + to −. They are dense between the charges.
  3. Like pair: lines bend away from each other. There is an empty spot in the middle where E = 0.
  4. Put arrows on every line.
- Worked example: Two protons side by side. Answer: **lines leave both protons, bend away from the midpoint and go to infinity; no line joins the two; E = 0 at the midpoint**.
- Traps: Forgetting arrows loses points. For unequal charges, draw more lines on the bigger one.
- Calculator: look up — type: field line sketch.

9. Read a field-line picture: signs, sizes and strength ranking

- Scope: ✅ IN.
- Where seen: UT Invitational 2020 B (INV) Q20 (rank 4 points by field strength from line density → D, C, A, B). UF Northern Regional Jan 2019 C (REG) Q12 (label charges from field-line pictures). I_bank Q27.
- How to solve:
  1. Lines leaving = +. Lines entering = −.
  2. Charge ratio = ratio of the line counts.
  3. Strength ranks with line density: count how close the lines are.
- Worked example: 8 lines leave charge A. 16 lines enter charge B. Answer: **A is positive, B is negative, and B has twice the size of A (Q_B = −2Q_A)**.
- Traps: Do not rank by distance from a charge alone. Use the spacing of the lines in the picture.
- Calculator: calculate — type: line count ratio.

10. Equipotentials and work along them

- Scope: ✅ IN.
- Where seen: BullSO Jan 2026 B (INV) Q16 (field lines vs equipotentials → **perpendicular**) and Q17 (select-all about work to return a charge to the same equipotential). GullSO Dec 2025 (INV) Q6c (equipotentials of a point charge → circles) and Q6d (work along an equipotential → **0**). ATX Regional 2020 B (REG) Q53 (E vs equipotential surface → perpendicular). BullSO Jan 2026 C (INV) Q16-17.
- How to solve:
  1. Equipotentials are at right angles to the field lines.
  2. Point charge: circles (spheres in 3D) centred on the charge. Uniform field: flat lines parallel to the plates.
  3. $W = q\Delta V$. On one equipotential, $\Delta V = 0$, so $W = 0$.
- Worked example (BullSO 2026 B Q17): A charge drifts away from a point charge, then work brings it back to the same equipotential. The net change in potential energy is **0**, so choice A is false. The charge drifted away, so the

two charges repel: they are **alike in sign (B is true)**. The final point can be anywhere on the circle (C false). The work put in was not stored, because the net change in U is zero (D false). Best answer: **B**.
- Traps: "Same equipotential" does not mean "same point". Note: the BullSO key letter was not checked by me (see "Not confirmed").
- Calculator: look up — type: equipotential work.

11. Unit of potential; potential of the ground

- Scope: ✅ IN.
- Where seen: Wichita Heights 2026 C (INV) Q29 (unit of electric potential → volt) and Q30 (potential of the ground → zero). Pembroke Hill Dec 2025 B (INV) (unit of potential, the volt; from phi26_test.txt line 152).
- How to solve:
  1. 1 V = 1 J/C.
  2. Ground is the reference: 0 V by choice.
- Worked example: How much energy does 3 C gain crossing 12 V? W = 3 × 12 = **36 J**. So 12 V means 12 J for each coulomb.
- Traps: Only potential **differences** are physical. "0 V at ground" is a choice of reference.
- Calculator: look up — type: volt definition.

Medium

12. Field of a point charge (E = kQ/r²)

- Scope: ✅ IN.
- Where seen: GullSO Dec 2025 (INV) Q6b (+1 nC at 0.7 m → 1.84×10^{1} N/C, choice B). Wichita Heights 2026 C (INV) Q15 (2.5×10^{-2} C at 1.28 cm → 1.37×10^{12} N/C). I_bank Q25.
- How to solve:
  1. Convert to C and m.
  2. $E = k \times |Q| \div r^2$.
  3. Direction: away for +, toward for −.
- Worked example (GullSO Q6b): $E = 8.99 \times 10^{9} \times 1 \times 10^{-9} / 0.7^2 = 8.99/0.49 =$ **18.3 N/C** → choice **B (1.84×10^{1} N/C)**.
- Worked example 2 (Wichita Q15): $E = 8.99 \times 10^{9} \times 0.025/0.0128^2 =$ **1.37×10^{12} N/C**.
- Traps: Square r. 1.28 cm = 0.0128 m. The answer does not use the test charge.
- Calculator: calculate — type: point charge field.

13. Uniform field between two parallel plates (E = V/d)

- Scope: ✅ IN as a uniform field. ⚠️ Borderline if the question calls the plates a capacitor or asks about stored charge (rule 3.d excludes capacitance).
- Where seen: MIT Jan 2019 C (INV) Q3 (field in a capacitor with d = 0.10 mm → 5.0×10^{4} V/m). Wichita Heights 2026 C (INV) Q26 (field direction in a capacitor → straight from one plate to the other). I_bank Q29.
- How to solve:
  1. $E = V \div d$, with d in metres.
  2. Direction: from the + plate to the − plate, straight across.
- Worked example: 120 V across plates 4.0 mm apart. $E = 120/0.0040 =$ **3.0×10^{4} V/m**. Force on an electron = $1.602 \times 10^{-19} \times 3.0 \times 10^{4} =$ **4.8×10^{-15} N**, toward the + plate.
- Traps: The field is the same everywhere between the plates (away from the edges). It does not get weaker in the middle.
- Calculator: calculate — type: E from V and d.

14. Acceleration of an electron or proton in a field

- Scope: ✅ IN (F = ma with algebra).
- Where seen: ATX Regional 2026 C (REG) MC22 (electron released near an infinite charged sheet → **constant acceleration**). MIT Jan 2021 C (INV) Scenario A Q14 (acceleration of the central charge → 1.2×10^{13} m/s²). I_bank Q30.

- How to solve:
  1. F = qE.
  2. a = F/m.
  3. A uniform field gives a constant acceleration.
- Worked example: An electron in 3.0×10^4 N/C. a = $1.602 \times 10^{-19} \times 3.0 \times 10^4 / 9.11 \times 10^{-31}$ = **5.3×10^{15} m/s²**. A proton in the same field: **2.9×10^{12} m/s²** (1836 times smaller).
- Traps: Electron and proton feel the **same force** in the same field. The electron accelerates more because it is lighter.
- Calculator: calculate — type: acceleration in field.

15. Work and energy when a charge crosses a voltage (W = qΔV)

- Scope: ✅ IN.
- Where seen: Princeton 2021 C (INV) Q32 (2 mC between plates with 10 V, moves to one plate → key 10 mJ; see "Mistakes"). I_bank Q33. North Pocono Jan 2020 C (INV) Q32 (key 20 J; question text not checked by me).
- How to solve:
  1. W = q × ΔV.
  2. A + charge loses potential energy going from high V to low V. That energy becomes kinetic energy.
- Worked example: Moving 2.0 μC from 0 V to 150 V needs W = $2.0 \times 10^{-6} \times 150$ = **3.0×10^{-4} J**.
- Worked example 2 (Princeton 2021 Q32): If the particle crosses the full 10 V, W = 0.002 × 10 = **20 mJ**. If it starts at the midpoint, it crosses 5 V → **10 mJ** (the key).
- Traps: mC is 10^{-3} C. Read where the charge starts.
- Calculator: calculate — type: work q delta V.

16. Electron-volt conversions

- Scope: ✅ IN (energy units, rule 3.c.iv).
- Where seen: not seen as its own question in a Div B test. eV appears in ut-c and ut-b photoelectric questions (UT Invitational 2019 B/C Q22; ❌ out as photoelectric). I_bank Q34.
- How to solve:
  1. eV → J: multiply by 1.602×10^{-19}.
  2. J → eV: divide by 1.602×10^{-19}.
  3. An electron (or proton) crossing V volts gains **V eV**.
- Worked example: An electron crosses 5.0 kV. Energy = 5000 eV = $5000 \times 1.602 \times 10^{-19}$ = **8.0×10^{-16} J**.
- Worked example 2: 3.2×10^{-17} J = $3.2 \times 10^{-17} / 1.602 \times 10^{-19}$ = **200 eV**.
- Traps: An alpha particle (charge 2e) crossing 100 V gains 200 eV, not 100 eV.
- Calculator: calculate — type: eV joule.

17. Potential of one or more point charges (scalar sum)

- Scope: ✅ IN.
- Where seen: ATX Regional 2026 C (REG) MC33 (midpoint between two protons L apart → **4ke/L**). GullSO Dec 2025 (INV) Q5b (square: two + on top, two − below → V = **0**). UT Invitational 2019 B (INV) Q41 part 2 (→ 0 V). Berkeley Feb 2026 B (INV) §2 2.1i-j (potentials in a circuit; that is circuit voltage, see the DC notes). I_bank Q32.
- How to solve:
  1. V = kQ/r for each charge. Keep the sign of Q.
  2. Add the numbers. There are no directions.
- Worked example (ATX 2026 C MC33): Each proton is L/2 away. V = ke/(L/2) + ke/(L/2) = **4ke/L**.
- Worked example 2: 5.0 nC at 0.10 m. V = $8.99 \times 10^9 \times 5 \times 10^{-9} / 0.10$ = **450 V**.
- Traps: Do not square r (that is the field). Do not drop the minus sign of a negative charge.
- Calculator: calculate — type: potential sum.

18. Potential energy of a pair of charges

- Scope: ✅ IN.
- Where seen: Berkeley Feb 2026 B (INV) §3 2b (two +2 C charges 4 m apart → U = **k**). I_bank Q35.
- How to solve:

1. $U = k\, q_1\, q_2 / r$ with signs.
2. $U > 0$: like charges; you had to push them together. $U < 0$: unlike charges; they are bound.

- Worked example (Berkeley 2b): $U = k \times 2 \times 2/4 =$ **$k = 8.99 \times 10^9$ J**.
- Traps: Not squared. In Berkeley 2a, the **force** on the left charge is $k \cdot 2 \cdot 2/4^2 =$ **k/4, to the left**. Do not mix the two answers.
- Calculator: calculate — type: pair potential energy.

19. Field and potential inside a conductor

- Scope: ✅ IN.
- Where seen: Berkeley Feb 2026 B (INV) Q14 (solid conducting sphere with net + charge → **zero everywhere inside**, C). GullSO Dec 2025 (INV) Q3 (conducting sphere, $r < R$ → **0**). UF Northern Regional Jan 2019 B (REG) Q6 (potential of a charged solid conductor → **constant**). ATX Regional 2020 B (REG) Q28 (E inside a conductor around a cavity charge → 0) and Q29 (potential there → constant, nonzero). Wichita Heights 2026 C (INV) Q28 (hollow conducting tube → zero). ATX Regional 2026 C (REG) MC21 (point charge q at 2R from a solid conducting sphere; E at the centre → **0**; the key is wrong). Princeton Feb 2026 C (INV) MC1 (shell in an outside field: surface charge rearranges, field inside is zero → C).
- How to solve:
  1. In a conductor, free electrons move until they feel no force.
  2. So E = 0 inside the metal. This is true even when an outside charge is nearby.
  3. E = 0 means no work to move charge inside. So V is the same everywhere in it. It is not zero unless the conductor is grounded or neutral and far from other charges.
- Worked example (ATX 2026 C MC21): A conducting sphere sits near a point charge. The induced surface charge cancels the point charge's field inside. Answer: **E = 0 at the centre** (choice a).
- Traps: E = 0 inside does **not** mean V = 0 inside (ATX 2020 B Q29). The outside charge still changes the **surface** charge pattern (Princeton 2026 MC1: non-uniform).
- Calculator: look up — type: conductor inside field.

20. Conductor surfaces and shielding

- Scope: ✅ IN. The history question "who invented the cage" (Faraday) is ❌ out (rule 3.c.i list). Shielding of radio waves and hole size vs wavelength is ❌ out (EM waves and frequency, rules 3.c and 3.d).
- Where seen: UF Northern Regional Jan 2019 B (REG) Q26 (field meets a conductor surface at a right angle → True). UF Northern Regional Jan 2020 B (REG) Q3 (who found that a metal container blocks fields → Faraday). Berkeley Feb 2026 B (INV) Q2 (same history). Princeton Feb 2020 C (INV) Q25 and Rickards Oct 2025 C (INV) Q5 (Faraday cage behaviour; full method in Charging Methods and Electroscopes [A2] type 13). ATX Regional 2026 C (REG) SA4 (teardrop conductor: highest charge density at the point; see Charging Methods and Electroscopes [A2] type 14).
- How to solve:
  1. Just outside a conductor, E is perpendicular to the surface. A sideways part would push charges along the surface.
  2. Charge crowds at sharp points, so E is strongest there.
  3. A closed metal shell has E = 0 in its hollow, whatever is outside.
- Worked example: Why is a person inside a car safe in a lightning strike? Answer: **the charge flows over the outside of the metal body to the ground; the field inside the metal shell stays near zero. The rubber tyres are not the reason.**
- Traps: A cage does not stop the field of a charge **inside** it from reaching outside, unless the cage is grounded.
- Calculator: look up — type: shielding.

21. Net field on a line from several charges (1D superposition)

- Scope: ✅ IN.
- Where seen: UT Invitational 2019 B (INV) Q41 and UT Austin Invitational 2019 C (INV) Q42 (four ±2 mC charges at 30 m and 40 m east and west of P → **17.5 kN/C west**, V = 0). Berkeley Feb 2026 B (INV) §3 2c (two +2 C at ±2 m; E at x = 0 → 0), 2e (E at x = 1 → **16k/9 to the left**), 3b (left charge made −2 C; E at x = 1 → **20k/9 to the left**). UF Northern Regional Jan 2019 B (REG) Q37 (4.74×10^7 N/C left).

- How to solve:
  1. Choose right (or east) = +.
  2. For each charge, find $|E| = k|Q|/r^2$.
  3. Direction: away from +, toward –. Give it a sign.
  4. Add the signed values.
- Worked example (UT 2019 B Q41): +2 mC 40 m west: 11 240 N/C east. –2 mC 30 m west: 19 980 N/C west. +2 mC 30 m east: 19 980 N/C west. –2 mC 40 m east: 11 240 N/C east. Net = 2 × (19 980 – 11 240) = **17 500 N/C = 17.5 kN/C west**. Potential = $k \times 2 \times 10^{-3} \times (1/40 - 1/30 + 1/30 - 1/40)$ = **0 V**.
- Worked example 2 (Berkeley 2e): At x = 1, the left +2 C is 3 m away: k·2/9 to the right. The right +2 C is 1 m away: 2k to the left. Net = 2k – 2k/9 = **16k/9 ≈ 1.6 × 10^{10} N/C to the left**.
- Traps: Each field uses its own distance. Draw an arrow for each charge before you add.
- Calculator: calculate — type: 1d net field.

Hard

22. Zero-field point (E = 0) on a line

- Scope: ✅ IN.
- Where seen: Berkeley Feb 2026 B (INV) §3 4b (–4 C at x = 0 and +1 C at x = 3 per the figure → **x = 6**). Berkeley Feb 2026 B (INV) §3 2c-2d (two equal + charges: E = 0 at the midpoint, so a –2 C charge there does not move). I_bank Q20-21, Q28.
- How to solve:
  1. Like charges: the point is **between** them, nearer the smaller one.
  2. Unlike charges: the point is **outside**, beyond the **smaller** one. Between unlike charges both fields point the same way.
  3. Set $|Q_1|/r_1^2 = |Q_2|/r_2^2$. Take square roots, or solve the quadratic.
  4. Throw away any root that is not in the region you assumed.
- Worked example (Berkeley 4b, figure numbers): For x > 3: $4/x^2 = 1/(x-3)^2 \rightarrow 4(x-3)^2 = x^2 \rightarrow 3x^2 - 24x + 36 = 0 \rightarrow x^2 - 8x + 12 = 0 \rightarrow (x-6)(x-2) = 0$. Only **x = 6 m** is in x > 3. (x = 2 is where the two field **sizes** match, but there both point left, so they add.)
- Worked example 2 (like charges): +9 nC at x = 0 and +1 nC at x = 0.40 m. $\sqrt{9}\,(0.40 - x) = \sqrt{1} \cdot x \rightarrow 1.2 - 3x = x \rightarrow$ **x = 0.30 m**.
- Worked example 3 (Berkeley text numbers): –3 C at x = –2 and +1 C at x = +2 → **x = 4 + 2√3 ≈ 7.46 m** (see Coulombs Law A2 type 15).
- Traps: The extra root from the quadratic is the most common trap. A point where E = 0 is also where any test charge feels no force.
- Calculator: calculate — type: zero field point.

23. Zero-potential points (V = 0) on a line

- Scope: ✅ IN.
- Where seen: Berkeley Feb 2026 B (INV) §3 4c (–4 C at 0, +1 C at 3 → key **x = 2.4 only**; correct: **x = 2.4 and x = 4**). Berkeley Feb 2026 B (INV) §3 3d (–2 C at –2, +2 C at +2: where does a negative charge have U = 0 → **x = 0**).
- How to solve:
  1. V = 0 needs unlike charges.
  2. Write $Q_1/r_1 + Q_2/r_2 = 0$ with signs. Use distances (always positive).
  3. Check **each** region: left of both, between, right of both.
  4. There are usually **two** answers on the line: one between the charges and one outside, on the side of the smaller charge.
- Worked example (Berkeley 4c): Between (0 < x < 3): $-4/x + 1/(3-x) = 0 \rightarrow x = 12 - 4x \rightarrow$ **x = 2.4**. Right (x > 3): $-4/x + 1/(x-3) = 0 \rightarrow x = 4x - 12 \rightarrow$ **x = 4**. Check: –4/4 + 1/1 = 0 ✓. Left (x < 0): the distances are –x and 3 – x, so $-4/(-x) + 1/(3-x) = 0 \rightarrow -x = 12 - 4x \rightarrow x = 4$. That is not in x < 0, so there is no answer on the left. Answer: **x = 2.4 m and x = 4 m**.
- Traps: V = 0 and E = 0 are different points. At x = 2.4, E is not zero. At x = 6, E = 0 but V = –4/6 + 1/3 = –⅓ k ≠ 0.

- Calculator: calculate — type: zero potential points.

24. Centre of a symmetric charge set: field vs potential

- Scope: ✅ IN.
- Where seen: GullSO Dec 2025 (INV) Q5a (square: + + on top, − − below; field direction at the centre → **down**) and Q5b (potential → **0**). UT Invitational 2019 B (INV) Q41 (V = 0 but E ≠ 0). I_bank Q19, Q32.
- How to solve:
  1. Field: draw four arrows. Away from each +, toward each −. Cancel matching parts.
  2. Potential: add kQ/r. All r are the same, so V ∝ the sum of the charges.
- Worked example (GullSO Q5): The two top + charges push the field down and out. The two bottom − charges pull it down and in. Sideways parts cancel. Answer: **E points down (↓); V = k(q + q − q − q)/r = 0**.
- Traps: V = 0 does **not** mean E = 0. Four equal + charges give E = 0 but V = 4kq/r ≠ 0.
- Calculator: look up — type: symmetric centre field potential.

25. Field above the midpoint of two charges (direction by symmetry)

- Scope: ✅ IN (direction only, or size by similar triangles).
- Where seen: Berkeley Feb 2026 B (INV) §3 2f (two +2 C at ±2 m; a + charge 1 m above x = 0 → **straight up**) and 3c (−2 C left, +2 C right → **straight left**). ATX Regional 2021 B (REG) Q57-59 (same idea for force; see Coulombs Law A2 type 20).
- How to solve:
  1. Draw the field from each charge at the point.
  2. Equal charges: the sideways parts cancel; the result is along the bisector.
  3. Opposite charges: the up/down parts cancel; the result is parallel to the line of charges, pointing from the + side toward the − side.
- Worked example (Berkeley 3c): The +2 C on the right pushes the field up-left. The −2 C on the left pulls it down-left. Up and down cancel. Answer: **straight left**, so a + charge starts moving **left**.
- Worked example 2 (size, Berkeley 2f, my extension): Each charge is $\sqrt{(2^2 + 1^2)} = \sqrt{5}$ m away. Vertical part of each = k·2/5 × (1/√5). Total = 2 × that = **0.358k ≈ 3.2 × 10^9 N/C up**.
- Traps: The size step needs √5 and the ratio 1/√5. That is Pythagoras and similar triangles, not trig. A writer may stop at the direction.
- Calculator: calculate — type: midpoint field.

26. Two fields at right angles (Pythagoras, no trig)

- Scope: ✅ IN. The direction may be a quadrant, an axis or 45°. An exact angle like 36.9° needs arctan → ❌ out (rule 3.e.v).
- Where seen: not seen in a test. Same method as Coulombs Law A2 type 13. I_bank Q28.
- How to solve:
  1. Find E from the charge on the x-axis (it lies along x).
  2. Find E from the charge on the y-axis (it lies along y).
  3. $|E| = \sqrt{(E_x^2 + E_y^2)}$. Name the quadrant from the signs.
- Worked example: Find E at the origin. +3.0 nC at (0.30 m, 0). −4.0 nC at (0, 0.40 m). From the +3 nC: 8.99 × 10^9 × 3 × 10^{-9}/0.09 = 300 N/C toward −x (away from it). From the −4 nC: 8.99 × 10^9 × 4 × 10^{-9}/0.16 = 225 N/C toward +y (toward it). $|E| = \sqrt{(300^2 + 225^2)}$ = **375 N/C, up and to the left (second quadrant)**.
- Traps: Never add 300 + 225. Get each direction before combining.
- Calculator: calculate — type: 2d perpendicular field.

27. Where is the potential highest or lowest near a dipole?

- Scope: ✅ IN (concept).
- Where seen: North Pocono Jan 2020 C (INV) Q9 (proton and electron; where is the potential greatest → key "between"; see "Mistakes").
- How to solve:
  1. Near the + charge, V is large and positive. Near the − charge, V is large and negative.
  2. Far away, V → 0.

3. On the line, V = 0 at the midpoint between equal and opposite charges.

- Worked example: Proton at x = 0, electron at x = 1 (in units of k e). V(x) = 1/|x| − 1/|x − 1|. At x = −1: V = +0.5. At x = 0.5 (midpoint): V = **0**. At x = 0.9: V = −8.9. Answer: **the potential is greatest close to the proton, on the side away from the electron (left of the proton); the region between holds both very high and very low values.**
- Traps: "Between" is half positive and half negative. The question is badly posed. Say the point nearest the proton.
- Calculator: look up — type: dipole potential.

28. Field inside a charged insulator (trend only)

- Scope: ⚠️ Borderline. It uses Gauss's law ideas, which the rules do not name. No calculus is needed for the trend.
- Where seen: GullSO Dec 2025 (INV) Q4 (uniform insulating sphere, r < R → **increases in proportion to r**, A). ATX Regional 2020 B (REG) Q30 (E inside an insulator with a cavity charge → depends on r). ATX Regional 2026 B (REG) MC33 (field around a uniformly charged straight wire → radial).
- How to solve:
  1. Conductor: E = 0 inside (type 19).
  2. Uniform insulating sphere: inside, E grows in a straight line with r. Outside, E = kQ/r².
  3. The peak is at the surface.
- Worked example: A uniform insulating sphere has E = 400 N/C at its surface (r = R). Find E at r = R/2 and at r = 2R. Inside: **200 N/C**. Outside: 400/4 = **100 N/C**.
- Traps: Do not mix up conductor (zero) and insulator (∝ r).
- Calculator: calculate — type: insulator sphere field.

29. Energy-to-speed: release a charge and find its final speed

- Scope: ✅ IN (energy conservation, square roots).
- Where seen: MIT Jan 2021 C (INV) Scenario A Q12-15 (11 charges q = 18 μC on 11 of 12 corners of a dodecagon, r = 82 cm; centre charge Q = 2.2 μC made of alpha particles; Q13 mass → 4.6×10^{-14} kg; Q15 final speed when it flies far away → **1.4×10^{7} m/s**).
- How to solve:
  1. Starting energy: U = sum of k q Q / r over all the fixed charges.
  2. Far away, U = 0. All of U became kinetic energy.
  3. ½ m v² = U → v = √(2U/m).
- Worked example (MIT 2021): Mass: number of alphas = Q/(2e) = 6.87×10^{12}. m = $6.87 \times 10^{12} \times 6.645 \times 10^{-27}$ = **4.6×10^{-14} kg**. U = $11 \times 8.99 \times 10^{9} \times 18 \times 10^{-6} \times 2.2 \times 10^{-6}/0.82$ = 4.78 J. v = √(2 × 4.78/4.56×10^{-14}) = **1.45×10^{7} m/s** (key 1.4×10^{7}).
- Traps: Use energy, not force. The force changes as Q moves away. Potential energy adds as plain numbers, so all 11 charges count, even though their forces almost cancel.
- Calculator: calculate — type: release speed.

Nationals

30. Potential energy of three or more charges (assembly energy)

- Scope: ✅ IN (algebra only).
- Where seen: not seen in a test. Extension of Berkeley Feb 2026 B (INV) §3 2b. I_bank Q35. OpenStax College Physics 2e §19.1-19.3 style.
- How to solve:
  1. List every pair (3 charges → 3 pairs; 4 charges → 6 pairs).
  2. U = k q_i q_j / r_ij for each pair, with signs.
  3. Add them.
- Worked example: +1 μC, +2 μC and −3 μC at the corners of an equilateral triangle with side 0.10 m. U = $8.99 \times 10^{9} \times (1 \cdot 2 + 1 \cdot (-3) + 2 \cdot (-3)) \times 10^{-12}/0.10$ = **−0.63 J**.
- Traps: Count each pair once. A negative total means the group holds itself together; you must add 0.63 J to pull it apart.

- Calculator: calculate — type: assembly energy.

31. Speed of a particle accelerated through a voltage

- Scope: ✅ IN.
- Where seen: not seen in a Div B test. OpenStax College Physics 2e §19.1 (electron gun). I_bank Q34.
- How to solve:
  1. Kinetic energy gained = $|q|\,\Delta V$.
  2. $v = \sqrt{(2|q|\Delta V/m)}$.
- Worked example: An electron starts at rest and crosses 100 V. KE = 100 eV = 1.60×10^{-17} J. $v = \sqrt{(2 \times 1.60 \times 10^{-17}/9.11 \times 10^{-31})}$ = **5.9×10^{6} m/s**. A proton through the same 100 V: **1.4×10^{5} m/s**.
- Traps: Same energy, very different speeds. Speed grows with √V: four times the voltage gives twice the speed.
- Calculator: calculate — type: speed from voltage.

32. Electric flux and Gauss's law (statement and q/ε_0)

- Scope: ⚠️ Borderline. Gauss's law is not named in the 2027 rules. The statement and $\Phi = Q/\varepsilon_0$ need only algebra. Cylinders, disks and integrals are ❌ out (rule 3.e.v).
- Where seen: Princeton 2021 C (INV) Q41 (flux of a 2 nC charge → **226 V·m**). Princeton Feb 2026 C (INV) MC2 (best statement of Gauss's law → net flux ∝ charge enclosed, B). UF Northern Regional Jan 2019 B (REG) Q41 (state Gauss's law → flux = Q/ε_0). ATX Regional 2020 B (REG) Q37 (charge at a cube corner: flux through the faces → 0 and $q/24\varepsilon_0$; ⚠️ 3D symmetry). ATX Regional 2026 B (REG) FR1b-c (charge enclosed inside a shell → 0).
- How to solve:
  1. Total flux out of any closed surface = Q_inside / ε_0.
  2. Charges outside the surface add zero net flux.
- Worked example (Princeton 2021 Q41): $\Phi = 2 \times 10^{-9}/8.854 \times 10^{-12}$ = **226 V·m**.
- Traps: The size and shape of the closed surface do not matter. Only the charge inside counts.
- Calculator: calculate — type: flux q over eps0.

33. Stacked plates with a battery: which statement must be true?

- Scope: ⚠️ Borderline. Field reasoning is IN (3.c.ii). Plates joined to a battery are close to capacitance (3.d).
- Where seen: USC Invitational Jan 2026 B (INV) Q18 (three identical charged plates, battery across top and bottom; points A and B above and below the middle plate → **D: a potential difference exists between A and B**).
- How to solve:
  1. The battery holds the top plate at a higher potential than the bottom one.
  2. Potential falls steadily as you go down through a field that points down.
  3. So any two points at different heights (A above B) have different potentials.
  4. Sign claims about plate charges need more information.
- Worked example: Is "A and B are at the same potential" possible? Answer: **No. A is nearer the high-potential top plate, so V_A > V_B.**
- Traps: Do not assume each plate's field "adds up" without knowing the charges. Use the potential difference set by the battery.
- Calculator: look up — type: plates battery potential.

34. Millikan oil drop: charge of a hovering drop

- Scope: ⚠️ Borderline. Millikan is not named in the rules. The balance qE = mg uses only algebra and shows charge quantization (3.c.ii). A history question about Millikan would be out (3.c.i list).
- Where seen: not seen in a test. I_bank Q31. OpenStax College Physics 2e §18.3-18.5 problems.
- How to solve:
  1. A hovering drop: electric force up = weight down. qE = mg.
  2. q = mg/E.
  3. n = q/e must be a whole number.

- Worked example: A drop of mass 1.0×10^{-15} kg hovers in $E = 2.04 \times 10^{4}$ N/C. $q = 1.0 \times 10^{-15} \times 9.8/2.04 \times 10^{4} = 4.80 \times 10^{-19}$ C. $n = 4.80 \times 10^{-19}/1.602 \times 10^{-19} =$ **3.0 → 3e**.
- Traps: If the field points up, a hovering drop is positive. If the field points down, the drop is negative (it has extra electrons).
- Calculator: calculate — type: millikan.

35. Charge density on a plate and the field of a sheet

- Scope: ⚠ Borderline. $\sigma = Q/A$ is plain algebra. $E = \sigma/(2\varepsilon_0)$ for a sheet comes from Gauss's law (not named).
- Where seen: Wichita Heights 2026 C (INV) Q27 (1.0 μC on a 1 mm² plate; graders told to grade charge density only → key 1×10^{6} C/m; see "Mistakes").
- How to solve:
  1. Surface charge density $\sigma = Q \div A$, in C/m².
  2. For one large sheet: $E = \sigma/(2\varepsilon_0)$, on both sides, pointing away for +.
- Worked example (Wichita Q27): $A = 1$ mm² $= 1 \times 10^{-6}$ m². $\sigma = 1.0 \times 10^{-6}/1 \times 10^{-6} =$ **1.0 C/m²**. Sheet field $= 1.0/(2 \times 8.854 \times 10^{-12}) =$ **5.6×10^{10} N/C**.
- Traps: 1 mm² $= 10^{-6}$ m², not 10^{-3} m². The unit is C/m², not C/m.
- Calculator: calculate — type: charge density.

36. Stable or unstable equilibrium of a charge

- Scope: ⚠ Borderline (concept only; no named rule, no math).
- Where seen: not seen in a test. I_bank Q22. ATX Regional 2021 B (REG) Q57 (equilibrium at the origin; stability follows from Q59).
- How to solve:
  1. Push the charge a little. Does the net force push it back (stable) or further away (unstable)?
  2. A + charge midway between two fixed + charges: stable along the line, unstable sideways.
  3. A – charge midway between two fixed + charges: unstable along the line, stable sideways.
  4. Earnshaw's theorem: no set of fixed charges can hold a charge stable in every direction.
- Worked example: A −q sits at the midpoint between two fixed +Q charges. It moves a little to the right. Answer: **it is now closer to the right +Q, so it is pulled further right: unstable along the line.**
- Traps: Zero net force does not mean stable.
- Calculator: look up — type: equilibrium stability.

Out of scope (skip)

- ✗ Field of a charged disk on its axis, from integration. Rule 3.e.v (no calculus). Seen: Florida State 2021 B (STATE) Q31.
- ✗ E as the gradient of V, and Poisson's or Laplace's equation. Rule 3.e.v. Seen: Florida State 2021 B (STATE) Q32 and Q39 (metal sphere in a uniform field).
- ✗ Maxwell's equations in divergence form. Rule 3.e.v; magnetism parts also break rule 3.c. Seen: Florida State 2021 B (STATE) Q14-16.
- ✗ Gauss's law for a coaxial cable (cylinders). Rule 3.e.v. Seen: Florida State 2021 B (STATE) Q38.
- ✗ Torque between two dipoles. Trig and vector calculus (3.e.v). Seen: Florida State 2021 B (STATE) Q40.
- ✗ Capacitor fields with fixed charge as the plates move; fringing fields; stored energy. Rule 3.d (capacitance). Seen: F-M Invitational 2018 (INV) Q2; UT Invitational Fall 2025 B/C (INV) §B 2f.
- ✗ Fields at angles other than 90° or 45° that need sin, cos or arctan. Rule 3.e.v.
- ✗ Motion of a charge in a magnetic field, Lorentz force, Poynting vector. Magnetism is not a listed area (rule 3.c). Seen: ATX Regional 2020 B (REG) Q22, Q52; BullSO Jan 2026 B (INV) Q25-27.
- ✗ Photoelectric effect (eV energy of photons). Not in any listed area. Seen: UT Invitational 2019 B and C (INV) Q22.
- ✗ Faraday cage hole size vs wavelength (5G). EM waves and frequency (rules 3.c and 3.d). Seen: ATX Regional 2026 C (REG) MC35, SA2.
- ✗ Potential of a log-shaped field (4-point probe). Calculus and logs. Seen: USC Invitational Jan 2026 C (INV) Q39.

Mistakes in tests and keys

- **Berkeley Feb 2026 B §3 4c**: the key says V = 0 only at x = 2.4 and "no other solutions". There is a second point at **x = 4**. Check: −4/4 + 1/(4 − 3) = −1 + 1 = 0. Python confirms V(4) = 0. With the text's numbers (–3 C at –2, +1 C at +2), V = 0 at x = 1 and x = 4.
- **Berkeley Feb 2026 B §3 Q4**: the text gives –3 C at x = –2 and +1 C at x = 2. The figure and key use –4 C at x = 0 and +1 C at x = 3. Use the figure. E = 0 at x = 6 (figure) or x = 4 + 2√3 (text).
- **ATX Regional 2026 C MC21**: key k q/(2R)². Correct: **0**. The induced charge on the conducting sphere cancels q's field inside (from tests/ut-a.md).
- **North Pocono Jan 2020 C Q9**: key "between the particles". Near the proton V → +∞ on both sides. Between the charges V also goes to −∞ near the electron. The best answer is **d, to the left of the proton** (from tests/misc-a.md; checked with V(x) in Python).
- **Princeton 2021 C Q32**: key 10 mJ. W = qV = 2 mC × 10 V = **20 mJ** if the particle crosses the whole gap. 10 mJ is right only if it starts at the midpoint, which the question does not say.
- **Wichita Heights 2026 C Q27**: key "1 × 10^6 C/m". Correct: **1.0 C/m²** (1.0 μC ÷ 1 mm²). The key has the wrong power of ten and the wrong unit.
- **UF Northern Regional Jan 2019 B Q7**: key A (vector field). Choice E ("zero inside a spherical conducting shell") is also true. Two right answers.
- **ATX Regional 2026 C MC17** (drift field, other area): key 1.28 × 10^{-9} V/m; correct 1.28 × 10^{-6} V/m (from tests/ut-a.md).

Not confirmed

- GullSO Dec 2025 MC answers (Q2-Q6) are images in the key. The key text did not show them. The answers above (D, E, A, ↓, 0, B, circles, 0) are my own physics. Where I looked: dlG/gull26_key.txt.
- BullSO Jan 2026 B Q15-17 key letters: not visible in bullso26_key.txt text. My answers: Q15 A, Q16 B, Q17 B. Not rendered.
- North Pocono 2020 C Q32 "20 J": I did not read the question text.
- MIT Jan 2019 C Q3: I used the extraction's E = 5.0 × 10^4 V/m. I did not read the voltage from the test.
- Berkeley 2f size (0.358k) is my extension. The test asked only the direction.
- Pembroke Hill Dec 2025 B unit-of-potential question number: seen at line 152 of phi26_test.txt; I did not read its number.

Sources

- Rules: 2027 Div B Circuit Lab rule 3.c.i, 3.c.ii, 3.c.iv, 3.d, 3.e.i, 3.e.ii, 3.e.v.
- Tests (extractions in /Users/kai/Daniel/circuitlab/research/tests/ and prior/G_tests.md): Berkeley Feb 2026 B; BullSO Jan 2026 B and C; GullSO Dec 2025; Pembroke Hill Dec 2025 B; Wichita Heights 2026 C; Princeton 2020, 2021 and Feb 2026 C; MIT Jan 2019 C and Jan 2021 C; UT Invitational 2019 B/C and 2020 B; ATX Regional 2020 B, 2021 B, 2026 B and C; UF Northern Regional 2019 B/C and 2020 B/C; North Pocono Jan 2020 C; F-M 2018 and Fall 2019; Cobra 2019 B; USC Invitational Jan 2026 B/C; Florida State 2021 B (STATE).
- Raw texts read: downloads/dlB/tests/berk26_test.txt and berk26_key.txt (§3 solutions; key pages 1-2 rendered as images), bullso26_test.txt; downloads/dlG/gull26_test.txt, gull26_key.txt, wich26C_test.txt, wich26C_key.txt, mit21_test.txt, mit21_key.txt, ut19inv_Btest.txt, puso26_test.txt, pz/p21 (Princeton 2021 exam and key), atx26reg_Ctest.txt; downloads/dlFind/usc26_Btest.txt, np20inv_test.txt, nfl19reg_Btest.txt.
- Textbooks: OpenStax College Physics 2e ch. 18 (§18.4-18.8: electric field, field lines, conductors) and ch. 19 (§19.1-19.4: potential, eV, equipotentials). Kuphaldt, Lessons in Electric Circuits Vol. I ch. 1.
- Numbers checked in Python (scratchpad a2/chk.py and a2/chk2.py).

Static Hazards

This note covers the dangers of static electricity and how people control them. It covers lightning, electrostatic discharge (ESD), fuel and dust fires, Van de Graaff generators, humidity, lightning rods and Faraday cages.

Everything here is ✅ **IN at every level**. Rule 3.c.ii names "sources/hazards of static electricity". Most questions are facts and short explanations. A few are short calculations (thunder distance, Q = It, spark length).

Hazards of **current electricity** (shock, "current kills", GFCIs, birds on wires) belong to the DC/AC hazards and control-device notes. They are named only where a test mixes them in.

Key facts

- **Static electricity:** charge that builds up on an object and stays there. It is "static" because it does not flow in a circuit.
- **Electrostatic discharge (ESD):** the sudden flow of built-up charge from one object to another. A spark is one kind of ESD.
- **Spark:** the air between two objects breaks down and conducts. Dry air breaks down at about **3×10^6 V/m** (3 kV per mm). That is about 30 kV per cm.
- **Lightning** is a huge electrostatic discharge between a cloud and the ground, or inside a cloud.
- In a thunderstorm, collisions of ice crystals and soft hail separate charge. The top of the cloud becomes **positive**. The middle and lower parts become **negative**. The ground below often becomes **positive** (National Weather Service).
- A typical lightning flash is about **300 million volts** and about **30 000 A** (National Weather Service).
- Lightning heats the air to about **50 000 °F**, about **5 times hotter than the surface of the Sun** (National Weather Service). Thunder is the sound of this air expanding.
- **Flash-to-bang rule:** sound travels about **1 mile every 5 seconds** (about 1 km every 3 s). Lightning can strike **10 miles** from a storm (National Weather Service).
- **Lightning rod:** a pointed metal rod on top of a building, joined by a thick wire to a metal rod in the ground. It gives the strike a safe path to ground. Benjamin Franklin invented it (around 1750). Franklin is not on the 2027 history list.
- A **car** protects you from lightning because the metal body carries the current around you (a Faraday cage effect). The rubber tyres are **not** the reason.
- **Faraday cage:** a closed metal shell or mesh. The field inside stays near zero, whatever happens outside.
- **Humidity:** water in the air. Humid air lets charge leak away. So static is **worse on dry days** (winter, heated indoor air) and better on humid days.
- **Grounding:** joining an object to the Earth with a conductor so its charge drains away.
- **Bonding:** joining two objects to each other with a conductor so they stay at the same potential. Then no spark can jump between them.
- **ESD and electronics:** a person feels a static zap only above about **3000 V**. Much smaller discharges, below what you can feel, can still destroy chips (CMOS and MOSFET parts are very sensitive).
- **ESD protection:** wrist straps, grounded mats, anti-static bags, conductive shoes, ionizers and humidity control.
- **ESD wrist strap:** a strap on the wrist joined to ground through a large **resistor** (often quoted as 1 MΩ). The resistor protects the wearer and slows the discharge.
- **Fuel and dust:** a spark can ignite petrol vapour, propane, hydrogen, flour dust, grain dust or coal dust. Fuel trucks are bonded and grounded before they unload.
- **Van de Graaff generator:** a machine where a moving rubber belt carries charge up to a hollow metal dome. It makes hundreds of thousands of volts. It is mostly safe because the **current and total charge are tiny**.
- **Types of static discharge** (weakest to most dangerous): **corona** (a faint glow at sharp points), **brush** (short branching sparks from a curved conductor to an insulator), **spark** (a single bright channel between conductors) and **propagating brush** (a powerful discharge from a thin insulating layer on a conductor). Corona and brush are the least dangerous. Spark and propagating brush can ignite most fuels.
- **St. Elmo's fire:** a corona glow on ship masts and aircraft in a storm.

- **Uses of static** (the good side): photocopiers and laser printers, electrostatic precipitators (smoke-stack cleaners), spray painting and powder coating, dust cloths.

Formulas

What	Formula	Notes
Thunder distance (miles)	d = t ÷ 5	t in seconds
Thunder distance (km)	d = t ÷ 3	or d = 343 × t metres
Charge in a stroke	Q = I × t	I in A, t in s
Energy of a discharge	E = Q × V	an upper estimate; see the traps
Energy in kWh	kWh = J ÷ 3.6 × 10^6	
Spark gap (dry air)	V ≈ 3 × 10^6 V/m × gap	gap in metres
Gap a voltage can jump	gap ≈ V ÷ (3 × 10^6)	
Van de Graaff dome limit	V_max ≈ 3 × 10^6 × r	r = dome radius in m
Charge on the dome	Q = V·r / k	k = 8.99 × 10^9
Current through a strap	I = V ÷ R	Ohm's law

Question types

Easy

1. Lightning is static electricity (an electrostatic discharge)

- Scope: ✅ IN.
- Where seen: UT Invitational 2020 B (INV) Q50 (true or false: lightning is caused by static electricity → **True**). "Shock Value" re-tryout B, about 2012 (practice) I-21 (lightning is an example of a(n) ___ → **electrostatic discharge**). Cobra 2019 B (INV) stations sheet (sort "Lightning Bolt" into AC or DC; answer not in the text I read).
- How to solve:
  1. Charge builds up in the cloud.
  2. The air breaks down, and the charge rushes to the ground.
- Worked example: Fill in: "Lightning is a giant _____." Answer: **electrostatic discharge (a spark)**.
- Traps: Lightning is not AC. It is one-way pulses of current (each stroke is a burst of DC). Sorting it as "DC" is the usual expected answer, but confirm with the test's wording.
- Calculator: look up — type: lightning is static.

2. Dry or humid days: when does static build up?

- Scope: ✅ IN.
- Where seen: UT Invitational 2020 B (INV) Q6 (static builds up better on humid, dry, cold or hot days → **dry**). Cobra 2019 B (INV) Q11 (appliance that cuts static in a Michigan winter → **humidifier**).
- How to solve:
  1. Water molecules on surfaces let charge leak away.
  2. Dry air does not. So charge builds up on dry days.
  3. Winter air is dry, especially when heated indoors.
- Worked example (Cobra Q11): What appliance helps? Answer: **a humidifier** (it adds water to the air).
- Traps: "Cold" is not the cause. Cold air holds less water, so cold days are often dry. Pick "dry" when it is a choice.
- Calculator: look up — type: humidity static.

3. Name the hazards of static electricity

- Scope: ✅ IN.
- Where seen: Fall 2019 F-M Invitational (INV) Q3 (two hazards → lightning injuring people; fires near chemicals or in high-oxygen places; static damaging electronics). Princeton 2019 C (INV) Part 2c (greatest hazard of ESD → **ignition of flammable substances**, e.g. fuel transported in trucks).
- How to solve: List three groups.

1. Fire and explosion (fuel vapour, dust, oxygen).
2. Damage to electronics (ESD).
3. Injury (lightning; a startle that makes a worker fall).

- Worked example: Give the greatest hazard of ESD. Answer: **ignition of flammable gas, vapour or dust**.
- Traps: A normal static zap from a doorknob does not hurt a healthy person. The danger is what the spark ignites.
- Calculator: look up — type: static hazards list.

4. Ways to prevent or remove static

- Scope: ✅ IN.
- Where seen: F_slides deck 2 homework (STAT-REMOVE-PREVENT: two ways to remove or prevent static). Cobra 2019 B (INV) Q11 (humidifier).
- How to solve: Pick from this list.
  1. Add humidity.
  2. Ground yourself or the object (touch grounded metal, wrist strap).
  3. Bond objects together.
  4. Use anti-static sprays, dryer sheets, conductive mats or shoes.
  5. Use an ionizer (it sprays both signs of ions to neutralize surfaces).
- Worked example: Give two ways to stop shocks from a carpet in winter. Answer: **run a humidifier; touch a grounded metal object with a key before your finger** (the spark jumps to the key, not your skin).
- Traps: Rubber-soled shoes make static worse. They insulate you from the floor.
- Calculator: look up — type: prevent static.

5. Why is a Van de Graaff generator safe to touch?

- Scope: ✅ IN.
- Where seen: Berkeley Feb 2026 B (INV) Q16 (why is it not dangerous → **A, it has a low current**). Hawk and Hornet Invitational 2026 C (INV) Q19 (long arcs but safe to touch → key **B, high voltage but low current**). Rickards Oct 2025 C (INV) and GullSO Dec 2025 (INV) list "Van de Graaff safety" with other current-safety questions (G_tests.md).
- How to solve:
  1. The dome can reach hundreds of thousands of volts.
  2. But it stores only a few microcoulombs. The discharge lasts a tiny time.
  3. So the current through your body is small and very short. Harm needs a large current for some time.
- Worked example: Choose: (a) high current, low voltage; (b) high voltage, low current. Answer: **(b)**.
- Traps: Choice "it is grounded" is false; a grounded dome could not charge up. "The field is confined inside the dome" is false; the charge and field are on the **outside**.
- Calculator: look up — type: van de graaff safe.

6. Name the static machine in the picture

- Scope: ✅ IN (static sources).
- Where seen: MIT Jan 2019 C (INV) Q5 (pictured device → **Van de Graaff generator**) and Q6 (pictured device → **Wimshurst machine**).
- How to solve:
  1. Tall column with a round metal dome on top → Van de Graaff (belt inside the column).
  2. Two glass or plastic discs that spin opposite ways, with metal sectors and two brass balls → Wimshurst machine.
  3. Glass jar with metal foil inside and out → Leyden jar (an early charge store; ⚠️ capacitance).
- Worked example: A machine has two counter-rotating discs with foil strips and a spark gap between two balls. Answer: **Wimshurst machine** (an induction machine).
- Traps: Wimshurst, Van de Graaff and Leyden are not on the history list. Ask only "what is it" and "how does it make charge".
- Calculator: look up — type: static machine id.

7. What does a lightning rod do?

- Scope: ✅ IN.
- Where seen: not seen in a test. I_bank Q39. OpenStax College Physics 2e §18.8 (Applications of electrostatics; lightning rods).
- How to solve:
  1. The rod is pointed and higher than the building.
  2. The field is strongest at its sharp tip, so the strike attaches there.
  3. A thick wire carries the current to a rod in the ground, around the building.
- Worked example: Why must the ground wire be thick and continuous? Answer: **so the huge current reaches the Earth through low resistance, instead of jumping through the building**.
- Traps: A lightning rod does not "attract" lightning from far away. It gives a strike that would hit the building a safe path.
- Calculator: look up — type: lightning rod.

8. Lightning safety: the car, the house, the open field

- Scope: ✅ IN.
- Where seen: not seen in a test. National Weather Service lightning safety pages. I_bank Q39.
- How to solve:
  1. Safe: a building with wiring and plumbing, or a hard-topped metal car with windows shut.
  2. Not safe: open fields, under a lone tree, near water, in an open shelter.
  3. "When thunder roars, go indoors." Wait 30 minutes after the last thunder.
- Worked example: Why is a hard-topped car safe? Answer: **the metal shell carries the current around the people inside to the ground; it is not the rubber tyres**.
- Traps: A convertible or a fibreglass car does not protect you.
- Calculator: look up — type: lightning safety.

9. Uses of static electricity

- Scope: ✅ IN (sources and uses of static, rule 3.c.ii).
- Where seen: not seen in a test. I_bank Q42. OpenStax College Physics 2e §18.8 (Xerography, laser printers, ink-jet printers, electrostatic precipitators).
- How to solve: Match the use to the idea.
  1. Photocopier and laser printer: charged drum attracts toner where the image is.
  2. Electrostatic precipitator: charges smoke particles, then collects them on plates.
  3. Spray painting and powder coating: charged paint drops are pulled to the grounded part.
- Worked example: How does a smoke-stack precipitator work? Answer: **it gives the smoke particles a charge, then oppositely charged plates attract and hold them**.
- Traps: A precipitator does not "burn" smoke.
- Calculator: look up — type: static uses.

Medium

10. How a thundercloud separates charge

- Scope: ✅ IN.
- Where seen: UT Invitational fall 2018 B/C (INV) Q19 (effect of ice-crystal and hail collisions → charge separation, top +, bottom −, leading to lightning; no key).
- How to solve:
  1. Updrafts carry small ice crystals up. Heavier soft hail (graupel) falls.
  2. When they collide, the small crystals become + and the hail becomes −.
  3. The light + crystals go to the top. The − hail stays in the middle and lower cloud.
  4. The − cloud base induces + charge on the ground below.
- Worked example: Which part of a thundercloud is usually positive? Answer: **the top**.
- Traps: The ground's + charge is **induced** (see Charging Methods and Electroscopes A2). It is not rubbed on.
- Calculator: look up — type: thundercloud charge.

11. Fuel and ignition: which situation is most dangerous?

- Scope: ✅ IN.
- Where seen: ATX Regional 2026 B (REG) MC26 (greatest risk of static ignition → **fuel flowing through a plastic hose**). Cobra 2019 B (INV) Q9 (static in gas pipes starts outside or inside the pipe → **inside**). Princeton Feb 2020 C (INV) Q23 (where triboelectric precautions are taken, select all → **B, D: CMOS and MOSFET chips; bulk fuel deliveries**).
- How to solve:
  1. Look for three things together: charge build-up, an insulator that keeps the charge, and a fuel.
  2. Flowing liquids and powders rub on pipe walls and charge up (friction).
  3. A grounded metal pipe drains the charge. A plastic hose does not.
- Worked example (ATX 2026 B MC26): Choices: removing a wool sweater indoors; fuel in a grounded metal pipe; fuel in a plastic hose; touching a door handle after walking. Answer: **fuel flowing through a plastic hose** (charge builds and there is fuel).
- Traps: The sweater and the door handle make sparks but have no fuel. The grounded metal pipe has fuel but no charge build-up.
- Calculator: look up — type: ignition risk.

12. Bonding vs grounding (tanker trucks, fuel pumps)

- Scope: ✅ IN.
- Where seen: not seen in a test. I_bank Q41. Princeton 2019 C (INV) Part 2c key mentions fuel trucks.
- How to solve:
  1. **Bonding** joins two objects so they have the same potential. No spark can jump between them.
  2. **Grounding** joins an object to the Earth so its charge drains away.
  3. A tanker is bonded to the tank it fills, and both are grounded.
- Worked example: At the petrol pump, why touch the car's metal body before the nozzle? Answer: **it discharges you away from the fuel vapour, so no spark happens at the filler opening**.
- Traps: Getting back into the car while fuelling recharges you (seat friction).
- Calculator: look up — type: bonding grounding.

13. ESD and electronics: who needs protection and how?

- Scope: ✅ IN.
- Where seen: Princeton Feb 2020 C (INV) Q23 (CMOS integrated circuits and MOSFETs need triboelectric precautions → B). MIT Jan 2021 C (INV) Q91 (why ESD wrist straps have a resistor; see type 14). I_bank Q40.
- How to solve:
  1. Chips have very thin insulating layers. A few hundred volts can punch through them.
  2. You feel a zap only above about 3000 V. So you can kill a chip without feeling anything.
  3. Use a grounded wrist strap, a grounded mat, and anti-static bags.
- Worked example: A technician feels no zap. Can the chip still be damaged? Answer: **yes; discharges below about 3000 V cannot be felt but can still destroy sensitive chips**.
- Traps: Traction motors and charging an electric car (Princeton 2020 Q23 choices A and C) are not triboelectric precautions.
- Calculator: look up — type: esd electronics.

14. Why does an ESD wrist strap have a resistor?

- Scope: ✅ IN (static hazards plus Ohm's law).
- Where seen: MIT Jan 2021 C (INV) Q91 (give two reasons; key: **safety**, the resistor limits current if you touch mains; **components**, it slows the discharge so the current spike does not damage parts).
- How to solve:
  1. Safety: if your other hand touches a live wire, the resistor keeps the current through your body tiny.
  2. Parts: a direct short would discharge a charged part in a sudden, large current. The resistor makes the discharge slow and gentle.

- Worked example: A strap has a 1.0 MΩ resistor. You touch 120 V. I = 120/1.0 × 10^6 = **1.2 × 10^{-4} A = 0.12 mA**. That is below the about 1 mA you can feel.
- Traps: The strap must still be a conductor. A strap with an open circuit gives no protection.
- Calculator: calculate — type: strap current.

15. What causes a static shock?

- Scope: ✅ IN.
- Where seen: ATX Regional 2020 B (REG) Q49 and ATX Regional 2020 C Q53 (static shock is due to → key **b, conservation of charge**; doubtful).
- How to solve:
  1. You pick up extra charge (for example, from a carpet).
  2. Near a conductor, the field in the air gap becomes strong. The air breaks down.
  3. The charge rushes across until both objects share it. That rush is the shock.
- Worked example: Best description: **a discharge: built-up charge flows quickly to a conductor to even out the potential**.
- Traps: Charge is conserved during the shock, but conservation does not **cause** it. Choose "discharge" or "potential difference" if it is offered.
- Calculator: look up — type: static shock cause.

16. Types of static discharge (corona, brush, propagating brush, spark)

- Scope: ✅ IN (hard trivia).
- Where seen: Princeton 2021 C (INV) Q30 (most dangerous form → **C, propagating brush**, per G_tests.md). UT Invitational 2019 B (INV) Q13 and UT Austin Invitational 2019 C (INV) Q13 (US deaths from corona discharge 2005-2015 → key **0**).
- How to solve: Learn the order.
  1. Corona: faint glow at sharp points. Lowest energy.
  2. Brush: short sparks to an insulator. Can ignite gases like hydrogen.
  3. Spark: between two conductors. Can ignite most fuels.
  4. Propagating brush: from a thin charged insulating film on metal. Highest energy. Can ignite dusts.
- Worked example: Which is least likely to cause a fire? Answer: **corona**.
- Traps: "Spark" sounds the most dangerous. Princeton keyed propagating brush.
- Calculator: look up — type: discharge types.

17. Lightning by the numbers (order of magnitude)

- Scope: ✅ IN.
- Where seen: USC Invitational Jan 2026 B and C (INV) Q19 (about how much current does the average strike carry → **B, 10 000 A**; the typical value is about 30 000 A, so B is only the closest choice).
- How to solve: Memorize round numbers.
  1. Current: about 30 000 A (range about 5 000 to 200 000 A).
  2. Voltage: about 300 million V.
  3. Temperature: about 50 000 °F, 5 times the Sun's surface.
  4. Strike reach: up to 10 miles from the storm.
- Worked example: Which is closest to a typical lightning current: 100 A, 10 000 A, 1 000 000 A? Answer: **10 000 A** (closest to 30 000 A on a log scale).
- Traps: Compare on a log scale. 30 000 is 3 times 10 000 but 33 times smaller than 1 000 000.
- Calculator: look up — type: lightning numbers.

18. Flash-to-bang: how far away is the lightning?

- Scope: ✅ IN (static hazards; a speed calculation).
- Where seen: not seen in a test. National Weather Service ("count the seconds, divide by 5"). I_bank Q39.
- How to solve:
  1. Light arrives almost at once. Sound travels about 343 m/s.
  2. Miles = seconds ÷ 5. Kilometres = seconds ÷ 3.

- Worked example: You count 6.0 s. Distance = 343 × 6.0 = **2.1 km** (about 1.2 miles).
- Worked example 2: 10 s → 10 ÷ 5 = **2 miles**.
- Traps: Do not use the speed of light. Under 10 miles, you are in range: go indoors.
- Calculator: calculate — type: flash to bang.

19. Charge carried by a stroke (Q = I × t)

- Scope: ✅ IN (current and charge, rule 3.c.iv).
- Where seen: not seen in a test. I_bank Q39.
- How to solve:
  1. Q = I × t.
  2. Change μs to s (× 10^{-6}).
- Worked example: A stroke carries 30 kA for 30 μs. Q = 30 000 × 30 × 10^{-6} = **0.90 C**.
- Traps: This assumes a steady current. Real strokes rise fast and fade slowly, so this is an estimate.
- Calculator: calculate — type: charge from current time.

Hard

20. Energy of a lightning flash

- Scope: ✅ IN (energy, rule 3.c.iv).
- Where seen: not seen in a test. I_bank Q39.
- How to solve:
  1. E = Q × V.
  2. Change to kWh: divide by 3.6 × 10^6.
- Worked example: A flash moves 5 C through 3 × 10^8 V. E = 5 × 3 × 10^8 = **1.5 × 10^9 J**. In kWh: 1.5 × 10^9/3.6 × 10^6 = **417 kWh**.
- Traps: Q × V is an upper estimate. The voltage falls as the charge flows, and most energy becomes heat, light and sound in the air.
- Calculator: calculate — type: lightning energy.

21. Spark length and breakdown voltage

- Scope: ✅ IN (static field and hazards).
- Where seen: not seen in a test. OpenStax College Physics 2e §18.7-18.8 (air breakdown at about 3 × 10^6 V/m).
- How to solve:
  1. Breakdown field of dry air ≈ 3 × 10^6 V/m.
  2. V needed = 3 × 10^6 × gap. Gap reached = V ÷ 3 × 10^6.
- Worked example: Your finger gets a 5 mm spark from a doorknob. Voltage ≈ 3 × 10^6 × 0.005 = **15 000 V**.
- Worked example 2: A 1.0 cm spark needs about **30 000 V**.
- Traps: mm → m (5 mm = 0.005 m). Humid air and sharp points break down at lower voltages.
- Calculator: calculate — type: spark gap voltage.

22. Van de Graaff dome: maximum voltage and charge

- Scope: ✅ IN (field and potential of a sphere, algebra only).
- Where seen: not seen in a test. I_bank Q38.
- How to solve:
  1. At the surface of a sphere, E = kQ/r² and V = kQ/r. So E = V/r.
  2. The dome sparks when E reaches 3 × 10^6 V/m. So V_max = 3 × 10^6 × r.
  3. Q = V × r ÷ k.
- Worked example: Dome radius 0.20 m. V_max = 3 × 10^6 × 0.20 = **6.0 × 10^5 V**. Q = 6.0 × 10^5 × 0.20/8.99 × 10^9 = **1.3 × 10^{-5} C (13 μC)**.
- Traps: A bigger dome reaches a higher voltage. That is why big machines have big round domes; sharp points would leak charge.
- Calculator: calculate — type: van de graaff max.

23. Corona and charge at sharp points (St. Elmo's fire)

- Scope: ✅ IN.
- Where seen: ATX Regional 2026 C (REG) SA4 (teardrop conductor: highest charge density at the pointed end; see Charging Methods and Electroscopes A2 type 14). UT Invitational 2019 B (INV) Q13 (corona deaths, key 0).
- How to solve:
  1. Charge crowds where a conductor curves most sharply.
  2. The field is strongest there. The air near the tip breaks down first and glows.
- Worked example: Why do high-voltage lines use smooth, wide fittings instead of sharp ends? Answer: **sharp ends make strong fields that cause corona, wasting power and making noise**.
- Traps: Corona is a slow leak, not one big spark.
- Calculator: look up — type: corona sharp points.

Nationals

24. Will a spark ignite the vapour? (compare energies)

- Scope: ✅ IN as an energy comparison. Finding the spark energy from body **capacitance** (½CV²) is ❌ out (rule 3.d).
- Where seen: not seen in a test. Princeton 2019 C (INV) Part 2c (ignition is the greatest ESD hazard).
- How to solve:
  1. Find the spark energy from what is given (often E = qV or a stated value).
  2. Compare it with the **minimum ignition energy** of the fuel (the smallest spark energy that lights it).
  3. Spark energy greater than the minimum → it can ignite.
- Worked example: A discharge releases 10 mJ. Petrol vapour has a minimum ignition energy of about 0.2 mJ. Answer: **10 mJ is 50 times the minimum, so it can ignite the vapour**.
- Traps: The 0.2 mJ value is a commonly quoted figure; the test will give its own value. A shock you can barely feel can still light fuel.
- Calculator: calculate — type: ignition energy compare.

25. Open-response: does the armour or cage protect you?

- Scope: ⚠️ Borderline. Static and hazards are IN. The Faraday-cage argument is a field idea that the rules do not name, but it fits "properties of fields" (3.c.ii).
- Where seen: North Pocono Jan 2020 C (INV) Q34 (a fantasy "shocking grasp" spell against metal armour: does it make sense? Key: credit for a real comparison, such as a car in lightning, plus either a "makes sense" reason about a large potential difference causing an arc, or a "does not make sense" reason about the armour acting as a Faraday cage).
- How to solve:
  1. Give a real-life comparison (a car in lightning).
  2. Pick a side and explain with fields: a closed metal shell carries charge around the inside; a spark can still jump to the metal from outside.
- Worked example: Does metal armour protect a knight from lightning? Answer: **mostly yes if it is a closed shell, like a car: the current flows over the outside metal. Gaps and skin contact weaken the protection.**
- Traps: Give both parts the key asks for: a comparison and a reason.
- Calculator: look up — type: faraday cage argument.

26. Combustible dust and oxygen-rich places

- Scope: ✅ IN.
- Where seen: Fall 2019 F-M Invitational (INV) Q3 key ("starting fires near chemicals, high-oxygen environment"). I_bank Q41.
- How to solve:
  1. A dust explosion needs fine dust in the air, oxygen, a closed space and a spark.
  2. Pure oxygen (hospitals, welding) makes things burn far more easily, so a small spark is enough.
  3. Control: ground the machines, keep humidity up, clean up dust, ban static-making clothes.
- Worked example: Why do grain silos and flour mills ground their machines? Answer: **moving grain and flour build up static; a spark in the dusty air can set off an explosion**.

- Traps: A solid lump of flour or grain burns slowly. The danger is fine dust in the air.
- Calculator: look up — type: dust explosion.

Out of scope (skip)

- ❌ Discharge time of a person through a wrist strap (RC time constant). Capacitance is excluded (rule 3.d), and exponentials need more math (3.e.v).
- ❌ Spark energy from body capacitance, $E = \frac{1}{2}CV^2$. Rule 3.d (capacitance).
- ❌ Faraday cage hole size vs radio wavelength (5G). EM waves and frequency (rules 3.c and 3.d). Seen: ATX Regional 2026 C (REG) MC35 and SA2.
- ❌ Who invented the Faraday cage (Faraday) or the lightning rod (Franklin) as history. Rule 3.c.i lists only Ampère, Coulomb, Kirchhoff, Volta, Ohm, Hertz and Boole. Seen: Berkeley Feb 2026 B (INV) Q2; UF Northern Regional Jan 2020 B (REG) Q3.
- ❌ Lightning's magnetic effects and induced currents in wiring. Magnetism is excluded.
- Note: "current kills, not voltage" (Rickards Oct 2025 C Q73), the clown on a wire (Nevada 2026 C Q29) and birds on power lines (GullSO Dec 2025 Q14) are **current** hazards. They are IN under DC/AC hazards, not this note.

Mistakes in tests and keys

- **ATX Regional 2020 B Q49 / C Q53**: key "conservation of charge" as the cause of a static shock. This is doubtful. The cause is a discharge across a potential difference. Charge is conserved, but that is not why the shock happens (from tests/atx-reg.md).
- **USC Invitational Jan 2026 B/C Q19**: key 10 000 A. The typical current is about 30 000 A (National Weather Service). 10 000 A is only the closest choice, so the key is acceptable.
- **Berkeley Feb 2026 B Q16**: the key "low current" is right. Choice D ("the field is confined inside the metal dome") is physically wrong: the charge sits on the outside of the dome.

Not confirmed

- UT Invitational 2019 B Q13 key "0 corona deaths 2005-2015": I found no source. Where I looked: the test and key text only.
- Princeton 2021 C Q30 key letter (propagating brush): taken from prior/G_tests.md. The key text repeats the choices without a mark.
- "1 MΩ" for wrist-strap resistors and "up to 35 kV" from walking on carpet: common figures (I_bank Q40), but a web search did not find an ESD Association page that confirms them. One lecture source gives 250 kΩ as the minimum grounding resistance for a person.
- About 3000 V to feel a zap: from a Desco Industries Q&A page (feel above about 3 kV, see above about 5 kV, hear above about 6 kV). Not checked against the ESD Association.
- Petrol vapour minimum ignition energy 0.2 mJ: a commonly quoted value. I did not check a primary source.
- Lightning charge per flash (5 C used above) is an example value. I did not verify it.
- Cobra 2019 B stations "Lightning Bolt: AC/DC" answer: not in the text I read.

Sources

- Rules: 2027 Div B Circuit Lab rule 3.c.i, 3.c.ii, 3.c.iv, 3.d, 3.e.v.
- Tests (extractions in /Users/kai/Daniel/circuitlab/research/tests/ and prior/G_tests.md): UT Invitational fall 2018, 2019 B/C and 2020 B; ATX Regional 2020 B/C and 2026 B/C; Cobra 2019 B; Fall 2019 F-M Invitational; North Pocono Jan 2020 C; Princeton 2019, 2020 and 2021 C; MIT Jan 2019 C and Jan 2021 C; Berkeley Feb 2026 B; Hawk and Hornet 2026 C; USC Invitational Jan 2026 B/C; "Shock Value" re-tryout B (about 2012); Rickards Oct 2025 C; GullSO Dec 2025; Nevada 2026 C.
- Raw texts read: downloads/dlG/ut20inv_Btest.txt (Q6, Q50), hh26_test.txt and hh26_key.txt (Q19), mit21_test.txt and mit21_key.txt (Q91), atx26reg_Btest.txt (MC26), pz/p19 Princeton 2019 test and key (Part 2), pz/p20 Princeton 2020 exam and key (Q23), pz/p21 Princeton 2021 exam (Q30), ut19inv_Btest.txt (Q13);

downloads/dlFind/usc26_Btest.txt (Q19), fm19inv_key.txt (Q3), np20inv_key.txt (Q34), cobra19_Btheory.txt; downloads/dlB/tests/berk26_key.pdf page 2 (Q16, rendered).

- Web: National Weather Service, Lightning power (300 million V, 30 000 A) and Lightning science overview (50 000 °F, 5 times the Sun's surface, cloud charge layers, 5 s per mile, 10 miles). Desco Industries Q&A (feel above about 3000 V). Wroclaw University EMC lecture (250 kΩ minimum grounding).
- Textbooks: OpenStax College Physics 2e §18.7-18.8 (conductors, corona, applications of electrostatics). Kuphaldt, Lessons in Electric Circuits Vol. I ch. 1.
- Numbers checked in Python (scratchpad a2/chk.py and a2/chk2.py).

TAB A3

DC and AC

Batteries and Cells 219

Other DC Sources, Converters and DC Uses 230

AC Characteristics 240

Household Wiring 251

Shock and Safety 262

Each note: key facts, formulas, then question types from Easy to Nationals with worked examples.

Batteries and Cells

Batteries are the main **DC source** (DC = direct current, current that flows one way only). Rule area **3.c.iii** says "Characteristics, sources, uses, and hazards of direct current (DC) and alternating current (AC)". So battery facts, battery math and battery hazards are ✅ **IN at every level**. Volta made the first battery, and Volta is on the history list (rule 3.c.i). The hands-on list uses batteries too: "Given some wires, batteries, resistors, and 2 LEDs..." (rule II.c.iii) and "calculate how long it would take to charge a specific battery" (rule II.c.iv).

Two parts are different:

- Electrode-potential math (E°cell from a chemistry table) is ⚠️ **borderline**. It is chemistry, but it is only subtraction.
- Networks with two batteries **and** a load need KVL/KCL. They are ✅ **States/Nationals only** (rule 3.c.ix).

Key facts

Words

- A **cell** is one unit that turns chemical energy into electrical energy.
- A **battery** is two or more cells joined together. People also call one cell a "battery" (an AA "battery" is one cell).
- An **electrode** is a solid conductor where the chemical reaction happens. Each cell has two.
- The **electrolyte** is the material between the electrodes. It lets **ions** (charged atoms) move inside the cell.
- The **separator** is a thin porous layer. It keeps the electrodes from touching (a short circuit) but lets ions pass.
- **Oxidation** = losing electrons. **Reduction** = gaining electrons. Memory aid: "OIL RIG" (Oxidation Is Loss, Reduction Is Gain).
- The **anode** is where oxidation happens. In a cell that is giving power (discharging), the anode is the **negative (–) terminal**.
- The **cathode** is where reduction happens. In a discharging cell, the cathode is the **positive (+) terminal**. Memory aid: "An Ox, Red Cat".
- Outside the cell, electrons flow from – to + through the wire. Conventional current flows from + to – outside the cell.
- Inside the cell, the chemistry pushes charge "uphill" from – to +. This push is the **EMF** (electromotive force), measured in volts.

Types

- A **primary cell** is single-use. You cannot safely recharge it. Examples: alkaline, carbon-zinc (zinc-carbon), lithium coin, silver oxide, zinc-air.
- A **secondary cell** is rechargeable. Examples: lead-acid, NiCd, NiMH, lithium-ion (Li-ion), lithium-polymer (LiPo), $LiFePO_4$.
- A **dry cell** has a paste electrolyte. Examples: carbon-zinc, alkaline. It does not spill.
- A **wet cell** has a liquid electrolyte. Example: a flooded lead-acid car battery (sulfuric acid). Volta's pile and the Daniell cell were wet cells.

Common voltages (nominal = the typical working value)

Chemistry	Volts per cell	Primary or secondary	Where used
Carbon-zinc (Leclanché dry cell)	1.5 V	Primary	Cheap AA, clocks
Alkaline (Zn / MnO_2, KOH electrolyte)	1.5 V	Primary	AA, AAA, C, D, 9 V
Silver oxide	1.55 V	Primary	Watch button cells
Zinc-air	1.4 V	Primary	Hearing aids
Lithium coin (CR2032)	3.0 V	Primary	Watches, key fobs, motherboards
Lead-acid	2.0 V (2.1 V full)	Secondary	Cars (6 cells = 12 V), UPS
NiCd	1.2 V	Secondary	Old cordless tools
NiMH	1.2 V	Secondary	Rechargeable AA

Chemistry	Volts per cell	Primary or secondary	Where used
Li-ion / LiPo	3.6-3.7 V (4.2 V full, about 3.0 V empty)	Secondary	Phones, laptops, EVs, drones
$LiFePO_4$ (lithium iron phosphate)	3.2 V	Secondary	Solar storage, some EVs
Lemon or potato cell (Zn / Cu)	about 0.9 V (range 0.7-1.2 V)	Primary (it uses up the zinc)	Science demos

- AA, AAA, C and D alkaline cells all give **1.5 V**. Size changes the **capacity**, not the voltage.
- A rectangular **9 V** battery has **six 1.5 V cells** in series inside.
- A **12 V car battery** has **six 2 V lead-acid cells** in series. Fully charged, it reads about 12.6 V (6 × 2.1).
- Heavy trucks and many diesel engines use **24 V** (two 12 V batteries in series). Cars use 12 V.
- A rechargeable NiMH AA gives only 1.2 V. Most devices still work, because a "1.5 V" alkaline falls to about 1.2 V for much of its life.

Capacity and energy

- **Capacity** is how much charge the battery can deliver. Units: ampere-hours (Ah) or milliampere-hours (mAh).
- 1 Ah = 1 A for 1 hour = 3600 coulombs (C). 1 mAh = 3.6 C.
- **Energy** = capacity × voltage. Units: watt-hours (Wh). 1 Wh = 3600 joules (J). 1 kWh = 3.6 MJ.
- Typical capacities (approximate; they depend on brand and current): AA alkaline 2000-3000 mAh; AAA alkaline about 1000-1200 mAh; 9 V alkaline about 500-600 mAh; NiMH AA 1900-2800 mAh; CR2032 about 220-240 mAh; phone Li-ion 3000-5000 mAh; car battery 40-100 Ah.
- **C-rate** describes current compared with capacity. 1C means the current that empties the battery in 1 hour. For a 2000 mAh cell, 1C = 2000 mA and 0.5C = 1000 mA.
- High current gives less usable capacity than the label (rate effect). Cold also lowers capacity.
- **Specific energy** (energy per kilogram, approximate): lead-acid about 30-40 Wh/kg; NiMH about 60-120 Wh/kg; alkaline about 100-150 Wh/kg; Li-ion about 150-270 Wh/kg. This is why phones and EVs use Li-ion.

Packs

- Cells in **series** (+ to –): voltages add. Capacity (mAh) stays the same as one cell.
- Cells in **parallel** (+ to +, – to –): voltage stays the same. Capacities add.
- Energy (Wh) adds either way.
- Pack names: "3S2P" means 3 cells in series, and 2 of those strings in parallel.
- One cell put in backward in a series string **subtracts** its voltage. The total drops by 2 × (one cell voltage).
- Only put **equal** cells in parallel. A strong cell pushes current into a weak one and heats it.

Internal resistance (short version; full note: Internal Resistance and Power Transfer A5)

- A real battery acts like an ideal EMF ε in series with a small resistor r.
- Terminal voltage V_t = ε – I·r when the battery gives current.
- Short-circuit current I_sc = ε/r. All the power then turns into heat inside the battery.
- r rises when a battery is old, cold or nearly empty. A weak battery can read full voltage with no load, then drop under load.

History (Volta is IN as history; the others are battery facts)

- 1780s: Luigi **Galvani** saw frog legs twitch with two metals. He called it "animal electricity".
- **1800: Alessandro Volta** built the **voltaic pile**. It was zinc and copper (or silver) discs stacked with brine-soaked cardboard. It was the first battery, the first steady source of current. The unit **volt** honors him.
- 1836: John Frederic **Daniell**: the Daniell cell (Zn/Cu, about 1.1 V). It gave a steadier voltage. Telegraphs used it.
- 1859: Gaston **Planté**: the lead-acid cell, the first rechargeable battery.
- 1866: Georges **Leclanché**: the zinc-carbon cell. Carl Gassner made a "dry" paste version in 1886.
- 1899: Waldemar **Jungner**: the NiCd cell.
- 1950s: Lewis **Urry** (Eveready): the long-life alkaline cell.
- 1991: Sony sold the first commercial Li-ion battery. The 2019 Nobel Prize in Chemistry went to Goodenough, Whittingham and Yoshino for Li-ion.

Hazards

- **Short circuit:** a wire, coin, key or wrench across the terminals. Very high current, heat, burns, fire. Never carry loose batteries with coins.
- **Li-ion thermal runaway:** damage, overcharge or a short heats the cell, which makes more heat. It can swell, vent, catch fire or explode. Never use a swollen battery.
- **Lead-acid:** sulfuric acid burns skin and eyes. Charging makes **hydrogen gas**, which a spark can explode. Remove metal rings and watches before car-battery work.
- **Button-cell ingestion:** a swallowed coin cell can burn through a child's throat in about 2 hours. Get emergency help at once.
- **Do not** recharge primary cells. Do not mix old and new cells, or different types, in one device. Do not throw batteries in fire.
- A 12 V car battery rarely gives a dangerous **shock** through dry skin, but it can deliver hundreds of amps into a short. The danger is heat, sparks and gas. See Shock and Safety A3.

Formulas

What	Formula	Notes
Charge from current	Q = I·t	C = A·s
Capacity to coulombs	Q (C) = mAh × 3.6	1 Ah = 3600 C
Energy stored	E (Wh) = capacity (Ah) × V	E (J) = Wh × 3600
Runtime	t (h) = capacity (mAh) ÷ current (mA)	Or Wh ÷ W
C-rate current	I = C-rate × capacity	1C empties in 1 h
Series pack	V = n·V_cell, mAh = cell mAh	r_total = n·r
Parallel pack	V = V_cell, mAh = m × cell mAh	r_total = r/m
nSmP pack	V = n·V_cell, mAh = m × cell mAh, cells = n·m	Wh = cells × cell Wh
One cell reversed in series	V = (n − 2)·V_cell	Same-voltage cells
Terminal voltage	V_t = ε − I·r; I = ε/(R + r)	Discharging
Charging	V_t = ε + I·r	Current into +
Short-circuit current	I_sc = ε/r; heat P = I_sc²·r	
Cell voltage from electrodes	E°cell = E°(cathode) − E°(anode)	⚠ Borderline
Specific energy	Wh/kg = Wh ÷ mass (kg)	
Cells needed	n = target V ÷ V_cell, round **up**	

Standard reduction potentials (25 °C, volts; chemistry tables): Li −3.04, Mg −2.37, Al −1.66, Zn −0.76, Fe −0.44, Ni −0.25, Pb −0.13, H_2 0.00, Cu +0.34, Ag +0.80. The metal with the **lower** (more negative) value is the anode (−).

Question types

Easy

1. What does a battery convert? (energy conversion)

- Scope: ✅ IN.
- Where seen: SO MYSO Div B Practice Test, Nov 2020, Div B, practice, Q7 (converts chemical to electrical energy → battery). JC Booth Invitational "Shock Value", 2011, Div B, INV, Q4-6 (battery, motor, generator: chem → elec, elec → mech, mech → elec). "Shock Value" tryout test, about 2012, Div B, practice, VI-6 (same three).
- How to solve:
  1. A battery: chemical → electrical.
  2. A motor: electrical → mechanical. A generator: mechanical → electrical. (Motors and generators use magnetism, which is OUT, but naming the energy change is common.)
  3. A solar cell: light → electrical. A thermocouple: heat → electrical. A fuel cell: chemical (fuel) → electrical.
- Worked example: "Which device turns chemical energy into electrical energy: motor, battery, generator, bulb?" **Answer: battery.**

- Traps: A battery does not "store electricity" or "store charge" in the way a capacitor does. It stores **chemical energy**. A battery that is "charged" holds chemicals ready to react.
- Calculator: look up — type: energy conversion battery.

2. Cell or battery? (definitions)

- Scope: ✅ IN.
- Where seen: not seen as a stand-alone test question. B_topics §3b and F_slides deck 3 teach it.
- How to solve:
  1. One electrochemical unit = cell.
  2. Several cells joined = battery.
- Worked example: "How many cells are inside a 9 V alkaline battery?" 9 ÷ 1.5 = **6 cells**. "How many in a 12 V car battery?" 12 ÷ 2 = **6 cells**.
- Traps: "AA battery" is really one cell. A 9 V battery is a true battery.
- Calculator: look up — type: cells in 9V / car battery.

3. Nominal voltage by chemistry

- Scope: ✅ IN.
- Where seen: not seen as a direct chemistry-to-voltage question. Official practice deck 3 (F_slides, BATT-IDENTIFY-POLARITY: common voltages 1.5, 3, 6, 9, 12, 24 V).
- How to solve: use the voltage table in Key facts.
- Worked example: "What voltage does one NiMH AA cell give?" **1.2 V.** "One Li-ion phone cell?" **3.7 V nominal (4.2 V full).** "A CR2032?" **3 V.**
- Traps: Size does not change voltage. A D cell and an AAA cell are both 1.5 V. Li-ion "3.6 V" and "3.7 V" are the same kind of cell. Brands label them differently.
- Calculator: look up — type: chemistry name (alkaline, NiMH, Li-ion, lead-acid, CR2032).

4. Battery voltage by use (car, truck, devices)

- Scope: ✅ IN.
- Where seen: UT Austin Invitational 2019, Div B, INV, Q11 (diesel-engine battery → key 24 V). Same question in UT Austin Invitational 2019, Div C, INV, Q11. Northview Sylvania Invitational, Dec 2025, Div C, INV, Q47 (truck battery → 24 V).
- How to solve:
  1. Cars: 12 V. Heavy trucks, buses, many diesel engines, some boats: 24 V. Small aircraft: often 24 V or 28 V systems (the 28 V figure is the charging bus; not confirmed for every plane).
  2. Phones: 3.7 V Li-ion cell. Laptops: about 11-15 V packs (3 or 4 Li-ion cells in series). EVs: about 400 V or 800 V packs.
- Worked example: "What voltage does a heavy diesel truck's starting system use?" **24 V** (two 12 V batteries in series).
- Traps: The UT question said "diesel engine". Diesel **cars** and pickups use 12 V. Only the heavy trucks use 24 V. See Mistakes.
- Calculator: look up — type: battery voltage car / truck / laptop / EV.

5. Primary or secondary? (single-use or rechargeable)

- Scope: ✅ IN.
- Where seen: Cobra Invitational, 2019-20, Div B, INV, Q15 (laptop, CR2032, phone, AA alkaline, "longer service per charge, good retention" → secondary, primary, secondary, primary, primary). UF Northern Regional, Jan 2020, Div B, REG, Q40 (secondary = rechargeable). "Shock Value" tryout, about 2012, Div B, practice, VI-9. Princeton Invitational, Mar 2021, Div C, INV, Q26. cyanophycean314 practice test, about 2013, Div C, practice, Q7 (→ alkaline).
- How to solve:
  1. Can it be recharged safely? Yes → secondary. No → primary.
  2. Phone, laptop, car, power tool, EV → secondary. Alkaline AA, carbon-zinc, coin cell, hearing-aid zinc-air → primary.

- Worked example: "Classify: (a) CR2032 coin, (b) car battery, (c) NiMH AA, (d) alkaline 9 V." **(a) primary, (b) secondary, (c) secondary, (d) primary.**
- Traps: "Rechargeable alkaline" cells exist but are rare. The test answer for "alkaline" is primary. Primary cells usually hold their charge longer on a shelf (low self-discharge). That is why Cobra's "good retention" item is primary.
- Calculator: look up — type: primary / secondary + chemistry.

6. Dry cell or wet cell?

- Scope: ✅ IN.
- Where seen: JC Booth Invitational "Shock Value", 2011, Div B, INV, Q21-22 (dry = paste electrolyte; wet = liquid electrolyte). Official practice deck 3 (F_slides, BATT-DRY-VS-WET; the deck also labels the parts of a dry cell: brass cap, seal, zinc can, carbon rod, chemical mix).
- How to solve:
  1. Paste or gel electrolyte → dry. Liquid → wet.
  2. Dry examples: carbon-zinc, alkaline. Wet examples: flooded lead-acid car battery, Volta's pile, Daniell cell.
- Worked example: "Name the part of a zinc-carbon dry cell that is the negative electrode." **The zinc can.** The center **carbon rod** is the positive current collector.
- Traps: A "dry" cell is not dry inside. It has a moist paste. Without water, ions cannot move.
- Calculator: look up — type: dry cell / wet cell.

7. Anode or cathode? Which is positive? Where does oxidation happen?

- Scope: ✅ IN.
- Where seen: Pembroke Hill Invitational, Dec 6 2025 (2026 season), Div B, INV, Q18 (where does oxidation occur → anode). JC Booth "Shock Value", 2011, Div B, INV, Q30 (anode = negative electrode) and Q32 (cathode = positive electrode). Rickards Invitational, Oct 2025 (2026 season), Div C, INV, Q30 (cell parts). Northview Sylvania, Dec 2025, Div C, INV, Q32.
- How to solve:
  1. Anode = oxidation (loses electrons). Cathode = reduction (gains electrons).
  2. In a cell giving power: anode = –, cathode = +.
  3. Electrons leave the anode, go through the outside wire, and enter the cathode.
- Worked example: "In a zinc-copper cell, which metal is oxidized, and which terminal is it?" **Zinc is oxidized. It is the anode, the – terminal.**
- Traps: While **charging** a rechargeable cell, the reactions reverse. Chemists then call the + electrode the anode. Tests almost always mean a discharging cell. Say "in a discharging cell" if you write a free answer.
- Calculator: look up — type: anode / cathode / oxidation.

8. Parts of a cell (electrodes, electrolyte, separator)

- Scope: ✅ IN.
- Where seen: "Shock Value" tryout, about 2012, Div B, practice, I-8 (ions gather at the → electrodes) and I-14 (what keeps electrons from going straight from anode to cathode → separator/electrolyte). JC Booth "Shock Value", 2011, Div B, INV, Q16 (battery electrodes → two dissimilar metals). "Shock Value" tryout VI-12 (home-made battery electrodes → zinc and copper).
- How to solve:
  1. Two **different** conductors (electrodes) + an electrolyte = a cell.
  2. The electrolyte carries ions, not electrons. Electrons must go through the outside circuit.
  3. The separator stops the electrodes touching.
- Worked example: "Two copper nails in a lemon. What voltage?" **About 0 V.** Both electrodes are the same metal, so there is no difference in their pull on electrons.
- Traps: Same metal + same metal gives no voltage. The electrolyte must not conduct electrons (if it did, the cell would short itself).
- Calculator: look up — type: parts of a cell.

9. Battery symbol and polarity

- Scope: ✅ IN.
- Where seen: Official practice deck (F_slides, BATT-IDENTIFY-POLARITY, hands-on: read voltage and polarity by inspection). BullSO Invitational 2026, Div B, INV, short answer 1c (identify the battery symbol).
- How to solve:
  1. In the symbol, the **long thin line is +**. The **short thick line is –**.
  2. Several pairs of lines = a battery of several cells.
  3. On real batteries: + is marked, often with a bump (AA) or the smaller snap (9 V). Wires: red = +, black = –.
- Worked example: "On a 9 V battery, which snap is positive?" **The smaller round snap is +. The larger hexagonal snap is –.** (The battery clip is the reverse shape.)
- Traps: Do not judge the sign by the length of the whole drawing. Look at each pair of lines.
- Calculator: look up — type: battery symbol.

10. Can a battery give AC? (battery = DC)

- Scope: ✅ IN.
- Where seen: BullSO Invitational 2026, Div B, INV, Q21 ("Are batteries able to output AC?" → No).
- How to solve:
  1. A battery has a fixed + and a fixed – terminal. So its current goes one way: DC.
  2. To get AC from a battery, you need an **inverter** (DC → AC). See Other DC Sources A3.
- Worked example: "A camping kit runs a 120 V AC lamp from a 12 V battery. What device is between them?" **An inverter.**
- Traps: A switch that flips a battery's leads back and forth makes a crude AC square wave. The battery itself still gives DC.
- Calculator: look up — type: battery AC.

Medium

11. Cells in series and in parallel (voltage and capacity)

- Scope: ✅ IN.
- Where seen: official practice deck 3 (F_slides, BATT-CELLS-IN-SERIES: lemon cells at 0.7 V; how many to pass 2 V → 3). Question bank I_bank A12. Hands-on task II.c.iii uses batteries in packs.
- How to solve:
  1. Series: add the voltages. Capacity = one cell's capacity.
  2. Parallel: voltage = one cell. Add the capacities.
  3. Energy (Wh) = total cells × one cell's Wh, either way.
- Worked example: four 1.5 V, 2500 mAh AA cells. All in series (4S): **6.0 V, 2500 mAh, 15 Wh.** Two strings of two (2S2P): **3.0 V, 5000 mAh, 15 Wh.**
- Traps: Parallel does **not** add voltage. Series does **not** add mAh. Students mix these up.
- Calculator: calculate — type: n series, m parallel, cell V, cell mAh.

12. One cell put in backward

- Scope: ✅ IN.
- Where seen: not seen in a test. Question bank I_bank A12.
- How to solve:
  1. Add the voltages of the cells that point forward.
  2. Subtract the voltage of each reversed cell.
  3. For equal cells: $V = (n - 2 \cdot k) \cdot V_{cell}$, where k = reversed cells.
- Worked example: a flashlight takes four 1.5 V cells in series. One is put in backward. $V = 6.0 - 2 \times 1.5 =$ **3.0 V**. The bulb is dim.
- Traps: The reversed cell does not just "add zero". It subtracts. It also gets pushed backward (charged), which can make it leak.
- Calculator: calculate — type: n cells, reversed count, cell V.

13. Runtime from capacity and current

- Scope: ✅ IN.
- Where seen: UT Austin Invitational fall 2018, Div B/C, INV (no key) (battery mWh and W → hours; listed in ut-c as "Battery mWh / W -> hours"). Question bank I_bank A13.
- How to solve:
  1. Put capacity and current in the same units (mAh with mA, or Ah with A).
  2. t = capacity ÷ current. Or t = energy (Wh) ÷ power (W).
- Worked example: a 2500 mAh AA cell runs a 500 mA toy. t = 2500 ÷ 500 = **5.0 h**. A 50 Wh laptop battery runs a 10 W load: t = **5.0 h**.
- Traps: Watch the units. 2.5 Ah ÷ 500 mA is not 0.005 h. Convert first. Real runtime is often less at high current.
- Calculator: calculate — type: capacity, current (or Wh, W).

14. Energy and charge stored (Wh, J, C)

- Scope: ✅ IN.
- Where seen: MIT Invitational, Jan 2020, Div C, INV (top), Q11-12 (battery charge from a current-time graph over 15 ks → 110 kC; energy → 1.29 MJ). Question bank I_bank A13 (3000 mAh, 3.7 V → 11.1 Wh; 2000 mAh → 7200 C).
- How to solve:
  1. Charge: Q (C) = mAh × 3.6. Or Q = I × t with t in seconds.
  2. Energy: Wh = Ah × V. Then J = Wh × 3600.
  3. From a graph of I against t: charge = area under the line. From V × I against t: energy = area.
- Worked example: a phone cell is 3000 mAh at 3.7 V. Energy = 3.0 Ah × 3.7 V = **11.1 Wh = 39 960 J (about 40 kJ)**. Charge = 3000 × 3.6 = **10 800 C**.
- Traps: mAh is **charge**, not energy. Two batteries with the same mAh at different voltages hold different energy.
- Calculator: calculate — type: mAh, V.

15. Kitchen (lemon or potato) battery

- Scope: ✅ IN.
- Where seen: official practice deck 3 (F_slides, BATT-KITCHEN-CELL: Cu-Zn potato about 1.2 V, Cu-steel lemon about 0.7 V; which terminal is +, which is the anode, what is the electrolyte). "Shock Value" tryout, about 2012, Div B, practice, VI-12 (home-made battery electrodes → zinc and copper). JC Booth "Shock Value", 2011, Div B, INV, Q16.
- How to solve:
  1. Electrodes: two different metals. Common: galvanized (zinc-coated) nail and a copper coin or wire.
  2. Copper = + (cathode). Zinc = – (anode). The fruit juice (acid) = electrolyte.
  3. Voltage per cell is about 0.9 V (Zn/Cu). Current is tiny (well under 1 mA in most fruit), because the internal resistance is high.
  4. For more voltage, join cells in series: + of one to – of the next.
- Worked example: an LED needs about 3 V. Each lemon cell gives 0.9 V. Cells needed = 3 ÷ 0.9 = 3.33 → round up to **4 cells (3.6 V)**. The LED will be very dim, because the current is tiny.
- Traps: The energy comes from the **zinc** being used up, not from the lemon's "juice power". A US penny made after 1982 is mostly zinc with a thin copper coat, so scratch-through can change the result (fact from US Mint composition; not confirmed in a test).
- Calculator: calculate — type: target V, cell V (round up).

16. Battery history (Volta and the battery timeline)

- Scope: ✅ IN (Volta is on rule 3.c.i). Other inventors (Daniell, Planté, Leclanché) are ⚠️ borderline trivia: they are battery facts, not listed scientists.
- Where seen: BullSO Invitational 2026, Div B, INV, Q2 ("Who invented the electric battery?" → Volta). Rickards Invitational 2020 (2021 season), Div B/C, INV, Q20-25 (Volta facts). Northview Sylvania, Dec 2025, Div C, INV, Q49 (methane → Volta; the key letter looks wrong, see G_tests).
- How to solve: match the person to the date and the device (see the History list in Key facts). For Volta also know: he discovered methane gas (1776-78), invented the electrophorus (a static-charge device, 1775), and found that $V \propto Q$ for a conductor.

- Worked example: "Who built the first device to give a steady electric current, and when?" **Alessandro Volta, 1800, the voltaic pile.**
- Traps: Galvani found "animal electricity". He did not invent the battery. Volta showed the twitching came from the two metals. Benjamin Franklin used the word "battery" (for a row of Leyden jars) before Volta's pile. See CL 1 - History [T].
- Calculator: look up — type: Volta / battery history.

17. Battery hazards and safe use

- Scope: ✅ IN (rule 3.c.iii "hazards").
- Where seen: "Shock Value" tryout, about 2012, Div B, practice, VI-7 (why lead-acid and LiPo batteries are dangerous → very high current, acid, hydrogen gas). Cobra Invitational, 2019-20, Div B, INV, Q14 (dry fingers on a car battery: shock? → No). ATX Regional 2020, Div B, REG, Q54 (T/F "a 12 V battery can kill" → key True; debatable).
- How to solve: name the hazard, then the cause.
  1. Short circuit → huge current → heat, fire, burns.
  2. Li-ion damage/overcharge → thermal runaway → fire.
  3. Lead-acid → acid and hydrogen gas → explosion near sparks.
  4. Button cell swallowed → internal burns.
  5. Mixing old and new cells, or charging a primary cell → leaks, rupture.
- Worked example: "Why remove a metal ring before working on a car battery?" **A ring touching the + terminal and the car body makes a short circuit. Hundreds of amps flow through the ring. It heats red-hot and burns the finger.**
- Traps: The car battery's danger is **current into a short**, not a shock through skin. 12 V across dry skin (about 100 kΩ) gives only about 0.12 mA.
- Calculator: look up — type: battery hazard.

18. Terminal voltage and internal resistance

- Scope: ✅ IN.
- Where seen: Pembroke Hill Invitational, Dec 2025, Div B, INV, Q4 (ε = 9.0 V, V_t = 8.5 V across 10 Ω → r = 0.59 Ω). Full list of sightings in Internal Resistance and Power Transfer [A5].
- How to solve:
  1. I = V_t ÷ R_load.
  2. r = (ε − V_t) ÷ I.
- Worked example: Pembroke Q4. I = 8.5 ÷ 10 = 0.85 A. r = 0.5 ÷ 0.85 = **0.59 Ω**.
- Traps: Use V_t (not ε) to find the current through the load.
- Calculator: calculate — type: emf, V terminal, R load.

19. Short-circuit current and heating

- Scope: ✅ IN.
- Where seen: not seen as a number question in a test. Question bank I_bank A16. Hazard questions above ask the idea.
- How to solve:
  1. I_sc = ε ÷ r.
  2. Heat in the cell P = $I_{sc}^2 \times r$ (= ε × I_sc).
- Worked example: an AA cell has ε = 1.5 V, r = 0.10 Ω. I_sc = **15 A**. Heat = $15^2 \times 0.10$ = **22.5 W** inside a tiny cell. It gets hot fast.
- Traps: A short is not "zero current". It is the **maximum** current.
- Calculator: calculate — type: emf, r.

20. Car battery under load and on charge

- Scope: ✅ IN.
- Where seen: not seen in a test. Built from the internal-resistance types and car-battery facts.
- How to solve:

1. Cranking: V_t = ε – I·r.
2. Charging: I = (V_charger – ε) ÷ r.

- Worked example: a car battery has ε = 12.6 V and r = 0.010 Ω. The starter draws 200 A. V_t = 12.6 – 200 × 0.01 = **10.6 V**. The headlights dim while cranking. Later the alternator holds 14.4 V across it with r = 0.05 Ω: I = (14.4 – 12.6) ÷ 0.05 = **36 A** into the battery.
- Traps: The charging voltage must be **above** the EMF. If it is equal, no current flows in.
- Calculator: calculate — type: emf, r, I (or charger V).

Hard

21. Design a pack for a target voltage and capacity

- Scope: ✅ IN.
- Where seen: not seen in a test. It fits hands-on task II.c.iii (batteries) and the pack facts.
- How to solve:
  1. Series count n = target V ÷ cell V. Round to the nearest standard pack (or up if the load needs at least that voltage).
  2. Parallel count m = target capacity ÷ cell capacity, rounded up.
  3. Cells = n × m. Energy = n·V_cell × m·capacity.
- Worked example: Li-ion cells are 3.7 V, 2000 mAh. You need about 11 V and 6 Ah. n = 3 (11.1 V). m = 6 ÷ 2 = 3. Pack = **3S3P, 9 cells, 11.1 V, 6.0 Ah, 66.6 Wh**.
- Traps: Do not add parallel cells to raise voltage. Each parallel string must have the same number of cells.
- Calculator: calculate — type: target V, target Ah, cell V, cell Ah.

22. C-rate, charge time and discharge time

- Scope: ✅ IN (it is capacity ÷ current).
- Where seen: not seen in a test.
- How to solve:
  1. I (at 1C) = capacity per hour. I = C-rate × capacity.
  2. Ideal time (h) = 1 ÷ C-rate.
- Worked example: a 2000 mAh cell charges at 0.5C. I = **1000 mA**. Ideal time = **2 h**. At 2C discharge, it lasts about **30 min**.
- Traps: Real Li-ion charging slows near full (constant current, then constant voltage). So real charge time is longer than the ideal.
- Calculator: calculate — type: capacity, C-rate.

23. Compare batteries by energy per kilogram or per dollar

- Scope: ✅ IN (energy and unit math, rule 3.c.iv).
- Where seen: not seen in a test.
- How to solve:
  1. Wh = V × Ah.
  2. Specific energy = Wh ÷ kg. Cost per kWh = price ÷ (Wh ÷ 1000).
- Worked example: an 18650 Li-ion cell: 3.6 V, 3.0 Ah, 45 g. Wh = 10.8. Specific energy = 10.8 ÷ 0.045 = **240 Wh/kg**. An AA alkaline: 1.5 V × 2.5 Ah = 3.75 Wh. At $0.50 each, cost = 0.50 ÷ 0.00375 = **$133 per kWh**, about **890 times** a $0.15/kWh home rate.
- Traps: Use kWh, not Wh, when you compare with an electric bill.
- Calculator: calculate — type: V, Ah, mass or price.

24. Coin-cell or low-drain runtime in years

- Scope: ✅ IN.
- Where seen: not seen in a test.
- How to solve: t (h) = mAh ÷ mA. Years = h ÷ 8760.
- Worked example: a CR2032 (225 mAh) runs a watch at 0.010 mA. t = 22 500 h = **2.6 years**.
- Traps: 1 year = 8760 h (8784 in a leap year). Shelf self-discharge also matters for multi-year runtimes.

- Calculator: calculate — type: mAh, mA (years).

25. Two unequal cells in parallel with no load (circulating current)

- Scope: ✅ IN. It is one loop, so Ohm's law is enough.
- Where seen: not seen in a test.
- How to solve:
  1. Around the loop, the net EMF = $\varepsilon_1 - \varepsilon_2$ (they face each other).
  2. Total resistance = $r_1 + r_2$.
  3. $I = (\varepsilon_1 - \varepsilon_2) \div (r_1 + r_2)$. It flows out of the stronger cell and into the weaker one.
- Worked example: a 1.5 V cell (r = 0.10 Ω) is put in parallel with a 1.2 V cell (r = 0.10 Ω). I = 0.3 ÷ 0.2 = **1.5 A**. That heats both cells with no useful work. This is why you only put equal cells in parallel.
- Traps: The EMFs subtract, because + faces + in a parallel loop.
- Calculator: calculate — type: emf1, r1, emf2, r2.

Nationals

26. Two real batteries in parallel feeding a load (KVL/KCL)

- Scope: ✅ States/Nationals (rule 3.c.ix). It needs node or loop equations. Keep it to two unknowns.
- Where seen: not seen with batteries and internal resistance in a Div B test. KVL two-source networks appear in many tests; see Kirchhoff Analysis (States-Nationals) A5.
- How to solve:
  1. Call the top node voltage V.
  2. KCL: $(\varepsilon_1 - V)/r_1 + (\varepsilon_2 - V)/r_2 = V/R$.
  3. Solve for V. Then $I_1 = (\varepsilon_1 - V)/r_1$ and $I_2 = (\varepsilon_2 - V)/r_2$. A negative current means that battery is being charged.
- Worked example: ε_1 = 12 V, r_1 = 1 Ω; ε_2 = 10 V, r_2 = 1 Ω; load R = 2 Ω. (12 – V) + (10 – V) = V/2 → 22 = 2.5V → **V = 8.8 V**. I_1 = **3.2 A**, I_2 = **1.2 A**, load current = **4.4 A** (3.2 + 1.2 ✓).
- Traps: With a lighter load, the weaker battery's current can turn negative. Then it is charging, not helping.
- Calculator: calculate — type: emf1, r1, emf2, r2, R (node V).

27. Electrode potentials → cell voltage

- Scope: ⚠️ Borderline. Batteries are IN, and the math is subtraction. But a standard-potential table is chemistry, not listed in rule 3.c. A supervisor would have to give the table.
- Where seen: MIT Invitational, Jan 2020, Div C, INV (top), Q42 (Cu/Ag cell → 0.46 V, voltaic). Rickards Invitational, Oct 2025 (2026 season), Div C, INV, Q30. Northview Sylvania, Dec 2025, Div C, INV, Q32.
- How to solve:
  1. Find both reduction potentials.
  2. The more positive one is the cathode (+).
  3. E°cell = E°cathode – E°anode. A positive E°cell means the cell works on its own (voltaic).
- Worked example: Zn/Cu (Daniell cell): 0.34 – (–0.76) = **1.10 V**. Cu/Ag: 0.80 – 0.34 = **0.46 V** (MIT Q42). Mg/Cu: 0.34 – (–2.37) = **2.71 V**.
- Traps: Do not multiply the potential by the number of electrons. Voltage does not scale with the amount of chemical. Real fruit cells give less than E°, because they are not at standard conditions.
- Calculator: calculate — type: metal 1, metal 2 (looks up E° and subtracts).

28. Will this pack light the LED? (cells + LED + resistor)

- Scope: ✅ IN (links to the 2027 hands-on task II.c.iii and LEDs A7).
- Where seen: GullSO Invitational, Dec 13 2025, Div B/C, INV, Q50i (battery tolerance 2.9-3.2 V with typical/maximum Vf and 47 Ω → LED current 6.4 mA to 23 mA; G_tests).
- How to solve:
  1. Pack voltage = n × V_cell (use the range if given).
  2. LED current = (V_pack – V_f) ÷ R. If V_pack < V_f, the LED stays off.
- Worked example: two 1.5 V cells (3.0 V), red LED V_f = 2.0 V, 47 Ω. I = (3.0 – 2.0) ÷ 47 = **21 mA**. With two tired cells at 1.2 V each (2.4 V): I = 0.4 ÷ 47 = **8.5 mA**, visibly dimmer.

- Traps: A blue or white LED (V_f ≈ 3 V) may not light at all from two tired cells.
- Calculator: calculate — type: n cells, cell V, Vf, R.

Out of scope (skip)

- Electrochemistry beyond simple cell voltage: Nernst equation, Gibbs energy, equilibrium constants. Not in rule 3.c (and the Nernst equation uses logarithms, beyond the algebra limit, rule 3.e).
- Battery equivalent circuits with capacitors (Randles model). ❌ OUT: capacitance (rule 3.d).
- Battery charging curves as calculus (integrating I over time with a changing current). ❌ OUT: calculus (rule 3.e). Graph **area** of a straight-line shape is fine.
- Supercapacitors as energy storage. ❌ OUT: capacitance (rule 3.d).

Mistakes in tests and keys

- **UT Austin Invitational 2019 B and C, Q11:** "diesel engine battery" → key 24 V. Diesel cars and light trucks use 12 V. Only heavy trucks and some large engines use 24 V. Answer 24 V if the question says "heavy truck"; otherwise point out the ambiguity.
- **ATX Regional 2020 B, Q54:** "A 12 V battery can kill" → key True. It is debatable. Through intact dry skin, 12 V gives far too little current. Through wet or broken skin, or an internal path (a medical device), it can be dangerous. The key accepts True.
- **Northview Sylvania Dec 2025 C, Q49:** "methane → Volta". The key letter looks wrong (G_tests). Volta did discover methane.

Not confirmed

- Typical capacities (AA, AAA, 9 V, CR2032) and specific-energy ranges come from general battery references from memory. Brands differ. I did not check a datasheet.
- History dates (Planté 1859, Leclanché 1866, Gassner 1886, Jungner 1899, Sony 1991) are from memory. I did not check them against a biography in this run.
- The "button cell burns in about 2 hours" figure is from the question bank (I_bank S5, marked [G] = general knowledge). I did not check a medical source.
- The penny composition (copper-plated zinc since 1982) is from memory.
- Small-aircraft 24/28 V systems: from memory; not checked.
- I did not find any battery question from a real State or National test. No State/Nationals test was in the downloads, apart from Florida State 2021 (which had only the source ID question).

Sources

- Rules: `Science_Olympiad_Div_B_Rules_2027_for_Web_Secured.pdf`, Circuit Lab (areas 3.c.i, 3.c.iii, 3.c.iv, 3.c.ix; hands-on II.c.iii-iv; exclusions 3.d; math 3.e).
- Tests (extractions in `circuitlab/research/tests/` and `prior/G_tests.md`): Pembroke Hill Invitational Dec 2025 B; BullSO Invitational 2026 B; MIT Invitational 2020 C; Cobra Invitational 2019-20 B; UF Northern Regional 2020 B; JC Booth "Shock Value" 2011 B; "Shock Value" tryout about 2012 B; UT Austin Invitational fall 2018 B/C and 2019 B/C; ATX Regional 2020 B; Princeton 2021 C; Rickards 2020 and Oct 2025 C; Northview Sylvania Dec 2025 C; GullSO Dec 2025 B/C; cyanophycean314 practice C; SO MYSO practice Nov 2020 B.
- Official practice decks (soinc.org Circuit Lab B, deck 3 "Kitchen batteries, series, parallel"), summarized in `prior/F_slides.md`.
- Prior research: `prior/B_topics.md` §3b, `prior/I_bank.md` A10-A16 and S5.
- Textbooks (general knowledge, not re-read in this run): OpenStax *College Physics 2e* ch. 21.2 (EMF and terminal voltage); Kuphaldt, *Lessons in Electric Circuits* vol. 1 ch. 11 (Batteries and power systems); OpenStax *Chemistry 2e* ch. 17 (standard reduction potentials).

Other DC Sources, Converters and DC Uses

Note: Other DC Sources

This note covers every DC source that is not a battery: solar cells, thermocouples, fuel cells, generators, wall adapters, USB chargers and power supplies. It also covers converters (rectifier, inverter, DC-DC) and the **uses** of DC, including HVDC (high-voltage DC) power lines.

Scope:

- Rule area **3.c.iii** names the "sources, uses, and hazards" of DC and AC. So identifying sources and uses is ✅ **IN at every level**.
- The **USB charger** is an official hands-on task (rule II.c.iv): "Given a USB charger, read the label and provide details of the various output power levels it can provide and calculate how long it would take to charge a specific battery." ✅ IN, and very likely.
- A **solar cell** is a silicon PN junction, a single-junction device. So it is also IN under rule 3.c.vii.
- **How** a generator, motor or transformer works is magnetism or inductance. ❌ OUT (rule 3.c lists no magnetism; rule 3.d excludes inductance). You can still be asked **what** they produce (AC or DC).
- Inside parts of an adapter: the diodes are IN (3.c.vii). The transformer (inductance) and the smoothing capacitor (capacitance) are ❌ OUT (rule 3.d). So a "name the blocks" question is ⚠️ borderline.

Key facts

AC or DC?

- A source is **DC** if its output never changes sign (polarity). It can still vary in size ("pulsating DC").
- A source is **AC** if its output reverses direction again and again.

Source	Output	Why
Battery, any chemistry	DC	Fixed + and – terminals
Solar (photovoltaic, PV) cell or panel	DC	A PN junction pushes current one way
Thermocouple	DC	A fixed temperature difference gives a fixed polarity
Fuel cell	DC	Electrochemical, like a battery
USB port, phone charger output	DC	It is a rectified and regulated output
Wall adapter ("wall wart") output	DC (most)	A few old ones say "AC output"; read the label
Car alternator (after its built-in diodes)	DC	Inside it makes AC; diodes rectify it
DC generator / dynamo with a commutator	DC (pulsating)	The commutator flips the connection each half turn
Dynamo + bridge rectifier	DC	The rectifier makes it one-way
Lightning	DC (one huge one-way pulse)	Charge flows one way, cloud to ground (usually)
Static electricity spark	DC pulse	One-way discharge
Wall outlet	AC	Utility grid
Power-plant generator, alternator before diodes, wind turbine generator	AC	Rotating magnet makes alternating EMF
Inverter output	AC	It converts DC to AC

Converters (name and direction)

- **Rectifier**: AC → DC. It is made of diodes. A **bridge rectifier** uses 4 diodes (full-wave).
- **Inverter**: DC → AC. Used with solar panels, car "power inverters", UPS units (uninterruptible power supplies) and EV motors.
- Rectifier and inverter are **inverse devices**. They do opposite jobs.
- **DC-DC converter**: changes one DC voltage to another. A **buck** converter steps **down**. A **boost** converter steps **up**. A buck-boost does both.

- **Transformer**: changes AC voltage up or down. It does **not** work on steady DC. (How it works is OUT.)
- **Linear regulator**: holds a lower steady DC voltage by wasting the extra voltage as heat (example: 7805 → 5 V).
- **Switch-mode power supply (SMPS)**: modern chargers. It switches on and off very fast with a small high-frequency transformer. It is small, light and about 80-95% efficient.

Wall adapters and power supplies

- Labels show input and output. "Input: 100-240 V ~ 50/60 Hz" means it works on any country's mains (universal input).
- Symbol **~** = AC. Symbol ⎓ (a solid line over a dashed line) = DC.
- "Output: 12 V ⎓ 2 A" means 12 V DC, up to 2 A. Maximum output power = 12 × 2 = 24 W.
- The **polarity symbol** (a dot inside a C, with + and − labels) shows if the barrel plug's center pin is + ("center positive") or −.
- The output current on the label is a **maximum**. The device draws only what it needs.
- A **bench power supply** has two limits. In **CV** (constant voltage) mode it holds the set voltage. If the load tries to draw more than the current limit, it switches to **CC** (constant current) mode and lowers the voltage.

USB (Universal Serial Bus) power levels (USB-IF specifications; values from general references)

Standard	Voltage	Max current	Max power
USB 2.0 port	5 V	0.5 A	2.5 W
USB 3.x port	5 V	0.9 A	4.5 W
USB Battery Charging 1.2 (BC 1.2)	5 V	1.5 A	7.5 W
USB Type-C current	5 V	1.5 A or 3 A	7.5 W or 15 W
USB Power Delivery (PD), standard range	5, 9, 15, 20 V	3 A (5 A with an e-marked cable)	60 W (100 W)
USB PD 3.1 extended range (EPR)	adds 28, 36, 48 V	5 A	240 W

- The charger **offers** levels. The phone **negotiates** and picks one. A charger never forces extra current into a device.
- USB wires (classic USB-A cable): red = VBUS (+5 V), black = ground, green = D+, white = D− (data).
- Phone cells are 3.7 V (some 3.85 V) Li-ion. The phone's own charging circuit converts the USB voltage down to the cell voltage.

Solar (photovoltaic) cells

- A solar cell is a large silicon **PN junction**. Light frees electrons, and the junction's built-in field pushes them one way. Output: DC.
- One silicon cell gives about **0.5-0.6 V** open circuit, whatever its size. Bigger cells give more **current**, not more voltage.
- Current is roughly proportional to the light intensity (irradiance).
- Panels join cells in series for voltage. A "36-cell" panel (about 18-21 V open circuit) charges 12 V batteries. A "60-cell" home panel gives about 36-40 V open circuit.
- **Standard test conditions (STC):** 1000 W/m² of light, 25 °C cell temperature. Panel ratings (watts) are at STC.
- Typical panel efficiency: about 15-23%. Hot cells give slightly less voltage.
- **Peak sun hours**: the number of hours of 1000 W/m² light that equal a day's sunlight. Daily energy ≈ panel W × peak sun hours.
- Words: **V_oc** (open-circuit voltage, no load), **I_sc** (short-circuit current), **maximum power point** (the V and I where V × I is largest).
- Grid-tied systems use an **inverter** to make AC. Off-grid systems charge batteries through a **charge controller**.
- History: Edmond **Becquerel** found the photovoltaic effect in 1839. Bell Labs made the first practical silicon solar cell in 1954 (about 6% efficient).

Thermocouples (Seebeck effect)

- Two different metals joined at one end. When the joined end is hotter than the open ends, a small DC voltage appears. This is the **Seebeck effect** (Thomas Seebeck, 1821).
- The voltage is tiny: tens of microvolts per °C. A type K thermocouple (chromel-alumel) gives about **41 µV/°C**.

- Uses: temperature sensors in ovens, furnaces and gas water heaters (a flame-safety sensor). A **thermopile** is many thermocouples in series.
- **RTG** (radioisotope thermoelectric generator): heat from decaying plutonium-238 drives thermocouples. Voyager and the Curiosity rover use RTGs.
- The reverse effect (current makes one junction hot and the other cold) is the **Peltier effect**. It runs small "thermoelectric" coolers.

Fuel cells

- A fuel cell turns fuel (usually hydrogen) and oxygen into electricity, water and heat. Output: DC.
- Unlike a battery, it does not run down while fuel keeps flowing.
- One hydrogen cell gives about 1.23 V in theory and about **0.6-0.7 V** under load. Stacks put many cells in series.
- History: William **Grove**, 1839 ("gas voltaic battery"). NASA Gemini and Apollo spacecraft used fuel cells for power and drinking water.

Generators (as sources only)

- A **generator** turns mechanical energy into electrical energy. A **motor** does the reverse.
- A rotating generator naturally makes AC. An **alternator** gives AC. A **DC generator (dynamo)** uses a **commutator** (a split ring) to flip the connection each half turn, so its output is pulsating DC.
- Power plants (coal, gas, nuclear, hydro, wind turbines) spin AC generators. Solar PV, batteries and fuel cells are DC sources.
- The physics (Faraday's law, magnetic fields) is ❌ OUT.

Uses of DC

- All electronics run on DC inside: phones, computers, TVs, LED drivers, logic circuits.
- Vehicles: cars 12 V, heavy trucks 24 V, EV traction packs about 400-800 V.
- Telephone exchanges and cell towers: −48 V DC (approximate standard).
- Electrolysis and **electroplating** need DC, because the chemical change must go one way.
- Many subways and trams: about 600-750 V DC (third rail or overhead). Values differ by city.
- **HVDC** (high-voltage DC) transmission: very long overhead lines, undersea and underground cables, and links between grids that run at different frequencies or are not in step (for example Japan's 50 Hz and 60 Hz halves). Converter stations at each end turn AC to DC and back. HVDC has no "charging current" loss in long cables and no skin effect. It needs expensive converter stations, so it pays only over long distances.

Formulas

What	Formula	Notes
Max output power on a label	$P = V \times I$	Per output level
Charge time (same-voltage method)	t (h) = capacity (mAh) ÷ charge current (mA)	Simple
Charge time (energy method)	t (h) = (mAh × V_cell ÷ 1000) ÷ (P_charger × η)	η = efficiency (0.8-0.9)
Battery energy	Wh = Ah × V_cell	
Power-bank charges	n = (bank Wh × η) ÷ phone Wh	
Supply sizing	P_supply ≥ P_load × (1 + margin)	Then pick the next size up
Solar panel power	P = irradiance × area × efficiency	W/m² × m²
Solar daily energy	E (Wh) = P_rated × peak sun hours	
Cells for a voltage	n = V_target ÷ V_cell, round up	Solar 0.5-0.6 V; fuel cell 0.6-0.7 V
Thermocouple voltage	$V = S \times \Delta T$	S ≈ 41 µV/°C for type K
Line current	$I = P \div V$	Same for DC or AC (resistive)
Line loss	$P_{loss} = I^2 \times R_{line}$	
Bench supply mode	if V_set ÷ R > I_limit → CC mode, V = I_limit × R	

Question types

Easy

1. Classify a source as AC or DC

- Scope: ✅ IN.
- Where seen: Cobra Invitational, 2019-20, Div B, INV, Q13 (lead-acid, solar cell, generator, dynamo + full-bridge rectifier, inverter, lightning → DC, DC, AC, DC, AC, DC). ATX Regional 2021, Div B, REG, Q22 ("Which is NOT a source of AC?" → solar panel). UT Austin Invitational 2020 (2021 season), Div B, INV, Q9 ("Which primarily generates AC: battery, oscilloscope, synthesizer, alternator?" → alternator). UF Northern Regional, Jan 2019, Div C, REG, Q18 (common household DC source → batteries), Q21 (wall sockets give AC → True), Q22 (photocells produce AC → False). SO MYSO practice, Nov 2020, Div B, Q7.
- How to solve:
  1. Ask: does the output ever reverse polarity?
  2. Chemical, light and heat sources (battery, PV cell, thermocouple, fuel cell) → DC.
  3. Spinning generators without a commutator or rectifier, the wall outlet, and inverters → AC.
- Worked example: "Which is NOT an AC source: wall outlet, alternator, inverter, solar panel?" **Solar panel** (DC).
- Traps: "Generator" alone is ambiguous. An alternator gives AC; a dynamo with a commutator gives DC. Cobra's key did not mark this. Lightning is a one-way pulse, so call it DC.
- Calculator: look up — type: source name → AC/DC.

2. Which source gives the steadiest (flattest) DC?

- Scope: ✅ IN.
- Where seen: F-M (Fayetteville-Manlius) Invitational 2018, "Circuits Lab", INV, Q3 (which source gives a flat DC waveform → photovoltaic cell; division not printed).
- How to solve:
  1. Batteries, solar cells under steady light and fuel cells give flat DC.
  2. A dynamo with a commutator, or a rectifier without a filter, gives **pulsating** DC (one direction, changing size).
- Worked example: "Dynamo, half-wave rectifier, photovoltaic cell, alternator: which gives flat DC?" **Photovoltaic cell.**
- Traps: Pulsating DC is still DC. It never reverses.
- Calculator: look up — type: flat DC source.

3. Name the converter (rectifier, inverter, DC-DC, transformer)

- Scope: ✅ IN (naming). Transformer physics is ❌ OUT.
- Where seen: BullSO Invitational 2026, Div B, INV, Q13 ("Which device converts DC to AC?" → inverter). ATX Regional 2020, Div B, REG, Q26 (inverse devices → rectifier and inverter). UT Austin Invitational fall 2018, Div B/C, INV, Q12 (fill-in: inverter converts DC to AC; no key). BullSO Invitational 2026, Div C, INV, Q13. Northview Sylvania, Dec 2025, Div C, INV, Q6. Hawk & Hornet Invitational 2026, Div C, INV, FR2 (buck vs boost converter).
- How to solve:
  1. AC → DC = rectifier. DC → AC = inverter.
  2. DC → lower DC = buck. DC → higher DC = boost.
  3. AC → different AC voltage = transformer.
- Worked example: "A solar panel feeds a house's AC outlets. Which device does that?" **An inverter.** "A USB-C laptop charger makes 20 V from 120 V AC. Which two jobs happen?" **Rectifying (AC → DC) and converting the DC voltage (switch-mode).**
- Traps: A transformer cannot change DC to AC, and it does not work on steady DC (Pembroke Hill Dec 2025 B, Q34 tests "true about an ideal transformer" → power in = power out; this is ⚠️ borderline because transformers need inductance).
- Calculator: look up — type: converter name / direction.

4. Solar-cell facts

- Scope: ✅ IN (DC source; a PN junction under rule 3.c.vii).

- Where seen: not seen as a facts question beyond source classification (Cobra Q13, ATX 2021 B Q22, UF NR 2019 C Q22, F-M 2018 Q3). Question bank I_bank A17.
- How to solve: recall the Key facts. One cell ≈ 0.5-0.6 V. Light → DC. More area → more current, same voltage.
- Worked example: "Doubling the area of a solar cell in the same light does what?" **Doubles the current; the voltage stays about 0.5-0.6 V.**
- Traps: Solar panels do not produce AC. Rooftop systems need an inverter for AC.
- Calculator: look up — type: solar cell.

5. Thermocouple facts (Seebeck effect)

- Scope: ⚠️ Borderline. It is a DC source (3.c.iii), but it is not named in the rules or any Div B test I read.
- Where seen: not seen in a test. Question bank I_bank A19.
- How to solve: two different metals + a temperature difference → small DC voltage. Name: Seebeck effect.
- Worked example: "What is a thermocouple used for in a gas water heater?" **It senses the pilot flame. If the flame goes out, the voltage drops and the gas valve closes.**
- Traps: A thermocouple needs a temperature **difference**. Both ends at the same temperature give 0 V.
- Calculator: look up — type: thermocouple / Seebeck.

6. Fuel-cell facts

- Scope: ⚠️ Borderline (not named, but a DC source).
- Where seen: not seen in a test. Question bank I_bank A19.
- How to solve: hydrogen + oxygen → water + DC electricity + heat. About 0.7 V per cell under load.
- Worked example: "What waste product does a hydrogen fuel cell make?" **Water.**
- Traps: A fuel cell is not "burning" the hydrogen in a flame. It is an electrochemical reaction, like a battery that is refueled.
- Calculator: look up — type: fuel cell.

7. Generator, motor or dynamo? (energy direction and output)

- Scope: ⚠️ Borderline. Naming the energy change is common. The commutator and magnetic physics are magnetism (❌ OUT).
- Where seen: UF Northern Regional, Jan 2020, Div B, REG, Q29 ("A motor turns mechanical into electrical energy" → key True; **key error**: that is a generator). UT Austin Invitational 2019, Div B, INV, Q15 ("What reverses current every half cycle in a motor?" → commutator). ATX Regional 2021, Div B, REG, Q36 (role of a commutator; OUT). UT Austin Invitational fall 2018, Div B/C, INV, Q12 (commutator fill-in). JC Booth "Shock Value", 2011, Div B, INV, Q4-6.
- How to solve:
  1. Motor: electrical → mechanical.
  2. Generator: mechanical → electrical.
  3. Alternator → AC. Dynamo with commutator → DC.
- Worked example: "A wind turbine turns a generator. What energy change?" **Mechanical (kinetic) → electrical.**
- Traps: Motor and generator are the same machine run in opposite directions. Read which way the energy goes.
- Calculator: look up — type: motor / generator / dynamo.

8. DC uses: which devices or systems use DC?

- Scope: ✅ IN.
- Where seen: UT Austin Invitational 2020, Div B, INV, Q53-57 (electronic circuits, home outlets, long-distance lines, car batteries, phones → DC, AC, AC, DC, DC). Pembroke Hill Dec 2025 B and Berkeley 2026 B Q41 (why AC and DC are each used; see AC Characteristics A3).
- How to solve:
  1. Electronics, batteries, cars, phones, LEDs, logic → DC.
  2. Outlets, the grid, most long-distance lines → AC (HVDC is the exception).
- Worked example: "Electroplating a spoon with silver needs which current?" **DC.** The metal must move one way only.

- Traps: Long-distance lines are "AC" in most keys (UT 2020 Q55). HVDC lines exist, but answer AC unless the question names HVDC.
- Calculator: look up — type: use → AC/DC.

Medium

9. Read a wall-adapter or power-supply label

- Scope: ✅ IN.
- Where seen: not seen in a test as a label question. B_topics §9d (charger labels). The official USB task (rule II.c.iv) uses the same skill.
- How to solve:
  1. Input side: ~ means AC. "100-240 V" means universal. The input current is a maximum at the lowest voltage.
  2. Output side: ⎓ means DC. P_max = V × I.
  3. Check the polarity symbol before plugging into a device.
- Worked example: label "Input 100-240 V ~ 50/60 Hz 0.6 A; Output 12 V ⎓ 2.0 A, center +". The output is **12 V DC**, up to **2.0 A**, so up to **24 W**. It works in the US and in Europe.
- Traps: The output current is a **limit**, not what it always gives. A 2 A adapter running a 0.3 A device gives 0.3 A.
- Calculator: calculate — type: V out, I out.

10. Name the blocks of an AC-to-DC supply

- Scope: ⚠️ Borderline. The diodes (rectifier) are IN. The transformer is ❌ OUT (inductance). The smoothing capacitor is ❌ OUT (capacitance). The regulator is borderline.
- Where seen: Fall 2019 F-M Invitational, Circuit Lab, INV, Q5b (three parts in a 5 V DC wall supply → transformer, rectifier/diodes, capacitor). North Pocono Invitational, Jan 2020, Div C, INV, Q17 (rectifier input/output current sketch, I_rms = 2 A → I_max = 2.83 A).
- How to solve: in order: transformer (lower the AC voltage) → rectifier (one way) → filter capacitor (smooth) → regulator (hold steady).
- Worked example: "Which block makes the current flow one way?" **The rectifier (diodes).** For a diode-only question, see Diode Applications [A7].
- Traps: The rectifier alone gives pulsating DC, not flat DC.
- Calculator: look up — type: AC adapter blocks.

11. USB charger: list the output power levels

- Scope: ✅ IN. This is the official hands-on task II.c.iv.
- Where seen: not seen in any test I read (B_topics §9d says no USB question in the 2026 Div B tests). It is named in both the 2026 and 2027 rules.
- How to solve:
  1. Find each output level on the label: V and I.
  2. P = V × I for each.
  3. Note the overall maximum (often printed as "65 W max"). Some levels share power.
- Worked example: label "Output: 5 V ⎓ 3 A, 9 V ⎓ 3 A, 15 V ⎓ 3 A, 20 V ⎓ 3.25 A (65 W max)". Levels: **15 W, 27 W, 45 W, 65 W.**
- Traps: Do not add the levels together. The charger gives **one** level at a time per port. A "PPS" range (for example 3.3-21 V ⎓ 3 A) means a variable voltage; its top power is 21 × 3 = 63 W.
- Calculator: calculate — type: list of V, I pairs.

12. USB charge time: simple capacity method

- Scope: ✅ IN (hands-on II.c.iv).
- Where seen: not seen in a test. B_topics §9d.
- How to solve:
  1. Use this only when the charge current goes straight into the cell at the cell's voltage (or the question says to ignore voltage conversion).
  2. t = capacity (mAh) ÷ charging current (mA).

- Worked example: a 3000 mAh battery charges at 1000 mA. t = **3.0 h**.
- Traps: Real charging is slower near full (the CC/CV taper). State your assumption.
- Calculator: calculate — type: mAh, mA.

13. USB charge time: energy method with efficiency

- Scope: ✅ IN (hands-on II.c.iv).
- Where seen: not seen in a test. B_topics §9d; I_bank A14.
- How to solve:
  1. Battery energy: Wh = Ah × V_cell (3.7 V for Li-ion if not given).
  2. Charger power: P = V × I at the level used.
  3. t = Wh ÷ (P × efficiency).
- Worked example: phone battery 5000 mAh at 3.7 V → 18.5 Wh. Charger 9 V × 2 A = 18 W, 85% efficient. t = 18.5 ÷ (18 × 0.85) = **1.21 h (about 1 h 13 min)**. On a 5 V 2 A (10 W) charger: t = 18.5 ÷ 8.5 = **2.18 h**.
- Traps: **Big trap:** dividing 5000 mAh by the charger's 3 A at 9 V gives 1.67 h, which is wrong. The 3 A is at 9 V, but the cell is at 3.7 V. Convert to Wh first.
- Calculator: calculate — type: mAh, V cell, V charger, A charger, efficiency.

14. Power bank: how many phone charges?

- Scope: ✅ IN.
- Where seen: not seen in a test. B_topics §9d (power-bank label "10000 mAh 3.7 V 37 Wh").
- How to solve:
  1. Bank energy (Wh) = mAh × 3.7 V ÷ 1000 (the label usually prints Wh).
  2. Usable energy = Wh × efficiency (about 85%).
  3. Charges = usable Wh ÷ phone Wh.
- Worked example: a 10 000 mAh (37 Wh) bank, 85% efficient, charges a 3000 mAh, 3.7 V (11.1 Wh) phone. 37 × 0.85 ÷ 11.1 = **2.8 full charges**. Its "rated capacity at 5 V" is 37 × 0.85 ÷ 5 = **6290 mAh** (some banks print this).
- Traps: 10 000 mAh of bank does **not** give 10 000 mAh to the phone. The voltage and the losses change it.
- Calculator: calculate — type: bank mAh, phone mAh, efficiency.

15. Size a DC power supply with a safety margin

- Scope: ✅ IN.
- Where seen: Cobra Invitational, 2019-20, Div B, INV, Q20 (LED strip: 10 LEDs/ft × 0.5 W × 20 ft, at least 20% margin → choose 125 W from 100/125/80 W).
- How to solve:
  1. Load power = number of parts × watts each.
  2. Needed = load × (1 + margin).
  3. Choose the smallest supply that is at least that.
- Worked example: Cobra Q20. Load = 10 × 20 × 0.5 = 100 W. Needed = 100 × 1.2 = 120 W. **Choose 125 W.**
- Traps: Do not pick 100 W "because it matches". A supply run at 100% gets hot and may shut down.
- Calculator: calculate — type: load W, margin %.

16. Solar panel power and daily energy

- Scope: ✅ IN.
- Where seen: not seen in a test. I_bank A17.
- How to solve:
  1. P = irradiance × area × efficiency.
  2. Daily energy = P × peak sun hours.
- Worked example: a 1.6 m² panel, 20% efficient, in 1000 W/m² light. P = **320 W**. With 5 peak sun hours, E = **1.6 kWh per day**.
- Traps: Efficiency in % must become a decimal (20% = 0.20).
- Calculator: calculate — type: W/m², area, efficiency, sun hours.

17. How many solar (or fuel) cells in series for a voltage?

- Scope: ✅ IN.
- Where seen: not seen in a test. Same skill as the lemon-cell question (official deck 3).
- How to solve: n = target V ÷ voltage per cell. Round up.
- Worked example: solar cells at 0.5 V each must charge a 12 V battery (needs about 18 V to push current in). n = 18 ÷ 0.5 = **36 cells** (the classic "36-cell" panel). A 48 V fuel-cell stack at 0.7 V per cell: 48 ÷ 0.7 = 68.6 → **69 cells**.
- Traps: To charge a battery, the source must be **above** the battery's voltage.
- Calculator: calculate — type: target V, cell V.

18. HVDC uses

- Scope: ✅ IN as a "use of DC" (3.c.iii). Some notes call it borderline because HVDC is not named.
- Where seen: UT Austin Invitational fall 2018, Div B/C, INV, Q17 ("What is HVDC useful for?" → long-distance and undersea transmission, linking grids of different frequency; no key). B_topics §3a.
- How to solve: list the three uses: very long lines, undersea/underground cables, linking unsynchronized grids (different frequency).
- Worked example: "Why does Japan need converter stations between its east and west grids?" **East Japan runs at 50 Hz and west Japan at 60 Hz. AC grids at different frequencies cannot be joined directly. An HVDC link (back-to-back converter) connects them.**
- Traps: HVDC does not mean "DC is better for everything". AC is still cheaper for short and medium distances because transformers are cheap.
- Calculator: look up — type: HVDC.

Hard

19. Bench power supply in CV or CC mode

- Scope: ✅ IN (hands-on "calculate the power supplied", rule II.c.ii; Ohm's law).
- Where seen: not seen in a test.
- How to solve:
  1. Find the current the load would draw at the set voltage: I = V_set ÷ R.
  2. If I ≤ current limit → CV mode. V = V_set.
  3. If I > current limit → CC mode. I = limit, V = limit × R.
  4. Power = V × I.
- Worked example: set 12 V, limit 0.50 A. Load 100 Ω: I = 0.12 A < 0.5 → CV, **12 V, 0.12 A, 1.44 W**. Load 10 Ω: 12 ÷ 10 = 1.2 A > 0.5 → CC, **V = 5.0 V, I = 0.50 A, P = 2.5 W**.
- Traps: In CC mode the display voltage falls below the set value. That is normal, not a fault.
- Calculator: calculate — type: V set, I limit, R.

20. Solar panel charging a battery: how long?

- Scope: ✅ IN.
- Where seen: not seen in a test.
- How to solve:
  1. Battery energy = Ah × V.
  2. Panel energy per day = W × sun hours × efficiency of charging.
  3. Days = battery energy ÷ daily energy.
- Worked example: 12 V, 100 Ah battery = 1200 Wh. 100 W panel, 80% charging efficiency, 5 peak sun hours. Sun hours needed = 1200 ÷ (100 × 0.8) = **15 h**, so **3 days**.
- Traps: Use the panel's real output (with efficiency), not its label alone.
- Calculator: calculate — type: Ah, V, panel W, efficiency, sun hours.

21. Solar cell maximum power point (V × I)

- Scope: ✅ IN (a single-junction device, 3.c.vii; P = V·I). Fill factor is ⚠️ borderline vocabulary.
- Where seen: not seen in a test.
- How to solve:

1. From a table or graph of (V, I) points, compute P = V × I for each.
2. The largest P is the maximum power point.
3. Fill factor (optional) = P_max ÷ (V_oc × I_sc).

- Worked example: a panel has V_oc = 22 V, I_sc = 6.0 A, and its best point is 18 V at 5.56 A. P_max = **100 W**. Fill factor = 100 ÷ (22 × 6) = **0.76**.
- Traps: V_oc × I_sc is not the real power. At open circuit I = 0; at short circuit V = 0. Both give zero power.
- Calculator: calculate — type: list of V, I pairs (max P).

22. Thermocouple voltage from a temperature difference

- Scope: ⚠️ Borderline (thermocouples are not named). The math is one multiplication.
- Where seen: not seen in a test.
- How to solve: V = S × ΔT. Or ΔT = V ÷ S.
- Worked example: type K, S = 41 μV/°C. The hot end is 250 °C above the cold end. V = 41 × 250 = **10.25 mV**. A reading of 4.1 mV means ΔT = **100 °C**.
- Traps: Convert μV to mV (÷ 1000). The voltage depends on the **difference**, not the hot temperature alone.
- Calculator: calculate — type: S (μV/°C), ΔT.

23. HVDC (or any) line current and loss

- Scope: ✅ IN (I = P/V and P = I^2R; rule 3.c.iv).
- Where seen: not seen for HVDC. The AC version is common (see AC Characteristics A3 types on transmission loss).
- How to solve:
  1. I = P ÷ V.
  2. Loss = I^2 × R_line. Percent = loss ÷ P × 100.
- Worked example: 2000 MW sent at 500 kV over a line of 5 Ω total. I = **4000 A**. Loss = 4000^2 × 5 = **80 MW (4%)**. At 800 kV: I = 2500 A, loss = **31.25 MW (1.6%)**.
- Traps: P ÷ V uses the **sending** voltage, not the voltage drop on the line.
- Calculator: calculate — type: P, V, R line.

24. Car alternator and charging system

- Scope: ✅ IN (source ID, P = V·I). The alternator's physics is OUT.
- Where seen: not seen in a test as a number question. I_bank A8 ("A car alternator's output becomes DC through what?" → diode rectifier).
- How to solve:
  1. The alternator makes AC. Built-in diodes rectify it to DC at about 13.8-14.4 V.
  2. Power = V × I.
- Worked example: an alternator gives 14 V at 100 A. P = **1400 W**. The DC is made by **a diode rectifier inside the alternator**.
- Traps: The car's 12 V system actually runs at about 14 V while the engine runs.
- Calculator: calculate — type: V, I.

Nationals

No States/Nationals-only (KVL/KCL) question type was found for this note. A supply charging a battery while running a load is a KVL network; it is in Batteries and Cells A3 type 26.

Out of scope (skip)

- How generators, alternators, motors and transformers work (Faraday's law, flux, turns ratios). ❌ OUT: magnetism (not in 3.c) and inductance (3.d).
- Smoothing capacitors and ripple voltage in power supplies. ❌ OUT: capacitance (3.d).
- Switch-mode converter design (duty cycle with inductors). ❌ OUT: inductance (3.d).
- Transistors, thyristors and IGBTs inside inverters and HVDC stations. ❌ OUT: semiconductors beyond PN junctions (3.d).

- Solar-cell diode equation (exponential). ❌ OUT: beyond the algebra limit (3.e) and non-linear device analysis (3.d).

Mistakes in tests and keys

- **UF Northern Regional 2020 B, Q29:** "A motor turns mechanical energy into electrical energy" → key True. **Wrong.** A generator does that. A motor turns electrical into mechanical.
- **Cobra 2019-20 B, Q13:** "generator" with no type. The key is not marked. An alternator gives AC; a commutated dynamo gives DC. Ask the supervisor or write both.
- **ATX Regional 2020 B, Q26 and ATX 2021 B, Q22:** fine. But note that in Q22 a transformer is "not a source" either, because it only changes AC.

Not confirmed

- USB current and power limits are from USB-IF specifications as general knowledge. I did not open the USB PD or BC 1.2 documents in this run.
- Solar cell voltage (0.5-0.6 V), panel efficiencies (15-23%), and the 1954 Bell Labs figure (about 6%) are from general references, from memory.
- Telecom −48 V, transit 600-750 V and EV 400/800 V are approximate typical values from memory.
- Type K sensitivity (about 41 µV/°C) is from memory of thermocouple tables. It varies with temperature.
- Fuel-cell history (Grove 1839) and NASA Gemini/Apollo use are from memory.
- I found no USB charger question in any test, including Florida State 2021 and the USC 2026 test. I looked in all 8 extraction files and G_tests.md.

Sources

- Rules: 2027 Div B Circuit Lab, areas 3.c.iii, 3.c.vii; hands-on II.c.ii and II.c.iv; exclusions 3.d.
- Tests: Cobra Invitational 2019-20 B; ATX Regional 2020 B and 2021 B; UT Austin Invitational fall 2018 B/C, 2019 B, 2020 B; UF Northern Regional 2019 C and 2020 B; F-M Invitational 2018 and fall 2019; North Pocono 2020 C; BullSO 2026 B and C; Pembroke Hill Dec 2025 B; Northview Sylvania Dec 2025 C; Hawk & Hornet 2026 C; JC Booth 2011 B; SO MYSO practice 2020 B.
- Prior research: `prior/B_topics.md` §3a and §9d; `prior/I_bank.md` A8-A9, A14, A17-A19, A21; `prior/G_tests.md` (DC & AC section).
- Textbooks and references (general knowledge, not re-read in this run): Kuphaldt, *Lessons in Electric Circuits* vol. 1 ch. 11 and vol. 3 (power supplies); OpenStax *College Physics 2e* ch. 23 (generators, as background only); USB-IF specifications (USB 2.0, USB 3.x, BC 1.2, USB PD 3.1).

AC Characteristics

AC (alternating current) is current that reverses direction again and again. **DC** (direct current) flows one way only. Rule area **3.c.iii** says "Characteristics, sources, uses, and hazards of direct current (DC) and alternating current (AC)". So AC facts, AC uses and AC hazards are ✅ **IN at every level**.

The hard line is rule **3.d**: "Topics not included ... AC circuit theory, frequency analysis, capacitance, inductance, ... 3 phase power, ... oscilloscopes". Rule **3.e.v** also says "No questions may require the use of trigonometry".

How to sort a question:

- ✅ IN: what AC is, mains voltage and frequency, period T = 1/f, why the grid uses AC, line loss (I^2R), the war of the currents as an AC/DC "uses" fact, AC power in a plain resistor (heater, bulb).
- ⚠️ Borderline: RMS, peak and peak-to-peak. They use $\sqrt{2}$, which is not trigonometry. Many 2026 tests still ask them (BullSO 2026 B Q22, GullSO Dec 2025 Q12-13, Hawk & Hornet 2026 Q8). Learn them.
- ❌ OUT: phase, impedance, reactance, power factor, phasors, average-value math, skin effect, transformer turns ratios, filters, 3-phase, oscilloscope traces, the voltage of a generator at a given coil angle (that needs sine of an angle).
- Hands-on: rule II.d says "Participants must not be exposed to ... AC circuits". So AC is written-test only.

Key facts

Words

- **Waveform**: a graph of voltage (or current) against time.
- **Cycle**: one full repeat of the waveform (up, down, and back to the start).
- **Period (T)**: the time for one cycle. Unit: seconds (s).
- **Frequency (f)**: cycles per second. Unit: **hertz (Hz)**. 1 Hz = 1 cycle per second. The unit honors **Heinrich Hertz** (on the 2027 history list).
- **Amplitude** or **peak (V_p)**: the largest voltage, measured from zero.
- **Peak-to-peak (V_pp)**: from the most positive value to the most negative value. For a sine wave centered on zero, V_pp = 2 × V_p.
- **RMS** (root mean square): the "effective" value. An AC voltage with an RMS value of 120 V heats a resistor exactly as much as 120 V DC.
- **Mains**: the AC supply from wall outlets.

AC versus DC

- DC: one direction, fixed polarity (+ stays +). The voltage can still change in size (a dying battery, pulsating DC). It is DC as long as it never reverses.
- AC: the direction reverses. The usual shape is a **sine wave**. Square, triangle and "modified sine" (stepped) AC also exist.
- Not all AC is a sine wave (UT Austin 2020 B Q52: "all AC is a sine wave" → False).
- A wave that goes up and down but never crosses zero is **pulsating DC** or DC with **ripple**. Example: the output of a rectifier.
- A wave that crosses zero is AC, even if it has a DC offset (it is "AC plus DC").
- At 60 Hz the current reverses direction **120 times per second** (two reversals per cycle). It passes through zero 120 times per second.
- Symbols: ~ = AC, ⎓ = DC (solid line over a dashed line). AC source in a schematic: a circle with a sine wave inside. A meter's AC volts setting is V~ (or VAC).

Mains around the world (nominal RMS values)

Country or region	Voltage	Frequency
United States, Canada	120 V (240 V split-phase for big appliances)	60 Hz
Mexico	127 V	60 Hz

Country or region	Voltage	Frequency
Most of Europe, UK, India, Australia, most of Africa	230 V	50 Hz
China	220 V	50 Hz
South Korea	220 V	60 Hz
Japan	100 V	**50 Hz in the east (Tokyo), 60 Hz in the west (Osaka)**
Brazil	127 V or 220 V by region	60 Hz

- US outlets are called "110", "115", "117" or "120" V. The modern standard is **120 V**. ANSI C84.1 allows about 114-126 V at the outlet.
- Europe moved from 220 V and 240 V to one "harmonized" **230 V** value.
- Japan has two frequencies because, in the 1890s, Tokyo bought German 50 Hz generators and Osaka bought American 60 Hz generators. The two halves link through converter stations (see HVDC in Other DC Sources A3).
- A US outlet at 120 V RMS has a peak of about **170 V** and a peak-to-peak of about **340 V**.

Why the grid uses AC

- A **transformer** can step AC voltage up or down easily and cheaply. It does not work on steady DC. (How a transformer works is magnetism and inductance: ❌ OUT. The "why AC" fact is IN.)
- Power lost in a line is **$P_{loss} = I^2R$** (R = line resistance).
- For the same power P = V·I, a higher voltage means a lower current.
- So: step up to a very high voltage (hundreds of kV) for long lines, then step down to 120/240 V for homes.
- Raise the voltage **n** times → current falls n times → loss falls **n^2** times.
- AC generators (alternators) are simpler than DC dynamos. They need no commutator.
- AC current passes through zero 120 times per second (at 60 Hz). An arc in a switch tends to go out at each zero. DC arcs do not get this help, so DC is harder to switch off and DC arc fires are harder to stop. Switches show this in their ratings, for example "10 A 250 VAC, 10 A 30 VDC".
- DC is still better for some uses: electronics, batteries, very long or undersea **HVDC** lines (see Other DC Sources A3).

War of the currents (1880s-1890s)

- **Thomas Edison** backed **DC**. His **Pearl Street Station** in New York City (1882) sent about 110 V DC to customers within about a mile.
- DC could not change voltage easily then. Low-voltage DC lost too much power in long wires, so DC needed a power station every mile or so.
- **Nikola Tesla** designed AC (polyphase) motors and systems. **George Westinghouse** bought Tesla's AC patents (1888) and built AC systems.
- Edison's main argument: **AC is dangerous** (ATX Regional 2020 B Q20). His side publicized AC deaths. The first electric chair (1890) used AC.
- AC won. Westinghouse lit the **1893 Chicago World's Fair** with AC. The **Niagara Falls** AC power plant (1895-96) sent power to Buffalo, about 26 miles away.
- Tesla, Edison and Westinghouse are **not** on the 2027 history list (rule 3.c.i lists only Ampère, Coulomb, Kirchhoff, Volta, Ohm, Hertz and Boole). So "who did it" questions are ⚠ borderline. But "which current won and why" fits area 3.c.iii.
- Nobody won a Nobel Prize for inventing AC. Tests use this as a trick (Northview Sylvania Dec 2025 C Q46; UT Austin 2019 B Q8: answer "Nobody").

RMS, peak and peak-to-peak (⚠ borderline but common)

- Sine wave: $V_{rms} = V_p \div \sqrt{2} \approx 0.707 \times V_p$. And $V_p = \sqrt{2} \times V_{rms} \approx 1.414 \times V_{rms}$.
- Square wave that switches between +A and −A: $V_{rms} = A$ (it is always at full size).
- All mains values (120 V, 230 V) are RMS values unless the question says "peak".
- Meters read RMS on the AC setting.
- AC power in a resistor uses RMS: $P = V_{rms} \times I_{rms} = V_{rms}^2/R$. This is ordinary Ohm's law power.

- The plain average of a sine wave over a full cycle is **0**. That fact is AC theory (ATX Regional 2020 B Q18): ❌ OUT for 2027.

Uses of AC

- Homes and buildings (outlets, lights, heaters, motors, fridges, air conditioners).
- The power grid (generation, transmission, distribution).
- Heaters, toasters and incandescent bulbs work equally well on AC or DC of the same RMS value. They are plain resistors.
- Electronics (phones, laptops, LED drivers) run on **DC** inside. Their chargers convert AC to DC.

Hazards of AC (short version; full note: Shock and Safety A3)

- 50-60 Hz AC is especially good at causing **ventricular fibrillation** (the heart quivers instead of pumping).
- AC can cause muscles to clamp, so the person "cannot let go".
- Mains voltage (120 V) can push a deadly current through wet skin.

Formulas

What	Formula	Notes
Frequency and period	f = 1/T, T = 1/f	60 Hz → 16.7 ms; 50 Hz → 20 ms
Time for n cycles	t = n × T = n/f	
Reversals per second	2f	60 Hz → 120
Peak from RMS (sine)	$V_p = \sqrt{2} \times V_{rms}$	120 V → 169.7 V
RMS from peak (sine)	$V_{rms} = V_p/\sqrt{2}$	120 V peak → 84.9 V
Peak-to-peak (sine)	$V_{pp} = 2 \times V_p = 2\sqrt{2} \times V_{rms}$	120 V → 339.4 V
RMS of a ±A square wave	$V_{rms} = A$	
Resistive AC power	$P = V_{rms} \cdot I_{rms} = I_{rms}^2 R = V_{rms}^2/R$	Same as DC
Line current	I = P/V	
Line loss	$P_{loss} = I^2 R_{line}$	
Loss after raising V by n times	$P_{loss,new} = P_{loss,old} \div n^2$	
Energy	E (kWh) = P (kW) × t (h)	Bills are in kWh

Question types

Easy

1. Define AC and DC

- Scope: ✅ IN.
- Where seen: "Shock Value Test" re-tryout, about 2012, Div B, practice, I-1 ("charge changes direction in ___" → AC) and I-10 ("unidirectional flow" → DC). UF Northern Regional 2020, Div C, REG, Q9 (AC vs DC difference → direction). UF Northern Regional 2019, Div B, REG, Q21 ("DC makes a collapsing sine wave" → False). UT Austin Invitational Fall 2025, Div B/C, INV, 1c.vi α (what does AC stand for → alternating current). Official practice decks (F_slides ACDC-DEFINE).
- How to solve:
  1. Ask: does the current change **direction**? Yes → AC. No → DC.
  2. Size changes do not matter. Only direction matters.
- Worked example: "In which type of current do charges move back and forth?" **Answer: AC.** "What does DC stand for?" **Direct current.**
- Traps: A battery whose voltage slowly drops is still DC. "Alternating" means direction, not size.
- Calculator: look up — type: AC DC definition.

2. Classify a waveform graph as AC or DC

- Scope: ✅ IN (reading a graph; no AC theory).

- Where seen: Cobra Invitational 2019-20, Div B, INV, Q16 (8 voltage-time graphs: sawtooth 0-2 V → DC; sine 1.2-3.6 V → DC; triangle ±1 → AC; sine ±1 → AC; flat → DC; square ±1 → AC; steps 0-5-10 V → DC; sine –0.6 to +3.4 V → AC). University of Michigan (UMSO) Invitational, about 2019, Div not stated, INV, Q6-9 (triangle, flat, square ±, sine → AC, DC, AC, AC). UF Northern Regional 2020, Div C, REG, Q10 (label DC and AC graphs). UT Austin Invitational 2020, Div B, INV, Q52 ("all AC is a sine wave" → False).
- How to solve:
  1. Find the zero line on the graph.
  2. Does the curve ever go below zero (change sign)? Yes → AC (or AC with a DC offset). No → DC (maybe pulsating DC).
  3. Shape (sine, square, triangle) does not decide it.
- Worked example: "A sine wave goes from 1.2 V up to 3.6 V and back, over and over. AC or DC?" It never goes below 0 V, so current never reverses. **Answer: DC (pulsating, with a DC offset).**
- Traps: A wave from –0.6 V to +3.4 V is mostly positive, but it does reverse. Cobra's key calls it AC. A strict reader may call it "DC plus AC". Write "AC (it reverses)" and add the offset note if there is space.
- Calculator: look up — type: waveform AC DC rule (crosses zero?).

3. AC and DC symbols

- Scope: ✅ IN (schematics, rule 3.c.v; labels, rule II.c.iv).
- Where seen: UF Northern Regional 2020, Div B, REG, Q21f (a circle with a sine wave inside; key says "variable voltage source", which is wrong: it is an **AC voltage source**). Label symbols ~ and ⎓ on chargers are taught in Other DC Sources A3 type 9.
- How to solve: ~ means AC. ⎓ means DC. A circle with a sine wave = AC source. A circle with + and – (or a cell symbol) = DC source.
- Worked example: "A charger says INPUT 100-240 V ~ 50/60 Hz, OUTPUT 5 V ⎓ 2 A. Which side is AC?" **Answer: the input (~).**
- Traps: A circle with an arrow inside is a **current source**, not AC. See Mistakes for the UF key.
- Calculator: look up — type: AC symbol, DC symbol.

4. US mains voltage and frequency

- Scope: ✅ IN.
- Where seen: Science Olympiad @ Berkeley Invitational, Feb 2026, Div B, INV, Q12 (options 50/60 Hz with 100/120 V → **60 Hz, 120 V**). UT Austin (ATX) Regional 2021, Div B, REG, Q23 (US outlet → "110 V 60 Hz", the closest option). UT Austin Invitational Fall 2025, Div B/C, INV, 1c.vi δ (US grid 60 Hz).
- How to solve: US and Canada: **120 V, 60 Hz**. Big appliances: 240 V.
- Worked example: "Which is the US household standard: 50 Hz 100 V, 50 Hz 120 V, 60 Hz 100 V, 60 Hz 120 V?" **Answer: 60 Hz, 120 V.**
- Traps: 110 V and 115 V are older names for the same supply. 100 V is Japan. 50 Hz is Europe and most of Asia.
- Calculator: look up — type: US mains.

5. Mains in other countries

- Scope: ✅ IN (AC characteristics).
- Where seen: not seen in a test. Prior bank I_bank A3.
- How to solve: use the table in Key facts. Europe 230 V 50 Hz. Japan 100 V with two frequencies.
- Worked example: "A traveler from the US takes a 120 V-only hair dryer to France. What happens?" France is 230 V. The dryer gets almost twice its rated voltage. Power = V^2/R rises by $(230/120)^2$ = **3.7 times**. **Answer: it overheats and can burn out or catch fire. Use a converter or a dual-voltage dryer.**
- Traps: A plug adapter only changes the plug shape. It does not change the voltage. A charger labeled "100-240 V ~" works everywhere with just an adapter.
- Calculator: look up — type: mains by country (country name).

6. Frequency to period, period to frequency

- Scope: ✅ IN (T = 1/f is algebra, not "frequency analysis").

- Where seen: UT Austin Invitational Fall 2025, Div B/C, INV, 1c.vi δ (period for 60 Hz → 1/60 = 0.0167 s). F-M Invitational fall 2019, Div not stated, INV, Q5a (120 V 60 Hz → T = 0.017 s). Science Olympiad @ Berkeley Feb 2026, Div B, INV, Q21 (20 Hz, how long do 10 cycles take → **0.50 s**).
- How to solve:
  1. T = 1/f. Put f in Hz, get T in seconds.
  2. For n cycles: t = n/f.
  3. Convert: 1 ms = 0.001 s.
- Worked example: "What is the period of 50 Hz mains?" T = 1/50 = **0.020 s = 20 ms**. "How long do 10 cycles at 20 Hz take?" t = 10/20 = **0.50 s**. "One cycle takes 4 ms. Frequency?" f = 1/0.004 = **250 Hz**.
- Traps: Forgetting ms → s. 1/60 = 0.0167 s = 16.7 ms, not 16.7 s.
- Calculator: calculate — type: f=60 → T; or n=10, f=20 → t.

7. How many times per second does 60 Hz current reverse?

- Scope: ✅ IN.
- Where seen: not seen in a test. Prior bank I_bank A2.
- How to solve: each cycle has two reversals (+ to −, and − to +). Reversals per second = 2f.
- Worked example: "How many times per second does the current in a US outlet change direction?" 2 × 60 = **120 times**.
- Traps: The answer is not 60. Sixty is the number of full cycles.
- Calculator: calculate — type: f=60 → 2f.

8. The unit hertz and Heinrich Hertz

- Scope: ✅ IN (history list, rule 3.c.i).
- Where seen: F-M Invitational fall 2019, Div not stated, INV, Q7 (Hertz proved EM waves; unit of frequency). UF Northern Regional 2019, Div B, REG, Q4 (who found EM waves experimentally → Hertz). Full facts in Hertz A1.
- How to solve: 1 Hz = 1 cycle per second. The unit honors Heinrich Hertz (1857-1894), who proved radio waves exist (1886-1889).
- Worked example: "The unit of frequency is named after whom?" **Answer: Heinrich Hertz.**
- Traps: Hertz did not invent AC power. He studied electromagnetic waves.
- Calculator: look up — type: hertz unit.

9. Why the grid uses AC (concept)

- Scope: ✅ IN (the reason). Transformer physics is ❌ OUT.
- Where seen: Science Olympiad @ Berkeley Feb 2026, Div B, INV, Q41 (which statement is NOT true → "DC is easier to step up and down with a transformer"). UT Austin (ATX) Regional 2021, Div B, REG, Q30 (advantage of AC → long-distance transmission). UT Austin (ATX) Regional 2020, Div B, REG, Q16 (less loss over long distance). UF Northern Regional 2020, Div B, REG, Q28 (why AC for power lines: transformers and P = I^2R). NY Lake Erie/Niagara Invitational Dec 2018, Div C, INV, Q5 (why AC for the grid → transformers). Rickards Invitational Oct 2025, Div C, INV, Q42 (high voltage → lower current → less power loss). Princeton Invitational Feb 2026, Div C, INV, SA5a (one advantage of AC). Official practice decks (F_slides ACDC-WHY-AC-TRANSMISSION).
- How to solve:
  1. Transformers change AC voltage easily; they do not work on steady DC.
  2. High voltage → low current for the same power.
  3. Low current → low I^2R loss in the wires.
- Worked example: "Why is power sent across the country at very high AC voltage and then stepped down?" **Answer: high voltage means low current, so less power is lost as heat (I^2R) in the lines. Transformers can only step AC up and down.**
- Traps: "AC flows with less resistance" is false. The wire resistance is the same. The **current** is smaller. "DC is easier to step up" is false.
- Calculator: look up — type: why AC grid.

10. War of the currents

- Scope: ✅ IN as an AC/DC uses-and-hazards fact. ⚠️ Borderline as "history", because Edison, Tesla and Westinghouse are not on the rule 3.c.i list.
- Where seen: Pembroke Hill Invitational Dec 2025, Div B, INV, Q15 (who championed DC → Thomas Edison). UT Austin (ATX) Regional 2020, Div B, REG, Q20 (Edison's main argument against AC → not safe). Science Olympiad @ Berkeley Feb 2026, Div B, INV, Q1 (credited with AC power generation and transmission → Tesla). UF Northern Regional 2020, Div B, REG, Q6 and UF 2020 Div C Q2 (AC pioneer / who invented AC → Tesla). UT Austin (ATX) Regional 2020, Div B, REG, Q5 (Westinghouse bought his patents → Tesla). Northview Sylvania Invitational Dec 2025, Div C, INV, Q46 and UT Austin Invitational 2019, Div B, INV, Q8 (Nobel prize for inventing AC → Nobody).
- How to solve: Edison = DC (Pearl Street, 1882). Tesla + Westinghouse = AC. AC won (1893 World's Fair, Niagara Falls 1895-96) because transformers let it travel far.
- Worked example: "Who championed DC during the War of the Currents, and what was his main argument against AC?" **Answer: Thomas Edison; he said AC was too dangerous.**
- Traps: "Who won a Nobel Prize for AC?" → **Nobody**. Tesla never won a Nobel Prize. For 2027, a "who" question about Tesla is technically outside the history list; answer it anyway.
- Calculator: look up — type: war of currents (Edison, Tesla, Westinghouse).

11. AC or DC: which systems use which?

- Scope: ✅ IN.
- Where seen: UT Austin Invitational 2020, Div B, INV, Q53-57 (electronic circuits, home outlets, long-distance lines, car batteries, phones → DC, AC, AC, DC, DC). UF Northern Regional 2019, Div C, REG, Q21 (AC from wall sockets → True). Full type in Other DC Sources A3 type 8.
- How to solve: outlets and grid = AC. Batteries, electronics, cars, phones, USB = DC. Long lines are usually AC (HVDC for special cases).
- Worked example: "Does a phone's circuit board run on AC or DC?" **Answer: DC.** The charger turns 120 V AC into 5 V DC.
- Traps: "Long-distance lines" is usually keyed AC. HVDC lines exist, but pick AC unless the question says HVDC.
- Calculator: look up — type: device AC or DC.

Medium

12. RMS to peak (mains peak voltage)

- Scope: ⚠️ Borderline. RMS is close to "AC circuit theory" (3.d), but it uses only $\sqrt{2}$ (no trig). It is still asked in 2026.
- Where seen: F-M Invitational fall 2019, Div not stated, INV, Q5a (sketch 120 V 60 Hz; give V_p and T → 170 V, 0.017 s). GullSO Invitational Dec 2025, Div C (B/C), INV, Q13 (sine with amplitude 120 V → closest RMS 85 V). MIT Invitational Jan 2019, Div C, INV, Q18 (RMS of a 120 V peak sine → 84.9 V).
- How to solve:
  1. Decide which value you have. Mains values are RMS.
  2. $V_p = \sqrt{2} \times V_{rms}$. $V_{rms} = V_p \div \sqrt{2}$.
- Worked example: "What is the peak voltage of a 120 V RMS outlet?" $120 \times 1.414 =$ **169.7 V ≈ 170 V**. "What is the RMS value of a sine wave with a 120 V peak?" $120 \div 1.414 =$ **84.9 V**. "Peak of European 230 V?" **325 V.**
- Traps: Multiplying when you should divide. Check: peak is always **bigger** than RMS for a sine.
- Calculator: calculate — type: Vrms=120 → Vp; or Vp=120 → Vrms.

13. Peak-to-peak voltage of an outlet

- Scope: ⚠️ Borderline (same reason as type 12).
- Where seen: Hawk & Hornet Invitational 2026, Div C, INV, Q8 (peak-to-peak of a US outlet → **340 V**, options 120/240/340/170). Northview Sylvania Invitational Dec 2025, Div C, INV, Q38 (same question, key C = 340 V). Prior bank I_bank A4.
- How to solve: $V_{pp} = 2 \times V_p = 2 \times \sqrt{2} \times V_{rms}$.
- Worked example: "Peak-to-peak voltage of a 120 V outlet?" $2 \times 169.7 =$ **339.4 V ≈ 340 V**.
- Traps: 170 V is the peak, not the peak-to-peak. 240 V is the US split-phase voltage, a distractor.

- Calculator: calculate — type: Vrms=120 → Vpp.

14. Equal-heating DC voltage (what RMS means)

- Scope: ⚠️ Borderline.
- Where seen: BullSO Invitational Jan 2026, Div B, INV, Q22 (DC voltage that gives the same power as a 100 V sinusoidal source → **100/√2**). UT Austin Invitational fall 2018, Div B/C, INV, Q16 (how is an AC voltage value agreed on? → the RMS value; no key). BullSO 2026 Div C Q22 (same item, per G_tests).
- How to solve:
  1. RMS = the DC value that makes the same heat in a resistor.
  2. For a sine with peak V_p: V_DC = V_p/√2.
- Worked example: "What DC voltage heats a toaster the same as an AC sine with a 100 V peak?" 100/√2 = **70.7 V**.
- Traps: The "100 V" in a question can be a peak or an RMS value. Read the words "sinusoidal ... 100 V". BullSO treats it as the peak. If a question says "100 V RMS", the answer is 100 V.
- Calculator: calculate — type: Vp=100 → Vdc equivalent.

15. RMS of a square wave

- Scope: ⚠️ Borderline.
- Where seen: GullSO Invitational Dec 2025, Div C (B/C), INV, Q12 (square wave between +120 V and −120 V → **120 V**).
- How to solve: a square wave sits at full size the whole time. Its square is always A^2. So RMS = A.
- Worked example: "A square wave switches between +12 V and −12 V. RMS?" **12 V.** "A sine with the same 12 V peak?" 12/√2 = **8.49 V**.
- Traps: Do not use √2 for a square wave. √2 is only for a sine wave.
- Calculator: look up — type: RMS square wave = amplitude.

16. AC appliance on mains: current, resistance, power

- Scope: ✅ IN (resistive power with RMS values is ordinary Ohm's law).
- Where seen: MIT Invitational Jan 2019, Div C, INV, Q11 (microwave 1.25 kW at 120 V → 10.4 A). UT Austin (ATX) Regional 2019, Div B/C, REG, Q12 (a 22 kΩ "toaster" on 120 V → 0.655 W). Science Olympiad @ Berkeley Feb 2026, Div B, INV, Q22 (60 W bulb at 120 V → 240 Ω). JC Booth Invitational "Shock Value" 2011, Div B, INV, Q55 (240 V across 24 Ω → 10 A).
- How to solve:
  1. Treat 120 V as if it were DC (it is RMS).
  2. I = P/V. R = V^2/P. P = V^2/R.
- Worked example: "A 1500 W space heater runs on 120 V. Find the current and its resistance." I = 1500/120 = **12.5 A**. R = 120^2/1500 = **9.6 Ω**.
- Traps: Using the peak (170 V) instead of 120 V. Rated resistance is for the hot filament or element; a cold bulb has lower resistance (non-linear).
- Calculator: calculate — type: P=1500, V=120 → I, R.

17. Transmission line loss (I^2R)

- Scope: ✅ IN.
- Where seen: MIT Invitational Jan 2019, Div C, INV, Q36 (line R = 15 Ω, 10.0 A → **1500 W**) and Q37 (same line at 240 V, 5.0 A → **375 W**). Northview Sylvania Invitational Dec 2025, Div C, INV, Q45 (2000 A in a 2.3 Ω line → **9.2 MW**; the key says 9.2 kW, see Mistakes). Rickards Invitational Oct 2025, Div C, INV, Q43 (equation that explains why high voltage lowers loss).
- How to solve:
  1. Find the line current: I = P_delivered/V (or it is given).
  2. $P_{loss} = I^2 \times R_{line}$.
  3. Do not use V^2/R with the supply voltage. The supply voltage is not across the line.
- Worked example: "A 15 Ω line carries 10 A. Loss?" $10^2 \times 15$ = **1500 W**. "The same power is sent at twice the voltage, so 5 A. Loss?" $5^2 \times 15$ = **375 W**, one quarter.

- Traps: $P = V^2/R$ with V = 120 V gives a silly answer. The 120 V is across the **load**, not the line. Units: $2000^2 \times 2.3$ = 9 200 000 W = 9.2 **M**W.
- Calculator: calculate — type: I=10, R=15 → Ploss.

18. Do resistor rules work for AC?

- Scope: ⚠️ Borderline. For pure resistors, yes (IN). Anything with "impedance" is ❌ OUT (3.d).
- Where seen: UT Austin Invitational Fall 2025, Div B/C, INV, 1a.viii (does a resistive voltage divider work for AC? → Yes). UT Austin (ATX) Regional 2020, Div B, REG, Q27 ("DC techniques apply to AC" → True, with impedance: OUT). UT Austin (ATX) Regional 2020, Div B, REG, Q19 ("AC equivalent of resistance" → impedance: OUT).
- How to solve: Ohm's law, series and parallel rules and dividers all work for **resistors** with AC, using RMS values (or using instantaneous values at each moment).
- Worked example: "A 120 V RMS outlet feeds 30 Ω and 10 Ω in series. Voltage across the 10 Ω?" 120 × 10/40 = **30 V RMS**.
- Traps: Capacitors and inductors do not follow these rules with AC. They are OUT for 2027, so a Div B question should only use resistors.
- Calculator: calculate — type: divider Vin=120, R1=30, R2=10.

19. Is this a good AC source? (shape of a changing signal)

- Scope: ✅ IN (judgement from a graph; no trig).
- Where seen: UT Austin Invitational Fall 2025, Div B/C, INV, 1c.vi ε-ζ (a divider with a resistor that changes in time makes a rounded, "not quite sinusoidal" output; is it a good AC source? → No).
- How to solve:
  1. A good AC source gives a steady, repeating, symmetric wave around zero with a known size and frequency.
  2. A divider with a changing resistor never goes negative, its shape is distorted, and it cannot deliver much power.
- Worked example: "V_{out} = 10 V × R1/(R1 + R2) where R2 swings from 0 to R1 and back. Does V_{out} reverse?" V_{out} goes between 10 V (R2 = 0) and 5 V (R2 = R1). **It never reverses, so it is pulsating DC, not AC.**
- Traps: "Goes up and down" is not the same as "alternates".
- Calculator: calculate — type: divider with R2 range → Vmin, Vmax.

20. Rectified AC: sketch input and output

- Scope: ⚠️ Borderline. A rectifier is a diode use (IN, rule 3.c.vii). The RMS-to-peak step is borderline. Diode details are in Diode Applications A7.
- Where seen: North Pocono Invitational Jan 2020, Div C, INV, Q17 (current source + rectifier + R, I_{rms} = 2 A; draw input and output → I_{max} = 2.8 A; input two-way, output one-way). Rickards Invitational Oct 2025, Div C, INV, Q62-64 (half-wave vs full-wave; connect a bridge).
- How to solve:
  1. Input: a sine that goes + and −.
  2. Half-wave output: only the + halves; the − halves become 0.
  3. Full-wave (bridge) output: the − halves are flipped up. Same peak (minus diode drops).
  4. Peak current = $\sqrt{2} \times I_{rms}$ for a sine.
- Worked example: "I_{rms} = 2 A sine into a full-wave rectifier. Peak of the output current?" 2 × 1.414 = **2.83 A**. The output never goes below 0.
- Traps: The output is pulsating DC, not steady DC. Smoothing with a capacitor is ❌ OUT.
- Calculator: calculate — type: Irms=2 → Ipeak.

21. Why are DC arcs harder to stop than AC arcs?

- Scope: ✅ IN (AC characteristic and hazard).
- Where seen: Princeton Invitational Feb 2026, Div C, INV, SA5c ("Why are DC fires harder to put out than AC fires?" → AC crosses zero, so arcs go out). Rickards Invitational Dec 2020 (2021 season), INV, Q16 (why fuses have a voltage rating → arcing; division not recorded in G_tests). Prior bank I_bank D-section (switch AC vs DC ratings).
- How to solve:

1. AC passes through zero 2f times per second (120 at 60 Hz).
2. At each zero the arc loses its push and tends to go out.
3. DC never drops to zero, so an arc can keep burning.

- Worked example: "A switch is rated 10 A at 250 VAC but only 10 A at 30 VDC. Why is the DC voltage rating so much lower?" **Answer: DC has no zero crossings, so the arc that forms when the contacts open does not go out by itself. The switch can only break a low DC voltage safely.**
- Traps: It is not because DC is "stronger". It is about the zero crossings.
- Calculator: look up — type: DC arc vs AC arc.

Hard

22. Raise the voltage: how much less loss? (percent loss)

- Scope: ✅ IN.
- Where seen: not seen with these numbers. MIT Jan 2019 C Q36-37 is the simple version (type 17). Prior bank I_bank A20.
- How to solve:
  1. I = P/V at each voltage.
  2. $P_{loss} = I^2R$.
  3. Percent loss = $P_{loss}/P \times 100\%$.
  4. Shortcut: n times the voltage → $1/n^2$ the loss.
- Worked example: "100 MW is sent through a line with 10 Ω total resistance. Find the loss at 100 kV and at 500 kV." At 100 kV: $I = 10^8/10^5 = 1000$ A, loss = $1000^2 \times 10$ = **10 MW (10%)**. At 500 kV: I = 200 A, loss = **0.40 MW (0.4%)**. The voltage is 5 times higher, so the loss is **25 times** lower.
- Traps: Loss falls with the **square** of the voltage, not in proportion. Use the line resistance only, not the load.
- Calculator: calculate — type: P=1e8, R=10, V=1e5 and V=5e5 → I, Ploss, %.

23. Electric bill chain: dollars → kWh → W → A → Ω

- Scope: ✅ IN (power and energy on AC mains).
- Where seen: "Shock Value Test" re-tryout, about 2012, Div B, practice, V-2 (V-2a: $5324.18 at 9.4 ¢/kWh → 56 640 kWh; then W, then A at 120 V → 54.6 A in V-2d). Cobra Invitational 2019-20, Div B, INV, Q21 (5 × 15 W bulbs, 5 h a day, February 2020, $0.07/kWh → $0.76; trap: Feb 2020 has 29 days).
- How to solve:
  1. kWh = dollars ÷ price per kWh.
  2. Average power (kW) = kWh ÷ hours.
  3. I = P/V. R = V/I.
- Worked example: "A device runs all day, every day. The 30-day bill for it is $18 at $0.15 per kWh, on 120 V. Find its power, current and resistance." kWh = 18/0.15 = 120 kWh. Hours = 30 × 24 = 720 h. P = 120/720 = 0.1667 kW = **167 W**. I = 166.7/120 = **1.39 A**. R = 120/1.389 = **86.4 Ω**.
- Traps: kW vs W (× 1000). Days vs hours. The bill is energy (kWh), not power. Count the days in the month (February 2020 had 29). Check: 5 × 15 W × 5 h × 29 days = 10.875 kWh → $0.761.
- Calculator: calculate — type: dollars=18, price=0.15, days=30, hours/day=24, V=120 → kWh, W, A, Ω.

24. Charging an electric car from an outlet

- Scope: ✅ IN (energy and power).
- Where seen: UT Austin Invitational 2020, Div B, INV, Q63 (Tesla Model 3: 8 h at 120 V / 15 A; uses 13 kW at 60 mph → **66.46 miles**).
- How to solve:
  1. Energy in = V × I × t (kWh, using kW and h).
  2. Driving time = energy ÷ driving power.
  3. Distance = speed × time.
- Worked example: Energy = 120 × 15 × 8 = 14 400 Wh = 14.4 kWh. Time = 14.4/13 = 1.108 h. Distance = 60 × 1.108 = **66.5 miles**.
- Traps: Charging is not 100% efficient in real life; the question assumed it was. Keep the units in kW and h.

- Calculator: calculate — type: V=120, I=15, t=8, Pdrive=13000, speed=60 → miles.

25. Rated bulbs (W at 120 V) in series and in parallel on mains

- Scope: ✅ IN. More practice in Series and Parallel Resistance A5.
- Where seen: UT Austin Invitational 2019, Div B, INV, Q36 and UT Austin Invitational 2019, Div C, INV, Q32 (15 W, 35 W, 40 W bulbs on 120 V; key gives 0.750 A in series, which is wrong; see Mistakes). F-M Invitational 2018, Div not stated, INV, Q5 (50 W bulb in series with 60 W || 40 W on 120 V; 60 W burns out → 50 W bulb is dimmer but lit).
- How to solve:
  1. Find each bulb's resistance from its rating: $R = V_{rated}^2/P_{rated}$.
  2. Parallel on 120 V: each bulb gets 120 V, so $I = P/V$.
  3. Series: add the resistances, $I = 120/R_{total}$, the same in every bulb.
- Worked example: $R = 120^2/15 = 960\ \Omega$, $120^2/35 = 411.4\ \Omega$, $120^2/40 = 360\ \Omega$. Parallel currents: **0.125 A, 0.292 A, 0.333 A**. Series: $R_{total} = 1731.4\ \Omega$, $I = 120/1731.4$ = **0.0693 A** in each bulb.
- Traps: In series, the rated watts do not apply. The 15 W bulb is the **brightest** in series (largest R, so largest I^2R). Real filaments have lower resistance when cold, so real answers differ a little.
- Calculator: calculate — type: ratings 15,35,40 W at 120 V → R each; series I; parallel I.

Nationals

No States/Nationals-only (KVL/KCL) question type was found in a test for this note. AC networks with two unknowns are in Household Wiring A3 type 25 (split-phase neutral current) and in Shock and Safety A3 type 26 (a person bridging a network).

Out of scope (skip)

- Phase angle, impedance, reactance, power factor, phasors. ❌ OUT: rule 3.d "AC circuit theory". Seen: UT Austin (ATX) Regional 2020 B Q19 (impedance).
- Average value of AC = 0 as a calculation. ❌ OUT: AC circuit theory (3.d). Seen: UT Austin (ATX) Regional 2020 B Q18.
- Skin effect. ❌ OUT: AC circuit theory / frequency analysis (3.d). Seen: BullSO Jan 2026 B Q14; North Pocono Jan 2020 C Q3.
- Corona discharge on power lines. ⚠️ Borderline (a static/discharge fact, not AC theory). Seen: Rickards Oct 2025 C Q27.
- Transformer turns ratios and currents. ❌ OUT: inductance (3.d) and magnetism (not in 3.c). Seen: Science Olympiad @ Berkeley Feb 2026 B Q9-10; Pembroke Hill Dec 2025 B Q11 and Q34; ATX Regional 2026 C MC25; ATX Regional 2020 B Q39 and Q42.
- Generator voltage at a given coil position. ❌ OUT: magnetism and trigonometry (3.e.v). Seen: Northview Sylvania Dec 2025 C Q48.
- Filters (low-pass, high-pass). ❌ OUT: frequency analysis and capacitance (3.d). Seen: ATX Regional 2021 B Q20.
- 3-phase power (line vs phase voltage, 208 V, 480 V). ❌ OUT: 3-phase power (3.d).
- Oscilloscope traces (volts/div, time/div). ❌ OUT: oscilloscopes (3.d).
- Lenz's law and the slide-wire generator. ❌ OUT: magnetism. Seen: ATX Regional 2020 B Q12 and C Q68.

Mistakes in tests and keys

- **UF Northern Regional 2020 B, Q21f:** the key calls a circle with a sine wave a "variable voltage source". It is an **AC voltage source**.
- **Northview Sylvania Dec 2025 C, Q45:** the key gives B (9.2 kW). The correct loss is $2000^2 \times 2.3$ = **9.2 MW** (choice D). (Checked in Python. I read the key from the text layer of `nv26C_key.pdf`.)
- **UT Austin 2019 B Q36 and UT 2019 C Q32:** the key gives 0.750 A in each bulb in series. That is the total **parallel** current (90 W ÷ 120 V). The correct series current is **0.0693 A**.
- **ATX Regional 2021 B Q23:** the best option was "110 V 60 Hz". The modern US nominal value is 120 V. Pick the 60 Hz option.

- **Cobra 2019-20 B Q16, graph H:** a sine from −0.6 V to +3.4 V is keyed AC. It is AC with a DC offset. A strict grader could accept "DC with ripple" only if the curve stayed above zero, which it does not.
- **Berkeley Feb 2026 B Q1** asks who is "credited with" AC → Tesla. Many people built early AC systems (Gaulard and Gibbs, Stanley, Ferraris). The intended answer is still Tesla.

Not confirmed

- The world mains table, the ANSI 114-126 V range and the Japan 50/60 Hz origin story are from general references, from memory. I did not open IEC 60038 or ANSI C84.1 in this run.
- War-of-the-currents dates (1882, 1888, 1890, 1893, 1895-96) and the "26 miles" to Buffalo are from memory of standard histories.
- The switch rating example (10 A 250 VAC / 10 A 30 VDC) is a typical label, not a specific part.
- The Northview Q45 key letter: I read "45. B" from the key text layer. If the text layer shifted columns, the key may be right. I did not render the key page.
- I found no AC question in the Florida State 2021 test beyond the outlet item (see Household Wiring A3 and Shock and Safety A3). I looked in `tests/state.md` and grepped all 8 extraction files and `prior/G_tests.md`.

Sources

- Rules: 2027 Div B Circuit Lab (manual pages B12-B13): areas 3.c.i, 3.c.iii, 3.c.v, 3.c.vii; exclusions 3.d; math 3.e.v; hands-on safety II.d.
- Tests (extractions in `circuitlab/research/tests/` and `prior/G_tests.md`): Berkeley Feb 2026 B; BullSO Jan 2026 B; Pembroke Hill Dec 2025 B; UT Austin Fall 2025 B/C; Northview Sylvania Dec 2025 C; Hawk & Hornet 2026 C; GullSO Dec 2025 B/C; Rickards Oct 2025 C; Princeton Feb 2026 C; ATX Regional 2020 B, 2021 B, 2026 C; UT Austin Invitational fall 2018 B/C, 2019 B/C, 2020 B; MIT Jan 2019 C; UF Northern Regional 2019 B, 2020 B/C; NY Lake Erie/Niagara Dec 2018 C; Cobra 2019-20 B; F-M 2018 and fall 2019; North Pocono Jan 2020 C; UMSO about 2019; JC Booth 2011 B; "Shock Value" re-tryout about 2012 B.
- Prior research: `prior/B_topics.md` §3c; `prior/I_bank.md` A1-A7, A20, A22-A23; `prior/F_slides.md` (ACDC-DEFINE, ACDC-WHY-AC-TRANSMISSION, ACDC-HAZARDS); `prior/G_tests.md` (DC & AC section).
- Textbooks (general knowledge, not re-read in this run): OpenStax *College Physics 2e* ch. 20.5 (Alternating current versus direct current: RMS, peak); Kuphaldt, *Lessons in Electric Circuits* vol. 2 ch. 1 (AC: "What is alternating current?", "Measurements of AC magnitude"); All About Circuits, "AC Waveforms".

Household Wiring

This note covers the AC wiring in a house: hot, neutral and ground wires, outlets and plugs, 120 V and 240 V circuits, wire gauge, and basic **NEC** rules. The NEC (National Electrical Code, also called NFPA 70) is the US rulebook for safe wiring.

Scope:

- Household wiring is how AC is **used**, and most of it is about **hazards**. Rule area **3.c.iii** ("uses, and hazards of ... AC") covers it. So it is ✅ **IN**.
- The 2026 rules said "common household AC components". The 2027 rules dropped that phrase. Tests still ask it (USC Invitational Jan 2026 B Q6; Berkeley Feb 2026 B Q23; BullSO Jan 2026 B Q33). So treat it as IN. A few code details (exact NEC article numbers, NEMA shapes) are ⚠️ borderline trivia.
- Breakers, fuses and GFCIs are rule area **3.c.vi**. Their full question types are in CL 6 - Switches, Relays, Fuses, Breakers, GFCI 🅃. This note covers only how they fit into house wiring.
- Wiring math with Ohm's law (wire resistance, voltage drop, fault current) is ✅ IN (rules 3.c.iv and 3.c.v).
- Split-phase networks that need two loop equations are ✅ **States/Nationals only** (rule 3.c.ix, KVL/KCL).
- 3-phase power is ❌ OUT (rule 3.d).
- Hands-on: no AC (rule II.d). This is written-test only.

Key facts

The three wires

- **Hot** (also "live" or "line"): carries the voltage from the panel. It is 120 V away from ground. US colors: **black** or **red** (sometimes blue or other colors, but never white, gray or green).
- **Neutral**: the normal return path for current. It is close to 0 V. US colors: **white** or **gray**.
- **Ground** (equipment grounding conductor): a safety path. It carries current **only during a fault**. US colors: **green**, green with a yellow stripe, or **bare copper**.
- Neutral and ground are joined (**bonded**) only at the **main service panel**. They are kept separate everywhere else.
- At the panel, the ground connects to a **grounding electrode**, usually a copper rod about 8 ft (2.4 m) long driven into the earth, and to the metal water pipe.
- Europe and IEC colors: **brown** = live, **blue** = neutral, **green-yellow** = earth (ground). Old UK colors: red = live, black = neutral.
- Trap: in old US wiring, a white wire can be used as a hot in a switch loop. It should then be marked with black tape.

What the ground wire does

- Suppose the hot wire touches a metal appliance case (a **ground fault**).
- With a ground wire: a large current flows from hot → case → ground wire → panel. The breaker trips at once. The case stays near 0 V.
- Without a ground wire: the case sits at 120 V. Nothing trips. The next person to touch the case and a grounded object (sink, pipe, wet floor) becomes the path.
- So the ground wire protects **people** by making the breaker trip. The breaker itself protects the **wires**.

Outlets (receptacles) and plugs

- A **receptacle** is the outlet in the wall. A **plug** is on the cord.
- Standard US outlet, **NEMA 5-15R**: two vertical slots and a round (D-shaped) hole.
  - **Short (narrow) slot = hot.**
  - **Long (wide) slot = neutral.**
  - **Round hole = ground.**
  - Memory aid: "Narrow = hot (the danger is narrow)". Or "wide = white = neutral".
- **NEMA 5-20R**: 20 A version. The neutral slot is T-shaped, so it also takes 15 A plugs.
- NEMA name code: the **first number** is the wiring type and voltage (1 and 5 = 125 V; 6 = 250 V; 10 and 14 = 125/250 V). The **second number** is the current rating in amps. **R** = receptacle, **P** = plug. **L** in front = a twist-

locking type (for example L5-30).

- Common US receptacles:

NEMA	Rating	Use
1-15	15 A, 125 V, 2 prongs, no ground	Old outlets, double-insulated devices
5-15	15 A, 125 V, grounded	Standard outlet
5-20	20 A, 125 V, T-slot neutral	Kitchens, garages
6-15, 6-20	15 A or 20 A, 250 V	Window air conditioners, shop tools
10-30	30 A, 125/250 V, no ground (old)	Old dryers
14-30	30 A, 125/250 V, 4 prongs	Electric dryers
14-50	50 A, 125/250 V, 4 prongs	Ranges, RVs, EV chargers

- Outlet ratings are 125 V and 250 V. Supply voltages are 120 V and 240 V. The rating is the maximum, so it is a little higher.
- **Tamper-resistant (TR)** outlets have shutters that keep children from pushing objects into one slot. **WR** = weather-resistant (outdoors).

Polarized plugs

- A **polarized plug** has one blade wider than the other. The wide blade is **neutral**. It fits only one way.
- Purpose: the appliance's switch and fuse are put on the **hot** side. When the switch is off, the parts inside are not live.
- In a lamp, the **screw shell** of the socket goes to **neutral** and the small center tab goes to **hot**. So touching the shell while changing a bulb is safer.
- A polarized plug does **not** ground the case. It is the weakest of the shock protections (BullSO Jan 2026 B Q33).

Three-prong plugs and double insulation

- A **three-prong** (grounded) plug connects the metal case to ground. This is a "Class I" appliance.
- A **double-insulated** ("Class II") appliance has two layers of insulation and usually a plastic case. It needs no ground, so it has a 2-prong plug.
- The double-insulation symbol is a **square inside a square** (⧈).
- Never cut off the ground pin. Never use a "cheater" 3-to-2 adapter without grounding its tab.

120 V and 240 V (split-phase)

- The pole or pad transformer outside a US house gives **two hots** (called legs L1 and L2) and a **neutral** from the center of its coil.
- Each hot is **120 V** to neutral. The two hots are opposite (one is + when the other is –). So **hot to hot = 240 V**.
- This is called **split-phase** (or "single-phase three-wire"). It is not 3-phase.
- 120 V circuits: outlets and lights. 240 V circuits: dryers, ranges, water heaters, central air conditioners, EV chargers.
- A 240 V circuit uses a **double-pole breaker** (two linked handles), one for each hot.
- Why 240 V for big loads: the same power needs **half the current**. Half the current allows thinner wire and gives less I^2R loss.
- Current on the neutral = the **difference** between the two leg currents (when loads are on opposite legs). If the legs are balanced, the neutral carries almost nothing.
- A US service panel is usually rated **100 A, 150 A or 200 A** at 240 V.

Wire gauge (AWG)

- **AWG** (American Wire Gauge) is the US wire size system. A **bigger** gauge number means a **thinner** wire.
- Thinner wire has more resistance, so it heats more at the same current. The current a wire can carry safely (its **ampacity**) depends mostly on its **gauge** (Hawk & Hornet 2026 Q2).
- The voltage a wire can withstand depends on its **insulation** (Hawk & Hornet 2026 Q3).
- Rules of thumb: every **3 gauges** ≈ half the area (double the resistance). Every **6 gauges** ≈ half the diameter. 10 AWG ≈ **1 Ω per 1000 ft**.
- Diameter formula: d (mm) = 0.127 × 92^((36 – n)/39).

AWG	Diameter	Resistance (copper)	Typical breaker (NEC 240.4(D))	Use
22	0.64 mm	16.1 Ω/1000 ft	—	Breadboard jumper wire
18	1.02 mm	6.4 Ω/1000 ft	—	Lamp cords
16	1.29 mm	4.0 Ω/1000 ft	—	Light extension cords
14	1.63 mm	2.5 Ω/1000 ft	**15 A**	General outlets and lights
12	2.05 mm	1.6 Ω/1000 ft	**20 A**	Kitchen, bathroom, garage outlets
10	2.59 mm	1.0 Ω/1000 ft	**30 A**	Dryers, water heaters

- Memory aid: "14-15, 12-20, 10-30".
- **Never** put a bigger breaker on a thinner wire (for example a 20 A breaker on 14 AWG). The wire could overheat before the breaker trips.
- House cable ("NM" cable, often called Romex) is named like "14/2 with ground": 14 AWG, 2 insulated wires (black, white) plus a bare ground. "12/3 with ground" has black, red, white and bare.
- US copper wire is common in homes. Aluminum wire (1960s-70s) needs special connectors because it loosens and overheats.

Branch circuits and loads

- A **branch circuit** is one breaker and the outlets or lights it feeds.
- Maximum power on a circuit = V × breaker current. 15 A at 120 V = **1800 W**. 20 A at 120 V = **2400 W**.
- **Continuous loads** (on for 3 hours or more) should be at most **80%** of the breaker rating (NEC 210.20(A) and 210.19(A)). 15 A → 1440 W. 20 A → 1920 W.
- All outlets on one breaker share its current. Adding loads in parallel adds current.

Where the NEC requires protection (2020-2023 NEC, simplified)

- **GFCI** (ground-fault circuit interrupter, NEC 210.8): bathrooms, kitchens (countertops and near sinks), garages, outdoors, crawl spaces, unfinished basements, laundry areas, within 6 ft of sinks, tubs and showers, pools and spas, boathouses. Details in CL 6 - Switches, Relays, Fuses, Breakers, GFCI [T].
- **AFCI** (arc-fault circuit interrupter, NEC 210.12): most 120 V, 15 A and 20 A circuits in living areas (bedrooms, living rooms, kitchens, hallways, closets, laundry). An AFCI detects the electrical "noise" of an arc (a loose or damaged wire) and prevents fires.
- **Tamper-resistant** outlets (NEC 406.12) in homes.
- Switches go in the **hot** wire, never the neutral.
- A **3-way** switch pair controls one light from two places. Wiring is in 3-Way and 4-Way Switching [A6].

Testing an outlet (written questions only)

- Hot to neutral: about **120 V**. Hot to ground: about **120 V**. Neutral to ground: about **0 V** (under 2 V).
- **Reversed polarity**: hot and neutral swapped. Narrow slot to ground reads 0 V; wide slot to ground reads 120 V. The appliance works, but its switch now breaks the neutral, so parts stay live when "off".
- **Open ground**: hot to ground reads 0 V (or a low "ghost" value on a digital meter).
- **Open neutral**: the outlet gives no power to a load, but the slots can still be live.
- A plug-in **outlet tester** has three lights that show these faults.

Formulas

What	Formula	Notes
Current from power	$I = P/V$	1500 W at 120 V → 12.5 A
Circuit maximum power	$P_{max} = V \times I_{breaker}$	15 A × 120 V = 1800 W
Continuous-load limit	$P_{cont} = 0.8 \times V \times I_{breaker}$	15 A → 1440 W
Total current on a circuit	$I_{total} = \Sigma P \div V$	Loads are in parallel
Wire resistance	$R = \rho L/A$, $\rho_{copper} = 1.68 \times 10^{-8}$ Ω·m	L = total length of both wires
Wire resistance from a table	R = (Ω per 1000 ft) × (round-trip feet) ÷ 1000	Round trip = 2 × cord length
Voltage drop in wires	$V_{drop} = I \times R_{wires}$	Percent = $V_{drop} \div V \times 100\%$
Power lost in wires	$P = I^2 \times R_{wires}$	

What	Formula	Notes
AWG diameter	d (mm) = 0.127 × 92^((36 – n)/39)	
Area of a round wire	A = π d²/4	
Ground-fault current	I_fault = V ÷ (R_hot + R_ground)	Usually hundreds of amps
Neutral current (opposite legs)	I_N =	I_L1 – I_L2
Neutral current (same leg)	I_N = I_1 + I_2	
240 V load	I = P/240	Half the 120 V current

Question types

Easy

1. Wire colors: hot, neutral, ground

- Scope: ✅ IN.
- Where seen: Science Olympiad @ Berkeley Feb 2026, Div B, INV, Q23 (US hot wire color → **black**; options gray, green, black, ...). Prior topic notes (B_topics §3c).
- How to solve: US: black or red = hot; white or gray = neutral; green or bare = ground. Europe: brown = live, blue = neutral, green-yellow = earth.
- Worked example: "In US house wiring, which wire is the neutral: black, red, white or green?" **Answer: white.**
- Traps: Green is ground, not neutral. Gray is a neutral color in the US (an allowed option), not a hot. Blue is neutral in Europe but can be a hot in US commercial wiring.
- Calculator: look up — type: wire colors US / EU.

2. Outlet slots: which is hot?

- Scope: ✅ IN.
- Where seen: UT Austin Invitational 2019, Div B, INV, Q12 and UT Austin Invitational 2019, Div C, INV, Q12 (polarized receptacle: which slot is hot → the **shorter (narrow)** slot; the options were oddly worded). MIT Invitational Jan 2021, Div C, INV, Q94 (receptacle A: which pin is neutral → pin 1, from a drawing).
- How to solve: on a US outlet: narrow slot = hot; wide slot = neutral; round hole = ground.
- Worked example: "Looking at a standard US outlet, which opening connects to the black wire?" **Answer: the short (narrow) slot.**
- Traps: Pictures of a **plug** are the mirror image of the receptacle. MIT 2021 warned about this. Find the wide blade first (neutral), then the ground pin.
- Calculator: look up — type: NEMA 5-15 slots.

3. True statements about three-prong and 240 V outlets

- Scope: ✅ IN.
- Where seen: USC Invitational Jan 2026, Div B, INV (top tier), Q6. Statements: I. three-prong outlets include a hot wire, neutral wire and ground wire; II. three-prong outlets are more prone to shocks and fires than two-prong ones; III. for electronics, a 240 V three-prong outlet is safer than a 120 V one; IV. in a 240 V outlet, each hot slot has 60 V and the neutral/ground slot has 120 V. Key: **I only.**
- How to solve:
  1. Check each statement against Key facts.
  2. II is false: the ground makes three-prong outlets **safer**.
  3. III is false: 240 V is not "safer"; it is a higher voltage.
  4. IV is false: each hot is **120 V** to neutral, and hot to hot is 240 V.
- Worked example: "In a US 240 V circuit, what is the voltage from each hot to neutral, and from hot to hot?" **Answer: 120 V each; 240 V hot to hot.**
- Traps: "60 + 60 = 120" is a made-up split. The 240 V comes from two 120 V legs that are opposite.
- Calculator: look up — type: split-phase 120/240.

4. Purpose of the ground wire

- Scope: ✅ IN.

- Where seen: UT Austin (ATX) Regional 2019, Div B/C, REG, Q21 (purpose of a grounding wire → a low-resistance fault path back to the panel so the breaker or fuse trips). BullSO Jan 2026, Div B, INV, Q33 option A (three-prong plugs ground the case).
- How to solve:
  1. Normal use: no current in the ground wire.
  2. Fault (hot touches the metal case): huge current flows in the ground wire.
  3. The breaker trips. The case stays near 0 V, so a person is not shocked.
- Worked example: "Why is the metal case of a washing machine connected to the ground pin?" **Answer: if a hot wire touches the case, the fault current flows through the ground wire instead of a person, and the large current trips the breaker.**
- Traps: The ground wire is not the normal return path. That is the neutral. Neutral and ground carry different currents and must not be swapped.
- Calculator: look up — type: ground wire purpose.

5. Neutral versus ground

- Scope: ✅ IN.
- Where seen: not seen as a stand-alone question. Prior bank I_bank W3. USC Jan 2026 B Q6 (statement I) touches it.
- How to solve: neutral = normal return current. Ground = fault current only. Both connect to the earth at the main panel only.
- Worked example: "A lamp draws 0.5 A from the hot wire. With no fault, how much current flows in the neutral and in the ground?" **Answer: neutral 0.5 A; ground 0 A.**
- Traps: "The neutral is safe to touch" is false. The neutral can carry voltage if it is broken or if the circuit is miswired. Treat every wire as live.
- Calculator: look up — type: neutral vs ground.

6. Polarized plugs: what do they do?

- Scope: ✅ IN.
- Where seen: BullSO Jan 2026, Div B, INV, Q33 (least practical way to prevent shock with metal-cased appliances → **polarized plugs** "to control which wire is hot"). BullSO Jan 2026, Div C, INV, Q33 (same, per G_tests). UT Austin 2019 B Q12 and C Q12 (polarized receptacle).
- How to solve:
  1. Wide blade = neutral. The plug fits only one way.
  2. The device's switch then breaks the hot wire.
  3. It does not connect the case to ground, so it gives less protection than a ground pin or double insulation.
- Worked example: "A lamp has a polarized plug. Which part of the bulb socket connects to the wide blade?" **Answer: the screw shell (neutral).**
- Traps: Polarized does not mean grounded. A 2-prong polarized plug has no ground.
- Calculator: look up — type: polarized plug.

7. Double insulation and the square-in-square symbol

- Scope: ✅ IN.
- Where seen: not seen in a test. Prior topic notes (B_topics §6) and prior bank I_bank W2. BullSO Jan 2026 B Q33 option D ("an insulator in extra space in the case") is the same idea.
- How to solve: two layers of insulation (or an insulating case) mean no exposed metal can become live. No ground pin is needed.
- Worked example: "A hair dryer has a 2-prong plug and a ⧈ symbol. Is it unsafe?" **Answer: no. The symbol means double-insulated (Class II), so it does not need a ground.**
- Traps: Double insulation does not protect you if the device is dropped in water. Use a GFCI near water.
- Calculator: look up — type: double insulated symbol.

8. What sets a wire's current rating and its voltage rating?

- Scope: ✅ IN.

- Where seen: Hawk & Hornet Invitational 2026, Div C, INV, Q2 (current-carrying capacity most depends on → **gauge**) and Q3 (voltage rating depends on → **insulation material**).
- How to solve:
  1. Current → heating → set by the wire's thickness (gauge) and material.
  2. Voltage → breakdown of the insulation → set by the insulation.
- Worked example: "Two wires have the same gauge. One has thick rubber insulation, the other thin PVC. Which has the higher voltage rating?" **Answer: the thick rubber one. The gauge (and so the current rating) is the same.**
- Traps: Length changes the voltage **drop**, not the current rating. Material matters (copper vs aluminum) but "gauge" is the best single answer.
- Calculator: look up — type: wire ampacity vs voltage rating.

9. Bigger AWG number: thicker or thinner?

- Scope: ✅ IN (resistance, rule 3.c.iv).
- Where seen: not seen in a test. Prior bank I_bank U19.
- How to solve: bigger number = thinner = more resistance = less current allowed.
- Worked example: "Which can safely carry more current: 12 AWG or 14 AWG?" **Answer: 12 AWG (thicker; 20 A vs 15 A).**
- Traps: It works backward from what you expect. 22 AWG breadboard wire is very thin.
- Calculator: look up — type: AWG table (gauge).

10. Which appliances use 240 V, and why?

- Scope: ✅ IN.
- Where seen: USC Jan 2026 B Q6 (statements III and IV). Not seen as a stand-alone question.
- How to solve:
  1. Big heating or motor loads (dryer, range, water heater, central AC, EV charger) use 240 V.
  2. Same power at double voltage → half the current → thinner wire, less heat in wires.
- Worked example: "A 5400 W dryer. Current at 240 V? At 120 V?" 5400/240 = **22.5 A** (fits a 30 A circuit with 10 AWG). At 120 V it would need **45 A**.
- Traps: 240 V is not "safer". It is half the current for the same power, but a shock is worse.
- Calculator: calculate — type: P=5400, V=240 and V=120 → I.

Medium

11. Maximum watts on a branch circuit (and the 80% rule)

- Scope: ✅ IN.
- Where seen: not seen with mains numbers. The DC version is Princeton Invitational Mar 2021, Div C, INV, Q37 (0.75 A fuse, 10 V, 100 Ω bulbs in parallel → at most **7** bulbs). Prior bank I_bank W4.
- How to solve:
  1. P_max = V × I_breaker.
  2. Continuous load (3 h or more): use 80%.
- Worked example: "How many watts can a 15 A, 120 V circuit supply? How much for a continuous load?" 15 × 120 = **1800 W**. 0.8 × 1800 = **1440 W**.
- Traps: The 80% rule is for continuous loads only. For a quick question with no "continuous", use the full value.
- Calculator: calculate — type: I=15, V=120 → Pmax, 0.8 Pmax.

12. Will the breaker trip? (adding appliance currents)

- Scope: ✅ IN.
- Where seen: not seen with mains numbers. Same idea as Princeton Mar 2021 C Q37 (above).
- How to solve:
  1. Outlets on one circuit are in parallel. Add the powers (or currents).
  2. I_total = P_total/V.
  3. Compare with the breaker rating.

- Worked example: "A 20 A, 120 V kitchen circuit runs a 900 W toaster, a 1100 W microwave and an 800 W coffee maker at the same time. Does the breaker trip?" Total 2800 W. I = 2800/120 = **23.3 A > 20 A. Yes, it trips** (after a short delay).
- Traps: Using series rules. House loads are in **parallel**. Each gets 120 V, and the currents add.
- Calculator: calculate — type: powers 900,1100,800; V=120; breaker=20 → I, trip?

13. Match the wire gauge to the breaker

- Scope: ✅ IN (as a hazard: an overloaded wire is a fire risk). ⚠️ The exact NEC table is borderline trivia.
- Where seen: not seen in a test. Prior bank I_bank W4 and D-section (NEC 240.4(D)).
- How to solve: 14 AWG → 15 A. 12 AWG → 20 A. 10 AWG → 30 A.
- Worked example: "Someone replaced a tripping 15 A breaker on 14 AWG wire with a 20 A breaker. Why is this dangerous?" **Answer: 14 AWG wire is only safe to 15 A. Up to 20 A can now flow without a trip, so the wire can overheat inside the wall and start a fire.**
- Traps: The breaker protects the **wire**, so it is chosen to match the wire, not the load.
- Calculator: look up — type: AWG to breaker (gauge).

14. Voltage drop in an extension cord

- Scope: ✅ IN.
- Where seen: not seen in a test with AWG tables. The SI version is Princeton Feb 2026 C Comp 5 (type 19). Prior bank I_bank U19 (50 ft of 14 AWG at 15 A → 3.8 V).
- How to solve:
  1. Round-trip length = 2 × cord length (current goes out on hot and back on neutral).
  2. R = (Ω per 1000 ft) × round-trip feet ÷ 1000.
  3. V_drop = I × R. Percent = V_drop/120 × 100%.
  4. Power wasted in the cord = I^2R.
- Worked example: "A 50 ft, 16 AWG cord (4.016 Ω per 1000 ft) feeds a 12 A saw at 120 V. Drop and power lost?" R = 4.016 × 100/1000 = 0.402 Ω. V_drop = 12 × 0.402 = **4.82 V (4.0%)**. P = 12^2 × 0.402 = **57.8 W** heating the cord.
- Traps: Using 50 ft instead of 100 ft. A thin, long cord gets warm; a coiled cord under heavy load can overheat.
- Calculator: calculate — type: length_ft=50, ohm_per_kft=4.016, I=12, V=120 → R, Vdrop, %, P.

15. Read a NEMA receptacle

- Scope: ⚠️ Borderline (rating facts and safety fit 3.c.iii and 3.c.vi; the shapes are trivia).
- Where seen: MIT Invitational Jan 2021, Div C, INV (top tier), Q92-97, Scenario M (two receptacle drawings: A → **20 A, 125 V, neutral = pin 1**; B → **30 A, 347 V**, which pin is hot).
- How to solve:
  1. Look for a T-shaped slot → 20 A (5-20).
  2. Two horizontal/angled hot blades and no neutral → 250 V type (6-series).
  3. Four openings (two hots, an L-shaped neutral, ground) → 14-series, 125/250 V.
  4. The second number in the NEMA name is the amp rating.
  5. Remember: a receptacle drawing is the mirror image of the plug.
- Worked example: "A receptacle is labeled NEMA 14-50. What are its current and voltage ratings, and what is it for?" **Answer: 50 A, 125/250 V (two hots, neutral, ground); ranges, RVs and EV chargers.**
- Traps: 347 V is a Canadian commercial voltage (NEMA 24-series). Do not expect it in a house. The voltage rating is 125 V or 250 V, not 120 V or 240 V.
- Calculator: look up — type: NEMA code (for example 5-20, 14-50).

16. Outlet meter readings and wiring faults

- Scope: ✅ IN (a written question; no AC hands-on).
- Where seen: not seen in a test. Kuphaldt, *Lessons in Electric Circuits* vol. 1 ch. 3 (electrical safety, meter use).
- How to solve: compare the three readings with the normal ones: hot-neutral 120 V, hot-ground 120 V, neutral-ground about 0 V.

- Worked example: "A meter reads: narrow slot to ground 0 V, wide slot to ground 120 V, narrow to wide 120 V. What is wrong?" **Answer: reversed polarity. Hot and neutral are swapped.**
- Traps: Reversed polarity still runs the appliance. The fault is hidden, so it needs a tester.
- Calculator: look up — type: outlet readings (three values).

17. What happens to the case in a ground fault? (with and without ground)

- Scope: ✅ IN (series circuit with Ohm's law).
- Where seen: not seen in a test with numbers. UT Austin (ATX) Regional 2019 B/C Q21 is the concept (type 4).
- How to solve:
  1. With ground: fault current I = 120/(R_hot + R_ground). The case voltage = I × R_ground.
  2. Without ground: the case is at 120 V. A person becomes the path: I = 120/(R_hot + R_person).
- Worked example: "Hot wire 0.3 Ω, ground wire 0.5 Ω. The hot touches the case. Fault current and case voltage?" I = 120/0.8 = **150 A**, so the 20 A breaker trips in a fraction of a second. Case voltage during the fault = 150 × 0.5 = **75 V**. "Same fault, no ground wire, a wet person (1000 Ω) touches the case and a pipe." I = 120/1000.3 = **0.120 A = 120 mA** (deadly; see Shock and Safety A3).
- Traps: The case is not exactly 0 V during a fault. That is why the breaker must trip fast, and why GFCIs are added in wet places.
- Calculator: calculate — type: V=120, Rhot=0.3, Rground=0.5 → Ifault, Vcase; Rperson=1000 → Iperson.

18. Neutral current with two loads (KCL idea)

- Scope: ✅ IN as a simple sum or difference. The full two-loop version is type 25 (States/Nationals).
- Where seen: not seen in a test.
- How to solve:
  1. Loads on **opposite** legs (one on L1, one on L2): their neutral currents flow in opposite directions. I_N = difference.
  2. Loads on the **same** leg: I_N = sum.
- Worked example: "A 15 A load is on leg L1 and a 10 A load is on leg L2. Neutral current?" **5 A.** "Both on L1?" **25 A.**
- Traps: Adding when the legs are opposite. This is why balancing loads across the two legs keeps the neutral current low.
- Calculator: calculate — type: I1=15, I2=10, same leg? → In.

Hard

19. Size a wire from a voltage-drop budget

- Scope: ✅ IN (R = ρL/A and Ohm's law).
- Where seen: Princeton Invitational Feb 2026, Div C, INV, Comp 5 (2 m copper cable, 12 V supply, the circuit needs at least 10 V at 2.2 A → max R = **0.91 Ω**, min area = **3.69×10^{-8} m²**, power lost = **4.4 W**).
- How to solve:
  1. Allowed drop = supply − needed voltage.
  2. R_max = drop ÷ I.
  3. A_min = ρ L ÷ R_max (L = total conductor length; read the question to see if it means one wire or the round trip).
  4. Power lost = drop × I.
- Worked example: Drop = 12 − 10 = 2 V. R_max = 2/2.2 = 0.909 Ω. A_min = $1.68 \times 10^{-8} \times 2 \div 0.909$ = **3.70×10^{-8} m²** (0.037 mm²; Princeton rounds to 3.69). P = 2 × 2.2 = **4.4 W**.
- Traps: Princeton used L = 2 m. A real cable has two conductors, so the round trip would be 4 m and the area must double. State which you used.
- Calculator: calculate — type: Vs=12, Vmin=10, I=2.2, L=2, rho=1.68e-8 → Rmax, Amin, Ploss.

20. AWG from a diameter, or resistance from a gauge

- Scope: ✅ IN (resistance from size, rule 3.c.iv).
- Where seen: not seen in a test. Prior bank I_bank U19.

- How to solve:
  1. d = 0.127 × 92^((36 – n)/39) mm.
  2. A = π d²/4.
  3. R = ρ L/A.
- Worked example: "Find the resistance of 1000 ft (304.8 m) of 14 AWG copper (ρ = 1.724 × 10^{-8} Ω·m)." d = 1.628 mm. A = 2.081 mm² = 2.081 × 10^{-6} m². R = 1.724 × 10^{-8} × 304.8 ÷ 2.081 × 10^{-6} = **2.53 Ω**.
- Traps: mm² → m² is × 10^{-6}. Use the radius (d/2), not the diameter, if you use πr².
- Calculator: calculate — type: n=14, L=304.8, rho=1.724e-8 → d, A, R.

21. Open neutral on a split-phase service

- Scope: ✅ IN (two loads in series across 240 V; Ohm's law).
- Where seen: not seen in a test. Kuphaldt, *Lessons in Electric Circuits* vol. 2 ch. 10 (split-phase systems), from memory.
- How to solve:
  1. With the neutral broken, the two 120 V loads on opposite legs end up **in series across 240 V**.
  2. Find each load's resistance from its rating: R = 120²/P.
  3. I = 240/(R1 + R2). V_each = I × R.
- Worked example: "A 60 W bulb is on L1 and a 1200 W heater is on L2. The shared neutral breaks." R_bulb = 240 Ω, R_heater = 12 Ω. I = 240/252 = 0.952 A. Bulb voltage = **228.6 V** (it flashes very bright and burns out). Heater voltage = **11.4 V** (it barely warms).
- Traps: The **small** load gets almost all of the 240 V. People report "lights getting very bright" when the neutral fails. It is a fire and equipment hazard.
- Calculator: calculate — type: P1=60, P2=1200, V=120 → R1, R2, I, V1, V2.

22. Wire a 3-way light including hot, neutral and ground

- Scope: ✅ IN (rule 3.c.vi and hands-on II.c.vii; the DC hands-on version uses SPDT switches and an LED).
- Where seen: UT Austin Invitational 2019, Div B, INV, Q44 (wire a 3-way light with hot, neutral, ground and terminals A/B/C/G). Northview Sylvania Dec 2025 C lab (build a 3-way circuit). Full method in 3-Way and 4-Way Switching A6.
- How to solve:
  1. Hot from the panel → the common terminal of switch 1.
  2. Two travelers join the traveler terminals of switch 1 and switch 2.
  3. The common of switch 2 → the lamp's center tab. The lamp's shell → neutral.
  4. Every ground wire joins at each box and at every switch's green screw.
- Worked example: "Where does the neutral go in a 3-way light circuit?" **Answer: straight to the lamp (shell). It never goes through a switch.**
- Traps: The switches must be in the hot. Ground wires are not travelers.
- Calculator: look up — type: 3-way wiring steps.

23. Choose the safest protection for a situation

- Scope: ✅ IN (rules 3.c.iii and 3.c.vi).
- Where seen: BullSO Jan 2026, Div B, INV, Q33 (least practical shock prevention → polarized plugs). Princeton Invitational Feb 2026, Div C, INV, SA3d (a hair dryer runs normally but trips a GFCI near a sink → leakage current to ground). Pembroke Hill Dec 2025, Div B, INV, Q16 (protects people in wet areas by detecting live/neutral imbalance → GFCI).
- How to solve: match the danger to the device:
  - Wire overload or short → fuse or breaker.
  - Shock near water (current leaking to ground) → GFCI.
  - Arcing in a damaged wire (fire) → AFCI.
  - Live metal case → ground wire (and breaker).
  - No exposed metal → double insulation.

- Worked example: "Which device protects a person who drops a plugged-in radio into a bathtub?" **Answer: a GFCI.** A breaker would not trip, because the leak current (tens of mA) is far below 15 A.
- Traps: Breakers do not protect people from shock. They protect wires.
- Calculator: look up — type: hazard → protection device.

24. Power wasted in the house wiring for a load

- Scope: ✅ IN.
- Where seen: not seen in a test with house numbers. MIT Jan 2019 C Q36-37 is the power-line version (AC Characteristics A3 type 17).
- How to solve:
  1. Load current I = P/V.
  2. Wire resistance R = (Ω per 1000 ft) × round trip ÷ 1000.
  3. Loss = I^2R. Voltage at the load ≈ 120 − I·R.
- Worked example: "A 1500 W heater is 50 ft (100 ft round trip) from the panel on 14 AWG wire (2.525 Ω per 1000 ft)." I = 12.5 A. R = 0.2525 Ω. Drop = 12.5 × 0.2525 = **3.16 V (2.6%)**. Loss = $12.5^2 \times 0.2525$ = **39.5 W**.
- Traps: The NEC suggests (in an informational note, not a rule) keeping branch-circuit drop under about 3%.
- Calculator: calculate — type: P=1500, V=120, ohm_per_kft=2.525, roundtrip_ft=100 → I, Vdrop, %, Ploss.

Nationals

25. Split-phase loads with a resistive neutral (two loop equations)

- Scope: ✅ States/Nationals only (KVL/KCL, rule 3.c.ix). Two equations, so it fits the math limit (rule 3.e.i).
- Where seen: not seen in a test.
- How to solve:
  1. Draw two loops: leg L1 (120 V) through load 1 and the neutral; leg L2 (120 V) through load 2 and the neutral.
  2. The neutral carries I_1 − I_2 (the two loop currents flow in opposite directions in it).
  3. Write KVL for each loop.
  4. Solve the two equations (substitution or Cramer's rule).
- Worked example: "Each leg is 120 V. Load 1 = 10 Ω on L1. Load 2 = 20 Ω on L2. The shared neutral has 0.5 Ω. Hot wires have no resistance. Find each load's voltage and the neutral current."
  - Loop 1: $120 = 10\ I_1 + 0.5\ (I_1 - I_2) \rightarrow 10.5\ I_1 - 0.5\ I_2 = 120$.
  - Loop 2: $120 = 20\ I_2 + 0.5\ (I_2 - I_1) \rightarrow -0.5\ I_1 + 20.5\ I_2 = 120$.
  - Determinant = 10.5 × 20.5 − 0.25 = 215. I_1 = (120 × 20.5 + 0.5 × 120)/215 = **11.72 A**. I_2 = (10.5 × 120 + 0.5 × 120)/215 = **6.14 A**.
  - Neutral current = **5.58 A**. Load 1 gets **117.2 V**. Load 2 gets **122.8 V**.
- Traps: The lighter load gets **more** than 120 V. A loose (high-resistance) neutral makes this worse. Type 21 is the limit when the neutral resistance becomes infinite.
- Calculator: calculate — type: V=120, R1=10, R2=20, Rn=0.5 → I1, I2, In, V1, V2 (2×2 solve).

Out of scope (skip)

- 3-phase power: 208 V and 480 V systems, wye and delta, line vs phase voltage. ❌ OUT: "3 phase power" (3.d).
- How the center-tapped pole transformer works (turns ratios). ❌ OUT: inductance (3.d) and magnetism.
- Power factor of motors, reactive power, kVA vs kW. ❌ OUT: AC circuit theory (3.d).
- How an AFCI senses arc "signatures" (frequency content). ❌ OUT: frequency analysis (3.d). The fact that an AFCI exists and stops arc fires is ⚠️ borderline (it is not named in rule 3.c.vi, but it is a breaker).
- Exact NEC article numbers, box-fill and conduit-fill tables, ampacity correction factors. ⚠️ Borderline trivia; not seen in any test.
- Hands-on mains wiring. ❌ Not allowed (rule II.d: no AC circuits).

Mistakes in tests and keys

- **USC Jan 2026 B Q6:** key I only. That is right. Statement IV ("each hot slot has 60 V") is a deliberate distractor. Each hot is 120 V to neutral.
- **UT Austin 2019 B Q12 and C Q12:** the keyed choice says the hot slot is "shorter" and "swings (+)". "Shorter" (narrow) is right. AC swings both + and −, so the "+" wording is odd. Pick the narrow/short slot.
- **BullSO Jan 2026 B Q33:** keyed C (polarized plugs) is fine. But D ("an insulator in extra space in the case") is vague. Double insulation is a real, strong protection.
- No arithmetic key errors were found in the wiring questions.

Not confirmed

- NEC article numbers (210.8, 210.12, 210.19-20, 240.4(D), 406.12) and the GFCI/AFCI location lists are from memory of the 2020-2023 NEC. Requirements change every 3 years and by state. I did not open the NEC in this run.
- The NEMA table (1-15, 5-15, 5-20, 6-15/20, 10-30, 14-30, 14-50) and the 347 V Canadian 24-series are from memory of NEMA WD-6. I did not render the MIT 2021 receptacle drawings, so I could not confirm the pin numbers beyond the key's "pin 1 = neutral" for A.
- The AWG diameters and resistances were computed in Python from the AWG formula with $\rho = 1.724 \times 10^{-8}$ Ω·m (annealed copper). Published tables differ in the last digit.
- The 8 ft ground rod, the 100-200 A panel sizes and the NM-cable color code are typical US practice, from memory.
- The "3% voltage drop" figure is an NEC informational note, not a requirement (from memory).
- I found no household-wiring question in the Florida State 2021 test except the toddler-in-the-hot-slot item (it is in Shock and Safety A3). I looked in all 8 extraction files and `prior/G_tests.md`.

Sources

- Rules: 2027 Div B Circuit Lab (manual pages B12-B13): areas 3.c.iii, 3.c.iv, 3.c.v, 3.c.vi, 3.c.ix; exclusions 3.d; math 3.e.i; hands-on II.c.vii and II.d.
- Tests (extractions in `circuitlab/research/tests/` and `prior/G_tests.md`): USC Invitational Jan 2026 B; Berkeley Feb 2026 B; BullSO Jan 2026 B and C; Pembroke Hill Dec 2025 B; Princeton Feb 2026 C and Mar 2021 C; Hawk & Hornet 2026 C; Northview Sylvania Dec 2025 C; MIT Jan 2021 C; UT Austin Invitational 2019 B/C; UT Austin (ATX) Regional 2019 B/C.
- Prior research: `prior/B_topics.md` §3c and §6; `prior/I_bank.md` W1-W6, U19, D-section (breakers, GFCI, AFCI); `prior/A_rules.md` (2026 vs 2027 wording).
- References (general knowledge, not re-read in this run): NFPA 70 *National Electrical Code* (2020/2023); NEMA WD-6 (receptacle configurations); Kuphaldt, *Lessons in Electric Circuits* vol. 1 ch. 3 (electrical safety) and vol. 2 ch. 10 (polyphase / split-phase); OpenStax *College Physics 2e* ch. 20.6 (Electric hazards and the human body: grounding, three-wire systems).

Shock and Safety

An **electric shock** is current flowing through the body. This note covers how much current hurts or kills, body resistance, AC versus DC danger, the path and time of a shock, power-line safety and first aid.

Scope:

- Rule area **3.c.iii** names the "hazards of direct current (DC) and alternating current (AC)". So all shock facts and shock math are ✅ **IN at every level**.
- Static shocks and lightning are rule area **3.c.ii** ("sources/hazards of static electricity"). They are in the static notes; this note only links to them.
- GFCIs, fuses and breakers are rule area **3.c.vi**. This note shows how they protect people; their full question types are in CL 6 - Switches, Relays, Fuses, Breakers, GFCI 🅃.
- A person in a network that needs two loop equations is ✅ **States/Nationals only** (rule 3.c.ix).
- There is no shock hands-on. Rule II.d says "Participants must not be exposed to hazardous conditions including soldering, high power circuits, and AC circuits."

Key facts

Words

- **Shock**: current through the body.
- **Electrocution**: death from an electric shock.
- **Let-go current**: the largest current at which a person can still open their hand and let go of the wire. Above it, muscles clamp (**tetanus** or **tetany**: a muscle held tight by the current).
- **Ventricular fibrillation**: the lower chambers of the heart quiver out of rhythm instead of pumping. Blood stops moving. Death follows in minutes unless a **defibrillator** resets the heart.
- **Respiratory arrest**: breathing stops because the chest muscles are clamped or the brain's breathing center is hit.
- **Arc flash**: a blast of heat and light from an electric arc through air. It causes burns at a distance.
- **Step voltage** (step potential): the voltage between your two feet near a fallen power line, caused by current spreading through the ground.
- **Touch voltage**: the voltage between your hand and your feet when you touch something live.

"Current kills, voltage drives it"

- The current (I) does the damage. The voltage (V) pushes the current through the body's resistance (R): **I = V/R**.
- High voltage is dangerous because it can push a large current through the skin. Warning signs say "high voltage" because voltage is what you can measure and see on equipment (Rickards Oct 2025 C Q73).
- A **Van de Graaff** generator or a static spark can be 10 000 V or more, but it is safe for most people. It can push only a tiny current for a very short time (very little charge and energy).

Effects of 60 Hz AC on the body (about 1 second, hand to hand or hand to foot)

Current	Effect (OpenStax *College Physics 2e* Table 20.3, from memory)
1 mA	Threshold of feeling (a tingle)
5 mA	Maximum "harmless" current. GFCIs trip at 4-6 mA.
10-20 mA	Muscles clamp. "Cannot let go." Chest muscles may stop breathing during the shock.
50 mA	Severe pain
100-300 mA	Ventricular fibrillation possible; often fatal
300 mA and up	Burns, depending on how concentrated the current is
6 A (6000 mA)	The whole heart muscle clamps and breathing stops. When the current stops, the heart may restart normally. This is how defibrillators work.

Other tables give slightly different numbers. Tests have used these values:

- Let-go ("paralysis") range **10-16 mA** (ATX Regional 2021 B Q37).

- Lethal range **100-200 mA** (ATX Regional 2020 B Q40).
- Women and children have lower thresholds than men (about two-thirds).

Dalziel's AC and DC thresholds (Kuphaldt, *Lessons in Electric Circuits* vol. 1 ch. 3, from memory)

Effect	DC, men	DC, women	60 Hz AC, men	60 Hz AC, women
Slight sensation	1 mA	0.6 mA	0.4 mA	0.3 mA
Threshold of perception	5.2 mA	3.5 mA	1.1 mA	0.7 mA
Painful, but can still let go	62 mA	41 mA	9 mA	6 mA
Painful, **cannot let go**	76 mA	51 mA	16 mA	10.5 mA
Severe pain, hard to breathe	90 mA	60 mA	23 mA	15 mA
Possible fibrillation (3 s)	500 mA	500 mA	100 mA	100 mA

- Look at the ratio: AC causes each effect at about **one-fourth to one-fifth** of the DC current. So 60 Hz AC is about 4-5 times as dangerous as DC at the same current.
- F-M Invitational 2018 Q4 used "22 mA AC or 88 mA DC" → "cannot let go". These match the "severe pain" row closely.

AC versus DC

- At the same voltage, 50-60 Hz AC is **more dangerous** than DC. This is the standard test answer (Berkeley Feb 2026 B Q15; USC Jan 2026 B Q28; UT Austin 2020 B Q10; Cobra 2019-20 B Q14).
- Reason 1: 50-60 Hz is close to the frequency that best upsets the heart's rhythm. AC causes **ventricular fibrillation** at lower currents.
- Reason 2: AC makes muscles clamp at lower currents, so the person **cannot let go** and the shock lasts longer.
- DC tends to cause **one strong jerk** that can throw the person off. It can also cause the heart to stop (cardiac arrest) at high current. DC also causes electrolytic (chemical) burns.
- Sources disagree on "let-go". Kuphaldt's text says DC is "more likely to cause muscle tetanus" (freezing), and AC is more likely to cause fibrillation. Dalziel's numbers above say AC freezes muscles at a lower current. Tests have keyed both:
  - "AC: cannot let go" (USC Jan 2026 B Q28; Berkeley Feb 2026 B Q15 rejects "AC can let go more easily").
  - "High-voltage DC → muscle tetanus; high-voltage AC → fibrillation" (UF Northern Regional 2019 C Q17 and Q20; Princeton Feb 2026 C SA5b).
  - Read all the choices. Pick the one that matches the question's pairing.
- Most of all: "AC is more dangerous than DC" → true. "DC is more dangerous than AC" → false (UF Northern Regional 2019 B Q22).

Body resistance

- Most of the resistance is in the **skin**.
- Dry skin: about **100 kΩ** (can be higher). This is the "maximum dry-skin resistance" in ATX Regional 2020 B Q41.
- Wet or sweaty skin: about **1 kΩ**. Broken skin (a cut) is lower still.
- Inside the body (blood, muscle, salty fluids): only about **300-500 Ω** from hand to foot.
- So water lowers the resistance about **100 times** and makes the current 100 times larger.
- Body resistance also falls as the voltage rises. High voltage can puncture the skin, then only the low internal resistance is left.

What decides how bad a shock is

- The **current** (set by the voltage and the resistance).
- The **path**. A path through the **chest** (hand to hand, or hand to foot) crosses the heart. A path through one finger to another on the same hand mostly does not.
- The **time** (duration). Longer shocks are worse. A GFCI cuts power in about 25 ms (1/40 s), before most hearts fibrillate.
- The **frequency**. 50-60 Hz is the worst. Very high frequencies (above about 10 kHz) tend to stay near the skin and cause burns instead of heart effects.
- The person's health, size and the contact area.

- Hawk & Hornet 2026 Q5 asked "voltage, current, frequency, contact duration, or all of the above?" → **all of the above**.

Effects of current in the body (all real)

- Tingling and pain.
- Muscle clamping (tetany), so the person cannot let go.
- Breathing stops (respiratory arrest).
- Ventricular fibrillation or cardiac arrest.
- **Burns**, both on the skin and **inside** (tissues heat up: $P = I^2R$). Internal burns can be much worse than they look.
- Nerve damage. Current interferes with nerve signals and causes clenching.
- Falls. Many deaths come from a jerk that makes a person fall off a ladder.

No complete path, no shock

- Current needs a **closed loop**. A bird on one power line does not get shocked. Both feet are on the same wire, so there is (almost) **no voltage difference** between them, and there is no path to ground (GullSO Dec 2025 Q14; Northview Sylvania Dec 2025 C Q29; UF Northern Regional 2020 B Q39; "Shock Value" re-tryout about 2012 VI-2).
- A person touching only the hot wire is safe **only if** nothing else they touch is grounded or connected to another wire. Standing on a wet floor, a sink or a pipe completes the path. Never test this.
- A bird that touches two wires (or a wire and a grounded pole) at once is killed.

Battery and low-voltage shocks

- A 12 V car battery through dry skin: 12/100 000 = 0.12 mA. You cannot feel it. "Can you get a dangerous shock from touching both car-battery terminals with dry fingers?" → **No** (Cobra 2019-20 B Q14).
- The car battery is still dangerous: a wrench or ring across its terminals carries hundreds of amps, gets red hot and causes burns. Charging makes hydrogen gas that a spark can explode. See Batteries and Cells A3.
- "Can 12 V kill?" ATX Regional 2020 B Q54 keyed **True**. It is possible only with a very low-resistance path (wet, broken skin, or a probe inside the body). Normal contact: no.
- Voltages under about 50 V AC or 120 V smooth DC are called **extra-low voltage** (ELV) and are thought safe to touch in dry places. This is why toys, doorbells and USB use low voltages.

Power lines and outdoor safety

- Stay at least **10 ft (3 m)** from overhead lines; utilities say 35 ft (about 10 m) from a downed line.
- Treat every downed line as live.
- If a line falls on your car: **stay inside**. Call 911. The car body is all at one voltage, so you are safe inside.
- If you must leave (fire): **jump** clear with **feet together**, so you never touch the car and the ground at the same time. Then **shuffle** or hop away with your feet together (UT Austin Invitational 2019 C Q25).
- Why feet together: the ground near the line has a voltage that falls off with distance. A long step puts your feet at very different voltages (step voltage).
- Do not fly kites or drones near lines. Look up before raising a ladder.
- Lightning: go indoors or into a hard-topped car. Avoid tall objects and water. See the static notes.

Working safely

- Turn off the breaker and **test** the wire is dead before you touch it. Lock the panel if others are near (lockout/tagout).
- Use the **one-hand rule** with high voltage: keep one hand in your pocket, so current cannot go hand to hand across your chest.
- Keep hands, floors and tools **dry**. Stand on a rubber mat.
- Use insulated tools, and do not wear rings, watches or metal jewelry.
- Never use a damaged cord. Never remove a ground pin. Use GFCIs near water.
- Never put metal objects in outlets. Use tamper-resistant outlets with small children.
- Unplug appliances by the plug, not the cord.
- Ceramic, glass, rubber and plastic are **insulators**. That is why power-line holders are ceramic or glass ("Shock Value" re-tryout about 2012 VI-3).

First aid for electric shock

1. **Do not touch** the person while they may still be in contact with the source. You would become part of the circuit.
2. **Turn off the power** at the breaker, switch or plug.
3. If you cannot turn it off and it is **low voltage** (household), push the source away with a **dry non-conducting** object (wood, plastic). Never do this with high-voltage lines; wait for the utility.
4. **Call 911** (or the local emergency number).
5. Once the person is free: check breathing. If not breathing, start **CPR** and use an **AED** (automated external defibrillator) if one is near.
6. Cover burns with a clean dry cloth. Do not use ice or butter. Keep the person warm and lying down.
7. Everyone who had a real shock should **see a doctor**. Internal burns and heart rhythm problems can show up hours later.

Formulas

What	Formula	Notes
Body current	$I = V \div R_{body}$	Use RMS volts for AC
Body resistance in a path	$R = R_{skin,in} + R_{internal} + R_{skin,out}$	Series
Voltage needed for a current	$V = I \times R_{body}$	
Ratio wet to dry	$I_{wet}/I_{dry} = R_{dry}/R_{wet}$	About 100
Energy delivered	$E = V \cdot I \cdot t = I^2 R \cdot t = V^2 t / R$	Defibrillators, shock energy
Charge delivered	$Q = I \times t$	
Step voltage	$V_{step} \approx$ (ground voltage gradient, V/m) × (step length, m)	Simplified
Current split: person in parallel with a ground wire	$I_{person} = V_{case} \div R_{person}$	Same voltage across parallel paths
Current divider (two paths)	$I_1 = I_{total} \times R_2/(R_1 + R_2)$	

Question types

Easy

1. Does current or voltage kill?

- Scope: ✅ IN.
- Where seen: "Shock Value Test" re-tryout, about 2012, Div B, practice, VI-1 (voltage or current kills → current). Rickards Invitational Oct 2025, Div C, INV, Q73 ("current kills, not voltage", so why do signs say "high voltage"?). Official practice decks (F_slides ACDC-HAZARDS).
- How to solve:
  1. Current does the damage.
  2. Voltage decides how much current can flow through the body's resistance ($I = V/R$).
  3. Signs say "high voltage" because high voltage can push a deadly current through skin, and voltage is the property of the equipment you can know in advance. The current depends on the person.
- Worked example: "Why is a 10 000 V static spark harmless but 120 V from an outlet can kill?" **Answer: the spark carries a tiny charge for a very short time, so the current and energy are small. The outlet can keep pushing a large current (over 100 mA through wet skin) for as long as you touch it.**
- Traps: "Voltage kills" alone is not the best answer. "Current kills" alone also misses why voltage matters. Give both parts in a free-response.
- Calculator: look up — type: current kills.

2. Effect of a given current (threshold table)

- Scope: ✅ IN.
- Where seen: ATX Regional 2021, Div B, REG, Q37 (AC current that causes "paralysis", cannot let go → **10-16 mA**). ATX Regional 2020, Div B, REG, Q40 (lethal range → **100-200 mA**). F-M Invitational 2018, Div not stated, INV, Q4 (22 mA AC or 88 mA DC → **cannot let go**). MIT Invitational Jan 2019, Div C, INV, Q41 (heart condition

from a shock → fibrillation). UF Northern Regional 2019, Div C, REG, Q20 (high-voltage AC shock, effect on the heart → fibrillation).

- How to solve: match the current to the table: 1 mA feel; 5 mA harmless maximum; 10-20 mA cannot let go; 100-300 mA fibrillation; 6 A whole-heart clamp.
- Worked example: "About 15 mA of 60 Hz AC flows from hand to hand. What is the most likely effect?" **Answer: muscles clamp and the person cannot let go.**
- Traps: Numbers vary by table. Choose the closest range. DC thresholds are about 4-5 times higher than AC.
- Calculator: look up — type: current mA → effect (AC or DC).

3. Body resistance: dry and wet

- Scope: ✅ IN.
- Where seen: ATX Regional 2020, Div B, REG, Q41 (maximum dry-skin resistance → **100 kΩ**).
- How to solve: dry skin about 100 kΩ; wet skin about 1 kΩ; inside the body about 300-500 Ω.
- Worked example: "Roughly how many times lower is wet-skin resistance than dry-skin resistance?" 100 000/1000 = **100 times**.
- Traps: Body resistance is not one fixed number. It depends on moisture, contact area and voltage.
- Calculator: look up — type: body resistance.

4. Why is AC more dangerous than DC?

- Scope: ✅ IN.
- Where seen: Science Olympiad @ Berkeley Feb 2026, Div B, INV, Q15 (why is AC more dangerous than DC at the same RMS voltage → **the frequency is effective at causing ventricular fibrillation**). USC Invitational Jan 2026, Div B, INV, Q28 (what makes 120 V AC more dangerous than 120 V DC → **muscle contractions that prevent letting go**). UT Austin Invitational 2020, Div B, INV, Q10-11 (which is more dangerous → AC; give two reasons; the key has no model answer). Cobra Invitational 2019-20, Div B, INV, Q14 (AC or DC worse at the same voltage → AC). UF Northern Regional 2019, Div B, REG, Q22 ("DC more dangerous than AC" → False). Princeton Invitational Feb 2026, Div C, INV, SA5b (main hazard of DC and of AC).
- How to solve: give two reasons: (1) 50-60 Hz easily triggers ventricular fibrillation; (2) AC makes muscles clamp, so the victim cannot let go and the shock lasts longer.
- Worked example: "Give two reasons 120 V AC is more dangerous than 120 V DC." **Answer: AC at 60 Hz triggers ventricular fibrillation at lower currents, and it causes sustained muscle contraction so the person cannot let go.**
- Traps: "AC causes greater heating" is false at the same RMS value (RMS means equal heating). "AC delivers more average power" is false for the same reason. "DC does not pass through skin" is false.
- Calculator: look up — type: AC vs DC danger.

5. Heart effect: fibrillation or tetanus?

- Scope: ✅ IN.
- Where seen: UF Northern Regional 2019, Div C, REG, Q17 (high-voltage DC shock, effect → **muscle tetanus**) and Q20 (high-voltage AC shock → **fibrillation**). MIT Invitational Jan 2019, Div C, INV, Q41 (→ fibrillation). Princeton Feb 2026 C SA5b (key per G_tests: DC → cannot let go; AC → fibrillation).
- How to solve: when a question pairs them, AC → fibrillation; DC → tetanus / one big contraction / cardiac arrest.
- Worked example: "Which condition, where the heart quivers instead of pumping, is the main danger of a 60 Hz shock?" **Answer: ventricular fibrillation.**
- Traps: See Key facts. Some keys say AC causes "cannot let go". If both "fibrillation" and "cannot let go" appear for AC, both are true; pick the one the question's words point to.
- Calculator: look up — type: fibrillation vs tetanus.

6. Bird on a wire (no path, no voltage difference)

- Scope: ✅ IN.
- Where seen: GullSO Invitational Dec 2025, Div C (B/C), INV, Q14 (why birds are not electrocuted → **no voltage difference between their feet**). Northview Sylvania Invitational Dec 2025, Div C, INV, Q29 (a barefoot clown walks on a line → **no return path**). UF Northern Regional 2020, Div B, REG, Q39 (walking on a power line: why alive → no path to ground). "Shock Value Test" re-tryout, about 2012, Div B, practice, VI-2 (same).

- How to solve:
  1. Current needs a closed loop and a voltage difference.
  2. Both feet on one wire → nearly the same voltage → nearly zero current.
- Worked example: "A wire carries 100 A and has 0.0001 Ω per cm. A bird's feet are 4 cm apart. What voltage is across the bird?" V = 100 × 0.0004 = **0.04 V**. Too small to matter.
- Traps: "Bird feet are good insulators" is wrong. "Birds can handle more current" is wrong. A line worker on a pole touches the line and the grounded pole, so there is a path.
- Calculator: calculate — type: I=100, R per cm=1e-4, d=4 → V.

7. Touching only the hot slot (Florida State)

- Scope: ✅ IN.
- Where seen: Florida State Tournament 2021, Div B (= C), STATE, Q13 (a toddler puts two fingers in only the hot slot → key **D, none of the above**; no path through the body unless the child is grounded; the key is debatable).
- How to solve:
  1. Is there a second contact at a different voltage (neutral, ground, a wet floor, a pipe)?
  2. If yes → a current flows through the body.
  3. If the child is fully insulated from ground, the two fingers are at the same voltage, so very little current flows.
- Worked example: "A child touches only the hot slot with two fingers and stands on a dry wooden floor. Is there a dangerous current?" **Answer: probably not. Both fingers are at the same voltage and the floor insulates. On a wet tile floor, current flows to ground and it is dangerous.**
- Traps: Real floors and bodies are not perfect insulators. "Safe" is never the advice. Tamper-resistant outlets prevent this.
- Calculator: look up — type: single-slot contact.

8. Factors that decide how dangerous a shock is

- Scope: ✅ IN.
- Where seen: Hawk & Hornet Invitational 2026, Div C, INV, Q5 (voltage, current, frequency, contact duration → **all of the above**).
- How to solve: current, voltage (it drives current), body resistance (wet or dry), path (through the heart?), duration, frequency, the person's health.
- Worked example: "Name three things that make a shock more dangerous." **Answer: a higher current (from higher voltage or wet skin), a path through the chest, and a longer time.**
- Traps: Frequency matters. 50-60 Hz is the worst range.
- Calculator: look up — type: shock factors.

9. Which effect is NOT possible?

- Scope: ✅ IN.
- Where seen: BullSO Invitational Jan 2026, Div B, INV, Q34 (which is not a possible effect of current through the body: A "weakening of muscles in the brain"; B "internal tissue burning due to the electrical resistance of the body"; C "fibrillation of the heart due to shock to the diaphragm muscle"; D "nerve interference causing clenching" → **key B**, which looks wrong; see Mistakes). BullSO Jan 2026 Div C Q34 (same, per G_tests).
- How to solve: real effects are burns (outside and inside), clenching, breathing stops, fibrillation, nerve damage. Look for a choice that is not biology (the brain has no muscles) or has the wrong cause (fibrillation is in the heart, not from the diaphragm).
- Worked example: "Which is not a real effect of a shock: (a) internal burns, (b) the heart quivering, (c) muscles clenching, (d) the bones becoming magnetic?" **Answer: (d).**
- Traps: Internal burns are **real**. Current heats tissues ($P = I^2R$).
- Calculator: look up — type: shock effects list.

10. Can a 12 V car battery shock you?

- Scope: ✅ IN.
- Where seen: Cobra Invitational 2019-20, Div B, INV, Q14 (dry fingers on a car battery: dangerous shock? → **No**; key not marked). ATX Regional 2020, Div B, REG, Q54 (T/F "a 12 V battery can kill" → key **True**; debatable).

- How to solve: I = 12 V ÷ 100 kΩ (dry) = 0.12 mA. Below the feeling level.
- Worked example: "Touching both terminals of a 12 V battery with dry hands (100 kΩ). Current?" 12/100 000 = **0.00012 A = 0.12 mA**. **Not felt.** With wet hands (1 kΩ): 12 mA DC, which is felt but is below the DC let-go level (about 76 mA for men).
- Traps: "Can 12 V **ever** kill?" Yes, in very rare cases (wet, broken skin, or a path inside the body). Read the question's wording. The real car-battery danger is a short (burns) and hydrogen gas.
- Calculator: calculate — type: V=12, R=100000 and R=1000 → I.

11. Why is a Van de Graaff generator safe to touch?

- Scope: ✅ IN (static hazard and the current-vs-voltage idea).
- Where seen: Hawk & Hornet Invitational 2026, Div C, INV, Q19 (long arcs but can be touched → **high voltage, low current**). Science Olympiad @ Berkeley Feb 2026, Div B, INV, Q16 (why a Van de Graaff is not dangerous, per B_topics). Full static facts in Charge Basics A2.
- How to solve: the dome holds only a small charge. It can give only a tiny current for a tiny time, so the energy is small.
- Worked example: "A Van de Graaff at 200 000 V holds 2 μC. How much energy can it deliver?" E = ½QV is capacitor math (OUT). Use the idea instead: **the charge is tiny, so the current lasts only microseconds. Answer: high voltage, low current and low energy.**
- Traps: It is not safe for people with pacemakers or heart problems. A big machine can still give a painful spark.
- Calculator: look up — type: Van de Graaff safety.

12. Power line on your car: what to do

- Scope: ✅ IN.
- Where seen: UT Austin Invitational 2019, Div C, INV, Q25 (how to escape → **jump, feet together**).
- How to solve: stay inside if you can. If there is a fire, jump clear without touching the car and the ground together. Land with feet together and shuffle away.
- Worked example: "A power line falls across your car. There is no fire. What should you do?" **Answer: stay inside, keep still, call 911, and wait for the utility to cut the power.**
- Traps: Do not step out normally. Your hand on the car and foot on the ground would be at different voltages.
- Calculator: look up — type: downed line car.

13. First aid for an electric shock

- Scope: ✅ IN (hazards).
- Where seen: not seen in a test.
- How to solve: follow the steps in Key facts: do not touch, cut the power, push away with a dry non-conductor only at low voltage, call 911, CPR and AED, see a doctor.
- Worked example: "Your friend is stuck to a frayed lamp cord and cannot let go. What do you do first?" **Answer: unplug the lamp or switch off the breaker. Do not grab your friend.**
- Traps: Grabbing the victim puts you in the circuit too. Water is not used on electrical fires; use a Class C (CO_2 or dry chemical) extinguisher after the power is off.
- Calculator: look up — type: shock first aid.

14. Safety practices and the least effective protection

- Scope: ✅ IN.
- Where seen: BullSO Invitational Jan 2026, Div B, INV, Q33 (least practical protection for metal-cased appliances → **polarized plugs**). Official practice decks (F_slides ACDC-HAZARDS: why fuses and breakers exist). "Shock Value Test" re-tryout, about 2012, Div B, practice, VI-3 (why ceramic is used → insulator).
- How to solve: strong protections: ground pin + breaker, GFCI, double insulation, keeping hands dry, one-hand rule, power off. Weak: polarized plug alone (the case is not grounded).
- Worked example: "Which gives the most protection against shock in a bathroom: a 20 A breaker, a polarized plug, or a GFCI?" **Answer: a GFCI.**
- Traps: Breakers and fuses protect wiring from fire. They do not trip at the small currents that kill people.
- Calculator: look up — type: shock protection ranking.

Medium

15. Current through the body from voltage and resistance

- Scope: ✅ IN.
- Where seen: not seen as a numeric test question. Prior bank I_bank S2. Cobra 2019-20 B Q14 is the concept.
- How to solve:
  1. Pick R: dry about 100 kΩ, wet about 1 kΩ.
  2. I = V/R.
  3. Compare with the threshold table.
- Worked example: "A person with wet hands (1.0 kΩ) touches 120 V. Current and effect?" I = 120/1000 = **0.12 A = 120 mA**: ventricular fibrillation is possible. "Dry hands (100 kΩ)?" I = **1.2 mA**: a tingle.
- Traps: Give the current in mA to compare with the table. Use 120 V (RMS), not the 170 V peak.
- Calculator: calculate — type: V=120, R=1000 → I (mA) → effect.

16. Minimum voltage to cause a given effect

- Scope: ✅ IN.
- Where seen: not seen in a test.
- How to solve: V = I × R. Use the threshold current for I.
- Worked example: "Wet skin is 1.0 kΩ. What voltage drives 30 mA (enough to stop breathing)?" V = 0.030 × 1000 = **30 V**. "With dry skin (100 kΩ), what voltage drives the 16 mA let-go current?" V = 0.016 × 100 000 = **1600 V**.
- Traps: This is why about 50 V AC is treated as the touch-safety limit in wet places, and why dry skin protects you at 120 V.
- Calculator: calculate — type: I=0.030, R=1000 → V.

17. Body path as resistors in series

- Scope: ✅ IN (series resistance).
- Where seen: not seen in a test. OpenStax *College Physics 2e* ch. 20.6 uses the idea.
- How to solve:
  1. Model the path: skin where the current enters + inside of the body + skin where it leaves.
  2. Add them (series).
  3. I = V/R_total.
- Worked example: "Wet-hand contact 500 Ω, internal 500 Ω, wet-foot contact 500 Ω. A person touches 120 V. Current?" R = 1500 Ω. I = 120/1500 = **80 mA**: dangerous.
- Traps: Shoes and floors add a large resistance in series. Rubber soles help; bare wet feet do not.
- Calculator: calculate — type: Rs=500,500,500; V=120 → I.

18. Wet versus dry: how many times more current?

- Scope: ✅ IN.
- Where seen: not seen in a test. ATX Regional 2020 B Q41 gives the dry value.
- How to solve: at the same voltage, I is inversely proportional to R. Ratio = R_dry/R_wet.
- Worked example: "Dry 100 kΩ, wet 1 kΩ. How many times more current with wet hands?" 100 000/1000 = **100 times**.
- Traps: The ratio is not "1.01". Divide the resistances, do not subtract them.
- Calculator: calculate — type: Rdry=100000, Rwet=1000 → ratio.

19. Defibrillator energy

- Scope: ✅ IN (energy, rule 3.c.iv). (The capacitor inside a defibrillator is ❌ OUT; the energy formula with V, R and t is IN.)
- Where seen: North Pocono Invitational Jan 2020, Div C, INV, Q32 (1000 V, 500 Ω, 10 ms → **20 J**).
- How to solve: $P = V^2/R$. E = P × t.
- Worked example: $P = 1000^2/500 = 2000$ W. E = 2000 × 0.010 = **20 J**.
- Traps: ms → s. A real defibrillator delivers 120-360 J (from memory), using a large, short current. That is the "6 A clamps the whole heart" row of the table.

- Calculator: calculate — type: V=1000, R=500, t=0.010 → E.

20. How a GFCI saves a life (time and energy)

- Scope: ✅ IN (rules 3.c.iii and 3.c.vi). Full GFCI types in CL 6 - Switches, Relays, Fuses, Breakers, GFCI T.
- Where seen: UT Austin (ATX) Regional 2026, Div B, REG, MC28 (GFCI stands for → ground fault circuit interrupter) and MC30 (GFCIs prevent → shocks). Pembroke Hill Dec 2025 B Q16. Princeton Feb 2026 C SA3a-d. Prior bank I_bank D32.
- How to solve:
  1. Find the body current: I = V/R.
  2. The GFCI sees that much current missing from the neutral. Above about 5 mA it trips in about 25 ms.
  3. Energy delivered E = I^2R t. Charge Q = I t.
- Worked example: "120 mA flows through a wet person (1 kΩ). A GFCI cuts it after 25 ms. Energy and charge delivered?" E = $0.12^2 \times 1000 \times 0.025$ = **0.36 J**. Q = 0.12 × 0.025 = **0.003 C (3 mC)**. A 15 A breaker would never trip at 0.12 A.
- Traps: The GFCI does not limit the current. It limits the **time**. It does not trip if a person touches hot and neutral only (the currents still balance).
- Calculator: calculate — type: V=120, R=1000, t=0.025 → I, E, Q.

21. Shock duration and the IEC zones

- Scope: ✅ IN (hazards). ⚠️ The IEC 60479 chart is trivia.
- Where seen: UT Austin Invitational 2019, Div C, INV, Q18 (200 mA hand to feet for 20 ms → key **B, tingling**; not verified).
- How to solve:
  1. Short shocks are much less dangerous than long ones at the same current.
  2. The IEC 60479-1 chart plots current against time. Zones: AC-1 (not felt), AC-2 (felt, usually no harm), AC-3 (strong muscle effects, breathing trouble, reversible heart effects), AC-4 (fibrillation likely).
  3. A shock under about 10 ms can be survived at a few hundred mA. Over 1 s, about 40-50 mA can cause fibrillation.
- Worked example: "Which is more dangerous: 50 mA for 2 s or 200 mA for 10 ms?" **Answer: 50 mA for 2 s.** The long shock lasts over a full heartbeat, so it can hit the heart's sensitive moment.
- Traps: Without the chart, use the rule "longer = worse". The UT key's "tingling" for 200 mA may be too mild (see Not confirmed).
- Calculator: look up — type: IEC zone (I mA, t ms).

22. Escaping a shock: one-hand rule and path

- Scope: ✅ IN.
- Where seen: not seen in a test. Prior topic notes (B_topics §3d safety rules).
- How to solve: a path from one hand to the other crosses the heart. One hand in your pocket means current can only go hand to foot or finger to finger.
- Worked example: "Why do technicians keep one hand in their pocket near high voltage?" **Answer: so a shock cannot travel from one hand to the other through the chest and heart.**
- Traps: Hand to foot still crosses the chest. Rubber-soled shoes and a rubber mat help block that path.
- Calculator: look up — type: one-hand rule.

Hard

23. Person in parallel with the ground wire (touching a faulted case)

- Scope: ✅ IN (parallel resistors).
- Where seen: not seen in a test. Built from Household Wiring A3 type 17.
- How to solve:
  1. Find the case voltage during the fault: V_case = I_fault × R_ground.
  2. The person (case to earth) is in parallel with the ground wire, so the person has the same voltage.
  3. I_person = V_case/R_person.

- Worked example: "Hot wire 0.3 Ω, ground wire 0.5 Ω, 120 V. A wet person (1000 Ω) is touching the case and a grounded pipe when the fault happens." I_fault ≈ 120/0.8 = 150 A. V_case = 150 × 0.5 = **75 V**. I_person = 75/1000 = **75 mA**, until the breaker trips. The person's branch barely changes the 150 A (1000 Ω ≫ 0.5 Ω).
- Traps: The person is not "safe because the ground wire takes it all". The ground wire takes most of the current, but the person still has 75 V across them. A fast trip (or a GFCI) is what makes it safe.
- Calculator: calculate — type: V=120, Rhot=0.3, Rg=0.5, Rp=1000 → Ifault, Vcase, Ip.

24. Step voltage near a fallen line

- Scope: ✅ IN (voltage difference and Ohm's law).
- Where seen: UT Austin Invitational 2019, Div C, INV, Q25 is the concept (feet together). Numeric version not seen.
- How to solve:
  1. Near the line, the ground voltage falls steadily with distance (simplified: a constant gradient in V/m).
  2. V_step = gradient × distance between the feet.
  3. I = V_step/R_leg-to-leg.
- Worked example: "The ground voltage falls by 100 V per meter. A person's leg-to-leg resistance is 1000 Ω. Compare a 0.8 m stride with shuffling (feet 0.05 m apart)." Stride: V = 100 × 0.8 = 80 V, I = **80 mA** (dangerous). Shuffle: V = 5 V, I = **5 mA** (harmless level).
- Traps: The real gradient is not constant; it is steepest near the wire. The idea is the same: small steps, feet together.
- Calculator: calculate — type: gradient=100, d=0.8 and 0.05, R=1000 → V, I.

25. Why do warning signs say "high voltage"? (free response)

- Scope: ✅ IN.
- Where seen: Rickards Invitational Oct 2025, Div C, INV, Q73 (2 points; current as small as 0.2 A can be lethal, so why do signs say "high voltage"?). UT Austin Invitational 2019, Div C, INV, Q13 (corona discharge deaths; trivia, not verified).
- How to solve: write a short argument in three steps:
  1. The current through a person is I = V/R_body.
  2. R_body is roughly fixed (about 1-100 kΩ). So a high voltage guarantees a high current.
  3. The source's voltage is known in advance; the current is not, because it depends on the victim. High voltage can also arc through air to you, and it can break down the skin.
- Worked example: "Body resistance is 2000 Ω. What voltage gives 0.2 A?" V = 0.2 × 2000 = **400 V**. **Answer: any source above a few hundred volts can push a lethal current, so the voltage is what warns you.**
- Traps: Do not say "voltage is what kills". Say voltage **causes** the deadly current.
- Calculator: calculate — type: I=0.2, R=2000 → V.

Nationals

26. A person bridging two points of a network (two node equations)

- Scope: ✅ States/Nationals only (KVL/KCL, rule 3.c.ix). Two unknowns, so it fits the math limit (rule 3.e.i).
- Where seen: not seen in a test. Built like an unbalanced Wheatstone bridge (GullSO Dec 2025 Q43 and Princeton Feb 2026 Comp 1 are balanced bridges).
- How to solve:
  1. Label the two node voltages V_1 and V_2 (bottom of the source = 0 V).
  2. Write KCL at each node: current in = current out.
  3. Solve the two equations.
  4. I_person = $(V_1 - V_2)$/R_person.
- Worked example: "A 120 V source feeds two branches. Left branch: 100 Ω (top) then 200 Ω (bottom), meeting at node 1. Right branch: 300 Ω (top) then 100 Ω (bottom), meeting at node 2. A person (1000 Ω) touches node 1 and node 2. Current through the person?"
  - Without the person: V_1 = 120 × 200/300 = 80 V. V_2 = 120 × 100/400 = 30 V. A 50 V difference.
  - Node 1: $(120 - V_1)/100 = V_1/200 + (V_1 - V_2)/1000$.

  - Node 2: $(120 - V_2)/300 + (V_1 - V_2)/1000 = V_2/100$.
  - Solve: V_1 = **77.1 V**, V_2 = **33.3 V**. I_person = 43.8/1000 = **43.8 mA** (severe pain, trouble breathing).
- Traps: The person loads the circuit, so the voltage they feel (43.8 V) is less than the open 50 V. A balanced bridge (equal ratios) would give 0 mA.
- Calculator: calculate — type: V=120, R1=100, R2=200, R3=300, R4=100, Rp=1000 → V1, V2, Ip (2×2 solve).

Out of scope (skip)

- Defibrillator capacitor charge, energy ½CV² and RC discharge time. ❌ OUT: capacitance (3.d).
- How a GFCI senses imbalance with a toroid coil. ❌ OUT: magnetism and inductance. (That a GFCI compares hot and neutral current is IN.)
- Frequency dependence of body impedance and the skin capacitance model. ❌ OUT: AC circuit theory and frequency analysis (3.d).
- Lightning physics (stepped leaders, return strokes). Belongs in the static notes (rule 3.c.ii), not here.
- Arc-flash incident energy (cal/cm²) calculations from NFPA 70E. ⚠️ Borderline: the hazard is IN, the industrial formulas are not in any test and are beyond the rules' scope.

Mistakes in tests and keys

- **BullSO Jan 2026 B Q34:** the key says **B** ("internal tissue burning due to the electrical resistance of the body") is not a possible effect. That is wrong: internal burns from I²R heating are a real and serious effect. Choice A (brain "muscles") is not biology, and choice C names the wrong cause (fibrillation is not caused by the diaphragm). The best answer is **A**; C is also flawed.
- **ATX Regional 2020 B Q54:** "a 12 V battery can kill" keyed True. It is true only in rare cases. A normal-contact answer would be False. Expect either; read the wording.
- **UT Austin 2020 B Q11:** the key has only a placeholder for "two reasons AC is more dangerous". Use: fibrillation at 50-60 Hz, and muscle clamping (cannot let go).
- **Florida State 2021 Q13 (toddler in the hot slot):** key D (none of the above). Debatable: it depends on whether the child touches anything grounded.
- **UT Austin 2019 C Q18:** key "tingling" for 200 mA over 20 ms. This may understate it. 200 mA is far above the let-go level, even if 20 ms is too short for fibrillation (not confirmed).
- **AC "let-go" versus DC "tetanus":** keys disagree between tests (USC Jan 2026 B Q28 vs UF 2019 C Q17). Not an error in either key; the sources disagree. See Key facts.

Not confirmed

- OpenStax Table 20.3 values and Dalziel's AC/DC table (via Kuphaldt) are from memory. I did not re-open either book in this run. The numbers match the test keys above (10-16 mA let-go, 100-200 mA lethal, 22/88 mA).
- The GFCI trip values (4-6 mA, about 25 ms) are from general knowledge of UL 943, not checked here.
- The extra-low-voltage limits (50 V AC, 120 V ripple-free DC) are from memory of IEC 61140 / IEC 60364.
- Power-line distances (10 ft, 35 ft) are typical utility advice, from memory.
- The IEC 60479-1 zone boundaries and the zone for 200 mA at 20 ms (UT 2019 C Q18) were not checked against the chart.
- The defibrillator energy range (120-360 J) is from memory.
- First-aid steps follow the usual Red Cross advice, from memory. I did not open a first-aid source in this run.
- The Berkeley Feb 2026 B Q15-16 answers come from the prior topic notes (B_topics), because the circled key answers are not in the key's text layer.

Sources

- Rules: 2027 Div B Circuit Lab (manual pages B12-B13): areas 3.c.ii, 3.c.iii, 3.c.iv, 3.c.vi, 3.c.ix; exclusions 3.d; math 3.e.i; hands-on safety II.d.
- Tests (extractions in `circuitlab/research/tests/` and `prior/G_tests.md`): Berkeley Feb 2026 B; USC Invitational Jan 2026 B; BullSO Jan 2026 B and C; Pembroke Hill Dec 2025 B; ATX Regional 2026 B, 2021 B,

2020 B; Florida State 2021 B; GullSO Dec 2025 B/C; Northview Sylvania Dec 2025 C; Hawk & Hornet 2026 C; Rickards Oct 2025 C; Princeton Feb 2026 C; UT Austin Invitational 2019 C, 2020 B; MIT Jan 2019 C; UF Northern Regional 2019 B and C, 2020 B; Cobra 2019-20 B; F-M 2018; North Pocono Jan 2020 C; "Shock Value" re-tryout about 2012 B.

- Prior research: `prior/B_topics.md` §3a, §3c, §3d; `prior/I_bank.md` S1-S6, D32; `prior/F_slides.md` (ACDC-HAZARDS); `prior/G_tests.md` (DC & AC hazards).
- Textbooks and references (general knowledge, not re-read in this run): OpenStax *College Physics 2e* ch. 20.6 "Electric Hazards and the Human Body" (Table 20.3); Kuphaldt, *Lessons in Electric Circuits* vol. 1 ch. 3 "Electrical Safety" (Dalziel's table, "Shock current path", "Ohm's Law (again!)", "Safe practices", "Emergency response"); IEC 60479-1 (effects of current on humans); OSHA "Controlling Electrical Hazards" (OSHA 3075); UL 943 (GFCIs).

TAB A4

Current, Voltage, Resistance, Power, Energy

Units, Prefixes and Significant Figures 277
Current and Charge 286
Voltage and EMF 293
Resistance and Resistivity 300
Power, Energy and Cost 313

Each note: key facts, formulas, then question types from Easy to Nationals with worked examples.

Units, Prefixes and Significant Figures

Note: Units Prefixes and Sig Figs

This note covers SI units, metric prefixes, unit conversions, unit analysis and significant figures for Circuit Lab.

Two rules make this note important:

- Rule 3.b: "Unless otherwise requested, answers must be in **metric units** with **appropriate significant figures**." So every numeric answer on every test uses this note. Keys often take off points for a missing unit.
- Rule 3.c.iv names "concepts, calculations, and **units** of current, voltage, resistance, power, and energy".

So all of this note is ✅ **IN at every level**, except:

- Units of excluded quantities (farad, henry, tesla, weber) are ❌ OUT as test content. Knowing their names helps you spot an out-of-scope question.
- Building a formula by dimensional analysis is ⚠️ **borderline** depth. Checking a formula by its units is IN.

The scientists behind the units are in Units Named After People [A1].

Key facts

- **SI** (Système international) is the modern metric system.
- SI has **7 base units**: second (s, time), metre (m, length), kilogram (kg, mass), **ampere (A, current)**, kelvin (K, temperature), mole (mol, amount) and candela (cd, light intensity).
- The **ampere is a base unit**. The coulomb, volt, ohm, watt and joule are **derived units**. They are built from base units.
- Since 20 May 2019, the SI fixes exact values for some constants. The elementary charge e = 1.602176634 × 10^{-19} C is **exact**. The ampere is now defined from e.
- A unit named after a person has a **capital** symbol (A, V, Ω, W, J, C, S, Hz, K). The written name is **lowercase** (ampere, volt, ohm).
- **Electrical units table:**

Quantity	Symbol	Unit	Unit symbol	Equals	SI base units
Current	I	ampere	A	C/s	A
Charge	Q, q	coulomb	C	A·s	A·s
Voltage, EMF	V, E	volt	V	J/C = W/A	$kg·m^2·s^{-3}·A^{-1}$
Resistance	R	ohm	Ω	V/A = W/A^2 = V^2/W	$kg·m^2·s^{-3}·A^{-2}$
Conductance	G	siemens (mho)	S (℧)	A/V = 1/Ω	$kg^{-1}·m^{-2}·s^3·A^2$
Power	P	watt	W	J/s = V·A = A^2·Ω = V^2/Ω	$kg·m^2·s^{-3}$
Energy, work	E, W	joule	J	N·m = W·s = V·C	$kg·m^2·s^{-2}$
Resistivity	ρ	ohm-metre	Ω·m	V·m/A	$kg·m^3·s^{-3}·A^{-2}$
Conductivity	σ	siemens per metre	S/m	1/(Ω·m)	$kg^{-1}·m^{-3}·s^3·A^2$
Electric field	E	volt per metre	V/m	N/C	$kg·m·s^{-3}·A^{-1}$
Force	F	newton	N	kg·m/s^2	$kg·m·s^{-2}$
Frequency	f	hertz	Hz	1/s	s^{-1}
Temperature coefficient	α	per degree	/°C or /K		K^{-1}
Current density	J	ampere per square metre	A/m^2		$A·m^{-2}$

- Units for **excluded** quantities (❌ OUT, rule 3.d and the 3.c list): farad (F, capacitance), henry (H, inductance), tesla (T) and gauss (magnetic field), weber (Wb, magnetic flux). If a question needs one, it is probably out of scope. You may ask the supervisor.
- **Metric prefixes:**

Prefix	Symbol	Factor	Example
tera	T	10^{12}	3 TW (world electric power, roughly)
giga	G	10^9	1 GW (a large power plant)

Prefix	Symbol	Factor	Example
mega	M	10^{6}	1 MΩ (a meter's input resistance)
kilo	k	10^{3}	4.7 kΩ, 1 kWh
(none)		$10^{0} = 1$	1 V
centi	c	10^{-2}	1 cm (rare in circuits)
milli	m	10^{-3}	20 mA, 55 mΩ
micro	μ	10^{-6}	10 μA, 1 μs
nano	n	10^{-9}	0.2 ns
pico	p	10^{-12}	22 pF (OUT topic, but the prefix is common)
femto	f	10^{-15}	rare

- The newest prefixes (added in 2022) are ronna (R, 10^{27}), quetta (Q, 10^{30}), ronto (r, 10^{-27}) and quecto (q, 10^{-30}). This is trivia only.
- **Case matters.** m = milli (10^{-3}). M = mega (10^{6}). The factor between them is 10^{9}. Kilo is a **lowercase** k. A capital K is the kelvin.
- μ is the Greek letter mu. Some printers write "u" (uA = μA).
- **Engineering notation** uses powers of 10 that are multiples of 3 (10^{3}, 10^{-6}). It matches the prefixes. 0.0047 A = 4.7×10^{-3} A = 4.7 mA.
- A calculator shows 4.7×10^{3} as **4.7E3**.
- **Squared and cubed prefixes:** 1 mm^2 = $(10^{-3}\ m)^2$ = **$10^{-6}\ m^2$**. 1 cm^2 = **$10^{-4}\ m^2$**. 1 cm^3 = $10^{-6}\ m^3$.
- **Common non-SI units and conversions:**

Unit	SI value	Used for
1 kilowatt-hour (kWh)	3.6×10^{6} J	electricity bills
1 watt-hour (Wh)	3600 J	batteries
1 milliamp-hour (mAh)	3.6 C	battery capacity (charge)
1 amp-hour (Ah)	3600 C	car batteries
1 electron volt (eV)	1.602×10^{-19} J	energy of one electron (see Voltage and EMF A4)
1 horsepower (hp)	745.7 W	motors, cars
1 calorie (cal)	4.184 J	food and heat (1 food Calorie = 1000 cal)
1 BTU	1055 J (0.293 Wh)	heaters, air conditioners
1 inch	2.54 cm (exact)	
1 foot	0.3048 m (exact)	wire length (Ω per 1000 ft)
1 mile	1.609 km	
0 °C	273.15 K	temperature

- A **temperature change** of 1 °C is the same as 1 K. Only the zero point is different. So α in /°C equals α in /K.
- **Significant figures (sig figs, s.f.)** are the digits that carry meaning in a measured number.
- **Rules for counting sig figs:**
  1. All non-zero digits count. 4.7 has 2.
  2. Zeros **between** non-zero digits count. 1002 has 4.
  3. **Leading** zeros never count. 0.00450 has 3 (4, 5 and the last 0).
  4. **Trailing** zeros after a decimal point count. 2.50 has 3. 100.0 has 4.
  5. Trailing zeros in a whole number with no decimal point are **unclear**. 1200 may have 2, 3 or 4. Write 1.20×10^{3} to show 3.
  6. **Exact numbers** have unlimited sig figs. Examples: counted items (3 bulbs), defined values (60 s per min, 1000 mA per A, 3600 s per h, 2.54 cm per inch), and the exact SI constant e.
- **Rules for calculating:**
  - **Multiply or divide:** the answer has as many sig figs as the input with the **fewest sig figs**.
  - **Add or subtract:** the answer has as many **decimal places** as the input with the **fewest decimal places**.
  - Keep extra digits in the middle steps. Round only the final answer.
  - Round 5 and above up. 2.553 → 2.6 (2 s.f.).

- **Percent error** = |measured − accepted| / accepted × 100%.
- **Percent difference** between two measurements = |a − b| / ((a + b)/2) × 100%.

Formulas

What	Formula	Note
Prefix to base unit	value × factor	4.7 kΩ = 4.7 × 10^3 Ω
Base unit to prefix	value ÷ factor	0.020 A ÷ 10^{-3} = 20 mA
Area prefix	(factor)²	1 mm^2 = 10^{-6} m^2
Volume prefix	(factor)³	1 cm^3 = 10^{-6} m^3
Kelvin from Celsius	K = °C + 273.15	
kWh to J	× 3.6 × 10^6	
mAh to C	× 3.6	
eV to J	× 1.602 × 10^{-19}	
hp to W	× 745.7	
Percent error	|meas − true| / true × 100%	
Sig figs, × ÷	fewest sig figs	
Sig figs, + −	fewest decimal places	

Question types

Easy

1. Name the unit of a quantity (or the quantity of a unit) — ✅ IN

- Where seen: UF Northern Regional 2020, B, REG, Q4 (SI unit of charge → coulomb). Florida MYSO practice Nov 2020, B, Q8 (unit of resistance → ohm). UT Invitational fall 2018, B/C, INV, Q2 (unit of "electrical pressure" or EMF → volt). JC Booth 2011 (Shock Value), B, INV, Q14 (units of I, V, R, P → amperes, volts, ohms, watts; the B part is OUT). UF Northern Regional 2019, B, REG, Q33-36 (term matching).
- How to solve: Learn the electrical units table in Key facts.
- Worked example: "What is the unit of power?" → **watt (W)**. "What quantity is measured in ohms?" → **resistance**.
- Traps: "Electrical pressure" is an old name for voltage. The answer is the volt, not the pascal.
- Calculator: look up — type: unit, quantity.

2. Unit named after a scientist — ✅ IN (history, rule 3.c.i)

- Where seen: ATX Regional 2021, B, REG, Q2 ("the unit of charge bears my name" → Coulomb). ATX Regional 2026, B, REG, MC2 (unit of resistance named after → Ohm). F-M Invitational 2018, INV, Q7 (Hertz proved EM waves; his unit is for frequency). BullSO Jan 2026, B, INV, Q5 (law relating V, I, R → Ohm).
- How to solve: Match: ampere (A) → Ampère; coulomb (C) → Coulomb; volt (V) → Volta; ohm (Ω) → Ohm; hertz (Hz) → Hertz; siemens (S) → Werner von Siemens; watt (W) → James Watt; joule (J) → James Joule. See Units Named After People A1.
- Worked example: "Which of the seven 2027 scientists has a unit of frequency?" → **Hertz**.
- Traps: Kirchhoff and Boole have **no** SI units. "Boolean" is a name, not a unit.
- Calculator: look up — type: unit, scientist.

3. Symbols and capital letters (m vs M, W vs W, S vs s) — ✅ IN

- Where seen: not seen as its own question. It causes the prefix traps in MIT 2019, C, INV, Q7-11. Keys deduct for unit errors (BullSO Jan 2026, B, INV, SA5c: "−1 pt if units are missing/incorrect").
- How to solve: Learn the pairs that look alike:
  - m (milli) vs M (mega).
  - k (kilo) vs K (kelvin).
  - W (the unit watt) vs W (the quantity work or energy).
  - C (the unit coulomb) vs C (capacitance, OUT).

  - S (siemens) vs s (second).
  - A (the unit ampere) vs A (area in R = ρL/A).
  - Ω (ohm) vs Ʊ (mho, the old name for siemens).
- Worked example: "What is 2 MΩ in ohms?" → **2 000 000 Ω**. "What is 2 mΩ?" → **0.002 Ω**.
- Traps: "mA" and "MA" differ by 10^9. Write symbols carefully on the answer sheet.
- Calculator: look up — type: symbol, prefix, case.

4. Convert between prefixes — ✅ IN

- Where seen: soinc practice deck 2 ("UNIT-METRIC-PREFIX", 12 V across 3 kΩ → 4 mA). Prior I_bank U3. Every test uses it.
- How to solve:
  1. Change to the base unit: multiply by the prefix factor.
  2. Change to the new prefix: divide by its factor.
  3. Check: a smaller prefix gives a bigger number.
- Worked example: 250 μA = 250 × 10^{-6} A = 2.50 × 10^{-4} A = **0.250 mA**. 0.047 MΩ = **47 kΩ** = 47 000 Ω. 3.3 nA = **3300 pA**.
- Traps: Count the steps of 1000. From μ to m is one step (÷ 1000). From μ to k is three steps (÷ 10^9).
- Calculator: calculate — type: prefix, convert.

5. Which combination equals a unit (unit equivalences) — ✅ IN

- Where seen: Pembroke Hill Dec 2025, B, INV, Q33 (which defines the volt → J/C; W/A is also correct, see Mistakes). Prior I_bank U2 (which equals a watt; which equals an ohm). The volt part is in Voltage and EMF A4 type 1.
- How to solve:
  1. Write each unit as a formula: W = J/s, A = C/s, V = J/C, Ω = V/A.
  2. Substitute until both sides match.
- Worked example: Which equal a watt? A·V ✓ (P = IV). J/s ✓. V²/Ω ✓ (P = V²/R). A²·Ω ✓ (P = I²R). N·m/s ✓. **All five.** Which equals an ohm? V/A ✓, W/A² ✓, V²/W ✓.
- Traps: C/s is an ampere, not a volt. A/s means nothing in a DC circuit test.
- Calculator: look up — type: unit equivalence, watt, ohm, volt.

6. Count significant figures — ✅ IN (rule 3.b)

- Where seen: not seen as its own question. Rule 3.b requires sig figs in every answer.
- How to solve: Use the 6 counting rules in Key facts.
- Worked example: 0.00450 A → **3** s.f. 1002 Ω → **4**. 2.50 V → **3**. 1.20 × 10^3 Ω → **3**. 1200 Ω → **2 to 4** (unclear).
- Traps: Leading zeros only hold the place. 0.0047 has 2 sig figs, not 5.
- Calculator: look up — type: sig figs, count.

7. Scientific and engineering notation; calculator E notation — ✅ IN

- Where seen: not seen as its own question. All drift-velocity and Coulomb questions use it (for example Florida State 2021, B, ST, Q12 answers in powers of 10).
- How to solve:
  1. Scientific notation: one digit before the decimal point (4.7 × 10^3).
  2. Engineering notation: the exponent is a multiple of 3 (47 × 10^3 = 47 k).
  3. On the calculator, type 4.7E3 for 4.7 × 10^3 (use the EE key, not "× 10 ^").
- Worked example: 0.000 073 m/s = **7.3 × 10^{-5} m/s** (scientific) = **73 × 10^{-6} m/s** = **73 μm/s** (engineering).
- Traps: Typing 1.6 × 10 ^ −19 without brackets can go wrong when you divide. The EE key avoids this.
- Calculator: look up — type: scientific notation, engineering notation.

Medium

8. Prefix traps inside a calculation — ✅ IN

- Where seen: MIT 2019, C, INV, Q8 (5.0 mA in 0.75 Ω → 3.75 mV), Q9 (1.5 V across 55 mΩ → 27.3 A) and Q10 (5.0 V, 560 mA → 8.93 Ω). Prior I_bank U3 (4.7 kΩ × 2.2 mA). More in Resistance and Resistivity A4 type 3.

- How to solve:
  1. Change every input to base units.
  2. Calculate.
  3. Choose a prefix for the answer. Round to the right sig figs.
- Worked example: 4.7 kΩ × 2.2 mA = 4700 × 0.0022 = 10.34 V → **10 V** (2 s.f.; write "10. V" or "1.0 × 10^1 V" to show 2). Short cut: k × m = 10^3 × 10^{-3} = 1, so kΩ × mA = V.
- Traps: Useful pairs: kΩ × mA = V. V ÷ kΩ = mA. MΩ × μA = V. mA × V = mW.
- Calculator: calculate — type: prefix, Ohm's law, pairs.

9. Squared prefixes: mm² and cm² to m² — ✅ IN

- Where seen: JC Booth 2011 (Shock Value), B, INV, Q48 (iron rod, A = 65 cm² → 1.79 × 10^{-5} Ω). Florida State 2021, B, ST, Q12 (1 mm² copper; drift speed). Rickards Oct 2025, C, INV, Q55 (a unit trap in a resistivity question; I did not see the text).
- How to solve:
  1. Square the prefix factor: $(10^{-2})^2 = 10^{-4}$; $(10^{-3})^2 = 10^{-6}$.
  2. Multiply.
- Worked example: 65 cm² = 65 × 10^{-4} m² = **6.5 × 10^{-3} m²**. 1.5 mm² = **1.5 × 10^{-6} m²**.
- Traps: 65 cm² is **not** 0.65 m². That uses 10^{-2} instead of 10^{-4}. The resistance would be 100 times too small.
- Calculator: calculate — type: area, mm2, cm2.

10. Sig figs when you multiply or divide — ✅ IN (rule 3.b)

- Where seen: not seen as its own question. Prior I_bank U4. Every test key rounds answers (for example USC 2026 B Q11 key 0.017 W, from 0.0174).
- How to solve:
  1. Count the sig figs of each measured input.
  2. Round the answer to the smallest count.
- Worked example: 12.0 V / 4.7 Ω = 2.553 A. 12.0 has 3 s.f., 4.7 has 2. Answer: **2.6 A**. A second example: 1.5 V / 2200 Ω = 6.818 × 10^{-4} A → **6.8 × 10^{-4} A** (0.68 mA). Here 2200 is taken as 2 s.f.
- Traps: Do not count exact numbers. In P = I²R, the "2" is exact. In 1 h = 3600 s, 3600 is exact.
- Calculator: look up — type: sig figs, multiply, divide.

11. Sig figs when you add or subtract; mixed steps — ✅ IN (rule 3.b)

- Where seen: not seen as its own question. Series resistance totals use it.
- How to solve:
  1. Line up the decimal points.
  2. Round the answer to the fewest decimal places.
  3. In a multi-step problem, keep all digits until the end.
- Worked example: 12.11 Ω + 0.3 Ω = 12.41 Ω → **12.4 Ω** (1 decimal place). A second example: 4.7 kΩ + 330 Ω. Change to the same unit first: 4700 + 330 = 5030 Ω. 4.7 kΩ is known only to the hundreds place, so the answer is **5.0 kΩ**.
- Traps: Adding gives decimal places, not sig figs. 100.0 + 0.25 = 100.25 → **100.3** (still 4 s.f.).
- Calculator: look up — type: sig figs, add, subtract.

12. Non-SI conversions (kWh, mAh, eV, hp, cal, ft, mph) — ✅ IN

- Where seen: MIT 2019, C, INV, Q12 (350 hp → W → current). UT Invitational 2020, B, INV, Q63 (60 mph, miles). Pembroke Hill Dec 2025, B, INV, Q7 (kWh). UT Invitational fall 2018, Q8 (mWh). Electron volts: see Voltage and EMF A4 type 4. mAh: see Current and Charge A4 type 9.
- How to solve:
  1. Write the conversion as a fraction equal to 1 (for example 745.7 W / 1 hp).
  2. Multiply so the old unit cancels.
- Worked example: 2.5 kWh × 3.6 × 10^6 J/kWh = **9.0 × 10^6 J**. 2400 mAh × 3.6 C/mAh = **8640 C**. 60 mph × 1.609 km/mi = **96.6 km/h**. 100 ft × 0.3048 m/ft = **30.5 m**.
- Traps: Wh is energy. Ah is charge. Do not convert one into the other without a voltage.

- Calculator: calculate — type: convert, kWh, mAh, hp, eV, feet, miles.

13. Express a derived unit in SI base units — ✅ IN

- Where seen: UF Northern Regional 2019, C, REG, Q23-27 (A, V, Ω, W, J in base units → A; $kg \cdot m^2 \cdot s^{-3} \cdot A^{-1}$; $kg \cdot m^2 \cdot s^{-3} \cdot A^{-2}$; $kg \cdot m^2 \cdot s^{-3}$; $kg \cdot m^2 \cdot s^{-2}$). UMSO 2019, INV, Q11-15 (C, V, Ω, W, J). North Pocono Jan 2020, C, INV, Q20 (W, m, A, F, V, S; key errors in 3 of 6). Rickards Dec 2020, C, INV, Q14 (conductance → $s^3 \cdot A^2 \cdot m^{-2} \cdot kg^{-1}$). PA Central Regional about 2013, C, REG (the ampere in base SI).
- How to solve:
  1. Start from mechanics: N = $kg \cdot m \cdot s^{-2}$. J = N·m = $kg \cdot m^2 \cdot s^{-2}$. W = J/s = $kg \cdot m^2 \cdot s^{-3}$.
  2. C = A·s.
  3. V = W/A = $kg \cdot m^2 \cdot s^{-3} \cdot A^{-1}$.
  4. Ω = V/A = $kg \cdot m^2 \cdot s^{-3} \cdot A^{-2}$.
  5. S = 1/Ω = $kg^{-1} \cdot m^{-2} \cdot s^3 \cdot A^2$.
- Worked example: The siemens in base units. Ω = $kg \cdot m^2 \cdot s^{-3} \cdot A^{-2}$. Flip every exponent: S = **$kg^{-1} \cdot m^{-2} \cdot s^3 \cdot A^2$**.
- Traps: The ampere is already a base unit. Its answer is just "A". The farad (F) part of North Pocono Q20 is ❌ OUT.
- Calculator: look up — type: base units, derived unit.

14. Units of a constant from its formula (ρ, α, σ, n) — ✅ IN

- Where seen: USC Invitational Jan 2026, B, INV, Q35a (units of resistivity in V, A and m → V·m/A). Rickards Oct 2025, C, INV, Q55 (a unit trap with resistivity).
- How to solve:
  1. Solve the formula for the constant.
  2. Put in the units of the other quantities.
- Worked example: ρ = RA/L. Units = Ω × m^2 / m = **Ω·m**. In V, A and m: Ω = V/A, so ρ is **V·m/A**. A second example: α from $R = R_0(1 + \alpha\Delta T)$. αΔT has no unit, so α is in **1/°C** (or K^{-1}). A third example: n in I = nqAv. n = I/(qAv) → $A/(C \cdot m^2 \cdot m/s)$ = **m^{-3}**.
- Traps: Ω/m is the unit of resistance **per length** (a wire table), not resistivity. Ω·m is resistivity.
- Calculator: look up — type: unit of constant, resistivity, alpha.

15. Temperature units (°C and K) — ✅ IN

- Where seen: not seen as its own question. Superconductor and temperature-coefficient questions use it. See Resistance and Resistivity A4 types 20, 21 and 25.
- How to solve:
  1. K = °C + 273.15.
  2. A change of 1 °C = a change of 1 K.
- Worked example: Liquid nitrogen boils at 77 K = 77 − 273.15 = **−196 °C**. A 300 °C oven is **573 K**.
- Traps: Kelvin has no degree sign. Write "77 K", not "77 °K".
- Calculator: calculate — type: kelvin, celsius.

16. Metric answers and units for points — ✅ IN (rule 3.b)

- Where seen: BullSO Jan 2026, B, INV, SA5c ("−1 pt if units are missing/incorrect") and FR Q6-7 ("max final score of 3 pts if ... missing/incorrect units"). BullSO Jan 2026, B, INV, SA2a-c and SA3 ("include units"). UT Invitational 2020, B, INV, Q28 key wrote "N/m" for torque (should be N·m).
- How to solve:
  1. Always write a unit after every number.
  2. Give metric (SI) units unless the question asks for others.
  3. Convert feet, inches and miles to metres or kilometres.
- Worked example: "The car goes 66.46 miles." In metric: 66.46 × 1.609 = **107 km**.
- Traps: N·m (torque or energy) and N/m (spring constant) are different. The dot matters.
- Calculator: look up — type: metric answer, units.

17. Percent error and tolerance in a measurement — ✅ IN

- Where seen: ATX Regional 2026, B, REG, MC27 (1520 Ω ± 8% measured at 1420 Ω → not faulty). Hands-on tasks compare a measured value with a calculated one (rule Part II.c.i, II.c.viii).
- How to solve:
  1. Percent error = |measured − expected| / expected × 100%.
  2. Compare with the tolerance.
- Worked example: A 100 Ω resistor (± 5%) reads 98.2 Ω. Error = 1.8/100 × 100% = **1.8%**. It is inside ± 5%, so it is **OK**. A second example: 1420 vs 1520 Ω is a **6.6%** error, inside ± 8%.
- Traps: Divide by the **expected** (nominal) value, not the measured one.
- Calculator: calculate — type: percent error, tolerance.

Hard

18. Check a formula with units (dimensional analysis) — ✅ IN

- Where seen: not seen as its own question. USC Invitational Jan 2026, B, INV, Q35 uses units to build a formula (type 21). OpenStax College Physics 2e, ch. 1.
- How to solve:
  1. Write the units of each side.
  2. Simplify. Both sides must have the same units.
  3. If they do not match, the formula is wrong.
- Worked example: Is P = V²/R right? V²/Ω = V²/(V/A) = V·A = **W** ✓. Is E = V·R·t right? V·Ω·s = V·(V/A)·s = V²·s/A. A joule is V·A·s. They do not match, so **E = VRt is wrong** (the right one is E = V²t/R).
- Traps: Matching units do not prove a formula is right. A missing number like ½ or 2π cannot be found this way.
- Calculator: look up — type: dimensional analysis, check formula.

19. Units written in unusual combinations (C, J, s, V, A, m) — ✅ IN (⚠️ borderline when the quantity is an AC idea)

- Where seen: Hawk & Hornet 2026, C, INV, FR4 (impedance in C, J, s → J·s/C²; impedance is an AC idea, ⚠️ borderline, but the result equals the ohm). USC Invitational Jan 2026, B, INV, Q35a (V·m/A).
- How to solve:
  1. Write the target unit in base units.
  2. Write each allowed unit in base units.
  3. Find the powers that match. Or substitute formulas: Ω = V/A, V = J/C, A = C/s.
- Worked example: The ohm in J, C and s. Ω = V/A = (J/C) / (C/s) = **J·s/C²**.
- Traps: Go one step at a time. Write each substitution on its own line.
- Calculator: look up — type: unit combination, ohm in J C s.

20. Order-of-magnitude estimates — ✅ IN

- Where seen: Florida State 2021, B, ST, Q12 (drift speed in a phone charger cable → about 10^{-3} m/s was the nearest choice), Q25 (US per-citizen power; the key is wrong) and Q26 (LED bulb → 10 W). Hawk & Hornet 2026, C, INV, Q20 (answer in mm/s).
- How to solve:
  1. Round each input to one digit and a power of 10.
  2. Calculate with the powers of 10.
  3. Pick the choice with the nearest exponent.
- Worked example: A house uses about 1 kW on average. 1 kW × 8760 h a year ≈ 9 × 10^3 kWh, so about **10^4 kWh a year**.
- Traps: Read "per person" vs "per country", "per day" vs "per year".
- Calculator: look up — type: estimate, order of magnitude.

Nationals

21. Build a formula by dimensional analysis — ⚠️ Borderline

- Why borderline: units are IN (rule 3.c.iv). Building a new physics formula from units goes past the list.

- Where seen: USC Invitational Jan 2026, B, INV, Q35c (four-point probe on a thin sheet: ρ from V, I, t and spacing a → ρ ~ Vt/I). The worked method is in Resistance and Resistivity A4 type 33.
- How to solve:
  1. List the quantities you may use and their units.
  2. Write ρ = V^a × I^b × t^c.
  3. Match the units of each base unit. Solve for a, b and c (keep to at most two equations, rule 3.e.i).
- Worked example: Ω·m = (V/A)·m. So a = 1, b = −1, c = 1. **ρ ~ V·t/I**.
- Traps: Pure numbers (2π, ln 2) cannot be found. If two lengths are given (t and a), the physics decides which one is used.
- Calculator: look up — type: dimensional analysis, build formula.

22. SI definitions and exact constants (2019 SI) — ✅ IN (as unit facts)

- Where seen: not seen in a test. Possible at Nationals as trivia. Source: BIPM SI Brochure, 9th edition (2019).
- How to solve: Learn the facts:
  1. There are 7 base units. The ampere is one of them.
  2. Since 2019, the ampere is set by fixing e = 1.602176634 × 10^{-19} C exactly.
  3. So 1 A is exactly 1/(1.602176634 × 10^{-19}) elementary charges per second, about 6.24 × 10^{18}.
  4. The kilogram is now set by the Planck constant. It no longer uses a metal block.
- Worked example: "Which SI base unit belongs to electricity?" → **the ampere**. "Is the elementary charge a measured or an exact value in SI?" → **exact (since 2019)**.
- Traps: The old (1948) ampere definition used the force between two parallel wires. That definition is magnetism, so it is ❌ OUT as a calculation.
- Calculator: look up — type: SI definition, base unit, exact constant.

Out of scope (skip)

- **Units of capacitance, inductance and magnetism** (farad, henry, tesla, gauss, weber). ❌ OUT: rule 3.d (capacitance, inductance) and the "areas only" list in rule 3.c (no magnetism). ATX Regional 2020, B, Q55 (unit before the tesla → gauss), JC Booth 2011, Q25b-26 (farad, henry), ATX Regional 2026, C, MC34 (unit of the RC time constant).
- **Units of AC quantities** (VA, VAR, power factor, reactance). ❌ OUT: AC circuit theory, rule 3.d.
- **Logarithmic units (decibels).** ❌ OUT: frequency analysis and amplifier gain are not in the list. They also need logarithms, which rule 3.e does not list.
- **Floating-point and negative binary numbers.** ❌ OUT: rule 3.e.iii. (These are number formats, not units, but tests sometimes mix them in. See Binary Numbers and Addition A8.)

Mistakes in tests and keys

- **Pembroke Hill Dec 2025, B, INV, Q33.** The key is "Joule per Coulomb". Choice B, "Watt per Ampere", is also a volt (W/A = (J/s)/(C/s) = J/C). Two choices are correct.
- **North Pocono Jan 2020, C, INV, Q20.** Key errors in 3 of 6 base-unit answers. Correct: V = kg·m^2·s^{-3}·A^{-1} (the key has A to the +1). S = kg^{-1}·m^{-2}·s^3·A^2. F = kg^{-1}·m^{-2}·s^4·A^2 (F is OUT).
- **UT Invitational 2020, B, INV, Q28 (= C, Q29).** The key writes torque in "N/m". It should be N·m. (The question itself is OUT: magnetism.)
- **UT Invitational fall 2025, B/C, INV, FR1d.** The key gives a capacitance in "C". It should be F. (The question is OUT: capacitance.)
- **ATX Regional 2020, B, REG, Q64.** The key gives a voltage in "A" (186.8 A should be volts). The problem also has no solution as drawn.
- **Florida State 2021, B, ST, Q25.** The key's order of magnitude is the US total, not per citizen.

Not confirmed

- Rickards Oct 2025, C, Q55 "unit trap" and Rickards Dec 2020, C, Q14 come from prior/G_tests.md. I did not re-read the question texts.

- Hawk & Hornet 2026 FR4 ($J \cdot s/C^2$) comes from prior/G_tests.md. I checked the unit algebra, not the question text.
- The newest SI prefixes (ronna, quetta, ronto, quecto, adopted in 2022) and the 2019 redefinition details are general knowledge. I did not re-open the BIPM brochure in this run.
- Conversion factors (hp 745.7 W, cal 4.184 J, BTU 1055 J) are standard values from general knowledge.
- The sig-fig rules are the standard textbook rules (OpenStax College Physics 2e, ch. 1.3). Science Olympiad does not publish its own sig-fig rules. Supervisors may differ on "1200".
- No real **Nationals** Circuit Lab test was available. The "USC 2026" files are the USC Invitational. So the Nationals types here come from top invitationals and the rules.

Sources

- 2027 Div B rules, Circuit Lab (pages B12-B13), rules 3.b, 3.c.i, 3.c.iv, 3.d, 3.e.
- Test extractions in `circuitlab/research/tests/`: atx-reg (ATX Regional 2020 B/C, 2021 B), misc-a (F-M 2018, North Pocono 2020, UMSO 2019), misc-b (MIT 2019, UF Northern Regional 2019 B/C and 2020 B, JC Booth 2011), nat26 (USC Invitational 2026 B), state (Florida State 2021, MYSO 2020), ut-a (UT Fall 2025, ATX Regional 2026 B/C), ut-b (UT 2020 B), ut-c (UT 2018).
- 2026 Div B tests from `downloads/dlB/tests/`: BullSO (Jan 24, 2026) Q5, SA2-SA5, FR 6-7 and key; Pembroke Hill (Dec 6, 2025) Q7, Q33.
- prior/G_tests.md (Rickards 2020 and 2026, HH 2026, PA Central), prior/I_bank.md (U1-U4), prior/B_topics.md §4, prior/F_slides.md §4 (soinc deck 2).
- BIPM, The International System of Units (SI Brochure), 9th edition, 2019: base units and exact constants.
- OpenStax College Physics 2e, ch. 1 (units, sig figs, dimensional analysis): general knowledge of the book; I did not re-open it in this run.
- All worked examples were checked in Python.

Current and Charge

This note covers electric charge, electric current, battery capacity in mAh, and how fast electrons really move.

Rule 3.c.iv names "concepts, calculations, and units of current". So almost all of this note is ✅ **IN at every level**. Two parts are different:

- **Drift velocity math** (I = nqAv) is ⚠️ **borderline**. The rule does not name it. It needs only algebra, so a writer may still ask it. Top invitationals did ask it.
- **Kirchhoff's current law** in a real circuit analysis is ✅ **States/Nationals only** (rule 3.c.ix). The full method is in Kirchhoff Analysis (States-Nationals) [A5].

Key facts

- **Charge (Q or q)** is a property of matter that makes electric forces. Its unit is the **coulomb (C)**.
- One electron has a charge of −e. One proton has +e.
- **e = 1.602176634 × 10^{-19} C.** This value has been exact since the SI redefinition of 20 May 2019.
- 1 C is the charge of about **6.24 × 10^{18} electrons**.
- Charge is **conserved**. It is never made or destroyed. It only moves.
- **Current (I)** is the rate at which charge flows past a point. I = Q/t.
- The unit of current is the **ampere (A)**. 1 A = 1 C/s. The unit is named after André-Marie Ampère.
- The ampere is one of the seven SI base units. The coulomb is a derived unit: 1 C = 1 A·s.
- Small currents use milliamps (mA, 10^{-3} A) and microamps (μA, 10^{-6} A).
- **Conventional current** flows from + to − through the outside circuit. Circuit diagrams and formulas use it.
- **Electron flow** goes the other way, from − to +, because electrons are negative.
- Both pictures give the same answers for every calculation. Only the arrow direction changes.
- Inside a battery, conventional current goes from − to +. The battery lifts the charge "uphill".
- **Current is not used up.** The same current leaves a bulb as enters it. The bulb uses energy, not charge.
- In a series loop, the current is the same at every point.
- At a junction, current in = current out. This is **Kirchhoff's current law (KCL)**. It comes from conservation of charge.
- An **ammeter** measures current. It goes **in series**, inside the path of the current. An ideal ammeter has 0 Ω.
- **DC** (direct current) flows one way. **AC** (alternating current) reverses direction many times per second. See CL 3 - DC and AC [T].
- **Battery capacity** is often given in **mAh** (milliamp-hours) or **Ah**. It is an amount of **charge**, not energy.
- **1 mAh = 3.6 C.** 1 Ah = 3600 C.
- A battery rated 2000 mAh can give 2000 mA for 1 h, or 200 mA for 10 h (in the simple model).
- **Drift velocity (v_d)** is the slow average speed of the free electrons along a wire. In household wires it is about 0.01 to 1 mm/s.
- Electrons crash into metal atoms all the time. They move in a random zig-zag at about 10^{6} m/s. Their **net** forward speed is tiny.
- A light still turns on at once. The electric **signal** (the field) travels along the wire at a large fraction of the speed of light. Electrons in the bulb start moving almost at once.
- **Free-electron density (n)** is the number of free electrons per cubic metre. It is a property of the material. For copper, n ≈ 8.5 × 10^{28} m^{-3} (some books use 8.34 × 10^{28}).
- **Current density (J)** is current per unit of cross-section area: J = I/A, in A/m^2.
- Typical currents: an indicator LED 10-20 mA; a phone charging 1-3 A; a house branch circuit up to 15 or 20 A; a car starter motor about 100-400 A; a lightning stroke about 30 kA at its peak.

Formulas

What	Formula	Units
Current	I = Q/t	A = C/s

What	Formula	Units
Charge	Q = I·t	C = A·s
Time	t = Q/I	s
Number of electrons	N = Q/e	(no unit)
Charge of N electrons	Q = N·e	C
mAh to coulombs	Q (C) = mAh × 3.6	C
Battery life (simple)	t (h) = capacity (mAh) / current (mA)	h
Charge from a current-time graph	Q = area under the I-t graph	C
Drift velocity ⚠️	I = n·q·A·v_d, so v_d = I/(n·q·A)	m/s
Current density ⚠️	J = I/A = n·q·v_d	A/m²
Ohm's law, micro form ⚠️	J = σE = E/ρ	A/m²
Junction (KCL)	ΣI_in = ΣI_out	A

Question types

Easy

1. Charge from current and time (Q = I·t) — ✅ IN

- Where seen: Berkeley 2026, B, INV, Q20 (30 mA for 2 minutes → 3.6 C). MIT 2020, C, INV, Q26 uses the same step inside an energy question.
- How to solve:
  1. Change the current to amperes (mA ÷ 1000).
  2. Change the time to seconds (min × 60, h × 3600).
  3. Multiply: Q = I·t.
- Worked example: 30 mA flows for 2.0 minutes. Q = 0.030 A × 120 s = **3.6 C**. A second example: 2.5 A for 3.0 min gives 2.5 × 180 = **450 C**.
- Traps: Using minutes instead of seconds gives 0.06 C, which is 60 times too small. Using 30 instead of 0.030 gives 3600 C.
- Calculator: calculate — type: current, time.

2. Current from charge and time (I = Q/t) — ✅ IN

- Where seen: Cobra Invitational 2019-20, B, INV, Q22 (2 C in 1 s → 2 A). ATX Regional 2019, B/C, REG, Q13 (30 C in 90 s → 0.333 A; a "6 mm" length is a distractor).
- How to solve:
  1. Change the time to seconds.
  2. Divide: I = Q/t.
  3. Ignore numbers that do not matter, such as the wire length.
- Worked example: 30 C passes a point in 90 s. I = 30/90 = **0.33 A** (333 mA).
- Traps: The test may add a wire length or a diameter. Current does not depend on them here.
- Calculator: calculate — type: charge, time.

3. Time for a given charge (t = Q/I) — ✅ IN

- Where seen: MIT 2019, C, INV, Q17 (12.0 C drained at 5.0 mA → 2400 s).
- How to solve:
  1. Change mA to A.
  2. Divide: t = Q/I.
  3. Change seconds to minutes or hours if asked.
- Worked example: 12.0 C at 5.0 mA. t = 12.0/0.0050 = **2400 s = 40 min**.
- Traps: 12.0/5.0 = 2.4 s is wrong. The milli prefix makes the time 1000 times longer.
- Calculator: calculate — type: charge, current.

4. Conventional current vs electron flow (direction) — ✅ IN

- Where seen: Pembroke Hill Dec 2025 (2026 season), B, INV, Q8 (T/F "conventional current flows the same way as electrons" → False). NY Lake Erie/Niagara Dec 2018, C, INV, Q4. JC Booth 2011 (Shock Value), B, INV, Q31 ("current from – to +" → electron current) and Q52 (conventional = plus to minus). Shock Value tryout about 2012, B, practice, I-5. UF Northern Regional 2020, B, REG, Q20 (draw the electron-flow direction on a figure).
- How to solve:
  1. Find the + terminal of the battery (the long line in the symbol).
  2. Conventional current leaves + and goes around the outside circuit to –.
  3. Electrons go the opposite way: out of –, around, into +.
- Worked example: A battery has + on the top. The wire goes from + to the right, down through a bulb, and back to –. Conventional current goes **clockwise**. Electrons go **counterclockwise**.
- Traps: Inside the battery, conventional current goes from – to +. A diode or LED arrow points in the conventional direction.
- Calculator: look up — type: conventional current, electron flow.

5. Number of electrons in a charge (N = Q/e) — ✅ IN

- Where seen: not seen in a test for this area. It is common in OpenStax College Physics 2e, chapters 18 and 20. It also appears in static questions (see CL 2 - Static Electricity and Coulomb's Law [T]).
- How to solve:
  1. Find the charge Q (often Q = I·t first).
  2. Divide by e = 1.602×10^{-19} C.
- Worked example: 2.0 mA flows for 10 s. Q = 0.020 C. N = $0.020/1.602 \times 10^{-19}$ = **1.25×10^{17} electrons**. A second fact: 1 A is about 6.24×10^{18} electrons each second.
- Traps: Multiplying by e instead of dividing gives a tiny number. The answer must be huge.
- Calculator: calculate — type: charge, electrons.

6. Name the unit or quantity — ✅ IN

- Where seen: UF Northern Regional 2020, B, REG, Q4 (SI unit of charge → coulomb). JC Booth 2011, B, INV, Q14 (units of I, V, R, P → amperes, volts, ohms, watts; the B part is OUT). MYSO practice Nov 2020, B, Q8 (unit of resistance).
- How to solve: Learn the table in Units Prefixes and Sig Figs [A4].
- Worked example: "What is the SI unit of charge?" → **coulomb (C)**. "What is 1 C/s?" → **1 ampere**.
- Traps: The ampere is the base unit. The coulomb is derived (A·s). Some tests ask which one is "base".
- Calculator: look up — type: unit, charge, current.

7. Current is the same everywhere in a series loop ("current is not used up") — ✅ IN

- Where seen: not seen as its own question in these tests. It is a classic misconception item in Kuphaldt's Socratic worksheets and in the soinc practice deck 1.
- How to solve:
  1. Trace the loop. If there is only one path, the current is the same at every point.
  2. Bulbs and resistors change the **voltage** and use **energy**. They do not use up charge.
- Worked example: Three bulbs in series. An ammeter before bulb 1 reads 0.25 A. An ammeter after bulb 3 reads **0.25 A**.
- Traps: "The first bulb uses up some current, so the last bulb is dimmer." This is false for identical bulbs in series.
- Calculator: look up — type: series current.

8. Simple junction: current in = current out — ✅ IN as a concept; ✅ States-Nationals for full KVL/KCL analysis

- Where seen: Berkeley 2026, B, INV, Q19 (two wires in, one is 4 A; one wire out with 7 A → the other is 3 A). F-M Invitational 2018, INV, Q10 (KCL = conservation of charge). GullSO Dec 2025, C, INV, Q38 (KCL junction → 7 A, 9 A).
- How to solve:
  1. Add all currents going in.
  2. Set it equal to the sum of all currents going out.
  3. Solve for the one unknown.

- Worked example: 4 A and I_x go in. 7 A goes out. 4 + I_x = 7, so I_x = **3 A**.
- Traps: A negative answer means the current really goes the other way. It is not an error.
- Calculator: calculate — type: junction, currents in, currents out.

Medium

9. mAh and Ah to coulombs (and back) — ✅ IN

- Where seen: not seen as a stand-alone question. It sits under the 2027 hands-on task "USB charger … how long to charge a battery" (rule Part II.c.iv). The prior topic note lists 1 mAh = 3.6 C.
- How to solve:
  1. mAh ÷ 1000 = Ah.
  2. Ah × 3600 s = coulombs.
  3. Short cut: C = mAh × 3.6.
- Worked example: a 3000 mAh phone battery holds 3.000 Ah × 3600 s = **10,800 C**.
- Traps: mAh is charge, not energy. To get energy you also need the voltage (Wh = Ah × V). See Power Energy and Cost [A4].
- Calculator: calculate — type: mAh, coulomb.

10. Battery life from capacity and current draw — ✅ IN

- Where seen: not seen in a test with mAh and current only. Close relatives: UT Invitational 2019, B, INV, Q28 (3000 mAh, 4.4 V, 6 h → power) and UT Invitational fall 2018, B/C, INV, Q8 (42,000 mWh at 15 W). The diode research also has an LED example (500 mAh at 20 mA → 25 h).
- How to solve:
  1. Make the units match: capacity in mAh and current in mA.
  2. t (h) = capacity ÷ current.
  3. If the test gives a usable fraction (for example 80%), multiply the capacity by it first.
- Worked example: a 2500 mAh battery runs a 20 mA LED circuit. t = 2500/20 = **125 h**. If only 80% is usable: **100 h**.
- Traps: Do not divide by the voltage. Voltage is not needed for this method when the battery runs the load directly.
- Calculator: calculate — type: mAh, mA.

11. Charge-time with a rectifier: half-wave vs full-wave — ✅ IN (diode use, rule 3.c.vii)

- Where seen: UT Invitational fall 2019, C, INV, Q45 part 1 (full-wave takes 3 h → half-wave takes 6 h).
- How to solve:
  1. A half-wave rectifier passes current for only half of each AC cycle.
  2. So the average charging current is about half.
  3. So the time is about twice as long.
- Worked example: a full-wave charger fills a battery in 3 h. The same battery on a half-wave charger takes about **6 h**.
- Traps: The capacitor-smoothing part of the same question is OUT (capacitance, rule 3.d). See Diode Applications [A7].
- Calculator: look up — type: half wave, full wave, charge time.

12. Charge from a current-vs-time graph (area under the line) — ⚠️ Borderline

- Why borderline: the area of rectangles and triangles is geometry, which is allowed. A curved graph would need calculus, which is OUT (rule 3.e.v).
- Where seen: MIT 2020, C, INV, Q11-12 (battery current over 15 ks → 110 kC; energy 1.29 MJ). I could not re-check those numbers because I did not see the graph.
- How to solve:
  1. Split the graph into rectangles and triangles.
  2. Area of a rectangle = I × t. Area of a triangle = ½ × I × t.
  3. Add the areas. The unit is A·s = C.

- Worked example: a current is 2.0 A for 10 s. Then it falls in a straight line to 0 over the next 10 s. Q = 2.0 × 10 + ½ × 2.0 × 10 = **30 C**.
- Traps: Use seconds on the time axis. If the axis is in hours, the area is in A·h. Multiply by 3600 to get coulombs.
- Calculator: calculate — type: current, time, triangle, rectangle.

13. Drift speed: order of magnitude and "why the light turns on at once" — ✅ IN (as a concept)

- Where seen: Florida State 2021, B, ST, Q12 (drift speed in a phone charger cable → about 10^{-3} m/s was the nearest choice). MYSO practice Nov 2020, B, Q5 (average electron speed when a bulb lights at once → less than 1 cm/s). ATX Regional 2020, B, REG, Q43 (drift velocity is the velocity in an E field). Hawk & Hornet 2026, C, INV, Q20 (answer in mm/s).
- How to solve:
  1. Remember the size: about 0.01 to 1 mm/s in normal wires.
  2. The signal moves at a large fraction of the speed of light. The electrons themselves move very slowly.
  3. The wire is already full of free electrons. They all start to drift at almost the same time.
- Worked example: "Electrons drift at about 0.1 mm/s. How long to cross 1 m of wire?" t = 1/0.0001 = 10,000 s, which is **about 3 hours**. The bulb still lights in **far less than a second**.
- Traps: "Electrons move at the speed of light" is false. "Electrons must travel from the switch to the bulb first" is false.
- Calculator: look up — type: drift speed, signal speed.

Hard

14. Drift velocity calculation (I = nqAv_d) — ⚠️ Borderline

- Why borderline: the rule says "concepts, calculations … of current", but it does not name drift velocity. The math is only algebra.
- Where seen: Rickards Oct 2025 (2026 season), C, INV, Q51-52 and Q59 (one answer is 2.51 A). Florida State 2021, B, ST, Q12 (my check: 2 A in 1 mm² copper → 1.5 × 10^{-4} m/s). ATX Regional 2026, C, REG, MC17 (a drift formula with collision time; the key is wrong, see below).
- How to solve:
  1. Find the area A. For a round wire, A = π(d/2)². Change mm² to m² (× 10^{-6}).
  2. Use n for the metal (copper 8.5 × 10^{28} m^{-3} unless given).
  3. v_d = I/(n·e·A).
  4. For the reverse problem, I = n·e·A·v_d.
- Worked example: 2.0 A flows in copper wire with A = 1.0 mm². v_d = 2.0/(8.5 × 10^{28} × 1.602 × 10^{-19} × 1.0 × 10^{-6}) = **1.5 × 10^{-4} m/s** (0.15 mm/s). To move 1 m takes 6800 s, about **1.9 h**.
- A reverse example: v_d = 1.0 × 10^{-4} m/s in 1.0 mm² copper gives I = **1.4 A**.
- Traps: Forgetting to change mm² to m² gives an answer 10^{6} times too big. Using the diameter as the radius gives an area 4 times too big.
- Calculator: calculate — type: current, n, area, drift velocity.

15. Drift velocity scaling (thicker wire, more current, other metal) — ⚠️ Borderline

- Where seen: ATX Regional 2026, C, REG, MC16 (wire A has twice B's radius: electron density ratio; the key is wrong, see below).
- How to solve:
  1. n depends only on the material. A thick copper wire and a thin copper wire have the **same n**.
  2. With the same current, v_d is proportional to 1/A, so to 1/r².
  3. With the same area, v_d is proportional to I.
- Worked example: two copper wires carry the same current. Wire A has twice the radius of wire B. Then n_A = n_B, and v_A = **¼ v_B**.
- Traps: "Bigger wire, more electrons per volume" is false. It has more electrons in total, but the same number per cubic metre.
- Calculator: look up — type: drift, radius, ratio.

16. Current density (J = I/A) and J = σE — ⚠️ Borderline

- Why borderline: not named in the rule; it is still only algebra.
- Where seen: Rickards Oct 2025 (2026 season), C, INV, Q60 (ρ = E/J = 0.4 Ω·m). Princeton 2021, C, INV, Q34 (σ = 0.083).
- How to solve:
  1. J = I/A.
  2. J = σE, where σ (sigma) is conductivity, σ = 1/ρ.
  3. So E = J·ρ, or ρ = E/J.
- Worked example: 2.0 A in 1.0 mm² copper. J = 2.0/1.0 × 10^{-6} = **2.0 × 10^{6} A/m²**. The field in the wire is E = J·ρ = 2.0 × 10^{6} × 1.68 × 10^{-8} = **0.034 V/m**.
- Traps: Area must be in m². E is in V/m, which is the same as N/C.
- Calculator: calculate — type: current density, conductivity, field.

Nationals

17. Junction currents inside a full circuit (KCL with KVL) — ✅ States-Nationals

- Where seen: BullSO Jan 2026, C, INV, SA6-8 (mesh currents 5 A and 2.5 A; nodal voltages). BullSO Jan 2026, B, INV, free response Q6 (one loop, 20 V + 12 V sources → 0.8 A, V1 = 14.4 V, V2 = 17.6 V) and Q7 (nodal analysis). GullSO Dec 2025, C, INV, Q38. Full list in Kirchhoff Analysis (States-Nationals) A5.
- How to solve:
  1. Label each branch current with an arrow.
  2. Write KCL at one node: in = out.
  3. Write KVL around loops. Keep to two unknowns (rule 3.e.i).
  4. Solve the two equations.
- Worked example: a node has I_1 = 2.0 A coming in from the left. I_2 leaves down. I_3 leaves to the right. A loop equation gives I_3 = 0.5 A. KCL: 2.0 = I_2 + 0.5, so I_2 = **1.5 A**.
- Traps: Pick a direction for each current and keep it. A negative result is fine.
- Calculator: calculate — type: KCL, two equations.

18. Drift-velocity model with collision time (v_d = eEτ/m) — ⚠️ Borderline (probably above Div B)

- Why borderline: the formula is algebra, but the Drude model goes past "concepts of current". Treat it as rare.
- Where seen: ATX Regional 2026, C, REG, MC17 (v_d = 4.5 × 10^{-5} m/s, τ = 0.2 ns → E).
- How to solve:
  1. E = m·v_d/(e·τ). Use m = 9.109 × 10^{-31} kg.
  2. Change ns to s (× 10^{-9}).
- Worked example: v_d = 4.5 × 10^{-5} m/s and τ = 0.2 ns. E = 9.109 × 10^{-31} × 4.5 × 10^{-5} / (1.602 × 10^{-19} × 0.2 × 10^{-9}) = **1.28 × 10^{-6} V/m**.
- Traps: The test's key gave 1.28 × 10^{-9} V/m, which needs τ = 0.2 μs. Check prefixes.
- Calculator: calculate — type: drift velocity, tau, field.

Out of scope (skip)

- **Current in a capacitor while it charges** (RC circuits, time constants). ❌ OUT: capacitance, rule 3.d.
- **Current induced by a moving magnet or changing flux** (Faraday's law, motional EMF). ❌ OUT: magnetism is not in the 3.c list; inductance is in 3.d.
- **Force on a current-carrying wire** (F = ILB) and the field around a wire. ❌ OUT: magnetism.
- **AC current with phase angles, RMS from a sine with trig, or impedance.** ❌ OUT: AC circuit theory and trig (rules 3.d and 3.e.v). The simple fact "RMS = peak/√2" is in CL 3 - DC and AC T.
- **Transistor currents (β = I_C/I_B).** ❌ OUT: semiconductors beyond the PN junction (rule 3.d). Princeton 2021, C, Q33 is an example.
- **Radial current density with ln r** (USC Jan 2026, C, Q39). ❌ OUT: calculus (rule 3.e.v).
- **Current from a curved I-t graph by integration.** ❌ OUT: calculus.

Mistakes in tests and keys

- **ATX Regional 2026, C, REG, MC16.** The key says n_B = 4 n_A for a wire with half the radius. Free-electron density is a material property, so n_B = n_A. Even "per unit length" would give ¼, not 4.
- **ATX Regional 2026, C, REG, MC17.** The key says $E = 1.28 \times 10^{-9}$ V/m. With $\tau = 0.2$ ns the right answer is 1.28×10^{-6} V/m (checked in Python). No option was correct.
- **Florida State 2021, B, ST, Q12.** The keyed answer (10^{-3} m/s) is only the nearest option. A real 2 A charger cable with 1 mm² copper gives about 1.5×10^{-4} m/s.

Not confirmed

- MIT 2020, C, Q11-12 (110 kC, 1.29 MJ): I did not see the graph, so I could not re-check these numbers. Source: prior/G_tests.md.
- Rickards 2026, C, Q51-52 and Q59 (2.51 A) and Princeton 2021 Q34 ($\sigma = 0.083$): numbers come from prior/G_tests.md. I did not re-read the question texts.
- Hawk & Hornet 2026 Q20: only "answer in mm/s" is recorded. I did not see the full question.
- The copper n value: 8.5×10^{28} m^{-3} is the common textbook value. OpenStax uses 8.342×10^{28} m^{-3}. Answers differ by about 2%.
- Typical current values (car starter, lightning) are general knowledge. I did not check them in a source in this run.
- No real **Nationals** Circuit Lab test was available. The "USC 2026" files are the USC Invitational, not the 2026 Nationals. So the Nationals types here come from the rules and from top invitationals.

Sources

- 2027 Div B rules, Circuit Lab, rules 3.b, 3.c.iv, 3.c.ix, 3.d, 3.e, Part II.c.iv.
- Test extractions in `circuitlab/research/tests/`: misc-a (Cobra, F-M), misc-b (MIT 2019, UF Northern Regional, NY Lake Erie, JC Booth, Shock Value tryout), atx-reg (ATX Regional 2020-21), ut-a (UT Fall 2025, ATX Regional 2026), ut-c (UT 2018-19, ATX Regional 2019), state (Florida State 2021, MYSO 2020), nat26 (USC Invitational 2026).
- 2026 Div B tests: Berkeley (Feb 28, 2026), BullSO (Jan 24, 2026), Pembroke Hill (Dec 6, 2025), from `downloads/dlB/tests/`.
- prior/G_tests.md (Rickards, Princeton, MIT 2020, GullSO, HH), prior/I_bank.md (U5, U17, U18), prior/B_topics.md §4 and §9d, prior/H_logic_diodes.md (B19).
- OpenStax College Physics 2e, chapter 20.1 (current, drift velocity) and 18 (charge): general knowledge of the book; I did not re-open it in this run.
- Kuphaldt, *Lessons in Electric Circuits*, Vol. I, ch. 1-2, and the Socratic worksheets: general knowledge.
- BIPM, SI Brochure 9th edition (2019): exact value of e.

Voltage and EMF

This note covers voltage (potential difference), EMF, the electron volt, and common source voltages.

Rule 3.c.iv names "concepts, calculations, and units of … voltage". So this note is ✅ **IN at every level**, with three exceptions:

- Voltage loops with two or more sources need KVL, so they are ✅ **States/Nationals only** (rule 3.c.ix).
- Electrochemistry numbers (standard electrode potentials) are ⚠️ **borderline**. Batteries are DC sources (rule 3.c.iii), but the chemistry is not named.
- Speed of a charge after it crosses a voltage is ⚠️ **borderline**. It mixes static electricity (rule 3.c.ii) with mechanics.

Key facts

- **Voltage (V)** is the energy given to, or taken from, each coulomb of charge. Another name is **potential difference**.
- **1 volt = 1 joule per coulomb** (1 V = 1 J/C). Also 1 V = 1 W/A.
- The unit is named after Alessandro Volta. He built the first battery (the voltaic pile) in 1800.
- Voltage is always **between two points**. "The voltage at A" means the voltage between A and a chosen reference point.
- The reference point is called **ground** or **common**. It is set to 0 V.
- Voltage is a **scalar**. It has a size and a sign, but no direction in space.
- **Electric potential** at a point is the energy per charge there. Voltage between A and B is V_AB = V_A – V_B.
- If V_AB is positive, A is at the higher potential. Positive charge "falls" from A to B and gives up energy.
- **EMF (ε, electromotive force)** is the voltage a source makes when no current flows. It is a voltage, not a force.
- A real battery has a small **internal resistance r**. Under load, the terminal voltage is V_t = ε – I·r. Details are in Internal Resistance and Power Transfer [A5].
- A **voltmeter** measures voltage. It goes **in parallel**, across the part. An ideal voltmeter has infinite resistance.
- A real voltmeter with low resistance "loads" the circuit. It reads **lower** than the true voltage.
- Voltage across an ideal wire (or a closed switch) is 0 V. Voltage across an **open switch** in a working series circuit equals the full supply voltage.
- Parts in **parallel** all have the **same voltage**.
- In a **series** loop, the voltage drops add up to the source voltage. This is **Kirchhoff's voltage law (KVL)**. It comes from conservation of energy.
- **Water analogy:** voltage is like water pressure, current like flow, resistance like a narrow pipe, and a battery like a pump.
- **Electron volt (eV)** is an energy unit. It is the energy one electron gains when it crosses 1 V. 1 eV = 1.602176634 × 10^{-19} J (exact since 2019).
- The energy a charge gains or loses is W = q·V.
- Cells in **series** add their voltages. Cells in **parallel** keep the same voltage, but add their capacity.
- A cell put in **backwards** in a series string subtracts its voltage.
- "Dead", "0 V" and "ground" are words for zero voltage. "Live" or "hot" means a wire at a dangerous voltage.

Common voltages (good to know for estimation questions):

Source	Voltage
Alkaline or carbon-zinc cell (AA, AAA, C, D)	1.5 V (the size changes capacity, not voltage)
NiMH or NiCd rechargeable cell	1.2 V
Lithium coin cell (CR2032)	3 V
Li-ion cell (phones)	3.6-3.7 V nominal, 4.2 V full
9 V battery	9 V (six 1.5 V cells inside)
Lead-acid cell	about 2.0-2.1 V
Car battery	12 V (six lead-acid cells)
Large truck system	24 V

Source	Voltage
USB port	5 V (USB-C PD also 9, 15, 20 V)
US home outlet	120 V AC (240 V for dryers and ovens)
Potato or lemon cell (Cu and Zn)	about 0.7-1.2 V
Nerve impulse	about 0.07 V
Transmission lines	thousands to hundreds of thousands of volts

Formulas

What	Formula	Units
Definition	$V = W/q$	V = J/C
Energy gained or lost by a charge	$W = q \cdot V$	J
Ohm's law	$V = I \cdot R$	V
From power	$V = P/I$, $V = \sqrt{(P \cdot R)}$	V
Potential difference	$V_{AB} = V_A - V_B$	V
Terminal voltage (discharging)	$V_t = \varepsilon - I \cdot r$	V
Cells in series	$V_{total} = V_1 + V_2 + \ldots$	V
eV to joules	$E\,(J) = E\,(eV) \times 1.602 \times 10^{-19}$	J
Speed after crossing V ⚠	$\frac{1}{2}mv^2 = qV$, so $v = \sqrt{(2qV/m)}$	m/s
Field between plates (static note)	$E = V/d$	V/m

Question types

Easy

1. Define the volt; which units equal a volt — ✅ IN

- Where seen: Pembroke Hill Dec 2025 (2026 season), B, INV, Q33 (the volt = joule per coulomb; distractors C/s, W per A... note W/A is also a volt, see Mistakes). UT Invitational fall 2018, B/C, INV, Q2 (unit of "electrical pressure" or EMF → volt). UF Northern Regional 2019, C, REG, Q23-27 (volt in SI base units).
- How to solve:
  1. Know V = J/C.
  2. Know V = W/A (because W = J/s and A = C/s).
  3. Know V = A·Ω (from Ohm's law).
- Worked example: "Which is a volt: C/s, A/s, J/C?" → **J/C**. C/s is an ampere.
- Traps: Watt per ampere (W/A) is also equal to a volt. If two choices are both correct, tell the supervisor.
- Calculator: look up — type: volt, unit.

2. Voltage is a scalar (T/F) — ✅ IN

- Where seen: ATX Regional 2020, B, REG, Q23 (T/F "voltage is a vector" → False).
- How to solve: Voltage is energy per charge. Energy has no direction. So voltage is a scalar.
- Worked example: "Voltage is a vector." → **False**. The electric **field** is a vector. Voltage is not.
- Traps: Voltage can be negative. A sign is not a direction.
- Calculator: look up — type: scalar, vector, voltage.

3. Energy from charge and voltage (W = qV) — ✅ IN

- Where seen: Princeton 2021, C, INV, Q32 (2 mC between plates at 10 V → key 10 mJ; see Mistakes). ATX Regional 2019, B/C, REG, Q15 (2 C moves with 10 J → V = 5 V). The prior bank has 9.0 V moving 3.0 C → 27 J.
- How to solve:
  1. Change mC or μC to C.
  2. W = q·V.
  3. To find V instead: V = W/q.

- Worked example: a 9.0 V battery moves 3.0 C of charge. W = 3.0 × 9.0 = **27 J**. Another example: 10 J is used to move 2 C, so V = 10/2 = **5 V**.
- Traps: If the charge starts **between** two plates, it only crosses part of the voltage.
- Calculator: calculate — type: charge, voltage, energy.

4. Electron volt conversions — ✅ IN (energy unit, rule 3.c.iv)

- Where seen: BullSO Jan 2026, B, INV, Q27 (answers given in J and in eV; the physics there is magnetism, which is OUT). The diode research uses eV for LED photon energy (see LEDs A7). Not seen as a stand-alone conversion.
- How to solve:
  1. An electron (or proton) that crosses V volts gains V electron volts.
  2. A charge of 2e (an alpha particle) that crosses V gains 2V eV.
  3. To get joules, multiply eV by 1.602×10^{-19}.
- Worked example: an electron crosses 5.0 V. It gains **5.0 eV = 8.0×10^{-19} J**. A proton crosses 1000 V: **1000 eV = 1.6×10^{-16} J**. An alpha particle crosses 100 V: **200 eV**.
- Traps: The eV is energy, not voltage. "1 eV" is not "1 V".
- Calculator: calculate — type: eV, joule.

5. Cells in series and parallel: total voltage — ✅ IN

- Where seen: soinc practice deck 3 (lemon cells 0.7 V each; how many in series to pass 2 V → 3). Northview Dec 2025, C, INV, Q47 (truck battery voltage → 24 V). prior/B_topics §3b (a 9 V battery is six 1.5 V cells).
- How to solve:
  1. Series: add the voltages. A reversed cell subtracts.
  2. Parallel (identical cells): the voltage stays the same as one cell.
  3. To reach a target, divide and round **up**.
- Worked example: four 1.5 V AA cells in series give **6.0 V**. The same four in parallel give **1.5 V**. Four in series with one reversed give 1.5 + 1.5 + 1.5 − 1.5 = **3.0 V**. Lemon cells at 0.7 V: 2/0.7 = 2.86, so **3 cells** (2.1 V).
- Traps: Rounding 2.86 down to 2 gives only 1.4 V. Always round up for "at least".
- Calculator: calculate — type: cells, series, parallel, voltage.

6. Vocabulary: ground, dead, 0 V, hot — ✅ IN

- Where seen: Florida State 2021, B, ST, Q22 (odd one out: 0 V, ground, dead, equilibrium → key "dead"; this is debatable, see Mistakes).
- How to solve:
  1. "Ground" is the 0 V reference.
  2. "Dead" in trade slang means no voltage (safe to touch).
  3. "Hot" or "live" means at a dangerous voltage.
- Worked example: "Which word does NOT mean zero voltage: ground, dead, hot?" → **hot**.
- Traps: Read the choices for slang meanings before you answer.
- Calculator: look up — type: ground, dead, hot.

7. Water analogy — ✅ IN

- Where seen: soinc practice deck 1 (UNIT-WATER-ANALOGY). Not seen in these tests.
- How to solve: Map each part.
  1. Voltage → water pressure (or height difference).
  2. Current → flow rate.
  3. Resistance → a narrow pipe.
  4. Battery → pump.
- Worked example: "What does a narrow pipe stand for?" → **a resistor (resistance)**.
- Traps: Pressure is not flow. A high voltage with an open switch gives no current.
- Calculator: look up — type: water analogy.

8. Typical voltage of a source (estimation) — ✅ IN

- Where seen: Northview Dec 2025, C, INV, Q47 (truck battery 24 V). JC Booth 2011, B, INV, Q12 (a light string on 24 V, 0.78 A).
- How to solve: Learn the common-voltages table above.
- Worked example: "A typical phone battery is about …" → **3.7 V**. "A car battery?" → **12 V**.
- Traps: AA and D cells have the same voltage. Bigger means more capacity, not more volts.
- Calculator: look up — type: battery voltage.

Medium

9. Potential difference between two points (V_AB = V_A – V_B) — ✅ IN

- Where seen: not seen as its own question in these tests. It is used inside Wheatstone-bridge questions (see CL 5 - DC Circuits T). Kuphaldt Socratic worksheets use it often.
- How to solve:
  1. Find the voltage of each point relative to ground.
  2. Subtract: V_AB = V_A – V_B.
  3. The sign tells which point is higher.
- Worked example: V_A = 9 V and V_B = 3 V. V_AB = **+6 V**. V_BA = **–6 V**.
- Traps: A voltmeter with its red lead on B and black lead on A reads –6 V.
- Calculator: calculate — type: node voltage, difference.

10. Voltage across an open switch, a closed switch, a blown fuse — ✅ IN

- Where seen: not seen with this exact wording in these tests. Pembroke Hill Dec 2025, B, INV, Q24 asks open vs short from a 0.1 Ω reading. It is a core troubleshooting skill (hands-on task Part II.c.vi). Kuphaldt, *Lessons in Electric Circuits* Vol. I ch. 5.
- How to solve:
  1. No current flows in a series loop with an open switch.
  2. So every resistor has 0 V across it (V = I·R = 0).
  3. So the whole source voltage appears across the open gap.
  4. A closed switch or a good wire has 0 V across it.
- Worked example: a 9 V battery, a switch and a lamp in series. With the switch open, a voltmeter across the switch reads **9 V**. Across the lamp it reads **0 V**.
- Traps: "No current means no voltage anywhere" is false. The gap holds the full voltage.
- Calculator: look up — type: open switch, voltage.

11. Energy, charge and time through a branch — ✅ IN

- Where seen: ATX Regional 2019, B/C, REG, Q15 (30 Ω ‖ 10 Ω on a source; 2 C moves from A to B with 10 J → 3 s).
- How to solve:
  1. Find V from V = W/q.
  2. Find the current along the path the charge takes (I = V/R).
  3. Find the time from t = q/I.
- Worked example: 2 C moves with 10 J, so V = 5 V. The source branch feeds 30 Ω ‖ 10 Ω = 7.5 Ω, so I = 5/7.5 = 0.667 A. Time = 2/0.667 = **3.0 s**. Through only the 10 Ω branch it would be 4 s. Through the 30 Ω branch it would be 12 s.
- Traps: The question did not say which branch. Write your assumption.
- Calculator: calculate — type: energy, charge, voltage, time.

12. Terminal voltage, EMF and internal resistance — ✅ IN

- Where seen: Pembroke Hill Dec 2025, B, INV, Q4 (9.0 V EMF, 8.5 V across 10 Ω → r = 0.59 Ω). UF Northern Regional 2020, B, REG, Q38 (12 V, 11.9 V across 50 Ω → 0.420 Ω). UT Invitational 2020 (2021 season), B, INV, Q65 (EMF 3.00 V, r = 0.24 Ω, terminal 2.91 V → R_L = 7.76 Ω). UT Invitational 2019 (2020 season), B, INV, Q24 (data table → r ≈ 0.51 Ω).
- How to solve:

1. Lost voltage = ε – V_t.
2. Current: I = (ε – V_t)/r, or I = V_t/R.
3. Then find the unknown (r or R).

- Worked example: ε = 3.00 V, r = 0.24 Ω, V_t = 2.91 V. I = 0.09/0.24 = 0.375 A. R_L = 2.91/0.375 = **7.76 Ω**.
- Traps: A voltmeter on a battery with no load reads the EMF, not the working voltage.
- More types: see Internal Resistance and Power Transfer [A5] (all 4 forms, data-table slope, cells in series and parallel).
- Calculator: calculate — type: emf, terminal voltage, internal resistance.

13. Voltmeter loading (a real meter reads low) — ✅ IN

- Where seen: Pembroke Hill Dec 2025, B, INV, Q27 (low-resistance voltmeter → reads lower).
- How to solve:
  1. The meter's resistance is in parallel with the part.
  2. The parallel pair has a lower resistance than the part alone.
  3. So it takes a smaller share of the voltage.
- Worked example: 10 V across two 10 kΩ resistors in series. The true voltage across the lower one is 5 V. A 10 kΩ voltmeter across it makes 10 kΩ ‖ 10 kΩ = 5 kΩ. It reads 10 × 5/15 = **3.3 V**.
- Traps: An ammeter has the opposite problem. Its resistance must be near 0 Ω.
- Calculator: calculate — type: voltmeter, loading, divider.

Hard

14. Speed of a charge after it crosses a voltage — ⚠️ Borderline

- Why borderline: it uses W = qV (IN) and kinetic energy ½mv² (mechanics, not named). Only algebra and a square root are needed.
- Where seen: MIT 2021, C, INV, Scenario A Q15 (final speed from potential energy → kinetic energy). Not seen in a Div B test.
- How to solve:
  1. Energy gained: W = q·V.
  2. Set W = ½mv².
  3. v = √(2qV/m).
- Worked example: an electron starts at rest and crosses 100 V. v = $\sqrt{(2 \times 1.602 \times 10^{-19} \times 100 / 9.109 \times 10^{-31})}$ = **5.9×10^{6} m/s**.
- Traps: Use the electron mass 9.109×10^{-31} kg, not the proton mass. The answer must be less than the speed of light (3×10^{8} m/s).
- Calculator: calculate — type: charge, voltage, mass, speed.

15. Voltage of an electrochemical cell (E°cell) — ⚠️ Borderline

- Why borderline: batteries are DC sources (rule 3.c.iii), and Volta is a listed scientist. Standard electrode potentials are chemistry, not named in the rule.
- Where seen: MIT 2020, C, INV, Q42 (Cu/Ag cell → 0.46 V, voltaic). Rickards Oct 2025, C, INV, Q30 and Northview Dec 2025, C, INV, Q32 (cell questions).
- How to solve:
  1. Find the two reduction potentials (Ag^{+}/Ag = +0.80 V, Cu^{2+}/Cu = +0.34 V, Zn^{2+}/Zn = –0.76 V).
  2. E°cell = E°(cathode) – E°(anode). The cathode has the higher value.
  3. Positive E°cell means the cell runs on its own (voltaic).
- Worked example: a Cu/Ag cell. E°cell = 0.80 – 0.34 = **0.46 V**. Silver is the cathode (+). Copper is the anode (–).
- Traps: Do not multiply the potential by the number of electrons. Voltage does not scale with the amount.
- Calculator: look up — type: electrode potential, cell voltage.

16. Potato, lemon or kitchen battery: polarity and parts — ✅ IN

- Where seen: soinc practice deck 3 (Cu-Zn potato about 1.2 V; Cu-steel lemon about 0.7 V). Pembroke Hill Dec 2025, B, INV, Q18 (oxidation happens at the anode).

- How to solve:
  1. The more reactive metal (zinc) is the **anode**, the – terminal. It is oxidized (loses electrons).
  2. The less reactive metal (copper) is the **cathode**, the + terminal.
  3. The fruit juice is the **electrolyte**.
- Worked example: copper and zinc in a lemon. The + terminal is **copper**. The – terminal is **zinc**.
- Traps: One fruit cell gives very little current. A string of cells can light an LED only if the total voltage passes the LED's forward voltage.
- Calculator: look up — type: kitchen battery, anode, cathode.

Nationals

17. Loop with two sources (voltages add or oppose) — ✅ States-Nationals

- Where seen: soinc homework generator Level 7 (one loop, three sources: V_eq = $V_1 + V_2 – V_3$). BullSO Jan 2026, B, INV, free response Q6 (one loop with a 20 V and a 12 V source that add: 32 V over 22 Ω + 18 Ω → 0.8 A, V1 = 14.4 V, V2 = 17.6 V). Full types in Kirchhoff Analysis (States-Nationals) [A5].
- How to solve:
  1. Pick a direction around the loop.
  2. Add the source voltages that push that way. Subtract the ones that push the other way.
  3. I = V_net / R_total.
- Worked example: a 12 V and a 9 V battery face each other in one loop with 1.5 Ω in total. V_net = 12 – 9 = 3 V. I = 3/1.5 = **2.0 A**. The 12 V battery is charging the 9 V battery.
- Traps: Two sources that face the same way add. Facing opposite ways, they subtract.
- Calculator: calculate — type: two sources, loop, KVL.

Out of scope (skip)

- **Induced EMF from a changing magnetic flux** (Faraday's law, Lenz's law, motional EMF ε = BLv). ❌ OUT: magnetism (not in the 3.c list) and inductance (rule 3.d).
- **Transformer voltage ratios** ($V_s/V_p = N_s/N_p$). ❌ OUT: transformers work by magnetism and inductance. The plain fact "transformers step AC voltage up or down" is in CL 3 - DC and AC [T].
- **Voltage on a capacitor** (Q = CV, RC charging). ❌ OUT: capacitance, rule 3.d.
- **Op-amp output voltages** (gain, buffers). ❌ OUT: op-amps are semiconductors beyond the PN junction (rule 3.d).
- **AC peak, RMS and phase with trig or phasors.** ❌ OUT: AC circuit theory and trig (rules 3.d, 3.e.v).
- **Photoelectric-effect voltages** (work function, stopping voltage). ❌ OUT: not in any listed area. UT Invitational 2019, B, Q22 is an example.
- **Thevenin voltage of a network.** ❌ OUT: rule 3.d.

Mistakes in tests and keys

- **Princeton 2021, C, INV, Q32.** A 2 mC charge "is placed between the plates" of a capacitor with 10 V across it. It moves to one plate. The key says 10 mJ. That is only true if the charge starts at the midpoint (it crosses 5 V). If it starts at the other plate, the answer is 2 mC × 10 V = 20 mJ (checked in Python). The question does not say where it starts. The prior file G_tests.md copied it as "2 mC × 10 V → 10 mJ", which is wrong as written.
- **Pembroke Hill Dec 2025, B, INV, Q33.** The key is "Joule per Coulomb". Choice B, "Watt per Ampere", is also a correct definition of the volt (W/A = (J/s)/(C/s) = J/C). Two choices are correct.
- **Florida State 2021, B, ST, Q22.** The key's "odd one out" is "dead". In trade slang "dead" also means 0 V. The question is debatable.
- **ATX Regional 2019, B/C, REG, Q15.** The question does not say which branch the charge goes through. The answer can be 3 s, 4 s or 12 s.

Not confirmed

- MIT 2021 Scenario A Q15, MIT 2020 Q42, Rickards 2026 C Q30 and Northview Q32, Q47: details come from prior/G_tests.md. I did not re-read those test texts.
- Kitchen-cell voltages (0.7-1.2 V) come from the soinc practice deck 3. Real values depend on the metals and the fruit.
- The table of common voltages is general knowledge. I did not check each line in a source in this run.
- Standard electrode potentials (Ag +0.80 V, Cu +0.34 V, Zn −0.76 V) are general chemistry values. I did not re-check them in a source in this run.
- No real **Nationals** Circuit Lab test was available. The "USC 2026" files are the USC Invitational.

Sources

- 2027 Div B rules, Circuit Lab, rules 3.c.ii-v, 3.c.ix, 3.d, 3.e, Part II.c.vi.
- 2026 Div B tests: Pembroke Hill (Dec 6, 2025), BullSO (Jan 24, 2026), Berkeley (Feb 28, 2026), with keys, in `downloads/dlB/tests/`.
- Test extractions in `circuitlab/research/tests/` (atx-reg, misc-b, ut-b, ut-c, state).
- `downloads/dlG/pz/p21/` Princeton 2021 Circuit Lab exam and key (read for Q32).
- prior/G_tests.md, prior/I_bank.md (U1, U2, U10), prior/B_topics.md §3b and §4, prior/F_slides.md §4 (soinc decks 1 and 3, homework generator).
- OpenStax College Physics 2e, ch. 19 (electric potential, eV) and ch. 21 (EMF, terminal voltage): general knowledge of the book.
- Kuphaldt, *Lessons in Electric Circuits*, Vol. I, ch. 1, 5 and 6: general knowledge.
- BIPM SI Brochure (2019): exact value of e, so of the eV.

Resistance and Resistivity

This note covers Ohm's law, resistance, resistivity, wire size, temperature effects, superconductors, conductance and wire gauge.

Rule 3.c.iv names "concepts, calculations, and units of ... resistance". Rule 3.c.v names "Ohm's Law". So almost all of this note is ✅ **IN at every level**. Some parts are different:

- **Wire gauge (AWG)** is ⚠️ **borderline**. The rule does not name it. Top invitationals still asked it.
- **The temperature coefficient formula** is ✅ IN as algebra. A non-linear bulb curve is only a concept question.
- **Superconductors** are ⚠️ **borderline**. "R = 0 below a critical temperature" is a resistance fact. Any magnet part (levitation, flux) is ❌ OUT.
- **Bridge circuits and four-point probes** are ⚠️ **borderline** instrument trivia.

Series and parallel resistor networks are in Series and Parallel Resistance [A5] and Hard Resistor Networks [A5]. Resistor color codes are in Resistor Color Codes [A5].

Key facts

- **Resistance (R)** is how much a part opposes the flow of current. Its unit is the **ohm (Ω)**. 1 Ω = 1 V/A.
- The ohm is named after **Georg Simon Ohm**. He published his law in 1827. See Ohm [A1].
- **Ohm's law:** V = I·R. The current through a resistor is proportional to the voltage across it.
- A part is **ohmic** when its R stays the same for all voltages. Its V-I graph is a straight line through the origin.
- A part is **non-ohmic** when R changes. A light bulb filament and a diode are non-ohmic.
- On a **V vs I** graph (V up, I across), the slope is R. On an **I vs V** graph (I up, V across), the slope is 1/R.
- **Resistivity (ρ, the Greek letter rho)** is a property of the material. It tells how strongly the material opposes current. Its unit is the **ohm-metre (Ω·m)**.
- **R = ρL/A.** L is the length. A is the cross-section area (the area of the cut end).
- A longer wire has more resistance. A thicker wire has less resistance.
- For a round wire, $A = \pi(d/2)^2 = \pi r^2$. So R is proportional to $1/d^2$. Double the diameter, and R becomes ¼.
- **Conductance (G)** is 1/R. Its unit is the **siemens (S)**. An old name is the **mho** (ohm spelled backwards). Its symbol is an upside-down Ω (℧).
- **Conductivity (σ, sigma)** is 1/ρ. Its unit is S/m.
- **Resistivity values at 20 °C** (Ω·m). These are the values in the national supervisor practice decks and common textbooks:

Material	ρ (Ω·m)	Type
Silver (Ag)	1.59×10^{-8}	best metal conductor
Copper (Cu)	1.68×10^{-8}	house wiring
Gold (Au)	2.44×10^{-8}	contacts (does not corrode)
Aluminum (Al)	2.65×10^{-8}	power lines (light)
Tungsten (W)	5.6×10^{-8}	bulb filaments
Iron (Fe)	9.7×10^{-8} to 1.0×10^{-7}	
Platinum (Pt)	1.06×10^{-7}	
Lead (Pb)	2.2×10^{-7}	
Nichrome	about 1.1×10^{-6}	heater wire
Graphite (carbon)	about 10^{-5} to 10^{-6}	pencil lead
Sea water	about 0.2	conducts (salt ions)
Silicon (pure)	about 6.4×10^{2}	semiconductor
Distilled (pure) water	about 1.8×10^{5}	poor conductor
Glass	10^{10} to 10^{14} (some tables 10^{11} to 10^{15})	insulator
Air	about 10^{16}	insulator
Teflon (PTFE)	10^{23} to 10^{25}	very good insulator

- **Conductors** have low ρ (metals). **Insulators** have very high ρ (glass, rubber, plastic, air). **Semiconductors** are in between (silicon, germanium).
- Pure (distilled) water is a poor conductor. Tap water and sea water conduct because they hold dissolved ions.
- **Temperature and resistance:**
  - In a **metal**, R rises as temperature rises. The atoms shake more, so electrons crash more.
  - In a **semiconductor**, R falls as temperature rises. Heat frees more charge carriers.
  - A **thermistor** is a resistor whose R changes with temperature. An **NTC** thermistor falls with heat. A **PTC** thermistor rises with heat.
  - Carbon (graphite) also has a slightly negative temperature effect.
- **Temperature coefficient (α, alpha)** tells how fast R changes with temperature. $R = R_0[1 + \alpha(T - T_0)]$. Its unit is 1/°C (same as 1/K).
  - Copper α ≈ 0.0039 /°C (some tables 0.00386 or 0.00393). Aluminum ≈ 0.0039 to 0.0043. Tungsten ≈ 0.0045. Nichrome ≈ 0.0004 (very small, which is why heaters use it). Carbon ≈ –0.0005.
  - A 6-band resistor color code shows the temperature coefficient in ppm/K (parts per million per kelvin). See Resistor Color Codes A5.
- A **light bulb filament** is tungsten. It runs at about 2500 to 3000 °C. Its hot R is about 10 to 15 times its cold R. So a bulb draws a large **inrush current** when it is first switched on.
- A **superconductor** has exactly zero resistance below its **critical temperature (T_c)**.
  - Heike Kamerlingh Onnes found it in 1911. Mercury lost all resistance at about 4.2 K.
  - "High-temperature" superconductors such as YBCO (yttrium barium copper oxide) work at about 92 K. This is above the boiling point of liquid nitrogen (77 K).
  - A current in a superconducting loop keeps flowing without a source. This is a resistance fact. Anything about the magnetic field of the loop is ❌ OUT (magnetism).
- **Variable resistors:** a **potentiometer** has three terminals (two ends and a wiper). A **rheostat** uses two terminals to vary the current. A **photoresistor (LDR)** falls in R when light shines on it. A **varistor** falls in R at high voltage (surge protection). Only the plain resistor and the potentiometer are clearly IN. The others are "non-linear devices" (rule 3.d), so treat them only as names.
- **Real vs ideal:** a schematic assumes wires have 0 Ω. Real wires, contacts and meters have some resistance.
- **Wire gauge (AWG, American Wire Gauge):** a **bigger** gauge number means a **thinner** wire.
  - Every 3 gauge numbers: area halves, so R doubles.
  - Every 6 gauge numbers: diameter halves, so R becomes 4 times bigger.
  - Every 10 gauge numbers: R becomes about 10 times bigger.
  - House circuits: **14 AWG** copper on a **15 A** breaker, **12 AWG** on a **20 A** breaker, 10 AWG on 30 A.
  - Breadboard jumper wire is usually **22 AWG** (solid).
- **AWG table (copper, about 20 °C):**

AWG	Diameter (mm)	R (Ω per 1000 ft)	R (Ω per m)
10	2.59	about 1.0	about 0.0033
12	2.05	about 1.6	about 0.0052
14	1.63	about 2.5	about 0.0083
22	0.64	about 16	about 0.053

- **Resistor tolerance** gives the allowed range. A 1520 Ω ± 8% resistor may be anywhere from 1398.4 Ω to 1641.6 Ω.
- **Sheet resistance (R_s)** is ρ divided by the thickness t of a thin film, in "ohms per square" (Ω/□). A strip of length L and width w has $R = R_s \times (L/w)$. This is ⚠ borderline depth.

Formulas

What	Formula	Units
Ohm's law	V = I·R, I = V/R, R = V/I	V, A, Ω
Resistance from a material	R = ρL/A	Ω (ρ in Ω·m, L in m, A in m²)

What	Formula	Units
Area of a round wire	$A = \pi(d/2)^2 = \pi r^2$	m^2
Area of a flat strip or trace	A = w × t (width × thickness)	m^2
Length or area for a target R	L = RA/ρ; A = ρL/R	m; m^2
Resistivity from a measurement	ρ = RA/L	Ω·m
Scaling (same material)	$R_2/R_1 = (L_2/L_1) \times (A_1/A_2)$	(ratio)
Stretch at constant volume (length × n)	R becomes $n^2 \cdot R$	
Cut into n pieces, bundle in parallel	R becomes R/n^2	
Temperature coefficient	$R = R_0[1 + \alpha(T - T_0)]$	Ω
Temperature from R	$T = T_0 + (R/R_0 - 1)/\alpha$	°C
Conductance	G = 1/R = I/V	S
Conductivity	σ = 1/ρ	S/m
Parallel conductances	G_total = $G_1 + G_2 + \ldots$	S
Sheet resistance ⚠️	R_s = ρ/t; R = R_s·(L/w)	Ω/□
Tolerance range	R_nominal × (1 ± tolerance)	Ω
AWG diameter ⚠️	d = 0.127 mm × 92^((36 – n)/39)	mm

Question types

Easy

1. Ohm's law: find V, I or R — ✅ IN

- Where seen: ATX Regional 2021, B, REG, Q14 (100 V across 25 Ω → 4 A). UT Invitational 2020, B, INV, Q38 (6 V across 2 kΩ → 3 mA). JC Booth 2011 (Shock Value), B, INV, Q54 (0.5 A in 20 Ω → 10 V) and Q55 (240 V across 24 Ω → 10 A). UMSO 2019, INV, Q21 (0.5 A from 9 V → 18 Ω). BullSO Jan 2026, B, INV, SA2c (3 mA wanted → replace 5 kΩ with 3 kΩ). Every test has at least one.
- How to solve:
  1. Write down the two values you know, with units.
  2. Change prefixes to base units (kΩ × 1000, mA ÷ 1000).
  3. Use V = IR, I = V/R or R = V/I.
  4. Put the prefix back if it makes a nicer answer.
- Worked example: 9.0 V across 450 Ω. I = 9.0/450 = 0.020 A = **20 mA**. A second example: 0.50 A from a 9.0 V battery. R = 9.0/0.50 = **18 Ω**.
- Traps: A "triangle" helps you remember: V on top, I and R on the bottom. Cover the one you want.
- Calculator: calculate — type: V, I, R.

2. State Ohm's law; match the law to its words — ✅ IN

- Where seen: UF Northern Regional 2019, C, REG, Q28 (Ohm's law in words → V = IR). UF Northern Regional 2020, B, REG, Q17 (Ohm's law equation). UF Northern Regional 2019, B, REG, Q33 (matching "Ohm's law"). NY Lake Erie Dec 2018, C, INV, Q6 (opposition to current → resistance). Pembroke Hill Dec 2025, B, INV, Q31 lists "Ohm's Law" as a wrong choice for the principle behind KCL (the answer is conservation of charge).
- How to solve: Know the sentence. "The current through a conductor is proportional to the voltage across it, when the temperature stays the same."
- Worked example: "What is the opposition to the flow of current called?" → **resistance**.
- Traps: Ohm's law is not true for every part. A bulb filament and a diode do not follow it. The law needs a constant temperature.
- Calculator: look up — type: Ohm's law, definition.

3. Prefix traps in Ohm's law (mΩ, kΩ, MΩ, mA, μA) — ✅ IN

- Where seen: MIT 2019, C, INV, Q8 (5.0 mA in 0.75 Ω → 3.75 mV), Q9 (1.5 V across 55 mΩ → 27.3 A) and Q10 (5.0 V, 560 mA → 8.93 Ω). Soinc practice deck 2 (12 V across 3 kΩ → 4 mA).
- How to solve:

1. Change every value to base units first.
2. Watch for the lowercase **m** (milli, 10^{-3}) and the capital **M** (mega, 10^{6}).
3. Do the division or multiplication.

- Worked example: 1.5 V across 55 mΩ. R = 0.055 Ω. I = 1.5/0.055 = **27 A** (27.3 A). A second example: 5.0 V and 560 mA. R = 5.0/0.560 = **8.9 Ω** (8.93 Ω).
- Traps: Reading 55 mΩ as 55 MΩ gives 2.7×10^{-8} A. That is wrong by a factor of 10^{9}. See Units Prefixes and Sig Figs A4.
- Calculator: calculate — type: V, I, R, prefix.

4. Proportional reasoning with Ohm's law (R fixed) — ✅ IN

- Where seen: UT Invitational 2020, B, INV, Q14 (current halves, R fixed → voltage halves).
- How to solve:
  1. Keep R the same. Then V and I change by the same factor.
  2. If V doubles, I doubles. If I halves, V halves.
  3. If V stays the same and R doubles, I halves.
- Worked example: A resistor has 6.0 V and 2.0 mA. The voltage is raised to 18 V. The current becomes 3 × 2.0 = **6.0 mA**.
- Traps: Do not mix this with power. Power changes by the **square** of the factor (see Power Energy and Cost A4).
- Calculator: calculate — type: ratio, V, I.

5. Resistance from a V-I graph (slope) — ✅ IN

- Where seen: F-M Invitational fall 2019, INV, Q2 (V-I graph with 200 V at 2 A → 100 Ω). Valparaiso 2013, B, INV, Q9 (slope; least R). NC Shock Value practice 2010, B, Q3 (→ 2 Ω).
- How to solve:
  1. Check which quantity is on which axis.
  2. V up and I across: R = slope = ΔV/ΔI.
  3. I up and V across: slope = ΔI/ΔV, so R = 1/slope.
  4. On an I-V graph, the steepest line has the **least** resistance.
- Worked example: A V-I line passes through (2.0 A, 200 V). R = 200/2.0 = **100 Ω**. A second example: an I vs V line has slope 0.020 A/V. R = 1/0.020 = **50 Ω**.
- Traps: Read the axis units (mA vs A). A steep line means small R only when current is on the vertical axis.
- Calculator: calculate — type: slope, graph, V, I.

6. Ohmic vs non-ohmic (why a bulb is non-ohmic) — ✅ IN

- Where seen: UT Invitational fall 2018, B/C, INV, Q9 (why is an incandescent bulb non-ohmic → the filament heats and its R rises). Florida MYSO practice Nov 2020, B, Q3 (true about a light bulb → R rises as it heats; ⚠️ borderline as a non-linear device, but fine as a concept). Shock Value tryout about 2012, B, practice, II-5 (bulb R rises as it heats → True).
- How to solve:
  1. Ohmic part: straight V-I line through the origin. R is constant.
  2. Bulb: the I-V curve bends over (current grows more slowly at high V). The filament gets hotter, so R rises.
  3. Diode: almost no current until about 0.6-0.7 V (silicon), then current rises fast. See PN Junction and I-V Curve A7.
- Worked example: A bulb reads 0.10 A at 1.0 V and 0.25 A at 6.0 V. R is 10 Ω, then 24 Ω. R is not constant, so the bulb is **non-ohmic**.
- Traps: "A bulb obeys Ohm's law at every instant" is a trick wording. V = IR still gives the R at that moment. But R is not constant, so the part is not ohmic.
- Calculator: look up — type: ohmic, non-ohmic, bulb.

7. Factors that change resistance (length, area, temperature, material) — ✅ IN

- Where seen: Cobra Invitational 2019-20, B, INV, Q23 (↑L ↑R; ↓A ↑R; ↑T ↑R). UF Northern Regional 2019, B, REG, Q16 (thick to thin wires → R increasing) and Q27 (R varies with temperature → True). Shock Value tryout about

2012, B, practice, II-4 (thicker filament → higher R → False). ATX Regional 2021, B, REG, Q35 (lower the R of a copper bar: decrease temperature, decrease length, increase area).

- How to solve:
  1. Use R = ρL/A in words.
  2. Longer → more R. Thicker → less R. Hotter metal → more R. Better conductor (lower ρ) → less R.
- Worked example: "How do you lower the resistance of a copper bar?" → **cool it, make it shorter, make it thicker**.
- Traps: Temperature works the other way for semiconductors and NTC thermistors.
- Calculator: look up — type: resistance, length, area, temperature.

8. Rank materials by resistivity; conductor, insulator or semiconductor — ✅ IN

- Where seen: Cobra Invitational 2019-20, B, INV, Q24 (rank 9 materials: graphite, air, silver, platinum, copper, glass, iron, distilled water, gold; the key has an error, see below). F-M Invitational fall 2019, INV, Q5 (least resistive wire: Ag, Cu, Au, Pt → silver). UF Northern Regional 2019, B, REG, Q15 (poorest conductor → nichrome). Florida MYSO practice Nov 2020, B, Q1 (not a conductor: Al, Cu, distilled water, sea water → distilled water) and Q6 (same size wire, highest R: Al, Cu, Au, Pb → lead). Soinc practice decks 2 and 6 (name 10 conductors, 10 insulators, 2 semiconductors).
- How to solve:
  1. Learn the metal order: **Ag < Cu < Au < Al < W < Fe < Pt < Pb < nichrome** (lowest ρ first).
  2. Then graphite, then semiconductors, then water, then insulators.
  3. Insulator order (most resistive first): air ≈ 10^{16}, glass 10^{10}-10^{14}, distilled water ≈ 10^{5}.
- Worked example: Rank by resistance, most to least: glass, silver, copper, distilled water, air, gold. Answer: **air, glass, distilled water, gold, copper, silver**.
- Traps: Gold is **not** the best conductor. Silver is best and copper is second. Gold is used because it does not corrode.
- Calculator: look up — type: resistivity, material, rank.

9. Temperature direction: metal vs semiconductor vs thermistor — ✅ IN

- Where seen: ATX Regional 2020, B, REG, Q44 (heating a semiconductor vs a conductor; the key is wrong, see below). Berkeley Feb 2026, B, INV, Q11 (resistor that changes R with temperature → thermistor). UF Northern Regional 2019, B, REG, Q27.
- How to solve:
  1. Metal: R goes **up** with temperature (positive α).
  2. Semiconductor and NTC thermistor: R goes **down** with temperature (negative α).
  3. Superconductor: R drops to **zero** below T_c.
- Worked example: "A silicon sample and a copper wire are both heated. What happens?" → **silicon R falls, copper R rises**.
- Traps: The names are similar. **Thermistor** = temperature. **Varistor** = voltage. **Photoresistor** = light. **Rheostat** = a variable resistor turned by hand.
- Calculator: look up — type: temperature, semiconductor, thermistor.

10. Conductance and the siemens (mho) — ✅ IN (unit of resistance, rule 3.c.iv)

- Where seen: Rickards Dec 2020, C, INV, Q14 (conductance in SI base units → $s^3 \cdot A^2 \cdot m^{-2} \cdot kg^{-1}$). PA Central Regional about 2013, C, REG ("what is a mho"). North Pocono Jan 2020, C, INV, Q20 (siemens in base units; the key is wrong, see below).
- How to solve:
  1. G = 1/R. The unit is the siemens (S). 1 S = 1 A/V.
  2. Conductances in parallel add: $G_{total} = G_1 + G_2$.
- Worked example: R = 25 Ω, so G = 1/25 = **0.040 S** (40 mS). A second example: 10 Ω and 40 Ω in parallel. G = 0.10 + 0.025 = 0.125 S, so R = **8.0 Ω**.
- Traps: Do not confuse S (siemens) with s (second).
- Calculator: calculate — type: conductance, siemens.

11. Real circuits vs ideal schematics — ✅ IN

- Where seen: ATX Regional 2021, B, REG, Q28 (which is NOT a reason diagrams differ from reality → "laws are only theoretical").
- How to solve:
  1. Real reasons: wires have small resistance; batteries have internal resistance; meters draw a little current; parts have tolerance; parts heat up.
  2. Ohm's law and Kirchhoff's laws are not "only theoretical". They work for real circuits.
- Worked example: "Why does a real circuit read 0.2 V less than the schematic predicts?" → **the wires and the battery have some resistance**.
- Traps: "Ohm's law is only a theory" is the wrong choice.
- Calculator: look up — type: ideal, real circuit.

12. Which conductor can carry more current (wider trace or thicker wire) — ⚠️ Borderline

- Why borderline: the idea (R = ρL/A) is IN. The UT setting was a capacitance figure, which is OUT.
- Where seen: UT Invitational fall 2025, B/C, INV, Q2e (vsig1 or vDD can carry more current → vDD, the wider copper).
- How to solve: The wider or thicker copper has the larger A. So it has less R and heats less for the same current.
- Worked example: Two copper traces are the same length. One is 2 mm wide, one is 0.5 mm wide. The 2 mm trace has **¼ of the resistance**, so it carries more current safely.
- Traps: Length matters too. Compare L/A, not width alone.
- Calculator: look up — type: trace width, current.

Medium

13. R = ρL/A calculation for a round wire — ✅ IN

- Where seen: JC Booth 2011 (Shock Value), B, INV, Q48 (iron rod, $\rho = 9.71 \times 10^{-8}$ Ω·m, L = 1.2 m, A = 65 cm² → 1.79×10^{-5} Ω). Soinc practice decks 1 and 9 and the homework generator ("Material Datasheets" tab). Princeton Feb 2026, C, INV, Comp 5 (wire sizing).
- How to solve:
  1. Find A. For a round wire, $A = \pi(d/2)^2$. Change mm to m first (÷ 1000).
  2. Change L to metres.
  3. R = ρL/A.
- Worked example: 100 m of copper wire, diameter 1.0 mm. $A = \pi(0.50 \times 10^{-3})^2 = 7.85 \times 10^{-7}$ m². $R = 1.68 \times 10^{-8} \times 100 / 7.85 \times 10^{-7}$ = **2.1 Ω** (2.14 Ω). A second example: the iron rod. A = 65 cm² = 65×10^{-4} m² = 0.0065 m². R = $9.71 \times 10^{-8} \times 1.2 / 0.0065$ = **1.8×10^{-5} Ω**.
- Traps: 1 cm² = 10^{-4} m², not 10^{-2} m². 1 mm² = 10^{-6} m². Using the diameter as the radius makes A four times too big.
- Calculator: calculate — type: resistivity, length, diameter.

14. R = ρL/A for a flat strip or PCB trace — ✅ IN

- Where seen: MIT 2019, C, INV, Q49 (copper PCB trace 5 mm × 1 mm × 0.035 mm → 2.4×10^{-3} Ω). USC Invitational Jan 2026, B, INV, Q35b (thin strip, length L, width w, thickness t → R = ρL/(wt)).
- How to solve:
  1. The current flows along the length L.
  2. The cross-section is width × thickness: A = w·t.
  3. R = ρL/(w·t).
- Worked example: A copper trace is 5 mm long, 1 mm wide and 0.035 mm thick. $A = 1 \times 10^{-3} \times 3.5 \times 10^{-5} = 3.5 \times 10^{-8}$ m². $R = 1.68 \times 10^{-8} \times 0.005 / 3.5 \times 10^{-8}$ = **2.4×10^{-3} Ω** (2.4 mΩ).
- Traps: Pick the right dimension as L. It is the direction of the current, not always the longest side.
- Calculator: calculate — type: resistivity, length, width, thickness.

15. Solve R = ρL/A for length, area or diameter — ✅ IN

- Where seen: not seen as a stand-alone test question. It sits inside Princeton Feb 2026, C, INV, Comp 5 (find the minimum area). Soinc deck "MAT-RESISTIVITY-CALC" asks "solve for L or A".
- How to solve:
  1. $L = R \cdot A/\rho$, or $A = \rho L/R$.
  2. For a diameter: $d = 2\sqrt{(A/\pi)}$.
- Worked example: You need a 10 Ω heater from nichrome wire with d = 0.50 mm ($\rho = 1.10 \times 10^{-6}$ Ω·m). $A = \pi(0.25 \times 10^{-3})^2 = 1.96 \times 10^{-7}$ m². $L = 10 \times 1.96 \times 10^{-7} / 1.10 \times 10^{-6} =$ **1.8 m**.
- Traps: Keep A in m². Square the radius, not the diameter.
- Calculator: calculate — type: resistivity, find length, find area.

16. Find ρ from a measurement and name the metal — ✅ IN

- Where seen: not seen in a test. It fits the hands-on rule (measure R with a multimeter) and OpenStax College Physics 2e, ch. 20.3.
- How to solve:
  1. Measure or read R, L and d.
  2. $\rho = R \cdot A/L$.
  3. Compare with the table.
- Worked example: A 2.0 m wire with d = 0.32 mm reads 0.43 Ω. $A = \pi(0.16 \times 10^{-3})^2 = 8.04 \times 10^{-8}$ m². $\rho = 0.43 \times 8.04 \times 10^{-8} / 2.0 =$ **1.7×10^{-8} Ω·m**, so it is **copper**.
- Traps: Meter leads add about 0.1-0.5 Ω. For a small R, short the leads first and subtract that reading.
- Calculator: calculate — type: resistivity, from measurement.

17. Geometry scaling (L and A change) — ✅ IN

- Where seen: BullSO Jan 2026, B, INV, Q9 (length tripled, area halved → 6R). Prior I_bank U13 (diameter doubled → R/4).
- How to solve:
  1. Write $R_2/R_1 = (L_2/L_1) \times (A_1/A_2)$.
  2. If the diameter (or radius) changes by k, the area changes by k^2.
- Worked example: L × 3 and A × ½. $R_2/R_1 = 3 \times 2 =$ **6**, so $R_2 = 6R$. A second example: the same length, diameter × 2. A × 4, so $R_2 =$ **R/4**.
- Traps: "Diameter doubled" is not "area doubled". Square it.
- Calculator: calculate — type: scaling, length factor, area factor.

18. Rank wires by L/r^2 (same material) — ✅ IN

- Where seen: UT Invitational fall 2018, B/C, INV, Q7 (5 wires, same material, given radius and length → most IV, least II).
- How to solve:
  1. For the same metal, R is proportional to L/r^2 (or L/d^2).
  2. Work out L/r^2 for each wire. No ρ is needed.
  3. The largest value has the most R.
- Worked example: Wires (L, r): A (1, 1), B (2, 1), C (1, 2), D (4, 2), E (3, 0.5). $L/r^2 = 1, 2, 0.25, 1, 12$. Most R: **E**. Least R: **C**. A and D tie.
- Traps: Doubling L and doubling r does not cancel. It halves R (2/4).
- Calculator: calculate — type: rank, length, radius.

19. Thicker filament and bulb wattage — ✅ IN

- Where seen: Florida MYSO practice Nov 2020, B, Q2 (four bulbs, same filament length; thickest filament → the 120 W bulb; the stem says "different voltages" but means different wattages at one voltage). State test level: practice.
- How to solve:
  1. At a fixed voltage, $R = V^2/P$. A higher-wattage bulb has a **lower** R.
  2. With the same length and metal, lower R means a larger area.
  3. So the highest-wattage bulb has the thickest filament.

- Worked example: A 25 W and a 100 W bulb, both 120 V, same filament length. R = 576 Ω and 144 Ω. The 100 W filament has **4 times the area** (2 times the diameter).
- Traps: Many students think "more watts = more resistance". It is the reverse at a fixed voltage.
- Calculator: calculate — type: bulb, wattage, filament.

20. Temperature coefficient: find the new R — ✅ IN (algebra)

- Where seen: not seen as a calculation in a test. The 6-band color code with a tempco appears in Rickards Oct 2025, C, INV, Q16 (1 ppm/K). The formula is in OpenStax College Physics 2e, ch. 20.3, and Kuphaldt *Lessons in Electric Circuits* Vol. I, ch. 12.
- How to solve:
  1. $\Delta T = T - T_0$.
  2. $R = R_0(1 + \alpha\Delta T)$.
- Worked example: A tungsten filament is 24 Ω at 20 °C. It runs at 2520 °C. $\alpha = 0.0045$ /°C. $R = 24 \times (1 + 0.0045 \times 2500) =$ **294 Ω** (about 12 times the cold value).
- Traps: Use ΔT, not T. For ppm/K, change to a decimal: 100 ppm/K = $100 \times 10^{-6} = 0.0001$ /K. The linear formula is only an estimate for big temperature changes.
- Calculator: calculate — type: tempco, alpha, R0, delta T.

21. Temperature coefficient: find the temperature (a resistance thermometer) — ✅ IN (algebra)

- Where seen: not seen in a test. It is a standard Kuphaldt Socratic worksheet item ("RTD").
- How to solve:
  1. $R/R_0 - 1 = \alpha\Delta T$.
  2. $\Delta T = (R/R_0 - 1)/\alpha$.
  3. $T = T_0 + \Delta T$.
- Worked example: A copper coil is 10.0 Ω at 20 °C. It reads 11.2 Ω after use. $\Delta T = (1.12 - 1)/0.0039 = 30.8$ °C. T = **51 °C**.
- Traps: The answer is the new temperature, not ΔT. Read what the question asks.
- Calculator: calculate — type: tempco, find temperature.

22. Bulb inrush current (cold vs hot resistance) — ✅ IN (as a concept and Ohm's law)

- Where seen: not seen as a calculation in a test. The concept is in UT Invitational fall 2018, Q9, and MYSO Nov 2020, Q3.
- How to solve:
  1. Hot $R = V^2/P$ (from the rating).
  2. Cold R is about 1/10 to 1/15 of the hot R (or it is given).
  3. Inrush current = V / cold R.
- Worked example: A 60 W, 120 V bulb has a hot R of 240 Ω. Its cold R is 20 Ω. The running current is 0.50 A. The first current is 120/20 = **6.0 A**, which is 12 times bigger.
- Traps: This is why bulbs often burn out the moment they are switched on.
- Calculator: calculate — type: inrush, cold resistance.

23. Resistor tolerance: is the part within spec? — ✅ IN

- Where seen: ATX Regional 2026, B, REG, MC27 (1520 Ω ± 8% measured as 1420 Ω: faulty? → No).
- How to solve:
  1. Low limit = R × (1 − tol). High limit = R × (1 + tol).
  2. If the reading is inside, the part is OK.
- Worked example: 1520 Ω ± 8%. Low = 1398.4 Ω. High = 1641.6 Ω. 1420 Ω is inside, so it is **not faulty**.
- Traps: 8% of 1520 is 121.6 Ω, not 8 Ω. Gold band = ±5%, silver = ±10%. See Resistor Color Codes [A5].
- Calculator: calculate — type: tolerance, range.

24. Potentiometer: wiper position ↔ resistance — ✅ IN

- Where seen: UT Invitational fall 2025, B/C, INV, Q1c.i-iv (wiper at x = L gives R_min; x = 0 gives R_max; R(x) = ρ(L − x)/A; where it overheats at constant V → near R_min, because $P = V^2/R$).
- How to solve:

1. The resistance between one end and the wiper is proportional to the track length between them.
2. R(x) = R_total × (track length used) / (full length).
3. At constant V, the smallest R gets the biggest power ($P = V^2/R$).

- Worked example: A 10 kΩ pot has a 50 mm track. The wiper is 20 mm from the far end, so 30 mm of track is used. R = 10 000 × 30/50 = **6.0 kΩ**.
- Traps: Check which end the problem measures x from. Near-zero R at constant V can burn the track.
- Calculator: calculate — type: potentiometer, position.

25. Superconductor facts — ⚠️ Borderline

- Why borderline: R = 0 below T_c is a resistance fact (rule 3.c.iv). The Meissner effect, levitation and any field are ❌ OUT (magnetism).
- Where seen: BullSO Jan 2026, B, INV, Q23 (a superconducting electromagnet with a current: sustain the field forever? → D, yes, because the current does not decay). That question has a magnetism context. USC Invitational Jan 2026, C, Q38 and Q42 (superconducting disk and loop, magnetism → ❌ OUT).
- How to solve:
  1. Below T_c, R = 0 exactly. A current keeps flowing with no source.
  2. Above T_c, the material has normal resistance.
  3. Know the names and numbers: Onnes, 1911, mercury, 4.2 K; YBCO about 92 K; liquid nitrogen 77 K.
- Worked example: "Why are YBCO superconductors called high-temperature?" → **they work above 77 K, so cheap liquid nitrogen can cool them**.
- Traps: "Superconductors have very low resistance" is not exact. They have **zero** resistance.
- Calculator: look up — type: superconductor, critical temperature.

26. Wire gauge (AWG) facts — ⚠️ Borderline

- Why borderline: not named in the rule. It fits "resistance" and "hazards" (overheating wires).
- Where seen: Hawk & Hornet 2026, C, INV, Q2 (wire gauge vs current → gauge) and Q3 (insulation vs voltage rating). B_topics notes "household 15 A and 20 A circuits use 14 and 12 AWG".
- How to solve:
  1. Bigger number = thinner wire = more R per metre = less safe current.
  2. 3 gauges → R × 2. 6 gauges → R × 4. 10 gauges → R × 10.
  3. The gauge sets the current limit. The insulation sets the voltage limit.
- Worked example: 14 AWG has about 2.5 Ω per 1000 ft. 20 AWG (6 gauges higher) has about **4 times** that, about 10 Ω per 1000 ft.
- Traps: "A 10 AWG wire is thinner than a 14 AWG wire" is false.
- Calculator: look up — type: AWG, gauge, ampacity.

27. Compare two wires of different metals and sizes — ✅ IN

- Where seen: not seen in a test in this exact form. It combines Q8 types (material) with Q17 types (geometry). OpenStax College Physics 2e, ch. 20 problems use it.
- How to solve:
  1. $R_2/R_1 = (\rho_2/\rho_1) \times (L_2/L_1) \times (A_1/A_2)$.
  2. Area ratio = (diameter ratio)².
- Worked example: An aluminum wire has twice the diameter of a copper wire. Same length. R_Al/R_Cu = (2.65/1.68) × (1/4) = **0.39**. The aluminum wire has less resistance.
- Traps: This is why power lines use thick aluminum: it is lighter for the same R.
- Calculator: calculate — type: compare wires, material, diameter.

Hard

28. Stretch a wire, or cut it and bundle the pieces — ✅ IN

- Where seen: not seen in these tests. It is a classic textbook item (I_bank U13; OpenStax ch. 20 problems).
- How to solve:
  1. Stretching keeps the **volume** the same. Volume = L × A.

2. If L becomes n·L, then A becomes A/n. So R becomes n^2 R.
3. Cutting into n equal pieces gives R/n each. Putting them in parallel gives (R/n)/n = R/n^2.

- Worked example: A 12 Ω wire is stretched to twice its length. R = $2^2 \times 12$ = **48 Ω**. A second example: a 16 Ω wire is cut into 4 equal pieces and bundled side by side. R = $16/4^2$ = **1.0 Ω**.
- Traps: "Twice as long, so twice R" forgets that the wire also got thinner.
- Calculator: calculate — type: stretch, cut, bundle.

29. Reshape a conductor and keep the same R — ✅ IN (algebra)

- Where seen: ATX Regional 2026, C, REG, MC27 (a cylinder with length 2r is reshaped into a cube of side π cm with the same R → r = 2 cm).
- How to solve:
  1. Write R for each shape with R = ρL/A.
  2. Cylinder of length 2r: R = ρ(2r)/(πr^2) = 2ρ/(πr).
  3. Cube of side a (current face to face): R = ρa/a^2 = ρ/a.
  4. Set them equal and solve: 2/(πr) = 1/a, so r = 2a/π.
- Worked example: a = π cm. r = 2π/π = **2 cm**.
- Traps: ρ cancels. Do not look up a material.
- Calculator: calculate — type: reshape, cylinder, cube.

30. Wire sizing from a voltage-drop budget — ✅ IN

- Where seen: Princeton Feb 2026, C, INV, Comp 5 (12 V supply, the load needs at least 10 V, 2.2 A, 2 m of copper → 0.91 Ω, 3.69 × 10^{-8} m^2, 4.4 W lost). Kuphaldt *Lessons in Electric Circuits* Vol. I, ch. 12 (wire size).
- How to solve:
  1. Allowed drop = supply V − minimum load V.
  2. R_max = drop / I.
  3. A_min = ρ × L / R_max. Count the **total** wire length (out and back) unless the test says the length already includes both.
  4. Power lost in the wire = I^2 × R.
- Worked example: Drop = 12 − 10 = 2 V. R_max = 2/2.2 = 0.91 Ω. With 2 m of copper in total, A_min = 1.68 × 10^{-8} × 2 / 0.909 = **3.7 × 10^{-8} m^2** (0.037 mm^2). Power lost = 2.2^2 × 0.909 = **4.4 W**.
- Traps: If 2 m is one way only, the loop is 4 m and A_min doubles to 7.4 × 10^{-8} m^2. State your reading.
- Calculator: calculate — type: wire size, voltage drop, current, length.

31. Voltage drop in a house wire run (AWG table) — ⚠️ Borderline

- Why borderline: AWG is not named. The math is only Ohm's law.
- Where seen: not seen in a test. Prior I_bank U19. Kuphaldt *Lessons in Electric Circuits* Vol. I, ch. 12.
- How to solve:
  1. Round trip = 2 × the one-way distance.
  2. R_wire = (Ω per 1000 ft) × length/1000.
  3. Drop = I × R_wire. Percent = drop / supply × 100.
- Worked example: A 50 ft run of 14 AWG copper carries 15 A from 120 V. Round trip 100 ft. R = 2.46 × 0.1 = 0.246 Ω. Drop = 15 × 0.246 = **3.7 V**, which is **3.1%** of 120 V.
- Traps: Forgetting the return wire halves the answer. Answers must be metric: 100 ft = 30.5 m.
- Calculator: calculate — type: AWG, length, current, drop.

32. Sheet resistance and ohms per square — ⚠️ Borderline

- Why borderline: it is R = ρL/A in a new form. The name is not in the rule.
- Where seen: USC Invitational Jan 2026, B, INV, Q35b (strip R = ρL/(wt)). Not seen with "ohms per square".
- How to solve:
  1. R_s = ρ/t.
  2. Count squares: n = L/w.
  3. R = R_s × n.

- Worked example: 35 µm copper foil. $R_s = 1.68 \times 10^{-8} / 35 \times 10^{-6} = 4.8 \times 10^{-4}$ Ω/□. A trace 100 mm long and 0.5 mm wide is 200 squares. R = **0.096 Ω**.
- Traps: The size of a square does not matter. A 1 mm square and a 1 m square of the same film have the same R.
- Calculator: calculate — type: sheet resistance, squares.

Nationals

33. Dimensional analysis to build a resistivity formula (four-point probe) — ⚠️ Borderline

- Why borderline: units and resistance are IN. The four-point probe method goes past the list.
- Where seen: USC Invitational Jan 2026, B, INV, Q35c (thin sheet, probe spacing a, thickness t, V and I → ρ ~ Vt/I). The exact result 2πtV/(I ln 2) needs calculus (❌ OUT, USC C Q39).
- How to solve:
  1. ρ has units Ω·m = V·m/A.
  2. V/I gives ohms. Multiply by one length to get Ω·m.
  3. For a thin sheet, the thickness t is the length that matters. So ρ ~ (V/I)·t.
- Worked example: V = 2.0 mV, I = 10 mA, t = 0.50 mm. ρ ~ (0.002/0.010) × 0.0005 = **1.0×10^{-4} Ω·m** (up to a number factor).
- Traps: Dimensional analysis cannot find pure numbers like 2π or ln 2.
- Calculator: calculate — type: four point, sheet, dimensional.

34. Four-wire (Kelvin) measurement of a very small resistance — ⚠️ Borderline

- Why borderline: bridges and four-wire meters are DC resistor circuits (rule 3.c.v), but this is instrument trivia.
- Where seen: USC Invitational Jan 2026, B, INV, Q23 (advantage of a Kelvin bridge over a Wheatstone bridge → measures very low resistances). UT Invitational 2020, B, INV, Q16 (a Wheatstone bridge measures → unknown resistance).
- How to solve:
  1. Two wires carry the current. Two other wires measure the voltage.
  2. The voltage wires carry almost no current, so their resistance does not add error.
  3. R = V/I for the part alone.
- Worked example: 1.00 A flows through a short copper bar. The inner probes read 0.25 mV. R = 0.00025/1.00 = **0.25 mΩ**. A two-wire meter would add its lead resistance (often 0.1-0.5 Ω), which is 400 times bigger.
- Traps: The Wheatstone bridge itself is in Hard Resistor Networks [A5].
- Calculator: calculate — type: four wire, Kelvin, small resistance.

35. Design a resistor from material and size limits (heater or shunt) — ✅ IN

- Where seen: not seen in a test. It combines $R = V^2/P$ (see Power Energy and Cost [A4]) with R = ρL/A. A Nationals writer could chain them.
- How to solve:
  1. Find the R you need: $R = V^2/P$, or R = V/I.
  2. Choose the material and the diameter.
  3. L = R·A/ρ.
- Worked example: A 1000 W heater on 120 V. $R = 120^2/1000 = 14.4$ Ω. Nichrome, d = 0.50 mm ($A = 1.96 \times 10^{-7}$ m²). $L = 14.4 \times 1.96 \times 10^{-7} / 1.10 \times 10^{-6}$ = **2.6 m**.
- Traps: Use the hot R if the problem gives it. Nichrome's tempco is small, so the change is small.
- Calculator: calculate — type: heater design, power, voltage, length.

Out of scope (skip)

- **Skin effect** (AC current crowds to the outside of a wire). ❌ OUT: AC circuit theory and frequency analysis, rule 3.d. BullSO 2026, B, Q14 asked it.
- **Hall effect**. ❌ OUT: magnetism (not in the 3.c list).
- **Superconductor levitation, Meissner effect, flux trapping.** ❌ OUT: magnetism. USC Invitational 2026, C, Q38, Q42.

- **Impedance and reactance** (resistance for AC with capacitors or inductors). ❌ OUT: AC circuit theory, capacitance and inductance, rule 3.d.
- **Transistor resistance (r_e, Early voltage), op-amp input resistance.** ❌ OUT: semiconductors beyond the PN junction, rule 3.d.
- **Radial resistance of a coaxial cable or a sphere (needs ln r or integration).** ❌ OUT: calculus, rule 3.e.v. USC 2026, C, Q39.
- **Non-linear thermistor (Steinhart-Hart, exponential) or varistor curves.** ❌ OUT: non-linear devices beyond those listed, rule 3.d. The names and directions are fine as facts.
- **Delta-Y conversion and Thevenin resistance.** ❌ OUT or ⚠ borderline: Thevenin is excluded in rule 3.d, and delta-Y needs three equations (rule 3.e.i).

Mistakes in tests and keys

- **ATX Regional 2020, B, REG, Q44.** The key says choice a. When heated, a semiconductor's R falls and a metal's R rises. That is choice b.
- **Cobra Invitational 2019-20, B, INV, Q24.** The key puts glass above air. The key's own source (a Wikipedia table) gives air about 1.3-3.3 × 10^{16} Ω·m and glass 10^{11}-10^{15} Ω·m. So air should be first. The metal order is right.
- **North Pocono Jan 2020, C, INV, Q20.** The key gives the wrong base units for the volt, siemens and farad. Correct: V = kg·m²·s^{-3}·A^{-1}; S = kg^{-1}·m^{-2}·s³·A². (The farad part is OUT.)
- **Florida MYSO practice Nov 2020, B, Q2.** The stem says "different voltages". It must mean different wattages at the same voltage. Key D (120 W) is right for that reading.
- **Shock Value student test key, about 2011, B, practice.** Its bulb ranking is in the wrong order. At a fixed voltage, the higher-wattage bulb has **less** resistance.

Not confirmed

- Rickards 2020 C Q14 (conductance in base units) and PA Central "what is a mho" come from prior/G_tests.md. I did not re-read the question text.
- Princeton 2026 Comp 5: the key gives 3.69 × 10^{-8} m², which matches a 2 m **total** length. I did not see the drawing, so I could not check whether 2 m is one way.
- Resistivity values for graphite, sea water, distilled water, glass, air and Teflon vary a lot between tables. Use the order, not the exact numbers. The metal values are from the national supervisor practice slides (prior/B_topics.md §4) and common textbooks.
- Temperature coefficients (Cu 0.0039, W 0.0045, nichrome 0.0004, carbon −0.0005) are common textbook values. I did not re-open a table in this run.
- The AWG table values are from the standard AWG formula and ρ_Cu = 1.68 × 10^{-8} Ω·m, checked in Python. Published tables use annealed copper (1.72 × 10^{-8}), so they are about 2% higher (14 AWG: 2.525 Ω per 1000 ft).
- Superconductor dates and temperatures (Onnes 1911, Hg 4.2 K, YBCO 92 K, LN_2 77 K) are general knowledge. I did not check a source in this run.
- No real **Nationals** Circuit Lab test was available. The "USC 2026" files are the USC Invitational. So the Nationals types here come from top invitationals and the rules.

Sources

- 2027 Div B rules, Circuit Lab (pages B12-B13), rules 3.b, 3.c.iv, 3.c.v, 3.d, 3.e, Part II.b-c.
- Test extractions in `circuitlab/research/tests/`: atx-reg (ATX Regional 2020 B, 2021 B), misc-a (Cobra 2019-20, F-M 2019, North Pocono 2020, UMSO 2019), misc-b (MIT 2019, UF Northern Regional 2019 B/C and 2020 B, NY Lake Erie 2018, JC Booth 2011, Shock Value tryout), nat26 (USC Invitational 2026 B/C), state (Florida MYSO 2020), ut-a (UT Fall 2025, ATX Regional 2026 B/C), ut-b (UT 2020 B), ut-c (UT 2018).
- 2026 Div B tests from `downloads/dlB/tests/`: Berkeley (Feb 28, 2026) Q11; BullSO (Jan 24, 2026) Q9, Q14, Q23, SA2; Pembroke Hill (Dec 6, 2025) Q31.
- prior/G_tests.md (Princeton 2026, Rickards 2020 and 2026, HH 2026, PA Central, Valparaiso, NC 2010), prior/I_bank.md (U12-U16, U19), prior/B_topics.md §4, §5f, §10, prior/F_slides.md §4 (soinc decks 1, 2, 6, 9 and

the homework generator).

- OpenStax College Physics 2e, ch. 20.2-20.3 (Ohm's law, resistivity, temperature coefficient): general knowledge of the book; I did not re-open it in this run.
- Kuphaldt, *Lessons in Electric Circuits*, Vol. I, ch. 2 (Ohm's law) and ch. 12 (physics of conductors and insulators, wire size, temperature coefficient, superconductivity), and the Socratic worksheets: general knowledge.
- All worked examples were checked in Python.

Power, Energy and Cost

Note: Power Energy and Cost

This note covers electric power, energy, the kilowatt-hour, electricity cost, efficiency, power ratings, light-bulb puzzles, battery energy and USB charging time.

Rule 3.c.iv names "concepts, calculations, and units of ... power, and energy". So almost all of this note is ✅ **IN at every level**. Two hands-on tasks also use it: "Calculate the power supplied to a circuit" (Part II.c.ii) and "Given a USB charger, read the label ... and calculate how long it would take to charge a specific battery" (Part II.c.iv). Some parts are different:

- **Heating water** needs the specific heat of water. It is ⚠️ **borderline** (a thermodynamics idea).
- **Maximum power transfer** is ⚠️ **borderline**. The simple series-resistor form is only algebra. The general rule uses Thevenin equivalents, which are ❌ OUT (rule 3.d).
- **AC power with RMS values** is in CL 3 - DC and AC [T]. Power factor and 3-phase power are ❌ OUT.

The full power-transfer method is in Internal Resistance and Power Transfer [A5]. Bulb brightness in networks is in Series and Parallel Resistance [A5].

Key facts

- **Energy (E or W)** is the ability to do work. Its SI unit is the **joule (J)**.
- **Power (P)** is the rate of using or giving energy. P = E/t. Its unit is the **watt (W)**. 1 W = 1 J/s.
- The watt is named after James Watt. The joule is named after James Prescott Joule.
- **Electric power:** P = V·I. 1 W = 1 V × 1 A.
- For a resistor, Ohm's law gives two more forms: **$P = I^2R$** and **$P = V^2/R$**.
- **Joule heating (Joule's law):** the power in a resistor turns into heat. Heat = I^2Rt.
- A resistor **dissipates** (uses up) power. A battery or supply **supplies** power.
- **Conservation of energy:** total power supplied = total power used by all parts.
- **The kilowatt-hour (kWh)** is the energy unit on electricity bills. 1 kWh = 1000 W × 3600 s = **3.6×10^6 J** (3.6 MJ).
- 1 Wh = 3600 J. 1 mWh = 3.6 J.
- A typical US electricity price is about $0.10 to $0.20 per kWh. (The 2026 tests used $0.07, $0.094 and $0.15.)
- **Battery energy:** Wh = Ah × V. A 3000 mAh, 3.7 V phone cell holds 3.0 × 3.7 = 11.1 Wh.
- **mAh is charge, not energy.** You need the voltage to get energy. See Current and Charge [A4].
- **Efficiency (η, eta)** = useful power out / power in. It is always less than 100% (1.0). The rest becomes heat.
- Efficiencies in a chain **multiply**: 90% × 80% = 72%.
- **Power rating** is the most power a part can safely dissipate. Common resistor ratings are 1/8 W, 1/4 W, 1/2 W, 1 W.
- A light bulb's **rating** (for example 60 W, 120 V) is its power **at that voltage only**. Use it to find the bulb's resistance: $R = V^2/P$.
- A bulb's **brightness** depends on the **power it actually uses**, not on its rating.
- At a fixed voltage, a higher-wattage bulb has **lower** resistance.
- In **series**, the bulb with the **higher R** (the lower wattage rating) gets more power and glows brighter.
- In **parallel** (each across the full rated voltage), each bulb gets its rated power. The higher-wattage bulb is brighter.
- **Incandescent bulbs** make light by heating a filament. Only about 2-5% of the energy becomes visible light. **LEDs** are about 5 to 8 times more efficient. A 9-10 W LED gives about the light of a 60 W incandescent bulb.
- A **CFL** (compact fluorescent lamp) is in between. A 13-15 W CFL replaces a 60 W bulb.
- **Horsepower:** 1 hp ≈ 745.7 W (mechanical horsepower). Some tests round to 746 W.
- **Transmission lines:** for the same power, a higher voltage means a lower current. Line loss is I^2R, so it falls with the **square** of the voltage.
- **Typical powers:** LED indicator 0.02-0.06 W; LED bulb 9-10 W; laptop 45-65 W; incandescent bulb 60-100 W; microwave 1000-1250 W; hair dryer or kettle 1500 W; electric car charging at home 1.4 kW (120 V × 12 A) to 11 kW; average US home about 1.2 kW averaged over a day.

- **Typical USB levels** (from the USB-IF specifications, as general knowledge): USB 2.0 port 5 V, 0.5 A (2.5 W); USB 3.x port 5 V, 0.9 A (4.5 W); USB Battery Charging 1.2: 5 V, 1.5 A (7.5 W); USB-C: 5 V at 1.5 A or 3 A (7.5 or 15 W); USB Power Delivery: 5, 9, 15, 20 V, up to 3 A (60 W) or 5 A with an e-marked cable (100 W); PD 3.1 adds 28, 36, 48 V, up to 240 W.
- **On a charger label,** "~" means AC and "⎓" means DC. "Input" is the wall (AC) side. "Output" is the USB (DC) side.
- The charger offers levels. The device **asks for** (negotiates) the level it uses.
- Li-ion cells charge with **constant current**, then **constant voltage** (CC/CV). The current tapers near full, so real charging is slower than the simple estimate.

Formulas

What	Formula	Units
Power (any part)	$P = V \cdot I$	W
Power (resistor)	$P = I^2R = V^2/R$	W
Current from power	$I = P/V$; $I = \sqrt{(P/R)}$	A
Resistance from a rating	$R = V^2/P$	Ω
Voltage from power	$V = P/I$; $V = \sqrt{(P \cdot R)}$	V
Energy	$E = P \cdot t = V \cdot I \cdot t$	J (t in s) or kWh (P in kW, t in h)
Energy in a resistor	$E = I^2Rt = V^2t/R$	J
Energy of a charge	$E = qV$	J
kWh to joules	$1\ \text{kWh} = 3.6 \times 10^6$ J	
Cost	cost = energy (kWh) × price ($/kWh)	$
Battery energy	Wh = Ah × V = (mAh/1000) × V	Wh
Run time from energy	t (h) = Wh / W	h
Average power from a battery	P = Wh / t	W
Charge time (energy method)	t = battery Wh / (charger W × η)	h
Efficiency	$\eta = P_{out}/P_{in} = E_{out}/E_{in}$	(no unit or %)
Chain efficiency	$\eta_{total} = \eta_1 \times \eta_2 \times \ldots$	
Mechanical power (lifting)	$P = mgh/t$, with $g = 9.8$ m/s²	W
Horsepower	1 hp = 745.7 W	W
Line loss	$P_{loss} = I^2R_{line}$, with $I = P/V$	W
Loss ratio at two voltages (same power)	$P_{loss1}/P_{loss2} = (V_2/V_1)^2$	
Rated bulb at another voltage	$P_{new} = P_{rated} \times (V_{new}/V_{rated})^2$	W
Total power	$P_{total} = V_{source} \times I_{total}$ = sum of all parts	W
Heating water ⚠️	$E = m \cdot c \cdot \Delta T$, $c = 4186$ J/(kg·°C)	J

Question types

Easy

1. Power from voltage and current (P = VI) — ✅ IN

- Where seen: JC Booth 2011 (Shock Value), B, INV, Q12 (0.78 A through a light string → 19 W). UMSO 2019, INV, Q22 (9 V × 0.5 A → 4.5 W). F-M Invitational 2018, INV, Q6c (total power → 0.135 W). Berkeley Feb 2026, B, INV, FR 2.1c (5 V, 0.5 A → 2.5 W). MIT 2021, C, INV, Q4 (a device that is not a resistor: P = VI still works → 1.1 W). BullSO Jan 2026, B, INV, SA2b (power delivered by the 9 V source → 16.2 mW).
- How to solve:
  1. Change mA to A and kV to V.
  2. Multiply: P = V × I.
- Worked example: A 5.0 V source drives 0.50 A. P = 5.0 × 0.50 = **2.5 W**. A second example: a source gives 9.0 V and 1.8 mA. P = 9.0 × 0.0018 = **0.016 W** (16 mW).

- Traps: P = VI works for every part, even a motor or an LED. P = I²R and V²/R work only for resistors.
- Calculator: calculate — type: power, V, I.

2. Current from power and voltage (I = P/V) — ✅ IN

- Where seen: MIT 2019, C, INV, Q11 (1.25 kW microwave at 120 V → 10.4 A). Pembroke Hill Dec 2025, B, INV, Q29 (5 A fuse, 600 W device at 120 V → 5 A, right at the rating; the key says it does not blow).
- How to solve:
  1. Change kW to W.
  2. I = P/V.
  3. For a fuse question, compare I with the fuse rating.
- Worked example: A 1250 W microwave on 120 V. I = 1250/120 = **10.4 A**. A second example: a 1500 W heater on 120 V draws 12.5 A, so it fits on a 15 A circuit alone.
- Traps: A fuse rated 5 A may run at 5 A for a long time. Fuses blow **above** their rating. The Pembroke key accepts "no, but it is close". See CL 6 - Switches, Relays, Fuses, Breakers, GFCI 🅃.
- Calculator: calculate — type: current, power, voltage.

3. Resistance from a power rating (R = V²/P) — ✅ IN

- Where seen: Berkeley Feb 2026, B, INV, Q22 (60 W bulb at 120 V → 240 Ω). UF Northern Regional 2019, C, REG, Q14 (50 W bulb, 20 V → 8 Ω).
- How to solve:
  1. R = V²/P.
  2. Or find I = P/V first, then R = V/I.
- Worked example: 60 W at 120 V. R = 120²/60 = 14 400/60 = **240 Ω**. Check: I = 0.50 A and 120/0.50 = 240 Ω.
- Traps: R = P/V² is upside down. It gives 0.0042, which has no sense as ohms.
- Calculator: calculate — type: rating, resistance, V, P.

4. Power in a resistor (P = I²R or V²/R) — ✅ IN

- Where seen: MIT 2019, C, INV, Q7 (1.0 A in 0.25 Ω → 0.25 W). USC Invitational Jan 2026, B, INV, Q11 (9.0 V, 1.0 kΩ + 2.2 kΩ in series: power in 2.2 kΩ → 0.017 W). UT Invitational fall 2019, B, INV, Q26 (2000 A line, 2.3 Ω → 9.2 MW). ATX Regional 2019, B/C, REG, Q12 (22 kΩ "toaster" on 120 V → 0.65 W). Pembroke Hill Dec 2025, B, INV, Q3 (4 V across 3 Ω → 5.33 W). Berkeley Feb 2026, B, INV, FR 2.1g-h (power in R3, R4 → 178 mW).
- How to solve:
  1. If you know I and R: P = I²R.
  2. If you know V across the resistor and R: P = V²/R.
  3. In a series circuit, find I first (I = V_total/R_total).
- Worked example: 9.0 V across 1.0 kΩ + 2.2 kΩ in series. I = 9.0/3200 = 2.81 mA. P(2.2 kΩ) = (0.00281)² × 2200 = **0.017 W** (17.4 mW).
- Traps: In P = V²/R, use the voltage **across that resistor**, not the battery voltage.
- Calculator: calculate — type: power, I squared R, V squared over R.

5. Joule heating: name the law; energy conversions — ✅ IN

- Where seen: ATX Regional 2020, B, REG, Q17 (law for electrical heat → Joule). UF Northern Regional 2019, B, REG, Q33-36 (matching: "motors use electrical energy", "potential energy"). Florida MYSO practice Nov 2020, B, Q7 (converts chemical energy to electrical → battery). JC Booth 2011, B, INV, Q4-6 (battery chem → elec; motor elec → mech; generator mech → elec; motor and generator are ⚠️ borderline because they use magnetism).
- How to solve: Learn the pairs. Battery: chemical → electrical. Resistor or heater: electrical → heat. Bulb: electrical → light and heat. Solar cell: light → electrical. Motor: electrical → motion.
- Worked example: "Which law says the heat in a wire is I²Rt?" → **Joule's law** (Joule's first law).
- Traps: A **motor** turns electrical energy into mechanical energy. A **generator** does the reverse. UF Northern Regional 2020, B, Q29 had this wrong in its key.
- Calculator: look up — type: Joule, energy conversion.

6. Energy cost in kWh (simple) — ✅ IN

- Where seen: Pembroke Hill Dec 2025, B, INV, Q7 (100 W bulb for 24 h at $0.15/kWh → $0.36). Rickards Oct 2025, B, INV, Q18 (100 W, 20 h a month, $0.15 → $3.60 a year).
- How to solve:
  1. Change W to kW (÷ 1000).
  2. Energy (kWh) = kW × hours.
  3. Cost = kWh × price.
- Worked example: 100 W for 24 h. 0.100 kW × 24 h = 2.4 kWh. Cost = 2.4 × $0.15 = **$0.36**. A second example: 100 W for 20 h each month for a year. 0.1 × 20 × 12 = 24 kWh → **$3.60**.
- Traps: Using 100 instead of 0.100 gives $360. Read "per day", "per month" and "per year" carefully.
- Calculator: calculate — type: kWh, cost, hours, price.

7. Energy in joules (E = Pt) — ✅ IN

- Where seen: MIT 2020, C, INV, Q26 (12 V, 100 mA, 4 h → 17.3 kJ). Berkeley Feb 2026, B, INV, FR 2.2e (circuit runs 2 hours → 21.5 kJ). North Pocono Jan 2020, C, INV, Q32 (defibrillator 1000 V, 500 Ω, 10 ms → 20 J).
- How to solve:
  1. Find P ($P = VI$, I^2R or V^2/R).
  2. Change the time to seconds.
  3. $E = P \times t$.
- Worked example: 12 V and 100 mA for 4.0 h. P = 1.2 W. t = 14 400 s. E = 1.2 × 14 400 = **17 kJ** (17 280 J). A second example: the defibrillator. P = $1000^2/500$ = 2000 W. E = 2000 × 0.010 = **20 J**.
- Traps: Joules need seconds. kWh need kW and hours. Do not mix them.
- Calculator: calculate — type: energy, joules, power, time.

8. Estimate the wattage of everyday things — ✅ IN

- Where seen: Florida State 2021, B, ST, Q26 (average LED bulb wattage → 10 W) and Q25 (power used by the average US citizen in 2010; the key is wrong, see below).
- How to solve: Learn the "Typical powers" list in Key facts. Pick the closest order of magnitude.
- Worked example: "About how much power does a household LED bulb use?" → **about 10 W**.
- Traps: Watch "per person" vs "whole country". The US total is about 3×10^{12} W. Per person it is about 1×10^4 W.
- Calculator: look up — type: typical power, estimate.

9. Brightness: series vs parallel (identical bulbs) — ✅ IN

- Where seen: UF Northern Regional 2020, B, REG, Q25 (3 bulbs on 12 V: series or parallel for max brightness → parallel). Shock Value tryout about 2012, B, practice, VI-18 (series vs parallel bulbs: brighter → parallel). More network versions are in Series and Parallel Resistance A5.
- How to solve:
  1. Brightness follows the power in each bulb.
  2. In parallel, each bulb gets the full voltage.
  3. In series, the voltage is shared, so each gets less.
- Worked example: Three identical 4 Ω bulbs on 12 V. Parallel: each gets $12^2/4$ = **36 W**. Series: each gets 4 V, so $4^2/4$ = **4 W**. Parallel bulbs are 9 times brighter.
- Traps: Parallel bulbs also drain the battery faster (3 × 36 = 108 W in total).
- Calculator: calculate — type: brightness, series, parallel.

Medium

10. Ratio and scaling of power — ✅ IN

- Where seen: Pembroke Hill Dec 2025, B, INV, Q32 (V doubled, R halved → P × 8). UT Invitational 2020, B, INV, Q15 (R fixed, current doubles → P × 4). UF Northern Regional 2019, B, REG, Q23 (constant V, P doubled → R halves; True).
- How to solve:
  1. Pick the form that has the quantities that change.
  2. Put in the factors: $P = V^2/R \rightarrow (2)^2/(½)$ = 8.

- Worked example: V × 2 and R × ½. P_new/P_old = $2^2 \div ½ =$ **8**. A second example: I × 2 with R fixed. $P = I^2R \rightarrow$ **× 4**.
- Traps: Do not use P = VI when both V and I change. That works, but you must change both. Using only one gives × 2.
- Calculator: calculate — type: power ratio, scaling.

11. Power rating: maximum current or voltage; will it burn? — ✅ IN

- Where seen: ATX Regional 2019, B/C, REG, Q14 (470 Ω, 9 W rating → max current 0.138 A). North Pocono Jan 2020, C, INV, Q30 (9 V across 200, 100, 50 Ω in parallel, each 0.5 W → B and C burn). Soinc deck 4 ("POW-RATING-CHECK", 100 Ω, 1/4 W across 9 V → 0.81 W, overheats).
- How to solve:
  1. Max current: $I = \sqrt{(P/R)}$. Max voltage: $V = \sqrt{(P \cdot R)}$.
  2. To check, compute the real power and compare with the rating.
- Worked example: 470 Ω, 9 W. $I_{max} = \sqrt{(9/470)} =$ **0.138 A**. A second example: 100 Ω, 1/4 W across 9 V. P = 81/100 = 0.81 W > 0.25 W, so it **overheats**. Its safe voltage is $\sqrt{(0.25 \times 100)} = 5.0$ V.
- Traps: The square root is easy to forget. 9/470 = 0.019 is not the current.
- Calculator: calculate — type: power rating, max current, max voltage.

12. Rated bulb on a different voltage — ✅ IN

- Where seen: not seen as its own question. It is the first step of type 18. Prior I_bank U20.
- How to solve:
  1. $R = V_{rated}^2/P_{rated}$ (assume R stays the same).
  2. $P_{new} = V_{new}^2/R$.
  3. Short cut: $P_{new} = P_{rated} \times (V_{new}/V_{rated})^2$.
- Worked example: A 60 W, 120 V bulb on 60 V. R = 240 Ω. $P = 60^2/240 =$ **15 W** (¼ of the rating).
- Traps: A real filament is cooler at low voltage, so its R is lower. Its real power is a bit more than 15 W. Say "assuming constant resistance".
- Calculator: calculate — type: rated bulb, new voltage.

13. Electricity bill with days, months and a calendar trap — ✅ IN

- Where seen: Cobra Invitational 2019-20, B, INV, Q21 (5 × 15 W bulbs, 5 h a day, February 2020, $0.07/kWh → $0.76; 2020 is a leap year, so February has 29 days). Prior I_bank U9.
- How to solve:
  1. Total power = number of devices × power each.
  2. Hours = hours per day × number of days.
  3. kWh = kW × hours. Cost = kWh × price.
- Worked example: 5 × 15 W = 75 W. 5 h × 29 days = 145 h. E = 0.075 × 145 = 10.875 kWh. Cost = 10.875 × 0.07 = **$0.76**. A second example: a 1500 W heater, 4.0 h a day for 30 days at $0.16 → 180 kWh → **$28.80**.
- Traps: Leap years: 2020, 2024, 2028. February has 29 days in them. 30 days is not always "a month".
- Calculator: calculate — type: bill, devices, hours per day, days, price.

14. Bill backwards: cost → kWh → W → A → Ω — ✅ IN

- Where seen: Shock Value tryout about 2012, B, practice, V-2a-e ($5324.18 at 9.4 ¢/kWh → 56 640 kWh; over 360 days → 6556 W; at 120 V → 54.6 A; R = 2.20 Ω; saving per day off → about $14.79).
- How to solve:
  1. kWh = cost / price.
  2. Average W = kWh × 1000 / hours.
  3. I = P/V. R = V/I.
- Worked example: $5324.18 at $0.094/kWh. E = **56 640 kWh**. Over 360 days (8640 h): P = 56 640 000 / 8640 = **6556 W**. I = 6556/120 = **54.6 A**. R = 120/54.6 = **2.20 Ω**. Cost per day = 5324.18/360 = **$14.79**.
- Traps: Change cents to dollars (9.4 ¢ = $0.094). The "days off" part was unclear in the original test.
- Calculator: calculate — type: bill backwards, cost, price, days, voltage.

15. LED vs incandescent: savings and efficiency — ✅ IN

- Where seen: Shock Value tryout about 2012, B, practice, VI-13 (CFL vs incandescent: which burns you more → incandescent). Florida State 2021, B, ST, Q26 (LED bulb wattage). Not seen as a savings calculation; prior I_bank U9.
- How to solve:
  1. Saved power = P_old − P_new.
  2. Saved energy = saved kW × hours. Saved money = kWh × price.
- Worked example: A 9 W LED replaces a 60 W bulb for 1000 h at $0.15/kWh. Saved = 51 W × 1000 h = 51 kWh → **$7.65**.
- Traps: Watts measure power used, not light given. Light output is in lumens (borderline trivia).
- Calculator: calculate — type: savings, LED, incandescent.

16. Efficiency (η = P_out/P_in) — ✅ IN

- Where seen: Princeton 2020, C, INV, Q33 (motor: 7.5 mW in, 20 N·mm torque at 0.05 rad/s out → 13%). Prior I_bank U11. BullSO Jan 2026, B, INV, Q20 lists "the ratio of power output to input" as a choice (that question is about transformers, ❌ OUT).
- How to solve:
  1. η = useful power out / power in.
  2. Wasted power = P_in − P_out (usually heat).
  3. For a chain, multiply the efficiencies.
- Worked example: A motor draws 120 V × 2.0 A = 240 W. It gives 180 W of motion. η = 180/240 = **75%**. Wasted: **60 W** of heat. A second example: a 90% charger feeds an 80% battery. Overall η = **72%**.
- Traps: η can never be more than 100%. If you get 133%, you divided the wrong way.
- Calculator: calculate — type: efficiency, power in, power out.

17. Battery energy and run time (Wh and mWh) — ✅ IN

- Where seen: UT Invitational fall 2018, B/C, INV, Q8 (15 W processor, 42 000 mWh battery → 2.8 h). UT Invitational fall 2019, B, INV, Q28 (phone 3000 mAh, 4.4 V, 6 h life → average power 2.2 W). Also UT Invitational fall 2019, C, INV, TB#6 (same item).
- How to solve:
  1. If capacity is in mAh, change to Wh: Wh = Ah × V.
  2. Run time (h) = Wh / W.
  3. Average power = Wh / hours.
- Worked example: 42 000 mWh = 42 Wh. t = 42/15 = **2.8 h**. A second example: 3000 mAh at 4.4 V = 13.2 Wh. Over 6 h: P = 13.2/6 = **2.2 W**.
- Traps: Do not divide mAh by watts. Change it to Wh first.
- Calculator: calculate — type: battery, Wh, mAh, run time.

18. USB charger label: list the power levels — ✅ IN (hands-on rule II.c.iv)

- Where seen: not seen in a test. It is an official 2027 hands-on example task (rule Part II.c.iv), also in 2026.
- How to solve:
  1. Find the "Output" line. Ignore the "Input" line (that is the AC side).
  2. For each level, P = V × I.
  3. The "max" W on the label may be lower than V × I at the top level.
- Worked example: Output "5 V ⎓ 3 A, 9 V ⎓ 3 A, 15 V ⎓ 3 A, 20 V ⎓ 3.25 A". Powers: **15 W, 27 W, 45 W, 65 W**.
- Traps: "100-240 V ~ 1.5 A" is the input. It does not mean the charger gives 240 V × 1.5 A = 360 W.
- Calculator: calculate — type: USB, label, output levels.

19. USB charge time for a battery — ✅ IN (hands-on rule II.c.iv)

- Where seen: not seen in a test. Official 2027 hands-on task (rule Part II.c.iv). Prior B_topics §9d.
- How to solve:
  1. Simple method (same voltage): t = capacity (mAh) / charging current (mA).
  2. Energy method (different voltages): battery Wh = Ah × cell voltage.
  3. t = Wh / (charger W × efficiency).

4. State your efficiency (for example 85%) and that charging slows near full.

- Worked example: A 5000 mAh, 3.7 V phone battery on an 18 W charger at 85% efficiency. E = 5.0 × 3.7 = 18.5 Wh. t = 18.5 / (18 × 0.85) = **1.2 h** (about 73 min). A second example (simple method): 3000 mAh at 1000 mA → **3.0 h**.
- Traps: Do not divide 5000 mAh by the charger's current at 9 V. The cell is at 3.7 V, not 9 V. Use Wh.
- Calculator: calculate — type: USB, charge time, mAh, cell voltage, charger W, efficiency.

20. Total power = sum of the parts; power supplied vs absorbed — ✅ IN

- Where seen: UT Invitational fall 2018, B/C, INV, Q25b (power of each part, dissipated or generated → source 0.125 W; 5 Ω 78.1 mW; 4 Ω 35.2 mW; 10 Ω 9.77 mW; 2 Ω 1.95 mW). BullSO Jan 2026, B, INV, SA2b (power delivered by the 9 V source). Soinc deck "POW-TOTAL-VS-SUM". Hands-on rule Part II.c.ii.
- How to solve:
  1. Find the total current from the source.
  2. P_source = V_source × I_total.
  3. Find each resistor's power.
  4. Check: the sum of the resistor powers = P_source.
- Worked example: 12 V feeds 2 Ω in series with (6 Ω || 3 Ω). R_total = 2 + 2 = 4 Ω. I = 3.0 A. P_source = **36 W**. Parts: 2 Ω → 18 W; parallel group has 6 V: 6 Ω → 6 W, 3 Ω → 12 W. Sum = 18 + 6 + 12 = **36 W** ✓.
- Traps: Use each resistor's own voltage or current. The check catches most mistakes.
- Calculator: calculate — type: total power, each resistor, check sum.

21. Power-supply sizing with a safety margin — ✅ IN

- Where seen: Cobra Invitational 2019-20, B, INV, Q20 (LED strip: 10 LEDs per foot × 0.5 W × 20 ft, at least 20% margin → choose the 125 W supply from 100, 125, 80 W).
- How to solve:
  1. Load power = number of parts × power each.
  2. Needed rating = load × (1 + margin).
  3. Pick the smallest supply at or above that.
- Worked example: 10 × 0.5 W × 20 = 100 W. With 20% margin: 120 W. Choose **125 W**.
- Traps: 100 W "just fits" but has no margin.
- Calculator: calculate — type: supply sizing, margin.

22. Transmission line current and I^2R loss — ✅ IN

- Where seen: UT Invitational fall 2018, B/C, INV, Q18 (90 kW at 22 000 V → line current 4.09 A; loss in 1.3 Ω lines → 21.8 W). UT Invitational fall 2019, B, INV, Q26 (2000 A, 2.3 Ω → 9.2 MW). Rickards Oct 2025, C, INV, Q42-43 and NY Lake Erie 2018, C, Q5 (why AC for transmission).
- How to solve:
  1. Line current I = P/V.
  2. Loss = I^2 × R_line.
- Worked example: 90 kW at 22 000 V. I = 90 000/22 000 = **4.09 A**. Loss = 4.09^2 × 1.3 = **21.8 W**.
- Traps: The loss is **not** V^2/R with the 22 000 V. Most of that voltage is across the load, not the line.
- Calculator: calculate — type: transmission, line current, loss.

23. Horsepower to watts to current — ✅ IN

- Where seen: MIT 2019, C, INV, Q12 (350 hp from 12 V batteries → 21 750 A).
- How to solve:
  1. P (W) = hp × 745.7.
  2. I = P/V.
- Worked example: 350 hp = 350 × 745.7 = 260 995 W. At 12 V: I = 260 995/12 = **2.2×10^4 A** (21 750 A).
- Traps: Some tests use 746 W per hp. The answer changes in the 4th digit only.
- Calculator: calculate — type: horsepower, watts, current.

Hard

24. Horsepower chain: energy and run time — ✅ IN

- Where seen: MIT 2019, C, INV, Q13 (the 350 hp car runs 3.0 h: battery energy in kWh; the key is wrong) and Q14 (85 kWh battery at 250 hp: hours; the key is wrong). See "Mistakes" below.
- How to solve:
  1. P (kW) = hp × 0.7457.
  2. Energy (kWh) = kW × hours.
  3. Time (h) = kWh / kW.
- Worked example: 350 hp × 0.7457 = 261 kW. For 3.0 h: E = **783 kWh**. A second example: 85 kWh at 250 hp (186.4 kW). t = 85/186.4 = **0.46 h** (about 27 min).
- Traps: The MIT key gave 65 kWh, which is really 65.25 kAh (21 750 A × 3 h). Amp-hours are not kWh. The key for Q14 put the ratio upside down.
- Calculator: calculate — type: horsepower, energy, time.

25. Electric car charging chain (outlet → kWh → hours → miles) — ✅ IN

- Where seen: UT Invitational 2020, B, INV, Q63 (Tesla Model 3: 8 h at 120 V and 15 A; it uses 13 kW at 60 mph → 66.46 miles).
- How to solve:
  1. Charging power = V × I.
  2. Energy = power × hours (kWh).
  3. Driving time = energy / driving power.
  4. Distance = speed × time.
- Worked example: 120 × 15 = 1800 W. 1.8 kW × 8 h = 14.4 kWh. Time = 14.4/13 = 1.108 h. Distance = 60 × 1.108 = **66.5 miles** (107 km).
- Traps: Answers should be metric (rule 3.b) unless the question asks for miles.
- Calculator: calculate — type: EV, charging, distance.

26. Transmission efficiency ratio at two voltages — ✅ IN

- Where seen: UT Invitational 2020, B, INV, Q64 (same power at 25 kV vs 2 kV: how many times more efficient (line loss) → 156.25).
- How to solve:
  1. Same power, so I is proportional to 1/V.
  2. Loss is I^2R, so it is proportional to $1/V^2$.
  3. Loss ratio = $(V_{high}/V_{low})^2$.
- Worked example: $(25\,000/2000)^2 = 12.5^2$ = **156.25**. The 25 kV line loses 156 times less.
- Traps: The ratio is not 12.5. Square it.
- Calculator: calculate — type: transmission ratio, two voltages.

27. Rated bulbs in series vs in parallel — ✅ IN

- Where seen: UT Invitational fall 2019, B, INV, Q36 and UT Invitational fall 2019, C, INV, Q32 (15 W, 35 W, 40 W bulbs rated at 120 V, in series and in parallel on 120 V: currents; the series key is wrong). F-M Invitational 2018, INV, Q5 (120 V: 50 W bulb A in series with 60 W B || 40 W C; B burns out → A becomes dimmer but stays lit). Shock Value student key about 2011 (bulb ranking; the key is wrong).
- How to solve:
  1. Find each bulb's $R = V_{rated}^2/P_{rated}$.
  2. Parallel on the rated voltage: each gets its rated power. I = P/V.
  3. Series: R_{total} = sum. I = V/R_{total}. Each bulb's power = I^2R.
  4. In series, the **lowest-wattage** bulb (highest R) is the **brightest**.
- Worked example: R = 960, 411, 360 Ω. Series: I = 120/1731 = **0.0693 A** in all three. Powers: 4.6 W, 2.0 W, 1.7 W, so the 15 W bulb is brightest. Parallel: I = **0.125, 0.292, 0.333 A**. A second example (F-M): R_A = 288, R_B = 240, R_C = 360 Ω. Before: P_A = 22.2 W. After B burns out: P_A = **9.9 W**, so A is dimmer.

- Traps: The UT key added the rated watts (90 W) and got 0.750 A. Rated watts do not apply in series. When B burns out, the parallel group's R goes **up**, so the current through A goes down.
- Calculator: calculate — type: rated bulbs, series, parallel, burnout.

28. Protect parallel resistors with a series resistor (power ratings) — ✅ IN

- Where seen: North Pocono Jan 2020, C, INV, Q31 (add a series R_x so no resistor burns: minimum R_x; the key is wrong).
- How to solve:
  1. Find the weakest resistor: the one that reaches its rating at the lowest voltage.
  2. Its safe voltage: V = √(P·R). This is the most the parallel group may have.
  3. Group current: I = V_safe / R_group.
  4. R_x = (V_source − V_safe) / I.
  5. If R_x has its own rating, check I^2R_x too.
- Worked example: 9 V; 200, 100, 50 Ω in parallel; each 0.5 W. The 50 Ω part is safe up to √(0.5 × 50) = 5.0 V. R_group = 28.6 Ω. I = 5.0/28.6 = 0.175 A. R_x = 4.0/0.175 = **22.9 Ω**. But then R_x gets 0.70 W. If R_x is also 0.5 W, you need R_x ≥ **96.4 Ω**.
- Traps: The test key gave 35.7 Ω. That value fits neither reading. At 35.7 Ω, R_x still burns 0.70 W.
- Calculator: calculate — type: protect resistor, series resistor, power rating.

29. Electrical energy into mechanical work (lifting) — ✅ IN

- Where seen: not seen in a test in this form. Princeton 2020, C, Q33 is the closest (motor efficiency). Prior I_bank U22. OpenStax College Physics 2e, ch. 7 and 20.
- How to solve:
  1. Useful power = mgh/t (g = 9.8 m/s²).
  2. Input power = useful power / η.
  3. I = P_in / V.
- Worked example: A 12 V motor lifts 10 kg by 2.0 m in 4.0 s at 50% efficiency. P_out = 10 × 9.8 × 2.0/4.0 = 49 W. P_in = 98 W. I = 98/12 = **8.2 A**.
- Traps: The motor's physics (magnets) is ❌ OUT. The energy accounting is IN.
- Calculator: calculate — type: lift, motor, efficiency, current.

30. Heating water with an electric kettle — ⚠️ Borderline

- Why borderline: it needs the specific heat of water, which is a thermodynamics idea. The electric part (E = Pt) is IN.
- Where seen: not seen in a test. Prior I_bank U23. OpenStax College Physics 2e, ch. 14.
- How to solve:
  1. Heat needed: E = m × c × ΔT, with c = 4186 J/(kg·°C).
  2. Time = E / (P × η).
- Worked example: A 1000 W kettle heats 1.0 kg of water from 20 °C to 100 °C at 100% efficiency. E = 1.0 × 4186 × 80 = 334 880 J. t = **335 s** (5.6 min).
- Traps: 1 L of water has a mass of about 1 kg.
- Calculator: calculate — type: kettle, water, heat, time.

31. Average power over a duty cycle; standby power — ✅ IN

- Where seen: not seen in a test. Kuphaldt Socratic worksheets and home-energy problems use it.
- How to solve:
  1. Energy per day = P × hours actually on.
  2. Average power = energy per day / 24 h.
  3. Cost = kWh × price.
- Worked example: A 150 W fridge compressor runs 8 h a day. E = 1.2 kWh a day = **438 kWh a year**. A second example: a 5 W standby load on all year uses 5 × 24 × 365 = 43.8 kWh → **$6.57** at $0.15.
- Traps: "Rated 150 W" is not "uses 150 W all day". Use the hours it is really on.
- Calculator: calculate — type: duty cycle, standby, kWh per year.

32. Compare loads for maximum power; simple maximum power transfer — ⚠️ Borderline

- Why borderline: the series-resistor version is only algebra. The general theorem uses Thevenin equivalents (❌ OUT, rule 3.d).
- Where seen: ATX Regional 2026, B, REG, MC38 (largest power: 50 Ω, 1.2 nF, two 100 Ω in parallel, or a short → the short; the wording is contradictory). Soinc deck 4, quiz 4 (12 V, R_L = 16 Ω, source R = 16 Ω → 0.375 A, 2.25 W in the load, 4.5 W total). Full method: Internal Resistance and Power Transfer [A5].
- How to solve:
  1. With a fixed source resistance r, the load gets the most power when R_L = r.
  2. Then I = V/(2r) and P_load = $V^2/(4r)$.
  3. Half of the total power is wasted in r.
- Worked example: 12 V source with r = 16 Ω and R_L = 16 Ω. I = 12/32 = **0.375 A**. P_load = $0.375^2 \times 16$ = **2.25 W**. P_total = **4.5 W**.
- Traps: If you may choose the **source** resistance for a fixed load, the best choice is r = 0, not r = R_L. Read which one you may change.
- Calculator: calculate — type: max power, matched load.

Nationals

33. Power sign convention with mixed sources — ✅ IN (P = VI); ✅ States-Nationals when KVL/KCL is needed

- Where seen: MIT 2021, C, INV, Scenario B Q19-33 (V = 2.3 V, R = 120 Ω, I = 46 mA current source: resistor 0.25 W consumed; current source supplies 0.36 W). BullSO Jan 2026, B, INV, FR Q6 (two sources, 20 V and 12 V, in one loop). See Kirchhoff Analysis (States-Nationals) [A5].
- How to solve:
  1. Use the "passive sign" rule. If current **enters the + terminal**, the part **absorbs** power (+P).
  2. If current **leaves the + terminal**, the part **supplies** power (–P absorbed).
  3. The sum of all absorbed powers is 0.
- Worked example: A 20 V and a 12 V battery in series aid each other in one loop with 22 Ω and 18 Ω. I = 32/40 = 0.80 A. The 20 V source supplies 16 W. The 12 V source supplies 9.6 W. The resistors absorb $0.8^2 \times 22$ = 14.08 W and $0.8^2 \times 18$ = 11.52 W. Total 25.6 W = 25.6 W ✓.
- Traps: A battery being charged **absorbs** power. Its current goes into its + terminal.
- Calculator: calculate — type: power sign, absorbed, supplied.

34. Per-capita and national power estimation (orders of magnitude) — ✅ IN

- Where seen: Florida State 2021, B, ST, Q25 (power used by the average US citizen in 2010; the key gives the national total).
- How to solve:
  1. National energy per year → divide by seconds per year (3.15×10^7 s) → watts.
  2. Divide by the population for "per person".
- Worked example: About 98 quads a year is about 3.3×10^{12} W for the US. With 3.09×10^8 people: 3.3×10^{12} / 3.09×10^8 = **1.1×10^4 W per person** (about 11 kW, all energy uses, not only electricity).
- Traps: Electricity alone per person is much smaller, about 1.3-1.5 kW. Say which one you used.
- Calculator: look up — type: per capita power, estimate.

35. Chained hands-on: measure, then compute power supplied — ✅ IN (rule Part II.c.ii)

- Where seen: Berkeley Feb 2026, B, INV, FR 2 (the full circuit: R_eq, currents, power, energy in 2 h → 21.5 kJ). BullSO Jan 2026, B, INV, SA2. The 2027 rule lists "Calculate the power supplied to a circuit" as a hands-on task.
- How to solve:
  1. Measure the source voltage with the circuit **on** (the loaded voltage).
  2. Measure the total current (meter in series) or the voltage across a known resistor.
  3. P = V × I. Or add up V^2/R for each part.
  4. Give units and sensible significant figures.
- Worked example: You measure 8.92 V at the battery and 0.334 V across a 10.0 Ω series resistor. I = 0.334/10.0 = 33.4 mA. P = 8.92 × 0.0334 = **0.298 W**.

- Traps: The 9 V label is not the real voltage. Measure it under load. See Multimeter Use [A9].
- Calculator: calculate — type: power supplied, measured V, measured I.

Out of scope (skip)

- **Power factor, real/reactive/apparent power (VA, VAR).** ✗ OUT: AC circuit theory, rule 3.d.
- **Three-phase power (√3 × V × I).** ✗ OUT: 3-phase power is named in rule 3.d.
- **Energy stored in a capacitor (½CV²) or an inductor (½LI²).** ✗ OUT: capacitance and inductance, rule 3.d. BullSO 2026, B, Q32 asked where a capacitor stores energy.
- **Transformer power (P_primary = P_secondary).** ✗ OUT: magnetism. Pembroke 2026 Q34, BullSO 2026 Q19-20.
- **Op-amp power gain.** ✗ OUT: op-amps are not in rule 3.c. ATX Regional 2021, C, Q52.
- **Motor torque equations.** ✗ OUT: magnetism. UT 2020, B, Q28.
- **Power from a curved P-t graph by integration, or RMS of a sine with trig.** ✗ OUT: calculus and trig, rule 3.e.v. The simple fact "100 V RMS AC gives the same power as 100 V DC" is in CL 3 - DC and AC [T] (BullSO 2026 B Q22).

Mistakes in tests and keys

- **MIT 2019, C, INV, Q13.** Key: 65 kWh. Correct: 350 hp × 3 h = **783 kWh**. 65.25 is kAh (21 750 A × 3 h), not kWh.
- **MIT 2019, C, INV, Q14.** Key: 2.2 h. Correct: 85 kWh / 186.4 kW = **0.456 h**. The key turned the ratio upside down.
- **UT Invitational fall 2019, B, Q36 (= C, Q32).** Series key: 0.750 A each. Correct: **0.0693 A** each (use R = V²/P). The key added the rated watts. The labels A and B are also swapped between the text and the figure.
- **North Pocono Jan 2020, C, INV, Q31.** Key: 35.7 Ω. Correct: 22.9 Ω (only the original three count) or 96.4 Ω (R_x's own 0.5 W rating also counts).
- **Florida State 2021, B, ST, Q25.** Key: 3×10^{12} W, which is the **US total**, not per citizen. Per citizen is about 1.1×10^{4} W. No choice fits.
- **UF Northern Regional 2020, B, REG, Q29.** Key: "a motor turns mechanical into electrical energy" is True. That is a generator, so the answer is False.
- **Pembroke Hill Dec 2025, B, INV, Q29.** 600 W/120 V = 5 A exactly on a 5 A fuse. The key says "no, but it's close". It is a fair answer, but the question has no clear "right" side.
- **ATX Regional 2021, B, REG, Q43.** Total power key 390.62 W. The drawing gives **384.00 W**. The key paired the resistors in a way the drawing does not show.
- **UMSO 2019, Q22.** The question says "the circuit in problem 20". It means problem 21 (9 V × 0.5 A = 4.5 W).

Not confirmed

- MIT 2020, C, Q26 (17.3 kJ) and Rickards 2026 B Q18 ($3.60 a year) come from prior/G_tests.md. I did not re-read the question texts. Both numbers check in Python.
- Princeton 2020 Q33 (13%): 20 N·mm × 0.05 rad/s = 1.0 mW; 1.0/7.5 = 13%. I did not re-read the question.
- USB levels, typical appliance powers, electricity prices and the LED/incandescent efficiency numbers are general knowledge. I did not check them in a source in this run. Prior B_topics §9d also marks the USB table as "verify in a reference".
- US energy use (98 quads in 2010, 3.09×10^{8} people) is from the state.md extraction. I did not re-check the EIA figure.
- ATX Regional 2021, B, Q43 (384 W) is the extraction's reading of the figure. I did not render that page again.
- No real **Nationals** Circuit Lab test was available. The "USC 2026" files are the USC Invitational. So the Nationals types here come from top invitationals and the rules.

Sources

- 2027 Div B rules, Circuit Lab (pages B12-B13), rules 3.b, 3.c.iv, 3.c.ix, 3.d, 3.e, Part II.c.ii, Part II.c.iv.
- Test extractions in `circuitlab/research/tests/`: atx-reg (ATX Regional 2020 B, 2021 B/C), misc-a (Cobra 2019-20, F-M 2018, North Pocono 2020, UMSO 2019), misc-b (MIT 2019, UF Northern Regional 2019 B/C and

2020 B, NY Lake Erie 2018, JC Booth 2011, Shock Value tryout), nat26 (USC Invitational 2026 B), state (Florida State 2021, MYSO 2020), ut-a (ATX Regional 2026 B), ut-b (UT 2019 B, UT 2020 B), ut-c (UT 2018, UT 2019 C, ATX Regional 2019).

- 2026 Div B tests from `downloads/dlB/tests/`: Berkeley (Feb 28, 2026) Q22, FR 2; BullSO (Jan 24, 2026) Q20, Q22, Q32, SA2, FR 6; Pembroke Hill (Dec 6, 2025) Q3, Q7, Q29, Q32, Q34.
- prior/G_tests.md (MIT 2020 and 2021, Princeton 2020, Rickards 2026), prior/I_bank.md (U7-U11, U20-U23), prior/B_topics.md §4 and §9d, prior/F_slides.md §5 (soinc deck 4 and the homework generator).
- OpenStax College Physics 2e, ch. 20.4 (electric power and energy) and ch. 14 (specific heat): general knowledge of the book; I did not re-open it in this run.
- Kuphaldt, *Lessons in Electric Circuits*, Vol. I, ch. 2 (power) and the Socratic worksheets: general knowledge.
- USB-IF specifications (USB 2.0, USB 3.x, BC 1.2, Type-C, Power Delivery 3.1): general knowledge, not re-checked.
- All worked examples were checked in Python.

TAB A5

DC Circuits

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Each note: key facts, formulas, then question types from Easy to Nationals with worked examples.

Schematic Symbols and Reading Diagrams

Note: Schematic Symbols

Rule area **3.c.v** names "circuit schematics". Area **3.c.vi** adds switches, relays, fuses, GFCIs and breakers. Area **3.c.vii** adds diodes and LEDs. Area **3.c.viii** adds logic gates. Every hands-on station also asks you to "draw the schematic of the circuit you built". Symbol questions are the easiest points on the test. Some tests ask 10 or more of them.

Key facts

- A **schematic** = a drawing that shows how parts connect, using standard symbols. It does not show where the parts sit in real life.
- A **wiring (pictorial) diagram** shows the real parts and the real wire routes. A **one-line diagram** shows a power system with one line for all the wires of a circuit.
- Two symbol standards exist:
  - **ANSI / IEEE 315** (American). Resistor = zigzag.
  - **IEC 60617** (international, used in Europe). Resistor = plain rectangle.
  - A test can use either. Learn both.
- **Wires:** a straight line = a wire with 0 Ω. A **dot** where lines meet = connected (a **junction** or **node**). Two lines that cross with **no dot** = not connected. A small **hop** (half circle) on one line also means "not connected".
- **Cell** = one long line and one short line. The **long line is +**. The short (often thicker) line is –.
- **Battery** = two or more cells in series. Count the long/short pairs to count the cells. Some drawings show two pairs with a dashed line between them for "many cells".
- **DC source** = a circle with + and –. **AC source** = a circle with a sine wave (~) inside. **Current source** = a circle with an arrow inside.
- **Variable source** = a source symbol with a diagonal arrow through it.
- **Ground** symbols:
  - **Earth ground** = three horizontal lines that get shorter (a downward "triangle" of lines). It connects to the real earth.
  - **Chassis (frame) ground** = a line with three slanted "rake" teeth. It connects to the metal case.
  - **Signal (common) ground** = a small hollow triangle pointing down. It is the 0 V reference of the circuit.
  - **Protective earth** = the earth symbol inside a circle. It is the safety ground of mains equipment.
  - Some books swap these names. Read every option.
- **Resistor family:**
  - **Fixed resistor**: zigzag (ANSI) or rectangle (IEC).
  - **Variable resistor**: a resistor with a diagonal arrow through it.
  - **Potentiometer** ("pot") = a resistor with an arrow that points **at the middle** of it. That arrow is the **wiper** (the sliding contact). It has three terminals.
  - **Rheostat** = a variable resistor with two terminals. It is often drawn as a pot with the wiper tied to one end.
  - **Thermistor** = a resistor with a diagonal line that has a short flat "hockey stick" end, often with "T" or "t°". Its resistance changes with temperature.
  - **Photoresistor (LDR)** = a resistor with two arrows pointing **in** (light coming in).
- **Lamp (bulb):** a circle with an X inside (IEC), or a circle with a small loop (filament) inside (ANSI).
- **Switches:**
  - **SPST** = single pole, single throw: one moving blade, one contact (on/off).
  - **SPDT** = single pole, double throw: one blade that picks one of two contacts.
  - **DPST / DPDT** = two blades that move together, joined by a dashed line.
  - **Push button NO** (normally open) = a bar above two contacts. Pushing closes it. **NC** (normally closed) = a bar touching the contacts from below. Pushing opens it.
  - **Rotary switch** = one blade with many contacts in an arc.
  - **Reed switch** = switch contacts inside a capsule (a magnet closes it; ⚠ magnetism is out, the symbol is OK).

- **Fuse:** a rectangle with a line through it (IEC), or an S-shaped wire between two terminals (ANSI). **Circuit breaker:** a switch-like arc, often with a small curved hook or a square.
- **Relay** = a coil (a rectangle or a row of loops) plus a set of switch contacts. A dashed line joins the coil and the contacts. Reference letter **K**.
- **Diodes:** triangle + bar. The triangle points in the direction of conventional current. The **bar is the cathode (–)**. LED = diode + two arrows pointing **out**. Photodiode = arrows pointing **in**. Zener = bar with bent ends (Z shape). Schottky = bar with S-shaped hooks. See LEDs A7 and Testing Diodes A7.
- **Meters:** a circle with a letter. **V** = voltmeter. **A** = ammeter. **Ω** = ohmmeter. **G** (or an arrow) = galvanometer. **W** = wattmeter.
- **Loads: M** in a circle = motor. A bell or half-dome shape = buzzer. A small box with a cone = speaker. (Their physics is out, but a test can use them as loads.)
- **Logic gates** have their own symbols. See Logic Gates and Notation A8. A triangle with no bubble = buffer. A bubble (small circle) = NOT.
- **Reference designators** = the letter code next to each part: R (resistor), BT or B (battery), SW or S (switch), D (diode), LED, F (fuse), CB (breaker), K (relay), L (lamp or inductor), M (motor), U or IC (chip), J/P (connector). Numbers make each one unique (R1, R2...).
- A schematic should show each part's **value and unit** (for example, R1 = 330 Ω, BT1 = 9 V).
- **Real circuits differ from schematics** because real wires have some resistance, real sources have internal resistance, and real parts have tolerances. The laws (Ohm, Kirchhoff) are not "only theory". (ATX 2021 B Q28.)
- **PCB** = printed circuit board: a board with copper traces that replace wires. The usual material is woven glass and epoxy (FR-4).

Formulas

- Cells in series: **V_battery = n × V_cell** (for example, 1.5 V per alkaline cell).
- Two elements are **in series** if one current must pass through both, with no junction between them.
- Two elements are **in parallel** if both of their ends touch the **same two nodes**.
- Count nodes: every set of points joined only by wires is **one** node, even if it looks long or bent.

Question types

Easy

1. Name the symbol (basic set) ✅ IN

- Seen: UF Northern Regional 2020, Div B, REG, Q21a-j (name 10 symbols: resistor, ammeter, diode, voltmeter, switch, AC source, capacitor, inductor, LED, variable resistor). BullSO 2026, Div B, INV, Short Answer 1a-e (resistor, capacitor, battery, diode, switch). SO MYSO practice Nov 2020, Div B, practice, Q24 ("What is A?" → ammeter). VA Regional 2013, Div C, REG, Q26 (→ diode). National supervisor practice deck 9 (symbol chart). ATX Regional 2026, Div B, REG, MC12-MC25 (lettered image sheet).
- How to solve:
  1. Look at the outer shape first (circle = source or meter; triangle + bar = diode; zigzag/box = resistor).
  2. Then look at what is inside or attached (letter, arrow, wave, bubble).
  3. Name the exact part, not just the family ("LED", not "diode").
- Example: A circle with a ~ wave inside. **Answer: an AC voltage source.**
- Traps: Arrows **out** = LED; arrows **in** = photodiode or LDR. A sine wave in a circle is AC, not "variable" (the UF 2020 key got this wrong).
- Calculator: look up — type: `symbol <word>` (for example `symbol rake`)

2. Draw the symbol for a named part ✅ IN

- Seen: JC Booth Invitational 2011, Div B (Shock Value), INV, Q1 (resistor, variable R, capacitor, battery, switch, ammeter, push button, coil). Shock Value student test ~2011, Div B, practice, Section IV (variable resistor, coil, capacitor, AC source, push button, diode, inductor, fuse, heater, DPDT switch, galvanometer, cell). Hawk and Hornet 2026, Div C, INV, FR10a-d (relay, NPN BJT, potentiometer, LED). Northview 2025, Div C, INV, FR5 (relay,

diode, variable resistor, DPDT). Shock Value re-tryout ~2012, Div B, practice, IV-2 (breaker, diode, fuse, battery, lamp, push switch, variable R, capacitor, AC supply, junction, cell).

- How to solve: Draw the standard ANSI shape. Label its terminals if they matter (+ on a cell; anode/cathode on a diode; wiper on a pot).
- Example: Draw a potentiometer. **Answer: a zigzag resistor with an arrow pointing at its middle from the side. Three terminals: two ends and the wiper.**
- Traps: A pot's arrow points **at** the resistor. A variable resistor's arrow goes **through** it at a slant. Hawk and Hornet's key says symbol variations exist; draw the most common one and label it.
- Calculator: look up — type: `draw <part>`

3. Cell vs battery; which end is +; count cells ✅ IN

- Seen: SO MYSO practice Nov 2020, Div B, practice, Q23 (bulb circuit: how many batteries? → 2). ATX Regional 2026, Div B, REG, MC24 (single cell → D) and MC25 (multi-cell battery → E). Shock Value student test, Div B, practice, Section IV ("cell").
- How to solve:
  1. Each long/short line pair = one cell.
  2. The long line is +.
  3. Multiply by the cell voltage if you need the total.
- Example: A battery symbol has 4 long/short pairs. Each cell is 1.5 V. **Answer: 4 cells, 6.0 V. The long line at the end is the + terminal.**
- Traps: Do not count lines; count **pairs**. Two pairs with a dashed gap still means "several cells", not exactly two, unless the label says so.
- Calculator: calculate — type: `n, Vcell`

4. Wire junction (dot) vs crossing (no dot) ✅ IN

- Seen: Pembroke Hill 2025-26, Div B, INV, Q26 ("two wires cross with a dot... they electrically ___" → connected).
- How to solve: Dot = connected. No dot = not connected. Hop = not connected.
- Example: Two wires cross in a plus sign with no dot. **Answer: they are not connected.**
- Traps: Old drawings sometimes show a "+" crossing as connected. Modern practice avoids 4-way junctions; it uses two T-junctions with dots instead.
- Calculator: look up — type: `junction`

5. Ground symbols: earth, chassis, signal, protective ✅ IN

- Seen: ATX Regional 2026, Div B, REG, MC19 (earth ground; the key picks the chassis "rake" symbol — see Mistakes) and MC20 (low-noise / clean ground → B). Rickards 2026, Div C, INV, Q6 TB#1 (earth symbol in a circle → protective ground, D).
- How to solve: Three shrinking lines = earth. Rake = chassis. Hollow triangle = signal. Earth in a circle = protective earth.
- Example: A symbol shows three horizontal lines that get shorter, inside a circle. **Answer: protective (safety) earth ground.**
- Traps: Tests disagree on names. Pick the option that matches the shape, and write a note if two options both fit.
- Calculator: look up — type: `ground`

6. Resistor family symbols: fixed, variable, pot, rheostat ✅ IN

- Seen: Pembroke Hill 2025-26, Div B, INV, Q21 (zigzag with a diagonal arrow → variable resistor). Rickards 2026, Div C, INV, Q3 (arrow pointing at the middle → potentiometer, B). ATX Regional 2026, Div B, REG, MC17 (potentiometer symbol). Cobra Invitational 2019-20, Div B, INV, Q28 (rheostat and potentiometer among 6 symbols).
- How to solve: Arrow through at a slant = variable. Arrow pointing at the side = pot (wiper). Two terminals with an adjustable part = rheostat.
- Example: A zigzag has an arrow touching its middle from above. It has 3 leads. **Answer: a potentiometer.**
- Traps: Rickards Q3 offers "variable resistor" and "potentiometer". The arrow **at** the body (with a third lead) makes it a pot.

- Calculator: look up — type: `pot`

7. Switch symbols: SPST, SPDT, DPST, DPDT, push buttons, rotary ✅ IN

- Seen: Princeton 2026, Div C, INV, MC8 (switch symbol → SPDT, A). Rickards 2026, Div C, INV, Q25 (→ push button, D). ATX Regional 2026, Div B, REG, MC21 (reed), MC22 (rotary), MC23 (push button). Cobra Invitational 2019-20, Div B, INV, Q28 (DPST, thermal switch). Northview 2025, Div C, INV, FR5D (draw DPDT).
- How to solve:
  1. Count the blades (poles). One = SP. Two joined by a dashed line = DP.
  2. Count the contacts each blade can touch (throws). One = ST. Two = DT.
  3. Push buttons have a T-shaped plunger. Above the contacts = NO. Touching from below = NC.
- Example: Two blades joined by a dashed line, each with two contacts. **Answer: DPDT (double pole, double throw). It has 6 terminals.**
- Traps: SPDT has 3 terminals. DPST has 4. DPDT has 6. A 3-way household switch is an SPDT. See CL 6 - Switches, Relays, Fuses, Breakers, GFCI [T].
- Calculator: look up — type: `switch poles throws`

8. Protection and control symbols: fuse, breaker, relay, thermal switch ✅ IN

- Seen: ATX Regional 2026, Div B, REG, MC11 (molded-case breaker; key doubtful — see Mistakes). Cobra Invitational 2019-20, Div B, INV, Q28 (SPST NO relay, thermal switch, fuse). Hawk and Hornet 2026, Div C, INV, FR10a (draw a relay). Northview 2025, Div C, INV, FR5A (draw a relay). Shock Value re-tryout ~2012, Div B, practice, IV-2 (breaker, fuse).
- How to solve: Fuse = rectangle with a line through it, or an S-wire. Breaker = an arc contact. Relay = a coil + contacts, joined by a dashed line.
- Example: A rectangle labeled "K1" is joined by a dashed line to an open switch. **Answer: a relay with a normally open (NO) contact. The coil (K1) closes the switch when current flows in it.**
- Traps: A relay coil drawn as loops looks like an inductor. The dashed link to contacts tells you it is a relay.
- Calculator: look up — type: `fuse` / `relay` / `breaker`

9. Diode family symbols (diode, LED, photodiode, Zener, Schottky) ✅ IN

- Seen: UF Northern Regional 2019, Div C, REG, Q37-38 (circle the LED symbol; what LED means). UF Northern Regional 2020, Div C, REG, Q18 (box the LED). ATX Regional 2026, Div B, REG, MC14 (LED) and MC18 (photodiode). Rickards 2026, Div C, INV, Q9 TB#2 (→ Schottky, C). MIT Invitational 2019, Div C, INV, Q33 (diode in a matching set). VA Regional 2013, Div C, REG, Q26 (→ diode).
- How to solve: Find the triangle and bar. Then look at the bar shape and the arrows.
- Example: A diode bar has small hooks at each end that curl in opposite directions, like an S. **Answer: a Schottky diode.**
- Traps: Zener = bent ends (Z). Schottky = curled ends (S). The bar is always the cathode. See Special Diodes [A7].
- Calculator: look up — type: `diode symbol`

10. Meter symbols and the "Ampmeter" trap ✅ IN

- Seen: SO MYSO practice Nov 2020, Div B, practice, Q9 ("Ampmeter" distractor → ammeter) and Q24 ("A" → ammeter). UF Northern Regional 2020, Div B, REG, Q21 (ammeter, voltmeter). Shock Value student test ~2011, Div B, practice, Section IV (galvanometer).
- How to solve: Read the letter in the circle. V = voltmeter. A = ammeter. Ω = ohmmeter. G = galvanometer.
- Example: Which device measures current: ampmeter, ammeter, voltmeter or ohmmeter? **Answer: ammeter. "Ampmeter" is not a real name.**
- Traps: A galvanometer measures small currents (it is the moving part inside old analog meters). It is not a voltmeter.
- Calculator: look up — type: `meter`

Medium

11. ANSI vs IEC versions of the same part ✅ IN

- Seen: National supervisor practice deck 9 (resistor in zigzag and box forms). Not seen as a direct question in a test.
- How to solve: Learn these pairs:
  - Resistor: zigzag (ANSI) vs rectangle (IEC).
  - Lamp: circle with a filament loop (ANSI) vs circle with an X (IEC).
  - Fuse: S-wire (ANSI) vs rectangle with a line through it (IEC).
  - Logic gates: curved shapes (ANSI) vs rectangles with "&", "≥1", "=1", "1" (IEC). See Logic Gates and Notation A8.
- Example: A plain rectangle with two leads, labeled 220 Ω. **Answer: a resistor (IEC style). It is not a fuse, because a fuse has a line through it and is rated in amps.**
- Traps: A plain rectangle can also be a relay coil or a "generic load". Use the label and the unit to decide.
- Calculator: look up — type: `iec <part>`

12. Sensor symbols: thermistor, LDR, photovoltaic cell ⚠️ Borderline (these parts are not named in 3.c; the symbol question is light)

- Seen: Berkeley 2026, Div B, INV, Q11 (resistor that changes with temperature → thermistor). ATX Regional 2026, Div B, REG, MC12 (PV cell → F) and MC13 (photoresistor → J). GullSO 2025, Div C, INV, Q26 (manually adjusted resistor → rheostat; photoresistor as a distractor).
- How to solve: Arrows in + resistor = LDR. "T" or a bent line through a resistor = thermistor. A cell symbol in a circle with arrows in = PV (solar) cell.
- Example: Which part changes resistance with light: varistor, photoresistor, rheostat, thermistor? **Answer: photoresistor (LDR). Its resistance goes down when light gets brighter.**
- Traps: Varistor = changes with voltage. Thermistor = changes with temperature. Rheostat = changes by hand.
- Calculator: look up — type: `sensor`

13. Reject out-of-scope symbols used as distractors ⚠️ Borderline (the parts are ❌ out by 3.d, but tests still show them)

- Seen: Florida State 2021, Div B, STATE, Q18 (photo → capacitor), Q19 (current mirror → transistor), Q20 (op-amp). BullSO 2026, Div B, INV, SA1b (capacitor). MIT Invitational 2019, Div C, INV, Q33 (capacitor, variable capacitor). Rickards 2026, Div C, INV, Q3 (inductor as a distractor). Hawk and Hornet 2026, Div C, INV, FR10b (NPN BJT). UF Northern Regional 2020, Div C, REG, Q25-26 (draw PNP, NPN).
- How to solve: Know them only well enough to say "that is not a resistor".
  - Capacitor = two parallel plates (one curved if polarized).
  - Inductor = a row of loops (a coil).
  - Transformer = two coils side by side with lines between them.
  - Transistor = a circle with three leads, one with an arrow.
  - Op-amp = a triangle with + and − inputs.
- Example: Two short parallel lines of equal length, with a gap. **Answer: a capacitor (out of scope for 2027).** A cell has one long and one short line.
- Traps: The capacitor looks like a cell. Equal lines = capacitor. Unequal lines = cell.
- Calculator: look up — type: `out symbol`

14. Draw a schematic from a word description ✅ IN

- Seen: Cobra Invitational 2019-20, Div B, INV, Q31 (draw the LED circuit with standard symbols; label values and units). Shock Value re-tryout ~2012, Div B, practice, IV-1-3 (bulb, diode, DC source, switch, fuse). UT Invitational fall 2025, Div B/C, INV, Q5b (source, resistor, SPST switch, LED in series). National supervisor practice deck 1 homework (draw the circuit of one room and of a flashlight).
- How to solve:
  1. List the parts in the order the current meets them.
  2. Draw the source on the left, with + on top.
  3. Draw one closed loop through every series part.
  4. Put the fuse right next to the + terminal.
  5. Draw the diode/LED with its triangle pointing in the direction of current (from + toward −).

6. Label each part with a designator, a value and a unit.

- Example: 9 V battery, SPST switch, 1 A fuse, 330 Ω resistor, red LED, all in series. **Answer: BT1 (9 V) + → F1 (1 A) → SW1 → R1 (330 Ω) → LED1 (triangle pointing away from R1) → back to BT1 –. LED current ≈ (9 – 2.0)/330 = 21 mA.**
- Traps: A missing label loses points. An LED drawn backwards is a wrong circuit.
- Calculator: calculate — type: `Vs, Vf, R` (for the LED current)

15. Draw the schematic of a breadboard circuit you built ✅ IN

- Seen: GullSO 2025, Div C, INV, Q49b ("draw the schematic for the circuit that you built; label all known values") and Q50d. MIT Invitational 2020, Div C, INV, Lab 2 (identify a pre-built gate, draw its schematic). Princeton 2020, Div C, INV, Lab 2 (gate circuit → schematic).
- How to solve:
  1. Each 5-hole row of the breadboard is **one node**.
  2. Each long rail is one node (check for a split in the middle).
  3. Give each node a dot on paper. Then draw each part between the two nodes its legs sit in.
  4. Tidy the drawing so series parts line up.
- Example: A resistor sits in rows 5 and 10. An LED sits in rows 10 and 14. Row 5 has a wire to the + rail. Row 14 has a wire to the – rail. **Answer: + rail → resistor → LED → – rail. It is one series loop with node "row 10" between the resistor and the LED.**
- Traps: Both legs in the same row = the part is shorted. A part that straddles the center trench connects two different rows.
- Calculator: look up — type: `breadboard`

16. Read a schematic: which parts are in series, which in parallel ✅ IN

- Seen: SO MYSO practice Nov 2020, Div B, practice, Q23 (Bulb 1 in series with a parallel group). National supervisor practice deck 4 (series and parallel). JC Booth Invitational 2011, Div B, INV, Q42 (longest series path → L1, L4, L5).
- How to solve:
  1. Color each node (every wire-connected set) with its own color.
  2. Two parts that share **both** colors are in parallel.
  3. Two parts that share one node with **nothing else** attached there are in series.
- Example: A battery feeds bulb 1. After bulb 1, the wire splits to bulb 2 and bulb 3, which rejoin and return to the battery. **Answer: bulbs 2 and 3 are in parallel. That pair is in series with bulb 1.**
- Traps: A part drawn at a slant or upside down can still be in parallel. Only the nodes matter.
- Calculator: look up — type: `series parallel`

17. Reference designators and labels ✅ IN

- Seen: not seen in a test as its own question. It is needed to read every schematic (R1, SW1, F1, K1).
- How to solve: Learn the letters in Key facts.
- Example: A schematic lists "K2: 12 V coil". **Answer: K = relay. K2 is the second relay, with a 12 V coil.**
- Traps: L can mean a lamp or an inductor. Check the symbol.
- Calculator: look up — type: `designator <letter>`

18. Schematic vs reality; schematic vs wiring diagram; PCB ✅ IN

- Seen: ATX Regional 2021, Div B, REG, Q28 ("which is NOT a reason diagrams differ from reality" → E, "laws are only theoretical"). Florida State 2021, Div B, STATE, Q35 (PCB stands for → printed circuit board). Rickards 2026, Div C, INV, Q17 (least common PCB material; key doubtful).
- How to solve: Real wires have resistance. Real sources have internal resistance. Real parts have tolerance. The laws still hold exactly.
- Example: Why does a real 9 V battery read 8.6 V when it powers a load? **Answer: its internal resistance drops some voltage. The schematic draws it as an ideal 9 V source.** See Internal Resistance and Power Transfer A5.
- Traps: "The laws are only theoretical" is never a correct reason.
- Calculator: look up — type: `pcb`

19. Trace current paths for each switch setting ✅ IN

- Seen: UT Invitational 2019, Div B, INV, Q43 (3 SPDT switches and 3 lamps: 8-row on/off table). SO MYSO practice Nov 2020, Div B, practice, Q26-Q30 (open SW1, close SW2: what each bulb does). Shock Value re-tryout ~2012, Div B, practice, IV-1-1 (3-way switch) and IV-1-2 (DPDT reverses a motor). National generator Level 9 (switch-state table).
- How to solve:
  1. Redraw the circuit once for each switch setting.
  2. Remove open switches. Replace closed switches with wires.
  3. Find each lamp's path. No closed loop through the lamp = it is off.
- Example: Two SPDT switches wired as a 3-way (travelers joined). Switch A up, switch B up = lamp on. **Answer: flipping either switch turns the lamp off. Flipping both turns it on again. The lamp is on when A and B match (XNOR of the two positions).**
- Traps: Some drawings short the battery in one switch position (UT 2019 B Q43 with S1 OFF). Say so in your answer.
- Calculator: look up — type: `3way`

Hard

20. Spot a part that is shorted by a wire in the drawing ✅ IN

- Seen: ATX Regional 2026, Div B, REG, FR2a (100 Ω R4 with a plain wire in parallel → 0 A). USC Invitational 2026, Div B, INV, Q10 (rank 4 circuits; one hides a shorted resistor). UT Austin (ATX) Regional 2019, Div B/C, REG, Q19 (188.3 Ω shorted). JC Booth Invitational 2011, Div B, INV, Q41 (darkest bulb → L6, shorted by a wire). Full treatment: Open and Short Circuits A5 type 9.
- How to solve: Find any part whose two ends are on the same node. Cross it out. It carries no current.
- Example: R1 = 100 Ω and R2 = 100 Ω in series on 10 V, with a wire drawn from one end of R2 to the other. **Answer: R2 is shorted. I = 10/100 = 0.10 A. V_R2 = 0.**
- Traps: The wire can be drawn far from the resistor, around a corner. Trace nodes, do not trust the look.
- Calculator: calculate — type: `V, R list` (after removing the shorted part)

21. Count nodes, branches and loops in a schematic ✅ IN (✅ States-Nationals when it feeds KVL/KCL)

- Seen: covered with Kirchhoff problems. See Kirchhoff Analysis (States-Nationals) A5 type 5. MIT Invitational 2019, Div C, INV, lab S1-5 ("can circuit 3 be reduced with series/parallel only?" → no).
- How to solve: Merge every wire-joined region into one node. Branch = a path between two nodes with parts. Loop = any closed path.
- Example: A battery feeds two resistors in parallel. **Answer: 2 nodes, 3 branches (battery, R1, R2), 3 loops (but only 2 independent loops, called meshes).**
- Traps: Corners and long wires are not extra nodes.
- Calculator: look up — type: `nodes`

22. Name an unfamiliar circuit from its schematic ⚠️ Borderline (general reading skill; the parts are switches, which are IN)

- Seen: Rickards 2026, Div C, INV, Q47 (a 4 × 4 grid of push buttons wired to 8 pins → matrix keypad, A). USC Invitational 2026, Div B, INV, hands-on part a (map the internal wiring of a 4-pin push button; draw it).
- How to solve: Look for a pattern (a grid, a bridge, a ladder). Ask: what do the parts do together?
- Example: 16 push buttons in 4 rows and 4 columns. Each button joins one row wire to one column wire. **Answer: a matrix keypad. A chip finds the pressed key by its row and column.**
- Traps: "Crossbar switch" is a telephone exchange term. The keypad is the common answer.
- Calculator: look up — type: `matrix`

Nationals

23. Read a schematic and set up the math (with the 2-equation limit) ✅ States-Nationals

- Seen: USC Invitational 2026, Div B, INV, Q10 (rank circuits by R_eq). See Kirchhoff Analysis (States-Nationals) A5 type 24.

- How to solve: Count unknown nodes. Two or fewer → nodal analysis is allowed (rule 3.e.i). Three or more → look for symmetry, a balanced bridge, or a shorted part that removes a node.
- Example: A bridge has 5 resistors and one battery. It is balanced (R1/R2 = R3/R4). **Answer: the middle resistor carries no current. Remove it. The circuit becomes two series pairs in parallel. No equations needed.**
- Traps: A test writer must keep to 2 equations. If you need 3, you have missed a short cut.
- Calculator: calculate — type: `R1, R2, R3, R4` (balance check)

Out of scope (skip)

- ❌ Drawing or naming **transistor** (BJT, MOSFET) and **op-amp** symbols as working parts: Hawk and Hornet 2026 C FR10b; UF Northern Regional 2020 C Q25-26; Florida State 2021 Q19-20. Rule 3.d: semiconductors beyond those listed.
- ❌ **Capacitor, inductor and transformer** as topics (current and voltage transformer symbols in ATX Regional 2026 B MC15-16). Rule 3.d: capacitance, inductance. Only reject them as distractors.
- ❌ **3-phase** one-line diagrams. Rule 3.d: 3-phase power.
- ❌ **Oscilloscope** symbols and traces. Rule 3.d: oscilloscopes.
- ❌ **Flip-flops and clock** symbols. Rule 3.d: sequential logic, clock signals.

Mistakes in tests and keys

- **UF Northern Regional 2020, Div B, Q21(f):** the key says "variable voltage source". The symbol is a sine wave in a circle, so it is an **AC voltage source**.
- **ATX Regional 2026, Div B, MC19:** the key picks C (the rake). The rake is **chassis** ground. The standard earth ground (three shrinking lines, image A) is not an option. The question is flawed.
- **ATX Regional 2026, Div B, MC17:** the answer "B) C" is confusing. Capital C on the sheet is the chassis ground; the pot is image "I" (sub-label "c"). The intended answer is the pot image.
- **ATX Regional 2026, Div B, MC11:** the key picks V for the molded-case breaker. In one-line diagrams the plain arc (W) is usually the low-voltage molded-case breaker; the chevrons on V mean "draw-out". Probably W. Not confirmed.
- **Rickards 2026, Div C, Q17:** the key marks "woven glass" as the **least** common PCB material. Woven glass + epoxy (FR-4) is the **most** common. The key looks reversed. (Not confirmed against a standard; this is general knowledge.)
- **UT Invitational 2019, Div B, Q43:** the drawn circuit shorts the battery when S1 is OFF. The key does not mention it.

Not confirmed

- ATX Regional 2026 B MC20 (low-noise ground = B): plausible, but I found no standard that names a "low-noise ground" symbol. I did not search the web.
- Ground symbol names vary between books (signal vs chassis vs earth). The names here follow IEC 60417 / IEEE 315 as I know them from memory; I did not open the standards.
- MIT Invitational 2019 C Q33 matching letters come from a garbled text layer. The earlier extraction did not re-check them against the symbols.
- The USC Invitational 2026 tests sit in the "NATIONALS 2026" research group, but every file says "USC Invitational, January 17, 2026". I tagged them INV. The real 2026 Nationals tests are sold only in the soinc test packet; I did not see them.
- No STATE or NATIONAL test in the downloads had a pure symbol-drawing question. Florida State 2021 had photo/schematic ID (Q18-21).

Sources

- 2027 Div B rules, Circuit Lab, 3.c.v-viii, 3.d, Part II b (local PDF, manual pages B12-B13).
- Test extractions: `research/tests/atx-reg.md`, `ut-a.md`, `ut-b.md`, `ut-c.md`, `misc-a.md`, `misc-b.md`, `state.md`, `nat26.md`; `prior/G_tests.md`.

- Raw tests and keys read this run: `downloads/dlB/tests/phi26_test.txt` + key (Pembroke Q21, Q26), `bullso26_key.txt` (SA1), `berk26_test.txt` (Q11); `downloads/dlG/rsoi26C_key.pdf` pages 2-5 and 10 (rendered: Q3, Q6, Q9, Q17, Q25, Q47), `puso26_test.txt` + key (MC8), `hh26_test.txt` (FR10), `nv26C_test.txt` (FR5), `wiki/CircuitLabRegTest2013.txt` + key (Q26), `wiki/Shock_value_test_.txt` (Section IV).
- `prior/B_topics.md` §5d, §5g; `prior/I_bank.md` C41-C43; `prior/F_slides.md` §6 (CIR-SYMBOLS, deck 9).
- IEEE 315 / ANSI Y32.2 and IEC 60617 (symbol standards, from memory). Kuphaldt, *Lessons in Electric Circuits* Vol. I ch. 1 and Vol. IV ch. 3. All About Circuits, "Schematic symbols" worksheet.
- All numbers checked in Python.

Resistor Color Codes, Tolerance, Power Ratings and Potentiometers

Note: Resistor Color Codes

Rule area **3.c.v** covers "individual components" of DC circuits, and **3.c.iv** covers resistance and power. The hands-on list (Part II c.i, c.iii, c.viii) hands you real resistors. You must read their bands fast and correctly. Color-code questions appear in almost every test, from regionals to top invitationals.

Key facts

- The **color code** = colored bands painted on a resistor. They give its value, tolerance and (sometimes) its temperature coefficient. The standard is **IEC 60062**.
- **Digit colors** (memorize in order, 0 to 9): **black 0, brown 1, red 2, orange 3, yellow 4, green 5, blue 6, violet (purple) 7, gray 8, white 9.**
- Memory sentence: "**B**ad **B**eer **R**ots **O**ur **Y**oung **G**uts **B**ut **V**odka **G**oes **W**ell" (black, brown, red, orange, yellow, green, blue, violet, gray, white).
- **Multiplier** band = "times 10 to the power of the digit". Black ×1, brown ×10, red ×100, orange ×1 k, yellow ×10 k, green ×100 k, blue ×1 M, violet ×10 M, gray ×100 M, white ×1 G. **Gold ×0.1. Silver ×0.01.**
- **Tolerance** band = how far the real value may be from the printed value:
  - brown ±1%, red ±2%, green ±0.5%, blue ±0.25%, violet ±0.1%, gold ±5%, silver ±10%, no band ±20%.
- **Temperature coefficient** (6th band, in ppm/K = parts per million per kelvin): black 250, brown 100, red 50, orange 15, yellow 25, green 20, blue 10, violet 5, gray 1.
- **How many bands:**
  - **4-band:** digit, digit, multiplier, tolerance.
  - **5-band:** digit, digit, digit, multiplier, tolerance (precision resistors, usually ±1% or better).
  - **6-band:** 5-band + temperature coefficient.
  - **3-band:** digit, digit, multiplier; no tolerance band means ±20%.
  - **1 black band** = a **zero-ohm resistor** (a jumper wire in a resistor body).
- **Which end to start from:** the tolerance band (gold or silver, often a bit wider and set apart by a gap) is on the **right**. The first band is usually closer to the end of the body. Gold and silver are never first digits.
- **Preferred values (E-series, IEC 60063):** resistors come only in standard values.
  - **E6 (±20%):** 10, 15, 22, 33, 47, 68.
  - **E12 (±10%):** 10, 12, 15, 18, 22, 27, 33, 39, 47, 56, 68, 82.
  - **E24 (±5%):** E12 plus 11, 13, 16, 20, 24, 30, 36, 43, 51, 62, 75, 91.
  - **E96 (±1%):** 96 values per decade (100, 102, 105, 107, 110 ...).
  - Each value is then multiplied by 1, 10, 100, 1 k and so on (for example, 4.7 Ω, 47 Ω, 470 Ω, 4.7 kΩ).
  - The steps are about equal ratios (E12 ≈ ×1.21 each step), so tolerance ranges of neighbors touch or overlap.
- **Power rating** = the most power (in watts) the resistor can turn into heat without damage. Common through-hole sizes: 1/8 W, 1/4 W, 1/2 W, 1 W, 2 W. Bigger body = higher rating. **The color code does not show the power rating.**
- Good design keeps the real power at **half the rating or less** (a safety margin, called **derating**).
- A resistor that overheats usually **fails open** (burns through). It may first darken or smell.
- **SMD (surface-mount) codes** (⚠ borderline: not seen in a 2027-style test):
  - 3-digit: two digits + number of zeros. **103 = 10 × 10^3 = 10 kΩ. 471 = 470 Ω. 000 = 0 Ω.**
  - 4-digit: three digits + number of zeros. **1002 = 100 × 10^2 = 10.0 kΩ.**
  - **R** = the decimal point: **4R7 = 4.7 Ω, R10 = 0.10 Ω.**
  - EIA-96: two-digit code number (01 = 100, 02 = 102 ... 68 = 499 ... 96 = 976) + a letter multiplier (Z ×0.001, Y or R ×0.01, X or S ×0.1, A ×1, B ×10, C ×100, D ×1 k, E ×10 k, F ×100 k). **01C = 10 kΩ. 68X = 49.9 Ω.**
- **Potentiometer (pot)** = a resistor with three terminals: two fixed ends and a sliding **wiper**. The resistance end-to-end is fixed. The wiper splits it into two parts that always add up to the total.

- **Rheostat** = a variable resistor used with only **two** terminals (the wiper and one end). It controls **current** in series with a load. It often carries the full load current, so it is built big.
- A pot used with all three terminals is a **variable voltage divider**. A pot with the wiper tied to one end works as a rheostat.
- **Taper** = how resistance changes as you turn the knob.
  - **Linear taper:** resistance changes evenly with rotation (half-turn = half resistance).
  - **Logarithmic (audio) taper:** resistance changes slowly at first, then fast. It matches how ears hear loudness. It is used for **volume** controls.
  - Marking letters vary by maker ("A" often = audio/log, "B" = linear on many Asian pots). Check the datasheet.
- **Trimmer (trimpot)** = a small pot set once with a screwdriver.
- Uses: dimmer → pot or rheostat (Princeton 2026 matching); volume → log pot; sensor calibration → trimmer.

Formulas

- 4-band value: $\mathbf{R = (10 \cdot d_1 + d_2) \times multiplier}$.
- 5-band value: $\mathbf{R = (100 \cdot d_1 + 10 \cdot d_2 + d_3) \times multiplier}$.
- Tolerance range: $\mathbf{R_{min} = R(1 - t)}$, $\mathbf{R_{max} = R(1 + t)}$, with t as a decimal (5% → 0.05).
- Temperature change: $\mathbf{R(T) = R_0\,(1 + \alpha \cdot \Delta T)}$, with α in ppm/K × 10^{-6}.
- Power: $\mathbf{P = I^2R = V^2/R = VI}$.
- Most current for a rating: $\mathbf{I_{max} = \sqrt{(P_{rated} / R)}}$. Most voltage: $\mathbf{V_{max} = \sqrt{(P_{rated} \cdot R)}}$.
- Pot divider (no load): $\mathbf{V_{out} = V_{in} \times (R_{lower} / R_{total})}$, where R_lower = the part between the wiper and the grounded end.
- Pot divider with a load R_L: $\mathbf{V_{out} = V_{in} \times (R_{lower} \parallel R_L) / (R_{upper} + R_{lower} \parallel R_L)}$.
- Rheostat in series with a load: $\mathbf{I = V / (R_{load} + R_{set})}$.
- Parallel: $\mathbf{a \parallel b = ab / (a + b)}$.

Question types

Easy

1. Color → digit; the order of the colors ✅ IN

- Seen: in every color-code question below. National homework generator Level 10a (24 color-band resistors).
- How to solve: Recite the 10 colors from black (0) to white (9).
- Example: Which digit is violet? Which is gray? **Answer: violet = 7, gray = 8.**
- Traps: Brown (1) and red (2), and orange (3) and red, look alike under bad light. Tilt the resistor toward a white light.
- Calculator: look up — type: `color <name>`

2. Read a 4-band resistor ✅ IN

- Seen: UT Austin (ATX) Regional 2021, Div B, REG, Q19 (photo: brown-green-brown-gold → 150 Ω, B). UF Northern Regional 2020, Div B, REG, Q31 (brown-black-red-gold → 1 kΩ). Rickards 2021 season, Div B, INV, Q27 (red-blue-orange-gold → 26 000 Ω). cyanophycean practice test 1 ~2013, Div C, practice, Q4 (orange-orange-red-gold → 3.3 kΩ). PA Central Regional ~2013, Div C, REG (green-brown-violet-gold). National practice deck 9 (red-violet-green-gold → 2.7 MΩ). F-M Invitational fall 2019, Div not stated, INV, hands-on HO-13 (two visible resistors → 1 kΩ, 4.7 kΩ).
- How to solve:
  1. Put the gold/silver band on the right.
  2. Write the first two digits.
  3. Multiply by 10^(third band).
  4. Read the tolerance from the fourth band.
- Example: yellow-violet-red-gold. **Answer: 47 × 100 = 4700 Ω = 4.7 kΩ ± 5%.**
- Traps: The third band is a multiplier, not a digit. Brown-black-red is 1 kΩ, not 102 Ω. PA Central's green-brown-violet-gold works out to 51 × 10^7 = 510 MΩ, an unusual value (no key; not checked).

- Calculator: calculate — type: `c1, c2, c3, c4` (colors)

3. Tolerance band: what it means; read it ✅ IN

- Seen: VA Regional 2013, Div C, REG, Q31 ("what does the last colored bar indicate?" → tolerance). GullSO 2025, Div C, INV, Q11 (red-yellow-black-gold → ±5%, C). Rickards 2026, Div B, INV, Q9 (tolerance of green-black-black-gold → ±5%).
- How to solve: Gold 5%, silver 10%, brown 1%, red 2%, none 20%.
- Example: What tolerance does a silver last band show? **Answer: ±10%.**
- Traps: Gold is ±5% as the **last** band but ×0.1 as the **third** band. Position decides meaning.
- Calculator: look up — type: `tol <color>`

4. Gold or silver as the multiplier (values under 10 Ω) ✅ IN

- Seen: not seen as its own test item. The generator and deck tables include gold ×0.1 and silver ×0.01.
- How to solve: Gold multiplier = ×0.1. Silver multiplier = ×0.01.
- Example: yellow-violet-gold-gold. **Answer: 47 × 0.1 = 4.7 Ω ± 5%.** Red-red-silver-gold = 22 × 0.01 = **0.22 Ω ± 5%**.
- Traps: The scioly wiki table once listed gold and silver as "values" 0.1 and 0.01 instead of multipliers.
- Calculator: calculate — type: `c1, c2, c3, c4`

5. Value → colors (reverse code) ✅ IN

- Seen: NY Lake Erie/Niagara Invitational 2018, Div C, INV, Q12 (6.8 kΩ ± 5% → blue-gray-red-gold, D).
- How to solve:
  1. Write the value as two digits × 10^n.
  2. Digit colors, then the color for n, then the tolerance color.
- Example: 330 Ω ± 5%. **Answer: 33 × 10^1 → orange-orange-brown-gold.** For a 5-band 4.7 kΩ ± 1%: 470 × 10^1 → **yellow-violet-black-brown-brown**.
- Traps: For a 5-band code, the multiplier is one power lower (470 × 10, not 47 × 100).
- Calculator: calculate — type: `R, tol, bands`

6. Which kind of part is it? (component ID from a photo) ✅ IN

- Seen: Florida State 2021, Div B, STATE, Q21 (photo → potentiometer, B). Cobra Invitational 2019-20, Div B, INV, station S3 (real parts: name, symbol, rating, use). Florida State 2021, Div B, STATE, Q18 (photo → capacitor; ❌ out as a topic).
- How to solve: Resistor = small tan/blue body with bands and two leads. Pot = round body with a shaft or knob and three pins. Trimmer = tiny square with a screw slot.
- Example: A part has a round metal body, a turning shaft and three pins in a row. **Answer: a potentiometer (rotary pot).**
- Traps: A blue body with bands can be a 1% metal-film resistor, not a capacitor. Capacitors have printed numbers and µF, not bands.
- Calculator: look up — type: `photo pot`

Medium

7. Read a 5-band (precision) resistor ✅ IN

- Seen: North Pocono Invitational 2020, Div C, INV, Q19 (yellow-white-violet-black-gold → 497 Ω, with max 522 Ω and min 472 Ω). Northview 2025, Div C, INV, Q27 (blue-violet-black-brown-gold → 6.7 kΩ 5%, B).
- How to solve: Three digits, then the multiplier (fourth band), then tolerance.
- Example: brown-black-black-red-brown. **Answer: 100 × 100 = 10 000 Ω = 10 kΩ ± 1%.**
- Traps: Reading a 5-band resistor as 4-band gives nonsense (brown-black-black-red → "10 × 1" then "red tolerance"). Count the bands first.
- Calculator: calculate — type: `c1..c5`

8. Tolerance range: smallest and largest true value ✅ IN

- Seen: North Pocono Invitational 2020, Div C, INV, Q19 (497 Ω ± 5% → 472.15 to 521.85 Ω). National practice deck (RES-COLOR-TOLERANCE-RANGE).

- How to solve: R_min = R(1 − t). R_max = R(1 + t).
- Example: 4.7 kΩ ± 5%. **Answer: 4.465 kΩ to 4.935 kΩ.**
- Traps: Use the decimal (0.05), not 5. Keep the unit.
- Calculator: calculate — type: `R, tol`

9. In spec or faulty? (measured value vs tolerance) ✅ IN

- Seen: ATX Regional 2026, Div B, REG, MC27 (1520 Ω ± 8% measured 1420 Ω → not faulty; range 1398.4-1641.6 Ω). I_bank C37.
- How to solve: Find the range. Inside = good. Outside = out of spec.
- Example: A 220 Ω ± 5% resistor measures 228 Ω. **Answer: range 209 to 231 Ω, so it is in spec.**
- Traps: ±8% is not a real color-code tolerance; the math works the same. Subtract meter lead resistance first if it is given (see Meters and Measurement A5 type 8).
- Calculator: calculate — type: `R, tol, Rmeas`

10. Read the code from the correct end ✅ IN

- Seen: not seen as its own question. Every photo question needs it (ATX 2021 B Q19).
- How to solve:
  1. Gold/silver goes on the right.
  2. If there is no gold/silver, read both ways. The reading that gives a standard E-series value is right.
  3. The wider gap sits before the tolerance band.
- Example: You see the bands gold-red-violet-yellow from left to right. **Answer: gold cannot be a first digit, so turn it around: yellow-violet-red-gold = 47 × 100 = 4.7 kΩ ± 5%.**
- Traps: Some cases are truly ambiguous. Then measure with an ohmmeter.
- Calculator: calculate — type: `c1..c4, reverse`

11. Is it a standard (E-series) value? ✅ IN (E-series facts are a component detail; ⚠️ the series names are trivia)

- Seen: National homework generator Level 10a (digits drawn from the E12 list). I_bank C39. Not seen as a direct question in a test.
- How to solve: Strip the power of ten. Check the two digits against E12 (or E24).
- Example: Which is not an E12 value: 4.7 kΩ, 5.0 kΩ, 68 Ω, 820 Ω? **Answer: 5.0 kΩ. E12 has 47 and 56, not 50. (E24 has 51, still not 50.)**
- Traps: Test writers sometimes use made-up values. Rickards 2026 B used 50 Ω and 40 Ω (green-black-black, yellow-black-black). They read correctly, but they are not E24 values.
- Calculator: look up — type: `e12` / `e24`

12. Choose the nearest standard resistor for a design ✅ IN

- Seen: links to LED and divider design problems (Hawk and Hornet 2026, Div C, INV, FR6; GullSO 2025, Div C, INV, Q50-51). See LEDs A7.
- How to solve:
  1. Compute the ideal R.
  2. For a current limiter, pick the **next higher** standard value (less current = safer).
  3. Recompute the real current.
- Example: 9 V supply, red LED (V_F = 2.0 V), target 20 mA. **Answer: ideal R = 7.0/0.020 = 350 Ω. Next higher E12 value = 390 Ω. Real I = 7.0/390 = 17.9 mA. (330 Ω would give 21.2 mA, above the target.)**
- Traps: Rounding down gives more current than the spec.
- Calculator: calculate — type: `Vs, Vf, I`

13. Power rating: largest safe current or voltage ✅ IN

- Seen: Hawk and Hornet 2026, Div C, INV, FR6 (blue LED 3.0 V, 20 mA, 1/4 W resistor → max supply 15.5 V). I_bank C38. UT Austin (ATX) Regional 2019, Div B, REG, Q14 (470 Ω, 9 W rating → max current 0.138 A).
- How to solve: I_max = √(P/R). V_max = √(P·R). For an LED circuit with a fixed current: V_R,max = P/I, then add V_F.
- Example: A 1/4 W, 100 Ω resistor. **Answer: I_max = √(0.25/100) = 0.050 A = 50 mA. V_max = √(0.25 × 100) = 5.0 V.** HH FR6 check: V_R = 0.25/0.020 = 12.5 V, so V_supply,max = 12.5 + 3.0 = **15.5 V**.

- Traps: Watts are not volts. A 1/4 W resistor can take any voltage as long as $V^2/R \le 0.25$ W.
- Calculator: calculate — type: `P, R` (or `P, I, Vf`)

14. Does the resistor overheat? Pick a rating ✅ IN

- Seen: USC Invitational 2026, Div B, INV, hands-on part c (9 V across 510 Ω → 0.16 W, safe for a 1/4 W part). Rickards 2026, Div B, INV, Q8-13 (four resistors on 9 V: currents and powers).
- How to solve: Compute $P = V^2/R$ or I^2R. Compare with the rating. Prefer P ≤ half the rating.
- Example: 470 Ω directly across 12 V. **Answer: P = 144/470 = 0.306 W. That is more than 1/4 W, so use a 1/2 W (or better, 1 W) resistor.**
- Traps: In parallel, the **smallest** resistor gets the most power. In series, the **largest** resistor gets the most power.
- Calculator: calculate — type: `V, R, Prated`

15. Color code + circuit: build a target R_eq from coded resistors ✅ IN

- Seen: UT Austin (ATX) Regional 2021, Div B, REG, Q24 (X = 15 Ω brown-green-black, Y = 30 Ω, Z = 100 Ω 5-band; which arrangement gives 9.09 Ω → all three in parallel, D). GullSO 2025, Div C, INV, Q42a (red-yellow-black-gold = 24 Ω; which resistor in the circuit can it replace?). Official hands-on task Part II c.viii (target equivalent resistance). National generator Level 10b ("make this R_eq" puzzles).
- How to solve:
  1. Decode every resistor.
  2. Try all-series (sum), all-parallel, and mixed combinations.
  3. Check the target in Python or on the calculator.
- Example: 15 Ω, 30 Ω, 100 Ω. Which arrangement gives 9.09 Ω? **Answer: all three in parallel: 1/(1/15 + 1/30 + 1/100) = 9.09 Ω.**
- Traps: The parallel result is always smaller than the smallest resistor. Use that to rule out options fast.
- Calculator: calculate — type: `R list, target`

16. Color code + circuit: currents and powers on a battery ✅ IN

- Seen: Rickards 2026, Div B, INV, Q8-13 (brown-red-brown-gold, green-black-black-gold, yellow-black-black-gold, violet-green-black-gold + 9 V battery). Rickards 2026, Div B, INV, Q6-7 (voltmeter across R4; wire broken at A).
- How to solve: Decode, then solve the circuit (series: one current; parallel: one voltage).
- Example: The four Rickards resistors (120, 50, 40, 75 Ω) in series on 9 V. **Answer: R = 285 Ω. I = 9/285 = 31.6 mA. Powers: 120 Ω 0.120 W, 50 Ω 0.050 W, 40 Ω 0.040 W, 75 Ω 0.075 W. Total 0.284 W.** In parallel: R = 15.0 Ω, the 40 Ω resistor takes 81/40 = **2.03 W** (it would burn a 1/4 W part).
- Traps: Parallel on a battery can overheat small resistors. Say so if the question asks "what happens".
- Calculator: calculate — type: `R list, V, series/parallel`

17. Potentiometer basics: terminals and wiper position ✅ IN

- Seen: UT Invitational fall 2025, Div B/C, INV, Q1c.i (potentiometer ring: wiper position for R_min → x = L). Princeton 2026, Div C, INV, Matching Part 2 (dimmer → potentiometer). Rickards 2026, Div C, INV, Q3 (pot symbol).
- How to solve: R_end-to-end is fixed. R_A-wiper + R_wiper-B = R_total. Moving the wiper toward A makes R_A-wiper smaller.
- Example: A 10 kΩ pot. The wiper is 30% of the way from terminal A. **Answer: R_A-W = 3.0 kΩ, R_W-B = 7.0 kΩ, R_A-B = 10 kΩ (always).**
- Traps: The two outer pins never change resistance. A meter across them always reads the full value.
- Calculator: calculate — type: `Rtotal, fraction`

18. Potentiometer vs rheostat ✅ IN

- Seen: UF Northern Regional 2019, Div B, REG, Q38 (2 vs 3 terminals; the rheostat carries the load current). GullSO 2025, Div C, INV, Q26 (manually adjusted resistor to control current → rheostat, B). Berkeley 2026, Div B, INV, Q11 (rheostat as a distractor). Cobra Invitational 2019-20, Div B, INV, Q28 (rheostat and pot symbols).
- How to solve: Pot = 3 terminals, divides voltage. Rheostat = 2 terminals, controls current in series.
- Example: An old stage-light dimmer sits in series with the lamp and carries all of its current. **Answer: a rheostat.**

- Traps: The same part can be either. It depends on how it is wired.
- Calculator: look up — type: `rheostat`

19. Pot as a voltage divider (no load) ✅ IN

- Seen: national practice materials (divider section); GullSO 2025, Div C, INV, Q50 (divider design). Not seen as a pot-only test item.
- How to solve: V_out = V_in × R_lower / R_total.
- Example: A 10 kΩ pot across 9.0 V. The wiper is 30% of the way up from the grounded end. **Answer: V_out = 9.0 × 0.30 = 2.7 V.**
- Traps: "From the top" vs "from the bottom" flips the answer (2.7 V vs 6.3 V). Draw it.
- Calculator: calculate — type: `Vin, fraction`

20. Pot taper: linear vs logarithmic ✅ IN

- Seen: MIT Invitational 2019, Div C, INV, Q34 (volume-control pot: log or linear → logarithmic).
- How to solve: Volume (audio) → log taper. Most other controls (sensor set points, dividers) → linear.
- Example: Why does a volume knob use a log taper? **Answer: ears sense loudness on a ratio scale, so a log taper makes each turn sound like an equal step.**
- Traps: Taper letter codes (A/B) differ between makers.
- Calculator: look up — type: `taper`

Hard

21. Six-band resistor with temperature coefficient ✅ IN

- Seen: Rickards 2026, Div C, INV, Q16 (violet-red-blue-black-green-gray; key B "736 Ω ± 0.5%, 1 ppm/K" — the true value is 726 Ω, see Mistakes).
- How to solve:
  1. Bands 1-3 = digits. Band 4 = multiplier. Band 5 = tolerance. Band 6 = tempco.
  2. For a change in temperature: $\Delta R = R_0 \cdot \alpha \cdot \Delta T$.
- Example: A 1.00 kΩ resistor has a brown 6th band (100 ppm/K). It warms by 50 K. **Answer: $\Delta R = 1000 \times 100\times10^{-6} \times 50 = 5.0\ \Omega$, so R = 1005 Ω.**
- Traps: ppm = parts per million = $\times10^{-6}$. One kelvin of change = one °C of change.
- Calculator: calculate — type: `R0, ppm, dT`

22. Loaded pot divider ✅ IN

- Seen: GullSO 2025, Div C, INV, Q50-51 (loaded voltage-divider design to a spec; measure loaded voltages).
- How to solve: Put R_L in parallel with the lower part. Then use the divider formula.
- Example: 10 kΩ pot at mid-travel (5 kΩ + 5 kΩ) across 9.0 V. A 10 kΩ load hangs on the wiper. **Answer: lower = 5 ‖ 10 = 3.33 kΩ. V_out = 9.0 × 3.33/(5 + 3.33) = 3.6 V (not 4.5 V).**
- Traps: A load always pulls the output **down**. Use a pot much smaller than the load to keep the drop small.
- Calculator: calculate — type: `Vin, Rup, Rlow, RL`

23. Rheostat: current range and where it gets hot ✅ IN

- Seen: UT Invitational fall 2025, Div B/C, INV, Q1c (pot ring: where it overheats, $P = V^2/R$; $R(x) = \rho(L - x)/A$).
- How to solve:
  1. I_max at R_set = 0: I = V/R_load. I_min at R_set = max.
  2. Rheostat power: $P = I^2 \cdot R_{set}$.
  3. A pot set near one end puts the full current (or voltage) on a small piece of track. That piece gets hot.
- Example: 12 V, 24 Ω lamp, 0-48 Ω rheostat in series. **Answer: I goes from 12/24 = 0.50 A down to 12/72 = 0.167 A. Rheostat power at 24 Ω setting = $(12/48)^2 \times 24 = 1.5$ W (the largest value in the 0/12/24/36/48 Ω table).**
- Traps: A pot used as a rheostat with its wiper turned to 0 Ω gives no protection. Add a fixed series resistor.
- Calculator: calculate — type: `V, Rload, Rset`

24. SMD resistor codes ⚠️ Borderline (not named in the rule; not seen in a test; SMD parts are still DC components)

- Seen: not seen in a test. I_bank C40. UT Invitational 2020, Div C, INV, Q34 asks about the SMD **diode** mark (see Testing Diodes A7).
- How to solve: 3-digit = 2 digits + zeros. 4-digit = 3 digits + zeros. R = decimal point. EIA-96 = code number + letter.
- Example: Decode 472, 4R7, 1002 and 01C. **Answer: 4.7 kΩ; 4.7 Ω; 10.0 kΩ; 10 kΩ (100 × 100).**
- Traps: "100" is 10 Ω (10 + 0 zeros), not 100 Ω.
- Calculator: calculate — type: `smd <code>`

Nationals

25. Pot track geometry and a target resistance over time ✅ IN (⚠️ the "sketch over time" part uses functions only, rule 3.e.i)

- Seen: UT Invitational fall 2025, Div B/C, INV, Q1c (ring pot: R(x) = ρ(L – x)/A; draw the wiper position x(t) that gives a target R(t)).
- How to solve:
  1. Write R as a function of wiper position: R(x) = ρ(L – x)/A (a straight line).
  2. Solve for x: x = L – R·A/ρ.
  3. Put the target R(t) in to get x(t).
- Example: Track length 20 mm, full R = 10 kΩ (so 0.5 kΩ per mm). You need R to fall from 10 kΩ to 0 at a steady rate in 4 s. **Answer: x(t) = 5 mm/s × t, from 0 to 20 mm. R(t) = 10 kΩ – 2.5 kΩ/s × t.**
- Traps: x is measured from the end where R is largest. Keep units in mm and kΩ consistently.
- Calculator: calculate — type: `L, Rtotal, t`

26. Tolerance stack-up in series and parallel ✅ IN

- Seen: not seen in a test. Allowed by 3.c.v (calculations with components) and 3.e.i (algebra).
- How to solve: Worst case in series: add all minimum values, then all maximum values. Worst case in parallel: combine all minimums, then all maximums.
- Example: Two 1.0 kΩ ± 5% resistors in series. **Answer: R_min = 950 + 950 = 1900 Ω. R_max = 1050 + 1050 = 2100 Ω. So 2.0 kΩ ± 5%.** In parallel: 475 Ω to 525 Ω (500 Ω ± 5%).
- Traps: The percent stays the same for equal-tolerance parts in series or parallel. It does not add up to 10%.
- Calculator: calculate — type: `R list, tol`

Out of scope (skip)

- ❌ **Capacitor and inductor color codes** and the capacitor number code (104 = 100 nF). Rule 3.d: capacitance, inductance.
- ❌ **Varistor (MOV)** behavior beyond naming. Rule 3.d: non-linear devices beyond those listed.
- ❌ **Thermistor** curve math (β equation, exponentials). Rule 3.d: non-linear devices; rule 3.e.v (no calculus; the β formula uses exponentials). Naming a thermistor is ⚠️ borderline (Berkeley 2026 B Q11).
- ❌ **Noise, frequency response, inductance of wirewound resistors.** Rule 3.d: frequency analysis, inductance.

Mistakes in tests and keys

- **Rickards 2026, Div C, Q16:** violet-red-blue-black-green-gray = 7, 2, 6 × 1 = **726 Ω ± 0.5%, 1 ppm/K**. The key's choice B says **736 Ω**. No option is right; B is the closest (a digit typo). Checked in Python and on the rendered key page 4.
- **scioly.org wiki color table:** it lists gold and silver as "values" 0.1 and 0.01. They are **multipliers** (noted by the earlier research run, `prior/B_topics.md` §5f).
- **North Pocono 2020, Div C, Q31:** the key's minimum series resistor (35.7 Ω) is wrong. If only the three original resistors count: 22.9 Ω. If the new resistor's own 0.5 W rating also counts: 96.4 Ω. At 35.7 Ω the new resistor takes 0.70 W (from `tests/misc-a.md`, checked there in Python).
- **ATX Regional 2026, Div B, MC27:** ±8% is not a color-code tolerance. The math is fine; the value is unusual.

Not confirmed

- Tolerance colors **orange (±0.05%), yellow (±0.02%) and gray (±0.01% or ±0.05%)** appear in some IEC 60062 tables. Sources disagree. I did not open the standard. Use them only if the test gives them.
- **Pink = ×0.001** multiplier (IEC 60062:2016). From memory; not checked.
- The tempco color table is from memory. The Rickards 2026 C key agrees that gray = 1 ppm/K. The other colors are not confirmed against the standard.
- Pot taper letter codes (A = log, B = linear) vary by maker. From memory.
- PA Central Regional ~2013 (green-brown-violet-gold) has no key in the downloads. My 510 MΩ reading is not confirmed by a key.
- GullSO 2025 Q11 and Q42a answers: the MC key is on page 1 as an image; I did not render it. ±5% (C) follows from the code itself.

Sources

- 2027 Div B rules, Circuit Lab, 3.c.iv-v, 3.d, Part II c (local PDF, manual pages B12-B13).
- Raw tests and keys read this run: `downloads/dlG/rsoi26C_key.pdf` p. 4 (rendered), `rsoi26B_test.txt` (Q6-13), `rsoiOldB_test.txt` + key (Q27), `nv26C_test.txt` + key (Q27), `gull26_test.txt` (Q8-11, Q26, Q42), `wiki/central.txt`, `wiki/Circuit_Lab_cyanophycean314*.txt`, `wiki/Circuit_Lab_2_cyanophycean314*.txt` (Q13-14: blue-violet-yellow-gold → 670 kΩ ± 5%; green-gray-orange-silver → 58 kΩ ± 10%), `wiki/CircuitLabRegTest2013.txt` + key (Q31), `dlB/tests/berk26_test.txt` (Q11).
- Test extractions: `research/tests/atx-reg.md`, `misc-a.md`, `misc-b.md`, `state.md`, `ut-a.md`, `nat26.md`; `prior/G_tests.md`.
- `prior/B_topics.md` §5f; `prior/I_bank.md` C35-C40; `prior/F_slides.md` §11 (generator Level 10).
- IEC 60062 (color and SMD codes) and IEC 60063 (E-series), from memory. Kuphaldt, *Lessons in Electric Circuits* Vol. I ch. 2 and Vol. V (resistor color codes). All About Circuits, "Resistor color codes" worksheet. OpenStax *College Physics 2e* §20.3 (resistance and temperature coefficient).
- All numbers checked in Python.

Series and Parallel Resistance

Rule area **3.c.v** says "calculations, constructions, and configurations of DC circuits ... including series and parallel circuits, open and short circuits, Ohm's Law, and circuit schematics". Hands-on task **II.c.viii** says "Create a specified equivalent resistance from given individual resistors". So this is IN at every level. It is the biggest point block on most tests. Every Div B test in the research folder had at least 3 of these questions.

Key facts

- **Equivalent resistance (R_eq)** = the one resistor that could replace a whole group and draw the same current from the source.
- **Series** = parts in one line, with no branch point between them. The **same current** flows through each part.
- **Parallel** = parts connected across the **same two nodes**. Each part has the **same voltage**.
- A **node** = all the wire that is joined with no part in between. A long wire with several dots is still one node.
- Series R_eq is always **bigger than the biggest** part. Parallel R_eq is always **smaller than the smallest** part.
- Adding any resistor in series raises R_eq. Adding any resistor in parallel lowers R_eq.
- A **short circuit** = a path with (almost) 0 Ω. A resistor with a wire across it carries no current. Remove it.
- An **open circuit** = a break in the path. No current flows in that path.
- A resistor with one end not connected to anything carries no current. Remove it.
- **Conductance (G)** = 1/R. Its unit is the **siemens (S)**, also called the mho. In parallel, conductances add.
- **Reduce from the far end.** Start at the part farthest from the source. Work back toward the source one step at a time.
- After you find the total current, **go back out** to find each branch: series parts share the current, parallel parts share the voltage.

Formulas

- Series: **$R_{eq} = R_1 + R_2 + R_3 + \ldots$**
- Parallel: **$1/R_{eq} = 1/R_1 + 1/R_2 + 1/R_3 + \ldots$**
- Two in parallel: **$R_{eq} = R_1R_2/(R_1 + R_2)$** ("product over sum").
- n equal resistors R: series **nR**, parallel **R/n**. Series ÷ parallel = **n^2**.
- Ohm's law: **$V = IR$**. Power: **$P = VI = I^2R = V^2/R$**.
- Total power = sum of all part powers = V_source × I_source.
- Current split between two parallel parts: **$I_1 = I \cdot R_2/(R_1 + R_2)$** (see Dividers and Bulb Brightness A5).
- Max safe current of a resistor: **$I_{max} = \sqrt{(P_{rating}/R)}$**. Max safe voltage: **$V_{max} = \sqrt{(P_{rating} \cdot R)}$**.

Question types

Easy

1. Series or parallel? Say what is shared (concept)

- Scope: ✅ IN.
- Where seen: BullSO 2026 B (INV) Q12 (parallel parts share voltage; series parts share current). UF Northern Regional 2020 B (REG) Q12 (three 2 Ω: series or parallel → parallel). UT Invitational fall 2018 B/C (INV) Q3 (break in the path → open circuit) and Q4 (T/F: parallel has only one path → false). Shock Value re-tryout ~2012 B (practice) I-11 ("a circuit is a special ___" → network), I-17 (two circuits are "equivalent" if their V-I relations match), II-2 (series current is the same everywhere → true), VI-17 (why parallel lowers R). National supervisor practice deck 3 homework (same V on all parts → parallel; same I → series).
- How to solve:
  1. Trace each part. Mark its two end nodes.
  2. Two parts with the **same two nodes** are in parallel.
  3. Two parts joined end to end, with **nothing else** at the joint, are in series.
  4. Parallel = same voltage. Series = same current.

- Worked example: "Two lamps are connected so each lamp has its own path to the battery. Which quantity is the same for both?" Answer: **the voltage** (they are in parallel).
- Traps:
  - Parts drawn side by side are not always in parallel. Check the nodes, not the picture.
  - Parts drawn in a line are not in series if a third wire joins at the middle point.
  - "Why does parallel lower R?" More paths for the current. "Why does series raise R?" A longer path.
- Calculator: look up — type: series, parallel, shared, same current, same voltage.

2. Series circuit: R_eq, current, each drop, each power

- Scope: ✅ IN.
- Where seen: UT Invitational fall 2018 B/C (INV) Q6 (240 V; 12, 20, 16 Ω → 5 A; 60, 100, 80 V). UF Northern Regional 2020 B (REG) Q8 (10 V, 2 + 3 Ω → 2 A), Q10-11 (5 V, 2 + 3 Ω → 1 A; 3 V across the 3 Ω). Berkeley 2026 B (INV) FRQ 2.1a-c (5 V, two 5 Ω → 0.5 A each, 2.5 W). USC Invitational 2026 B (INV) Q11 (9.0 V, 1.0 kΩ + 2.2 kΩ → 0.017 W in the 2.2 kΩ). National supervisor homework generator Level 1.
- How to solve:
  1. R_eq = sum of all resistors.
  2. I = V/R_eq. This one current flows through every part.
  3. Each drop: V_n = I·R_n. The drops add up to the source voltage.
  4. Each power: P_n = I^2·R_n. The powers add up to V·I.
- Worked example (UT fall 2018 Q6): 240 V across 12 Ω, 20 Ω and 16 Ω in series.
  - R_eq = 12 + 20 + 16 = 48 Ω. I = 240/48 = **5.0 A**.
  - Drops: 5 × 12 = **60 V**, 5 × 20 = **100 V**, 5 × 16 = **80 V**. Check: 60 + 100 + 80 = 240 V.
  - Powers: 300 W, 500 W, 400 W. Total 1200 W = 240 V × 5 A.
- Traps:
  - The biggest resistor gets the biggest drop and the biggest power in series.
  - Do not divide the source voltage equally unless the resistors are equal.
  - Give units. Many keys take off points for a missing unit (BullSO 2026 B short answers).
- Calculator: calculate — type: V, list of series R; outputs R_eq, I, each V, each P.

3. Missing value in a series loop (third resistor or third drop)

- Scope: ✅ IN.
- Where seen: UF Northern Regional 2020 B (REG) Q32 (20 V, 10 A, 1 Ω + 0.75 Ω + ? → 0.25 Ω). UF Northern Regional 2019 B (REG) Q40 (13 V, drops 3.3 V and 4.8 V → 4.9 V). MYSO Div B practice test 2020 (practice) Q18-22 (30 V loop; R_2 = 8 Ω has 16 V → I = 2 A, V_3 = 8 V, V_1 = 6 V, R_total = 15 Ω).
- How to solve:
  1. Total R = V/I. Missing R = total − the known ones.
  2. Missing drop = source − the known drops.
  3. If one part's V and R are known, I = V/R. That I is the same everywhere in the loop.
- Worked example (UF 2020 B Q32): 20 V drives 10 A. Total R = 20/10 = 2 Ω. Missing R = 2 − 1 − 0.75 = **0.25 Ω**.
- Traps: Watch the units. A test can mix Ω and kΩ, or V and mV, in one loop.
- Calculator: calculate — type: V, I (or known drops), known R list.

4. Parallel circuit: R_eq, total current, branch currents, powers

- Scope: ✅ IN.
- Where seen: UF Northern Regional 2020 C (REG) Q11 (8 Ω || 4 Ω, 0.2 A in the 8 Ω → 1.6 V, 0.4 A). UF Northern Regional 2020 B (REG) Q19 (formula for 3 parallel R). Shock Value re-tryout ~2012 B (practice) I-6 (method for more than 2 parallel R → reciprocals). National supervisor homework generator Level 2 (60 Ω || 30 Ω on 120 V).
- How to solve:
  1. Every branch has the full source voltage (if nothing is in series with the group).
  2. Branch current: I_n = V/R_n.
  3. Total current = sum of branch currents. R_eq = V/I_total (or use the reciprocal formula).
  4. The **smallest** resistor carries the **most** current and uses the **most** power.

- Worked example (generator Level 2): 120 V across 60 Ω || 30 Ω.
  - I_1 = 120/60 = **2 A**. I_2 = 120/30 = **4 A**. I_total = **6 A**.
  - R_eq = 120/6 = **20 Ω** (also 60·30/90 = 20 Ω).
  - P_1 = 240 W, P_2 = 480 W, total 720 W.
- Traps:
  - Product over sum works for **two** resistors only. For three, use reciprocals, or pair them up two at a time.
  - Forgetting the last "1/x" step: 1/R_eq = 0.05 means R_eq = 20 Ω, not 0.05 Ω.
- Calculator: calculate — type: V, list of parallel R; outputs R_eq, each I, each P.

5. n equal resistors; series-to-parallel ratio; infinitely many in parallel

- Scope: ✅ IN.
- Where seen: 2026 ATX Regional B (REG) MC36 (two equal resistors: series R is how many times parallel R? → 4). GullSO 2025 C (INV) Q37g iv (n identical m-ohm resistors in parallel → m/n). USC Invitational 2026 B (INV) Q2 (infinitely many R in parallel, measured across one → 0). National supervisor deck 4 (three 3 kΩ in parallel → 1 kΩ).
- How to solve:
  1. n equal in series: nR. n equal in parallel: R/n.
  2. Ratio series/parallel = nR ÷ (R/n) = **n^2**.
  3. As n grows, R/n goes to 0.
- Worked example: five 10 kΩ resistors. Series = **50 kΩ**. Parallel = **2 kΩ**. Ratio = 50/2 = **25** = 5^2.
- Traps: "How many times larger" for two equal resistors is 4, not 2.
- Calculator: calculate — type: R, n.

6. One resistor in series with a parallel pair: R_eq, I, V, P of each part

- Scope: ✅ IN.
- Where seen: Pembroke Hill 2026 B (INV) Q1-3 (12 V; 4 Ω + (6 Ω || 3 Ω) → 2.0 A, 4 V, 5.33 W). ATX Regional 2021 B (REG) Q16 (10 Ω + (3 + 2) || 10 → 13.3 Ω). MIT Invitational 2019 C (INV) Q24 (5 V, R + (R || R) → 3.3 V across R). UF Northern Regional 2019 C (REG) Q16 (10 V, 5 Ω + (10 || 10) → 5 W in the 5 Ω). NY Lake Erie/Niagara 2018 C (INV) Q24-25 (12 V, 3.3 kΩ + (4.7 kΩ || 6.8 kΩ) → 5.49 V, 1.17 mA). JC Booth Invitational 2011 B (INV) Q2 ((10 || 20) + 13 → 19.7 Ω) and Q56 (24 V, 15 Ω + (18 || 18) → 1, 0.5, 0.5 A). National supervisor generator Level 3 (50 + (60 || 30), 140 V).
- How to solve:
  1. Combine the parallel pair first.
  2. Add the series resistor. That is R_eq.
  3. I_total = V/R_eq. It flows through the series resistor.
  4. V across the pair = I_total × (pair R_eq). Each branch: I = V_pair/R.
- Worked example (Pembroke Hill 2026 B Q1-3): 12 V; A = 4 Ω in series with B = 6 Ω || C = 3 Ω.
  - B || C = 18/9 = 2 Ω. R_eq = 4 + 2 = 6 Ω. I = 12/6 = **2.0 A**.
  - V_B = V_C = 2.0 × 2 = **4.0 V**. V_A = 8 V.
  - P_C = 4^2/3 = **5.33 W**. I_B = 0.667 A, I_C = 1.33 A.
- Traps:
  - The voltage on the pair is **not** the source voltage. The series resistor takes its share first.
  - In MIT 2019 Q24, "V across R" means the series R (3.33 V), not the pair (1.67 V). Read which part the question names.
- Calculator: calculate — type: V, R_series, R_a, R_b.

7. Fill the circuit table (V, I, R, P for each part)

- Scope: ✅ IN.
- Where seen: MYSO Div B practice test 2020 (practice) Q14-17 (3 parallel resistors in series with R_4; 24 V, 7 A; V_1 = 16 V, I_2 = 2 A, I_3 = 1 A → I_1 = 4 A, V_3 = 16 V, I_4 = 7 A, V_4 = 8 V). National supervisor generator Levels 1-8 (every level asks for a full V, I, P table). Virginia Regional 2013 C (REG) Station 1 and Whiting Invitational 2013 B (INV) lab (measure V and I, fill a V-I-R table).

- How to solve:
  1. Write a table: one row per part, columns V, I, R, P.
  2. Fill what you know. Use rules to spread values: series rows share I; parallel rows share V.
  3. Each row needs only 2 known values. Then V = IR and P = VI give the other two.
  4. Check: the drops around the loop add to the source. The branch currents add to the total.
- Worked example (MYSO Q14-17): the source gives 24 V and 7 A. The parallel group has V = 16 V.
  - $I_1 = 7 - 2 - 1 =$ **4 A**. $V_3 =$ **16 V** (parallel). $I_4 =$ **7 A** (series with the source). $V_4 = 24 - 16 =$ **8 V**.
  - Then $R_1 = 4\ \Omega$, $R_2 = 8\ \Omega$, $R_3 = 16\ \Omega$, $R_4 = 8/7 = 1.14\ \Omega$.
- Traps: Do not mix up a row's V with the source V.
- Calculator: calculate — type: table rows with known V or I or R; solver fills the rest.

8. Which arrangement gives the lowest or highest R? (minimum from a set)

- Scope: ✅ IN.
- Where seen: Berkeley 2026 B (INV) Q27 (from 1, 2, 3, 4, 5 Ω, use 3 → minimum = 1 || 2 || 3 = 0.55 Ω). ATX Regional 2021 B (REG) Q26 (lowest overall R → small resistors in parallel) and Q34 (resistors in series for max R; the capacitor half is ❌ OUT). Shock Value re-tryout ~2012 B (practice) V-1-4 (minimum + maximum of 243, 223, 556, 919 Ω → 87.06 + 1941 = 2028.1 Ω).
- How to solve:
  1. **Maximum**: put the parts in series. Choose the biggest ones if you must pick some.
  2. **Minimum**: put the parts in parallel. Choose the smallest ones if you must pick some.
- Worked example (Berkeley 2026 B Q27): use 3 of {1, 2, 3, 4, 5 Ω}. Minimum = 1 || 2 || 3: 1/R = 1 + 0.5 + 0.333 = 1.833. R = **0.545 Ω ≈ 0.55 Ω**.
- Traps:
  - If you must use 3 parts, a lone 1 Ω does not count. You must use the three smallest in parallel.
  - The minimum parallel value is always below the smallest part (here, below 1 Ω).
- Calculator: calculate — type: list of R, how many to use, min or max.

Medium

9. Multi-level (nested) combination: total current, then each branch

- Scope: ✅ IN.
- Where seen: UT Invitational 2020 B (INV, 2021 season) Q39 (6 V; (2 kΩ + 4 kΩ) || 3 kΩ → 3 mA) and Q42 (6 V; 2 kΩ || [3 kΩ + (2 kΩ || 2 kΩ)] → 4.5 mA) and Q67 (10 V ladder → 5.73 Ω, 8.94 V, 0.53 V). Berkeley 2026 B (INV) FRQ 2.1d-i (add two 6 Ω → 8.53 Ω, 586 mA) and FRQ 2.2a (add three 7 Ω → 8.40 Ω). ATX Regional 2021 B (REG) Q38-40 (20 V, 100 Ω + (20 || 40 || (10 + 50)) + 50 Ω → 120 mA, 6.2 V, 140 mW). 2026 ATX Regional B (REG) FR2b-c (9 V; (10 || 5) + 20 → 0.129 A in the 10 Ω, 1.286 V). UT Invitational fall 2018 B/C (INV) Q20-21 ((R_1 || R_2) + (R_3 || R_4) → 1.5 Ω; 2.43 A, 1.39 A, 1.04 A). UF Northern Regional 2020 B (REG) Q23 (12 V, 3 + 2 + (5 || 2 || (2 + 4)) → 1.95 A). UF Northern Regional 2020 C (REG) Q12 (12 V; 5 + 2 + (3 || (4 + 1)) → 9.14 W in R_1). F-M fall 2019 Invitational (INV) Q6a-b (9 V; 200 + 300 + (150 || 300) → 3, 4.5, 1.5, 1.5 V; 15, 15, 10, 5 mA). National supervisor generator Levels 4 and 8, deck 4 examples.
- How to solve:
  1. Redraw the circuit as boxes inside boxes.
  2. Start at the innermost (farthest) group. Combine it. Replace it with one resistor.
  3. Repeat until one R_eq is left. Find I_total.
  4. Go back out. At each series step, the current stays the same. At each parallel step, the voltage stays the same.
- Worked example (Berkeley 2026 B FRQ 2.1d-g): 5 V; $R_1 = 5\ \Omega$ in series with [$R_2 = 5\ \Omega$ || ($R_3 + R_4 = 6 + 6\ \Omega$)].
  - $R_3 + R_4 = 12\ \Omega$. 5 || 12 = 60/17 = 3.53 Ω. R_eq = 5 + 3.53 = **8.53 Ω**.
  - $I_1 = 5/8.53 =$ **0.586 A**. V on the parallel part = 0.586 × 3.53 = 2.07 V.
  - $I_3 = 2.07/12 =$ **0.172 A**. $P_3 = 0.172^2 \times 6 =$ **0.178 W**. $P_4 =$ **0.178 W** (same current, same R).
- Traps:
  - Do not round in the middle. Keep 4 digits until the end.

  - The current in one branch is **not** the total current. The 2026 ATX Regional B key for FR2b gave the total (0.3857 A) instead of the branch (0.1286 A).
  - "Power between A and B" can be ambiguous if another branch also joins A and B (ATX 2021 B Q40, see Mistakes).
- Calculator: calculate — type: nested expression with "+" for series and "||" for parallel (for example 5+(5||(6+6))), V.

10. Hidden short: a wire across a resistor; a dangling resistor

- Scope: ✅ IN (rule 3.c.v says "open and short circuits").
- Where seen: USC Invitational 2026 B (INV) Q10 (rank 4 circuits; one resistor is shorted by a wire). 2026 ATX Regional B (REG) FR2a (100 Ω with a plain wire in parallel → 0 A). National supervisor deck 4 quizzes 2-3 (a 0 Ω branch takes all the current). UF Northern Regional 2020 C (REG) Q17 (five resistors, a wire merges nodes → 125 Ω). GullSO 2025 C (INV) Q39a (short a resistor → new currents).
- How to solve:
  1. Find every plain wire. Give both ends of the wire the same node name.
  2. A resistor with both ends on the same node is shorted. It has 0 V and 0 A. Erase it.
  3. A resistor with one end on nothing carries 0 A. Erase it.
  4. Then reduce the rest as normal.
- Worked example: three 10 Ω resistors in series. A wire joins the two ends of the middle one. R_eq = 10 + 10 = **20 Ω**. The middle resistor carries **0 A**.
- Traps:
  - A shorted resistor is still drawn on the page. Tests use it as a decoy.
  - A short in **parallel** with a group shorts the whole group.
- Calculator: calculate — type: the nested expression with the shorted parts left out.

11. Redraw a tangled network (merged nodes, crossing wires)

- Scope: ✅ IN.
- Where seen: ATX Regional 2021 B (REG) Q41-43 (24 V across two 3-part blocks → 1.50 Ω, 8.00 A, 384 W; key error, see Mistakes). UT Invitational 2020 B (INV) Q66 (8 × 100 Ω with a bridged node → 163 Ω). JC Booth Invitational 2011 B (INV) tiebreaker (80 V, 8 resistors → 800 Ω). UMSO 2019 (INV, division not stated) Q29 (nine 5 Ω → 4.2 Ω). UF Northern Regional 2020 C (REG) Q17 (→ 125 Ω).
- How to solve:
  1. Give every node a letter. All points joined by plain wire get the **same** letter.
  2. Write each resistor as "letter to letter" (for example A-C, 4 Ω).
  3. Redraw: put the source terminals at the top and bottom. Place each node once. Draw each resistor between its two letters.
  4. Now series and parallel groups are easy to see.
- Worked example (ATX Regional 2021 B Q41-42): 24 V across two blocks in parallel. Left block: 0.6 Ω + (1 + 3) || (4 + 2). Right block: 0.5 Ω + (1 + 4) || (3 + 2).
  - Left: 4 || 6 = 2.4 Ω. 0.6 + 2.4 = 3.0 Ω. Right: 5 || 5 = 2.5 Ω. 0.5 + 2.5 = 3.0 Ω.
  - R_eq = 3.0 || 3.0 = **1.50 Ω**. Total current = 24/1.5 = 16 A. Current in the 0.6 Ω = 24/3.0 = **8.00 A**. Total power = $24^2/1.5$ = **384 W**.
- Traps:
  - Two wires that cross with **no dot** are not joined. Two wires with a dot are joined.
  - Read which resistors pair up in each block. The ATX key paired them wrongly.
- Calculator: calculate — type: edge list "node node R" (solver finds R_eq between two nodes).

12. Rank several circuits by R_eq (or by current, or by power)

- Scope: ✅ IN.
- Where seen: USC Invitational 2026 B (INV) Q10 (4 circuits of equal R: III > II = IV > I; circuit IV hides a shorted resistor). ATX Regional 2021 B (REG) Q26.
- How to solve:
  1. Write each R_eq as a multiple of R (for example 1.5R).

2. Watch for shorts and dangling parts in each circuit.
3. Rank the numbers. For the same source, a bigger R_eq means a smaller current and a smaller power.

- Worked example (USC 2026 B Q10 values): I = R || 2R = **0.667R**. II = R + (R || R) = **1.5R**. III = R + (R || 2R) = **1.667R**. IV = 1.5R (it has an extra R that a wire shorts). Ranking: **III > II = IV > I**.
- Traps: Equal answers are common. Look for an "=" choice.
- Calculator: calculate — type: one nested expression per circuit, with R = 1.

13. Back-solve: find an unknown resistor from R_eq or from the total current

- Scope: ✅ IN (one unknown; algebra, rule 3.e.i).
- Where seen: BullSO 2026 B (INV) SA4 (4 equal resistors in the SA3 network; R_eq = 45 Ω → R = 27 Ω) and SA2c (replace a 5 kΩ so the current is 3 mA). BullSO 2026 C (INV) SA4 (→ 27 Ω). UT Invitational 2019 B (INV, 2020 season) Q32 = UT Invitational fall 2019 C (INV) Q28 (12 V, 3.18 A total; 8 || (5 + 6 || (1 + 10 || R_4)) → R_4 = 3.04 Ω).
- How to solve:
  1. Find the R_eq you need: R_eq = V/I_total.
  2. If all parts are equal, write R_eq = k·R. Then R = R_eq/k.
  3. If one part is unknown, peel the network from the outside. Each step undoes one series or parallel step.
  4. To undo a parallel step: 1/R_unknown = 1/R_eq − 1/R_known.
- Worked example 1 (BullSO 2026 B SA3-4): the network is 2 Ω + 7 Ω || (3 Ω + 4 Ω).
  - SA3: 7 || 7 = 3.5 Ω. R_eq = 2 + 3.5 = **5.5 Ω**.
  - SA4: with 4 equal resistors R, R_eq = R + R || 2R = R + 2R/3 = 5R/3. Set 5R/3 = 45 Ω. R = **27 Ω**.
- Worked example 2 (UT 2019 B Q32): 12 V/3.18 A = 3.774 Ω.
  - 8 || X = 3.774 → X = 7.143 Ω. X = 5 + Y → Y = 2.143 Ω.
  - 6 || Z = 2.143 → Z = 3.333 Ω. Z = 1 + W → W = 2.333 Ω.
  - 10 || R_4 = 2.333 → **R_4 = 3.04 Ω**.
- Traps:
  - Rounding the total current (3.18 A) makes each step drift. Keep all digits.
  - If the needed R_eq is below a parallel partner's value, check it is still possible (a parallel group can never exceed its smallest part).
- Calculator: calculate — type: expression with one unknown x, target R_eq (solver).

14. Build a target R_eq from a given set (hands-on task viii)

- Scope: ✅ IN (official hands-on task II.c.viii).
- Where seen: BullSO 2026 B (INV) lab (100, 200, 50, 50 Ω; use all four; at least one parallel group and one series part → 100 Ω). BullSO 2026 C (INV) lab (key: 133.33 Ω). ATX Regional 2021 B (REG) Q24 (color-coded 15, 30, 100 Ω → 9.09 Ω = all three in parallel). Shock Value re-tryout ~2012 B (practice) V-1-1 to V-1-3 (243, 223, 556, 919 Ω → 466.4, 483.9, 354.1 Ω). Shock Value student test ~2011 B (practice) VI (16.7, 3.9, 3.5 Ω from 16, 4, 56, 45 Ω). F-M Invitational 2018 (INV) hands-on (≈620 kΩ from 5 resistors, scored by % error). UT Invitational fall 2018 B/C (INV) Q25a (1 V, 2, 4, 5, 10 Ω → 125 mA → ((2 + 10) || 4) + 5 = 8 Ω). National supervisor generator Level 10b. Rickards 2026 B (INV) hands-on Q8 (read four color codes → 120, 50, 40, 75 Ω, then series/parallel I and P).
- How to solve:
  1. If the target is **above** every part, you need some series. If it is **below** every part, you need some parallel.
  2. Look for easy pairs: two equal parts in parallel give half. Two parts summing to a third part, in parallel with it, give half of it.
  3. Try "one part || (all the others in series)" and "one part + (all the others in parallel)".
  4. Write the steps and each intermediate R_eq. BullSO gave 8 of 29 points for the steps.
  5. Build it. Measure with the ohmmeter, power off. Report % error = |measured − target| / target × 100.
- Worked example (BullSO 2026 B lab): parts 100, 200, 50, 50 Ω. Target 100 Ω, use all four.
  - 100 + 50 + 50 = 200 Ω in series. Put it in parallel with the 200 Ω: 200 || 200 = **100 Ω**.
  - It has a parallel group and series parts, so it meets both rules.
- Traps:
  - Real resistors have tolerance (±5% gold band). The measured value can differ. Report the measured value.
  - Measure R only with the battery removed.

  - Check the breadboard rows: two leads in the same 5-hole row are joined.
  - Read the color codes first if the values are not printed (ATX 2021 B Q24). See CL 9 - Meters, Breadboards and Hands-On Tasks [T].
- Calculator: calculate — type: list of R values, target, "use all" yes/no (brute-force search over series-parallel trees).

15. How many values can a set make? Equal-resistor puzzles

- Scope: ✅ IN.
- Where seen: not seen in a test as a "count the values" question. Close relatives: Shock Value re-tryout ~2012 B (practice) V-1 (several targets from one set), BullSO 2026 B lab. Textbook: Kuphaldt, *Lessons in Electric Circuits* Vol. I, ch. 7 and the Socratic "series-parallel" worksheets.
- How to solve:
  1. List the shapes: all series, all parallel, one + (the rest ||), one || (the rest series), pairs.
  2. Compute each one as a multiple of R.
  3. To reach a value with the fewest parts, try fractions: R/2 = two in parallel; 3R/2 = R + R/2; 2R/3 = R || 2R.
- Worked example: three equal resistors R, all used. The 4 possible values are **R/3, 2R/3, 3R/2, 3R**. With four equal resistors, all used, there are **9** values (R/4, 2R/5, 3R/5, 3R/4, R, 4R/3, 5R/3, 5R/2, 4R). Example: make 600 Ω from 1 kΩ resistors. 600 = 1000 || 1500, and 1500 = 1000 + (1000 || 1000). So you need **4 resistors**: 1 k || (1 k + 1 k || 1 k).
- Traps: R itself can be made from 4 parts as (R + R) || (R + R). A puzzle may ask for that.
- Calculator: calculate — type: R, number of parts; lists all values.

16. Design to a target current or power using all the given resistors

- Scope: ✅ IN.
- Where seen: ATX Regional 2019 B/C (REG) Q23 (use all of 10, 10, 1, 5, 6, 8 Ω so 12 V gives 48 W; key error). UT Invitational fall 2018 B/C (INV) Q25a (1 V source, 125 mA → 8 Ω).
- How to solve:
  1. Turn the target into an R_eq: R = V/I, or R = V²/P.
  2. Then do type 14.
- Worked example (ATX 2019 Q23): 12 V and 48 W. R_eq = 12²/48 = **3 Ω**. One answer that uses all six: 10 || 5 = 3.33; + 10 = 13.33; || 8 = 5.0; + 1 = 6.0; || 6 = **3.0 Ω**. So 6 || (1 + 8 || (10 + 10 || 5)) = 3 Ω gives 48 W.
- Traps: The ATX key drawing is 8 || (5 + 6 || (10 || 10 + 1)) = 4 Ω. That gives 36 W, not 48 W.
- Calculator: calculate — type: V, target I or P, list of R.

17. Work backward from one known current, voltage or power

- Scope: ✅ IN.
- Where seen: UF Northern Regional 2020 B (REG) Q33 (15 A through 3.2 + (4.1 + 2.4) || 2.7 + 3.6 → 130.6 V). ATX Regional 2019 B/C (REG) Q25 (40 V + V_2 in series; the 8 Ω uses 184.6 W → V_2 ≈ 90 V). UF Northern Regional 2020 C (REG) Q11 (I_1 = 0.2 A in 8 Ω → I_2 = 0.4 A in 4 Ω).
- How to solve:
  1. From the known part: V = √(P·R), or V = IR, or I = √(P/R).
  2. Use "same V in parallel" to get the partner's current.
  3. Add currents to get the series current. Add drops to get the source.
- Worked example (not seen in a test): a 2 Ω resistor in series with 6 Ω || 3 Ω. The 6 Ω uses 6 W.
  - V_6 = √(6 × 6) = 6 V. I_6 = 1 A. I_3 = 6/3 = 2 A. Total = **3 A**.
  - V_2 = 3 × 2 = 6 V. Source = 6 + 6 = **12 V**. Check: R_eq = 2 + 2 = 4 Ω, 12/4 = 3 A.
- Worked example (UF 2020 B Q33): 15 × (3.2 + 6.5 || 2.7 + 3.6) = 15 × (3.2 + 1.908 + 3.6) = **130.6 V**.
- Traps: √ gives two signs. Voltage size is the positive root.
- Calculator: calculate — type: known part R and its P (or I or V), network expression.

18. Power ratings: largest safe voltage or current for a combination

- Scope: ✅ IN (power, rule 3.c.iv, with series-parallel rules).

- Where seen: not seen in a test for a resistor network. Close relatives: Hawk & Hornet 2026 C (INV) FR6 (largest supply before an LED's series resistor passes 1/4 W → 15.5 V); North Pocono 2020 C (INV) Q31 (a resistor's rating also counts). Textbook: Kuphaldt Vol. I ch. 2 (power ratings); OpenStax *College Physics 2e* ch. 21 problems.
- How to solve:
  1. For each resistor, find its limit: I_max = √(P/R).
  2. In series, the **smallest** I_max limits the whole string.
  3. In parallel, find each V_max = √(P·R). The smallest V_max limits the group.
  4. n equal resistors in a square (series-parallel) shape share the power equally. The rating multiplies by n.
- Worked example (not seen in a test): a 100 Ω, 1/2 W resistor in series with a 200 Ω, 1/4 W resistor.
  - I_max(100 Ω) = √(0.5/100) = 70.7 mA. I_max(200 Ω) = √(0.25/200) = 35.4 mA.
  - The string can carry only **35.4 mA**. V_max = 0.0354 × 300 = **10.6 V**.
  - Second example: four 100 Ω, 1/4 W resistors as two series pairs in parallel. R_eq = 100 Ω. Rating = 4 × 1/4 = 1 W. V_max = √(1 × 100) = **10 V**.
- Traps: Adding the ratings is wrong for unequal resistors. The part with the smallest limit sets the limit.
- Calculator: calculate — type: list of (R, P_rating), arrangement.

19. Adding a resistor: which one changes the circuit least or most?

- Scope: ✅ IN.
- Where seen: GullSO 2025 C (INV) Q37f (add 1 μΩ, 1 Ω or 1 MΩ in parallel; which changes the current least → 1 MΩ). National supervisor deck 4 (load and source).
- How to solve:
  1. A **huge** resistor in **parallel** changes almost nothing (almost no current goes through it).
  2. A **tiny** resistor in **series** changes almost nothing.
  3. A tiny resistor in parallel shorts the part. A huge resistor in series nearly opens the circuit.
- Worked example: 10 V across 100 Ω draws 0.1000 A. Add 1 MΩ in parallel. New current = 0.10001 A. The change is only **0.01%**.
- Traps: The reverse is the "biggest change" question. A 1 μΩ part in parallel is almost a short.
- Calculator: calculate — type: V, R, added R, series or parallel.

Hard

20. Long multi-part FRQ: R_eq → currents → powers → energy → cost → what-if

- Scope: ✅ IN.
- Where seen: Berkeley 2026 B (INV) FRQ section 2 "DC Circuit Analysis" (33 points: current arrows, currents, power, add resistors, new R_eq, branch currents, powers, potential at a point; then three 7 Ω more → 8.40 Ω). 2026 ATX Regional B (REG) FR2. UT Invitational 2020 B (INV) Q66-67. Question bank "mega-question" template (prior research I_bank).
- How to solve:
  1. Draw the current arrows first: out of the + terminal, through the parts, back to −.
  2. Do each part of the question in order. Later parts use earlier answers, so keep 4 digits.
  3. When the question adds parts, start a new R_eq from the start. Do not reuse old currents.
- Worked example (Berkeley 2026 B FRQ 2.2a): the circuit from type 9, with R_4 = 6 Ω now in parallel with R_5 + R_6 + R_7 = 21 Ω.
  - 6 || 21 = 4.67 Ω. + R_3 (6 Ω) = 10.67 Ω. || R_2 (5 Ω) = 3.40 Ω. + R_1 (5 Ω) = **8.40 Ω**.
- Traps:
  - Conventional current goes from + to −. Electron flow goes the other way. Draw conventional current unless told otherwise.
  - A "potential at a point" needs a reference (ground). See Dividers and Bulb Brightness A5 type 17.
- Calculator: calculate — type: nested expression, V; then each branch.

Nationals

21. Network puzzles with a twist: very many parts, a hidden pattern, or "all the given parts"

- Scope: ✅ IN when it is series-parallel. The general-N or counting versions are ❌ OUT (see Out of scope).
- Where seen: Florida State 2021 B (STATE) Q41-45 (M batteries and N resistors; counting wires, probability; mostly ❌ OUT or ill-posed). USC Invitational 2026 B (INV) Q2 (infinitely many in parallel). UT Invitational 2020 B (INV) Q66 (8 × 100 Ω, 163 Ω).
- How to solve:
  1. Look for a repeating block. Find the R_eq of one block.
  2. Replace each copy by its R_eq, then combine the copies.
  3. If the network goes on forever, see the ladder types in Hard Resistor Networks [A5].
- Worked example (not seen in a test): six blocks in series. Each block is (2 Ω + 2 Ω) || 4 Ω. One block = 4 || 4 = 2 Ω. Six blocks = **12 Ω**.
- Traps: The Florida State 2021 questions on counting wires use combinatorics. That is outside the 2027 math limits. Skip them.
- Calculator: calculate — type: block expression, number of copies.

Out of scope (skip)

- ❌ **Capacitors in series and parallel** (rule 3.d: "capacitance"). Seen: ATX Regional 2021 B Q34 (capacitor half), Berkeley 2026 B Q28 (max C from three capacitors), Pembroke Hill 2026 B Q5 (parallel capacitors' charge), ATX 2021 B FRQ1 (infinite capacitor ladder).
- ❌ **Thevenin and Norton equivalents** (rule 3.d). Seen: ATX Regional 2021 B Q18 (Norton current), Q47-49 (Thevenin). Use plain series-parallel instead.
- ❌ **Op-amp circuits** drawn with resistors (rule 3.d "semiconductors beyond those listed"). Seen: ATX Regional 2021 B Q25.
- ❌ **Counting and probability puzzles** about networks (rule 3.e.i allows only functions, algebra and two equations). Seen: Florida State 2021 B Q41 (wires in a complete graph) and Q45 (probability of a current).
- ❌ **Impedance** as an "AC resistance" (rule 3.d "AC circuit theory"). Seen: ATX Regional 2020 B Q19.

Mistakes in tests and keys

- **ATX Regional 2021 B/C (REG) Q41 and Q43:** key 1.47 Ω and 390.62 W. The drawing gives **1.50 Ω** and **384 W**. The key paired (3 + 1) || (2 + 4) in the right block too. Q42 (8.00 A) is right.
- **ATX Regional 2021 B (REG) Q40:** key 140 mW is the power in the 20 Ω and 40 Ω only. If the (10 + 50) branch also counts as "between A and B", the answer is **168.5 mW**. The wording is ambiguous.
- **2026 ATX Regional B (REG) FR2b:** key 0.3857 A is the total current. The current in R_1 (10 Ω) is **0.1286 A**.
- **UF Northern Regional 2020 B (REG) Q13-16:** key 1.67 A is the R_eq (5/3 Ω), not the current. Correct: **6.0 A** total and **2.0 A** in each 2 Ω branch.
- **UF Northern Regional 2020 B (REG) Q23:** key 2.05 A. Correct: **1.95 A**.
- **UF Northern Regional 2020 B (REG) Q30a-c:** key 0.4 A, 10.38 Ω, 0.2 A. From the figure as read: **12.61 Ω**, 0.331 A, 0.166 A. No reading of the figure gives 10.38 Ω.
- **ATX Regional 2019 B/C (REG) Q23:** the key's drawing is 4 Ω (36 W), not 3 Ω (48 W).
- **MIT Invitational 2019 C (INV) lab S1-4:** key 2.4 kΩ ± 0.4. Exact: **2.59 kΩ** (still inside the key's tolerance).
- **BullSO 2026 B and C (INV) lab:** the task text says 100 Ω. The key says 133.333 Ω and gives an example (100 in series with 300 || 50 || 150) that uses resistors not in the kit. With 100, 200, 50, 50 Ω, 133.33 Ω is not possible using all four. 100 Ω is (checked by brute force).
- **Prior research file I_bank C7** guessed the BullSO SA4 network as giving R = 45 Ω. The figure (rendered) is 2 + 7 || (3 + 4), so the answer is **27 Ω**, as the key says.
- **UMSO 2019 (INV) Q19:** key 19.9 Ω. Correct: **19.7 Ω**.

Not confirmed

- UT Invitational 2020 B Q66 (163 Ω), JC Booth tiebreaker (800 Ω), UMSO Q29 (4.17 Ω), UF 2020 C Q17 (125 Ω): I used the answers checked in the extraction files. I did not re-render these figures.
- ATX Regional 2019 Q25 ($V_2 \approx 90$ V): the extraction gives the answer but not a clear topology, so I did not use it as a worked example.
- ATX 2021 B Q41: the block layout comes from the extraction's reading of the figure. I did not re-render it.
- F-M Invitational 2018 hands-on (620 kΩ): the resistor values are not in the test, so the task could not be checked.
- Whether any 2027 State or National test uses a "count the values" puzzle (type 15): not seen. Where I looked: all 8 extraction files in `research/tests/`, `prior/G_tests.md`, `prior/I_bank.md`.
- No real 2026 National Tournament test was found. The "nat26" files are the USC **Invitational** (January 2026).

Sources

- Rules: 2027 Science Olympiad Div B Rules Manual, Circuit Lab, rules 3.c.v, 3.d, 3.e and Part II c.viii (page B12-B13). See Circuit Lab - Rules 2027 Ⓡ.
- Tests (extraction files in `~/Daniel/circuitlab/research/tests/` and `prior/G_tests.md`): Pembroke Hill 2026 B, Berkeley 2026 B, BullSO 2026 B and C (figure for SA3-4 rendered), USC Invitational 2026 B; ATX Regionals 2019, 2020, 2021, 2026; UT Invitationals fall 2018, 2019 B, fall 2019 C, 2020 B; UF Northern Regionals 2019-2020 B/C; MIT 2019 C; NY Lake Erie 2018 C; JC Booth 2011 B; Shock Value tests ~2011-2012 B; MYSO 2020 B; Florida State 2021 B; F-M 2018 and fall 2019; UMSO 2019; GullSO 2025 C; Rickards 2026 B; Hawk & Hornet 2026 C; North Pocono 2020 C.
- National supervisor practice decks 3 and 4 and the homework generator Levels 1-10 (`prior/F_slides.md`).
- Textbooks: OpenStax *College Physics 2e*, ch. 21.1 "Resistors in Series and Parallel"; Kuphaldt, *Lessons in Electric Circuits* Vol. I, ch. 5 (series and parallel) and ch. 7 (series-parallel combination); All About Circuits, "Series and Parallel Circuits" chapter.
- All numbers in this note were checked in Python on 2026-10-16 (including a brute-force search of every series-parallel arrangement for types 14-16).

Open and Short Circuits, and Written Fault Diagnosis

Note: Open and Short Circuits

Rule area **3.c.v** names "open and short circuits" word for word. Hands-on task **Part II c.vi** is "Troubleshoot a faulty circuit." So you must know what an open and a short do, and how to find one from meter readings. Rule area **3.c.iii** adds the hazards of DC (a short can start a fire).

Key facts

- **Circuit** = a closed path for current. **Closed circuit** = the path is complete; current flows.
- **Open circuit (an "open")** = a break in the path. Examples: an open switch, a burned-out bulb filament, a blown fuse, a cut or loose wire, a part pulled out.
  - Current through an open = **0 A**.
  - Resistance of an open = **infinite** (an ohmmeter shows **OL**).
  - Voltage across an open can be **anything**. In a single loop, the **full source voltage** appears across the open.
- **Short circuit (a "short")** = a path with (almost) **zero resistance** that goes around a part, or straight across the source. Examples: two bare wires touching, a wire across a resistor, a part with both legs in the same breadboard row, a coin or keys across battery terminals.
  - Voltage across a short = **0 V** ($V = I \times 0$).
  - Resistance of a short = **0 Ω** (an ohmmeter shows about 0, for example 0.1 Ω).
  - Current through a short can be **very large**. Only the wires, the battery's internal resistance and the fuse limit it.
- **Memory pair:** open → **I = 0**. Short → **V = 0**. (Valparaiso 2013 Q6-7 asks exactly this.)
- **Current takes the short.** A part in parallel with a short gets 0 V and 0 A. It is "out of the circuit".
- **Series rules:**
  - One open → **everything** in that loop goes out. The open part shows the full source voltage. Good parts show 0 V.
  - One short (of one part) → the **other** parts get more voltage and more current. Bulbs get **brighter**.
- **Parallel rules:**
  - One branch opens → **only that branch** goes out. The others stay the same (with an ideal source). The total current goes down.
  - One branch shorts → the **whole parallel group** is shorted. Every branch in that group goes out. With nothing in series, this shorts the source.
- **Dangers of a short:**
  - Huge current → heat ($P = I^2R$ in wires and in the battery) → melted insulation, burns, fire.
  - Batteries can get hot, leak, or (lithium) catch fire.
  - That is why a **fuse** or **breaker** goes **in series** at the start of the circuit. It opens the path when the current is too big. See CL 6 - Switches, Relays, Fuses, Breakers, GFCI T.
- **Overload vs short:**
  - **Overload** = too many normal loads. The current is a bit too high (for example 1.2 × the rating).
  - **Short** = a near-zero path. The current is many times the rating.
  - Both trip a breaker or blow a fuse.
- **An open with high voltage across it can arc.** A blown fuse in a high-voltage circuit can keep a spark (arc) across its gap. That is why fuses have a voltage rating (Rickards 2021 B Q16 → arcing).
- **Faults in parts:**
  - Resistors usually **fail open** (they burn through).
  - Incandescent bulbs fail open (the filament breaks).
  - LEDs and diodes can fail **open or short**.
  - Fuses are designed to fail open.
- **Troubleshooting order** (national supervisor slide 11):
  1. Check the obvious: power on, battery voltage, switch on, fuse or breaker OK, burned part, loose wire.
  2. Draw the circuit. Write the expected voltage at each node.

3. Measure V across each part with power **on**. Compare with your numbers.
4. Measure R with power **off** and the part isolated.

- **What the voltmeter shows:**
  - Across an open in a dead series loop → the full source voltage.
  - Across a good resistor in a dead loop → 0 V (no current, no drop).
  - Across a short → 0 V.
  - Across a good closed switch or a good wire → about 0 V. A wire or joint that shows a voltage has **resistance** (a bad connection).

Formulas

- Ohm's law: **V = I R**. Open: R = ∞, I = 0. Short: R = 0, V = 0.
- Series string of n equal parts on V_s: each gets **V_s / n**. One open: that part gets **V_s**, the others get **0**.
- Shorting one of n equal series bulbs: each remaining bulb's power goes up by **(n / (n – 1))²**.
- Parallel: with an ideal source, each branch current stays **V / R_branch** when another branch opens.
- Short-circuit current of a real source: **I_sc = ε / r** (r = internal resistance). Heat inside the battery: **P = I_sc² r = ε² / r**.
- Fuse check: **I_total = Σ V / R_i** must stay below the fuse rating.
- Divider V_out (R_1 on top, R_2 at the bottom): **V_out = V_s R_2 / (R_1 + R_2)**. R_1 open → 0. R_1 shorted → V_s. R_2 open → V_s (no load). R_2 shorted → 0.

Question types

Easy

1. Define open and short (I = 0 vs V = 0) ✅ IN

- Seen: Valparaiso University Invitational 2013, Div B, INV, Q6-7 ("current defined as 0 A" → open circuit, key D; "voltage drop defined as 0 V" → short circuit, key A). UT Invitational fall 2018, Div B/C, INV, Q3 (name for a circuit with a break → open circuit). Wichita Heights Invitational 2026, Div C, INV, Q37 ("define an open circuit"). National supervisor practice deck (CIR-OPEN-SHORT-DEFINE).
- How to solve: Open = a break; no current flows. Short = a path with almost no resistance; no voltage across it; current can be huge.
- Example: "An area of a circuit where the voltage across two terminals is 0 V." **Answer: a short circuit.**
- Traps: A **closed** switch makes a **closed** (working) circuit. An **open** switch makes an **open** circuit. In everyday talk "open" sounds like "on". In circuits it means "off".
- Calculator: look up — type: `open short`

2. Name the fault from a description ✅ IN

- Seen: UF Northern Regional 2020, Div B, REG, Q35 (wires touching → a short circuit). JC Booth Invitational 2011, Div B, INV, Q51 ("two wires sometimes touch, creating an unsafe path" → short circuit). Pembroke Hill Invitational 2026, Div B, INV, Q22 key ("they all go out (open circuit)").
- How to solve: A path that is added and bypasses a part = short. A path that is broken = open.
- Example: A bulb's filament breaks. **Answer: an open circuit.** Bare wires touch each other. **Answer: a short circuit.**
- Traps: "Series circuit" and "parallel circuit" are distractors. They describe the layout, not a fault.
- Calculator: look up — type: `fault name`

3. Why is a short dangerous? Spot the shorted source ✅ IN

- Seen: UT Invitational 2020, Div B, INV, Q37 (6 V source with a bare wire across it: "no resistance, so it is short circuiting and potentially very dangerous"). UF Northern Regional 2019, Div C, REG, Q15 (circuit shorted on the right side → may catch fire or melt the wire). Shock Value student test ~2011, Div B, practice, Section IV Q4 ("why are short circuits dangerous?"). Shock Value re-tryout ~2012, Div B, practice, VI-8 (smell of burning skin; explain a short).
- How to solve:

1. Find a wire that joins the two ends of the source (or of a part) with nothing in between.
2. Say: R ≈ 0, so I = V/R is very large. Large I heats the wire ($P = I^2R$). Heat melts insulation and can start a fire. The battery can overheat.

- Example: A 9 V battery has a 0.05 Ω wire across it. Its internal resistance is about 1.5 Ω. **Answer: I ≈ 9/1.55 = 5.8 A. Most of the heat (about 50 W) is inside the battery, so it gets hot fast.**
- Traps: Say **why** (low R → high I → heat). "It is bad" scores nothing.
- Calculator: calculate — type: `V, r, R_wire`

4. Series string: one bulb burns out ✅ IN

- Seen: Pembroke Hill Invitational 2026, Div B, INV, Q22 (holiday lights in series; one burns out → they all go out). ATX (UT Austin) Regional 2026, Div C, REG, SA5 key (fuses: in series one burns out and the current stops). National supervisor deck (SER-RULES-CONCEPT: one open lamp turns all lamps off).
- How to solve: Series = one path. Any break stops the current everywhere in the loop.
- Example: 50 equal bulbs in series on 120 V. One burns out. **Answer: before, each bulb had 120/50 = 2.4 V. After, all go out. The dead socket has the full 120 V across it. Every good bulb shows 0 V.**
- Traps: Some real holiday bulbs have a **shunt** wire that closes when the filament breaks, so the rest stay lit. A test expects the simple answer unless it says so.
- Calculator: calculate — type: `V, n`

5. Parallel: one branch opens ✅ IN

- Seen: Valparaiso University Invitational 2013, Div B, INV, Q16 (bulb B removed → bulb C goes out; key D; figure not rendered). JC Booth Invitational 2011, Div B, INV, Q45 (L1 burns out → L4 and L5 stop). ATX (UT Austin) Regional 2026, Div C, REG, SA5 key (in parallel the current still goes through the others).
- How to solve:
  1. Find which branch the failed part is in.
  2. Only parts in **series with it, inside that branch**, go out.
  3. Other branches keep their voltage. Total current drops by that branch's current.
- Example: Three 12 Ω bulbs in parallel on 6.0 V. One burns out. **Answer: each bulb took 0.50 A, total 1.5 A. After: the other two still take 0.50 A each. Total = 1.0 A.**
- Traps: In a real battery with internal resistance, the others get slightly **brighter** when a branch opens (less current → less drop inside). With an ideal source, they stay the same.
- Calculator: calculate — type: `V, R list, which open`

6. Ohmmeter reading → open or shorted ✅ IN

- Seen: Pembroke Hill Invitational 2026, Div B, INV, Q24 (reads 0.1 Ω → shorted). National deck (METER-OL-AND-LEAD-ZERO: OL = open). See also Multimeter Use A9 type 7.
- How to solve: About 0 Ω → short. **OL** → open. Near the color-code value → good.
- Example: A fuse reads OL. A 470 Ω resistor reads 0.2 Ω. **Answer: the fuse is blown (open). The resistor is shorted (or it was measured with a wire across it).**
- Traps: A good wire, switch or fuse **should** read about 0 Ω. Know what the part should read before you call it a fault.
- Calculator: look up — type: `R_read, R_expected`

7. Voltage and current at an open and at a short (numbers) ✅ IN

- Seen: I_bank C26 and `prior/B_topics.md` §5b (open: I = 0 and full source V across the gap; short: V = 0). Valparaiso 2013 B Q6-7 (concept). Rickards 2021 B Q16 (full voltage across a blown fuse → arcing).
- How to solve:
  1. Open in a series loop: I = 0. V across the open = V_s. V across every other part = 0.
  2. Short of one part: remove it (R = 0). Solve the rest. V across the short = 0.
- Example: 9 V on R1 = 1 kΩ, R2 = 2 kΩ, R3 = 3 kΩ in series. **Answer: (a) R2 opens: I = 0; V_R2 = 9 V; V_R1 = V_R3 = 0. (b) R2 is shorted: I = 9/4000 = 2.25 mA; V_R1 = 2.25 V; V_R3 = 6.75 V; V_R2 = 0.**
- Traps: "Voltage across an open is 0" is wrong. The current is 0. The voltage is the source voltage.
- Calculator: calculate — type: `V, R list, which part, open/short`

Medium

8. An open in a series-parallel network: what goes out, what gets brighter ✅ IN

- Seen: F-M Invitational 2018, division not stated, INV, Q5 (120 V; 50 W bulb A in series with 60 W B || 40 W C; B burns out → A dimmer but lit). SO MYSO practice test 2020, Div B, official practice, Q26-28 (only SW1 open: Bulb 1 current decreases, Bulb 2 zero, Bulb 3 greater). Rickards Invitational 2026, Div B, INV, Q7 (wire broken at point A → which resistors still carry current). UT Invitational fall 2018, Div B/C, INV, Q24c (L3 removed → L4 power ×2.78, L5 ×0.694). JC Booth Invitational 2011, Div B, INV, Q45. I_bank C23.
- How to solve:
  1. Erase the open part and everything in series with it inside its branch.
  2. Recompute R_eq and the total current.
  3. Find the new voltage of each remaining part. Brightness follows power.
- Example: Identical bulbs: A in series with (B || C). B burns out. **Answer: before, $P_A = 4/9$ and $P_B = P_C = 1/9$ (in units of V^2/R). After, A and C are in series: each gets 1/4. A gets dimmer (0.444 → 0.25). C gets brighter (0.111 → 0.25).**
- Example 2 (F-M 2018 Q5): $R = V^2/P$ from the 120 V ratings: A = 288 Ω, B = 240 Ω, C = 360 Ω. **Answer: before, $P_A = 22.2$ W, $P_C = 4.44$ W. After B opens: $P_A = 9.88$ W (dimmer), $P_C = 12.3$ W (brighter).**
- Traps: The bulb in **series** with the group dims. The **partner** bulb in parallel brightens. Students often say "everything dims".
- Calculator: calculate — type: `V, R_A, R_B, R_C, open part`

9. Hidden short: a wire across a part or across a group ✅ IN

- Seen: ATX (UT Austin) Regional 2026, Div B, REG, FR2a (100 Ω R4 with a plain wire in parallel → 0 A). USC Invitational 2026, Div B, INV, Q10 (rank four circuits; one hides a shorted resistor). UT Austin (ATX) Regional 2019, Div B/C, REG, Q19 (a 188.3 Ω resistor is shorted; the odd value is a decoy). JC Booth Invitational 2011, Div B, INV, Q41 (darkest bulb → L6, shorted by a wire). National deck 4 quizzes 2-3 (PAR-SHORTED-BRANCH). Also Series and Parallel Resistance [A5] type 10 and Schematic Symbols [A5] type 20.
- How to solve:
  1. Give every plain wire's two ends the same node name.
  2. Any part with both ends on one node is shorted: 0 V, 0 A. Erase it.
  3. A wire across a **group** shorts the whole group.
  4. Solve what is left.
- Example: 12 V. R1 = 100 Ω in series with (R2 = 200 Ω || R3 = 300 Ω). A wire runs across R3. **Answer: the wire shorts the whole parallel group. $R_{eq} = 100$ Ω. $I = 0.12$ A. R2 and R3 both carry 0 A.**
- Traps: The wire may be drawn far away, around a corner. Trace nodes; do not judge by the picture. A strange resistor value (188.3 Ω) is often the shorted one.
- Calculator: calculate — type: `V, R list` (after removing shorted parts)

10. Short one part in series; a shorting switch ✅ IN

- Seen: JC Booth Invitational 2011, Div B, INV, Q46 (L2 shorted → no bulb stops). SO MYSO practice test 2020, Div B, official practice, Q29-30 (only SW2 closed; SW2 shorts the parallel group → Bulb 1 brighter, Bulb 3 goes out). USC Invitational 2026, Div B/C, INV, hands-on parts c-e (push buttons wired across an LED short it; a NOR made of shorting buttons). I_bank C23 (short B → A at full V).
- How to solve:
  1. Remove the shorted part.
  2. The rest of the series loop now has less resistance. Current goes up.
  3. Each remaining part gets more voltage and more power.
- Example: Three identical 10 Ω bulbs in series on 9.0 V. One is shorted. **Answer: before, $I = 0.30$ A and each bulb takes 0.90 W. After, $I = 0.45$ A and each of the two bulbs takes 2.03 W. That is 2.25 × as much ($= (3/2)^2$).**
- Traps: The other bulbs may now take **more than their rating** and burn out next.
- Calculator: calculate — type: `V, R list, shorted part`

11. Short-circuit current of a real battery (hazard) ✅ IN

- Seen: re-tryout ~2012 B VI-8 and UT 2020 B Q37 (concept). Numbers: see Internal Resistance and Power Transfer [A5] types 5 and 12. Not seen with numbers in this area's tests.
- How to solve:
  1. With only a wire across the battery, the only resistance is r (internal).
  2. I_sc = ε / r. Heat inside = ε² / r.
- Example: A 9.0 V battery with r = 1.5 Ω is shorted. **Answer: I_sc = 9.0/1.5 = 6.0 A. Heat in the battery = $6.0^2 \times 1.5 = 54$ W.** An AA cell (1.5 V, about 0.15 Ω): I_sc ≈ 10 A.
- Traps: The terminal voltage of a shorted battery is about 0 V. The voltage "disappears" inside the battery.
- Calculator: calculate — type: `emf, r`

12. Which load draws the most power? (a short) ✅ IN

- Seen: ATX (UT Austin) Regional 2026, Div B, REG, MC38 (choices: 50 Ω, 1.2 nF capacitor, two 100 Ω in parallel, a short → key: short circuit). See Internal Resistance and Power Transfer [A5] type 20.
- How to solve:
  1. Ideal source: P = V²/R. The smallest R wins. A short (R = 0) "draws" unlimited power from an ideal source.
  2. Real source: the **source** gives the most total power into a short (V²/r). But the **load** (the wire) gets almost none. It all heats the battery.
- Example: 9.0 V source with r = 0.5 Ω. Loads: 50 Ω; 100 Ω || 100 Ω; a short. **Answer: 50 Ω and 100 || 100 = 50 Ω tie: I = 0.178 A, P_load = 1.59 W, total from the source = 1.60 W. The short: I = 18 A, total from the source = 162 W, but almost 0 W in the wire.**
- Traps: A capacitor takes **no DC power** after it charges (and capacitance is ❌ out anyway). "Most power" can mean "from the source" or "in the load". Say which you mean.
- Calculator: calculate — type: `V, r, R list`

13. Overload vs short; why fuses go in series; how many loads before the fuse blows ✅ IN

- Seen: ATX (UT Austin) Regional 2026, Div C, REG, SA5 (why fuses are in series, not parallel). Northview Sylvania Invitational 2025, Div C, INV, Q20 (what can trip a breaker: overload, short, fault; "V = IR" is the odd one out). Rickards Invitational 2021 season, Div B, INV, Q16 (blown fuse with a voltage above its rating → arcing). Princeton Invitational 2021, Div C, INV, Q37 (0.75 A fuse, 10 V, 100 Ω bulbs in parallel → 7 bulbs). Also CL 6 - Switches, Relays, Fuses, Breakers, GFCI [T].
- How to solve:
  1. A fuse must carry the **whole** load current, so it goes in **series**, next to the source.
  2. In parallel with a load, a blown fuse would not stop the load. In parallel with the source, a fuse would itself be a short.
  3. Count loads: add the branch currents until the sum passes the rating.
- Example: A 0.75 A fuse feeds bulbs in parallel on 10 V. Each bulb is 100 Ω. **Answer: each bulb takes 0.10 A. 7 bulbs = 0.70 A (OK). 8 bulbs = 0.80 A (blows). Maximum = 7.**
- Example 2: A short with 0.20 Ω of wiring on 120 V. **Answer: I = 600 A, far above a 15 A breaker. It trips at once.**
- Traps: A fuse does not lower the current. It only opens the circuit when the current is too high. Fix the fault before you replace the fuse.
- Calculator: calculate — type: `V, R, I_fuse`

14. Troubleshoot a dead series string with a voltmeter ✅ IN

- Seen: national supervisor practice deck 11 (TROUBLE-SERIES-OPEN: six 12 Ω lamps in series on 6 V; all out; the lamp that reads the full voltage is open). I_bank C26 (three bulbs on 9 V; bulb 2 reads 9 V → bulb 2 is open). Rule Part II c.vi.
- How to solve:
  1. Write the expected readings with all parts good.
  2. Power on. Measure V across each part.
  3. The part with the **full source voltage** across it is the open one. All others read 0.
- Example: Six 12 Ω lamps in series on 6.0 V. All are dark. **Answer: expected: 1.0 V per lamp and I = 6/72 = 83 mA. Measured: L5 reads 6.0 V; the others read 0 V. So L5 (or its socket) is open.**

- Traps: If **every** part reads 0 V, the open is somewhere else: the battery, the switch, the fuse or a wire. Measure those next.
- Calculator: calculate — type: `V, R list`

15. Troubleshoot a parallel circuit: one branch is dead ✅ IN

- Seen: national supervisor practice deck 11 (TROUBLE-PARALLEL-OPEN: six 12 Ω lamps in parallel on 6 V; after the obvious fix, L5 stays out).
- How to solve:
  1. Measure V across the dead branch. If it has the full voltage, the branch has power, so the open is **in** the branch.
  2. Power off. Measure the lamp's R. OL → the filament is broken. Normal → check its socket and wires.
- Example: Six 12 Ω lamps in parallel on 6.0 V. L5 is dark and reads 6.0 V. **Answer: each good lamp takes 0.50 A; the total should be 3.0 A but is 2.5 A. L5 has voltage but no current, so L5 or its own wire is open.**
- Traps: A dark lamp that reads **0 V** has no power reaching it (an open in its feed wire). A dark lamp that reads **full V** is itself open.
- Calculator: calculate — type: `V, R, n`

16. Node voltages around an open or a short (voltmeter to ground) ✅ IN

- Seen: not seen as its own test question. `prior/B_topics.md` §9c ("no current means no IR drops"). Kuphaldt, *Lessons in Electric Circuits* Vol. I ch. 5 "Building series circuits" and Socratic troubleshooting worksheets.
- How to solve:
  1. Put the black lead on ground (the – terminal). Touch the red lead to each node.
  2. In a dead loop (open somewhere), every node on the + side of the open reads V_s. Every node on the – side reads 0.
  3. The open is between the last node that reads V_s and the first node that reads 0.
- Example: 9.0 V. + → R1 (1 kΩ) → node X → R2 (2 kΩ) → –. Normal V_X = 6.0 V. **Answer: R2 open → V_X = 9.0 V. R1 open → V_X = 0 V. R1 shorted → V_X = 9.0 V. R2 shorted → V_X = 0 V.**
- Traps: "R2 open" and "R1 shorted" give the **same** V_X (9 V). To tell them apart, measure the current or measure R with power off.
- Calculator: calculate — type: `Vs, R1, R2, fault`

17. Diagnose a circuit from its symptoms (LED off, a switch that does nothing) ✅ IN

- Seen: Princeton Invitational 2020, Div C, INV, Lab 1 Q42 (LED does not light; use only a voltmeter at labeled leads; give two one-change fixes) and Q43 (LEDs unequal → redraw). UT Invitational 2020, Div B, INV, Q32 (one LED lights, the other does not → reversed, or the blue LED needs a higher V_F). UT Invitational 2020, Div B, INV, Q30 (switch A works only when switch B is down → a traveler wire is broken; see 3-Way and 4-Way Switching A6 type 8). Cobra Invitational 2019-20, Div B, INV, Station 1 (LED never lights; the switch is wired wrong, so the circuit is always open; prove the rest works by bridging the switch with a coin). UMSO 2019 lab note ("the battery may have died, the LED may have burnt out").
- How to solve:
  1. List the possible causes: dead battery, open switch, open wire, reversed LED, LED V_F too high, wrong resistor, part shorted.
  2. Test the cheapest first: measure the battery. Then measure across each part.
  3. A voltage ≈ V_s across an LED that is off → the LED is reversed or open. ≈ 0 V across the LED and the resistor → no current reaches them.
  4. **Jumper test:** bridge a suspect switch with a wire (or coin). If the LED lights, the switch is the open.
- Example: 9 V, switch, 470 Ω, red LED in series. LED off. Readings: battery 9.0 V; switch 9.0 V (switch closed); resistor 0 V; LED 0 V. **Answer: the full voltage sits across the closed switch, so the switch is open inside (broken or wired to the wrong pins). A coin across its terminals lights the LED.**
- Traps: Do not short the battery with the coin. Bridge only the switch, which is in series with the resistor.
- Calculator: look up — type: `symptom`

Hard

18. Find the failed part from a set of readings (series-parallel) ✅ IN

- Seen: not seen as a written question with numbers. Allowed by Part II c.vi ("troubleshoot a faulty circuit") and 3.c.v. Princeton 2020 C Lab 1 is the closest hands-on. Kuphaldt Socratic worksheets "Troubleshooting".
- How to solve:
  1. Compute the normal readings.
  2. For each possible single fault (each part open, each part shorted), compute what the meters would read.
  3. Pick the fault that matches all the readings.
- Example: 12 V. R1 = 100 Ω in series with (R2 = 200 Ω || R3 = 200 Ω). Normal: V_R1 = 6.0 V, V_par = 6.0 V. Measured: V_R1 = 4.0 V, V_par = 8.0 V. **Answer: R3 open gives R_eq = 300 Ω, I = 40 mA, V_R1 = 4.0 V, V_par = 8.0 V. That matches. (R2 open gives the same numbers; check which branch has current.) Other faults: R1 shorted → 0 V and 12 V; R2 or R3 shorted → 12 V and 0 V; R1 open → 12 V and 0 V.**
- Traps: Two different faults can give the same readings (R2 open vs R3 open here). Then you need one more measurement (a branch current, or R with power off).
- Calculator: calculate — type: `Vs, R list, readings`

19. Partial faults: a drifted resistor or a bad (high-resistance) joint ✅ IN

- Seen: not seen in a test. Kuphaldt, *Lessons in Electric Circuits* Vol. I ch. 5-7 troubleshooting sections; All About Circuits "Troubleshooting" worksheets.
- How to solve:
  1. A part that "drifts" changes value but still conducts. Recompute with the new value.
  2. A bad joint acts like a small resistor in series. A voltage across a wire or a closed switch is the sign.
- Example: Same circuit as type 18. A loose joint adds 50 Ω in series with R1. **Answer: R_eq = 250 Ω, I = 48 mA. V_R1 = 4.8 V, V_par = 4.8 V, and the joint shows 2.4 V (a good joint shows about 0).** If R2 drifts to 400 Ω instead: V_R1 = 5.14 V, V_par = 6.86 V.
- Traps: A partial fault does not make anything go fully dark. Look for readings that are "off" by a few volts.
- Calculator: calculate — type: `Vs, R list, extra R`

20. Voltage-divider fault table (with and without a load) ✅ IN

- Seen: not seen in a test. Kuphaldt Socratic worksheet "Voltage divider circuits" (failure analysis). Linked hands-on: UT Invitational 2019, Div B, INV, Q47-49 (build and measure a divider).
- How to solve: For each fault, redraw and recompute V_out.
- Example: 12 V. R1 = 10 kΩ (top), R2 = 20 kΩ (bottom). **Answer: normal V_out = 8.0 V. R1 open → 0 V. R1 shorted → 12 V. R2 open → 12 V. R2 shorted → 0 V.** With a 20 kΩ load on the output: normal V_out = 6.0 V; R2 open → 8.0 V (the load alone forms the divider).
- Traps: "R2 open" gives V_s only with **no load**. With a load, the load takes R2's place.
- Calculator: calculate — type: `Vs, R1, R2, RL, fault`

21. Breadboard faults: same row, split rail, wrong row ✅ IN

- Seen: USC Invitational 2026, Div B/C, INV, hands-on part a (find the internal wiring of a 4-pin push button with a battery-LED-resistor continuity tester). `prior/B_topics.md` §9b (common errors). National deck (BREADBOARD-CONNECTIVITY). Also Breadboards A9 types 2, 10, 11.
- How to solve:
  1. Both legs of a part in one 5-hole row → the part is **shorted** by the strip.
  2. Two parts that should touch but sit in different rows → an **open**.
  3. A long board with a split power rail → the far half has **no power** (open). Bridge it with a jumper.
  4. Check with the continuity beep (power off).
- Example: A 470 Ω resistor has both legs in row 12. It feeds a red LED from 9 V. **Answer: the resistor is shorted. Only the battery's internal resistance limits the LED current, so the LED takes far too much current and burns out (it may fail open or short).**
- Traps: Ask the supervisor for the board's internal wiring. Rule Part II b allows it.
- Calculator: look up — type: `breadboard fault`

22. Stuck-at faults in a logic circuit ✅ IN (logic, rule 3.c.viii; the fault idea is troubleshooting)

- Seen: USC Invitational 2026, Div B, INV, Q22 (one AND input permanently grounded = stuck-at-0 → new output E = NOT(AB' + A'B), key C). `prior/H_logic_diodes.md` A50. See CL 8 - Digital Logic T.
- How to solve:
  1. "Stuck-at-0" (shorted to ground) → put 0 for that wire. "Stuck-at-1" (shorted to +V) → put 1.
  2. Simplify the Boolean expression. Check with a truth table.
- Example: Y = A·B + C. Input B is shorted to +V (stuck-at-1). **Answer: Y = A·1 + C = A + C (checked for all four A, C rows).**
- Traps: An **open** (floating) input on many real logic chips acts like a 1 (TTL). A test should tell you. If not, say your assumption.
- Calculator: calculate — type: `expression, wire, 0/1`

Nationals

23. An open or a short in a bridge (non-series-parallel network) ✅ States-Nationals (KVL/KCL, rule 3.c.ix; two unknown node voltages, so it fits rule 3.e.i)

- Seen: not seen as a fault question. Bridge networks: Florida State 2021, Div B, ST, Q49 (symmetric 9 kΩ network → 10 kΩ); MIT Invitational 2019, Div C, INV, Station 1-1c (hidden R in a bridge). See Kirchhoff Analysis (States-Nationals) A5 and Hard Resistor Networks A5.
- How to solve:
  1. Solve the normal bridge with two node equations (or note a balance).
  2. A **shorted** middle resistor merges the two middle nodes. The bridge becomes two parallel pairs in series.
  3. An **open** middle resistor removes the bridge. Two separate dividers remain.
  4. An **open** outer arm often turns the bridge into a plain series-parallel circuit.
- Example: 10 V on top, 0 V at the bottom. R1 = 100 Ω (top left), R2 = 200 Ω (bottom left), R3 = 200 Ω (top right), R4 = 100 Ω (bottom right), R5 = 100 Ω across the middle. **Answer:**
  - **Normal:** V_a = 5.71 V, V_b = 4.29 V. R_eq = 140 Ω, I = 71.4 mA. R5 carries 14.3 mA from a to b.
  - **R5 open:** V_a = 6.67 V, V_b = 3.33 V. R_eq = 150 Ω, I = 66.7 mA.
  - **R5 shorted:** (100 || 200) + (200 || 100) = 133 Ω. I = 75 mA. The middle node is 5.0 V. The short carries 25 mA.
  - **R4 open:** R3 + R5 = 300 Ω in parallel with R1 = 75 Ω, then + R2 → R_eq = 275 Ω, I = 36.4 mA. R5 now carries 9.09 mA from b to a (the direction **reverses**).
- Traps: A fault can **reverse** the current in the bridge resistor. Keep your sign convention.
- Calculator: calculate — type: `Vs, R1, R2, R3, R4, R5, fault`

Out of scope (skip)

- ❌ **Capacitor "acts like an open/short"** questions (UT Invitational 2020 B Q34-35: a capacitor at t = 0 acts like a wire; at t = ∞ like an open). Rule 3.d: capacitance.
- ❌ **"Replace voltage sources with shorts and current sources with opens"** (Princeton 2020 C key, for Thevenin/superposition). Rule 3.d: Norton/Thevenin. (Superposition by itself is not named; treat it as ⚠️ borderline.)
- ❌ **Op-amp faults** (Princeton 2020 C Lab 1 Q44: op-amp saturated at the 5 V rail). Rule 3.d: semiconductors and non-linear devices beyond those listed.
- ❌ **Probability that parallel switches conduct** (Florida State 2021 Q33). Rule 3.e: the math limits do not include probability.
- ❌ **Short-circuit faults in 3-phase or AC power systems** (fault current in transformers, line-to-line faults). Rule 3.d: 3-phase power, AC circuit theory.

Mistakes in tests and keys

- **Shock Value re-tryout ~2012, Div B, VI-8:** "connect the anode and the cathode on my finger … why I smell burning skin." A finger has a high resistance (tens of kΩ when dry). A small battery cannot push enough current

through it to burn. A burn happens when **metal** (a ring, keys, a coin) shorts the battery and gets hot. No key exists; answer with the metal-short idea and say why a finger is not a short.

- **UT Invitational 2019, Div B, Q43:** the drawn switch-lamp circuit **shorts the battery** whenever S1 is OFF (from `research/tests/ut-b.md`). A test drawing can contain a fault by mistake. Point it out in your answer.
- **Pembroke Hill Invitational 2026, Div B, Q22:** the key "they all go out" is right for a simple series string. Some real holiday bulbs have shunts (see type 4 traps). The question does not mention them, so the key stands.
- **National supervisor deck 11 (TROUBLE-SERIES-OPEN):** it says the open lamp reads "10 V" on a 6 V source. It should be **6 V** (from `prior/F_slides.md`; a typo).
- **ATX (UT Austin) Regional 2026, Div B, MC38:** "largest power output" with an ideal short has no finite answer; the key "short" is right only in the limit. With any internal resistance, the short delivers the most source power but 0 W to the load (type 12).
- **Florida State 2021, Div B, Q22:** "dead" (de-energized, 0 V) and "ground" overlap, so the odd-one-out is not clean (from `research/tests/state.md`).

Not confirmed

- Valparaiso 2013 Q16 (bulb B removed → bulb C out): I did not render Figure 4. The key says D. The extraction cannot check the wiring.
- Rickards 2021 B Q16 (arcing): the key is highlighted in the PDF. I read the choices from the text layer only. "Arcing" is the physics answer.
- Rickards 2026 B Q7 (wire broken at A): the figure was not rendered in this run. The answer depends on where point A is.
- Battery internal resistances (9 V ≈ 1.5 Ω, AA ≈ 0.15 Ω) are typical values from memory, not from a datasheet.
- Holiday-bulb shunts and "LEDs fail open or short" are general knowledge. I did not open a manufacturer source.
- I found **no 2026 Nationals test** in the downloads (the "nat26" files are the USC Invitational 2026). No Nationals-level open/short question was seen; type 23 is built from the rules.

Sources

- 2027 Div B rules, Circuit Lab, 3.c.iii, 3.c.v, 3.c.vi, 3.c.ix, 3.d, 3.e, Part II b, c.vi (local PDF, manual pages B12-B13).
- Raw tests and keys read this run: `downloads/dlB/tests/phi26_test.txt` + key (Pembroke Hill Q22, Q24); `downloads/dlG/circuit_lab_valparaiso_university_2013_invitational.txt` + key (Q6-7, Q16); `ut20inv_Btest.txt` + key (Q30, Q34-38); `atx26reg_Bkey.txt` (FRQ 2a), `atx26reg_Ckey.txt` (SA5); `nv26C_test.txt` (Q19-21); `rsoi26B_test.txt` (Q6-7); `rsoiOldB_test.txt` (Q16); `wich26C_test.txt` (Q37); `wiki/Shock_value_test.txt` (JC Booth Q41-51); `wiki/Shock_value_test_.txt` (Section IV Q4); `wiki/Shock_Retryout_test_no_key.txt` (VI-8); `pz/p20/Circuit Lab/2020 Princeton Circuit Lab Exam.txt` + key (Lab 1 Q42-44); `downloads/dlFind/cobra19_Bstations.txt` (Station 1); `umso19_test.txt` (lab note); `nfl20reg_Btest.txt` (Q35); `nfl19reg_Ckey.txt` (Q15).
- Test extractions: `research/tests/atx-reg.md`, `misc-a.md`, `misc-b.md`, `nat26.md`, `state.md`, `ut-a.md`, `ut-b.md`, `ut-c.md`; `prior/G_tests.md`.
- `prior/B_topics.md` §5b, §9b, §9c; `prior/I_bank.md` C23-C26, D20-D22; `prior/F_slides.md` §6-7 (CIR-OPEN-SHORT-DEFINE, PAR-SHORTED-BRANCH, SER-RULES-CONCEPT, TROUBLE-SERIES-OPEN, TROUBLE-PARALLEL-OPEN, TROUBLE-CHECKLIST, BREADBOARD-CONNECTIVITY); `prior/H_logic_diodes.md` A50.
- Textbooks: OpenStax *College Physics 2e* §21.1 (series and parallel) and §20.5-20.6 (electrical safety, shorts and fuses); Kuphaldt, *Lessons in Electric Circuits* Vol. I ch. 5-7 (series, parallel, series-parallel; "Component failure analysis") and ch. 10 (network analysis); Kuphaldt Socratic Electronics worksheets ("Troubleshooting", "Voltage divider circuits"); All About Circuits, DC textbook (same text online).
- All numbers in this note were checked in Python on 2026-10-16.

Dividers and Bulb Brightness

Rule area **3.c.v** says "calculations, constructions, and configurations of DC circuits ... including series and parallel circuits, open and short circuits, Ohm's Law, and circuit schematics". Dividers, bulb brightness and "what happens if" questions all live here. So they are IN at every level. Hands-on tasks **II.c.i** (mystery resistor from voltages), **II.c.iii** (two LEDs equally bright), **II.c.vi** (troubleshoot) and **II.c.vii** (3-way switch) all use the same ideas.

Key facts

- A **voltage divider** = two or more resistors in series. The source voltage splits between them in proportion to their resistance.
- The **output** of a divider = the voltage across one resistor (or one group). It is always less than the input.
- A **current divider** = two or more parallel branches. The total current splits between them. The **smaller** resistance gets the **bigger** current.
- A **load** = whatever you connect to the divider output. A load is in parallel with the bottom resistor. So it **lowers** the output voltage.
- A divider is **"stiff"** when the load is much bigger than the divider resistors (about 10 times or more). Then the load hardly changes the output.
- A **potentiometer (pot)** = a resistor with a sliding contact called the **wiper**. It has 3 terminals. Used with all 3, it is an adjustable voltage divider.
- A **rheostat** = a variable resistor used with only 2 terminals (one end and the wiper). It sets current in series with a load.
- **Potential** at a point = its voltage measured from a reference point. The reference is called **ground** (0 V). Potential can be negative.
- **Brightness of an incandescent bulb follows its power.** More power = brighter. For identical bulbs, more current (or more voltage) = brighter.
- Treat each bulb as a **fixed resistor** unless the question says otherwise. Real filaments get hotter and their resistance rises (see type 33).
- A bulb's **rating** (for example "60 W, 120 V") tells you its resistance: $R = V^2/P$. It does **not** tell you its power in a different circuit.
- At the **same voltage**, a higher-watt bulb has **lower** resistance (a thicker filament).
- In **series**, the bulb with the **higher resistance** gets more power. So a lower-watt bulb glows brighter in series with a higher-watt bulb.
- In **parallel**, the bulb with the **lower resistance** gets more power.
- An **open** bulb (broken filament) carries no current. Its branch goes dark.
- A **shorted** bulb (wire across it) has 0 V across it. It goes dark. The rest of the circuit gets more voltage.
- An **ideal source** keeps its voltage fixed. Adding or removing a branch across an ideal source does not change the other branches across it.
- **Closing a switch** that adds a parallel branch always lowers total R and raises total current.
- **Voltage across an open part = the full voltage of the path. Voltage across a closed switch or a short = 0.**
- **Ideal voltmeter** = infinite resistance (takes no current). **Ideal ammeter** = zero resistance (a wire).

Formulas

- Voltage divider (two resistors): **$V_{out} = V_{in} \cdot R_2/(R_1 + R_2)$** ($V_{out}$ across R_2).
- Any series string: **$V_k = V_{in} \cdot R_k/R_{total}$**. Ratio form: $V_1 : V_2 : V_3 = R_1 : R_2 : R_3$.
- Current divider (two branches): **$I_1 = I_{total} \cdot R_2/(R_1 + R_2)$** (note: the **other** resistor goes on top).
- Current divider (any number): **$I_k = I_{total} \cdot R_{par}/R_k$**, where R_{par} = all branches in parallel. Or **$I_k = I_{total} \cdot G_k/G_{total}$** with $G = 1/R$.
- Loaded divider: **$V_{out} = V_{in} \cdot (R_2 \| R_L)/(R_1 + R_2 \| R_L)$**, with $R_2 \| R_L = R_2 R_L/(R_2 + R_L)$.
- Design ratio: for $V_{out}/V_{in} = k$, **$R_1/R_2 = (1 - k)/k$**.
- Potentiometer of total R_p, wiper a fraction x from the bottom (no load): **$V_{out} = x \cdot V_{in}$**.

- Bulb resistance from its rating: **$R = V_{rated}^2/P_{rated}$**. Rated current: **$I = P_{rated}/V_{rated}$**.
- Same bulb on a new voltage (fixed R): **$P_{new} = P_{rated} \cdot (V_{new}/V_{rated})^2$**.
- Power in a part: **$P = I^2R$** (good for series) or **$P = V^2/R$** (good for parallel).
- Potential at a point: start at a known point, then **subtract each drop** going with the current (or add going against it).
- Divider power: **$P = V_{in}^2/(R_1 + R_2)$**.

Question types

Easy

1. Two-resistor voltage divider: voltage across one resistor

- Scope: ✅ IN.
- Where seen: Pembroke Hill 2026 B (INV) Q6 (15 V, 1 kΩ + 4 kΩ → 12 V across 4 kΩ). Rickards 2026 C (INV) Q44 (7.5 V, $R_1 = 5$, $R_2 = 8$ → 4.62 V). UF Northern Regional 2020 B (REG) Q10-11 (5 V, 2 + 3 Ω → 3 V across 3 Ω). UT Invitational fall 2025 B/C (INV) 1a.i-ii (Vin = $V_1 + V_2$; $V_2 = Vin \cdot R_2/(R_1 + R_2)$), 1a.vi (divider power $Vin^2/(R_1 + R_2)$), 1a.viii (does a resistor divider work for AC? yes). National supervisor deck 5 (voltage dividers). Homework generator `d` page (`prior/E_build.md`).
- How to solve:
  1. Add the series resistors: R_total.
  2. V_out = V_in × (the resistor you want)/R_total.
  3. Check: the drops add up to V_in.
- Worked example (Pembroke Hill 2026 B Q6): 15 V across 1 kΩ and 4 kΩ in series. V(4 kΩ) = 15 × 4/5 = **12 V**. V(1 kΩ) = 3 V. Check: 12 + 3 = 15 V.
- Second example (Rickards 2026 C Q44): 7.5 × 8/13 = **4.62 V**.
- Traps:
  - Use the resistor the voltage is **across**, not the other one.
  - kΩ on top and bottom cancel. You do not need to convert to Ω.
  - A resistor divider works the same for AC and DC (UT fall 2025 1a.viii). Only capacitor or inductor dividers depend on frequency, and those are OUT.
- Calculator: calculate — type: V_in, R_1, R_2; outputs V_1, V_2, I, P.

2. Series string by ratio (no source current needed)

- Scope: ✅ IN.
- Where seen: JC Booth Invitational 2011 B (INV, Shock Value) Q11 (24 V string of bulbs X, 2X, 3X, two of each → 2, 4, 6, 2, 4, 6 V) and Q12 (0.78 A through the string → 18.7 W). UF Northern Regional 2020 B (REG) Q11.
- How to solve:
  1. Write the resistances as multiples of one unit (X, 2X, 3X).
  2. Add the multiples. Each part gets (its multiple ÷ total) of the voltage.
- Worked example (JC Booth 2011 Q11): X + 2X + 3X + X + 2X + 3X = 12X. The X bulbs get 24 × 1/12 = **2 V**. The 2X bulbs get **4 V**. The 3X bulbs get **6 V**. Check: 2(2 + 4 + 6) = 24 V. Q12: P = 24 V × 0.78 A = **18.7 W** (key rounds to 19 W).
- Traps: Do not split the voltage equally. Equal split only happens for equal resistors.
- Calculator: calculate — type: V, list of ratios.

3. Two-branch current divider

- Scope: ✅ IN.
- Where seen: UF Northern Regional 2020 C (REG) Q11 (8 Ω || 4 Ω, 0.2 A in the 8 Ω → 1.6 V, 0.4 A in the 4 Ω). Berkeley 2026 B (INV) DC 2.1(f) (current in R_3 by current division → 172 mA) and 2.2(b)-(c). UF Northern Regional 2020 B (REG) Q14-16 (equal split in three 2 Ω branches). National supervisor deck 5. `prior/I_bank.md` C16 (6.0 A into 2 Ω || 4 Ω → 4.0 A in the 2 Ω).
- How to solve:

1. If you know one branch current: V = I × R for that branch. All branches share this V. Then I = V/R for each other branch.
2. If you know the total current: $I_1 = I_total \times R_2/(R_1 + R_2)$.
3. Check: the branch currents add up to the total.

- Worked example (UF 2020 C Q11): 0.2 A flows in 8 Ω. V = 0.2 × 8 = **1.6 V**. The 4 Ω has the same 1.6 V, so I_2 = 1.6/4 = **0.4 A**. Total 0.6 A.
- Traps:
  - The current divider puts the **other** resistor on top. Students often swap them.
  - The smaller resistor takes the bigger current.
  - Equal branches split the current equally (UF 2020 B Q14-16: 6.0 A → **2.0 A** each; the key's 0.56 A is wrong, see Mistakes).
- Calculator: calculate — type: I_total, R_1, R_2 (or known I_1, R_1, R_2).

4. Series or parallel bulbs: which is brighter, and which drains the battery faster

- Scope: ✅ IN.
- Where seen: UF Northern Regional 2020 B (REG) Q25 (3 bulbs on 12 V: series or parallel for max brightness → parallel) and Q34 (3 bulbs, series vs parallel, separate 20 V batteries: which battery dies first → parallel, 9× the current). Shock Value re-tryout ~2012 B (practice) VI-18 (parallel is brighter).
- How to solve:
  1. Parallel: each bulb gets the full source voltage.
  2. Series: the voltage splits. Each of n equal bulbs gets V/n, so P = $1/n^2$ of its parallel power.
  3. Total current in parallel is n^2 times the series current. The battery drains faster.
- Worked example (not seen in a test): three 10 Ω bulbs on 12 V. Series: I = 12/30 = 0.40 A, each P = $0.4^2 \times 10$ = **1.6 W**. Parallel: each P = $12^2/10$ = **14.4 W** (9 times more). Total current: 0.40 A vs 3.6 A (9 times more).
- Traps: Brighter bulbs cost energy. The parallel set runs the battery down 9 times faster.
- Calculator: calculate — type: V, R per bulb, n.

5. One bulb burns out: series string vs parallel set

- Scope: ✅ IN.
- Where seen: Pembroke Hill 2026 B (INV) Q22 (holiday lights in series, one burns out → all go out). JC Booth 2011 B (INV) Q45 (L1 burns out → L4, L5 stop). National supervisor deck 3 (one open lamp turns all lamps off). `prior/I_bank.md` C22.
- How to solve:
  1. Series: one break stops the current everywhere. All bulbs go out.
  2. Parallel on an ideal source: only the broken branch goes out. The others stay the same.
  3. Mixed: only bulbs whose every path runs through the broken bulb go out.
- Worked example: 10 bulbs in series on 120 V. One filament breaks. Answer: **all 10 go out**. The broken bulb now has the full **120 V** across it.
- Traps:
  - Some real holiday strings have a **shunt** inside each bulb. It closes when the filament breaks, so the rest stay lit (slightly brighter). Only answer this way if the question mentions shunts.
  - In parallel, the other bulbs do not get brighter (ideal source).
- Calculator: look up — type: burn out, series, parallel, open.

6. Complete-circuit pictures: which bulb lights? Does polarity matter?

- Scope: ✅ IN.
- Where seen: MYSO November 2020 B (official practice) Q4 (battery, bulb and one wire: which of 4 pictures lights → the bulb tip on one terminal and the wire from the other terminal to the bulb's screw side). Shock Value re-tryout ~2012 B VI-11 (bulb wired backwards → no change; a filament has no polarity).
- How to solve:
  1. A bulb has two contacts: the **tip** at the bottom and the **screw shell** on the side.
  2. Trace a loop: battery + → one bulb contact → filament → other contact → battery –.

3. It lights only if the loop is complete and both bulb contacts are used.

- Worked example: Picture: the bulb tip touches the battery's + end, and a wire goes from the – end to the bulb's tip too. Answer: **no light**. Both wire ends touch the same contact, so the filament is bypassed (a short across the battery).
- Traps:
  - Touching the bulb's glass or the battery's side does nothing.
  - A filament bulb has no polarity. An LED **does** (see LEDs A7).
- Calculator: look up — type: bulb contacts, tip, screw, complete loop.

7. Bulb ratings: resistance, current, and filament thickness

- Scope: ✅ IN.
- Where seen: Berkeley 2026 B (INV) Q22 (60 W bulb at 120 V → 240 Ω). MYSO 2020 B (official practice) Q2 (four bulbs, same filament length: thickest → the 120 W bulb). Shock Value re-tryout ~2012 B (practice) bulb ranking by wattage (`prior/G_tests.md`; the student key's order was wrong).
- How to solve:
  1. $R = V^2/P$. $I = P/V$.
  2. At the same voltage, more watts = less R = thicker (or shorter) filament.
- Worked example (not seen in a test): 40 W, 60 W and 100 W bulbs, all rated 120 V. $R = 120^2/P$ = **360 Ω, 240 Ω, 144 Ω**. Currents 0.33 A, 0.50 A, 0.83 A. Thickest filament: **100 W**. Berkeley Q22: $120^2/60$ = **240 Ω**.
- Traps:
  - "More watts means more resistance" is wrong at a fixed voltage.
  - MYSO Q2 says "different voltages" but means different wattages at the same voltage.
- Calculator: calculate — type: P_rated, V_rated; outputs R, I.

8. Potentiometer vs rheostat; wiper position; taper

- Scope: ✅ IN (a variable resistor is a DC component).
- Where seen: UF Northern Regional 2019 B (REG) Q38 (rheostat vs potentiometer: 2 vs 3 terminals; a rheostat carries the load current). MIT Invitational 2019 C (INV) Q34 (volume-control pot: log or linear → logarithmic, "audio taper"). UT Invitational fall 2025 B/C (INV) 1c.i-iv (wiper position for R_min and R_max; where it overheats; $R(x) = \rho(L - x)/A$). Florida State 2021 B (STATE) Q21 (photo of a potentiometer). Princeton 2026 C (INV) matching (dimmer → potentiometer). `prior/I_bank.md` C17.
- How to solve:
  1. Three terminals used → it is a divider. The wiper voltage is between the two end voltages.
  2. Two terminals used → it is a rheostat. Its resistance is set by the length of track between the end and the wiper. $R = \rho L/A$.
  3. Wiper a fraction x up from the bottom (no load): $V_{out} = x \cdot V_{in}$.
- Worked example (`I_bank` C17): a 10 kΩ pot across 9.0 V, wiper 30% up from the grounded end. $V_{out} = 0.30 \times 9.0$ = **2.7 V**.
- Traps:
  - In a rheostat, a very short track (wiper near the end) has very low R. With a fixed voltage, $P = V^2/R$ is then very large. That end burns first (UT fall 2025 1c.iii).
  - **Linear taper** = R changes evenly with rotation. **Log (audio) taper** = R changes slowly at first, then fast. Ears hear loudness on a log scale.
- Calculator: calculate — type: V_in, fraction x (or R_top, R_bottom).

Medium

9. Divider inside a series-parallel circuit (voltage across one part)

- Scope: ✅ IN.
- Where seen: MIT Invitational 2019 C (INV) Q24 (5 V, R in series with two equal R in parallel: V across the single R → 3.33 V). NY Lake Erie/Niagara 2018 C (INV) Q24 (12 V, 3.3 kΩ + (4.7 kΩ || 6.8 kΩ): V across the parallel pair → 5.49 V). F-M fall 2019 (INV) Q6a (9 V; 200 + 300 + (150 || 300) → 3, 4.5, 1.5, 1.5 V). ATX Regional 2021 B (REG) Q16-17 (6 V, 10 Ω + (3 + 2) || 10 → R_eq 13.3 Ω, V at A = 1.5 V). UT Invitational fall 2018 B/C (INV) Q20-21 ((R_1 ||

R_2) + (R_3 || R_4) on 6 V). UT Invitational 2020 B (INV) Q67 (10 V ladder; V_R2 = 8.94 V, V_R5 = 0.53 V). 2026 ATX Regional B (REG) FR2b-c (9 V, R_3 = 20 Ω + (10 Ω || 5 Ω)). Berkeley 2026 B (INV) DC 2.1 and 2.2.

- How to solve:
  1. Replace each parallel group by its R_eq.
  2. Now the circuit is a plain series string. Use the divider rule on it.
  3. The voltage across a parallel group is the voltage across **each** of its branches.
  4. Go one level deeper if a branch has more parts inside.
- Worked example (NY Lake Erie 2018 C Q24): 4.7 || 6.8 = **2.779 kΩ**. V = 12 × 2.779/(3.3 + 2.779) = **5.49 V**.
- Second example (2026 ATX Regional B FR2): 10 || 5 = 3.333 Ω. R_eq = 23.33 Ω. I = 9/23.33 = 0.3857 A. V across the pair = 0.3857 × 3.333 = **1.286 V**. Current in the 10 Ω = **0.1286 A**. Current in the 5 Ω = 0.2571 A.
- Third example (F-M fall 2019 Q6a): R_eq = 200 + 300 + 100 = 600 Ω. Drops: 9 × 200/600 = **3 V**, 9 × 300/600 = **4.5 V**, and **1.5 V** across each of 150 Ω and 300 Ω.
- Traps:
  - The divider rule only works on parts that are **in series**. A parallel group must be combined first.
  - 2026 ATX FR2b key gave the total current (0.3857 A) as the current in R_1. The answer is 0.1286 A.
- Calculator: calculate — type: V, series list with parallel groups; outputs the V across each group.

10. Potential at a point (relative to ground); negative potentials

- Scope: ✅ IN.
- Where seen: Berkeley 2026 B (INV) DC 2.1(i)-(j) (potential at point 1 → 2.07 V; at point 2 → 1.035 V) and 2.2(d) (potential at point 1 → 2.02 V). ATX Regional 2021 B (REG) Q17 (voltage at A → 1.5 V).
- How to solve:
  1. Find where 0 V (ground) is. If none is drawn, use the battery's – terminal.
  2. Start at the battery's + terminal (its potential = V_source).
  3. Walk to the point along the current. Subtract I·R for each resistor you pass.
- Worked example (Berkeley 2026 B 2.1): 5 V battery. R_1 = 5 Ω in series with R_2 = 5 Ω || (R_3 + R_4 = 6 + 6 Ω). R_eq = 5 + 60/17 = 8.53 Ω. I = 5/8.53 = 0.586 A. Point 1 (after R_1): 5 – 0.586 × 5 = **2.07 V**. Point 2 (between R_3 and R_4): half of 2.07 = **1.03 V** (key 1.035 V).
- Traps:
  - "Potential at a point" and "voltage across a resistor" are different questions. The potential at the far end of R_1 is V_source minus the drop.
  - If ground is in the middle of a string, points below it are **negative**.
- Calculator: calculate — type: V, series list; outputs the potential at each node.

11. Current divider with three or more branches

- Scope: ✅ IN.
- Where seen: National supervisor deck 5 quiz (10 A into 680 Ω || 330 Ω || 220 Ω → 1.63 A, 3.35 A, 5.02 A). Homework generator Level 4 (forced equal split). UF Northern Regional 2020 B (REG) Q13-16.
- How to solve:
  1. Find R_par of all the branches.
  2. V = I_total × R_par.
  3. Each branch: I_k = V/R_k.
- Worked example (deck 5 quiz): R_par = 1/(1/680 + 1/330 + 1/220) = **110.5 Ω**. V = 10 × 110.5 = 1105 V. Branch currents: **1.63 A** (680 Ω), **3.35 A** (330 Ω), **5.02 A** (220 Ω). Sum 10.0 A.
- Traps:
  - The two-resistor formula $R_2/(R_1 + R_2)$ does not work for three branches. Use V first.
  - Deck 5 printed the formula as I_R2 = I_total·R_total/R_1. It should be /R_2 (see Mistakes).
- Calculator: calculate — type: I_total, list of branch R.

12. Loaded divider: the load pulls the output down

- Scope: ✅ IN.

- Where seen: GullSO 2025 C (INV) Q51(g) (add a 220 Ω load and re-measure). `prior/I_bank.md` C15. National supervisor homework generator Level 4 ("Rload" in a ladder). Not seen as a plain numeric question in a Div B test.
- How to solve:
  1. The load is in parallel with the bottom resistor. Combine them: $R_2 \| R_L$.
  2. Use the divider rule with R_1 and $R_2 \| R_L$.
- Worked example (`I_bank` C15): 12 V, $R_1 = R_2 = 10$ kΩ. No load: 6.0 V. A 10 kΩ load: 10 || 10 = 5 kΩ. V_out = 12 × 5/15 = **4.0 V**, not 6 V.
- Traps:
  - The output **always drops** when a load is added. It never rises.
  - Rule of thumb: keep R_L at least 10 times R_2 for less than about 10% drop.
- Calculator: calculate — type: V_in, R_1, R_2, R_L.

13. Voltmeter loading (a real meter changes what it measures)

- Scope: ✅ IN.
- Where seen: North Pocono Invitational 2020 C (INV) Q25 (5 V across 50 kΩ + 25 kΩ: ideal voltmeter on the 25 kΩ → 1.67 V; a 2.5 MΩ meter → 1.66 V). `prior/I_bank.md` C30 (1 MΩ + 1 MΩ, 10 MΩ meter → 4.76 V).
- How to solve:
  1. The meter is a resistor in parallel with the part it measures.
  2. Combine the part with the meter's resistance.
  3. Use the divider rule again.
- Worked example (North Pocono 2020 C Q25): ideal: 5 × 25/75 = **1.67 V**. With 2.5 MΩ: 25 kΩ || 2.5 MΩ = 24.75 kΩ. V = 5 × 24.75/74.75 = **1.66 V**.
- Traps:
  - Loading matters most with **large** resistors (MΩ). With small resistors (Ω, kΩ), a 10 MΩ meter changes almost nothing.
  - An ammeter has a small resistance. In series it lowers the current a little (`I_bank` C31: 1.5 V, 10 Ω, 1.0 Ω meter → 0.136 A, not 0.150 A).
- Calculator: calculate — type: V_in, R_1, R_2, R_meter.

14. Design a divider: ratio for a target output; best pair from standard values; build from a kit

- Scope: ✅ IN (hands-on task viii is related).
- Where seen: UT Invitational fall 2025 B/C (INV) 1a.vii (5 V in, ADC maximum 3.3 V → R_1:R_2 = 17:33) and 1a.ix α-δ (from {1k, 3.3k, 10k, 22k, 51k, 100k}, pick about 1:50, 1:100, 1:10, 1:3 with the fewest parts and the lowest current). UT Invitational 2019 B (INV) Q47 and UT Invitational fall 2019 C (INV) H47 (hands-on: about 5 V to about 1 V with 1 kΩ and 2.2 kΩ parts → 2.2k + 1k + 2.2k, V_out = 0.926 V).
- How to solve:
  1. Choose the ratio k = V_out/V_in.
  2. $R_1/R_2 = (1 - k)/k$.
  3. Pick real parts closest to that ratio. Series and parallel combinations count.
  4. "Lowest current" → pick the larger pair with the same ratio.
- Worked example (UT fall 2025 1a.vii): k = 3.3/5 = 0.66. R_1/R_2 = 0.34/0.66 = **17:33** (0.515). Check: 5 × 33/50 = 3.30 V.
- Second example (UT 2019 B Q47): kit has 1 kΩ and 2.2 kΩ. Put 2.2 kΩ + 2.2 kΩ on top and 1 kΩ on the bottom. V_out = 5 × 1/5.4 = **0.926 V** (inside 10% of 1 V).
- Third example (UT fall 2025 1a.ix, "1:50"): 1k with 51k gives V_out/V_in = 1/52. This is the closest pair (**1k : 51k**).
- Traps:
  - "1:50" can mean V_out:V_in or R_1:R_2. Read it as the test uses it. UT used V_out:V_in.
  - The 1:3 key (10k:22k = 1/3.2) is not the best pair; 51k:100k gives 1/2.96 with less current (see Mistakes).
- Calculator: calculate — type: V_in, V_out target, list of available R; outputs best pairs.

15. Rheostat and potentiometer ranges (min and max current or voltage)

- Scope: ✅ IN.
- Where seen: UT Invitational fall 2025 B/C (INV) 1c.i-ii (wiper positions for R_min and R_max). `prior/I_bank.md` C17 (rheostat 0-50 Ω with a 10 Ω lamp on 12 V → 0.20 A to 1.2 A). End-resistor pot circuits: not seen in a test (Kuphaldt, *Lessons in Electric Circuits* Vol. I, potentiometer section).
- How to solve:
  1. Do the circuit twice: once with the variable part at its minimum, once at its maximum.
  2. The answer is the range between the two results.
- Worked example (`I_bank` C17): rheostat at 0 Ω: I = 12/10 = **1.2 A**. At 50 Ω: I = 12/60 = **0.20 A**.
- Second example (not seen in a test): 12 V across 1 kΩ, a 10 kΩ pot and 1 kΩ in series. The wiper can reach from 12 × 1/12 = **1.0 V** to 12 × 11/12 = **11.0 V**.
- Traps: Fixed resistors at the ends of a pot stop the output from reaching 0 V or the full supply.
- Calculator: calculate — type: V, R_fixed, R_var min, R_var max.

16. Rank identical bulbs in a network by brightness

- Scope: ✅ IN.
- Where seen: JC Booth 2011 B (INV) Q3 (six identical bulbs: circle the 3 brightest → the 3 in series with the source), Q40 (seven-bulb network: brightest → L7, directly across the source), Q42 (longest series path), Q43-44 (most and least current → L7, L6), Q47 (1 A in L2 → 1 A in L3, same series path). MYSO 2020 B Q25 (brightest → Bulb 1, it carries the sum of both branch currents). UT Invitational fall 2018 B/C (INV) Q24a. `prior/I_bank.md` C23. Physics-education classic (Arons, AP Physics 1).
- How to solve:
  1. A bulb directly across the source gets the full voltage. It is the brightest.
  2. A bulb that carries the **total** current is brighter than bulbs on branches that share it.
  3. Two equal branches split the current in half. Each gets 1/4 of the power of the full current.
  4. A bulb with a wire across it is dark.
- Worked example (`I_bank` C23): identical bulbs. A in series with (B || C). R_total = 1.5R. I_A = V/1.5R. P_A = (4/9)·V²/R. B and C each carry I_A/2, so P = (1/9)·V²/R. Answer: **A brightest; B = C**.
- Traps:
  - Current is not "used up". Bulbs in the same series path have the same brightness.
  - Brightness depends on the bulb's power, not on its place in the drawing (first or last).
- Calculator: look up — type: rank bulbs, brightest, series path, branch.

17. Different-watt bulbs in series (the lower-watt bulb is brighter)

- Scope: ✅ IN.
- Where seen: UT Invitational 2019 B (INV) Q36 and UT Invitational fall 2019 C (INV) Q32 (15 W, 35 W, 40 W bulbs rated at 120 V, in series and in parallel on 120 V: find each current). `prior/I_bank.md` C24 (60 W and 100 W in series). B_topics "Borderline: a higher-wattage bulb is dimmer in series".
- How to solve:
  1. Find each bulb's R from its rating: R = V²/P.
  2. Series: I = V/(sum of R). The same I flows in each bulb.
  3. Each power: P = I²R. The biggest R (lowest rating) gets the most power.
  4. Parallel: each bulb gets its rated voltage, so it makes its rated power. I = P/V.
- Worked example (UT 2019 B Q36): R = 960 Ω, 411.4 Ω, 360 Ω. Total 1731 Ω. Series current = 120/1731 = **0.0693 A** in each. Powers: 4.61 W, 1.98 W, 1.73 W. So the **15 W bulb is brightest**. Parallel currents: **0.125 A, 0.292 A, 0.333 A**.
- Second example (`I_bank` C24): 60 W (240 Ω) and 100 W (144 Ω) in series on 120 V. I = 0.3125 A. P_{60} = **23.4 W**, P_{100} = **14.1 W**. The 60 W bulb is brighter.
- Traps:
  - The UT key gave 0.750 A for the series current. That is the **parallel total** (90 W/120 V). See Mistakes.
  - Rated watts are only true at the rated voltage.
- Calculator: calculate — type: V_rated, list of P_rated, V_source, series or parallel.

18. A rated bulb on a different voltage

- Scope: ✅ IN.
- Where seen: `prior/I_bank.md` U20 (60 W/120 V bulb on 60 V → 15 W) and the Pembroke Hill 2026 B item noted there (V doubled, R halved → P × 8). Not seen as a bulb question in an extraction file.
- How to solve:
  1. R = V_rated²/P_rated (assume it does not change).
  2. P_new = V_new²/R. Or P_new = P_rated × (V_new/V_rated)².
- Worked example: 60 W, 120 V bulb on 60 V. R = 240 Ω. P = 60²/240 = **15 W** (one quarter, not one half).
- Traps:
  - Half the voltage gives one quarter of the power (fixed R).
  - A real filament runs cooler at low voltage. Its R drops, so the real power is a little more than 15 W (BORDERLINE detail; see type 33).
- Calculator: calculate — type: P_rated, V_rated, V_new.

19. What changes when a switch opens or closes (up, down, same)

- Scope: ✅ IN.
- Where seen: MYSO 2020 B (official practice) Q26-30 (Bulb 1 in series with three parallel paths: Bulb 2 + SW1, Bulb 3 + ammeter, SW2. Only SW1 open → Bulb 1 current decreases, Bulb 2 zero, Bulb 3 greater. Only SW2 closed → Bulb 1 brighter, Bulb 3 goes out). UT Invitational 2019 B (INV) Q33 and UT fall 2019 C (INV) Q29 (S1 adds R_2 in parallel with R_1: power in R_1 → same across an ideal source; decreases when R_3 is in series). `prior/I_bank.md` C25.
- How to solve:
  1. Find the total R before and after. A closed parallel switch lowers R. An opened branch raises R.
  2. Total current goes the opposite way to total R.
  3. Parts in series with the source follow the total current.
  4. Parts in parallel get "what is left": V_parallel = V_source – (drop on the series parts).
- Worked example (MYSO Q26-28, identical bulbs): before (SW1 closed, SW2 open): R_total = R + R/2 = 1.5R. $P_1 = (4/9)V^2/R$, $P_3 = (1/9)V^2/R$. Open SW1: R_total = 2R. $P_1 = (1/4)V^2/R$ (**decreases**). Bulb 2 = **0**. $P_3 = (1/4)V^2/R$ (**greater**, from 1/9 to 1/4).
- Second example (MYSO Q29-30): close SW2 (a wire across the parallel group). Bulb 1 gets the full V: $P_1 = V^2/R$ (**brighter**). Bulbs 2 and 3 **go out**.
- Traps:
  - A new branch across an **ideal** source changes nothing in the other branches (UT 2019 B Q33 part 1). With a series resistor (or internal resistance), it lowers their voltage.
  - Only one switch changes at a time. Reset to the starting state for each part.
- Calculator: look up — type: switch closes, switch opens, increase, decrease, same.

20. A bulb fails open or shorted inside a network

- Scope: ✅ IN.
- Where seen: JC Booth 2011 B (INV) Q45 (L1 burns out → L4 and L5 stop) and Q46 (L2 shorted → none stop). F-M Invitational 2018 (INV) Q5 (120 V: 50 W bulb A in series with (60 W B ‖ 40 W C); B burns out → A dims but stays lit). `prior/I_bank.md` C22-C23 (remove B → A dims from 4/9 to 1/4, C brightens from 1/9 to 1/4; short B → A at full V, B and C out).
- How to solve:
  1. Open: delete that bulb and its branch. Recompute R_total.
  2. Short: replace that bulb with a wire. Anything in parallel with the wire also goes dark.
  3. Use type 19 steps to say what goes up and down.
- Worked example (F-M 2018 Q5): R = V²/P: A = 288 Ω, B = 240 Ω, C = 360 Ω. Before: B ‖ C = 144 Ω, I = 120/432 = 0.278 A, P_A = **22.2 W**. B opens: I = 120/648 = 0.185 A, P_A = **9.88 W**. So A **dims but stays lit**. C rises from 4.44 W to 12.3 W.
- Traps:
  - "Burns out" means **open**, not short.
  - A short in a **parallel** set shorts the whole source. The fuse or breaker opens.

- Calculator: calculate — type: list of bulbs with links, which one opens or shorts.

21. Spot the shorted (darkest) bulb

- Scope: ✅ IN.
- Where seen: JC Booth 2011 B (INV) Q41 (darkest → L6, shorted by a wire) and Q44 (least current → L6). MYSO 2020 B Q29-30 (SW2 closed shorts the parallel group; Bulb 3 goes out).
- How to solve:
  1. Look for a wire (or a closed switch, or an ideal ammeter) that joins the two ends of a bulb, with no part in it.
  2. That bulb has 0 V across it. It is dark.
- Worked example: a bulb drawn with a plain wire from its left end to its right end. Answer: **0 V, 0 A, dark**. The wire carries the current.
- Traps: An ideal ammeter is a wire. An ammeter drawn across a bulb shorts it.
- Calculator: look up — type: shorted bulb, darkest, 0 V.

22. How many bulbs in parallel before the fuse blows

- Scope: ✅ IN.
- Where seen: Princeton Invitational 2021 C (INV) Q37 (0.75 A fuse, 10 V, 100 Ω bulbs → 7 bulbs). `prior/I_bank.md` (15 A, 120 V circuit, 100 W bulbs → 18 bulbs).
- How to solve:
  1. Current per bulb: I = V/R (or P/V).
  2. Number = fuse rating ÷ current per bulb. **Round down.**
- Worked example (Princeton 2021 Q37): each bulb 10/100 = 0.10 A. 0.75/0.10 = 7.5. So **7 bulbs** (0.70 A). An 8th gives 0.80 A and blows the fuse.
- Traps: Always round down, even at 7.9.
- Calculator: calculate — type: fuse A, V, R or P per load.

23. Find the dead bulb from meter readings

- Scope: ✅ IN (hands-on task vi).
- Where seen: National supervisor deck 11 (six 12 Ω lamps in series on 6 V, all out → the open lamp shows the full source voltage; six in parallel, L5 out → L5 has 6 V but no current). `prior/I_bank.md` C26. See also Multimeter Use [A9].
- How to solve:
  1. Series string, all dark: measure V across each bulb. The good bulbs read **0 V**. The open bulb reads the **full** source V.
  2. Parallel set, one dark: it has full V but no current. Its filament or socket is open.
  3. Check with an ohmmeter, power **off**: OL = open.
- Worked example (deck 11): good string: each lamp 6/6 = 1 V, I = 6/72 = **83 mA**. After the fault: L5 reads **6 V**, the others 0 V. L5 is open. Parallel case: each good lamp takes 6/12 = **0.5 A**.
- Traps: The deck printed "voltage at L5 = 10 V" on a 6 V source. It should be 6 V.
- Calculator: look up — type: troubleshoot, open bulb, full voltage, OL.

Hard

24. Before-and-after numbers: by what factor does a bulb's power change?

- Scope: ✅ IN.
- Where seen: UT Invitational fall 2018 B/C (INV) Q24c (L3 removed: power in L4 rises × 2.78; L5 falls × 0.694). F-M 2018 (INV) Q5. `prior/I_bank.md` C23.
- How to solve:
  1. Work with R = 1 and V = 1 (identical bulbs). Only ratios matter.
  2. Find each bulb's current before and after.
  3. Power factor = (I_after/I_before)².
- Worked example (UT fall 2018 Q24c, figure rendered): with S1 at W, S2 at Z, S3 at A, the circuit is L1 + (L3 || L4) + L5. R_total = 2.5. I = 0.4. L4 gets 0.2, L5 gets 0.4. Remove L3: L1 + L4 + L5 in series, I = 1/3. L4: (0.333/0.2)² = **×**

2.78 (brighter). L5: $(0.333/0.4)^2$ = **× 0.694** (dimmer).

- Traps: Say the direction **and** the reason ("L4 now carries the whole current").
- Calculator: calculate — type: before network, after network, bulb name.

25. Bulb network with SPDT switches: brightness for each setting; find a setting

- Scope: ✅ IN.
- Where seen: UT Invitational fall 2018 B/C (INV) Q24a-b (5 bulbs, 3 switches: compare brightness for one setting; which setting lights all 5). UT Invitational 2019 B (INV) Q43 and UT fall 2019 C (INV) Q44 (3 SPDT switches cascading 3 lamps: 8-row on/off table, L1 = S1, L2 = S1·S2, L3 = S1·S2·S3). UMSO 2019 lab (two SPDT switches as an OFF/LOW/HIGH selector). See also 3-Way and 4-Way Switching [A6].
- How to solve:
  1. Redraw the circuit for one switch setting. Erase every unused throw.
  2. Mark bulbs with an open end. They are dark.
  3. Reduce the rest to series and parallel. Rank them (type 16).
  4. For "light all bulbs", look for one long series path through all of them.
- Worked example (UT fall 2018 Q24a, figure rendered): S1 = W, S2 = Y, S3 = A. L2's top goes to the unused X throw, so **L2 is off**. L1 feeds node B. L3 (B to C) and L4 (B to the right rail) are joined at the bottom by S3. So the circuit is L1 + (L3 || L4) + L5. Answer: **L1 = L5 > L3 = L4 > L2 (off)**.
- Q24b: S1 = X, S2 = Y, S3 open. Path: L1 → L2 → C → L3 → B → L4 → L5. All 5 in **series**, all lit, all equal.
- Traps:
  - An SPDT switch connects its common to **one** throw only. The other throw is a dead end.
  - Some settings short the battery (UT 2019 B Q43 with S1 off). Say so.
- Calculator: look up — type: SPDT, bulb network, setting.

26. Design a bulb arrangement for a brightness pattern

- Scope: ✅ IN.
- Where seen: JC Booth 2011 B (INV) Q10 (draw 6 bulbs: 3 very bright, 1 less, 2 even less).
- How to solve:
  1. Bulbs carrying the full current must be in the main series path.
  2. To make bulbs dimmer, put them on branches that share the current.
  3. A branch with more bulbs in series gets less current.
- Worked example (JC Booth Q10): three bulbs in series, then a parallel pair of branches: branch 1 = one bulb, branch 2 = two bulbs in series. The main current I splits 2:1. Powers relative to the main bulbs: **1, 4/9, 1/9**. So 3 very bright, 1 less bright, 2 dimmest.
- Traps: Check the ranking with numbers. A guess from the drawing often fails.
- Calculator: calculate — type: proposed network; outputs relative power of each bulb.

27. Dim an LED with a switched shunt resistor; OFF/LOW/HIGH selector

- Scope: ✅ IN (LED from area vii; divider and shunt from area v).
- Where seen: USC Invitational 2026 B (INV) hands-on "Light Switch" part e (one button dims the LED but keeps it on → push button + a second 510 Ω in parallel with the LED). UMSO 2019 lab (two SPDT switches, 9 V, LED, resistors → OFF / LOW / HIGH).
- How to solve:
  1. Treat the LED as a fixed drop V_f (about 2.0 V red, 3.0 V blue).
  2. Series resistor current: $I_R = (V_s - V_f)/R$.
  3. A shunt resistor across the LED takes V_f/R_{shunt}. The LED gets the rest.
  4. For OFF/LOW/HIGH: one switch opens the circuit (OFF); the other picks a large R (LOW) or a small R (HIGH).
- Worked example (USC 2026 e, red LED): 9 V, 510 Ω. LED alone: (9 – 2)/510 = **13.7 mA**. Press the button: shunt 2.0/510 = 3.9 mA. LED = **9.8 mA** (29% less, still on). Blue LED (3.0 V): 11.8 mA → **5.9 mA**.
- Traps:
  - A plain wire as the shunt turns the LED **off** (that is the NOR part of the same task).
  - Resistor power: $(9 - 2)^2/510 = 0.096$ W. A 1/4 W part is safe.

- Calculator: calculate — type: V_s, V_f, R_series, R_shunt.

28. Map a two-filament bulb (car headlight) with an ohmmeter

- Scope: ✅ IN.
- Where seen: Cobra Invitational 2020 B (INV) station S2 (car headlight bulb, 2 filaments, 3 terminals: draw the filaments as resistors and find the high beam → terminal B).
- How to solve:
  1. Measure all 3 pairs: AB, AC, BC.
  2. The largest reading is the two filaments in series. The terminal **not** in that pair is the common.
  3. The two readings with the common are the two filaments.
  4. The high beam uses more power at the same 12 V, so it has the **lower** R.
- Worked example (not the real Cobra values): AB = 0.6 Ω, AC = 0.9 Ω, BC = 1.5 Ω. BC is largest, so **A is common**. Filament A-B (0.6 Ω) is the **high beam**.
- Traps: Ohmmeter readings are **cold**. A hot filament has about 10 times more R. Use the readings only to compare.
- Calculator: calculate — type: R_AB, R_AC, R_BC; outputs common terminal and each filament.

29. Zero-current condition: pick the source so one branch carries no current

- Scope: ✅ IN (one equation; divider reasoning, no KVL system).
- Where seen: UF Northern Regional 2020 C (REG) Q16 (12 V, 100 Ω, 300 Ω shunt, 200 Ω to a second source E: E for zero current in 200 Ω → 9 V).
- How to solve:
  1. If a branch carries no current, it has no drop. Both its ends are at the same potential.
  2. So the rest of the circuit acts as if that branch is not there.
  3. Find the node voltage with the divider rule. Set E equal to it.
- Worked example (UF 2020 C Q16): with no current in the 200 Ω, the 12 V source drives 100 Ω + 300 Ω. Node = 12 × 300/400 = 9 V. So **E = 9.0 V**.
- Traps: Watch the battery's direction. Its + side must face the node.
- Calculator: calculate — type: V_s, R_top, R_bottom; outputs the matching E.

30. Sensor in a divider: output limits and sketches

- Scope: ⚠️ Borderline. Thermistors and light sensors are not listed devices (rule 3.d "non-linear devices beyond those listed"). But when the test gives R as a formula, it is just a variable resistor and algebra (3.e.i "functions").
- Where seen: UT Invitational fall 2025 B/C (INV) 1a.iii-v (sketch V_2 vs R_1, vs R_2, vs Vin with limits 0 and ∞), 1b.i-iii (R_2 = PTC thermistor with R = AT + B: V_2 as T rises and falls; V_2 in terms of T).
- How to solve:
  1. Write V_out = V_in·R_2/(R_1 + R_2).
  2. Put in the limits: R_2 → 0 gives 0 V. R_2 → very large gives V_in.
  3. Put the sensor formula in for R_2 if one is given.
- Worked example (not seen in a test): 5 V, R_1 = 10 kΩ on top. A light sensor on the bottom goes from 1 kΩ (bright) to 100 kΩ (dark). V_out goes from 5 × 1/11 = **0.45 V** (bright) to 5 × 100/110 = **4.55 V** (dark).
- Traps:
  - UT 1b.ii key said V_2 → 0 as T falls. With R = AT + B and T in kelvin, R_2 → B, so V_2 → Vin·B/(R_1 + B) (see Mistakes).
  - V_out vs R_2 is a curve, not a straight line. V_out vs V_in is a straight line.
- Calculator: calculate — type: V_in, R_1, R_2 min, R_2 max.

31. Potentiometer with a load: the output is not x·V_in

- Scope: ✅ IN.
- Where seen: not seen in a test. Source: Kuphaldt, *Lessons in Electric Circuits* Vol. I (potentiometer as a voltage divider) and the Socratic "voltage divider" worksheets.
- How to solve:
  1. Split the pot into R_top = (1 − x)R_p and R_bottom = x·R_p.

2. The load is in parallel with R_bottom.
3. Use the loaded-divider formula (type 12).

- Worked example: 10 kΩ pot across 10 V, wiper in the middle, 10 kΩ load. Bottom: 5 || 10 = 3.33 kΩ. V = 10 × 3.33/(5 + 3.33) = **4.0 V**, not 5.0 V.
- Traps: The bigger the load resistance, the closer the output gets to x·V_in.
- Calculator: calculate — type: V_in, R_pot, x, R_L.

32. Voltage between two points (two dividers side by side)

- Scope: ✅ IN. Full details in Hard Resistor Networks A5 type 5.
- Where seen: Princeton Invitational 2026 C (INV) Comp 1d (unbalanced bridge voltmeter → 0.4 V). Northview Sylvania 2025 C (INV) FR6 (→ 1.1 V). National supervisor deck 10 and homework generator Level 11 (V_DB).
- How to solve:
  1. Each side of the bridge is its own divider across the same source.
  2. Find the potential of each middle point.
  3. Subtract.
- Worked example (generator Level 11): 12 V; left side R_1 = 100 Ω over R_2 = 200 Ω; right side R_3 = 300 Ω over R_x = 300 Ω. V_D = 12 × 200/300 = 8 V. V_B = 12 × 300/600 = 6 V. **V_DB = +2 V**.
- Traps: An ideal voltmeter between the points takes no current, so each side stays a plain divider.
- Calculator: calculate — type: V, R_1, R_2, R_3, R_4.

33. Real bulbs: resistance rises with temperature

- Scope: ⚠️ Borderline. A filament is a non-linear device not in the list (rule 3.d). Concept questions still appear.
- Where seen: UT Invitational fall 2018 B/C (INV) Q9 (why is a bulb non-ohmic → the filament heats and its R rises). MYSO 2020 B Q3 (true about a light bulb → R rises as it heats). UT Invitational 2020 B (INV) Q60 (match I-V curves: copper line, filament lamp bends over, diode). Shock Value re-tryout ~2012 B II-5. `prior/I_bank.md` (tungsten 24 Ω at 20 °C → 294 Ω hot).
- How to solve:
  1. Say: higher voltage → more current → hotter filament → higher R.
  2. On an I-V graph, the bulb's curve bends toward the V axis (less extra current per volt).
  3. A cold bulb draws a large **inrush** current at switch-on.
- Worked example (`I_bank`): R_hot = 24 × (1 + 0.0045 × 2500) = **294 Ω**. So the cold bulb draws about **12 times** its running current at switch-on.
- Traps: In calculations, still treat bulbs as fixed resistors unless the test gives a curve.
- Calculator: look up — type: non-ohmic, filament, inrush, temperature.

Nationals

34. Loaded divider design to a spec (State and National hands-on level)

- Scope: ✅ IN (the hardest divider type seen).
- Where seen: GullSO 2025 C (INV) Q51 (3.00 V in; unloaded output 1.25 V; at least 1.20 V with a 220 Ω load; design, derive the formula, pick ideal values, approximate with the kit, build, measure).
- How to solve:
  1. Unloaded ratio: R_1/R_2 = (V_in − V_out)/V_out.
  2. Loaded condition: $V_{in} \cdot (R_2 \| R_L)/(R_1 + R_2 \| R_L) \geq V_{min}$.
  3. Keep the ratio from step 1. Solve step 2 for the **largest** R_2 allowed (one equation, one unknown).
  4. Build R_2 from series and parallel combinations of kit parts.
- Worked example (GullSO Q51): R_1/R_2 = 1.75/1.25 = **7:5**. With $R_1 = 1.4R_2$ and R_L = 220 Ω, the output is exactly 1.20 V when R_2 = **15.7 Ω** (R_1 = 22 Ω). So $R_2 \leq$ 15.7 Ω. Kit build: R_1 = 22 Ω, R_2 = 22 || 56 = **15.8 Ω**. Unloaded 1.254 V; loaded **1.203 V** (passes).
- Traps:
  - Small divider resistors make a stiff divider but waste power. Here: $3^2/37.8 \approx 0.24$ W. Check the resistor ratings.
  - Measure the loaded output with the load connected (part g).
- Calculator: calculate — type: V_in, V_out unloaded, V_out min, R_L; outputs ratio and R_2 max.

35. Multi-tap divider: several outputs from one string

- Scope: ✅ IN.
- Where seen: not seen in a test. Source: Kuphaldt, *Lessons in Electric Circuits* Vol. I ("Voltage divider circuits"); All About Circuits voltage divider worksheet.
- How to solve:
  1. Pick the string current, I. Then R_total = V_in/I.
  2. Each tap's potential sets the resistance below it: R_below = V_tap/I.
  3. Each resistor = difference between taps ÷ I.
- Worked example: 12 V in, taps at 9 V and 5 V, string current 1.0 mA. R_total = 12 kΩ. Bottom 5 kΩ (0 to 5 V), middle 4 kΩ (5 to 9 V), top 3 kΩ (9 to 12 V). Check: 12 × 5/12 = **5 V**, 12 × 9/12 = **9 V**.
- Traps: Loads on the taps change every tap, not only their own.
- Calculator: calculate — type: V_in, list of tap voltages, I.

36. Divider swing design (largest output change for a changing resistor)

- Scope: ✅ IN (the curve-shape part is ⚠️ Borderline).
- Where seen: UT Invitational fall 2025 B/C (INV) 1c.vi β-ε (R_min ≪ R_1 ≪ R_max for the largest V_1 swing; sketch V_1 when R_2 changes like a sine → not a scaled sine).
- How to solve:
  1. Output across the fixed R_1: $V_1 = V_{in} \cdot R_1/(R_1 + R_2)$.
  2. For a big swing, R_1 must be much larger than R_min and much smaller than R_max.
  3. A good middle choice is $R_1 = \sqrt{(R_{min} \cdot R_{max})}$ (a known result; deriving it needs calculus, which is OUT).
- Worked example (not seen in a test): R_2 swings from 100 Ω to 1 MΩ. $R_1 = \sqrt{(100 \times 10^6)}$ = **10 kΩ**. V_1/V_{in} goes from 10/10.1 = **0.990** to 10/1010 = **0.0099**. Almost the full range.
- Traps: The UT key wrote "R_min ≪ R_2". It should be R_1 (see Mistakes).
- Calculator: calculate — type: V_in, R_min, R_max; outputs R_1 and the swing.

37. Two-rail divider (+ and – supplies); potentials below ground

- Scope: ✅ IN.
- Where seen: not seen in a test. Source: Kuphaldt, *Lessons in Electric Circuits* Vol. I ("Voltage and current" polarity, "Ground" reference); All About Circuits.
- How to solve:
  1. Total voltage across the string = V_top – V_bottom (for +12 and –12: 24 V).
  2. Current = total voltage ÷ total R.
  3. Node potential = V_top – I·R_above.
- Worked example: +12 V → 10 kΩ → node → 20 kΩ → –12 V. I = 24/30 kΩ = 0.8 mA. Node = 12 – 8 = **+4 V**. Swap the resistors: node = 12 – 16 = **–4 V**.
- Traps: Do not use only the + supply voltage. The string sees the difference, 24 V.
- Calculator: calculate — type: V_top, V_bottom, R_top, R_bottom.

Out of scope (skip)

- ❌ **Capacitor dividers and RC timing** (rule 3.d "capacitance"). Seen: UF Northern Regional 2020 C (REG) Q43 (switch closes on 10 Ω || 1 mF); North Pocono 2020 C (INV) Q13 (which capacitor charges fastest).
- ❌ **Thevenin method for a loaded divider** (rule 3.d "Norton/Thevenin equivalents"). Seen: UT fall 2019 C (INV) Q36 (load design via Thevenin). Use plain series-parallel instead (type 12).
- ❌ **Op-amp buffer after a divider** (rule 3.d "semiconductors beyond those listed"). Seen: UT fall 2019 C (INV) Q38.
- ❌ **Inductive lamp transient** (rule 3.d "inductance"). Seen: Cobra 2020 B (INV) Q27 (transformer + battery + switch: when does the lamp light).
- ❌ **Motor start dims the lights** (magnetism; rule 3.c lists areas "only"). Seen: UT fall 2025 (INV) SA3. Battery-sag version with a plain resistor is IN: see Internal Resistance and Power Transfer A5.

- ⚠ **Dimmer internals with a triac** (rule 3.d "semiconductors beyond those listed"). Seen: MIT 2019 C (INV) Q35. The variable-resistor part is IN.
- ⚠ **Dependent source acting as a resistor** (not in 3.c). Seen: USC Invitational 2026 B (INV) Q33b (VCCS of 4 m℧ = 250 Ω → divider gives 1.07 V).
- ✗ **Three-equation node analysis** of a bulb or divider network (rule 3.e.i "no more than two equations"). Seen: UT fall 2019 C (INV) Q33.

Mistakes in tests and keys

- **UT Invitational 2019 B (INV) Q36 and UT fall 2019 C (INV) Q32:** key gives 0.750 A in each series bulb. That is the parallel total (90 W ÷ 120 V). Correct series current: **0.0693 A** (R = 960, 411, 360 Ω). The parallel values in the key are right.
- **2026 ATX Regional B (REG) FR2b:** key 0.3857 A is the total current. Current in R_1 (10 Ω) = **0.1286 A**.
- **UF Northern Regional 2020 B (REG) Q13-16:** key 1.67 A total and 0.56 A per branch. 5/3 is the R_eq in ohms. Correct: **6.0 A** total, **2.0 A** in each 2 Ω.
- **UT Invitational fall 2025 (INV) 1a.ix δ:** key 10k:22k for "about 1:3". 51k:100k gives 1/2.96 (closer) with less current. If "1:3" means the resistor ratio, 3.3k:10k (1:3.03) is best.
- **UT Invitational fall 2025 (INV) 1b.ii:** key says $V_2 \to 0$ as T falls. With R = AT + B and T ≥ 0 K, $V_2 \to V_{in} \cdot B/(R_1 + B)$. It is 0 only if B = 0.
- **UT Invitational fall 2025 (INV) 1c.vi β-γ:** key "R_min ≪ R_2" and "R_max ≫ R_2". The question compares with R_1. Typo.
- **UT Invitational fall 2025 (INV) 1a.vii:** key "$33R_1 : 17R_2$" reads like 33:17. It means $33R_1 = 17R_2$, so $R_1{:}R_2$ = **17:33**. Sloppy notation, right value.
- **National supervisor deck 5:** current divider lines print I_R2 = I_total·R_total/R_1 and I_R3 = .../R_1. They should end in /R_2 and /R_3.
- **National supervisor deck 11:** "voltage at L5 = 10 V" on a 6 V source. It should read 6 V.
- **Shock Value student key (~2012 B, practice):** bulb ranking by wattage has the resistance order backward. At a fixed voltage, the higher-watt bulb has more current and **less** R.
- **MYSO 2020 B Q2:** stem says "different voltages" but means different wattages at one voltage. The answer (120 W = thickest) is right.
- **Berkeley 2026 B (INV) DC 2.2(e):** key energy 2.15 × 10^4 J. From the key's own numbers, 5 V × 0.595 A × 7200 s = **2.14 × 10^4 J**. Rounding only.

Not confirmed

- **F-M fall 2019 Q6a** circuit: the figures are EMF images that I could not render. I checked the key's numbers against its own stated circuit ($R_1 + R_2$ in series with $R_3 \parallel R_4$).
- **ATX Regional 2021 B Q17** ("voltage at A" = 1.5 V): I used the extraction's reading that A is the top of the parallel group. I did not re-render the figure.
- **JC Booth 2011 B Q3, Q10, Q40-47:** the extraction traced the scanned figure at 150 dpi. Some wire crossings are hard to read. I did not re-render it.
- **MYSO 2020 B Q4 and Q23-30:** the figure is low resolution. The topology (Bulb 1 in series with three parallel paths) comes from the extraction file.
- **Cobra 2020 B S2:** the real ohmmeter values are not in the extraction. My worked example uses made-up values.
- **UT 2019 B Q43 / fall 2019 C Q44** (3 SPDT cascade): I used the extraction's answer. I did not render the figure.
- **Princeton 2021 C Q37** (7 bulbs): taken from `prior/G_tests.md`. I did not open the Princeton 2021 key.
- **Type 36 rule $R_1 = \sqrt{(R_{min} \cdot R_{max})}$:** a standard result (it maximizes the swing), from memory. Its proof needs calculus.
- **Florida State 2021 B (STATE):** no divider or bulb-brightness question except the potentiometer photo (Q21). Where I looked: `research/tests/state.md`.
- **No real 2026 National Tournament test was found.** The "nat26" files are the USC Invitational (January 2026). So I cannot confirm what Nationals asks on dividers. Types 34-37 are my best guess at Nationals difficulty.

- Textbook items (types 31, 35, 37): from general knowledge of Kuphaldt and All About Circuits. I did not re-open those pages this run.

Sources

- Rules: 2027 Science Olympiad Div B Rules Manual, Circuit Lab, rules 3.c.v, 3.d, 3.e and Part II c.i, c.iii, c.vi, c.vii (pages B12-B13). See Circuit Lab - Rules 2027 Ⓡ.
- Test extractions in `~/Daniel/circuitlab/research/tests/`: `misc-a.md` (Cobra 2020 B, F-M 2018, F-M fall 2019, North Pocono 2020 C, UMSO 2019), `misc-b.md` (MIT 2019 C, UF Northern Regionals 2019-2020 B/C, NY Lake Erie 2018 C, JC Booth 2011 B, Shock Value ~2012 B), `ut-a.md` (UT fall 2025, 2026 ATX Regional B/C), `ut-b.md` (UT 2020 B/C, UT 2019 B), `ut-c.md` (UT fall 2018, UT fall 2019 C, ATX Regional 2019), `atx-reg.md` (ATX Regionals 2020-2021), `state.md` (Florida State 2021 B, MYSO 2020 B), `nat26.md` (USC Invitational 2026 B/C).
- `prior/G_tests.md` (GullSO 2025 C Q51, Rickards 2026 C Q44, Princeton 2021 and 2026 C, Northview 2025 C); `prior/I_bank.md` §5.2-5.4 (C14-C31, U20); `prior/B_topics.md` §5; `prior/F_slides.md` (decks 3, 5, 10, 11; generator Levels 4 and 11).
- Raw tests read this run: `downloads/dlB/tests/phi26_test.txt` + key (Pembroke Q6, Q22), `berk26_test.txt` + key (Q22; DC 2.1-2.2), `bullso26_test.txt`; `downloads/dlG/gull26_test.txt` + key (Q51); `downloads/dlG/ut18inv_C.pdf` page 6 (rendered: Q24 figure).
- Textbooks: OpenStax *College Physics 2e*, ch. 21.1 (series and parallel; bulbs) and 21.2 (terminal voltage); Kuphaldt, *Lessons in Electric Circuits* Vol. I, ch. 6 ("Divider circuits and Kirchhoff's laws": voltage divider, current divider, potentiometer) and ch. 8 (meter loading); All About Circuits, "Voltage Divider Circuits" and "Current Divider Circuits".
- All numbers in this note were checked in Python on 2026-10-16.

Meters and Measurement (Meter Theory)

Note: Meters and Measurement

Rule **Part II b** says teams "must be familiar with the operation of breadboards and multimeters and how to use them." Rule area **3.c.v** covers DC circuits and components, and **3.c.iv** covers current, voltage and resistance. So meters show up in two ways. The written test asks meter theory (where a meter goes, what an ideal meter is, loading). The hands-on asks you to measure. This note is the theory side. Multimeter Use [A9] is the button-and-jack side.

Key facts

- **Ammeter** = a meter that measures **current** (unit: ampere, A). It goes **in series**. You must break the circuit and let the current flow through the meter.
- **Voltmeter** = a meter that measures **voltage** (potential difference, unit: volt, V). It goes **in parallel**, across the part. The circuit must be **on**.
- **Ohmmeter** = a meter that measures **resistance** (unit: ohm, Ω). The circuit must be **off**. The part should be **isolated** (lift one leg out of the circuit).
- **Multimeter** = one meter that does all three (plus continuity and diode test). A digital one is a **DMM** (digital multimeter).
- **Galvanometer** = a very sensitive current meter. It moves a needle for a tiny current (μA to mA). It is the heart of an **analog** (needle) meter.
  - Ampère suggested the name "galvanometer" (after Galvani). Ohm used a **torsion galvanometer** in his 1820s experiments. See CL 1 - History [T].
- **Wattmeter** = measures power. It is not on a basic multimeter. On a test, find power with **P = V × I** from two meters.
- **Ideal meters:**
  - Ideal ammeter: **0 Ω**. It acts like a plain wire. It drops no voltage.
  - Ideal voltmeter: **infinite (∞) Ω**. It acts like a break (an open). It takes no current.
  - Why: a meter must not change the circuit it measures.
- **Real meters:**
  - A DMM voltmeter has about **10 MΩ** input resistance on DC volts (typical; check the manual).
  - An analog voltmeter has a **sensitivity** in **Ω/V** (ohms per volt). A 20 kΩ/V meter on its 10 V range = 20 000 × 10 = **200 kΩ**.
  - A DMM ammeter has a small **shunt** resistor inside (often a fraction of an ohm up to a few ohms, more on the μA/mA ranges). The voltage it drops is called **burden voltage**.
- **Loading effect** = a real meter changes the circuit.
  - A voltmeter with too low a resistance, across a big resistor, reads **too low**.
  - An ammeter with too high a resistance reads **too low** (it adds series resistance).
- **Misplaced ideal meters:**
  - Voltmeter in **series**: no current flows. The voltmeter reads (almost) the **full source voltage**. Everything else reads 0.
  - Ammeter in **parallel** with a part: it **shorts** that part. Across a source, it is a short circuit. A big current flows and the meter's **fuse blows** (or the meter is damaged).
- **In analysis, replace an ideal ammeter with a wire. Replace an ideal voltmeter with a gap.**
- **Polarity:** current enters the ammeter at **+** (red). A voltmeter reads **V(red) – V(black)**. Swap the leads and a DMM shows a **minus sign**. An analog needle tries to go below zero (it can be damaged).
- **"OL"** on a DMM = over-limit. On ohms it means an **open** (or more resistance than the range). On volts it means the voltage is above the range.
- **Lead resistance:** the probe wires have a little resistance (often 0.1 to 0.5 Ω). Touch the tips together, read it, and subtract it from small readings.
- **Range and resolution:**

  - A "3½-digit" meter counts to 1999 (a "2000-count" meter). On the 20 V range it shows 0.01 V steps.
  - Pick the **lowest range that is still above** the value. Start high if you do not know.
  - **Accuracy** is written as ±(% of reading + counts). Example: ±(0.5% + 2).
- **Indirect measurements:**
  - **Current from a voltage:** I = V_R / R across a known resistor.
  - **Resistance from two meters (voltmeter-ammeter method):** R = V / I.
  - **Mystery resistor with only a voltmeter:** put it in series with a known R. R_x = R_known × V_x / V_known. See Task - Mystery Resistor A9.
  - **Power supplied:** P = V_source × I_total.
- **Galvanometer conversions (meter design):**
  - **Voltmeter** = galvanometer + a big **series** resistor (a **multiplier**).
  - **Ammeter** = galvanometer + a tiny **parallel** resistor (a **shunt**).
  - **Full-scale current** I_g = the current that moves the needle all the way.
- **Analog ohmmeter** = a battery + a galvanometer + an adjust resistor. The "zero" is on the **right** (full current). The scale is **not even** (it is crowded at the left, high-R end). Half scale = the meter's own internal resistance.
- **Wheatstone bridge** = four resistors in a diamond with a meter across the middle. It measures an unknown R very accurately by a **null** (zero reading). Invented by **Samuel Hunter Christie (1833)**; Charles Wheatstone made it popular (1843).
- **Kelvin (double) bridge** and **four-wire (Kelvin) measurement** = ways to measure **very low** resistances. They keep lead and contact resistance out of the result.

Formulas

- Ohm's law for meters: **V = I R**, **R = V / I**, **I = V / R**.
- Voltmeter loading (meter R_m across R_2 in a divider with R_1): **$R_2' = R_2 \parallel R_m = R_2 R_m / (R_2 + R_m)$**, then **$V_{read} = V_s \times R_2' / (R_1 + R_2')$**.
- Ammeter loading: **I_read = V / (R + R_A)**.
- Percent error: **(measured – true) / true × 100%**.
- Analog voltmeter resistance: **R_m = (Ω/V) × range**. Sensitivity: **Ω/V = 1 / I_g**.
- Multiplier (series) for a voltmeter of full scale V_fs: **R_s = V_fs / I_g – R_g**.
- Shunt (parallel) for an ammeter of full scale I_fs: **R_sh = I_g R_g / (I_fs – I_g)**.
- Analog ohmmeter: **I = V_b / (R_int + R_x)**, **R_int = V_b / I_g**. Fraction of full scale f: **R_x = R_int (1/f – 1)**.
- Wheatstone balance: **$R_1 / R_2 = R_3 / R_x$** (ratio of the two arms on each side is the same).
- Unbalanced bridge, ideal voltmeter across the middle: **$V = V_s (R_4/(R_3+R_4) – R_2/(R_1+R_2))$** (two voltage dividers).
- Meter resistance from one loaded reading: **$R_{par} = R_1 V / (V_s – V)$**, then **$R_m = 1 / (1/R_{par} – 1/R_2)$**.
- DMM accuracy: **± (p% × reading + n × resolution)**.

Question types

Easy

1. Name the meter for a quantity (and its unit) ✅ IN

- Seen: Valparaiso University Invitational 2013, Div B, INV, Q1-4 (matching: current → ammeter, potential → voltmeter, resistance → ohmmeter, all → multimeter; key C, D, A, B). SO MYSO practice test 2020, Div B, official practice, Q9-10 (current → ammeter; the distractor "Ampmeter" is fake; potential difference → voltmeter) and Q24 ("A" in a circle = ammeter). JC Booth Invitational 2011, Div B, INV, Q7-9 (R, I, V → ohmmeter, ammeter, voltmeter). GullSO Invitational 2025, Div C, INV, Q8 (voltage across a DC source → voltmeter; distractors oscilloscope, clamp meter, ammeter, ohmmeter). Shock Value student test ~2011, Div B, practice, Section VII (what an ammeter, ohmmeter, voltmeter and galvanometer measure, with units).
- How to solve: Match the quantity to the meter. Current → ammeter (A). Voltage → voltmeter (V). Resistance → ohmmeter (Ω). All three → multimeter. Tiny current → galvanometer.

- Example: Which device measures potential difference: ammeter, ohmmeter, voltmeter, "Ampmeter"? **Answer: voltmeter.**
- Traps: "Ampmeter" is not a word. "Clamp meter" measures current with a magnetic jaw (magnetism is ❌ out). "Oscilloscope" is ❌ out (rule 3.d), so it is never the intended answer.
- Calculator: look up — type: `meter quantity`

2. Where does the meter go? Series or parallel ✅ IN

- Seen: Pembroke Hill Invitational 2026, Div B, INV, Q13 (ideal voltmeter → parallel). F-M Invitational 2018, division not stated, INV, Q8 (picture: which meter is placed correctly → the ammeter in series). UF Northern Regional 2020, Div B, REG, Q9 (identify the ammeter and voltmeter in a schematic; say series or parallel; explain). NY Lake Erie/Niagara Invitational 2018, Div C, INV, Q10 (voltmeter → in parallel). MIT Invitational 2019, Div C, INV, Station 1 Q6 (ammeter: series; low R; in parallel it shorts the circuit and blows the meter fuse).
- How to solve:
  1. Current flows **through** the ammeter. So put it **in the path** (series).
  2. Voltage is **between two points**. So put the voltmeter **across** the part (parallel).
- Example: You want the current through R1 and the voltage across R2. **Answer: break the wire next to R1 and insert the ammeter there (series). Put the voltmeter leads on the two ends of R2 (parallel).**
- Traps: "Measure the current **across** two points" is wrong wording (Pembroke Hill 2026 B Q39 uses it). Current is measured **through** a branch.
- Calculator: look up — type: `meter placement`

3. Ideal meter resistance ✅ IN

- Seen: Pembroke Hill Invitational 2026, Div B, INV, Q12 (ideal ammeter → zero). Virginia Regional 2013, Div C, REG, Q27-28 (ideal voltmeter ∞; ideal ammeter 0). GullSO Invitational 2025, Div C, INV, Q9 (ideal ammeter → 0; distractors ∞, "matched", 50 Ω). ATX (UT Austin) Regional 2026, Div C, REG, MC19-20 (ammeter "none (0)"; voltmeter infinite). Shock Value re-tryout ~2012, Div B, practice, Section II-3 (true/false "ammeter resistance is 0" → false for a real meter; it is small).
- How to solve: Ammeter → 0 Ω. Voltmeter → ∞ Ω. Ohmmeter → it is a source, so "ideal" is not asked.
- Example: What is the internal resistance of an ideal voltmeter? **Answer: infinite.**
- Traps: Do not mix them up. Memory trick: "**A**mmeter = **A**lmost nothing; **V**oltmeter = **V**ery large."
- Calculator: look up — type: `ideal meter`

4. Why must the ideal values be 0 and ∞? (explain) ✅ IN

- Seen: North Pocono Invitational 2020, Div C, INV, Q18 (why ammeter internal R near zero → it adds series R and lowers the current). MIT Invitational 2019, Div C, INV, Station 1 Q6 (why low R). Pembroke Hill Invitational 2026, Div B, INV, Q27 key reasoning (a low-R voltmeter acts as a parallel resistor).
- How to solve: Say what the meter adds to the circuit.
  1. An ammeter is in series. Any R it has adds to the loop and lowers the current it is trying to measure.
  2. A voltmeter is in parallel. Any current it takes lowers the parallel resistance and the voltage it is trying to measure.
- Example: A 1.5 V cell drives 10 Ω. An ammeter with 1.0 Ω is put in series. **Answer: true I = 0.150 A. Read I = 1.5/11 = 0.136 A. The meter's own R made it 9.1% low.**
- Traps: Give the reason, not only the value. Points are for "it changes the circuit".
- Calculator: calculate — type: `V, R, R_A`

5. Draw meters into a schematic ✅ IN

- Seen: cyanophycean314 Circuit Lab 2 (~2013), Div C, practice, Q12 ("illustrate how to measure voltage across R3 and current through R1"; X out wires to break them). UF Northern Regional 2020, Div B, REG, Q9 (identify them in a drawing).
- How to solve:
  1. Voltmeter: draw a circle with V. Connect its two wires to the two ends of the part.
  2. Ammeter: draw an X on one wire next to the part. Put a circle with A in the gap.
  3. Mark + and – on each meter if asked. + goes toward the higher potential (the battery's + side).

- Example: A battery, R1, then R2 || R3. Show the current through R3 only. **Answer: break the wire inside R3's branch (not the main wire) and put the ammeter there.**
- Traps: An ammeter in the main wire reads the **total** current, not one branch.
- Calculator: look up — type: `meter symbols`

6. Read an ideal meter in a simple circuit ✅ IN

- Seen: UF Northern Regional 2020, Div B, REG, Q10-11 (5 V on 2 Ω + 3 Ω: ammeter 1 A; voltmeter across 3 Ω reads 3 V). Rickards Invitational 2026, Div B, INV, Q6 (voltmeter across R4 in a series-parallel circuit). ATX (UT Austin) Regional 2021, Div C, REG, Q51 ("what does the voltmeter read?").
- How to solve:
  1. Replace each ideal ammeter with a wire. Remove each ideal voltmeter.
  2. Solve the circuit (R_eq, then I, then each drop).
  3. Ammeter reading = current in its wire. Voltmeter reading = V between its two points.
- Example: 5 V across 2 Ω and 3 Ω in series. **Answer: ammeter 1.0 A. Voltmeter across the 3 Ω reads 3.0 V.**
- Traps: A voltmeter across a **wire** reads 0 V (no resistance, no drop). A voltmeter across an **open switch** in a series loop reads the full source voltage.
- Calculator: calculate — type: `V, R list`

7. Lead polarity: a negative reading ✅ IN

- Seen: Valparaiso University Invitational 2013, Div B, INV, Q13 (6.0 Ω bulb, 0.57 A; + lead at B, – lead at A, where current goes A → B through the bulb → **–3.42 V**, key A; figure checked). ATX (UT Austin) Regional 2020, Div B, REG, hands-on H6 (meter leads flipped → the graph reflects; the magnet part is ❌ out, the polarity idea is in).
- How to solve:
  1. Find the drop: V = I R.
  2. Current flows from high to low potential through a resistor.
  3. Reading = V(red lead) – V(black lead). If red is on the low side, the sign is minus.
- Example: The 6.0 Ω bulb carries 0.57 A from A to B. Red on B, black on A. **Answer: V = 6.0 × 0.57 = 3.42 V, and B is lower, so the meter reads –3.42 V.**
- Traps: A DMM is not damaged by a minus reading. An analog needle can be.
- Calculator: calculate — type: `I, R, sign`

8. OL and lead resistance (zero the leads) ✅ IN

- Seen: national supervisor practice deck 2 (METER-OL-AND-LEAD-ZERO: leads read 0.493 Ω, resistor reads 61.77 Ω). Not seen in a test. Linked from Resistor Color Codes A5 type 9 and Multimeter Use A9 type 8.
- How to solve:
  1. Touch the probe tips. Read R_leads.
  2. Measure the part. Subtract: R_true = R_read – R_leads.
  3. "OL" on ohms = open (or above range). "OL" on volts = above range; pick a higher range.
- Example: leads 0.493 Ω, resistor reads 61.77 Ω. **Answer: R = 61.77 – 0.493 = 61.28 Ω (61.277, rounded to the meter's 0.01 Ω).**
- Traps: Lead resistance matters only for small R. For 10 kΩ it is far below the tolerance.
- Calculator: calculate — type: `R_read, R_leads`

Medium

9. What you can and cannot measure in a live circuit ✅ IN

- Seen: "Circuit Lab Test (Key)" ~2013, Div C, practice, Q21-23 (on a powered breadboard: voltage across line 18 → yes, 20 V DC range; resistance → cannot, with power on it is wrong and can damage the meter; current through a branch → cannot without a break, because the ammeter must be in series). Valparaiso 2013 B Q10 (resistance "as seen by an ohmmeter" between two terminals = R_eq).
- How to solve:
  1. Voltage: yes, any time. Leads across the two points. Power on.
  2. Resistance: only with power **off**. Isolate the part, or you read the parallel paths too.

3. Current: only if you can **break** the path and insert the meter. Otherwise measure V across a known R and use I = V/R.

- Example: A powered breadboard. You may not unplug anything. Can you find the current in the left branch? **Answer: not with the ammeter. But if the branch has a known 1.00 kΩ resistor reading 2.35 V, then I = 2.35/1000 = 2.35 mA.**
- Traps: An ohmmeter on a live circuit gives nonsense, because the circuit's own voltage adds to the meter's test voltage.
- Calculator: calculate — type: `V_R, R`

10. What a misplaced meter reads (and what it does) ✅ IN

- Seen: ATX (UT Austin) Regional 2020, Div B, REG, Q60 (12 V; voltmeter in series with R1; ammeter in the return; key: voltmeter 12.00 V, ammeter 0.00 A, V_R1 = 0.00 V). MIT Invitational 2019, Div C, INV, Station 1 Q6 (ammeter in parallel → shorts, blows the fuse). See also Multimeter Use [A9] type 10.
- How to solve:
  1. Voltmeter in series = a near-open. Current ≈ 0. All of the source voltage sits across the voltmeter.
  2. Ammeter in parallel with a part = a wire across it. The part is shorted. The ammeter reads the current that the rest of the circuit lets through.
  3. Ammeter straight across a source = a short. Huge current. Fuse blows.
- Example: 9 V battery, 1.00 kΩ resistor, and a 10 MΩ DMM voltmeter in series. **Answer: I = 9/(10 000 000 + 1000) = 0.90 µA. The voltmeter reads 8.999 V (about 9.00 V). The resistor drops only 0.9 mV.**
- Example 2: 9 V on R1 = R2 = 1.00 kΩ in series. An ideal ammeter is put across R2. **Answer: R2 is shorted. The ammeter reads 9/1000 = 9.0 mA. R1 now takes 81 mW (it was 20 mW).**
- Traps: A misplaced voltmeter is safe but useless. A misplaced ammeter can be dangerous.
- Calculator: calculate — type: `V, R list, R_meter`

11. Ideal meters as wires and gaps in a circuit problem ✅ IN (✅ States-Nationals when it needs KVL)

- Seen: UMSO (University of Michigan) Invitational 2019, division not stated (probably C), INV, Q26 (200 V and 160 V sources, ideal ammeters A1-A5; A1 = 16 A → R = 12.5 Ω; 4 A and 8 A in the 40 Ω and 20 Ω; ammeter A4 shorts R's node to the common rail, so the loops split). SO MYSO practice test 2020, Div B, official practice, Q23-24 (a bulb circuit with an ammeter; count the cells; name "A").
- How to solve:
  1. Redraw. Every ideal ammeter becomes a wire. Every ideal voltmeter is removed.
  2. Look for nodes that the ammeter wires merge. This often splits the circuit into simple loops.
  3. Solve. Then read each ammeter as the current in its wire.
- Example: 12 V, R1 = 4 Ω in series with R2 = 6 Ω. An ideal ammeter is wired across R2. **Answer: R2 is shorted. I = 12/4 = 3.0 A. The ammeter reads 3.0 A. R2 carries 0 A.**
- Traps: An ammeter drawn "in a branch" may in fact join two nodes and short something. Trace nodes.
- Calculator: calculate — type: `V, R list` (after shorting the ammeter wires)

12. R from a voltmeter and an ammeter (and I = V_R/R) ✅ IN

- Seen: NC Shock Value practice test 2010, Div B, state practice, Station 4 (battery, resistor and current meter in series, voltmeter across the resistor; R = V/I) and Station 1 (6 V on 1 kΩ + hidden 2 kΩ; voltmeter only; 2 V and 4 V → 2 mA → 2 kΩ). UT Invitational Fall 2025, Div B/C, INV, Section D (I = V_R/R from a measured voltage). Pembroke Hill Invitational 2026, Div B, INV, hands-on Q38-39 (measure V and I with correct units).
- How to solve:
  1. Measure V across the part and I through it.
  2. R = V/I. Keep units (V/mA = kΩ).
- Example: The voltmeter reads 4.50 V. The ammeter reads 2.25 mA. **Answer: R = 4.50 / 0.00225 = 2000 Ω = 2.00 kΩ.**
- Traps: V in volts, I in amperes. 4.50 V / 2.25 mA = 2.00 **k**Ω, not 2.00 Ω.
- Calculator: calculate — type: `V, I`

13. Voltmeter loading ✅ IN

- Seen: Pembroke Hill Invitational 2026, Div B, INV, Q27 (low-R voltmeter on a high-R resistor → reads lower). North Pocono Invitational 2020, Div C, INV, Q25 (5 V on 50 kΩ + 25 kΩ: ideal meter on 25 kΩ → 1.67 V; 2.5 MΩ meter → 1.66 V). Also Multimeter Use [A9] type 13 and Task - Mystery Resistor [A9] type 10.
- How to solve:
  1. Put R_m in parallel with the part: R' = R || R_m.
  2. Redo the divider with R'.
- Example: 10 V across two 1.00 MΩ resistors in series. A 10 MΩ DMM reads the bottom one. **Answer: R' = 1 ‖ 10 = 0.909 MΩ. V = 10 × 0.909/1.909 = 4.76 V. True value is 5.00 V, so it reads 4.8% low.**
- Example 2 (analog meter): 12 V on 100 kΩ + 100 kΩ. A 20 kΩ/V meter on its 10 V range (200 kΩ). **Answer: R' = 66.7 kΩ. V = 12 × 66.7/166.7 = 4.80 V instead of 6.00 V.**
- Traps: Loading matters only when the meter R is not much bigger than the circuit R. A rule of thumb: meter R ≥ 100 × circuit R → under 1% error.
- Calculator: calculate — type: `Vs, R1, R2, Rm`

14. Ammeter loading (burden) ✅ IN

- Seen: not seen in a test with numbers (North Pocono 2020 C Q18 asks the concept). I_bank C31. Kuphaldt, *Lessons in Electric Circuits* Vol. I ch. 8 "Ammeter impact on measured circuit".
- How to solve: Add R_A in series. I_read = V / (R + R_A).
- Example: 1.5 V cell, 10 Ω load, ammeter 1.0 Ω. **Answer: I_read = 1.5/11 = 0.136 A. True 0.150 A. Error –9.1%.**
- Traps: The error is large only when the circuit R is small (a few ohms). On a 10 kΩ circuit, a 1 Ω ammeter is invisible.
- Calculator: calculate — type: `V, R, R_A`

15. Which hookup: voltmeter across the part only, or across part + ammeter? ✅ IN

- Seen: not seen in a test. Kuphaldt, *Lessons in Electric Circuits* Vol. I ch. 8; OpenStax *College Physics 2e* §21.4 (voltmeter and ammeter effects).
- How to solve:
  1. Hookup A: voltmeter right across R (ammeter outside). The ammeter also counts the voltmeter's current. R_calc is a little **low** (you get R || R_V).
  2. Hookup B: voltmeter across R and the ammeter. The voltmeter also counts the ammeter's drop. R_calc is a little **high** (you get R + R_A).
  3. Small R → use A. Large R → use B.
- Example: 10 V ideal source, R = 100 Ω, ammeter 1.0 Ω, voltmeter 10 kΩ. **Answer: Hookup A: I = 99.99 mA, V = 9.90 V, R_calc = 99.0 Ω. Hookup B: I = 99.0 mA, V = 10.0 V, R_calc = 101 Ω. Each is 1% off here; for R = 100 Ω the two errors are about equal.**
- Traps: Neither hookup is perfect. Say which way the error goes.
- Calculator: calculate — type: `V, R, R_A, R_V`

16. Range, resolution, accuracy and significant figures ✅ IN

- Seen: rule 3.b ("answers must be in metric units with appropriate significant figures"). Pembroke Hill Invitational 2026, Div B, INV, Q38-39 ("record it with correct units"). UT Invitational 2019, Div B, INV, Q48 (measure V_i and V_o to 3 significant figures). GullSO Invitational 2025, Div C, INV, hands-on (f), (h). Not seen as a written range question.
- How to solve:
  1. Choose the lowest range above the value (9 V battery → 20 V range, not 2 V).
  2. Resolution = the smallest step the display shows (20 V range on a 2000-count meter → 0.01 V).
  3. Accuracy ±(p% + n counts): error = p% × reading + n × resolution.
  4. Write the reading with its unit and the right number of digits.
- Example: A meter rated ±(0.5% + 2 counts) on its 20 V range shows 9.37 V. **Answer: error = 0.005 × 9.37 + 2 × 0.01 = 0.067 V. True value is 9.30 V to 9.44 V.**
- Traps: "mV" vs "V" on the display. 450 mV = 0.450 V. An autoranging meter moves the decimal point for you; read the unit symbol.
- Calculator: calculate — type: `reading, p%, counts, resolution`

17. Power supplied from meter readings ✅ IN

- Seen: rule Part II c.ii (hands-on task "calculate the power supplied to a circuit"). UT Invitational Fall 2025, Div B/C, INV, Section D lab (measure V_R, V_SW, V_LED; I = V_R/R). Full treatment in Internal Resistance and Power Transfer [A5] type 16.
- How to solve:
  1. Measure the source voltage **under load** (the circuit on).
  2. Measure the total current (ammeter in the main wire), or V across a known series R.
  3. P = V × I.
- Example: The battery reads 9.0 V under load. The main current is 12.5 mA. **Answer: P = 9.0 × 0.0125 = 0.1125 W ≈ 0.11 W (113 mW).**
- Traps: Use the voltage under load, not the label voltage. Do not use a branch current.
- Calculator: calculate — type: `V, I`

Hard

18. Galvanometer → voltmeter (series multiplier) ⚠️ Borderline (the galvanometer works by magnetism, which is ❌ out; the math is pure DC Ohm's law, rule 3.c.iv-v)

- Seen: PA Central Regional ~2013, Div C, REG, Q21 part 1 (1 mA full scale, 250 Ω → 15 V voltmeter; total R = 15 kΩ). I_bank C33. OpenStax *College Physics 2e* §21.4. See also Multimeter Use [A9] type 20.
- How to solve:
  1. Full-scale: the whole meter carries I_g when V_fs is across it.
  2. Total R = V_fs / I_g.
  3. Series resistor R_s = V_fs / I_g − R_g.
- Example: Galvanometer 1.0 mA full scale, R_g = 250 Ω. Make a 15 V meter. **Answer: total R = 15/0.001 = 15 000 Ω = 15 kΩ. R_s = 15 000 − 250 = 14 750 Ω in series. Sensitivity = 1/I_g = 1000 Ω/V.**
- Traps: The question may ask the **total** resistance (15 kΩ) or the **added** resistor (14.75 kΩ). Read which.
- Calculator: calculate — type: `Ig, Rg, Vfs`

19. Galvanometer → ammeter (parallel shunt) ⚠️ Borderline (same reason as type 18)

- Seen: PA Central Regional ~2013, Div C, REG, Q21 part 2 (same galvanometer → 5 A ammeter; total resistance asked). I_bank C32.
- How to solve:
  1. At full scale, I_g goes through the galvanometer. The rest, I_fs − I_g, goes through the shunt.
  2. Both share the same voltage: I_g R_g = (I_fs − I_g) R_sh.
  3. R_sh = I_g R_g / (I_fs − I_g). Total R = R_g || R_sh = I_g R_g / I_fs.
- Example: Same 1.0 mA, 250 Ω galvanometer → 5.0 A ammeter. **Answer: R_sh = 0.001 × 250 / 4.999 = 0.05001 Ω in parallel. Total meter R = 0.25 V / 5 A = 0.0500 Ω.**
- Example 2: 50 Ω, 1.0 mA → 1.0 A meter. **Answer: R_sh = 0.05/0.999 = 0.0501 Ω.**
- Traps: The shunt goes in **parallel**, the multiplier in **series**. Swapping them is the classic mistake.
- Calculator: calculate — type: `Ig, Rg, Ifs`

20. Multi-range meters and ohms-per-volt ⚠️ Borderline (same reason; not seen in a test)

- Seen: not seen in a test. Kuphaldt, *Lessons in Electric Circuits* Vol. I ch. 8 "Voltmeter design" and "Ammeter design" (multi-range switch, Ayrton shunt).
- How to solve:
  1. Each range needs its own multiplier: R_s = V_range / I_g − R_g.
  2. Sensitivity Ω/V = 1/I_g. Meter R on a range = (Ω/V) × range.
- Example: 100 µA, 1.0 kΩ movement. Ranges 1 V, 10 V, 100 V. **Answer: R_s = 9.0 kΩ, 99 kΩ, 999 kΩ. Sensitivity = 1/0.0001 = 10 000 Ω/V. On the 10 V range the meter is 100 kΩ.**
- Traps: A higher range has a **higher** meter resistance, so it loads the circuit **less** (but reads with fewer digits).
- Calculator: calculate — type: `Ig, Rg, ranges`

21. Analog ohmmeter: the backwards, uneven scale ⚠️ Borderline (analog ohmmeters are not named; the math is Ohm's law; not seen in a test)

- Seen: not seen in a test. Kuphaldt, *Lessons in Electric Circuits* Vol. I ch. 8 "Ohmmeter design".
- How to solve:
  1. Short the leads. Adjust to full scale. Then R_int = V_b / I_g.
  2. With R_x: I = V_b / (R_int + R_x).
  3. At fraction f of full scale: R_x = R_int (1/f – 1). Half scale → R_x = R_int.
- Example: 1.5 V cell, 1.0 mA movement. **Answer: R_int = 1500 Ω. A 1500 Ω resistor reads half scale (0.50 mA). A 4500 Ω resistor reads quarter scale (0.25 mA). At 3/4 scale, R_x = 500 Ω.**
- Traps: 0 Ω is at the **right** end. ∞ is at the left. The marks crowd together at high R.
- Calculator: calculate — type: `Vb, Ig, f`

22. Wheatstone bridge as an instrument (balance, both answers, unbalanced reading) ✅ IN (a DC resistor circuit, rule 3.c.v; the unbalanced case uses two dividers, not KVL)

- Seen: UT Invitational 2020, Div B, INV, Q16 (bridge measures an unknown resistance). Rickards Invitational 2020, Div B, INV, Q32 (100 and 200 on the left, 500 on the right → R_x = 250 or 1000 Ω, depending on which arm). Princeton Invitational 2026, Div C, INV, Comp 1a-d (function; meaning of balance; R_x = 34.3 Ω; unbalanced with 40 Ω → voltmeter 0.4 V). GullSO Invitational 2025, Div C, INV, Q43 (which bridge is balanced → 0 V across the middle). Northview Sylvania Invitational 2025, Div C, INV, FR6 (unbalanced → 1.1 V). ATX (UT Austin) Regional 2020, Div B, REG, Q56 (balanced bridge on 120 V → the middle carries no current → 48.0 W). National supervisor practice deck 10.
- How to solve:
  1. Balanced: the meter reads 0. Then $R_1/R_2 = R_3/R_x$ (top/bottom on the left = top/bottom on the right).
  2. Balanced: remove the middle part (it carries no current). The rest is series-parallel.
  3. Unbalanced, ideal voltmeter: find each middle-node voltage with a divider. Subtract.
- Example: Left arm: R_1 = 100 Ω on top, R_2 = 200 Ω below. Right arm: R_3 = 150 Ω on top, R_x below. Balanced. **Answer: $R_x = R_3 R_2 / R_1 = 150 \times 200/100 = 300$ Ω.**
- Example 2 (unbalanced): 10 V. Left: 100 Ω over 100 Ω. Right: 100 Ω over 150 Ω. **Answer: left middle = 5.0 V; right middle = 10 × 150/250 = 6.0 V. The voltmeter reads 1.0 V.**
- Traps: "Both possible values": if you do not know which side R_x is on, there are two answers (Rickards 2020: 250 Ω or 1000 Ω). With a **real** galvanometer in the middle, an unbalanced bridge needs KVL (type 24).
- Calculator: calculate — type: `R1, R2, R3` (balance) or `Vs, R1, R2, R3, R4` (unbalanced)

23. Very low resistance: Kelvin bridge and four-wire measurement ⚠️ Borderline (instrument trivia; the idea is lead resistance, which is in)

- Seen: USC Invitational 2026, Div B, INV, Q23 (advantage of a Kelvin bridge over a Wheatstone bridge → it measures very low resistances; key C). Not seen with numbers.
- How to solve:
  1. Two-wire: the meter adds the leads and contacts to the reading.
  2. Four-wire: one pair of wires carries a known current. A second pair measures only the voltage across the part. R = V / I.
- Example: A 0.050 Ω shunt. Each lead plus contact is 0.20 Ω. **Answer: two-wire reads 0.050 + 0.40 = 0.45 Ω (9× too high). Four-wire with 1.000 A reads 50.0 mV across the part → 0.0500 Ω.**
- Traps: The voltage wires carry (almost) no current, so their resistance does not matter.
- Calculator: calculate — type: `V, I`

Nationals

24. Bridge with a real galvanometer (two node equations) ✅ States-Nationals (KVL/KCL, rule 3.c.ix; exactly two unknown node voltages, so it fits rule 3.e.i)

- Seen: not seen in a test with a non-ideal meter. Princeton Invitational 2026, Div C, INV, Comp 1d and Northview Sylvania 2025 C FR6 use an ideal voltmeter (type 22). Kuphaldt, *Lessons in Electric Circuits* Vol. I ch. 10 "Network Analysis". See Kirchhoff Analysis (States-Nationals) A5.

- How to solve:
  1. Ground the bottom. The top node is V_s. Call the two middle nodes a and b.
  2. KCL at a: $(V_s - V_a)/R_1 = V_a/R_2 + (V_a - V_b)/R_g$.
  3. KCL at b: $(V_s - V_b)/R_3 = V_b/R_4 + (V_b - V_a)/R_g$.
  4. Solve the two equations. Meter current = $(V_b - V_a)/R_g$.
- Example: 10 V. R_1 = 100 Ω (top left), R_2 = 100 Ω (bottom left), R_3 = 100 Ω (top right), R_4 = 150 Ω (bottom right), galvanometer R_g = 50 Ω. **Answer: V_a = 5.31 V, V_b = 5.63 V. The meter drops 0.3125 V and carries 6.25 mA from b to a. (An ideal voltmeter would read 1.0 V.)**
- Traps: A real meter **pulls the two nodes together**, so its reading is less than the ideal 1.0 V.
- Calculator: calculate — type: `Vs, R1, R2, R3, R4, Rg`

25. Find a meter's own resistance from a loaded reading ✅ IN (one equation; ✅ States-Nationals style when combined with KVL)

- Seen: not seen in a test. Kuphaldt, *Lessons in Electric Circuits* Vol. I ch. 8. See also Multimeter Use A9 type 22.
- How to solve:
  1. Use the reading to find the loaded bottom resistance: $R_{par} = R_1 V / (V_s - V)$.
  2. $R_{par} = R_2 \parallel R_m$. So $R_m = 1 / (1/R_{par} - 1/R_2)$.
- Example: 10.0 V on two 100 kΩ resistors in series. A meter on the bottom one reads 4.00 V (not 5.00 V). **Answer: R_par = 100 k × 4/6 = 66.7 kΩ. R_m = 1/(1/66.7 k − 1/100 k) = 200 kΩ.**
- Traps: Use the source voltage, not the "expected" 5 V, in step 1.
- Calculator: calculate — type: `Vs, R1, R2, Vread`

Out of scope (skip)

- ❌ **Oscilloscopes** (rule 3.d names them). GullSO 2025 Q8 uses one only as a distractor.
- ❌ **AC meters, true-RMS, frequency, capacitance and inductance modes** on a DMM. Rule 3.d: AC circuit theory, frequency analysis, capacitance, inductance.
- ❌ **How a galvanometer or clamp meter works inside** (coil, magnet, torque, Hall sensor). Rule 3.c says "areas only"; magnetism is not one of them.
- ❌ **Transistor (hFE) test socket.** Rule 3.d: semiconductors beyond those listed.
- ❌ **Meter math with log or calculus** (for example USC 2026 C Q39c, a voltmeter across a log potential). Rule 3.e.v.
- ❌ **Norton or Thevenin model of the meter's circuit** (Valparaiso 2013 B Q23, Q28; Princeton 2026 Comp 2a). Rule 3.d.

Mistakes in tests and keys

- **Pembroke Hill Invitational 2026, Div B, Q39:** "Measure the current **across** two points." Current is measured **through** a branch, with the ammeter in series. Write that in your answer.
- **ATX (UT Austin) Regional 2026 B and C hands-on:** the key uses the multimeter on DC volts as a **power source**. A meter on DC volts does not supply power. The key wiring also has no current-limiting resistors (from `research/tests/ut-a.md`).
- **ATX (UT Austin) Regional 2026, Div C, MC7:** says Ohm used Cu and Bi as a substitute "ammeter". The Bi-Cu thermocouple was his **voltage source**. His current meter was a **torsion galvanometer**.
- **USC Invitational 2026 hands-on:** it lets you test switches with an LED but **never uses a multimeter**, which the 2027 rules require teams to know (from `research/tests/nat26.md`).
- **I_bank C32 (prior research):** gives R_sh = 0.050 Ω for 50 Ω, 1 mA → 1 A. The exact value is 0.0501 Ω. The rounding is fine.
- **"Ampmeter"** in MYSO Q9 is a deliberate fake distractor, not a typo.

Not confirmed

- DMM input resistance "about 10 MΩ" and typical ammeter shunt values are general knowledge, not from a manual. Check the meter you use.

- PA Central ~2013 Q21 has no key in the downloads. My 15 kΩ and 0.0500 Ω answers follow from Ohm's law.
- Princeton 2026 Comp 1c-d (34.3 Ω, 0.4 V) and Northview FR6 (1.1 V) are from the earlier extraction (`prior/G_tests.md`). I did not re-render their figures.
- Valparaiso 2013 Q13: the figure is low resolution. I read the current arrow as A → B through the bulb. The key (A, –3.42 V) agrees.
- Christie (1833) as the bridge's first inventor is from memory (it is common knowledge in textbooks). I did not open a primary source.
- I found **no 2026 Nationals test**. The file group called "nat26" is the USC Invitational 2026 (a top invitational held where Nationals was later held). No Nationals-level meter question was seen.

Sources

- 2027 Div B rules, Circuit Lab, 3.b, 3.c.iv-v, 3.c.ix, 3.d, 3.e, Part II b-c (local PDF, manual pages B12-B13).
- Raw tests and keys read this run: `downloads/dlB/tests/phi26_test.txt` + key (Pembroke Hill Q12, Q13, Q24, Q27, Q38-39); `downloads/dlG/circuit_lab_valparaiso_university_2013_invitational.pdf` (page 5 rendered: Q10-13) + `circuit_lab_answer_key_vu_2013–1.txt`; `gull26_test.txt` (Q8-9); `rsoi26B_test.txt` (Q6-7); `puso26_test.txt` (Comp 1-2); `wiki/central.txt` (Q21); `wiki/Circuit_Lab_test_key_fix.txt` (Q21-23); `wiki/Shock_Value_practice_test_key_NC_2010.txt` (Stations 1, 4); `wiki/Shock_value_test_.txt` (Section VII); `wiki/Circuit_Lab_2_cyanophycean314.txt` (Q12); `wiki/CircuitLabTest2019I.txt` (Q10); `wiki/CircuitLabRegTest2013.txt` (Q27-28); `atx20reg_Btest.txt` (Q60-61); `downloads/dlFind/nfl20reg_Btest.txt` (Q9-11); `np20inv_test.txt` (Q18); `fm18inv_test.txt`.
- Test extractions: `research/tests/atx–reg.md`, `misc–a.md`, `misc–b.md`, `nat26.md`, `state.md`, `ut–a.md`, `ut–b.md`, `ut–c.md`; `prior/G_tests.md`.
- `prior/B_topics.md` §9a, §9c; `prior/I_bank.md` C26-C34, D22; `prior/F_slides.md` §7 (METER-HOW-TO-MEASURE, METER-OL-AND-LEAD-ZERO, METER-I-FROM-V-AND-R); `prior/A_rules.md` (2.e, Part II b).
- Textbooks: OpenStax *College Physics 2e* §21.4 "DC Voltmeters and Ammeters" and §21.5 "Null Measurements" (Wheatstone bridge); Kuphaldt, *Lessons in Electric Circuits* Vol. I ch. 8 "DC Metering Circuits" (meter design, meter impact, ohmmeter design, Kelvin four-wire, bridge circuits); All About Circuits, DC textbook ch. 8 (same text online).
- All numbers in this note were checked in Python on 2026-10-16.

Internal Resistance, Terminal Voltage and Power Transfer

Note: Internal Resistance and Power Transfer

A real battery is not perfect. Inside it there is a small resistance. This note covers what that resistance does to the voltage, the current and the power.

Most of this note is ✅ **IN at every level**. It uses only Ohm's law and series rules (rules 3.c.iii, 3.c.iv, 3.c.v). Two parts are different:

- Networks with two real batteries need KVL/KCL, so they are ✅ **States/Nationals only** (rule 3.c.ix).
- **Maximum power transfer** is ⚠️ **borderline**. Its usual proof uses calculus or Thevenin equivalents. Both are excluded (rules 3.e.v and 3.d). The plain fact "the load gets the most power when R_load = r" needs only algebra.

Key facts

- **EMF (ε, "electromotive force")** is the voltage a battery makes inside itself. It is a voltage, measured in volts, not a force.
- **Internal resistance (r)** is the resistance inside the battery: its chemicals, plates and connections.
- A real battery acts like an **ideal battery ε in series with a resistor r**.
- **Terminal voltage (V_t)** is the voltage you measure across the battery's two terminals.
- When no current flows (open circuit), V_t = ε. An ideal voltmeter on a battery alone reads the EMF.
- When current flows out of the battery, part of ε is lost inside: V_t = ε – I·r. The more current, the lower V_t.
- When a battery is **being charged**, current flows into its + terminal. Then V_t = ε + I·r. The charger must push harder than the EMF.
- **Short-circuit current (I_sc)** is the current when a wire joins the terminals: I_sc = ε/r. It is the most current the battery can give.
- A short circuit puts all the power into r. The battery gets hot. This is why shorted batteries can burn, leak, swell or explode.
- Internal resistance rises as a battery gets old, cold or flat. So a weak battery can still read nearly full EMF with no load, then sag under load.
- The load gets the **most power** when R_load = r. At that point the efficiency is only 50%. Power companies and batteries are run far from this point. Audio amplifiers and antennas are sometimes matched to it.
- **Efficiency** = (power to the load) ÷ (total power made) = R/(R + r) = V_t/ε.

Formulas

What	Formula	Notes
Current in a simple circuit	$I = \varepsilon/(R + r)$	R = load
Terminal voltage (discharging)	$V_t = \varepsilon - I \cdot r = \varepsilon \cdot R/(R + r)$	Divider between R and r
Terminal voltage (charging)	$V_t = \varepsilon + I \cdot r$	Current into +
Internal resistance	$r = (\varepsilon - V_t)/I = (\varepsilon - V_t) \cdot R/V_t$	From one load
Two loads	$r = (V_2 - V_1)/(I_1 - I_2)$, then $\varepsilon = V_1 + I_1 \cdot r$	Two equations
Data table	r = –(slope of V_t against I); ε = intercept at I = 0	Straight-line fit
Short-circuit current	$I_{sc} = \varepsilon/r$	Also $r = \varepsilon/I_{sc}$
Power made by the EMF	$P = \varepsilon \cdot I$	Total
Power to the load	$P = I^2R = V_t \cdot I$	Useful
Power lost inside	$P = I^2r$	Heat
Efficiency	$\eta = R/(R + r) = V_t/\varepsilon$	Fraction
Maximum power	$R = r \rightarrow P_{max} = \varepsilon^2/(4r)$, η = 50%	⚠️
Same power at two loads	$R_1 \cdot R_2 = r^2$	⚠️
n identical cells in series	$\varepsilon_{total} = n \cdot \varepsilon$, $r_{total} = n \cdot r$	
n identical cells in parallel	$\varepsilon_{total} = \varepsilon$, $r_{total} = r/n$	

Question types

Easy

1. Terminal voltage from ε, r and the load

- Scope: ✅ IN.
- Where seen: not seen as a stand-alone test question. It is step 1 of every question below. Prior topics note B_topics §3 (V_terminal = ε − I·r).
- How to solve:
  1. I = ε/(R + r).
  2. V_t = ε − I·r (or I·R).
- Worked example: ε = 9.0 V, r = 0.50 Ω, R = 4.0 Ω. I = 9/4.5 = **2.0 A**. V_t = 9 − 2 × 0.5 = **8.0 V**.
- Traps: A voltmeter across the battery while it runs reads V_t, not ε.
- Calculator: calculate — type: emf, r, R.

2. Open-circuit voltage: what a voltmeter reads on a battery alone

- Scope: ✅ IN.
- Where seen: not seen in a test. It is a common concept trap behind type 3.
- How to solve:
  1. An ideal voltmeter draws no current.
  2. With I = 0, the drop I·r is 0. So the reading = ε.
- Worked example: a battery with ε = 1.50 V and r = 0.30 Ω. An ideal voltmeter on it alone reads **1.50 V**. After you add a 2.70 Ω load, it reads 1.50 × 2.70/3.00 = **1.35 V**.
- Traps: "The battery reads 1.5 V, so it is good" can be false. Test it under load.
- Calculator: look up — type: open circuit, emf.

3. Internal resistance from the EMF and the terminal voltage under a load

- Scope: ✅ IN.
- Where seen: Pembroke Hill 2025 B (INV) Q4 (ε = 9.0 V, V_t = 8.5 V across 10 Ω → r = 0.59 Ω). UF Northern Regional 2020 B (REG) Q38 (ε = 12 V, 11.9 V across 50 Ω → r = 0.420 Ω).
- How to solve:
  1. Current from the load: I = V_t/R.
  2. Lost voltage = ε − V_t.
  3. r = (ε − V_t)/I.
- Worked example (Pembroke Hill 2025 B Q4): I = 8.5/10 = 0.85 A. Lost = 9.0 − 8.5 = 0.5 V. r = 0.5/0.85 = **0.59 Ω**.
- Worked example (UF Northern Regional 2020 B Q38): I = 11.9/50 = 0.238 A. r = 0.1/0.238 = **0.420 Ω**.
- Traps: Use V_t (not ε) to find the current through the load.
- Calculator: calculate — type: emf, V_t, R.

4. Find the load resistance from ε, r and the terminal voltage

- Scope: ✅ IN.
- Where seen: UT Invitational 2020 B (INV) Q65 (r = 0.24 Ω, ε = 3.00 V, V_t = 2.91 V → R_L = 7.76 Ω).
- How to solve:
  1. I = (ε − V_t)/r.
  2. R_L = V_t/I.
- Worked example (UT 2020 B Q65): I = (3.00 − 2.91)/0.24 = 0.375 A. R_L = 2.91/0.375 = **7.76 Ω**.
- Traps: Small differences (0.09 V) need full precision. Do not round I early.
- Calculator: calculate — type: emf, r, V_t.

5. Short-circuit current and heating (hazard)

- Scope: ✅ IN (DC hazards, rule 3.c.iii; open and short circuits, rule 3.c.v).
- Where seen: Shock Value re-tryout ~2012 B (practice) VI-8 (finger across battery terminals → burning). UT Invitational 2020 B (INV) Q37 (a 6 V source shorted by a wire is dangerous). Prior question bank A16.

- How to solve:
  1. $I_{sc} = \varepsilon/r$.
  2. Heat inside the battery = $I_{sc}^2 \cdot r = \varepsilon^2/r$.
- Worked example (question bank A16): $\varepsilon = 1.5$ V, $r = 0.10\ \Omega$. I_{sc} = **15 A**. Heat = $1.5^2/0.10$ = **22.5 W**. That is why a shorted AA cell gets hot.
- Traps:
  - The heat is **inside** the battery (and in the wire), not "in the short."
  - An ideal battery ($r = 0$) gives "infinite" current. Real answers always use r.
- Calculator: calculate — type: emf, r.

Medium

6. Internal resistance and EMF from two different loads

- Scope: ✅ IN (2 equations, rule 3.e.i).
- Where seen: not seen in a test. Prior question bank C19. OpenStax *College Physics 2e* §21.2 (emf and internal resistance) uses the same model.
- How to solve:
  1. For each load: $I = V_t/R$.
  2. Write $\varepsilon = V_1 + I_1 \cdot r$ and $\varepsilon = V_2 + I_2 \cdot r$.
  3. Subtract: $r = (V_2 - V_1)/(I_1 - I_2)$. Then ε from either equation.
- Worked example (question bank C19): 4 Ω → 1.20 V; 9 Ω → 1.35 V. $I_1 = 0.30$ A, $I_2 = 0.15$ A. $r = 0.15/0.15$ = **1.0 Ω**. $\varepsilon = 1.20 + 0.30 \times 1.0$ = **1.5 V**.
- Traps: The larger load resistance gives the **higher** terminal voltage. If yours is lower, you swapped them.
- Calculator: calculate — type: R1, V1, R2, V2.

7. Internal resistance from a table of (I, V) data

- Scope: ✅ IN.
- Where seen: UT Invitational 2019 B (INV) Q24, tiebreaker 7, and the same question in UT Invitational fall 2019 C (INV) Q24 (four points: 0.1 A 14.1 V; 1.5 A 13.3 V; 3.1 A 12.6 V; 4.6 A 11.8 V → choices 0.38, 0.44, 0.47, 0.51 Ω → **D 0.51 Ω**).
- How to solve:
  1. $V_t = \varepsilon - r \cdot I$ is a straight line. Its slope is $-r$.
  2. Use the two end points: $r = (V_{first} - V_{last})/(I_{last} - I_{first})$.
  3. Or fit a line on the calculator. The intercept at $I = 0$ is ε.
- Worked example (UT 2019 B Q24): end points: $r = (14.1 - 11.8)/(4.6 - 0.1) = 2.3/4.5$ = **0.511 Ω**. A best-fit line gives 0.503 Ω and $\varepsilon = 14.1$ V. The closest choice is **D (0.51 Ω)**.
- Traps: Neighbouring points give 0.57, 0.44 and 0.53 Ω. Do not use just one pair. That is how the wrong choices were made.
- Calculator: calculate — type: list of I, list of V (linear fit).

8. LED on a battery with internal resistance

- Scope: ✅ IN (LEDs, rule 3.c.vii).
- Where seen: MIT Invitational 2019 C (INV) Q42 (3 V supply, $r = 0.3\ \Omega$, red LED $V_f = 1.8$ V, no resistor → 4.0 A) and Q43 (add 220 Ω → 5.45 mA).
- How to solve:
  1. KVL: $\varepsilon = V_f + I \cdot (R + r)$.
  2. $I = (\varepsilon - V_f)/(R + r)$.
- Worked example (MIT 2019 C Q42-43): with no resistor, $I = (3 - 1.8)/0.3$ = **4.0 A** (the LED burns out at once). With 220 Ω, $I = 1.2/220.3$ = **5.45 mA**.
- Traps: The internal r is the only thing limiting the current without a resistor. That is why "LED straight on a battery" is a bad idea.
- Calculator: calculate — type: emf, r, V_f, R.

9. Cells in series and in parallel

- Scope: ✅ IN.
- Where seen: not seen with internal r in a test. Prior question bank §6 (choose series or parallel cells for voltage and run time). Kuphaldt, *Lessons in Electric Circuits* Vol. I, chapter "Batteries and Power Systems" (series and parallel cells).
- How to solve:
  1. In series, EMFs add and internal resistances add.
  2. Identical cells in parallel: same EMF, r divided by n.
  3. Then use I = ε_total/(R + r_total).
- Worked example: four AA cells in series, each 1.5 V and 0.15 Ω, on a 10 Ω load. ε = 6.0 V, r = 0.60 Ω. I = 6/10.6 = **0.566 A**.
- Worked example: three of those cells in parallel on 1.0 Ω. r = 0.05 Ω. I = 1.5/1.05 = **1.43 A**.
- Traps:
  - Never put unequal cells in parallel without resistors. The stronger one charges the weaker one (type 14).
  - Parallel cells give more current and longer run time, not more voltage.
  - The 0.15 Ω for an AA cell is my example value, not a measured fact.
- Calculator: calculate — type: n, emf, r, series or parallel, R.

10. Power to the load, power lost inside, efficiency

- Scope: ✅ IN.
- Where seen: not seen with internal r in a test. The total-power check is in every national supervisor generator level (P_total = V_bat × I_total). MIT 2021 C (INV) Scenario B (which parts absorb and which supply).
- How to solve:
  1. I = ε/(R + r).
  2. P_load = I^2R. P_lost = I^2r. P_total = ε·I.
  3. η = P_load/P_total = R/(R + r).
- Worked example: ε = 12 V, r = 2 Ω, R = 10 Ω. I = 1.0 A. P_load = **10 W**. P_lost = **2 W**. P_total = **12 W**. η = **83%**.
- Traps: "Power supplied by the battery" can mean ε·I (total) or V_t·I (delivered to the outside). Read the question. Say which one you used.
- Calculator: calculate — type: emf, r, R.

11. Voltage sag when a big load switches on (lights dim)

- Scope: ✅ IN.
- Where seen: not seen in a test. Prior question bank C25 (adding a parallel branch lowers the bulb voltage when there is a series or internal r). OpenStax *College Physics 2e* §21.2 (starter motor example).
- How to solve:
  1. Find the new total current with the extra load.
  2. V_t = ε − I_total·r.
  3. Every part across the terminals now gets the lower V_t.
- Worked example: a car battery, ε = 12.6 V, r = 0.020 Ω (an example value). Headlights alone (1.2 Ω): V_t = **12.39 V**. The starter draws 150 A: V_t = 12.6 − 150 × 0.02 = **9.6 V**. The headlights dim.
- Traps: With an ideal battery (r = 0), adding a parallel load changes nothing for the other loads. The dimming only happens because of r.
- Calculator: calculate — type: emf, r, each load current or R.

12. Find r from the open-circuit voltage and the short-circuit current

- Scope: ✅ IN for one battery. ❌ OUT for a whole network (that is a Thevenin equivalent, rule 3.d).
- Where seen: not seen in a test. Solar-cell data sheets give V_oc and I_sc (but a solar cell is not a simple ε + r, so this is only rough).
- How to solve: r = V_oc/I_sc. ε = V_oc.
- Worked example: a battery reads 9.0 V open. A (brief) short gives 18 A. r = 9/18 = **0.50 Ω**.
- Traps: Do not short a real battery to measure this. Use two loads instead (type 6).
- Calculator: calculate — type: V_oc, I_sc.

Hard

13. Charging a battery: charger voltage and the power split

- Scope: ✅ IN (one loop) / ✅ States-Nationals if inside a network.
- Where seen: not seen as a test question. National supervisor generator Level 13 shows a battery absorbing power. OpenStax *College Physics 2e* §21.2 (charging a battery).
- How to solve:
  1. While charging, current flows into +. $V_t = \varepsilon + I \cdot r$.
  2. Power in = $V_t \cdot I$. Power stored as chemical energy = $\varepsilon \cdot I$. Heat = $I^2 r$.
  3. If a charger with its own resistance R_c and voltage V_c is used: $I = (V_c - \varepsilon)/(R_c + r)$.
- Worked example: a 12.0 V battery, r = 0.050 Ω, charged at 10 A. $V_t = 12.0 + 10 \times 0.05$ = **12.5 V**. Power in = **125 W**. Stored = **120 W**. Heat = **5 W**.
- Worked example: a 14.0 V charger with 0.10 Ω leads, same battery. $I = (14 - 12)/(0.10 + 0.05)$ = **13.3 A**. $V_t = 12 + 13.3 \times 0.05$ = **12.67 V**.
- Traps: When charging, the terminal voltage is **above** ε, not below.
- Calculator: calculate — type: emf, r, I (or charger V and R).

14. Two unequal batteries in parallel (one charges the other)

- Scope: ✅ States-Nationals (KCL with one node).
- Where seen: MIT Invitational 2019 C (INV) Q50 (12 V + 330 Ω and E + 100 Ω feed a 220 Ω load; the E branch current is –3.03 mA, so E is being charged). Prior question bank C21 and K3. Not seen as "two real batteries with internal r" in a test.
- How to solve:
  1. Treat each battery as ε in series with its r.
  2. One-node KCL: $V = (\varepsilon_1/r_1 + \varepsilon_2/r_2) \div (1/r_1 + 1/r_2 + 1/R)$.
  3. Branch current = $(\varepsilon - V)/r$. Negative means that battery is being charged.
- Worked example: battery 1: 12.6 V, 0.020 Ω. Battery 2: 12.0 V, 0.050 Ω. Load 1.0 Ω.
  - $V = (630 + 240) \div (50 + 20 + 1)$ = **12.25 V**.
  - $I_1 = (12.6 - 12.25)/0.02$ = **17.3 A** out. $I_2 = (12.0 - 12.25)/0.05$ = **–5.07 A**, so 5.07 A flows **into** battery 2. It is being charged even while the load runs.
  - With no load: V = 12.43 V and **8.57 A** circulates from battery 1 into battery 2.
- Traps: The circulating current wastes energy and can overheat both batteries. That is why you never mix old and new cells.
- Calculator: calculate — type: emf1, r1, emf2, r2, R_load (one-node solver).

15. Design: choose a series resistor when the source has internal resistance

- Scope: ✅ IN.
- Where seen: not seen with internal r in a test. GullSO 2025 C (INV) Q51 (loaded divider design to a spec) is the same kind of design task.
- How to solve:
  1. Find the current the load needs: $I = V_{load}/R_{load}$.
  2. Total resistance needed = ε/I.
  3. $R_{series} = \varepsilon/I - R_{load} - r$.
- Worked example: ε = 9 V, r = 2 Ω. A 100 Ω load needs 3.0 V, so I = 30 mA. $R_{series} = 9/0.03 - 100 - 2$ = **198 Ω**.
- Traps: Forgetting r gives 200 Ω, which is 1% off. Supervisors may accept both, so show your work.
- Calculator: calculate — type: emf, r, V_load, R_load.

16. Hands-on: power supplied to a circuit from measurements

- Scope: ✅ IN (rule II.c.ii "Calculate the power supplied to a circuit").
- Where seen: listed in the 2027 rules as a hands-on task. National supervisor deck 4 practical (measure V_bat and each drop). Not seen as a scored station in the tests I read.
- How to solve:
  1. Measure the battery's terminal voltage **while the circuit runs** (V_t).

2. Find the total current: measure it with an ammeter in series, or measure the voltage across a known resistor and use I = V/R.
3. P_supplied = V_t × I.
4. Optional: measure V_t with the circuit off to get ε, then r = (ε − V_t)/I.

- Worked example: V_t = 8.70 V under load. A 220 Ω resistor in series with everything shows 4.40 V. I = 4.40/220 = 20.0 mA. P = 8.70 × 0.020 = **0.174 W** (174 mW).
- Traps:
  - Measure V_t under load, not with the circuit switched off.
  - An ammeter must go in series. Across a battery it is a short and can blow the meter's fuse.
- Calculator: calculate — type: V_t, V across known R, known R (or I).

Nationals

17. Maximum power transfer: the best load resistance

- Scope: ⚠ Borderline. The fact "R_load = r gives the most load power" is algebra. Its proof needs calculus, and its general form needs Thevenin, both excluded (rules 3.e.v, 3.d).
- Where seen: national supervisor practice deck 4, quiz 4 (12 V, R_L = 16 Ω → R_1 = 16 Ω, I = 0.375 A, P_load = 2.25 W, P_total = 4.5 W). Deck 4 impedance-matching concept (8 Ω speaker on an 8 Ω source). ✗ OUT versions: ATX Regional 2019 B/C (REG) Q17 (Norton), UT Invitational fall 2019 C (INV) Q36 (Thevenin), ATX Regional 2019 B Q30 = C Q33 (REG) (dependent source + max power), MIT 2020 C (INV) Q41. Prior question bank C20.
- How to solve:
  1. The load gets the most power when R = r.
  2. Then I = ε/(2r) and P_max = $\varepsilon^2/(4r)$.
  3. Efficiency at that point = 50%.
  4. To show it without calculus, make a small table of P for several R.
- Worked example (question bank C20): ε = 12 V, r = 2 Ω. Best R = **2 Ω**. P_max = 144/8 = **18 W**. Efficiency = **50%**.
  - Table: R = 1 Ω → 16 W; 2 Ω → **18 W**; 3 Ω → 17.3 W; 4 Ω → 16 W; 8 Ω → 11.5 W. The peak is at R = r.
- Traps:
  - Maximum **power** is at R = r. Maximum **current** is at R = 0. Maximum **efficiency** and maximum **voltage** are at very large R.
  - Matching does not mean "best." It wastes half the power.
- Calculator: calculate — type: emf, r (and a list of R for the table).

18. Two loads give the same power: find r

- Scope: ⚠ Borderline (algebra only, but it builds on type 17).
- Where seen: not seen in a test. It is a standard textbook result: if two loads get equal power, $R_1 \cdot R_2 = r^2$.
- How to solve:
  1. Set $\varepsilon^2 R_1/(R_1 + r)^2 = \varepsilon^2 R_2/(R_2 + r)^2$.
  2. The algebra gives $r^2 = R_1 R_2$, so $r = \sqrt{(R_1 R_2)}$.
- Worked example: a battery gives the same power to 2 Ω and to 8 Ω. r = √16 = **4 Ω**. Check with ε = 6 V: 2 Ω → 2.0 W; 8 Ω → 2.0 W; and the 4 Ω load gets 2.25 W, the most.
- Traps: Taking the average (5 Ω) is wrong. It is the **geometric** mean.
- Calculator: calculate — type: R1, R2.

19. The "choose the source resistance" trap

- Scope: ⚠ Borderline (same reason as type 17).
- Where seen: national supervisor practice deck 4, quiz 4 (asks for the series "source" resistor R_1 that gives maximum power to a fixed 16 Ω load; the deck answers 16 Ω). UT Invitational fall 2019 C (INV) Q36 part 2 (asks for maximum **current**; the key gives the maximum-power answer).
- How to solve:
  1. If the **load** can change and the source is fixed: best R_load = r.

2. If the **source resistance** can change and the load is fixed: best source resistance = **0**. Less source resistance always gives the load more power.

- Worked example (deck 4 numbers): 12 V, fixed load 16 Ω. With R_1 = 16 Ω, P_load = $0.375^2 \times 16$ = **2.25 W**. With R_1 = 0, P_load = $0.75^2 \times 16$ = **9.0 W**. So R_1 = 0 is better.
- Traps: Ask yourself which part you are allowed to choose. If the test clearly wants "match the resistances," give R_1 = R_L, and add one line noting that R_1 = 0 gives more load power.
- Calculator: calculate — type: emf, fixed R, choice range.

20. Which load takes the most power (compare loads)

- Scope: ✅ IN (the capacitor choice is ❌ OUT as a topic, rule 3.d, but you only need "no DC current").
- Where seen: ATX Regional 2026 B (REG) MC38 (50 Ω, 1.2 nF, two 100 Ω in parallel, a short → the key says the short).
- How to solve:
  1. With an ideal voltage source, P = V^2/R. Lower R takes more power.
  2. A capacitor takes no steady DC current, so no DC power.
  3. With a real source (internal r), the most load power is at R = r. A short then gets **zero** load power, because V across a wire is 0.
- Worked example (ATX 2026 B MC38): 50 Ω and 100 || 100 = 50 Ω tie. The capacitor gets 0 W. With an ideal source, the short draws unlimited current, so the key picks the **short**.
- Traps: The question's wording ("same current and voltage source") contradicts itself. With a real battery, the short's own power is 0 (all the heat is inside the battery). Write your assumption next to your answer.
- Calculator: look up — type: compare loads, ideal or real source.

Out of scope (skip)

- ❌ **Thevenin/Norton equivalents** and maximum power transfer from a whole network (rule 3.d: "Norton/Thevenin equivalents"). Seen: ATX Regional 2019 B/C Q17, UT Invitational fall 2019 C Q34 and Q36, MIT 2020 C Q41, GullSO 2025 C Q42/44, VA Regional 2013 C Station 2.
- ❌ **Proving** R = r with calculus (rule 3.e.v: no calculus).
- ❌ **Capacitor charging** and battery-capacitor RC questions (rule 3.d: capacitance). Seen in many old tests (UF Northern Regional 2019 C Q19, USC 2026 B Q8, Q12).
- ❌ **Impedance matching with AC** or complex impedance (rule 3.d: AC circuit theory; 3.e.v: complex numbers).
- ❌ **Electrochemistry details** past anode/cathode and cell voltages are not in this area. Battery chemistry and hazards belong in CL 3 - DC and AC 🅃.
- ❌ **Galvanometer internal resistance and meter design** are not internal-resistance-of-a-battery questions. They belong in the meters note (⚠️ there).

Mistakes in tests and keys

- **UT Invitational fall 2019 C (INV) Q36 part 2:** asks for the R_1 that gives maximum **current**. The answer is R_1 = 0, giving 5.143/37.14 = 138 mA. The key writes R_1 = R_th = 37.1 Ω (the maximum-power answer) but keeps the 138 mA. The key does not agree with itself. (Thevenin, ❌ OUT anyway.)
- **National supervisor deck 4, quiz 4:** asks for the series source resistor that gives maximum power to a fixed load and answers "R_1 = R_L." For a fixed load the real answer is R_1 = 0 (9.0 W instead of 2.25 W). See type 19.
- **ATX Regional 2026 B (REG) MC38:** the wording "same current and voltage source" is contradictory. The keyed "short circuit" only works with an ideal voltage source.
- **UT Invitational 2019 B (INV) Q24:** not an error, but the best-fit slope (0.503 Ω) is a little nearer 0.51 than 0.47. Choice D is right either way.

Not confirmed

- Typical internal resistances (AA alkaline about 0.15 Ω, car battery about 0.02 Ω) are example values I chose. I did not check a datasheet for this note.

- Pembroke Hill 2025 B Q4 and UF Northern Regional 2020 B Q38 come from the earlier extraction files (B_topics, misc-b). I did not re-read those tests here. The numbers check in Python.
- ATX Regional 2026 B MC38: from the ut-a extraction. I did not render that page.
- No STATE or NATIONAL test with an internal-resistance question was found. The real 2026 Nationals tests are only in the soinc "Test Packet 2027" (not downloaded).
- Whether a 2027 supervisor treats maximum power transfer as IN: the rules do not say. I found no rules clarification.

Sources

- Rules: 2027 Science Olympiad Div B Rules Manual, Circuit Lab, rules 3.c.iii-v, 3.c.ix, 3.d, 3.e (page B12), and II.c.ii (hands-on). See Circuit Lab - Rules 2027 Ⓡ.
- Tests (extraction files in `~/Daniel/circuitlab/research/tests/` and `prior/`): Pembroke Hill 2025 B; UF Northern Regional 2020 B; UT Invitational 2020 B; UT Invitational 2019 B and fall 2019 C; MIT Invitational 2019 C and 2021 C; ATX Regionals 2019 C and 2026 B; GullSO 2025 C; Shock Value re-tryout ~2012 B; MIT 2020 C; VA Regional 2013 C.
- Raw test text checked for this note: UT Invitational 2019 B Q24 data table; UT Invitational 2020 B Q65 wording.
- National supervisor practice decks 4, 10, 11 and the homework generator (`prior/F_slides.md`).
- Textbooks: OpenStax *College Physics 2e*, §21.2 "Electromotive Force: Terminal Voltage" (internal resistance, charging, starter-motor sag); Kuphaldt, *Lessons in Electric Circuits*, Vol. I (DC), "Batteries and Power Systems" (series and parallel cells) and the maximum power transfer section; All About Circuits, DC textbook (same material online).
- All numbers in this note were checked in Python on 2026-10-16.

Hard Resistor Networks

These are the hardest resistor questions on top invitational and State tests. They use rule area **3.c.v** (series and parallel). Some need KCL, so they are **States/Nationals only** (rule 3.c.ix). The math limit is rule **3.e.i**: "functions, algebraic manipulations, and systems of no more than two equations". A quadratic equation is algebra, so the infinite ladder is allowed. Rule **3.e.v** bans trigonometry, matrices and calculus.

The main trick is **symmetry**. Symmetry turns a network that is not series-parallel into one that is. Then no equations are needed.

Key facts

- **Equipotential points** = points at the same voltage. No current flows in a resistor between two equipotential points.
- So you may **join** two equipotential points with a wire, or **cut** a resistor between them. Both moves leave R_eq the same.
- **Symmetry rule:** if you can turn or flip the network so that it looks the same, and A and B stay in place, then points that swap places have the same voltage.
- **Anti-symmetry rule:** if a flip swaps A and B, then mirror points have voltages that add up to V_AB. Points on the mirror line sit at exactly half the voltage.
- **Wheatstone bridge** = two voltage dividers side by side, with a "bridge" resistor (or meter) between their middles.
- A bridge is **balanced** when $R_1/R_2 = R_3/R_4$ (top/bottom on the left = top/bottom on the right). Then the bridge resistor carries **0 A**. You can remove it.
- **Ladder network** = series resistors along the top with "rungs" (shunt resistors) across. Reduce it from the far end.
- **Infinite ladder:** adding one more section does not change R_eq. So R_eq appears on both sides of the equation. This gives a quadratic.
- **Delta (Δ)** = three resistors in a triangle. **Wye (Y)** = three resistors meeting at one center point. A Δ can be replaced by an equal Y (⚠️ borderline for 2027 B).
- A network that is not series-parallel and has no symmetry needs KCL (States/Nationals) or Δ-Y (⚠️).

Formulas

- Balanced bridge: $\mathbf{R_1/R_2 = R_3/R_4}$, so $\mathbf{R_x = R_2R_3/R_1}$ (match the letters to the drawing).
- Unbalanced bridge, ideal voltmeter: $\mathbf{V_meter = V{\cdot}R_2/(R_1 + R_2) - V{\cdot}R_4/(R_3 + R_4)}$.
- Cube of 12 equal R: body diagonal **5R/6**; face diagonal **3R/4**; one edge **7R/12**.
- Tetrahedron (6 equal R), across one edge: **R/2**. Octahedron (12 equal R), opposite corners: **R/2**.
- Complete network of N nodes, every pair joined by R: **2R/N** (❌ general N is OUT; small N by symmetry is ✅).
- Infinite ladder, series R first, then shunt R: $\mathbf{R_eq = R(1 + \sqrt{5})/2 \approx 1.618R}$.
- Infinite ladder, shunt R first: $\mathbf{R_eq = R(\sqrt{5} - 1)/2 \approx 0.618R}$.
- Infinite ladder with series R_1 and shunt R_2: $\mathbf{R_eq = [R_1 + \sqrt{(R_1^2 + 4R_1R_2)}]/2}$.
- Quadratic formula (for ladders): $x^2 + bx + c = 0 \rightarrow \mathbf{x = [-b + \sqrt{(b^2 - 4c)}]/2}$ (keep the positive root).
- Δ → Y: $\mathbf{R_a = R_ab{\cdot}R_ac/(R_ab + R_bc + R_ac)}$ (the two Δ sides that touch corner a, over the sum).
- Y → Δ: $\mathbf{R_ab = (R_aR_b + R_bR_c + R_cR_a)/R_c}$ (sum of pair products, over the opposite Y arm).
- 3-terminal Y box from pair readings: $\mathbf{R_a = (R_AB + R_AC - R_BC)/2}$.

Question types

Easy

1. Wheatstone bridge: purpose, history, variants

- Scope: ⚠️ Borderline. The bridge is a plain DC resistor circuit (✅ 3.c.v). The Kelvin bridge is instrument trivia.
- Where seen: UT Invitational 2020 B (INV, 2021 season) Q16 (a Wheatstone bridge measures → an unknown resistance). USC Invitational 2026 B (INV) Q23 (advantage of a Kelvin bridge → measures very low resistances). National supervisor practice deck 10 (who invented the Wheatstone bridge).

- How to solve: Know three facts.
  1. It measures an **unknown resistance** by balancing it against known ones.
  2. Samuel Hunter **Christie** invented it in **1833**. Charles **Wheatstone** made it popular in **1843**.
  3. A **Kelvin (double) bridge** removes the effect of lead and contact resistance. So it measures very small resistances (below about 1 Ω).
- Worked example: "Why is the meter in a balanced bridge reading 0?" Answer: **both middle points are at the same voltage**, so no current flows through the meter.
- Traps: The "bridge" in a **bridge rectifier** is four diodes, not this circuit. See Diode Applications A7.
- Calculator: look up — type: wheatstone, christie, kelvin bridge.

2. Spot a network that cannot be reduced with series and parallel

- Scope: ✅ IN.
- Where seen: MIT Invitational 2019 C (INV) lab S1-5 (can circuit 3, a bridge, be reduced with series/parallel only? → No).
- How to solve:
  1. Look for any pair in series (one shared node with nothing else on it) or in parallel (same two nodes).
  2. If no such pair exists anywhere, the network is not series-parallel.
  3. Then check for balance (type 3) or symmetry (types 9-10). If neither works, you need KCL (States/Nationals).
- Worked example: five resistors A-C, C-B, A-D, D-B and C-D. No two share both nodes. No node has just two resistors. Answer: **not series-parallel** (it is a bridge).
- Traps: A bridge that is balanced **can** be reduced after you remove the middle resistor. The question "with series/parallel only" usually means "without that trick".
- Calculator: look up — type: series-parallel test, bridge.

Medium

3. Balanced bridge: spot it, remove the middle, find R_eq or power

- Scope: ✅ IN.
- Where seen: ATX Regional 2020 B (REG) Q56 (120 V; four arms 300 Ω, middle 100 Ω → 48 W). GullSO 2025 C (INV) Q43 (which bridge is balanced → 0 V across the middle). Prior research I_bank C9.
- How to solve:
  1. Check the ratio: left top/left bottom = right top/right bottom?
  2. If yes, the middle resistor carries 0 A. Erase it (or replace it with a wire; both give the same answer).
  3. Now the network is (left top + left bottom) || (right top + right bottom).
- Worked example (ATX 2020 B Q56): all four arms are 300 Ω. The middle is 100 Ω.
  - 300/300 = 300/300, so it is balanced. Erase the 100 Ω.
  - R_eq = (300 + 300) || (300 + 300) = **300 Ω**. P = $120^2/300$ = **48.0 W**.
- Traps: The middle resistor value does not matter when the bridge is balanced. It is a decoy.
- Calculator: calculate — type: four arm resistors (and the middle); checks balance, gives R_eq.

4. Balanced bridge: find the unknown resistor (and both possible values)

- Scope: ✅ IN.
- Where seen: Rickards 2020 B (INV, 2021 season) Q32 (100 Ω and 200 Ω on the left, 500 Ω on the right → 250 Ω or 1000 Ω). Princeton 2026 C (INV) Computation 1c (→ 34.3 Ω). National supervisor deck 10 and homework generator Level 11 (R_1 = 100, R_2 = 300, R_3 = 200 → R_x = 600 Ω).
- How to solve:
  1. Write balance as top/bottom (left) = top/bottom (right).
  2. Put the unknown in its place. Solve one equation.
  3. If the drawing does not say which arm is which, there are **two** answers. Give both if asked.
- Worked example (Rickards 2020 B Q32): left arms 100 Ω and 200 Ω. Right arms 500 Ω and R_x.
  - Case 1: 100/200 = 500/R_x → R_x = **1000 Ω**.
  - Case 2: 200/100 = 500/R_x → R_x = **250 Ω**.
- Traps: Cross-multiplying the wrong pair is the top error. Write the ratio with words: "top over bottom".

- Calculator: calculate — type: three known arms and the arm position of R_x.

5. Unbalanced bridge with an ideal voltmeter (two dividers)

- Scope: ✅ IN (an ideal voltmeter draws no current, so this is just two voltage dividers). When a resistor or real meter joins the middles, it is ✅ States-Nationals (see Kirchhoff Analysis (States-Nationals) A5 type 18).
- Where seen: Princeton 2026 C (INV) Computation 1d (R_x = 40 Ω → 0.40 V). Northview Sylvania 2025 C (INV) FR6 (→ 1.1 V). National supervisor homework generator Level 11 (V_DB, sign +, – or 0).
- How to solve:
  1. Left divider: V_left = V × (bottom-left)/(left total).
  2. Right divider: V_right = V × (bottom-right)/(right total).
  3. Meter reading = V_left – V_right. The sign tells which side is higher.
- Worked example (generator Level 11): 12 V. Left: R_1 = 100 Ω on top, R_2 = 200 Ω on the bottom. Right: R_3 = 300 Ω on top, R_x = 300 Ω on the bottom.
  - V_D = 12 × 200/300 = 8 V. V_B = 12 × 300/600 = 6 V. V_DB = **+2 V**. Not balanced.
- Traps: The ratio check (2 vs 1) tells you "not balanced" before you compute.
- Calculator: calculate — type: V, four arm resistors; outputs both middle voltages and the difference.

6. Finite ladder: reduce from the far end (Fibonacci pattern)

- Scope: ✅ IN.
- Where seen: USC Invitational 2026 B (INV) Q34a (build-up ladder with R = 1 Ω → 1.00, 0.667, 0.625, 0.619 Ω). UT Invitational 2020 B (INV) Q67 (10 V ladder → 5.73 Ω, 8.94 V, 0.53 V). UF Northern Regional 2020 B (REG) Q30 (12 V ladder; key error). National supervisor generator Level 4 and deck 4 example 2 (20 + [40 || (25 + 100 + 25)] + 20 → 71.58 Ω). Prior research I_bank C4 (3 rungs of 1 Ω → 13/8 Ω).
- How to solve:
  1. Start at the end farthest from the source.
  2. Combine the last rung with the series resistor next to it (series).
  3. Combine that with the next rung (parallel).
  4. Repeat until you reach the source.
- Worked example (USC 2026 B Q34a): start with one 1 Ω shunt (R_a = 1 Ω). Each new section adds a 1 Ω in series and then a 1 Ω shunt across the front.
  - R_b = 1 || (1 + 1) = **0.667 Ω**. R_c = 1 || (1 + 0.667) = **0.625 Ω**. R_d = 1 || (1 + 0.625) = **0.619 Ω**.
  - The values are ratios of Fibonacci numbers: 1/1, 2/3, 5/8, 13/21.
- Traps: Reducing from the source end does not work. You must start at the far end.
- Calculator: calculate — type: series R, shunt R, number of sections, which comes first.

7. Switched ladder: close switches to add rungs

- Scope: ✅ IN.
- Where seen: Cobra Invitational 2020 B (INV, 2019-20 season) Q17-19 (120 V; current with S1 only → 3 A; S1 and S2 → 3.75 A; all three → 3.9375 A). UT Invitational fall 2018 B/C (INV) Q24a-c (bulb network with three SPDT switches).
- How to solve:
  1. An open switch removes its rung. A closed switch puts its rung in.
  2. For each switch setting, redraw only the live parts.
  3. Reduce from the far end (type 6). I = V/R_eq.
- Worked example (Cobra Q17-19, as I read the numbers): a 20 Ω lead resistor, then rungs of 20 Ω, with 10 Ω links between rungs. 120 V.
  - S1 only: 20 + 20 = 40 Ω → **3.00 A**.
  - S1, S2: 20 + 20 || (10 + 20) = 20 + 12 = 32 Ω → **3.75 A**.
  - All three: 20 + 20 || (10 + 20 || (10 + 20)) = 20 + 20 || 22 = 30.48 Ω → **3.9375 A**.
- Traps: Each extra rung adds less and less current. The current grows toward a limit.
- Calculator: calculate — type: list of rungs and links, which switches are closed.

8. Current pushed into two corners of a network (third corner floating)

- Scope: ✅ IN.
- Where seen: USC Invitational 2026 B (INV) Q13 (triangle: A-C 50 Ω, A-B 45 Ω, B-C 5 Ω; 5 A in at A, out at B → V_AB = 123.75 V).
- How to solve:
  1. The current enters at one node and leaves at another. Only those two nodes are the "terminals".
  2. A third node with no outside wire is just a joint. Find the paths between the two terminals.
  3. V = I × R_eq between the terminals.
- Worked example (USC 2026 B Q13): between A and B there are two paths: 45 Ω directly, and 50 + 5 = 55 Ω through C.
  - R_AB = 45 || 55 = 24.75 Ω. V_AB = 5 × 24.75 = **123.75 V**.
- Traps: Do not treat C as a terminal. Nothing enters or leaves there.
- Calculator: calculate — type: triangle sides, entry node, exit node, current.

Hard

9. Symmetry: join equal-voltage points or cut resistors between them

- Scope: ✅ IN (pure reasoning; no equations).
- Where seen: USC Invitational 2026 B and C (INV) Q1 (regular octahedron, R on each of 12 edges, opposite corners → R/2). Rickards 2026 C (INV) Q54 (8 × 1 Ω network → 1.615 Ω). Hawk & Hornet 2026 C (INV) FR7 (derive a general expression, 25 points). Prior research I_bank C11 (triangle, tetrahedron, square with a diagonal).
- How to solve:
  1. Put A at the top and B at the bottom. Imagine current flowing from A to B.
  2. Find points that the symmetry swaps. Give them the same voltage.
  3. Join them with a wire (or cut any resistor between them).
  4. The network becomes series-parallel. Reduce it.
- Worked example 1 (USC 2026 B Q1): an octahedron has a top corner A, a bottom corner B and 4 corners around the middle.
  - By symmetry, the 4 middle corners have the same voltage. The 4 ring edges between them carry 0 A. Cut them.
  - A to the ring: 4 edges in parallel = R/4. Ring to B: R/4. R_AB = R/4 + R/4 = **R/2**.
- Worked example 2: a tetrahedron (4 corners, 6 equal R), across one edge A-B. The other two corners C and D are mirror images, so V_C = V_D. Cut C-D. Then R_AB = R || 2R || 2R = R || R = **R/2**.
- Worked example 3: a square of 4 equal R, with a diagonal R between the two corners that are not A or B. Those two corners are mirror images, so the diagonal carries 0 A. R = 2R || 2R = **R**.
- Traps:
  - The symmetry must keep A and B where they are. A flip that swaps A and B is anti-symmetry (type 15).
  - Joining points is safe **only** if they really have the same voltage.
- Calculator: look up — type: octahedron, tetrahedron, square diagonal; calculate — type: edge list (solver).

10. Resistor cube: body diagonal, face diagonal, one edge

- Scope: ✅ IN (symmetry; no equations).
- Where seen: F-M Invitational 2018 (INV, division not stated) Q15 (12 × 3 Ω, opposite corners → 2.5 Ω). UMSO 2019 (INV, division not stated) Q31A-C (a to c, one edge → 7R/12; a to d, face diagonal → 3R/4; a to b, body diagonal → 5R/6; the key is wrong on A and B). Rickards 2020 B (INV, 2021 season) Q37 (key 0.833R; the wording says "opposite edges"). Prior research I_bank C12.
- How to solve (body diagonal, A to the opposite corner B):
  1. From A, 3 edges go out. By symmetry they carry equal current. Their far ends are at one voltage. Join them.
  2. From those 3 corners, 6 edges go to the 3 corners next to B. Those 3 corners are also at one voltage.
  3. Then 3 edges go into B.
  4. R = R/3 + R/6 + R/3 = **5R/6**.
- How to solve (face diagonal and one edge): use the mirror plane through A and B to join equal corners. Then reduce. The answers are **3R/4** and **7R/12**.

- Worked example (F-M 2018 Q15): 12 resistors of 3 Ω. Body diagonal = 5 × 3/6 = **2.5 Ω**. Face diagonal = 3 × 3/4 = 2.25 Ω. One edge = 7 × 3/12 = 1.75 Ω.
- Traps:
  - Know which two corners the question uses. Body diagonal = through the inside of the cube. Face diagonal = across one face. Edge = next-door corners.
  - The order is edge < face < body: 7/12 < 3/4 < 5/6.
  - "Opposite edges" in the Rickards key is unclear. Its answer (0.833R) is the body-diagonal value.
- Calculator: look up — type: cube, R, which pair (edge / face / body).

11. Infinite ladder, series resistor first (the 1.618R ladder)

- Scope: ✅ IN (a quadratic is algebra, rule 3.e.i). Some extraction files tag it ⚠️ because Div B rarely asks it. Treat it as a State/National type.
- Where seen: Florida State 2021 B (STATE) Q34 (every R → 1.618R). Rickards 2020 B (INV, 2021 season) Q38 (R = 1 Ω → 1.618 Ω). Princeton 2019 C (INV) Part 4 (R(1 + √5)/2). PA Central Regional ~2013 C (REG) Q25. ATX Regional 2021 B (REG) FRQ1 (same ladder made of capacitors; ❌ OUT as capacitors).
- How to solve:
  1. Call the whole ladder's resistance x.
  2. Cut off the first section. What is left is the same infinite ladder, so it is also x.
  3. First section = R in series, then R in parallel with the rest: $x = R + (R \parallel x) = R + Rx/(R + x)$.
  4. Multiply out: $x^2 - Rx - R^2 = 0$.
  5. Quadratic formula: $x = R(1 + \sqrt{5})/2$. Keep the positive root.
- Worked example: R = 10 kΩ. x = 10 × 1.618 = **16.18 kΩ**. With R = 1 Ω, x = **1.618 Ω** (the "golden ratio").
- Traps:
  - The negative root (–0.618R) is not a resistance. Drop it.
  - Check: R_eq must be more than R (there is a series R in front) and less than 2R.
- Calculator: calculate — type: R (series), R (shunt), which comes first.

12. Infinite ladder, shunt resistor first; recurrence and calculator iteration

- Scope: ✅ IN.
- Where seen: USC Invitational 2026 B (INV) Q34a-d (build R_a to R_d; write R_next in terms of R_prev; iterate on the calculator with "Ans" → 0.618 Ω; exact value (√5 – 1)/2 Ω). USC Invitational 2026 C (INV) Q38b (exact form, (√5 – 1)/2·R).
- How to solve:
  1. The recurrence: **R_next = R || (R + R_prev) = 1/(1/R + 1/(R + R_prev))**.
  2. Calculator: type 1, press enter. Then type 1/(1/1 + 1/(1 + Ans)) and press enter again and again. The number stops changing at 0.6180.
  3. Exact: set R_next = R_prev = x. Then $x = R(R + x)/(2R + x)$. So $x^2 + Rx - R^2 = 0$.
  4. $x = R(-1 + \sqrt{5})/2$ = **0.618R**.
- Worked example: R = 1 Ω. Iterates: 1, 0.6667, 0.6250, 0.6190, 0.6182, 0.6180 … → **0.618 Ω**.
- Traps: Shunt-first gives 0.618R. Series-first gives 1.618R. They differ by exactly R (the front series resistor).
- Calculator: calculate — type: R, start value, number of steps ("Ans" loop).

13. Infinite ladder with two different resistor values

- Scope: ✅ IN (quadratic).
- Where seen: not seen in a test. Textbook: Kuphaldt Socratic worksheets ("ladder networks"); common physics-olympiad drill.
- How to solve:
  1. Series value R_1, shunt value R_2. Call the ladder x.
  2. $x = R_1 + (R_2 \parallel x) = R_1 + R_2x/(R_2 + x)$.
  3. Multiply out: $x^2 - R_1x - R_1R_2 = 0$.
  4. $x = [R_1 + \sqrt{(R_1^2 + 4R_1R_2)}]/2$.
- Worked example: R_1 = 2 Ω, R_2 = 12 Ω. $x = [2 + \sqrt{(4 + 96)}]/2 = (2 + 10)/2$ = **6 Ω**. Check: 2 + (12 || 6) = 2 + 4 = 6 Ω. ✓

- Traps: Put R_1 and R_2 in the right places. Swapping them gives a different answer.
- Calculator: calculate — type: R_1 (series), R_2 (shunt).

14. Infinite parallel and infinite "stack" networks (sums that settle)

- Scope: ✅ IN (self-similar algebra; no calculus needed).
- Where seen: USC Invitational 2026 B (INV) Q2 (infinitely many R in parallel → 0). USC Invitational 2026 C (INV) Q38c (row k has 2^k resistors in parallel; rows in series to infinity → 2R). Princeton 2019 C (INV) Part 4 (parallel-ladder limit → 0).
- How to solve:
  1. Write the first few terms: R, R/2, R/4, … for the stack.
  2. Each term is half the last one. The sum gets closer and closer to 2R.
  3. Or use self-similarity: the rows after the first look like the whole stack with every resistor halved. So x = R + x/2. Then x = **2R**.
- Worked example (USC 2026 C Q38c): rows of 1, 2, 4, 8, … resistors R in parallel, stacked in series. x = R + R/2 + R/4 + … = **2R**.
- Traps: Infinitely many resistors in **series** give infinite R. Infinitely many in **parallel** give 0. A stack can settle at a finite value.
- Calculator: calculate — type: R, ratio between rows.

15. Anti-symmetry: mirror points add up to V; the middle line sits at V/2

- Scope: ✅ States-Nationals when you still need one KCL equation; ✅ IN when the answer falls out with no equation.
- Where seen: Florida State 2021 B (STATE) Q49 (ten 9 kΩ resistors in a symmetric triangle-bridge → 10 kΩ; mirror symmetry leaves one unknown node). Solved in detail in Kirchhoff Analysis (States-Nationals) A5 type 20.
- How to solve:
  1. Put 1 V on A and 0 V on B.
  2. Find a flip that swaps A and B. A point P and its mirror P′ then have V_P + V_P′ = 1 V.
  3. Any point that lies on the mirror line is at exactly 0.5 V.
  4. Use this to cut the unknowns to one. Write one KCL equation. R = 1 V ÷ (current out of A).
- Worked example (not seen in a test): bridge A-C = 1 Ω, C-B = 3 Ω, A-D = 3 Ω, D-B = 1 Ω, middle C-D = 2 Ω.
  - Flipping swaps A with B and C with D. So V_D = 1 − V_C.
  - KCL at C: (1 − V_C)/1 = V_C/3 + (V_C − V_D)/2. This gives 1.5 = (7/3)V_C, so V_C = 9/14 V.
  - Current out of A = (1 − 9/14)/1 + (1 − 5/14)/3 = 5/14 + 3/14 = 8/14 A. R_AB = 14/8 = **1.75 Ω**. (Checked with a full network solve.)
- Traps: This bridge is not balanced (1/3 ≠ 3/1). So the middle resistor does carry current.
- Calculator: calculate — type: edge list (one-node solver with the symmetry), or the full solver.

Nationals

16. Unbalanced bridge R_eq (no symmetry)

- Scope: ✅ States-Nationals (two unknown node voltages, two equations). ⚠️ Borderline if solved with Δ-Y (type 17).
- Where seen: Prior research I_bank C10 (unbalanced bridge R_eq; no test example found). MIT Invitational 2019 C (INV) lab S1-1c (mystery resistor inside a non-series-parallel bridge). Close relatives: Princeton 2026 C (INV) 1d and Northview Sylvania 2025 C (INV) FR6 (voltages only).
- How to solve:
  1. Put 1 V on A and 0 V on B. Call the middle nodes C and D.
  2. KCL at C and at D. That is two equations in V_C and V_D (the limit is two).
  3. Solve. Current out of A = (1 − V_C)/R_AC + (1 − V_D)/R_AD. R_AB = 1/I.
- Worked example (not seen in a test): A-C = 10 Ω, C-B = 20 Ω, A-D = 30 Ω, D-B = 40 Ω, C-D = 50 Ω.
  - Solving the two KCL equations gives **R_AB = 20.95 Ω**. Type 17 gets the same value with Δ-Y.
- Traps: The ratios 10/20 and 30/40 differ, so it is not balanced. Do not erase the 50 Ω.
- Calculator: calculate — type: five resistors (two-node solver).

17. Delta-Y (Δ-Y) and Y-Δ conversion

- Scope: ⚠ Borderline. It is not in rule 3.d's exclusion list. It uses only algebra. But it is not named in rule 3.c either. A 2027 test would likely give the formula.
- Where seen: UF Northern Regional 2019 C (REG) Q35 (Δ of 100, 250, 150 Ω → Y of 75, 30, 50 Ω). MIT Invitational 2020 C (INV) Q38 (→ 80.0 Ω, 2.00 kW). GullSO 2025 C (INV) Q45. 2014 scioly.org question marathon (one Δ-Y question).
- How to solve (Δ → Y):
  1. Add the three Δ sides: S.
  2. Each Y arm = (product of the two Δ sides that touch that corner) / S.
  3. Replace the Δ by the Y. The network often becomes series-parallel.
- Worked example 1 (UF 2019 C Q35): Δ sides 100, 250 and 150 Ω. S = 500 Ω.
  - Arms: 100 × 150/500 = **30 Ω**. 100 × 250/500 = **50 Ω**. 250 × 150/500 = **75 Ω**. Which arm goes to which corner depends on the figure.
- Worked example 2 (the bridge of type 16): the Δ is A-C = 10, A-D = 30, C-D = 50 Ω. S = 90 Ω.
  - Arms: at A = 10 × 30/90 = 3.33 Ω. At C = 10 × 50/90 = 5.56 Ω. At D = 30 × 50/90 = 16.67 Ω.
  - Now: 3.33 + [(5.56 + 20) || (16.67 + 40)] = 3.33 + 17.61 = **20.95 Ω**. ✓
- Traps:
  - A Y arm uses the two Δ sides that **touch** its corner, not the opposite side.
  - Three equal Δ sides R give three equal Y arms R/3.
- Calculator: calculate — type: three Δ sides (or three Y arms), direction.

18. Three-terminal black box: find the hidden Y from pair readings

- Scope: ⚠ Borderline. Three unknowns break the two-equation limit (rule 3.e.i), unless you use the sum trick below, which needs no system.
- Where seen: PA Central Regional ~2013 C (REG) Q4 (AB = 35, AC = 25, BC = 50 Ω → find the Y). NY Lake Erie/Niagara 2018 C (INV) Station II (three-terminal box → deduce the network).
- How to solve:
  1. Each reading is two Y arms in series: AB = a + b, AC = a + c, BC = b + c.
  2. Add the two readings that touch A, and subtract the one that does not: a = (AB + AC − BC)/2.
  3. Do the same for b and c.
- Worked example (PA Central Q4): a = (35 + 25 − 50)/2 = **5 Ω**. b = (35 + 50 − 25)/2 = **30 Ω**. c = (25 + 50 − 35)/2 = **20 Ω**. Check: 5 + 30 = 35, 5 + 20 = 25, 30 + 20 = 50. ✓
- Traps: A Δ inside the box gives the same three readings as some Y. Without more data you cannot tell them apart.
- Calculator: calculate — type: AB, AC, BC readings.

19. Infinite grid or lattice: resistance between next-door points

- Scope: ⚠ Borderline. It uses only Ohm's law and symmetry, but the superposition idea goes past the Div B list.
- Where seen: USC Invitational 2026 C (INV) Q2 (infinite 3D cubic lattice → R/3). Prior research I_bank C13 (2D square grid → R/2).
- How to solve (2D square grid):
  1. Push current I into point A. By symmetry it splits equally into A's 4 edges: I/4 in the edge to B.
  2. Separately, pull current I out of B. By symmetry, I/4 comes into B along the same edge.
  3. Add the two cases (superposition): I/2 flows in edge A-B. $V_{AB} = (I/2) \cdot R$.
  4. $R_{AB} = V/I$ = **R/2**. In 3D each point has 6 edges, so the answer is **R/3**.
- Worked example: an infinite grid of 10 Ω resistors. R between next-door points = **5 Ω**. (A 41 × 41 grid in Python gives 5.003 Ω.)
- Traps: This only works for **next-door** points. Diagonal points need much harder math (❌ OUT).
- Calculator: look up — type: infinite grid, 2D or 3D, R.

Out of scope (skip)

- ❌ **Complete graph of N nodes in general** (Florida State 2021 B Q42 → 2R/N). The general-N proof is past the two-equation limit (rule 3.e.i). Small cases by symmetry (N = 3: 2R/3; N = 4: R/2) are ✅ IN.
- ❌ **Networks that need 3 or more simultaneous equations** (rule 3.e.i) or matrices (rule 3.e.v). Seen: MIT 2020 C Q37-43, Florida State 2021 B Q48.
- ❌ **Thevenin or Norton** to simplify a network (rule 3.d). Use series-parallel, symmetry or two-node KCL instead.
- ❌ **Infinite capacitor ladders** (rule 3.d "capacitance"). Seen: ATX Regional 2021 B FRQ1 (C_eq = 0.618 F). The resistor version is ✅ .
- ❌ **Diagonal points on an infinite grid** (needs calculus-level math, rule 3.e.v).
- ❌ **Gauss's law flux through a cube** (ATX Regional 2020 B Q37). This is a charge question, not a resistor cube. It needs 3D geometry beyond rule 3.e.ii.

Mistakes in tests and keys

- **UMSO 2019 (INV) Q31A and Q31B:** the key gives R for both. Correct: one edge **7R/12**, face diagonal **3R/4**. Q31C (5R/6) is right.
- **Rickards 2020 B (INV) Q37:** the key says "opposite edges" but gives 0.833R. That is the body-diagonal value (opposite **corners**).
- **UF Northern Regional 2020 B (REG) Q30a-c:** ladder key 0.4 A, 10.38 Ω, 0.2 A. The figure gives 12.61 Ω, 0.331 A, 0.166 A.
- **MIT Invitational 2019 C (INV) lab S1-1c:** the key says the hidden bridge resistor is 10.7 kΩ. The setup sheet says 4.7 kΩ. It cannot be settled without the real board.
- **Prior research:** an earlier file tagged the infinite ladder ⚠️ Borderline. The rules allow algebra, and a quadratic is algebra. Florida **State** 2021 asked it. I tag it ✅ IN, hard.

Not confirmed

- Florida State 2021 B Q49 network shape: I used the extraction's description and its nodal check (10 kΩ). I did not re-render the figure.
- Rickards 2026 C Q54 (8 × 1 Ω → 1.615 Ω) and Hawk & Hornet 2026 C FR7: from prior/G_tests.md only. I did not re-solve them.
- Cobra 2020 B Q17-19 topology: I rebuilt it from the three keyed currents (40, 32, 30.48 Ω). It fits all three, but I did not re-render the figure.
- The Christie (1833) and Wheatstone (1843) dates come from the prior research files (I_bank, F_slides). I did not re-check them online.
- Whether a 2027 State or National supervisor will accept Δ-Y as in scope: the rules are silent. I found no rules clarification on soinc.org.
- No real 2026 National Tournament test was found. The "nat26" files are the USC **Invitational** (January 2026). Where I looked: every `usc26_*` file header and the soinc Test Packet note in prior/G_tests.md §4.

Sources

- Rules: 2027 Science Olympiad Div B Rules Manual, Circuit Lab, rules 3.c.v, 3.c.ix, 3.d, 3.e (page B12). See Circuit Lab - Rules 2027 Ⓡ.
- Tests (extraction files in `~/Daniel/circuitlab/research/tests/` and `prior/G_tests.md`): USC Invitational 2026 B and C; Florida State 2021 B; ATX Regionals 2020 B, 2021 B; UT Invitationals fall 2018, 2020 B; UF Northern Regionals 2019 C, 2020 B; MIT 2019 C and 2020 C; F-M 2018; UMSO 2019; Cobra 2020 B; Rickards 2020 B and 2026 C; Princeton 2019 C and 2026 C; Northview Sylvania 2025 C; GullSO 2025 C; Hawk & Hornet 2026 C; PA Central ~2013 C; NY Lake Erie 2018 C.
- National supervisor practice decks 4 and 10, and the homework generator Levels 4 and 11 (`prior/F_slides.md`).
- Textbooks: OpenStax *College Physics 2e*, ch. 21.1 and 21.3; Kuphaldt, *Lessons in Electric Circuits* Vol. I, ch. 7 (series-parallel) and ch. 10 ("Δ-Y and Y-Δ conversions", "Bridge circuits" in ch. 8); All About Circuits, "Δ-Y

Conversions" and "Bridge Circuits" pages.

- All numbers in this note were checked in Python on 2026-10-16 with a full network solver (Laplacian matrix): cube, tetrahedron, octahedron, bridges, ladders, grid.

Kirchhoff Analysis (States and Nationals only)

Note: Kirchhoff Analysis (States-Nationals)

The 2027 rules list this area as "ix. States/Nationals only: Circuit analysis using Kirchhoff's Voltage & Current Laws" (rule 3.c.ix). So a **regional** test should not ask you to do KVL or KCL analysis. A **State** or **National** test can. Old regional tests (ATX, UF Northern Regional) still used it, because older rules allowed it.

The math limit is strict. Rule 3.e.i allows "systems of no more than two equations". Rule 3.e.v bans "matrix algebra". So every Kirchhoff problem on a 2027 B test has at most **2 unknowns**. You solve them by substitution or elimination. Your calculator can still solve the pair for you.

This is a **high-point area**. At Nationals-level invitationals (MIT, USC, Princeton, UT) the Kirchhoff free-response questions are often worth 8 to 16 points each.

Key facts

- A **node** is a point where two or more parts connect. All wire joined together without a part in between is one node.
- An **essential node** is a node where three or more parts connect.
- A **branch** is one path between two essential nodes. One current flows through every part of a branch.
- A **loop** is any closed path through the circuit. A **mesh** is a loop with no other loop inside it ("a window" of the drawing).
- **KCL (Kirchhoff's Current Law, the junction rule):** the current into a node equals the current out of the node. It comes from **conservation of charge**. Charge cannot pile up in a wire.
- **KVL (Kirchhoff's Voltage Law, the loop rule):** the voltage changes around any closed loop add up to zero. It comes from **conservation of energy**. A charge that goes around a loop comes back to the same potential.
- Kirchhoff wrote these laws in 1845, as a student in Königsberg. The history questions about him are IN at every level (rule 3.c.i).
- **Ground** is the node you call 0 V. You may pick any node. The usual choice is the node with the most connections, or the battery's – terminal.
- A **node voltage** is the voltage of a node above ground.
- **Nodal analysis** means: pick ground, name the unknown node voltages, write KCL at each unknown node.
- **Mesh analysis** means: give each mesh a loop current (all clockwise), write KVL around each mesh.
- A **supernode** is two nodes joined by a voltage source. You treat them as one big node in KCL. The source gives you the second equation.
- A **supermesh** is two meshes that share a current source. You write KVL around the outside of both. The source gives you the second equation.
- An **ideal voltage source** (battery) keeps the same voltage for any current. An **ideal current source** keeps the same current for any voltage. Its symbol is a circle with an arrow.
- A **dependent source** (controlled source) is a source whose value depends on a voltage or current somewhere else in the circuit. It is ⚠ borderline for 2027 B.
- A current can come out **negative**. That only means the real current flows opposite to the arrow you drew. Keep the minus sign. Do not redo the problem.
- **Passive sign convention:** if current enters a part at its + side, the part **absorbs** power. If current leaves at its + side, the part **delivers** power.
- In any circuit, the total power delivered equals the total power absorbed.
- A battery with current going **into its + terminal** is **being charged**. It absorbs power.

Formulas

What	Formula	Notes
KCL	$\Sigma I_{in} = \Sigma I_{out}$, or $\Sigma I_{out} = 0$ with signs	Charge conservation
KVL	$\Sigma \Delta V = 0$ around a loop	Energy conservation

What	Formula	Notes
Resistor current, nodal	$I = (V_a - V_b)/R$	Current from a to b
Resistor drop, KVL	$-IR$ when you walk **with** the current	$+IR$ when you walk against it
Battery, KVL	$+\varepsilon$ when you walk from − to +	$-\varepsilon$ when you walk from + to −
One-node formula (Millman)	$V = (E_1/R_1 + E_2/R_2 + \ldots + I_{in}) \div (1/R_1 + 1/R_2 + \ldots)$	Every branch from the node to ground; a plain resistor branch has $E = 0$
Mesh, shared resistor	Voltage $= R(I_1 - I_2)$	Use $I_1 - I_2$ in mesh 1's equation, $I_2 - I_1$ in mesh 2's
Supernode	$V_P - V_Q = E$	Plus one KCL around both nodes
Supermesh	$I_2 - I_1 = I_{source}$	Plus one KVL around the outside
Power, any part	$P = VI$	Absorbed if current enters at +
Power balance	$\Sigma P_{delivered} = \Sigma P_{absorbed}$	Use it to check your answer
Two equations	$ax + by = c$; $dx + ey = f \rightarrow x = (ce - bf)/(ae - bd)$, $y = (af - cd)/(ae - bd)$	Cramer's rule for 2 unknowns. You use it; the test can't require matrices
Balanced bridge	$R_1/R_2 = R_3/R_4$	No current in the middle
Unbalanced bridge, ideal voltmeter	$V = V_s \cdot R_2/(R_1+R_2) - V_s \cdot R_4/(R_3+R_4)$	Two dividers

Question types

Easy

1. Name the law, or say what it conserves

- Scope: ✅ IN (as history and law statements, rule 3.c.i) / ✅ States-Nationals when you use it.
- Where seen: UT Invitational 2020 B (2021 season, INV) Q12 (nodal analysis is based on KCL) and Q13 (KCL conserves charge; the key wrongly says energy). F-M Invitational 2018 (INV) Q10. Berkeley 2026 B (INV) Q17 and Q18. Pembroke Hill 2025 B (INV) Q31. NY Lake Erie/Niagara 2018 C (INV) Q14 (sum of voltages in a loop = 0) and Q15 (nodal uses KCL). UF Northern Regional 2020 B (REG) Q18 (state KCL). UF Northern Regional 2019 B (REG) Q24 (true/false: "the product of the drops equals the applied voltage" → false, it is the sum). UF Northern Regional 2020 C (REG) Q1. ATX Regional 2021 B (REG) Q5 ("algebraic sum of currents at a point is zero" → Kirchhoff) and Q46 (which law helps → KVL). ATX Regional 2020 B (REG) Q6 (Kirchhoff generalized Ohm's work).
- How to solve:
  1. KCL = current law = junction rule = conservation of **charge**.
  2. KVL = voltage law = loop rule = conservation of **energy**.
  3. Nodal analysis uses KCL. Mesh analysis uses KVL.
- Worked example: "KCL is a statement of the conservation of what?" Answer: **electric charge**.
- Traps:
  - The UT 2020 B key says "energy" for KCL. That is wrong. If you see this on a test, answer "charge" and file an appeal.
  - "Sum," not "product." "Zero," not "the source voltage" (unless the source is counted as a separate term).
  - A question can ask when KCL or KVL "fails." In a changing magnetic field, KVL fails (Florida State 2021 B Q17). Magnetism is OUT for 2027, so this question is OUT too.
- Calculator: look up — type: kcl, kvl, conserves.

2. KCL: find a missing current at a node (size and direction)

- Scope: ✅ States-Nationals.
- Where seen: MIT Invitational 2019 C (INV) Q22-23 (2 A in, 10 A out, 3 A in, 8 A out → I_4 = 13 A, in). Berkeley 2026 B (INV) Q19 (4 A in, 7 A out → 3 A). UT Invitational 2019 B (INV) Q39 (I_1 = 2 A, I_2 = 3 A in → I_3 = −5 A). GullSO 2025 C (INV) Q38 (→ 7 A, 9 A). Rickards 2026 B (INV) Q27 (→ i_5). National supervisor practice deck 10 (9 A splits into 3 A and 6 A).
- How to solve:
  1. Add all the currents that go in. Add all the currents that come out.
  2. The missing current makes the two sums equal.

3. If the missing current must flow in to balance, its direction is "in."

- Worked example (MIT 2019 C Q22-23): 2 A in, 10 A out, 3 A in, 8 A out. In = 5 A. Out = 18 A. The missing current is 18 − 5 = **13 A, flowing in**.
- Traps:
  - If every arrow points in, the answer has a minus sign. "$I_3 = -5$ A" means 5 A flows out (UT 2019 B Q39).
  - Count every wire at the node. A node drawn as a long wire with several dots is still one node.
- Calculator: calculate — type: list of currents with + for in and − for out; answer = −(sum).

3. KCL sign reasoning: all arrows point into one node

- Scope: ✅ States-Nationals.
- Where seen: ATX Regional 2021 B (REG) Q15 (three currents, all arrows into one node; which condition is possible? → only the choice where they add to zero).
- How to solve:
  1. With all arrows in, KCL says $I_1 + I_2 + I_3 = 0$.
  2. So at least one current must be negative, or all must be zero.
  3. Test each choice by adding the three values.
- Worked example: $I_1 = 2$ A, $I_2 = -5$ A, $I_3 = 3$ A. Sum = 2 − 5 + 3 = **0**. This set is possible. I_2 really flows out.
- Traps: "All positive" is never possible. "All zero" is possible.
- Calculator: calculate — type: I1 I2 I3; check sum = 0.

4. KVL: find a missing voltage in one loop

- Scope: ✅ States-Nationals.
- Where seen: National supervisor practice deck 10 (12 V battery, drops 5 V and 3 V → 4 V). Deck 4 practical: measure V_bat, V_R1, V_R2, V_R3 and "notice the relationship." BullSO 2026 C (INV) SA6 (KVL: $-20 + V_1 - 12 + V_2 = 0$).
- How to solve:
  1. Walk once around the loop.
  2. Add rises (battery − to +). Subtract drops (resistors, walking with the current).
  3. Set the total to zero and solve for the missing drop.
- Worked example (deck 10): 12 V battery, V_R1 = 5 V, V_R2 = 3 V, all in series. V_R3 = 12 − 5 − 3 = **4 V**.
- Traps: Two batteries in one loop can oppose each other. Then one battery counts as a drop.
- Calculator: calculate — type: source voltages (+ or −) and known drops.

5. Vocabulary: node, branch, loop, mesh, supernode; count them

- Scope: ✅ States-Nationals.
- Where seen: ATX Regional 2020 B (REG) Q34 (definition of a supernode → two nodes joined by a voltage source). ATX Regional 2019 B/C (REG) Q18 (ladder: count essential nodes and meshes, then pick the method; key: 3 nodes, 5 meshes, node-voltage method).
- How to solve:
  1. Colour each piece of connected wire. Each colour is one node.
  2. Essential nodes have 3 or more parts.
  3. Meshes = the "windows" in the drawing.
  4. Unknowns for nodal = essential nodes − 1 (minus 1 more for each voltage source to ground). Unknowns for mesh = meshes (minus 1 for each current source on the outside).
  5. Pick the method with fewer unknowns.
- Worked example: A circuit has 3 essential nodes and 5 meshes. Nodal needs 3 − 1 = **2 equations**. Mesh needs 5. Pick **nodal**.
- Traps:
  - Two dots joined by a plain wire are the **same** node.
  - The ATX 2019 ladder figure is unclear. I could not confirm the key's count (see "Not confirmed").
- Calculator: look up — type: node, mesh, supernode, supermesh.

Medium

6. One loop with several batteries (aiding and opposing)

- Scope: ✅ IN at every level if you only use series rules / ✅ States-Nationals if the test says "use KVL."
- Where seen: BullSO 2026 C (INV) SA6 (20 V and 12 V in one loop, 22 Ω and 18 Ω → I = 0.8 A, V_1 = 14.4 V, V_2 = 17.6 V). National supervisor generator Level 7 "Multi-Source" (V_eq = V_1 + V_2 − V_3). Prior question bank C21 (opposing batteries).
- How to solve:
  1. Pick a direction for the current.
  2. Batteries that push that way count +. Batteries that push the other way count −.
  3. I = (net emf) ÷ (total resistance).
  4. Each drop = I × R.
- Worked example (BullSO 2026 C SA6): two aiding sources, 20 V + 12 V = 32 V. Total R = 22 + 18 = 40 Ω. I = 32/40 = **0.8 A**. V across 18 Ω = **14.4 V**. V across 22 Ω = **17.6 V**.
- Worked example (opposing): 12 V and 6 V face each other with 2 Ω and 4 Ω. I = (12 − 6)/(2 + 4) = **1.0 A**. It flows out of the 12 V battery's + terminal. The 6 V battery is being charged.
- Traps: If the net emf comes out negative, the current flows the other way. Check the battery symbol: the long line is +.
- Calculator: calculate — type: emf list with signs, resistor list.

7. Node voltages from a ground; a battery to ground fixes a node

- Scope: ✅ States-Nationals.
- Where seen: ATX Regional 2020 B (REG) Q59a (24 V source, R_1 = 10 Ω to a 16 V source → V_R1 = 8.00 V). USC Invitational 2026 B (INV, top tier) Q32c (a current source pushes into an ideal 4 V battery → the node stays at 4.00 V). Prior question bank K5.
- How to solve:
  1. Mark ground. Every node joined to ground by a battery has a **known** voltage: ±E.
  2. Voltage across a resistor = difference of its two node voltages.
  3. Only nodes with no battery to ground are unknowns.
- Worked example (ATX 2020 B Q59a, figure checked): V_1 = 24 V holds the top-left node at 24 V. V_2 = 16 V holds the bottom of R_1 at 16 V. V_R1 = 24 − 16 = **8.00 V**. No equation is needed.
- Traps:
  - Many students write a KCL equation at a node that a battery already fixes. That wastes time and can give a wrong extra equation.
  - An ideal battery wins against a current source. The node stays at the battery voltage. The battery absorbs the extra current.
- Calculator: calculate — type: node voltages a and b; V = a − b.

8. Current source in series with a resistor

- Scope: ✅ States-Nationals.
- Where seen: USC Invitational 2026 B (INV) Q32a (1 mA source, 140 Ω in series, then 20 Ω || 50 Ω → 0.0143 V; the 140 Ω does not matter). ATX Regional 2020 B (REG) Q59b (6 A source in series with R_5 = 5 Ω). UT Invitational 2020 B (INV) Q69 (5 A source in series with 16 Ω).
- How to solve:
  1. A current source sets the current in its branch. A resistor in series with it changes nothing outside.
  2. Put the source current straight into the KCL equation of the node it feeds.
  3. The series resistor matters only if they ask for its voltage or power, or the source's voltage.
- Worked example (USC 2026 B Q32a): 1 mA flows into 20 Ω || 50 Ω = 14.29 Ω. V_A = 0.001 × 14.29 = **0.0143 V**.
- Traps: Do not add the series resistor into a parallel combination. Do not use it in nodal equations.
- Calculator: calculate — type: source current, resistors the current flows into.

9. One-unknown nodal: two or three batteries feeding one node (the "two-loop" circuit)

- Scope: ✅ States-Nationals. One equation, so it fits the math limit easily.

- Where seen: ATX Regional 2021 B (REG) FRQ2, Q44-45 (24 V with A = 3 Ω; shared C = 3 Ω; B = 3 Ω + D = 4 Ω with 29 V → I_C = 5 A, V_A = 9 V). UT Invitational fall 2018 B/C (INV) Q22 (28 V and 7 V with 4, 2, 1 Ω → node 8 V, V_R3 = 1 V). UF Northern Regional 2020 C (REG) Q27-28 (E_1 = 10 V, E_2 = 5 V, three 3.57 Ω → I_R2 = 0 A, I_R1 = 1.40 A). NY Lake Erie/Niagara 2018 C (INV) Q26-27 (three emfs → node 10 V, V_R1 = 2.0 V, I_R2 = 1.0 A). North Pocono 2020 C (INV) Q28 (9 V, 3 V, R_1 = 5, R_2 = 3, R_3 = 7 → key wrong). UT Invitational 2020 B (INV) Q68 (several stacked sources → V_10Ω = 8.95 V). Prior question bank K3.
- How to solve:
  1. Pick the bottom wire as ground. Call the top middle node V.
  2. For each branch, write (E – V)/R for current flowing **into** the node. A branch with no battery has E = 0.
  3. Set the sum of all currents into the node to zero. Or use the one-node formula.
  4. Find each branch current from V.
- Worked example 1 (ATX 2021 B FRQ2): left branch 24 V + 3 Ω; middle 3 Ω; right 29 V + (3 + 4) Ω.
  - (24 – V)/3 + (29 – V)/7 = V/3.
  - V = (8 + 4.143) ÷ (1/3 + 1/3 + 1/7) = **15.0 V**.
  - I_C = 15/3 = **5.00 A**. I_A = (24 – 15)/3 = 3.00 A, so V_A = **9.00 V**. Right branch: (29 – 15)/7 = 2.00 A. Check: 3 + 2 = 5 ✓.
- Worked example 2 (UF Northern Regional 2020 C Q27-28, figure checked): E_1 = 10 V in the left branch with R_1. E_2 = 5 V in the right branch with R_2. R_3 in the middle. All R = 3.57 Ω.
  - V = (10/3.57 + 5/3.57) ÷ (3/3.57) = **5.00 V**.
  - I_R2 = (5 – 5)/3.57 = **0 A**. I_R1 = (10 – 5)/3.57 = **1.40 A**.
- Worked example 3 (UT 2020 B Q68, figure checked; the test said "using only Kirchhoff's Laws"): on the left, V3 = 12 V (+ up), then 6 V (+ down), then 16 V (+ up). The middle branch leaves between the 6 V and V3 and holds 20 V (+ on the left) and 10 Ω. The outer path holds 5 + 6 + 12 Ω on top and 16 + 16 + 15 Ω on the bottom.
  - Ground the bottom of V3. Top-left = 12 – 6 + 16 = 22 V. Middle branch start = 12 – 20 = –8 V.
  - KCL at the right node X: (22 – X)/23 + (–8 – X)/10 = X/47 → X = 0.950 V.
  - V_10Ω = |–8 – 0.950| = **8.95 V**. (The key says 8.94 V; rounding.)
- Traps:
  - Do not assume each battery drives "its own" resistor. That is exactly the North Pocono 2020 key error.
  - The sign of (E – V) tells the current direction. Negative means it flows out of the node into that battery.
- Calculator: calculate — type: for each branch E and R (E = 0 for a plain resistor), plus any source current into the node.

10. Power absorbed or delivered (sign convention); the "box" problem

- Scope: ✅ States-Nationals (P = VI itself is ✅ IN at every level).
- Where seen: UF Northern Regional 2020 C (REG) Q29-30 (section of a circuit with a "Box": R_1 = 2.4 Ω, R_2 = 4.4 Ω, R_3 = 6.2 Ω, i = 6.0 A, V_A – V_B = 74 V → box **provides** 36.1 W). MIT Invitational 2021 C (INV) Scenario B Q19-33 (does each part supply or absorb? → resistor absorbs 0.25 W, current source supplies 0.36 W). MIT 2021 C Q4 (a device that is not a resistor: P = VI still holds → 1.1 W). National supervisor generator Level 13 (a battery absorbs power).
- How to solve:
  1. Find the voltage across the part and the current through it.
  2. If current enters the + side, the part absorbs P = VI. If it leaves the + side, the part delivers P = VI.
  3. Check: total delivered = total absorbed.
- Worked example (UF Northern Regional 2020 C Q29-30, figure checked): A → R_1 → two parallel paths → B. The lower path is R_3 = 6.2 Ω and carries i = 6.0 A. The upper path is R_2 = 4.4 Ω and the box.
  - V across the parallel part = 6.0 × 6.2 = 37.2 V.
  - V across R_1 = 74 – 37.2 = 36.8 V, so I_total = 36.8/2.4 = 15.33 A.
  - Upper current = 15.33 – 6.0 = 9.33 A. Drop on R_2 = 9.33 × 4.4 = 41.07 V.
  - That is more than 37.2 V. So the box must **raise** the potential by 41.07 – 37.2 = 3.87 V.
  - The current goes through the box from – to +. The box **provides** energy at 3.87 × 9.33 = **36.1 W**.
- Traps:

- In the figure both lower and upper resistors look like "R_2" at low resolution. The numbers only work if the lower one is $R_3 = 6.2\ \Omega$.
- A resistor always absorbs. A battery can absorb (being charged) or deliver.
- A current source can absorb power too. Look at the voltage across it.

- Calculator: calculate — type: V across part, I through part, "enters at +?" yes/no.

11. A battery being charged

- Scope: ✅ States-Nationals (when it needs KVL/KCL) / ✅ IN for a single loop.
- Where seen: national supervisor generator Level 13 ("I_bat = −1 A, the battery absorbs power"). Prior question bank K3. It is not asked as its own question in the tests I read. It appears as a negative branch current in two-source problems (MIT 2019 C Q50 below gives −3.03 mA in the E branch).
- How to solve:
  1. Solve the circuit (type 9 or 12).
  2. If the current flows **into** a battery's + terminal, that battery is being charged.
  3. Power stored in the battery = E × I. Heat in its resistor = I^2R.
- Worked example (question bank K3): a 10 V battery with 2 Ω and a 5 V battery with 1 Ω both feed a node. A 4 Ω load goes from the node to ground.
  - $(10 - V)/2 + (5 - V)/1 = V/4 \rightarrow V = 10 \div 1.75 =$ **5.71 V**.
  - I from the 10 V branch = 2.14 A. I from the 5 V branch = (5 − 5.71)/1 = **−0.71 A**. So 0.71 A flows **into** the 5 V battery's + terminal. It is being charged.
  - Load current = 1.43 A. Check: 2.14 − 0.71 = 1.43 ✓.
  - Power stored in the 5 V battery = 5 × 0.714 = **3.57 W**. Heat in its 1 Ω = 0.51 W.
- Traps: A negative answer is not a mistake. It is the whole point of the question.
- Calculator: calculate — type: same as type 9; then E·I and I^2R.

Hard

12. Two-node nodal analysis (2 equations)

- Scope: ✅ States-Nationals. Two equations is the maximum allowed.
- Where seen: BullSO 2026 C (INV) SA7 ($\rightarrow V_1 = 28$ V, $V_2 = 30$ V). UT Invitational 2019 B (INV) Q31 and the same question in UT Invitational fall 2019 C (INV) Q27 (12 V, 2 Ω, 3 Ω, 4 Ω, 2 A source → V_R3 = 8.00 V, P on 3 Ω = 30.7 W). USC Invitational 2026 B (INV) Q32b (two current sources and a 3 V battery → V_B = 1.56 V) and Q32d (two current sources, 2 unknown nodes → V_D = 0.851 V). UT Invitational 2020 B (INV) Q71 (back-solve R → 658 Ω).
- How to solve:
  1. Pick ground. Name the two unknown nodes V_1 and V_2.
  2. KCL at node 1: (sum of currents leaving through resistors) = (source currents in).
  3. KCL at node 2: same.
  4. Tidy each into the form $a \cdot V_1 + b \cdot V_2 = c$.
  5. Solve the pair by elimination, or with the 2-equation formula.
- Worked example 1 (BullSO 2026 C SA7; the figure is not in the text file, so I used the key's equations):
  - Node 1: $V_1/4 + (V_1 - V_2)/2 = 6 \rightarrow 3V_1 - 2V_2 = 24$.
  - Node 2: $V_2/6 + (V_2 - V_1)/2 = 6 \rightarrow -3V_1 + 4V_2 = 36$.
  - Add them: $2V_2 = 60 \rightarrow V_2 =$ **30 V**. Then $V_1 = (24 + 60)/3 =$ **28 V**.
- Worked example 2 (UT 2019 B Q31; circuit rebuilt from the node voltages, see "Not confirmed"): 12 V source with 2 Ω into node 1. 3 Ω from node 1 to ground. $R_3 = 4\ \Omega$ from node 1 to node 2. A 2 A source pushes into node 2.
  - Node 2: $2 = (V_2 - V_1)/4 \rightarrow V_2 - V_1 = 8$.
  - Node 1: $(12 - V_1)/2 + (V_2 - V_1)/4 = V_1/3 \rightarrow (12 - V_1)/2 + 2 = V_1/3 \rightarrow V_1 =$ **9.60 V**, $V_2 =$ **17.6 V**.
  - V across $R_3 =$ **8.00 V**. Power in the 3 Ω = $9.6^2/3 =$ **30.7 W**.
- Traps:
  - Use the same sign rule at both nodes. "Current leaving through resistors = source current in" is the safest habit.

  - The shared resistor appears in **both** equations, with opposite signs.
  - The UT 2019 question writes "R3" (the 4 Ω) and "the 3 Ω" in the same question. Read which resistor they really mean.
- Calculator: calculate — type: a b c d e f for $a \cdot V_1 + b \cdot V_2 = c$ and $d \cdot V_1 + e \cdot V_2 = f$.

13. Two-mesh analysis

- Scope: ✅ States-Nationals.
- Where seen: BullSO 2026 C (INV) SA8 (25 V, 4 Ω, shared 2 Ω, 8 Ω, 15 V → $I_1 = 5$ A, $I_2 = 2.5$ A). ATX Regional 2021 B (REG) FRQ2 also works by mesh. Prior question bank K4.
- How to solve:
  1. Draw a clockwise current in each window: I_1 and I_2.
  2. KVL in mesh 1: battery rises = (own resistors)·I_1 + (shared R)·($I_1 - I_2$).
  3. KVL in mesh 2: battery rises = (own resistors)·I_2 + (shared R)·($I_2 - I_1$).
  4. Solve the pair. The shared resistor carries $I_1 - I_2$.
- Worked example (BullSO 2026 C SA8):
  - Mesh 1: $-25 + 4I_1 + 2(I_1 - I_2) = 0 \rightarrow 6I_1 - 2I_2 = 25$.
  - Mesh 2: $-15 + 8I_2 + 2(I_2 - I_1) = 0 \rightarrow -2I_1 + 10I_2 =$ **15**.
  - Solve: $I_1 =$ **5.0 A**, $I_2 =$ **2.5 A**. The shared 2 Ω carries 2.5 A.
  - Note: the printed key simplifies mesh 2 to "= 25". That is a typo. With 25 you get 5.36 A and 3.57 A. The key's final answers (5 A, 2.5 A) are right.
- Worked example 2 (ATX 2021 B FRQ2 by mesh): mesh 1: $6I_1 - 3I_2 = 24$. Mesh 2 (29 V opposing the clockwise direction): $-3I_1 + 10I_2 = -29$. So $I_1 = 3$ A, $I_2 = -2$ A, and the shared 3 Ω carries $3 - (-2) =$ **5 A** ✓ (same as nodal).
- Traps:
  - The sign of a battery in mesh 2 depends on its direction relative to clockwise. Check each one.
  - A negative mesh current is fine. It just runs counter-clockwise.
- Calculator: calculate — type: the six coefficients, as in type 12.

14. Supernode (a battery between two unknown nodes)

- Scope: ✅ States-Nationals.
- Where seen: ATX Regional 2020 B (REG) Q61 ($V_2 = 5$ V between two nodes → V_R5 = 2.33 V) and Q34 (definition). USC Invitational 2026 B (INV) Q33d (supernode with two dependent sources, ⚠️). UF Northern Regional 2019 C (REG) Q40 (needs 3 unknowns → ❌ OUT).
- How to solve:
  1. Draw a balloon around both nodes and the battery between them.
  2. Equation 1: V_P − V_Q = E (the + side is higher).
  3. Equation 2: KCL for the whole balloon. All currents out through resistors = all currents in. Ignore the battery's own current.
  4. Substitute and solve one equation.
  5. Any resistor directly across the battery does not change the node voltages.
- Worked example (ATX 2020 B Q61, figure checked): $V_1 = 16$ V through $R_1 = 24$ Ω into node P. $R_2 = 10$ Ω from P to ground. $V_2 = 5$ V from P (+) to Q. $R_3 + R_4 = 5 + 3$ Ω also join P and Q. $R_5 = 6$ Ω and $R_7 = 3$ Ω go from Q to ground. $R_8 = 5$ Ω goes from Q to $V_3 = 10$ V.
  - V_P = V_Q + 5.
  - KCL for the balloon: (16 − V_P)/24 + (10 − V_Q)/5 = V_P/10 + V_Q/6 + V_Q/3.
  - Put V_P = V_Q + 5: 235 = 101·V_Q → V_Q = **2.33 V**, V_P = 7.33 V.
  - V_R5 = V_Q = **2.33 V**.
- Traps:
  - The 8 Ω ($R_3 + R_4$) sits straight across the 5 V battery. It carries 5/8 A but does not enter the balloon equation.
  - Do not write KCL at P alone. You do not know the battery's current.
- Calculator: calculate — type: E between nodes; branch list for P; branch list for Q.

15. Supermesh (a current source shared by two meshes)

- Scope: ✅ States-Nationals.
- Where seen: UT Invitational 2020 B (INV) Q70 (10 V, 16 V, 2 A source, "mesh analysis" → 3.64 V on the 3 Ω).
- How to solve:
  1. Give each mesh a clockwise current.
  2. The shared current source gives: (current up the shared branch) = I_right − I_left = I_source (check the arrow).
  3. Write one KVL around the outside of both meshes. Skip the current source.
  4. Solve the pair.
- Worked example (UT 2020 B Q70, figure checked): top-left mesh: 10 V (+ up) and 15 Ω. Top-right mesh: 3 Ω, 5 Ω, 10 Ω. The 2 A source (arrow up) sits between them. A lower mesh holds 16 V with 16 + 10 + 15 Ω. It shares only a plain wire with the top meshes.
  - Constraint: I_b − I_a = 2.
  - Outer KVL: 10 = 15·I_a + (3 + 5 + 10)·I_b.
  - 10 = 15·I_a + 18(I_a + 2) → I_a = −0.788 A, I_b = **1.212 A**.
  - V on the 3 Ω = 3 × 1.212 = **3.64 V**. The lower loop is separate: 16/41 = 0.390 A.
- Traps:
  - The question says "R1," but no part in the figure has that label. The key means the 3 Ω.
  - A wire shared by two meshes carries no voltage. It does not couple the loops.
- Calculator: calculate — type: I_source, outer loop emf, resistances in each mesh.

16. Back-solve: find an unknown emf or resistor from one measured value

- Scope: ✅ States-Nationals.
- Where seen: MIT Invitational 2019 C (INV) Q50 (12 V + 330 Ω and E + 100 Ω branches; the 220 Ω load carries 20 mA → E = 4.10 V). UT Invitational 2020 B (INV) Q71 (V = 1.8 V → R = 658 Ω). UT Invitational 2019 B (INV) Q34 (6 A source, V_1 = 20 V across 10 Ω → V_s = 12 V). UF Northern Regional 2020 C (REG) Q16 (choose the emf so one branch carries zero current → 9 V). UF Northern Regional 2019 C (REG) Q41 (I_x = 3 A and $V_1 = V_2$ → both 50 V).
- How to solve:
  1. Use the measurement to find a node voltage first (V = IR on the measured part).
  2. Find every current you can from that node voltage.
  3. Use KCL to get the unknown branch current.
  4. Use KVL (or Ohm's law) in that branch to get the unknown E or R.
- Worked example (MIT 2019 C Q50):
  - Node voltage = 0.020 × 220 = 4.40 V.
  - 12 V branch: (12 − 4.40)/330 = 23.03 mA into the node.
  - KCL: the E branch must give 20 − 23.03 = −3.03 mA. So 3.03 mA flows **into** E's + side.
  - KVL in the E branch: E = 4.40 + 100 × (−0.00303) = **4.10 V**. Battery E is being charged.
- Worked example (zero-current condition): 12 V feeds 2 Ω and 4 Ω in series to ground. The middle node joins a battery E through 3 Ω. For zero current in the 3 Ω, E must equal the divider voltage: E = 12 × 4/6 = **8.0 V**.
- Traps: Find the node voltage first. Students who start by guessing currents usually need three unknowns.
- Calculator: calculate — type: measured I and R (or V); other branch E and R; unknown branch R (or E).

17. Superposition

- Scope: ⚠️ Borderline. The rules do not name it and do not exclude it. In practice it is a States/Nationals method, because it is a way to solve multi-source circuits. Every answer can also be found with KCL.
- Where seen: UT Invitational 2020 B (INV) Q69, tiebreaker 2 ("using the superposition principle" → 77.06 V on the 10 Ω). National supervisor generator Level 13 (I_s + V_bat + R_1-R_4; "superposition does not work for power"). USC Invitational 2026 C (INV) MC2 (infinite cubic lattice by superposition → R/3, ⚠️). Prior topics note B_topics §5e.
- How to solve:
  1. Keep one source. Turn off the others: a battery becomes a **wire**; a current source becomes an **open** (a gap).
  2. Solve this one-source circuit with series and parallel rules.
  3. Repeat for each source.
  4. Add the voltages (or currents) with signs.

5. Find power only from the final total. Never add powers.

- Worked example (UT 2020 B Q69, figure checked): one top node. Branches to ground: 5 Ω + 16 A source (up); 12 Ω + 10 V (+ up); 10 Ω + 12 Ω; and 16 Ω + 5 A source (up).
  - 12 Ω || 22 Ω = 7.765 Ω (the current-source branches are open when turned off).
  - 16 A alone: 16 × 7.765 = 124.24 V. 5 A alone: 5 × 7.765 = 38.82 V. 10 V alone: 10 × 22/34 = 6.47 V.
  - Total node voltage = **169.53 V**. Check with KCL: 21 = (V − 10)/12 + V/22 → V = 169.53 V ✓.
  - V on the 10 Ω = 169.53 × 10/22 = **77.06 V**.
- Power trap (same circuit): power in the 10 Ω from the three parts added = 351 W. True power = $77.06^2/10$ = **594 W**. Power is not linear.
- Traps:
  - A turned-off current source is an **open**, not a wire. A turned-off battery is a **wire**, not an open.
  - The national supervisor generator Level 13 key has bugs in I_3, I_4, I_bat and V_ls. Do not trust it.
- Calculator: calculate — type: sources one at a time; or just use the one-node solver (type 9).

18. Unbalanced Wheatstone bridge

- Scope: ✅ IN with an ideal voltmeter (two dividers, no KVL needed) / ✅ States-Nationals when a meter or resistor across the middle carries current (2 unknown nodes).
- Where seen: Princeton 2026 C (INV) Computation 1c-d (balanced → R_x = 34.3 Ω; then R_x = 40 Ω → voltmeter reads 0.4 V). Northview Sylvania 2025 C (INV) FR6 (→ 1.1 V). National supervisor generator Level 11 (V_DB). ATX Regional 2020 B (REG) Q56 (balanced bridge → 48 W). UT Invitational 2020 B (INV) Q16 (bridge measures an unknown resistance). USC Invitational 2026 B (INV) Q23 (Kelvin bridge, ⚠️ trivia).
- How to solve (ideal voltmeter):
  1. Each side of the bridge is a voltage divider.
  2. Find the middle voltage of each side.
  3. Meter reading = the difference.
- How to solve (meter or resistor R_g across the middle):
  1. Name the two middle nodes a and b. The top is the battery voltage; the bottom is ground.
  2. KCL at a and at b. Each has a term (V_a − V_b)/R_g.
  3. Solve the two equations.
- Worked example 1 (Princeton 2026 C 1c-d): 12 V. Left side: 80 Ω on top, R_x on the bottom. Right side: 140 Ω on top, 60 Ω on the bottom.
  - Balanced: 140/60 = 80/R_x → R_x = **34.3 Ω**.
  - Now R_x = 40 Ω. Left middle = 12 − 0.1 × 80 = 4.0 V. Right middle = 12 − 0.06 × 140 = 3.6 V. Reading = **0.40 V**.
- Worked example 2 (Nationals version, not seen in a test): same bridge, but a 100 Ω galvanometer joins the middles.
  - Node a: (12 − V_a)/80 = V_a/40 + (V_a − V_b)/100.
  - Node b: (12 − V_b)/140 = V_b/60 + (V_b − V_a)/100.
  - V_a = 3.937 V, V_b = 3.700 V. Meter current = **2.37 mA**. Meter voltage = 0.237 V (less than the 0.40 V of an ideal meter).
- Traps:
  - The balance rule compares **top/bottom on one side** with **top/bottom on the other side**. Mixing them up gives the wrong R_x.
  - A balanced bridge has zero middle current. You can remove the middle part, then use series-parallel.
- Calculator: calculate — type: V_s, four arm resistors, optional R_g.

19. Ideal ammeters and wires that split a circuit

- Scope: ✅ States-Nationals (it can also be solved with single loops).
- Where seen: UMSO 2019 (INV, division not stated) Q26 (200 V and 160 V sources, ideal ammeters A1-A5; A1 = 16 A → R = 12.5 Ω; currents in 40 Ω and 20 Ω = 4 A, 8 A). North Pocono 2020 C (INV) Q18 (why an ammeter's resistance is near zero).
- How to solve:

1. An ideal ammeter is a wire (0 Ω). An ideal voltmeter is a gap (infinite Ω).
2. Merge the nodes that an ammeter joins.
3. Often the circuit falls apart into separate single loops.
4. The ammeter reads the current through that wire, from KCL.

- Worked example (not seen in a test): a 10 V battery with 5 Ω and a 6 V battery with 3 Ω both feed a node. An ideal ammeter joins that node to ground. A 10 Ω resistor is also there, in parallel with the ammeter.
  - The ammeter holds the node at 0 V. Each battery drives its own resistor: 2 A and 2 A.
  - Ammeter reading = **4 A**. The 10 Ω carries **0 A**.
- Traps: A resistor in parallel with an ideal ammeter is shorted. It carries no current.
- Calculator: calculate — type: each battery E and R; ammeter = sum of currents.

Nationals

20. Symmetric networks that are not series-parallel (one unknown by symmetry)

- Scope: ✅ States-Nationals (KCL with 1-2 unknowns). Pure symmetry tricks with series-parallel are ✅ IN.
- Where seen: Florida State 2021 B (STATE) Q49 (ten 9 kΩ resistors in a symmetric triangle-bridge → 10 kΩ). USC Invitational 2026 B (INV) MC1 (octahedron with R on 12 edges, opposite vertices → R/2) and MC13 (current injected at two nodes of a triangle → 123.75 V). Florida State 2021 B (STATE) Q42 (N-node complete graph → 2R/N; ❌ OUT, general N). Rickards 2020 B (INV) Q37 (resistor cube). The series-parallel versions belong in the series-parallel note.
- How to solve:
  1. Put 1 V on terminal A and 0 V on terminal B.
  2. Look for symmetry. Points that are mirror images often have equal voltages, or voltages that add to 1 V.
  3. Use symmetry to cut the unknowns to 1 or 2.
  4. KCL at the unknown nodes. Total current I out of A. R = 1 V ÷ I.
- Worked example (not seen in a test): A-C = 1 Ω, A-D = 2 Ω, C-B = 2 Ω, D-B = 1 Ω, C-D = 1 Ω. Find R_AB.
  - Symmetry (turn the drawing over): V_D = 1 − V_C.
  - KCL at C: (1 − V_C)/1 = V_C/2 + (V_C − V_D)/1 → 2 = 3.5·V_C → V_C = 4/7 V, V_D = 3/7 V.
  - Current from A = (1 − 4/7)/1 + (1 − 3/7)/2 = 5/7 A. R_AB = **1.4 Ω** (checked with a full 2-node solve).
- Traps: If the network is balanced, the middle resistor carries no current. Then it is just series-parallel.
- Calculator: calculate — type: edge list; one-node or two-node solver.

21. Dependent (controlled) sources

- Scope: ⚠️ Borderline. Dependent sources are not in the rule 3.c list and not in the 3.d exclusion list. They need KVL and KCL, so they are States/Nationals at most. Many are within two equations.
- Where seen: USC Invitational 2026 B (INV) Q33a-d and C Q37 (VCCS circuits → −200 V, 1.07 V, −4.09 V (key; I get −15.0 V), −1.67 V). UT Invitational 2020 B (INV) Q72 (2V_x source → 4.61 V) and Q73 (0.6i_x source → 0.084 W). UT Invitational 2019 B (INV) Q38 = fall 2019 C Q37 (5V_o source → −10.9 V, −100 V). UT Invitational fall 2019 C (INV) Q30 (CCCS I_x → 1.33 A). ATX Regional 2019 B/C (REG) Q19-20 (2I_x → 1 A; 4I_x → 6 Ω). ATX Regional 2020 B (REG) Q64 (3I_x; no solution as drawn). UT Invitational fall 2025 B/C (INV) Section C ("magic box" amplifier model, KVL/KCL, then A → ∞). National supervisor deck 11 and generator Level 12. Florida State 2021 B (STATE) Q47-48 (❌ OUT: Norton; 4 unknowns).
- Names: VCVS = voltage-controlled voltage source. VCCS = voltage-controlled current source. CCVS = current-controlled voltage source. CCCS = current-controlled current source. The symbol is a **diamond**.
- How to solve:
  1. Treat the dependent source like a normal source, with its value written in terms of the control (for example $2 \cdot i_1$).
  2. Write the KVL or KCL equations.
  3. Add one more equation: the control variable in terms of your unknowns.
  4. Substitute so you have at most 2 unknowns. Solve.
- Worked example 1 (national supervisor deck 11): 10 V, then 10 kΩ carrying i_1, to node A. From A: 2 kΩ carrying i_2, and a dependent current source $2i_1$ (pointing into A).

  - KVL: $10 = 10\,000 \cdot i_1 + 2000 \cdot i_2$.
  - KCL at A (with the source arrow as the deck draws it): $i_1 = i_2 + 2i_1$. So $i_2 = -i_1$.
  - $10 = 10\,000 \cdot i_1 - 2000 \cdot i_1 = 8000 \cdot i_1 \rightarrow i_1 =$ **1.25 mA**, $i_2 =$ **−1.25 mA**, source = 2.5 mA.
- Worked example 2 (USC 2026 B Q33b): 1.5 V, 100 Ω, and a VCCS of 4 mS that senses its own terminal voltage. A source that passes I = 0.004·V across itself acts like a resistor of 1/0.004 = 250 Ω. Divider: V = 1.5 × 250/350 = **1.07 V**.
- Worked example 3 (USC 2026 B Q33a): a VCCS with k = 2 S is controlled by a 4 V battery. It pulls 2 × 4 = 8 A down out of node A through 25 Ω. V_A = −8 × 25 = **−200 V**.
- Traps:
  - Never turn off a dependent source in superposition.
  - "S" (siemens) and "℧" (mho) are the same unit: A/V.
  - Watch the sign of the control voltage. The USC 2026 Q33c key seems to flip it (see "Mistakes").
- Calculator: calculate — type: gain k, control (V or I), KVL/KCL coefficients.

22. Hands-on: mystery resistor in a bridge from voltages only

- Scope: ✅ States-Nationals (the series and parallel versions are ✅ IN at every level; rule II.c.i "Determine the value of a mystery resistor in a circuit using only voltage measurements").
- Where seen: MIT Invitational 2019 C (INV) lab S1-1c (hidden R_{10} across the middle of a bridge with 10 k / 1 k / 1 k / 10 k arms; key 10.7 kΩ, setup sheet says 4.7 kΩ).
- How to solve:
  1. Measure the battery voltage and the two middle node voltages V_a and V_b.
  2. Find the current in a known arm from Ohm's law.
  3. KCL at node a gives the current through the mystery resistor.
  4. R_x = (V_b − V_a) ÷ that current.
- Worked example (not seen in a test, built like the MIT board): 9 V top. Left: 10 kΩ from the top to a, 1 kΩ from a to ground. Right: 1 kΩ from the top to b, 10 kΩ from b to ground. R_x joins a and b. You read V_a = 1.845 V and V_b = 7.155 V.
  - Current down the left 1 kΩ = 1.845 mA. Current into a from the top = (9 − 1.845)/10 000 = 0.7155 mA.
  - KCL at a: current in from R_x = 1.845 − 0.7155 = 1.1295 mA.
  - R_x = (7.155 − 1.845)/0.0011295 = **4.70 kΩ**.
- Traps: Small rounding in the meter readings changes R_x a lot. Keep 3 decimals.
- Calculator: calculate — type: V_s, V_a, V_b, known arm resistors.

23. Diode or LED inside a KCL/KVL problem

- Scope: ✅ States-Nationals (diodes and LEDs are ✅ IN, rule 3.c.vii; KCL makes it S/N).
- Where seen: not seen in a test as a nodal problem. Single-loop LED questions are common (MIT Invitational 2019 C (INV) Q42-43). The constant-drop model is in prior topics B_topics §7.
- How to solve:
  1. Guess which diodes are on. An "on" diode is a fixed voltage drop (0.7 V for silicon, about 1.8-3.3 V for an LED).
  2. Solve the circuit with KCL/KVL.
  3. Check: each "on" diode must carry forward current. If not, redo the guess with it off (an open).
- Worked example: 5 V through 100 Ω to node N. From N: an LED (2.0 V drop) + 150 Ω to ground, and 300 Ω to ground.
  - KCL: (5 − N)/100 = (N − 2)/150 + N/300 → N = **3.17 V**.
  - LED current = (3.17 − 2)/150 = **7.8 mA** (positive, so the LED is on ✓). 300 Ω current = 10.6 mA. Source current = 18.3 mA.
- Traps: Never let an LED carry "negative" current. That means it is off.
- Calculator: calculate — type: source, resistors, diode drop; then check the sign.

24. Choose the method and check the two-equation limit

- Scope: ✅ States-Nationals (choice); ❌ OUT if the circuit really needs 3 or more equations (rule 3.e.i).

- Where seen: ATX Regional 2019 B/C (REG) Q18 (count nodes and meshes; pick the method). UT Invitational 2019 B (INV) Q37 = fall 2019 C Q33 (3-node nodal, ❌ OUT; key also wrong). UF Northern Regional 2019 C (REG) Q40 (3 unknowns, ❌ OUT). MIT 2020 C (INV) Q37, Q39, Q40, Q43 (3+ unknowns, ❌ OUT). Florida State 2021 B (STATE) Q48 (4 unknowns, ❌ OUT).
- How to solve:
  1. Count nodal unknowns: essential nodes − 1 − (nodes fixed by batteries) − (one per supernode).
  2. Count mesh unknowns: meshes − (one per current source).
  3. Pick the smaller number. A 2027 B test must give a circuit where one of them is 2 or fewer.
  4. Look for shortcuts: a current source in series with a resistor, a balanced bridge, symmetry.
- Worked example: a circuit has 4 essential nodes, and one battery goes from a node to ground. Nodal unknowns = 4 − 1 − 1 = **2**. That is allowed. Mesh analysis needs 3 meshes. Pick nodal.
- Traps: If your setup has 3 unknowns, look again. There is usually a fixed node or a supernode you missed.
- Calculator: look up — type: count nodes, count meshes.

Out of scope (skip)

- ❌ **Thevenin and Norton equivalents**, and **source transformation** (rule 3.d: "Norton/Thevenin equivalents"). Seen: ATX Regional 2020 B Q62 (Norton), ATX Regional 2019 B/C Q17 and UT fall 2019 C Q34 and Q36, Florida State 2021 B Q47, MIT 2020 C Q41, GullSO 2025 C Q42/44. USC 2026 B Q32d's key uses source transformation, but plain nodal analysis gives the same answer, so the question itself is ✅.
- ❌ **Three or more simultaneous equations** (rule 3.e.i). Seen: UT Invitational 2019 B Q37, UF Northern Regional 2019 C Q40, MIT 2020 C Q37/39/40/43, Florida State 2021 B Q48.
- ❌ **Matrix methods** for solving (rule 3.e.v bans matrix algebra as a requirement). Use substitution. Your calculator may still solve the pair.
- ❌ **Op-amp circuits** (not a listed device; rule 3.d "non-linear devices (beyond those listed)" and "semiconductors (beyond those listed)"). Seen: Florida State 2021 B Q46, ATX Regional 2021 C Q49 (KCL "fails" around an op-amp because of hidden supply pins). The idea "every terminal counts in KCL" is still useful.
- ❌ **Capacitors or inductors in the circuit** (rule 3.d: capacitance, inductance). Seen: USC 2026 B Q32c had a 1 μF capacitor in parallel; after a long time the capacitor carries no current, so the question still reduces to type 7 (⚠️).
- ❌ **When does KVL fail (changing magnetic flux)** (Florida State 2021 B Q17): magnetism and inductance are not in 3.c.
- ❌ **Delta-Y (Δ-Y) transformation** when it needs 3 equations (MIT 2020 C Q38, PA Central Q4). ⚠️ only if a test gives the formulas.
- ❌ **AC circuit analysis** with phasors (rule 3.d: AC circuit theory, frequency analysis; 3.e.v: complex numbers).

Mistakes in tests and keys

- **UT Invitational 2020 B (INV) Q13:** the key says KCL conserves energy. Correct: **charge**.
- **North Pocono 2020 C (INV) Q28:** key 1.8, 2.2, 0.4 A. The printed figure has the batteries facing **opposite ways** (9 V: + at the bottom; 3 V: + at the top; checked at 400 dpi on 2026-10-19). Then **I(R1) = 1.394 A, I(R2) = 0.676 A, I(R3) = 0.718 A**. App: `N V1–0 9, 1–2 5, 2–0 3, V0–3 3, 3–2 7`. The key looks like 9/5 and 3/7, as if each battery drove only its own resistor. No battery direction gives the key.
- **BullSO 2026 C (INV) SA8:** the key simplifies mesh 2 to "$-2I_1 + 10I_2 = 25$". It should be **= 15**. The key's final answers (5 A, 2.5 A) are right.
- **USC Invitational 2026 B (INV) Q33c / C Q37c:** key −4.09 V. I read the figure at 600 dpi and get **−15.0 V** (I = −0.100 A). The key seems to flip the sign of the control voltage. PLAUSIBLE, not proven.
- **ATX Regional 2020 B (REG) Q64 / C Q72:** as drawn, the two node equations contradict each other. There is no solution. The key's "186.8 A" for a voltage should be in V. If the 3I_x arrow pointed down, I_x = 2.73 A and V_R3 = 13.7 V.
- **UT Invitational 2019 B (INV) Q37 / fall 2019 C Q33:** key 2.99 A, 38.1 V. Correct: **3.24 A, 38.2 V**. The key's V_B fails its own node equation. (3 unknowns, ❌ OUT anyway.)
- **UT Invitational 2019 B (INV) Q31:** the question mixes "R3" (4 Ω) and "the 3 Ω." The key answers both.

- **UT Invitational 2020 B (INV) Q70:** "voltage across R1," but no part is labelled R1. The key means the 3 Ω.
- **ATX Regional 2021 B (REG) Q46:** the key says KVL; KCL also works for that circuit.
- **National supervisor generator Level 13 (superposition):** wrong formula in the current-source half. With all 10 Ω it gives I_3 = 0.6 A, I_4 = 1.6 A, I_bat = −0.4 A instead of 0, 1.0 and −1.0 A. It also adds V_bat into V_Is (22 V instead of 20 V).
- **National supervisor deck 10:** the KVL slide is titled "Kirchhoff's Voltage Law (KCL)."
- **UT Invitational 2020 B (INV) Q68:** key 8.94 V; exact 8.950 V (rounding only).

Not confirmed

- **No real 2026 National Tournament Circuit Lab test was found.** The "nat26" files are the USC **Invitational** of January 2026, not the May 2026 Nationals. Where I looked: every `usc26_*` file header and the soinc store note (Test Packet 2027 Div B, US$14). The only STATE test found is Florida State 2021 B.
- BullSO 2026 C SA6-SA8: the figures are images that are not in the text file. I used the key's equations. I did not render the figures.
- UT Invitational 2019 B Q31: I rebuilt the circuit from the extraction's node voltages (9.6 V, 17.6 V). It matches both key numbers, but I did not render the figure.
- ATX Regional 2021 B FRQ2: the circuit comes from the extraction's description; I did not render the figure. The node answer (15 V) matches the key exactly.
- USC 2026 B Q32b and Q32d, UT Invitational 2020 B Q71-73, UT Invitational 2019 B Q34 and Q38, NY Lake Erie 2018 Q26-27, UMSO 2019 Q26, Florida State 2021 B Q49, Northview FR6: the answers come from the earlier extraction files, which checked them in Python. I did not re-solve them from the figures here.
- ATX Regional 2019 Q18 (count nodes and meshes): the figure is too unclear to confirm "3 nodes, 5 meshes."
- MIT 2019 lab S1-1c: the key (10.7 kΩ) and the setup sheet (4.7 kΩ) disagree. It needs the real board.
- Whether a 2027 States/Nationals supervisor will use dependent sources or superposition: the rules are silent. I found no rules clarification about them on soinc.org.
- Kirchhoff's 1845 date and Königsberg: from the prior history file (I_bank, B_topics), not re-checked online here.

Sources

- Rules: 2027 Science Olympiad Div B Rules Manual, Circuit Lab, rules 3.c.ix, 3.d, 3.e (page B12), and II.c.i (hands-on). See Circuit Lab - Rules 2027 Ⓡ.
- Tests (extraction files in `~/Daniel/circuitlab/research/tests/` and `prior/G_tests.md`): USC Invitational 2026 B and C; ATX (UT Austin) Regionals 2019 B/C, 2020 B/C, 2021 B/C; UT Invitationals fall 2018, 2019 B, fall 2019 C, 2020 B, fall 2025; Florida State 2021 B; MIT Invitationals 2019 C, 2020 C, 2021 C; UF Northern Regionals 2019 B/C, 2020 B/C; NY Lake Erie/Niagara 2018 C; North Pocono 2020 C; UMSO 2019; F-M 2018; BullSO 2026 C; Berkeley 2026 B; Pembroke Hill 2025 B; Princeton 2026 C; Northview Sylvania 2025 C; GullSO 2025 C; Rickards 2020 B and 2026 B.
- Figures I rendered and checked for this note: UF Northern Regional 2020 C page 7 (Q27-30); UT Invitational 2020 B pages 13-14 (Q66-71); ATX Regional 2020 B pages 11-12 (Q59-62).
- National supervisor practice decks 4, 10, 11 and the homework generator Levels 7, 11, 12, 13 (`prior/F_slides.md`).
- Textbook methods: OpenStax *College Physics 2e*, chapter 21 (Kirchhoff's rules, sign conventions); Kuphaldt, *Lessons in Electric Circuits*, Vol. I (DC), chapters on network analysis (branch current, mesh current, node voltage, superposition); All About Circuits, "Network Analysis" chapter (same material online).
- All numbers in this note were checked in Python on 2026-10-16.

TAB A6

Switches, Relays, Fuses, Breakers, GFCI

Switch Types 421
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Circuit Breakers 451
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Each note: key facts, formulas, then question types from Easy to Nationals with worked examples.

Switch Types

A **switch** is a part that opens or closes a path for current. This note covers how switches are named, drawn and used in circuits.

Almost all of this note is ✅ **IN at every level**. Rule 3.c.vi names "switches". Rule II.c.vii (hands-on) asks you to wire a 3-way switch with SPDT switches. Switch logic also ties to rule 3.c.viii (logic).

Key facts

- **Closed** switch = the contacts touch. Current can flow. An ideal closed switch has 0 Ω and 0 V across it.
- **Open** switch = the contacts are apart. No current flows. An open switch in a working circuit has the full "missing" voltage across it.
- A **pole** is one separate circuit that the switch controls. Each pole has its own moving contact, called the **common (COM)**.
- A **throw** is one position that a pole can connect to.
- **SPST** (single pole, single throw): a plain on/off switch. 2 terminals.
- **SPDT** (single pole, double throw): one common that connects to one of two terminals. 3 terminals. An electrician calls it a **"3-way" switch**.
- **DPST** (double pole, single throw): two SPSTs moved by one lever. 4 terminals. Used to cut both wires of a supply at once.
- **DPDT** (double pole, double throw): two SPDTs moved by one lever. 6 terminals. Used to reverse a motor or swap polarity. Wired as a crossover, it becomes a **"4-way" switch**.
- Terminal count for a switch with p poles and t throws: $p \times (t + 1)$ when $t \geq 2$, and $2p$ when $t = 1$. So SP3T has 4 and TPTT (triple pole, triple throw) has 12.
- **Latching** (maintained) switches stay where you put them. Examples: toggle, rocker, slide, rotary, DIP switch.
- **Momentary** switches go back by spring when you let go. Example: a push button.
- **Normally open (NO)**: the contacts are open until you press. A doorbell button is NO.
- **Normally closed (NC)**: the contacts are closed until you press. A fridge-door light switch is NC. The door pushes the button when closed, so the light goes off.
- "Normal" means the rest state, with nobody pressing.
- A **push-button symbol** shows the contact bar above (NO) or touching (NC) the two terminals, with a small "plunger" line.
- Other types you may be shown: **rotary** (selector with many positions), **DIP** (tiny on/off switches in a row on a circuit board), **limit switch** (pressed by a moving machine part), **micro switch** (small snap-action switch), **reed switch** (two metal reeds in a glass tube that close near a magnet), **thermal switch** (a bimetal strip that bends with heat; a thermostat).
- A common **4-pin tactile push button** (breadboard type) has its pins joined in pairs inside. The two pins on each side are always connected. Pressing the button joins the two pairs. Electrically it is SPST-NO.
- **Switch logic** (Claude Shannon, 1937): switches in **series** act like **AND**. Switches in **parallel** act like **OR**. An **NC** contact acts like **NOT** of its button.
- A switch that **shorts** a part (in parallel with it) turns that part **off** when closed. This gives NOT-type behavior.
- **Ratings:** a switch has a maximum current and a maximum voltage. The DC rating is often lower than the AC rating. A DC arc does not die by itself at a zero crossing, so it is harder to break.
- **Safety:** a switch (and a fuse) belongs on the **hot** wire. Then "off" removes voltage from the load.
- **Break-before-make** SPDT switches open the old contact before closing the new one. **Make-before-break** switches close the new one first. ⚠️ Not named in the rules.

Formulas

What	Formula	Notes
Closed ideal switch	$R = 0$, $V = 0$	Acts as a wire
Open ideal switch	$R = \infty$, $I = 0$	Gets the full gap voltage

What	Formula	Notes
Terminals	p(t + 1) for t ≥ 2; 2p for t = 1	p poles, t throws
Switch positions	2^n for n two-position switches	Rows in a truth table
Series switches	L = A · B	AND
Parallel switches	L = A + B	OR
NC contact or shorting switch	L = A′	NOT
Rating check	I_load = P/V ≤ switch rating	Same for the voltage
Pull-down current when pressed	I = V_supply / R_pull	⚠️

Question types

Easy

1. Name the switch from a description

- Scope: ✅ IN.
- Where seen: Berkeley 2026 B (INV) Q24 (red LED off, green LED on with one push → SPDT). UF Northern Regional 2020 C (REG) Q15 (one common, two throws → SPDT / "3-way"). Rickards 2021-season B (INV, Dec 2020) Q8 and C Q10 (3-way switches use → SPDT). National supervisor practice deck (CTRL-SWITCH-TYPES: "turns on one thing or another, not both" → SPDT).
- How to solve:
  1. Count how many separate circuits it controls. That is the pole count.
  2. Count how many places each pole can go. That is the throw count.
  3. "One OR the other" = double throw. "On or off" = single throw.
  4. "Two circuits at once" = double pole.
- Worked example: "One lever turns a motor on in forward or in reverse, with both motor wires swapped." Two wires switch together (2 poles). Each goes to one of two places (2 throws). Answer: **DPDT**.
- Traps: "3-way" does not mean 3 positions. It is an SPDT switch with 3 terminals. "4-way" is a DPDT wired as a crossover.
- Calculator: look up — type: description words (poles, throws, "or", "both").

2. Identify a switch from its symbol or picture

- Scope: ✅ IN. Reed switch symbol ⚠️ (it is worked by a magnet; the symbol alone is fine).
- Where seen: Princeton 2026 C (INV) MC8. MIT 2021 C (INV) Q11 (symbol → DPTT, key D). GullSO 2026 C (INV, Dec 2025) Q25. Rickards 2026 C (INV) Q25 (reed / rotary / toggle / push button). ATX Regional 2026 B (REG) MC21 (reed switch) and MC22 (rotary switch). Cobra Invitational 2020 B (INV) Q28 (SPST-NO relay, DPST, rheostat, potentiometer, thermal switch, fuse). F-M Invitational 2018 (INV, division not stated) Q7 (glass capsule picture → reed switch). UF Northern Regional 2020 B (REG) Q21e (switch). JC Booth Invitational 2011 B (INV, Shock Value) Q1 (draw switch and push button). BullSO 2026 B (INV) SA 1e (open switch).
- How to solve:
  1. Count the moving blades. Each blade is one pole.
  2. Count the fixed contacts each blade can reach. That is the throw count.
  3. A dashed line joining blades means they move together (more poles).
  4. A small T-shaped plunger means a push button. Bar above = NO. Bar touching = NC.
  5. A circle of contacts with one arm = rotary. A blade inside a tube or oval = reed.
- Worked example: two blades joined by a dashed line, each with three fixed contacts. Poles = 2, throws = 3. Answer: **DPTT** (MIT 2021 Q11).
- Traps: Do not count the common terminal as a throw. A relay symbol has a coil box next to the contacts; without the coil it is a plain switch.
- Calculator: look up — type: blades, contacts per blade, plunger yes/no.

3. Count the terminals, or list the possible switch names

- Scope: ✅ IN.

- Where seen: Rickards 2026 C (INV) Q8 ("which are possible switches": SPST, DPDT, SPDT, TPTT → all of the above). Prior bank (I_bank D1) for terminal counts.
- How to solve:
  1. For each pole there is one common.
  2. Add one terminal per throw (single throw adds 1 contact, so 2 per pole).
  3. Multiply by the number of poles.
- Worked example: how many terminals on a TPTT switch? Each pole: 1 common + 3 throws = 4. Three poles: **12 terminals**. Check: SPST 2, SPDT 3, DPST 4, DPDT 6, SP3T 4.
- Traps: Any pole/throw count is "possible" (triple pole, triple throw exists). Do not reject a strange name just because it is rare.
- Calculator: calculate — type: poles, throws.

4. Momentary NO/NC: what happens when you press it

- Scope: ✅ IN.
- Where seen: MIT 2021 C (INV) Q10 (momentary NC DPST pressed → connections broken: 2; key B).
- How to solve:
  1. NC means closed at rest. Pressing opens it.
  2. NO means open at rest. Pressing closes it.
  3. Multiply by the number of poles.
- Worked example: a momentary NC DPST switch is pressed. Both poles open. **2 connections break.** When released, both close again.
- Traps: "Momentary" means it returns. The change lasts only while pressed.
- Calculator: look up — type: NO/NC, poles.

5. Switches as logic: series = AND, parallel = OR

- Scope: ✅ IN (rule 3.c.vi and 3.c.viii).
- Where seen: national supervisor practice deck (LOGIC-SWITCH-GATE: lamp in series with switches A and B → AND). Princeton 2020 C (INV) Q15 and MIT 2020 C (INV) Q19-20 (switch pictures → gates). See also 3-Way and 4-Way Switching A6.
- How to solve:
  1. Let 1 = switch closed, 1 = lamp on.
  2. A path through series switches needs all of them closed: AND.
  3. Parallel branches need any one closed: OR.
  4. Mixed networks: write series parts as products and parallel parts as sums.
- Worked example: switches A and B in parallel, and that pair in series with C. Lamp = **(A + B) · C**. With A = 0, B = 1, C = 1, the lamp is **on**.
- Traps: A switch across the lamp (a shorting switch) gives NOT. Pressing it turns the lamp off.
- Calculator: look up — type: network as series/parallel groups.

6. Match a switch to its job

- Scope: ✅ IN.
- Where seen: Princeton 2026 C (INV) Matching Part 2 (microwave door → switch). Cobra Invitational 2020 B (INV) station S3 (real part → limit switch, 5 A 250 V).
- How to solve:
  1. A door or a moving part pressing a contact = limit switch or micro switch.
  2. A "turns off when pushed" job = NC button.
  3. Choosing one of many settings = rotary switch.
  4. Board settings set once = DIP switch.
- Worked example: "A microwave must stop when the door opens." Use a door **switch** (a limit/interlock switch) in series with the power. Door open → switch open → no current.
- Traps: If the choices include "relay", only pick it when an electric signal (not a hand or a door) works the switch.
- Calculator: look up — type: job keyword.

Medium

7. Open or close one switch: what happens to each bulb

- Scope: ✅ IN.
- Where seen: SO MYSO Div B practice test, Nov 2020 (practice, about REG level) Q25-30 (SW1 open → bulb 1 dimmer, bulb 2 zero, bulb 3 brighter; SW2 shorts a parallel group → bulb 1 brighter, bulb 3 out). UT Invitational 2019 B (INV, 2020 season) Q33 and UT Invitational fall 2019 C (INV) Q29 (power in R1 when S1 adds a branch: "same" with an ideal source and no series R; "decreases" with a series R3). For networks with two sources, see Kirchhoff Analysis (States-Nationals) [A5].
- How to solve:
  1. Redraw the circuit for the new switch state. Replace a closed switch with a wire and an open switch with a gap.
  2. Find the new total resistance. Lower total R → more total current.
  3. Parts in series with the source see the total current change.
  4. Parts in parallel with a closed switch are shorted: 0 V, 0 A, off.
  5. Parts in an opened branch get 0 A.
- Worked example: 12 V ideal source. Bulb A (6 Ω) in series with two 6 Ω bulbs B and C in parallel. Closing a switch across B || C shorts them. Before: R = 6 + 3 = 9 Ω, I = 1.33 A, P_A = 10.7 W. After: R = 6 Ω, I = 2.0 A, P_A = **24 W (brighter)**. B and C go **out**.
- Traps: With an ideal source, a branch directly across the source does not change when you add another parallel branch. Only a series resistor makes the other parts change.
- Calculator: calculate — type: circuit for each switch state (series/parallel builder).

8. Calculate currents for every switch state

- Scope: ✅ IN.
- Where seen: Cobra Invitational 2020 B (INV) Q17-19 (switched ladder, E = 120 V, R1 = 20 Ω, R2 = 10 Ω → 3 A, 3.75 A, 3.9375 A). National supervisor homework generator "Level 9 Switches" (four S1/S2 states).
- How to solve:
  1. List every switch state (2^n states for n switches).
  2. Redraw and find R_eq for each.
  3. I_total = V / R_eq. Then split currents with Ohm's law.
- Worked example (Cobra Q17-19): E = 120 V. A 20 Ω resistor feeds rung 1 (20 Ω, switch S1). Then 10 Ω leads to rung 2 (20 Ω, S2), then 10 Ω to rung 3 (20 Ω, S3).
  - S1 only: R = 20 + 20 = 40 Ω, I = **3.00 A**.
  - S1 and S2: R = 20 + 20 || 30 = 32 Ω, I = **3.75 A**.
  - All three: R = 20 + 20 || (10 + 20 || 30) = 30.476 Ω, I = **3.9375 A**.
- Second example (generator Level 9): V = 12 V, R1 = 100, R2 = R3 = 200, R4 = 300 Ω. S1 adds R2 || R3; S2 adds R4 across the branch. States: both open 300 Ω, 40 mA; S1 only 200 Ω, 60 mA; S2 only 150 Ω, 80 mA; both 120 Ω, **100 mA**.
- Traps: Each closed switch adds a path, so the total current only goes up. If your answer goes down, check the drawing again.
- Calculator: calculate — type: V, resistor list, network for each state.

9. Fill a switch-lamp ON/OFF table

- Scope: ✅ IN (truth tables, rule 3.c.viii).
- Where seen: UT Invitational 2019 B (INV, 2020 season) Q43 (3 SPDT switches, 3 lamps, 12 V 10 W: L1 = S1, L2 = S1·S2, L3 = S1·S2·S3). UT Invitational fall 2019 C (INV) Q44 (S1 feeds all; S2 feeds L2, L3; S3 feeds L3). UT Invitational 2019 B (INV) Q49 hands-on (3 push buttons, 4 LEDs from a breadboard photo).
- How to solve:
  1. Make 2^n rows. Count in binary: 000, 001, ... 111.
  2. For each row, trace the path from + to each lamp.
  3. A lamp is on only if a closed path reaches it and returns.

- Worked example: S1 feeds all lamps, S2 is after S1 and feeds L2 and L3, S3 is after S2 and feeds L3. Row S1 S2 S3 = 1 1 0: L1 on, L2 on, **L3 off**. Row 0 1 1: **all off** (S1 cuts everything).
- Traps: In UT 2019 B Q43, one switch position as drawn shorts the battery. Write "short" if you see a path from + to – with no load.
- Calculator: calculate — type: number of switches, lamp expressions.

10. Check a switch rating against a load

- Scope: ✅ IN.
- Where seen: not seen in a test. Prior bank I_bank D8. Real part labels: Cobra Invitational 2020 B (INV) S3 (limit switch 5 A 250 V).
- How to solve:
  1. Find the load current: I = P / V.
  2. Compare with the switch current rating.
  3. Check the voltage rating too. Use the DC rating for DC circuits.
- Worked example: a switch is rated 6 A at 125 VAC. It controls a 1000 W, 120 V heater. I = 1000/120 = **8.33 A > 6 A → overloaded**. Use a 10 A or 15 A switch.
- Traps: A "250 VAC" rating does not mean "250 VDC". DC ratings are often much lower.
- Calculator: calculate — type: P, V, switch amps.

11. Where to put the switch; draw it in a schematic

- Scope: ✅ IN.
- Where seen: Shock Value team re-tryout ~2012 B (practice) IV-1-3 (bulb, diode, source, switch, fuse in a loop). UT Invitational fall 2025 B/C (INV) lab 5b (source, resistor, SPST, LED in series). Prior bank I_bank D7 (hot side).
- How to solve:
  1. Draw one loop: source → fuse → switch → load → back.
  2. Put the fuse nearest the source. Put the switch on the + (hot) side.
  3. Use the right symbols (see CL 9 - Meters, Breadboards and Hands-On Tasks Ⓣ).
- Worked example: "Draw a 9 V battery, a switch, a 470 Ω resistor and an LED in series." Answer: + → switch → 470 Ω → LED anode; LED cathode → –. Switch open: I = 0. Switch closed with a red LED (2.0 V): I = (9 – 2)/470 = **14.9 mA**.
- Traps: In a series loop the order does not change the current. It does change safety: a switch on the neutral side leaves the load "live" when off.
- Calculator: calculate — type: V, R, Vf.

12. Find a push button's internal wiring

- Scope: ✅ IN (switches; hands-on II.b skills).
- Where seen: USC Invitational 2026 B and C (INV, top tier) lab "Light Switch" part a (4-pin button, use a battery-LED-resistor as a continuity tester). UT Invitational fall 2025 B/C (INV) lab 5e ("SPST", "pins on either side").
- How to solve:
  1. Build a tester: battery → resistor → LED → two probe wires.
  2. Touch each pair of pins with the button released. Then repeat with it pressed.
  3. Pairs that light when released are always joined.
  4. Pairs that light only when pressed are switched.
- Worked example: on a 4-pin tactile button, pins 1-2 always light and pins 3-4 always light. Pins 1-3 light only when pressed. Answer: **1-2 and 3-4 are joined inside; the button bridges the two pairs (SPST-NO)**.
- Traps: If you use two pins that are always joined, the "button" never does anything. Put the button across the breadboard's center gap and use pins on opposite pairs.
- Calculator: look up — type: continuity results per pin pair.

13. Voltage across a switch, and the coin (jumper) test

- Scope: ✅ IN (troubleshooting, rule II.c.vi).
- Where seen: UT Invitational fall 2025 B/C (INV) lab (measure V_R, V_SW, V_LED and check they add to the supply). Cobra Invitational 2020 B (INV) station S1 (broken LED circuit; the switch is wired wrong; bridge its

terminals with a coin → LED on).

- How to solve:
  1. A closed good switch reads about 0 V.
  2. An open switch (or a broken one) reads the full gap voltage. In a series loop that is the source voltage.
  3. Jumper the switch with a wire or coin. If the circuit now works, the switch (or its wiring) is the fault.
- Worked example: 9 V battery, switch, 1 kΩ, red LED (2.0 V). The switch is "on" but the LED is dark. You read 9.0 V across the switch. Answer: **the switch is open inside (or wired to the wrong pins)**. With a coin across it, I = 7/1000 = 7.0 mA and the LED lights.
- Traps: Kirchhoff's voltage idea: the drops still add to 9 V. An open part takes all of it.
- Calculator: calculate — type: supply V, measured drops.

Hard

14. DPDT motor reversing (polarity swap)

- Scope: ✅ IN (the motor is only a load; how it spins is magnetism and ❌ OUT).
- Where seen: Shock Value team re-tryout ~2012 B (practice) IV-1-2 (2 × 9 V, motor, DPDT reverses the motor).
- How to solve:
  1. Connect the motor to the two commons.
  2. Connect + to throw 1 of pole A and throw 2 of pole B.
  3. Connect – to throw 2 of pole A and throw 1 of pole B (the cross wires).
  4. Position 1: + reaches motor terminal A. Position 2: + reaches motor terminal B.
- Worked example: commons C_A, C_B to motor. Up position: C_A → +, C_B → –. Down position: C_A → –, C_B → +. **Current reverses, so the motor reverses.** The cross wires make it work.
- Traps: Without the cross wires, both positions give the same polarity. Check that no position joins + and – directly (a short).
- Calculator: look up — type: commons, up/down terminals.

15. Switch-and-LED logic in a hands-on (shorting switches)

- Scope: ✅ IN (rule II.c.v: logic from a truth table).
- Where seen: USC Invitational 2026 B and C (INV) lab "Light Switch" parts c-e (LED on only when both buttons released = NOR; LED on iff A pressed and B released = A·B′; one button dims the LED).
- How to solve:
  1. A button in series with the LED = "on when pressed" (A).
  2. A button across the LED (after the series resistor) = "off when pressed" (A′).
  3. Two shorting buttons in parallel = NOR. Two series buttons = AND.
  4. Always keep a resistor in series with the battery, so a shorting button shorts only the LED.
- Worked example (USC part c): 9 V, 510 Ω, red LED. Both buttons are in parallel across the LED. Pressing either one shorts the LED. Then the resistor carries 9/510 = **17.6 mA** and makes $9^2/510$ = **0.159 W**. That is safe for a 1/4 W resistor.
- Second example (part e, dimming): add a 510 Ω shunt through a button across the red LED (2.0 V). Without the shunt the LED gets (9 – 2)/510 = 13.7 mA. With it, the shunt takes 2/510 = 3.9 mA, so the LED gets **9.8 mA (about 29% less)**. A blue LED (3.0 V) drops from 11.8 mA to 5.9 mA.
- Traps: A shorting button with no series resistor shorts the battery. Check the resistor power in the shorted state.
- Calculator: calculate — type: V, R_series, Vf, R_shunt.

16. One button toggles two LEDs

- Scope: ✅ IN.
- Where seen: ATX Regional 2026 B (REG) hands-on (red on and green off when up; green on and red off when pressed). Berkeley 2026 B (INV) Q24 (same idea as MC → SPDT).
- How to solve:
  1. Use an SPDT (3-pin) switch. + goes to COM.
  2. NC → R1 → red LED → –.
  3. NO → R2 → green LED → –.

- Worked example: 9 V, R1 = R2 = 390 Ω, red 2.0 V, green 2.1 V. Released: red gets (9 – 2)/390 = **17.9 mA**; green gets 0. Pressed: green gets (9 – 2.1)/390 = 17.7 mA; red gets 0.
- Traps: The ATX 2026 key "solution" does not work (see Mistakes). With only an SPST-NO button there is no clean answer.
- Calculator: calculate — type: V, R, Vf of each LED.

17. Diode-steered lamps: which lamps light for each switch

- Scope: ✅ IN (diode use, rule 3.c.vii, plus switches).
- Where seen: UF Northern Regional 2019 B (REG) Q1-3 (SW1 closed → all 3 lamps; only LAMP1 and LAMP2 on → SW2 and SW3).
- How to solve:
  1. Treat each diode as a one-way valve (arrow = allowed direction).
  2. For each closed switch, trace every path that goes with the arrows.
  3. For the reverse question, list which switch reaches each lamp, then pick the set.
- Worked example: SW1 feeds lamps 1, 2, 3 through three diodes. SW2 feeds only lamp 2. A diode stops SW2's current from going back to lamps 1 and 3. Close SW2 only: **only lamp 2 lights**.
- Traps: Without the diodes, every switch would light every lamp. The diodes are what make the switches separate. The drawing is needed; look at the arrows.
- Calculator: look up — type: switch → lamp list.

18. Bulb network with SPDT switches

- Scope: ✅ IN.
- Where seen: UT Invitational fall 2018 B/C (INV, no key) Q24a-b (compare brightness for S1 = W, S2 = Y, S3 = A; find positions that light all 5 bulbs in series).
- How to solve:
  1. For each SPDT, draw only the throw it is set to. Erase the other.
  2. Find series and parallel groups. Rank brightness by current (or power) in each bulb.
  3. For "make all bulbs light", look for a path that goes through every bulb.
- Worked example: identical bulbs. One position puts L1 and L5 in series with the source and L3 || L4 after them, with L2 bypassed. Brightness: **L1 = L5 > L3 = L4 > L2 (off)**.
- Traps: A bulb with both ends on the same node is shorted (off), even if it is not "cut out".
- Calculator: calculate — type: network for the chosen positions.

19. Pull-down (or pull-up) resistor for a switch input

- Scope: ⚠️ Borderline. Pull-down resistors are not named in the rules. The math is a plain DC divider, which is IN.
- Where seen: UT Invitational fall 2025 B/C (INV) lab 5g (Supply+ → switch → node Vo → resistor → Supply–).
- How to solve:
  1. Switch open: no current in the resistor, so Vo = 0 V (pulled down).
  2. Switch closed: Vo = supply (the switch is 0 Ω).
  3. Current when closed: I = V / R.
- Worked example: 5 V, 10 kΩ pull-down. Pressed: Vo = 5 V, I = **0.5 mA**. Released: Vo = **0 V**.
- Traps: A logic input with no pull resistor "floats" and reads at random. Without the resistor, pressing does nothing useful and releasing leaves an unknown value.
- Calculator: calculate — type: V, R_pull.

Nationals

20. Break-before-make vs make-before-break

- Scope: ⚠️ Borderline (not named; it is a switch property, so a top-tier writer could ask).
- Where seen: not seen in a test. Prior topics note B_topics §6a.
- How to solve:
  1. Break-before-make: for an instant, neither throw is connected.
  2. Make-before-break: for an instant, both throws are connected.

3. Ask: what happens to the circuit during that instant?

- Worked example: an SPDT selects one of two batteries for a load. A make-before-break switch joins both batteries for an instant. If they differ, a large current flows between them. Answer: **use break-before-make**.
- Traps: Rotary selector switches for meters use break-before-make for this reason.
- Calculator: look up — type: BBM/MBB, what the throws connect.

21. Contact bounce

- Scope: ⚠ Borderline. The fact is about switches (IN). Debounce circuits use capacitors or flip-flops, which are ❌ OUT (rule 3.d: capacitance, sequential logic).
- Where seen: not seen in a test. Kuphaldt, *Lessons in Electric Circuits* Vol. IV, "Switches" chapter (contact bounce).
- How to solve:
  1. Metal contacts spring apart a few times when they close.
  2. A slow lamp does not notice. A fast logic counter sees many presses.
- Worked example: "A push button feeds a counter. One press adds 3." Answer: **contact bounce**. A fix exists but is out of scope.
- Traps: Do not design a debounce circuit at Div B. Name the cause only.
- Calculator: look up — type: symptom.

Out of scope (skip)

- ❌ **Probability that parallel switches conduct** (Florida State 2021 B, ST, Q33: $1 - (1 - P)^N$). Probability is not in the math list (rule 3.e).
- ❌ **Transistor switches** (BJT or MOSFET as a switch). Rule 3.d: semiconductors beyond those listed.
- ❌ **Debounce circuits** with capacitors or latches. Rule 3.d: capacitance, sequential logic.
- ❌ **Capacitor networks with switches** (UT Invitational 2019 B Q30) and **RC switch transients** (USC 2026 B MC12; UT 2020 B Q75; UT fall 2019 C Q27). Rule 3.d: capacitance.
- ❌ **Transformer + switch "when does the lamp light"** (Cobra 2020 B Q27). Rule 3.d: inductance; rule 3.c lists no magnetism.
- ⚠ **How a reed switch's magnet works**: magnetism is not in rule 3.c. Naming the switch is fine.

Mistakes in tests and keys

- **ATX Regional 2026 B (REG) hands-on key:** the key powers the circuit from "a multimeter set to DC voltage" (a meter on DCV supplies no power). Its wiring keeps red always on and never lights green. It has no current-limiting resistors. The test says 30 points; the key says 40.
- **UT Invitational 2019 B (INV) Q43:** with S1 OFF, the drawing shorts the battery. The key ignores this.
- **USC Invitational 2026 lab:** not a key error, but note that the lab never uses a multimeter, although 2027 rule II.b says teams must know multimeters.

Not confirmed

- MIT 2021 Q10: the key letter is B. I took "B = 2" from the earlier extraction (G_tests.md). I did not see the option list (it is not in the text layer).
- Princeton 2026 MC8 and GullSO 2026 Q25: the symbols are images. I used the earlier extraction and did not render them.
- UT 2018 Q24 has no key. The answer is the extractor's own.
- Florida State 2021 has no switch-network question in this area except the OUT probability item. No STATE or NATIONAL test with these types was found. The real 2026 Nationals tests were not in the downloads.
- The contact-bounce source is from memory of Kuphaldt Vol. IV. I did not fetch the page this run.

Sources

- Rules: 2027 Div B Rules Manual, Circuit Lab, rules 3.c.vi, 3.c.viii, 3.d, 3.e and II.c.v-vii. See Circuit Lab - Rules 2027 [R].
- Test extraction files: `~/Daniel/circuitlab/research/tests/` (misc-a: Cobra, F-M 2018, North Pocono; misc-b: UF Northern Regional, JC Booth, Shock Value re-tryout; ut-a: UT fall 2025, ATX Regional 2026; ut-b and ut-c: UT 2018-2020, ATX Regional 2019; nat26: USC 2026; state: Florida State 2021, MYSO) and `prior/G_tests.md` (Princeton, MIT, GullSO, Rickards, Northview).
- Raw text checked this run: MIT 2021 C test (Q10, Q11), Berkeley 2026 B (Q24), Rickards 2026 C (Q8, Q25), Cobra 2020 B theory (Q17-19).
- National supervisor practice decks and homework generator (`prior/F_slides.md`).
- Textbooks: Kuphaldt, *Lessons in Electric Circuits*, Vol. IV (Digital), chapter "Switches" (pole/throw, NO/NC, contact bounce) and Vol. IV "Ladder Logic"; All About Circuits, "Switch Types"; OpenStax *College Physics 2e* ch. 21 (series/parallel circuits).
- Claude Shannon, "A Symbolic Analysis of Relay and Switching Circuits" (MIT master's thesis, 1937), for switch logic.
- All numbers checked in Python on 2026-10-16.

3-Way and 4-Way Switching

Multiway switching lets you turn one light on or off from two or more places. A hallway or a staircase is the usual example.

This whole note is ✅ **IN at every level**. Rule 3.c.vi names switches. Rule II.c.vii is a named hands-on task: "Wire a 3-way switch using SPDT switches." So expect it as a station.

Key facts

- A **3-way switch** is an SPDT switch: one **common** terminal and two **traveler** terminals. Electricians call it "3-way" because it has 3 terminals, not 3 positions.
- A **4-way switch** is a DPDT switch wired as a **crossover**. It has 4 traveler terminals. In one position it passes the two travelers straight through. In the other it swaps (crosses) them.
- **Two locations:** two 3-way switches. The common of switch 1 goes to the source. The common of switch 2 goes to the lamp. Two **traveler wires** join the matching throws of the two switches.
- The lamp is **on when both commons select the same traveler**. It is off when they select different travelers.
- Flipping **either** switch changes which traveler it selects. So every flip toggles the lamp.
- As logic, the two-switch circuit is **XNOR** (lamp on when A = B) or **XOR** (on when A ≠ B). Which one depends on how you label "up" = 1 on each switch.
- **Three or more locations:** always **two 3-way switches at the ends** plus **(n – 2) 4-way switches** in the middle. Any number of 4-way switches can be added.
- With n switches there are 2^n position combinations. The lamp is on for exactly half of them: 2^{n-1}.
- For any n, the lamp state is the **parity** of the switches. Each flip toggles it. It is the XOR (or XNOR) of all n switch positions.
- In house wiring, travelers run in a 3-wire cable (plus a bare ground). The switch goes on the hot wire. The neutral goes straight to the lamp.
- **Carter system:** an old way of wiring a 3-way where each switch connects a lamp wire to either hot or neutral. The lamp lights when the switches differ. When it is "off", both lamp terminals can be at hot voltage. The lamp looks off but is still live. The US National Electrical Code (NEC) has banned it since **1923**.
- **California (or "coast") 3-way:** a variant that runs hot and neutral to both switch boxes. It lets you add switched and unswitched loads near both switches without many extra wires. It is noted as useful for **long hallways**.
- The **traveler system** is the normal modern system described above.
- Hands-on version: use a battery, an LED with a series resistor, and two SPDT slide switches on a breadboard. No AC is allowed in hands-on (rule II.d).

Formulas

What	Formula	Notes
Switches for n locations	2 three-way + (n – 2) four-way	n ≥ 2
Total switches	n	One per location
Position combinations	2^n	Rows in the table
Combinations with the lamp on	2^{n-1}	Half
Two-switch logic	L = A ⊙ B (XNOR) or A ⊕ B (XOR)	By labeling
n-switch logic	L = A ⊕ B ⊕ C ⊕ ... (or its NOT)	Parity
Traveler wires	2 between each neighboring pair of switches	Plus common wires at the ends
LED resistor for a 3-way station	R = (V – V_f) / I	See LEDs A7

Question types

Easy

1. What switches do you need for two locations?

- Scope: ✅ IN.
- Where seen: Pembroke Hill 2026 B (INV) Q25 (two-location hallway light → SPDT). Northview Sylvania 2026 C (INV, Dec 2025) Q16 (two locations → 2 SPDT). MIT Invitational 2019 C (INV) Q38 (parts for two-switch light control → SPDT, key D). Rickards 2021-season B (INV, Dec 2020) Q8 and C Q10 (3-way switches use → SPDT). UF Northern Regional 2020 C (REG) Q15 (one common, two throws → "3-way switch").
- How to solve:
  1. Two locations → two 3-way switches.
  2. A 3-way switch = SPDT.
  3. More locations → add one 4-way (DPDT) per extra location.
- Worked example: "A light must be switched from both ends of a hall. What do you need?" Answer: **two SPDT (3-way) switches and two traveler wires**.
- Traps: DPDT is a distractor for two locations. SPST cannot work: two SPSTs in series = AND, in parallel = OR, and neither lets each switch toggle the light.
- Calculator: look up — type: number of locations.

2. What does a 3-way setup do?

- Scope: ✅ IN.
- Where seen: BullSO 2026 B (INV) Q24 (key C: "two switches, each able to change the state of a lightbulb"). BullSO 2026 C (INV) has the same item.
- How to solve:
  1. "3-way" names the switch type, not the number of switches.
  2. The setup has two switches and one light.
  3. Either switch can change the light.
- Worked example: choices "three positions" / "three switches" / "two switches, each able to change the bulb" / "three wires". Answer: **two switches, each able to change the bulb's state**.
- Traps: "Three switches" and "three positions" are the traps. The name comes from the 3 terminals.
- Calculator: look up — type: "3-way meaning".

3. Why does flipping either switch toggle the light?

- Scope: ✅ IN.
- Where seen: Berkeley 2026 B (INV) Q25 (key B: "the light is on only when both switches connect to the same traveler wire").
- How to solve:
  1. A path exists only if both commons are on the same traveler.
  2. One flip moves one common to the other traveler.
  3. That always changes "same" to "different" or back.
- Worked example: S1 on traveler 1, S2 on traveler 1 → on. Flip S2 → S2 on traveler 2 → **off**. Flip S1 too → both on traveler 2 → **on again**.
- Traps: "Each switch independently completes the circuit" is wrong. Neither switch alone controls the path.
- Calculator: look up — type: S1 traveler, S2 traveler.

Medium

4. Pick the correct 3-way wiring diagram

- Scope: ✅ IN.
- Where seen: ATX Regional 2021 B (REG) Q32 (pick the switch pair so either switch toggles the bulb → two SPDT with two travelers, key A). NY Lake Erie/Niagara 2018 C (INV, 2019 season) Q9 (4 drawings → travelers to travelers, key D).
- How to solve:
  1. Find each switch's common (the pivot of the blade).
  2. One common must go to the source. The other must go to the lamp.
  3. The two throws of switch 1 must go to the two throws of switch 2.
  4. Check all 4 positions. Exactly 2 must light the lamp, and no position may short the source.

- Worked example: a drawing joins common 1 to common 2 and puts the source across the throws of switch 1. In one position this shorts the source through the blade. **Reject it.** The right drawing has commons at the ends and travelers in the middle.
- Traps: A drawing where the lamp is in series with both commons and the travelers are crossed is still correct. Crossing the travelers only swaps XOR and XNOR.
- Calculator: look up — type: what each common and throw connects to.

5. Draw the 3-way schematic or design it in words

- Scope: ✅ IN.
- Where seen: Shock Value team re-tryout ~2012 B (practice) IV-1-1 (2 SPDT control one bulb). F-M fall 2019 Invitational (INV, division not stated) Q4 (room with 2 doors and 1 light). National supervisor practice deck 1 names "3 way light switch circuit" on its rules slide but does not teach it.
- How to solve:
  1. Draw the battery. From + draw a wire to common 1.
  2. Draw two wires (travelers) from the throws of switch 1 to the throws of switch 2.
  3. From common 2 go through the resistor and the lamp (or LED) back to –.
  4. Label S1, S2 and the travelers.
- Worked example: 9 V battery, red LED (2.0 V), target 20 mA. R = (9 – 2)/0.020 = 350 Ω. Use the next standard size, 390 Ω. Then I = 7/390 = **17.9 mA**.
- Traps: Put the resistor in the common-2 wire (or at the battery), not in one traveler. Otherwise the LED is bright in one "on" state and dim in the other.
- Calculator: calculate — type: V, Vf, I target.

6. A 3-way switch is most like which gate?

- Scope: ✅ IN (logic, rule 3.c.viii).
- Where seen: North Pocono Invitational 2020 C (INV) Q4 (key: XNOR). Princeton 2020 C (INV) Q15 (staircase light with 2 switches → XOR).
- How to solve:
  1. Write the 4-row table for S1, S2.
  2. Mark the lamp on when both select the same traveler.
  3. If "up" on both means the same traveler, the lamp = A XNOR B. If you label so "up" on one is the opposite traveler, it is XOR.
- Worked example: S1 S2 = 00 → on, 01 → off, 10 → off, 11 → on. That is **XNOR**. Relabel one switch and it becomes XOR.
- Traps: Both XOR and XNOR are correct, depending on the labels. If only one is offered, pick it. If both are offered, read the drawing to see which labels are used.
- Calculator: look up — type: lamp state for each row.

7. How many switches (and of which kind) for n locations?

- Scope: ✅ IN.
- Where seen: Northview Sylvania 2026 C (INV) Q16 (two locations). Prior bank I_bank D5 (three locations → two 3-way + one 4-way).
- How to solve:
  1. Ends: always two 3-way switches.
  2. Middle: n – 2 four-way switches.
  3. If asked for position counts: 2^n combinations; half light the lamp.
- Worked example: a light controlled from **5** places. Switches = 2 three-way + **3 four-way**. Combinations = 2^5 = **32**. Lamp on in **16**.
- Traps: Do not answer "five 3-way switches". A 3-way switch has only 3 terminals and cannot pass two travelers through.
- Calculator: calculate — type: n.

8. Troubleshoot a broken 3-way circuit

- Scope: ✅ IN (rule II.c.vi: troubleshoot a faulty circuit).
- Where seen: UT Invitational 2020 B (INV, 2021 season) Q30 ("switch A works only when B is down" → a traveler wire is broken).
- How to solve:
  1. List the four positions and which ones should light.
  2. Compare with what the lamp does.
  3. One broken traveler: only one of the two "on" states works. Each switch "works" only when the other is on the good traveler.
  4. Broken common wire: the lamp never lights.
  5. Shorted travelers (joined together): the lamp is always on (if powered).
- Worked example: traveler 2 is cut. On-states were (T1, T1) and (T2, T2). Only (T1, T1) still lights. With S2 on T1, S1 turns the lamp on and off. With S2 on T2, S1 does nothing. Answer: **a broken traveler**.
- Traps: A dead bulb gives "never lights", like a broken common. Test the bulb first.
- Calculator: look up — type: symptom.

9. HANDS-ON: wire a 3-way with two SPDT switches

- Scope: ✅ IN. This is official task II.c.vii.
- Where seen: Northview Sylvania 2026 C (INV) lab (2 SPDT, LED, resistor). 2027 rule II.c.vii.
- How to solve:
  1. Find each slide switch's middle pin. That is usually the common. Check with a continuity test if unsure.
  2. ▪ → resistor → common of S1.
  3. Outer pins of S1 → outer pins of S2 (two jumpers).
  4. Common of S2 → LED anode (long leg). LED cathode → –.
  5. Test all 4 positions. You should see on, off, off, on (or off, on, on, off).
- Worked example: 9 V, 390 Ω, red LED. Both "on" states give about **17.9 mA**. Both "off" states give **0 mA**.
- Traps: Check the LED direction first. Do not plug a slide switch's three pins into one breadboard column (they would all be shorted). Turn it so each pin is in its own column.
- Calculator: calculate — type: V, Vf, R.

10. What does a 4-way switch do inside?

- Scope: ✅ IN.
- Where seen: not seen in a test as its own question. Prior bank I_bank D5. Wikipedia "Multiway switching".
- How to solve:
  1. A 4-way switch is a DPDT with the throws cross-wired.
  2. Position 1: traveler in A → out A, in B → out B (straight).
  3. Position 2: in A → out B, in B → out A (crossed).
- Worked example: "Why does flipping the middle switch toggle the lamp?" Because swapping the travelers changes "same traveler" to "different traveler". **Each flip inverts the path.**
- Traps: A 4-way is not a fourth switch type. It is a DPDT. You can build one from a DPDT with two cross wires.
- Calculator: look up — type: position, input traveler.

Hard

11. Counting puzzle: the hotel windows (216)

- Scope: ✅ IN.
- Where seen: Northview Sylvania 2026 C (INV, Dec 2025) Q26 (6 rows of 5 rooms; any switch in a row toggles the whole row; A SPDT and B DPDT; find A × B; key D = 216).
- How to solve:
  1. Each row is one multiway circuit with 5 locations.
  2. Each row needs 2 three-way (SPDT) and 5 – 2 = 3 four-way (DPDT).
  3. Multiply by the number of rows.
- Worked example: A = 6 × 2 = 12. B = 6 × 3 = 18. **A × B = 216.**

- Traps: Choice 225 (15 × 15) and 144 (12 × 12) are traps. Rows are separate circuits, so do not try one giant circuit.
- Calculator: calculate — type: rows, switches per row.

12. Carter system: which positions light, and why it is banned

- Scope: ✅ IN (switch logic and hazard). The drawing uses mains wiring, but no AC theory is needed.
- Where seen: UT Invitational 2020 B (INV, 2021 season) Q29 (banned by the NEC in 1923 → Carter). North Pocono Invitational 2020 C (INV) Q21 ((a) lit when one switch is "on" and one "off"; (b) the lamp can look off but still be live).
- How to solve:
  1. Each Carter switch connects its lamp wire to hot or to neutral.
  2. The lamp lights only when one end is hot and the other neutral (switches differ).
  3. When both select hot, no current flows, but both lamp terminals are at hot voltage.
- Worked example: S1 → hot, S2 → hot. Lamp is dark. A person changing the bulb touches the socket shell and a grounded pipe. **They get shocked**, because the shell is at 120 V. This is why the NEC banned it in 1923.
- Traps: "Off" does not mean "safe". A safe circuit puts the switch on the hot wire only.
- Calculator: look up — type: S1 (hot/neutral), S2 (hot/neutral).

13. Name the multiway system

- Scope: ✅ IN (trivia about switches).
- Where seen: UT Invitational 2019 B (INV, 2020 season) Q14 and UT Invitational fall 2019 C (INV) Q14 ("which multiway switching system is notable for its usefulness for long hallways?"; choices Alternative, California, Carter, Traveler; key A "Alternative").
- How to solve:
  1. Carter = old, banned 1923, socket can stay live.
  2. Traveler = the normal modern system.
  3. California / coast / "alternative" = runs hot and neutral to both ends; good for hallways with extra loads.
- Worked example: "Long hallways" → the key says **Alternative**. Wikipedia describes the California ("coast") 3-way as the long-hallway system.
- Traps: The key and Wikipedia may use different names for the same idea. See Not confirmed.
- Calculator: look up — type: system name.

14. Wire a household 3-way with hot, neutral and ground

- Scope: ✅ IN (as a wiring diagram; no AC theory and no hands-on AC).
- Where seen: UT Invitational 2019 B (INV, 2020 season) Q44 (A/B/C/G terminals; add ground wires).
- How to solve:
  1. Hot (black) → common of switch 1.
  2. Travelers (red and black in the 3-wire cable) → the two traveler screws on both switches.
  3. Common of switch 2 → lamp. Neutral (white) → straight to the other lamp terminal.
  4. Bare/green ground → the green screw of each switch and the fixture box.
- Worked example: terminal letters A = common, B and C = travelers, G = ground. Answer: **source hot → A_1; B_1–B_2 and C_1–C_2; A_2 → lamp; neutral → lamp; G to G to ground**.
- Traps: Never switch the neutral. The ground never carries normal current.
- Calculator: look up — type: terminal labels.

15. SPDT selector: OFF / LOW / HIGH

- Scope: ✅ IN (close to II.c.vii).
- Where seen: UMSO 2nd Annual Invitational (INV, file labeled 2019, division not stated) lab (two 3-way switches, 9 V, LED, resistors; build OFF/LOW/HIGH; "many solutions").
- How to solve:
  1. S1 common to +9 V. One S1 throw is left empty (OFF).
  2. The other S1 throw feeds S2's common.
  3. S2's throws go through a small R (HIGH) or a large R (LOW) to the LED, then to −.

- Worked example: red LED (2.0 V). HIGH via 330 Ω: I = 7/330 = **21.2 mA**. LOW via 2.2 kΩ: I = 7/2200 = **3.2 mA**. OFF: **0**.
- Traps: Check that no position shorts + to −.
- Calculator: calculate — type: V, Vf, R_low, R_high.

Nationals

16. Three-location truth table (parity)

- Scope: ✅ IN.
- Where seen: Princeton 2020 C (INV) Q40 (one light, three switches; L = 1 when an even number of switches are on).
- How to solve:
  1. Make the 8-row table.
  2. Start at the given row (often 000 → on).
  3. Each row that differs by one switch flips the lamp.
  4. So the lamp = NOT (A ⊕ B ⊕ C) if 000 is on.
- Worked example: 000 on, 001 off, 010 off, **011 on**, 100 off, 101 on, 110 on, **111 off**. The circuit is 3-way, 4-way, 3-way.
- Traps: The lamp is not "on when 2 or more are up". It is parity, not majority.
- Calculator: calculate — type: n switches, starting state.

17. Design a switch network from a truth table (3-way + 4-way, two lights)

- Scope: ✅ IN (but very hard).
- Where seen: MIT Invitational 2020 C (INV, top tier) Q49 (three 3-way and one 4-way switch, 5 V DC, lights L1 and L2; 12 points).
- How to solve:
  1. Look for parity patterns: rows that differ in one switch and flip a light.
  2. A light that is pure parity of k switches needs a 3-way, (k – 2) 4-way, 3-way chain.
  3. A light that also needs a switch "off" to be on (an AND) can be fed through a throw of another switch.
  4. Check every row of the table at the end.
- Worked example (a smaller version you can check): S1 is an SPDT with its common on +. Its "0" throw feeds the common of S2. S2 and S3 are a 3-way pair, and S3's common feeds L1. S1's "1" throw feeds L2. Result: L2 = S1. L1 = S1′ · (S2 XNOR S3). Rows S1 S2 S3: 000 → L1 on, L2 off; 001 → both off; 011 → L1 on; 1xx → **L1 off, L2 on**. This is the "feed one network through another switch's throw" trick that MIT Q49 needs.
- Traps: The MIT table lists 15 rows and leaves out row 0101. It does not match a simple parity rule. See Not confirmed.
- Calculator: calculate — type: truth table (search over switch networks is too hard for a calculator; check rows only).

18. Hands-on: add a middle location with a DPDT

- Scope: ✅ IN.
- Where seen: not seen in a test. It extends rule II.c.vii.
- How to solve:
  1. Build the two-switch 3-way first and check it.
  2. Cut both travelers. Put a DPDT in the middle.
  3. Travelers from S1 → the two commons of the DPDT.
  4. Wire the DPDT throws straight to S2 in one position and crossed in the other: throw 1 of pole A and throw 2 of pole B go to traveler X; throw 2 of pole A and throw 1 of pole B go to traveler Y.
  5. Test all 8 positions: 4 on, 4 off.
- Worked example: with 9 V and 390 Ω, every "on" state gives **17.9 mA** and every "off" state gives 0.
- Traps: If one switch stops working, two of the cross wires are swapped.
- Calculator: calculate — type: V, Vf, R.

Out of scope (skip)

- ❌ **AC mains hands-on wiring.** Rule II.d forbids AC circuits and high power at stations. Diagrams of house wiring are fine on the written test.
- ❌ **Smart/electronic multiway switches** (transistor or triac dimmers). Rule 3.d: semiconductors beyond those listed.
- ❌ **Latching relays or flip-flops used as "push-button anywhere" controls** (impulse relays). Rule 3.d: sequential logic.

Mistakes in tests and keys

- **North Pocono 2020 C (INV) Q4:** the key accepts only XNOR. XOR is also correct with a different labeling.
- **MIT 2020 C (INV) Q49:** the truth table has 15 rows. Row S1 S2 S3 S4 = 0101 is missing. I checked in Python.
- **UT 2019 B and UT fall 2019 C Q14:** the key "Alternative" is a name I could not match exactly. See Not confirmed.

Not confirmed

- **"Alternative" system for long hallways (UT Q14):** Wikipedia "Multiway switching" (fetched 2026-10-16, via a summary tool) describes the California/coast 3-way as useful for long corridors. The summary called it the "alternative" system, but I did not read that word on the page myself.
- **MIT 2020 Q49:** I did not solve the full design, and I did not see the key's drawing. The table may have a typo.
- **Northview 2026 lab:** the details come from the earlier extraction (G_tests.md). I did not read the lab sheet this run.
- No STATE or NATIONAL test with a 3-way item was found. The real 2026 Nationals tests were not in the downloads.

Sources

- Rules: 2027 Div B Rules Manual, Circuit Lab, rules 3.c.vi, 3.c.viii, II.c.vii and II.d. See Circuit Lab - Rules 2027 Ⓡ.
- Tests (extraction files in `~/Daniel/circuitlab/research/tests/` and `prior/G_tests.md`): Pembroke Hill 2026 B, Berkeley 2026 B, BullSO 2026 B/C, Northview 2026 C, MIT 2019 C, MIT 2020 C, Princeton 2020 C, Rickards 2021-season B/C, ATX Regional 2021 B, NY Lake Erie 2018 C, UF Northern Regional 2020 C, UT 2019 B, UT fall 2019 C, UT 2020 B, North Pocono 2020 C, F-M fall 2019, UMSO, Shock Value re-tryout.
- Raw text checked this run: Northview 2026 C Q26 (hotel), MIT 2020 Q49 (table), UT 2019 B/C Q14 (choices), Berkeley 2026 B Q25, BullSO 2026 B Q24 and key.
- Wikipedia, "Multiway switching" (traveler, California/coast, Carter systems; Carter prohibited by the NEC since 1923; any number of intermediate switches).
- Kuphaldt, *Lessons in Electric Circuits*, Vol. IV, "Switches" and "Ladder Logic"; All About Circuits (same chapters online).
- All numbers checked in Python on 2026-10-16.

Relays

A **relay** is a switch that is worked by electricity instead of a finger. A small current in a coil moves the contacts. The contacts can switch a separate, larger circuit.

This note is ✅ **IN at every level**. Rule 3.c.vi names relays. Relay logic ties to rule 3.c.viii (logic gates). The flyback diode ties to rule 3.c.vii (diode uses).

Two limits apply:

- **How the coil makes a magnetic pull is magnetism.** Rule 3.c lists no magnetism, and rule 3.d excludes inductance. Know that "the coil becomes a magnet and pulls the armature". Do not calculate fields or inductance.
- **Latching relays used as memory** are sequential logic. Rule 3.d excludes that. ❌ if asked as a flip-flop.

Key facts

- A relay has two separate circuits: the **control (coil) circuit** and the **load (contact) circuit**. They are not connected electrically. This is called **isolation**.
- Parts: **coil** (wire wound on an iron core), **armature** (the moving iron arm), **spring** (pulls the armature back), **contacts** (the switch part).
- Contact terminals: **COM** (common), **NO** (normally open: closes when the coil is on), **NC** (normally closed: opens when the coil is on).
- "Normal" means the coil is **off** (de-energized).
- Relays come as SPST, SPDT, DPDT and more, just like hand switches.
- **Why use one:** a small, low-power signal (a thermostat, a microcontroller, a car's dashboard switch) can control a large, high-power load (a heater, a motor, headlights). The low-power side stays safe and isolated.
- **Coil rating:** the coil voltage (for example 5 V, 12 V) and the coil resistance. Coil current = V / R_coil.
- **Contact rating:** the largest current and voltage the contacts can switch. Example: "10 A 250 VAC, 10 A 30 VDC".
- **Pick-up (pull-in) voltage**: the lowest coil voltage that surely turns the relay on. **Drop-out voltage**: the highest coil voltage at which it surely turns off. Pick-up is higher than drop-out. Between them the state is not guaranteed.
- **Flyback diode** (also called a snubber, freewheeling or suppression diode): a diode across the coil, **reverse-biased** in normal use (cathode to the + side). When the coil is switched off, the coil's stored energy tries to keep the current flowing. That makes a large voltage spike. The diode gives the current a safe loop and protects the switch or transistor that drives the coil.
- **Relay logic:** NO contacts in **series** = **AND**. NO contacts in **parallel** = **OR**. An **NC** contact = **NOT**. Early computers (for example Konrad Zuse's Z3, 1941, and Howard Aiken's Harvard Mark I, 1944) used relays. They were slow, which is why vacuum tubes replaced them.
- **Claude Shannon** (1937, MIT master's thesis) showed that relay and switch circuits follow **Boolean algebra** (George Boole's algebra of logic).
- **Automotive relay pins** (DIN 72552 numbers): **85 and 86** = coil; **30** = common; **87** = normally open; **87a** = normally closed.
- **Solid-state relay (SSR)**: no moving parts; it uses semiconductors. ⚠️ Borderline (rule 3.d: semiconductors beyond those listed).
- **Reed relay**: a reed switch inside a coil. ⚠️ (magnetic, but still a relay).
- **Latching relay**: it stays in its last position with no coil power. ⚠️ to ❌ (memory = sequential logic).

Formulas

What	Formula	Notes
Coil current	I_coil = V_coil / R_coil	Ohm's law
Coil power	P = V² / R_coil = I² · R_coil	Heat in the coil
Minimum current to surely turn on	I_min = V_pick-up / R_coil	MIT 2021 Q70

What	Formula	Notes
Minimum power when surely on	$P_{min} = V_{pick\text{-}up}^2 / R_{coil}$	MIT 2021 Q71
Load current check	$I_{load} = P_{load} / V_{load} \le$ contact rating	
Two coils in parallel	$R_{in} = R_{coil} / 2$	Two relays share one input
Contact chain output	$V_{out} = V \cdot R_{load} / (R_{load} + R_{on})$	Divider
Fan-out (n identical inputs)	$R_{load} = R_{in} / n$	Then the divider above

Question types

Easy

1. What is a relay? Name its parts

- Scope: ✅ IN.
- Where seen: UF Northern Regional 2019 C (REG) Q34 (identify a relay and its electromagnet). Princeton 2026 C (INV) SA1 a-b (what switches and relays are used for; how they differ in structure).
- How to solve:
  1. Say it is an electrically operated switch.
  2. Name the coil, armature, spring and contacts.
  3. Say how it differs from a switch: a coil, not a hand, moves the contacts.
- Worked example (Princeton SA1 b): "How do a switch and a relay differ in structure?" Answer: **a switch is moved by hand; a relay has a coil (electromagnet) that pulls an armature to move its contacts, and the coil circuit is separate from the contact circuit.**
- Traps: The coil and the contacts are **not** connected to each other. Do not draw current flowing from the coil into the load.
- Calculator: look up — type: "relay parts".

2. Read a relay symbol

- Scope: ✅ IN.
- Where seen: Cobra Invitational 2020 B (INV) Q28 (symbol → SPST NO relay). Princeton 2026 C (INV) symbol list (item K "Relay").
- How to solve:
  1. Find the coil: a rectangle (or a zigzag loop) with two wires.
  2. Find the contacts next to it. A dashed line links the coil to the contacts.
  3. Read the contacts like a switch: poles, throws, NO or NC.
- Worked example: a box with two leads, a dashed line, and one open blade with two terminals. Answer: **SPST normally-open relay**.
- Traps: A drawn open blade is "open when the coil is off". It closes when the coil is on.
- Calculator: look up — type: coil present, poles, throws, NO/NC.

3. When should you use a relay?

- Scope: ✅ IN.
- Where seen: ATX Regional 2026 B (REG) MC39 ("where is a relay best used" → key: high power, high voltage). Princeton 2026 C (INV) SA1 c (why a relay is preferred when a low-power control circuit must control a high-power load). Princeton 2026 C (INV) Matching Part 2 (turn on a heater from a thermostat → relay).
- How to solve:
  1. Look for a small control signal and a big load.
  2. Look for a need to keep the two circuits apart (safety or different voltages).
- Worked example (Princeton SA1 c): Answer: **the small control current only has to power the coil. The heavy load current flows through the relay's contacts, which are rated for it. The control circuit stays isolated from the high voltage.**
- Traps: A relay is not used to "make more power". It only switches power from another source.
- Calculator: look up — type: control vs load.

4. COM, NO and NC: what is connected when?

- Scope: ✅ IN.
- Where seen: not seen in a test as its own item. Prior topics note B_topics §6b.
- How to solve:
  1. Coil off: COM touches NC.
  2. Coil on: COM touches NO.
- Worked example: a lamp is wired COM → NC. A buzzer is wired COM → NO. Coil off: **lamp on, buzzer off**. Coil on: **lamp off, buzzer on**.
- Traps: "Normally" is about the coil, not the load.
- Calculator: look up — type: coil on/off, terminal.

5. Read a real relay's label

- Scope: ✅ IN.
- Where seen: Cobra Invitational 2020 B (INV) station S3 (identify a real part: SPDT relay, 12 V, 60 A).
- How to solve:
  1. The coil voltage is often printed as "12 VDC" or "COIL 12V".
  2. The contact rating is printed as amps at a voltage.
  3. Count pins or read the diagram on the case for the contact type.
- Worked example: case says "12VDC, 60A, 30/85/86/87/87a". Answer: **12 V coil (pins 85, 86); SPDT contacts rated 60 A: 30 = common, 87 = NO, 87a = NC.**
- Traps: The "12 V" is the coil voltage, not the largest load voltage.
- Calculator: look up — type: label text.

Medium

6. Coil current and coil power

- Scope: ✅ IN.
- Where seen: MIT 2021 C (INV) Q70-72 (V_{IH} = 9.0 V, V_{IL} = 3.0 V, R_i = 120 Ω → minimum on-current 75 mA; minimum on-power 680 mW; minimum off-power 0). Prior bank I_bank D10.
- How to solve:
  1. $I = V / R_{coil}$.
  2. $P = V^2 / R_{coil}$.
  3. For "guaranteed on", use the pick-up voltage V_{IH}. For "off", the input could be 0 V.
- Worked example (MIT Q70-72): I_{min} = 9.0/120 = **75 mA**. P_{min} = 9.0^2/120 = **675 mW** (the key rounds to 680 mW). Off: the voltage can be 0, so P_{min} = **0 W**.
- Second example: 12 V coil, 400 Ω → I = **30 mA**, P = **0.36 W**.
- Traps: The coil is a resistor for DC steady state. Do not use inductance (OUT).
- Calculator: calculate — type: V, R_{coil}.

7. Can the contacts switch this load?

- Scope: ✅ IN.
- Where seen: not seen in a test. Prior bank I_bank D8 (same check for switches).
- How to solve:
  1. Find $I_{load} = P / V$.
  2. Compare with the contact rating at that type of voltage (AC or DC).
- Worked example: relay contacts rated 10 A 250 VAC, 10 A 30 VDC. A 2000 W, 120 V heater draws 2000/120 = **16.7 A > 10 A → no**. Use a 20 A or 30 A relay.
- Traps: The coil rating has nothing to do with the load. A 5 V coil relay can switch 120 V.
- Calculator: calculate — type: P, V, contact amps.

8. Can a small output drive the coil directly?

- Scope: ✅ IN (the transistor driver itself is ❌ OUT; only the yes/no is IN).
- Where seen: not seen in a test. Prior bank I_bank D11.

- How to solve:
  1. Coil current = V / R_coil.
  2. Compare with the most current the output can give.
- Worked example: a microcontroller pin gives at most 20 mA at 5 V. The relay coil is 5 V, 71 Ω. Coil current = 5/71 = **70 mA > 20 mA → it cannot drive it directly**. A driver is needed.
- Traps: Do not design the transistor driver at Div B.
- Calculator: calculate — type: V, R_coil, output max current.

9. Flyback diode: where and why

- Scope: ✅ IN (diode use). The energy-in-a-coil idea is ⚠️, because inductance is excluded. The level "it stops a voltage spike" is fine.
- Where seen: Princeton 2026 C (INV) SA9 (TB1: a transistor switches a relay coil; what happens to the coil's stored energy at turn-off and to the transistor voltage → a large spike). Princeton 2020 C (INV) Q26 (purpose of a flyback diode → suppress the spike).
- How to solve:
  1. Draw the diode across the coil.
  2. Point it so it is **reverse-biased** in normal use: cathode to the + end of the coil.
  3. Explain: at turn-off the coil keeps the current going for a moment. The diode gives it a loop, so no spike reaches the switch.
- Worked example: "Without the diode, what happens when the switch opens?" Answer: **the coil makes a high voltage spike across the switch or transistor. It can arc the switch or destroy the transistor.** With the diode, the spike is clamped to about 0.7 V above the supply.
- Traps: A forward-biased diode across the coil shorts the supply when the relay is on. Check the direction.
- Calculator: look up — type: diode direction.

10. Relay logic: which gate is it?

- Scope: ✅ IN (rule 3.c.vi with 3.c.viii).
- Where seen: MIT 2021 C (INV) Q75-78 (four relay circuits A-D → gates; key A, I, E, D from a letter list). Prior H_logic_diodes A20 (Shannon's mapping).
- How to solve:
  1. Treat each relay's contact as a switch worked by its input (1 = coil on).
  2. NO contacts in series = AND. In parallel = OR.
  3. An NC contact = NOT of its input.
  4. A contact to ground that "pulls the output low" inverts the logic (for example a NOR from parallel NO contacts to ground with a pull-up).
  5. Fill the 4-row table to check.
- Worked example: two NO contacts in series from 12 V to the output. A pull-down holds the output low otherwise. Output is high only when both coils are on. Answer: **AND**. Make them parallel and it is **OR**. Use NC contacts in parallel and it is **NAND**.
- Traps: Check what the output does when no contact connects it. Floating output is not "0".
- Calculator: look up — type: contact network (series/parallel, NO/NC).

Hard

11. Input and output resistance of a relay gate

- Scope: ✅ IN.
- Where seen: MIT 2021 C (INV) Q73-74 (relay NOT gate from two relays: input resistance 60 Ω; output resistance 0.80 Ω).
- How to solve:
  1. Input: the input wire feeds both coils in parallel. R_in = 120 || 120.
  2. Output: in each state, one relay is on and connects the output to 12 V or ground through R_on. The other is off (150 kΩ, so it barely matters).
- Worked example: R_in = 120 × 120 / 240 = **60 Ω**. R_out ≈ R_on = **0.80 Ω** (R_on || 150 kΩ = 0.79999 Ω).

- Traps: Do not add the two coil resistances in series; they share the same input node.
- Calculator: calculate — type: R_coil, number of coils, R_on, R_off.

12. A bad relay design: the output fights itself

- Scope: ✅ IN.
- Where seen: MIT 2021 C (INV) Q79 (circuit B: when inputs differ, it joins 12 V to ground through two R_on → output 6 V, which is neither a valid 0 nor a valid 1).
- How to solve:
  1. Find the state where both an "up" path and a "down" path are closed.
  2. The output is a voltage divider between them.
  3. Compare with V_IL and V_IH.
- Worked example: 12 V → 0.80 Ω → output → 0.80 Ω → ground. V_out = 12 × 0.8 / 1.6 = **6.0 V**. That is between 3.0 V and 9.0 V, so the next relay's state is **not guaranteed**. The current is 12/1.6 = **7.5 A** (90 W) — a short.
- Traps: "Output = 6 V" is not a logic 1. Always compare with both thresholds.
- Calculator: calculate — type: V, R_up, R_down, V_IL, V_IH.

13. Pick-up and drop-out thresholds

- Scope: ✅ IN.
- Where seen: MIT 2021 C (INV) Q70-72 context (V_IH = 9.0 V, V_IL = 3.0 V). The "static discipline" follow-up Q68-69 is ⚠️ (noise margins, not named in the rules).
- How to solve:
  1. Above V_IH: surely on.
  2. Below V_IL: surely off.
  3. Between: not guaranteed (it may be on or off, depending on what happened before).
- Worked example: the coil sees 6.0 V. Answer: **not guaranteed**. At 9.5 V: on. At 2.0 V: off.
- Traps: The "between" region is why bad designs (type 12) fail.
- Calculator: look up — type: V_in, V_IL, V_IH.

14. Automotive relay pins (30, 85, 86, 87, 87a)

- Scope: ✅ IN (relays).
- Where seen: not seen in a test. DIN 72552 terminal numbering.
- How to solve:
  1. 85 and 86 = coil.
  2. 30 = common (often from the battery through a fuse).
  3. 87 = NO (to the load). 87a = NC.
- Worked example: "Headlights must come on when the dashboard switch powers the coil." Wire **battery → fuse → 30; 87 → headlights → ground; switch → 86; 85 → ground**.
- Traps: Pin 87a is often missing on SPST relays.
- Calculator: look up — type: pin number.

Nationals

15. Fan-out: how many relay inputs can one output drive?

- Scope: ✅ IN (series/parallel and divider math).
- Where seen: not seen in a test. It extends MIT 2021 Scenario J.
- How to solve:
  1. n inputs in parallel: R_load = R_in / n.
  2. V_out = V × R_load / (R_load + R_on).
  3. Find the largest n with V_out > V_IH.
- Worked example: 12 V through R_on = 0.80 Ω, each input 120 Ω, V_IH = 9.0 V. Need R_load ≥ 3 × 0.80 = 2.4 Ω, so n ≤ 50. At n = 50, V_out is exactly 9.0 V, which is not "greater than". So the answer is **49 inputs** (V_out = 9.05 V).
- Traps: Watch "greater than" vs "at least" at the exact edge.

- Calculator: calculate — type: V, R_on, R_in, V_IH.

16. Relay reversing or changeover circuits

- Scope: ✅ IN.
- Where seen: not seen in a test. It combines Switch Types A6 type 14 with a relay.
- How to solve:
  1. Replace the DPDT hand switch with a DPDT relay.
  2. Coil off: the NC side sets forward. Coil on: the NO side sets reverse.
  3. Check no state shorts the supply.
- Worked example: a DPDT relay with motor across the two commons. NC1 → +, NC2 → –, NO1 → –, NO2 → +. Coil off: **forward**. Coil on: **reverse**.
- Traps: The cross wires on the NO side are what make it reverse.
- Calculator: look up — type: coil state, contact map.

Out of scope (skip)

- ❌ **Magnetic force and field of the coil** ($B = \mu_0 nI$ and the like; solenoid questions appear in many old tests). Rule 3.c lists no magnetism; rule 3.d excludes inductance.
- ❌ **Coil inductance, L·di/dt spike size, and RL time constants.** Rule 3.d: inductance; rule 3.e.v: calculus.
- ❌ **Transistor relay drivers** (base resistor, hFE). Rule 3.d: semiconductors beyond those listed.
- ❌ **Latching relays used as memory, self-holding (seal-in) circuits as flip-flops, relay counters.** Rule 3.d: sequential logic.
- ⚠️ **Solid-state relays**: name only. Rule 3.d: semiconductors beyond those listed.

Mistakes in tests and keys

- **MIT 2021 C (INV) Q71:** $9.0^2/120 = 675$ mW. The key gives 680 mW (rounded to 2 significant figures). Not an error, but write 675 mW (or 0.68 W).
- **ATX Regional 2026 B (REG) MC39:** the key "high power, high voltage" is a loose answer. The usual teaching is "a low-power control circuit switching a high-power load".

Not confirmed

- MIT 2021 Q75-78: the circuits are images. I did not render them. The gate letters come from the key (A, I, E, D) and the earlier extraction.
- UF Northern Regional 2019 C Q34 and Cobra S3: from the extraction files. I did not render the pictures.
- Zuse Z3 (1941) and Harvard Mark I (1944) dates are from memory. DIN 72552 pin numbers are from memory. I did not fetch a source this run.
- No STATE or NATIONAL test with a relay item was found.

Sources

- Rules: 2027 Div B Rules Manual, Circuit Lab, rules 3.c.vi-viii, 3.d, 3.e. See Circuit Lab - Rules 2027 R.
- Tests: MIT 2021 C (raw text and key checked this run: Scenario J, Q70-79); Princeton 2026 C (raw text and key checked: SA1, SA9); Princeton 2020 C; ATX Regional 2026 B; Cobra 2020 B; UF Northern Regional 2019 C. Extraction files in `~/Daniel/circuitlab/research/tests/` and `prior/`.
- Prior research: `prior/B_topics.md` §6b, `prior/I_bank.md` D9-D14, `prior/H_logic_diodes.md` A20.
- Textbooks: Kuphaldt, *Lessons in Electric Circuits*, Vol. IV (Digital), chapters "Electromechanical Relays" and "Ladder Logic"; All About Circuits (same chapters online).
- Claude Shannon, "A Symbolic Analysis of Relay and Switching Circuits" (MIT thesis, 1937).
- All numbers checked in Python on 2026-10-16.

Fuses

A **fuse** is a safety part that stops the current when it gets too big. Inside is a thin metal strip or wire called the **element**. Too much current heats the element until it melts. The melted gap opens the circuit.

This note is ✅ **IN at every level**. Rule 3.c.vi names fuses. Fuse calculations use Ohm's law and power (rules 3.c.iv and 3.c.v). Fuse problems with two sources or two loops are ✅ States/Nationals (rule 3.c.ix, KVL and KCL).

Hands-on limit: rule Part II.d says no high-power or AC circuits at the station. A hands-on fuse task would use a small DC circuit only.

Key facts

- **Overcurrent** means more current than the circuit is designed to carry. A fuse is an **overcurrent protection device** (Shock Value tryout I-22).
- The fuse detects excess current by **heat**. The heat is I^2R in the thin element. Big current means fast melting.
- A fuse goes in **series** with the load. All the load current must pass through it. Then, when it opens, the whole current stops.
- A fuse goes on the **hot** wire, as close to the source as possible. Then a blown fuse removes voltage from everything after it.
- A fuse is **one-time use**. After it blows ("burns out"), you must replace it. It never resets (GullSO 2026 Q27).
- Replace a fuse with the **same rating** only. A bigger fuse lets the wire overheat. Find and fix the cause first.
- A fuse protects the **wire and the equipment** from fire and damage. It does **not** protect people from shock. A shock current (about 0.1 A) is far too small to melt a 15 A fuse.
- **Ratings on a fuse:**
  - **Current rating** (amps): the current it carries without blowing.
  - **Voltage rating**: the largest voltage it can safely interrupt. Above it, an **arc** (a spark that keeps burning across the gap) can form after the element melts (Rickards 2021 Q16).
  - **Breaking capacity** (interrupting rating): the largest fault current it can safely stop, for example 10 kA.
- A real fuse carries its rated current with no time limit. It blows at a current somewhat **above** the rating. Small overloads take a long time. Big shorts take milliseconds.
- **Fast-blow (fast-acting)** fuses open quickly. They protect delicate electronics.
- **Slow-blow (time-delay)** fuses ride through a short **inrush** (a brief surge of current at switch-on). Motors and lamps have inrush. A cold tungsten filament has a much lower resistance than a hot one.
- **Types:**
  - Glass cartridge (you can see the element; common sizes 5 × 20 mm and 1/4 × 1-1/4 in).
  - Ceramic cartridge (higher breaking capacity; often filled with sand to quench the arc).
  - Automotive blade fuse (color-coded by amps).
  - Plug fuse (screw-in, old homes).
  - Resettable **PTC fuse** ("polyfuse"): its resistance rises sharply when it gets hot. ⚠️ Borderline: non-linear device (rule 3.d).
  - Thermal fuse (thermal cutoff): opens on **temperature**, not current. Used in hair dryers and coffee makers.
- **Meter fuses:** a multimeter's current jacks have fuses. The mA jack has a small fuse (about 200-500 mA). The 10 A jack has a 10 A or larger fuse. Measuring current across a battery blows the meter fuse.
- **Fuse versus breaker:** a fuse is usually faster and cheaper. A breaker can be reset. See Circuit Breakers A6.
- **Symbol:** a rectangle with a line through it, or an S-shaped wave between two terminals. A breaker symbol is a curved arc (see Circuit Breakers A6).
- **Test a fuse with an ohmmeter** (out of the circuit). A good fuse reads about 0 Ω (continuity beep). A blown fuse reads "OL" (open).

Formulas

What	Formula	Notes
Current through the fuse	I = V / R_eq	Use the equivalent resistance of everything after the fuse
Current from power	I = P / V	Appliance labels give watts
Several loads in parallel	I_total = I_1 + I_2 + ...	KCL: all branch currents pass the main fuse
Number of equal loads	n_max = whole-number part of (I_fuse / I_one)	Round **down**
Blows?	blows if I > I_rating	Test keys treat I = I_rating as "does not blow"
Heat in the fuse element	P = I² · R_fuse	
Voltage across a good fuse	V = I · R_fuse	Very small
Voltage across a blown fuse	V = source voltage	The fuse is now an open switch
Choose a rating	next standard size **above** the normal current, and **not above** the wire's rating	

Standard small fuse sizes (amps): 0.5, 1, 2, 3, 5, 7.5, 10, 15, 20, 25, 30. Household plug fuses: 15, 20, 30 A.

Question types

Easy

1. What does a fuse do, and how does it detect excess current?

- Scope: ✅ IN.
- Where seen: USC Invitational 2026 B (INV) Q31 and USC 2026 C Q32 ("how a fuse detects excess current" → D, a thin wire melts). Shock Value re-tryout (B, about 2012, practice) VI-16 (what a fuse does) and I-22 (a fuse is a type of ___ → overcurrent protection device). Princeton 2026 C (INV) SA3a (fuse part).
- How to solve:
  1. Say the fuse is in series with the load.
  2. Say too much current heats the thin element (I²R).
  3. Say the element melts and opens the circuit, so all current stops.
- Worked example: "How does a fuse detect excess current?" Answer: **a thin wire in the fuse heats up and melts when the current is too large**.
- Traps: A fuse does not "measure" current with a sensor. It is just heat. A fuse does not detect voltage.
- Calculator: look up — type: "fuse how it works".

2. Identify the fuse symbol or a real fuse

- Scope: ✅ IN.
- Where seen: Cobra Invitational 2020 B (INV; 2019-20 season) Q28 (six symbols, one is a fuse). Shock Value re-tryout IV-2 (symbol chart: breaker, fuse, and others). Princeton 2026 C (INV) symbol list, item I "Fuse". ATX Regional 2026 B (REG) image sheet (draw-out fuse symbol shown).
- How to solve:
  1. A rectangle with a line through it, or an S-wave between two dots, is a fuse.
  2. A curved arc over a gap is a breaker.
  3. A thermal switch shows a bent (bimetal) strip; do not confuse it with a fuse.
- Worked example: a small rectangle on a wire with a line through its long axis. Answer: **fuse**.
- Traps: The resistor box (IEC style) is a plain rectangle with no line. The fuse box has a line through it.
- Calculator: look up — type: symbol shape.

3. Can a fuse be reset? How long until it works again?

- Scope: ✅ IN.
- Where seen: GullSO Invitational 2026 (C, INV) Q27 ("How long does it take a typical fuse to reset?" → D, never). Shock Value re-tryout I-2 (a fuse differs from a ___ because it cannot be reset → circuit breaker).
- How to solve:
  1. Remember: a fuse destroys itself to open the circuit.
  2. It must be replaced. It never resets.
- Worked example: GullSO Q27. Answer: **D, never**.
- Traps: A PTC "resettable fuse" does reset (after it cools). Unless the question names a PTC or polyfuse, the answer is "never".
- Calculator: look up — type: "fuse reset".

4. Where does the fuse go in the circuit, and why in series?

- Scope: ✅ IN.
- Where seen: ATX Regional 2026 C (REG) FRQ 5 ("Why are fuses always connected in series and not in parallel?"). Shock Value re-tryout IV-1-3 (draw a schematic with bulb, diode, DC source, switch and fuse → series loop, fuse next to the source).
- How to solve:
  1. The fuse must carry the full load current. So it is in series.
  2. When it opens, the series path breaks and all current stops.
  3. In parallel with a load, a blown fuse would not stop the load current.
  4. In parallel with the source, the fuse would be a short circuit.
  5. Put it on the hot side, close to the source.
- Worked example: "Why in series?" Answer: **in series the fuse carries all the current, so when it melts the current stops. In parallel the current would go around it, or the fuse would short the source.**
- Traps: The ATX key talks about "several fuses in parallel". The cleaner reason is the one above. A fuse after the load (on the return side) still stops current, but it leaves the load connected to the hot wire. That is a shock hazard.
- Calculator: look up — type: "fuse series why".

5. Match a device to a job: overcurrent → fuse

- Scope: ✅ IN.
- Where seen: Princeton 2026 C (INV) Matching Part 2 (13 items; "protect a circuit from overcurrent" → fuse).
- How to solve:
  1. Overcurrent or short → fuse or breaker.
  2. Leakage to ground or shock near water → GFCI.
  3. Small signal controls a big load → relay.
- Worked example: "Stop a short circuit from overheating the wires." Answer: **fuse** (or breaker if the list has no fuse).
- Traps: "Protect a person from shock in a bathroom" is a GFCI, not a fuse.
- Calculator: look up — type: job keyword.

6. Automotive blade fuse colors and fuse types ⚠️

- Scope: ⚠️ Borderline. It is fuse trivia, so it fits rule 3.c.vi. But the colors are an industry code, not a science idea.
- Where seen: not seen in a test. Prior bank I_bank D19.
- How to solve: recall the color code.
  - 2 A gray, 3 A violet, 5 A tan, 7.5 A brown, 10 A red, 15 A blue, 20 A yellow, 25 A clear (natural), 30 A green.
- Worked example: "A red blade fuse is rated at what?" Answer: **10 A**.
- Traps: Mini, standard (ATO) and maxi blades share most colors, but maxi fuses use different colors for some ratings.
- Calculator: look up — type: color → amps.

Medium

7. Will the fuse blow? (one load or a parallel group)

- Scope: ✅ IN.
- Where seen: Pembroke Hill 2026 B (INV) Q29 (5 A fuse, 600 W device on 120 V → key: no). scioly.org forum Shock Value / Circuit Lab question marathon 2014 (three 3 Ω in parallel on 5 V with a 4 A fuse → blows). Prior bank I_bank D16.
- How to solve:
  1. Find the current through the fuse: I = P/V, or I = V/R_eq.
  2. Compare with the rating.
  3. I > rating → blows. I ≤ rating → does not blow (test-key convention).
- Worked example 1 (Pembroke Q29): I = 600/120 = **5.0 A**. The fuse is rated 5 A. Answer: **no, it does not blow** (key: No).
- Worked example 2 (forum): R_eq = 1/(1/3 + 1/3 + 1/3) = 1 Ω. I = 5/1 = **5 A > 4 A → blows**.
- Traps: Exactly at the rating is "no" on keys. In real life a fuse running at 100% gets hot and may blow over time. If the question asks "is this a good choice", say no: leave a margin.
- Calculator: calculate — type: P and V (or V and R_eq), fuse amps.

8. How many loads can you add in parallel before the fuse blows?

- Scope: ✅ IN.
- Where seen: Princeton 2021 C (INV) Q37 (0.75 A fuse, 10 V battery, 100 Ω bulbs → 7).
- How to solve:
  1. Find the current of one load: I_1 = V/R (or P/V).
  2. Divide the fuse rating by I_1.
  3. Round **down** to a whole number.
- Worked example (Princeton 2021 Q37): I_1 = 10/100 = 0.1 A. 0.75/0.1 = 7.5. Answer: **7 bulbs**.
- Traps: Do not round 7.5 up to 8. Eight bulbs draw 0.8 A > 0.75 A. If the question says "how many more can be added", subtract the bulb already there.
- Calculator: calculate — type: V, R (or P), fuse amps, bulbs already in place.

9. Choose the right fuse rating

- Scope: ✅ IN.
- Where seen: not seen in a test. Prior bank I_bank D17. Source: Kuphaldt, *Lessons in Electric Circuits*, Vol. I, "Fuses".
- How to solve:
  1. Find the normal current: I = P/V.
  2. Pick the next standard size **above** it, so normal use does not blow it.
  3. Check it is **not above** the wire's rating, so the wire is protected.
- Worked example: a 12 V, 60 W car accessory. I = 60/12 = **5 A**. Choose the next size up: **7.5 A**. If the wire is rated 10 A, 7.5 A is fine.
- Traps: A 5 A fuse at 5 A has no margin. A 30 A fuse on thin wire lets the wire burn before the fuse opens.
- Calculator: calculate — type: P, V, wire amps; look up — standard sizes.

10. Fuse voltage rating and arcing

- Scope: ✅ IN.
- Where seen: Rickards 2021-season B (INV) Q16 and Rickards 2021-season C Q19 ("a fuse burns out, but the voltage is greater than its voltage rating" → C, arcing). GullSO 2026 (C, INV) FRQ 37e (replace the 3 Ω resistor with a fuse; is a fuse rated 1 V enough? → no).
- How to solve:
  1. The voltage rating is about **after** the fuse blows.
  2. Then the full source voltage appears across the gap.
  3. If that voltage is above the rating, an arc can jump the gap and keep current flowing.

- Worked example (GullSO 37): 3 V source, 2 Ω, 3 Ω and 1 Ω in series. I = 3/6 = **0.5 A**. The 3 Ω resistor drops 0.5 × 3 = **1.5 V**. Answer: **no, a 1 V rating is not enough**. Better reason: once the fuse opens, the full **3 V** sits across it. The fuse needs a rating of at least 3 V.
- Traps: A working fuse drops almost 0 V (it is nearly a wire). The GullSO key uses the 1.5 V resistor drop. The fuse-maker's meaning is the open-circuit voltage (3 V here). Both give "no".
- Calculator: calculate — type: V source, series R values, fuse V rating.

11. Fast-blow or slow-blow? (inrush)

- Scope: ✅ IN. The inrush idea uses only Ohm's law with a hot and a cold resistance.
- Where seen: not seen in a test. Prior bank I_bank D18. Source: Kuphaldt, *Lessons in Electric Circuits*, Vol. I, "Fuses"; Littelfuse fuse selection guides (general knowledge, not fetched).
- How to solve:
  1. Find the normal current with the hot resistance.
  2. Find the starting current with the cold resistance.
  3. If the start is much bigger, use a slow-blow fuse. For delicate electronics, use fast-blow.
- Worked example: a 100 W, 120 V bulb. Hot R = $120^2/100$ = **144 Ω**, so I = **0.83 A**. Cold R is about 1/12 of hot: 12 Ω. Inrush = 120/12 = **10 A**. A 1 A fast-blow fuse may blow at switch-on. Answer: **use a slow-blow (time-delay) fuse**.
- Traps: The "12 times" factor is a typical value, not a law. Use the numbers the question gives.
- Calculator: calculate — type: P, V, cold-to-hot ratio (or cold R).

12. Meter fuse: why the current reading is zero, or why it blew

- Scope: ✅ IN (meters and faults, rule 3.c.v; hands-on Part II.b).
- Where seen: MIT 2019 C (INV) lab S1-6 (ammeter in series or parallel; parallel shorts the circuit and blows the meter fuse). Prior notes B_topics §9 (blown meter fuse reads 0).
- How to solve:
  1. An ammeter has nearly 0 Ω.
  2. Across a source it makes a short circuit.
  3. Find the short-circuit current. Compare with the meter fuse.
- Worked example: a 9 V battery with 1 Ω internal resistance. Ammeter (0.1 Ω) placed across it. I = 9/(1 + 0.1) = **8.2 A**. The mA jack fuse (0.4 A) **blows**. Afterwards the meter reads **0 mA** in every circuit.
- Traps: On the 10 A jack the fuse may survive, but the battery gets hot. Always start current checks on the 10 A jack.
- Calculator: calculate — type: V, R_internal, R_meter, meter fuse amps.

13. Which fuse blows: branch fuse or main fuse?

- Scope: ✅ IN (parallel circuits and KCL at a single node; one equation).
- Where seen: not seen in a test. Source: Kuphaldt, *Lessons in Electric Circuits*, Vol. I, "Fuses"; OpenStax *College Physics 2e* §20.6 "Electric Hazards and the Human Body" (fuses and breakers).
- How to solve:
  1. Find each branch current. Compare it with its branch fuse.
  2. Add the branch currents for the main fuse (KCL).
  3. Compare the total with the main fuse.
- Worked example: 120 V. Main fuse 20 A. Branch A: 1500 W heater, 15 A fuse. Branch B: 1200 W microwave, 15 A fuse. I_A = **12.5 A** (ok). I_B = **10 A** (ok). Total = **22.5 A > 20 A**. Answer: **the main fuse blows; both branches go dark**.
- Traps: Each branch can be fine while the total is too much. Check both levels.
- Calculator: calculate — type: V, each P, each branch fuse, main fuse.

14. A short or an open somewhere: does the fuse blow, and what goes dark?

- Scope: ✅ IN (open and short circuits, rule 3.c.v).
- Where seen: not seen as a fuse question in a test. Related: Shock Value re-tryout VI-8 (finger across battery terminals burns → short circuit). Prior bank I_bank (series and parallel, one open or shorted).

- How to solve:
  1. A **short** across a load in parallel with the source makes R_eq ≈ 0. The current is huge. The fuse blows. Everything after it goes dark.
  2. A short across one bulb in a series string makes the current **rise**. Check it against the fuse.
  3. An **open** in one parallel branch lowers the total current. The fuse does not blow.
- Worked example: 12 V, three 6 Ω bulbs in series, 1.5 A fuse. Normal: I = 12/18 = **0.67 A**. One bulb shorts: I = 12/12 = **1.0 A** (fuse ok; two bulbs brighter). Two bulbs short: I = 12/6 = **2.0 A > 1.5 A → blows**.
- Traps: An open never blows a fuse. A blown fuse looks like an open.
- Calculator: calculate — type: V, each R, which are shorted or open, fuse amps.

Hard

15. Heat and voltage drop in the fuse itself

- Scope: ✅ IN (power, rule 3.c.iv).
- Where seen: not seen in a test. Prior bank I_bank D20.
- How to solve:
  1. $P = I^2 \cdot R_{fuse}$.
  2. $V = I \cdot R_{fuse}$.
- Worked example: a fuse element of 0.010 Ω carries 30 A. $P = 30^2 \times 0.010 =$ **9.0 W**. $V = 30 \times 0.010 =$ **0.30 V**.
- Traps: Doubling the current makes **four times** the heat. That is why a fuse reacts so strongly to big currents.
- Calculator: calculate — type: I, R_fuse.

16. The fuse must protect the wire (wire gauge and wire rating)

- Scope: ✅ IN (hazards; resistance of wires, rule 3.c.iv).
- Where seen: HH 2026 Invitational (INV) Q2 (current-carrying capacity of a wire most depends on → gauge) and Q3 (voltage rating of a wire depends on → insulation material). Prior bank I_bank W4 and D25.
- How to solve:
  1. Thicker wire (smaller AWG number) carries more current.
  2. The fuse rating must be **at or below** the wire's current rating.
  3. The wire's voltage rating comes from its insulation.
- Worked example: household 14 AWG copper is rated 15 A. Answer: **use a 15 A fuse or smaller; never 20 A**. A 12 AWG wire takes 20 A.
- Traps: The AWG number goes **down** as the wire gets **thicker**. Length changes the voltage drop, not the current rating.
- Calculator: look up — type: AWG → amps (14 → 15, 12 → 20, 10 → 30).

17. Reverse-polarity protection: a diode plus a fuse

- Scope: ✅ IN (diode uses, rule 3.c.vii, with a fuse, rule 3.c.vi).
- Where seen: not seen in a test. Prior notes B_topics §7 ("reverse-polarity protection: a series diode, or a shunt diode plus fuse").
- How to solve:
  1. A diode is placed across the input, pointing so that it is reverse-biased in normal use.
  2. If the battery goes in backwards, the diode conducts. It acts as a near-short.
  3. The large current blows the fuse. The fuse protects the circuit and the diode.
- Worked example: 12 V battery reversed. The diode drops 0.7 V. The wires and battery have 0.05 Ω. I = (12 – 0.7)/0.05 = **226 A**. A 5 A fuse **blows** at once.
- Traps: A series diode (the other method) needs no fuse but always wastes 0.7 V.
- Calculator: calculate — type: V, diode drop, path R, fuse amps.

18. Troubleshoot: is the fuse the problem? (hands-on)

- Scope: ✅ IN (hands-on Part II.c.vi, troubleshoot a faulty circuit).
- Where seen: not seen as a fuse station in a test. Related: Cobra 2020 B (INV) station S1 (broken LED circuit, find the fault with a coin jumper). Prior notes B_topics §9 (troubleshooting checklist: power, battery, switch, fuse or

breaker, parts, wires).

- How to solve:
  1. With power on, measure the voltage across the fuse. About 0 V → fuse good. Full source voltage → fuse open.
  2. With power off, take it out and measure resistance. About 0 Ω → good. "OL" → blown.
  3. If it is blown, find the short (measure R of the load) before you replace it.
- Worked example: 9 V circuit, LED dark. Voltmeter across the fuse reads **9.0 V**. Answer: **the fuse is open (blown)**. All of the source voltage appears across the open part.
- Traps: Across a good fuse you read 0 V even when the circuit is working. Do not call that "dead".
- Calculator: look up — type: voltage across fuse (0 or full).

Nationals

19. Fuse in a two-source circuit (KCL/KVL)

- Scope: ✅ States-Nationals (rule 3.c.ix). Two unknowns at most (rule 3.e.i).
- Where seen: not seen in a test with a fuse. Source: OpenStax *College Physics 2e* §21.3 "Kirchhoff's Rules" (method).
- How to solve:
  1. Name the node voltage V (or two loop currents).
  2. Write KCL at the node: current in = current out.
  3. Solve for V. Find the current through the fuse's branch.
  4. Compare with the fuse rating.
- Worked example: 12 V source through 4 Ω and 6 V source through 2 Ω both feed node N. From N, 3 Ω goes to ground through a 2 A fuse. KCL: (12 – V)/4 + (6 – V)/2 = V/3. Multiply by 12: 36 – 3V + 36 – 6V = 4V, so V = 72/13 = **5.54 V**. Fuse current = 5.54/3 = **1.85 A < 2 A** → **does not blow**. Check: I_1 = 1.62 A, I_2 = 0.23 A, sum 1.85 A ✓.
- Traps: Do not just add 12 + 6. The sources share the node. Check the sign of each branch current.
- Calculator: calculate — type: E1, R1, E2, R2, R3, fuse amps.

20. Resettable PTC fuse (polyfuse) ⚠️

- Scope: ⚠️ Borderline. A PTC is a non-linear device not listed in rule 3.c (rule 3.d excludes "non-linear devices beyond those listed"). Know the name and the idea only.
- Where seen: not seen in a test. Prior notes B_topics §6c, I_bank D21.
- How to solve:
  1. PTC means "positive temperature coefficient": resistance rises with temperature.
  2. Overcurrent heats it. Its resistance jumps very high. The current drops to a tiny value.
  3. Remove the fault and power. It cools and conducts again.
- Worked example: "Which fuse type resets itself after it cools?" Answer: **a PTC (polyfuse) resettable fuse**.
- Traps: A PTC never fully opens. A small "holding" current still flows while it is tripped.
- Calculator: look up — type: "PTC fuse".

Out of scope (skip)

- ❌ **Melting-time math** (I^2t "let-through energy", time-current curves on log scales, adiabatic heating). Rule 3.e.v (calculus) and 3.e (logarithms are not on the math list).
- ❌ **Fuses in AC circuits with phase angle, power factor or reactance.** Rule 3.d: AC circuit theory.
- ❌ **Transformer primary fuse sizing** (for example the neon-sign transformer in UF Northern Regional 2020 C Q19). Rule 3.d: inductance; magnetism is not in rule 3.c.
- ❌ **Fuses in 3-phase systems.** Rule 3.d: 3-phase power.
- ⚠️ PTC polyfuse beyond its name and idea (type 20).

Mistakes in tests and keys

- **Pembroke Hill 2026 B (INV) Q29:** the key says "No" for 5.0 A on a 5 A fuse. That fits the test-key rule (blow only above the rating). In practice a fuse at 100% runs hot. Answer "No" on the test, and mention "no margin" if there is a free-response line.
- **GullSO 2026 (C, INV) FRQ 37e:** the key reason uses the 1.5 V drop on the resistor. A fuse's voltage rating is about the open-circuit voltage after it blows (3 V here). Both reasons give "no".
- **ATX Regional 2026 C (REG) FRQ 5:** the key explains with "several fuses in parallel". The clean physics reason is "in series the fuse carries all the current, so opening it stops all current". The key is not wrong, just narrow.
- **Rickards 2026 C (INV) Q48** (fuse preferred to a breaker because it does NOT ___): the extraction file lists the key as "cost as much". A fuse also responds faster, but "does NOT respond faster" is false, so D is right. See Circuit Breakers [A6].

Not confirmed

- The rule "a fuse carries 100% indefinitely and blows at about 135% within an hour" (UL 248) is from memory. A web search this run did not find the UL 248 text. Treat it as approximate.
- The blade-fuse color code is from memory (prior bank I_bank D19). I did not fetch a manufacturer chart.
- The "cold filament is about 1/12 of hot" factor is a typical value from prior bank I_bank U14. It changes with the bulb.
- The scioly.org forum 2014 fuse question comes from prior notes (B_topics §6c). I did not reopen the forum page.
- No STATE or NATIONAL test with a fuse calculation was found. Florida State 2021 (B/C) and the USC 2026 tests have only concept fuse items or none. (The "nat26" file is the USC Invitational 2026, not Nationals.)

Sources

- Rules: 2027 Div B Rules Manual, Circuit Lab, rules 3.c.iv-vi, 3.c.ix, 3.d, 3.e, Part II.c-d (pages B11-B12). See Circuit Lab - Rules 2027 [R].
- Tests (raw text checked this run): Pembroke Hill 2026 B Q29 and key; Princeton 2021 C Q37; Princeton 2026 C SA3 and key; GullSO 2026 Q27 and FRQ 37; Rickards 2021-season B Q16 and C Q19; Rickards 2026 C Q48; ATX Regional 2026 C FRQ 5 and key; HH 2026 Q2-Q3. From extraction files only: USC 2026 B Q31 / C Q32; Cobra 2020 B Q28; Shock Value re-tryout I-2, I-22, IV-1-3, IV-2, VI-8, VI-16; MIT 2019 C lab S1-6; UF Northern Regional 2020 C Q19.
- Extraction files: `~/Daniel/circuitlab/research/tests/` (nat26, misc-a, misc-b, ut-a, state) and `prior/G_tests.md`.
- Prior research: `prior/B_topics.md` §6c, `prior/I_bank.md` D15-D22, W4; `prior/F_slides.md` (deck 11, fuse vs breaker).
- Textbooks: Kuphaldt, *Lessons in Electric Circuits*, Vol. I, chapter 12 "Physics of Conductors and Insulators", section "Fuses"; OpenStax *College Physics 2e* §20.6 and §21.3; All About Circuits, "Fuses".
- All numbers checked in Python on 2026-10-16.

Circuit Breakers

A **circuit breaker** is an automatic switch that opens when the current is too big. Unlike a fuse, it is not destroyed. You fix the problem and flip it back on. This is called **resetting** it.

This note is ✅ **IN at every level**. Rule 3.c.vi names breakers. Load and current calculations use rules 3.c.iv and 3.c.v. A panel problem that needs KCL with two unknowns is ✅ States/Nationals (rule 3.c.ix).

Two limits apply:

- **The magnetic trip works by magnetism.** Rule 3.c lists no magnetism, and rule 3.d excludes inductance. Know only "a big current makes a strong electromagnet that trips the latch fast". Do not calculate fields.
- **AFCI (arc-fault) breakers** are a breaker type, so they fit "breakers". The rule does not name them. ⚠️ Borderline. They were asked once (ATX 2026 B MC31).

Key facts

- A breaker is a switch plus a **trip unit**. The trip unit senses overcurrent and releases a spring-loaded latch. The contacts snap open.
- **Overload** = a little too much current for a long time (too many appliances). **Short circuit** = a huge current right away (hot touches neutral or ground).
- **Thermal trip:** a **bimetallic strip** (two metals bonded together that expand by different amounts). I^2R heating bends it. It slowly pushes the latch open. It handles **overloads**. A bigger overload trips faster.
- **Magnetic trip:** a coil in series with the load. A very large current pulls an iron piece and trips the latch in a few milliseconds. It handles **short circuits**. (Mechanism: name only, see the limits above.)
- Most home breakers are **thermal-magnetic**: they have both trips.
- A tripped breaker handle usually sits in the **middle** position. To reset, push it fully **OFF**, then **ON**.
- **Ratings:**
  - **Current rating** (amps, printed on the handle): 15 A and 20 A are the usual home circuits. 30 A for a dryer or a water heater.
  - **Voltage rating**: 120 V (single-pole) or 120/240 V (double-pole).
  - **Interrupting rating** (AIC): the biggest fault current it can safely stop, for example 10 kA.
- **Single-pole** breaker: one hot wire, 120 V. **Double-pole** breaker: two hot wires, 240 V (dryers, ovens, water heaters, air conditioners). The two handles are tied, so both hots open together.
- The **main breaker** sits at the top of the panel. It carries the sum of all branch currents. Typical home service: 100 A or 200 A.
- Breakers protect the **wiring** from overheating and fire. They do **not** protect people from shock. A deadly shock is about 0.1 A. A 15 A breaker ignores it. A GFCI protects people.
- **Fuse versus breaker:**

	Fuse	Breaker
After a fault	Melts; replace it	Trips; reset it
Speed	Usually faster	Usually a little slower
Cost to buy	Cheap	More expensive
Cost over time	New fuse each time	Reusable
Overcurrent protection	Yes	Yes
Short-circuit (fault) interruption	Yes	Yes
Works on AC and DC	Yes (check the voltage rating)	Yes (check the rating; DC is harder to interrupt)
Shock protection for people	No	No (unless it is a GFCI breaker)

- **Breaker size and wire size** go together (US NEC 240.4(D)): 14 AWG copper → 15 A; 12 AWG → 20 A; 10 AWG → 30 A. Never put a bigger breaker on a thinner wire.
- **80% rule:** a load that runs 3 hours or more is a **continuous load**. It should use at most 80% of the breaker rating (NEC 210.20(A)).

- **AFCI** (arc-fault circuit interrupter) detects the electrical "signature" of arcing (sparking) in damaged wires and loose connections. It prevents **fires**. An ordinary breaker can miss an arc, because the arc current may be small.
- **Dual-function** AFCI/GFCI breakers do both jobs.
- **Arc suppression:** opening a large current makes an arc. Big switchgear uses **mineral oil**, vacuum, or a special gas (SF_6) to put out the arc (ATX 2026 B MC40). Home breakers use **arc chutes** (metal plates that split and cool the arc).
- **Symbols:** a curved arc over the switch gap (IEC/ANSI one-line style); a switch with a square trip box; "CB" label. In ANSI one-line diagrams the chevrons mean a **draw-out** breaker (it slides out of its cubicle).

Formulas

What	Formula	Notes
Current of one appliance	I = P / V	
Total current on a branch	I_total = sum of branch currents	KCL
Largest power on a branch	P_max = V × I_breaker	15 A × 120 V = 1800 W
Continuous-load limit	I_cont ≤ 0.8 × I_breaker	80% rule
Breaker for a continuous load	I_breaker ≥ 1.25 × I_load	Same rule seen the other way
Number of equal loads	n = whole-number part of (I_limit / I_one)	Round down
Fault current	I_fault = V / R_path	R_path = wire resistance there and back
Double-pole load	I = P / 240	

Question types

Easy

1. Fuse versus breaker: which statement is true?

- Scope: ✅ IN.
- Where seen: Berkeley 2026 B (INV) Q13 (true difference → C, a fuse must be replaced; a breaker can be reset). BullSO 2026 B Q29 and BullSO 2026 C Q29 (select all → key B, C). Shock Value re-tryout I-2 (a fuse differs from a ___ → circuit breaker). National supervisor practice deck 11 (CTRL-FUSE-VS-BREAKER).
- How to solve:
  1. Remember the one big difference: fuse = replace, breaker = reset.
  2. Both protect against overcurrent and shorts.
  3. Both work on AC and DC.
  4. A fuse is usually faster, but not "in all cases".
- Worked example (Berkeley Q13): "A. A fuse responds faster in all cases. B. Breakers protect against voltage surges, fuses against current surges. C. A fuse must be replaced after it blows, a breaker can be reset. D. Breakers melt a wire, fuses use a mechanical switch." Answer: **C**.
- Traps: Watch for "always" and "in all cases". Option D swaps the two devices. Neither device is a surge protector.
- Calculator: look up — type: "fuse vs breaker".

2. Why choose a fuse over a breaker (or the reverse)?

- Scope: ✅ IN.
- Where seen: Rickards 2026 C (INV) Q48 ("A fuse may be preferred to a breaker because it does NOT ___": A respond faster, B have reusability potential, C pose a touching hazard, D cost as much, E none → key D).
- How to solve:
  1. List what a fuse has: faster, cheaper, simple.
  2. List what a breaker has: reusable, easy reset.
  3. Read "does NOT" carefully. Pick the thing a fuse lacks that makes it better.
- Worked example (Rickards Q48): a fuse does NOT cost as much. Answer: **D**.
- Traps: B ("does not have reusability") is true, but it is a disadvantage, not a reason to prefer a fuse.

- Calculator: look up — type: "fuse advantage".

3. What does a breaker protect? Breaker versus GFCI

- Scope: ✅ IN.
- Where seen: ATX Regional 2026 C (REG) FRQ 6 (difference between a standard breaker and a GFCI → breaker prevents fires from too much current; GFCI prevents shock from leakage). Princeton 2026 C (INV) SA3a.
- How to solve:
  1. Breaker: watches the **amount** of current. Protects wires from fire.
  2. GFCI: watches the **balance** of hot and neutral. Protects people from shock.
- Worked example: "Why does a 15 A breaker not save a person from a 100 mA shock?" Answer: **0.1 A is far below 15 A, so the breaker never sees an overload. Only a GFCI notices the 0.1 A leaking to ground.**
- Traps: "Breakers protect people" is a common wrong choice.
- Calculator: look up — type: "breaker vs GFCI".

4. Identify a breaker symbol or a real breaker and its rating

- Scope: ✅ IN.
- Where seen: ATX Regional 2026 B (REG) MC11 ("molded case, industry circuit breaker" from symbols T/U/V/W → key V; see Mistakes). Shock Value re-tryout IV-2 (symbol chart includes a breaker). Cobra 2020 B (INV) station S3 (real part → circuit breaker, 30 A; give name, symbol, rating, use).
- How to solve:
  1. Look for a curved arc over the gap, or a switch with a "CB" or trip box.
  2. On a real breaker, read the number on the handle: that is the amp rating.
  3. State its use: overcurrent protection of a branch circuit.
- Worked example (Cobra S3): a black block with a toggle marked "30". Answer: **circuit breaker, 30 A, protects a circuit (for example a dryer) from overcurrent**.
- Traps: A fuse symbol is a rectangle with a line or an S-wave. A breaker is an arc or a switch with a latch.
- Calculator: look up — type: symbol shape, handle number.

5. How do you reset a tripped breaker?

- Scope: ✅ IN.
- Where seen: not seen in a test. Source: manufacturer instructions (general knowledge); OpenStax *College Physics 2e* §20.6.
- How to solve:
  1. Unplug or switch off the load that caused the trip.
  2. Push the handle fully to OFF (it is stuck in the middle).
  3. Push it to ON.
  4. If it trips again at once, there is a short. Do not keep resetting.
- Worked example: "A breaker trips every time you plug in a toaster, even alone." Answer: **the toaster (or its cord) likely has a short; repair it, do not upsize the breaker**.
- Traps: Never replace a breaker with a bigger one to stop trips.
- Calculator: look up — type: "reset breaker".

Medium

6. Thermal trip versus magnetic trip

- Scope: ✅ IN for the thermal trip and the overload/short difference. ⚠️ Borderline for the magnetic trip mechanism (magnetism is not in rule 3.c).
- Where seen: not seen in a test. Prior notes B_topics §6d, I_bank D23. Source: Kuphaldt, *Lessons in Electric Circuits*, Vol. I, "Fuses" and Vol. IV, "Electromechanical relays" (thermal overloads); All About Circuits.
- How to solve:
  1. Slow overload → heat → bimetal bends → **thermal** trip (seconds to minutes).
  2. Sudden huge current → **magnetic** trip (milliseconds).

- Worked example: "A space heater on an extension cord trips the breaker after 10 minutes." Answer: **thermal trip (overload)**. "A screwdriver shorts hot to neutral and the breaker trips at once." Answer: **magnetic trip (short circuit)**.
- Traps: A bimetal strip bends because the two metals expand by **different** amounts, not because one melts.
- Calculator: look up — type: fault type → trip type.

7. Largest power a branch circuit can supply

- Scope: ✅ IN.
- Where seen: not seen in a test. Prior bank I_bank W4. Source: OpenStax *College Physics 2e* §20.4 "Electric Power and Energy".
- How to solve:
  1. P_max = V × I_breaker.
- Worked example: 15 A at 120 V → **1800 W**. 20 A at 120 V → **2400 W**.
- Traps: Use the circuit voltage (120 V), not 110 V, unless the question says 110 V.
- Calculator: calculate — type: V, breaker amps.

8. Will the breaker trip? (several appliances)

- Scope: ✅ IN.
- Where seen: not seen in a test for a breaker. Same method as Pembroke Hill 2026 B Q29 (fuse). Prior bank I_bank D24.
- How to solve:
  1. Find each appliance current: I = P/V.
  2. Add them (parallel loads on one circuit).
  3. Compare with the breaker rating.
- Worked example: a 20 A, 120 V kitchen circuit. A 1500 W hair dryer and a 1200 W microwave run together. I = 12.5 + 10 = **22.5 A > 20 A → trips**. A 1500 W dryer and a 900 W toaster: I = **20 A**, exactly at the limit; on a test "does not trip", but any extra load or startup surge trips it.
- Traps: Appliances on one circuit are in **parallel**, so currents add. Add the powers first if that is easier: 2700 W / 120 V = 22.5 A.
- Calculator: calculate — type: V, list of P, breaker amps.

9. How many identical loads before the breaker trips?

- Scope: ✅ IN.
- Where seen: not seen in a test for a breaker. Same method as Princeton 2021 C Q37 (fuse). Prior bank I_bank D24.
- How to solve:
  1. I_one = P/V.
  2. n = I_breaker / I_one. Round **down**.
  3. If the question says "continuous" or "safely", use 80% of the rating.
- Worked example: 100 W bulbs on a 15 A, 120 V circuit. I_one = 0.833 A. 1800/100 = **18 bulbs**. With the 80% rule: 1440/100 = 14.4 → **14 bulbs**.
- Traps: Do not round 14.4 up.
- Calculator: calculate — type: V, P each, breaker amps, continuous (yes/no).

10. Match breaker size and wire size

- Scope: ✅ IN (hazards; resistance and gauge).
- Where seen: HH 2026 Invitational (INV) Q2 (current capacity depends on gauge) and Q3 (voltage rating depends on insulation). Prior bank I_bank D25.
- How to solve:
  1. Recall: 14 AWG ↔ 15 A, 12 AWG ↔ 20 A, 10 AWG ↔ 30 A (copper, US homes).
  2. The breaker must not be bigger than the wire allows.
- Worked example: "A 20 A breaker feeds 14 AWG wire. Is that safe?" Answer: **no. 14 AWG is rated 15 A. At 18 A the wire overheats but the breaker never trips.**

- Traps: A smaller AWG number means a **thicker** wire.
- Calculator: look up — type: AWG or breaker amps.

11. The breaker keeps tripping: overload, short, or ground fault?

- Scope: ✅ IN (troubleshooting, hands-on Part II.c.vi idea; open and short circuits, rule 3.c.v).
- Where seen: not seen in a test. Princeton 2026 C SA3d is the GFCI version (see GFCI A6). Source: OpenStax *College Physics 2e* §20.6.
- How to solve:
  1. Trips after a while with many loads on → **overload**. Move loads.
  2. Trips instantly when one device is plugged in → **short** in that device.
  3. Trips instantly with nothing plugged in → short in the wiring.
  4. A GFCI (not the breaker) trips near water → **ground fault**.
- Worked example: "The breaker trips as soon as you flip it on, with every plug pulled." Answer: **a short in the fixed wiring of that circuit**.
- Traps: A weak breaker can also trip early, but tests expect the circuit causes above.
- Calculator: look up — type: when it trips → cause.

Hard

12. The 80% continuous-load rule

- Scope: ✅ IN (power and current calculation). The rule itself is a code detail, so the question must state it or ask "safely".
- Where seen: not seen in a test. Prior bank I_bank W4 (NEC 210.20(A)).
- How to solve:
  1. Continuous limit = 0.8 × breaker rating.
  2. Or: breaker ≥ 1.25 × continuous load current.
- Worked example: a 20 A, 120 V circuit runs a load all night. Limit = 0.8 × 20 = 16 A, so P = 16 × 120 = **1920 W**.
- Traps: Use the 80% limit only for continuous loads (3 hours or more) unless told otherwise.
- Calculator: calculate — type: breaker amps, V.

13. Fault current: why a short trips the breaker so fast

- Scope: ✅ IN (Ohm's law, resistance of wires; short circuits).
- Where seen: not seen in a test. Source: Kuphaldt, *Lessons in Electric Circuits*, Vol. I, "Electrical Safety"; wire resistance from standard copper tables (14 AWG about 2.525 Ω per 1000 ft).
- How to solve:
  1. Find the path resistance: wire out and back (total length × Ω per length).
  2. I_fault = V / R_path.
  3. Compare with the breaker: many times the rating → magnetic trip at once.
- Worked example: a short 50 ft from the panel on 14 AWG. Path = 100 ft × 2.525 Ω/1000 ft = **0.25 Ω**. I = 120/0.2525 = **475 A** (about 32 × 15 A) → **instant magnetic trip**. Compare: a poor contact with 20 Ω makes only **6 A** → **no trip**, but it can still arc and heat. That is why AFCIs exist.
- Traps: The 475 A is an upper estimate. Real faults also include the panel and utility resistance. Do not forget the return wire (double the length).
- Calculator: calculate — type: V, length, Ω per 1000 ft, breaker amps.

14. AFCI: what it prevents and where it is used ⚠️

- Scope: ⚠️ Borderline. An AFCI is a breaker type (rule 3.c.vi), but the rule does not name it.
- Where seen: ATX Regional 2026 B (REG) MC31 ("What do AFCIs prevent?" → B, fires).
- How to solve:
  1. AFCI = **arc-fault circuit interrupter**.
  2. It senses the noisy current pattern of sparking.
  3. It prevents **fires** (GFCI prevents **shocks**).

4. US code requires them on most 120 V, 15 A and 20 A home circuits: bedrooms, living rooms, family rooms, dining rooms, kitchens, laundry areas, hallways, closets (NEC 210.12; edition-dependent).

- Worked example (ATX MC31): Answer: **fires**.
- Traps: Do not mix up AFCI (fire) and GFCI (shock). Memory hook: **A**rc → **A**blaze; **G**round → **G**uards people.
- Calculator: look up — type: "AFCI".

15. What helps put out an arc? (switchgear)

- Scope: ✅ IN (breakers; hazards).
- Where seen: ATX Regional 2026 B (REG) MC40 ("What material may help prevent arcing?" water / air / mineral oil / graphite → C, mineral oil).
- How to solve:
  1. An arc needs a conducting path through hot, ionized gas.
  2. An insulating liquid (mineral oil) cools and smothers it.
  3. Water and graphite conduct. Air is where arcs form.
- Worked example: Answer: **mineral oil**.
- Traps: Pure water is a poor conductor, but real water has ions and conducts. Never pick water near electricity.
- Calculator: look up — type: "arc quench".

16. Double-pole (240 V) breaker sizing

- Scope: ✅ IN for P = VI. ⚠️ Borderline for split-phase details (two 120 V legs). It is not 3-phase power, so it is not excluded.
- Where seen: not seen in a test with a breaker. Related: USC Invitational 2026 B (INV) Q6 statement IV (240 V outlet hot slots; see Surge Protection and Grounding A6).
- How to solve:
  1. I = P / 240.
  2. For a continuous load, multiply by 1.25.
  3. Pick the next standard size (15, 20, 25, 30, 40, 50 A).
- Worked example: a 4500 W, 240 V water heater. I = 4500/240 = **18.75 A**. Continuous: 18.75 × 1.25 = **23.4 A** → **25 A double-pole breaker** (10 AWG wire is often used, rated 30 A).
- Traps: Using 120 V doubles the current by mistake. Both hots must open together, so it is one tied breaker, not two separate ones.
- Calculator: calculate — type: P, V = 240, continuous (yes/no).

17. Trip-curve classes (B, C, D) ⚠️

- Scope: ⚠️ Borderline. Reading a time-current graph is fine. Log-scale math is not in rule 3.e.
- Where seen: not seen in a test. Prior bank I_bank D26 (IEC 60898).
- How to solve:
  1. Know the magnetic trip bands: **B** trips at 3-5 × rating, **C** at 5-10 ×, **D** at 10-20 ×.
  2. Multiply the rating by the band to get the instant-trip current.
- Worked example: a 16 A type C breaker trips instantly between 5 × 16 = **80 A** and 10 × 16 = **160 A**.
- Traps: Motors with big inrush need C or D. B suits lighting and plain resistive loads.
- Calculator: calculate — type: rating, class.

Nationals

18. Panel problem: branches, main breaker and KCL

- Scope: ✅ IN for adding branch currents (single node). ✅ States-Nationals if the panel needs two-unknown KCL/KVL (for example wire resistances in two branches).
- Where seen: not seen in a test. Prior bank I_bank "whole-house" scenario (loads → current → breaker check → wire gauge → GFCI need → cost).
- How to solve:
  1. Find each branch current (I = P/V).
  2. Check each branch against its breaker.

3. Add all branch currents for the main (KCL at the bus).
4. For 240 V loads, each leg carries the full 240 V load current.

- Worked example: 120 V branches carry 12.5 A, 10 A and 8 A. Total = **30.5 A**. A 100 A main **does not trip**. Each branch is under its 15 A or 20 A breaker.
- Traps: Add currents, not breaker ratings. Ten 20 A breakers on a 100 A main is normal, because they are not all fully loaded.
- Calculator: calculate — type: list of P and V, branch ratings, main rating.

19. Wire resistance in a branch: voltage at the load and breaker check (two equations)

- Scope: ✅ States-Nationals (rule 3.c.ix; two unknowns).
- Where seen: not seen in a test. Related wire-sizing item: Princeton 2026 C (INV) Comp 5 (12 V supply, at least 10 V at the load, 2.2 A, 2 m of copper → 0.91 Ω, 3.69 × 10^{-8} m², 4.4 W).
- How to solve:
  1. Model each wire as a resistor in series with the load.
  2. Write KVL around the loop: V = I(R_wire + R_load).
  3. Find I. Check the breaker. Find the load voltage V_load = I × R_load.
- Worked example (Princeton Comp 5 numbers): the wires may drop 12 − 10 = 2 V at 2.2 A. Largest wire resistance = 2/2.2 = **0.91 Ω**. Power lost in the wires = 2 × 2.2 = **4.4 W**.
- Traps: Count both wires (out and back) if the question gives one-way length.
- Calculator: calculate — type: V supply, V needed, I, (length, resistivity).

Out of scope (skip)

- ❌ **Magnetic-trip field strength, solenoid force, coil inductance.** Magnetism is not in rule 3.c; rule 3.d excludes inductance.
- ❌ **3-phase breakers and panels** (three-pole breakers on 208 V or 480 V). Rule 3.d: 3-phase power.
- ❌ **Time-current curve math with logs or I²t.** Rule 3.e (only algebra, two equations; rule 3.e.v no calculus).
- ❌ **AC fault current with impedance, X/R ratio, power factor.** Rule 3.d: AC circuit theory.
- ❌ **Current and voltage transformers in switchgear** (shown on the ATX 2026 image sheet). Rule 3.d: inductance; magnetism not in rule 3.c.
- ⚠️ AFCI details beyond "detects arcs, prevents fires".

Mistakes in tests and keys

- **BullSO 2026 B Q29 and BullSO 2026 C Q29** (select all): the key is **B and C**. C ("only one can be reset without replacement") is right. B ("only one is suited for fault interruption") is **doubtful**. Both fuses and breakers interrupt faults. The correct answer is **C only**. If this appears, answer C and be ready to protest B.
- **ATX Regional 2026 B (REG) MC11** (molded-case breaker symbol): the key is V. The earlier extraction thinks **W** is the plain molded-case symbol and V (with chevrons) is a draw-out breaker. Not confirmed (I did not see the image sheet this run).
- **BullSO 2026 Q28** and **ATX 2026 B MC28** spell it "Interruptor". The standard spelling is "Interrupter". See GFCI A6.

Not confirmed

- The NEC numbers (240.4(D) wire sizes, 210.20(A) 80% rule, 210.12 AFCI locations) are from prior notes and memory. I did not fetch the NEC this run. The AFCI room list changes with the code edition.
- UL 489 test points: manufacturer datasheets (found by web search this run) show "must trip in under 1 hour at 135%" and "must not trip within 1 hour at 105%". I did not read the UL 489 standard itself.
- IEC B/C/D trip bands are from prior bank I_bank D26 (IEC 60898), not re-checked.
- The 14 AWG resistance (2.525 Ω per 1000 ft) is the standard copper table value from memory.
- No STATE or NATIONAL test with a breaker calculation was found. The Florida State 2021 test has no breaker item.

Sources

- Rules: 2027 Div B Rules Manual, Circuit Lab, rules 3.c.iv-vi, 3.c.ix, 3.d, 3.e (pages B11-B12). See Circuit Lab - Rules 2027 R.
- Tests (raw text checked this run): Berkeley 2026 B Q13; BullSO 2026 B Q29 and key; BullSO 2026 C Q29 and key; Rickards 2026 C Q48; ATX Regional 2026 B MC11, MC31, MC40; ATX Regional 2026 C FRQ 6 and key; HH 2026 Q2-Q3; Princeton 2026 C SA3. From extraction files only: Cobra 2020 B station S3; Shock Value re-tryout I-2, IV-2; Princeton 2026 Comp 5.
- National supervisor practice deck 11 (`prior/F_slides.md`, CTRL-FUSE-VS-BREAKER).
- Prior research: `prior/B_topics.md` §6d, `prior/I_bank.md` D23-D27, D34, W4.
- Web (this run): UL 489 datasheet values (Mouser-hosted UL 489 breaker datasheets, for example https://www.mouser.com/datasheet/2/16/1CUL_489-AC-221374.pdf).
- Textbooks: Kuphaldt, *Lessons in Electric Circuits*, Vol. I, "Electrical Safety" and "Fuses"; OpenStax *College Physics 2e* §20.4 and §20.6; All About Circuits.
- All numbers checked in Python on 2026-10-16.

GFCI (Ground Fault Circuit Interrupter)

Note: GFCI

A **GFCI** is a safety device that protects **people** from electric shock. It compares the current going out on the hot wire with the current coming back on the neutral wire. If they differ by a few milliamps, it shuts off the power in a fraction of a second. Another name is **GFI** (ground fault interrupter).

This note is ✅ **IN at every level**. Rule 3.c.vi names "ground fault circuit interrupters". Shock currents tie to rule 3.c.iii (AC hazards). Leakage problems that need KCL with two unknowns are ✅ States/Nationals (rule 3.c.ix).

One limit applies: inside, the GFCI senses the difference with a **toroid** (a ring-shaped coil). That works by magnetism. Rule 3.c lists no magnetism, and rule 3.d excludes inductance. Know "a sensing ring compares the two currents". Do not explain the magnetic induction.

Key facts

- **Ground fault** = current leaving the circuit by an unintended path to ground. The path may be a wet floor, a metal pipe, or a **person**.
- **Normal circuit:** every electron that goes out on the hot comes back on the neutral. So I_hot = I_neutral. This is KCL for the whole appliance.
- **Ground fault:** some current goes to ground instead. Then I_hot > I_neutral. The difference is the leakage current.
- **How it trips:** the hot and neutral wires both pass through the sensing ring. If the currents are equal, they cancel. If they are not equal, the ring makes a signal. A small circuit fires a trip coil, and the contacts open.
- **Trip level (US Class A, UL 943):** must trip at **6 mA** or more. Must not trip below **4 mA**. Tests use **5 mA** (ATX 2026 B MC29).
- **Trip time:** it must trip faster when the leakage is bigger. Typical shock-level faults are cleared in about **25 ms** (1/40 s). See type 16 for the time rule.
- **What it prevents: shocks** (ATX 2026 B MC30). A breaker or fuse prevents **fires** from overcurrent.
- **Where (US code, NEC 210.8):** bathrooms, kitchen countertops, garages, outdoors, crawl spaces, unfinished basements, laundry areas, within 6 ft of a sink, bathtub or shower, boathouses, pools and spas. In short: **near water or the ground**.
- **Buttons: TEST** puts a small, known imbalance through a built-in resistor. The GFCI should trip. **RESET** turns it back on. Test it monthly.
- **LINE and LOAD terminals:** power from the panel connects to **LINE**. Outlets further along the circuit connect to **LOAD**. One GFCI outlet then protects every outlet downstream. Wired backwards, the outlet may have power but give no protection (newer models refuse to reset).
- **Forms:** GFCI receptacle (outlet), GFCI breaker (protects a whole circuit), portable GFCI (in a cord, for tools), and built-in cord units on hair dryers.
- **It does NOT trip:**
  - on an **overload** (too much current, but balanced). That is the breaker's job.
  - on a **hot-to-neutral short** (balanced). The breaker handles it.
  - when a person touches **hot and neutral** together while insulated from ground. The current through the person goes out on hot and back on neutral, so it stays balanced.
- **It works without a ground wire.** The code allows a GFCI to replace an old two-prong outlet. It must be labeled "No Equipment Ground".
- **It does not limit the current size.** It only cuts the power quickly. A person still feels a short, sharp shock.
- **Inventor:** Charles Dalziel (University of California, Berkeley) developed the GFCI around 1961. (Not a rule-listed scientist; trivia only.)
- **Other countries:** the same device is called an **RCD** (residual current device) or **RCCB**. Home RCDs usually trip at **30 mA**. US industrial equipment-protection devices (GFPE) trip at about 30 mA or more. ⚠️ Borderline trivia.
- **Spelling:** "Interrupter" is standard. Some tests print "Interruptor" (BullSO 2026, ATX 2026 B).

Formulas

What	Formula	Notes
Leakage (imbalance)	I_leak = I_hot − I_neutral	KCL
Trips? (US Class A)	trips if I_leak ≥ 6 mA; does not if < 4 mA	4-6 mA: may or may not; tests use 5 mA
Current through a person	I = V / R_body	Dry skin about 100 kΩ; wet about 1 kΩ
Hot current with a person in the path	I_hot = I_load + I_person	
Energy of a short shock	E = V × I × t	
Several small leaks	I_leak = sum of each leak	Nuisance tripping
UL 943 maximum trip time ⚠️	T = (20 / I)^1.43 seconds, I in mA	Exponent is outside the math rules

Question types

Easy

1. What does GFCI stand for?

- Scope: ✅ IN.
- Where seen: BullSO 2026 B Q28 and BullSO 2026 C (INV) Q28 (→ B, Ground Fault Circuit Interruptor). ATX Regional 2026 B (REG) MC28 (same). Pembroke Hill 2026 B (INV) Q16 (name the device; key: GFCI, Ground Fault Circuit Interrupter).
- How to solve: Ground, Fault, Circuit, Interrupter.
- Worked example: Answer: **Ground Fault Circuit Interrupter**.
- Traps: The fake choices use the same letters ("Global Fault Control Interface"). Pick the one with "ground" and "fault".
- Calculator: look up — type: "GFCI".

2. What does a GFCI protect against?

- Scope: ✅ IN.
- Where seen: ATX Regional 2026 B (REG) MC30 (shocks / high energy bills / maintenance / fires → A, shocks). Pembroke Hill 2026 B (INV) Q16 ("protects humans from shock in wet areas by detecting imbalances between live and neutral" → GFCI).
- How to solve:
  1. GFCI → people → shock.
  2. Breaker or fuse → wires → fire from overcurrent.
  3. AFCI → arcs → fire.
- Worked example (ATX MC30): Answer: **shocks**.
- Traps: "Fires" is the AFCI answer (ATX MC31, the very next question).
- Calculator: look up — type: device → hazard.

3. Where are GFCIs required?

- Scope: ✅ IN.
- Where seen: ATX Regional 2026 B (REG) MC32 (bedroom / living room / kitchen / basement → key C, kitchen).
- How to solve:
  1. Look for water or a grounded floor: kitchen, bathroom, garage, outdoors, basement, laundry, near a sink or pool.
  2. Dry living spaces (bedroom, living room) do not need GFCIs in older codes. (They need AFCIs.)
- Worked example (ATX MC32): Answer: **kitchen** (countertops).
- Traps: Basements also need GFCIs under the US code (unfinished basements in older editions; all basements in newer ones). See Mistakes.
- Calculator: look up — type: room.

4. What leakage makes a GFCI trip?

- Scope: ✅ IN.

- Where seen: ATX Regional 2026 B (REG) MC29 ("What is the change in current needed for a GFCI to trigger?" 1 A / 5 A / 1 mA / 5 mA → D, 5 mA).
- How to solve: recall about **5 mA** (Class A: 4-6 mA).
- Worked example: Answer: **5 mA**.
- Traps: 5 A is a fuse-size number. 1 mA is too small (normal appliances leak a little). Watch "mA" versus "A".
- Calculator: look up — type: "GFCI trip current".

5. TEST and RESET buttons

- Scope: ✅ IN.
- Where seen: not seen in a test. Prior notes B_topics §6e, I_bank D31. Source: manufacturer instructions; OpenStax *College Physics 2e* §20.6.
- How to solve:
  1. TEST: makes a small imbalance through a resistor, so the GFCI should trip.
  2. RESET: closes the contacts again.
  3. If TEST does not trip it, replace the GFCI.
- Worked example: "You press TEST and the outlet stays on." Answer: **the GFCI is not protecting; replace it**.
- Traps: A GFCI that trips on TEST but is wired LINE/LOAD backwards can still leave the outlet live (older models).
- Calculator: look up — type: "GFCI test reset".

Medium

6. How does a GFCI know when to trip?

- Scope: ✅ IN (principle only; the magnetism detail is OUT).
- Where seen: Princeton 2026 C (INV) SA3b ("How does a ground fault circuit know when to trip?" → key: hot and neutral pass through a small toroid; it checks if the currents are equal; if not, current is leaking to ground and it trips). Pembroke Hill 2026 B (INV) Q16.
- How to solve:
  1. Say it compares hot current and neutral current.
  2. Say equal means all current returns on the neutral (safe).
  3. Say unequal (by about 5 mA) means current leaks to ground, maybe through a person. It trips.
- Worked example: Answer: **it measures the difference between the current on the hot wire and the current on the neutral wire; a difference of a few mA means leakage to ground, so it opens the circuit**.
- Traps: It does **not** measure the total current. It does **not** need the ground wire to sense a fault.
- Calculator: look up — type: "GFCI principle".

7. Fuse versus breaker versus GFCI (compare all three)

- Scope: ✅ IN.
- Where seen: Princeton 2026 C (INV) SA3a ("main difference between a fuse, a circuit breaker and a GFCI" → key: fuse melts permanently; breaker trips mechanically and resets; GFCI trips on leakage to ground, not necessarily excess current). ATX Regional 2026 C (REG) FRQ 6 (breaker vs GFCI). BullSO 2026 C Q29 (fuse vs breaker).
- How to solve:
  1. Fuse: overcurrent; melts; replace.
  2. Breaker: overcurrent; trips; reset.
  3. GFCI: hot/neutral imbalance (leakage); trips; reset. Protects people.
- Worked example: Answer: **a fuse and a breaker respond to too much current (the fuse melts, the breaker resets); a GFCI responds to a small difference between hot and neutral current, so it protects people from shock**.
- Traps: A GFCI can trip when the total current is tiny. A breaker can carry 14 A of perfectly balanced current with no GFCI trip.
- Calculator: look up — type: "fuse breaker GFCI".

8. Why does the fuse never blow while the GFCI trips again and again?

- Scope: ✅ IN.

- Where seen: Princeton 2026 C (INV) SA3c (key: a fuse responds to large overcurrent; a GFCI responds to leakage of a few mA, far below the current needed to melt a fuse).
- How to solve:
  1. Compare sizes: leakage ≈ 0.005 A; fuse ≈ 15 A.
  2. The leak is about 3000 times too small to blow the fuse.
- Worked example: 5 mA leak on a 15 A fuse. 15/0.005 = **3000**. Answer: **the leakage is thousands of times smaller than the fuse rating; only the GFCI notices it**.
- Traps: Do not say "the fuse is broken".
- Calculator: calculate — type: leak mA, fuse A.

9. A hair dryer near a sink trips the GFCI but not the breaker

- Scope: ✅ IN.
- Where seen: Princeton 2026 C (INV) SA3d (key: moisture lets a small current leak to ground; the GFCI detects it; total current stays below the breaker trip).
- How to solve:
  1. Water gives a new path to ground.
  2. A few mA leak away, so hot ≠ neutral.
  3. The total current (about 12 A for 1500 W) is still under the 15 A or 20 A breaker.
- Worked example: 1500 W at 120 V draws **12.5 A**, under a 20 A breaker. A 6 mA leak through water trips the GFCI. Answer: **the GFCI sees the 6 mA imbalance; the breaker only sees 12.5 A, which is normal**.
- Traps: The dryer is not "using too much power".
- Calculator: calculate — type: P, V, breaker amps, leak mA.

10. Imbalance calculation: hot and neutral currents given

- Scope: ✅ IN (KCL at one point, subtraction).
- Where seen: not seen in a test. Prior bank I_bank D29.
- How to solve:
  1. I_leak = I_hot − I_neutral.
  2. Convert to mA.
  3. ≥ 6 mA → trips. < 4 mA → does not. 4-6 mA → may trip (on a test using "5 mA", ≥ 5 mA → trips).
- Worked example 1: hot 10.000 A, neutral 9.994 A. Leak = **6 mA → trips**.
- Worked example 2: hot 5.000 A, neutral 4.998 A. Leak = **2 mA → does not trip**.
- Traps: Keep enough decimal places. 9.994 A is 6 mA less than 10.000 A, not 0.6 mA.
- Calculator: calculate — type: I_hot, I_neutral.

11. Does it trip? (scenario list)

- Scope: ✅ IN.
- Where seen: not seen in a test. Prior bank I_bank D30. Source: OpenStax *College Physics 2e* §20.6; Kuphaldt, *Lessons in Electric Circuits*, Vol. I, "Electrical Safety" ("Ground-fault detection and interruption").
- How to solve: ask one question. **Does any current avoid the neutral?**
  - Person touches hot, stands on a wet floor → yes → **trips**.
  - Hot wire touches a grounded metal case → yes → **trips** (the breaker may trip too).
  - Too many appliances, 25 A total, balanced → no → **no GFCI trip** (the breaker trips).
  - Hot touches neutral (short) → no → **no GFCI trip** (the breaker trips).
  - Person on a rubber mat holds hot in one hand and neutral in the other → no → **does not trip**, and the person is shocked.
- Worked example: "A worker touches hot and neutral at once while standing on rubber." Answer: **the GFCI does not trip, because the current returns on the neutral**.
- Traps: The last case is the classic trick. A GFCI is not perfect protection.
- Calculator: look up — type: current path.

12. GFCI on an old two-prong (ungrounded) outlet

- Scope: ✅ IN (grounding and hazards).

- Where seen: not seen in a test. Prior notes B_topics §6e, I_bank D30.
- How to solve:
  1. The GFCI senses the hot/neutral difference. It does not need a ground wire.
  2. A person touching a live case leaks current to earth. The GFCI still sees the difference and trips.
  3. Label the outlet "No Equipment Ground".
- Worked example: "Can you install a GFCI where there is no ground wire?" Answer: **yes; it still protects people, but the outlet has no equipment ground, so it must be labeled**.
- Traps: A surge protector plugged into it will not work fully (it needs ground). See Surge Protection and Grounding [A6].
- Calculator: look up — type: "GFCI no ground".

Hard

13. Body current with and without a GFCI

- Scope: ✅ IN (Ohm's law, rule 3.c.iv; AC hazards, rule 3.c.iii). Use the given resistance; no AC theory.
- Where seen: not seen in a GFCI test item. Related: ATX Regional 2020 B (REG) Q40 (lethal shock range → 100-200 mA); UT Invitational fall 2019 C Q18 (200 mA hand-to-feet for 20 ms → effect). Prior bank I_bank S2, D32.
- How to solve:
  1. I = V / R_body.
  2. Compare with the effect levels: 1 mA felt; about 10-20 mA cannot let go; about 100 mA heart fibrillation possible.
  3. Compare with the GFCI trip level. If above, it trips in a fraction of a second.
  4. Energy: E = V × I × t.
- Worked example: 120 V, wet skin 1 kΩ. I = 120/1000 = **120 mA** (very dangerous). A GFCI trips in about 25 ms. Energy = 120 × 0.12 × 0.025 = **0.36 J**. Dry skin 100 kΩ: I = **1.2 mA** (a tingle; below the 4 mA level, so no trip).
- Traps: The GFCI does not lower the 120 mA. It shortens the time. Time matters for the heart.
- Calculator: calculate — type: V, R_body, trip time.

14. LINE versus LOAD: which outlets are protected?

- Scope: ✅ IN (series/parallel wiring ideas; schematics).
- Where seen: not seen in a test. Prior bank I_bank D31. Source: manufacturer instructions (general knowledge).
- How to solve:
  1. Outlets fed from the GFCI's **LOAD** terminals are protected.
  2. Outlets fed **before** the GFCI (upstream) or from its LINE terminals are not.
  3. A GFCI breaker protects the whole circuit.
- Worked example: a bathroom circuit runs panel → outlet A (GFCI, wired LINE in, LOAD out) → outlet B → outlet C. A fault at C. Answer: **GFCI A trips; B and C both lose power, because they are on its LOAD side**.
- Traps: When a "dead" outlet has no buttons, look for a tripped GFCI upstream (often in another room or the garage).
- Calculator: look up — type: outlet order, which terminals.

15. Why a GFCI may trip with no one touching anything (nuisance tripping)

- Scope: ✅ IN (adding currents, KCL).
- Where seen: not seen in a test. Source: manufacturer application notes (general knowledge).
- How to solve:
  1. Every appliance leaks a little current to ground (through moisture or insulation).
  2. The GFCI sees the **sum** of all leaks on its LOAD side.
  3. If the sum reaches the trip level, it trips.
- Worked example: seven devices on one GFCI each leak **1 mA**. Total = **7 mA ≥ 6 mA → trips**, even though no single device is faulty.
- Traps: Moisture in an outdoor outlet box is the other common cause.
- Calculator: calculate — type: list of leak mA.

16. GFCI trip time rule (UL 943) ⚠️

- Scope: ⚠️ Borderline. The formula uses a non-integer power (exponent 1.43). Rule 3.e allows "functions and algebraic manipulations", so a supervisor could give it with a calculator. Deriving it is OUT.
- Where seen: not seen in a test. Formula from patent filings that quote UL 943 (web search this run).
- How to solve:
  1. Use T = (20/I)^1.43, with I in mA and T in seconds.
  2. This is the **longest** time allowed. Real GFCIs are usually faster.
- Worked example: at **5 mA**, T = 4^1.43 = **7.26 s**. At **20 mA**, T = **1.0 s**. At **264 mA**, T = **0.025 s** (25 ms). So the famous "25 ms" belongs to a large fault, not to 5 mA.
- Traps: Do not say "a GFCI always trips in 25 ms". It trips faster for bigger faults.
- Calculator: calculate — type: I in mA.

Nationals

17. Find the leakage through a person (KCL with the load)

- Scope: ✅ States-Nationals (rule 3.c.ix, KCL). One or two unknowns.
- Where seen: not seen in a test. Source: OpenStax *College Physics 2e* §20.6 and §21.3 (Kirchhoff's rules).
- How to solve:
  1. The appliance current goes out on hot and back on neutral.
  2. The person current goes out on hot and back through ground.
  3. KCL at the hot wire: I_hot = I_appliance + I_person.
  4. Imbalance = I_person. Compare with the trip level.
- Worked example: a 5.00 A appliance runs. A person (2000 Ω, hand to wet floor) touches the hot. I_person = 120/2000 = **60 mA**. I_hot = **5.060 A**, I_neutral = **5.000 A**. Imbalance = **60 mA** → **trips**.
- Traps: The appliance current does not matter for the GFCI. Only the difference does.
- Calculator: calculate — type: V, I_appliance, R_person.

18. Two leakage paths in parallel (current divider to ground)

- Scope: ✅ States-Nationals (KCL with two unknowns, rule 3.e.i).
- Where seen: not seen in a test. Source: method from OpenStax *College Physics 2e* §21.3.
- How to solve:
  1. Each leakage path from hot to ground is a resistor. They are in parallel.
  2. Find each path current: I = V/R.
  3. Add them for the imbalance.
- Worked example: 120 V. A damp cord leaks through 60 kΩ (I = **2 mA**). A person touches a live case and leaks through 40 kΩ (I = **3 mA**). Total = **5 mA**: at the Class A gray zone (4-6 mA), so it **may trip**; on a "5 mA" test it **trips**.
- Traps: Neither leak alone is above 4 mA. The GFCI responds to the total.
- Calculator: calculate — type: V, list of leakage R.

Out of scope (skip)

- ❌ **How the toroid works** (magnetic flux from the current difference, induced voltage in the sense coil). Magnetism is not in rule 3.c; rule 3.d excludes inductance.
- ❌ **The GFCI's internal electronics** (the sense amplifier chip and the SCR that fires the trip coil). Rule 3.d: semiconductors beyond those listed.
- ❌ **Phase angle and capacitive leakage in long cables.** Rule 3.d: AC circuit theory and capacitance.
- ❌ **3-phase RCDs.** Rule 3.d: 3-phase power.
- ⚠️ The UL 943 time formula (type 16): only if given.

Mistakes in tests and keys

- **ATX Regional 2026 B (REG) MC32** ("Where would you find GFCIs?" bedroom / living room / kitchen / **basement**): the key is C, kitchen. Basement (D) is also correct under the US code (unfinished basements; newer editions cover basements generally). If a protest is allowed, point this out.
- **BullSO 2026 B and C Q28**, **ATX 2026 B MC28:** "Interruptor" in the choice. The standard spelling is "Interrupter". Pick that choice anyway.
- **Florida State 2021 B/C Q13** (toddler puts two fingers in only the hot slot → key D, none of the above): see Surge Protection and Grounding [A6]. If the child touches anything grounded, current leaks to ground, and a GFCI would trip. The key assumes no path to ground.

Not confirmed

- UL 943 numbers (6 mA must trip, under 4 mA must not, T = (20/I)^1.43): from patent texts and forum posts found by web search this run. I did not read UL 943 itself.
- NEC 210.8 location list: from prior notes and memory. The list changes with the code edition (2017, 2020, 2023). I did not fetch the NEC.
- Charles Dalziel as inventor (about 1961): from memory. I did not fetch a source.
- The 30 mA RCD value (IEC countries): from memory and prior bank I_bank D33.
- No STATE or NATIONAL test with a GFCI item was found. Every GFCI item seen is from an invitational or a regional.

Sources

- Rules: 2027 Div B Rules Manual, Circuit Lab, rules 3.c.iii, 3.c.vi, 3.c.ix, 3.d, 3.e (pages B11-B12). See Circuit Lab - Rules 2027 [R].
- Tests (raw text checked this run): Princeton 2026 C SA3a-d and key; BullSO 2026 B Q28 and C Q28 with keys; ATX Regional 2026 B MC28-MC32; ATX Regional 2026 C FRQ 6 and key; Pembroke Hill 2026 B Q16 and key; Florida State 2021 B/C Q13. From extraction files only: ATX Regional 2020 B Q40; UT Invitational fall 2019 C Q18.
- Extraction files: `~/Daniel/circuitlab/research/tests/` (ut-a, ut-c, atx-reg, state) and `prior/G_tests.md`.
- Prior research: `prior/B_topics.md` §6e, `prior/I_bank.md` D28-D33, S2-S4.
- Web (this run): patent texts quoting UL 943 (for example US 9640971 "Self-calibrating circuit interrupter", https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/9640971); Mike Holt forum threads on GFCI trip time.
- Textbooks: OpenStax *College Physics 2e* §20.6 "Electric Hazards and the Human Body" and §21.3; Kuphaldt, *Lessons in Electric Circuits*, Vol. I, chapter 3 "Electrical Safety".
- All numbers checked in Python on 2026-10-16.

Surge Protection and Grounding

Grounding means connecting a metal part to the earth with a low-resistance wire. Then a fault current has an easy path that is not a person. **Surge protection** means stopping short, very high voltage spikes from reaching equipment.

Scope of this note:

- **Grounding, outlets, wire colors and polarized plugs:** ✅ IN. They are hazards of AC (rule 3.c.iii) and they make fuses, breakers and GFCIs work (rule 3.c.vi). Tests ask them often.
- **Surge protectors:** ⚠️ Borderline. The rule does not name them. Their main part, the **MOV** (metal-oxide varistor), is a non-linear device not listed (rule 3.d). Know what they do and how they are rated. Do not analyze the MOV.
- **Grounding calculations with parallel paths:** ✅ IN with series/parallel rules. ✅ States-Nationals when you need KCL/KVL with two unknowns (rule 3.c.ix).
- **Ground symbols** in schematics: ✅ IN (rule 3.c.v, schematics).

Key facts

Wires and outlets (US)

- **Hot** (also called **live** or line): carries the voltage, 120 V AC from the panel. Insulation is **black** (or red for a second hot).
- **Neutral:** the normal return path. It is near 0 V. Insulation is **white** (or gray).
- **Ground** (equipment grounding conductor): a safety wire. It carries current **only during a fault**. Insulation is **green**, or the wire is **bare** copper.
- Neutral and ground are tied together at **one place only**: the main panel. There, they connect to a **ground rod** in the earth.
- Other countries (IEC): brown = live, blue = neutral, green-and-yellow = earth.
- **Standard US outlet (NEMA 5-15R):** the **narrow (short) slot is hot**, the **wide (tall) slot is neutral**, and the round (D-shaped) hole is **ground**.
- **NEMA codes:** the number before the dash is the configuration (voltage and grounding type). The number after the dash is the current rating. 5-15 = 125 V, 15 A. 5-20 = 125 V, 20 A (one slot is T-shaped). Letter R = receptacle (outlet); P = plug.
- **Polarized plug:** one blade is wider. It fits only one way. That keeps the appliance's switch and fuse on the **hot** side, and keeps a lamp's screw shell on the neutral. It does **not** ground anything.
- **Three-prong (grounded) plug:** the third pin connects the metal case to ground.
- **Double-insulated** appliances have two layers of insulation and no ground pin. Their symbol is a **square inside a square**.
- **240 V circuits** (US homes): two hots, each 120 V from neutral, on opposite halves of the supply. Between the two hots there is 240 V. This is called **split-phase**. It is not 3-phase.

Why grounding works

- If a hot wire touches a grounded metal case, a **huge** current flows through the ground wire. The breaker or fuse trips at once. The case does not stay live.
- If the case is **not** grounded, the case sits at 120 V. Nothing trips. The next person who touches it and a grounded object becomes the path.
- **Bonding** means connecting metal parts together, so they are at the same voltage. **Grounding** means connecting them to the earth.
- **No path, no current:** a bird on one power line is safe. Both feet are on the same wire, at nearly the same voltage. Touching a line and the ground (or a second wire) makes a path.
- **Downed power line on a car:** stay inside if you can. If you must leave (fire), **jump clear with feet together** and shuffle or hop away. Do not touch the car and the ground at the same time.

Ground symbols

- **Earth ground:** three horizontal lines that get shorter, like an upside-down triangle made of lines.
- **Chassis (frame) ground:** a line with three slanted lines below it (a "rake").
- **Signal (common, or clean) ground:** a hollow or filled triangle. Some sheets add a mark for a "noiseless" ground.
- In a schematic, "ground" also means the **0 V reference node**. All points with a ground symbol are connected.

Surges and surge protectors ⚠️

- A **surge** (or transient, or spike) is a very short rise in voltage. Causes: lightning, utility switching, big motors turning off.
- A **surge protector** (surge protective device, SPD) diverts the extra energy to the ground (and neutral) wire. Its main part is a **MOV** (metal-oxide varistor). A MOV has a very high resistance at normal voltage. Its resistance drops sharply above its **clamping voltage**.
- **Ratings:**
  - **Clamping voltage** (UL 1449 "voltage protection rating", VPR): the voltage it lets through. Lower is better. 330 V is the lowest common rating for 120 V.
  - **Energy rating** (joules): how much surge energy it can absorb over its life. Higher is better.
  - **Response time** (nanoseconds).
  - An indicator light shows it still protects. MOVs wear out after big surges.
- A plain **power strip** (extra outlets plus a breaker) is **not** a surge protector unless it says so.
- A surge protector needs a **ground** to work properly.
- A surge protector does **not** stop overcurrent (that is a breaker) or shock (that is a GFCI).
- **Lightning rods** (Benjamin Franklin, 1750s) give lightning a low-resistance path to ground outside the building. They protect the structure, not the wiring inside.

Formulas

What	Formula	Notes
Fault current through the ground wire	I_fault = V / (R_hot + R_ground)	Big → breaker trips
Current through a person	I = V / R_body	Dry about 100 kΩ; wet about 1 kΩ
Case voltage (ground path and person in parallel)	V_case = V × R_p / (R_hot + R_p), with R_p = R_ground ‖ R_body	Divider
Parallel resistance	R_p = R_1 × R_2 / (R_1 + R_2)	
Voltage across a short piece of wire	V = I × R_wire	"Bird on a wire"
Surge energy (rough)	E = V × I × t	Joule rating check
240 V split-phase	V_hot1-to-hot2 = 120 + 120 = 240 V	Each hot is 120 V from neutral

Question types

Easy

1. Which wire color is hot?

- Scope: ✅ IN.
- Where seen: Berkeley 2026 B (INV) Q23 (US household hot wire color: gray / green / black / blue → C, black).
- How to solve: black (or red) = hot; white = neutral; green or bare = ground.
- Worked example (Berkeley Q23): Answer: **black**.
- Traps: Blue is neutral in the IEC (European) code. Green is always ground; never hot.
- Calculator: look up — type: color, country.

2. Which outlet slot is hot? What does a polarized plug do?

- Scope: ✅ IN.

- Where seen: UT Invitational 2019 B (INV; 2020 season) Q12 and UT Invitational fall 2019 C (INV) Q12 (polarized receptacle: which slot is hot → the shorter, narrow slot).
- How to solve:
  1. Narrow (short) slot = hot.
  2. Wide (tall) slot = neutral.
  3. Round hole = ground.
  4. The wide blade of a polarized plug fits only the neutral slot. So the appliance switch stays on the hot side.
- Worked example: Answer: **the narrow slot is hot; the polarized plug makes sure the switch disconnects the hot wire**.
- Traps: The UT choices add "swings +". AC hot swings both + and − about 60 times a second. Ignore the odd wording; pick "shorter/narrow".
- Calculator: look up — type: slot shape.

3. Identify ground symbols (earth, chassis, signal)

- Scope: ✅ IN.
- Where seen: ATX Regional 2026 B (REG) MC19 (earth ground symbol from options C, T, V, E → key C; see Mistakes). ATX 2026 B image sheet A-C (earth, clean/noiseless, chassis). Prior notes B_topics §5c.
- How to solve:
  1. Three shrinking horizontal lines = **earth**.
  2. A line with slanted "rake" teeth = **chassis**.
  3. A triangle = **signal/common**.
- Worked example: three horizontal lines, each shorter than the one above. Answer: **earth ground**.
- Traps: Many old tests use "ground" for any of them. Read the exact name asked.
- Calculator: look up — type: symbol shape.

4. What is the grounding wire for?

- Scope: ✅ IN.
- Where seen: ATX Regional 2019 B (REG) Q21 (purpose of a grounding wire → a low-resistance fault path back to the panel so the breaker or fuse trips).
- How to solve:
  1. Say it connects the metal case to earth.
  2. Say a fault current flows through it, not through a person.
  3. Say the big fault current trips the breaker or blows the fuse.
- Worked example: Answer: **it gives fault current a low-resistance path to ground, so the case does not become live and the breaker trips**.
- Traps: The ground wire does **not** carry the normal return current. That is the neutral.
- Calculator: look up — type: "ground wire purpose".

5. Neutral versus ground: which carries current normally?

- Scope: ✅ IN.
- Where seen: not seen in a test. Prior bank I_bank W3.
- How to solve:
  1. Neutral: carries the normal return current all the time.
  2. Ground: carries current only during a fault.
- Worked example: "A lamp draws 0.5 A. What current is in each wire?" Answer: **hot 0.5 A, neutral 0.5 A, ground 0 A**.
- Traps: Both connect to earth at the panel, but they have different jobs. Never use the ground as a return.
- Calculator: look up — type: wire name.

6. No path, no shock (bird on a wire, one slot only)

- Scope: ✅ IN (AC hazards; open circuits).
- Where seen: Florida State 2021 B and C (STATE) Q13 (toddler puts two fingers into only the hot slot → key D, none of the above). Shock Value re-tryout VI-2 (walking on a power line → no path to ground).

- How to solve:
  1. Current needs a closed loop.
  2. Touching only one wire, while insulated from everything else, gives no loop.
  3. Ask: is there a second contact (ground, neutral, another wire)?
- Worked example: a bird on a wire carrying 100 A. Wire resistance 0.1 Ω per km; the bird's feet are 5 cm apart. V = 100 × (0.1/1000) × 0.05 = **0.5 mV**. Answer: **almost no voltage across the bird, so almost no current**.
- Traps: A toddler is rarely fully insulated. Bare feet on a floor can give a path. See Mistakes.
- Calculator: calculate — type: I, Ω per km, distance.

7. Downed power line on a car: how to escape

- Scope: ✅ IN (AC hazards).
- Where seen: UT Invitational 2019 B (INV) Q25 and UT Invitational fall 2019 C (INV) Q25 (best escape → jump out with feet together).
- How to solve:
  1. Stay in the car if there is no fire. Call for help.
  2. If you must leave, jump clear so you never touch the car and the ground at once.
  3. Land with feet together and shuffle or hop away. The ground near the line has a voltage gradient (different voltage at each foot).
- Worked example: Answer: **jump out with feet together, without touching the car, and shuffle away**.
- Traps: The rubber tires are not what keeps you safe. The metal car body carries the current around you.
- Calculator: look up — type: "power line car".

Medium

8. Least useful shock protection (polarized plug versus grounding)

- Scope: ✅ IN.
- Where seen: BullSO 2026 B Q33 and BullSO 2026 C (INV) Q33 ("least practical at preventing shocks from metal-case appliances": A three-prong plug grounds the case; B keep hot away from the case; C polarized plugs control which wire is hot; D insulator in the case → key C).
- How to solve:
  1. Ask which choice stops a live case from shocking someone.
  2. Grounding (A), good design (B) and extra insulation (D) all help.
  3. A polarized plug only chooses which wire is hot. If hot touches the case, the case is still live.
- Worked example: Answer: **C, polarized plugs**.
- Traps: Polarized plugs do improve safety in lamps (the shell is on neutral). They do not protect against a hot-to-case fault.
- Calculator: look up — type: protection method.

9. True statements about three-prong and 240 V outlets

- Scope: ✅ IN for three-prong outlets. ⚠️ Borderline for split-phase details.
- Where seen: USC Invitational 2026 B (INV) Q6 and USC 2026 C Q6 (statements I-IV → key A, I only).
- How to solve: test each statement.
  1. "Three-prong outlets have hot, neutral and ground." **True.**
  2. "Three-prong outlets are more prone to shocks and fires than two-prong." **False** (they are safer).
  3. "For electronics, a 240 V outlet is safer than a 120 V one." **False** (higher voltage is more dangerous; most electronics are made for 120 V).
  4. "In a 240 V outlet each hot slot has 60 V and the neutral/ground slot is 120 V." **False**. Each hot is 120 V from neutral; 240 V between the hots; neutral and ground are about 0 V.
- Worked example: Answer: **A, I only**.
- Traps: 120 + 120 = 240, not 60 + 60 + 120.
- Calculator: look up — type: statement keyword.

10. Read a NEMA receptacle: current, voltage, which pin

- Scope: ✅ IN (hazards; ratings). The NEMA chart itself is reference material, so put it in your notes.
- Where seen: MIT 2021 C (INV) Q92-Q97 (receptacle A → 20 A, 125 V, neutral = pin 1; receptacle B → 30 A, 347 V, hot = pin 2; key notes B is a NEMA 24-30 receptacle).
- How to solve:
  1. Match the slot shapes to a NEMA chart. The chart gives the code, for example 5-20.
  2. The first number is the configuration (5 = 125 V, grounded; 6 = 250 V; 14 = 125/250 V; 24 = 347 V).
  3. The second number is the amps.
  4. The wide (or T-shaped) slot is neutral. The round or U-shaped hole is ground.
- Worked example (MIT Q92-94): a 5-20R (one slot T-shaped). Answer: **20 A, 125 V; the wide/T slot is neutral**.
- Traps: The test shows the outlet face. A plug seen face-on is the **mirror image**. 125 V is the rating, not the 120 V nominal (the key gives part credit for 110-125 V).
- Calculator: look up — type: NEMA code → V, A.

11. Double insulation versus grounding

- Scope: ✅ IN.
- Where seen: not seen in a test. Prior bank I_bank W2. Source: OpenStax *College Physics 2e* §20.6.
- How to solve:
  1. A grounded appliance has a metal case and a ground pin.
  2. A double-insulated appliance has a plastic (or doubly insulated) case and only two prongs.
  3. Look for the square-in-a-square symbol.
- Worked example: "A hair dryer has a two-prong plug and a square-in-a-square mark. Is it unsafe?" Answer: **no; it is double-insulated, so it needs no ground pin**.
- Traps: Never cut off a ground pin to fit a two-slot outlet.
- Calculator: look up — type: "double insulated".

12. What does a surge protector do, and how do you compare two? ⚠️

- Scope: ⚠️ Borderline. Not named in rule 3.c. MOV analysis is OUT (rule 3.d, non-linear devices).
- Where seen: not seen in a test. Prior bank I_bank D35. Source: UL 1449 (rating names), manufacturer guides (general knowledge).
- How to solve:
  1. It sends spike energy to ground, so the equipment sees a lower peak.
  2. Compare: **lower clamping voltage** is better; **higher joule rating** is better.
- Worked example: protector X: 330 V clamp, 2000 J. Protector Y: 500 V clamp, 1000 J. Answer: **X is better on both**. Rough check: a spike of 100 A held at 330 V for 50 µs puts E = 330 × 100 × 0.00005 = **1.65 J** into the MOV.
- Traps: A surge protector is not a breaker or a GFCI. A power strip without a joule rating gives no surge protection.
- Calculator: calculate — type: clamp V, I, time; look up — compare ratings.

13. Which device for which hazard? (surge, overload, shock, arc)

- Scope: ✅ IN (the GFCI, fuse and breaker parts); ⚠️ for surge and AFCI.
- Where seen: Berkeley 2026 B (INV) Q13 (choice B, "breakers protect against voltage surges", is a distractor). ATX Regional 2026 B (REG) MC30-MC31. Princeton 2026 C (INV) Matching Part 2.
- How to solve:
  - Voltage spike → **surge protector**.
  - Too much current (overload, short) → **fuse or breaker**.
  - Leakage to ground, shock → **GFCI**.
  - Sparking in damaged wiring → **AFCI**.
  - Live metal case → **grounding**.
- Worked example: "Which protects a TV from lightning-induced spikes?" Answer: **a surge protector**.
- Traps: A breaker does not protect against voltage surges (Berkeley Q13 choice B is false).
- Calculator: look up — type: hazard.

Hard

14. Fault current through the ground wire: does the breaker trip?

- Scope: ✅ IN (Ohm's law, series circuit).
- Where seen: not seen in a test. Prior bank I_bank W3. Source: Kuphaldt, *Lessons in Electric Circuits*, Vol. I, chapter 3 "Electrical Safety" ("Safe circuit design").
- How to solve:
  1. Add the resistances around the fault loop: hot wire + ground wire.
  2. I_fault = V / R_loop.
  3. Compare with the breaker.
- Worked example: 120 V. Hot wire 0.25 Ω, ground wire 0.25 Ω. I = 120/0.5 = **240 A**. A 15 A breaker **trips at once** (magnetic trip).
- Traps: Without the ground wire, the loop is open. I = 0, nothing trips, and the case stays at 120 V.
- Calculator: calculate — type: V, R_hot, R_ground, breaker amps.

15. Ungrounded versus grounded case: current through a person

- Scope: ✅ IN (series and parallel resistors, divider).
- Where seen: not seen in a test. Source: Kuphaldt, *Lessons in Electric Circuits*, Vol. I, chapter 3 "Electrical Safety"; OpenStax *College Physics 2e* §20.6 (grounding figure).
- How to solve:
  1. Ungrounded: the person is the only path. I = V / (R_hot + R_body).
  2. Grounded: the ground wire and the person are in **parallel** from the case to earth.
  3. Find R_p = R_ground || R_body. Then V_case = V × R_p / (R_hot + R_p). Person current = V_case / R_body.
- Worked example: 120 V, R_hot = 0.25 Ω, wet person 1000 Ω.
  - Ungrounded: I_person = 120/1000.25 = **120 mA** (dangerous, and no breaker trips).
  - Grounded (R_ground = 0.25 Ω): R_p ≈ 0.25 Ω. V_case ≈ 120 × 0.25/0.50 = **60 V**. I_person ≈ **60 mA**, but the fault current is about **240 A**, so the breaker trips in milliseconds.
- Traps: Grounding does not make the case 0 V during the fault. It makes the fault **short**, by tripping the breaker. A GFCI adds a second layer.
- Calculator: calculate — type: V, R_hot, R_ground, R_body.

16. Static grounding and bonding (fuel, ESD)

- Scope: ✅ IN (rule 3.c.ii, static hazards). Main coverage is in CL 2 - Static Electricity and Coulomb's Law 🅃.
- Where seen: ATX Regional 2026 B (REG) MC26 (greatest static ignition risk → fuel flowing through a plastic hose, not a grounded metal pipe). Cobra 2020 B (INV) Q9 (static in gas pipes builds up inside).
- How to solve:
  1. Charge builds where it cannot drain (insulators, ungrounded parts).
  2. Grounding drains it to earth. Bonding keeps parts at equal voltage, so no spark jumps.
- Worked example (ATX MC26): Answer: **fuel through a plastic hose** (plastic cannot drain the charge).
- Traps: A grounded metal pipe is the **safe** choice.
- Calculator: look up — type: scenario.

Nationals

17. Two ground paths with resistance: who carries what (KCL)

- Scope: ✅ States-Nationals (rule 3.c.ix; two unknowns).
- Where seen: not seen in a test. Source: method from OpenStax *College Physics 2e* §21.3.
- How to solve:
  1. Draw the fault: source → hot wire → case → (ground wire to earth) and (person to earth), both back to the source.
  2. Unknowns: the case voltage V_c (one node). Write KCL at the case: (V – V_c)/R_hot = V_c/R_ground + V_c/R_body.
  3. Solve for V_c. Find each branch current.

- Worked example: V = 120 V, R_hot = 0.25 Ω, R_ground = 0.25 Ω, R_body = 1000 Ω. KCL gives V_c = **59.99 V**. Ground wire current = 59.99/0.25 = **240 A**. Person current = **60 mA**. Check: total from the hot = (120 – 59.99)/0.25 = **240.03 A** = 240 A + 0.06 A ✓.
- Traps: Keep enough decimals; the person's 60 mA disappears if you round the 240.03 A.
- Calculator: calculate — type: V, R_hot, R_ground, R_body.

18. 240 V split-phase: voltages between any two wires ⚠

- Scope: ⚠ Borderline. It is a single-phase AC source (rule 3.c.iii). It is not 3-phase (rule 3.d), but deep phase talk is AC circuit theory (OUT).
- Where seen: USC Invitational 2026 B and C (INV) Q6 statement IV.
- How to solve:
  1. Hot 1 to neutral = 120 V. Hot 2 to neutral = 120 V.
  2. The two hots are opposite halves, so hot 1 to hot 2 = 240 V.
  3. Neutral to ground ≈ 0 V.
- Worked example: "What is the voltage between the two hot slots of a dryer outlet?" Answer: **240 V**; each hot to neutral is **120 V**.
- Traps: Do not split 240 V into 60 + 60 + 120. Do not use 208 V (that is 3-phase, OUT).
- Calculator: look up — type: wire pair.

Out of scope (skip)

- ❌ **MOV current-voltage curves, varistor equations, clamping math.** Rule 3.d: non-linear devices beyond those listed.
- ❌ **Lightning surge wave shapes (8/20 µs, 1.2/50 µs) as time functions.** Rule 3.d: frequency analysis; rule 3.e.v: calculus.
- ❌ **Ground-rod resistance formulas** (with logarithms) and soil resistivity math. Rule 3.e: math limits.
- ❌ **3-phase grounding (wye, delta), 208 V and 480 V systems.** Rule 3.d: 3-phase power.
- ❌ **Faraday cage field theory and the person "Faraday".** Faraday is not in rule 3.c.i. Field shielding beyond rule 3.c.ii is not tested.
- ❌ **Transformers in the service drop** (center-tapped secondary). Rule 3.d: inductance.

Mistakes in tests and keys

- **ATX Regional 2026 B (REG) MC19** (earth ground symbol): the key is C. Image C is the **chassis** (rake) ground. The true earth-ground symbol (image A) is not among the choices. No choice is fully correct.
- **Florida State 2021 B/C (STATE) Q13** (toddler, two fingers in only the hot slot → key D, none of the above): the key assumes no path to ground. A child standing on a floor may have a path, so "possible electrocution if fingers are wet" (A) is also defensible. The key is debatable.
- **UT Invitational 2019 B Q12 and UT fall 2019 C Q12:** the choices say the hot "swings (+)". AC hot swings both ways. The "shorter slot" part is right.
- **Berkeley 2026 B Q13 choice B:** a distractor that says breakers protect against voltage surges. It is false; keep this in mind as a fact, not a key error.

Not confirmed

- NEMA configuration numbers (5, 6, 14, 24) and the 24-30 = 347 V receptacle: from the MIT 2021 key and memory. I did not see the MIT receptacle drawings (they are images) and did not fetch a NEMA chart.
- The square-in-a-square double-insulation symbol and the UL 1449 330 V lowest VPR: from prior bank I_bank and memory. Not fetched this run.
- Wire resistance values in the examples (0.1 Ω/km for a power line; 0.25 Ω for a branch) are made-up round numbers for practice.
- Cobra 2020 B Q9 and ATX 2019 Regional B Q21: from extraction files; I did not reopen the raw text.
- No NATIONAL test with a grounding or surge item was found. The only STATE item is Florida State 2021 Q13.

Sources

- Rules: 2027 Div B Rules Manual, Circuit Lab, rules 3.c.ii-iii, 3.c.v-vi, 3.c.ix, 3.d, 3.e (pages B11-B12). See Circuit Lab - Rules 2027 [R].
- Tests (raw text checked this run): Berkeley 2026 B Q13, Q23; BullSO 2026 B Q33 and C Q33 with keys; MIT 2021 C Q92-Q97 and key; USC Invitational 2026 B and C Q6; Florida State 2021 B/C Q13; ATX Regional 2026 B MC26 and MC30-MC31. From extraction files only: ATX Regional 2026 B MC19 (image sheet); ATX Regional 2019 B Q21; UT Invitational 2019 B Q12, Q25; UT Invitational fall 2019 C Q12, Q25; Shock Value re-tryout VI-2; Cobra 2020 B Q9; Princeton 2026 C Matching Part 2.
- Extraction files: `~/Daniel/circuitlab/research/tests/` (nat26, state, ut-a, ut-b, ut-c, misc-a, misc-b) and `prior/G_tests.md`.
- Prior research: `prior/B_topics.md` §3 (wiring), §5c, §6f; `prior/I_bank.md` W1-W5, D35-D36, Q39 (lightning).
- Textbooks: Kuphaldt, *Lessons in Electric Circuits*, Vol. I, chapter 3 "Electrical Safety"; OpenStax *College Physics 2e* §20.6 and §21.3; All About Circuits, "Electrical Safety".
- All numbers checked in Python on 2026-10-16.

TAB A7

Diodes and LEDs

Semiconductors and Doping 477
PN Junction and I-V Curve 482
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Each note: key facts, formulas, then question types from Easy to Nationals with worked examples.

Semiconductors and Doping

Rule area **3.c.vii**: "Fundamental electrical characteristics of silicon PN junctions and uses of single-junction devices (e.g., diodes, LEDs)". This note covers the material science you need to explain a PN junction. It is tested at **every level** in 2027.

Key facts

- A **semiconductor** is a material whose resistance is between a conductor and an insulator. Examples: silicon (Si), germanium (Ge), gallium arsenide (GaAs).
- Silicon is in **group 14**. Each Si atom has **4 valence electrons** (outer electrons that make bonds). Si has atomic number 14.
- **Intrinsic** silicon means pure silicon. It has equal numbers of free electrons and holes.
- A **hole** is a missing electron in a bond. A hole acts like a **positive** charge that can move.
- **Doping** means adding a tiny amount of another element on purpose.
- **n-type:** dope with group 15 atoms (**phosphorus P, arsenic As, antimony Sb**). Each one gives one extra free electron. These atoms are called **donors**. The majority carriers are **electrons**.
- **p-type:** dope with group 13 atoms (**boron B, aluminum Al, gallium Ga, indium In**). Each one makes one hole. These atoms are called **acceptors**. The majority carriers are **holes**.
- A **majority carrier** is the more common type of moving charge. The other type is the **minority carrier**.
- Doped silicon is still **electrically neutral** overall. The dopant atoms bring their own protons.
- **Band gap (E_g):** the energy an electron needs to break free and conduct. It is measured in electron-volts (eV). 1 eV = 1.602×10^{-19} J.
- Band gaps at room temperature: **Ge 0.66-0.67 eV < Si 1.12 eV < GaAs 1.42 eV < GaN 3.4 eV**. Insulators have big gaps (more than about 4 eV). Metals have no gap.
- A semiconductor's resistance **goes down** when it gets hotter. Heat frees more carriers. A metal's resistance **goes up** when it gets hotter.
- Silicon is the most common chip material. It is cheap and plentiful (made from sand). It works at higher temperatures than germanium. Its oxide (SiO_2) is a good insulator layer.
- Silicon has an **indirect** band gap. So it makes heat, not light, when carriers recombine. LEDs use **direct-gap** compounds such as GaAs, GaP and GaN.
- III-V compounds (group 13 + group 15 elements, such as GaAs, GaN, InGaN) are the main LED materials.

Formulas

- Photon wavelength from band gap: **λ (nm) ≈ 1240 / E_g (eV)**. (Exact constant 1239.84.)
- Energy in joules: **E (J) = E (eV) × 1.602 × 10^{-19}**.
- Mass-action law (⚠️ Borderline): **n × p = n_i^2**. n = free electrons per cm^3, p = holes per cm^3, n_i = intrinsic carrier density. For Si, $n_i \approx 1 \times 10^{10}$ cm^{-3} (older books use 1.5×10^{10}).

Question types

Easy

1. Conductor, insulator or semiconductor? ✅ IN

- Seen: MIT Invitational 2019, Div C, INV, Q32 ("material engineered with a PN junction: silicon or carbon?" → silicon). National supervisor practice deck 6, "name 2 semiconductors" (soinc.org Circuit Lab B practice slides). Florida State 2021 and SO MYSO 2020 Div B ask the conductor/insulator half.
- How to solve:
  1. Metals (Cu, Ag, Au, Al) = conductors.
  2. Glass, rubber, plastic, pure (distilled) water = insulators.
  3. Si, Ge, GaAs, GaN, SiC = semiconductors.
- Example: Which one is a semiconductor: copper, germanium, rubber, silver? **Answer: germanium.**

- Traps: Carbon (graphite) conducts. Diamond insulates. Neither is the usual PN-junction material. Distilled water is an insulator; tap and sea water conduct.
- Calculator: look up — type: `semiconductor list`

2. Valence electrons and the periodic-table group ✅ IN

- Seen: not seen as its own question. It is the base fact behind every doping question.
- How to solve:
  1. Si and Ge are group 14 → 4 valence electrons.
  2. Group 15 → 5 valence electrons (one extra → n-type).
  3. Group 13 → 3 valence electrons (one short → p-type).
- Example: How many valence electrons does a phosphorus atom have? **Answer: 5.**
- Traps: Do not count all electrons. Si has 14 electrons but only 4 are valence electrons.
- Calculator: look up — type: `valence <element>`

3. Dopant element → n-type or p-type ✅ IN

- Seen: UT Austin (ATX) Regional 2021, Div C, REG, Q22 ("element on the n side of a p-n junction" → antimony). National supervisor practice deck 14 (PN-DOPING).
- How to solve:
  1. Find the element's group.
  2. Group 15 (P, As, Sb) → **n-type**.
  3. Group 13 (B, Al, Ga, In) → **p-type**.
- Example: Which dopant makes p-type silicon: phosphorus, arsenic, boron or antimony? **Answer: boron.**
- Traps: "N" does not mean "nitrogen". Nitrogen is not a normal Si dopant. Memory hook: **B**oron → **p** (both are "short" an electron); **P**hosphorus → n.
- Calculator: look up — type: `dopant <element>`

4. Majority carriers and their charge ✅ IN

- Seen: UF Northern Regional (Florida) 2020, Div C, REG, Q24 ("holes are ___, electrons are ___ charged" → positive, negative). Rickards Invitational 2020 (2021 season), Div C, INV, Q32 ("P-type is doped with … → holes"). Princeton Invitational 2019, Div C, INV, PN section.
- How to solve:
  1. n-type → majority carriers are electrons (negative).
  2. p-type → majority carriers are holes (positive).
  3. The other carrier is the minority carrier.
- Example: What are the minority carriers in n-type silicon? **Answer: holes.**
- Traps: A hole is not a proton. It is an empty spot that acts like a + charge.
- Calculator: look up — type: `carrier n-type`

5. Is doped silicon charged? ✅ IN

- Seen: not seen in a test. Common trap in Kuphaldt, *Lessons in Electric Circuits* Vol. III, ch. 2.
- How to solve: Each dopant atom is neutral when it enters the crystal. So the whole piece is neutral.
- Example: Is a block of n-type silicon negatively charged? **Answer: No. It is neutral overall. It only has more free electrons.**
- Traps: "n" means the carriers are negative, not the block.
- Calculator: look up — type: `doped neutral`

Medium

6. Temperature and resistance: semiconductor vs metal ✅ IN

- Seen: UT Austin (ATX) Regional 2020, Div B, REG, Q44 (and Div C Q48). **The key is wrong** (see Mistakes).
- How to solve:
  1. Metal: hotter → atoms shake more → electrons collide more → **R rises**.
  2. Semiconductor: hotter → more bonds break → more free carriers → **R falls**.

- Example: A silicon sample and a copper wire are both heated. What happens to each resistance? **Answer: silicon R falls; copper R rises.**
- Traps: The NTC thermistor (negative temperature coefficient) is a semiconductor sensor that uses this effect.
- Calculator: look up — type: `temp coeff semiconductor`

7. Why silicon instead of germanium (or other materials)? ✅ IN

- Seen: USC Invitational 2026, Div C, INV, Q29 (I. more dopants, II. high-temperature stability, III. low cost, IV. high insulation). Key B (I, II) is doubtful (see Mistakes).
- How to solve: List the real advantages of Si:
  1. Cheap and plentiful (sand).
  2. Works at higher temperature than Ge (bigger band gap → less leakage).
  3. Grows a good insulating oxide (SiO_2).
  4. Lower reverse leakage current than Ge.
- Example: Give two reasons silicon diodes replaced germanium diodes. **Answer: silicon is cheaper and works at higher temperatures with less leakage.**
- Traps: Ge has a **lower** forward drop (0.3 V). That is a Ge advantage, not a Si one.
- Calculator: look up — type: `si vs ge`

8. Band gap meaning and ranking ⚠️ Borderline

- Why borderline: "band gap" is device physics. It is not an electrical calculation. But tests use it to explain LED color, so expect it.
- Seen: Berkeley Invitational 2026, Div B, INV, Q40 (rank IR, red, blue LEDs by band gap → infrared < red < blue). National supervisor practice deck 14. Princeton Invitational 2026, Div C, INV, MC7.
- How to solve:
  1. Bigger gap → harder to free an electron → more insulating.
  2. Order to remember: metal (0) < Ge (0.67) < Si (1.12) < GaAs (1.42) < GaN (3.4) < insulator.
- Example: Rank Si, GaN, Ge by band gap, smallest first. **Answer: Ge, Si, GaN.**
- Traps: Do not confuse band gap (eV) with forward voltage (V). They are close in size for LEDs, but they are different quantities.
- Calculator: look up — type: `bandgap <material>`

Hard

9. Band gap → wavelength, or eV → joules ⚠️ Borderline

- Why borderline: it is photon physics ($E = hc/\lambda$). The math is legal (one division).
- Seen: Princeton Invitational 2026, Div C, INV, SA8 (the key uses $E = hc/\lambda$ to explain LED color). Not seen as a number question.
- How to solve:
  1. λ (nm) = 1240 / E_g (eV).
  2. Compare λ with visible light (about 380-750 nm). Longer than 750 nm = infrared.
  3. For joules: multiply eV by 1.602×10^{-19}.
- Example: What wavelength matches the band gap of silicon (1.12 eV)? Is it visible? **Answer: 1240/1.12 ≈ 1107 nm. Infrared, not visible.** (Ge 0.66 eV → 1879 nm; GaAs 1.42 eV → 873 nm; GaN 3.4 eV → 365 nm, ultraviolet.) In joules, 1.12 eV = **1.79×10^{-19} J**.
- Traps: Use nm with 1240. If you use metres, use $hc = 1.986 \times 10^{-25}$ J·m and E in joules.
- Calculator: calculate — type: `Eg (eV)` or `lambda (nm)`

10. Direct vs indirect band gap: why Si diodes do not glow ⚠️ Borderline

- Why borderline: band structure is beyond "fundamental electrical characteristics". Supervisors still like the question "why aren't LEDs made of silicon?"
- Seen: not seen in a test. Source: OpenStax *College Physics 2e* ch. 30 and Kuphaldt Vol. III ch. 3 (LEDs).
- How to solve: Give two reasons. (1) Si has an indirect gap, so recombination makes mostly heat. (2) Its gap (1.12 eV) is in the infrared anyway.

- Example: Why is a red LED not made of silicon? **Answer: silicon has an indirect band gap of 1.12 eV. It gives off heat, not visible light. Red light needs about 1.9-2.0 eV and a direct-gap material such as AlGaInP or GaAsP.**
- Traps: Silicon diodes do give off a very small amount of infrared. "No light at all" is a little too strong.
- Calculator: look up — type: `why not si led`

Nationals

11. Band diagram of a PN junction ⚠️ Borderline

- Why borderline: the national practice deck 14 homework says "research band diagrams". It is beyond plain electrical behavior.
- Seen: National supervisor practice deck 14 homework (PN-BAND-DIAGRAM). Not seen in a test.
- How to solve:
  1. Draw two lines for each side: conduction band (top) and valence band (bottom).
  2. On the p side the bands sit **higher**. On the n side they sit **lower**.
  3. The step between them equals the built-in voltage (about 0.7 eV for Si).
  4. Forward bias makes the step smaller. Reverse bias makes it bigger.
- Example: What happens to the band "hill" when the junction is forward-biased by 0.5 V? **Answer: the hill gets 0.5 eV lower, so more electrons can cross. Current rises.**
- Traps: Electrons roll "downhill" on a band diagram. Holes float "uphill".
- Calculator: look up — type: `band diagram`

12. Carrier density with the mass-action law ⚠️ Borderline

- Why borderline: it is quantitative semiconductor physics. The math (one division) is legal.
- Seen: not seen in a Div B test. MIT Invitational 2021, Div C, INV, Q45 used doping levels for a transistor (that one is ❌ OUT).
- How to solve:
  1. In n-type, n ≈ N_D (the donor density).
  2. p = n_i² / n.
- Example: Si with n_i = 1.0×10^{10} cm^{-3} is doped with N_D = 1.0×10^{16} cm^{-3}. Find the hole density. **Answer: p = $(10^{10})^2 / 10^{16} = 1.0 \times 10^4$ cm^{-3}.** (With n_i = 1.5×10^{10}: 2.25×10^4 cm^{-3}.)
- Traps: Doping raises one carrier and lowers the other. The product stays the same.
- Calculator: calculate — type: `ni, ND`

Out of scope (skip)

- ❌ Transistor doping, BJT/MOSFET regions, mobility and diffusion coefficients (MIT 2021 Div C Q42-51). Rule 3.d: "semiconductors (beyond those listed above)".
- ❌ "Why will Moore's law end?" (North Pocono Invitational 2020, Div C, Q22). Transistors, rule 3.d.
- ❌ Bardeen, Brattain and Shockley invented the transistor (ATX Regional 2020, Div C, Q11). Not in the history list (3.c.i) and transistors are 3.d.
- ❌ MOSFET facts (Rickards 2026 Div C Q10; Shock Value tryout ~2012 "MOSFET stands for"). Rule 3.d.
- ❌ Photoelectric effect with work function (UT Invitational 2019, Div B, Q22; ATX Regional 2019 Q1). It is in no 3.c area.

Mistakes in tests and keys

- **ATX Regional 2020, Div B Q44 / Div C Q48:** key says (a). Correct is **(b)**: semiconductor R falls with heat, metal R rises. Checked against OpenStax *College Physics 2e* §20.3 (temperature coefficients: carbon, Ge and Si are negative).
- **USC Invitational 2026, Div C Q29:** key B (I, II). "More possible dopants" is not a real reason. The strongest reasons are II (temperature) and III (cost). No choice lists only II and III. C (I, II, III) is the closest. Treat the key as **doubtful**.

Not confirmed

- n_i for Si: modern value about 1.0×10^{10} cm^{-3}; many textbooks print 1.5×10^{10}. Both are used. I did not check a primary source this run.
- Band-gap values come from standard textbook tables (general knowledge). I did not open a data handbook.
- No STATE or NATIONAL test in the downloads had a doping question. Florida State 2021 (Div B = C) had no diode or semiconductor questions. The 2026 Nationals test is sold only in the soinc test packet; I could not read it.

Sources

- 2027 Div B rules, Circuit Lab, 3.c.vii and 3.d (local PDF, manual pages B12-B13).
- Test extractions: `circuitlab/research/tests/atx-reg.md`, `misc-b.md`, `nat26.md`; `prior/G_tests.md`; `prior/H_logic_diodes.md` (B3, C5); `prior/F_slides.md` (deck 6, deck 14).
- OpenStax *College Physics 2e*, ch. 20 (resistivity and temperature) and ch. 30 (band gaps, LEDs).
- Kuphaldt, *Lessons in Electric Circuits* Vol. III (Semiconductors), ch. 2 "Solid-state device theory".
- All numbers checked in Python.

PN Junction and I-V Curve

Rule area **3.c.vii**: "Fundamental electrical characteristics of silicon PN junctions". This is the core of the area. Every level can ask it.

Key facts

- A **PN junction** is the boundary where p-type and n-type silicon meet. A **diode** is one PN junction with two leads.
- **Anode** = the p side. **Cathode** = the n side. In the symbol, the triangle points from anode to cathode. The bar is the cathode.
- **Depletion region:** near the junction, free electrons cross into the p side and fill holes. This leaves fixed charged atoms (ions) and **no free carriers**. So this thin zone acts like an insulator.
- The fixed ions make a **built-in electric field** (it points from the n side to the p side). They also make a **built-in voltage (barrier potential)**: about **0.7 V for Si** and **0.3 V for Ge**.
- **Forward bias:** anode (p) more positive than cathode (n). The depletion region gets **narrower**. Above about **0.6-0.7 V (Si)** current rises fast.
- **Reverse bias:** anode more negative than cathode. The depletion region gets **wider**. Only a tiny **leakage current** (reverse saturation current) flows. For Si it is nanoamps.
- **Zero bias:** no outside voltage. The built-in barrier stops net current.
- **Reverse breakdown:** if the reverse voltage is big enough, current suddenly flows. This is **avalanche** (high voltage) or **Zener** (heavy doping, low voltage) breakdown. A normal diode is often damaged by it. A Zener diode is built to use it.
- Typical ratings: 1N4001 = 1 A, 50 V peak reverse. 1N4007 = 1 A, 1000 V. 1N4148 = small signal diode, 100 V. A 5 mm LED breaks down at only about **5 V reverse**.
- **Conventional current** flows anode → cathode through a forward-biased diode. Electrons flow the other way, so **electrons enter at the cathode**.
- A diode is **non-linear**. Its I-V graph is not a straight line. So Ohm's law (R = V/I constant) does not describe it.
- **I-V curve shape:** forward side: almost flat near 0, then a sharp **knee** at about 0.7 V, then nearly vertical. Reverse side: almost flat just below zero, then a sharp plunge at the breakdown voltage.
- Temperature (⚠️): Si forward voltage falls about **2 mV per °C**. Leakage current about **doubles every 10 °C**.
- Forward current in a diode is mostly **diffusion** of majority carriers. Reverse leakage is **drift** of minority carriers.

Formulas

- Static (DC) resistance at a point: **R = V / I**.
- Dynamic (small-change) resistance: **r_d = ΔV / ΔI** between two close points on the curve.
- Load line for source V_S and series R: **I = (V_S – V_D) / R**. Ends: (V_D = V_S, I = 0) and (V_D = 0, I = V_S / R).
- Diode power: **P = V_D × I**.
- Temperature shift (⚠️): **V_F(T) ≈ V_F(25 °C) – 0.002 × (T – 25)**.
- Shockley equation (⚠️ Nationals only, borderline): **I = I_S (e^(V/(n·V_T)) – 1)**, V_T ≈ 25.9 mV at room temperature. For n = 1, **each ×10 in current adds about 60 mV**.

Question types

Easy

1. Forward vs reverse bias: definitions ✅ IN

- Seen: Pembroke Hill 2025 (2026 season), Div B, INV, Q17 ("current only flows when the anode is at a ___ potential than the cathode" → higher). F-M Invitational 2018, Div C, INV, Q13 ("voltage that lowers the PN barrier → forward bias"). UF Northern Regional 2019, Div C, REG, Q44 (describe forward vs reverse bias, 4 parts). NY Lake Erie/Niagara Invitational 2018, Div C, INV, Q20 ("a PN diode conducts in … forward bias"). Princeton Invitational 2026, Div C, INV, SA7b.
- How to solve:

1. Find the p side (anode) and the n side (cathode).
2. p to +, n to − → forward. Depletion narrows. Current flows above about 0.7 V.
3. p to −, n to + → reverse. Depletion widens. Almost no current.

- Example: A silicon diode has 3.0 V on its anode and 2.5 V on its cathode. Does it conduct? **Answer: it is forward-biased, but only by 0.5 V. That is below 0.7 V, so it is essentially off (only microamps).**
- Traps: "Forward bias" alone does not mean "on". You also need about 0.7 V.
- Calculator: look up — type: `forward bias`

2. The depletion region ✅ IN

- Seen: UT Invitational 2020, Div C, INV, Q32 ("while there is a large depletion region in a Si PN junction … current is blocked"). National supervisor practice deck 14 (PN-BIAS-DEPLETION). Princeton Invitational 2026, Div C, INV, SA7b.
- How to solve: Say three things. (1) It has no free carriers. (2) It has fixed + ions on the n side and fixed − ions on the p side. (3) It is wider in reverse bias and narrower in forward bias.
- Example: What happens to the depletion region when a diode is reverse-biased? **Answer: it gets wider, so current is blocked.**
- Traps: The + ions are on the **n** side and the − ions are on the **p** side. Many students swap them. (The n side lost electrons, so it is left positive.)
- Calculator: look up — type: `depletion region`

3. Barrier (threshold) voltage: name and value ✅ IN

- Seen: NY Lake Erie/Niagara Invitational 2018, Div C, INV, Q21 ("voltage needed before a diode conducts" → barrier voltage). Rickards Invitational 2026 (Oct 2025), Div C, INV, Q4 ("average voltage drop across a silicon diode" → 0.7 V; other choices 0.46, 1.1, 0.23 V). National practice deck 14 (Si 0.6-0.7 V, Ge 0.25-0.3 V).
- How to solve: Si → 0.7 V (some keys use 0.6 V). Ge → 0.3 V. Schottky → 0.2-0.4 V. LEDs → 1.6-3.5 V by color.
- Example: Typical forward drop of a silicon diode? **Answer: about 0.7 V.**
- Traps: 1.1 V is the Si **band gap** in eV, not the forward drop. It is a distractor.
- Calculator: look up — type: `vf si`

4. One-way device; terminal names; electron flow ✅ IN

- Seen: UF Northern Regional 2020, Div B, REG, Q26 ("an LED conducts one way only" → True). NY Lake Erie/Niagara Invitational 2018, Div C, INV, Q11 ("electrons enter a diode at the … cathode").
- How to solve:
  1. Conventional current: anode → cathode (the way the triangle points).
  2. Electrons move the opposite way, so they enter at the cathode.
- Example: In a forward-biased diode, which lead do electrons leave by? **Answer: the anode.**
- Traps: In a **battery**, the anode is the negative terminal while discharging. In a **diode**, the anode is the p side. Do not mix the two (Pembroke 2025 Q18 asks about battery oxidation at the anode).
- Calculator: look up — type: `anode cathode`

5. Is a diode linear? ✅ IN

- Seen: Northview Sylvania Invitational 2025 (Dec 6), Div C, INV, Q43 ("a diode is considered a linear device" → False). BullSO Invitational 2026, Div B, INV, Q7 (passive vs not-passive parts; the key treats the diode as not passive).
- How to solve: A linear device has a straight-line I-V graph through the origin (a resistor). A diode's graph bends at the knee. So it is **non-linear**.
- Example: True or false: doubling the voltage across a diode doubles its current. **Answer: False.** (Near 0.7 V, a small rise in V gives a big rise in I.)
- Traps: The rule 3.d excludes "non-linear devices (beyond those listed above)". Diodes and LEDs are listed, so they are allowed even though they are non-linear.
- Calculator: look up — type: `linear device`

Medium

6. Draw and label a PN junction ✅ IN

- Seen: University of Michigan Invitational (UMSO) 2019, Div C, INV, Q30 (label p, n, depletion region, electrons, holes, ions, bias arrows; which bias emits light). UT Austin (ATX) Regional 2019, Div C, REG, Q29 (draw a **reverse**-biased junction; label P, N, depletion region, terminals).
- How to solve:
  1. Draw a box: left half "P", right half "N".
  2. Shade a strip in the middle: "depletion region".
  3. Draw small circles (holes, +) on the P side and dots (electrons, –) on the N side, outside the strip.
  4. In the strip: "–" ions on the P side, "+" ions on the N side.
  5. Show the battery. Forward: + to P. Reverse: – to P, + to N, and draw the strip wider.
- Example: For a reverse-biased junction, which battery terminal goes to the P side? **Answer: the negative terminal.** For an LED, light comes out in **forward** bias.
- Traps: Label the anode and cathode too. Graders often give a point for each label.
- Calculator: look up — type: `draw pn`

7. When can current flow in reverse? (breakdown) ✅ IN

- Seen: UT Austin (ATX) Regional 2019, Div C, REG, Q30 ("condition for current under reverse bias" → exceed the breakdown voltage). National practice deck 14 (DIODE-REGIONS-ZENER).
- How to solve: Reverse current stays tiny until the reverse voltage reaches the **breakdown voltage** (V_{BR}). Then current rises sharply.
- Example: A 1N4001 (rated 50 V reverse) sees 30 V in reverse. Does it conduct? **Answer: No, only leakage. 30 V is below its 50 V rating.**
- Traps: Breakdown does not always destroy a diode. It does if the power ($V \times I$) is too high. Zener diodes work in breakdown on purpose.
- Calculator: look up — type: `breakdown`

8. Match an I-V graph to a part ✅ IN (⚠️ for the lamp)

- Why partly borderline: the filament lamp is a non-linear device not on the list. The diode and resistor curves are fully IN.
- Seen: UT Invitational 2020, Div B, INV, Q60 (match curves to copper, filament lamp, diode → copper straight line, lamp bends over, diode exponential knee).
- How to solve:
  1. Straight line through the origin → resistor or copper wire (ohmic).
  2. Line that bends toward the V axis (S-shape, symmetric) → filament lamp (R rises as it heats).
  3. Flat near zero, then a sharp rise only on the + side → diode.
  4. Same as 3 but the knee is at 1.6-3.5 V → LED.
- Example: A curve shows I = 0 for all negative V and a sharp rise after +0.7 V. What is it? **Answer: a silicon diode.**
- Traps: Check which axis is V. If V is on the y-axis, slope = R. If I is on the y-axis, slope = 1/R.
- Calculator: look up — type: `iv curve shapes`

9. Name the regions of the diode curve ✅ IN

- Seen: National supervisor practice deck 14 (DIODE-REGIONS-ZENER). Answer from the deck: $V > V_D$ → on (small resistance); $0 < V < V_D$ → off; $-V_Z < V < 0$ → off with leakage; $V < -V_Z$ → breakdown.
- How to solve: Draw the curve and mark 4 regions: forward conduction, forward cutoff (below knee), reverse leakage, reverse breakdown.
- Example: A silicon diode has –20 V across it and a breakdown voltage of 100 V. Which region? **Answer: reverse leakage (reverse-biased, below breakdown).**
- Traps: In region 2 (0 to 0.7 V) the diode is forward-biased but still almost off.
- Calculator: look up — type: `diode regions`

10. Read a graph or table: knee voltage, static R, dynamic r ✅ IN

- Seen: not seen in a test. Source: Kuphaldt Socratic worksheet "Diodes" (allaboutcircuits.com/worksheets) and OpenStax *College Physics 2e* ch. 20 non-ohmic curves.
- How to solve:
  1. Knee: where the curve turns sharply upward.
  2. Static R at a point: R = V / I.
  3. Dynamic r between two points: r = ΔV / ΔI.
- Example: A diode reads 5 mA at 0.65 V and 15 mA at 0.70 V. Find r_d and the static R at 15 mA. **Answer: r_d = 0.05 V / 0.010 A = 5 Ω. Static R = 0.70 / 0.015 = 46.7 Ω.**
- Traps: Static R changes at every point. So "the resistance of a diode" has no single value.
- Calculator: calculate — type: `V1, I1, V2, I2`

11. Ratings: maximum current, peak reverse voltage, power ✅ IN

- Seen: UF Northern Regional 2019, Div C, REG, Q39 (max current for a 5 mm LED → 20 mA). GullSO Invitational 2025, Div C, INV, Q50a (max continuous current from a datasheet → 50 mA).
- How to solve:
  1. Forward current must stay below I_F(max).
  2. Reverse voltage must stay below V_R(max) (also called PIV or V_RRM).
  3. Power P = V_F × I must stay below P(max).
- Example: A 1N4001 carries 0.8 A with a 0.9 V drop. What power does it waste? Is the current OK? **Answer: P = 0.9 × 0.8 = 0.72 W. 0.8 A is under the 1 A rating, so it is OK.**
- Traps: A diode at 1 A has a drop closer to 0.9-1.0 V, not 0.7 V. Use the given value.
- Calculator: calculate — type: `Vf, I`

12. Compare Si, Ge, Schottky and ideal diodes ✅ IN (⚠️ for Ge/Schottky detail)

- Why partly borderline: the rule names silicon. Ge and Schottky are compared with it, which tests usually allow.
- Seen: Rickards Invitational 2026, Div C, INV, Q4 (choices 0.7, 0.46, 1.1, 0.23 V). National practice deck 14.
- How to solve: Knee order: ideal (0 V) < Schottky (0.2-0.4 V) ≈ Ge (0.3 V) < Si (0.7 V) < red LED (1.8 V) < blue/white LED (3 V).
- Example: Three curves have knees at 0.3 V, 0.7 V and 3.1 V. Name them. **Answer: germanium diode, silicon diode, blue (or white) LED.**
- Traps: Ge and Schottky both sit near 0.3 V. A Schottky has much lower breakdown and is very fast; Ge leaks more.
- Calculator: look up — type: `vf compare`

Hard

13. Load line on a diode curve ✅ IN

- Seen: not seen in a test. Source: Kuphaldt *Lessons in Electric Circuits* Vol. III ch. 3; standard intro-electronics method. Legal math: one straight line and one graph.
- How to solve:
  1. Mark (V_S, 0) on the V axis.
  2. Mark (0, V_S/R) on the I axis.
  3. Draw the straight line between them.
  4. Where it crosses the diode curve = the operating point (Q-point).
  5. No graph? Use the constant-drop model: I = (V_S – 0.7)/R.
- Example: V_S = 2.0 V, R = 100 Ω. Find the load-line ends and estimate I. **Answer: ends at (2.0 V, 0) and (0, 20 mA). With 0.7 V, I ≈ 1.3/100 = 13 mA.**
- Traps: The I-axis end is V_S/R, not V_S/(R + something). The diode is "shorted" at that point in the drawing only.
- Calculator: calculate — type: `Vs, R`

14. Temperature effects ⚠️ Borderline

- Why borderline: device detail beyond the basics. It is useful for "why did the LED dim when hot?" questions.

- Seen: not seen in a test. NY Lake Erie/Niagara Invitational 2018, Div C, INV, Q13 asks LED efficiency vs temperature (key: "exponential decrease"; not verified).
- How to solve:
  1. V_F drops about 2 mV per °C.
  2. Leakage about doubles per 10 °C.
- Example: A Si diode drops 0.70 V at 25 °C. Estimate V_F at 75 °C. **Answer: 0.70 – 50 × 0.002 = 0.60 V.** Leakage 10 nA at 25 °C → at 55 °C: 10 × 2^3 = **80 nA**.
- Traps: Lower V_F at the same supply means **more** current. This can run away in a hot LED with no resistor.
- Calculator: calculate — type: `Vf, T1, T2`

Nationals

15. Shockley diode equation and the 60 mV rule ⚠️ Borderline

- Why borderline: an exponential is a "function" (math 3.e.i), so the math is legal. But it goes past "fundamental characteristics" for Div B. A top invitational or Nationals could still use it.
- Seen: not seen in a Div B test. Source: Kuphaldt Vol. III ch. 3 "Diode equation"; OpenStax *University Physics* Vol. 3 ch. 9.
- How to solve:
  1. I = I_S·e^(V/V_T) for forward bias (the "−1" can be ignored).
  2. Ratio form: $V_2 - V_1 = V_T \times \ln(I_2/I_1)$ (n = 1).
  3. Shortcut: ×10 current → +60 mV.
- Example: A diode drops 0.65 V at 1 mA (n = 1, V_T = 25.9 mV). Estimate V at 10 mA, and find I_S. **Answer: 0.65 + 0.0595 = 0.71 V. I_S = 0.001 / e^(0.65/0.02585) ≈ 1.2 × 10^{-14} A.**
- Traps: Use the natural log (ln), not $\log_{10}$. V_T is in volts (0.0259), not millivolts.
- Calculator: calculate — type: `V1, I1, I2` (and `n`)

Out of scope (skip)

- ❌ Tunnel (Esaki) diode, negative resistance (Rickards Invitational 2020, Div B Q29 and Div C Q36). Rule 3.d: "non-linear devices (beyond those listed above)".
- ❌ "A transistor can be thought of as 2 diodes" (UT Invitational 2020, Div C, Q17). Transistors are 3.d.
- ❌ Varactor (variable-capacitance diode) behavior. Rule 3.d: capacitance.
- ❌ Junction capacitance, reverse-recovery charge math. Rule 3.d: capacitance; frequency analysis.

Mistakes in tests and keys

- **Pembroke Hill 2025 Q17 vs Q18:** not a key error, but a trap. Q17 uses "anode" for a diode (p side, higher potential). Q18 uses "anode" for a battery (where oxidation happens). Keep the two meanings apart.
- **NY Lake Erie/Niagara 2018, Div C Q13** (LED efficiency vs temperature, key "exponential decrease"): I could not verify the key. LED light output does fall as temperature rises, but "exponential" is a modeling choice.

Not confirmed

- 1N4001/1N4007/1N4148 ratings are from memory of standard datasheets. I did not open the datasheets this run.
- The 2 mV/°C and "doubles per 10 °C" rules are textbook approximations (Kuphaldt Vol. III). Real parts vary.
- No STATE or NATIONAL test in the downloads had a PN junction question. Florida State 2021 had none.

Sources

- 2027 Div B rules, Circuit Lab, 3.c.vii, 3.d, 3.e (local PDF).
- Test extractions: `research/tests/misc-a.md`, `misc-b.md`, `ut-b.md`, `ut-c.md`, `nat26.md`; Pembroke Hill 2025 test text (`downloads/dlB/tests/phi26_test.txt`); Rickards 2026 C key (rendered page 2); `prior/F_slides.md` deck 14; `prior/H_logic_diodes.md` B2, B3, B20-B26, B30.
- Kuphaldt, *Lessons in Electric Circuits* Vol. III ch. 2-3; Socratic worksheet "Diodes".

- OpenStax *College Physics 2e* ch. 20.
- All numbers checked in Python.

Diode Circuit Analysis

Rule area **3.c.vii** (diodes) plus **3.c.v** (DC circuits). Circuits that need KCL/KVL with unknown node voltages are **States/Nationals only** (3.c.ix). The math limit is **at most two equations** (3.e.i). This is the most calculation-heavy part of the area. It is a high-point area at top tournaments.

Key facts

- **Diode model** = a simple stand-in for the real diode, so you can use normal circuit rules.
  - **Ideal model:** ON = a wire (0 V drop). OFF = an open switch.
  - **Constant-drop model:** ON = a 0.7 V battery that opposes current (Si). OFF = open. Use this unless the test says "ideal".
  - **Piecewise-linear model:** ON = 0.7 V plus a small resistor r_f (for example 1 Ω). OFF = open.
- A diode is ON only if (1) it is forward-biased **and** (2) the circuit can push at least V_F across it.
- In a series loop, **every** forward drop subtracts from the supply.
- In parallel branches, **the branch with the lowest turn-on voltage wins**. It holds the node voltage down, so the higher-voltage branches stay dark.
- **Assumed-state method** (for 2 or more diodes): guess ON/OFF, solve, then check. ON needs I > 0 in the forward direction. OFF needs V_anode − V_cathode < V_F. If a check fails, flip that diode and try again.
- With k diodes there are at most 2^k guesses. Start with the most likely guess (the diode pointing "downhill" from the biggest source is ON).

Formulas

- One diode in series: **I = (V_S − V_F) / R**, if V_S > V_F. Else I = 0.
- n diodes in series: **I = (V_S − n·V_F) / R**.
- Piecewise-linear: **I = (V_S − V_F) / (R + r_f)**.
- Voltage across an OFF diode in a series loop = the whole supply (no current, so no resistor drop).
- Clamp: a diode from node X to a reference V_ref holds **X ≤ V_ref + V_F**.
- Node equation (States/Nationals): sum of currents into a node = sum out.

Question types

Easy

1. One diode + resistor: current and voltages ✅ IN

- Seen: National supervisor practice deck 12 (DIODE-IDEAL-ON-OFF). Homework generator "Level 15 Diode", problem 1 (national practice materials on soinc.org).
- How to solve:
  1. Check orientation. Reversed → I = 0, V_D = V_S, V_R = 0.
  2. Forward and V_S > V_F → I = (V_S − V_F)/R.
  3. V_R = I·R. V_D = V_F. P_D = V_F·I.
- Example: 5.0 V, one Si diode (forward), 1.0 kΩ. **Answer: I = 4.3/1000 = 4.3 mA; V_R = 4.3 V; V_D = 0.7 V; P_D = 3.0 mW. If the diode is reversed: I = 0 and V_D = 5.0 V.**
- Traps: If V_S < 0.7 V, the constant-drop diode is OFF (I = 0, V_R = 0).
- Calculator: calculate — type: `Vs, Vf, R`

2. Several diodes in series ✅ IN

- Seen: Hawk and Hornet Invitational 2026, Div C, INV, Q26 (three diodes, V_F = 0.7 V, and 10 Ω in series with V_S: which equation? → **(V_S − 3·0.7)/10**, choice c).
- How to solve: Subtract every drop. I = (V_S − n·0.7)/R. If V_S < n·0.7, I = 0.
- Example: Same as HH Q26 with V_S = 5.0 V. **Answer: I = (5.0 − 2.1)/10 = 0.29 A.** With a red LED (2.0 V) + one Si diode + 100 Ω on 3.0 V: (3.0 − 2.7)/100 = **3 mA**.

- Traps: Choice (a) (V_S – 0.7)/10 forgets two diodes. Choice (d) subtracts a voltage from a current (unit error).
- Calculator: calculate — type: `Vs, n, Vf, R`

3. Ideal model vs constant-drop vs piecewise-linear ✅ IN

- Seen: Homework generator "Level 15 Diode", problem 1 (national practice materials). Rickards 2026 Div C Q4 tests the 0.7 V value.
- How to solve: Solve the same circuit three ways. Ideal: I = V_S/R. Constant drop: I = (V_S – 0.7)/R. PWL: I = (V_S – 0.7)/(R + r_f).
- Example: V_S = 3.0 V, R_L = 10 Ω, Si diode with r_f = 1 Ω. **Answer: ideal I = 0.30 A, V_L = 3.0 V. Real (0.7 V + 1 Ω): I = 2.3/11 = 0.209 A, V_L = 2.09 V.**
- Traps: Read the question for the model. Some keys use 0.6 V. Write your model on the answer sheet.
- Calculator: calculate — type: `Vs, R, Vf, rf`

4. Does the lamp or LED light? (orientation in a loop) ✅ IN

- Seen: F-M Invitational 2019, Div C, INV, Q12 (four diode + LED circuits; key: a lights, b no (both diodes reversed), c no (one reversed), d lights). UT Invitational 2020, Div B, INV, Q32 (red LED lights, Bob's does not → it is reversed, or it is blue and needs a higher V_F).
- How to solve:
  1. Trace the loop from + to –.
  2. Every diode/LED must point along that path (triangle points toward –).
  3. One reversed part anywhere in a series loop → nothing lights.
  4. Then check the voltage: V_S must exceed the sum of all V_F.
- Example: A 3 V battery, a red LED and a Si diode are in series, both pointing the same way. Does it light? **Answer: yes. 2.0 + 0.7 = 2.7 V < 3 V. Only about 0.3 V is left for the resistor, so it is dim.**
- Traps: A reversed diode in **parallel** with the LED does nothing (it stays off). Only a reversed part in **series** blocks the loop.
- Calculator: look up — type: `series loop check`

Medium

5. Parallel branches with different turn-on voltages ✅ IN

- Seen: not seen exactly in a test. Berkeley Invitational 2026, Div B, INV, Q34 lab uses the idea (keep a blue LED connected but unlit). Source: Kuphaldt Socratic worksheet "Diodes".
- How to solve:
  1. All parallel branches share one voltage.
  2. Find each branch's turn-on voltage.
  3. The lowest one turns on and clamps the node there.
  4. Every branch that needs more stays OFF.
- Example: 10 V → 1.0 kΩ → node. From node to ground: (a) a red LED (2.0 V), (b) two Si diodes in series (1.4 V). **Answer: node = 1.4 V, so the LED stays dark. I = (10 – 1.4)/1000 = 8.6 mA, all through the diodes.**
- Traps: The LED is not "broken". Remove the diodes and it lights.
- Calculator: calculate — type: `Vs, R, Von list`

6. Diode-steered switch network ✅ IN

- Seen: UF Northern Regional 2019, Div B, REG, Q1-3 (Image 1; I rendered it). A 1 V source. SW1 feeds a bus that reaches LAMP1-3 through D1-D3. SW2, SW3, SW4 each feed one lamp directly.
  - Q1: SW1 closed → **all three lamps** (e). Q2: SW2 closed → **LAMP1 only** (a); D1 is reverse-biased, so current cannot go back to the bus. Q3: only LAMP1 and LAMP2 on → **SW2 and SW3** (d).
- How to solve:
  1. Mark each closed switch's path.
  2. A diode passes current only along its arrow.
  3. Current that would go "backwards" through a diode is blocked.

- Example: In the same circuit, which lamps light with SW1 and SW4 closed? **Answer: all three (SW1 already lights all three; SW4 adds a second path to LAMP3).**
- Traps: The test treats the diodes as ideal. With real Si diodes and a 1 V source, a lamp fed through a diode gets only (1 – 0.7)/100 = 3 mA instead of 10 mA, so it is dim. Mention this if you have space.
- Calculator: look up — type: `diode steering`

7. Two diodes: one ON, one OFF (series-parallel only) ✅ IN

- Seen: National supervisor practice deck 12 (DIODE-MULTI-STATE-ENUM; picture checked by the earlier research run). 10 V source → 4 kΩ → D1 → node; node → 6 kΩ → ground; D2 from a 3 V source with its cathode at the node. Ideal diodes.
- How to solve:
  1. Guess D1 ON, D2 OFF. Then it is a plain divider.
  2. Node = 10 × 6/(4 + 6) = 6 V.
  3. Check D2: anode 3 V, cathode 6 V → reverse → OFF ✓.
- Example (deck 12): **Answer: D1 ON, D2 OFF, node = 6 V, I = 1 mA.**
- Traps: If you guess D2 ON, the node would be 3 V. Then D1 would carry (10 – 3)/4k = 1.75 mA in and 6 kΩ would take only 0.5 mA, so D2 would need –1.25 mA (backwards). The guess fails.
- Calculator: calculate — type: `circuit states` (try each guess)

8. Clamp a node to a DC reference ✅ IN

- Seen: not seen in a test. Source: Kuphaldt *Lessons in Electric Circuits* Vol. III ch. 3 "Clipper circuits" (DC version only).
- How to solve:
  1. Find the node voltage with the diode removed.
  2. If it is above V_ref + 0.7 (diode from node to V_ref), the diode turns ON.
  3. Then node = V_ref + 0.7.
- Example: 5 V → 1.0 kΩ → X. A diode has its anode at X and its cathode at a 2.0 V source. **Answer: X = 2.7 V. I = (5 – 2.7)/1000 = 2.3 mA flows into the 2 V source.**
- Traps: Current flows **into** the 2 V source. That is allowed. A source can absorb current.
- Calculator: calculate — type: `Vs, R, Vref`

9. Threshold design: light an LED only above a set voltage ✅ IN

- Seen: National supervisor practice decks 12 and 14, homework (DIODE-THRESHOLD-DESIGN: "light a green LED only when Vin > 3 V").
- How to solve:
  1. Add diodes (or a Zener) in series with the LED until the total turn-on equals the threshold.
  2. Add a series resistor to set the current.
- Example: Make a green LED (V_F ≈ 2.2 V) light only when V_in > about 3 V. **Answer: LED + one Si diode (0.7 V) + resistor in series. It turns on near 2.2 + 0.7 = 2.9 V.**
- Traps: Diodes give steps of 0.7 V only. A Zener gives any step you choose (see Special Diodes A7).
- Calculator: calculate — type: `Vthreshold, Vf_LED`

Hard

10. Diode in a shunt branch (real model) ✅ States-Nationals

- Why S/N: the real-diode version needs one node equation (KCL).
- Seen: Homework generator "Level 15 Diode", problem 2 (national practice materials). V_S = 10 V → 2 Ω → node. Branch 1: 2 Ω + diode. Branch 2: R_L = 2 Ω.
- How to solve:
  1. Ideal diode: it is a plain series-parallel circuit.
  2. Real diode (0.7 V, r_f = 1 Ω): write KCL at the node: (10 – V_o)/2 = V_o/2 + (V_o – 0.7)/3.
  3. Solve for V_o, then each current.

- Example: **Answer: ideal: R_eq = 2 + 2‖2 = 3 Ω, V_o = 3.33 V, I_o = 1.67 A, I_D = 1.67 A. Real: V_o = 3.925 V, I_o = 1.96 A, I_D = 1.075 A.**
- Traps: The generator's own key is **wrong** for the real case (see Mistakes).
- Calculator: calculate — type: `Vs, R1, R2, RL, Vf, rf`

11. Two diodes both ON, one node ✅ States-Nationals

- Seen: not seen in a test. Source: `prior/H_logic_diodes.md` B7; style of Kuphaldt Socratic worksheet "Diodes".
- How to solve:
  1. Guess both ON.
  2. Write KCL at the shared node N: (5 – 0.7 – N)/1k + (3 – 0.7 – N)/1k = N/1k.
  3. Solve and check both currents are positive.
- Example: 5 V → 1 kΩ → D1 → N. 3 V → 1 kΩ → D2 → N. N → 1 kΩ → ground. **Answer: N = 2.2 V. I1 = 2.1 mA, I2 = 0.1 mA (both > 0 ✓), I3 = 2.2 mA.**
- Traps: I2 is small but positive. If the 3 V source were 2.5 V, D2 would turn OFF. Always check.
- Calculator: calculate — type: `V1, R1, V2, R2, R3, Vf`

12. Infinite ladder of diodes ✅ IN

- Seen: USC Invitational 2026 (Jan 17), Div C, INV, FRQ 38d (infinite ladder of 0.7 V ideal-drop diodes, shunt + series: equivalent forward drop → **0.7 V**).
- How to solve:
  1. Look for the shortest path from the input to the output.
  2. The first shunt diode connects the two terminals directly with one 0.7 V drop.
  3. Every other path has 2 or more drops, so those diodes never turn on.
- Example (USC 38d): **Answer: 0.7 V.**
- Traps: Do not use the resistor-ladder trick (x = R + R‖x). Diodes do not add like resistors.
- Calculator: look up — type: `diode ladder`

Nationals

13. Two nodes, two diodes: a two-equation system ✅ States-Nationals

- Seen: not seen in a test. Built to the 3.e.i limit (two equations). Method from Kuphaldt Vol. III ch. 3 and `prior/H_logic_diodes.md` B7.
- Circuit: 10 V → 1 kΩ → X. X → 1 kΩ → ground. D1 (anode X) → 1 kΩ → Y. Y → 1 kΩ → ground. 5 V → 1 kΩ → D2 (anode side) → Y. Si diodes, 0.7 V.
- How to solve:
  1. Guess D1 ON, D2 ON.
  2. KCL at X: (10 – X) = X + (X – 0.7 – Y). (All in kΩ and mA.)
  3. KCL at Y: (X – 0.7 – Y) + (4.3 – Y) = Y.
  4. Solve the two equations. Check both diode currents are positive.
  5. If not, flip the bad diode and solve again.
- Example: **Answer: X = 4.46 V, Y = 2.69 V, I_D1 = 1.075 mA, I_D2 = 1.61 mA. Both > 0, so both ON ✓.** (Guess D1 OFF fails: X would be 5 V and Y 2.15 V, so D1 would have 2.85 V forward.)
- Traps: Keep units consistent (V, kΩ, mA). Write each diode as a 0.7 V step in the right direction.
- Calculator: calculate — type: `2x2 solve` (a·X + b·Y = c; d·X + e·Y = f)

14. Find which guess works when three diodes share one node ✅ States-Nationals

- Seen: not seen in a test. Diode-OR of three sources is a standard Kuphaldt worksheet problem.
- How to solve: The diode from the **highest** source is ON. It sets the node. Every other diode then has its cathode above its anode, so it is OFF.
- Example: 5 V, 3 V and 9 V each feed one node through their own Si diode. The node goes to ground through 1 kΩ. **Answer: only the 9 V diode is ON. Node = 8.3 V. I = 8.3 mA. The 5 V and 3 V diodes are reverse-biased.**
- Traps: If each source also has its own series resistor, two diodes can be ON at once (type 11). Check the resistors.

- Calculator: calculate — type: `source list, R`

Out of scope (skip)

- ❌ Thevenin equivalent of a circuit with an ideal diode and a dependent source (UT Invitational 2020, Div B, Q74 → 2.18 V, 1.45 Ω). Rule 3.d: Norton/Thevenin equivalents.
- ❌ Clampers (DC restorers), voltage doublers, peak detectors. Rule 3.d: capacitance.
- ❌ Any problem needing three or more simultaneous equations. Rule 3.e.i.
- ⚠ AC clipper with a sine input (waveform sketch). Peak-value reasoning is plain algebra, but sine waveforms touch "AC circuit theory" (3.d).

Mistakes in tests and keys

- **Homework generator "Level 15 Diode", problem 2 (real diode):** the key treats the diode branch as a plain 3 Ω resistor and drops the 0.7 V. It gives V_o = 3.75 V. Correct is **3.925 V** (checked in Python). Its text also says "(2 Ω + 5 Ω)" while the formula uses 2 + 1.
- **UF Northern Regional 2019, Div B, Q1-3:** the key assumes ideal diodes. With a 1 V source and real 0.7 V drops, lamps fed through a diode would be very dim. The keyed answers are still correct for ideal diodes.
- **F-M Invitational 2019, Div C, Q12:** I could not render the EMF figures. I checked only that the key's reasons agree with each other.

Not confirmed

- National practice deck 12 circuit values (10 V, 4 kΩ, 6 kΩ, 3 V) come from the earlier research run (`prior/F_slides.md`). I did not reopen the .pptx this run.
- No STATE or NATIONAL test in the downloads had a multi-diode analysis question. Types 10, 11, 13 and 14 are built from the rules and textbooks.

Sources

- 2027 Div B rules, Circuit Lab, 3.c.v, 3.c.vii, 3.c.ix, 3.d, 3.e (local PDF).
- UF Northern Regional 2019 Div B test and image packet (`downloads/dlFind/nfl19reg_Btest.txt`, `nfl19reg_Bimg.pdf`, page 2 rendered).
- Hawk and Hornet 2026 test and key (`downloads/dlG/hh26_test.txt`, `hh26_key.txt`).
- USC Invitational 2026 extraction (`research/tests/nat26.md`); F-M 2019 (`research/tests/misc-a.md`); UT 2020 (`research/tests/ut-b.md`).
- `prior/F_slides.md` §8 (decks 12, 14; generator Level 15); `prior/H_logic_diodes.md` B4-B8.
- Kuphaldt, *Lessons in Electric Circuits* Vol. III ch. 3; Socratic worksheet "Diodes" (allaboutcircuits.com/worksheets).
- All numbers checked in Python.

Special Diodes

Rule area **3.c.vii** says "uses of single-junction devices (e.g., diodes, LEDs)". The "e.g." means other single-junction parts can appear. Rule **3.d** excludes "semiconductors (beyond those listed above)" and "non-linear devices (beyond those listed above)". So each special diode gets its own scope tag below.

Scope summary

Device	Tag	Why
Zener diode	✅ IN (regulator design ⚠️)	One silicon PN junction. Princeton 2026 asked it. Deep regulator math is past "fundamental".
Germanium diode	✅ IN (as a comparison)	One PN junction, but not silicon. Tests compare it with Si.
Schottky diode	⚠️ Borderline	One junction, but metal-semiconductor, not PN. Rickards 2026 asked its symbol.
Photodiode	⚠️ Borderline	One PN junction used as a light sensor. Not named in the rules.
Solar (photovoltaic) cell	✅ as a DC source (3.c.iii); ⚠️ for its physics	A big PN junction. Tests ask it as a DC source.
Tunnel (Esaki) diode	❌ OUT	Negative resistance: a non-linear device beyond those listed (3.d).
Varactor (varicap)	❌ OUT	Used as a variable capacitor (3.d capacitance).
SCR, TRIAC, DIAC, Shockley 4-layer diode	❌ OUT	More than one junction (3.d semiconductors beyond those listed).

Key facts

- **Zener diode:** a silicon diode made to break down in reverse at an exact voltage, V_Z (for example 5.1 V). It is used **reverse-biased**. In breakdown it holds about V_Z across itself over a range of current. So it works as a voltage regulator, reference or limiter.
- Forward-biased, a Zener acts like a normal 0.7 V diode.
- Below about 5 V the "Zener effect" (tunneling through a thin, heavily doped junction) causes breakdown. Above about 5-6 V, **avalanche** breakdown dominates. Both are called "Zener diodes" in shops.
- Named after **Clarence Zener** (1934 theory of electrical breakdown).
- Zener symbol: a diode whose bar has bent ends (like a "Z").
- Common parts: 1N4733A = 5.1 V, 1 W. 1N4742A = 12 V, 1 W.
- **Germanium diode:** V_F ≈ 0.3 V. More reverse leakage than Si. Damaged at lower temperatures. Used in old crystal radios.
- **Schottky diode:** a metal touching n-type silicon. V_F ≈ 0.2-0.4 V. Switches very fast. Low reverse breakdown, more leakage. Symbol: the bar has S-shaped hooks.
- **Photodiode:** a PN junction with a window. It runs **reverse-biased**. Light makes electron-hole pairs, so the reverse current grows with light. Symbol: diode with arrows pointing **in**.
- **Solar cell:** a large PN junction with no bias. Light makes it a **DC source**. One Si cell gives about **0.5-0.6 V** open-circuit. Cells are wired in series for more voltage and in parallel for more current. Symbol: a cell (or diode) with arrows pointing in.
- **LED:** arrows pointing **out** (see LEDs A7). An LED can also work weakly as a light sensor.

Formulas

- Zener series resistor (no load): **R_S = (V_in – V_Z) / I_Z**.
- Zener regulator with load (⚠️):
  - First check: open-Zener voltage **V_open = V_in × R_L / (R_S + R_L)**. If V_open < V_Z, the Zener is OFF and V_out = V_open.
  - If ON: **V_out = V_Z**; **I_S = (V_in – V_Z)/R_S**; **I_L = V_Z/R_L**; **I_Z = I_S – I_L**; **P_Z = V_Z × I_Z**.
  - Smallest load resistor that still regulates: **R_L(min) = V_Z / I_S**.

  - Largest series resistor for load current I_L with minimum Zener current I_Z(min): **R_S(max) = (V_in – V_Z)/(I_L + I_Z(min))**.
  - Maximum Zener current: **I_Z(max) = P_Z(max) / V_Z**.
- Solar cells: **V_total = n × V_cell** (series). **P = V × I**. **Energy = P × t**.
- Photodiode (⚠️): **I_photo = responsivity (A/W) × light power (W)**.

Question types

Easy

1. Identify the symbol: Zener, Schottky, photodiode, LED, varactor ✅ IN

- Seen: Rickards Invitational 2026 (Oct 2025), Div C, INV, Q9 (TB#2; symbol with S-hooks → **Schottky**; I rendered the key page). UT Austin (ATX) Regional 2026, Div B, REG, MC12 (PV cell symbol), MC14 (LED symbol), MC18 (photodiode symbol). MIT Invitational 2019, Div C, INV, Q33 (match diode and other symbols).
- How to solve: Look at the bar and the arrows.
  1. Plain bar → rectifier diode.
  2. Bar with bent ends ("Z") → Zener.
  3. Bar with S-hooks → Schottky.
  4. Arrows out → LED. Arrows in → photodiode.
  5. Diode with a capacitor-like double bar → varactor (❌ OUT topic, but know the symbol).
- Example: A diode symbol has two arrows pointing toward it. What is it? **Answer: a photodiode.**
- Traps: Arrow direction is the whole difference between LED and photodiode.
- Calculator: look up — type: `symbol <name>`

2. What does a Zener do? ✅ IN

- Seen: Princeton Invitational 2026, Div C, INV, SA7c ("explain the behavior of a Zener diode; how are they useful?"; key: works in reverse, conducts once the Zener voltage is reached, then voltage stays constant → regulation or reference). National supervisor practice deck 14 (DIODE-REGIONS-ZENER).
- How to solve: Give three points. (1) It is used in reverse bias. (2) It breaks down at V_Z on purpose without damage. (3) It holds about V_Z, so it regulates or limits voltage.
- Example: Why is a Zener connected "backwards" (cathode to +)? **Answer: it must be reverse-biased to work in breakdown and hold V_Z.**
- Traps: A Zener connected forward just drops 0.7 V. It does not regulate at V_Z.
- Calculator: look up — type: `zener`

3. Germanium vs silicon ✅ IN

- Seen: USC Invitational 2026, Div C, INV, Q29 (why Si junctions are more common). Rickards 2026 Div C Q4 (choices include 0.23 V). National practice deck 14 (Ge 0.25-0.3 V).
- How to solve: Ge: lower V_F (0.3 V), more leakage, lower maximum temperature. Si: 0.7 V, less leakage, cheaper, works hotter.
- Example: A 3.0 V source drives 100 Ω through a diode. Find I for Si and for Ge. **Answer: Si: 2.3/100 = 23 mA. Ge: 2.7/100 = 27 mA.**
- Traps: The USC key (I, II) is doubtful. See Semiconductors and Doping A7.
- Calculator: calculate — type: `Vs, R, Vf`

4. Solar cell as a DC source ✅ IN

- Seen: UT Austin (ATX) Regional 2021, Div B, REG, Q22 ("not a source of AC power" → solar panel). F-M Invitational 2018, Div C, INV, Q3 ("source that gives a flat DC waveform" → photovoltaic cell). UF Northern Regional 2019, Div C, REG, Q22 ("photocells produce AC" → False). Cobra Invitational (Michigan), Div B, INV, Q13 (classify sources as AC or DC; solar cell → DC).
- How to solve: Solar cells, batteries and fuel cells give **DC**. Wall outlets and alternators give AC. An inverter turns solar DC into AC for a house.
- Example: Does a rooftop solar panel make AC or DC? **Answer: DC. A separate inverter changes it to AC.**
- Traps: "Solar power system" (with an inverter) delivers AC to the house. The **panel** itself is DC.

- Calculator: look up — type: `source ac dc`

Medium

5. Zener series resistor (no load) ✅ IN

- Seen: not seen as a number question in a test. Source: `prior/B_topics.md` §7d; Kuphaldt Vol. III ch. 3 "Zener diodes".
- How to solve: R_S = (V_in – V_Z) / I_Z.
- Example: A 5.1 V Zener on a 9.0 V supply should carry 20 mA. **Answer: R_S = (9.0 – 5.1)/0.020 = 195 Ω.** Use the next standard value up (200 Ω or 220 Ω) to keep the current at or below 20 mA.
- Traps: Use the Zener voltage, not 0.7 V.
- Calculator: calculate — type: `Vin, Vz, Iz`

6. Zener connected the wrong way (forward) ✅ IN

- Seen: not seen in a test. Source: `prior/H_logic_diodes.md` B14.
- How to solve: Forward-biased, a Zener is a normal diode. V = 0.7 V.
- Example: A 5.1 V Zener is placed forward-biased (anode to +) across the output of a 5 V supply through a resistor. What is the output voltage? **Answer: about 0.7 V.**
- Traps: The answer is not 5.1 V. Direction matters.
- Calculator: look up — type: `zener forward`

7. Schottky diode facts and uses ⚠️ Borderline

- Why borderline: it is a single junction, but metal-semiconductor (not PN).
- Seen: Rickards Invitational 2026, Div C, INV, Q9 (symbol). Not seen as a fact question.
- How to solve: Low V_F (0.2-0.4 V) → less power lost. Fast → used in switching power supplies. Low breakdown → not for high reverse voltage.
- Example: A 1.0 A load uses a series protection diode. How much power is lost with Si (0.7 V) vs Schottky (0.3 V)? **Answer: Si 0.70 W; Schottky 0.30 W.**
- Traps: A Schottky is **not** a PN junction. That is the main trick question.
- Calculator: calculate — type: `Vf, I`

8. Photodiode and light sensors ⚠️ Borderline

- Why borderline: it is a single PN junction but is not named in the rules.
- Seen: UT Austin (ATX) Regional 2026, Div B, REG, MC18 (photodiode symbol). No calculation seen.
- How to solve: Reverse bias. More light → more reverse current. To get a voltage, put a resistor in series and measure across it (V = I × R).
- Example: A photodiode has responsivity 0.5 A/W. 0.10 mW of light hits it. Its current goes through 10 kΩ. **Answer: I = 0.5 × 0.0001 = 50 µA. V = 50 µA × 10 kΩ = 0.5 V.**
- Traps: A photodiode is not an LED. It senses light; it does not make it.
- Calculator: calculate — type: `responsivity, P_light, R`

9. Solar cells in series and parallel ✅ IN (DC sources and circuits) / ⚠️ for cell physics

- Seen: not seen as a number question in a test. Source: `prior/H_logic_diodes.md` B31; OpenStax *College Physics 2e* ch. 21 (EMFs in series and parallel).
- How to solve: Series cells add voltage. Parallel strings add current. Power = V × I.
- Example: Si cells give 0.5 V each. How many in series for 6 V? **Answer: 12.**
- Traps: Parallel cells do not raise the voltage. Mismatched cells in parallel waste power.
- Calculator: calculate — type: `Vtarget, Vcell`

10. Solar power and charge time ✅ IN (power, energy: 3.c.iv)

- Seen: not seen in a test. Links to the USB-charger hands-on task (3.c.II.iv: "calculate how long it would take to charge a specific battery").
- How to solve:
  1. Panel power P = V × I.
  2. Battery energy = V × capacity (Ah) → Wh.

3. Time = energy / power.

- Example: An 18 V, 2.0 A panel charges a 12 V, 50 Ah battery (600 Wh). Ignore losses. **Answer: P = 36 W. t = 600/36 = 16.7 h.**
- Traps: Real charging loses 10-30 %. Say that losses make it longer.
- Calculator: calculate — type: `Vpanel, Ipanel, Vbat, Ah`

Hard

11. Zener regulator with a load: does it regulate? ⚠ Borderline

- Why borderline: the Zener is IN, but full regulator design is deeper than the "fundamental characteristics" level. Expect it only at a top invitational, States or Nationals.
- Seen: not seen in a test. Source: Kuphaldt Vol. III ch. 3; `prior/H_logic_diodes.md` B15.
- How to solve:
  1. Remove the Zener. Find V_open = V_in·R_L/(R_S + R_L).
  2. If V_open < V_Z → Zener OFF, V_out = V_open.
  3. Else V_out = V_Z. Find I_S, I_L, I_Z = I_S – I_L, P_Z.
- Example: V_in = 12 V, R_S = 220 Ω, V_Z = 5.1 V, R_L = 1.0 kΩ. **Answer: V_open = 9.84 V > 5.1 V → ON. I_S = 31.4 mA, I_L = 5.1 mA, I_Z = 26.3 mA, P_Z = 0.134 W.** Variant: R_S = 1.0 kΩ, R_L = 500 Ω → V_open = 4.0 V < 5.1 V → **Zener OFF, V_out = 4.0 V.**
- Traps: Never assume the Zener is ON. Always do step 1.
- Calculator: calculate — type: `Vin, Rs, Vz, RL`

12. Zener limits: minimum load, maximum series resistor, power rating ⚠ Borderline

- Why borderline: same as type 11.
- Seen: not seen in a test. Source: Kuphaldt Vol. III ch. 3; All About Circuits "Zener diodes".
- How to solve:
  1. R_L(min) = V_Z / I_S (the Zener current reaches zero).
  2. R_S(max) = (V_in – V_Z)/(I_L + I_Z(min)).
  3. I_Z(max) = P_Z(max)/V_Z. The worst case is no load: all of I_S goes through the Zener.
- Example: (a) Same as type 11: R_L(min)? (b) V_in = 9.0 V, V_Z = 5.1 V, load 20 mA, I_Z(min) = 5 mA: R_S(max)? (c) A 1 W, 5.1 V Zener: I_Z(max)? **Answer: (a) 5.1/0.03136 = 163 Ω. (b) 3.9/0.025 = 156 Ω. (c) 1/5.1 = 196 mA.**
- Traps: If the load is removed, the Zener takes all of I_S. Check its power at no load.
- Calculator: calculate — type: `Vin, Vz, IL, Izmin` or `Pz, Vz`

Nationals

13. Zener as an over-voltage threshold or clamp ⚠ Borderline

- Why borderline: same as type 11, but it is just a series-loop idea.
- Seen: National supervisor practice decks 12 and 14 homework (DIODE-THRESHOLD-DESIGN: "a circuit that passes Vin when above 3 V and gives about 0 below").
- How to solve:
  1. Zener in series with an LED and resistor: the LED lights only when V_in > V_Z + V_LED.
  2. Zener across a load: the load voltage cannot rise above V_Z.
- Example: A 9.1 V Zener (reverse) + red LED (2.0 V) + 1 kΩ in series. At what V_in does the LED start to light? What is the current at 15 V? **Answer: about 9.1 + 2.0 = 11.1 V. At 15 V: (15 – 11.1)/1000 = 3.9 mA.**
- Traps: The Zener must point "backwards" (cathode toward +) to use V_Z. Forward it adds only 0.7 V.
- Calculator: calculate — type: `Vz, Vf, R, Vin`

Out of scope (skip)

- ❌ Tunnel (Esaki) diode, "negative resistance" (Rickards Invitational 2020, Div B Q29 / Div C Q36; key "tunnel"). Rule 3.d: non-linear devices beyond those listed.
- ❌ Varactor / varicap ("Full Varcap" was a distractor in Rickards 2026 C Q9). Rule 3.d: capacitance.
- ❌ TRIAC in a home dimmer (MIT Invitational 2019, Div C, Q35). Rule 3.d: semiconductors beyond those listed.

- ❌ Photoelectric effect with a work function (UT Invitational 2019, Div B, Q22). In no 3.c area.
- ❌ Solid-state relays (optically coupled transistors/TRIACs). ⚠ The name may appear in a relay question, but the internals are 3.d.

Mistakes in tests and keys

- **USC Invitational 2026, Div C Q29:** key B is doubtful (see Semiconductors and Doping A7).
- **Rickards 2026 Div C Q9:** the choices say "Full Photodiode", "Full Zener Diode", "Full Schottky Diode". "Full" is not a real device name. The key (Schottky) is correct for the drawn symbol.

Not confirmed

- Zener part numbers (1N4733A, 1N4742A) and the 5-6 V crossover between Zener and avalanche breakdown are from general datasheet knowledge. I did not open a datasheet this run.
- Photodiode responsivity 0.5 A/W is a typical silicon value near 800-900 nm (general knowledge, not checked).
- Si solar cell 0.5-0.6 V open-circuit: general knowledge; matches `prior/B_topics.md` §7d.
- No STATE or NATIONAL test in the downloads asked a special-diode question.

Sources

- 2027 Div B rules, Circuit Lab, 3.c.iii, 3.c.vii, 3.d (local PDF).
- Rickards 2026 Div C key, page 3 rendered (`downloads/dlG/rsoi26C_key.pdf`). Princeton 2026 test and key (`downloads/dlG/puso26_test.txt`, `puso26_key.txt`). Rickards 2020 B key (`downloads/dlG/rsoiOldB_key.txt`).
- Test extractions: `research/tests/ut-a.md` (ATX 2026), `atx-reg.md`, `misc-a.md`, `misc-b.md`, `nat26.md`.
- `prior/F_slides.md` (decks 12, 14), `prior/H_logic_diodes.md` B14, B15, B31, B32; `prior/B_topics.md` §7d.
- Kuphaldt, *Lessons in Electric Circuits* Vol. III ch. 3 (Zener, Schottky, photodiode, solar cell sections). OpenStax *College Physics 2e* ch. 21.
- All numbers checked in Python.

LEDs

Rule area **3.c.vii** names LEDs directly. Hands-on task **II.c.iii** is "Given some wires, batteries, resistors, and 2 LEDs, hook them up so the LEDs are equally bright." In 2026, LED fundamentals were States/Nationals only. **In 2027 they are at every level.** LEDs were in the hands-on part of every 2026 Div B test read (Berkeley, Pembroke Hill, BullSO). This is the most-tested part of the area.

Key facts

- **LED** = light-emitting diode. It is a PN junction that gives off light when **forward-biased**.
- How it makes light: electrons cross the junction and fall into holes (**recombination**). Each one releases a **photon** (a packet of light). The photon's energy is about the **band gap** of the material.
- So the **material (its band gap) sets the color**. Not the current, not the voltage, not the resistor.
- **Shorter wavelength → higher photon energy → bigger band gap → higher forward voltage (V_F).** Order: IR < red < orange < yellow < green < blue < violet/UV.
- Polarity: **long leg = anode (+)**. **Short leg = cathode (–)**. The **flat edge** on the rim is the cathode. Inside, the bigger metal "anvil" is usually the cathode (not always reliable).
- Surface-mount LEDs and diodes mark the cathode with a line or a dot.
- Typical 5 mm indicator LED: **20 mA** normal, about 20-30 mA maximum. **Reverse breakdown only about 5 V.**
- An LED **must** have something that limits current (usually a series resistor). Without it, the current is set only by tiny internal resistances, so it is huge and the LED burns out.
- LEDs in **series** carry the same current. LEDs in **parallel** each need their **own** resistor.
- **Brightness** depends on current. Equal current gives equal brightness only for identical LEDs. Light output is roughly proportional to current, but it flattens at high current (read the datasheet graph).
- LED light is **narrow-band** (one main color, about 20-40 nm wide). It is **not** monochromatic and **not** coherent. Lasers are both.
- **White LED** = a blue InGaN LED plus a yellow phosphor coating.
- Efficiency: modern LEDs turn roughly 30-50 % of electric power into light. The rest becomes heat. Heat lowers efficiency and life, so power LEDs need **heatsinks**.
- History (⚠ trivia): Nick Holonyak Jr. made the first practical visible (red) LED in 1962. Akasaki, Amano and Nakamura won the 2014 Nobel Prize in Physics for efficient blue LEDs.

LED color table (typical values)

Color	Wavelength (nm)	V_F typical	Material
Infrared	> 760	1.2-1.6 V	GaAs, AlGaAs
Red	610-760	1.63-2.03 V (often "1.8-2.0")	AlGaAs, GaAsP, AlGaInP, GaP (doped)
Orange	590-610	2.03-2.10 V	GaAsP, AlGaInP, GaP
Yellow	570-590	2.10-2.18 V	GaAsP, AlGaInP, GaP
Green	500-570	1.9-4.0 V (old GaP ≈ 2.1; InGaN ≈ 3.0-3.4)	GaP, AlGaInP, InGaN
Blue	450-500	2.48-3.7 V (often "3.0-3.3")	InGaN, ZnSe, SiC
Violet	400-450	2.76-4.0 V	InGaN
UV	< 400	3.1-4.4 V	AlGaN, InGaN
White	broad	2.8-4.2 V (often "3.0-3.5")	blue InGaN + yellow phosphor

Source of ranges: the national homework generator "LED Datasheet" tab (same ranges as the Wikipedia "Light-emitting diode" color table). Real parts vary. Use the value the test gives.

Formulas

- Series resistor: **R = (V_S – V_F) / I**.
- With several LEDs in series: **R = (V_S – n·V_F) / I**.
- With source internal resistance r: **I = (V_S – V_F) / (R + r)**.

- LED power: **P_LED = V_F × I**. Resistor power: **P_R = (V_S – V_F) × I = I²R**. Total: **P = V_S × I**.
- Fraction of power that reaches the LED: **V_F / V_S**.
- Battery life (hours): **t = capacity (mAh) / I (mA)**.
- Photon energy (⚠️): **E (eV) ≈ 1240 / λ (nm)**.
- Junction temperature: **T_J = T_ambient + R_th × P** (R_th in °C/W).
- Piecewise-linear LED (Nationals): **V = V_0 + I × r**. From two readings: r = ΔV/ΔI, V_0 = $V_1 - I_1 \cdot r$.
- Standard E12 values (×1, ×10, ×100 …): 10, 12, 15, 18, 22, 27, 33, 39, 47, 56, 68, 82. **Round up** to keep the current at or below the target.

Question types

Easy

1. LED symbol and name ✅ IN

- Seen: UF Northern Regional 2019, Div C, REG, Q37-38 (circle the LED symbol; what "LED" means). UF Northern Regional 2020, Div C, REG, Q18 (box the LED symbol). UT Austin (ATX) Regional 2026, Div B, REG, MC14 (LED symbol → M).
- How to solve: LED = diode symbol with arrows pointing **out**. "LED" = light-emitting diode.
- Example: What does the "D" in LED stand for? **Answer: diode.**
- Traps: Arrows pointing **in** = photodiode.
- Calculator: look up — type: `led symbol`

2. Polarity: which leg is which ✅ IN

- Seen: Pembroke Hill 2025 (2026 season), Div B, INV, Q28 (what the short and long legs mean). Cobra Invitational (Michigan), Div B, INV, Q29 (label anode, cathode, +, –, current direction) and Q32 (which leg to which battery terminal). ATX Regional 2026, Div B, REG, hands-on (red/green LED toggle).
- How to solve: Long leg = anode = to +. Short leg = cathode = to –. Flat rim = cathode. Conventional current flows anode → cathode.
- Example: Which battery terminal does the short leg go to? **Answer: the negative (–) terminal.**
- Traps: Cut leads are the same length. Then use the flat edge, or test with a meter (see Testing Diodes A7).
- Calculator: look up — type: `led legs`

3. How an LED makes light ✅ IN

- Seen: Berkeley Invitational 2026, Div B, INV, Q39 ("an LED releases which particles as light" → **photons**; distractors electrons, polarons, positrons). UMSO 2019, Div C, INV, Q30 (which bias emits light → forward).
- How to solve: Forward bias → electrons and holes meet at the junction → they recombine → each gives off a photon.
- Example: Does an LED glow in reverse bias? **Answer: No. Light comes only from forward-bias recombination.**
- Traps: The electrons do not leave the LED. Only photons leave.
- Calculator: look up — type: `led how`

4. Why an LED needs a series resistor ✅ IN

- Seen: MIT Invitational 2019, Div C, INV, lab S2-3c ("purpose of the LED series resistors" → limit current). Princeton Invitational 2026, Div C, INV, Part 2 matching #11 ("limit current to prevent an LED from burning out" → **resistor**). GullSO Invitational 2025, Div C, INV, Q50c and e ("connect it directly to your battery pack?" → No; too much current).
- How to solve: An LED's voltage stays near V_F. Any extra supply voltage must drop somewhere. A resistor takes the extra voltage and sets the current.
- Example: Why can a 3 V battery burn out a red LED with no resistor? **Answer: the LED wants only about 2 V. The extra 1 V across tiny internal resistances drives a very large current.**
- Traps: "A potentiometer adjusts LED brightness" is also true (Princeton 2026 matching #3 → potentiometer). Read which job the question asks.
- Calculator: look up — type: `led resistor why`

5. Rank LED colors by forward voltage ✅ IN

- Seen: Berkeley Invitational 2026, Div B, INV, Q26 ("lowest voltage drop" → **red**). Princeton Invitational 2026, Div C, INV, SA8 (green, yellow, blue, red in increasing V_F → **red, yellow, green, blue**). MIT Invitational 2020, Div C, INV, Q27 (→ C: red < yellow < green < blue). MIT Invitational 2019, Div C, INV, Q44 (green or red higher? → **green**).
- How to solve: Follow the rainbow from long to short wavelength: IR < red < orange < yellow < green < blue < violet < UV.
- Example: Rank blue, red, yellow from lowest to highest V_F. **Answer: red, yellow, blue.**
- Traps: Old GaP green (≈2.1 V) is close to yellow. Modern InGaN green (≈3 V) is close to blue. Tests use the rainbow order.
- Calculator: look up — type: `led rank`

6. What sets an LED's color ✅ IN

- Seen: Princeton Invitational 2026, Div C, INV, MC7 (→ **the semiconductor band-gap energy**; distractors: current, voltage, series resistor). UT Austin (ATX) Regional 2019, Div B, REG, Q5 ("an LED's semiconductor compound mainly sets its …" → colour).
- How to solve: Color = photon energy = band gap = material.
- Example: Does a bigger resistor change a red LED to orange? **Answer: No. It only makes it dimmer.**
- Traps: Some cheap LEDs have colored plastic over a white chip. The question means the chip.
- Calculator: look up — type: `led color cause`

7. Typical LED current ✅ IN

- Seen: UF Northern Regional 2019, Div C, REG, Q39 (max current for a 5 mm LED → 20 mA). Pembroke Hill 2025 and Princeton 2026 labs use about 20 mA.
- How to solve: Indicator LEDs: 10-20 mA typical, about 20-30 mA maximum. Power LEDs: hundreds of mA (read the datasheet).
- Example: What current would you design for a standard 5 mm red LED? **Answer: about 20 mA (10-20 mA is safe).**
- Traps: "Maximum" and "typical" are different numbers on a datasheet (see type 27).
- Calculator: look up — type: `led current`

8. Basic series-resistor calculation (and its reverse) ✅ IN

- Seen (many):
  - Cobra Invitational, Div B, INV, Q30: 9 V, V_F = 1.8 V, 10 mA → **720 Ω**.
  - North Pocono Invitational 2020, Div C, INV, Q24: 9 V, V_F = 2.2 V, 15 mA, "maximum R that lights it" → key 450 Ω (exact **453 Ω**).
  - Northview Sylvania Invitational 2025, Div C, INV, Q42: V_F = 2.0 V, 20 mA, 5 V → **150 Ω** (choice B).
  - GullSO Invitational 2025, Div C, INV, Q50f: 3 V, V_F = 2.1 V, 20 mA → **45 Ω**.
  - Reverse (find I): Rickards Invitational 2026, Div C, INV, Q71: 9 V, red LED, resistor orange-blue-brown (360 Ω) → key **19 mA**, "reasonable? yes". MIT Invitational 2021, Div C, INV, Q90: 9.0 V, V_F = 1.2 V, 1 kΩ → **7.8 mA**.
- How to solve:
  1. Voltage left for the resistor: V_S – V_F.
  2. R = (V_S – V_F)/I. Or I = (V_S – V_F)/R.
- Example (Rickards Q71): **Answer: I = (9 – 2.0)/360 = 19.4 mA ≈ 19 mA. That is under 20 mA, so it is reasonable.** (With V_F = 1.8 V: 20.0 mA.)
- Traps: Convert mA to A (20 mA = 0.020 A). Do not forget to subtract V_F. A test that gives no V_F expects you to assume one (red ≈ 1.8-2.0 V); write your assumption.
- Calculator: calculate — type: `Vs, Vf, I` or `Vs, Vf, R`

Medium

9. Wavelength vs forward voltage ✅ IN

- Seen: MIT Invitational 2020, Div C, INV, Q28 (→ A: shorter λ has greater V_F).

- How to solve: Shorter λ → more energy per photon → more voltage needed.
- Example: An LED at 620 nm and one at 470 nm. Which has the higher V_F? **Answer: the 470 nm (blue) LED.**
- Traps: Frequency goes the other way: higher frequency → higher V_F.
- Calculator: look up — type: `lambda vf`

10. Color ↔ numeric V_F range (and V_F → color) ✅ IN

- Seen: UT Austin (ATX) Regional 2020, Div B, REG, Q51 (yellow LED V_F range; **key a (1.63-2.03 V) is wrong; correct is d, 2.10-2.18 V**). UT Invitational 2019, Div B, INV, Q16 (= UT 2019 Div C Q16; "LED with an ideal drop of 4.15 V: color" → key **white**). National homework generator "LED Datasheet" tab. USC Invitational 2026 lab part b (estimate V_F from color: red-green 2-3 V; blue/white 3-4 V).
- How to solve: Use the color table above. Match the number to the row.
- Example: An LED drops 1.8 V. What color is it most likely? **Answer: red.**
- Traps: Ranges overlap (green spans 1.9-4.0 V). If two rows fit, pick the one the test's own table uses. 4.15 V is above typical white (3.0-3.5 V); "white" is only the best of the choices.
- Calculator: look up — type: `vf <volts>` or `color <name>`

11. LED materials ✅ IN (trivia-level)

- Seen: UT Austin (ATX) Regional 2021, Div B, REG, Q27 (GaP NOT used for which color → **blue**). UF Northern Regional 2019, Div B, REG, Q11 (chemical for a red LED → **GaAs** family). MIT Invitational 2020, Div C, INV, Q29 (NOT a material for an orange LED → **D, gallium arsenide**; GaAs alone gives infrared). Rickards Invitational 2020 (2021 season), Div B, INV, Q19 (cannot make a red LED → **GaN**).
- How to solve: Remember three groups. (1) GaAs/AlGaAs → IR and red. (2) GaAsP, GaP, AlGaInP → red, orange, yellow, (old) green. (3) GaN/InGaN → blue, true green, violet, white.
- Example: Which material is in a blue LED: GaAsP, GaN, GaP? **Answer: GaN (InGaN).**
- Traps: GaAs alone is **infrared**. GaAs**P** is visible red-orange-yellow.
- Calculator: look up — type: `led material <color>`

12. Round to a standard resistor; check resistor power ✅ IN

- Seen: GullSO Invitational 2025, Div C, INV, Q50f-g (45 Ω exact → use **47 Ω** from the kit). Princeton Invitational 2026, Div C, INV, lab Part II.2 (series R for about 20 mA).
- How to solve:
  1. R = (V_S − V_F)/I.
  2. Round **up** to the next value you have (E12 list).
  3. New current: I = (V_S − V_F)/R_new.
  4. P_R = (V_S − V_F) × I. Pick a resistor rated at least 2× that (1/4 W is common).
- Example: 9 V, red V_F = 2.0 V, 20 mA. **Answer: R = 350 Ω → use 390 Ω. I = 7/390 = 17.9 mA. P_R = 7 × 0.0179 = 0.13 W, so a 1/4 W resistor is fine.** With a 470 Ω resistor: 14.9 mA.
- Traps: Rounding **down** pushes the current above the target.
- Calculator: calculate — type: `Vs, Vf, I` (gives R, E12, new I, P_R)

13. LEDs in series: total voltage, minimum battery, maximum count ✅ IN

- Seen: UT Invitational 2020, Div B, INV, Q31 (smallest battery for a 3 × 3 white-LED array → **12 V**). Pembroke Hill 2025, Div B, INV, hands-on Q37 (build 2 resistors and 2 LEDs all in series).
- How to solve:
  1. Total V_F = n × V_F.
  2. The supply must be bigger than that, with some left for the resistor.
  3. R = (V_S − n·V_F)/I.
- Example (UT Q31): A 3 × 3 array means 3 parallel strings of 3 white LEDs (about 3.0-3.3 V each). **Answer: each string needs about 9-9.9 V, so the smallest listed battery that works is 12 V.** Also: 9 V with two red LEDs at 20 mA → R = (9 − 4)/0.02 = **250 Ω**. 12 V with white 3.2 V LEDs → at most **3** in a string; R = 2.4/0.02 = **120 Ω**.
- Traps: A 9 V battery cannot light 3 white LEDs in series (needs > 9.6 V).
- Calculator: calculate — type: `Vs, n, Vf, I`

14. LEDs in parallel and current hogging ✅ IN

- Seen: Pembroke Hill 2025, Div B, INV, hands-on Q36 (2 resistors in parallel, in series with 2 LEDs in parallel). Berkeley Invitational 2026, Div B, INV, Q34 lab (blue LED connected but unlit).
- How to solve:
  1. Parallel LEDs share one voltage.
  2. On one shared resistor, the LED with the lowest V_F clamps the voltage and takes most (or all) of the current.
  3. Fix: give each LED its own resistor.
- Example: 9 V → 330 Ω → a red LED (2.0 V) in parallel with a blue LED (3.0 V). **Answer: the node sits at about 2.0 V. All the current (7/330 = 21.2 mA) goes through the red LED. The blue LED stays dark.**
- Traps: Even two same-color LEDs on one resistor can share unevenly, because real V_F values differ a little.
- Calculator: calculate — type: `Vs, R, Vf list`

15. Make two LEDs equally bright (official hands-on task) ✅ IN

- Seen: 2027 rules, Part II.c.iii (also in 2026 rules). Princeton Invitational 2026, Div C, INV, lab Part II (red, green and blue all equally bright: measure V_F, compute R for 20 mA, build, record current). BullSO 2026 and Berkeley 2026 labs use LEDs.
- How to solve:
  1. Same color: put both LEDs **in series** with one resistor. Same current → same brightness.
  2. If the supply is too low for series, use **two branches, each with its own equal resistor**.
  3. Different colors: give each its own resistor, sized for its own V_F, so both get the same current.
  4. Measure the current in each branch to prove it.
- Example: 9 V battery, two red LEDs (2.0 V), 15 mA. **Answer: series: R = (9 – 4)/0.015 = 333 Ω → use 330 Ω (I = 15.2 mA in both).** Red + blue on 9 V at 20 mA: R_red = 350 Ω, R_blue (3.0 V) = 300 Ω (3.2 V → 290 Ω).
- Traps: Equal current is not always equal **perceived** brightness for different colors. Say this on the answer sheet. Different LED types also differ in efficiency.
- Calculator: calculate — type: `Vs, Vf1, Vf2, I`

16. Why won't the LED light? ✅ IN

- Seen: UT Invitational 2020, Div B, INV, Q32 (A: it is reversed; B: blue needs a higher V_F). Princeton Invitational 2020, Div C, INV, lab 1 Q42-43 (blue LED does not light from a divider → use red or change R2). Princeton Invitational 2019, Div C, INV, Devices (b).
- How to solve: Check, in order: (1) polarity, (2) supply bigger than V_F (plus other drops), (3) open circuit (loose wire, wrong breadboard row), (4) LED burned out, (5) another LED in parallel with lower V_F.
- Example: A 3 V coin cell lights a red LED but not a blue one. Why? **Answer: blue needs about 3.0-3.3 V. A 3 V cell is not enough once any resistor drop is added.**
- Traps: A divider's output sags when the LED draws current. It can fall below V_F.
- Calculator: look up — type: `led not lit`

17. Anti-parallel LEDs: light for either polarity ✅ IN

- Seen: UT Austin (ATX) Regional 2021, Div B, REG, Q33 (arrangement that lights for either battery direction → **B, anti-parallel pair**). Cobra Invitational, Div B, INV, station S5 (two LEDs show current direction; the capacitor part is OUT).
- How to solve: Two LEDs in parallel, pointing opposite ways. One lights for each direction. This also protects each LED from reverse voltage (the other one clamps it at its V_F).
- Example: A red and a green LED are anti-parallel with a resistor on a battery. You flip the battery. What changes? **Answer: the other LED lights. It shows the current direction.**
- Traps: Both never light at the same time on DC.
- Calculator: look up — type: `anti-parallel`

18. LED power, efficiency and battery life ✅ IN

- Seen: Hawk and Hornet Invitational 2026, Div C, INV, Q11 ("approximately how efficient are typical LEDs?" → key **A, 30 %**). Not seen with numbers in a test. Source: `prior/H_logic_diodes.md` B19.
- How to solve: P_LED = V_F·I. P_total = V_S·I. LED share = V_F/V_S. Life = mAh / mA.

- Example: 9 V, 20 mA, red 2.0 V, 500 mAh battery. **Answer: P_LED = 40 mW. P_total = 180 mW. Only 22 % goes to the LED. Life = 500/20 = 25 h.**
- Traps: "Efficiency" can mean the share of power in the LED (circuit) or light out per power in (the LED itself, about 30-50 %).
- Calculator: calculate — type: `Vs, Vf, I, mAh`

19. LED vs incandescent: energy and cost ✅ IN (energy, 3.c.iv)

- Seen: Florida State 2021, Div B, ST, Q26 (average LED bulb wattage → **10 W**). `prior/I_bank.md` energy-cost example.
- How to solve: Energy (kWh) = power (kW) × hours. Cost = kWh × price.
- Example: A 9 W LED bulb replaces a 60 W bulb for 1000 h at $0.15/kWh. **Answer: saves 51 kWh = $7.65.**
- Traps: Convert W to kW (÷ 1000) before using a price per kWh.
- Calculator: calculate — type: `P1, P2, hours, price`

20. Coherent? Monochromatic? ✅ IN

- Seen: F-M Invitational 2018, Div C, INV, Q9 (LEDs are completely coherent / monochromatic / both / neither → **D, neither**).
- How to solve: An LED gives a narrow band of colors (not one wavelength). Its waves are not in step (not coherent). A laser is both.
- Example: Is a red LED a laser? **Answer: No. Its light is neither monochromatic nor coherent.**
- Traps: "Nearly one color" is not "monochromatic".
- Calculator: look up — type: `led coherent`

21. Heat and heatsinks ✅ IN

- Seen: USC Invitational 2026, Div B, INV, Q17 ("importance of an active heatsink in an LED" → **A, better heat dissipation**).
- How to solve: Heat lowers light output and life. A heatsink moves heat away. "Active" means it has a fan.
- Example: Why do LED bulbs have metal fins? **Answer: to carry heat away from the LED chip so it stays efficient and lasts longer.**
- Traps: LEDs make far less heat than filament bulbs, but they are more sensitive to heat.
- Calculator: look up — type: `heatsink`

22. Supply with internal resistance and no series resistor ✅ IN

- Seen: MIT Invitational 2019, Div C, INV, Q42 (3 V supply, internal R 0.3 Ω, red LED 1.8 V, no resistor → **4.0 A**) and Q43 (add 220 Ω → **5.4 mA**).
- How to solve: The internal resistance is the only resistor. I = (V_S – V_F)/r. Add any series R to r.
- Example (MIT 2019): **Answer: Q42: (3 – 1.8)/0.3 = 4.0 A (the LED would burn out). Q43: 1.2/220.3 = 5.45 mA.**
- Traps: In Q43 the 0.3 Ω barely matters. In Q42 it is everything.
- Calculator: calculate — type: `Vs, Vf, r, R`

23. LED strip power budget with a safety margin ✅ IN

- Seen: Cobra Invitational, Div B, INV, Q20 (10 LEDs/ft × 0.5 W × 20 ft, at least 20 % margin → **125 W** supply).
- How to solve: Total = count × power each. Multiply by (1 + margin). Pick the next supply size up.
- Example (Cobra Q20): **Answer: 10 × 0.5 × 20 = 100 W. 100 × 1.2 = 120 W. Choose 125 W (80 W and 100 W are too small).**
- Traps: Margin is added on top, not taken off.
- Calculator: calculate — type: `count, W each, margin`

Hard

24. LED at a target power → R, resistor power, how many LEDs ✅ IN

- Seen: Hawk and Hornet Invitational 2026, Div C, INV, FR1 (red V_F = 1.8 V at 35 mW from 3.7 V: (a) R → **98 Ω**; (b) resistor power → **37 mW**; (c) LEDs in parallel before a 1/4 W resistor burns → **6**).
- How to solve:
  1. I = P_LED / V_F.

2. R = (V_S – V_F)/I.
3. P_R = (V_S – V_F) × I.
4. For n LEDs, the resistor carries n × I, so P_R = n × (V_S – V_F) × I. Find the largest n with P_R ≤ 0.25 W.

- Example (HH FR1): **Answer: I = 0.035/1.8 = 19.4 mA. R = 1.9/0.0194 = 97.7 ≈ 98 Ω. P_R = 1.9 × 0.0194 = 36.9 mW ≈ 37 mW. n = 0.25/0.0369 = 6.8 → 6 LEDs.**
- Traps: Part (c) only works if the resistor is changed to keep each LED at 35 mW (R = 98/n). With the same 98 Ω and fixed V_F, the resistor current would not change (see Mistakes).
- Calculator: calculate — type: `Vs, Vf, P_LED, P_Rmax`

25. Largest supply before the resistor overheats ✅ IN

- Seen: Hawk and Hornet Invitational 2026, Div C, INV, FR6 (blue V_F = 3.0 V, a 1/4 W variable resistor holds 20 mA; largest V_S → **15.5 V**).
- How to solve:
  1. P = I²R → R_max = P/I².
  2. V_R = I × R_max.
  3. V_S = V_F + V_R.
- Example (HH FR6): **Answer: R_max = 0.25/0.020² = 625 Ω. V_R = 12.5 V. V_S = 15.5 V.**
- Traps: Use P = I²R (current is fixed), not V²/R.
- Calculator: calculate — type: `Vf, I, Pmax`

26. Battery tolerance → range of LED current ✅ IN

- Seen: GullSO Invitational 2025, Div C, INV, Q50i (battery 2.9-3.2 V, LED V_F typical 2.1 V and maximum 2.6 V, 47 Ω → **6.4 mA to 23 mA**).
- How to solve:
  1. Highest current: highest V_S with lowest V_F.
  2. Lowest current: lowest V_S with highest V_F.
- Example (GullSO Q50i): **Answer: I_max = (3.2 – 2.1)/47 = 23.4 mA. I_min = (2.9 – 2.6)/47 = 6.4 mA.**
- Traps: Pair the extremes the right way. Highest V_S with highest V_F does not give the extreme.
- Calculator: calculate — type: `Vs_min, Vs_max, Vf_typ, Vf_max, R`

27. Read an LED datasheet ✅ IN

- Seen: GullSO Invitational 2025, Div C, INV, Q50a-b, k (max continuous current → **50 mA**; V_F → **2.6 V** at **20 mA, 25 °C** test conditions; above 70 °C the maximum is **30 mA**, so 37.5 mA is **not OK**). MIT Invitational 2021, Div C, INV, Q83-87 (APTR3216CGCK: color → **green**; V_F → **2.1 V** (accept 2.0-2.2); max DC current at 25 °C → **30 mA**; at 65 °C → **15 mA** (accept 12-17); –3 dB beam width → **140°** (accept 130-150)).
- How to solve: Find these lines on the sheet:
  1. "Absolute maximum ratings": I_F(max), peak current, reverse voltage V_R, power, temperature range.
  2. "Electrical/optical characteristics": V_F (typ and max) **at** a test current and temperature; wavelength; viewing angle.
  3. Derating graph: maximum current vs ambient temperature.
- Example: A datasheet says "V_F: typ 2.1 V, max 2.6 V at I_F = 20 mA, T_a = 25 °C". What V_F do you design with? **Answer: 2.1 V for the normal current; check with 2.6 V for the worst case.**
- Traps: The "maximum" V_F is not the maximum allowed voltage. It is the worst-case drop at the test current.
- Calculator: look up — type: `datasheet terms`

28. Double the brightness from the luminous-intensity graph ✅ IN

- Seen: GullSO Invitational 2025, Div C, INV, Q50j and l (twice as bright → **about 37.5 mA** from the datasheet curve; R = (3 – 2.1)/0.0375 = **24 Ω** → **use 22 Ω**). Q50n: "is it actually twice as bright?"
- How to solve:
  1. Read the "relative luminous intensity vs forward current" graph. Find the current where intensity is 2× the value at your present current.
  2. R = (V_S – V_F)/I_new. Round to a value you have.
  3. Check I_new against the maximum rating (and the derating at high temperature).

- Example (GullSO): **Answer: about 37.5 mA; R = 24 Ω; the kit's 22 Ω is closest.** At 22 Ω, I = 0.9/22 = 41 mA, which is under the 50 mA maximum at 25 °C.
- Traps: Brightness is not exactly proportional to current. Doubling current gives a little less than double light. Our eyes see brightness on a log scale, so "twice as bright" also looks less than double.
- Calculator: look up — type: `intensity curve` (read the given graph)

29. Junction temperature from thermal resistance ✅ IN

- Seen: MIT Invitational 2021, Div C, INV, Q88 (20 mW of heat, 25 °C ambient, R_th = 630 °C/W → **37.6 ≈ 38 °C**; safe, well inside the limit).
- How to solve: T_J = T_ambient + R_th × P_heat.
- Example (MIT Q88): **Answer: 25 + 630 × 0.020 = 37.6 °C. It is safe.**
- Traps: Use watts (20 mW = 0.020 W). R_th is °C per watt.
- Calculator: calculate — type: `Ta, Rth, P`

30. Typical vs maximum V_F ✅ IN

- Seen: Rickards Invitational 2026, Div C, INV, Q18 ("average difference between LED maximum forward voltages and typical forward voltages" → key **A, 0.4-0.8 V**; I rendered the key page).
- How to solve: Datasheets usually list max V_F about 0.4-0.8 V above typical. Example: typ 2.0 V / max 2.6 V; typ 3.2 V / max 3.6-4.0 V.
- Example: GullSO's LED is typ 2.1 V, max 2.6 V. What is the gap? **Answer: 0.5 V.**
- Traps: This is a "rule of thumb" question. It depends on the parts the writer used.
- Calculator: look up — type: `vf typ max`

31. Dim an LED with a switched shunt resistor ✅ IN

- Seen: USC Invitational 2026, Div B and C, INV, hands-on part e ("one button dims the LED but keeps it on" → button + a second 510 Ω across the LED).
- How to solve:
  1. Without the shunt: I = (V_S – V_F)/R.
  2. With a shunt R_2 across the LED (one node, LED at V_F): resistor current = (V_S – V_F)/R. Shunt takes V_F/R_2. LED gets the rest.
- Example: 9 V, 510 Ω, red LED 2.0 V; button adds 510 Ω across the LED. **Answer: before: 7/510 = 13.7 mA. After: 13.7 – 2.0/510 (3.9 mA) = 9.8 mA (about 29 % dimmer).** A blue LED (3.0 V): 11.8 mA → 5.9 mA.
- Traps: A plain wire across the LED (no shunt resistor) turns it **off**, not dim (that is the NOR part of the same lab).
- Calculator: calculate — type: `Vs, R, Vf, Rshunt`

32. Berkeley lab: two dim and equal, one bright, blue connected but dark ✅ IN

- Seen: Berkeley Invitational 2026, Div B, INV, Q34 lab (9 V battery; resistors 10, 100, 2 × 220, 330, 1 k, 2 k, 10 k, 100 kΩ; 3 LEDs + 1 blue LED). No numeric key (design task).
- How to solve (one design):
  1. Bright LED: its own branch, 9 V → 330 Ω → LED → 21 mA.
  2. Two dim, equal LEDs: one series string, 9 V → 2 kΩ → LED → LED → 2.5 mA each.
  3. Blue LED: put it in parallel with one of the dim red LEDs. That node is about 2 V, below blue's 3 V, so blue stays dark. (Or reverse it; it then sees only a safe small reverse voltage.)
- Example: **Answer: bright branch (9 – 2)/330 = 21.2 mA; dim string (9 – 4)/2000 = 2.5 mA in both LEDs; blue unlit because it is clamped at about 2 V.**
- Traps: Do not reverse-bias the blue LED straight across 9 V. That is above its ~5 V reverse rating.
- Calculator: calculate — type: `Vs, Vf list, R list`

33. Photon energy from wavelength; V_F vs band gap ⚠️ Borderline

- Why borderline: E = hc/λ is photon physics, not an electrical topic. Princeton 2026 SA8's key uses it to explain V_F.
- Seen: Princeton Invitational 2026, Div C, INV, SA8 (key: "higher band-gap energy corresponds to a shorter wavelength (E = hc/λ) and requires a higher voltage (E = qV)"). Berkeley 2026 Div B Q40 (rank band gaps).
- How to solve: E (eV) = 1240/λ (nm). V_F is about E/e or a bit more.

- Example: Photon energy of a 630 nm red LED and a 470 nm blue LED? **Answer: 1240/630 = 1.97 eV; 1240/470 = 2.64 eV.** So blue needs at least about 2.6 V.
- Traps: Blue LEDs usually need 3.0-3.3 V, more than 2.64 V, because of extra internal drops.
- Calculator: calculate — type: `lambda (nm)`

34. Drive an LED from a logic gate (sink vs source) ✅ IN (LED use + logic, 3.c.viii)

- Seen: not seen in a test. Source: `prior/H_logic_diodes.md` A52; standard 74xx app notes.
- How to solve:
  1. **Sourcing:** gate output → resistor → LED → ground. LED on when output is HIGH (1).
  2. **Sinking:** +V → resistor → LED → gate output. LED on when output is LOW (0) (active-low).
  3. R = (V_supply − V_F − V_out,low)/I for sinking.
- Example: 5 V, red LED 2.0 V, 10 mA, sinking into an output that is 0.2 V when low. **Answer: R = (5 − 2.0 − 0.2)/0.010 = 280 Ω. The LED is ON when the output is LOW.**
- Traps: In the sinking circuit, logic 1 turns the LED **off**.
- Calculator: calculate — type: `Vcc, Vf, Vol, I`

Nationals

35. Piecewise-linear LED model from two measurements ✅ IN (two equations)

- Seen: not seen in a test. Built to the 3.e.i limit (two equations). Uses the hands-on skill "measure V and I".
- How to solve:
  1. Measure V and I at two currents.
  2. $r = \Delta V/\Delta I$. $V_0 = V_1 - I_1 \cdot r$.
  3. Predict the current for a new resistor: $I = (V_S - V_0)/(R + r)$.
- Example: An LED reads 2.00 V at 10 mA and 2.10 V at 20 mA. Find r and V_0. Predict I with 9 V and 330 Ω. **Answer: r = 0.10/0.010 = 10 Ω. V_0 = 2.00 − 0.010 × 10 = 1.90 V. I = (9 − 1.9)/340 = 20.9 mA.**
- Traps: This is still a straight-line model. It only holds near the measured currents.
- Calculator: calculate — type: `V1, I1, V2, I2, Vs, R`

36. Trace a breadboard photo of buttons and LEDs; find LED power ✅ IN

- Seen: UT Invitational 2019, Div B and C, INV, Q49 / H49 (3 push buttons, 4 LEDs on 9 V: which LEDs light for each button combination; actual LED power with buttons 1 + 2 → **7.08 mW** from the key's readings 1.88 V × 3.5 mA + 1.79 V × 0.28 mA; "theoretical" marked N/A because there is no series resistor).
- How to solve:
  1. Redraw the breadboard as a schematic (rows are joined; the center gap splits them).
  2. For each button combination, trace which LEDs have a complete forward path.
  3. P = Σ V_F × I for the lit LEDs.
- Example (UT 2019 Q49 part): **Answer: P = 1.88 × 3.5 mA + 1.79 × 0.28 mA = 6.58 + 0.50 = 7.08 mW.**
- Traps: With no series resistor, the current depends on the battery and button resistance, so you cannot predict it. That is why the key wrote "N/A".
- Calculator: calculate — type: `V1, I1, V2, I2`

Out of scope (skip)

- ❌ Capacitor across an LED: RC time constant and discharge time (Rickards 2026 Div C Q72). Rule 3.d: capacitance.
- ❌ Capacitor "mystery part" found by LED blinks (Cobra Invitational station S5). Rule 3.d: capacitance. (The anti-parallel LED idea itself is IN, type 17.)
- ❌ LED drivers made with transistors or op-amps (MIT 2019 lab S2-2 op-amp). Rule 3.d: semiconductors beyond those listed.
- ❌ Counter that drives the LEDs (MIT 2019 lab S2-3). Rule 3.d: sequential logic. Reading the LED truth table is IN.
- ⚠️ LED quantum efficiency vs temperature curves (NY Lake Erie/Niagara 2018, Div C, Q13). Device physics beyond the basics; key not verified.

Mistakes in tests and keys

- **UT Austin (ATX) Regional 2020, Div B Q51:** yellow LED V_F range. Key a (1.63-2.03 V) is the **red** range. Correct is **d, 2.10-2.18 V**.
- **UT Invitational 2019, Div B/C Q16:** "4.15 V ideal drop → white". 4.15 V is above typical white/blue values (3.0-3.5 V). "White" is the best of the choices, but the number is unusual. Key **doubtful**.
- **Hawk and Hornet 2026, Div C FR1(c):** "LEDs in parallel before burning out the resistor" (key 6). With the same 98 Ω and fixed V_F, the resistor current stays 19.4 mA no matter how many LEDs share it. The key's answer works only if each LED keeps its 35 mW (resistor changed to 98/n Ω). Write your assumption.
- **North Pocono 2020, Div C Q24:** "maximum R that lights it" (key 450 Ω; exact 453 Ω). A larger R still lights the LED, just more dimly. The question means "R for 15 mA".
- **Rickards 2026, Div C Q71:** no V_F is given. The key (19 mA) fits V_F ≈ 2.0 V; 1.8 V gives 20 mA.
- **GullSO 2025 Q50b:** the question says "maximum voltage at typical operating conditions"; the key gives 2.6 V (the max V_F) but then uses 2.1 V (typical) in part f. Both numbers are on the datasheet. Know which is which.

Not confirmed

- The color table ranges come from the national homework generator (`prior/F_slides.md` §8). They match the Wikipedia LED color table from memory. I did not re-open Wikipedia this run.
- LED efficiency (30-50 %) and the "20-40 nm" spectral width are general knowledge. HH 2026 Q11 key says 30 %.
- History dates (Holonyak 1962; Nobel 2014) are general knowledge (also in `prior/H_logic_diodes.md` C2).
- The GullSO and MIT 2021 datasheets were not in the downloads. I used the keys' numbers.
- UT 2019 Q49 LED table: earlier research could not trace the breadboard photo at the available resolution. I only checked the power arithmetic.
- No STATE or NATIONAL test in the downloads had LED questions. Florida State 2021 had only "average LED bulb wattage" (Q26).

Sources

- 2027 Div B rules, Circuit Lab, 3.c.vii, Part II.c.iii (local PDF). 2026 rules: LEDs were States/Nationals only (`prior/A_rules.md`).
- 2026 Div B tests: Berkeley (`downloads/dlB/tests/berk26_test.txt`), Pembroke Hill (`phi26_test.txt`), BullSO (`bullso26_test.txt`).
- Div C tests and keys in `downloads/dlG/`: `gull26_test.txt` / `gull26_key.txt`, `hh26_*`, `nv26C_*`, `puso26_*`, `rsoi26C_*` (pages 2-4 rendered), `mit20_*`, `mit21_*`, `rsoiOldB_*`.
- Test extractions: `research/tests/atx-reg.md`, `misc-a.md`, `misc-b.md`, `nat26.md`, `state.md`, `ut-a.md`, `ut-b.md`, `ut-c.md`; `prior/G_tests.md`.
- `prior/B_topics.md` §7e; `prior/F_slides.md` §8; `prior/H_logic_diodes.md` A52, B16-B19, B23-B24, B28.
- Kuphaldt, *Lessons in Electric Circuits* Vol. III ch. 3 "Light-emitting diodes". OpenStax *College Physics 2e* ch. 29-30 (photon energy).
- All numbers checked in Python.

Diode Applications

Rule area **3.c.vii** says "**uses** of single-junction devices". Area **3.c.iii** covers DC and AC sources and uses. But rule **3.d** excludes "AC circuit theory", "frequency analysis", "capacitance" and "inductance". So rectifiers are IN as a diode use, while waveform math, smoothing capacitors and coil physics are borderline or OUT.

Key facts

- **Rectifier** = a circuit that turns AC into DC. It is the main use of a diode.
- **Half-wave rectifier:** one diode. It passes one half of each AC cycle and blocks the other. Output: pulses with gaps.
- **Full-wave bridge rectifier:** four diodes in a diamond. Two diodes conduct in each half-cycle (an opposite pair). Both halves come out the same way. Output: pulses with no gaps, but it still falls to 0 at each AC zero crossing.
- **Center-tap full-wave rectifier:** two diodes plus a transformer with a center tap (⚠️ the transformer is an inductance topic).
- A **rectifier** turns AC into DC. An **inverter** turns DC into AC. They are opposite devices.
- A **smoothing capacitor** across the output fills the gaps (❌ capacitance: OUT for 2027).
- A wall adapter ("wall wart") = transformer (lowers voltage) + rectifier (diodes) + capacitor (smooths) + sometimes a regulator.
- A car **alternator** makes AC. Diodes inside rectify it to DC.
- **Reverse-polarity protection:** a series diode blocks current if the battery goes in backwards. It costs about 0.7 V (0.3 V for Schottky). A **shunt** diode + fuse is the other way: a reversed battery shorts through the diode and blows the fuse, with no loss in normal use.
- **Diode OR-ing (power):** two sources each feed the load through a diode. The higher source powers the load. The diodes stop one source from charging the other. Used for battery backup.
- **Flyback (freewheeling) diode:** a diode across a relay coil or motor, reversed (cathode to +). When the switch opens, the coil's voltage spike is absorbed by the diode. (⚠️ The coil physics is inductance.)
- **Diode logic (DL):** diodes and resistors can make **AND** and **OR** gates. They **cannot** make **NOT**, because diodes cannot invert or amplify. So diode logic is not a universal gate set.
- **Voltage dropping:** each Si diode in series drops about 0.7 V. Two diodes take 5 V down to about 3.6 V.
- **Steering/isolation:** diodes let several switches share an indicator without back-feeding each other (see Diode Circuit Analysis A7 type 6).

Formulas

- Half-wave peak output: **V_out,peak = V_peak – 0.7**. Peak reverse voltage on the diode ≈ **V_peak**.
- Bridge peak output: **V_out,peak = V_peak – 1.4** (two diodes in the path). Each diode sees ≈ **V_peak – 0.7** in reverse.
- Center-tap (⚠️): **V_out,peak = V_peak(half-winding) – 0.7**; diode PIV ≈ **2·V_peak – 0.7**.
- RMS to peak for a sine (⚠️, AC theory): **V_peak = √2 × V_rms**.
- Protection diode loss: **P = V_F × I**.
- Diode-OR output: **V_out = (highest source) – 0.7**.
- DL AND (pull-up R to V_CC, inputs on cathodes): output = **lowest input + 0.7** (or V_CC if all inputs are high).
- DL OR (pull-down R, inputs on anodes): output = **highest input – 0.7**.

Question types

Easy

1. What is rectification? Rectifier vs inverter ✅ IN

- Seen: Princeton Invitational 2026, Div C, INV, SA7a ("what is rectification and how does a diode perform it?" → AC to DC; the diode passes only one direction). UT Austin (ATX) Regional 2020, Div B, REG, Q26 ("inverse devices" → **c, rectifier and inverter**).

- How to solve: Rectifier = AC → DC (uses diodes). Inverter = DC → AC.
- Example: Which device turns a car battery's DC into 120 V AC for a laptop charger? **Answer: an inverter.**
- Traps: A transformer changes AC voltage. It does not make DC.
- Calculator: look up — type: `rectifier`

2. Identify a 4-diode bridge ✅ IN

- Seen: UT Austin (ATX) Regional 2021, Div B, REG, FRQ5 Q54 ("4-diode bridge: what is it for?" → **rectify AC to DC**, a full-wave rectifier).
- How to solve: Four diodes in a diamond, AC on two opposite corners, DC out of the other two → full-wave bridge rectifier.
- Example: In a bridge, how many diodes conduct at one moment? **Answer: two (one opposite pair).**
- Traps: The + output is the corner where two **cathodes** meet.
- Calculator: look up — type: `bridge`

3. Reverse-polarity protection ✅ IN

- Seen: Princeton Invitational 2026, Div C, INV, Part 2 matching #6 ("prevent reverse current from damaging a battery-powered device if the leads are connected incorrectly" → **D, diode**).
- How to solve: Put a diode in series with the + lead, pointing toward the circuit. Wrong battery direction → diode reverse-biased → no current.
- Example: A 9 V circuit has a series Si protection diode. What does it get with the battery in correctly, and reversed? **Answer: 8.3 V; 0 V.**
- Traps: The diode costs 0.7 V all the time. A Schottky costs only about 0.3 V (8.7 V left).
- Calculator: calculate — type: `Vs, Vf`

4. AC or DC after a rectifier? ✅ IN

- Seen: Cobra Invitational (Michigan), Div B, INV, Q13 (classify lead-acid, solar cell, generator, dynamo + full-bridge rectifier, inverter, lightning → a dynamo with a full-bridge rectifier gives **DC**).
- How to solve: Anything after a rectifier is DC (pulsing, but one direction). Anything after an inverter is AC.
- Example: Is a car alternator's output to the battery AC or DC? **Answer: DC. Built-in diodes rectify the alternator's AC.**
- Traps: "Pulsing DC" is still DC. It never reverses direction.
- Calculator: look up — type: `source ac dc`

Medium

5. Half-wave vs full-wave rectifier ✅ IN (⚠️ for waveform detail)

- Why partly borderline: comparing the two is a diode use (IN). Describing sine waveforms is close to "AC circuit theory" (3.d).
- Seen: Rickards Invitational 2026 (Oct 2025), Div C, INV, Q62 ("explain the difference between full-wave and half-wave rectifiers" → half-wave passes half the cycle; full-wave converts the whole waveform to DC). Princeton Invitational 2026, Div C, INV, SA7a. National supervisor practice deck 12 homework (DIODE-POSITIVE-ONLY-DESIGN: "circuit using diodes that only has positive outputs" → series diode or bridge).
- How to solve: Half-wave: 1 diode, uses half the cycle, gaps in the output. Full-wave: 4 diodes (bridge) or 2 with a center tap, uses both halves, no gaps.
- Example: Which wastes more of the input: half-wave or full-wave? **Answer: half-wave. It throws away every other half-cycle.**
- Traps: A bridge loses 1.4 V (two diodes in the path), a half-wave loses 0.7 V. The bridge still delivers more power overall.
- Calculator: look up — type: `half vs full`

6. Draw the bridge diodes in the right direction ✅ IN

- Seen: UT Invitational 2019, Div C, INV, Q45 Part 2 ("draw the 3 missing diode symbols in the boxes; direction is important"). Rickards Invitational 2026, Div C, INV, Q63 ("connect the parts to create a circuit that forces a unidirectional flow of current" → GA, BC, DH, EF).

- How to solve:
  1. Mark the + DC output corner and the – DC output corner.
  2. Both diodes at the + corner point **toward** it (cathodes at +).
  3. Both diodes at the – corner point **away** from it (anodes at –).
  4. Check one half-cycle: current goes AC top → diode → + out → load → – out → diode → AC bottom.
- Example: In a bridge, which diode ends touch the positive output? **Answer: two cathodes.**
- Traps: If one diode is drawn backwards, that half-cycle becomes a short circuit across the AC source.
- Calculator: look up — type: `draw bridge`

7. Rectifier output peak voltage and diode reverse voltage ⚠️ Borderline

- Why borderline: it uses peak AC values. The math is subtraction only, but it touches AC circuit theory (3.d).
- Seen: not seen as a number question. Source: `prior/H_logic_diodes.md` B11; Kuphaldt Vol. III ch. 3 "Rectifier circuits".
- How to solve: Half-wave: V_peak – 0.7. Bridge: V_peak – 1.4. PIV: half-wave ≈ V_peak; bridge ≈ V_peak – 0.7.
- Example: 12 V peak AC into a Si bridge. Then into a half-wave. **Answer: bridge 10.6 V peak; half-wave 11.3 V peak.** Half-wave diode PIV ≈ 12 V, so a 1N4001 (50 V) is fine.
- Traps: If the question gives RMS (for example "12 V AC"), the peak is 12 × √2 ≈ 17 V. That conversion is ⚠️ AC theory.
- Calculator: calculate — type: `Vpeak, type`

8. Why pulsing DC is "inefficient" ⚠️ Borderline

- Why borderline: waveform reasoning sits near AC circuit theory.
- Seen: UT Austin (ATX) Regional 2021, Div B, REG, FRQ5 Q55 ("why is it inefficient?" → the output falls to 0 at every AC zero crossing).
- How to solve: Even a bridge output drops to zero twice per cycle. The load gets no power in those moments.
- Example: Why does a bridge rectifier alone not make "flat" DC like a battery? **Answer: its output follows the AC shape and drops to 0 V at every zero crossing.**
- Traps: The fix (a smoothing capacitor) is ❌ OUT. You can name it, but you will not need to calculate it.
- Calculator: look up — type: `ripple why`

9. Half-wave vs full-wave charge time ⚠️ Borderline

- Why borderline: it assumes average current is half for half-wave, which is waveform (AC) reasoning.
- Seen: UT Invitational 2019, Div C, INV, Q45 Part 1 (3 h with a full-wave adapter → half-wave takes **6 h**).
- How to solve: Half-wave delivers about half the average current. Same charge needed → about twice the time.
- Example: A battery charges in 2.5 h with a full-wave charger. Estimate the time with a half-wave charger. **Answer: about 5 h.**
- Traps: Real chargers have regulators, so the true ratio may differ. Use the simple ×2 the test expects.
- Calculator: calculate — type: `t_full`

10. Rectifier faults: open or shorted diode ✅ IN (troubleshooting, II.c.vi)

- Seen: not seen in a test. Source: `prior/H_logic_diodes.md` B11 variant; Kuphaldt Socratic worksheet "Rectifier circuits".
- How to solve:
  1. One bridge diode **open** → that half-cycle is lost → the bridge acts like a half-wave rectifier.
  2. One bridge diode **shorted** → on one half-cycle the AC source is shorted through it → the fuse blows (or the diodes burn).
- Example: A full-wave bridge suddenly gives only half its usual average output. What is the likely fault? **Answer: one diode has failed open.**
- Traps: An open diode does not stop the output completely. Two diodes still work for the other half-cycle.
- Calculator: look up — type: `bridge fault`

11. Diode OR-ing and battery backup ✅ IN

- Seen: not seen in a test. Source: `prior/B_topics.md` §7f; `prior/H_logic_diodes.md` B13.

- How to solve: The source with the higher voltage forward-biases its diode. The other diode is reverse-biased. Output = higher source − 0.7.
- Example: A 9 V adapter and a 6 V backup battery feed one load through Si diodes. What does the load get? What if the adapter is unplugged? **Answer: 8.3 V from the adapter (the battery's diode is off). Unplugged: 5.3 V from the battery.**
- Traps: Without the diodes, the 9 V adapter would push current into the 6 V battery (charging it badly).
- Calculator: calculate — type: `V1, V2, Vf`

12. Protection diode power and choice ✅ IN

- Seen: not seen in a test. Source: Kuphaldt Vol. III ch. 3; All About Circuits "Reverse polarity protection".
- How to solve: P = $V_F \times I$. Check P and I against the diode ratings. Schottky wastes less.
- Example: A device draws 0.50 A through a series protection diode. Find the loss for Si (0.7 V) and Schottky (0.3 V). **Answer: Si 0.35 W; Schottky 0.15 W.**
- Traps: Do not forget the reverse rating. The diode must block the full battery voltage when reversed.
- Calculator: calculate — type: `Vf, I`

13. Drop a voltage with series diodes ✅ IN

- Seen: not seen in a test. Source: `prior/B_topics.md` §7f ("dropping 0.7 V per diode").
- How to solve: Each Si diode drops about 0.7 V. Count the diodes.
- Example: How do you get about 3.6 V from a 5 V USB supply with diodes only? **Answer: two Si diodes in series (5 − 1.4 = 3.6 V).**
- Traps: The drop changes a little with current. It is a rough method.
- Calculator: calculate — type: `Vs, n`

14. Diode logic gates (AND and OR) ✅ IN (diode use + logic 3.c.viii)

- Seen: National supervisor practice deck 15 (LOGIC-DIODE-GATE; picture checked by the earlier run: left = diode OR, right = diode AND with a pull-up to +E). MIT Invitational 2020, Div C, INV, Q30 ("which can NOT be used to construct a logic gate?" → **B, induction coils**; diodes CAN).
- How to solve:
  1. Diodes from inputs → output, resistor from output to ground → **OR**.
  2. Diodes from output → inputs (cathodes at inputs), resistor from output up to V_{CC} → **AND**.
  3. Output level: OR = highest input − 0.7. AND = lowest input + 0.7 (or V_{CC} when all inputs are high).
- Example: DL AND, V_{CC} = 5 V, 1 kΩ pull-up, A = 0 V, B = 5 V. And a DL OR with A = 5 V, B = 0 V, 1 kΩ pull-down. **Answer: AND output = 0.7 V (logic 0), 4.3 mA through D_A. OR output = 4.3 V (logic 1).**
- Traps: The anodes point toward the output in OR. In AND the cathodes point toward the inputs. Draw arrows to check.
- Calculator: calculate — type: `Vcc, inputs, R, type`

15. Why diodes cannot make a NOT gate ✅ IN

- Seen: National supervisor practice deck 15 (LOGIC-DIODE-GATE: "can diodes alone make NOT?" → no, a NOT needs an active part).
- How to solve: Diodes only pass or block. They cannot turn a HIGH into a LOW. An inverter needs gain (a transistor, relay or vacuum tube).
- Example: Is {diode AND, diode OR} a universal gate set? **Answer: No. Without NOT you cannot build every function.**
- Traps: Diode logic + one inverter (or a relay) **is** universal. The missing part is only NOT.
- Calculator: look up — type: `diode not`

Hard

16. Cascaded diode-logic stages lose voltage ✅ IN

- Seen: not seen in a test. Source: `prior/H_logic_diodes.md` B12 follow-ups; Kuphaldt Vol. IV ch. 3.
- How to solve: Each diode OR stage subtracts 0.7 V. Each diode AND stage adds 0.7 V to a LOW. Stack the drops.

- Example: Two DL OR stages in a row, with 5 V in. What is the final HIGH level? **Answer: 5 – 0.7 – 0.7 = 3.6 V.**
- Traps: After several stages, a HIGH may fall below what the next gate reads as 1. That is why real logic uses transistors.
- Calculator: calculate — type: `Vin, stages`

17. Flyback diode across a relay coil ⚠ Borderline

- Why borderline: the diode use is IN (3.c.vi relays + 3.c.vii). The coil's stored energy is inductance (3.d).
- Seen: Princeton Invitational 2026, Div C, INV, SA9 (a transistor switches a relay coil; what happens to the coil's energy at turn-off → a large voltage spike across the transistor). `prior/I_bank.md` D12.
- How to solve: Put a diode across the coil, cathode to the + side. In normal use it is reverse-biased (does nothing). When the switch opens, the coil's spike forward-biases it, and the current circulates and dies safely.
- Example: Which way does the flyback diode point? **Answer: cathode toward the positive supply (reverse-biased in normal operation).**
- Traps: If it is put in forward, it shorts the supply every time the switch closes.
- Calculator: look up — type: `flyback`

18. Wall adapter: name the blocks ⚠ Borderline

- Why borderline: the diodes are IN. The transformer (inductance) and capacitor (capacitance) are OUT as physics, but naming them is light.
- Seen: F-M Invitational 2019, Div C, INV, Q5b ("three parts in a 5 V DC wall supply" → transformer, rectifier/diodes, capacitor).
- How to solve: In order: transformer (steps down AC) → rectifier (diodes, AC → DC) → capacitor (smooths) → (regulator).
- Example: Which block in a wall adapter uses diodes? **Answer: the rectifier.**
- Traps: Modern USB chargers are "switch-mode" and use a small high-frequency transformer, but the rectifier idea is the same.
- Calculator: look up — type: `adapter blocks`

19. Rectifier current sketch from an RMS value ⚠ Borderline

- Why borderline: RMS ↔ peak is AC circuit theory (3.d). The sketch itself (input two-way, output one-way) is a diode use.
- Seen: North Pocono Invitational 2020, Div C, INV, Q17 (current source + rectifier + R; I_rms = 2 A; draw input and output I-t → **I_max = 2.8 A**; input two-way, output one-way).
- How to solve: I_peak = √2 × I_rms. Draw the input as a sine between +I_peak and –I_peak. Draw the output with the negative halves flipped up (full-wave) or removed (half-wave).
- Example (North Pocono Q17): **Answer: I_peak = 2√2 = 2.83 A.**
- Traps: √2 is not trig, but RMS is listed under AC theory, which is excluded. Expect it rarely.
- Calculator: calculate — type: `Irms`

Out of scope (skip)

- ❌ Smoothing capacitor: "simplest improvement" (ATX Regional 2021, Div B, FRQ5 Q56); "add a single component to smooth" (Rickards 2026 Div C Q64); "draw the capacitor" (UT 2019 Div C Q45 Part 3). Rule 3.d: capacitance.
- ❌ Ripple voltage, clampers, voltage doublers, peak detectors. Rule 3.d: capacitance.
- ❌ Transformer turns ratio (North Pocono 2020 Q27b; Princeton 2026 SA10). Rule 3.d: inductance.
- ❌ Op-amp NOT gate plus diodes (national practice deck 15 homework LOGIC-OPAMP-NOT). Rule 3.d: semiconductors beyond those listed.
- ❌ 3-phase rectifiers. Rule 3.d: 3-phase power.
- ❌ Output frequency of a full-wave rectifier (2 × input). Rule 3.d: frequency analysis.

Mistakes in tests and keys

- **ATX Regional 2021, Div B, FRQ5:** Q56's keyed fix (a capacitor) is fine for that year, but it is OUT for 2027.

- **Princeton 2026 SA9:** the key explains the spike but does not name the flyback diode. A complete answer should add "a flyback diode across the coil prevents this".
- No key errors found in the rectifier and diode-logic items checked.

Not confirmed

- National practice deck 15 diode-gate picture: checked by the earlier research run (`prior/F_slides.md`). I did not reopen the .pptx.
- The ATX 2021 FRQ5 figure was not re-rendered this run. I used the earlier extraction.
- Rickards 2026 C Q63's lettered parts (GA, BC, DH, EF) refer to a figure I did not render. I used the key text only.
- No STATE or NATIONAL test in the downloads had a diode-application question.

Sources

- 2027 Div B rules, Circuit Lab, 3.c.iii, 3.c.vi, 3.c.vii, 3.c.viii, 3.d (local PDF).
- Princeton 2026 test and key (`downloads/dlG/puso26_test.txt`, `puso26_key.txt`). Rickards 2026 C key (`downloads/dlG/rsoi26C_key.txt`). UT 2019 C test (`downloads/dlG/ut19inv_Ctest.txt`). MIT 2020 key (`downloads/dlG/mit20_key.txt`).
- Test extractions: `research/tests/atx-reg.md`, `misc-a.md`, `ut-c.md`; `prior/G_tests.md`.
- `prior/F_slides.md` §8-9 (decks 12, 15); `prior/H_logic_diodes.md` A53, B9-B13; `prior/B_topics.md` §7f; `prior/I_bank.md` D12.
- Kuphaldt, *Lessons in Electric Circuits* Vol. III ch. 3 (rectifiers, clippers) and Vol. IV ch. 3 (logic gates).
- All numbers checked in Python.

Testing Diodes

Rules: Part II.b says participants "must be familiar with the operation of breadboards and multimeters". Hands-on tasks include II.c.ii (power supplied), II.c.iii (2 LEDs equally bright) and II.c.vi (troubleshoot a faulty circuit). Hands-on is **25-50 %** of the score. This note covers meter skills with diodes and LEDs.

Key facts

- **Diode-test mode** (symbol: a diode, often shared with continuity): the meter pushes a small current (about 1 mA) and shows the **voltage** across the part.
  - Good Si diode, red probe on anode: about **0.5-0.7** (V). Reversed: **OL** (over limit = open).
  - Good LED forward: about **1.6-3.2** V, and it often **glows faintly**. Some meters cannot reach a blue/white LED's V_F, so it reads OL both ways.
  - **OL both ways** → open (dead) diode. **About 0 both ways** → shorted diode.
- In diode mode the **red** lead is + and the **black** lead (COM) is –. So the forward reading tells you the red lead is on the **anode**.
- **Ohms mode** on a diode is misleading. The reading depends on the range, because the diode is non-linear. Do not use it to find V_F.
- **Continuity mode** beeps below a few tens of ohms. A diode may beep or not, depending on the meter.
- To **measure an LED's V_F in a circuit**: DC volts, probes across the LED, while it is lit.
- To **measure current**: break the circuit and put the meter **in series**, red lead in the **mA** jack. Or measure V across a known resistor and use **I = V_R / R** (no rewiring).
- Never put the meter in current mode **across** (in parallel with) a part. It is nearly a short and can blow the meter fuse.
- **In-circuit** diode tests can be wrong. Other parts in parallel give other paths. Lift one lead to be sure.
- Breadboards: the 5-hole rows on each side are joined; the center gap separates them; the long side rails are joined along their length (sometimes split in the middle). Ask the supervisor (rules II.b allow it).
- A battery + resistor + LED makes a simple **continuity tester**: the LED lights when the probe ends are joined through a closed path.

Formulas

- Current from a resistor's voltage: **I = V_R / R**.
- Power of a lit LED: **P = V_LED × I**.
- Series check (KVL idea): **V_supply ≈ V_R + V_switch + V_LED**.
- Continuity-tester current: **I = (V_S – V_F) / R**.

Question types

Easy

1. Read a diode-test result ✅ IN

- Seen: not seen as its own test question. NY Lake Erie/Niagara Invitational 2018, Div C, INV, Station I (black box) used a multimeter to find a diode. Source: `prior/H_logic_diodes.md` B27; Fluke and Kuphaldt meter guides.
- How to solve:
  1. Forward reading 0.5-0.7 → good Si diode.
  2. Reverse reading OL → good (it blocks).
  3. OL both ways → open. ~0 both ways → shorted.
- Example: A diode reads 0.62 one way and OL the other way. Is it good? Where is the red probe for the 0.62 reading? **Answer: good silicon diode. The red probe is on the anode.**
- Traps: A reading of about 0.3 means Ge or Schottky, not a bad Si diode.
- Calculator: look up — type: `diode test reading`

2. Find the anode and cathode with a meter ✅ IN

- Seen: ATX Regional 2026, Div B, REG, hands-on (red/green LEDs must be placed with correct polarity). Cobra Invitational, Div B, INV, Q32 (which leg to which terminal).
- How to solve: Diode mode. Try both ways. The way that reads a voltage (and makes an LED glow) has red on the anode.
- Example: An LED's leads are cut to the same length. How do you find the cathode? **Answer: diode mode; when it glows (or reads about 1.8-3 V), the black probe is on the cathode. Also check the flat edge on the rim.**
- Traps: Some meters show "1" (not "OL") for an open reading.
- Calculator: look up — type: `find anode`

3. Ohmmeter on a diode ✅ IN

- Seen: not seen in a test. Source: `prior/B_topics.md` §7g; Kuphaldt Vol. III ch. 3 "Meter check of a diode".
- How to solve: The ohms reading of a diode changes with the meter range. It is not a real resistance. Use diode mode.
- Example: A diode reads 500 Ω on the 2 kΩ range and 3 kΩ on the 20 kΩ range. Is the meter broken? **Answer: no. A diode is non-linear, so its "resistance" depends on the test current of each range.**
- Traps: Old analog meters reverse the polarity in ohms mode (the black lead is +).
- Calculator: look up — type: `ohms on diode`

4. SMD (surface-mount) diode marking ✅ IN

- Seen: UT Invitational 2020, Div C, INV, Q34 ("how is the polarity of a surface-mount diode marked?" → **a line marks the cathode**).
- How to solve: Line or band = cathode, for both through-hole and surface-mount diodes. SMD LEDs often use a green dot, a notch or a "T" mark on the back.
- Example: A black 1N4007 has a silver band at one end. Which end is that? **Answer: the cathode.**
- Traps: The UT question had two identical answer choices (C and D). Pick the one that says "cathode".
- Calculator: look up — type: `diode marking`

Medium

5. Measure an LED's V_F and name its color ✅ IN

- Seen: North Pocono Invitational 2020, Div C, INV, hands-on HO-37 (9 V and a parts kit: light an LED, draw it, measure V_F). USC Invitational 2026, Div B and C, INV, hands-on part b (series battery-R-LED; name the color; estimate V_F → red-green 2-3 V, blue/white 3-4 V).
- How to solve:
  1. Build battery → resistor → LED.
  2. Meter on DC V, probes across the LED.
  3. Compare with the color table in LEDs A7.
- Example: A clear LED measures 3.1 V at 10 mA. What type is it? **Answer: blue, white or true green (InGaN). Not red.**
- Traps: V_F rises a little with current. Note the current when you record V_F.
- Calculator: look up — type: `vf <volts>`

6. Measure LED current and power ✅ IN (task II.c.ii "power supplied")

- Seen: Pembroke Hill 2025 (2026 season), Div B, INV, hands-on Q38-39 (measure voltage and current, record units). Princeton Invitational 2026, Div C, INV, lab Part II.4 (record the current through each LED). UT Invitational 2019, Div B, INV, Q49 (actual LED power → **7.08 mW**).
- How to solve:
  1. Measure V_R across the known resistor. I = V_R / R.
  2. Or put the meter in series on mA.
  3. P_LED = V_LED × I. P_total = V_supply × I.
- Example: In a 9 V circuit, the 330 Ω resistor reads 6.9 V and the LED reads 2.0 V. Find I and the LED power. **Answer: I = 6.9/330 = 20.9 mA. P_LED = 2.0 × 0.0209 = 42 mW.**
- Traps: Write units (mA, mW). Pembroke gave points for units.
- Calculator: calculate — type: `VR, R, Vled`

7. Battery + resistor + LED as a continuity tester ✅ IN

- Seen: USC Invitational 2026, Div B and C, INV, hands-on part a (find the internal wiring of a 4-pin push button using the battery/LED/resistor as a continuity tester; pins on each side are joined in pairs; the button bridges the two pairs).
- How to solve:
  1. Build 9 V → 510 Ω → LED, with two free wire ends.
  2. Touch the ends to two pins. LED lights → those pins are connected.
  3. Test every pair, with and without pressing the button.
- Example: What current flows when the tester's ends touch? **Answer: (9 – 2.0)/510 = 13.7 mA (red LED).**
- Traps: Put the LED the right way, or it never lights.
- Calculator: calculate — type: `Vs, R, Vf`

8. Black box: which part is the diode? ✅ IN (capacitor item ⚠️)

- Seen: NY Lake Erie/Niagara Invitational 2018, Div C, INV, Station I (5 terminal pairs → capacitor, battery, diode, insulator, resistor; key C, A, D, E, B).
- How to solve: Use the meter on each pair:
  1. DC V shows a voltage → battery.
  2. Diode mode reads 0.5-0.7 one way, OL the other → diode.
  3. Ohms is the same both ways → resistor.
  4. OL everything → insulator (open).
  5. Ohms climbs slowly from low to OL → capacitor (⚠️ capacitance topic).
- Example: A pair reads OL in ohms both ways, but diode mode shows 0.58 one way. What is it? **Answer: a diode.**
- Traps: A diode can look like an "insulator" on a high ohms range. Always try diode mode.
- Calculator: look up — type: `black box`

9. Troubleshoot an LED circuit from voltage readings ✅ IN (task II.c.vi)

- Seen: Cobra Invitational, Div B, INV, station S1 ("broken" LED circuit: the switch is wired wrong, so the circuit is always open; bridge the switch with a coin → LED on). UT Invitational Fall 2025, Div B/C, INV, Lab 5c (measure V_R, V_SW, V_LED and check they add up to the supply). Princeton Invitational 2020, Div C, INV, lab 1 (find the fault, fix with one change). `prior/H_logic_diodes.md` B29.
- How to solve:
  1. Measure the supply. Then measure across each part.
  2. **Full supply across one part, 0 V on the others** → that part is open (or the LED is backwards).
  3. **0 V across the LED, current flowing** → LED shorted (or bypassed by a wire).
  4. **Readings add up to the supply** → the loop is complete.
- Example: 5 V supply, LED + 330 Ω. You read 5.0 V across the LED and 0 V across the resistor. **Answer: no current flows. The LED is open or inserted backwards.**
- Traps: A backwards LED and a dead LED give the same readings. Flip it to tell them apart.
- Calculator: look up — type: `fault from voltages`

10. Check the series voltage sum ✅ IN (✅ States-Nationals if called KVL)

- Seen: UT Invitational Fall 2025, Div B/C, INV, Lab 5c (V_total = V_R + V_SW + V_LED).
- How to solve: Add the measured drops. Compare with the supply. A small difference is meter error or wire drops.
- Example: Supply 8.95 V. Readings: V_R = 6.92 V, V_SW = 0.01 V, V_LED = 2.01 V. Does it check? **Answer: 6.92 + 0.01 + 2.01 = 8.94 V ≈ 8.95 V. Yes.**
- Traps: A closed switch should read almost 0 V. If it reads the full supply, it is open.
- Calculator: calculate — type: `V list, Vs`

Hard

11. Trace a breadboard and predict LED states ✅ IN

- Seen: UT Invitational 2019, Div B and C, INV, Q49 / H49 (breadboard photo: 3 push buttons, 4 LEDs; fill which LEDs light for every button combination). ATX Regional 2019, Div C, REG, hands-on (3 mystery gates with LEDs).
- How to solve:
  1. Draw the board's hidden connections (rows of 5, rails).
  2. Redraw as a schematic.
  3. For each input combination, find which LEDs have a forward path from + to –.
- Example: An LED's two legs are in the same 5-hole row. Will it light? **Answer: No. Both legs are shorted together by the row.**
- Traps: Parts placed across the center gap are not shorted. Parts in one row are.
- Calculator: look up — type: `breadboard rows`

12. Why an in-circuit diode test reads wrong ✅ IN

- Seen: not seen in a test. Source: Kuphaldt Vol. III ch. 3 "Meter check of a diode"; Fluke application notes (general knowledge).
- How to solve: Look for parts in parallel with the diode. A resistor across it gives a reading in **both** directions. Lift one leg and test again.
- Example: A diode in a board reads 0.45 forward and 0.45 reverse. It has a 470 Ω resistor in parallel. Is the diode bad? **Answer: not necessarily. The resistor gives the reverse reading. Lift one diode leg and retest.**
- Traps: Do not replace a part based on one in-circuit reading.
- Calculator: look up — type: `in-circuit test`

Nationals

13. Measure two points, build a model, predict a new circuit ✅ IN (two equations)

- Seen: not seen in a test. Combines task II.c.ii (power), II.c.vi and math 3.e.i (two equations). See LEDs A7 type 35.
- How to solve:
  1. Measure (V, I) for the LED with two different resistors.
  2. $r = \Delta V/\Delta I$ and $V_0 = V_1 - I_1 \cdot r$.
  3. Predict I and P for a new resistor. Build it and compare.
- Example: With 1 kΩ the LED reads 1.92 V at 7.08 mA. With 330 Ω it reads 2.04 V at 21.1 mA. Predict I with 470 Ω on 9 V. **Answer: r = 0.12/0.01402 = 8.6 Ω. V_0 = 1.92 – 0.00708 × 8.6 = 1.86 V. I = (9 – 1.86)/(470 + 8.6) = 14.9 mA.**
- Traps: Include the meter's own burden (small) only if the question asks.
- Calculator: calculate — type: `V1, I1, V2, I2, Vs, R`

Out of scope (skip)

- ❌ Oscilloscope views of rectifier output. Rule 3.d: oscilloscopes.
- ❌ Capacitor ID by charging behavior (black-box capacitor; Cobra station S5 blinking LEDs). Rule 3.d: capacitance. Naming "capacitor" in a black box is ⚠️ Borderline.
- ❌ Transistor testing (hFE socket on a meter). Rule 3.d: semiconductors beyond those listed.
- ❌ Soldering or AC mains testing. Rule Part II.d forbids hazardous conditions, high-power circuits and AC circuits.

Mistakes in tests and keys

- **ATX Regional 2026, Div B hands-on (red/green LED toggle with one button):** the key's solution does not work. (1) A multimeter on DC V does not supply power. (2) With side A tied to –, red is always on, and pressing the button puts green's anode at –, so green never lights. (3) No resistors are used, so the LEDs would see too much current. Correct idea: use an SPDT action (or the button as a shunt across one LED), with a battery and a series resistor.
- **UT Invitational 2020, Div C Q34:** choices C and D are identical (typo). The intended answer is "a line marks the cathode".

- **UT Invitational 2019, Div B Q49:** "theoretical power N/A" is correct only because there is no series resistor. The 7.08 mW comes from the key's own readings.

Not confirmed

- Diode-test current (~1 mA) and display ranges vary by meter model. General knowledge (Fluke manuals); not checked this run.
- The Nationals example readings (type 13) are made up for practice. I checked the arithmetic only.
- NY 2018 Station I answer letters come from the earlier extraction (`research/tests/misc-b.md`); I did not see the physical box.
- No STATE or NATIONAL test in the downloads had a meter-on-diode question.

Sources

- 2027 Div B rules, Circuit Lab, Part II.b, II.c.ii, II.c.iii, II.c.vi, II.d (local PDF).
- Pembroke Hill 2025 test (`downloads/dlB/tests/phi26_test.txt`); USC 2026 extraction (`research/tests/nat26.md` §3); UT and ATX extractions (`research/tests/ut-a.md`, `ut-b.md`, `ut-c.md`); `misc-a.md` (Cobra, North Pocono); `misc-b.md` (NY 2018).
- `prior/H_logic_diodes.md` B27-B29; `prior/B_topics.md` §7g and meter section.
- Kuphaldt, *Lessons in Electric Circuits* Vol. III ch. 3 "Meter check of a diode"; Vol. I ch. 8 (meters).
- All numbers checked in Python.

TAB A8

Digital Logic

Logic Gates and Notation 521
Truth Tables and Boolean Expressions 529
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Each note: key facts, formulas, then question types from Easy to Nationals with worked examples.

Logic Gates and Notation

Rule text (2027 Div B, manual page B12, rule 3.c.viii): "Digital logic gates, truth tables, universal gate sets, boolean expressions, De Morgan's Law, and combinational logic circuits (e.g., multiplexers, adders)." Math rule 3.e.iv: "Logical operations as boolean expressions, including De Morgan's Law and truth tables."

Why this matters. Every 2027 test must have at least one logic question (rule 3.c says "at least one question from each of the following areas"). Logic is brand new for Div B. So the only real practice questions come from older **Div C** tests (2019-2021) and the USC 2026 test.

Key facts

- A **logic gate** is a small circuit. It takes inputs that are 0 or 1 and gives one output that is 0 or 1.
- **1** means TRUE, HIGH, or ON. **0** means FALSE, LOW, or OFF. This is called **positive logic**.
- The seven gates you must know are NOT, AND, OR, NAND, NOR, XOR and XNOR. Also know the **buffer**.
- A **bubble** (small circle) on a gate symbol means NOT. It flips 0 to 1 and 1 to 0.
- An **inverter** is another name for a NOT gate.
- A **buffer** is a triangle with no bubble. Its output equals its input. It does not flip anything.

The gate table (learn it cold)

The output column is listed for inputs AB = 00, 01, 10, 11, in that order.

Gate	Output is 1 when	Output column (00, 01, 10, 11)	Common ways to write it	ANSI shape (US)	IEC box label
Buffer	input is 1	(1 input: 0 → 0, 1 → 1)	A	triangle	"1"
NOT	input is 0	(1 input: 0 → 1, 1 → 0)	A', $\bar{A}$ (overbar), ¬A, ~A, !A	triangle + bubble	"1" + bubble
AND	all inputs are 1	0, 0, 0, 1	A·B, AB, A∧B, A&B, A*B	D shape (flat back)	"&"
OR	at least one input is 1	0, 1, 1, 1	A+B, A∨B, A|B	curved shield, pointed nose	"≥1"
NAND	not all inputs are 1	1, 1, 1, 0	(AB)', overbar on AB, A↑B	AND + bubble	"&" + bubble
NOR	all inputs are 0	1, 0, 0, 0	(A+B)', overbar on A+B, A↓B	OR + bubble	"≥1" + bubble
XOR	the inputs are different	0, 1, 1, 0	A⊕B, A'B + AB'	OR shield with a second curved line at the back	"=1"
XNOR	the inputs are the same	1, 0, 0, 1	(A⊕B)', A⊙B, AB + A'B'	XOR + bubble	"=1" + bubble (some books print "=")

Memory aids:

- AND is "all". OR is "any". XOR is "exactly one" (for 2 inputs). XNOR is "equal".
- The N in NAND, NOR, XNOR means "then flip the answer".
- XNOR is also called the **equivalence gate** (it says "A equals B"). 1-bit equality test = XNOR.

Multi-input gates

- AND with n inputs: 1 only when all n inputs are 1.
- OR with n inputs: 1 when any input is 1.
- NAND and NOR with n inputs: the flipped AND and OR.
- **XOR with n inputs: 1 when an ODD number of inputs are 1.** This is the **parity** rule.
- **XNOR with n inputs: 1 when an EVEN number of inputs are 1** (zero counts as even). It is NOT "all inputs equal".

Order of operations in an expression

1. Parentheses and long overbars first. An overbar works like parentheses over everything under it.
2. Then NOT on a single letter.
3. Then AND.
4. Then OR (and XOR).

- Example: A + BC' means A OR (B AND (NOT C)).

Switches as gates (Claude Shannon, 1937)

- Switches in **series** = AND. The lamp needs both closed.
- Switches in **parallel** = OR. The lamp needs either one closed.
- A **normally closed (NC)** switch or relay contact = NOT. Pressing it breaks the path.
- A button wired **across** (in parallel with) an LED shorts the LED. So pressing it turns the LED OFF. This acts like NOT.
- Two 3-way (SPDT) switches = XOR or XNOR. Which one depends on what you call "position 1".
- **SPDT** means single pole, double throw. It is a switch with one input and two possible outputs.

Diodes as gates

- A **diode OR**: each input goes through a diode to the output. A resistor pulls the output down to ground. Any HIGH input pulls the output HIGH (minus one diode drop, about 0.7 V).
- A **diode AND**: a resistor pulls the output up to the supply. Each input connects through a diode pointing toward the input. Any LOW input pulls the output LOW (plus about 0.7 V).
- **Diodes and resistors alone can NOT make a NOT gate.** Inverting needs an active part, such as a transistor or relay.

Formulas

- Rows in a truth table with n inputs: **2^n**.
- Number of different logic functions of n inputs: **$2^{(2^n)}$**. For n = 2 that is 16. For n = 3 it is 256.
- LED current limit resistor: **R = (V_supply – V_LED – V_out,low) / I**. Leave out V_out,low if the LED goes to ground.
- Diode gate output: diode OR HIGH output ≈ V_in – 0.7 V. Diode AND LOW output ≈ V_in,low + 0.7 V.
- Relay coil current: I = V / R_coil. Coil power: $P = V^2 / R_{coil}$.

Question types

Easy

1. Name the gate from its ANSI (US) symbol — ✅ IN

- Where seen: USC Invitational 2026, Div B, INV, MC20 (also Div C MC20): a triangle with no bubble, answer "Buffer; preserves the input". Also every "draw the gate" item in Princeton 2019 Div C INV "Logic" (a)-(f).
- How to solve:
  1. Look at the body. Flat back + round front = AND family. Curved back + pointed front = OR family. Extra curved line behind the back = XOR family. Triangle = buffer or NOT.
  2. Look at the output tip. A bubble = the inverted version (NAND, NOR, XNOR, NOT).
  3. If the bubbles sit on the inputs, see De Morgan and Bubble Pushing A8.
- Worked example: A triangle with no bubble. **Answer: buffer. Output = input** (USC 2026 B MC20, key D).
- Traps:
  - A buffer does NOT flip the input. Choice A on USC 2026 said "Buffer; flips the input". That is wrong.
  - The XOR back line is thin. Look closely. UF Northern Regional 2019 C Q42 key mixed up XOR and XNOR (see Truth Tables and Boolean Expressions A8).
- Calculator: look up — type: shape word + bubble yes/no (for example "shield back-line bubble" → XNOR).

2. Name the gate from its IEC (rectangular box) symbol — ✅ IN

- Where seen: not seen in a test. The national supervisor deck 15 and many textbooks show it.
- How to solve:
  1. Read the label in the box: "&" = AND. "≥1" = OR. "=1" = XOR. "1" = buffer.
  2. A bubble or small triangle (flag) at the output = inverted.
- Worked example: A box with "≥1" and an output bubble. **Answer: NOR.**
- Traps:
  - "=1" means "exactly one input is 1". That is XOR for 2 inputs. It is NOT "equal".
  - Some books print XNOR as a box with "=". Other books print "=1" plus a bubble. Both mean XNOR.
- Calculator: look up — type: box label + bubble (for example "≥1 b" → NOR).

3. Name the gate from a word description — ✅ IN

- Where seen: Pembroke Hill Invitational, 2026 season, Div B, INV, Q23: "Which digital logic gate outputs TRUE (1) only if both inputs are TRUE?" Answer: AND. This was the only Div B logic question found in the 2026 season.
- How to solve:
  1. Turn the words into an output column for 00, 01, 10, 11.
  2. Match the column to the gate table.
- Worked example: "Output is 0 only when both inputs are 1." Column = 1, 1, 1, 0. **Answer: NAND.**
- Traps:
  - "Only if both" (AND) vs "if either" (OR) vs "if exactly one" (XOR).
  - "Either A or B" in normal English can mean OR or XOR. Read the rest of the sentence.
- Calculator: look up — type: column of 4 bits (for example 1110 → NAND).

4. Translate between notations — ✅ IN

- Where seen: national supervisor practice deck 15 (Digital Logic), notation slide; ATX (UT Austin) Regional 2019, Div C, REG, Q31: diagram → ~(p AND (q OR ~r)); Rickards Invitational 2020 (2021 season), Div C, INV, Q33 uses ' for NOT and * for AND; Florida State 2021, Div B, STATE, Q27 uses the word XOR.
- How to solve:
  1. Rewrite every symbol in one style you trust, for example ' for NOT, · for AND, + for OR.
  2. Keep all parentheses. Turn each long overbar into ( )'.
- Worked example: Rewrite ¬(p ∧ (q ∨ ¬r)) and evaluate it at p = 1, q = 0, r = 1.
  - Rewrite: (p·(q + r'))'.
  - r' = 0. q + r' = 0 + 0 = 0. p·0 = 0. NOT 0 = 1.
  - **Answer: (p(q + r'))' = 1.** (Python check: 1.)
- Traps:
  - "+" means OR, not addition. 1 + 1 = 1 in Boolean algebra.
  - ∨ is OR and ∧ is AND. Deck 15 has a typo that writes AND as "X V Y". V is OR.
  - A bar over AB is (AB)'. That is NOT the same as A'B' (bars over each letter).
  - Programmers write & for AND, | for OR, ^ for XOR, ! or ~ for NOT.
- Calculator: look up — type: symbol (for example "∨" → OR, "↑" → NAND, "↓" → NOR, "⊙" → XNOR).

5. Count truth-table rows and possible functions — ✅ IN

- Where seen: not seen in a test.
- How to solve:
  1. Rows = 2^n, where n = number of inputs.
  2. Different functions = 2^(rows), because each row can be 0 or 1.
- Worked example: A circuit has 3 inputs. **Rows = 2^3 = 8. Functions = 2^8 = 256.** For 4 inputs: 16 rows and 65,536 functions. (Python check.)
- Traps: Do not count inputs + 1. Do not confuse rows with functions.
- Calculator: calculate — type: n.

6. Output of a multi-input gate — ✅ IN

- Where seen: USC Invitational 2026, Div B, INV, MC21 (a 3-input XNOR and a 3-input NOR with a repeated input).
- How to solve:

1. AND/NAND: are all inputs 1?
2. OR/NOR: is any input 1?
3. XOR/XNOR: count the 1s. Odd → XOR = 1. Even → XNOR = 1.
4. Flip the answer if the gate has an N.

- Worked example: Inputs 1, 1, 0.
  - 3-input XNOR: two 1s = even → **1**.
  - 4-input NAND with inputs 1, 1, 1, 0: not all 1 → **1**.
  - 3-input NOR with 0, 0, 0: no 1s → **1**.
  - 4-input XOR with 1, 1, 0, 1: three 1s = odd → **1**. (All Python checked.)
- Traps:
  - **A 3-input XNOR is NOT "all three equal".** XNOR(0, 0, 0) = 1, but XNOR(1, 1, 1) = 0 (three 1s is odd). Python column for 000 to 111: 1, 0, 0, 1, 0, 1, 1, 0.
  - A repeated input (NOR(C, B', B')) acts like one input. NOR(C, B', B') = (C + B')'.
- Calculator: calculate — type: gate name + list of input bits.

7. Switch circuit → gate — ✅ IN

- Where seen: national supervisor deck 15 (lamp with switches A and B in series → AND); MIT Invitational 2020, Div C, INV, Q19-20 (switch pictures; "what is this equivalent to?" with choices NOT/NOR/XOR/XNOR); North Pocono Invitational 2020, Div C, INV, Q4 ("a simple 3-way switch is most like which gate?", key XNOR).
- How to solve:
  1. Series = AND. Parallel = OR. Normally closed = NOT.
  2. For two 3-way switches: the lamp is ON when the positions match (XNOR) or differ (XOR). It depends on the labels.
- Worked example: Switches A and B in parallel. That pair is in series with switch C. **Lamp = (A + B)·C.**
- Traps:
  - NP 2020 Q4 accepted only XNOR. XOR is also defensible. Write a short reason on a free-response answer.
  - A switch in parallel with the lamp shorts the lamp. Pressing it turns the lamp OFF.
- Calculator: look up — type: "series", "parallel", "NC", "3-way".

8. Real-life decision → gate — ✅ IN

- Where seen: Rickards Invitational 2020 (2021 season), Div C, INV, Q46 ("Describe a real-life decision synonymous with AND, OR, XOR"). National supervisor deck 15 homework (mammal detector).
- How to solve:
  1. AND: you need every condition.
  2. OR: any one condition is enough.
  3. XOR: you can pick one, but not both.
- Worked example: "I eat dessert if I finish dinner AND it is a weekend" = AND. "I take an umbrella if it rains OR the forecast says rain" = OR. "The menu says cake or pie, not both" = **XOR**.
- Traps: Give a clear "not both" for XOR. Graders look for it.
- Calculator: look up — type: gate name → example sentence.

Medium

9. Diode gate → which gate (and why there is no diode NOT) — ✅ IN (diodes are rule 3.c.vii; gates are 3.c.viii)

- Where seen: national supervisor practice deck 15 (diode OR and diode AND pictures; "can diodes alone make NOT?").
- How to solve:
  1. Find the resistor. Resistor to ground = diode OR. Resistor to the supply (pull-up) = diode AND.
  2. Check the diode direction. Diodes point toward the output in an OR. Diodes point toward the inputs in an AND.
  3. Find the output voltage: add or subtract one diode drop (about 0.7 V for silicon).
- Worked example: 5 V logic, silicon diodes (0.7 V).
  - Diode OR, inputs 5 V and 0 V. **Output = 5 − 0.7 = 4.3 V (HIGH).**

  - Diode AND, pull-up to 5 V, one input at 0 V. **Output = 0 + 0.7 = 0.7 V (LOW).**
  - Can diodes alone make NOT? **No.** A NOT needs an active part (transistor, relay, or op-amp). (Numbers checked in Python.)
- Traps:
  - Each diode level in a chain loses another 0.7 V. Long diode chains stop working.
  - Diode logic cannot invert. So it is not a universal gate set.
- Calculator: calculate — type: gate type, V_high, V_low, V_diode.

10. Bitwise logic on whole numbers — ⚠️ Borderline (binary whole numbers are IN under 3.e.iii; bit-by-bit logic on numbers is not named)

- Where seen: Florida State 2021, Div B, STATE, Q27 (0101 XOR 0011 = 0110), Q28 (5 XOR 3 = 6), Q29 (0011 XOR 0100 = 0111 = 7, select all); Rickards 2020 (2021 season), Div C, INV, Q29 (108 ⊕ 57 = 85); Princeton 2021, Div C, INV, Q25 ("20 * 13 in Boolean algebra" = 20 AND 13 = 4); MIT Invitational 2019, Div C, INV, Q25 (0x09 & 0x07 = 1, hex is extra).
- How to solve:
  1. Change each number to binary. Line up the bits on the right. Pad with zeros.
  2. Apply the gate to each column of bits. There is no carry.
  3. Change the answer back to decimal if asked.
- Worked example: 108 XOR 57.
  - 108 = 1101100. 57 = 0111001.
  - Column by column: 1⊕0 = 1, 1⊕1 = 0, 0⊕1 = 1, 1⊕1 = 0, 1⊕0 = 1, 0⊕0 = 0, 0⊕1 = 1.
  - Result 1010101 = 64 + 16 + 4 + 1 = **85**. (Python: 108 ^ 57 = 85.)
- Traps:
  - Bitwise XOR is not addition. 5 XOR 3 = 6, but 5 + 3 = 8.
  - In Boolean algebra "*" means AND. So "20 * 13" = 20 AND 13 = 4, not 260.
  - Hex numbers (0x09) go past the 2027 math limits. Change to binary anyway.
- Calculator: calculate — type: number 1, gate, number 2 (decimal or binary).

11. Identify a mystery gate on a breadboard (hands-on) — ✅ IN (hands-on 3.c.v "construct a logic circuit"; meter use II.b)

- Where seen: MIT Invitational 2019, Div C, INV, lab station 2 part 3 (LEDs blink through all inputs; table 1, 1, 1, 0 → NAND); MIT Invitational 2020, Div C, INV, Lab 2 "Circuit 1" (pre-built gate → 2-input OR; then write the truth table); ATX (UT Austin) Regional 2019, Div C, REG, hands-on (three mystery gates with LEDs; no key).
- How to solve:
  1. Apply every input pattern: 00, 01, 10, 11.
  2. Read the output LED or meter. HIGH is near the supply voltage. LOW is near 0 V.
  3. Write the output column. Match it to the gate table.
- Worked example: The output LED is OFF only when both input LEDs are ON. Column = 1, 1, 1, 0. **Answer: NAND.**
- Traps:
  - Check whether the LED lights for HIGH or for LOW (see type 13). An active-low LED flips your column.
  - Floating (unconnected) TTL inputs read as HIGH. Floating CMOS inputs are random.
- Calculator: look up — type: observed 4-bit column.

12. Relay circuit → gate — ✅ IN (relays are rule 3.c.vi)

- Where seen: MIT Invitational 2021, Div C, INV, Scenario J Q70-79 (relay circuits A-D → gates; minimum coil on-current; coil power).
- How to solve:
  1. Each input drives a relay coil.
  2. A normally open (NO) contact closes when the coil is ON. It passes the input as-is.
  3. A normally closed (NC) contact opens when the coil is ON. It acts as NOT.
  4. Read the contacts with the switch rules: series = AND, parallel = OR.

- Worked example: A relay coil needs 9 V and has 120 Ω. **Minimum on-current = 9 / 120 = 0.075 A = 75 mA. Coil power = 9² / 120 = 0.675 W ≈ 0.68 W** (MIT 2021 key: 75 mA, 680 mW; Python check).
- Traps: A relay output can short the supply if the drawing is wrong. MIT 2021 has a "non-gate" that shorts 12 V through 1.6 Ω.
- Calculator: calculate — type: V, R_coil (gives I and P).

13. Logic levels and LED indicators — ⚠️ Borderline (useful for hands-on; chip internals are transistors, which are OUT)

- Where seen: not seen in a written test. MIT Invitational 2021, Div C, INV, Q68-69 (noise margins, "static discipline").
- How to solve:
  1. TTL (74xx chips) runs on 5 V. An input below 0.8 V is LOW. An input above 2.0 V is HIGH. Between them is undefined.
  2. An LED from the output to ground lights when the output is HIGH (the gate **sources** current).
  3. An LED from +5 V into the output lights when the output is LOW (the gate **sinks** current). This is **active-low**.
  4. Size the resistor: R = (V – V_LED – V_out) / I.
- Worked example: 5 V supply, red LED (2.0 V), 10 mA, output LOW at 0.2 V, LED wired from +5 V into the output. **R = (5 – 2.0 – 0.2) / 0.010 = 280 Ω. The LED is ON when the output is LOW.** (Python check.)
- Traps: An input at 1.4 V is neither 0 nor 1 for TTL. Do not guess.
- Calculator: calculate — type: V_supply, V_LED, V_out, I.

14. Transistor or CMOS drawing → gate — ⚠️ Borderline (the gate is IN; transistors are "semiconductors beyond those listed", 3.d)

- Where seen: F-M Invitational 2018, division not stated, INV, Q11 (CMOS pair with a truth table → NOT); MIT Invitational 2020, Div C, INV, Q50 (transistor circuit → NOT).
- How to solve:
  1. Use the truth table if the test gives one. Then it is just type 3.
  2. One transistor that pulls the output to ground when the input is HIGH = NOT.
- Worked example: The given table is 0 → 1, 1 → 0. **Answer: NOT (inverter).**
- Traps: Do not study transistor physics. It is out of scope.
- Calculator: look up — type: truth table column.

Hard

15. Gate properties: commutative, associative, "flips on any input change" — ✅ IN

- Where seen: not seen in a test in this form. Close relatives: MIT Invitational 2020, Div C, INV, Q31 (which gates are linear → XOR, XNOR), Q35 (which gate is reversible → NOT).
- How to solve:
  1. Commutative means swapping inputs gives the same output. All six 2-input gates are commutative.
  2. Associative means (A•B)•C = A•(B•C). Check one row where it might fail.
  3. "Flips when any one input flips" is the XOR family.
- Worked example: Which 2-input gates are NOT associative? Try NAND: NAND(NAND(1,1),0) = NAND(0,0) = 1, but NAND(1, NAND(1,0)) = NAND(1,1) = 0. **Answer: NAND and NOR are not associative. AND, OR, XOR and XNOR are.** (Python checked all 8 rows for each gate.)
- Traps:
  - This is why a "3-input NAND" is not two 2-input NANDs chained. NAND3(A,B,C) = (ABC)'.
  - Only XOR and XNOR change output every time any single input changes.
- Calculator: look up — type: property word.

16. Push-button logic design (hands-on, no chips) — ✅ IN (hands-on 3.c.v; switches 3.c.vi)

- Where seen: USC Invitational 2026, Div B and C, INV, lab "Light Switch" parts c and d (LED on only when both buttons are released = NOR; LED on when A pressed and B released = A·B').
- How to solve:

1. Write the truth table for the buttons.
2. Buttons in series with the LED make AND. Buttons in parallel make OR.
3. A button across the LED (shorting it) makes NOT.
4. Keep a resistor between the battery and any shorting button. Then a press never shorts the battery.

- Worked example: 9 V battery, 510 Ω resistor, LED. Both buttons are in parallel across the LED. Pressing either button shorts the LED. So **LED = NOR(A, B)**. With a button pressed, the resistor carries **9 / 510 = 17.6 mA** and burns **9^2 / 510 = 0.16 W**. A 1/4 W resistor is safe. (Python check.)
- Traps: Never put a shorting button straight across the battery with no resistor.
- Calculator: calculate — type: V, R (gives I and P).

Nationals

17. Trivia about special gate classes — ⚠ Borderline (not named in the rules; seen only at top Div C invitationals)

- Where seen: MIT Invitational 2020, Div C, INV, Q31 (linear gates: XOR, XNOR), Q32, Q35 (reversible: NOT); Princeton 2020, Div C, INV, Q18 (perceptron with weights 20 and bias –10 → OR); UT Invitational fall 2019, Div C, INV, Q26 ("most related to digital logic": neural networks).
- How to solve:
  1. "Linear" gates can be written with XOR and constants only: XOR, XNOR, NOT, buffer.
  2. "Reversible" means you can always recover the inputs from the outputs. A 1-input NOT is reversible. 2-input AND/OR/XOR are not (2 inputs → 1 output loses information).
  3. A perceptron outputs 1 when $w1 \cdot A + w2 \cdot B + b > 0$. Test all four rows.
- Worked example: w1 = w2 = 20, b = –10. Row 00: –10 → 0. Row 01: 10 → 1. Row 10: 10 → 1. Row 11: 30 → 1. **Column 0, 1, 1, 1 = OR.**
- Traps: XOR cannot be made by a single perceptron. No single straight line separates its 1s from its 0s.
- Calculator: calculate — type: w1, w2, b (gives the column and gate).

Out of scope (skip)

- ❌ Op-amp buffers and comparators. USC 2026 B MC15 ("op-amp buffer vs diode") is OUT: "semiconductors (beyond those listed above)" (3.d).
- ❌ Transistor-level gate design (CMOS, TTL totem pole, RTL/DTL). Same 3.d exclusion.
- ❌ Latches, flip-flops, counters, memory cells (MIT 2020 Q34, Princeton 2020 Q39, USC 2026 B FRQ 36b-c). Rule 3.d: "sequential logic and digital clock signals".
- ❌ Tri-state buffers and buses. The 2020 rules excluded "three-state logic gates". The 2027 rules do not name them. Treat them as out.
- ❌ Hex and octal as a main topic. Rule 3.e.iii allows only "whole numbers in binary".
- ❌ Arithmetic and logical bit shifts (ATX Regional 2020, Div C, Q60). Rule 3.e.iii allows no arithmetic other than addition and no negative numbers.

Mistakes in tests and keys

- **UF Northern Regional 2020, Div C, REG, Q21 and Q22:** the key swaps the answers. Column 0, 1, 1, 1 is **OR** (key: NAND). Column 1, 1, 1, 0 is **NAND** (key: OR).
- **North Pocono Invitational 2020, Div C, INV, Q4:** the key accepts only XNOR for a 3-way switch. XOR is also correct, depending on the position labels.
- **National supervisor deck 15:** it defines XNOR as "true only if A NOR B is true". That is wrong. XNOR is 1 when A = B. The deck also writes AND as "X V Y". The symbol V means OR.
- **UT Invitational fall 2019, Div C, INV, Q49:** the half-adder sum is drawn as XOR (correct) but labeled "A+B". In Boolean notation + means OR. It should read A ⊕ B.

Not confirmed

- Pembroke Hill 2026 Q23 was read by an earlier research pass (file `phi26_test.txt`, line 104). I re-read the line. I did not check its key in this pass.

- MIT 2020 Q19-20 (switch pictures) and Rickards 2020 C Q8: I did not re-render the figures. Their answers come from the earlier extraction (G_tests).
- The IEC "=" label for XNOR: textbooks differ. I found no Science Olympiad test that uses IEC symbols at all.
- Whether 2027 supervisors will ask TTL voltage thresholds. No 2027 test exists yet (Test Exchange checked 2026-10-16 by an earlier pass).
- No real States or Nationals test with logic was available. The file named "nat26" is the **USC Invitational 2026**, not the 2026 Nationals. The only State test with logic is Florida State 2021.

Sources

- 2027 Div B rules, Circuit Lab, manual page B12 (`/Users/kai/Daniel/Science_Olympiad_Div_B_Rules_2027_for_Web_Secured.pdf`).
- Test extractions in `circuitlab/research/tests/` (nat26 = USC 2026; atx-reg; misc-a; misc-b; state; ut-b; ut-c) and `research/prior/G_tests.md`, `H_logic_diodes.md` (types A1-A3, A20, A49-A53), `B_topics.md` §8, `F_slides.md` §9.
- USC Invitational 2026 B test page 4, rendered and read for MC20-22.
- National supervisor practice deck 15 "Digital Logic" (soinc.org Circuit Lab B resources).
- Kuphaldt, *Lessons in Electric Circuits*, Vol. IV (Digital), chapters "Logic Gates" and "Switches" (allaboutcircuits.com textbook).
- IEEE Std 91 / IEC 60617-12 for the two symbol styles (standard names only; not re-read here).
- Claude Shannon, *A Symbolic Analysis of Relay and Switching Circuits* (MIT thesis, 1937): series = AND, parallel = OR.
- Python truth-table checks of every worked example (gate columns, associativity of all six gates, bitwise numbers, diode and LED numbers), run 2026-10-16.

Truth Tables and Boolean Expressions

Rule text (2027 Div B, page B12): 3.c.viii "Digital logic gates, truth tables, ... boolean expressions"; 3.e.iv "Logical operations as boolean expressions, including De Morgan's Law and truth tables". Hands-on 3.c.v: "Construct a logic circuit to mimic a given truth table." Hands-on 3.c.vi: "Troubleshoot a faulty circuit."

This is the **core skill** of the logic area. Nearly every logic question in past tests is one of these types.

Key facts

- A **truth table** lists every possible input pattern and the output for each one.
- A **Boolean expression** is a formula made of letters, NOT ('), AND (·) and OR (+). For example F = AB + C'.
- Always list the rows in **binary counting order**: 000, 001, 010, ... 111. Then two people's tables line up.
- In this note, A is the leftmost (biggest) bit. For 3 inputs the row number is m = 4A + 2B + C.
- A **literal** is a letter or its NOT, such as A or A'.
- A **minterm** is an AND of every input, each plain or with NOT. It is 1 on exactly one row. Row 5 (101) has minterm AB'C.
- A **maxterm** is an OR of every input. It is 0 on exactly one row. Row 5 (101) has maxterm (A' + B + C').
- **SOP** means sum of products. It is an OR of AND terms, such as AB + A'C.
- **POS** means product of sums. It is an AND of OR terms, such as (A + B)(A' + C).
- **Σm(1, 3, 5)** is shorthand for "the SOP of minterms 1, 3 and 5". **ΠM(0, 2)** is shorthand for "the POS of maxterms 0 and 2".
- A **don't care** (X or d) is a row whose output does not matter. You may pick 0 or 1 for it.

Formulas

- Rows = 2^n for n inputs.
- Row number (3 inputs) m = 4A + 2B + C. (4 inputs: m = 8A + 4B + 2C + D.)
- Σm and ΠM use opposite row lists. If F = Σm(3, 5, 6, 7) with 3 inputs, then F = ΠM(0, 1, 2, 4).
- SOP rule: rows where F = 1. Write 1 as the letter, 0 as the letter with NOT. AND them. OR all the terms.
- POS rule: rows where F = 0. Write 0 as the letter, 1 as the letter with NOT. OR them. AND all the terms.

Question types

Easy

1. Truth table → gate name — ✅ IN

- Where seen: UT Austin (ATX) Regional 2021, Div C, REG, Q19 (TT→F, TF→T, FT→T, FF→F → XOR); NY Lake Erie/Niagara Invitational 2018 (2019 season), Div C, INV, Q16 (1, 0, 0, 0 → NOR); F-M Invitational fall 2019, INV, Q7 (T, T, T, F → NAND); UF Northern Regional 2020, Div C, REG, Q20-22; MIT Invitational 2019, Div C, INV, lab S2-3b (NAND); Princeton 2019, Div C, INV, "Logic" (a)-(f) (all six gates).
- How to solve:
  1. Put the rows in the order 00, 01, 10, 11. Some tests list TT first. Re-order if needed.
  2. Read the output column. Match it: 0001 AND, 0111 OR, 1110 NAND, 1000 NOR, 0110 XOR, 1001 XNOR.
- Worked example: ATX 2021 C Q19 lists TT → F, TF → T, FT → T, FF → F. In 00, 01, 10, 11 order the column is 0, 1, 1, 0. **Answer: XOR.**
- Traps:
  - **Row order.** A table that starts with TT (11) is upside down. NAND (1110 in normal order) reads 0111 from the top. That looks like OR.
  - UF 2020 C Q21-22: the key itself swapped OR and NAND (see "Mistakes").
- Calculator: look up — type: 4-bit column and "top row = 00" or "top row = 11".

2. Gate name → fill the truth table — ✅ IN

- Where seen: F-M Invitational fall 2019, INV, Q8 (fill in XOR: F, T, T, F); national supervisor deck 15 (fill tables for all 7 gates).
- How to solve: Write the four rows. Apply the gate rule to each row.
- Worked example: XNOR. **Column 1, 0, 0, 1.**
- Traps: XNOR is not NOR. XNOR(0, 0) = 1 and XNOR(1, 1) = 1.
- Calculator: look up — type: gate name.

3. Evaluate an expression for given inputs — ✅ IN

- Where seen: not seen as a pure expression in a test. It is the inner step of almost every logic item.
- How to solve:
  1. Put in the numbers.
  2. Work parentheses and overbars first, then NOT, then AND, then OR.
- Worked example: F = A'B + (B ⊕ C)', with A = 0, B = 1, C = 1.
  - A'B = 1·1 = 1. So F = 1 + (anything) = 1.
  - **Answer: F = 1.** (Python check.)
- Traps: Once one OR term is 1, the whole OR is 1. Once one AND input is 0, the whole AND is 0. Use this to save time.
- Calculator: calculate — type: expression, values of A, B, C.

4. Evaluate a gate diagram for given inputs — ✅ IN

- Where seen: Rickards Invitational 2020 (2021 season), Div C, INV, Q8 (A = True, B = False → output?); Shock Value / Circuit Lab question marathon 2014 on scioly.org (output = NOR(OR(A, B), NAND(C, D)) for three input sets).
- How to solve:
  1. Write the input values on the input wires.
  2. Go gate by gate from left to right. Write each gate's output on its wire.
  3. The last wire is the answer.
- Worked example: Output = NOR(OR(A, B), NAND(C, D)) with A = 1, B = 0, C = 0, D = 0.
  - OR(1, 0) = 1. NAND(0, 0) = 1. NOR(1, 1) = 0.
  - **Answer: 0.** With A = B = 0 and C = D = 1 the output is 1. (Python check.)
- Traps: A wire that splits carries the same value to both places.
- Calculator: calculate — type: gate list in order + input values.

5. Find the inputs that make the output 1 — ✅ IN

- Where seen: Princeton 2020, Div C, INV, Q12 (required inputs A, B, C for X = 1 → 0, 1, 1).
- How to solve:
  1. Start at the output. Ask what the last gate needs. An AND needs all inputs 1. A NOR needs all inputs 0.
  2. Walk back gate by gate.
  3. If a gate has many choices (OR = 1), try each one.
  4. Check your answer forward.
- Worked example: F = (A + B)'·C. F = 1 needs (A + B)' = 1 and C = 1. (A + B)' = 1 needs A = 0 and B = 0. **Answer: A = 0, B = 0, C = 1** (only one row).
- Traps: Some circuits have several answers. Some have none. Read whether the question wants "all" inputs that work.
- Calculator: calculate — type: expression → list of rows where F = 1.

Medium

6. Fill in every internal wire of a gate network — ✅ IN

- Where seen: UF Northern Regional 2019, Div C, REG, Q42 (10 points; A = 1, B = 0, C = 1; boxes on every wire); UMSO (U Michigan) Invitational ~2019, INV, Q16 (D = AB, E = (AC)', F = D + E, all 8 rows).
- How to solve: Same as type 4. Write a value in every box. Check each gate twice.
- Worked example (UF Northern Regional 2019 C Q42, figure rendered and read). Inputs A = 1, B = 0, C = 1.

  - NAND(A, B) = NAND(1, 0) = 1.
  - The middle gate is an **XNOR** (back line plus bubble). XNOR(B, C) = XNOR(0, 1) = 0.
  - NOT of that = 1.
  - NOT C = 0. AND(NOT C, C) = 0. (This AND is always 0, because C·C' = 0.)
  - OR(1, 0) = 1.
  - Final AND(NAND output 1, OR output 1) = **1**.
  - **Boxes: NAND 1, NOT 1, AND 0, OR 1, final 1.** (Python check.) The key gave NOT 0, OR 0, final 0. It read the gate as XOR.
- Traps:
  - Look for the bubble on XOR vs XNOR.
  - A gate fed by C and C' is a constant. AND(C, C') = 0 and OR(C, C') = 1.
- Calculator: calculate — type: gate list with wire names + inputs (prints every wire).

7. Gate circuit → full truth table — ✅ IN

- Where seen: North Pocono Invitational 2020, Div C, INV, Q26 (AND, OR, XOR, NOR, final NOR → output C; answer 0, 1, 0, 1, so C = B); UMSO Invitational ~2019, INV, Q16; MIT Invitational 2020, Div C, INV, Lab 2 (truth table template).
- How to solve:
  1. Make a column for every input, every internal wire, and the output.
  2. Fill row by row, or column by column.
  3. Look at the finished output column. It may equal a simple thing, such as one input.
- Worked example (UMSO Q16): D = A·B, E = (A·C)', F = D + E.
  - E = 0 only when A = C = 1. Then F = D = AB, which is 0 only when B = 0.
  - **F = 0 only on row A = 1, B = 0, C = 1. F = 1 on the other 7 rows.** (Python check.)
- Traps: When the final column looks like one input (NP Q26: C = B), the circuit has redundant gates. That is the point of the question.
- Calculator: calculate — type: expression → 2^n-row table.

8. Expression → truth table — ✅ IN

- Where seen: not seen alone. It is the check step in UT Austin (ATX) Regional 2021, Div C, REG, Q24 and Florida State 2021 Q36-37.
- How to solve:
  1. Write all 2^n rows in counting order.
  2. Add a column for each part of the expression.
  3. Combine into the output column.
- Worked example: F = AB + C. **F = 1 on rows 1, 3, 5, 6, 7.** Column for rows 0-7: 0, 1, 0, 1, 0, 1, 1, 1. (Python check.)
- Traps: Row 6 (110) is AB = 1 with C = 0. Do not miss it.
- Calculator: calculate — type: expression.

9. Gate diagram → Boolean expression — ✅ IN

- Where seen: UF Northern Regional 2020, Div C, REG, Q23 (chip-labeled diagram → A·[(B·C')' + C·D]); NY Lake Erie/Niagara Invitational 2018, Div C, INV, Q29 (choose the unsimplified F = AB' + AB + C); ATX Regional 2019, Div C, REG, Q31 (→ ~(p AND (q OR ~r))); PA Central Regional ~2013, Div C, REG, Q26 (write C and D, then simplify); MIT Invitational 2020, Div C, INV, Q48; USC Invitational 2026, Div B, INV, MC21.
- How to solve:
  1. Label each gate output with a short name (G1, G2, ...).
  2. Write each gate's expression using its input names.
  3. Substitute until only inputs remain.
  4. Put parentheses around each gate's inputs. Do not lose them.
  5. Simplify only if asked.

- Worked example (USC 2026 B MC21, figure rendered): A goes through a buffer. B goes through an inverter, giving B'. Gate C = AND(A, B'). Gate D = 3-input NOR(C, B', B'). Gate E = 3-input XNOR(A, C, D).
  - D = (AB' + B')' = (B')' = B. (Absorption: AB' + B' = B'.)
  - Rows AB = 00: C = 0, D = 0, E = XNOR(0, 0, 0) = 1. Row 01: C = 0, D = 1, E = XNOR(0, 0, 1) = 0. Row 10: C = 1, D = 0, E = XNOR(1, 1, 0) = 1. Row 11: C = 0, D = 1, E = XNOR(1, 0, 1) = 1.
  - Column 1, 0, 1, 1. That is 0 only at A = 0, B = 1. So **E = A + B'**, which also equals (A'B)'.
  - Answer: choice C (A + B') and choice D ((A'B)', the keyed answer) are **both correct**. (Python check.)
- Traps:
  - Simplify every answer choice. Two choices can be the same function.
  - A bar over a whole product is not a bar on each letter.
- Calculator: calculate — type: gate list → expression and table.

10. Expression → draw the gate diagram — ✅ IN

- Where seen: Princeton 2020, Div C, INV, Q38 (draw Z = A'·B'·(C + D) two ways); 2014 scioly.org marathon (draw with one 4-input gate plus inverters).
- How to solve:
  1. Find the last operation (the "outermost" one). That is the output gate.
  2. Draw its inputs as smaller gates. Repeat.
  3. Use one inverter per NOT literal.
- Worked example: Z = A'·B'·(C + D).
  - Way 1: NOT A, NOT B, OR(C, D), then a 3-input AND.
  - Way 2: A'B' = (A + B)'. So Z = NOR(A, B) AND OR(C, D). **That is 1 NOR, 1 OR and one 2-input AND (3 gates, no inverters).**
- Traps: The second way uses De Morgan. See De Morgan and Bubble Pushing A8.
- Calculator: look up — type: expression → gate list.

11. Truth table → SOP (sum of minterms) — ✅ IN

- Where seen: MIT Invitational 2019, Div C, INV, Q48 (circuit → SOP form); also needed for hands-on 3.c.v.
- How to solve:
  1. Find every row with output 1.
  2. For each row, write a product: the plain letter if its bit is 1, the letter with NOT if 0.
  3. OR all the products.
  4. Simplify (see Boolean Algebra and Simplification A8).
- Worked example: F = 1 on rows 001, 101, 111.
  - SOP = A'B'C + AB'C + ABC.
  - Simplify: A'B'C + AB'C = B'C. AB'C + ABC = AC. **F = B'C + AC = C(A + B').** (Python check.)
- Traps: You may reuse a minterm in two groups when simplifying (A + A = A).
- Calculator: calculate — type: list of rows where F = 1.

12. Truth table → POS (product of maxterms) — ✅ IN

- Where seen: MIT Invitational 2019, Div C, INV, Q46 (Q = 1 at 001, 101, 111 → POS (A + B')C).
- How to solve:
  1. Find every row with output **0**.
  2. For each row, write a sum: the plain letter if its bit is **0**, the letter with NOT if **1**.
  3. AND all the sums.
  4. Simplify.
- Worked example (MIT 2019 C Q46): Q = 1 at 001, 101, 111. So Q = 0 at 000, 010, 011, 100, 110.
  - Canonical POS: (A + B + C)(A + B' + C)(A + B' + C')(A' + B + C)(A' + B' + C).
  - Every row with C = 0 is a zero. That gives the factor C. Row 011 is the one extra zero with C = 1. It gives (A + B').
  - **Q = (A + B')·C.** (Python check of both forms.)
- Traps: POS uses the **zeros**, and the letter rule is the **opposite** of SOP. A 0 bit gives a plain letter.

- Calculator: calculate — type: list of rows where F = 0 (or where F = 1).

13. Σm and ΠM shorthand, and converting between them — ⚠️ Borderline (the rules say "truth tables" and "boolean expressions"; this notation is not named)

- Where seen: not seen in a test.
- How to solve:
  1. Σm lists the rows with output 1.
  2. ΠM lists the rows with output 0.
  3. To convert, list every row number 0 to $2^n - 1$ that is missing.
- Worked example: The 3-input **majority** function (1 when two or more inputs are 1) is **Σm(3, 5, 6, 7) = ΠM(0, 1, 2, 4)**. Minimal forms: AB + AC + BC = (A + B)(A + C)(B + C). (Python check.)
- Traps: Keep the bit order. If C is the big bit instead of A, every number changes.
- Calculator: calculate — type: n, list of 1-rows (gives the 0-rows).

14. Circuit → SOP with a NAND-made inverter — ✅ IN

- Where seen: MIT Invitational 2019, Div C, INV, Q47 and Q48 (P = NAND(C, C) = C'; Q = (A ⊕ B)·P → AB'C' + A'BC').
- How to solve:
  1. Replace NAND(X, X) by X'.
  2. Expand XOR as A'B + AB'.
  3. Distribute the AND over the OR.
- Worked example: Q = (A ⊕ B)·C' = (AB' + A'B)·C' = **AB'C' + A'BC'**. (Python check.)
- Traps: NAND(C, C) is C', not C.
- Calculator: calculate — type: expression → SOP.

15. Word problem → truth table → expression — ✅ IN

- Where seen: national supervisor deck 15 homework ("mammal detector" from 5 traits: M = A·B·C·D); Princeton 2020, Div C, INV, Q15 (staircase light → XOR).
- How to solve:
  1. Give each condition a letter. Say what 1 means for each.
  2. Write the truth table from the story.
  3. Write SOP. Simplify.
- Worked example: "The buzzer sounds when the key is in (K = 1) and the door is open (D = 1), or when the key is in and the seat belt is NOT fastened (S = 0)."
  - F = KD + KS'. Factor out K: **F = K(D + S')**. (Python check.)
  - Other standard stories: majority vote of 3 judges = AB + AC + BC. "Exactly one of three" = A'B'C + A'BC' + AB'C'. Alarm = armed AND door open, OR panic: Y = AD + P.
- Traps:
  - A condition that never changes the answer is **redundant**. In the mammal detector, input E (breathes oxygen) does not matter.
  - "Not fastened" means S'. Watch the polarity.
- Calculator: calculate — type: letters + the 1-rows.

Hard

16. Which of several expressions match a 3-input table (multi-select) — ✅ IN

- Where seen: UT Austin (ATX) Regional 2021, Div C, REG, Q24 (A = 1 for PQR = 101, 011, 001; which of I, II, III match?).
- How to solve:
  1. Simplify each expression, or build its truth table.
  2. Compare with the given table row by row.
  3. Mark every match. "I and III" type answers are common.

- Worked example (ATX 2021 C Q24): The table is 1 on rows 001, 011, 101. I = (¬(PQ) + P¬Q)(R + PQ). III = R·¬(PQ).
  - III: R = 1 and not (P = Q = 1). That is rows 001, 011, 101. It matches.
  - I: P¬Q is already inside ¬(PQ), so the first factor = ¬(PQ). Then ¬(PQ)·(R + PQ) = ¬(PQ)·R + 0 = R·¬(PQ). It matches.
  - **Answer: I and III (key D).** (Python check, all 8 rows.)
- Traps: On the printed test, II looks exactly like I. The overbars were probably lost when the PDF was made. As printed, "I, II and III" would also be true (see "Mistakes").
- Calculator: calculate — type: target 1-rows + each expression (reports match or no match).

17. Truth table of a non-gate operator → equivalent expression — ✅ IN

- Where seen: UT Invitational 2020 (2021 season), Div C, INV, C38 (implication p ⇒ q with table T, F, T, T; choices (p + q) NAND (pq), p + q', p' + q'; answer "none of the above").
- How to solve:
  1. Re-order the table to 00, 01, 10, 11 if needed.
  2. Write SOP or POS from the table.
  3. Simplify every answer choice. Compare.
- Worked example: "p implies q" is false only when p = 1 and q = 0. So the one maxterm is (p' + q). **p ⇒ q = p' + q.**
  - (p + q) NAND (pq) = ((p + q)pq)' = (pq)' = p' + q'. Not a match.
  - p + q' is false at p = 0, q = 1. Not a match.
  - **Answer: none of the above.** (Python check.)
- Traps: p + q' is the reverse implication (q ⇒ p). It is easy to pick by mistake.
- Calculator: calculate — type: table column + choice expressions.

18. Are two circuits (or expressions) the same? — ✅ IN

- Where seen: USC Invitational 2026, Div B, INV, MC21 (two choices are the same function); UT Invitational 2020, Div C, INV, C63 (multi-select of equal forms).
- How to solve:
  1. Build both truth tables.
  2. If every row matches, they are equal.
  3. If one row differs, that row is the **counterexample** (the input that proves they differ).
- Worked example: Is A ⊕ B the same as (A + B)(AB)'?
  - Rows 00: 0 and 0. 01: 1 and 1. 10: 1 and 1. 11: 0 and 0.
  - **Answer: yes, equal.** (Python check.)
- Traps: Testing only two or three rows proves nothing. Check all 2^n rows.
- Calculator: calculate — type: expression 1, expression 2.

19. Stuck-at fault → new expression — ✅ IN (hands-on 3.c.vi "troubleshoot"; logic 3.c.viii)

- Where seen: USC Invitational 2026, Div B, INV, MC22 (one input of AND gate C is "permanently grounded").
- How to solve:
  1. A **stuck-at-0** wire always reads 0. A **stuck-at-1** wire always reads 1.
  2. Put that constant into the expression.
  3. Simplify again.
- Worked example (USC 2026 B MC22, same circuit as type 9): Ground one input of AND gate C. So C = 0 always.
  - D = NOR(0, B', B') = (B')' = B.
  - E = XNOR(A, 0, B) = XNOR(A, B), because a 0 input does not change parity.
  - **E = (AB' + A'B)', choice C** (matches the key). (Python check.)
- Traps: Choice D (B') is a distractor. Re-check with a full table.
- Calculator: calculate — type: expression, wire name, stuck value.

20. Troubleshoot a logic circuit from a wrong truth table — ✅ IN (hands-on 3.c.vi)

- Where seen: not seen in a written test. The 2027 hands-on list names "Troubleshoot a faulty circuit".
- How to solve:

1. Compute what every internal wire should be on a failing row.
2. Measure (or reason) what each wire really is.
3. The first wire (from the inputs) that is wrong is the fault.
4. Try "stuck-at" guesses. A wire stuck at 0 or 1 makes the circuit act as if that wire were constant.

- Worked example: F should be AB + C. The measured output is F = C on all 8 rows. So the AB term never reaches the OR. **Fault: the AND output is stuck at 0** (or the AND chip has no power). (Python check: AB + C with AB = 0 gives C.)
- Traps: Two different faults can give the same wrong table. In F = (A + B)·C, "OR output stuck at 1" also gives F = C. Measure the inside wires to tell them apart.
- Calculator: calculate — type: expression + observed column (lists stuck-at faults that match).

21. Hands-on: build a circuit that matches a given truth table — ✅ IN (hands-on 3.c.v)

- Where seen: MIT Invitational 2020, Div C, INV, Lab 2 "Circuit 2" (build a 3-input NOR to match a table); USC Invitational 2026, Div B, INV, lab parts c-d (push buttons); Princeton 2020, Div C, INV, Lab 2.
- How to solve:
  1. Write SOP from the table. Simplify it.
  2. Rewrite it for the chips you have (NAND-only if you only have 7400 chips).
  3. Count gates. A 7400 chip has 4 two-input NAND gates.
  4. Power the chip: on a 14-pin 74xx chip, pin 14 = +5 V and pin 7 = ground.
  5. Use a pull-down (or pull-up) resistor on each switch input. Never leave an input floating.
  6. Put an LED with a resistor on the output.
  7. Test all 2^n rows before you call the supervisor.
- Worked example: The table is 1 only on rows 01 and 10. You have one 7400 chip. This is XOR. Build it from 4 NANDs: N1 = NAND(A, B); out = NAND(NAND(A, N1), NAND(B, N1)). **One 7400 chip (all 4 gates).** LED resistor for 5 V, 2.0 V LED, 10 mA: **R = (5 – 2) / 0.010 = 300 Ω.** (Python checks.)
- Traps:
  - The 7402 (NOR) chip has a different pinout from the 7400. Ask for the datasheet.
  - You may ask the supervisor how the breadboard is wired inside (rule Part II.b).
- Calculator: calculate — type: 1-rows + available gate type (gives a gate list).

Nationals

22. Design from a word spec with many inputs (and don't cares) — ✅ IN (combinational logic; no memory or clock)

- Where seen: MIT Invitational 2021, Div C, INV, Scenario F Q57-63 (output = index of the last 1 in a 2^n-bit input; n = 3, input 0111 0010 → 001; write expressions with AND/OR/NOT).
- How to solve:
  1. Decide the bit numbering. Write it down.
  2. Build the truth table. Mark impossible or "no answer" rows as don't care.
  3. Write each output bit as its own expression.
  4. Use don't cares to make the expressions smaller.
- Worked example: 4-bit input i3 i2 i1 i0. Output = the position of the lowest-order 1. Input 0000 is a don't care.
  - out1 = 1 when the lowest 1 is at position 2 or 3. That means i0 = 0 and i1 = 0. **out1 = i0'·i1'.**
  - out0 = 1 when the lowest 1 is at position 1 or 3. Position 1: i0 = 0, i1 = 1. Position 3: i0 = i1 = i2 = 0. Using the don't care, **out0 = i0'·(i1 + i2').**
  - Check: input 0010 → lowest 1 at position 1 → out = 01. Input 1000 → position 3 → out = 11. (Python checked all 15 rows.)
- Traps:
  - "Last 1" can mean leftmost or rightmost. The MIT key used the rightmost (lowest) bit: 0111 0010 → index 1.
  - This is called a **priority encoder**. It is combinational, so it is in scope.
- Calculator: calculate — type: output rule per row (or 1-rows and don't-care rows per output bit).

23. Count the rows that meet a condition — ✅ IN

- Where seen: not seen in a test.
- How to solve:
  1. Count with groups. "At least 3 of 4" = exactly 3 or exactly 4.
  2. Exactly k of n = the number of ways to choose k items from n.
- Worked example: A 4-input function is 1 when at least 3 inputs are 1. Exactly 3: 4 rows. Exactly 4: 1 row. **Answer: 5 rows.** (Python check.)
- Traps: Do not forget the all-ones row.
- Calculator: calculate — type: n, rule.

Out of scope (skip)

- ❌ Truth tables for latches and flip-flops (Princeton 2020 Q39 D flip-flop; USC 2026 B FRQ 36b cross-coupled NOR trace). Rule 3.d: "sequential logic and digital clock signals".
- ❌ "Is the circuit stateless?" (USC 2026 B FRQ 36c) is about memory, so it is out. But know one fact: **a circuit with feedback (an output wired back to an input) is not combinational.**
- ❌ Timing diagrams and propagation delay. Rule 3.d (clock signals).
- ❌ Probability of a switch network working (Florida State 2021 Q33). It is outside the math limits in rule 3.e.

Mistakes in tests and keys

- **UF Northern Regional 2019, Div C, REG, Q42:** the key gives NOT 0, OR 0, final 0. The figure shows an **XNOR** (I rendered it). The right boxes are NOT 1, OR 1, final 1. The key treated the gate as XOR.
- **UF Northern Regional 2020, Div C, REG, Q21-22:** the key swaps OR and NAND.
- **USC Invitational 2026, Div B (and Div C), INV, MC21:** two choices are correct. E = A + B' (choice C) = (A'B)' (choice D, keyed).
- **UT Austin (ATX) Regional 2021, Div C, REG, Q24:** expression II prints identically to I. So as printed, all three match (choice E). The key (D, "I and III") assumes II had different bars.

Not confirmed

- Rickards 2020 C Q8, Princeton 2020 Q12 and Q38, North Pocono 2020 Q26, UMSO Q16 and UF 2020 C Q23 depend on figures. I used the earlier extractions. I did not re-render those figures in this pass (I did render USC 2026 B MC21-22 and UF 2019 C Q42).
- UMSO's division is not printed. It is probably Div C.
- The 2014 marathon item is a forum post, not a real test. Its posted answers are partly unclear.
- No real States or Nationals logic test was available. The "nat26" file is the USC Invitational 2026.

Sources

- 2027 Div B rules, page B12 (rules 3.c.viii, 3.e.iv, hands-on c.v and c.vi).
- Extractions: `research/tests/atx-reg.md`, `misc-a.md`, `misc-b.md`, `nat26.md` (USC 2026), `state.md`, `ut-b.md`, `ut-c.md`; `research/prior/G_tests.md` (Digital logic section), `H_logic_diodes.md` (A5-A8, A19, A25, A40, A48-A50), `B_topics.md` §8f.
- Rendered figures: USC 2026 B test page 4 (MC20-22); UF Northern Regional 2019 C test page 10 (Q42).
- Kuphaldt, *Lessons in Electric Circuits* Vol. IV, "Boolean Algebra" and "Karnaugh Mapping" chapters; All About Circuits worksheets "Boolean Algebra" and "Sum-of-Products and Product-of-Sums".
- Python truth-table checks of every worked example (run 2026-10-16).

Boolean Algebra and Simplification

Rule text (2027 Div B, page B12): 3.c.viii "boolean expressions"; 3.e.iv "Logical operations as boolean expressions, including De Morgan's Law and truth tables". The rules do **not** name Boolean laws, simplification, or Karnaugh maps. But "boolean expressions" covers simple laws. Every Div C test with logic asked for simplification. So learn the laws fully. Learn K-maps as a fast checking tool.

History link. George Boole (1815-1864) is on the 2027 history list. He wrote *The Mathematical Analysis of Logic* (1847) and *An Investigation of the Laws of Thought* (1854). These books created Boolean algebra. Claude Shannon applied it to switch circuits in 1937.

Key facts

- **Boolean algebra** is algebra where every value is 0 or 1. It has three basic operations: NOT ('), AND (·) and OR (+).
- To **simplify** means to find a shorter expression with the same truth table. Shorter means fewer letters and fewer gates.
- A simplified answer must still match the original on **every** row. Always check with a truth table.
- Some laws look like normal algebra. Some do not. **A + A = A**, **A + 1 = 1**, and **A + BC = (A + B)(A + C)** are not true for normal numbers.
- The **dual** of an expression swaps AND with OR, and 0 with 1. Letters stay the same. Every law has a dual that is also a law.

The laws (each row shows the OR form and its dual AND form)

Law name	OR form	AND form (dual)
Identity	A + 0 = A	A · 1 = A
Null (dominance, annulment)	A + 1 = 1	A · 0 = 0
Idempotent	A + A = A	A · A = A
Complement (inverse)	A + A' = 1	A · A' = 0
Double negation (involution)	(A')' = A	(same)
Commutative	A + B = B + A	AB = BA
Associative	(A + B) + C = A + (B + C)	(AB)C = A(BC)
Distributive	A(B + C) = AB + AC	**A + BC = (A + B)(A + C)**
Absorption	A + AB = A	A(A + B) = A
Redundancy ("disappearing opposite")	A + A'B = A + B	A(A' + B) = AB
Combining (adjacency)	AB + AB' = A	(A + B)(A + B') = A
De Morgan	(A + B)' = A'B'	(AB)' = A' + B'
Consensus ⚠	AB + A'C + BC = AB + A'C	(A + B)(A' + C)(B + C) = (A + B)(A' + C)

XOR laws

- A ⊕ 0 = A. A ⊕ 1 = A'. A ⊕ A = 0. A ⊕ A' = 1.
- XOR is commutative and associative. You can drop any pair of equal letters.
- A ⊕ B = A'B + AB'. (A ⊕ B)' = AB + A'B' = A ⊕ B' = A' ⊕ B.

Order to try the laws

1. Remove double NOTs.
2. Use De Morgan to push NOTs down to single letters.
3. Multiply out (distribute) to get SOP.
4. Remove repeats (idempotent) and zero terms (A·A' = 0).
5. Look for absorption (A + AB = A).
6. Look for combining pairs (AB + AB' = A).
7. Look for redundancy (A + A'B = A + B).
8. Check the answer with a truth table.

Karnaugh map (K-map) basics ⚠️

- A **K-map** is a truth table drawn as a grid. Neighboring squares differ in exactly one input.
- The rows and columns use **Gray code order: 00, 01, 11, 10.** (Not 00, 01, 10, 11.)
- 2 inputs: 2 × 2 grid. 3 inputs: 2 × 4 grid (A down the side, BC across the top). 4 inputs: 4 × 4 grid (AB down the side, CD across the top).
- The map **wraps around**. The left edge touches the right edge. The top edge touches the bottom edge. The four corners all touch.
- Circle groups of 1s. Each group must be 1, 2, 4, 8 or 16 squares, in a rectangle.
- Make each group as big as possible. Use as few groups as possible.
- Each group gives one product term: keep only the letters that stay the same over the whole group.

Formulas

- A group of 2^k squares in an n-input K-map gives a term with n – k letters.
- Number of 2-input gates in an SOP (rough count): one AND per product with 2+ letters, plus ORs to join them, plus inverters for NOT letters.
- XOR as SOP: A ⊕ B = A'B + AB'. XNOR as SOP: AB + A'B'.

Question types

Easy

1. Name the law (or pick the true identity) — ✅ IN

- Where seen: NY Lake Erie/Niagara Invitational 2018 (2019 season), Div C, INV, Q28 (which rule is true → A + 1 = 1); Princeton 2021, Div C, INV, Q19 (pick the De Morgan statement → (A + B)' = A'B').
- How to solve:
  1. Test the identity with A = 0 and A = 1 (and the other letters).
  2. Or match it to the law table.
- Worked example: "A + 1 = 1" is the **null (dominance) law**. "A(A + B) = A" is **absorption**. "A + BC = (A + B)(A + C)" is the **second distributive law**.
- Traps: A + 1 = 1, not A + 1 = A. In Boolean algebra 1 + 1 = 1, not 2.
- Calculator: look up — type: identity → law name (or law name → identity).

2. One-step simplification — ✅ IN

- Where seen: F-M Invitational fall 2019, INV, Q9 (simplify A'·B + A·B → B).
- How to solve: Factor out the shared part. Then use A + A' = 1.
- Worked example: A'B + AB = (A' + A)B = 1·B = **B**. (Python check.)
- Traps: Do not "cancel" A' with A. They combine to 1, and 1·B = B.
- Calculator: calculate — type: expression.

3. Absorption and complement — ✅ IN

- Where seen: North Pocono Invitational 2020, Div C, INV, Q33 (simplify AA' + AB + A(A + B) → A).
- How to solve: Use A·A' = 0, then A(A + B) = A, then A + AB = A.
- Worked example: AA' + AB + A(A + B) = 0 + AB + A = **A** (absorption). (Python check.)
- Traps: A(A + B) = AA + AB = A + AB = A. It is not AB.
- Calculator: calculate — type: expression.

4. Use the laws to check an answer choice — ✅ IN

- Where seen: NY Lake Erie/Niagara Invitational 2018, Div C, INV, Q29 (choose the unsimplified F = AB' + AB + C; it simplifies to A + C).
- How to solve: Simplify each choice. Or test one or two rows to rule choices out fast. Then confirm with the full table.
- Worked example: AB' + AB + C = A(B' + B) + C = **A + C**. (Python check: combining law.)
- Traps: The question may want the **unsimplified** form. Read the word "unsimplified".

- Calculator: calculate — type: expression.

Medium

5. Multi-step simplification (SOP answer) — ✅ IN

- Where seen: Rickards Invitational 2020 (2021 season), Div C, INV, Q33 ((A' + B)·B + (C·B')·(A + C) → B + C); MIT Invitational 2020, Div C, INV, Q23 (X(Y + Z)(X + Y + Z) → XY + XZ).
- How to solve: Use the law order above. Multiply out first. Then absorb.
- Worked example 1 (Rickards 2020 C Q33):
  - (A' + B)B = A'B + BB = A'B + B = B (absorption).
  - (CB')(A + C) = AB'C + B'C = B'C (absorption).
  - B + B'C = **B + C** (redundancy).
  - (Python check: matches on all 8 rows. Key: B + C.)
- Worked example 2 (MIT 2020 C Q23):
  - X(X + Y + Z) = X (absorption).
  - So X(Y + Z)(X + Y + Z) = X(Y + Z) = **XY + XZ**. (Python check; key XY + XZ.)
- Traps: Write each step on its own line. One lost prime ruins the answer.
- Calculator: calculate — type: expression (gives minimal SOP).

6. Redundancy law (A + A'B = A + B) — ✅ IN

- Where seen: inside Rickards 2020 C Q33 (B + B'C = B + C). Not seen alone.
- How to solve: A + A'B = (A + A')(A + B) = 1·(A + B) = A + B. This uses the second distributive law.
- Worked example: X + X'YZ = **X + YZ**. The dual: A(A' + B) = **AB**. (Python check of A + A'B = A + B and A(A' + B) = AB.)
- Traps: The NOT letter must be the same letter as the lone term. A + B'C does not simplify this way.
- Calculator: calculate — type: expression.

7. The second distributive law — ✅ IN

- Where seen: not seen alone in a test. Florida State 2021 Q36 and Q37 need it.
- How to solve: A + BC = (A + B)(A + C). Read it in either direction.
- Worked example: (A + B)(A + C) = AA + AC + AB + BC = A + AC + AB + BC = **A + BC** (absorption). (Python check.)
- Traps: Normal algebra has no such rule. Do not "check" it with numbers like 2 and 3.
- Calculator: calculate — type: expression.

8. Dual of an expression — ✅ IN

- Where seen: not seen in a test.
- How to solve: Swap · and +. Swap 0 and 1. Keep every letter and every prime. Add parentheses to keep the grouping.
- Worked example: The dual of A + A'B = A + B is **A(A' + B) = AB**. The dual of F = A(B + C') is **A + BC'**.
- Traps: The dual is NOT the complement. The complement also flips every letter (see De Morgan and Bubble Pushing A8).
- Calculator: calculate — type: expression → dual.

9. Prove an identity (algebra or truth table) — ✅ IN

- Where seen: Shock Value / Circuit Lab question marathon 2014 on scioly.org ("show that its Boolean representation is equivalent to ..."). Princeton 2020, Div C, INV, Q37 (state De Morgan's theorem).
- How to solve:
  1. Way 1: simplify the left side until it equals the right side. Name each law.
  2. Way 2: build both truth tables. They must match on every row.
- Worked example: Prove (A + B)(A' + C) = AC + A'B.
  - Multiply out: AA' + AC + A'B + BC = 0 + AC + A'B + BC.
  - BC is the consensus term of AC and A'B. So it can be dropped.
  - **= AC + A'B.** (Python check, all 8 rows.)

- Traps: A truth table proof must show all 2^n rows.
- Calculator: calculate — type: left expression, right expression.

10. XOR simplification — ✅ IN

- Where seen: MIT Invitational 2019, Div C, INV, Q48 (expand A ⊕ B in SOP). Not seen as pure XOR algebra.
- How to solve: Re-order the XOR terms. Cancel pairs (A ⊕ A = 0). Then use A ⊕ 0 = A and A ⊕ 1 = A'.
- Worked example: A ⊕ B ⊕ 1 ⊕ A = (A ⊕ A) ⊕ B ⊕ 1 = 0 ⊕ B ⊕ 1 = **B'**. (Python check.)
- Traps: XOR does not distribute over OR. A ⊕ (B + C) is not (A ⊕ B) + (A ⊕ C).
- Calculator: calculate — type: expression with ⊕.

11. Fewest gates (simplify first) — ✅ IN

- Where seen: MIT Invitational 2020, Div C, INV, Q47 (minimum number of NAND gates for A + AB' + AB'C → 0).
- How to solve:
  1. Simplify fully first.
  2. Then count the gates for the simple form.
- Worked example (MIT 2020 C Q47): A + AB' + AB'C. Absorption: A + (anything with A) = A. So F = A. **Answer: 0 gates.** The output is just the wire A. (Python check.)
- Traps: The question is a trick. Students who build the original expression waste many gates.
- Calculator: calculate — type: expression → minimal form and gate count.

12. Combining many minterms — ✅ IN

- Where seen: not seen in a test in this form.
- How to solve: Group minterms that differ in one letter. Each pair loses that letter.
- Worked example: A'B'C' + A'BC' + AB'C' + ABC'. All four have C'. The A and B parts cover all four patterns. **F = C'.** (Python check.)
- Traps: Use each minterm as many times as you need (A + A = A).
- Calculator: calculate — type: minterm list.

Hard

13. POS → minimal SOP — ✅ IN

- Where seen: Florida State 2021, Div B (= C), STATE, Q36 (simplify (x + y + z)(x + y' + z)(x + y' + z') → x + y'z).
- How to solve:
  1. Combine two sums that differ in one letter: (X + y)(X + y') = X.
  2. Then use the second distributive law.
- Worked example (FL State 2021 Q36):
  - (x + y + z)(x + y' + z) = (x + z) + yy' = x + z.
  - Now (x + z)(x + y' + z') = x + z(y' + z') (second distributive law, with x pulled out).
  - z(y' + z') = y'z + zz' = y'z.
  - **Answer: x + y'z.** (Python check; matches the key.)
- Traps: Do not multiply all three sums out at once. It makes 27 terms.
- Calculator: calculate — type: POS expression → minimal SOP.

14. SOP → POS — ✅ IN

- Where seen: Florida State 2021, Div B, STATE, Q37 (convert b + a'c to POS → (b + c)(a' + b)); MIT Invitational 2020, Div C, INV, Q24 (X'Z' + XY'Z + XYZ' → simplified POS).
- How to solve:
  1. Way 1: use A + BC = (A + B)(A + C).
  2. Way 2: find the rows where F = 0. Write POS from those rows. Simplify (or use a K-map on the 0s).
- Worked example 1 (FL State 2021 Q37): b + a'c = (b + a')(b + c) by the second distributive law. **Answer: (a' + b)(b + c).** (Python check; matches the key.)
- Worked example 2 (MIT 2020 C Q24): F = X'Z' + XY'Z + XYZ'.
  - F = 1 on rows 0 (000), 2 (010), 5 (101), 6 (110). So F = 0 on rows **1, 3, 4, 7**.
  - Rows 1 (001) and 3 (011) share X = 0, Z = 1. Sum: (X + Z').

  - Rows 3 (011) and 7 (111) share Y = 1, Z = 1. Sum: (Y' + Z').
  - Row 4 (100) has no zero neighbor. Sum: (X' + Y + Z).
  - **Answer: F = (X + Z')(Y' + Z')(X' + Y + Z).** (Python check, all 8 rows.)
  - **The MIT key says (Y' + Z')(X + Y')(X + Z'). That is wrong.** It is 1 on rows 0, 4, 5, 6, not 0, 2, 5, 6. Row 010 shows it: F = X'Z' = 1 there, but (X + Y') = 0 there. I rendered the key page and confirmed the printed answer.
- Traps: POS from the zeros uses the opposite letter rule. A 0 bit gives the plain letter.
- Calculator: calculate — type: SOP expression → minimal POS.

15. Consensus theorem — ⚠ Borderline (a standard law, but beyond "boolean expressions" basics)

- Where seen: not seen in a test.
- How to solve: In AB + A'C + BC, the term BC is the **consensus** of AB and A'C. It is made by dropping the A/A' pair. It is redundant. Remove it.
- Worked example: XY + X'Z + YZ = **XY + X'Z**. Dual: (X + Y)(X' + Z)(Y + Z) = **(X + Y)(X' + Z)**. (Python check, both forms.)
- Traps: Only drop the consensus term. Both other terms must stay.
- Calculator: calculate — type: expression.

16. 3-variable K-map → minimal SOP — ⚠ Borderline (K-maps are not named in the rules)

- Where seen: not seen in a test by name. Earlier research lists MIT 2020 C Q24 as a K-map problem.
- How to solve:
  1. Draw a 2 × 4 grid. A on the side (0, 1). BC on top in Gray order: 00, 01, 11, 10.
  2. Put 1s in the grid from the minterm list.
  3. Circle the biggest groups (8, 4, 2, 1). Groups can wrap from the left edge to the right edge.
  4. Write one product per group.
- Worked example: F = Σm(1, 3, 5, 6, 7).
  - The column BC = 01 and the column BC = 11 are full. That 4-square group is "C = 1": term **C**.
  - Square 6 (110) is left. Pair it with square 7 (111): A = 1, B = 1: term **AB**.
  - **F = C + AB.** (Python check.)
- Worked example 2: F = Σm(0, 1, 2, 3, 4, 6). Group the whole top row (A = 0): A'. Group the 4 squares with C = 0 (wrapping left-right): C'. **F = A' + C'.** (Python check.)
- Traps: The top order is 00, 01, 11, 10. Writing 10 before 11 breaks the map.
- Calculator: calculate — type: n = 3, minterm list.

17. 4-variable K-map (corners and wrap-around) — ⚠ Borderline

- Where seen: not seen in a test.
- How to solve:
  1. Draw a 4 × 4 grid. AB on the side and CD on top, both in Gray order.
  2. Remember: the four corners make one group of 4.
  3. Find **essential** groups first. These are the only groups that cover some square.
  4. Then cover the rest with the fewest extra groups.
- Worked example 1: F = Σm(0, 2, 5, 7, 8, 10, 13, 15).
  - The four corners (0, 2, 8, 10) are a group: B'D'.
  - The centre four (5, 7, 13, 15) are a group: BD.
  - **F = B'D' + BD = XNOR(B, D).** (Python check.)
- Worked example 2: F = Σm(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14).
  - The two left columns (CD = 00 and 01) are full. That is C' (8 squares).
  - Squares 2 and 6 (A = 0, CD = 10) pair with 0 and 4 by wrap. The group 0, 2, 4, 6 gives A'D'.
  - Square 14 joins 12, 4, 6 (B = 1, D = 0): BD'.
  - **F = C' + A'D' + BD'.** (Python check.)
- Traps: A group must be a rectangle of 1, 2, 4, 8 or 16. An L-shape of 3 is not allowed.
- Calculator: calculate — type: n = 4, minterm list.

18. K-map with don't cares — ⚠ Borderline (don't cares are standard, but not named)

- Where seen: not seen in a test. MIT Invitational 2021, Div C, INV, Scenario F has an all-zeros input where the output does not matter.
- How to solve:
  1. Put X in each don't-care square.
  2. Use an X as a 1 only if it makes a group bigger.
  3. Never circle a group of only X squares.
- Worked example: F = Σm(1, 3, 7, 11, 15) + d(0, 2, 5).
  - Squares 3, 7, 11, 15 (C = D = 1) make a group: CD.
  - Square 1 still needs cover. Group 0, 1, 2, 3 (using d0 and d2) gives A'B'. Group 1, 3, 5, 7 (using d5) gives A'D.
  - **F = CD + A'B' or F = CD + A'D.** Both are minimal. Both are correct. (Python check of both against the on-set and don't-care set.)
- Traps: A key may print only one answer. Both forms earn credit if they match every "must" row.
- Calculator: calculate — type: n, minterm list, don't-care list.

19. Minimal POS from a K-map — ⚠️ Borderline

- Where seen: MIT Invitational 2020, Div C, INV, Q24 (simplified POS; see type 14).
- How to solve:
  1. Circle groups of **0s**.
  2. Each group gives a sum. A letter that is 0 over the group appears plain. A letter that is 1 appears with NOT.
  3. AND the sums.
- Worked example: Majority function, zeros at 0, 1, 2, 4. Groups {0, 1} → (A + B). {0, 2} → (A + C). {0, 4} → (B + C). **F = (A + B)(A + C)(B + C).** (Python check.)
- Traps: Square 0 is used three times. That is allowed.
- Calculator: calculate — type: n, zero-row list.

Nationals

20. Multi-select: which forms are all equal? — ✅ IN

- Where seen: UT Invitational 2020 (2021 season), Div C, INV, C63 (tiebreaker 4: which equal NOT[(AB)·NOT(A + B)]? → 1 and B' + B); USC Invitational 2026, Div B, INV, MC21 (two equal choices); Florida State 2021, Div B, STATE, Q29 (select all).
- How to solve:
  1. Simplify the given expression first.
  2. Simplify every choice.
  3. Mark every choice that ends in the same form. There may be 0, 1 or several.
- Worked example (UT 2020 C63): (AB)·(A + B)' = AB·A'B' = (AA')(BB') = 0. So NOT[0] = **1**. Every choice that equals 1 is right, including B' + B. (Python check: 1 on all 4 rows.)
- Traps: A choice like "B' + B" looks odd. It equals 1. Do not skip it.
- Calculator: calculate — type: target expression + each choice.

21. Simplify, then count gates for a given gate set — ✅ IN (links to universal gates)

- Where seen: MIT Invitational 2020, Div C, INV, Q47 (NAND count); see the universal-gates material in CL 8 - Digital Logic [T].
- How to solve:
  1. Simplify fully.
  2. Write the result as NAND-NAND (for SOP) or NOR-NOR (for POS). See De Morgan and Bubble Pushing [A8].
  3. Count gates. A single letter with NOT needs one NAND with tied inputs.
- Worked example: Minimum 2-input NAND gates (true inputs only): **AND 2, OR 3, NOR 4, XOR 4, XNOR 5.** (Python breadth-first search over all NAND circuits.)
- Traps: "Minimum" often means "after simplifying". Always simplify first.
- Calculator: look up — type: target gate + gate set.

Out of scope (skip)

- ❌ Algebra of more than 2 simultaneous equations does not apply here. Boolean simplification is allowed by 3.e.iv.
- ❌ Quine-McCluskey tables and K-maps with 5 or more inputs. These are not banned by name. But they go far past "boolean expressions" and no test has used them.
- ❌ Hazards and glitches (timing problems in simplified circuits). Rule 3.d: "digital clock signals"; they are about timing.
- ❌ Boolean "logic puzzles" with words (UT Invitational fall 2019 Div C tiebreaker on salaries). These are reasoning puzzles, not gate logic. ⚠️ Borderline at most.

Mistakes in tests and keys

- **MIT Invitational 2020, Div C, INV, Q24 (new finding):** the key answer (Y' + Z')(X + Y')(X + Z') does not equal X'Z' + XY'Z + XYZ'. The correct minimal POS is **(X + Z')(Y' + Z')(X' + Y + Z)**. Checked in Python. The key page was rendered and read. Earlier research (G_tests) copied the key answer as correct.
- **USC Invitational 2026, Div B, INV, MC21:** two equal choices (A + B' and (A'B)').
- **National supervisor deck 15:** it marks the Boolean-laws slide "Not allowed per rule 3.e". That note comes from an older rule set. The 2027 rule 3.e.iv allows Boolean expressions.

Not confirmed

- K-maps and don't cares: the rules do not name them. No Div B test exists yet to show whether supervisors use them.
- The 2014 marathon "show equivalent" problem: its original expression was an image. I could not see it in the saved text.
- Princeton 2021 Q19 and NY 2018 Q28 come from earlier extractions. I did not re-open those files in this pass.

Sources

- 2027 Div B rules, page B12 (3.c.viii, 3.e.iv).
- Extractions: `research/tests/state.md` (FL State 2021 Q36-37), `misc-a.md` (F-M 2019 Q9, NP 2020 Q33), `misc-b.md` (NY 2018 Q28-29), `ut-b.md` (UT 2020 C63), `nat26.md` (USC 2026); `research/prior/G_tests.md`; `H_logic_diodes.md` (A12-A18, A22, A24, A26); `B_topics.md` §8c.
- MIT 2020 test and key page 4 (rendered for Q23-24).
- Kuphaldt, *Lessons in Electric Circuits* Vol. IV, chapters 7 (Boolean Algebra) and 8 (Karnaugh Mapping).
- OpenStax does not cover digital logic. I used Kuphaldt and All About Circuits for the textbook-only types.
- Boole biography facts: from `research/subtopics/boole.md` and `prior/I_bank.md`.
- Python checks of every worked example (truth tables; sympy SOPform and POSform; NAND breadth-first search), run 2026-10-16.

De Morgan and Bubble Pushing

Rule text (2027 Div B, page B12): De Morgan's Law is named **twice**: in 3.c.viii ("boolean expressions, De Morgan's Law") and in 3.e.iv ("including De Morgan's Law and truth tables"). Expect at least one De Morgan question on every 2027 test. "Universal gate sets" (3.c.viii) also depend on De Morgan.

Key facts

- **De Morgan's laws** tell you how to take NOT of an AND or an OR.
- **Law 1:** (A · B)' = A' + B'. "NOT (A AND B)" = "(NOT A) OR (NOT B)". So **NAND = OR with inverted inputs.**
- **Law 2:** (A + B)' = A' · B'. "NOT (A OR B)" = "(NOT A) AND (NOT B)". So **NOR = AND with inverted inputs.**
- Memory aid: **"Break the bar, change the sign."** Split the long bar into short bars, and swap · with +.
- The laws work for any number of letters: (ABC)' = A' + B' + C', and (A + B + C)' = A'B'C'.
- The laws work on whole groups too: (XY + Z)' = (XY)'·Z' = (X' + Y')Z'.
- **Augustus De Morgan** (1806-1871) was an English mathematician and a friend of Boole. He stated these laws in *Formal Logic* (1847). He is **not** on the 2027 history list. Boole is.
- In set language: the complement of (X ∪ Y) is (X^c ∩ Y^c). The complement of (X ∩ Y) is (X^c ∪ Y^c).

Bubble pushing (De Morgan with pictures)

- A **bubble** is the small circle that means NOT.
- **Bubble pushing** means moving bubbles through a gate with De Morgan's law.
- The rule: **to push bubbles through a gate, change AND to OR (or OR to AND), and flip the bubble on every input and on the output.**
- Each gate has two drawings that do the same thing:

Gate	Normal drawing	De Morgan equivalent drawing
NAND	AND with output bubble	OR with bubbles on both inputs ("negative-OR")
NOR	OR with output bubble	AND with bubbles on both inputs ("negative-AND")
AND	AND	NOR with bubbles on both inputs
OR	OR	NAND with bubbles on both inputs
NOT	triangle, bubble at output	triangle, bubble at input

- **Two bubbles on one wire cancel** (double NOT: A'' = A). This is the key step in multi-level circuits.
- **XOR is special.** A bubble on one XOR input moves to the output: A' ⊕ B = (A ⊕ B)'. Bubbles on both inputs cancel: A' ⊕ B' = A ⊕ B. XOR does NOT change shape to AND or OR.

Complement vs dual

- The **complement** F' of F is the function that is 1 exactly where F is 0. Use De Morgan to write it.
- Short method for the complement: swap · and +, swap 0 and 1, **and** put NOT on every letter. Keep the grouping.
- The **dual** swaps · and +, and 0 and 1, but does **not** touch the letters. Do not mix them up.

Formulas

- (A · B · C · ...)' = A' + B' + C' + ...
- (A + B + C + ...)' = A' · B' · C' · ...
- NAND(A, B) = A' + B'. NOR(A, B) = A'B'.
- SOP → NAND-NAND: AB + CD = NAND(NAND(A, B), NAND(C, D)).
- POS → NOR-NOR: (A + B)(C + D) = NOR(NOR(A, B), NOR(C, D)).
- XOR bubbles: A' ⊕ B = A ⊕ B' = (A ⊕ B)'. A' ⊕ B' = A ⊕ B.

Question types

Easy

1. State De Morgan's laws — ✅ IN

- Where seen: Princeton 2021, Div C, INV, Q19 (pick the true statement → (A + B)' = A'B'); Princeton 2020, Div C, INV, Q37 (state De Morgan's theorem and draw the equivalent gates); national supervisor deck 15 (set and Boolean forms).
- How to solve: Write both laws. Then say each in words.
- Worked example: **(A + B)' = A'·B' and (A·B)' = A' + B'.** In words: "NOT (A OR B) equals (NOT A) AND (NOT B)". (Python check: both true on all 4 rows.)
- Traps:
  - (A + B)' is NOT A' + B'. That wrong form keeps the sign. You must change the sign.
  - (AB)' is NOT A'B'. Same mistake.
- Calculator: look up — type: "De Morgan".

2. One-step De Morgan — ✅ IN

- Where seen: national supervisor deck 15 (NOT(A AND B) = ?).
- How to solve: Break the bar. Change the sign.
- Worked example: NOT(A AND B) = **(NOT A) OR (NOT B)**. And (A + B + C)' = **A'B'C'**. (Python check.)
- Traps: A bar over three letters becomes three short bars. Do not drop one.
- Calculator: calculate — type: expression with a long bar.

3. Name a gate that has bubbles on its inputs — ✅ IN

- Where seen: not seen in a test as a separate item. It is needed to read circuits drawn in "mixed logic" style.
- How to solve: Use the table above. Or write the expression and use De Morgan.
- Worked example: An AND gate with bubbles on both inputs gives A'·B' = (A + B)'. **Answer: NOR.** An OR gate with bubbles on both inputs gives A' + B' = (AB)'. **Answer: NAND.** (Python check.)
- Traps: An AND with input bubbles is **not** NAND. NAND has the bubble at the output.
- Calculator: look up — type: body (AND/OR) + where the bubbles are.

4. History: who was De Morgan? — ⚠️ Borderline (De Morgan is not on the 2027 history list; Boole is)

- Where seen: not seen in a test.
- How to solve: Know a few facts. De Morgan (1806-1871) was English. He taught at University College London. He was a friend of Boole. His laws appear in *Formal Logic* (1847), the same year as Boole's *The Mathematical Analysis of Logic*.
- Worked example: "Which mathematician, a friend of Boole, gave his name to the law (AB)' = A' + B'?" **Answer: Augustus De Morgan.**
- Traps: Boole did not name De Morgan's laws. Do not swap the two names.
- Calculator: look up — type: name.

Medium

5. Complement of an expression (write F') — ✅ IN

- Where seen: MIT Invitational 2019, Div C, INV, Q45 (Q = (AB)' + C; write Q' in SOP → ABC').
- How to solve:
  1. Put a bar over the whole expression.
  2. Break the outermost bar first. Change that sign.
  3. Keep breaking bars until each bar sits on one letter.
  4. Remove double bars.
- Worked example 1 (MIT 2019 C Q45): Q' = ((AB)' + C)' = (AB)''·C' = **ABC'**. (Python check; matches the key.)
- Worked example 2: F = A(B + C'). F' = A' + (B + C')' = **A' + B'C**. (Python check.)
- Traps:
  - Keep the grouping. F = A(B + C') has two groups: A and (B + C'). F' must keep them as two groups.

  - Do not confuse the complement with the dual (dual of A(B + C') is A + BC').
- Calculator: calculate — type: expression → complement in SOP.

6. Nested De Morgan (bars inside bars) — ✅ IN

- Where seen: UT Invitational 2020 (2021 season), Div C, INV, C63 (NOT[(AB)·NOT(A + B)]).
- How to solve: Work from the outside in. One bar at a time. Write each step.
- Worked example 1: ((A + B')C)' = (A + B')' + C' = **A'B + C'**. (Python check.)
- Worked example 2: (A'(B + C'))' = A'' + (B + C')' = **A + B'C**. (Python check.)
- Worked example 3: (AB + C)' = (AB)'·C' = **(A' + B')C'**. (Python check.)
- Traps: (B')' = B. A lost double bar is the most common error.
- Calculator: calculate — type: expression.

7. Draw the De Morgan equivalent of one gate — ✅ IN

- Where seen: Princeton 2020, Div C, INV, Q37 ("draw the equivalent gates").
- How to solve: Swap the body shape (AND ↔ OR). Flip every bubble: add bubbles where none were, remove bubbles where they were.
- Worked example: NAND (AND body, output bubble) → **OR body with bubbles on both inputs and no output bubble.** NOR → **AND body with bubbles on both inputs.**
- Traps: Flip the output bubble too. Forgetting it turns NAND into AND-type nonsense.
- Calculator: look up — type: gate name → equivalent drawing words.

8. Use De Morgan to build a gate from NOR (or NAND) only — ✅ IN (universal gate sets, 3.c.viii)

- Where seen: USC Invitational 2026, Div B, INV, FRQ 36a (3 points; also Div C FRQ 40a: "using two NOR gates in inverter configuration and one normal NOR gate, construct an AND gate"); UT Invitational fall 2018, Div B/C, INV, Q29 (Div C: design a NOR gate from NAND gates only); PA Central Regional ~2013, Div C, REG, Q27-29 (OR from NOT + AND; AND from NOR only; AND from NAND only).
- How to solve:
  1. Write the target gate.
  2. Use De Morgan so that only the allowed gate appears.
  3. A NOR or NAND with both inputs tied together is a NOT.
- Worked example 1 (USC 2026 B FRQ 36a): AB = (A' + B')' (De Morgan). That is NOR(A', B'). A' = NOR(A, A). B' = NOR(B, B). **AND = NOR(NOR(A, A), NOR(B, B)). 3 NOR gates.** (Python check.)
- Worked example 2 (UT fall 2018 Q29): NOR = NOT(OR). OR = NAND(A', B') = NAND(NAND(A, A), NAND(B, B)). Then add a NAND-inverter. **NOR from 4 NANDs.** (Python check.)
- Traps: Do not use a plain NOT gate when the question says "only NOR". Make the NOT from a NOR.
- Calculator: look up — type: target gate + allowed gate (gives the build and the count).

Hard

9. NAND-NAND circuit → SOP (bubble pushing, 2 levels) — ✅ IN

- Where seen: not seen in a test as a direct question. It is the method behind MIT 2020 C Q47-48 and the hands-on "build from a truth table" with 7400 chips.
- How to solve:
  1. Redraw the **output** NAND as an OR with bubbles on its inputs.
  2. Now each internal wire has two bubbles: one at the first NAND's output, one at the OR's input. They cancel.
  3. Read the circuit as AND gates feeding an OR.
- Worked example: F = NAND(NAND(A, B), NAND(C, D)). Redraw the last gate as negative-OR. Bubbles cancel. **F = AB + CD.** (Python check.)
- Traps: A single letter that goes straight into the output NAND keeps its bubble. NAND(NAND(A, B), C) = AB + C'.
- Calculator: calculate — type: gate list.

10. NOR-NOR circuit → POS — ✅ IN

- Where seen: not seen in a test.

- How to solve: Redraw the output NOR as an AND with bubbles on its inputs. Cancel bubble pairs. Read OR gates feeding an AND.
- Worked example: F = NOR(NOR(A, B), NOR(C, D)). **F = (A + B)(C + D).** (Python check.)
- Traps: NOR-NOR gives POS, not SOP. NAND-NAND gives SOP.
- Calculator: calculate — type: gate list.

11. SOP → NAND-only drawing; POS → NOR-only drawing — ✅ IN

- Where seen: not seen in a test. Needed for the hands-on 3.c.v when the kit has only one chip type.
- How to solve:
  1. Simplify to SOP (for NAND) or POS (for NOR).
  2. Each product becomes a NAND. The final OR becomes a NAND.
  3. A single-letter term needs its bubble fixed: feed the letter's complement, or add a NAND-inverter.
- Worked example 1: F = AB + C'. **F = NAND(NAND(A, B), C).** The C'' = C cancels. 2 NAND gates. (Python check.)
- Worked example 2: F = (A + B')·C (MIT 2019 C Q46 answer). **F = NOR(NOR(A, B'), NOR(C, C))**. NOR(C, C) = C', and the outer NOR flips it back. (Python check.) B' still needs one more NOR-inverter if only true inputs are given.
- Traps: Count the inverters. They are gates too.
- Calculator: calculate — type: SOP or POS → NAND-only or NOR-only gate list.

12. Multi-level mixed circuit → one gate plus inverters — ✅ IN

- Where seen: Shock Value / Circuit Lab question marathon 2014 on scioly.org (output = NOR(OR(A, B), NAND(C, D)); "draw an equivalent logic gate circuit with one 4-input gate and any necessary inverters").
- How to solve:
  1. Write the expression.
  2. Use De Morgan on the outer gate.
  3. Cancel double bars.
  4. Match the result to one big gate.
- Worked example: F = NOR(A + B, (CD)').
  - F = (A + B)'·(CD)'' = A'B'·CD = **A'B'CD**.
  - One gate: **a 4-input AND with inverters on A and B.** Or: a 4-input NOR with inverters on C and D, since A'B'CD = (A + B + C' + D')'. (Python check of both.)
  - Values: F = 1 only when A = 0, B = 0, C = 1, D = 1.
- Traps: The "NOT of a NAND" is the AND. Do not leave (CD)'' as (CD)'.
- Calculator: calculate — type: gate list → expression.

13. XOR and XNOR with bubbles — ✅ IN

- Where seen: not seen in a test. UF Northern Regional 2019 C Q42 shows how easy it is to misread XOR vs XNOR.
- How to solve:
  1. Each input bubble on an XOR flips the output once.
  2. An even number of bubbles changes nothing. An odd number makes it XNOR.
- Worked example: XOR with a bubble on input A only: A' ⊕ B = **(A ⊕ B)' = XNOR(A, B).** XOR with bubbles on both inputs: A' ⊕ B' = **A ⊕ B.** (Python check.)
- Traps: Do not turn XOR into AND or OR when pushing bubbles. The XOR body never changes.
- Calculator: calculate — type: gate, bubble positions.

Nationals

14. Bubble push a 3-level NAND/NOR circuit — ✅ IN

- Where seen: not seen in a test. This is the hardest legal De Morgan question.
- How to solve:
  1. Start at the output gate. Pick the drawing (normal or De Morgan) that puts bubbles facing the gate before it.
  2. Move one level left. Again pick the drawing that matches bubbles.
  3. Cancel every pair. Any bubble left on an input letter becomes a NOT on that letter.

4. Write the expression. Check with a truth table.

- Worked example: F = NAND(NAND(A, NOR(B, C)), NAND(D, NOR(B, C))).
  - The output NAND becomes negative-OR. The two inner NAND bubbles cancel. So F = A·NOR(B, C) + D·NOR(B, C).
  - NOR(B, C) = B'C'. Factor: F = (A + D)·B'C'.
  - **F = (A + D)B'C'.** (Python check, all 16 rows.)
- Worked example 2: F = NOR(NAND(A, B), OR(C, D')).
  - De Morgan on the NOR: F = (NAND(A, B))'·(C + D')' = AB·C'D.
  - **F = ABC'D.** It is 1 on one row only: A = 1, B = 1, C = 0, D = 1. (Python check.)
- Worked example 3: F = NAND(NOR(A, B'), C'). F = (NOR(A, B'))' + C'' = **A + B' + C.** (Python check.)
- Traps:
  - A gate output that feeds two later gates must be pushed the same way for both. If not, add an inverter.
  - Go one level at a time. Do not try to push every bubble in one step.
- Calculator: calculate — type: gate list (gives the expression and the table).

15. Write a POS expression from De Morgan of the complement — ✅ IN

- Where seen: MIT Invitational 2020, Div C, INV, Q48 (Q = (X' + Y')(Z + W) → (XY)'Z + (XY)'W); MIT Invitational 2019, Div C, INV, Q45-46.
- How to solve:
  1. Write F' as an SOP (from the 0-rows of F).
  2. Take the complement with De Morgan. Each product turns into a sum. The OR turns into an AND.
  3. That gives F as a POS.
- Worked example: F = 0 on rows 011 and 100 (3 inputs). F' = A'BC + AB'C'. So F = (A'BC)'·(AB'C')' = **(A + B' + C')(A' + B + C)**.
- Worked example (MIT 2020 C Q48): Q = (X' + Y')(Z + W). X' + Y' = (XY)' by De Morgan. Distribute: **Q = (XY)'Z + (XY)'W**. (Python check, all 16 rows.)
- Traps: The rows for POS are the zeros of F, which are the ones of F'.
- Calculator: calculate — type: zero-rows → POS.

Out of scope (skip)

- ❌ De Morgan in transistor-level CMOS design (pull-up and pull-down networks). Rule 3.d: "semiconductors (beyond those listed above)".
- ❌ Active-low signals in sequential parts, such as SR latches built from NAND gates. Rule 3.d: "sequential logic".
- ❌ Set theory beyond two-set De Morgan (Venn diagrams with probability). Rule 3.e limits the math.

Mistakes in tests and keys

- **2014 scioly.org marathon:** the input values were images, so I could not check the posted answers. It is a forum post, not a test. Use it only as a practice idea.
- **USC Invitational 2026, Div B, INV, FRQ 36a:** the key says the NOR-based AND "is the only solution". With the stated parts (two NOR inverters and one NOR) that is true. In general other builds exist. I did not render the key drawing.
- No De Morgan key error found in the other tests.

Not confirmed

- Princeton 2020 Q37 and Princeton 2021 Q19: the answers come from the earlier extraction (G_tests). I did not re-open the PDFs in this pass.
- PA Central ~2013 Q27-29: from G_tests. The year is uncertain ("~2013").
- UT fall 2018 Q29 is marked "(C)" in the extraction. The B and C tests were combined in one file. I could not confirm whether Div B saw it.

- De Morgan's job title and *Formal Logic* date: from general reference (Wikipedia/MacTutor) via `prior/I_bank.md`. Not re-checked in this pass.

Sources

- 2027 Div B rules, page B12 (3.c.viii, 3.e.iv).
- Extractions: `research/tests/nat26.md` (USC 2026 FRQ 36a), `ut-c.md` (UT fall 2018 Q29), `misc-b.md` (MIT 2019 Q45-48), `ut-b.md` (UT 2020 C63); `research/prior/G_tests.md`; `H_logic_diodes.md` (A9-A11, A14, A22, A23); `B_topics.md` §8d-8e; `F_slides.md` §9 (deck 15).
- scioly.org forum thread "Shock Value B / Circuit Lab C Question Marathon" (2014), saved as `downloads/dlB/mar/m5022.txt`.
- Kuphaldt, *Lessons in Electric Circuits* Vol. IV, "DeMorgan's Theorems" and "Converting truth tables into Boolean expressions"; All About Circuits worksheet "DeMorgan's Theorems".
- Python checks of every worked example (truth tables over all rows), run 2026-10-16.

Universal Gates (NAND and NOR)

Rule area **3.c.viii** names "universal gate sets". So this note is ✅ IN at every level. A **universal gate set** (also called **functionally complete**) is a set of gates that can build every possible logic function. This is a high-point area. Supervisors like it because one drawing shows if you really understand gates.

Key facts

- A **logic gate** takes inputs that are 0 or 1 and gives one output that is 0 or 1.
- **NAND** = "NOT AND". Its output is 0 only when every input is 1.
- **NOR** = "NOT OR". Its output is 1 only when every input is 0.
- **NAND alone is universal. NOR alone is universal.** These are the two "universal gates".
- Other universal sets: {AND, NOT}, {OR, NOT}, {2:1 multiplexer with the constants 0 and 1}, {XOR, AND, constant 1}.
- **Not universal:** {AND, OR}. Neither gate can turn a 1 into a 0, so you can never build NOT.
- **Not universal:** {XOR, NOT}, {XOR, XNOR}, {XOR} alone. These only make "XOR-type" functions. They can never make AND.
- **Not universal:** {AND} alone, {OR} alone, {NOT} alone.
- **Diodes and resistors alone** can make AND and OR, but never NOT. So they are not universal. (See Logic Hardware and Chips A8.)
- **To prove a set is universal:** build NOT, and build AND or OR. Then De Morgan's Law gives the other one.
- **NOT from NAND:** tie both inputs together. NAND(A, A) = A′. (A′ means "NOT A".)
- **NOT from NAND, second way:** tie one input to 1. NAND(A, 1) = A′.
- **NOT from NOR:** tie both inputs together, NOR(A, A) = A′. Or tie one input to 0, NOR(A, 0) = A′.
- A **trap:** NAND(A, 0) = 1 always. NOR(A, 1) = 0 always. These are not inverters.
- **Bubble pushing:** a bubble (small circle) on a gate means NOT. A NAND equals an OR with bubbles on both inputs. A NOR equals an AND with bubbles on both inputs. This is De Morgan's Law drawn as pictures.
- **Two-level rule:** an AND-then-OR circuit (sum of products) becomes NAND-then-NAND. An OR-then-AND circuit (product of sums) becomes NOR-then-NOR.
- The 7400 chip has **four** 2-input NAND gates. The 7402 has four 2-input NOR gates. (See Logic Hardware and Chips A8.)
- George Boole made Boolean algebra (1847, 1854). Charles Sanders Peirce noticed around 1880 that NOR alone is enough (⚠ not confirmed in a primary source). Henry Sheffer published the NAND result in 1913. NAND is sometimes called the **Sheffer stroke** (A ↑ B). NOR is the **Peirce arrow** (A ↓ B).

Minimum gate counts (2-input gates only, plain inputs only)

I checked every count below with an exhaustive computer search. "Plain inputs" means A′ is **not** given for free. You must build it.

Build this	NAND only	NOR only
NOT	1	1
AND	2	3
OR	3	2
NAND	1	4
NOR	4	1
XOR	4	5
XNOR	5	4
A′B ("A NOT, B YES")	3	2
A + B′	2	3
3-input AND	4	6
3-input OR	6	4
3-input NAND	3	7

Build this	NAND only	NOR only
3-input NOR	7	3
C + AB	3	3
A(B + C)	3	3
2:1 multiplexer	4	4
3-input majority (AB + AC + BC)	6	6
3-input XOR (odd parity)	8	8
Half adder (sum and carry)	5	5
Full adder (sum and carry)	9	not checked

- **Memory trick:** NAND is "good at AND" (2) and "bad at OR" (3). NOR is the mirror image.
- **Duality trick:** swap NAND ↔ NOR and swap AND ↔ OR. The count stays the same. So NOR-only AND (3) = NAND-only OR (3).

Formulas

- NOT A = NAND(A, A) = NOR(A, A)
- AND: A·B = NAND(NAND(A,B), NAND(A,B))
- OR: A + B = NAND(NAND(A,A), NAND(B,B))
- NOR from NAND: NOR(A,B) = NAND(X, X), where X = NAND(NAND(A,A), NAND(B,B))
- XOR from 4 NANDs: n1 = NAND(A,B); n2 = NAND(A,n1); n3 = NAND(B,n1); **A ⊕ B = NAND(n2, n3)**
- XNOR from 5 NANDs: the XOR above, then one NAND-inverter.
- OR from NOR: A + B = NOR(NOR(A,B), NOR(A,B))
- AND from NOR: A·B = NOR(NOR(A,A), NOR(B,B))
- XNOR from 4 NORs: n1 = NOR(A,B); n2 = NOR(A,n1); n3 = NOR(B,n1); **(A ⊕ B)′ = NOR(n2, n3)**
- Sum of products → NAND-NAND: AB + CD = NAND(NAND(A,B), NAND(C,D))
- Product of sums → NOR-NOR: (A+B)(C+D) = NOR(NOR(A,B), NOR(C,D))
- De Morgan: (AB)′ = A′ + B′ and (A + B)′ = A′B′
- Chips needed = (gates needed) ÷ (gates per chip), rounded **up**. A 7400 has 4 gates.

Question types

Easy

1. Name the universal gates ✅ IN

- Seen: UT Austin Invitational 2020 (2021 season), Div C, INV, Q C37 ("the universal gate" → NAND). MIT Invitational 2020, Div C, INV, Q33 ("which exhibit universality", choose all → NOR, NAND). UT Austin Invitational fall 2019 (2020 season), Div C, INV, Q47 ("name the other universal gate" → NOR). Princeton Invitational 2020, Div C, INV, Q14 ("all combinational logic functions can be realized by" OR only / XOR only / AND only / NOR only → **NOR only**).
- How to solve:
  1. The two single gates that are universal are NAND and NOR.
  2. If it says "choose all", pick both.
  3. If only one answer is allowed and both NAND and NOR are choices, read the question again. Usually only one of them is listed.
- Example: "Which gate can build every logic function by itself: AND, OR, XOR, NOR?" **Answer: NOR.**
- Traps:
  - "The universal gate" on UT 2020 keys NAND, but NOR is also universal. If you must write one, write NAND. If you can explain, write "NAND (NOR is also universal)".
  - XOR and XNOR are **not** universal.
  - AND and OR are not universal on their own.
- Calculator: look up — type: `universal`

2. NAND or NOR with tied inputs = NOT ✅ IN

- Seen: MIT Invitational 2019, Div C, INV, Q47 (P = NAND(C, C) → C′). UT Austin Invitational fall 2019, Div C, INV, Q47 ("NAND with inputs tied = ?" → NOT).
- How to solve:
  1. Both inputs are the same value, A.
  2. NAND(A, A) = (A·A)′ = A′.
  3. NOR(A, A) = (A + A)′ = A′.
- Example: "A NOR gate has both inputs tied to A. What is the output when A = 1?" NOR(1, 1) = 0. **Answer: 0. The gate is an inverter.**
- Traps:
  - Tying one input to a constant also works, but the constant matters. NAND needs a 1. NOR needs a 0.
  - NAND(A, 0) is always 1. NOR(A, 1) is always 0.
  - An XOR with one input tied to 1 is also an inverter (A ⊕ 1 = A′). An XOR with tied inputs is always 0.
- Calculator: look up — type: `tied nand`

3. Which gates can be built from NOR only (or NAND only)? (select all) ✅ IN

- Seen: University of Michigan Invitational (UMSO) 2019, Circuit Lab, INV, Q10 (AND, OR, XOR, NOT, NAND from NOR only → **all of them**).
- How to solve:
  1. NOR is universal. So it can build **every** gate.
  2. The same is true for NAND.
  3. Pick every choice.
- Example: "Which can you build from NAND gates only: NOT, OR, XNOR, NOR?" **Answer: all four.**
- Traps:
  - Students often skip XOR or XNOR because they look "different". They can still be built.
  - If the question asks "from AND and OR only", the answer changes. You can build only AND and OR, never NOT.
- Calculator: look up — type: `universal`

4. Is this gate set universal (functionally complete)? ✅ IN

- Seen: not seen in a test in this form. MIT 2020 Q33 is the closest (single gates only). Rule 3.c.viii names "universal gate sets", so sets are fair game.
- How to solve:
  1. Try to build **NOT**. If the set cannot invert, it is **not** universal.
  2. Try to build **AND** (or OR).
  3. If you get NOT and one of AND or OR, the set **is** universal.
  4. Check for constants. A set may need a constant 1 or 0. Say so in your answer.
- Example: "Is {OR, NOT} universal?" NOT is given. AND: A·B = (A′ + B′)′, using De Morgan. **Answer: yes.**
- Example 2: "Is {AND, OR} universal?" Every output of AND or OR stays 0 when all inputs are 0. So you can never make NOT 0 = 1. **Answer: no.**
- Example 3: "Is {XOR, AND} universal if a constant 1 is available?" NOT A = A ⊕ 1. AND is given. **Answer: yes.**
- Traps:
  - {XOR, NOT} is **not** universal. XOR and NOT only make functions like A ⊕ B, A′, A ⊕ B ⊕ 1. I checked this: from A and B they make only 8 functions, and AND is not one of them.
  - {AND, NOT} **is** universal. Some students say no because OR is missing. De Morgan builds OR.
- Calculator: look up — type: `complete set`

5. Name a NAND-only or NOR-only circuit (diagram → gate) ✅ IN

- Seen: MIT Invitational 2019, Div C, INV, Q47 (NAND(C,C) → C′). Princeton Invitational 2019, Div C, INV, Logic (b) (c) (draw NOT, AND, OR from NAND and NOR). Not seen in a Div B test.
- How to solve:
  1. Label every gate output with a letter.

2. Write each output as an expression, starting from the inputs.
3. Remove double NOTs: (X′)′ = X.
4. Use De Morgan to clean up.
5. Check with a 4-row truth table.

- Example: Two NANDs make NAND(A, A) and NAND(B, B). A third NAND takes those two. What is it? NAND(A′, B′) = (A′B′)′ = A + B. **Answer: OR.**
- Traps:
  - Do not stop at (A′B′)′. Finish with De Morgan.
  - Check every row. A wrong guess often matches 3 of 4 rows.
- Calculator: calculate — type: `tt nand(nand(a,a),nand(b,b))`

Medium

6. Build NOT, AND or OR from NAND only ✅ IN

- Seen: PA Central Regional (about 2013), Div C, REG, Q29 ("create an AND gate using only NAND gates", 2 pts). Princeton Invitational 2019, Div C, INV, Logic (b)(c).
- How to solve:
  1. NOT = one NAND with both inputs tied.
  2. AND = NAND, then a NAND-inverter. (2 gates.)
  3. OR = invert A, invert B, then NAND them. (3 gates.)
  4. Draw every gate with the D shape and a bubble. Label inputs and the output.
- Example: "Make A AND B from NAND gates." Gate 1: NAND(A, B) = (AB)′. Gate 2: NAND(gate 1, gate 1) = AB. **Answer: 2 NAND gates.**
- Traps:
  - Forgetting the second NAND. One NAND gives NAND, not AND.
  - Drawing an AND symbol "made of NAND". Graders want the real NAND gates.
- Calculator: look up — type: `nand and`

7. Build AND or OR from NOR only ✅ IN

- Seen: PA Central Regional (about 2013), Div C, REG, Q28 ("AND using only NOR", 4 pts). USC Invitational 2026, Div B, INV, Q36a and the matching Div C test Q40a (build AND from 2 NOR inverters + 1 NOR → NOR(A′, B′)). (Note: earlier notes called these files "Nationals 2026". They are the USC **Invitational** of January 2026.)
- How to solve:
  1. NOT = one NOR with tied inputs.
  2. OR = NOR, then a NOR-inverter. (2 gates.)
  3. AND = invert A, invert B, then NOR them. (3 gates.)
- Example (USC 2026 B Q36a): NOR(A, A) = A′. NOR(B, B) = B′. NOR(A′, B′) = (A′ + B′)′ = AB. **Answer: AND from 3 NOR gates.**
- Traps:
  - With NOR, OR is the "easy" one (2 gates) and AND is the "hard" one (3 gates). This is the opposite of NAND.
  - The USC key says "this is the only solution". It is the only **3-gate** solution with plain inputs. Bigger circuits also work. ⚠️ I did not see the key's drawing.
- Calculator: look up — type: `nor and`

8. Build OR from AND and NOT ✅ IN

- Seen: PA Central Regional (about 2013), Div C, REG, Q27 ("create an OR gate using only NOT and AND gates", 4 pts).
- How to solve:
  1. De Morgan: A + B = (A′ · B′)′.
  2. Invert A. Invert B. AND them. Invert the result.
  3. That is 3 NOT gates and 1 AND gate.
- Example: Check A = 0, B = 1. A′ = 1, B′ = 0, AND = 0, NOT = 1. A + B = 1. **Answer: OR = NOT(AND(NOT A, NOT B)).**

- Traps:
  - Forgetting the last NOT. AND(A′, B′) alone is NOR.
  - The mirror question: "AND from OR and NOT" → A·B = (A′ + B′)′.
- Calculator: look up — type: `demorgan or`

9. Build NOR from NAND only (or NAND from NOR only) ✅ IN

- Seen: UT Austin Invitational fall 2019 (2020 season), Div C, INV, Q29 ("design a NOR gate from NAND gates only" → 4 NANDs).
- How to solve:
  1. First build OR from NAND (3 gates).
  2. Then add a NAND-inverter (1 gate).
  3. Total: 4 NANDs.
  4. For NAND from NOR: build AND from NOR (3 gates), then a NOR-inverter. Total 4 NORs.
- Example: A = 0, B = 0. NAND(0,0) = 1 for A′. NAND(0,0) = 1 for B′. NAND(1,1) = 0 (this is A + B = 0). NAND(0,0) = 1. NOR(0,0) = 1. ✓ **Answer: 4 NAND gates.**
- Traps:
  - 4 is the true minimum (checked by search). Do not try to do it in 3.
- Calculator: look up — type: `nand nor`

10. Build XOR or XNOR from NAND only (or NOR only) ✅ IN

- Seen: not seen in a test. The national supervisor practice slides (deck 15) build XOR from AND/OR/NOT, but not from NAND. The older research notes call "fewest NANDs for XOR" a likely question.
- How to solve (NAND XOR, 4 gates):
  1. n1 = NAND(A, B).
  2. n2 = NAND(A, n1).
  3. n3 = NAND(B, n1).
  4. XOR = NAND(n2, n3).
  5. For XNOR, add a fifth NAND as an inverter.
- How to solve (NOR XNOR, 4 gates): the same picture with NOR gates gives XNOR. Add a fifth NOR to get XOR.
- Example: A = 1, B = 0. n1 = NAND(1,0) = 1. n2 = NAND(1,1) = 0. n3 = NAND(0,1) = 1. XOR = NAND(0,1) = 1. ✓ **Answer: 4 NANDs for XOR, 5 for XNOR.**
- Traps:
  - The same 4-gate picture gives XOR with NAND but **XNOR** with NOR. Don't mix them up.
  - The "easy" way (build AND, OR, NOT, then XOR = AB′ + A′B) uses many more gates. It is correct, but it loses points if the question says "fewest".
- Calculator: look up — type: `nand xor`

11. Sum of products → NAND-NAND (and product of sums → NOR-NOR) ✅ IN

- Seen: not seen in a test. Standard textbook skill (Kuphaldt, Lessons in Electric Circuits Vol. IV, ch. 7; All About Circuits worksheets "Boolean Algebra").
- How to solve:
  1. Write F as a **sum of products** (SOP). SOP means AND terms joined by OR, for example AB + CD.
  2. Replace every AND with a NAND.
  3. Replace the final OR with a NAND.
  4. A term with one letter (like C′) goes straight into the final NAND, but inverted. So C′ becomes C. C becomes NAND(C, C).
  5. For a **product of sums** (POS), such as (A + B)(C + D), do the same with NOR gates.
- Example: F = AB + C′. NAND(A, B) = (AB)′. The single term C′ enters the last gate as C. F = NAND((AB)′, C) = AB + C′. **Answer: 2 NAND gates.**
- Example 2: F = (A + B)(C + D) = NOR(NOR(A,B), NOR(C,D)). **Answer: 3 NOR gates.**
- Traps:
  - A single-letter term flips. C′ in the SOP needs a plain C at the last NAND. A plain C needs an inverter.

  - The final NAND needs as many inputs as there are terms. With only 2-input gates you must split it. That costs extra gates.
- Calculator: calculate — type: `nandnand ab+c'`

12. Gate count → number of 74xx chips ✅ IN

- Seen: not seen in a test. Ties to the hands-on task "construct a logic circuit to mimic a given truth table" (rule Part II.c.v).
- How to solve:
  1. Count the gates of each type your design needs.
  2. Divide by the gates per chip (4 for a quad 2-input chip like the 7400, 6 for the 7404 hex inverter).
  3. Round **up**.
- Example: An all-NAND full adder needs 9 NANDs. 9 ÷ 4 = 2.25. **Answer: 3 chips of 7400 (3 gates left over).**
- Traps:
  - An XOR from NAND needs exactly 4. That is **one** 7400.
  - A NOT from a NAND uses one whole gate.
- Calculator: calculate — type: `chips 9 4`

Hard

13. Minimum NAND count: simplify first (redundancy trick) ✅ IN

- Seen: MIT Invitational 2020, Div C, INV, Q47 ("minimum number of NAND gates to implement A + AB′ + AB′C" → **0**, because it simplifies to A).
- How to solve:
  1. Simplify the expression **before** you count any gates.
  2. Use absorption: A + AX = A.
  3. If the result is a single plain input, you need **0 gates**. Just use the wire.
  4. If it is A′, you need 1 gate.
  5. Otherwise use the count table above.
- Example: A + AB′ + AB′C. The terms AB′ and AB′C both contain A. A + (anything × A) = A. **Answer: 0 NAND gates (the output is just wire A).**
- Traps:
  - Counting gates for the unsimplified expression. That gives 5 or more.
  - Writing 1 (a "buffer"). No gate is needed.
- Calculator: calculate — type: `simp a+ab'+ab'c`

14. Fewest gates for a 2-input function (NAND-only or NOR-only) ✅ IN

- Seen: not seen in a test. Rule 3.c.viii (universal gate sets) makes it fair.
- How to solve:
  1. Find which function it is (AND, OR, XOR, …).
  2. Use the count table. Learn the top 7 rows.
  3. For NOR, use the duality trick: swap AND ↔ OR and NAND ↔ NOR, then use the NAND column.
- Example: "Fewest 2-input NOR gates for XOR?" XOR is the dual of XNOR. NAND-only XNOR = 5. **Answer: 5.**
- Traps:
  - The A′B function costs 3 NANDs, not 2.
  - A + B′ costs only 2 NANDs: A + B′ = NAND(A′, B) = NAND(NAND(A,A), B). Write it out to be sure.
- Calculator: look up — type: `mincount xor nor`

15. Read a two-level NAND circuit as a sum of products ✅ IN

- Seen: not seen in a test in all-NAND form. UF Northern Regional 2020, Div C, REG, Q23 has a chip-labeled diagram → expression. NY Lake Erie Invitational 2018, Div C, INV, Q29 asks for the unsimplified expression.
- How to solve:
  1. Push bubbles. A NAND at the output = an OR with bubbles on its inputs.
  2. The bubbles cancel the bubbles on the first-level NANDs.

3. So the first-level NANDs act as ANDs. The last NAND acts as an OR.
4. Any input that goes straight into the last NAND gets inverted.

- Example: F = NAND(NAND(A,B), NAND(C, NAND(D,D))). NAND(D,D) = D′. The circuit is AND-OR: **Answer: F = AB + CD′.** (Checked on all 16 rows.)
- Traps:
  - A bubble on only one end does not cancel.
  - NAND(D, D) is D′, not D.
- Calculator: calculate — type: `tt nand(nand(a,b),nand(c,nand(d,d)))`

Nationals

16. Fewest 2-input NANDs for a 3-input function ✅ IN (hardest legal type)

- Seen: not seen in a test. The closest is MIT 2020 Q47.
- How to solve:
  1. Simplify the function.
  2. Look for a **factored** form. Factoring often beats SOP. Example: AB + AC = A(B + C).
  3. Convert to NAND-NAND.
  4. Look for shared pieces. One inverter can feed two gates.
  5. Know the proven minimums: C + AB = 3; A(B + C) = 3; 2:1 mux = 4; 3-input AND = 4; 3-input OR = 6; majority = 6; 3-input NOR = 7; 3-input XOR = 8; full adder = 9.
- Example (majority, 6 gates): g1 = NAND(C, C) = C′. g2 = NAND(g1, B) = B′ + C. g3 = NAND(g1, g2) = B + C. g4 = NAND(B, C). g5 = NAND(A, g3). F = NAND(g5, g4) = A(B + C) + BC = AB + AC + BC. **Answer: 6 NANDs.** (Search proved 5 is impossible.)
- Example 2 (2:1 multiplexer, output C when A = 1, B when A = 0): F = NAND(NAND(A, C), NAND(NAND(A,A), B)). **Answer: 4 NANDs.**
- Traps:
  - With **3-input** NANDs allowed, majority = NAND3(NAND(A,B), NAND(A,C), NAND(B,C)) = **4 gates**. Read which gate sizes are allowed.
  - "Free" complemented inputs change every count. Read whether A′ is supplied.
- Calculator: look up — type: `mincount maj`

17. All-NAND half adder and full adder ✅ IN

- Seen: not seen in a test. Classic textbook figure (Kuphaldt Vol. IV; All About Circuits "Full adder"). Links to Adders A8.
- How to solve (half adder, 5 NANDs):
  1. n1 = NAND(A, B).
  2. Use n1 for the 4-NAND XOR. That gives the sum.
  3. Carry = NAND(n1, n1) = AB. One more gate.
- How to solve (full adder, 9 NANDs):
  1. n1 = NAND(A, B); n2 = NAND(A, n1); n3 = NAND(B, n1); n4 = NAND(n2, n3). Now n4 = A ⊕ B.
  2. n5 = NAND(n4, Cin); n6 = NAND(n4, n5); n7 = NAND(Cin, n5); **Sum = NAND(n6, n7)**.
  3. **Cout = NAND(n1, n5).**
- Example: A = 1, B = 1, Cin = 1. n1 = 0, n2 = 1, n3 = 1, n4 = 0, n5 = 1, n6 = 1, n7 = 0, Sum = 1, Cout = NAND(0, 1) = 1. 1 + 1 + 1 = 11_2. ✓ **Answer: 9 NANDs; Sum = 1, Cout = 1.**
- Traps:
  - The carry reuses n1 and n5. That sharing is why 9 is enough. I checked by search that 8 is impossible.
  - Using 2 XORs (4 each) + 2 ANDs (2 each) + 1 OR (3) = 15 NANDs works, but it is not minimal.
- Calculator: look up — type: `nand fa`

18. Prove a set is NOT universal ⚠️ Borderline

- Seen: MIT Invitational 2020, Div C, INV, Q31 ("which gates are linear functions" → XOR, XNOR). That is the idea behind the proof. Borderline because the rules name "universal gate sets" but not proofs.
- How to solve:

1. Find a property that every gate in the set keeps.
2. AND and OR are **monotone**: changing an input from 0 to 1 never changes the output from 1 to 0. NOT breaks this. So {AND, OR} cannot make NOT.
3. AND and OR also **keep 0**: all-0 inputs give 0. NOT gives 1. So {AND, OR} fails.
4. XOR, XNOR and NOT are **linear** (they only add bits with no carry). AND is not linear. So {XOR, XNOR, NOT} cannot make AND.

- Example: "Why can't AND and OR gates make a NOT gate?" With every input at 0, every AND and every OR outputs 0. So the whole circuit outputs 0. NOT 0 must be 1. **Answer: they can never turn all-0 inputs into a 1.**
- Traps:
  - "Because there is no bubble" is not a proof. Give the 0-in, 0-out reason.
- Calculator: look up — type: `not universal`

Out of scope (skip)

- ❌ SR latch made of two cross-coupled NOR or NAND gates. Seen: USC Invitational 2026, Div B Q36b-c and Div C Q40b-d. **OUT: rule 3.d "sequential logic and digital clock signals".** Still know one fact: a circuit with feedback (an output wired back to an input) is **not** combinational.
- ❌ Building gates from transistors (CMOS NAND, TTL NAND insides). **OUT: rule 3.d "semiconductors (beyond those listed above)".** A truth table given with the drawing can still be answered (F-M Invitational 2018 Q11, CMOS pair → NOT).
- ❌ Op-amp NOT gates (national practice deck 15 homework). **OUT: op-amps are semiconductors beyond those listed (3.d).**
- ❌ Flip-flops, counters, memory from NAND. **OUT: 3.d sequential logic.** MIT 2020 Q34 ("which can store a bit: NANDs, capacitors…") is OUT (memory and capacitance).
- ⚠ Reversible gates (MIT 2020 Q35 → NOT) and perceptron gates (Princeton 2020 Q18 → OR). Not named in the rules. Treat as rare trivia.

Mistakes in tests and keys

- **UT Austin Invitational 2020 Div C Q C37:** the key says NAND is "the universal logic gate". NOR is also universal. The question is badly worded.
- **USC Invitational 2026 Q36a / C Q40a:** the key says the 3-NOR AND is "the only solution". It is only the unique minimum. Larger correct circuits exist.
- **National practice deck 15** defines XNOR as "true only if A NOR B is true". **Wrong.** XNOR is 1 when A and B are the same.
- **Earlier research files** called the USC 2026 tests "Nationals 2026". They are the USC Invitational (January 17, 2026). The real 2026 Nationals tests are only in the paid soinc.org test packet.

Not confirmed

- The minimum full-adder count with **NOR** gates only. I did not run that search.
- The minimum NAND count for **3-input XNOR**. It is at most 9 (XOR3 + inverter). My search stopped at 8 gates.
- Peirce's NOR result (about 1880) comes from general history sources, not a primary text.
- The USC 2026 Q36a answer drawing is an image. I did not render it.
- No 2027 Div B test exists yet. How often universal gates will appear is unknown.

Sources

- 2027 Div B rules PDF, Circuit Lab, rules 3.c.viii, 3.d, 3.e.iv and Part II.c.v (pages labeled B12-B13 in the PDF footer).
- Test extractions: `circuitlab/research/tests/nat26.md` (USC 2026), `ut-b.md` (UT 2020), `ut-c.md` (UT fall 2019), `misc-a.md` (UMSO 2019, F-M 2018), `misc-b.md` (MIT 2019, UF Northern 2020), `prior/G_tests.md` (MIT 2020, Princeton 2019/2020, PA Central).

- Original test files read for this note: `downloads/dlG/mit20_test.txt`, `mit20_key.txt`, `pz/p20/Circuit Lab/2020 Princeton Circuit Lab Exam.txt` and key, `wiki/central.txt`.
- Research bank: `prior/H_logic_diodes.md` A21-A24; `prior/B_topics.md` 8e; `prior/F_slides.md` deck 15.
- Kuphaldt, *Lessons in Electric Circuits*, Vol. IV (Digital), ch. 3 (logic gates) and ch. 7 (Boolean algebra).
- All About Circuits worksheets: "Boolean Algebra", "DeMorgan's Theorems".
- Exhaustive minimum-gate search and every example: my Python scripts (breadth-first search over sets of truth tables).

Binary Numbers and Addition

Math rule **3.e.iii** allows "representation of whole numbers in binary and the addition of binary numbers, not including negative numbers, floating point representations, or any arithmetic operations other than addition". So you must convert and add. You must **not** subtract, multiply or use negative numbers. This is ✅ IN at every level.

Key facts

- **Binary** (base 2) writes numbers with only two digits: 0 and 1.
- A **bit** is one binary digit. A **nibble** is 4 bits. A **byte** is 8 bits.
- Each place is worth twice the place to its right: 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024.
- The **LSB** (least significant bit) is the rightmost bit. It is worth 1.
- The **MSB** (most significant bit) is the leftmost bit. In an 8-bit number it is worth 128.
- A subscript shows the base: 1011_2 = eleven. 11_{10} = eleven in decimal.
- **n bits** can hold 2^n different values: 0 up to $2^n - 1$.
  - 4 bits: 16 values, 0 to 15.
  - 8 bits: 256 values, 0 to 255.
  - 10 bits: 1024 values, 0 to 1023.
- A number is **odd** when its LSB is 1. It is **even** when its LSB is 0.
- A **power of 2** has exactly one 1 bit: 1, 10, 100, 1000, …
- $2^n - 1$ is all 1s: 15 = 1111_2, 255 = 1111 1111_2.
- **Binary addition rules:**

Add	Sum bit	Carry to the next column
0 + 0	0	0
0 + 1	1	0
1 + 1	0	1 (because 1 + 1 = 10_2)
1 + 1 + 1	1	1 (because 1 + 1 + 1 = 11_2)

- **Carry** means the 1 that moves to the next column on the left.
- **Overflow** means the answer needs one more bit than the register (storage box) holds. For whole numbers, overflow = a carry out of the top column.
- Two n-bit numbers always add to at most n + 1 bits. Largest: $(2^n - 1) + (2^n - 1) = 2^{n+1} - 2$. For 4 bits: 15 + 15 = 30 = 11110_2.
- The number of rows in a truth table with n inputs is 2^n. (Same idea as counting values.)
- Gottfried Leibniz described binary arithmetic in 1703 (⚠️ general history, not checked in a primary source). Boole's algebra (1847, 1854) uses the same 0 and 1.

Formulas

- Binary → decimal: value = sum of (bit × place value). Example: 11010_2 = 16 + 8 + 2 = 26.
- Decimal → binary (method 1): subtract the biggest power of 2 that fits, again and again. Write 1 for each power you used, 0 for the others.
- Decimal → binary (method 2): divide by 2 again and again. Write down each remainder. Read the remainders from **bottom to top**.
- Values with n bits = 2^n. Largest value = $2^n - 1$.
- Bits needed for N = the smallest n with $2^n > N$. (Or: the number of digits of N in binary.)
- Max sum of two n-bit numbers = $2^{n+1} - 2$, which needs n + 1 bits.

Question types

Easy

1. Binary → decimal ✅ IN

- Seen: not seen in a test by itself. It is step 1 of every bitwise question below (Florida State 2021 Q27-29, Rickards 2021 C Q29).
- How to solve:
  1. Write the place values over the bits, starting with 1 on the right.
  2. Add the place values that sit over a 1.
- Example: 11010_2. Places 16, 8, 4, 2, 1. Bits 1, 1, 0, 1, 0. 16 + 8 + 2 = **26**.
- Traps:
  - Starting the place values from the left. Always start with 1 at the **right**.
  - Leading zeros do not change the value: 0011 = 11 = 3.
- Calculator: calculate — type: `bin>dec 11010`

2. Decimal → binary ✅ IN

- Seen: not seen in a test by itself. Needed for Florida State 2021 Q28 (5 XOR 3) and Rickards 2021 C Q29 (108 XOR 57).
- How to solve (subtract powers):
  1. Find the biggest power of 2 that fits.
  2. Subtract it. Write a 1 in that place.
  3. Repeat with what is left.
  4. Write 0 in every place you skipped.
- Example: 45. 32 fits → 13 left. 16 does not fit (0). 8 fits → 5 left. 4 fits → 1 left. 2 does not fit (0). 1 fits. Bits 1 0 1 1 0 1. **Answer: 101101_2.** Check: 32 + 8 + 4 + 1 = 45.
- Example 2: 200 → **11001000_2** (128 + 64 + 8).
- Traps:
  - Forgetting the 0s in skipped places. 45 is not 1111.
  - With the divide method, reading the remainders top to bottom gives the number backwards.
- Calculator: calculate — type: `dec>bin 45`

3. Bits, bytes, MSB and LSB vocabulary ✅ IN

- Seen: not seen in a test. MIT Invitational 2021, Div C, INV, Scenario F explains "most significant bit first" before Q57.
- How to solve:
  1. Bit = 1 digit. Nibble = 4 bits. Byte = 8 bits.
  2. MSB = leftmost = biggest place. LSB = rightmost = place 1.
- Example: "In 1000 0001_2, what is the MSB worth?" It is the 8th place from the right. **Answer: 128.**
- Traps:
  - "Position 0" is the LSB. The MSB of 8 bits is position 7, not 8.
- Calculator: look up — type: `bit words`

4. How many values, and the largest value, for n bits ✅ IN

- Seen: not seen in a test. Common in textbooks (Kuphaldt Vol. IV ch. 1, "Numeration systems").
- How to solve:
  1. Number of values = 2^n.
  2. Largest value = $2^n - 1$, because 0 uses one of the values.
- Example: 8 bits. 2^8 = 256 values. **Largest = 255.**
- Traps:
  - Answering 256 for "largest". 256 is the count, not the top value.
- Calculator: calculate — type: `nbits 8`

5. Odd or even, power of 2, compare two binary numbers ✅ IN

- Seen: not seen in a test.
- How to solve:
  1. Odd/even: look at the LSB only.
  2. Power of 2: exactly one bit is 1.

3. Compare: line the numbers up on the right. Go from the left. The first place where they differ decides. The one with the 1 is bigger.

- Example: Which is bigger, 10110_2 or 10011_2? They match in the first two places (1, 0). The third place is 1 vs 0. **Answer: 10110_2 (22) is bigger than 10011_2 (19).**
- Traps:
  - Comparing numbers of different lengths without lining them up. Add leading zeros first.
- Calculator: calculate — type: `cmp 10110 10011`

6. Count up in binary (what comes next?) ✅ IN

- Seen: not seen in a test. MIT 2021 Scenario F lists inputs like 0001 and 1110.
- How to solve:
  1. Adding 1 flips the rightmost 0 to 1.
  2. Every 1 to its right turns into 0.
- Example: What comes after 10111_2? The rightmost 0 is in the 8s place. It becomes 1. The three 1s to its right become 0. **Answer: 11000_2** (23 + 1 = 24).
- Traps:
  - Only flipping the last bit. 10111 + 1 is not 10110.
- Calculator: calculate — type: `binadd 10111 1`

Medium

7. Add two binary numbers ✅ IN

- Seen: not seen as a pure item in a test. The rule names it directly (3.e.iii). The old topic note uses 1011 + 0110.
- How to solve:
  1. Line the numbers up on the right.
  2. Add one column at a time, from the right.
  3. Use the table: 1 + 1 = 0 carry 1; 1 + 1 + 1 = 1 carry 1.
  4. Write the final carry on the far left.
  5. Check: convert both numbers to decimal, add, and compare.
- Example: $1011_2 + 0110_2$.
  - Column 1: 1 + 0 = 1.
  - Column 2: 1 + 1 = 0, carry 1.
  - Column 3: 0 + 1 + 1 = 0, carry 1.
  - Column 4: 1 + 0 + 1 = 0, carry 1.
  - Final carry: 1.
  - **Answer: $10001_2 = 17$** (11 + 6 = 17 ✓).
- Traps:
  - Writing "2" in a column. Binary has no digit 2.
  - Forgetting the last carry.
- Calculator: calculate — type: `binadd 1011 0110`

8. Count the carries ✅ IN

- Seen: not seen in a test. A natural follow-up for adder questions.
- How to solve:
  1. Do the addition column by column.
  2. Count each column that sends a 1 to the left.
- Example: $1011_2 + 0111_2$ (11 + 7). Columns from the right: 1+1 → carry; 1+1+1 → carry; 0+1+1 → carry; 1+0+1 → carry. **Answer: 4 carries. Sum = 10010_2 = 18.**
- Traps:
  - Missing a carry that comes from 0 + 1 + (carry 1).
- Calculator: calculate — type: `carries 1011 0111`

9. Overflow in an n-bit register ✅ IN

- Seen: not seen in a test. Textbook item (H_logic_diodes A46).
- How to solve:
  1. Add the numbers normally.
  2. If the answer has more bits than the register, there is overflow.
  3. The register keeps only the lowest n bits. The extra top 1 is the carry-out.
- Example: 8-bit register. 10110110_2 (182) + 01101101_2 (109) = 100100011_2 = 291. 291 > 255. **Answer: overflow. The register shows 00100011_2 (35) and the carry-out is 1.**
- Traps:
  - Saying "no overflow" because each number fits in 8 bits. The **sum** must fit.
  - Overflow in this rule set means only carry-out. Signed overflow (two's complement) is ❌ OUT.
- Calculator: calculate — type: `binadd8 10110110 01101101`

10. Bits needed to store a number (or a sum) ✅ IN

- Seen: not seen in a test.
- How to solve:
  1. Find the smallest power of 2 that is **bigger** than the number.
  2. Its exponent is the number of bits.
  3. For a sum of two n-bit numbers, the answer is n + 1 bits.
- Example: How many bits for 1000? 2^9 = 512 is too small. 2^{10} = 1024 > 1000. **Answer: 10 bits.** (1000 = 1111101000_2.)
- Example 2: How many bits for 256? 2^8 = 256 is not bigger than 256. **Answer: 9 bits.**
- Traps:
  - Exact powers of 2 need one more bit than you expect. 256 needs 9 bits.
- Calculator: calculate — type: `bitsfor 1000`

11. Bitwise XOR or AND of two numbers ⚠️ Borderline

- Seen: Florida State 2021, Div B, STATE, Q27 (0101 XOR 0011 → 0110), Q28 (decimal 5 XOR 3 → 6), Q29 (select all equal to 0011 XOR 0100 → 0111 and 7). Rickards Invitational 2021 season, Div C, INV, Q29 (108 ⊕ 57 → 85). Princeton Invitational 2021, Div C, INV, Q25 ("20 * 13 in Boolean algebra", bitwise AND → 4). MIT Invitational 2019, Div C, INV, Q25 (0x09 & 0x07 → 1, in hex).
- Why borderline: the rules allow binary numbers and truth tables. They do not name "bitwise" operations. Florida State **Div B** asked it, so prepare it.
- How to solve:
  1. Convert both numbers to binary. Give them the same number of bits.
  2. Line them up.
  3. Apply the gate to each column **separately**. No carries.
  4. Convert back if the question wants decimal.
- Example (Rickards 2021 C Q29): 108 = 1101100_2. 57 = 0111001_2. XOR each column: 1010101_2. **Answer: 85.**
- Example 2 (Princeton 2021 C Q25): 20 = 10100_2, 13 = 01101_2. AND each column: 00100_2. **Answer: 4.** ("·" or "*" in Boolean algebra means AND, not multiply.)
- Traps:
  - Carrying in XOR. XOR has **no** carry. It is "addition without carries".
  - Reading "*" as multiplication. In Boolean algebra it means AND.
- Calculator: calculate — type: `bxor 108 57` or `band 20 13`

12. Add three binary numbers ✅ IN

- Seen: not seen in a test.
- How to solve:
  1. Add the first two numbers.
  2. Add the third number to that answer.
  3. Or add a whole column at once. A column of three 1s plus a carry can make 100_2. Then you carry into **two** places.

- Example: $0111_2 + 0101_2 + 0011_2 = 7 + 5 + 3 =$ **$1111_2 = 15$**.
- Example 2: $111_2 + 111_2 + 111_2 = 21 =$ **10101_2**.
- Traps:
  - In one column, 1 + 1 + 1 + carry 1 = 4 = 100_2. Write 0 and carry 2 (a 1 into each of the next two columns).
  - Adding two at a time is slower but safer.
- Calculator: calculate — type: `binadd 0111 0101 0011`

Hard

13. Find a missing bit in an addition ⚠️ Borderline

- Seen: not seen in a test.
- Why borderline: you can solve it by trying 0 and 1, using only addition. A solution that subtracts is not needed.
- How to solve:
  1. Try the missing bit as 0. Add. Check.
  2. Try it as 1. Add. Check.
  3. Only one choice gives the shown sum.
- Example: $1?01_2 + 0110_2 = 10011_2$. Try ? = 0: 1001 + 0110 = 1111 (15). No. Try ? = 1: 1101 + 0110 = 10011 (19). Yes. **Answer: ? = 1.**
- Traps:
  - Do not subtract to find it. That is "another arithmetic operation" and may be marked out of scope by a careful supervisor. Trying both bits is always allowed.
- Calculator: calculate — type: `binadd 1101 0110`

14. Bit position puzzles (index of a bit in a long word) ✅ IN

- Seen: MIT Invitational 2021, Div C, INV, Scenario F Q57-59 ("give the index of the last 1"; positions count from 0 at the right).
- How to solve:
  1. Number the bits from the **right**, starting at 0.
  2. Find the bit the question wants.
  3. Write its position number in binary, with the asked number of bits.
- Example (MIT 2021 Q58): input 0011'0100'0000'0000 (16 bits, n = 4). The last 1 (the rightmost 1) is at position 10. 10 = 1010_2. **Answer: 1010.**
- Example 2 (MIT 2021 Q57): 0111'0010. The rightmost 1 is at position 1. With 3 output bits: **001**.
- Traps:
  - "Last" means the last one you read from left to right. That is the **rightmost** 1.
  - Pad to the right width: position 1 with 3 bits is 001, not 1.
  - The apostrophes are just spacers.
- Calculator: calculate — type: `lastone 0011010000000000`

15. Largest possible sum and range questions ✅ IN

- Seen: not seen in a test.
- How to solve:
  1. Largest n-bit number = $2^n - 1$.
  2. Largest sum of two = $2 \times (2^n - 1) = 2^{n+1} - 2$.
  3. It needs n + 1 bits.
- Example: Two 4-bit numbers. Largest sum = 15 + 15 = 30 = **11110_2 (5 bits)**.
- Traps:
  - The sum never needs n + 2 bits for two numbers. It can for three or more numbers.
- Calculator: calculate — type: `maxsum 4`

Nationals

16. Big conversions and long additions under time pressure ✅ IN

- Seen: not seen in a test at Div B. MIT 2021 (Div C) uses 16-bit inputs.
- How to solve:
  1. Group the bits in fours from the right. Each group of four is worth 0-15.
  2. Learn the four-bit groups by heart: 1000 = 8, 1010 = 10, 1100 = 12, 1111 = 15.
  3. For a 16-bit number, the groups are worth ×4096, ×256, ×16 and ×1.
  4. For long additions, do them column by column. Check in decimal.
- Example: 0011 0100 0000 0000_2. Groups 3, 4, 0, 0. Value = 3 × 4096 + 4 × 256 = 12288 + 1024 = **13312**.
- Traps:
  - Mixing up the group weights. The second group from the right is ×16, not ×10.
- Calculator: calculate — type: `bin>dec 0011010000000000`

17. Hexadecimal, octal, BCD and Gray code ⚠ Borderline

- Seen: MIT Invitational 2019, Div C, INV, Q25 (0x09 & 0x07 → 0x01). No Div B test seen.
- Why borderline: rule 3.e.iii says "whole numbers in **binary**". Hex and octal are just short ways to write binary. BCD and Gray code are binary codes. A supervisor might use them. They are not named.
- Definitions:
  - **Hexadecimal** (base 16): one hex digit = 4 bits. Digits 0-9 then A = 10 … F = 15. "0x" in front means hex.
  - **Octal** (base 8): one digit = 3 bits.
  - **BCD** (binary-coded decimal): each decimal digit gets its own 4 bits.
  - **Gray code**: counting where only one bit changes each step.
- How to solve:
  1. Hex → binary: replace each hex digit with 4 bits.
  2. Octal → binary: replace each digit with 3 bits.
  3. BCD: code each decimal digit separately.
  4. Binary → Gray: keep the MSB. Each next Gray bit = XOR of that binary bit and the one to its left.
- Example: 0x2F = 0010 1111_2 = **47**. 57_8 = 101 111_2 = **47**. 59 in BCD = **0101 1001**. Binary 1011 → Gray **1110**.
- Traps:
  - BCD of 59 is not 59 in binary (111011_2).
  - Codes 1010-1111 are not valid BCD digits.
- Calculator: calculate — type: `hex>dec 2F`, `bcd 59`, `gray 1011`

Out of scope (skip)

- ❌ Negative numbers, two's complement, sign bits, signed overflow. **Rule 3.e.iii: "not including negative numbers".**
- ❌ Binary subtraction, multiplication, division. **Rule 3.e.iii: "any arithmetic operations other than addition".**
- ❌ Shifts as multiply or divide, and arithmetic vs logical shift. Seen: ATX Regional 2020, Div C, REG, Q60. **OUT: 3.e.iii.**
- ❌ Binary fractions and floating point (for example 0.101_2). **Rule 3.e.iii: "floating point representations".**
- ❌ Clocks, counters, shift registers. **Rule 3.d: "sequential logic and digital clock signals".**

Mistakes in tests and keys

- No key errors found for binary items in the tests I read.
- **Florida State 2021 Q29** has a choice that uses mod (remainder) arithmetic. That choice goes past the 2027 math limits. The keyed answers (0111 and 7) are correct.
- **scioly.org wiki:** says a multiplexer needs √N select lines. Wrong; it needs $\log_2 N$. (This matters for bit counting. See Multiplexers and Decoders A8.)

Not confirmed

- The Leibniz 1703 date is from general history knowledge.

- I did not open the Princeton 2021 Q25 page myself. The answer (4) comes from `prior/G_tests.md`. I checked 20 AND 13 = 4 in Python.
- Whether a 2027 supervisor will count hex or BCD as "binary" is unknown.

Sources

- 2027 Div B rules PDF, rule 3.e.iii (page labeled B12 in the footer).
- Test extractions: `circuitlab/research/tests/state.md` (Florida State 2021 B), `atx-reg.md` (ATX 2020 C Q60), `misc-b.md` (MIT 2019 Q25), `prior/G_tests.md` (Rickards 2021 C, Princeton 2021 C).
- Original files read: `downloads/dlG/rsoiOldC_test.txt` (Q29), `downloads/dlG/mit21_test.txt` and `mit21_key.txt` (Scenario F).
- Kuphaldt, *Lessons in Electric Circuits*, Vol. IV, ch. 1 "Numeration Systems" and ch. 2 "Binary Arithmetic".
- OpenStax has no binary chapter in College Physics 2e. I used Kuphaldt instead.
- Every example: checked in Python.

Adders

Rule area **3.c.viii** names "combinational logic circuits (e.g., multiplexers, **adders**)". An **adder** is a logic circuit that adds binary numbers. **Combinational** means the output depends only on the inputs right now, with no memory. This is ✅ IN at every level and a high-point area.

Key facts

- A **half adder** adds two bits, A and B. It has two outputs: **Sum (S)** and **Carry (C)**.
  - S = A ⊕ B (XOR). C = A·B (AND).
  - So a half adder = **1 XOR + 1 AND**.
- A **full adder** adds three bits: A, B and a carry-in (**Cin**). Outputs: Sum and carry-out (**Cout**).
  - S = A ⊕ B ⊕ Cin. Sum is 1 when an **odd** number of inputs are 1.
  - Cout = AB + ACin + BCin. Cout is 1 when **two or more** inputs are 1. This is the **majority** function.
  - Another form: Cout = AB + Cin(A ⊕ B).
- A full adder = **2 half adders + 1 OR gate**. The OR joins the two carries. An OR is enough because the two carries are never both 1 (checked on all 8 rows).
- A full adder in gates = 2 XOR + 2 AND + 1 OR = 5 gates.
- A full adder from NAND only needs **9** gates (proven minimum). A half adder needs **5** NANDs. (See Universal Gates (NAND and NOR) A8.)
- A full adder is a **ones counter**: the 2-bit number Cout S tells how many of the 3 inputs are 1.
- The **sum bit** is the **parity** of the inputs. Parity tells if the number of 1s is odd or even.
- A **ripple-carry adder** chains full adders. Each Cout feeds the next Cin. An n-bit ripple adder uses n full adders. The first Cin is tied to 0. (Or: 1 half adder for bit 0 + n − 1 full adders.)
- The last Cout is the extra top bit of the answer. In an n-bit register it means **overflow**.
- "Ripple" means the carry moves through the stages one after another.
- The **74LS283** (and the older 7483) is a 4-bit adder chip. It has carry-in C0 and carry-out C4. (⚠️ Pin numbers not checked here; see Logic Hardware and Chips A8.)

Half adder truth table

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Full adder truth table

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

- Sum = 1 on rows 1, 2, 4, 7. In minterm form: **S = Σm(1, 2, 4, 7)**. (A **minterm** is one row where the output is 1, numbered by its binary input.)
- Cout = 1 on rows 3, 5, 6, 7: **Cout = Σm(3, 5, 6, 7)**.

Formulas

- Half adder: $S = A \oplus B = A'B + AB'$; $C = AB$.
- Full adder: $S = A \oplus B \oplus Cin$; $Cout = AB + ACin + BCin = AB + Cin(A \oplus B)$.
- Full adder SOP form of the sum: $S = A'B'Cin + A'BCin' + AB'Cin' + ABCin$.
- Ripple stage i: $s_i = a_i \oplus b_i \oplus c_i$; $c_{i+1} = \text{majority}(a_i, b_i, c_i)$; $c_0 = 0$.
- Gates for an n-bit ripple adder (XOR/AND/OR version): 5n. NAND-only version: 9n.
- Adder chips for n bits: $n \div 4$ chips of 74283 (round up).
- ⚠️ Borderline, carry lookahead: $G_i = a_i b_i$ (generate); $P_i = a_i \oplus b_i$ (propagate); $c_{i+1} = G_i + P_i c_i$.

Question types

Easy

1. Which gates make a half adder? ✅ IN

- Seen: ATX (UT Austin) Regional 2020, Div C, REG, Q17 ("half adder = which 2 gates" → AND + XOR).
- How to solve:
  1. Sum = XOR. Carry = AND.
  2. Pick the choice with XOR and AND.
- Example: "A half adder is made of which two gates?" **Answer: one XOR and one AND.**
- Traps:
  - "OR + AND" is a common wrong choice. OR gives 1 + 1 = 1, which is wrong for the sum.
  - The XOR gives the **sum**. The AND gives the **carry**. Do not swap them.
- Calculator: look up — type: `half adder`

2. The sum bit is which function? (parity) ✅ IN

- Seen: ATX Regional 2020, Div C, REG, Q42 ("the sum bit of 2 bits is which function" → PARITY).
- How to solve:
  1. Sum = XOR.
  2. XOR is 1 when an odd number of inputs are 1.
  3. That is the definition of odd **parity**.
- Example: "The sum output of a full adder for A = 1, B = 1, Cin = 0 is?" Two 1s is even. **Answer: 0.**
- Traps:
  - "Parity" may look strange as a choice. It is right.
  - The carry is the **majority** function, not parity.
- Calculator: look up — type: `parity`

3. Half adder outputs for given inputs ✅ IN

- Seen: not seen in a test as a separate item.
- How to solve:
  1. Add the two bits in your head.
  2. The answer is a 2-bit number C S.
- Example: A = 1, B = 1. $1 + 1 = 10_2$. **Answer: S = 0, C = 1.**
- Traps:
  - Writing S = 2. Binary has no 2.
- Calculator: calculate — type: `ha 1 1`

4. Full adder outputs for given inputs ✅ IN

- Seen: not seen in a test as a separate item.
- How to solve:
  1. Count the 1s among A, B and Cin. Call it k (0 to 3).
  2. Write k in 2 bits: Cout S.
- Example: A = 1, B = 0, Cin = 1. $k = 2 = 10_2$. **Answer: S = 0, Cout = 1.**

- Example 2: A = B = Cin = 1. k = 3 = 11_2. **Answer: S = 1, Cout = 1.**
- Traps:
  - Forgetting Cin. A full adder always adds three bits.
- Calculator: calculate — type: `fa 1 0 1`

Medium

5. Draw a half adder ✅ IN

- Seen: UT Austin Invitational fall 2019 (2020 season), Div C, INV, Q49 ("draw a half adder from 2 gates" → AND for carry, XOR for sum).
- How to solve:
  1. Draw inputs A and B on the left.
  2. Send both to an XOR gate. Label its output S.
  3. Send both to an AND gate. Label its output C.
  4. Write the truth table to prove it.
- Example: **Answer: S = A ⊕ B from an XOR gate; C = AB from an AND gate.**
- Traps:
  - Labeling the sum "A + B". In Boolean algebra "+" means OR. Write A ⊕ B. (The UT key made this mistake.)
  - Drawing an OR shape for XOR. The XOR has an extra curved line at the back.
- Calculator: look up — type: `half adder`

6. Full adder from two half adders ✅ IN

- Seen: not seen in a test. Textbook item (Kuphaldt Vol. IV ch. 9; H_logic_diodes A29).
- How to solve:
  1. Half adder 1 adds A and B. Outputs S1 = A ⊕ B and C1 = AB.
  2. Half adder 2 adds S1 and Cin. Outputs S = S1 ⊕ Cin and C2 = S1·Cin.
  3. Cout = C1 OR C2.
- Example: A = 1, B = 1, Cin = 1. S1 = 0, C1 = 1. S = 0 ⊕ 1 = 1, C2 = 0. Cout = 1 + 0 = 1. **Answer: S = 1, Cout = 1.** ✓
- Traps:
  - Using an AND to join the carries. It must be OR.
  - An XOR would also work in that spot, because C1 and C2 are never both 1. Most keys expect OR.
- Calculator: look up — type: `fa from ha`

7. Fill in the full adder truth table ✅ IN

- Seen: not seen in a test. Expect it as a FRQ.
- How to solve:
  1. Write the 8 input rows in counting order: 000 to 111.
  2. For each row, count the 1s.
  3. Write the count in binary as Cout S.
- Example: row A B Cin = 0 1 1. Two 1s → 10_2. **Answer: S = 0, Cout = 1.** (Full table above.)
- Traps:
  - Skipping rows. 3 inputs need $2^3 = 8$ rows.
- Calculator: look up — type: `fa table`

8. What is a ripple-carry adder? ✅ IN

- Seen: ATX Regional 2020, Div C, REG, Q41 ("a ripple-carry adder is" → keyed "a series of half-adders").
- How to solve:
  1. A ripple-carry adder is a chain of **full** adders.
  2. Each Cout goes into the next stage's Cin.
  3. If the choices only offer "half adders", pick it, but know the right answer.
- Example: "How is a 4-bit ripple-carry adder built?" **Answer: 4 full adders in a chain (or 1 half adder + 3 full adders), each carry-out feeding the next carry-in.**

- Traps:
  - The ATX key says "series of half-adders". That is loose wording. Half adders have no carry-in, so they cannot be chained alone.
- Calculator: look up — type: `ripple`

9. Trace a 4-bit ripple-carry addition ✅ IN

- Seen: not seen in a test. Rule names adders; this is the standard follow-up.
- How to solve:
  1. Start at bit 0 (the right). $c_0 = 0$.
  2. For each stage, add $a_i + b_i + c_i$. Write s_i and the carry.
  3. Pass the carry to the next stage.
  4. The final carry is the top bit of the answer.
- Example: $0111_2 + 0101_2$ (7 + 5).
  - Bit 0: $1 + 1 + 0 = 10 \rightarrow s_0 = 0, c_1 = 1$.
  - Bit 1: $1 + 0 + 1 = 10 \rightarrow s_1 = 0, c_2 = 1$.
  - Bit 2: $1 + 1 + 1 = 11 \rightarrow s_2 = 1, c_3 = 1$.
  - Bit 3: $0 + 0 + 1 = 01 \rightarrow s_3 = 1, c_4 = 0$.
  - **Answer: sum 1100_2 = 12; carries $c_1\ c_2\ c_3\ c_4$ = 1, 1, 1, 0; no overflow.**
- Traps:
  - Forgetting to pass the carry along.
  - Reading the sum bits in the wrong order. s_3 is the leftmost.
- Calculator: calculate — type: `ripple 0111 0101`

10. How many adders (or gates, or chips) for n bits? ✅ IN

- Seen: not seen in a test.
- How to solve:
  1. n-bit ripple adder = n full adders (or 1 half + n − 1 full).
  2. Gates = 5 per full adder (2 XOR, 2 AND, 1 OR).
  3. NAND-only = 9 per full adder.
  4. 74283 chips = n ÷ 4, rounded up.
- Example: 8-bit adder. **Answer: 8 full adders (or 1 half + 7 full), or 2 chips of 74283 with C4 of the low chip wired to C0 of the high chip.**
- Example 2: 4-bit adder from NAND only. 4 × 9 = 36 NANDs. 36 ÷ 4 = **9 chips of 7400.**
- Traps:
  - Forgetting to tie the first carry-in (C0) to 0 (ground).
- Calculator: calculate — type: `adders 8`

11. Carry-out and overflow of an adder ✅ IN

- Seen: not seen in a test. Old topic note uses 1011 + 0110 = 10001.
- How to solve:
  1. Add the two n-bit numbers.
  2. If the sum is bigger than $2^n - 1$, the last carry-out is 1. That is overflow.
- Example: 4-bit adder: $1011_2 + 0110_2 = 11 + 6 = 17 > 15$. Stage carry-outs (bits 0-3) are 0, 1, 1, 1. **Answer: sum bits 0001, carry-out 1 (overflow); the full answer is 10001_2 = 17.**
- Traps:
  - Saying the answer is 1. The register shows 0001, but the true sum is 17.
- Calculator: calculate — type: `ripple 1011 0110`

Hard

12. Write the adder outputs as Boolean expressions (SOP or minterms) ✅ IN

- Seen: not seen in a test. Boolean expressions are named in rule 3.c.viii.
- How to solve:

1. Make the truth table.
2. For each 1 in the output column, write a product term. Use the letter for a 1 and the letter with a prime for a 0.
3. OR all the terms.
4. Simplify if asked.

- Example: Cout. Rows 011, 101, 110, 111. Cout = A′BCin + AB′Cin + ABCin′ + ABCin. Simplify: **Cout = AB + ACin + BCin.**
- Traps:
  - Sum cannot be simplified in SOP. It stays as 4 terms (or A ⊕ B ⊕ Cin).
- Calculator: calculate — type: `sop fa cout`

13. Build a full adder from NAND gates only ✅ IN

- Seen: not seen in a test. Details and the 9-gate wiring are in Universal Gates (NAND and NOR) A8 type 17.
- How to solve: use the 9-NAND wiring. n1 = NAND(A,B) is shared by the sum and the carry.
- Example: A = 1, B = 0, Cin = 0. n1 = 1, n2 = 0, n3 = 1, n4 = 1, n5 = 1, n6 = 0, n7 = 1, S = 1, Cout = NAND(1, 1) = 0. **Answer: S = 1, Cout = 0.** ✓
- Traps:
  - 9 is the proven minimum. 8 is impossible (checked by search).
- Calculator: look up — type: `nand fa`

14. Design a 2-bit adder from a truth table ✅ IN

- Seen: not seen in a test. Fits the hands-on task "construct a logic circuit to mimic a given truth table" (Part II.c.v).
- How to solve:
  1. Inputs: A1 A0 and B1 B0 (4 inputs, 16 rows). Outputs: S2 S1 S0.
  2. S0 = A0 ⊕ B0 (a half adder).
  3. S1 = A1 ⊕ B1 ⊕ (A0B0).
  4. S2 = A1B1 + A0B0(A1 + B1). This is the carry-out.
  5. Or chain a half adder and a full adder.
- Example: How many of the 16 rows have S2 = 1? S2 = 1 when A + B ≥ 4. **Answer: 6 rows** (1+3, 2+2, 2+3, 3+1, 3+2, 3+3).
- Traps:
  - S1 needs the carry from bit 0. Do not use A1 ⊕ B1 alone.
- Calculator: calculate — type: `adder2 tt`

15. Find the broken stage (adder troubleshooting) ✅ IN

- Seen: not seen in a test for adders. USC Invitational 2026, Div B, INV, Q22 has a "stuck-at-0" input in a gate circuit. The rules list "troubleshoot a faulty circuit" as a hands-on task.
- How to solve:
  1. Compute the right answer for the test input.
  2. Compare with the shown output bit by bit.
  3. Find the lowest wrong bit. The fault is in that stage or in the carry coming into it.
  4. Test an input where only that carry matters.
- Example: In a 4-bit ripple adder, the carry from bit 1 to bit 2 is stuck at 0. What does 0011 + 0001 show? Bit 0: 1 + 1 = 0, carry 1. Bit 1: 1 + 0 + 1 = 0, carry 1, but it is stuck at 0. Bit 2: 0 + 0 + 0 = 0. **Answer: it shows 0000 instead of 0100.**
- Traps:
  - A stuck carry only shows up when that carry should be 1. Many inputs look fine. (Here 96 of the 256 input pairs give a wrong answer.)
- Calculator: calculate — type: `ripple 0011 0001 stuck1=0`

16. Full adder as a ones counter or parity circuit ✅ IN

- Seen: not seen in a test. ATX 2020 C Q42 (sum = parity) is related.
- How to solve:
  1. Cout S = the number of 1s among the three inputs, in binary.

2. S alone = odd parity of the three inputs.

- Example: A team of 3 votes. Each "yes" is a 1. Votes are 1, 0, 1. Cout S = 10_2. **Answer: 2 yes votes. Cout = 1 also means "majority yes".**
- Traps:
  - Cout is the majority vote. S is not.
- Calculator: calculate — type: `fa 1 0 1`

Nationals

17. Adder from a decoder or a multiplexer ✅ IN

- Seen: not seen in a test. Textbook item (H_logic_diodes A36, A39).
- How to solve (decoder):
  1. A 3-to-8 decoder gives all 8 minterms of A, B, Cin.
  2. S = OR of outputs 1, 2, 4, 7.
  3. Cout = OR of outputs 3, 5, 6, 7.
  4. So: 1 decoder + two 4-input OR gates.
- How to solve (8:1 mux): put A, B, Cin on the select lines. Put the S column (0,1,1,0,1,0,0,1) on the data inputs.
- Example: "A 3-to-8 decoder and OR gates make a full adder. Which outputs go to the Cout OR gate?" **Answer: 3, 5, 6 and 7.**
- Traps:
  - Active-low decoders (like the 74138) need NAND gates instead of OR.
- Calculator: look up — type: `fa decoder`

18. Carry generate and propagate (carry lookahead) ⚠️ Borderline

- Seen: not seen in a test.
- Why borderline: it is an adder, so it fits 3.c.viii. But its purpose is speed (timing), which drifts toward excluded clock topics. It is college-level.
- How to solve:
  1. $G_i = a_i b_i$. The stage makes a carry by itself.
  2. $P_i = a_i \oplus b_i$. The stage passes a carry through.
  3. $c_{i+1} = G_i + P_i c_i$.
- Example: A = 1011, B = 0110, $c_0 = 0$. G (bit 3 to bit 0) = 0010. P = 1101. $c_1 = 0$, $c_2 = 1$, $c_3 = 1$, $c_4 = 1$. **Answer: $c_4 = 1$** (same as the ripple trace).
- Traps:
  - Using OR for P. Some books do. Then the sum needs XOR anyway.
- Calculator: calculate — type: `cla 1011 0110`

19. Gate delay of a ripple adder ⚠️ Borderline

- Seen: not seen in a test.
- Why borderline: no clock is used, but timing is not named in the rules. Rule 3.d excludes "digital clock signals".
- How to solve:
  1. Find the longest path from an input to an output.
  2. Count gate levels on it.
  3. Multiply by the delay per gate.
- Example: 4-bit ripple adder. The carry path has 2 gate levels per stage. Each gate takes 10 ns. 4 × 2 × 10 = **80 ns**.
- Traps:
  - Forgetting that the carry passes through **every** stage.
- Calculator: calculate — type: `delay 4 2 10`

Out of scope (skip)

- ❌ Half and full **subtractors**, adder-subtractors. **Rule 3.e.iii: no arithmetic other than addition.**

- ❌ Two's complement addition and signed overflow. **Rule 3.e.iii: no negative numbers.**
- ❌ Multipliers and ALUs with multiply or subtract. **Rule 3.e.iii.**
- ❌ Serial adders with a flip-flop and a clock. **Rule 3.d: sequential logic and digital clock signals.**
- ❌ BCD adders that need "add 6" correction. ⚠ Borderline-to-out: the correction is still addition, but BCD is not named. Rare.
- ❌ Adder chips built from transistors. **Rule 3.d: semiconductors beyond those listed.**

Mistakes in tests and keys

- **ATX Regional 2020 Div C Q41:** the keyed answer says a ripple-carry adder is "a series of half-adders". It is a chain of **full** adders. The keyed choice is the closest option, but the wording is wrong.
- **UT Austin Invitational fall 2019 Div C Q49:** the key labels the half-adder sum "A + B". "+" means OR. It should read A ⊕ B. The gate drawn is correctly an XOR.

Not confirmed

- The 74283/7483 details (C0 and C4 names) come from the research bank, not a datasheet I opened.
- The NAND full-adder minimum (9) was checked by my own search. I did not find it stated in a textbook as a proven minimum.
- No Div B test has asked an adder question yet. All seen items are Div C.

Sources

- 2027 Div B rules PDF, rules 3.c.viii, 3.d, 3.e.iii, Part II.c.v (pages labeled B12-B13).
- Test extractions: `circuitlab/research/tests/atx-reg.md` (ATX 2020 C Q17, Q41, Q42), `ut-c.md` (UT fall 2019 C Q49), `nat26.md` (USC 2026 B Q22).
- Research bank: `prior/H_logic_diodes.md` A28-A33, A39; `prior/B_topics.md` 8g.
- Kuphaldt, *Lessons in Electric Circuits*, Vol. IV, ch. 9 "Combinational Logic Functions" (half adder, full adder).
- All About Circuits, "Full adder" and "Half adder" pages (Kuphaldt text).
- Every example: checked in Python.

Multiplexers and Decoders

Rule area **3.c.viii** names "combinational logic circuits (e.g., **multiplexers**, adders)". "e.g." means "for example". So other combinational blocks are also fair: demultiplexers, decoders, encoders, priority encoders, comparators, parity circuits and 7-segment decoders. Multiplexers are ✅ IN. The others are ✅ IN as combinational logic, but less likely.

Key facts

- A **multiplexer** (**mux**, "data selector") picks **one** of several data inputs and sends it to the single output.
- The **select lines** choose which input. A mux with N data inputs needs **$\log_2 N$** select lines.
  - 2:1 mux → 1 select line. 4:1 → 2. 8:1 → 3. 16:1 → 4. 32:1 → 5.
- The selected input number = the select lines read as a binary number. S1S0 = 10 picks input I2.
- A **demultiplexer** (**demux**) does the opposite. It sends one data input to **one** of several outputs. The other outputs are 0.
- A **decoder** (n-to-2^n) has n inputs and 2^n outputs. Exactly **one** output is active: the one whose number matches the input.
- A decoder with an "enable" input is the same circuit as a demux. The enable acts as the data input.
- **Active-low** means the active output is 0, not 1. A bar or a bubble on a pin name shows active-low. The 74LS138 3-to-8 decoder is active-low.
- An **encoder** does the reverse of a decoder. It turns "which one line is on" into a binary number. A 4-to-2 encoder: Y1 = D2 + D3, Y0 = D1 + D3.
- A **priority encoder** handles several inputs on at once. It outputs the number of the **highest-priority** input. It often has a "valid" output V that is 1 when any input is on.
- A **magnitude comparator** compares two binary numbers. Outputs: A > B, A = B, A < B.
  - 1-bit: A > B = AB′. A = B = (A ⊕ B)′ (XNOR). A < B = A′B.
- A **parity bit** is an extra bit that makes the number of 1s even (**even parity**) or odd (**odd parity**). A **parity generator** makes this bit. A **parity checker** tests a received word.
  - Even parity bit = XOR of all the data bits.
  - Parity finds any **odd** number of bit errors. It cannot find 2 errors. It cannot say which bit is wrong.
- A **7-segment display** shows a digit with 7 bars named a to g.
  - a = top, b = upper right, c = lower right, d = bottom, e = lower left, f = upper left, g = middle.
  - A **BCD-to-7-segment decoder** (7447, 7448, 4511) lights the right bars for a 4-bit digit.
- A mux is **universal**. A 2:1 mux with constants 0 and 1 can make NOT, AND and OR.
- A 2^n:1 mux can make **any** function of n variables. Put the truth-table column on the data inputs.

7-segment table (common font)

Digit	Lit segments	Count
0	a b c d e f	6
1	b c	2
2	a b d e g	5
3	a b c d g	5
4	b c f g	4
5	a c d f g	5
6	a c d e f g	6
7	a b c	3
8	a b c d e f g	7
9	a b c d f g	6

- ⚠️ Some fonts draw 6 without a, 7 with f, and 9 without d. A good question tells you the font.

Formulas

- Select lines = $\log_2$(number of data inputs).
- 2:1 mux: Y = S′I0 + S·I1.
- 4:1 mux: Y = S1′S0′I0 + S1′S0·I1 + S1·S0′I2 + S1·S0·I3.
- 2:4 decoder: Y0 = A′B′, Y1 = A′B, Y2 = AB′, Y3 = AB (A is the MSB).
- Mux tree: an N:1 mux from 2:1 muxes needs N − 1 of them.
- 2-bit comparator: (A > B) = A1B1′ + A1A0B0′ + A0B1′B0′.
- Equality of n-bit words: (A = B) = AND of all the bit XNORs.
- Even parity bit = d1 ⊕ d2 ⊕ … ⊕ dn.
- 2:1 mux as a gate: NOT A = mux(S = A, I0 = 1, I1 = 0); AB = mux(S = A, I0 = 0, I1 = B); A + B = mux(S = A, I0 = B, I1 = 1).

Question types

Easy

1. What does a multiplexer (or demux, decoder, encoder) do? ✅ IN

- Seen: not seen in a test as a definition item. The rules name multiplexers.
- How to solve:
  1. Mux: many in, one out. Select lines choose.
  2. Demux: one in, many out.
  3. Decoder: n bits in, one of 2^n lines on.
  4. Encoder: one of 2^n lines on, n bits out.
- Example: "Which device sends one of 8 inputs to a single output?" **Answer: an 8:1 multiplexer.**
- Traps:
  - Mixing up mux and demux. "Multi in" = mux.
  - Calling an encoder a decoder. A decoder **expands** a code into lines.
- Calculator: look up — type: `mux def`

2. How many select lines? ✅ IN

- Seen: not seen in a test. The scioly.org wiki gets this wrong (see Mistakes).
- How to solve:
  1. Select lines = $\log_2$N.
  2. Ask: "2 to what power gives N?"
- Example: 32:1 mux. 2^5 = 32. **Answer: 5 select lines.**
- Traps:
  - √N is wrong. For 16:1, both rules give 4, which hides the error. Try 64:1: $\log_2$ gives 6, √ gives 8. Only 6 is right.
  - If N is not a power of 2, round **up**. 10 inputs need 4 select lines.
- Calculator: calculate — type: `sel 32`

3. Mux output for given selects and data ✅ IN

- Seen: not seen in a Div B test. UT Austin Invitational fall 2019, Div C, INV, Q48 asks for the full 2:1 mux table.
- How to solve:
  1. Read the select lines as a binary number k.
  2. The output = data input Ik.
- Example: 4:1 mux. S1S0 = 10. Inputs I0..I3 = 0, 1, 1, 0. 10_2 = 2. **Answer: Y = I2 = 1.**
- Traps:
  - Reading S0S1 instead of S1S0. S1 is the MSB.
- Calculator: calculate — type: `mux 10 0110`

4. Demux or decoder: which output is active? ✅ IN

- Seen: not seen in a test.

- How to solve:
  1. Read the select (or input) bits as a number k.
  2. Output k is active. All others are inactive.
  3. Active-low: the active output is 0 and the others are 1.
- Example: 1-to-4 demux. S1S0 = 11, D = 1. **Answer: Y3 = 1; Y0, Y1, Y2 = 0.**
- Example 2: 74LS138, enabled, inputs CBA = 101. **Answer: output Y5 goes LOW; the other seven stay HIGH.**
- Traps:
  - Forgetting enable. If the chip is not enabled, no output is active.
- Calculator: calculate — type: `dec 101 low`

5. Which segments light for a digit? ✅ IN (as combinational logic)

- Seen: not seen in a test.
- How to solve:
  1. Draw an 8 and label the bars a to g, clockwise from the top, g in the middle.
  2. Shade the bars the digit uses.
- Example: digit 4. **Answer: b, c, f, g (4 segments).** Digit 7: **a, b, c.**
- Example 2: "Which digits light segment e?" **Answer: 0, 2, 6, 8.**
- Traps:
  - Forgetting segment f on 4. The left upper bar is part of the 4.
  - Different fonts for 6, 7 and 9.
- Calculator: look up — type: `seg 4`

Medium

6. Write the 2:1 mux truth table (8 rows) ✅ IN

- Seen: UT Austin Invitational fall 2019 (2020 season), Div C, INV, Q48 ("2:1 mux full truth table, inputs A, B, S; 8 rows" → Y = A when S = 0, B when S = 1).
- How to solve:
  1. List the 8 rows of A, B, S in counting order.
  2. If S = 0, copy A. If S = 1, copy B.
- Example:

A	B	S	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

Answer: Y = S′A + SB.

- Traps:
  - Using a 4-row table. A 2:1 mux has 3 inputs (2 data + 1 select), so 8 rows.
- Calculator: look up — type: `mux2 tt`

7. Write the mux equation (2:1 or 4:1) and draw it from gates ✅ IN

- Seen: not seen in a test.
- How to solve:
  1. Each data input gets one AND gate.
  2. Each AND also gets the select pattern that picks it (using NOT for 0s).
  3. OR all the AND outputs.

- Example: 2:1 mux: Y = S′I0 + S·I1. Gates: 1 NOT, 2 AND, 1 OR. NAND-only: **4 NANDs** (proven minimum).
- Traps:
  - A 4:1 mux needs 4 three-input ANDs, a 4-input OR and 2 NOTs.
- Calculator: look up — type: `mux4 eq`

8. Build a big mux from small muxes (mux tree) ✅ IN

- Seen: not seen in a test.
- How to solve:
  1. The first layer picks within groups using the low select bits.
  2. The next layer picks among the groups using the higher select bits.
  3. From 2:1 muxes, an N:1 mux needs N – 1.
- Example: 8:1 from 4:1 muxes. Two 4:1 muxes (selects S1 S0) feed one 2:1 mux (select S2). **Answer: 2 four-to-one muxes + 1 two-to-one mux.**
- Example 2: 16:1 from 2:1 muxes. **Answer: 15** (8 + 4 + 2 + 1). From 4:1 muxes: **5** (4 + 1).
- Traps:
  - Putting the MSB select on the first layer. It goes on the **last** layer.
- Calculator: calculate — type: `muxtree 16 2`

9. Encoder and priority encoder output ✅ IN

- Seen: MIT Invitational 2021, Div C, INV, Scenario F Q57-63 (a "last 1" encoder; see type 15). No plain encoder item seen.
- How to solve:
  1. Plain encoder: assume only one input is on. Output its number in binary.
  2. Priority encoder: find the highest-priority input that is on. Output its number.
  3. Valid V = 1 if any input is on.
- Example: 4-to-2 priority encoder, highest number wins. D3D2D1D0 = 0110. D2 is the highest one on. **Answer: Y1Y0 = 10, V = 1.**
- Traps:
  - A plain encoder with two inputs on gives garbage. Y1 = D2 + D3 = 1, Y0 = D1 + D3 = 1 → 11 (wrong). That is why priority encoders exist.
  - All inputs off: the plain encoder outputs 00, which looks like "D0 is on". V tells them apart.
- Calculator: calculate — type: `prienc 0110`

10. 1-bit and 2-bit comparators ✅ IN

- Seen: not seen in a test.
- How to solve:
  1. 1-bit: A > B only for A = 1, B = 0 → AB′. Equal → XNOR.
  2. Multi-bit: compare from the MSB. The first bit that differs decides.
  3. Equal = every bit pair equal = AND of XNORs.
- Example: A = 1010, B = 1001. Bits 3 and 2 match. Bit 1: A has 1, B has 0. **Answer: A > B** (10 > 9).
- Example 2: 2-bit A > B equation: **A1B1′ + A1A0B0′ + A0B1′B0′** (checked on all 16 rows).
- Traps:
  - Comparing from the LSB. Always start at the MSB.
  - A comparator is not subtraction, so it is allowed under rule 3.e.iii.
- Calculator: calculate — type: `cmp 1010 1001`

11. Parity bit and parity checking ✅ IN

- Seen: Princeton Invitational 2020, Div C, INV, Q40 (switch network that lights L1 when an **even** number of 3 switches are on = 3-input parity; see type 16). No pure parity-bit item seen.
- How to solve:
  1. Count the 1s in the data.
  2. Even parity: the parity bit makes the total even.
  3. Odd parity: the parity bit makes the total odd.

4. Checking: count the 1s in the whole received word (data + parity). Wrong total = error.

- Example: data 1011001 has four 1s. **Answer: even parity bit = 0; odd parity bit = 1.**
- Example 2: even parity word 1101 0110 arrives. It has five 1s, which is odd. **Answer: an error happened.**
- Traps:
  - Two flipped bits keep the parity right. The checker misses them.
- Calculator: calculate — type: `parity 1011001 even`

Hard

12. Use a mux to make a logic function ✅ IN

- Seen: not seen in a test. Standard textbook skill (H_logic_diodes A36).
- How to solve (2^n:1 mux, n variables):
  1. Put the variables on the select lines.
  2. Put the truth-table output column on the data inputs (0s and 1s).
- How to solve (2^{n-1}:1 mux, one size smaller):
  1. Put the first n − 1 variables on the selects.
  2. For each select pattern, look at the two rows where only the last variable (x) changes.
  3. Outputs 0, 0 → wire 0. Outputs 1, 1 → wire 1. Outputs 0, 1 → wire x. Outputs 1, 0 → wire x′.
- Example: F(A, B, C) = Σm(1, 2, 6, 7) on a 4:1 mux with selects A, B.
  - AB = 00: rows 0, 1 give 0, 1 → I0 = C.
  - AB = 01: rows 2, 3 give 1, 0 → I1 = C′.
  - AB = 10: rows 4, 5 give 0, 0 → I2 = 0.
  - AB = 11: rows 6, 7 give 1, 1 → I3 = 1.
  - **Answer: I0 = C, I1 = C′, I2 = 0, I3 = 1.**
- Example 2: majority on an 8:1 mux. **Answer: data inputs I0..I7 = 0, 0, 0, 1, 0, 1, 1, 1.**
- Traps:
  - Putting the variables on the select lines in the wrong order. A must be the MSB select.
- Calculator: calculate — type: `muxfn 1,2,6,7`

13. Show that a 2:1 mux is universal ✅ IN

- Seen: not seen in a test. Rule 3.c.viii names universal gate sets.
- How to solve:
  1. Build NOT: S = A, I0 = 1, I1 = 0.
  2. Build AND: S = A, I0 = 0, I1 = B.
  3. With NOT and AND you can build anything.
- Example: mux(S = A, I0 = B, I1 = 1). If A = 0, Y = B. If A = 1, Y = 1. **Answer: Y = A + B (OR).**
- Traps:
  - Without the constants 0 and 1, a mux cannot make NOT. Say that you need them.
- Calculator: look up — type: `mux universal`

14. Decoder plus OR gate makes a function ✅ IN

- Seen: not seen in a test.
- How to solve:
  1. A 3-to-8 decoder gives every minterm of 3 variables.
  2. OR the outputs whose numbers are in the function.
  3. For active-low outputs (74138), use a NAND instead of the OR.
- Example: F = Σm(3, 5, 6, 7) (the majority, also the full-adder carry). **Answer: OR together decoder outputs 3, 5, 6 and 7.**
- Example 2: make a 4-to-16 decoder from two 3-to-8 decoders. Use the 4th input (MSB) on the enables. One chip is enabled when the MSB is 0, the other when it is 1.
- Traps:
  - If more than half the outputs are needed, NOR the other ones instead. It uses a smaller gate.

- Calculator: calculate — type: `decfn 3,5,6,7`

Nationals

15. Design an encoder from a word specification (MIT "last 1") ✅ IN

- Seen: MIT Invitational 2021, Div C, INV, Scenario F Q57-63. Inputs: a 2^n-bit number. Output: the n-bit position of the **last** (rightmost) 1. All-0 input = "don't care". (A **don't care** is an input row whose output does not matter.)
- How to solve:
  1. Number the input bits from the right, starting at 0.
  2. Do the given examples first (Q57-59).
  3. Write the truth table. Put "X" (don't care) on the all-0 row.
  4. For each output bit, write an expression with AND, OR, NOT.
  5. Use the don't care to make the expression shorter.
- Answers (checked in Python against every row):
  - Q57: 0111'0010 → last 1 at position 1 → **001**.
  - Q58: 0011'0100'0000'0000 → position 10 → **1010**.
  - Q59: n = 1, input 11 → position 0 → **0**.
  - Q60 (n = 1): rows in[1] in[0] = 00 → X, 01 → 0, 10 → 1, 11 → 0.
  - Q62: **out[0] = NOT in[0]** (also accepted: in[1] AND NOT in[0]).
  - Q63 (n = 2): **out[1] = NOT (in[1] OR in[0])**; **out[0] = (NOT (in[2] OR in[1] OR in[0])) OR (in[1] AND NOT in[0])**.
- Example: n = 2, input 1100. in[1] = in[0] = 0, so out[1] = 1. in[2] = 1, so the first part of out[0] is 0, and in[1] AND NOT in[0] = 0. out[0] = 0. Output 10 = position 2. ✓ **Answer: 10.**
- Traps:
  - "Last" means rightmost. This is a **lowest-position** priority encoder, not the usual highest-position one.
  - Different correct answers can treat the all-0 row differently. The key accepts that.
- Calculator: calculate — type: `lastone 1100`

16. Parity with switches (3-way and 4-way) ✅ IN

- Seen: Princeton Invitational 2020, Div C, INV, Q40 (draw a network of 3-way SPDT and 4-way DPDT switches for one light). The given table lights L1 on rows 000, 110, 101, 011. That is **even parity** = (S1 ⊕ S2 ⊕ S3)′ (checked on all 8 rows). MIT Invitational 2020, Div C, INV, Q49 is a 4-switch, 2-light version.
- How to solve:
  1. Check the table. If flipping any one switch always toggles the light, it is a parity (XOR or XNOR) circuit.
  2. Use 3-way (SPDT) switches at the two ends.
  3. Put one 4-way (DPDT, crossover) switch in the middle for each extra switch.
  4. Each switch flip swaps which traveler wire is live, so it toggles the light.
- Example: "One light, three switches. Flipping any switch toggles the light. All switches at 0 = light on." **Answer: 3-way, 4-way, 3-way in a chain; L1 = (S1 ⊕ S2 ⊕ S3)′.**
- Traps:
  - With 3 switches you need a 4-way in the middle. Two 3-ways alone cannot do it.
  - XOR or XNOR depends on which position you call "1".
  - This is also in Logic Design Word Problems A8.
- Calculator: look up — type: `3way parity`

17. 7-segment decoder: write one segment's expression ✅ IN (⚠️ hard)

- Seen: not seen in a test.
- How to solve:
  1. Inputs are the BCD bits A B C D (A = MSB, D = LSB).
  2. List the digits (0-9) where the segment is lit.
  3. Codes 10-15 are don't cares.
  4. Simplify using the don't cares.

- Example: segment e is lit for 0, 2, 6, 8. All of these are even (D = 0). 4 is even but has e off. **Answer: e = B′D′ + CD′** (checked on digits 0-9).
- Traps:
  - Forgetting the don't cares. Without them the expression is longer.
  - Fonts: check whether 6 and 9 have tails.
- Calculator: calculate — type: `seg e expr`

18. Gray code converter ⚠ Borderline

- Seen: not seen in a test.
- Why borderline: Gray code is not named. It is binary-coded and built from XOR, so it is combinational logic. Karnaugh map axes use Gray order (00, 01, 11, 10).
- How to solve:
  1. Gray MSB = binary MSB.
  2. Each next Gray bit = XOR of that binary bit and the binary bit to its left.
- Example: binary 1011. G3 = 1. G2 = 1 ⊕ 0 = 1. G1 = 0 ⊕ 1 = 1. G0 = 1 ⊕ 1 = 0. **Answer: 1110.**
- Traps:
  - Going back (Gray → binary) uses the **new binary** bit each time, not the Gray bit.
- Calculator: calculate — type: `gray 1011`

Out of scope (skip)

- ❌ Muxes or decoders driven by a counter or clock (for example a scanning display). **Rule 3.d: sequential logic and digital clock signals.**
- ❌ Shift registers and registers that feed a mux. **Rule 3.d.**
- ⚠/❌ Tri-state buffers and buses. The 2020 rules excluded "three-state logic gates". The 2027 list does not name them, but they are not plain combinational logic. Treat as out.
- ❌ Analog multiplexers and transistor switch internals. **Rule 3.d: semiconductors beyond those listed.**
- ❌ Comparators that subtract (A – B). **Rule 3.e.iii: no arithmetic other than addition.** A logic comparator (compare bit by bit) is IN.
- ❌ Op-amp "comparators". **Rule 3.d (semiconductors beyond those listed).** Do not confuse them with digital comparators.

Mistakes in tests and keys

- **scioly.org Circuit Lab wiki:** says a multiplexer needs √N select lines. **Wrong.** It needs $\log_2 N$.
- **UT Austin Invitational fall 2019 Div C Q48:** key is correct (checked all 8 rows).
- **MIT 2021 Q63:** the key says correct answers may differ on the all-0 row. That is fair because that row is a don't care.

Not confirmed

- The MIT 2020 Q49 truth table (4 switches, 2 lights) has only 15 rows in the text file. Row 0101 is missing. I did not render the page to check it, and I did not solve that circuit.
- 74LS138, 7447, 7448 and 4511 details come from general knowledge and the research bank. I did not open datasheets.
- The 7-segment fonts for 6, 7 and 9 vary by chip. I did not check a datasheet.
- No Div B test has asked a mux, decoder, encoder or comparator question yet. All items seen are Div C.

Sources

- 2027 Div B rules PDF, rules 3.c.viii, 3.d, 3.e.iii (pages labeled B12-B13).
- Test extractions: `circuitlab/research/tests/ut-c.md` (UT fall 2019 C Q48), `prior/G_tests.md` (MIT 2021 Scenario F, MIT 2020 Q49, Princeton 2020 Q40).

- Original files read: `downloads/dlG/mit21_test.txt`, `mit21_key.txt`; `downloads/dlG/pz/p20/Circuit Lab/2020 Princeton Circuit Lab Exam.txt` (Q40 table); `downloads/dlG/mit20_test.txt` (Q49 table).
- Research bank: `prior/H_logic_diodes.md` A34-A43; `prior/B_topics.md` 8g.
- Kuphaldt, *Lessons in Electric Circuits*, Vol. IV, ch. 9 "Combinational Logic Functions" (decoder, encoder, mux, demux, 7-segment).
- All About Circuits worksheets ("Multiplexers and demultiplexers", "Encoders and decoders").
- Every example: checked in Python.

Logic Design Word Problems

Rule area **3.c.viii** names "combinational logic circuits". A **combinational** circuit has outputs that depend only on the inputs right now. It has no memory. A **word problem** gives you a story in English. You turn the story into a truth table, then an expression, then a circuit. Hands-on task **II.c.v** says "Construct a logic circuit to mimic a given truth table". Hands-on task **II.c.vii** says "Wire a 3-way switch using SPDT switches". Both use the skills in this note. This is a high-point area, so this note goes deep.

Related notes: Truth Tables and Boolean Expressions A8 (types 15, 21, 22), Logic Gates and Notation A8 (types 7, 8, 16), Universal Gates (NAND and NOR) A8, Multiplexers and Decoders A8 (types 15, 16), Logic Hardware and Chips A8.

Key facts

- **The design recipe** has six steps. Use it for every word problem.
  1. Name each input with a letter. Write what 1 means for each one (for example, "D = 1 means the door is open").
  2. Name each output. Write what 1 means.
  3. Make a truth table with 2^n rows for n inputs. Count in binary so you skip no row.
  4. Mark rows that can never happen as **don't care** (X). You may set an X to 0 or 1, whichever makes the circuit smaller.
  5. Write the expression. Use **SOP** (sum of products: OR of AND terms) from the 1-rows. Then simplify.
  6. Draw the circuit. Then test every row again.
- **Polarity** means which state you call 1. Changing the polarity of one input changes the gate name. XOR can become XNOR. Always write your polarity down.
- **English word → logic** (the most common translation list):

English words	Logic	Example
"and", "both", "all"	AND	"both doors closed" = A·B
"or", "either", "at least one"	OR	"A or B (or both)" = A + B
"not", "is off", "is released"	NOT	"B released" = B′
"but not"	AND NOT	"A but not B" = A·B′
"unless"	AND NOT	"H, unless W" = H·W′ (see type 6)
"neither A nor B"	NOR	(A + B)′ = A′·B′
"not both"	NAND	(A·B)′
"exactly one of two", "one or the other but not both"	XOR	A ⊕ B
"both the same", "match"	XNOR	(A ⊕ B)′
"majority", "at least 2 of 3"	majority	AB + BC + AC
"if A then B" (implication)	A′ + B	true unless A = 1 and B = 0

- **Standard functions** to memorize:
  - **Majority of 3** (3 judges vote, the majority wins): M = AB + BC + AC = (A + B)(A + C)(B + C).
  - **Exactly one of 3:** A′B′C + A′BC′ + AB′C′. It also equals (A ⊕ B ⊕ C)·(ABC)′.
  - **Odd parity of n** (an odd number of inputs are 1): A ⊕ B ⊕ C ⊕ …
  - **Even parity of n:** the NOT of the odd parity.
  - **Staircase (2-way) light:** XOR or XNOR of the two switches. It depends on polarity.
  - **Priority:** the highest-priority active input wins. Lower inputs need the NOT of every higher input.
- **Multiway switching** (one light, many switch places): use **2 three-way (SPDT) switches** at the ends and **(n – 2) four-way (DPDT crossover) switches** in the middle. The light is a parity function of all n switches.
  - **SPDT** = single pole, double throw. One common terminal connects to one of two others. Electricians call it a "3-way" switch.
  - **DPDT** = double pole, double throw. A **4-way** switch is a DPDT wired so that it swaps (crosses) two wires.
  - **Traveler** = one of the two wires that run between multiway switches.

- **Memory is out of scope.** A spec that needs the circuit to "remember" is **sequential logic** (rule 3.d). Examples: "press once for on, press again for off", "the first player to buzz locks out the others", "the pump starts at low and keeps running until full".
- **Counting rows** helps you check your table. "At least 2 of 4" has 11 one-rows. "Exactly 2 of 4" has 6. "Majority of 3" has 4.

Formulas

- Rows in a truth table = 2^n for n inputs.
- Majority (3 inputs): M = AB + BC + AC.
- Majority with a tie-breaking chair C (4 voters A, B, C, D; pass on 3 or more yes, or on exactly 2 yes when C votes yes): F = C(A + B + D) + ABD.
- At least 2 of 4: the OR of all 6 pairs: AB + AC + AD + BC + BD + CD.
- Exactly one of 3: A′B′C + A′BC′ + AB′C′.
- Priority (A highest, then B, then C): L_A = A, L_B = A′B, L_C = A′B′C.
- Combination lock with code 1011 (A B C D): open = A·B′·C·D.
- 3-bit prime detector (number = 4A + 2B + C; primes 2, 3, 5, 7): F = A′B + AC.
- BCD digit is 5 or more (A = MSB; codes 10-15 are don't cares): F = A + BC + BD.
- Staircase (2 switches): L = A ⊕ B (or (A ⊕ B)′).
- n switches, one light: L = A ⊕ B ⊕ … (or its NOT). Parts: 2 three-way + (n – 2) four-way switches.
- Threshold (weighted) gate: output = 1 when $W_1x_1 + W_2x_2 + b \geq 0$.

Question types

Easy

1. Real-life decision → gate (AND, OR, XOR) ✅ IN

- Seen: Rickards Invitational, 2021 season (test date Dec 5, 2020), Div C, INV, Q46 (9 pts: "Describe a real-life decision synonymous with the following logic gates: AND, OR, XOR"; the key says only "Placeholder"). Also listed in prior notes as deck 15 class material.
- How to solve:
  1. AND: you need **all** conditions. Give one sentence with "both".
  2. OR: **any one** condition is enough. Give one sentence with "either one, or both".
  3. XOR: **exactly one** condition, never both. Give a sentence where "both" is impossible or forbidden.
  4. Name the inputs and say what 1 means. Graders reward clear inputs.
- Example: "Describe a decision for AND, OR and XOR." **Answer:**
  - AND: "I go to the beach if it is sunny AND I finished my homework." (Beach = S·H.)
  - OR: "I take the bus if it rains OR my bike has a flat." (Bus = R + F.)
  - XOR: "At dinner I pick soup XOR salad: one or the other, not both." (Order = A ⊕ B.)
- Traps:
  - Everyday "or" is often exclusive ("tea or coffee?"). Logic OR includes "both". Say which one you mean.
  - Do not give a XOR example where "both" is harmless. Then it is really an OR.
- Calculator: look up — type: `decision gate`

2. Staircase light (two switches) → gate ✅ IN

- Seen: Princeton Invitational 2020, Div C, INV, Q15 (bulb with a switch at each floor; key **B: XOR**). North Pocono Invitational 2020, Div C, INV, Q4 ("a simple 3-way switch is most like which gate?"; key **XNOR** only). MIT Invitational 2020, Div C, INV, Q19 (3-way, 4-way, 3-way picture: "which switch turns on the light?"; key **D: any of them**).
- How to solve:
  1. Each switch can turn the light on or off, whatever the other switch does.
  2. So flipping **one** input always flips the output.
  3. Only XOR and XNOR do that.

4. Choose XOR if "both switches at 0" means the light is OFF. Choose XNOR if it means ON.
5. If only one of the two appears in the choices, pick it.

- Example: "Both switches down → light off. One switch up → light on. Both up → light off." Truth table 0, 1, 1, 0. **Answer: XOR.**
- Traps:
  - The test picks a polarity without telling you. North Pocono accepts only XNOR. Princeton accepts only XOR. Both are defensible.
  - AND and OR are wrong. With AND, the first switch cannot turn the light on alone.
- Calculator: look up — type: `staircase`

3. Sentence → expression (plain AND/OR/NOT) ✅ IN

- Seen: not seen as a Div B test item. Prior research lists it as a standard type (B_topics alarm example; H_logic_diodes A19). Kuphaldt, *Lessons in Electric Circuits* Vol. IV, ch. 7 and the All About Circuits worksheets use the same style.
- How to solve:
  1. Underline each condition. Give it a letter.
  2. Find the main joining word ("or" between two groups, "and" inside a group).
  3. Put brackets around each group.
  4. Write AND as a dot, OR as +, NOT as ′.
- Example: "The alarm sounds if the door is open (D) AND the system is armed (A), OR the panic button is pressed (P)." **Answer: Y = D·A + P.** The truth table (D A P from 000 to 111) is 0, 1, 0, 1, 0, 1, 1, 1 (Python check).
- Traps:
  - Wrong grouping. D·(A + P) is a different function. Check row D = 0, A = 0, P = 1: the panic button must still work.
  - A comma in English often marks the main split.
- Calculator: calculate — type: `expr from words` (enter the expression, get the truth table)

4. Classification detector (deck 15 "mammal" homework) ✅ IN

- Seen: national supervisor practice deck 15 (`15_CircuitLabSlides.Practice15.v1.pptx`, homework). Inputs: A has vertebrae, B has a neocortex, C is an animal, D has hair, E breathes oxygen. Design a gate circuit that outputs 1 for "mammal".
- How to solve:
  1. List the features that a mammal **must** have.
  2. Every required feature goes into one AND gate.
  3. Drop inputs that add nothing. An input is **redundant** when it is always 1 whenever the others are 1.
  4. Feature that must be absent → use its NOT.
- Example: **Answer: M = A·B·C·D.** E is redundant, because every animal with A, B, C and D breathes oxygen. M = A·B·C·D·E is also correct but uses a bigger gate.
- Traps:
  - Using OR. "Has hair OR a neocortex" lets in non-mammals.
  - The grader may want you to say **why** E can go. Write one sentence.
- Calculator: look up — type: `all required → AND`

5. "Light on only when …" with switches or buttons (push-button design) ✅ IN

- Seen: USC Invitational 2026 (Jan 17, 2026), Div B and C, INV (top tier), lab "Light Switch" parts c and d. Part c: "lights the LED if both buttons are released" (NOR). Part d: "lights the LED if and only if button A is pressed and button B is released" (A·B′).
- How to solve:
  1. Write the 4-row truth table for buttons A and B.
  2. Read the gate name from the output column.
  3. Build: buttons in series with the LED = AND. Buttons in parallel = OR.

4. A NOT needs a button that **shorts across** the LED. Keep a resistor in series with the battery so that a press never shorts the battery.
5. Test all 4 button combinations. The USC proctor tests all 4 before signing.

- Example (part d): A must be pressed and B released. **Answer: A in series feeding the LED, B across the LED.** Truth table (A B = 00, 01, 10, 11): 0, 0, 1, 0. With a 9 V battery and 510 Ω, a press of B makes the resistor carry 9 / 510 = **17.6 mA** and dissipate **0.16 W** (Python check). A 1/4 W resistor is safe.
- Traps:
  - "Released" means 0 for a normally open button. Write your polarity.
  - Never put a shorting button straight across the battery.
  - See Logic Hardware and Chips A8 type 22 for how the 4-pin button is wired inside.
- Calculator: calculate — type: `V, R` (gives short current and power)

Medium

6. Tricky English: "unless", "neither … nor", "only if", "but not" ✅ IN

- Seen: not seen as a separate test item. The implication form appears in UT Austin Invitational 2020, Div C, INV, Q38 (truth table of p ⇒ q; key "none of the above", because p ⇒ q = p′ + q). Textbook source: Kuphaldt Socratic worksheets, "Boolean Algebra".
- How to solve:
  1. "A unless B" = A·B′. The word "unless" means "and not".
  2. "Neither A nor B" = A′·B′ = (A + B)′ (a NOR).
  3. "A but not B" = A·B′ (an **inhibit** function).
  4. "Q only if A" means Q needs A. If Q is an output you design, use Q = A·(other conditions).
  5. "If A then B" as a true/false statement = A′ + B.
- Example: "The fan F runs when it is hot (H), unless a window is open (W). The fan never runs when the power switch (S) is off." **Answer: F = S·H·W′.** Only row S = 1, H = 1, W = 0 gives F = 1.
- Traps:
  - Reading "unless" as OR. That turns the fan on whenever a window opens.
  - "Either … or" in a word problem may mean XOR. Look for "but not both".
- Calculator: look up — type: `english word` (returns the gate)

7. Majority vote (3 judges, 3 sensors) ✅ IN

- Seen: not seen in a test. Prior research lists it as the standard voting circuit (B_topics; H_logic_diodes A19). Kuphaldt Vol. IV uses it as a K-map example.
- How to solve:
  1. Output 1 when 2 or 3 inputs are 1.
  2. The 1-rows are 011, 101, 110, 111.
  3. SOP: A′BC + AB′C + ABC′ + ABC.
  4. Simplify: each pair of judges that agree is enough. **M = AB + BC + AC.**
- Example: "Three judges press YES buttons. The light shows PASS on 2 or more YES." **Answer: PASS = AB + BC + AC.** It uses 3 AND gates and one 3-input OR. Python checked all 8 rows. The POS form (A + B)(A + C)(B + C) is the same function.
- Traps:
  - Forgetting row 111 when you list minterms.
  - "At least 2" and "exactly 2" differ only in row 111.
  - With NAND gates only, it needs 6 two-input NANDs, or 4 NANDs if 3-input NANDs are allowed (prior research search). See Universal Gates (NAND and NOR) A8.
- Calculator: calculate — type: `k of n` (n = 3, k = 2)

8. "Exactly one", "at least k of n", "exactly k of n" ✅ IN

- Seen: not seen in a test. Standard textbook variants (H_logic_diodes A19).
- How to solve:
  1. Count how many inputs are 1 in each row.

2. Mark the rows that meet the rule.
3. For "at least 2", OR together every pair of inputs.
4. For "exactly one", write each one-hot row: one input true, all others false.

- Example: "4 sensors. Sound the alarm when at least 2 sensors trip." **Answer: F = AB + AC + AD + BC + BD + CD (6 terms, 11 one-rows out of 16).** "Exactly 2 of 4" has only 6 one-rows. Python check.
- Traps:
  - The number of pairs of n inputs is n(n − 1)/2. For 4 inputs that is 6.
  - "Exactly one of 3" is not the same as 3-input XOR. XOR is also 1 at row 111. Exactly one = (A ⊕ B ⊕ C)·(ABC)′.
- Calculator: calculate — type: `k of n` (n, k, "at least" or "exactly")

9. Alarm or safety interlock with several inputs ✅ IN

- Seen: not seen in a test. Standard textbook type (B_topics; H_logic_diodes A19 seat-belt example).
- How to solve:
  1. Separate **enable** inputs (for example "armed", "key in") from **trigger** inputs (door, window, belt).
  2. OR the triggers together. AND the result with the enable.
  3. Add any override (panic button) with an OR at the end.
- Example: "A buzzer sounds when the key is in (K) and the door is open (D), or when the key is in and the seat belt is not fastened (S = 1 means fastened)." **Answer: F = K·D + K·S′ = K·(D + S′).** Python checked all 8 rows.
- Traps:
  - The belt input is "fastened", so you need S′. Read what 1 means.
  - Factoring out K saves a gate. Graders may ask for the simplest form.
- Calculator: calculate — type: `expr` (check two forms are equal)

10. Priority lights (highest input wins) ✅ IN

- Seen: not seen in this form. MIT Invitational 2021, Div C, INV, Scenario F Q57-63 is the hard cousin (index of the last 1; see Multiplexers and Decoders A8 type 15).
- How to solve:
  1. Rank the inputs. The top one needs nothing extra.
  2. Each lower output = its own input AND the NOT of every higher input.
  3. At most one output is on at a time.
- Example: "Three quiz buttons. A beats B. B beats C. Only one light may be on." **Answer: L_A = A, L_B = A′·B, L_C = A′·B′·C.** Python check: at most one light is on, and a light is on whenever any button is pressed.
- Traps:
  - "The first to press wins and locks out the others" needs memory. That is **sequential logic, rule 3.d: OUT.** The priority version above is combinational and IN.
- Calculator: calculate — type: `priority` (number of inputs)

11. Combination lock or code detector ✅ IN

- Seen: not seen in a test. Textbook type (Kuphaldt Vol. IV, "Combinational logic functions").
- How to solve:
  1. Write the secret code as bits.
  2. Each 1 bit passes straight into an AND gate.
  3. Each 0 bit passes through a NOT first.
- Example: "4 DIP switches A B C D. The lock opens only for 1011." **Answer: OPEN = A·B′·C·D.** Exactly one of the 16 rows is 1 (Python check).
- Traps:
  - A "lock" that needs switches in a time **order** has memory. That is OUT (rule 3.d).
  - Changing the code means moving NOT gates, not adding gates.
- Calculator: calculate — type: `code bits`

12. Number detector on a binary input (prime, at least N, multiple of 3) ✅ IN

- Seen: not seen in a test. Uses whole-number binary, which is IN (rule 3.e.iii).

- How to solve:
  1. Turn each row into its decimal value (A is the MSB).
  2. Mark rows where the number meets the rule.
  3. Write SOP and simplify (or use a K-map, which is ⚠️ Borderline as a method).
- Example: "3-bit number ABC. Output 1 if the number is prime." Primes are 2, 3, 5, 7 (1 is not prime). **Answer: F = A′B + AC.** Python check on all 8 rows.
- Traps:
  - 1 is not prime. 0 is not prime.
  - Get the bit order right. A = 4s place, B = 2s, C = 1s.
- Calculator: calculate — type: `number rule` (set of decimal values)

13. Design with a restricted parts kit (only NAND, only NOR, only AND + NOT) ✅ IN

- Seen: PA Central Regional (~2013), Div C, REG, Q27 (OR from NOT and AND only), Q28 (AND from NOR only), Q29 (AND from NAND only). USC Invitational 2026, Div B, INV, FRQ 36a (and Div C 40a): AND from 2 NOR inverters + 1 NOR. Hands-on kits often hold only one chip type.
- How to solve:
  1. Write the target as SOP.
  2. Use De Morgan to move to the allowed gate.
  3. OR from AND + NOT: A + B = (A′·B′)′.
  4. AND from NOR: AB = NOR(NOR(A, A), NOR(B, B)).
  5. AND from NAND: AB = NAND(NAND(A, B), NAND(A, B)).
- Example: "Build OR using only NOT and AND gates." **Answer: invert A and B, AND them, invert the result: (A′·B′)′ = A + B.** That is 3 NOT + 1 AND. Python checked all forms.
- Traps:
  - A NAND or NOR with tied inputs is a NOT. Count it as a gate.
  - More detail and gate counts: Universal Gates (NAND and NOR) A8.
- Calculator: look up — type: `gate from set`

14. Spot a spec that needs memory (combinational or not?) ✅ IN (recognition only)

- Seen: USC Invitational 2026, Div B, INV, FRQ 36c (Div C 40c): "is this circuit stateless?" (No: two steps had the same inputs but different outputs). The circuit itself (a NOR latch) is OUT for 2027 B.
- How to solve:
  1. Ask: can the same inputs ever need two different outputs?
  2. If yes, the circuit must remember. It is sequential (OUT by rule 3.d).
  3. If the output depends only on the present inputs, it is combinational (IN).
- Example: "One push button. Press once: light on. Press again: light off." The input "pressed" gives on one time and off the next. **Answer: needs memory → sequential → not combinational.** Contrast: "light is on while the button is held" = combinational (L = A).
- Traps:
  - A circuit with a wire fed back from the output to an input is usually sequential.
  - A test may still ask this recognition question. Do not try to design the latch.
- Calculator: look up — type: `memory?`

15. Reverse-engineer: "What does this design do?" ✅ IN

- Seen: PA Central Regional (~2013), Div C, REG, Q26 (four NAND gates and an inverter give outputs C and D; write the formulas unsimplified, then simplified). I read the figure from the original .docx.
- How to solve:
  1. Label every internal wire.
  2. Write each wire as an expression of the inputs.
  3. Make the truth table for the outputs.
  4. Match the table to a known block: XOR, half adder, mux, majority.

- Example (PA Central Q26): Gate 1: M = (AB)′. Gate 2: (A·M)′. Gate 3: (M·B)′. Gate 4: C = ((A·M)′·(M·B)′)′. The inverter: D = M′. **Answer: C = A ⊕ B and D = A·B. The circuit is a half adder made from NANDs** (C = sum, D = carry). Python checked all 4 rows.
- Traps:
  - Do not stop at the long expression. The "simplified" part wants the XOR.
  - The 4-NAND XOR is a classic shape. Learn to recognize it on sight.
- Calculator: calculate — type: `gate netlist` (gates and wires, gives truth table)

16. Truth table from a lab station → expression (analyze, then build) ✅ IN

- Seen: Princeton Invitational 2020, Div C, INV, Lab 2 Q45 (pre-built 3-input breadboard circuit; key table gives Q = 1 at 100, 001, 110, 101, 111; gates NOT, AND, OR). MIT Invitational 2020, Div C, INV, Lab 2 Q58 (Circuit 1 → 2-input OR, Q = A + B) and Q59 (build a circuit for a given table → 3-input NOR).
- How to solve:
  1. Try all 2^n input combinations. Record the output for each.
  2. Read off the 1-rows (or the 0-rows if there are fewer).
  3. Write SOP or POS. Simplify.
  4. Name the gate if it is a single gate.
- Example (Princeton 2020 Lab 2): 1-rows 100, 001, 110, 101, 111. Every row with A = 1 is a 1. Row 001 is the only extra one. **Answer: Q = A + B′C** (NOT on B, AND with C, OR with A). Python checked all 8 rows. (MIT 2020 Q59: only row 000 is 1 → **Q = (A + B + C)′, a 3-input NOR**.)
- Traps:
  - Princeton and MIT built their gates from NPN transistors. Transistors are OUT for 2027 B (rule 3.d), but the logic is IN.
  - Templates list the rows in a strange order (000, 100, 010, 001, …). Read each row's inputs; do not assume binary order.
- Calculator: calculate — type: `1-rows` (gives SOP and simplest form)

17. Threshold (weighted-sum) gate → which gate? ⚠️ Borderline

- Seen: Princeton Invitational 2020, Div C, INV, Q18 (h(x) = 1 if Wx + b ≥ 0; b = –10, $W_1 = W_2 = 20$; "choose all that apply"; key **A: OR**).
- Why borderline: neural networks are not in the rules. The math is only algebra, and the answer is a gate, so a supervisor could ask it.
- How to solve:
  1. Plug in all four input pairs (0 or 1).
  2. Compute $W_1x_1 + W_2x_2 + b$ for each.
  3. Output 1 when the sum is 0 or more.
  4. Match the column to a gate.
- Example (Princeton Q18): sums are –10, 10, 10, 30. Outputs 0, 1, 1, 1. **Answer: OR.** With b = –30 the outputs are 0, 0, 0, 1 → AND. With W = –20 and b = 10 → 1, 0, 0, 0 → NOR. Python check.
- Traps:
  - "≥ 0" vs "> 0" matters when a sum is exactly 0.
  - One threshold unit can never make XOR. XOR is not "linearly separable".
- Calculator: calculate — type: `W1, W2, b`

Hard

18. One light, many switches (multiway switching) ✅ IN

- Seen: Princeton Invitational 2020, Div C, INV, Q40 (10 pts: draw the switch diagram for a 3-switch table; L1 = 1 at 000, 110, 101, 011). MIT Invitational 2020, Div C, INV, Q19 (3-way + 4-way + 3-way picture). Related control-device items: ATX Regional 2021, Div B, REG, Q32 (pick the correct 3-way drawing); UT Austin Invitational 2020, Div B, INV, Q30 (broken traveler); North Pocono 2020, Div C, INV, Q21 (Carter system).
- How to solve:
  1. Check the table for **parity**. Flip one switch in any row. If the light always flips, it is parity.

2. All 0 → light on means even parity: L = (S1 ⊕ S2 ⊕ S3)′. All 0 → off means odd parity.
3. Draw: battery + → common of S1 (3-way). Two travelers → S2 (4-way crossover). Two travelers → common side of S3 (3-way). S3 common → light → battery –.
4. Choose which traveler is "position 0" so the all-0 row matches.
5. For n switches you need 2 three-way + (n – 2) four-way.

- Example (Princeton Q40): table rows 000 → 1, 100 → 0, 010 → 0, 001 → 0, 110 → 1, 101 → 1, 011 → 1, 111 → 0. **Answer: L1 = (S1 ⊕ S2 ⊕ S3)′ (even parity). Wiring: S1 (3-way) → S2 (4-way) → S3 (3-way) → L1, with the travelers lined up so all-0 lights the lamp.** Python checked all 8 rows.
- Traps:
  - Two 3-way switches alone cannot serve three places. The middle one must be a 4-way.
  - The Princeton prompt says "three 3-way (SPDT) and one 4-way" and "two lights". The table has 3 switches and 1 light. Follow the table (2 three-way + 1 four-way).
  - Carter-system 3-way wiring can leave a lamp "off but live". The NEC banned it in 1923 (UT 2020 B Q29).
- Calculator: calculate — type: `switch table` (tests for parity, prints the switch count)

19. Two lights from four switches (multi-output switch network) ✅ IN (Nationals level)

- Seen: MIT Invitational 2020, Div C, INV, Q49 (12 pts: "three 3-way (SPDT) and one 4-way (DPDT) switches that controls two lights in accordance to the truth table"; 5 V DC source and 5 V lights).
- How to solve:
  1. Split the table into one column per light.
  2. For each light, find which switches matter. Flip one switch and see if the light changes.
  3. A parity pattern → a 3-way / 4-way chain. A pattern like "on only when S1 = 0 and S4 = 0" → switches in series.
  4. Share wires. A second light can tap a traveler that is already live.
  5. Test every row of the table.
- Example (MIT 2020 key drawing, read at 200 dpi): S1 (3-way) feeds two travelers. S2 (4-way) crosses them. S3 (3-way) picks one traveler for L1. S4 (3-way) taps the upper traveler before S2 and feeds L2. **Answer from the drawing: L1 = (S1 ⊕ S2 ⊕ S3)′ and L2 = S1′·S4′.** Python: this matches 12 of the 15 printed rows. It differs at 0110, 0011 and 0111, and row 0101 is missing from the table (see Mistakes).
- Traps:
  - The light column may not depend on every switch. Check before you wire.
  - A lamp must never connect directly across the battery in any switch position.
  - The table has only 15 of 16 rows. Ask the supervisor about a missing row.
- Calculator: calculate — type: `switch table` (two output columns)

20. Multi-output design with impossible rows (water tank, sensors) ✅ IN

- Seen: not seen in a test. Textbook style (Kuphaldt Vol. IV; All About Circuits "Karnaugh mapping" worksheets use sensor examples).
- How to solve:
  1. Write one output column per output.
  2. Mark impossible sensor rows. Either use them as don't cares, or add a "fault" output for them.
  3. Write each output separately.
- Example: "Low float L = 1 when water is above the low mark. High float H = 1 when water is above the high mark. Pump P runs until the tank is full. Warning W lights when the water is below the low mark. Fault F lights when the sensors disagree in an impossible way." **Answer: P = H′, W = L′, F = H·L′.** The row L = 0, H = 1 is impossible, so it shows a sensor fault. Python check of all 4 rows.
- Traps:
  - "Start at low, run until full" needs memory between the marks. That is sequential (OUT). The version above uses only present inputs.
- Calculator: calculate — type: `multi-output table`

21. Design with don't cares (BCD input, only 10 legal codes) ✅ IN (⚠️ the K-map method is Borderline)

- Seen: not seen in a test. Textbook type (Kuphaldt Vol. IV; All About Circuits). The rules name truth tables and Boolean expressions, which covers it.
- How to solve:
  1. BCD (binary-coded decimal) uses 4 bits for one digit 0-9. Codes 10-15 never occur.
  2. Mark codes 10-15 as X.
  3. Group the 1s together with any helpful X cells.
- Example: "Light a lamp when the BCD digit ABCD (A = MSB) is 5 or more." 1-rows: 5, 6, 7, 8, 9. **Answer with don't cares: F = A + BC + BD.** Without don't cares you need F = AB′C′ + A′B(C + D), which is longer. Python checked digits 0-9 for both.
- Traps:
  - A don't care is not "always 0". You choose its value.
  - State that you used don't cares. Some keys mark the long form only.
- Calculator: calculate — type: `1-rows, dc-rows`

22. Weighted vote with a tie-breaker (4 inputs) ✅ IN (Nationals level)

- Seen: not seen in a test. Variant of the standard majority type (H_logic_diodes A19 "at least two of four").
- How to solve:
  1. Make the 16-row table. Count the YES votes in each row.
  2. Apply the rule row by row.
  3. Split on the chair's input C: write F when C = 1 and F when C = 0, then join them: F = C·(…) + C′·(…).
  4. Simplify.
- Example: "Four voters A, B, C, D. The motion passes with 3 or 4 YES. With exactly 2 YES, it passes only if chair C voted YES." If C = 1, any one more YES is enough: A + B + D. If C = 0, A, B and D must all vote YES. **Answer: F = C(A + B + D) + ABD (8 one-rows).** Python checked all 16 rows.
- Traps:
  - Splitting on one input (Shannon's expansion) is the fastest method here.
  - Do not forget the 4-of-4 row. It passes too.
- Calculator: calculate — type: `vote rule` (counts per row)

Nationals

23. Fewest gates for a word spec, with a given chip set ✅ IN

- Seen: MIT Invitational 2020, Div C, INV, Q47 (minimum NAND gates for A + AB′ + AB′C; key **0**, because it simplifies to A). Related: Universal Gates (NAND and NOR) A8 types 12-16.
- How to solve:
  1. Turn the words into SOP.
  2. Simplify fully first. A redundant term costs gates.
  3. Map SOP to NAND-NAND (or POS to NOR-NOR).
  4. Count the gates. Divide by 4 gates per quad chip and round up.
- Example: "Majority alarm from 3 sensors, using only 7400 (quad 2-input NAND) chips." **Answer: 6 two-input NANDs (prior exhaustive search) → 2 chips (6 / 4 rounded up), with 2 gates spare.** Python check of the chip count.
- Traps:
  - Simplify before you count. MIT Q47 needs no gate at all.
  - A 3-input NAND chip (7410) changes the count to 4 gates.
- Calculator: calculate — type: `gates, gates per chip`

24. Exactly-one / parity with switches only (no chips) ✅ IN

- Seen: not seen in a test. Combines type 18 with hands-on task II.c.vii.
- How to solve:
  1. Odd parity of 2 = two SPDT switches (a 3-way pair).
  2. Each extra input needs one DPDT (4-way) switch in the chain.

3. For a buzzer that sounds when the switches "disagree", use two SPDT switches wired so that matching positions break the circuit.

- Example: "Two players each set an SPDT switch to 'A' or 'B'. A buzzer sounds when they disagree." **Answer: buzzer = S1 ⊕ S2. Wire S1 common to +, S1's throws to S2's opposite throws (cross the travelers), S2 common → buzzer → –.** Python: outputs 0, 1, 1, 0.
- Traps:
  - Crossing or not crossing the travelers swaps XOR and XNOR.
- Calculator: look up — type: `parity switches`

Out of scope (skip)

- ❌ Toggle lights ("press once on, press again off"), quiz lockouts that remember the first press, pump hysteresis, counters, traffic-light sequences with timing. **Rule 3.d: sequential logic and digital clock signals.**
- ❌ Specs that need subtraction, negative numbers or multiplication. **Rule 3.e.iii: whole numbers in binary and addition only.**
- ❌ Probability of a switch network working. Florida State 2021, Div B, ST, Q33 $(1 – (1 – P)^N)$. **Rule 3.e: math limits** (outside algebra, binary and Boolean logic).
- ❌ Building the gates from NPN transistors (MIT 2020 Lab 2, Princeton 2020 Lab 2). **Rule 3.d: semiconductors beyond those listed.** The logic part is still IN.
- ⚠ K-maps as a required method. They are not named. Use them only as a tool (see Boolean Algebra and Simplification A8 types 16-19).

Mistakes in tests and keys

- **MIT Invitational 2020, Div C, Q49:** the table has 15 rows; row 0101 is missing. The key drawing gives L1 = (S1 ⊕ S2 ⊕ S3)′ and L2 = S1′·S4′. That matches 12 of 15 printed rows. It disagrees at 0110 (drawing 11, table 10), 0011 (drawing 00, table 10) and 0111 (drawing 10, table 00). I read the drawing at 200 dpi, so check it yourself.
- **Princeton Invitational 2020, Div C, Q40:** the prompt says "three 3-way (SPDT) and one 4-way (DPDT) switches" and "two lights". The table has only S1-S3 and one light. The correct parts are 2 three-way + 1 four-way.
- **North Pocono Invitational 2020, Div C, Q4:** the key accepts only XNOR for the 3-way switch. XOR is also defensible. It depends on polarity.
- **Rickards Invitational 2021 season, Div C, Q46:** the key says only "Placeholder". There is no official sample answer.
- **Deck 15 (national practice slides):** defines XNOR as "true only if A NOR B is true". Wrong. XNOR is 1 when A = B.

Not confirmed

- **Div B use:** I found no Div B test item that is a full word-problem design. All seen design items are Div C or a top invitational lab. The 2027 Div B rules make it fair game.
- **"USC 2026" is an invitational.** The research file `tests/nat26.md` shows that the files say "USC Invitational 2026, January 17, 2026". The real 2026 Nationals tests were not in the downloads.
- **MIT 2020 Q49:** my reading of the key drawing could be wrong at a wire crossing. I did not find the intended row 0101.
- **PA Central Q26:** the key is not in the downloads. My answer (half adder) comes from the figure in the .docx.
- **Princeton 2020 Lab 2** circuit: I used the key's truth table. I did not see the breadboard.
- The 6-NAND majority count comes from the prior research search (H_logic_diodes A24). I did not re-run that search.
- All other examples are my own designs. I checked each one in Python.

Sources

- 2027 Div B rules PDF: rules 3.c.viii, 3.d, 3.e.iii-iv, II.c.v, II.c.vii (pages labeled B12-B13).

- Original test files read: `downloads/dlG/pz/p20/Circuit Lab/2020 Princeton Circuit Lab Exam.txt` and `Answer Key.txt` (Q15, Q18, Q40, Lab 2 Q45); `downloads/dlG/mit20_test.txt`, `mit20_key.txt`, `mit20_key.pdf` page 10 rendered (Q19, Q47, Q49, Lab 2 Q58-59); `downloads/dlG/rsoiOldC_test.txt`, `rsoiOldC_key.txt` (Q46); `downloads/dlG/wiki/central.txt` and `Circuit_Lab_Test_Central.docx` image 15 (Q26-29); `downloads/dlFind/usc26_Btest.txt` (lab).
- Test extractions: `circuitlab/research/tests/nat26.md`, `misc-a.md`, `ut-b.md`, `ut-c.md`, `atx-reg.md`, `state.md`; `prior/G_tests.md`, `prior/F_slides.md` (deck 15), `prior/B_topics.md` §8, `prior/H_logic_diodes.md` A19-A24.
- Kuphaldt, *Lessons in Electric Circuits*, Vol. IV (Digital), ch. 7 "Boolean Algebra", ch. 8 "Karnaugh Mapping", ch. 9 "Combinational Logic Functions".
- All About Circuits Socratic worksheets: "Boolean algebra", "Karnaugh mapping", "Switch and relay logic".
- Every worked example: checked in Python (`check_logic.py`, `check_hw.py`, `q49.py` in this session's scratchpad).

Logic Hardware and Chips

This note covers the **real parts** that make logic: chips, breadboards, switches, relays, diodes and resistors. Rule **3.c.viii** covers the gates. Rule **3.c.vi** covers switches and relays. Rule **3.c.vii** covers diodes and LEDs. Rule **II.b** says "Participants must be familiar with the operation of breadboards and multimeters". Hands-on tasks **II.c.v** (mimic a truth table) and **II.c.vi** (troubleshoot) need this hardware. Chip numbers and voltage levels are not named in the rules. So written questions on them are ⚠ Borderline. They are still very likely at a hands-on station.

Related notes: Logic Gates and Notation A8 (types 7, 9, 11-14, 16), Universal Gates (NAND and NOR) A8 (type 12, chip counts), Logic Design Word Problems A8.

Key facts

Chips (integrated circuits, ICs)

- A **DIP** (dual in-line package) chip has two rows of pins. A 14-pin DIP is the usual gate chip.
- **Pin 1** is next to the notch or the dot. Pins count **counter-clockwise** when you look at the top. Pin 14 sits across from pin 1.
- On a 14-pin chip, pin n faces pin (15 – n). On a 16-pin chip, pin n faces pin (17 – n).
- **Power pins:** 14-pin 74xx: pin 14 = V_CC (+5 V), pin 7 = GND. 16-pin 74xx: pin 16 = V_CC, pin 8 = GND. 4000-series 14-pin: pin 14 = V_DD (+), pin 7 = V_SS (ground).
- A chip with no power does nothing. Gate chips have no power wire drawn in a logic diagram. You must add it.
- **"Quad 2-input"** means 4 gates, each with 2 inputs. **"Hex inverter"** means 6 NOT gates.

Chip (TTL / CMOS)	Contents	Notes
7400 / 4011	quad 2-input NAND	the "universal" chip
7402 / 4001	quad 2-input NOR	7402 pinout differs (output first)
7404 / 4069	hex inverter (NOT)	6 gates
7408 / 4081	quad 2-input AND	
7432 / 4071	quad 2-input OR	
7486 / 4070	quad 2-input XOR	
74266 / 4077	quad 2-input XNOR	74266 is open-collector
7410	triple 3-input NAND	
7411	triple 3-input AND	
7427	triple 3-input NOR	
7420	dual 4-input NAND	
7430	one 8-input NAND	
74283 (7483)	4-bit binary full adder	see Adders A8
74151 / 74153 / 74157	8:1 / dual 4:1 / quad 2:1 mux	see Multiplexers and Decoders A8
74138	3-to-8 decoder, active-low	

Pinouts (top view, pin 1 at the notch)

- **7400, 7408, 7432, 7486** (same layout): gate 1: in 1, 2 → out 3. Gate 2: in 4, 5 → out 6. Pin 7 GND. Gate 3: out 8 ← in 9, 10. Gate 4: out 11 ← in 12, 13. Pin 14 V_CC.
- **7402 (NOR) trap:** each gate has its **output first**. Gate 1: out 1 ← in 2, 3. Gate 2: out 4 ← in 5, 6. Pin 7 GND. Gate 3: in 8, 9 → out 10. Gate 4: in 11, 12 → out 13. Pin 14 V_CC.
- **7404 (hex NOT):** in 1 → out 2, in 3 → out 4, in 5 → out 6, pin 7 GND, out 8 ← in 9, out 10 ← in 11, out 12 ← in 13, pin 14 V_CC.
- **4011, 4001, 4081, 4071, 4070** (CMOS, same layout): in 1, 2 → out 3. Out 4 ← in 5, 6. Pin 7 V_SS. In 8, 9 → out 10. Out 11 ← in 12, 13. Pin 14 V_DD. This is **not** the 7400 layout.
- **74HC00** (high-speed CMOS) uses the 7400 layout, not the 4011 layout.

Families and voltage levels

- **TTL** (transistor-transistor logic) = the 74xx, 74LSxx chips. Supply 5 V ± 5% (4.75-5.25 V).
- **CMOS** (complementary metal-oxide semiconductor) = 4000 series (3-15 V), 74HC (2-6 V), 74HCT (5 V, TTL-level inputs).
- **Logic levels** for TTL:
  - Input LOW: V_IL ≤ 0.8 V. Input HIGH: V_IH ≥ 2.0 V. Between 0.8 V and 2.0 V the input is **undefined**.
  - Output LOW: V_OL ≤ 0.4 V. Output HIGH: V_OH ≥ 2.4 V (standard TTL), 2.7 V (LS).
- **Logic levels** for 4000-series CMOS at 5 V: input LOW ≤ 1.5 V, input HIGH ≥ 3.5 V (30% and 70% of V_DD). Outputs swing almost rail to rail (about 0.05 V and 4.95 V).
- **Noise margin** = how much noise a signal can take and still be read correctly.
  - NM_H = V_OH − V_IH. NM_L = V_IL − V_OL.
  - Standard TTL: NM_H = 2.4 − 2.0 = 0.4 V. NM_L = 0.8 − 0.4 = 0.4 V.
- **Static discipline:** a gate obeys it when its outputs are always inside the valid input ranges: V_OH ≥ V_IH and V_OL ≤ V_IL.
- **Floating input** = an input connected to nothing. A floating TTL input usually acts as HIGH, but it picks up noise. A floating CMOS input is undefined and can make the chip draw extra current. Never leave inputs floating.
- **Unused inputs:** tie an unused AND/NAND input HIGH (or to a used input). Tie an unused OR/NOR input LOW.
- **Sourcing** = the output pushes current out (into an LED to ground). **Sinking** = the output pulls current in (from an LED tied to +5 V). TTL sinks much more current than it sources. A 74LS output sinks up to 8 mA but sources only 0.4 mA.
- **Fan-out** = how many inputs one output can drive. Standard TTL: 10.

Breadboards

- A **breadboard** is a board with spring clips under the holes. Holes in one connected group share one clip.
- Each numbered row has two groups of 5 holes (a-e and f-j). The two groups are **not** joined.
- The **center trench** (gap) splits the two groups. A DIP chip **straddles** the trench, so each pin gets its own row.
- The long **power rails** (+ and –) run along the edges. On some boards each rail is split in the middle. Rule II.b lets you ask the supervisor about the internal wiring.

Switches, relays and diodes as logic

- **Switch logic** (Claude Shannon, 1937): contacts in **series** = AND. Contacts in **parallel** = OR. A **normally closed** (NC) contact = NOT of its control.
- **NO** (normally open) contact: closed only while pressed or energized. **NC:** open only while pressed or energized.
- A **switch across a load** (a shunt switch) turns the load off when pressed. That gives NOT.
- A **relay** is a switch worked by an electromagnet coil. A small coil current switches a separate, often larger, circuit. Relays were the gates of the first digital computers.
- **Diode logic** (DL): diodes and one resistor can make OR and AND. They **cannot** make NOT, because diodes cannot invert or amplify. So diode logic is not universal.
  - Diode OR: input diodes point **toward** the output; a pull-down resistor goes to ground. Output ≈ highest input − 0.7 V.
  - Diode AND: input diodes point **toward** the inputs; a pull-up resistor goes to +V. Output ≈ lowest input + 0.7 V.
- **Pull-up resistor:** a resistor from the input to +V. The input reads HIGH until a switch pulls it to ground. That makes an **active-low** input (pressed = 0).
- **Pull-down resistor:** a resistor from the input to ground. The input reads LOW until a switch connects it to +V (pressed = 1).

Formulas

- Chips needed = (gates needed) ÷ (gates per chip), rounded up.
- Pin across from pin n: 15 − n (14-pin), 17 − n (16-pin).
- Noise margins: NM_H = V_OH − V_IH; NM_L = V_IL − V_OL.
- Static discipline holds when V_OH ≥ V_IH and V_OL ≤ V_IL.

- Fan-out = the smaller of I_OL / I_IL and I_OH / I_IH.
- LED resistor, output sourcing: R = (V_OH − V_LED) / I. Output sinking: R = (V_CC − V_LED − V_OL) / I.
- Pull-down current when the switch is closed: I = V / R. Power: P = V² / R.
- Largest TTL pull-down that still reads LOW: R_max = V_IL / I_IL (0.8 V / 1.6 mA = 500 Ω for standard TTL).
- Relay: coil current I = V / R_coil. Coil power P = V² / R_coil.
- Diode OR output ≈ V_max − 0.7 V. Diode AND output ≈ V_min + 0.7 V (silicon diodes).
- A multi-stage diode gate loses 0.7 V per stage: two OR stages from 5 V → 5 − 0.7 − 0.7 = 3.6 V.

Question types

Easy

1. Chip number → gate (or gate → chip) ⚠️ Borderline (chip numbers are not named; likely in a hands-on kit)

- Seen: not seen in a test. Prior research lists the chip numbers (H_logic_diodes C4; B_topics §8).
- How to solve:
  1. Learn the six common numbers: 00 NAND, 02 NOR, 04 NOT, 08 AND, 32 OR, 86 XOR.
  2. Ignore the letters in the middle (LS, HC, HCT). They name the family, not the gate.
  3. For 4000 CMOS: 4011 NAND, 4001 NOR, 4069 NOT, 4081 AND, 4071 OR, 4070 XOR.
- Example: "A kit has a 74HC08 and a 74LS32. What gates do you have?" **Answer: 4 two-input AND gates and 4 two-input OR gates.**
- Traps:
  - 7402 is NOR, not "the second NAND".
  - "Quad" = 4 gates. "Hex" = 6 gates. "Triple 3-input" = three gates with 3 inputs each.
- Calculator: look up — type: `chip number`

2. Find pin 1 and the power pins ⚠️ Borderline (practical skill for II.c.v)

- Seen: not seen in a test. Prior research (H_logic_diodes A48, C4).
- How to solve:
  1. Find the notch or dot. Pin 1 is on its left when the notch points up (top view).
  2. Count down the left side, then up the right side (counter-clockwise).
  3. 14-pin 74xx: V_CC = 14, GND = 7. 16-pin: V_CC = 16, GND = 8.
- Example: "Which pin faces pin 3 on a 14-pin chip?" **Answer: pin 15 − 3 = 12.** "Which pin faces pin 1?" **Pin 14** (V_CC). Python check.
- Traps:
  - Counting clockwise when the chip is upside down. Always find the notch first.
  - On a 16-pin chip, GND is pin 8, not pin 7.
- Calculator: calculate — type: `pins, n`

3. Breadboard: which holes are connected? ✅ IN (rule II.b)

- Seen: national supervisor handout `Circuit_Board_Resistors.pdf` (F_slides: rails run the full length; A-E connected, F-J connected, the halves not joined). Every hands-on station uses a breadboard (USC Invitational 2026 lab, UT, MIT, Princeton labs).
- How to solve:
  1. Five holes in one half-row are one node.
  2. The two halves of a row are separate nodes.
  3. Each rail is one node (check for a split in the middle).
  4. Two parts in the same half-row are connected, even without a wire.
- Example: "A resistor goes from 10a to 15a. An LED goes from 10c to 10h. Are the resistor and LED connected?" **Answer: yes, at row 10 (holes a-e). The LED's other leg is at 10h, which is the other half and a different node.**
- Traps:
  - A part with both legs in the same half-row is **shorted out**.

  - A chip placed on one side of the trench shorts its opposite pins together.
- Calculator: look up — type: `breadboard`

4. Find the internal wiring of a 4-pin push button ✅ IN (rule 3.c.vi; troubleshooting skill)

- Seen: USC Invitational 2026, Div B and C, INV, lab part a (5 pts: use the battery, resistor and LED as a continuity tester; draw the inside of the button).
- How to solve:
  1. Build a tester: battery → resistor → LED → two probe wires.
  2. Touch every pair of pins. There are 6 pairs for 4 pins.
  3. Pairs that light the LED with the button released are **always joined**.
  4. Pairs that light only when pressed are joined **by the switch**.
- Example: **Answer: the pins on each side are joined in pairs inside (2 always-on pairs). The switch bridges the two pairs when pressed (4 more pairs light only then).** 2 + 4 = 6 pairs (Python check).
- Traps:
  - Using two pins that are always joined. Then the "button" is always on.
  - Use the diagonal pins. Diagonal pins are never the same always-joined pair.
- Calculator: look up — type: `4-pin button`

5. Relay basics: name it, say why it is used ✅ IN (rule 3.c.vi)

- Seen: UF Northern Regional 2019, Div C, REG, Q34 (identify a relay and its electromagnet). ATX Regional 2026, Div B, REG, MC39 ("where is a relay best used"; key: high power, high voltage). Cobra Invitational 2019, Div B, INV, Q28 (SPST NO relay symbol) and station S3 (real SPDT relay, 12 V 60 A).
- How to solve:
  1. A relay = coil + contacts. The coil current makes a magnet that moves the contacts.
  2. Its use: a small, low-power signal switches a separate, high-power circuit.
  3. The two circuits are **isolated**. No wire joins them.
- Example: "Why use a relay to start a car's starter motor from the key switch?" **Answer: the key switch carries only the small coil current. The relay's heavy contacts carry the large motor current.**
- Traps:
  - "Relay" is not an amplifier of voltage. It is a switch.
  - A **latching** relay holds its state with no coil power. That memory is sequential (OUT as a logic design topic).
- Calculator: look up — type: `relay`

6. Hardware history (Boole, Shannon, the integrated circuit) ⚠️ Borderline (only Boole is on the rule 3.c.i list)

- Seen: not seen in a hardware form. Boole is new on the 2027 history list.
- How to solve: match the person and the idea.
  - George Boole (1815-1864): Boolean algebra. *The Laws of Thought*, 1854.
  - Claude Shannon: 1937 MIT master's thesis, "A Symbolic Analysis of Relay and Switching Circuits". It applied Boole's algebra to relay switches.
  - Jack Kilby (Texas Instruments, 1958) and Robert Noyce (Fairchild, 1959): the integrated circuit.
  - The 7400 TTL family came from Texas Instruments in the mid-1960s.
- Example: "Who first showed that relay circuits follow Boolean algebra?" **Answer: Claude Shannon (1937).**
- Traps: Boole invented the algebra, not the circuits. Shannon linked them.
- Calculator: look up — type: `person`

Medium

7. Read a pinout: which pins form one gate? ⚠️ Borderline (needed for II.c.v)

- Seen: not seen in a test. Prior research flags the 7402 trap (B_topics §8; H_logic_diodes C4).
- How to solve:
  1. Get the datasheet or the supervisor's pin drawing. You may ask (rule II.b covers breadboard wiring).
  2. Find one gate's two inputs and one output.
  3. Check whether the chip lists the output first (7402) or last (7400).

- Example: "On a 7400, which pins are gate 3?" **Answer: inputs 9 and 10, output 8.** "On a 7402, which pins are gate 1?" **Output 1, inputs 2 and 3.**
- Traps:
  - Wiring a 7402 with the 7400 layout drives an output into an output.
  - A 4011 is a NAND like the 7400, but its pins differ (out 4 ← in 5, 6).
- Calculator: look up — type: `chip, gate number`

8. Switch network → lamp table (several lamps) ✅ IN (rule 3.c.vi)

- Seen: UT Invitational 2019, Div B (2020 season), INV, Q43 (3 SPDT switches and 3 lamps; L1 on iff S1; L2 iff S1·S2; L3 iff S1·S2·S3). UT Austin Invitational fall 2019, Div C, INV, Q44 (same idea: S1 feeds all; S2 feeds L2 and L3; S3 feeds L3). MIT Invitational 2020, Div C, INV, Q19 (3-way/4-way/3-way: any switch turns on the light).
- How to solve:
  1. Trace a path from + to each lamp to –.
  2. Write the condition for each path. Series = AND, parallel = OR.
  3. Fill the table one lamp at a time.
- Example (UT 2019 B Q43 shape): S1 feeds everything. S2 sits after S1. S3 sits after S2. **Answer: L1 = S1, L2 = S1·S2, L3 = S1·S2·S3. Only 1 of 8 rows lights all three.**
- Traps:
  - UT 2019 B Q43: with S1 off, the drawing shorts the battery (prior research note). Check every switch position for a short.
- Calculator: calculate — type: `paths` (switch list per lamp)

9. Diode-steered lamps (diode OR with switches) ✅ IN (rules 3.c.vi, 3.c.vii)

- Seen: UF Northern Regional 2019, Div B, REG, Q1-3 (Image 1: 1 V source; SW1 feeds three lamps through diodes D1-D3; SW2, SW3, SW4 each feed one lamp directly). Keys: Q1 SW1 closed → **all 3 lamps (E)**. Q2 SW2 closed → **LAMP1 (A)**. Q3 only LAMP1 and LAMP2 lit → **SW2 and SW3 (D)**.
- How to solve:
  1. Each lamp is fed by two paths: its own switch, or SW1 through a diode. So LAMP_k = SW1 + SW_own (an OR).
  2. The diodes stop current flowing **back** from one lamp's switch to the other lamps.
  3. For "which switches give this pattern?", find the pattern that needs no extra lamp.
- Example (Q3): LAMP1 = SW1 + SW2, LAMP2 = SW1 + SW3, LAMP3 = SW1 + SW4. SW1 would light LAMP3 too, so SW1 must be open. **Answer: SW2 and SW3 closed, SW1 and SW4 open** (the only pattern; Python check).
- Traps:
  - Without the diodes, SW2 would also light LAMP2 and LAMP3 through the shared bus.
  - The test uses a 1 V source with silicon diodes. Real diodes drop about 0.7 V, so lamps on the diode path would be almost dark. The test treats the diodes as ideal.
- Calculator: calculate — type: `lamp = OR of feeds`

10. Diode logic gates: which gate, what output voltage? ✅ IN (rules 3.c.vii, 3.c.viii)

- Seen: national practice deck 15 picture (diode OR: diodes from inputs to an output resistor to ground; diode AND: pull-up resistor to +E, diodes from output to inputs; "can diodes alone make NOT?" → no). Prior research H_logic_diodes B12.
- How to solve:
  1. Find the resistor. To ground = OR. To +V = AND.
  2. OR: the highest input wins. Output ≈ V_max – 0.7 V.
  3. AND: the lowest input wins. Output ≈ V_min + 0.7 V.
  4. Current = voltage across the resistor ÷ R.
- Example: "Diode AND, 5 V pull-up, 1 kΩ. A = 0 V, B = 5 V." **Answer: output = 0 + 0.7 = 0.7 V (logic 0). Current = (5 – 0.7) / 1000 = 4.3 mA, through diode A.** Diode OR with A = 5 V, B = 0 V, 1 kΩ pull-down: **output = 4.3 V (logic 1).** Python check.
- Traps:

  - Each stage loses 0.7 V. Two OR stages from 5 V give only 3.6 V.
  - Diodes cannot make NOT. A NOT needs an active part, such as a transistor or a relay.
- Calculator: calculate — type: `gate type, inputs, V, R`

11. Pull-up or pull-down resistor on a switch input ⚠️ Borderline (pull resistors are not named; the circuit is a plain DC divider)

- Seen: UT Austin Invitational 2025, Div B, INV, Section D Q5g (draw a pull-down switch input: supply + → switch → node V_o → resistor → supply –).
- How to solve:
  1. Pull-down: switch from +V to the input; resistor from the input to ground. Open = 0 V (LOW). Closed = V (HIGH).
  2. Pull-up: resistor from +V to the input; switch from the input to ground. Open = HIGH. Closed = LOW (active-low).
  3. Current only flows when the switch is closed: I = V / R.
- Example: "5 V pull-down input, 10 kΩ. Find the current and power when the switch is closed." **Answer: I = 5 / 10 000 = 0.5 mA. P = 25 / 10 000 = 2.5 mW.** Python check.
- Traps:
  - With no resistor, the open switch leaves the input **floating**. The gate may read noise.
  - With no resistor in the pull-up, closing the switch shorts the supply.
- Calculator: calculate — type: `V, R`

12. LED on a gate output: sourcing or sinking, resistor value ✅ IN (LED use, 3.c.vii)

- Seen: not seen in a written test. MIT Invitational 2019, Div C, INV, Station 2-3c ("purpose of the LED series resistors" → limit current). Prior research H_logic_diodes A52.
- How to solve:
  1. LED from the output to ground: lights when the output is HIGH (sourcing). R = (V_OH – V_LED) / I.
  2. LED from +V into the output: lights when the output is LOW (sinking). R = (V_CC – V_LED – V_OL) / I.
  3. TTL sinks well and sources poorly. Prefer sinking for TTL.
  4. Pick the next standard value up (for example 330 Ω).
- Example: "74LS gate, 5 V, red LED (2.0 V). Sink 8 mA at V_OL = 0.5 V." **Answer: R = (5 – 2.0 – 0.5) / 0.008 = 312.5 Ω. Use 330 Ω. Then I = 2.5 / 330 = 7.6 mA. The LED is ON when the output is LOW.** Python check.
- Traps:
  - A sinking LED shows the **opposite** of the logic value.
  - A 74LS output sources only 0.4 mA. An LED to ground would be very dim.
- Calculator: calculate — type: `V_CC, V_LED, V_out, I`

13. Is this voltage a 0, a 1 or undefined? ⚠️ Borderline

- Seen: not seen as a TTL item. MIT Invitational 2021, Div C, INV, Scenario H Q68-69 uses the same ideas (see type 19). Prior research H_logic_diodes A51.
- How to solve:
  1. Get the family's V_IL and V_IH.
  2. Below V_IL → 0. Above V_IH → 1. Between → undefined.
- Example: "A TTL input sits at 1.4 V." **Answer: undefined (between 0.8 V and 2.0 V).** "A 4000 CMOS input at 5 V supply sits at 3.0 V." **Undefined (between 1.5 V and 3.5 V).**
- Traps:
  - A meter reading in the gap often means a floating input or a bad connection.
  - CMOS thresholds scale with the supply voltage. TTL thresholds do not.
- Calculator: look up — type: `family, V`

14. Unused inputs: where do they go? ⚠️ Borderline

- Seen: not seen in a test. Prior research (B_topics §8; H_logic_diodes A48, A51).
- How to solve:
  1. AND/NAND: tie an unused input HIGH (or to a used input). HIGH does not change an AND.

2. OR/NOR: tie an unused input LOW (or to a used input). LOW does not change an OR.
3. Unused whole gates on a CMOS chip: tie their inputs to V_DD or ground.

- Example: "You have only a 7410 (3-input NAND) and need a 2-input NAND." **Answer: tie the third input HIGH. NAND(A, B, 1) = NAND(A, B).** For a 3-input NOR used as 2-input, tie the third input LOW. Python checked both.
- Traps:
  - Tying an unused NAND input LOW forces the output HIGH forever.
  - Leaving it floating "works" on TTL until noise hits. Do not rely on it.
- Calculator: look up — type: `gate type`

15. Identify a mystery gate or chip by probing (hands-on) ✅ IN (II.c.v, II.c.vi)

- Seen: UT Austin (ATX) Regional 2019, Div C, REG, hands-on (3 mystery gates on a breadboard with LEDs; identify them and give truth tables). MIT Invitational 2019, Div C, INV, Station 2-3 (LEDs blink through the inputs; key table 1, 1, 1, 0 → **NAND**). MIT Invitational 2020, Div C, INV, Lab 2 Q58 (Circuit 1 → **2-input OR**). Princeton Invitational 2020, Div C, INV, Lab 2 Q45.
- How to solve:
  1. Apply all 2^n input combinations (switches or jumper wires to + or –).
  2. Read the output with an LED or a multimeter. TTL: above about 2 V = 1, below 0.8 V = 0.
  3. Write the output column. Match it to a gate (see Truth Tables and Boolean Expressions A8 type 1).
- Example (MIT 2019 Station 2-3): outputs for 00, 01, 10, 11 are 1, 1, 1, 0. **Answer: NAND.**
- Traps:
  - Check whether the LED is sourcing or sinking. A sinking LED shows the opposite value (type 12).
  - In the MIT 2019 station, a counter drives the inputs. The counter is sequential (OUT), but reading the LEDs is IN.
- Calculator: look up — type: `output column`

16. Two-LED indicator with one button (red when up, green when pressed) ✅ IN (II.c.iii, II.c.vii relatives)

- Seen: ATX Regional 2026, Div B and C, REG, hands-on (2 LEDs, 1 button, 10 wires, breadboard, multimeter, 2 resistors; red ON and green OFF when up, the reverse when pressed).
- How to solve:
  1. You need a NOT between the two LEDs: green = button, red = button′.
  2. An **SPDT** (3-pin) button does it: + → common; NC throw → R1 → red → –; NO throw → R2 → green → –.
  3. With only an SPST normally-open button, use the shunt trick: the button shorts the red LED and feeds the green one. Keep a series resistor before the shorting button.
- Example: **Answer (SPDT): + → COM; NC → 470 Ω → red → –; NO → 470 Ω → green → –. Up: red only. Pressed: green only.**
- Traps:
  - The ATX key "solution" does not work (see Mistakes).
  - A multimeter on DC volts cannot be a power source.
- Calculator: look up — type: `2-LED toggle`

17. Logic with shorting buttons (NOR, A·B′) and LED current ✅ IN (II.c.v)

- Seen: USC Invitational 2026, Div B and C, INV, lab parts c, d and e (NOR with both buttons across the LED; A·B′ with A in series and B across; e: one button dims the LED with a second 510 Ω in parallel). UT Invitational 2019, Div B, INV, Q49 (= UT fall 2019 Div C H49): trace a breadboard photo of 3 buttons and 4 LEDs; key LED power 7.08 mW.
- How to solve:
  1. Series button = AND. Parallel buttons = OR. A button across the LED = NOT.
  2. Put the series resistor between the battery and everything else.
  3. LED current = (V – V_LED) / R when nothing shorts it.
- Example (USC part e, red LED with V_f = 2.0 V, 9 V, 510 Ω): normal current (9 – 2) / 510 = **13.7 mA**. Pressing adds 510 Ω across the LED, which takes 2.0 / 510 = 3.9 mA. **LED current = 13.7 – 3.9 = 9.8 mA (–29%), so the LED dims but stays on.** For a blue LED (3.0 V): 11.8 mA → 5.9 mA. UT 2019 power from the key's readings: 1.88 V × 3.5 mA + 1.79 V × 0.28 mA = **7.08 mW**. Python check.

- Traps:
  - Shorting the LED with no series resistor shorts the battery.
  - UT 2019 key: "theoretical power N/A" because the LEDs had no series resistor.
- Calculator: calculate — type: `V, V_LED, R, R_shunt`

Hard

18. Relay as a logic gate: coil current, coil power, gate resistance ✅ IN (rule 3.c.vi)

- Seen: MIT Invitational 2021, Div C, INV, Scenario J Q70-74 (relay V_IH = 9.0 V, V_IL = 3.0 V, R_i = 120 Ω, R_on = 0.80 Ω, R_off = 150 kΩ). Keys: Q70 **75 mA**, Q71 **680 mW**, Q72 **0**, Q73 **60 Ω**, Q74 **0.80 Ω**.
- How to solve:
  1. Minimum ON current = V_IH / R_i.
  2. Minimum ON power = V_{IH}^2 / R_i.
  3. Minimum OFF power = 0 (the input can be 0 V).
  4. The relay NOT gate has one coil from 12 V to the input and one from the input to ground. They are in parallel as seen from the input: R_in = 120 ‖ 120.
  5. Output resistance = one closed contact = R_on.
- Example: **Answer: I = 9.0 / 120 = 75 mA. P = 9.0^2 / 120 = 0.675 W ≈ 680 mW. R_in = 60 Ω. R_out = 0.80 Ω.** Python check.
- Traps:
  - "Guaranteed on" means use V_IH, not the 12 V supply.
  - Two coils hang on the NOT gate's input. Do not use 120 Ω.
- Calculator: calculate — type: `V_IH, R_i, R_on`

19. Static discipline and noise margins ⚠️ Borderline (logic levels are not named; MIT asked it)

- Seen: MIT Invitational 2021, Div C, INV, Scenario H Q68-69 (NOT gate: V_IL = 0.6 V, V_IH = 2.1 V, HIGH output ≥ 2.9 V, LOW output ≤ 0.5 V). Keys: Q68 **A: yes**; Q69 (V_IH = 2.3 V, V_IL = 0.4 V) **B: no**.
- How to solve:
  1. Check HIGH: is the worst HIGH output ≥ V_IH?
  2. Check LOW: is the worst LOW output ≤ V_IL?
  3. Both yes → it obeys the static discipline.
  4. Noise margins: NM_H = V_OH − V_IH, NM_L = V_IL − V_OL.
- Example (MIT Q68): 2.9 ≥ 2.1 and 0.5 ≤ 0.6. **Answer: yes. NM_H = 0.8 V, NM_L = 0.1 V.** Q69: 0.5 > 0.4, so NM_L = −0.1 V. **Answer: no.** TTL: NM_H = NM_L = 0.4 V. Python check.
- Traps:
  - MIT swaps the labels: it calls the HIGH output "V_OL" and the LOW output "V_OH". Use the numbers, not the labels.
  - A negative noise margin means the gate fails.
- Calculator: calculate — type: `V_OH, V_OL, V_IH, V_IL`

20. Which relay circuit is which gate? (and which is not a gate) ✅ IN (rules 3.c.vi, 3.c.viii)

- Seen: MIT Invitational 2021, Div C, INV, Scenario J Q75-79 (Circuits A-D; 12 V supply; one circuit is not a gate). Keys: A **AND**, B **none of the above**, C **XOR**, D **NOR**. Q79 key: in B, if the inputs differ, 12 V is shorted to ground through 1.6 Ω and the output sits at 6 V.
- How to solve:
  1. For each input combination, decide which relays are ON.
  2. A relay with its coil from the input to ground turns ON when the input is HIGH. A relay with its coil from 12 V to the input turns ON when the input is LOW.
  3. Trace from 12 V and from ground to the output through closed contacts.
  4. Output pulled to 12 V → 1. Pulled to ground → 0. Both at once → a short (not a gate). Neither → floating.
- Example (Circuit A): coils on A and B; contacts in series from 12 V to the output; 12 Ω pull-down to ground. **Answer: AND.** Both ON: V_out = 12 × 12 / (12 + 0.8 + 0.8) = **10.6 V ≥ 9 V (valid 1)**. Circuit B with unequal inputs: I = 12 / 1.6 = **7.5 A**, P = **90 W**, V_out = **6 V** (between 3 V and 9 V, invalid). Python check.

- Traps:
  - A series pair of contacts adds R_on twice (1.6 Ω).
  - Circuit D is a NOR built like CMOS: series "pull-up" relays that close on LOW inputs, parallel "pull-down" relays that close on HIGH inputs.
- Calculator: calculate — type: `R_on, R_pull, V`

21. Draw a relay or switch network for an expression ✅ IN (rule 3.c.vi)

- Seen: not seen as a drawing task. Prior research B_topics §6 (relay logic: series = AND, parallel = OR, NC = NOT) and H_logic_diodes A20.
- How to solve:
  1. Write the expression in SOP.
  2. Each product term = one series string of contacts.
  3. Put the strings in parallel.
  4. A plain letter = an NO contact. A primed letter = an NC contact.
  5. An XOR = two SPDT contacts wired like a 3-way switch.
- Example: "Draw a relay circuit for F = A·B′." **Answer: relay A's NO contact in series with relay B's NC contact, feeding the lamp.** Outputs for AB = 00, 01, 10, 11: 0, 0, 1, 0 (Python check).
- Traps:
  - One relay can have several contacts (for example DPDT). You may reuse a coil.
  - Draw each coil once. Label each contact with its coil's letter.
- Calculator: look up — type: `expr → contacts`

22. Pin-level wiring plan for a chip circuit ✅ IN (II.c.v)

- Seen: not seen as a written item. MIT Invitational 2020, Div C, INV, Lab 2 Q59 (build a circuit for a table, then draw the circuit diagram with component values; key: 3-input NOR).
- How to solve:
  1. Simplify the function. Choose the chip(s).
  2. Give each gate a number and list its pins.
  3. Write a wire list: "pin 3 → pin 5 and pin 9".
  4. Add power (pins 14 and 7), input switches with pull resistors, and an output LED with a resistor.
  5. Test all rows. The proctor tests all rows too.
- Example: "Build XOR from one 7400." **Answer:** gate 1: A → pin 1, B → pin 2, M at pin 3. Gate 2: A → pin 4, pin 3 → pin 5, output pin 6. Gate 3: pin 3 → pin 9, B → pin 10, output pin 8. Gate 4: pin 6 → pin 12, pin 8 → pin 13, **XOR out at pin 11**. Pin 14 → +5 V, pin 7 → GND. Python checked all 4 rows.
- Traps:
  - Forgetting power. Logic diagrams never show it.
  - Using the 7400 layout on a 4011 (CMOS NAND). Check the datasheet.
- Calculator: look up — type: `chip, gates` (prints the pin map)

23. Troubleshoot a chip circuit on a breadboard ✅ IN (II.c.vi)

- Seen: USC Invitational 2026, Div B, INV, MC22 (written stuck-at-0: one AND input grounded; key **C: XNOR(A, B)**). Rule II.c.vi lists "Troubleshoot a faulty circuit". Prior research H_logic_diodes A50.
- How to solve:
  1. Check power first: measure V_CC to GND **at the chip pins** (pin 14 to pin 7).
  2. Check the notch direction and the trench.
  3. Check each input with the meter for a clean 0 or 1 (no floating).
  4. Find the first wrong node, working from the inputs to the output.
  5. Check the LED direction (long leg = anode) and its resistor.

Symptom	Likely cause
Output never changes, chip cold	no power, or chip in backwards
Output always the same, chip hot	chip in backwards, or output shorted
Input reads 1.0-1.8 V	floating input or loose wire

Symptom	Likely cause
One row wrong	wrong pin on one gate, or wrong gate type
LED never lights	LED reversed, or no series path to ground
Output follows only one input	other input stuck at 0 or 1

- Example: "F should be AB + C. You measure F = C on every row." **Answer: the AB term never reaches the OR gate: the AND output is stuck at 0, or the AND chip has no power.**
- Traps:
  - Measure at the chip pin, not at the rail. A broken rail split cuts power to half the board.
- Calculator: look up — type: `symptom`

Nationals

24. Mixing TTL and CMOS; fan-out ⚠ Borderline (practical, not named)

- Seen: not seen in a test. Prior research H_logic_diodes A51, A54 (fan-out marked Borderline).
- How to solve:
  1. Compare the driver's V_OH with the receiver's V_IH. Compare V_OL with V_IL.
  2. If V_OH < V_IH, the HIGH is not guaranteed. Fix it with a pull-up resistor or a 74HCT receiver.
  3. Fan-out = the smaller of I_OL / I_IL and I_OH / I_IH.
- Example: "Can a standard TTL output (V_OH = 2.4 V) drive a 4000 CMOS input at 5 V (V_IH = 3.5 V)?" **Answer: no, 2.4 V < 3.5 V. Add a pull-up resistor or use 74HCT.** Standard TTL fan-out: I_OL = 16 mA, I_IL = 1.6 mA → 10; I_OH = 0.4 mA, I_IH = 40 µA → 10. **Fan-out = 10.** Python check.
- Traps:
  - CMOS → TTL works on voltage, but a 4000 output sinks very little current.
- Calculator: calculate — type: `V_OH, V_OL, V_IH, V_IL, currents`

25. Largest pull-down resistor for a TTL input ⚠ Borderline

- Seen: not seen in a test. Uses only Ohm's law (rule 3.c.iv).
- How to solve:
  1. A LOW TTL input pushes current **out**: I_IL = 1.6 mA (standard), 0.4 mA (LS).
  2. That current flows through the pull-down and raises the input voltage.
  3. The input must stay ≤ V_IL = 0.8 V. So R ≤ 0.8 V / I_IL.
- Example: **Answer: standard TTL: R ≤ 0.8 / 0.0016 = 500 Ω. 74LS: R ≤ 0.8 / 0.0004 = 2 kΩ.** A 10 kΩ pull-down fails on TTL. It works on CMOS, whose input current is almost zero. Python check.
- Traps:
  - This is why TTL circuits use pull-**ups** (switch to ground) for buttons.
- Calculator: calculate — type: `V_IL, I_IL`

Out of scope (skip)

- ❌ Transistor-level gates: NPN/PNP switches, CMOS transistor pairs, TTL totem-pole outputs. MIT Invitational 2020, Div C, Q50 and Lab 2 and Princeton 2020 Lab 2 used NPN transistors. F-M Invitational 2018 Q11 showed a CMOS pair. **Rule 3.d: semiconductors beyond those listed.** If a truth table is given, the gate name is still IN (Logic Gates and Notation A8 type 14).
- ❌ Op-amp gates and comparators (deck 15 homework "NOT gate from an op-amp"). **Rule 3.d.**
- ❌ Flyback diodes "because the coil is an inductor". **Rule 3.d: inductance.** "A diode across the coil protects the switch" is a fair one-liner.
- ❌ Latching relays, flip-flops, the 4511 latch, counters that drive a station's inputs. **Rule 3.d: sequential logic and digital clock signals.**
- ❌ Propagation delay, maximum clock rate, rise times. **Rule 3.d: clock signals; oscilloscopes.**
- ⚠/❌ Tri-state buffers, buses and open-collector "wired-AND". The 2020 rules excluded "three-state logic gates". The 2027 text does not name them. Treat them as out.
- ❌ Soldering and mains-powered relay boards at the station. **Rule II.d: no soldering, high power or AC.**

Mistakes in tests and keys

- **ATX Regional 2026, Div B and C, hands-on:** the key uses "the multimeter set to DC Voltage" as the power source. A meter on DC volts supplies no power. With the key's wiring, red is always on and green never lights. It also uses no resistors. A working build needs an SPDT button (type 16).
- **MIT Invitational 2021, Div C, Scenario H:** the text calls the HIGH output "V_OL = 2.9 V" and the LOW output "V_OH = 0.5 V". The labels are swapped. The keyed answers (A, B) are right for the numbers.
- **ATX Regional 2026, MC39:** key "high power, high voltage". The usual teaching is "a low-power signal switching a high-power load". The key is acceptable but loose.
- **UT Invitational 2019, Div B, Q43:** with S1 off, the drawing shorts the battery (prior research note).
- **UF Northern Regional 2019, Div B, Q1-3:** the 1 V source and silicon diodes would leave the diode-fed lamps almost dark. The key treats the diodes as ideal.
- **UT Invitational 2019, Div B, Q49 / Div C H49:** "theoretical power N/A" because the LEDs have no series resistor. The 7.08 mW comes from the key's own readings.

Not confirmed

- **Pinouts and electrical numbers** (7400, 7402, 7404, 4011 layouts; TTL and CMOS levels; I_OL, I_IL, fan-out; 74LS 8 mA sink) come from general datasheet knowledge and the prior research bank. I did not open a TI or other datasheet in this run. Check against the datasheet the supervisor gives you.
- **Dates:** the 7400 family date (mid-1960s) is approximate in the prior research.
- **MIT 2021 relay Circuits C and D:** I read the hand-drawn figure at 70 dpi. I trust the key (C = XOR, D = NOR) and did not trace every contact.
- **USC Invitational 2026** is an invitational, not the 2026 Nationals, although the computed task called this group "Nationals 2026" (`tests/nat26.md` §0).
- **UT 2025 Section D:** the lab handout is not in the downloads. The prompts are inferred from the key.
- **4000-series output current** ("sinks very little") is not checked against a datasheet.
- No Div B written test in the downloads asked about chip numbers, pinouts or TTL levels.

Sources

- 2027 Div B rules PDF: rules 3.c.i, 3.c.vi-viii, 3.d, II.b, II.c.v-vii, II.d (pages labeled B12-B13).
- Original test files read: `downloads/dlG/mit21_test.txt` and `mit21_key.txt` (Scenario H Q68-69, Scenario J Q70-79), pages 23-24 rendered; `downloads/dlG/mit20_test.txt` and `mit20_key.txt` (Q19, Lab 2 Q58-59); `downloads/dlFind/nfl19reg_Btest.txt` and `nfl19reg_Bimg.pdf` page 2 rendered (Q1-3); `downloads/dlFind/usc26_Btest.txt` (lab parts a-e).
- Test extractions: `circuitlab/research/tests/nat26.md`, `ut-a.md` (ATX 2026 hands-on, MC39; UT 2025 Section D), `ut-b.md` (UT 2019 B Q43, Q49), `ut-c.md` (UT fall 2019 C Q44, H49; ATX 2019 C hands-on), `misc-a.md` (Cobra 2019 B Q28, S3), `misc-b.md` (MIT 2019 Station 2-3; NFL 2019 C Q34).
- Prior research: `prior/B_topics.md` §6, §8, §9b; `prior/H_logic_diodes.md` A20, A48-A54, B12, C1, C4; `prior/F_slides.md` (deck 15, `Circuit_Board_Resistors.pdf`).
- Kuphaldt, *Lessons in Electric Circuits*, Vol. IV (Digital), ch. 3 "Logic Gates" (TTL/CMOS, logic levels), ch. 5 "Electromechanical Relays", ch. 6 "Ladder Logic"; Vol. III (Semiconductors), ch. 3 "Diodes" (diode gates).
- All About Circuits worksheets: "Logic gates", "Relay logic", "Diodes and rectifiers".
- Claude Shannon, "A Symbolic Analysis of Relay and Switching Circuits", MIT master's thesis, 1937.
- Every worked example: checked in Python (`check_hw.py`, `check_logic.py` in this session's scratchpad).

TAB A9

Hands-On and Meters

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Each note: key facts, formulas, then question types from Easy to Nationals with worked examples.

Breadboards

A **breadboard** is a plastic board full of holes. Metal clips under the holes join some holes together. You push wires and part legs into the holes, so you can build a circuit with no solder.

The 2027 rules say teams must know how breadboards work (Part II.b). The same rule says: **you may ask the Event Supervisor for the internal wiring of any breadboard used.** Ask when in doubt. Everything here is ✅ **IN at every level**.

Key facts

The layout of a standard (full or half-size) board

- **Terminal strips** are the short rows in the middle. Each row has 5 holes joined together (columns a-e on one side, f-j on the other).
- The **center trench** (gap) runs down the middle. It separates a-e from f-j. Row 12 a-e is **not** joined to row 12 f-j.
- **Power rails** (bus strips) are the long lines along the edges. Each rail is one long connection. Usually red (+) and blue or black (–).
- Some long boards have a **split rail**: the rail is broken in the middle. You must bridge it with a jumper. Check with the meter's continuity mode.
- The SO resource "Circuit Board or Bread Board" labels the long rails X and Y (X1 to X47 joined; Y1 to Y47 joined) and the short groups 1A-1E, 1F-1J.
- A **jumper** is a short wire that joins two holes.
- Board numbering differs between brands. Always look at the real board, or ask.

How parts go in

- A part's two legs must be in **different** rows. Both legs in one row shorts the part: no current goes through it.
- A **DIP chip** (dual in-line package, two rows of pins) straddles the trench. Then each pin gets its own row.
- On a 14-pin 74xx logic chip, pin 14 is V_cc (+5 V) and pin 7 is ground. Count pins counter-clockwise from the notch or dot, seen from the top.
- A **4-pin tactile push button** has two pairs of pins that are always joined. Pressing joins the two pairs. Put it across the trench so the pairs land in different rows.
- An **SPDT slide switch** has 3 pins in a line. Put each pin in its own row. The middle pin is usually the common (COM).
- The LED's long leg is the **anode** (+). The flat edge on the rim marks the **cathode** (–).
- Wire colors are a habit, not a rule: red for +, black for –.

Limits

- Breadboards are for low power. Typical rating is about 1 A and a few volts to tens of volts. Circuit Lab hands-on tasks must not use high power or AC (Part II.d).
- Loose contacts are common. A wire that "works when you press it" is a fault.

Formulas

There are no special formulas. Breadboard questions use the circuit rules:

What	Rule
Same row	Same node (same voltage)
Different rows, no part between	Not connected
Part with both legs in one row	Shorted, I_part = 0
Series	Parts share one row each, end to end, with nothing else in that row
Parallel	Both ends of each part share the same two rows (or rails)

Question types

Easy

1. Which holes are connected?

- Scope: ✅ IN.
- Where seen: SO resource "Circuit Board or Bread Board" (Circuit_Board_Resistors.pdf, 2019) and the national supervisor practice deck (BREADBOARD-CONNECTIVITY). Not seen as a numbered test question.
- How to solve:
  1. Same row and same side of the trench → connected.
  2. Same rail (and no split) → connected.
  3. Across the trench → not connected.
- Worked example: Are a3 and e3 connected? **Yes**. Are a3 and f3 connected? **No** (the trench). Are a3 and a4 connected? **No** (different rows).
- Traps: rails on a long board may be split in the middle. The rail's colored line often has a gap there.
- Calculator: look up — type: hole 1, hole 2.

2. A part with both legs in the same row

- Scope: ✅ IN.
- Where seen: not seen in a test. B_topics §9b and §9c (common errors).
- How to solve:
  1. Find each leg's row.
  2. If both legs are in one row, the row's metal clip shorts the part.
- Worked example: a 1 kΩ resistor has both legs in row 12 (12b and 12d). The meter reads **0 V** across it and **no current** goes through it. The rest of the circuit acts as if the resistor were a wire.
- Traps: this is the most common breadboard mistake. Look along the row, not only at the part.
- Calculator: look up — type: rows of each leg.

3. Use the power rails

- Scope: ✅ IN.
- Where seen: USC Invitational 2026, Div B and C, INV, Lab setup sheet (9 V battery with snap leads into the board). Northview Invitational 2025, Div C, INV, Lab sheets (+5 V and 0 V rails drawn on the board).
- How to solve:
  1. Battery + to the red rail. Battery – to the blue rail.
  2. Bridge a split rail with a jumper.
  3. Run short jumpers from the rails to the rows you need.
- Worked example: a long board has a split rail. Your LED on the right half does not light. Answer: **add a jumper across the rail split**.
- Traps: some boards put + on the outer line and – on the inner, others the opposite. Read the markings.
- Calculator: look up — type: rail layout.

Medium

4. Build a circuit from a schematic

- Scope: ✅ IN.
- Where seen: Pembroke Hill Invitational 2025-26, Div B, INV, hands-on (2 resistors in parallel in series with 2 LEDs in parallel; then all in series; measure V and I). North Pocono Invitational 2020, Div C, INV, HO-38 (8-resistor network on 9 V; find the largest drop). Whiting Invitational 2013, Div B, INV, Lab Practical circuits 1-3. UT Austin Invitational 2019, Div B, INV, Q48.
- How to solve:
  1. Number every node on the schematic.
  2. Give each node its own row (or rail).
  3. Place each part between its two node rows.

4. Check: every row should match one node, and every node one row (or rows joined by a jumper).

- Worked example: series circuit: + rail → row 5 → 330 Ω → row 9 → LED (anode row 9, cathode row 12) → row 12 → jumper → – rail. Three nodes besides the rails: rows 5, 9 and 12. ✓
- Traps: a jumper from a row to the rail that skips a part. The part is then bypassed.
- Calculator: look up — type: schematic nodes.

5. Draw the schematic of a built (or pictured) breadboard

- Scope: ✅ IN.
- Where seen: UT Austin Invitational Fall 2025, Div B/C, INV, Lab Q5b (draw the built source, resistor, SPST switch, LED circuit). I_bank C42. Cobra Invitational 2019-20, Div B, INV, Station 1 (draw the broken circuit, then the fix).
- How to solve:
  1. Give each used row a node name.
  2. Draw each part between the nodes its legs touch.
  3. Redraw neatly. Label values and units.
- Worked example: rows 3 and 8 each hold one leg of R1 (1 kΩ) and one leg of R2 (2.2 kΩ). Answer: **R1 and R2 are in parallel** between nodes 3 and 8.
- Traps: two parts in the same two rows are in parallel, even if they look far apart.
- Calculator: look up — type: rows for each leg.

6. Place a logic chip

- Scope: ✅ IN (the hardware is needed for the hands-on "logic circuit from a truth table" task). The chip's internal transistors are ❌ Out.
- Where seen: not seen in a test with a chip in Div B. Logic-gate identify/build labs are in MIT Invitational 2020, Div C, INV, Lab 2 and Princeton Invitational 2020, Div C, INV, Lab 2 (G_tests). H_logic_diodes A48.
- How to solve:
  1. Put the chip across the trench, notch up.
  2. Pin 14 to +5 V. Pin 7 to ground (for 7400, 7402, 7404, 7408, 7432, 7486).
  3. Find the gate pins on the datasheet. Ask the ES for it if you need it.
- Worked example: a 7408 (quad AND). Gate 1 inputs are pins 1 and 2; the output is pin 3. Feed pin 1 and pin 2 from switches; LED + resistor from pin 3 to ground.
- Traps: the **7402 (NOR)** has its outputs on pins 1, 4, 10 and 13, not 3, 6, 8, 11. Floating TTL inputs read HIGH.
- Calculator: look up — type: chip number, gate number.

7. Find the internal wiring of a push button

- Scope: ✅ IN.
- Where seen: USC Invitational 2026, Div B (and C), INV, Q39a (5 pts): map a 4-pin tactile button with the battery, LED and resistor as a tester; draw it.
- How to solve:
  1. Make a tester: battery → resistor → LED → two loose wire ends.
  2. Touch two pins. LED on = joined.
  3. Test every pair, with the button up and then pressed.
- Worked example: answer from the USC key: **the pins on each side are joined in pairs all the time; pressing the button joins the two pairs**. 4 pins give 6 pairs to test.
- Traps: if you put the button with the always-joined pins across one row, it shorts that row.
- Calculator: look up — type: pin pairs.

8. Power the board correctly (what is a source?)

- Scope: ✅ IN.
- Where seen: 2026 ATX Regional, Div B and C, REG, hands-on key (wrongly used the multimeter on DC volts as the source).
- How to solve:
  1. Only a battery, battery pack or bench supply is a source.

2. A multimeter on V, A or Ω does not power a circuit. (The ohmmeter sends a tiny test current; it is not a power source.)

- Worked example: the kit lists 2 LEDs, 1 button, 10 wires, breadboard, multimeter, 2 resistors and no battery. Answer: **ask the ES for a power source**. Do not try to run LEDs from the meter.
- Traps: an ohmmeter in diode mode may make an LED glow faintly. That is not a working circuit.
- Calculator: look up — type: kit list.

Hard

9. Trace a pre-built board (photo) and predict which LEDs light

- Scope: ✅ IN.
- Where seen: UT Austin Invitational 2019, Div B (and C), INV, Q49 (3 push buttons, 4 LEDs on 9 V; fill the table for each button combination; then LED power).
- How to solve:
  1. Turn the photo into a schematic (type 5).
  2. For each button combination, redraw with closed buttons as wires and open buttons removed.
  3. An LED lights if forward-biased and the supply is above its Vf (plus any series drops).
  4. Fill the table row by row.
- Worked example (power part, from the key's readings): lit LEDs read 1.88 V at 3.5 mA and 1.79 V at 0.28 mA. P = 1.88 × 0.0035 + 1.79 × 0.00028 = **7.08 mW** (Python: 7.0812 mW).
- Traps: the key's "theoretical" power is N/A because the board had **no series resistor**. Without a resistor you cannot predict the current.
- Calculator: calculate — type: list of V, list of I.

10. Check a rail or row with continuity

- Scope: ✅ IN.
- Where seen: not seen in a test. B_topics §9b.
- How to solve:
  1. Power off. Meter on continuity.
  2. Put probes (or wires) in two holes. A beep means joined.
- Worked example: probes in the top + rail at hole 1 and hole 50: no beep. Answer: **the rail is split**; add a jumper.
- Traps: thick probe tips may not reach the clips. Push a short wire into each hole and probe the wires.
- Calculator: look up — type: hole 1, hole 2.

Nationals

11. Find all faults on a board (more than one fault)

- Scope: ✅ IN.
- Where seen: not seen in a test as a multi-fault board. Related: Cobra Invitational 2019-20, Div B, INV, Station 1; Princeton Invitational 2020, Div C, INV, Lab 1 (find the fault, fix with one change) (G_tests).
- How to solve:
  1. Draw what the board really is (type 5), not what it should be.
  2. Compare it with the given schematic, node by node.
  3. Check part orientation (LED, chip notch), values (color code), rows (shorted parts) and rails (splits).
  4. Fix one fault at a time, then measure again.
- Worked example: an LED circuit does not light. You find (1) the 330 Ω resistor in one row (shorted), and (2) the LED reversed. Fixing only the LED lights it with no current limit; it may burn. Answer: **fix the LED and move the resistor so its legs are in two rows**.
- Traps: stop after the first fault and you lose points (or burn a part).
- Calculator: look up — type: list of checks.

Out of scope (skip)

- Soldering, perfboards and printed circuit boards (PCBs) as build tasks. ❌ Soldering is not allowed (Part II.d). (The term "PCB" itself appeared as vocabulary: Florida State 2021, Div B/C, ST, Q35.)
- PCB plane capacitance (UT Austin Invitational Fall 2025 Section B). ❌ Capacitance (rule 3.d).
- Breadboard stray capacitance and high-frequency limits. ❌ Frequency analysis (rule 3.d).
- Microcontroller (Arduino) breadboard builds. ❌ Sequential logic and clocks (rule 3.d).

Mistakes in tests and keys

- **2026 ATX Regional, Div B and C, REG, hands-on key:** the "solution" powers the circuit from a multimeter on DC volts, ties one button side to –, and uses no resistors. Red is always on and green never lights. See Task - 3-Way Switch A9 for the full check and a working build.
- **UT Austin Invitational 2019, Div B, INV, Q49 key:** it marks only red and yellow LEDs as lit in every row. I could not check this against the photo.
- **SO "Circuit Board or Bread Board" PDF:** its color-code sample range is rounded (3500-3900 Ω instead of 3515-3885 Ω).

Not confirmed

- Breadboard current rating (about 1 A) is a typical maker figure, not from SO.
- Rail layout and numbering vary by brand. The X/Y naming is from the SO PDF only.
- The UT 2019 Q49 LED table could not be read from the photo at the available resolution (ut-b extraction, 100 dpi).

Sources

- 2027 Div B Rules Manual, Circuit Lab B, Part II.b and II.d.
- SO resource: "Circuit Board or Bread Board" (https://www.soinc.org/sites/default/files/uploaded_files/Circuit_Board_Resistors.pdf), read from `downloads/dlF/Circuit_Board_Resistors.pdf`.
- USC Invitational 2026 ES notes (`downloads/dlFind/usc26_Bsetup.txt`) and lab (nat26.md §3).
- Northview 2025 lab sheet (`downloads/dlG/nv26C_lab.txt`); ATX 2026 lab score sheet (`dlG/atx26reg_lab.txt`).
- Test extractions: `tests/ut-a.md`, `tests/ut-b.md`, `tests/ut-c.md`, `tests/misc-a.md`, `tests/misc-b.md`; `prior/G_tests.md`; `prior/B_topics.md` §9b; `prior/H_logic_diodes.md` A48; `prior/I_bank.md` C42.
- Kuphaldt, *Lessons in Electric Circuits*, Vol. 6 (Experiments) ch. 1 "Breadboard".
- Numbers checked in Python (`check2.py`, 2026-10-16).

Multimeter Use

A **multimeter** is one tool that measures voltage, current and resistance. A **DMM** is a digital multimeter (it has a number display).

The 2027 rules say teams "must be familiar with the operation of breadboards and multimeters" (Part II.b). So every item in this note is ✅ **IN at every level**, unless it says otherwise.

You may bring your own "basic" multimeter if the Event Supervisor allows it (rule 2.e). The **Event Supervisor (ES)** is the adult who runs the event. Practice on the meter you will use.

Key facts

The jacks (holes for the leads)

- The black lead always goes in **COM**. COM means "common". It is the meter's negative side.
- For volts, ohms, continuity and diode test, the red lead goes in **VΩ** (some meters label it VΩmA).
- For current, the red lead goes in **mA** (small currents) or **10 A** (large currents).
- The current jacks have a **fuse** inside the meter. A blown fuse makes every current reading 0.
- Many 10 A jacks have no fuse or a slow fuse. Some say "10 A max, 10 s". Do not leave big currents on for long.

The dial

- **V⎓** (a straight line over a dashed line) is DC volts. Use this one in Circuit Lab.
- **V~** (a wavy line) is AC volts. Hands-on tasks do not use AC (rule 3 Part II.d).
- **A⎓** or **mA⎓** is DC current. **Ω** is resistance.
- The **continuity** symbol looks like sound waves. The meter beeps when the resistance is very low.
- The **diode test** symbol is a diode (triangle and bar). It shows a diode's forward drop in volts.
- A **range** is the largest value a setting can show. Examples: 200 mV, 2 V, 20 V, 200 V.
- An **autoranging** meter picks the range itself. Watch the unit on the screen (mV, V, kΩ, MΩ).

The display

- **OL** means "overload" or "open". The value is larger than the range, or there is no connection.
- Some older meters show only a "1" at the left for OL.
- A **minus sign** means the red lead is at the lower voltage. The size of the number is still right.
- A meter that shows up to 1999 is a "3½-digit" or "2000-count" meter.
- **Resolution** is the smallest step the display shows. On the 20 V range of a 2000-count meter it is 0.01 V.

How to connect each meter

- **Voltmeter:** across the part (in parallel), with power ON. An ideal voltmeter has infinite resistance. A real DMM has about 10 MΩ.
- **Ammeter:** break the circuit and put the meter in the gap (in series). An ideal ammeter has zero resistance.
- **Ohmmeter:** power OFF. Take the part out of the circuit, or lift one leg. Otherwise you also measure the parallel paths.
- An ohmmeter sends its own small current. Outside power ruins the reading and can damage the meter.

What goes wrong

- An ammeter across a source or across a part acts as a short circuit. A large current flows and blows the meter fuse.
- A voltmeter in series stops almost all current. It reads about the full source voltage.
- Your fingers on both probe tips add your body's resistance in parallel. This lowers big-resistance readings.
- **Loading:** a real meter changes the circuit a little. A voltmeter across a large resistor reads low. An ammeter in a low-resistance circuit lowers the current.
- The leads have a small resistance (often 0.1-0.5 Ω). Subtract it when you measure small resistances.

Indirect measurements (very common in hands-on tasks)

- Current from a voltage: I = V_R / R across a known resistor. No ammeter needed.

- Resistance from voltage and current: R = V / I.
- Power: P = V·I, P = I²R or P = V²/R.

Formulas

What	Formula	Notes
Ohm's law	V = I·R	All indirect methods use it
Current from a known resistor	I = V_R / R	No ammeter needed
Parallel resistance (loading)	R_p = R·R_m / (R + R_m)	R_m = meter resistance
Voltmeter reading with loading	V_read = V_s·R_p / (R_top + R_p)	Divider with the meter in it
Ammeter loading	I = V / (R + R_A)	R_A = ammeter resistance
Lead correction	R_true = R_read − R_leads	Short the leads to find R_leads
Meter spec uncertainty	±(p% × reading + n × resolution)	From the meter's manual
Galvanometer to voltmeter	R_series = V_full / I_g − R_g	⚠️ Borderline
Galvanometer to ammeter	R_shunt = I_g·R_g / (I_full − I_g)	⚠️ Borderline
Battery internal resistance	r = (V_open − V_loaded) / I	I = V_loaded / R_load

Question types

Easy

1. Name the meter for each quantity

- Scope: ✅ IN.
- Where seen: SO MYSO Div B practice test (Nov 2020), Div B, practice, Q9-10 ("Ampmeter" is a fake distractor). Shock Value re-tryout (about 2012), Div B, practice, Q7-9. Valparaiso Univ. Invitational 2013, Div B, INV, Q1-4 (matching).
- How to solve:
  1. Current → ammeter. Voltage → voltmeter. Resistance → ohmmeter.
  2. A multimeter is all three.
- Worked example: "Which tool measures potential difference?" Answer: **voltmeter**.
- Traps: "Ampmeter" and "voltometer" are not real names. A galvanometer detects tiny currents; it is not the answer for "voltage".
- Calculator: look up — type: meter, quantity.

2. Where does the meter go? (series or parallel)

- Scope: ✅ IN.
- Where seen: F-M Invitational 2018, Div B/C, INV, Q8 (picture: which meter is placed right). UF Northern Regional 2020, Div B, REG, Q9. NY Lake Erie/Niagara Invitational 2018 (2019 season), Div C, INV, Q10. MIT Invitational 2019, Div C, INV, Lab Station 1 Q6.
- How to solve:
  1. Voltmeter: its two leads go on the two ends of the part.
  2. Ammeter: open the loop. Put the meter in the gap so all the current goes through it.
  3. Say why: the ammeter must carry the same current; the voltmeter must see the same two nodes.
- Worked example: an ammeter with 0.1 Ω is put across a 5 V supply by mistake. I = 5/0.1 = **50 A**. The meter fuse blows (MIT 2019 key: "parallel shorts the circuit, blows the meter fuse").
- Traps: in a picture, look at where the leads touch. Two leads on the same wire is a meter reading 0 V, not a series ammeter.
- Calculator: calculate — type: V, R_meter.

3. Ideal meter resistance

- Scope: ✅ IN.
- Where seen: 2026 ATX Regional, Div C, REG, MC19-20. Virginia Regional 2013, Div C, REG, Q27-28. GullSO Invitational 2025, Div C, INV, Q9. North Pocono Invitational 2020, Div C, INV, Q18 (why near zero). Shock Value

re-tryout, Div B, practice, II-3 (true/false "ammeter R is 0").

- How to solve:
  1. Ideal voltmeter: R = ∞, so it takes no current.
  2. Ideal ammeter: R = 0, so it drops no voltage.
- Worked example: "Why is an ammeter's resistance near zero?" Answer: it sits in series. Any resistance would **add to the circuit and lower the current** it is trying to measure.
- Traps: "a real ammeter has R = 0" is false. It is small, not zero.
- Calculator: look up — type: ideal ammeter, ideal voltmeter.

4. Set up the meter (jacks and dial)

- Scope: ✅ IN.
- Where seen: not seen in a test. National supervisor practice deck (METER-HOW-TO-MEASURE). B_topics §9a.
- How to solve:
  1. Black lead in COM.
  2. Red lead in VΩ for V, Ω, diode, continuity. Red lead in mA or 10 A for current.
  3. Choose V⎓, A⎓ or Ω. Choose a range bigger than the expected value.
- Worked example: you expect 4.7 V. The ranges are 200 mV, 2 V, 20 V, 200 V. Choose **20 V**. It is the smallest range above 4.7 V, so it gives the best resolution (0.01 V).
- Traps: the red lead left in the mA jack after a current reading. The next voltage reading then shorts the circuit.
- Calculator: look up — type: expected value, list of ranges.

5. Read the display: OL, units, minus sign

- Scope: ✅ IN.
- Where seen: national supervisor practice deck (METER-OL-AND-LEAD-ZERO). Related: UT Austin Regional 2020, Div B/C, REG, hands-on H6 (meter leads flipped → the graph flips). The magnet part of that lab is ❌ Out.
- How to solve:
  1. OL = over range or open. Switch to a bigger range, or check the connection.
  2. Read the unit: 4.70 mV is not 4.70 V.
  3. A minus sign means the leads are reversed. Report the size, and note which point is higher.
- Worked example: the meter reads **–4.98 V** with red on point A and black on point B. So B is 4.98 V higher than A.
- Traps: "k" on the screen in Ω mode. 2.20 k means 2200 Ω.
- Calculator: look up — type: display text.

6. Ohmmeter rules: power off, part isolated

- Scope: ✅ IN.
- Where seen: "Circuit Lab Test (Key)" practice (about 2013), Div C, practice, Q21-23 (resistance in a powered circuit → cannot; current with no break → cannot). National supervisor slide #11.
- How to solve:
  1. Turn the power off.
  2. Lift one leg of the part, or pull it out of the breadboard.
  3. Then measure.
- Worked example: a 1.00 kΩ resistor still has a 2.20 kΩ resistor in parallel on the board. The meter reads 1.00 × 2.20 / 3.20 = **0.688 kΩ (687.5 Ω)**, not 1.00 kΩ.
- Traps: a battery left connected gives nonsense readings or a negative ohm value.
- Calculator: calculate — type: R_part, R_parallel.

7. Short or open from an ohmmeter reading

- Scope: ✅ IN.
- Where seen: Pembroke Hill Invitational 2025-26, Div B, INV, Q24 (reads 0.1 Ω → shorted). UF Northern Regional 2019, Div C, REG, Q15 (wire across R = short). Shock Value re-tryout, Div B, practice, Q51 ("wires touching").
- How to solve:
  1. About 0 Ω = a short (or a plain wire).
  2. OL = an open (a break).

3. A value near the color-code value = a good part.

- Worked example: a fuse reads OL. Answer: **the fuse is blown (open)**.
- Traps: a good 0.1 Ω wire and a shorted resistor both read about 0. You must know what the part should read.
- Calculator: look up — type: reading, expected R.

8. Subtract the lead resistance

- Scope: ✅ IN.
- Where seen: national supervisor practice deck 2 (METER-OL-AND-LEAD-ZERO). Not seen in a test.
- How to solve:
  1. Touch the probe tips together. Read R_leads.
  2. Measure the part. Subtract R_leads.
- Worked example: leads read 0.493 Ω. A resistor reads 61.77 Ω. True R = 61.77 − 0.493 = **61.28 Ω**.
- Traps: this matters only for small resistances. For 10 kΩ it is far below the tolerance.
- Calculator: calculate — type: R_read, R_leads.

Medium

9. Current without an ammeter (I = V_R / R)

- Scope: ✅ IN.
- Where seen: UT Austin Invitational Fall 2025, Div B/C, INV, Lab Q5d. JC Booth Invitational 2011 (Shock Value), Div B, INV, Station 1 (currents in 5 resistors from measured V). National supervisor deck (METER-I-FROM-V-AND-R).
- How to solve:
  1. Measure V across a resistor whose value you know.
  2. I = V/R. The same current flows in everything in series with it.
- Worked example: 2.35 V across 470 Ω. I = 2.35/470 = **5.00 mA**.
- Traps: use the measured R (ohmmeter, power off) if you are allowed. The color code is only ±5%.
- Calculator: calculate — type: V_R, R.

10. What a misplaced meter reads

- Scope: ✅ IN.
- Where seen: UT Austin Regional 2020, Div B, REG, Q60 (voltmeter in series with R1 on 12 V, ammeter in the return → 12.00 V, 0.00 A, 0.00 V across R1).
- How to solve:
  1. A voltmeter in series has huge R. So almost no current flows.
  2. All the source voltage appears across the voltmeter.
  3. The ammeter and every resistor in that loop read about 0.
- Worked example: 9 V battery, 1 kΩ, and a 10 MΩ voltmeter all in series. I = 9/10 001 000 ≈ **0.9 µA**. The voltmeter reads about **9.00 V**. The resistor drops about 0.9 mV.
- Traps: students expect the meter to "read the resistor". It reads the source.
- Calculator: calculate — type: V_s, R, R_meter.

11. Ammeter reads 0 but the LED is lit (blown meter fuse)

- Scope: ✅ IN.
- Where seen: not seen in a test. B_topics §9a and §9c (classic fault list).
- How to solve:
  1. If current flows somewhere (the LED glows), the ammeter path is not in series. Or its fuse is blown.
  2. A blown fuse makes the ammeter an open. Then the LED would go out if the meter were truly in series.
  3. So: LED on + ammeter 0 → the meter is not in the current path. LED off + ammeter in series → check the fuse with the ohmmeter (fuse jack to COM).
- Worked example: you insert the ammeter in series and the LED goes **out**. The meter shows 0.00 mA. Answer: **the meter fuse is blown** (or a lead is in the wrong jack).
- Traps: the 10 A jack often still works when the mA fuse is blown. Try it to check.

- Calculator: look up — type: symptoms.

12. Continuity and diode test (find the anode, Vf, a dead diode)

- Scope: ✅ IN.
- Where seen: North Pocono Invitational 2020, Div C, INV, HO-37 (light an LED, measure Vf). USC Invitational 2026, Div B and C, INV, Lab part a (map a push button, using battery + LED as a continuity tester). B_topics §7g.
- How to solve:
  1. Diode mode, red on one leg, black on the other.
  2. A number (Si about 0.5-0.7 V; LED about 1.6-3.3 V) means red is on the **anode**. OL means reversed.
  3. Reverse the leads. OL both ways = open diode. About 0 both ways = shorted diode.
- Worked example: an LED shows 1.82 V and glows faintly with red on the long leg. It shows OL reversed. Answer: **long leg = anode; good red LED (Vf ≈ 1.8 V)**.
- Traps: the ohmmeter mode gives odd, range-dependent numbers on diodes. Use diode mode. Some meters cannot light blue or white LEDs (test voltage too low).
- Calculator: look up — type: diode reading, LED color.

13. Voltmeter loading

- Scope: ✅ IN.
- Where seen: North Pocono Invitational 2020, Div C, INV, Q25 (5 V across 50 k + 25 k; ideal → 1.67 V; 2.5 MΩ meter → 1.66 V). Pembroke Hill Invitational 2025-26, Div B, INV, Q27 (a low-R voltmeter on a high-R part reads lower).
- How to solve:
  1. The meter resistance R_m is in parallel with the part.
  2. R_p = R·R_m/(R + R_m).
  3. Use the divider with R_p in place of R.
- Worked example (North Pocono Q25): ideal: 5 × 25/75 = **1.67 V**. With R_m = 2.5 MΩ: R_p = 24.75 kΩ, V = 5 × 24.75/74.75 = **1.66 V** (Python: 1.6556 V).
- Traps: loading always makes the voltmeter read **low**. It matters only when the part's R is not small next to R_m.
- Calculator: calculate — type: V_s, R_top, R_bottom, R_meter.

14. Ammeter loading (burden)

- Scope: ✅ IN.
- Where seen: not seen in a test. I_bank C31. OpenStax College Physics 2e §21.4.
- How to solve:
  1. Add the ammeter resistance R_A in series.
  2. I = V/(R + R_A).
- Worked example: 1.5 V cell, 10 Ω load, ammeter 1.0 Ω. True current 0.150 A. With the meter: 1.5/11 = **0.136 A** (9% low).
- Traps: the mA jack often has more resistance than the 10 A jack. For low-R circuits, use the 10 A jack.
- Calculator: calculate — type: V, R, R_ammeter.

15. Range, resolution and meter accuracy

- Scope: ✅ IN (part of "appropriate significant figures", rule 3 Part I.b).
- Where seen: not seen in a test. Meter manuals (Fluke, Keysight), Kuphaldt *Lessons in Electric Circuits* Vol. 1 ch. 8.
- How to solve:
  1. Resolution = the last digit's step on that range.
  2. Spec ±(p% of reading + n counts). One count = one resolution step.
  3. Uncertainty = p/100 × reading + n × resolution.
- Worked example: 2000-count meter, 20 V range, spec ±(0.5% + 2). It reads 9.00 V. Uncertainty = 0.045 + 0.02 = **±0.065 V** ≈ ±0.07 V. Report **9.00 ± 0.07 V**.
- Traps: more digits on screen is not the same as more accuracy.
- Calculator: calculate — type: reading, percent, counts, resolution.

16. Body resistance in parallel

- Scope: ✅ IN (static and shock topics also use body resistance).
- Where seen: not seen in a test. B_topics §9a.
- How to solve:
  1. Your hands holding both tips put your body in parallel.
  2. Use the parallel formula.
- Worked example: a 1.00 MΩ resistor; hand-to-hand body resistance about 500 kΩ (assumed, dry skin). Reading = 1 × 0.5/1.5 = **0.333 MΩ**.
- Traps: this only matters for big resistances (above about 100 kΩ).
- Calculator: calculate — type: R, R_body.

Hard

17. Black box: name the part from meter readings

- Scope: ✅ IN. The capacitor item is ⚠️ Borderline (capacitance is excluded, rule 3.d).
- Where seen: NY Lake Erie/Niagara Invitational 2018 (2019 season), Div C, INV, Q30 Station I (5 terminal pairs → capacitor, battery, diode, insulator, resistor). Cobra Invitational 2019-20, Div B, INV, Station 3 (real parts: breaker, relay, limit switch, capacitor).
- How to solve:
  1. Measure DC volts first. A steady voltage = a **battery**.
  2. Ohms both ways. Same value both ways = **resistor**.
  3. Diode mode: a drop one way, OL the other way = **diode**.
  4. OL both ways, no voltage = **insulator** (or an open).
  5. Ohms that climb over a few seconds, then OL = a **capacitor** charging.
- Worked example: pair C reads 0.62 V in diode mode one way and OL the other. Answer: **diode**. Pair A reads 1.58 V DC. Answer: **battery (a 1.5 V cell)**.
- Traps: do the voltage test first. An ohmmeter on a battery can damage the meter.
- Calculator: look up — type: readings for each pair.

18. Two-filament bulb with an ohmmeter

- Scope: ✅ IN.
- Where seen: Cobra Invitational 2019-20, Div B, INV, Station 2 (car headlight bulb, 3 terminals, find the high beam).
- How to solve:
  1. Measure all 3 pairs of terminals.
  2. The pair with the largest reading is the two filaments in series. The third terminal is the common one.
  3. At the same voltage, higher power means lower R ($R = V^2/P$). So the lower filament R is the high beam.
- Worked example: 12 V bulb, 55 W low, 60 W high. Hot R: 144/55 = **2.62 Ω** (low), 144/60 = **2.40 Ω** (high). The two in series: **5.02 Ω**. Cold readings are much lower (filament R rises with heat), but the order stays.
- Traps: cold filament R is about 1/10 of hot R. Do not expect V^2/P values on the meter. Use the order only.
- Calculator: calculate — type: V, P_low, P_high.

19. Battery internal resistance with a voltmeter

- Scope: ✅ IN.
- Where seen: Pembroke Hill Invitational 2025-26, Div B, INV (I_bank A15: 9.0 V open, 8.5 V on 10 Ω). See Internal Resistance and Power Transfer [A5].
- How to solve:
  1. Measure V with no load (V_open ≈ EMF).
  2. Connect a known load R. Measure V_loaded.
  3. I = V_loaded/R. r = (V_open − V_loaded)/I.
- Worked example: 9.0 V open, 8.5 V on 10 Ω. I = 0.85 A. r = 0.5/0.85 = **0.59 Ω**.
- Traps: do not leave a small load on a 9 V battery for long; it heats and drains.

- Calculator: calculate — type: V_open, V_loaded, R_load.

20. Galvanometer to voltmeter or ammeter

- Scope: ⚠️ Borderline. The math is plain DC (Ohm's law). But a galvanometer works by magnetism, which is excluded (rule 3.d).
- Where seen: PA Central Regional (about 2013), Div C, REG, Q21 (1 mA, 250 Ω → 15 V meter and 5 A meter). I_bank C32-C33. OpenStax College Physics 2e §21.4.
- How to solve:
  1. Voltmeter: add a big resistor in series. R_s = V_full/I_g − R_g.
  2. Ammeter: add a small resistor in parallel (a shunt). R_sh = I_g·R_g/(I_full − I_g).
- Worked example: I_g = 1.00 mA, R_g = 250 Ω. 15 V meter: R_s = 15 000 − 250 = **14 750 Ω**. 5 A meter: R_sh = 0.25/4.999 = **0.0500 Ω**.
- Traps: the shunt carries almost all the current. Its resistance is tiny.
- Calculator: calculate — type: I_g, R_g, full-scale V or I.

Nationals

21. Three-terminal black box (Δ or Y) from three readings

- Scope: ⚠️ Borderline. Finding the Δ or Y values needs three equations; the rules allow at most two (rule 3.e.i). The measuring part is IN.
- Where seen: NY Lake Erie/Niagara Invitational 2018 (2019 season), Div C, INV, Q31 Station II. PA Central Regional (about 2013), Div C, REG, Q4.
- How to solve:
  1. Measure R_AB, R_BC, R_CA with power off.
  2. For a Y (star) with arms a, b, c: R_AB = a + b, R_BC = b + c, R_CA = c + a.
  3. Add all three, halve: S = a + b + c. Then a = S − R_BC, and so on.
- Worked example: Δ of 33 k (AB), 56 k (BC), 39 k (CA). The meter reads R_AB = 33 || 95 = **24.5 kΩ**, R_BC = 56 || 72 = **31.5 kΩ**, R_CA = 39 || 89 = **27.1 kΩ**. The same box as a Y: 10.05 k (A), 14.4 k (B), 17.1 k (C). Check: 10.05 + 14.4 = 24.5 ✓.
- Traps: both a Δ and a Y fit the same three readings. The key may accept either (NY 2019 did).
- Calculator: calculate — type: R_AB, R_BC, R_CA.

22. Find a meter's resistance from a loaded reading

- Scope: ✅ IN (one equation, algebra only).
- Where seen: not seen in a test. Kuphaldt *Lessons in Electric Circuits* Vol. 1 ch. 8 ("voltmeter impact on measured circuit").
- How to solve:
  1. Write the divider with R_p = R·R_m/(R + R_m).
  2. Solve for R_p from the reading. Then R_m = R·R_p/(R − R_p).
- Worked example: 10.0 V across two 100 kΩ in series. The meter on the bottom reads 4.76 V. R_p/(100k + R_p) = 0.476 → R_p = 90.8 kΩ → R_m ≈ **0.99 MΩ ≈ 1 MΩ**.
- Traps: a reading of exactly half (5.00 V) means the meter's effect is too small to measure.
- Calculator: calculate — type: V_s, R_top, R_bottom, V_read.

Out of scope (skip)

- AC voltage and AC current readings (V~, A~), true-RMS meters. ❌ Hands-on must not use AC circuits (Part II.d). AC circuit theory is excluded (rule 3.d).
- Capacitance mode (F), frequency mode (Hz), duty cycle. ❌ Capacitance and frequency analysis are excluded (rule 3.d).
- Oscilloscopes. ❌ Excluded by name (rule 3.d).
- Clamp meters (they sense the magnetic field around a wire). ❌ Magnetism is not in the 2027 list.
- Transistor test sockets (hFE). ❌ Semiconductors beyond diodes and LEDs (rule 3.d).
- Magnet-through-a-coil voltage labs (UT Austin Regional 2020 hands-on H1-H7). ❌ Magnetism and induction.

- Meter safety categories (CAT I-IV) for mains work. ⚠ Borderline trivia; mains circuits are not used in the event.

Mistakes in tests and keys

- **2026 ATX Regional, Div B, REG, hands-on key:** it uses a multimeter "set to DC Voltage" as the power source. A meter on V⎓ does not supply power. (The full fault list is in Task - 3-Way Switch A9.)
- **UF Northern Regional 2020, Div B, REG, Q21f:** the key names a sine-in-a-circle symbol "variable voltage source". It is an **AC voltage source**.
- **Valparaiso 2013 key / Shock Value-era keys:** no meter errors found.
- **SO "Circuit Board or Bread Board" resource PDF:** the color-code sample gives 3700 Ω ± 5% as "3500-3900 Ω". The exact range is **3515-3885 Ω** (Python). It rounded 185 Ω up to 200 Ω.

Not confirmed

- Body resistance (500 kΩ dry, hand to hand) is a typical textbook value, not from a test. Real values vary a lot.
- The meter spec ±(0.5% + 2) is a typical value for a basic DMM. Check your own meter's manual.
- The 10 A jack fusing differs by model. I did not check a specific meter.
- I did not find what counts as a "basic" multimeter (rule 2.e). No official clarification was posted (checked soinc.org Circuit Lab B clarifications page, 2026-10-16).
- NY 2019 Station I black-box answers depend on the real box; I could not check them.

Sources

- 2027 Div B Rules Manual, Circuit Lab B, Parameters 2.d-2.e and Part II.b (manual pages B12-B13).
- Test extractions: `research/tests/misc-a.md` (F-M 2018, North Pocono 2020, Cobra 2019-20), `misc-b.md` (MIT 2019, UF Northern Regionals 2019-2020, NY 2018, JC Booth 2011, Shock Value tryout), `atx-reg.md` (UT Regional 2020), `ut-a.md` (UT Fall 2025, ATX 2026), `state.md` (MYSO), `nat26.md` (USC 2026).
- Prior research: `prior/G_tests.md` (VA 2013, GullSO 2025, PA Central, Circuit Lab Test Key, Valparaiso 2013), `prior/B_topics.md` §9a, §7g, `prior/F_slides.md` §7 (national supervisor practice decks), `prior/I_bank.md` C29-C34, A15.
- OpenStax, *College Physics 2e*, §21.4 "DC Voltmeters and Ammeters".
- Tony Kuphaldt, *Lessons in Electric Circuits*, Vol. 1 ch. 8 "DC Metering Circuits"; Vol. 6 ch. 2 "Voltmeter usage", "Ohmmeter usage", "Ammeter usage".
- All worked numbers checked in Python (scratch scripts `check1.py`, 2026-10-16).

Lab Skills for the Hands-On Part

Note: Lab Skills

These skills apply to every hands-on station. ✅ **IN at every level.**

The rules give the hands-on part **25-50% of the score** (rule 4.c). Points go to "correct answers, measurements, calculations, and data analysis" (rule 4.b). Answers must be in **metric units with appropriate significant figures** (rule 3, Part I.b). The supervisor supplies all parts and meters (rule 2.d). You may bring your own basic multimeter if the supervisor allows it (rule 2.e).

Key facts

- **Significant figures** (sig figs) are the digits that carry meaning in a measured number.
- × and ÷: keep as many sig figs as the **least** precise input.
- + and −: keep as many **decimal places** as the least precise input.
- Keep extra digits during the steps. Round only the final answer.
- **Uncertainty** is the "±" part of a result. It tells how far the true value may be from your number.
- Round the uncertainty to 1 sig fig (2 if it starts with 1). Round the value to the same decimal place.
- **Percent error** compares with a known ("actual") value. **Percent difference** compares two measurements.
- **Range** is the meter dial setting (for example 20 V). Use the lowest range that does not show OL ("overload").
- **Resolution** is the smallest step the display shows on that range.
- Write the **unit** on every number. Write the meter **range** next to every reading.
- Make a **data table** before you start: quantity, value, unit, range, notes.
- Draw a **schematic** of what you built. Many keys give points for a labeled sketch (BullSO 2026: 8 points).
- **Sign-off**: the supervisor checks your working circuit and signs or initials your sheet. Test every state yourself first. Then raise your hand.
- **Safety**: no AC, no soldering, no high power (Part II.d). Still, a shorted 9 V battery gets hot. An LED with no resistor burns out.
- Check resistor power before you connect: $P = V^2/R$ must be under 1/4 W for most kit resistors.
- **Explain** your method in words. Princeton 2026 gave 10 of 16 lab points for explanations.
- Two partners: one builds, one records and checks. Swap roles on the next station.

Formulas

What	Formula	Notes
Percent error	|measured − actual| / actual × 100%	"Actual" on the bottom
Percent difference	|a − b| / ((a + b)/2) × 100%	Two measurements
Mean of n trials	(x1 + … + xn) / n	
Spread (simple)	± (max − min) / 2	Half the range
Meter uncertainty	±(p% × reading + n × resolution)	From the meter spec
× or ÷ uncertainty	add the percent uncertainties	R = V/I, P = VI
+ or − uncertainty	add the absolute uncertainties	V_x = V_s − V_k
Minimum safe R (1/4 W)	R_min = V^2 / 0.25 W	Across one resistor
Time per station	50 min × (station points / total points)	Rough plan

Metric prefixes: m (milli) = 10^{-3}, μ (micro) = 10^{-6}, k (kilo) = 10^{3}, M (mega) = 10^{6}.

Question types

Easy

1. Units and prefixes

- Scope: ✅ IN (rule 3, Part I.b; rule 3.c.iv "units").

- Where seen: every hands-on key. GullSO Invitational 2025, Div C, INV, Q50f (mA and Ω in one formula). UT 2019 B and C, INV, Q47-48 (V to 3 s.f.).
- How to solve:
  1. Change every value to the base unit (V, A, Ω, W) before you calculate.
  2. Change the answer back to a handy prefix.
- Worked example: 15 mA through 470 Ω. V = 0.015 A × 470 Ω = **7.05 V**. A reading of 0.0175 A = **17.5 mA**. 2.2 kΩ = **2200 Ω**.
- Traps: 15 × 470 = 7050 is "mV", not "V". Always carry the prefix.
- Calculator: calculate — type: value, prefix.

2. Significant figures for × and ÷

- Scope: ✅ IN.
- Where seen: rule 3, Part I.b. Mystery-resistor ratio answers (Princeton 2026, GullSO 2025; see Task - Mystery Resistor A9).
- How to solve:
  1. Count the sig figs of each input.
  2. Round the answer to the smallest count.
- Worked example: R_x = 1.00 kΩ × 3.44 V / 1.56 V = 2.2051… → **2.21 kΩ** (3 sig figs).
- Traps: the calculator shows 2.205128. Writing all of it claims false precision.
- Calculator: calculate — type: inputs.

3. Significant figures for + and –

- Scope: ✅ IN.
- Where seen: KVL sum checks (UT Invitational 2025 lab: V_R + V_SW + V_LED = supply; ut-a extraction).
- How to solve:
  1. Find the input with the fewest decimal places.
  2. Round the answer to that many decimal places.
- Worked example: 9.10 V – 2.84 V = **6.26 V**. 12.0 V + 0.346 V = 12.346 → **12.3 V**.
- Traps: for subtraction, the decimal place matters, not the sig fig count.
- Calculator: calculate — type: inputs.

4. Choose the meter range

- Scope: ✅ IN (Part II.b; see Multimeter Use A9 type 15).
- Where seen: not asked alone in a test. It decides the digits of every reading.
- How to solve:
  1. Start on a high range if you do not know the size.
  2. Step down until the next lower range would show OL.
- Worked example: a 4.87 V battery. 2 V range → **OL**. 20 V range → **4.87 V** (best). 200 V range → **4.9 V** (one digit lost).
- Traps: auto-ranging meters change range by themselves. Check the unit symbol (mV or V) on the screen.
- Calculator: none.

5. Record data in a table

- Scope: ✅ IN (rule 4.b "measurements … data analysis").
- Where seen: ATX Regional 2020, Div B and C, REG, hands-on H1 (3 trials and the average; the physics is OUT, the method is IN). Northview 2025 lab ("Record your answer on the bottom left corner of your Freeform answer sheet").
- How to solve:
  1. Columns: what you measured, value, unit, range, trial number.
  2. Write each reading at once. Do not keep numbers in your head.
  3. Put the final answer where the sheet asks for it.
- Worked example: | V across R_k | 1.56 | V | 20 V | trial 1 |. One row per reading.
- Traps: an answer in the wrong box can score 0. Read where it goes.

- Calculator: none.

Medium

6. Meter uncertainty of one reading

- Scope: ✅ IN.
- Where seen: not seen in a test. MIT Invitational 2019, Div C, INV, Lab keys accept ±0.2 to ±0.6 kΩ (they expect uncertainty).
- How to solve:
  1. Read the spec: ±(p% of reading + n counts).
  2. One count = one resolution step on that range.
  3. Uncertainty = p/100 × reading + n × resolution.
- Worked example: 200 mA range, resolution 0.1 mA, spec ±(1.0% + 3). Reading 17.5 mA. Uncertainty = 0.175 + 0.3 = 0.475 → **17.5 ± 0.5 mA**.
- Traps: the current ranges are usually less accurate than the voltage ranges. Use V across a known R when you can.
- Calculator: calculate — type: reading, p, n, resolution.

7. Repeat trials: mean and spread

- Scope: ✅ IN.
- Where seen: ATX Regional 2020, Div B and C, REG, H1 (3 trials and the average).
- How to solve:
  1. Take 3 readings. Reconnect the probes each time.
  2. Mean = sum / 3.
  3. Spread = ± half of (max − min).
- Worked example: 9.12, 9.08, 9.10 V. Mean = **9.10 V**. Spread = (9.12 − 9.08)/2 = **±0.02 V**. Report **9.10 ± 0.02 V**.
- Traps: three identical readings do not mean zero error. The meter spec still applies (type 6).
- Calculator: calculate — type: list of readings.

8. Percent error and percent difference

- Scope: ✅ IN.
- Where seen: GullSO Invitational 2025, Div C, INV, Q49f ("|actual − predicted| / actual · 100%").
- How to solve:
  1. Known true value → percent error.
  2. Two measured values, no true value → percent difference.
- Worked example: predicted 54.8 Ω, actual 56 Ω. Error = 1.2/56 = **2.1%**. Partners read 2.18 kΩ and 2.22 kΩ. Difference = 0.04/2.20 = **1.8%**.
- Traps: do not divide by the predicted value in percent error.
- Calculator: calculate — type: actual, measured.

9. Carry uncertainty into R = V/I

- Scope: ✅ IN (algebra only).
- Where seen: not seen in a test. GullSO 2025 Q49g asks how to "increase the precision".
- How to solve:
  1. Change each ± into a percent.
  2. For × or ÷, add the percents.
  3. Change the result back into a ±.
- Worked example: V = 4.87 ± 0.05 V (1.03%), I = 2.20 ± 0.04 mA (1.82%). R = 4.87/0.00220 = 2213.6 Ω. Percent = 2.85% → ±63 Ω. Report **2.21 ± 0.06 kΩ**.
- Traps: for + and −, add the ± values themselves, not the percents.
- Calculator: calculate — type: V, ΔV, I, ΔI.

10. Safety check before you connect the battery

- Scope: ✅ IN (Part II.d; rule 3.c.iv power).

- Where seen: USC Invitational 2026, Div B/C, INV, Lab (the 510 Ω series resistor takes 0.16 W on 9 V when a button shorts the LED). 2026 ATX Regional key (no LED resistors; see Task - 3-Way Switch A9 type 5).
- How to solve:
  1. Trace the circuit for a direct + to – path (a short).
  2. Check each LED has a series resistor.
  3. Check P = V²/R for the resistor with the most voltage.
- Worked example: 1/4 W resistors on 9 V. R_min = 81/0.25 = **324 Ω**. On 5 V: R_min = 25/0.25 = **100 Ω**. A 100 Ω across 9 V would take 0.81 W and get hot.
- Traps: disconnect the battery while you rewire. Breadboard mistakes are fastest to fix with the power off.
- Calculator: calculate — type: V, P rating.

11. Get the supervisor sign-off

- Scope: ✅ IN.
- Where seen: BullSO Invitational 2026, Div B, INV, Lab (sign-off 9 of 29 points); Div C, INV, Lab (6 of 24). USC Invitational 2026, Div B/C, INV, Lab (a proctor signs each working circuit after testing all 4 button states; unlimited tries).
- How to solve:
  1. Test every input state yourself. Write the results.
  2. Raise your hand. Show the circuit and your table.
  3. Do not touch the board while the supervisor tests it.
  4. Ask the supervisor to sign or initial the right line.
- Worked example: BullSO B points: sketch 8 + steps 8 + explanation 4 + sign-off 9 = **29**. The sign-off alone is **31%**.
- Traps: a loose jumper can fail at sign-off. Push every wire in fully before you call.
- Calculator: none.

Hard

12. Plan your time

- Scope: ✅ IN (the event is about 50 minutes; rule 1).
- Where seen: MIT Invitational 2019, Div C, INV, Lab (two stations, 6 minutes each). USC Invitational 2026, Div B, INV: lab 25 of 184 points.
- How to solve:
  1. Read the whole packet first (2 minutes).
  2. Time per part ≈ 50 min × (its points / total points).
  3. Start the build early. A build can wait for sign-off while you do the written part.
- Worked example: hands-on worth 25% → about **12.5 minutes**. Worth 50% → about **25 minutes**. USC 2026 B lab = 25/184 = **14%** of the points (below the 2027 minimum of 25%).
- Traps: one stuck circuit can eat the whole event. Set a time limit. Then move on and come back.
- Calculator: calculate — type: points, total, minutes.

13. Explain sources of error

- Scope: ✅ IN.
- Where seen: GullSO Invitational 2025, Div C, INV, Q50i ("sources of error between the ideal model and your actual circuit"); Q49g ("How might you increase the precision of your results?"). Princeton Invitational 2026, Div C, INV, Lab (explain how you used the color chart; explain your circuits).
- How to solve:
  1. Name the cause. Say which way it moves the answer.
  2. Typical causes: resistor tolerance, meter accuracy, battery voltage drop under load, lead resistance, meter loading, LED V_f spread.
  3. Say how to reduce it: closer known resistor, better range, repeat trials, measure V_s under load.
- Worked example: "The known resistor is ±5%, so R_x is only good to about ±5%. Measuring R_k first with the ohmmeter makes it about ±1%."

- Traps: "human error" alone scores nothing. Name a physical cause.
- Calculator: none.

14. Common point-loss mistakes (checklist)

- Scope: ✅ IN.
- Where seen: patterns across the keys in this folder (BullSO, Princeton, GullSO, USC, ATX).
- How to solve: before you hand in, check each line.
  1. Every number has a unit and sensible sig figs.
  2. The schematic is labeled with values (BullSO: 3 of 8 sketch points).
  3. Steps are shown, not only the answer.
  4. The meter was on the right mode (ammeter in series, voltmeter in parallel, ohmmeter with power off).
  5. LED polarity: long leg (anode) toward +.
  6. The answer is in the right box, and the sign-off line is signed.
  7. The rules of the station were followed ("only voltage", "use all resistors once").
- Worked example: an ammeter placed across the battery blows its fuse. Every later current reading is then 0. (See Multimeter Use A9 type 11.)
- Traps: using the ohmmeter on a "voltage only" task can lose all the points (Princeton 2026).
- Calculator: none.

Nationals

15. Report a final result with its uncertainty

- Scope: ✅ IN.
- Where seen: MIT Invitational 2019, Div C, INV, Lab keys (answers with ± bands). Not seen at a State or Nationals station.
- How to solve:
  1. Find the total uncertainty (types 6, 7 and 9).
  2. Round it to 1 sig fig.
  3. Round the value to the same decimal place.
  4. Compare with the expected value: is the gap smaller than the uncertainty?
- Worked example: R = 2213.6 Ω ± 63 Ω → **2.21 ± 0.06 kΩ**. Expected 2.2 kΩ. The gap (0.01 kΩ) is smaller than ±0.06 kΩ, so they **agree**.
- Traps: "2.2136 ± 0.06" mixes precisions. The value and the ± must end at the same place.
- Calculator: calculate — type: value, uncertainty.

Out of scope (skip)

- Standard deviation and statistics beyond mean and range. ⚠️ Borderline: rule 3.e allows only algebra. Half-range is enough.
- Oscilloscope use. ❌ Oscilloscopes (rule 3.d).
- Soldering and mains wiring. ❌ Not allowed (Part II.d).
- Uncertainty with square roots (quadrature sums). ⚠️ Borderline: the simple "add the percents" rule is enough at this level.

Mistakes in tests and keys

- **USC Invitational 2026, Div B, INV:** the lab is 25 of 184 points (14%). The 2027 rules need 25-50% for the hands-on part. Expect bigger labs in 2027.
- **2026 ATX Regional, Div B, REG:** the test says the hands-on is 30 points. The key's rubric totals 40 (ut-a extraction).
- **BullSO 2026 C:** the test and the key give different points for the explanation (2 vs 8). See Task - Target Resistance A9.

Not confirmed

- The meter spec in type 6 (±(1.0% + 3) on 200 mA) is a typical budget meter value, not from a kit list.
- No 2026 State or Nationals hands-on was found. Where I looked: `research/tests/state.md` and `nat26.md` (USC Invitational, not Nationals).
- Supervisors may grade sig figs strictly or loosely. No key in this folder shows a sig-fig deduction.

Sources

- 2027 Div B Rules Manual, Circuit Lab B, rules 1, 2.d-e, 3 Part I.b, 3.c.iv, 3.d-e, Part II.b and II.d, 4.b-c.
- BullSO 2026 B and C labs and keys (`downloads/dlB/tests/bullso26_*.txt`, `dlG/bull26C_*.txt`); GullSO 2025 test (`dlG/gull26_test.txt`, Q49-50); Princeton 2026 lab (`dlG/puso26_lab.txt`); Northview 2025 lab (`dlG/nv26C_lab.txt`).
- Test extractions: `research/tests/nat26.md` (USC 2026 lab and points), `misc-b.md` (MIT 2019 lab), `atx-reg.md` (ATX 2020 H1), `ut-a.md` (UT 2025 lab, ATX 2026), `ut-b.md` (UT 2019 Q47-48), `state.md` §4; `prior/G_tests.md` §2, `prior/B_topics.md` §9.
- OpenStax, *College Physics 2e*, §1.3 "Accuracy, Precision, and Significant Figures".
- Kuphaldt, *Lessons in Electric Circuits*, Vol. 1 ch. 8 "DC Metering Circuits".
- All numbers checked in Python (`lab.py`, 2026-10-16).

Task: Find a Mystery Resistor Using Only Voltages

Note: Task - Mystery Resistor

Official task (2027 rules, Part II.c.i): "Determine the value of a mystery resistor in a circuit using only voltage measurements." ✅ **IN at every level.** The bridge version (type 10) needs KCL, so it is ✅ **States/Nationals**.

A **mystery resistor** is a resistor whose value is hidden (tape, marker, or a covered box). "Only voltage" means you may not use the ohmmeter or the ammeter. Princeton 2026 said so in bold. Touching the cover can mean disqualification.

Key facts

- In a **series** circuit, the same current flows through every part.
- So the voltage across each part is in proportion to its resistance: V_x / V_k = R_x / R_k.
- R_k is a **known resistor** (you know its value from the color code, the kit list, or the ES).
- Best precision: pick a known resistor close in size to the mystery one. Then both voltages are large and similar.
- A very small voltage is hard to read. One digit of meter error is then a big percent error.
- Measure the source voltage **with the circuit connected**. A real battery's voltage drops under load.
- A real voltmeter has about 10 MΩ. It changes the reading when the resistors are near 1 MΩ or more.
- Many kits use ±5% (gold band) or ±1% (brown band) resistors. Your answer can only be as good as your known resistor.
- After the calculation, some tests let you check with the ohmmeter and report a **percent error** (GullSO 2025).
- Explain your circuit on the answer sheet: Princeton 2026 gave 10 of 16 points for the explanation.

Formulas

What	Formula	Notes
Ratio method	R_x = R_k × V_x / V_k	Measure both drops
Source method	R_x = R_k × (V_s – V_k) / V_k	Measure the source and one drop
Current first	I = V_k / R_k, then R_x = V_x / I	Same as the ratio method
Parallel branch (series inside it)	R_x = R_k × V_x / V_k, for R_k in the same branch	Same branch = same current
Loading correction	Use R_k ‖ R_m in place of R_k when you measured across R_k	R_m ≈ 10 MΩ
Percent error	|actual – predicted| / actual × 100%	GullSO wording
Tolerance range	R_x range = (R_k × (1 ± t)) × V_x / V_k	t = 0.05 for gold
Bridge (one KCL)	R10 = (V_A – V_B) / [(V_s – V_A)/R_top – V_A/R_bottom]	IN-SN

Question types

Easy

1. Series with a known resistor (ratio method)

- Scope: ✅ IN.
- Where seen: MIT Invitational 2019, Div C, INV, Lab Station 1 Q1a (1 kΩ + hidden 2.2 kΩ on 5 V). Northview Invitational 2025, Div C, INV, Lab sheet 2 (5 V, 1 kΩ, R?). GullSO Invitational 2025, Div C, INV, Q49. NC Shock Value practice 2010, Div B, practice (G_tests). Princeton Invitational 2026, Div C, INV, Lab Part I.
- How to solve:
  1. Build: source → known R_k → mystery R_x → back to the source.
  2. Measure V_k and V_x with the voltmeter.
  3. R_x = R_k × V_x / V_k.
- Worked example: 5 V supply, R_k = 1.00 kΩ. You read V_k = 1.56 V and V_x = 3.44 V. R_x = 1.00 × 3.44/1.56 = **2.21 kΩ** (Python: 2.205 kΩ). Nearest standard value: 2.2 kΩ.

- Traps: do not divide the wrong way. The larger voltage belongs to the larger resistor.
- Calculator: calculate — type: R_k, V_k, V_x.

2. Source-and-one-drop method

- Scope: ✅ IN.
- Where seen: same tests as type 1 (any of the two methods is accepted). B_topics §9a.
- How to solve:
  1. Measure V_s across the source while the circuit is connected.
  2. Measure V_k.
  3. V_x = V_s − V_k (series sum). Then use type 1.
- Worked example: V_s = 5.00 V (loaded), V_k = 1.56 V, R_k = 1.00 kΩ. V_x = 3.44 V. R_x = **2.21 kΩ**.
- Traps: if you measure V_s with the battery disconnected, it is too high (see type 6).
- Calculator: calculate — type: R_k, V_s, V_k.

3. Read the known resistor's color code first

- Scope: ✅ IN.
- Where seen: Princeton Invitational 2026, Div C, INV, Lab Part I Q1-2 (list all values; explain how the chart gives the known value). GullSO 2025, Div C, INV (kit list gives 5-band codes). F-M Invitational 2019, Div B/C, INV, HO-13.
- How to solve:
  1. 4-band: digit, digit, multiplier, tolerance.
  2. 5-band: digit, digit, digit, multiplier, tolerance.
- Worked example: Brown-Black-Black-Red-Brown = 1, 0, 0 × 100 = **10.0 kΩ ± 1%**. Yellow-Violet-Black-Gold-Brown = 4, 7, 0 × 0.1 = **47.0 Ω ± 1%**.
- Traps: gold and silver as the **multiplier** mean × 0.1 and × 0.01. They are tolerance only in the last band.
- Calculator: look up — type: band colors.

Medium

4. Choose the known resistor for the best precision

- Scope: ✅ IN.
- Where seen: GullSO Invitational 2025, Div C, INV, Q49c (justify component values) and Q49g (how to increase precision). Kit: 22, 47, 220, 4.7 k, 10 k (two each); mystery = 56 Ω.
- How to solve:
  1. First do a quick rough test with any known R.
  2. Then rebuild with the known R closest to your rough answer.
  3. Explain: similar sizes give two large voltages, so a one-digit meter error is a small percent.
- Worked example: 3.00 V source, mystery 56 Ω. With R_k = 10 kΩ, V_x is only 0.0167 V. On a meter that shows 0.02 V, R_x = 10 000 × 0.02/2.98 = **67 Ω** (20% off). With R_k = 47 Ω, V_x = 1.63 V. R_x = 47 × 1.63/1.37 = **55.9 Ω** (0.2% off).
- Traps: do not pick the biggest known resistor "to be safe". Also do not pick a tiny one: a 22 Ω + 56 Ω on 3 V draws 38 mA, which drains the cells fast.
- Calculator: calculate — type: V_s, R_k list, R_x estimate.

5. Percent error against the ohmmeter value

- Scope: ✅ IN.
- Where seen: GullSO Invitational 2025, Div C, INV, Q49f ("|actual − predicted| / actual · 100%"; key: actual 56 Ω).
- How to solve:
  1. Measure R_x with the ohmmeter after the voltage method (only when the task allows it).
  2. Percent error = |actual − predicted| / actual × 100%.
- Worked example: predicted 54.8 Ω, actual 56 Ω. Error = 1.2/56 × 100 = **2.1%**.
- Traps: "actual" goes on the bottom. Keep the sign out (absolute value).
- Calculator: calculate — type: actual, predicted.

6. Non-ideal battery: measure V_s under load

- Scope: ✅ IN.
- Where seen: GullSO Invitational 2025, Div C, INV, Q49g and Q50i (battery voltage 2.9-3.2 V; sources of error). Not seen as a direct question.
- How to solve:
  1. Measure V_s with the circuit connected.
  2. Or use the ratio method, which does not need V_s at all.
- Worked example: a 9 V battery reads 9.40 V alone but 9.10 V with the circuit on. V_k = 2.84 V across 1.00 kΩ. Right: R_x = 1000 × (9.10 − 2.84)/2.84 = **2.20 kΩ**. Wrong (open-circuit V_s): 1000 × (9.40 − 2.84)/2.84 = **2.31 kΩ** (5% high).
- Traps: the battery's internal resistance makes the "no load" voltage too high.
- Calculator: calculate — type: R_k, V_s loaded, V_k.

7. Mystery resistor inside a parallel branch

- Scope: ✅ IN.
- Where seen: MIT Invitational 2019, Div C, INV, Lab Station 1 Q1b (circuit 2: (1 k + 1 k) || (1 k + hidden 6.8 k), then 1 k, on 5 V). MIT Station 1 Q4 asked the R_eq (key 2.4 ± 0.4 kΩ).
- How to solve:
  1. Find a known resistor in **series with R_x inside the same branch**. They share one current.
  2. Measure V across both. R_x = R_k × V_x / V_k.
- Worked example (MIT circuit 2, true R5 = 6.8 kΩ): the branch voltage is 3.07 V. The meter reads V(R4 = 1 k) = **0.394 V** and V(R5) = **2.677 V**. R5 = 1000 × 2.677/0.394 = **6.80 kΩ**. R_eq = 2k || 7.8k + 1k = **2.59 kΩ**; total current **1.93 mA**.
- Traps: a known resistor in a **different** branch does not carry the same current. Do not use it in the ratio.
- Calculator: calculate — type: R_k, V_k, V_x (same branch).

8. Several mystery resistors with one known resistor

- Scope: ✅ IN.
- Where seen: Princeton Invitational 2026, Div C, INV, Lab Part I (4 mystery + 1 known, no ohmmeter, battery, breadboard, multimeter).
- How to solve:
  1. Put the known resistor and all the mystery resistors in one series string. One current flows.
  2. Measure every drop.
  3. Each R = R_k × V_each / V_k.
  4. Or: find one mystery value, then use it as a new "known" resistor (but its error carries forward).
- Worked example: 9.00 V, string of 1.00 k (known) + A + B + C + D. Readings: V_k = 0.530 V, V_A = 1.167 V, V_B = 0.249 V, V_C = 1.750 V, V_D = 5.303 V. A = **2.20 kΩ**, B = **470 Ω**, C = **3.30 kΩ**, D = **10.0 kΩ** (Python, rounded readings give 2202, 470, 3302, 10 006 Ω).
- Traps: if one resistor is huge, the others get tiny voltages. Then measure them again in a second string with only similar resistors.
- Calculator: calculate — type: R_k, V_k, list of V.

9. Covered resistor from a total resistance (ohmmeter allowed)

- Scope: ✅ IN (it uses resistance, not voltage; so it is a cousin of the official task).
- Where seen: F-M Invitational 2019, Div B/C, INV, HO-14 and HO-15 (three resistors in parallel; total measured; find the covered one).
- How to solve:
  1. Measure R_total of the parallel group (power off).
  2. 1/R_x = 1/R_t − 1/R_1 − 1/R_2.
- Worked example: R_1 = 1.0 kΩ, R_2 = 4.7 kΩ, R_t = 0.600 kΩ. 1/R_x = 1.667 − 1.000 − 0.213 = 0.454 → R_x = **2.20 kΩ**.
- Traps: small rounding in R_t gives a big change in R_x (you subtract close numbers). Keep 3-4 significant figures.

- Calculator: calculate — type: R_total, list of known R.

Hard

10. Very large mystery resistor: correct for voltmeter loading

- Scope: ✅ IN.
- Where seen: UF Northern Regional 2019, Div C, REG, P3 (mystery resistor with a known **100 kΩ**). The loading correction itself is not seen in a key.
- How to solve:
  1. When you measure across R_k, the meter (R_m) is in parallel with it.
  2. Use R_k' = R_k || R_m in the formula: R_x = R_k' × (V_s − V_k) / V_k.
  3. Or measure across R_x and correct R_x in the same way.
- Worked example: 9.00 V, R_k = 1.00 MΩ, R_x (true) = 2.20 MΩ, meter 10 MΩ. The meter reads V_k = 2.632 V. Naive: R_x = 1.00 × 6.368/2.632 = **2.42 MΩ** (10% high). Corrected: R_k' = 0.909 MΩ → R_x = **2.20 MΩ**.
- Traps: with 100 kΩ the error is only about 1%. With 1 MΩ it is about 10%. Say this in your error discussion.
- Calculator: calculate — type: V_s, R_k, V_k, R_meter.

11. Report a range from the known resistor's tolerance

- Scope: ✅ IN.
- Where seen: GullSO Invitational 2025, Div C, INV, Q49e key ("known resistances are to the specified tolerance"). MIT Invitational 2019, Div C, INV, Lab Station 1 (key accepts ±0.2, ±0.4, ±0.6 kΩ).
- How to solve:
  1. R_k min = R_k × (1 − t). R_k max = R_k × (1 + t).
  2. R_x range = each R_k limit × V_x / V_k.
- Worked example: type 1 numbers with a ±5% known resistor: R_x between **2.09 kΩ and 2.32 kΩ**.
- Traps: the meter reading also has an uncertainty (see Multimeter Use A9 type 15). The tolerance usually dominates.
- Calculator: calculate — type: R_k, tolerance, V_k, V_x.

Nationals

12. Mystery resistor in a bridge (not series-parallel)

- Scope: ✅ States-Nationals (it needs KCL at a node, rule 3.c.ix).
- Where seen: MIT Invitational 2019, Div C, INV, Lab Station 1 Q1c (circuit 3: R8 10 k / R9 1 k on top, R11 1 k / R12 10 k on the bottom, hidden R10 across the middle, 5 V). Station 1 Q5 asked: can it be reduced with series/parallel rules? (**No**.)
- How to solve:
  1. Name the two middle nodes A and B. Measure V_A and V_B (to ground).
  2. KCL at A: current in from the top = current down + current across R10.
  3. (V_s − V_A)/R_topA = V_A/R_bottomA + (V_A − V_B)/R10.
  4. Solve this one equation for R10.
- Worked example (my reading of the topology: R8 from + to A, R11 from A to ground, R9 from + to B, R12 from B to ground; R10 = 4.7 kΩ as on the setup sheet): the meter reads V_A = **1.025 V**, V_B = **3.975 V**. Current in R8 = 3.975/10k = 0.3975 mA. Current in R11 = 1.025 mA. So 0.6275 mA flows **from B to A** through R10. R10 = 2.950 V / 0.6275 mA = **4.70 kΩ ✓**.
- Traps: the current in R10 may flow "backwards" (from B to A). Keep the signs. A balanced bridge (V_A = V_B) tells you nothing about R10.
- Calculator: calculate — type: V_s, R_top A, R_bottom A, V_A, V_B.

Out of scope (skip)

- Using the ohmmeter when the task says "only voltage" is against the task rules (and loses all points). Not a scope issue, but skip it.

- Thevenin "measure the open-circuit voltage and short-circuit current" methods (VA Regional 2013 Station 2). ❌ Norton/Thevenin are excluded (rule 3.d).
- Mystery capacitor from LED blinks (Cobra 2019-20 Station 5). ❌ Capacitance (rule 3.d).
- Wheatstone bridge with a galvanometer null as a meter method. ⚠️ Borderline: the bridge math is DC, but the galvanometer is magnetic.

Mistakes in tests and keys

- **MIT Invitational 2019, Div C, INV, Lab Station 1 Q1c:** the key says R10 = 10.7 kΩ ± 0.6, but the setup sheet schematic says **4.7 kΩ**. I could not tell which board was used. (If R10 were 10.7 kΩ, my topology gives V_A = 0.752 V and V_B = 4.248 V.)
- **MIT Invitational 2019, Div C, INV, Lab Station 1 Q4:** key R_eq 2.4 kΩ; computed **2.59 kΩ**. Mine is inside the key's ±0.4 tolerance. Station 1 Q3 key 1.6 mA vs computed 1.56 mA (rounding).
- **F-M Invitational 2019 hands-on key:** the header says R1 = 2.2 k, R2 = 4.7 k, Rx = 3.3 k. Its own answer and guide use 1 k, 4.7 k and Rx = 2.2 k.

Not confirmed

- MIT 2019 bridge topology: I used my reading of "R8/R9 on top, R11/R12 on the bottom". The real wiring may differ. I did not render the setup figure.
- UF Northern 2019 C P3 key says 100 kΩ for the mystery value; I could not check it without the board.
- GullSO 2025 Q49 actual mystery value 56 Ω is from the key; the teams' measured values are not known.

Sources

- 2027 Div B Rules Manual, Circuit Lab B, Part II.c.i.
- Princeton 2026 lab sheet (`downloads/dlG/puso26_lab.txt`); Northview 2025 lab sheet (`dlG/nv26C_lab.txt`); GullSO 2025 test and key (`dlG/gull26_test.txt`, `gull26_key.txt`).
- Test extractions: `tests/misc-b.md` (MIT 2019 Lab, UF Northern 2019 C), `tests/misc-a.md` (F-M 2019); `prior/G_tests.md` §2 "Hands-on station types seen"; `prior/B_topics.md` §9a; `prior/I_bank.md` C27; `prior/F_slides.md` §7.
- OpenStax, *College Physics 2e*, §21.1 (series resistors) and §21.4 (meters).
- All numbers checked in Python (`check1.py`, 2026-10-16).

Task: Calculate the Power Supplied to a Circuit

Note: Task - Power Supplied

Official task (2027 rules, Part II.c.ii): "Calculate the power supplied to a circuit." ✅ **IN at every level.** The two-source version (type 14) needs KVL/KCL, so it is ✅ **States/Nationals**.

Power is how fast energy moves. Its unit is the **watt (W)**. One watt is one joule per second.

Power supplied is the power that the source (battery or supply) gives to the circuit. It equals the source voltage times the current that leaves the source.

Key facts

- Power supplied by a source: P = V_source × I_source.
- Power used by one part: P = V_part × I_part.
- For a resistor you can also use $P = I^2R$ or $P = V^2/R$.
- **Conservation of energy:** power supplied = the sum of the powers used by all parts.
- You usually measure the current without an ammeter. Measure the voltage across a known series resistor. Then I = V_R / R.
- Measure the source voltage **with the circuit connected**. A real battery's voltage drops under load.
- A real battery loses some power inside itself. Power delivered to the circuit = V_terminal × I.
- An LED uses P = V_f × I. Here V_f is the LED's **forward voltage** (the voltage across it when it is on).
- Small resistors are rated 1/8 W (0.125 W) or 1/4 W (0.25 W). More power than the rating makes them hot. They can burn.
- Report power in W or mW (1 mW = 0.001 W). Use the right number of significant figures.
- Energy = power × time. E (J) = P (W) × t (s). E (Wh) = P (W) × t (h).
- The rules forbid high-power circuits and AC in the hands-on part (Part II.d). So expect batteries and small parts.

Formulas

What	Formula	Notes
Power supplied	$P = V_s \times I$	I leaves the + terminal
Current from a known resistor	$I = V_k / R_k$	No ammeter needed
Power supplied, voltages only	$P = V_s \times V_k / R_k$	Series circuit
Resistor power	$P = I^2R = V^2/R = V \times I$	Any of the three
Total from R_eq	$P = V_s^2 / R_{eq}$	Ideal source
LED power	$P_{LED} = V_f \times I$	
Efficiency to the LED	$\eta = P_{LED} / P_s = V_f / V_s$	Series LED circuit
Energy	$E = P \times t$	J or Wh
Battery: delivered and lost	$P_{out} = V_t \times I$; $P_{lost} = (\varepsilon - V_t) \times I$	ε = no-load voltage
Power check	$P_s = \Sigma P_{parts}$	Conservation

Question types

Easy

1. Power from the source voltage and a measured current

- Scope: ✅ IN.
- Where seen: Pembroke Hill Invitational 2026, Div B, INV, Q38-Q39 (measure a voltage and a current with units). UT Austin Invitational 2019, Div B, INV, Q49 (measure V and I, then P). JC Booth Invitational 2011, Div B, INV, Station 2.
- How to solve:
  1. Measure V_s across the source while the circuit runs.
  2. Measure I with the ammeter in series, at the source.

3. P = V_s × I.

- Worked example: V_s = 9.00 V, I = 12.5 mA. P = 9.00 × 0.0125 = **0.113 W** (112.5 mW).
- Traps: change mA to A first. 12.5 mA is 0.0125 A, not 12.5 A.
- Calculator: calculate — type: V_s, I.

2. Power supplied from voltages only (known series resistor)

- Scope: ✅ IN.
- Where seen: MIT Invitational 2019, Div C, INV, Lab Station 2 Q1a (12 V, 1 kΩ + hidden 3.3 kΩ; key **33.5 mW**).
- How to solve:
  1. Measure V_k across the known resistor R_k.
  2. I = V_k / R_k. (Series: the same current leaves the source.)
  3. Measure V_s. P = V_s × I.
- Worked example (MIT numbers): V_k = 2.79 V across 1.00 kΩ, so I = 2.79 mA. V_s = 12.0 V. P = 12.0 × 2.79 mA = **33.5 mW** (Python: 33.49 mW).
- Traps: the known resistor must be in series with the source. A resistor in one branch only carries part of the current.
- Calculator: calculate — type: V_s, V_k, R_k.

3. Power in each resistor from its measured voltage

- Scope: ✅ IN.
- Where seen: JC Booth Invitational 2011, Div B, INV, Station 2 ($P = V^2/R$ for each resistor; key gives ranges such as 0.08-0.11 W). GullSO Invitational 2025, Div C, INV, Q51i.
- How to solve:
  1. Measure V across each resistor.
  2. $P = V^2/R$ for each one.
  3. Add them for the total.
- Worked example: 220 Ω with 3.30 V across it. $P = 3.30^2/220$ = **0.0495 W** (49.5 mW).
- Traps: square the voltage, not the resistance.
- Calculator: calculate — type: V, R.

4. Power from the equivalent resistance

- Scope: ✅ IN.
- Where seen: Rickards Invitational 2025 (2026 season), Div B, INV, Simulated Hands-On Q10-Q13 (120, 50, 40, 75 Ω on 9 V; series 285 Ω, 31.6 mA, **284 mW**; parallel 15 Ω, 600 mA, **5.40 W**).
- How to solve:
  1. Find R_eq.
  2. $P = V_s^2 / R_{eq}$.
- Worked example: 9 V across 120 + 50 + 40 + 75 = 285 Ω. P = 81/285 = **0.284 W**. All four in parallel: R_eq = 15.0 Ω, P = 81/15 = **5.40 W**.
- Traps: see type 5. The parallel case is a burn hazard with small resistors.
- Calculator: calculate — type: V_s, list of R, series or parallel.

Medium

5. Check the power rating of each part

- Scope: ✅ IN.
- Where seen: GullSO Invitational 2025, Div C, INV, Q51i ("If your resistors were rated for 1/8 W, would any exceed their rating?"). USC Invitational 2026, Div B and C, INV, Lab Q39c (9 V across 510 Ω when a button shorts the LED).
- How to solve:
  1. Find P for each part.
  2. Compare with the rating (1/8 W = 0.125 W, 1/4 W = 0.25 W).

- Worked example: USC lab, both buttons pressed: 9 V across 510 Ω. P = 81/510 = **0.159 W**. That is over 1/8 W but under 1/4 W. Rickards parallel case: the 40 Ω resistor gets 81/40 = **2.03 W**. Every resistor is far over 1/4 W.
- Traps: a supervisor may give 1/8 W parts. Then a value that is "safe" for 1/4 W is not safe.
- Calculator: calculate — type: V, R, rating.

6. LED power from measured V and I

- Scope: ✅ IN.
- Where seen: UT Austin Invitational 2019, Div B, INV, Q49 (buttons 1 and 2 pressed; key **7.08 mW** = 1.88 V × 3.5 mA + 1.79 V × 0.28 mA; "theoretical: N/A" because the board had no series resistor).
- How to solve:
  1. Measure V across each lit LED.
  2. Measure its current (ammeter, or V across its series resistor ÷ R).
  3. P = V × I for each LED. Add them.
- Worked example: 1.88 V × 3.5 mA = 6.58 mW. 1.79 V × 0.28 mA = 0.50 mW. Total = **7.08 mW** (Python: 7.081 mW).
- Traps: do not use $P = V^2/R$ for an LED. An LED is not a resistor.
- Calculator: calculate — type: list of (V_f, I).

7. Check that power supplied = power used

- Scope: ✅ IN.
- Where seen: not seen in a test as its own question. It is the standard check in OpenStax *College Physics 2e* §21.1-21.2.
- How to solve:
  1. P_s = V_s × I.
  2. Add the power of every part.
  3. The two numbers should match (within measurement error).
- Worked example: 9.00 V, 390 Ω, red LED at 2.00 V. I = 7.00/390 = 17.9 mA. P_s = **0.162 W**. P_R = I^2R = 0.126 W. P_LED = 2.00 × 0.0179 = 0.036 W. Sum = **0.162 W** ✓.
- Traps: a big mismatch means a wrong reading or a hidden path (see Task - Troubleshoot A9).
- Calculator: calculate — type: V_s, I, list of parts.

8. Real battery: power delivered and power lost inside

- Scope: ✅ IN.
- Where seen: Pembroke Hill Invitational 2026, Div B, INV (internal resistance: 9.0 V no-load, 8.5 V across 10 Ω). GullSO Invitational 2025, Div C, INV, Q50i (battery between 2.9 and 3.2 V).
- How to solve:
  1. Measure ε with no load and V_t with the load.
  2. I = V_t / R_load.
  3. P_delivered = V_t × I. P_lost = (ε – V_t) × I. Total from the chemistry = ε × I.
- Worked example: ε = 9.0 V, V_t = 8.5 V across 10 Ω. I = 0.85 A. P_delivered = **7.23 W**. P_lost = 0.5 × 0.85 = **0.43 W**. Internal r = 0.5/0.85 = **0.59 Ω**.
- Traps: "power supplied to the circuit" uses V_t, not ε. Say which one you used.
- Calculator: calculate — type: ε, V_t, R_load.

9. Efficiency of an LED circuit

- Scope: ✅ IN.
- Where seen: not seen in a test. From prior research (H_logic_diodes B19).
- How to solve:
  1. P_s = V_s × I. P_LED = V_f × I.
  2. η = P_LED / P_s = V_f / V_s.
- Worked example: 9 V battery, red LED 2.0 V, 20 mA. P_LED = 40 mW. P_s = 180 mW. η = **22%**. The resistor wastes the other 78%.
- Traps: a higher battery voltage makes the circuit **less** efficient, even at the same LED current.

- Calculator: calculate — type: V_s, V_f.

10. Energy used in a given time

- Scope: ✅ IN.
- Where seen: Rickards Invitational 2025, Div B, INV, Q32 (15 W for 1 h → 15 Wh = 54 000 J). UT Austin Invitational 2019, Div B, INV, Q28 (battery life → average power). See Task - USB Charger [A9].
- How to solve:
  1. Find P.
  2. E = P × t. Use seconds for joules, hours for Wh.
- Worked example: the type 9 circuit (0.180 W) runs for 20 min = 1200 s. E = 0.180 × 1200 = **216 J** = **0.060 Wh**.
- Traps: minutes must become seconds (× 60) or hours (÷ 60).
- Calculator: calculate — type: P, t, unit.

Hard

11. Power in every switch state

- Scope: ✅ IN.
- Where seen: national supervisor homework generator, Level 9 "SW-STATE-TABLE" (F_slides §7). USC Invitational 2026, Div B, INV, Lab Q39c-e (power changes as buttons are pressed).
- How to solve:
  1. Redraw the circuit for each switch state.
  2. Find R_eq in each state.
  3. P = V_s^2 / R_eq.
- Worked example (generator: 12 V; R1 = 100, R2 = R3 = 200, R4 = 300 Ω): R_eq = 300, 200, 150, 120 Ω for the four states. P = **0.48, 0.72, 0.96, 1.20 W**.
- Traps: a closed switch across a part removes that part (short). An open switch removes its whole branch.
- Calculator: calculate — type: V_s, R_eq list.

12. Power used by a hidden part (from voltages only)

- Scope: ✅ IN.
- Where seen: MIT Invitational 2019, Div C, INV, Lab Station 2 Q1b (hidden 3.3 kΩ).
- How to solve:
  1. I = V_k / R_k from the known resistor.
  2. V_hidden = V_s − V_k (series).
  3. P_hidden = V_hidden × I.
- Worked example (MIT): V_k = 2.79 V, I = 2.79 mA, V_hidden = 9.21 V. P_hidden = **25.7 mW**. R_hidden = 9.21/2.79 mA = **3.30 kΩ**.
- Traps: do not use P = V^2/R with an unknown R. Use V × I.
- Calculator: calculate — type: V_s, V_k, R_k.

13. Power in a loaded voltage divider

- Scope: ✅ IN.
- Where seen: GullSO Invitational 2025, Div C, INV, Q51h-i (3.00 V divider, 220 Ω load; key's best build: 22 Ω on top, 22 Ω || 56 Ω on the bottom).
- How to solve:
  1. Bottom = R2 || R_load. Then the total R and the current.
  2. V_out across the bottom group.
  3. Each part: P = V^2/R. The top resistor carries all the current.
- Worked example (key's circuit): bottom = 22 || 56 || 220 = 14.73 Ω. I = 3.00/36.73 = 81.7 mA. V_out = **1.20 V** (meets the spec). P(top 22 Ω) = **0.147 W**. P(bottom 22 Ω) = 0.066 W. P(56 Ω) = 0.026 W. P(load) = 0.0066 W. Total **0.245 W**.
- Traps: the top 22 Ω is **over a 1/8 W rating** (0.147 > 0.125). The key does not say this (see "Mistakes").
- Calculator: calculate — type: V_in, R_top, R_bottom list, R_load.

Nationals

14. Power from each of two sources (one may absorb power)

- Scope: ✅ States-Nationals (KCL, rule 3.c.ix; one node equation).
- Where seen: not seen as a hands-on task. Two-source KVL/KCL circuits appear in written tests (USC Invitational 2026, Div B, INV, Q32b; UF Northern Regional 2019, Div C, REG, Q41).
- How to solve:
  1. Write one KCL equation at the shared node. Solve for its voltage.
  2. Find each source's current. Keep the sign.
  3. P_source = V × I. A negative value means that source **absorbs** power (it is being charged).
  4. Check: the sum of the source powers = the sum of the resistor powers.
- Worked example: 9 V through 100 Ω and 6 V through 200 Ω feed one node; 300 Ω from the node to ground. KCL: (9 – V)/100 + (6 – V)/200 = V/300 → V = **6.55 V**. I_9V = **+24.5 mA**, I_6V = **–2.73 mA**. P_9V = **+0.221 W**. P_6V = **–0.0164 W** (it absorbs). Resistors use **0.205 W** ✓.
- Traps: do not assume both sources supply power. Do not put unequal batteries straight in parallel.
- Calculator: calculate — type: V1, R1, V2, R2, R3.

Out of scope (skip)

- AC power, RMS values and power factor. ❌ AC circuit theory (rule 3.d). The hands-on part also bans AC circuits (Part II.d).
- Energy stored in a capacitor (½CV²). ❌ Capacitance (rule 3.d).
- "Maximum power transfer" with a Thevenin resistance. ❌ Norton/Thevenin (rule 3.d). (The simple fact "a battery wastes power in its internal resistance" is still IN.)
- Op-amp supply power (MIT Invitational 2019, Div C, Lab Station 2 Q2f, 0.152 W). ❌ Semiconductors beyond those listed (rule 3.d).
- Three-phase power. ❌ (rule 3.d).

Mistakes in tests and keys

- **GullSO Invitational 2025, Div C, INV, Q51i:** the key's own "best" divider (22 Ω on top of 22 Ω || 56 Ω, 220 Ω load) puts **0.147 W** in the top 22 Ω. That exceeds a 1/8 W rating. The key does not mention it. A team that says "yes, the top resistor exceeds 1/8 W" is correct.
- **UT Austin Invitational 2019, Div B, INV, Q49:** the key marks "theoretical power" as N/A. The board had no series resistor, so the LED currents are set by the battery and the LEDs. That is a design fault in the station, not in the key.
- **Rickards Invitational 2025, Div B, INV, Q13:** the key's 5.40 W is right. But building it with normal 1/4 W resistors would overheat them. The question is "simulated" (paper only), so no one built it.

Not confirmed

- MIT 2019 Station 2 readings: I used the setup values (1 kΩ, 3.3 kΩ, 12 V). Real teams read slightly different numbers.
- UT 2019 Q49: the LED readings (1.88 V, 3.5 mA, 1.79 V, 0.28 mA) are the key writer's readings. I could not check the board.
- Type 9 and type 14 numbers are my own examples. They are not from a test.

Sources

- 2027 Div B Rules Manual, Circuit Lab B, Part II.c.ii and II.d.
- MIT 2019 lab, setup and key (`downloads/dlFind/mit19_lab*.txt`); GullSO 2025 test and key (`dlG/gull26_test.txt`, `gull26_key.txt`); Rickards 2026 B test and key (`dlG/rsoi26B_*.txt`); USC 2026 setup (`dlFind/usc26_Bsetup.txt`); Pembroke Hill 2026 B (`dlB/tests/phi26_*.txt`).

- Test extractions: `research/tests/misc-b.md` (MIT 2019, JC Booth 2011, UF Northern 2019), `ut-b.md` (UT 2019 B), `nat26.md` (USC 2026); `prior/F_slides.md` §7; `prior/H_logic_diodes.md` B19; `prior/I_bank.md` A15.
- OpenStax, *College Physics 2e*, §20.4 "Electric Power and Energy" and §21.1-21.2.
- Kuphaldt, *Lessons in Electric Circuits*, Vol. 1 ch. 2 "Calculating electric power".
- All numbers checked in Python (`check3.py`, 2026-10-16).

Task: Make Two LEDs Equally Bright

Note: Task - Equal LEDs

Official task (2027 rules, Part II.c.iii): "Given some wires, batteries, resistors, and 2 LEDs, hook them up so the LEDs are equally bright." ✅ **IN at every level.**

An **LED** (light-emitting diode) is a diode that gives off light when current flows through it the "forward" way. The **anode** (+, long leg) must face the + side of the battery. The **cathode** (–, short leg, flat edge on the rim) faces the – side.

Brightness depends on current, not on voltage. Two LEDs of the same type with the same current are equally bright.

Key facts

- An LED that is on has an almost fixed voltage across it. This is the **forward voltage V_f**.
- Typical V_f: infrared 1.2-1.5 V; red 1.8-2.1 V; yellow and orange 2.0-2.2 V; old green 2.0-2.2 V; true green, blue and white 2.7-3.5 V.
- Shorter wavelength (blue end) → bigger band gap → higher V_f.
- A 5 mm indicator LED works at about 10-20 mA. The maximum is about 20-30 mA (read the datasheet).
- Always put a **current-limiting resistor** in series. Without it the current is too high, and the LED burns out.
- **Series LEDs** carry the same current. Two LEDs of the same type in series are always equally bright.
- **Parallel LEDs** need **one resistor each**. With one shared resistor, the LED with the lower V_f takes most of the current (**current hogging**).
- A red LED and a blue LED in parallel on one resistor: the red one lights; the blue one stays dark.
- Equal current in two **different colors** does not give exactly the same look. Your eye is most sensitive to green-yellow. Explain this on the answer sheet.
- Reverse voltage over about 5 V can damage an LED.
- You can measure V_f with the meter's **diode test** mode, or across a lit LED (see Testing Diodes A7).

Formulas

What	Formula	Notes
Series resistor	R = (V_s – V_f) / I	Round **up** to a standard value
Current with a chosen R	I = (V_s – V_f) / R	
n LEDs in series	R = (V_s – n·V_f) / I	Need V_s > n·V_f
Max LEDs in series	n_max = largest n with n·V_f < V_s	Leave headroom
Parallel branches	R_each = (V_s – V_f) / I; I_total = n·I	One R per LED
LED power	P = V_f × I	
Resistor power	P = (V_s – V_f) × I	Check 1/4 W
Current range from a battery range	I_min = (V_s,min – V_f,max)/R; I_max = (V_s,max – V_f,min)/R	GullSO Q50i

Standard E12 values: 10, 12, 15, 18, 22, 27, 33, 39, 47, 56, 68, 82 (× 1, 10, 100, 1000 …).

Question types

Easy

1. Size the series resistor for one LED

- Scope: ✅ IN.
- Where seen: Princeton Invitational 2026, Div C, INV, Lab Part II Q2 (~20 mA through each LED). GullSO Invitational 2025, Div C, INV, Q50f (3 V, 2.1 V, 20 mA → **45 Ω**). Fall 2019 F-M Invitational, Div B/C, INV, Q30 (9 V, 1.8 V, 10 mA → **720 Ω**). North Pocono Invitational 2020, Div C, INV, Q24 (9 V, 2.2 V, 15 mA → 453 Ω).

- How to solve:
  1. R = (V_s – V_f) / I.
  2. Round **up** to a value you have, so the current stays at or below the target.
- Worked example: 9 V, red V_f = 2.0 V, 20 mA. R = 7.0/0.020 = **350 Ω**. Use **390 Ω** → 17.9 mA. With 470 Ω → 14.9 mA.
- Traps: use V_s – V_f, not V_s. 9/0.020 = 450 Ω is wrong.
- Calculator: calculate — type: V_s, V_f, I.

2. Two LEDs of the same type in series

- Scope: ✅ IN.
- Where seen: Pembroke Hill Invitational 2026, Div B, INV, Q37 ("Build a circuit with 2 resistors and 2 LEDs in series"). Rules task II.c.iii.
- How to solve:
  1. Battery + → resistor → LED 1 → LED 2 → battery –. Both anodes face +.
  2. R = (V_s – 2·V_f) / I.
- Worked example: 9 V, two red LEDs (2.0 V each), 15 mA. R = 5.0/0.015 = **333 Ω**. Use 330 Ω → **15.2 mA** in both. They are equally bright.
- Traps: one LED backwards turns **both** off (series open).
- Calculator: calculate — type: V_s, n, V_f, I.

3. Find the LED's polarity

- Scope: ✅ IN.
- Where seen: Pembroke Hill Invitational 2026, Div B, INV, Q28 (long and short legs). 2026 ATX Regional, Div B and C, REG, hands-on (red/green LEDs). North Pocono Invitational 2020, Div C, INV, HO-37.
- How to solve:
  1. Long leg = anode (+). Flat edge = cathode (–).
  2. If the legs are cut, use diode test: the reading shows V_f when the red lead is on the anode.
- Worked example: diode test shows 1.82 V with the red lead on leg X, OL the other way. Leg X is the **anode**. The LED is probably **red**.
- Traps: the bigger metal piece inside is usually the cathode, but not always. Do not trust it.
- Calculator: look up — type: V_f reading.

4. Forward voltage by color (and why it matters)

- Scope: ✅ IN.
- Where seen: Princeton Invitational 2026, Div C, INV, Lab Part II Q1 ("What is the forward voltage of each LED color? Explain why knowing it is important"). USC Invitational 2026, Div B and C, INV, Lab Q39b (name the color; estimate V_f: red-green 2-3 V, blue/white 3-4 V). Berkeley Invitational 2026, Div B, INV, Q26 (lowest drop → red), Q40 (rank band gaps).
- How to solve:
  1. Measure V_f, or use the table in Key facts.
  2. Explain: the resistor must drop V_s – V_f. A wrong V_f gives a wrong current.
- Worked example: red 2.0 V, green 2.2 V, blue 3.0 V (typical). At 9 V and 20 mA: R = **350, 340, 300 Ω**.
- Traps: "green" can mean old GaP (≈2.2 V) or InGaN (≈3.0 V). Measure it.
- Calculator: look up — type: color.

Medium

5. Two LEDs in parallel, one resistor each

- Scope: ✅ IN.
- Where seen: rules task II.c.iii. Prior research H_logic_diodes B18. Not seen as a single test question.
- How to solve:
  1. Make two branches. Each branch = resistor + LED.
  2. Size each resistor for the same current: R = (V_s – V_f) / I.

- Worked example: 5 V, three red LEDs (2.0 V), 15 mA each. Series is impossible (6 V > 5 V). Use three branches of **200 Ω**. Total current = **45 mA**.
- Traps: series needs fewer parts. But it fails when n·V_f is near V_s.
- Calculator: calculate — type: V_s, V_f, I, n.

6. Shared resistor with parallel LEDs (current hogging)

- Scope: ✅ IN.
- Where seen: Pembroke Hill Invitational 2026, Div B, INV, Q36 ("2 resistors in parallel connected in series to two LEDs in parallel"). That build shares one resistance between two LEDs.
- How to solve:
  1. Both LEDs get the same voltage. The one with the lower V_f turns on first and takes most of the current.
  2. So use one resistor per LED, or put the LEDs in series.
- Worked example (simple LED model V = V_f + 10 Ω × I): 9 V, shared 330 Ω. LED A (2.00 V) and LED B (2.10 V): A gets **15.4 mA**, B gets **5.4 mA**. Not equal. A red (2.0 V) and a blue (3.0 V): red gets **20.6 mA**, blue gets **0 mA**.
- Traps: Pembroke's two 220 Ω in parallel act as one 110 Ω. The LEDs still share it.
- Calculator: calculate — type: V_s, R_shared, V_fA, V_fB.

7. Three colors equally bright (red, green, blue)

- Scope: ✅ IN.
- Where seen: Princeton Invitational 2026, Div C, INV, Lab Part II Q1-Q4 (R, G, B LEDs; series R for ~20 mA each; labeled diagram; build; record each current).
- How to solve:
  1. Measure each V_f.
  2. R_each = (V_s – V_f,each) / 20 mA. Pick the nearest resistor you have.
  3. Build three branches. Measure each current (V across its resistor ÷ R).
- Worked example: 9 V; red 2.0, green 2.2, blue 3.0 V. Ideal R = **350, 340, 300 Ω**. With three 330 Ω: **21.2, 20.6, 18.2 mA**. Close, but blue is lower. A 300 Ω for blue gives 20.0 mA.
- Traps: equal current is the target the key grades. Mention that perceived brightness can still differ by color.
- Calculator: calculate — type: V_s, list of V_f, I.

8. Pick resistors from a fixed kit (dim pair + bright one + blue unlit)

- Scope: ✅ IN.
- Where seen: Berkeley Invitational 2026, Div B, INV, Hands-On Q34 (40 pts). Kit: 9 V; 10, 100, 2 × 220, 330 Ω, 1 k, 2 k, 10 k, 100 kΩ; 3 LEDs + 1 blue LED. Two LEDs dim but equal, one bright, blue connected but unlit.
- How to solve:
  1. Dim and equal: put the two LEDs in series with a large resistor.
  2. Bright: a separate branch with a small resistor (keep I ≤ 20-25 mA).
  3. Blue unlit: put it in parallel with a lower-V_f LED (that LED holds the voltage near 2 V, below blue's turn-on), or put it in backwards across a low voltage (under 5 V).
- Worked example (three red LEDs, 2.0 V): dim pair + 1 kΩ → (9 – 4)/1000 = **5.0 mA** each. Bright LED + 330 Ω → **21.2 mA**. Blue across one dim red LED → blue sees 2.0 V → **dark**. (220 Ω would give 31.8 mA: too much.)
- Traps: putting the blue LED across a 9 V point lights it, or kills it if backwards (9 V reverse).
- Calculator: calculate — type: V_s, V_f list, resistor kit.

9. Battery range → LED current range

- Scope: ✅ IN.
- Where seen: GullSO Invitational 2025, Div C, INV, Q50i (battery 2.9-3.2 V; datasheet V_f typical 2.1 V, max 2.6 V; 47 Ω; key **23 mA** and **6.4 mA**).
- How to solve:
  1. Highest current: highest V_s and lowest V_f.
  2. Lowest current: lowest V_s and highest V_f.
- Worked example: I_max = (3.2 – 2.1)/47 = **23.4 mA**. I_min = (2.9 – 2.6)/47 = **6.4 mA**.

- Traps: with a low battery voltage, a small V_f change makes a huge current change. Use a higher V_s for stable brightness.
- Calculator: calculate — type: V_s min, V_s max, V_f min, V_f max, R.

10. Use the datasheet (max current, test conditions, temperature)

- Scope: ✅ IN.
- Where seen: GullSO Invitational 2025, Div C, INV, Q50a-c, k (max continuous current **50 mA**; V_f max **2.6 V at 20 mA, 25 °C**; connecting straight to the battery = no; above 70 °C the max is **30 mA**).
- How to solve:
  1. Read "absolute maximum ratings" for the current limit.
  2. Read "electrical characteristics" for V_f and its test current and temperature.
  3. Check the derating line for hot conditions.
- Worked example: GullSO asks: is 37.5 mA OK above 70 °C? Max at 70 °C is 30 mA, so **no**.
- Traps: "typical" is not "maximum". Design with the worst case.
- Calculator: look up — type: datasheet values.

Hard

11. Twice as bright

- Scope: ✅ IN.
- Where seen: GullSO Invitational 2025, Div C, INV, Q50j-n (key: **≈37.5 mA**; ideal R = 24 Ω; build with **22 Ω**; then measure and discuss).
- How to solve:
  1. Read the datasheet graph of "relative luminous intensity vs. forward current". Find the current for 2× intensity.
  2. If there is no graph, brightness ≈ proportional to current: double the current.
  3. New R = (V_s – V_f) / I_new. Check the max current and temperature.
- Worked example: 3.0 V, V_f = 2.1 V, I_new = 37.5 mA. R = 0.9/0.0375 = **24 Ω**. Use 22 Ω → **40.9 mA** (still under the 50 mA max at 25 °C).
- Traps: your eye does not see 2× light as "twice as bright". Say so in the discussion (Q50n).
- Calculator: calculate — type: V_s, V_f, I_new.

12. Dim an LED but keep it on (switched shunt resistor)

- Scope: ✅ IN.
- Where seen: USC Invitational 2026, Div B and C, INV, Lab Q39e (a button adds a second 510 Ω in parallel with the LED; the LED dims but stays on).
- How to solve:
  1. Series resistor R_s first. The shunt resistor R_sh sits across the LED.
  2. Current in R_s = (V_s – V_f)/R_s. The shunt takes V_f/R_sh. The LED gets the rest.
- Worked example: 9 V, R_s = R_sh = 510 Ω. Red (2.0 V): 13.7 mA → **9.8 mA** when pressed (–29%). Blue (3.0 V): 11.8 mA → **5.9 mA**.
- Traps: a plain wire across the LED (no resistor) turns it **off**. That is the NOR part (Q39c), not the dim part.
- Calculator: calculate — type: V_s, V_f, R_s, R_sh.

13. Most LEDs in one series string

- Scope: ✅ IN.
- Where seen: not seen in a test. Prior research H_logic_diodes B17.
- How to solve:
  1. n_max = largest n with n·V_f < V_s (leave at least about 1 V for the resistor).
  2. R = (V_s – n·V_f)/I.
- Worked example: 12 V, white LEDs (3.2 V), 20 mA. 3 LEDs = 9.6 V; 4 = 12.8 V (too much). R = 2.4/0.020 = **120 Ω**.
- Traps: with very little headroom, the current changes a lot with V_f and battery voltage (see type 9).
- Calculator: calculate — type: V_s, V_f, I.

14. Keep an LED connected but unlit

- Scope: ✅ IN.
- Where seen: Berkeley Invitational 2026, Div B, INV, Hands-On (blue LED "connected but unlit").
- How to solve (three ways):
  1. **Reverse bias:** put it in backwards across a part with less than 5 V.
  2. **Clamp:** put it in parallel with a lower-V_f LED. The voltage stays below its turn-on.
  3. **No current path:** connect both legs into one circuit node (both ends in the same breadboard row). Ask the ES if this counts as "connected".
- Worked example: blue (turn-on ≈ 2.6-3.0 V) across a lit red LED (2.0 V) → **unlit**. Blue backwards across a red LED → –2.0 V reverse → **unlit and safe**.
- Traps: backwards across 9 V can damage it (over 5 V reverse).
- Calculator: look up — type: V_f of both LEDs.

Nationals

15. Equal currents in two branches after a shared series resistor

- Scope: ✅ IN (series-parallel). It is States/Nationals difficulty.
- Where seen: not seen in a test. It combines Princeton 2026 Part II with a shared resistor.
- How to solve:
  1. Total current = sum of both branch currents.
  2. Node voltage after the shared resistor: V_n = V_s – I_total × R_0.
  3. Each branch: R = (V_n – V_f) / I.
- Worked example: 9 V, shared R_0 = 100 Ω, red (2.0 V) and blue (3.0 V) branches, 15 mA each. I_total = 30 mA. V_n = 9 – 3.0 = **6.0 V**. R_red = 4.0/0.015 = **267 Ω**. R_blue = 3.0/0.015 = **200 Ω**.
- Traps: if one LED fails open, the other branch current rises (less drop in R_0).
- Calculator: calculate — type: V_s, R_0, V_f1, V_f2, I.

Out of scope (skip)

- LED driver chips, PWM dimming and transistor drivers. ❌ Semiconductors beyond those listed; digital clock signals (rule 3.d).
- Mystery capacitor shown by LED blinks (Cobra Invitational 2019-20, Div B, INV, Station 5). ❌ Capacitance (rule 3.d). The idea "two LEDs in opposite directions show current direction" is still IN.
- Photon energy E = hc/λ. ⚠️ Borderline: it explains color and V_f, but the rules do not name it.
- Lux, lumens and candela conversions. ⚠️ Borderline: not in the rule list; only "brightness" is needed.

Mistakes in tests and keys

- **GullSO Invitational 2025, Div C, INV, Q50f vs Q50i:** part (f) uses V_f = 2.1 V (typical). Part (b) calls 2.6 V the "maximum voltage … at typical operating conditions". The two words mix "typical" and "maximum". Read the datasheet table yourself.
- **GullSO Q50j:** the key gives ≈37.5 mA for "twice as bright" with no work. It probably comes from the datasheet's intensity graph. I could not see that datasheet.
- **2026 ATX Regional, Div B and C, REG, hands-on:** the key wires the LEDs with no resistors and uses a multimeter on DC volts as the "power source". See Task - 3-Way Switch [A9] (type 5) for the full error.
- **North Pocono Invitational 2020, Div C, INV, Q24:** "maximum R that lights it" (453 Ω) is badly worded. A larger R still lights the LED, only dimmer.

Not confirmed

- The 10 Ω LED model in type 6 is my simple model. Real LEDs differ. The direction of the effect (lower V_f takes more current) is certain.
- Berkeley 2026: the colors of the "3 LEDs" are not stated. I assumed red. The key has no answer for the hands-on.
- Typical V_f values are datasheet-typical ranges. Your kit's LEDs may differ by 0.2 V or more.

Sources

- 2027 Div B Rules Manual, Circuit Lab B, Part II.c.iii.
- Princeton 2026 lab (`downloads/dlG/puso26_lab.txt`); GullSO 2025 test and key (`dlG/gull26_*.txt`); Berkeley 2026 test (`dlB/tests/berk26_test.txt`); Pembroke Hill 2026 (`dlB/tests/phi26_test.txt`); USC 2026 setup (`dlFind/usc26_Bsetup.txt`).
- Test extractions: `research/tests/nat26.md` §3, `misc-a.md` (F-M 2019, North Pocono 2020, Cobra), `ut-a.md` (ATX 2026); `prior/B_topics.md` §7e; `prior/H_logic_diodes.md` B16-B19, B27-B28; `prior/G_tests.md` §2.
- Kuphaldt, *Lessons in Electric Circuits*, Vol. 3 ch. 3 "Light-emitting diodes"; Vol. 6 "LED experiments".
- All About Circuits, "LEDs" (textbook chapter).
- All numbers checked in Python (`check3.py`, 2026-10-16).

Task: Read a USB Charger Label and Find the Charge Time

Note: Task - USB Charger

Official task (2027 rules, Part II.c.iv): "Given a USB charger, read the label and provide details of the various output power levels it can provide and calculate how long it would take to charge a specific battery." ✅ **IN at every level.** It was also in the 2026 rules.

A **USB charger** turns wall AC into low-voltage DC. Its **label** lists the input (AC side) and the outputs (DC side). You only read the label. You do not plug it in: the rules ban AC circuits in the hands-on part (Part II.d).

Key facts

- **Input** line, for example "100-240 V ~ 50/60 Hz 1.5 A". This is the AC side. "100-240 V" means it works in any country.
- **Output** line, for example "5 V ⎓ 3 A, 9 V ⎓ 3 A, 15 V ⎓ 3 A, 20 V ⎓ 3.25 A (65 W max)".
- Symbols: ⎓ (a solid line over a dashed line) = DC. **~** = AC.
- Each output level has a power: P = V × I.
- "65 W max" can be a total for all ports together. A two-port charger may split it.
- The **device** asks for the level it wants (this is called **negotiation**). The charger does not push extra current into the device.
- **mAh** (milliamp-hour) measures charge. 1 mAh = 3.6 C (coulombs). 1000 mAh = 1 Ah.
- **Wh** (watt-hour) measures energy. 1 Wh = 3600 J.
- Battery energy (Wh) = capacity (Ah) × nominal voltage (V).
- A phone Li-ion cell has a nominal voltage of 3.7 V (some say 3.85 V).
- Real charging loses energy as heat. Efficiency is often 80-90%.
- Li-ion charging is **CC/CV**: constant current first, then constant voltage. The current gets small near full, so the last part is slow.
- Common USB levels (from the USB-IF specifications; general knowledge, not in an SO document):

Standard	Output
USB 2.0 port	5 V, 0.5 A (2.5 W)
USB 3.x port	5 V, 0.9 A (4.5 W)
USB Battery Charging 1.2	5 V, 1.5 A (7.5 W)
Old phone cube	5 V, 1 A (5 W)
Tablet charger	5 V, 2.1-2.4 A (about 10-12 W)
USB Type-C current	5 V at 1.5 A (7.5 W) or 3 A (15 W)
USB Power Delivery (PD)	5, 9, 15, 20 V; up to 3 A (60 W) with a normal cable; 5 A (100 W) with an e-marked cable
USB PD 3.1 EPR	adds 28, 36, 48 V; up to 240 W

Formulas

What	Formula	Notes
Output power per level	P = V × I	
Simple charge time	t (h) = capacity (mAh) / current (mA)	Same voltage only
Battery energy	E (Wh) = capacity (Ah) × V_nominal	
Energy charge time	t (h) = E (Wh) / (P_charger × η)	η = efficiency
Energy from power and time	E = P × t	Wh or J (× 3600)
Charge in coulombs	Q (C) = capacity (mAh) × 3.6	
Average power from battery life	P = E / t	
Input power	P_in = P_out / η	
Charge while using	t = E / (P_charger × η − P_use)	

Question types

Easy

1. Output power of each level on the label

- Scope: ✅ IN.
- Where seen: Rickards Invitational 2025 (2026 season), Div B, INV, Q31 ("A USB charger label shows an output of 5.0 V and 3.0 A. How much power?" key **15 W**). Rules task II.c.iv.
- How to solve:
  1. Find each "V ⎓ A" pair on the Output line.
  2. P = V × I for each pair.
- Worked example: "5 V ⎓ 3 A, 9 V ⎓ 3 A, 15 V ⎓ 3 A, 20 V ⎓ 3.25 A". P = **15, 27, 45, 65 W**.
- Traps: do not use the **Input** current. That is the AC side.
- Calculator: calculate — type: list of (V, I).

2. Energy given in a time

- Scope: ✅ IN.
- Where seen: Rickards Invitational 2025, Div B, INV, Q32 (15 W for 1 hour; key **15 Wh / 54 000 J**).
- How to solve:
  1. E (Wh) = P (W) × t (h).
  2. E (J) = Wh × 3600.
- Worked example: 15 W for 1 h = **15 Wh** = 15 × 3600 = **54 000 J**.
- Traps: Rickards says "stored". A real battery stores less (efficiency). Write your assumption.
- Calculator: calculate — type: P, t.

3. Read the symbols on the label

- Scope: ✅ IN (AC and DC characteristics, rule 3.c.iii). Safety marks are ⚠️ Borderline trivia.
- Where seen: not seen in a test. Rules task II.c.iv ("details of the various output power levels").
- How to solve:
  1. ⎓ = DC. ~ = AC. "50/60 Hz" = AC frequency.
  2. "Input 100-240 V" = universal (any country).
  3. A square inside a square = **double insulated** (Class II). UL, CE, FCC = safety or radio marks (Borderline).
- Worked example: "Input: 100-240 V~ 1.5 A 50/60 Hz; Output: 5 V ⎓ 2.4 A". AC in, DC out. One level: **12 W**.
- Traps: the "1.5 A" input is a maximum. It is not the normal current.
- Calculator: look up — type: symbol.

4. Simple charge time (mAh ÷ mA)

- Scope: ✅ IN.
- Where seen: not seen in a test. B_topics §9d; I_bank A14.
- How to solve:
  1. Change both to the same unit (mAh and mA).
  2. t (h) = capacity / current.
- Worked example: 3000 mAh battery at 1000 mA = **3.0 h**.
- Traps: this only works if the current is at the **battery** voltage. A 9 V charger current is not the battery current (see type 5).
- Calculator: calculate — type: mAh, mA.

Medium

5. Charge time with the energy method (different voltages)

- Scope: ✅ IN.
- Where seen: not seen in a test. This is the method the rules task needs when the label shows 9 V or 20 V.
- How to solve:
  1. Battery energy E = Ah × V_nominal.

   2. Usable charger power = P × η.
   3. t = E / (P × η).
- Worked example: 5000 mAh × 3.7 V = **18.5 Wh**. Charger 9 V ⎓ 2 A = 18 W, 85% efficient. t = 18.5 / (18 × 0.85) = **1.21 h** (about 73 min).
- Traps: 5000 mAh ÷ 2000 mA = 2.5 h is **wrong** here. The 2 A is at 9 V, not at 3.7 V.
- Calculator: calculate — type: mAh, V_nominal, V_out, I_out, efficiency.

6. Average power from battery life

- Scope: ✅ IN.
- Where seen: UT Austin Invitational 2019, Div B, INV, Q28 (3000 mAh, 4.4 V battery; battery life 6 h; find the average power).
- How to solve:
   1. E = Ah × V.
   2. P = E / t.
- Worked example: 3.0 Ah × 4.4 V = 13.2 Wh. P = 13.2 / 6 = **2.2 W**.
- Traps: 3000 mAh is 3.0 Ah, not 3000 Ah.
- Calculator: calculate — type: mAh, V, hours.

7. Battery capacity in coulombs and joules

- Scope: ✅ IN.
- Where seen: not seen in a test. I_bank A14 and Area 4.
- How to solve:
   1. Q (C) = mAh × 3.6.
   2. E (J) = Q × V.
- Worked example: 2000 mAh = **7200 C**. At 3.7 V: E = 26 640 J = **7.4 Wh**.
- Traps: mAh is charge, not energy. You need the voltage to get energy.
- Calculator: calculate — type: mAh, V.

8. Which output level will the device use?

- Scope: ✅ IN (uses of DC).
- Where seen: not seen in a test. Rules task II.c.iv ("various output power levels").
- How to solve:
   1. A plain 5 V device (no fast charging) uses the 5 V level.
   2. A USB-PD phone or laptop asks for a higher level (9, 15 or 20 V).
   3. The device draws only the current it needs, up to the level's limit.
- Worked example: a 65 W PD charger and a phone that supports only 5 V ⎓ 2 A. The phone uses **5 V and at most 2 A = 10 W**. The charger's 65 W does not matter.
- Traps: "65 W charger" does not mean every device charges at 65 W.
- Calculator: look up — type: device level, charger levels.

9. Power bank: how much it can really give

- Scope: ✅ IN.
- Where seen: not seen in a test. B_topics §9d.
- How to solve:
   1. Bank energy = mAh × 3.7 V (the cell voltage).
   2. Usable = energy × efficiency.
   3. Number of phone charges = usable ÷ phone battery energy.
- Worked example: 10 000 mAh, 3.7 V bank = **37 Wh**. At 85%: 31.45 Wh. At 5 V that is **6290 mAh**. Phone 3000 mAh × 3.85 V = 11.55 Wh → **2.7 charges** (ignoring the phone's own charging loss).
- Traps: the bank's "10 000 mAh" is at 3.7 V, not at 5 V.
- Calculator: calculate — type: bank mAh, V_cell, efficiency, phone mAh, phone V.

10. Input current from output power

- Scope: ✅ IN (AC uses and characteristics; only P = V × I, no AC theory).
- Where seen: not seen in a test.
- How to solve:
  1. P_in = P_out / η.
  2. I_in = P_in / V_in.
- Worked example: 65 W out, 90% efficient → P_in = **72.2 W**. At 120 V: I_in = **0.60 A**. At 100 V: 0.72 A. The label's "1.5 A" is a safe maximum.
- Traps: AC power with power factor is ❌ OUT. Use P = V × I only.
- Calculator: calculate — type: P_out, efficiency, V_in.

Hard

11. CC/CV charging takes longer than the simple estimate

- Scope: ⚠️ Borderline (needs a model that the question must give).
- Where seen: not seen in a test. B_topics §9d.
- How to solve:
  1. Phase 1 (CC): constant current up to about 80%.
  2. Phase 2 (CV): the current falls. Use the average current the question gives.
  3. Add the two times.
- Worked example: 3000 mAh, 1.5 A CC to 80%: 2400 mAh ÷ 1500 mA = **1.6 h**. Last 600 mAh at an average 0.5 A = **1.2 h**. Total **2.8 h** (the simple estimate says 2.0 h).
- Traps: state your model. A grader will accept a clear assumption.
- Calculator: calculate — type: mAh, CC current, CC fraction, CV average current.

12. Compare two chargers

- Scope: ✅ IN.
- Where seen: not seen in a test.
- How to solve:
  1. Find the usable power of each charger (P × η).
  2. Use the energy method for each.
  3. Ratio of times = inverse ratio of powers.
- Worked example: 18.5 Wh battery, 85% efficiency. 5 V ⎓ 1 A (5 W) → **4.35 h**. 9 V ⎓ 2 A (18 W) → **1.21 h**. The fast charger is **3.6 times** faster.
- Traps: the phone must support the faster level (type 8).
- Calculator: calculate — type: E, list of chargers, efficiency.

13. Slow port: USB 2.0 limit

- Scope: ✅ IN.
- Where seen: not seen in a test.
- How to solve:
  1. The port's power limit sets the speed (5 V × 0.5 A = 2.5 W).
  2. t = E / (P × η).
- Worked example: 18.5 Wh on a USB 2.0 port: 18.5/2.5 = **7.4 h** (100%); **8.7 h** at 85%.
- Traps: a phone that is in use may use most of that 2.5 W itself (type 15).
- Calculator: calculate — type: E, port V, port I, efficiency.

14. Cost of the energy

- Scope: ✅ IN (energy, rule 3.c.iv).
- Where seen: not seen in a test.
- How to solve:
  1. Wall energy per charge = E_battery / η.
  2. kWh per year = Wh per charge × charges ÷ 1000.
  3. Cost = kWh × price.

- Worked example: 18.5 Wh / 0.85 = 21.8 Wh per charge. 365 charges = **7.94 kWh**. At $0.15 per kWh = **$1.19 per year**.
- Traps: kWh, not kW.
- Calculator: calculate — type: E, efficiency, charges, price.

Nationals

15. Charging while the device is in use

- Scope: ✅ IN.
- Where seen: not seen in a test.
- How to solve:
  1. Net charging power = P_charger × η – P_use.
  2. t = E / net power.
  3. If the net power is ≤ 0, the battery never fills.
- Worked example: 18 W charger, 85% → 15.3 W. Phone uses 2 W. Net **13.3 W**. 18.5 Wh → **1.39 h** (not 1.21 h). On a USB 2.0 port: 2.5 × 0.85 – 2 = 0.125 W → **148 h** (it barely charges).
- Traps: watch the sign. Net power can be negative; then the battery drains while plugged in.
- Calculator: calculate — type: E, P_charger, efficiency, P_use.

16. Laptop charge time at different levels

- Scope: ✅ IN.
- Where seen: not seen in a test.
- How to solve:
  1. Battery Wh is often printed on the laptop battery.
  2. Use each output level that the laptop accepts.
- Worked example: 60 Wh laptop battery, 90% efficiency. 20 V ⎓ 3.25 A (65 W) → **1.03 h**. 5 V ⎓ 3 A (15 W) → **4.44 h** (if the laptop even accepts 5 V).
- Traps: many laptops refuse 5 V. Say "if the device accepts it".
- Calculator: calculate — type: Wh, list of levels, efficiency.

Out of scope (skip)

- Plugging in or opening the charger. ❌ Hands-on AC circuits are banned (Part II.d).
- How the switching converter works inside (transformer, transistors). ❌ Inductance, magnetism and semiconductors beyond those listed (rule 3.d).
- AC power factor and RMS. ❌ AC circuit theory (rule 3.d).
- Wireless (Qi) charging. ❌ Magnetism/induction (rule 3.c only).
- Airline 100 Wh limit and battery chemistry trivia. ⚠️ Borderline: "hazards of DC" could include battery fires, but the rules do not name these facts.

Mistakes in tests and keys

- **Rickards Invitational 2025, Div B, INV, Q32:** the key says "15 Wh would be stored". A real battery stores less, because charging is not 100% efficient. The number is right for "energy delivered by the charger".
- No other USB question found in the tests I searched. So I found no other key errors.

Not confirmed

- I found **no hands-on USB station** in any test in the downloads. Where I looked: every `.txt` in `downloads/dlG`, `dlFind`, `dlB/tests` (grep "USB" and "mAh"). Only Rickards 2026 B (Q14 option, Q31-Q32) and UT 2019 B (Q28) mention it.
- USB level values come from the USB-IF specifications as general knowledge. I did not open the USB-IF documents in this run.
- Efficiency 80-90% and the CC/CV 80% point are typical values, not official numbers.

Sources

- 2027 Div B Rules Manual, Circuit Lab B, Part II.c.iv and II.d.
- Rickards 2026 B test and key (`downloads/dlG/rsoi26B_test.txt`, `rsoi26B_key.txt`); UT 2019 B test and key (`dlG/ut19inv_Btest.txt`, `ut19inv_Bkey.txt`).
- Prior research: `prior/B_topics.md` §9d; `prior/I_bank.md` A14; `prior/F_slides.md` §5 and §13 (no deck covers energy or USB).
- USB Implementers Forum, USB 2.0, USB 3.x, Battery Charging 1.2, Type-C and USB Power Delivery specifications (general knowledge).
- OpenStax, *College Physics 2e*, §20.4 "Electric Power and Energy".
- All numbers checked in Python (`check4.py`, 2026-10-16).

Task: Build a Logic Circuit That Matches a Truth Table

Note: Task - Logic from Truth Table

Official task (2027 rules, Part II.c.v): "Construct a logic circuit to mimic a given truth table." ✅ **IN at every level.** It is **new for 2027** in Div B.

A **truth table** lists every input combination and the output for each one. With n inputs there are 2^n rows.

A **logic circuit** can be built three ways in a lab:

- **Switches** (push buttons, slide switches) and an LED. Series = AND. Parallel = OR.
- **Logic chips** (for example the 7400 family) with switches as inputs and an LED as the output.
- **Diodes and resistors** (diode logic). This makes AND and OR, but never NOT.

Key facts

- 1 = HIGH = on = switch pressed (unless the task says otherwise). 0 = LOW = off.
- **Switches in series → AND.** The LED lights only if every switch is closed.
- **Switches in parallel → OR.** The LED lights if any switch is closed.
- **A switch across the LED (a shunt) → NOT.** Pressing it shorts the LED, so it goes off. A series resistor must stay in the circuit.
- A **normally closed** contact gives a NOT input (A').
- Common chips (14-pin DIP): **7400** quad 2-input NAND; **7402** quad NOR; **7404** hex NOT; **7408** quad AND; **7432** quad OR; **7486** quad XOR.
- On these chips **pin 14 = V_cc (+5 V)** and **pin 7 = GND**. Count pins counter-clockwise from the notch.
- The **7402** has a different pinout: its outputs are pins 1, 4, 10, 13.
- A chip must straddle the breadboard's center gap (see Breadboards A9).
- Never leave an input **floating** (connected to nothing). Use a pull-up or pull-down resistor. A floating TTL input acts as HIGH. A floating CMOS input is random.
- An LED on a gate output needs a series resistor (about 330-470 Ω at 5 V).
- **NAND alone** or **NOR alone** can build every function. These are **universal gates**.
- Always test **every row** before you call the supervisor. The USC 2026 proctors tested all four button combinations.

Formulas

What	Rule	Notes
Sum of products (SOP)	OR together one AND term per output-1 row	Complement inputs that are 0
Product of sums (POS)	AND together one OR term per output-0 row	⚠ Borderline (not named)
Switch AND	S1 in series with S2	
Switch OR	S1 in parallel with S2	
Switch NOT (shunt)	switch across the LED, series R kept	
De Morgan	(AB)' = A' + B'; (A + B)' = A'B'	
NOT from NAND	A' = NAND(A, A)	
AND from NOR	AB = NOR(A', B') = NOR(NOR(A,A), NOR(B,B))	3 NOR gates
XOR from 4 NAND	n = NAND(A,B); out = NAND(NAND(A,n), NAND(B,n))	
XNOR from 4 NOR	n = NOR(A,B); out = NOR(NOR(A,n), NOR(B,n))	
Half adder	S = A ⊕ B; C = AB	
Full adder	S = A ⊕ B ⊕ C_in; C_out = AB + C_in(A ⊕ B)	
2:1 MUX	$Y = S'D_0 + SD_1$	
LED resistor	R = (5 V – V_f) / I	

Question types

Easy

1. Name the gate from a 4-row truth table

- Scope: ✅ IN.
- Where seen: NY Lake Erie/Niagara Invitational 2018, Div C, INV, Q16 (1, 0, 0, 0 → **NOR**). Princeton Invitational 2019, Div C, INV, "Logic" (a)-(f). MIT Invitational 2019, Div C, INV, Lab Station 2 Q3a (read blinking LEDs: 1, 1, 1, 0 → **NAND**).
- How to solve:
  1. List the output column in the order 00, 01, 10, 11.
  2. Match: AND 0001, OR 0111, NAND 1110, NOR 1000, XOR 0110, XNOR 1001.
- Worked example: outputs 1, 1, 1, 0 → **NAND** (Python check).
- Traps: check the row order of the table. Some tables list 00, 10, 01, 11.
- Calculator: look up — type: 4 output bits.

2. Identify a pre-built (mystery) gate by testing

- Scope: ✅ IN.
- Where seen: MIT Invitational 2020, Div C, INV, Lab 2 Q58 (draw Circuit 1, name it, write its truth table; key: 2-input **OR**). 2019 UT Austin Regional, Div C, REG, hands-on (3 mystery gates with LEDs). Shock Value tryout ~2012, Div B, practice. Princeton Invitational 2020, Div C, INV, Lab 2 (3-input transistor gate → truth table).
- How to solve:
  1. Apply all 2^n input combinations (switches or jumpers to +5 V and GND).
  2. Record the output (LED on/off, or meter: HIGH > about 2 V, LOW < about 0.8 V for TTL).
  3. Match the column (type 1).
- Worked example: output LOW only when both inputs are HIGH → **NAND**.
- Traps: a floating input gives a wrong row. Tie every input to a level.
- Calculator: look up — type: output bits.

3. AND or OR with two switches

- Scope: ✅ IN.
- Where seen: USC Invitational 2026, Div B and C, INV, Lab Q39 (push buttons as logic). Prior research H_logic_diodes A20 (Shannon's switch mapping).
- How to solve:
  1. AND: battery → resistor → S1 → S2 → LED → battery.
  2. OR: put S1 and S2 side by side (parallel) in the same spot.
- Worked example: 9 V, 470 Ω, red LED, S1 and S2 in series. The LED lights only for "both pressed". Current = 7.0/470 = **14.9 mA**.
- Traps: a 4-pin push button has two pins joined on each side. Use pins from **opposite** pairs (see Breadboards A9 type 7).
- Calculator: look up — type: gate name.

Medium

4. NOR with two buttons that short the LED

- Scope: ✅ IN.
- Where seen: USC Invitational 2026, Div B and C, INV, Lab Q39c (LED on only when both buttons are released).
- How to solve:
  1. Battery → 510 Ω → LED → battery.
  2. Put both buttons in parallel **across the LED**.
  3. Pressing either button shorts the LED. The LED is on only when neither is pressed.
- Worked example: truth table on = 1, 0, 0, 0 → **NOR** (Python check). A button press puts 9 V across 510 Ω: **17.6 mA**, 0.159 W in the resistor.

- Traps: never short the LED without the series resistor. Then you short the battery.
- Calculator: look up — type: gate name.

5. A AND NOT B (one series button, one shunt button)

- Scope: ✅ IN.
- Where seen: USC Invitational 2026, Div B and C, INV, Lab Q39d (LED on only if the left button is pressed and the right one is released).
- How to solve:
  1. Button A in series (it feeds the LED).
  2. Button B across the LED (it shorts the LED).
- Worked example: table for (A, B) = 00, 01, 10, 11 → 0, 0, 1, 0 = **AB'**.
- Traps: if B is in series and A is the shunt, you build A'B instead. Read which button is "specific".
- Calculator: look up — type: expression.

6. Truth table → SOP → simplify → build with chips

- Scope: ✅ IN.
- Where seen: MIT Invitational 2020, Div C, INV, Lab 2 Q59 (build a gate that matches a 3-input table; "there may be multiple solutions"; proctor initials 10 pts). Prior research H_logic_diodes A48.
- How to solve:
  1. Write one AND term for each row with output 1.
  2. OR the terms. Simplify with Boolean rules.
  3. Count gates. Pick chips (4 two-input gates per chip).
  4. Wire V_cc and GND first. Then the gates. Test every row.
- Worked example: output 1 on rows 001, 011, 101, 110, 111 (A, B, C). SOP simplifies to **F = C + AB**. Build: one 7408 AND gate, one 7432 OR gate (Python check).
- Traps: forgetting pin 14 and pin 7. A chip with no power gives no output at all.
- Calculator: calculate — type: output column (8 bits).

7. Three-input NOR from two-input chips

- Scope: ✅ IN.
- Where seen: MIT Invitational 2020, Div C, INV, Lab 2 Q59 (table 1, 0, 0, 0, 0, 0, 0, 0 → **3-input NOR**).
- How to solve:
  1. Q = (A + B + C)'.
  2. With a 7432 and a 7402: X = OR(A, B), then Q = NOR(X, C).
  3. With only a 7402: X = NOR(A, B); X' = NOR(X, X); Q = NOR(X', C). That is 3 NOR gates.
- Worked example: Q = 1 only for A = B = C = 0 (Python check for both builds).
- Traps: NOR(NOR(A, B), C) is **not** a 3-input NOR. You must invert the first NOR.
- Calculator: look up — type: gate list.

8. Only NAND chips: build XOR (or any function)

- Scope: ✅ IN (universal gates, rule 3.c.viii).
- Where seen: PA Central Regional ~2013, Div C, REG, Q27-29 (build OR, AND from one gate type). Princeton Invitational 2019, Div C, INV, (b)-(c) (NAND/NOR → NOT, AND, OR). Prior research H_logic_diodes A48 example.
- How to solve:
  1. Write the function with AND, OR, NOT.
  2. Replace each with its NAND form, or use a known circuit (XOR = 4 NAND).
  3. Cancel double inverters.
- Worked example: F = 1 on rows 01 and 10 → XOR. n = NAND(A, B); F = NAND(NAND(A, n), NAND(B, n)). That is **4 gates = exactly one 7400** (Python check).
- Traps: a NAND with both inputs tied is a NOT. Do not leave the second input floating.
- Calculator: look up — type: target gate, allowed gate.

9. AND from NOR gates only

- Scope: ✅ IN.
- Where seen: USC Invitational 2026, Div B, INV, Q36a (key: NOR(NOT A, NOT B), with NOT from NOR). PA Central Regional ~2013, Div C, REG, Q28.
- How to solve:
  1. A' = NOR(A, A). B' = NOR(B, B).
  2. AB = NOR(A', B') (De Morgan).
- Worked example: three NOR gates → table 0, 0, 0, 1 = **AND** (Python check).
- Traps: one NOR on its own gives (A + B)', not AB.
- Calculator: look up — type: target gate, allowed gate.

10. Drive the output LED the right way

- Scope: ✅ IN (LED use + logic).
- Where seen: not seen in a test. Prior research H_logic_diodes A52.
- How to solve:
  1. **Sourcing:** gate output → resistor → LED → GND. The LED is on when the output is HIGH.
  2. **Sinking:** +5 V → resistor → LED → gate output. The LED is on when the output is LOW (an inverted display).
- Worked example: 5 V output, red LED 2.0 V, 330 Ω → **9.1 mA**.
- Traps: with sinking, the LED shows the **opposite** of the table. Say this, or add a NOT gate.
- Calculator: calculate — type: V_out, V_f, R.

Hard

11. Majority vote of three switches

- Scope: ✅ IN.
- Where seen: not seen in a hands-on. Majority-of-3 is a standard word problem (prior research H_logic_diodes A19).
- How to solve:
  1. SOP: F = AB + BC + AC.
  2. Factor for fewer parts: F = A(B + C) + BC.
  3. With switches: (B || C) in series with A, all of that in parallel with (B in series with C). B and C each appear twice, so you need **double-pole** switches (or chips).
- Worked example: table 0, 0, 0, 1, 0, 1, 1, 1 for both forms (Python check). Chips: 3 AND (7408) + 2 OR (7432).
- Traps: a single-pole switch cannot appear in two places. That is why chips are easier here.
- Calculator: calculate — type: output column.

12. Diode logic (no chips)

- Scope: ✅ IN (uses of single-junction devices, rule 3.c.vii).
- Where seen: not seen in a hands-on. National supervisor deck 15 asks: can you make NOT from diodes? (No.)
- How to solve:
  1. **Diode OR:** inputs → diode anodes; cathodes joined → output, with a resistor to GND.
  2. **Diode AND:** inputs → diode cathodes; anodes joined → output, with a resistor to +5 V.
- Worked example: diode AND with one input at 0 V gives an output of about **0.7 V** (LOW). Diode OR with one input at 5 V gives about **4.3 V** (HIGH).
- Traps: each diode stage loses 0.7 V. Several stages in a row lose too much.
- Calculator: calculate — type: V_in, number of diode drops.

13. Half adder or full adder on the breadboard

- Scope: ✅ IN (adders, rule 3.c.viii).
- Where seen: not seen as a hands-on. UT Austin Invitational 2019, Div C, INV, Q49 (half adder drawing).
- How to solve:
  1. Half adder: S = A ⊕ B (7486), C = AB (7408). Two output LEDs.
  2. Full adder: two XOR, two AND, one OR.
  3. Check: S + 2C = A + B (+ C_in) for every row.

- Worked example: full adder rows all pass S + 2·C_out = A + B + C_in (Python check, 8 rows).
- Traps: label the LEDs "S" and "C". The carry LED is the 2s place.
- Calculator: look up — type: inputs A, B, C_in.

14. 2:1 multiplexer

- Scope: ✅ IN (multiplexers, rule 3.c.viii).
- Where seen: not seen in a test as a hands-on.
- How to solve:
  1. $Y = S'D_0 + SD_1$.
  2. Parts: 1 NOT (7404), 2 AND (7408), 1 OR (7432).
- Worked example: S = 0 → Y follows D_0. S = 1 → Y follows D_1 (Python check, 8 rows).
- Traps: the select input S must go to both AND gates (one through the NOT).
- Calculator: look up — type: S, D_0, D_1.

Nationals

15. Fewest gates of one type (for example XNOR from NOR)

- Scope: ✅ IN.
- Where seen: not seen in a test. Prior research H_logic_diodes (minimum-gate search).
- How to solve:
  1. Use the dual of a known NAND circuit: swap NAND ↔ NOR and the function becomes its dual.
  2. XOR from 4 NAND ↔ **XNOR from 4 NOR**.
- Worked example: n = NOR(A, B); Q = NOR(NOR(A, n), NOR(B, n)) → 1, 0, 0, 1 = **XNOR** (Python check). That is 4 gates = one 7402.
- Traps: XOR from NOR alone needs **5** gates (XNOR + an inverter).
- Calculator: look up — type: target, allowed gate.

16. Match a table that has "don't care" rows or missing rows

- Scope: ✅ IN.
- Where seen: MIT Invitational 2020, Div C, INV, Q49 (a switch truth table with 15 of 16 rows given; see Task - 3-Way Switch A9 type 11).
- How to solve:
  1. Mark the missing rows as "don't care" (X).
  2. Choose X = 0 or 1, whichever gives the simpler expression.
  3. Say your choice on the answer sheet.
- Worked example: outputs 0, 1, 1, X → choosing X = 1 gives **A + B** (one OR gate). X = 0 gives A ⊕ B.
- Traps: a supervisor tests only the given rows. Do not lose time on the X row.
- Calculator: calculate — type: output column with X.

Out of scope (skip)

- Latches, flip-flops, counters and anything with a clock. ❌ Sequential logic and digital clock signals (rule 3.d). Example: USC Invitational 2026, Div B, Q36b-d (cross-coupled NOR latch). MIT 2019 Station 2 used a counter to drive the inputs; you only read the LEDs, which is IN.
- Building gates from transistors (Princeton Invitational 2020, Lab 2 internals). ❌ Semiconductors beyond those listed (rule 3.d). Reading the resulting truth table is IN.
- TTL/CMOS voltage-level specs (V_IL, V_IH, noise margin). ⚠️ Borderline: useful for a meter check, but not named in the rules.
- Karnaugh maps. ⚠️ Borderline: not named, but a good tool to simplify.

Mistakes in tests and keys

- **2026 ATX Regional, Div B and C, REG, hands-on:** the key's wiring does not work (red always on, green never on, no resistors). See Task - 3-Way Switch A9 type 5.

- **UT Austin Invitational 2019, Div C, INV, Q49:** the half-adder sum gate is drawn as XOR (right), but labeled "A + B" (wrong: + means OR). It should be A ⊕ B.
- **USC Invitational 2026, Div B and C, INV, MC21:** two answer choices are the same function (A + B' and NOT(A'B)). Simplify every choice before you pick.

Not confirmed

- MIT 2020 Lab 2 Q58 Circuit 1 (OR) is from the key text. I did not see the board.
- The 2019 UT Regional C mystery-gate key depends on the kit. No answers are in the file.
- The USC 2026 Q36a answer drawing is an image. I did not render it; the text says NOR(NOT A, NOT B).
- Chip pinouts are from standard 74xx datasheets (general knowledge). Ask the supervisor for the datasheet at the event.

Sources

- 2027 Div B Rules Manual, Circuit Lab B, Part II.c.v and 3.c.viii, 3.d.
- MIT 2020 test and key (`downloads/dlG/mit20_test.txt`, `mit20_key.txt`); USC 2026 setup (`dlFind/usc26_Bsetup.txt`); MIT 2019 lab key (`dlFind/mit19_lab_key.txt`).
- Test extractions: `research/tests/nat26.md` §3-4, `misc-b.md` (MIT 2019, NY 2018), `ut-c.md` (UT 2019 C, ATX 2019 C); `prior/G_tests.md` §2 (logic); `prior/H_logic_diodes.md` A19-A24, A48-A52; `prior/B_topics.md` §8f-8g.
- Kuphaldt, *Lessons in Electric Circuits*, Vol. 4 (Digital) ch. 3 "Logic Gates", ch. 7 "Boolean Algebra", ch. 9 "Combinational Logic Functions".
- All About Circuits, Digital textbook, "Switch and relay logic".
- All truth tables checked in Python (`check5.py`, 2026-10-16).

Task: Troubleshoot a Faulty Circuit

Note: Task - Troubleshoot

Official task (2027 rules, Part II.c.vi): "Troubleshoot a faulty circuit." ✅ **IN at every level.** It is **new for 2027**. The KVL/KCL version (type 16) is ✅ **States/Nationals**.

Troubleshooting means finding why a circuit does not work, and fixing it. A **fault** is the thing that is wrong.

The two basic faults:

- An **open** is a break in the path. No current flows through it.
- A **short** is an unwanted path with almost no resistance. Current goes around a part instead of through it.

Key facts

- **Check the obvious first:** power connected, battery good (measure it), switch on, fuse or breaker OK, part burned out, loose wire. (National supervisor deck 11.)
- Draw the circuit. Write the **expected** voltage at each point. Then measure and compare.
- **Series circuit with one open part:** everything goes out. The **open part has the full source voltage** across it. Every good part reads 0 V.
- **Parallel circuit with one open branch:** only that branch goes out. The others work normally.
- **Shorted part:** 0 V across it. In series, the other parts get more voltage and get brighter.
- No current → no voltage drop across a resistor. Both ends read the same voltage.
- Ohmmeter checks: power **off**, part isolated. OL = open. About 0 Ω = short (or a wire).
- Common faults in a hands-on:
  - LED in backwards.
  - Wrong resistor value (color code misread).
  - Both legs of a part in the same breadboard row (it is shorted).
  - Split power rail not bridged.
  - Loose jumper or a jumper in the wrong row.
  - Dead battery.
  - Blown fuse inside the meter (current reads 0).
  - Chip with no power (pin 14 and pin 7).
- Change **one thing at a time**. Then test again.
- Write down what you measured. Many points come from the explanation.

Formulas

Situation	What the meter shows
Good series part	V = I × R (a share of V_s)
Open part in series	V_open = V_s; all others 0 V
Shorted part	V = 0; R ≈ 0 Ω
Open part (ohmmeter, power off)	OL
Open LED (or reversed) in series	V_LED = V_s; V_R = 0
Good LED	V ≈ V_f (1.8-3.3 V)
Divider node	V_node = V_s × R_below / R_total
KCL check	Σ I_in = Σ I_out at a node
KVL check	Σ drops = V_s around a loop
Leak resistance	R_leak = V_node / (I_in − I_known)

Question types

Easy

1. The troubleshooting checklist

- Scope: ✅ IN.
- Where seen: national supervisor practice deck 11 ("TROUBLE-CHECKLIST", F_slides §7).
- How to solve:
  1. Power connected? Battery voltage OK under load?
  2. Switch on? Fuse or breaker OK?
  3. Any burned part or loose wire?
  4. Then compare measured node voltages with calculated ones.
- Worked example: "the LED circuit does nothing". First reading: the battery shows 9.4 V alone but **0.3 V** when connected. The battery is dead (or there is a short pulling it down).
- Traps: do not rebuild the whole circuit first. Measure first.
- Calculator: look up — type: symptom.

2. Series string, all out: find the open part

- Scope: ✅ IN.
- Where seen: national supervisor deck 11 ("TROUBLE-SERIES-OPEN": six 12 Ω lamps on 6 V; L5 is the open one). Prior research I_bank C26.
- How to solve:
  1. Measure V across each part with the power on.
  2. The part that shows the **full source voltage** is open.
- Worked example: six 12 Ω lamps on 6 V. Good circuit: 1.0 V per lamp, 83 mA. Faulty: L1-L4 and L6 read **0 V**; L5 reads **6 V** → L5 is open.
- Traps: deck 11 says "voltage at L5 = 10 V" with a 6 V source. That is a typo; it cannot exceed 6 V.
- Calculator: calculate — type: V_s, list of V readings.

3. Parallel circuit, one branch out

- Scope: ✅ IN.
- Where seen: national supervisor deck 11 ("TROUBLE-PARALLEL-OPEN": six 12 Ω lamps in parallel on 6 V; L5 stays dark).
- How to solve:
  1. Measure V across the dark lamp. If it is 6 V, power reaches it.
  2. So the lamp itself (or its socket) is open. Check its R with power off: OL = broken filament.
- Worked example: good lamps take 6/12 = **0.5 A** each. L5 has 6 V but 0 A → **L5 open**.
- Traps: in parallel, a dark lamp with **0 V** across it means its wire to the supply is broken, not the lamp.
- Calculator: calculate — type: V_s, R_lamp.

4. LED does not light: reversed or open

- Scope: ✅ IN.
- Where seen: Princeton Invitational 2019, Div C, INV, Devices (b). 2026 ATX Regional, Div B and C, REG, hands-on. Pembroke Hill Invitational 2026, Div B, INV, Q28 (leg polarity). Prior research H_logic_diodes B29.
- How to solve:
  1. Measure V across the LED and across its resistor.
  2. V_LED = V_s and V_R = 0 → the LED is open or backwards.
  3. Flip it. If it still fails, test it in diode mode.
- Worked example: 5 V, 330 Ω, red LED. Readings: V_LED = **5.0 V**, V_R = **0 V**. The LED is reversed (or dead).
- Traps: a reversed LED looks exactly like an open one on a voltmeter.
- Calculator: look up — type: V_LED, V_R, V_s.

5. Broken switch: prove the rest works with a jumper

- Scope: ✅ IN.
- Where seen: Cobra Invitational 2019-20, Div B, INV, Station 1 ("broken" LED circuit; find the fault; prove the rest works using only a coin. Key: the switch is wired wrong, so the circuit is always open; bridge the switch with the coin → the LED lights).
- How to solve:
  1. Suspect the part that should close the path (switch, fuse).
  2. Bridge its two terminals with a wire or coin.
  3. If the LED now lights, that part is the fault.
- Worked example: LED off with the switch "on". A coin across the switch terminals → LED on. **The switch is the fault.**
- Traps: never bridge a **resistor** or a fuse with a wire in a live circuit. That can short the battery.
- Calculator: look up — type: symptom.

Medium

6. Short circuit vs open circuit effects

- Scope: ✅ IN (open and short circuits, rule 3.c.v).
- Where seen: JC Booth Invitational 2011, Div B, INV, Q45 (L1 burns out: which stop? key L4, L5), Q46 (L2 shorted: which stop? key none), Q51 (wires touching → short circuit).
- How to solve:
  1. Open in a branch: everything in series with it stops.
  2. Short across a part: that part goes out; parts in series with it get more voltage.
- Worked example: three equal bulbs in series on 9 V (3 V each). Short bulb 2 → bulbs 1 and 3 get **4.5 V** each (brighter); bulb 2 is out.
- Traps: a short across the **whole** source is the dangerous one. The battery gets hot.
- Calculator: calculate — type: V_s, list of R, which part is shorted.

7. A wire breaks at point A: what still carries current?

- Scope: ✅ IN.
- Where seen: Rickards Invitational 2025 (2026 season), Div B, INV, Simulated Hands-On Q7 (wire at point A broken → R1, R2, R3 still carry current).
- How to solve:
  1. Erase the broken wire from the drawing.
  2. Trace every loop that still goes from + back to −.
  3. Parts on a complete loop still carry current.
- Worked example (Rickards): the break at A cuts only the R4 branch. **R1, R2 and R3** still carry current.
- Traps: a part with one end dangling carries no current, even if it touches the circuit at one end.
- Calculator: look up — type: circuit, break point.

8. Switch what-ifs in a bulb network

- Scope: ✅ IN.
- Where seen: SO MYSO Division B Practice Test, Nov 2020, Div B, practice, Q25-30 (only SW1 open: bulb 1 current decreases, bulb 2 zero, bulb 3 greater; only SW2 closed: bulb 1 brighter, bulb 3 out). Florida State 2021 had no hands-on.
- How to solve:
  1. Redraw the circuit for the new switch state.
  2. An open switch removes its branch. A closed switch across parts shorts them.
  3. Find the new R_total and the new currents. Compare.
- Worked example: bulb 1 in series with (bulb 2 || bulb 3), equal bulbs, 9 V. Open bulb 2's branch → R_total goes from 1.5R to 2R → bulb 1 current **drops**; bulb 3 voltage goes from 3 V to **4.5 V** (brighter).
- Traps: "bulb 3 gets brighter when bulb 2 goes out" surprises many people. Calculate it.
- Calculator: calculate — type: circuit, switch states.

9. LED does not light from a divider (V_f too high)

- Scope: ✅ IN.
- Where seen: Princeton Invitational 2020, Div C, INV, Lab 1 Q42-43 (blue LED does not light from a divider → use a red LED or change R2; unequal brightness → redraw).
- How to solve:
  1. Find the divider's output voltage with the LED **not** connected.
  2. If it is below the LED's V_f, the LED stays dark.
  3. Fix: a lower-V_f LED, or a larger bottom resistor, or feed the LED from the full supply with its own resistor.
- Worked example: 5 V divider with 2.2 kΩ on top and 1 kΩ on the bottom gives **1.56 V**. A red LED (1.8 V or more) and a blue LED (≈3 V) both stay dark.
- Traps: the LED also loads the divider. The output drops when the LED conducts.
- Calculator: calculate — type: V_s, R_top, R_bottom, V_f.

10. Wrong resistor value (misread color code)

- Scope: ✅ IN.
- Where seen: not seen as its own question. Classic fault in prior research (B_topics §9c).
- How to solve:
  1. The LED is very dim or very bright → check the resistor.
  2. Read its bands again. Measure it with the ohmmeter (power off).
- Worked example: 9 V, red LED, should be 1 kΩ (brown-black-**red**) → 7.0 mA. A 10 kΩ (brown-black-**orange**) was used → **0.7 mA**, very dim.
- Traps: a red and an orange band look alike under bad light.
- Calculator: look up — type: band colors.

11. Meter or battery is the fault

- Scope: ✅ IN.
- Where seen: GullSO Invitational 2025, Div C, INV, Q50i (battery 2.9-3.2 V). Prior research B_topics §9a (fused current jacks). See Multimeter Use A9 type 11.
- How to solve:
  1. Current reads 0, but the LED is lit → the meter's current fuse is blown (or the lead is in the wrong jack).
  2. Battery fine alone, but low under load → weak battery (high internal resistance).
- Worked example: a 9 V battery reads 9.3 V open but **6.1 V** with a 100 Ω load. The battery is weak.
- Traps: test the meter on a known battery first.
- Calculator: look up — type: symptom.

Hard

12. Find the fault from a list of node voltages

- Scope: ✅ IN.
- Where seen: national supervisor slide #11 method (compare expected and measured node voltages). North Pocono Invitational 2020, Div C, INV, HO-38 (8-resistor network; find the largest drop).
- How to solve:
  1. Calculate the expected voltage at every node.
  2. Find where measured and expected first disagree.
  3. Decide: open (full voltage jumps across one part) or short (a drop is missing).
- Worked example: 9 V, three 1 kΩ in series (R1 top, R2, R3). Expected: N1 = **6 V**, N2 = **3 V**. Measured N1 = 9 V, N2 = 0 V → **R2 open**. Measured N1 = 4.5 V, N2 = 4.5 V → **R2 shorted**. Measured N1 = 4.5 V, N2 = 0 V → **R3 shorted**.
- Traps: "R2 shorted" and "R1 open" give different patterns. Check every node, not just one.
- Calculator: calculate — type: V_s, list of R, measured node voltages.

13. A hidden short inside a network

- Scope: ✅ IN.

- Where seen: USC Invitational 2026, Div B, INV, MC10 (rank four circuits; one resistor is shorted by a wire, so II = IV = 1.5R).
- How to solve:
  1. Look for a wire that joins both ends of a part.
  2. Remove that part from the drawing (it carries no current).
  3. Recalculate.
- Worked example: R in series with (R || R) = 1.5R. Add a wire across one parallel R → that pair becomes 0 → total **R** (not 1.5R).
- Traps: a wire across a parallel group shorts the **whole** group, not one part.
- Calculator: calculate — type: circuit, shorted part.

14. Stuck-at fault in a logic circuit

- Scope: ✅ IN.
- Where seen: USC Invitational 2026, Div B and C, INV, MC22 (one AND input grounded → new output XNOR(A, B)). Prior research H_logic_diodes A50.
- How to solve:
  1. Write the expected output for every input row.
  2. Compare with what you see.
  3. Find the one wire that, stuck at 0 or 1, explains every wrong row.
- Worked example: F should be AB + C. The LED shows **F = C** in every row. So the AB term never reaches the OR gate → the AND output is **stuck at 0** (or the AND chip has no power).
- Traps: check chip power (pin 14, pin 7) before you blame a gate.
- Calculator: calculate — type: expected table, observed table.

15. More than one fault on a breadboard

- Scope: ✅ IN.
- Where seen: not seen in a test. See Breadboards A9 type 11.
- How to solve:
  1. Fix the first fault you find.
  2. Test again. The symptom may change.
  3. Repeat until every row or state works.
- Worked example: an LED stays dark. Fault 1: LED backwards → flip it; still dark. Fault 2: the resistor and LED legs are in the same row → move one leg. Now it lights.
- Traps: do not stop at the first fix. The task may hide two faults.
- Calculator: look up — type: symptom list.

Nationals

16. Find a leak path with KCL

- Scope: ✅ States-Nationals (KCL, rule 3.c.ix; one equation).
- Where seen: not seen in a test. Generator "KVL-MEASURE-AND-NOTICE" (F_slides §7) uses the same idea.
- How to solve:
  1. Measure a node voltage. Compare with the expected one.
  2. KCL: current in = sum of currents out. Find the "missing" current.
  3. R_leak = V_node / I_missing.
- Worked example: 12 V, 1 kΩ (top) to node A, 2 kΩ from A to ground. Expected V_A = **8 V**. Measured **6 V**. Current in = 6/1k = 6 mA. The 2 kΩ takes 3 mA. Missing **3 mA** → R_leak = 6/0.003 = **2 kΩ** from A to ground (for example a wrong part in the same row).
- Traps: the leak can also go to another node, not to ground. Measure that node too.
- Calculator: calculate — type: V_s, R_top, R_bottom, V_measured.

17. KVL check: drops do not add up

- Scope: ✅ States-Nationals (KVL by name). The idea "series drops add to the source" is ✅ IN.

- Where seen: UT Austin Invitational Fall 2025, Div B/C, INV, Lab 5c (measure V_R, V_SW, V_LED; check V_total = V_R + V_SW + V_LED).
- How to solve:
  1. Measure every drop around the loop.
  2. Add them. Compare with the source voltage.
  3. A big gap means a part you did not measure (a bad contact, a jumper with resistance).
- Worked example: V_s = 9.02 V; V_R = 6.95 V, V_LED = 1.98 V, V_SW = 0.04 V. Sum = 8.97 V. Gap = **0.05 V** (0.6%): normal meter error. A gap of 1 V would mean a bad contact.
- Traps: the switch drop is small but not zero. Include it.
- Calculator: calculate — type: V_s, list of drops.

Out of scope (skip)

- Thevenin "measure open-circuit voltage and short-circuit current" methods (Virginia Regional 2013, Div C, REG, Station 2). ❌ Norton/Thevenin (rule 3.d).
- Faults in capacitor or op-amp circuits (MIT 2019 Station 2 op-amp; Cobra Station 5 capacitor). ❌ Capacitance; semiconductors beyond those listed (rule 3.d).
- Faults in house AC wiring with a live outlet. ❌ No AC circuits in the hands-on (Part II.d). Written questions on breakers and GFCIs are still IN.
- Using an oscilloscope to find a fault. ❌ Oscilloscopes (rule 3.d).

Mistakes in tests and keys

- **National supervisor deck 11 (series troubleshooting):** says "voltage at L5 = 10 V" on a 6 V circuit. It should be 6 V. The logic (L5 is open) still holds.
- **Cobra Invitational 2019-20, Div B, INV, Station 1:** the key text says only that the switch was "wired wrong". It does not say which wire. I could not see the board.
- **2026 ATX Regional, Div B and C, REG, hands-on:** the key's own circuit is faulty (see Task - 3-Way Switch A9 type 5). It is a good troubleshooting exercise: find why green never lights.

Not confirmed

- Rickards 2026 B Q7: I did not render the circuit figure. I used the key's answer (R1, R2, R3) to say the break cuts the R4 branch.
- Princeton 2020 Lab 1 resistor values are not in the downloads (I only have the prior summary). My type 9 numbers are my own example.
- Cobra Station 1: no board, so the exact fault is from the key's words only.

Sources

- 2027 Div B Rules Manual, Circuit Lab B, Part II.c.vi and II.d.
- National supervisor practice deck 11 (`downloads/dlB/slides`), summarized in `prior/F_slides.md` §7 and §12.
- Rickards 2026 B key (`downloads/dlG/rsoi26B_key.txt`); USC 2026 B/C (`dlFind/usc26_*`); MYSO practice (`dlFind/myso_B.txt`).
- Test extractions: `research/tests/misc-a.md` (Cobra, North Pocono), `misc-b.md` (JC Booth 2011), `state.md` (MYSO, Florida State 2021), `ut-a.md` (UT 2025, ATX 2026), `nat26.md`; `prior/B_topics.md` §9c; `prior/I_bank.md` C22-C26; `prior/H_logic_diodes.md` A50, B29; `prior/G_tests.md` §2.
- Kuphaldt, *Lessons in Electric Circuits*, Vol. 1 ch. 5-7 (series, parallel, troubleshooting); Socratic worksheet "Troubleshooting series and parallel circuits".
- All numbers checked in Python (`check5.py`, 2026-10-16).

Task: Wire a 3-Way Switch with SPDT Switches

Note: Task - 3-Way Switch

Official task (2027 rules, Part II.c.vii): "Wire a 3-way switch using SPDT switches." ✅ **IN at every level.** It is **new for 2027**. Switches are also a written topic (rule 3.c.vi, "electrical control devices").

An **SPDT switch** (single pole, double throw) has 3 pins. The middle pin is the **common** (C). The lever joins the common to one of the two outer pins, the **travelers** (T1 or T2). In house wiring an SPDT switch is called a **"3-way switch"**.

A **3-way circuit** lets you turn one light on or off from two places. A hallway or a staircase with a switch at each end is the classic use.

Key facts

- You need **two SPDT switches** and **two traveler wires**.
- Switch A: common to the source (+). Switch B: common to the load (resistor and LED). The load then goes to –.
- Wire T1 of A to T1 of B. Wire T2 of A to T2 of B. These two wires are the travelers.
- The light is on when both commons pick the **same** traveler. Flipping either switch changes the state.
- This is the **XNOR** of the two switch positions. If you cross the travelers (T1 to T2), it is **XOR**. Both versions work as a 3-way circuit.
- Which gate you name depends on which position you call "1". Say your convention on the answer sheet.
- An LED needs a **series resistor**. Put it anywhere in the one path: before switch A, or after switch B.
- One resistor is enough. Only one path can carry current at a time.
- For three or more places, add a **4-way switch** between the two 3-ways. A 4-way switch is a **DPDT** (double pole, double throw) wired as a **crossover**: "straight through" in one position and "crossed" in the other.
- n places need 2 three-way switches and (n – 2) four-way switches.
- Find the common pin with the meter's **continuity** mode. The common is the pin that beeps with one outer pin in one position and with the other outer pin in the other position.
- Many breadboard slide switches have the common in the middle. Do not trust this. Test it.
- The rules forbid AC and high power (Part II.d). Build with a 5 V or 9 V battery and an LED, never mains.

Formulas

What	Formula	Notes
Light state (standard wiring)	L = A XNOR B = (A·B) + (A'·B')	1 = on
Light state (crossed travelers)	L = A ⊕ B	Same circuit, other labels
Three places (3-way, 4-way, 3-way)	L = NOT(A ⊕ M ⊕ B)	On when an even number of switches are "1"
Switches for n places	2 SPDT + (n – 2) DPDT	n ≥ 2
Pins	SPST 2, SPDT 3, DPST 4, DPDT 6	
LED resistor	R = (V_s – V_f) / I	See Task - Equal LEDs A9
Resistor power	P = (V_s – V_f)² / R	Check 1/4 W

Truth table for the standard wiring (0 = T1, 1 = T2):

A	B	Light
0	0	on
0	1	off
1	0	off
1	1	on

Question types

Easy

1. Name the switch type and count its pins

- Scope: ✅ IN.
- Where seen: Pembroke Hill Invitational 2026, Div B, INV, Q25 (key: SPDT, "also called three-way switches"). Princeton Invitational 2026, Div C, INV, MC8. GullSO Invitational 2025, Div C, INV, Q25. UF Northern Regional 2020, Div C, REG, Q15 ("one common pole, two throws" → SPDT). Rickards Invitational 2026, Div C, INV, Q8.
- How to solve:
  1. Pole = how many separate circuits the switch controls.
  2. Throw = how many positions each pole can connect to.
  3. Pins = poles × (throws + 1) for these simple switches.
- Worked example: DPDT = 2 poles, 2 throws. Pins = 2 × (2 + 1) = **6**. SPDT = 1 × 3 = **3**.
- Traps: a "3-way switch" has **3 pins**, not 3 positions. It has only 2 positions.
- Calculator: none.

2. How many switches, and what kind, for two places?

- Scope: ✅ IN.
- Where seen: Northview Invitational 2025, Div C, INV, Q16 (→ 2 SPDT). MIT Invitational 2019, Div C, INV, Q38 (→ SPDT). F-M Invitational 2019, Div B/C, INV, Q4 (room with 2 doors, 1 light: design it). Rickards Invitational 2020, Div C, INV, Q8.
- How to solve:
  1. Two places → two SPDT switches.
  2. Three or more places → add DPDT (4-way) switches in the middle.
- Worked example: a hall light with switches at both ends and one in the middle needs **2 SPDT + 1 DPDT**.
- Traps: two SPST switches in parallel do not work. Then the light can only be turned off when both are off.
- Calculator: none.

3. Find the common pin with a continuity test

- Scope: ✅ IN (meter use, Part II.b).
- Where seen: not seen for an SPDT. USC Invitational 2026, Div B/C, INV, Lab part a asked the same skill for a 4-pin push button.
- How to solve:
  1. Power off. Set the meter to continuity (beep) or Ω.
  2. Label the pins 1, 2, 3. Test the pairs 1-2, 2-3 and 1-3 in each switch position.
  3. The common is in the beeping pair in **both** positions.
- Worked example: position X beeps on pins 1-3. Position Y beeps on pins 2-3. Pin **3** is in both pairs, so pin 3 is the common.
- Traps: never test continuity with the battery connected. The reading is wrong and can harm the meter.
- Calculator: none.

4. Build the basic 3-way light on a breadboard

- Scope: ✅ IN. This is the official task.
- Where seen: Northview Invitational 2025, Div C, INV, Hands-On Lab Project (25 points): "Utilizing 2 SPDT switches, jumper wires, a resistor and an LED, wire a 3-way light circuit where the light can be controlled from two locations." Power: +5 V rails. Shock Value re-tryout (about 2012), Div B, practice, IV-1-1 (draw it).
- How to solve:
  1. ▪ rail → common of switch A.
  2. T1 of A → T1 of B. T2 of A → T2 of B.
  3. Common of B → resistor → LED anode (long leg). LED cathode → – rail.
  4. Test all 4 position pairs. Write the truth table on your sheet.

- Worked example: 5 V supply, red LED (V_f = 2.0 V), 150 Ω resistor. I = (5 – 2.0)/150 = **20 mA**. Resistor power = $3.0^2/150$ = **0.06 W** (safe for 1/4 W). With 220 Ω, I = **13.6 mA** (dimmer, also fine).
- Traps: putting the two switch bodies in the same breadboard rows shorts their pins together. Put each switch across separate rows (see Breadboards A9).
- Calculator: calculate — type: V_s, V_f, R.

5. Two LEDs that swap with one switch (red/green toggle)

- Scope: ✅ IN (switches, LEDs).
- Where seen: 2026 ATX Regional, Div B and C, REG, hands-on (red LED on and green off with the button up; green on and red off when pressed; kit: 2 LEDs, 1 button, 10 wires, breadboard, multimeter, 2 resistors). Berkeley Invitational 2026, Div B, INV, Q24 ("one push turns off a red LED and turns on a green LED" → SPDT).
- How to solve:
  1. ▪ → common of the SPDT (or the COM pin of a 3-pin button).
  2. NC (normally closed) pin → R1 → red LED → –.
  3. NO (normally open) pin → R2 → green LED → –.
  4. Or use one resistor between + and the common. Only one LED is on at a time, so one resistor is enough.
- Worked example: 5 V, R1 = R2 = 150 Ω. Red (V_f = 2.0 V): I = **20 mA**. Green (V_f = 2.2 V): I = (5 – 2.2)/150 = **18.7 mA**. Python check of the switch states: up → red on, green off; pressed → red off, green on.
- Traps: a plain SPST button (2 functional pins) cannot do a clean swap. You need an SPDT action. If the kit only has an SPST button, ask the supervisor.
- Calculator: calculate — type: V_s, V_f red, V_f green, R.
- **Key error (2026 ATX Regional):** the key powers the circuit from "the multimeter set to DC Voltage". A meter on V⎓ does not supply power. The key also ties button side A to –, with the red LED from + to side A. So red is always on (with no resistor), and pressing the button only puts green's anode at –. Green never lights. Python check: red = on, green = off in both button states. The key uses neither of the 2 resistors.

Medium

6. Explain why either switch toggles the light

- Scope: ✅ IN.
- Where seen: Berkeley Invitational 2026, Div B, INV, Q25 (key B: "The light is on only when both switches connect to the same traveler wire"). BullSO Invitational 2026, Div B, INV, Q24, and Div C, INV, Q24 ("How does a three-way switch setup operate?").
- How to solve:
  1. Trace the path: + → common A → one traveler → common B → light.
  2. The path is complete only when both commons sit on the same traveler.
  3. Flipping one switch moves its common to the other traveler. The path breaks (or closes).
- Worked example: both on T1 → light on. Flip B → B is on T2, A is on T1 → no path → **off**. Flip A → both on T2 → **on** again.
- Traps: "the switches are in parallel" is wrong (Berkeley choice C). They are in series, with a choice of two paths.
- Calculator: none.

7. The 3-way switch as a logic gate (XOR/XNOR)

- Scope: ✅ IN (rule 3.c.viii, truth tables).
- Where seen: North Pocono Invitational 2020, Div C, INV, Q4 (key: XNOR). H_logic_diodes A20 (staircase → XOR).
- How to solve:
  1. Choose a label: position T1 = 0, T2 = 1.
  2. Fill the 4-row truth table from the wiring.
  3. Light on for equal inputs → XNOR. Light on for different inputs → XOR.
- Worked example: standard wiring (T1-T1, T2-T2): table on, off, off, on → **XNOR**. Crossed travelers (T1-T2, T2-T1): off, on, on, off → **XOR** (Python check of both).
- Traps: both answers are right with the matching label. North Pocono's key accepts only XNOR. Write your convention.

- Calculator: none.

8. Pick (or draw) the correct wiring diagram

- Scope: ✅ IN.
- Where seen: UT Austin (ATX) Regional 2021, Div B, REG, Q32 (pick from 4 drawings; key A, two SPDT + 2 travelers). NY Lake Erie/Niagara Invitational 2018, Div C, INV, Q9 (key D, travelers to travelers). UT Austin Invitational 2019, Div B, INV, Q44 (draw a house 3-way with hot, neutral and ground).
- How to solve:
  1. Find the two commons. One goes to the source, one to the load.
  2. Check that both travelers run switch to switch.
  3. Check that no position joins + to – directly (a short).
- Worked example: a drawing with the source on a traveler pin of A and the light on a traveler pin of B. Python check: the light is on in only **1 of 4** states. So that drawing is wrong.
- Traps: in house wiring (UT 2019 Q44) the hot goes to a common, the neutral goes straight to the lamp, and the ground goes to every box. Do not switch the neutral.
- Calculator: none.

9. Troubleshoot a miswired or broken 3-way circuit

- Scope: ✅ IN (also official task II.c.vi; see Task - Troubleshoot A9).
- Where seen: UT Austin Invitational 2020, Div B, INV, Q30 ("switch A works only when B is down" → a broken traveler).
- How to solve:
  1. Test all 4 states. Write the table.
  2. Compare it with the right table (on, off, off, on).
  3. Use the voltmeter (black lead on –) at each pin to find where + stops.
- Worked example (Python check of each fault, states 00, 01, 10, 11):
  - Correct: on, off, off, on.
  - Travelers crossed: off, on, on, off. **Not a fault.** It still works from both places.
  - Traveler 2 open: on, off, off, **off**. A works only when B is on T1.
  - Source on a traveler pin of A (common and T1 swapped): on, off, off, off.
  - Load on a traveler pin of B: on, off, off, off.
- The last three faults give the same table. Tell them apart with the voltmeter. With A and B both on T2: a broken traveler shows +V at A's T2 pin, but 0 V at B's T2 pin.
- Traps: a light that works in 2 of 4 states with crossed travelers is fine. Do not "fix" it.
- Calculator: none.

10. SPDT switches as an OFF / LOW / HIGH selector

- Scope: ✅ IN (close to task vii).
- Where seen: University of Michigan Invitational (UMSO) 2019-20, Div C (probably), INV, Lab (two 3-way switches, 9 V, LED, resistors, clips: build OFF, LOW and HIGH brightness; "many solutions").
- How to solve:
  1. S1 common → +9 V. S1 position X → nothing (OFF).
  2. S1 position Y → S2 common.
  3. S2 position 1 → small R → LED. S2 position 2 → large R → same LED.
  4. LED cathode → –.
- Worked example: 9 V, red LED 2.0 V. HIGH with 470 Ω: I = 7.0/470 = **14.9 mA**. LOW with 2.2 kΩ: I = 7.0/2200 = **3.2 mA**.
- Traps: do not use a 0 Ω "HIGH" path. The LED then has no current limit.
- Calculator: calculate — type: V_s, V_f, R_low, R_high.

Hard

11. Switch network from a 16-row truth table (MIT 2020)

- Scope: ✅ IN (very hard).
- Where seen: MIT Invitational 2020, Div C, INV, Q49 (12 points): three 3-way (SPDT) and one 4-way (DPDT) switches, two lights, 5 V DC. Only **15 of 16 rows** are given. The row S1 S2 S3 S4 = 0 1 0 1 is missing.
- How to solve:
  1. Split the table on one switch (Shannon split). Look at the S1 = 0 rows and the S1 = 1 rows apart.
  2. Name each half as a simple gate.
  3. S1 is then an SPDT that sends power to one of two sub-networks.
- Worked example (Python check of the given rows):
  - S1 = 1: L2 is always off. L1 = S2 ⊕ S3 (a plain 3-way pair). S4 does not matter.
  - S1 = 0 and S4 = 0: L1 = S2 XNOR S3.
  - S1 = 0: L2 = S4' · NOT(S2·S3).
  - So S1 picks the branch, and S2-S3 act as a 3-way pair in each branch.
- Traps: treat the missing row as a "don't care" (see Task - Logic from Truth Table A9 type 16). The key's drawing is an image. I did not get the full network from it.
- Calculator: none.

12. Three places: add a 4-way (DPDT crossover) switch

- Scope: ✅ IN.
- Where seen: Princeton Invitational 2020, Div C, INV, Q40 (10 points; one light; L1 = 1 when an even number of switches are on). I_bank D5. B_topics §6a.
- How to solve:
  1. ▪ → common of 3-way A. A's T1 and T2 → the two inputs of the 4-way.
  2. 4-way outputs → T1 and T2 of 3-way B. B's common → load → –.
  3. Wire the DPDT as a crossover: inputs to the poles; one throw pair straight, the other throw pair crossed.
- Worked example (Python check, 8 states): the light is on for A M B = 000, 011, 101, 110. That is "even number of 1s" = NOT(A ⊕ M ⊕ B). It matches the Princeton 2020 table exactly.
- Traps: Princeton 2020 says "three 3-way and one 4-way", but labels only S1-S3. Three places need only **two 3-way + one 4-way**.
- Calculator: none.

13. Count the switches in a big multi-way system

- Scope: ✅ IN.
- Where seen: Northview Invitational 2025, Div C, INV, Q26 ("hotel": 6 rows × 5 windows; each switch toggles its whole row; A SPDT and B DPDT; find A × B; choices 0, 144, 200, 216, 225).
- How to solve:
  1. Per row of n switches: 2 SPDT and (n – 2) DPDT.
  2. Multiply by the number of rows.
- Worked example: per row 2 SPDT + 3 DPDT. A = 6 × 2 = **12**. B = 6 × 3 = **18**. A × B = **216** (choice D).
- Traps: the number of windows per row is the number of switches per row. Do not count lights.
- Calculator: calculate — type: rows, switches per row.

14. The Carter system (an unsafe 3-way)

- Scope: ✅ IN (switch logic and hazard). The test drew it on AC mains, but only the logic is asked.
- Where seen: North Pocono Invitational 2020, Div C, INV, Q21 (which positions light the lamp; why banned since 1923). UT Austin Invitational 2020, Div B, INV, Q29 ("Which 3-way type was banned by the NEC in 1923?" → Carter).
- How to solve:
  1. In the Carter system the travelers go to hot and neutral. The commons go to the two ends of the lamp.
  2. The lamp is on when the two commons sit on different supply wires.
  3. When both commons pick hot, the lamp is off, but both lamp wires are **live**.
- Worked example (Python check): with a bulb, the lamp is on in states 01 and 10 only. With a DC LED (polar), it is on in **only 1 state**. The other "different" state reverses the LED.

- Traps: "off" does not mean "safe". This is why the system was banned.
- Calculator: none.

15. Switch tables with three SPDTs (cascade and bulb networks)

- Scope: ✅ IN.
- Where seen: UT Austin Invitational 2019, Div B, INV, Q43 (3 SPDT switches + 3 lamps, 12 V, 10 W; fill the 8-row table; key: L1 = S1, L2 = S1·S2, L3 = S1·S2·S3). UT Invitational 2018, Div B/C, INV, Q24 (5 bulbs, 3 SPDT switches: compare brightness; positions that light all 5).
- How to solve:
  1. For each row, trace the path from + for each lamp.
  2. A lamp is on only when every switch on its path is in the right position.
- Worked example: S1 = 1, S2 = 1, S3 = 0 → L1 on, L2 on, **L3 off** (key rule L3 = S1·S2·S3).
- Traps: UT 2019's drawing shorts the battery when S1 is off. Look for a direct + to – path in every row.
- Calculator: none.

16. DPDT polarity reverser

- Scope: ✅ IN (switches; the motor is only a load).
- Where seen: Shock Value re-tryout (about 2012), Div B, practice, IV-1-2 (2 × 9 V, motor, DPDT reverses the motor).
- How to solve:
  1. Battery + and – go to the two poles (commons).
  2. Throw pair 1 goes straight to the load. Throw pair 2 goes to the load crossed.
  3. Flip the switch → the load sees the opposite polarity.
- Worked example: show the direction with two LEDs back to back (anti-parallel) plus one 470 Ω. 9 V, red LEDs (2.0 V): I = 7.0/470 = **14.9 mA**. One LED lights per position.
- Traps: crossing the wires on the same throw pair shorts the battery in that position.
- Calculator: calculate — type: V_s, V_f, R.

Nationals

17. Push buttons as logic (USC 2026 "Light Switch" station)

- Scope: ✅ IN (switches; logic from a truth table).
- Where seen: USC Invitational 2026 (top invitational at the Nationals site), Div B and C, INV, Lab "Light Switch" (25 points). Kit: 2 × 510 Ω, 2 four-pin push buttons, 1 LED, 9 V. A proctor signs each working circuit after testing all 4 button combinations. Parts: (a) find the button's internal wiring, (b) LED color and V_f, (c) LED on only when both buttons are up (NOR), (d) on when A pressed and B up (A·B'), (e) one button dims the LED.
- How to solve:
  1. NOR: 510 Ω in series, then both buttons in parallel across the LED. A pressed button shorts the LED.
  2. A·B': button A in series to feed the LED; button B across the LED.
  3. Dim: button + second 510 Ω in parallel with the LED.
- Worked example: NOR with a button pressed: the resistor carries 9/510 = **17.6 mA** and 0.16 W (safe for 1/4 W). Dimming, red LED (2.0 V): 13.7 mA → **9.8 mA**. Blue (3.0 V): 11.8 mA → **5.9 mA**.
- Traps: a shorting button with **no** series resistor shorts the battery. Always keep the 510 Ω first in line.
- Calculator: calculate — type: V_s, V_f, R.

Out of scope (skip)

- Wiring real 120 V house switches. ❌ AC circuits and high power are not allowed at the station (Part II.d). Drawing a house 3-way (UT 2019 Q44) is still IN as a control device.
- Smart switches, dimmers with triacs, and timer switches. ❌ Semiconductors beyond those listed; sequential logic (rule 3.d).
- Latching relays and flip-flop "toggle" circuits for one push button. ❌ Sequential logic (rule 3.d).
- Relay-based 3-way circuits. ⚠️ Borderline: relays are IN (3.c.vi), but the coil is magnetism. Know only "a relay is a switch moved by a coil".

Mistakes in tests and keys

- **2026 ATX Regional, Div B and C, REG, hands-on key:** see type 5. The meter on DC volts is not a power source. Red is always on, green never lights, and no resistor is used. The B score sheet totals 40 points; the B test says 30.
- **North Pocono Invitational 2020, Div C, INV, Q4:** the key accepts only XNOR. XOR is also right with the other label.
- **Princeton Invitational 2020, Div C, INV, Q40:** it asks for "three 3-way and one 4-way" switches, but the table has only S1-S3. Two 3-way + one 4-way is enough.
- **UT Austin Invitational 2019, Div B, INV, Q43:** the drawing shorts the battery when S1 is off (UT-b extraction note).

Not confirmed

- No 2026 State or Nationals hands-on station was found. Where I looked: `research/tests/state.md` (Florida 2021, MYSO: no stations) and `nat26.md` (these are the USC **Invitational** 2026 tests, not Nationals). The real Nationals tests are only in the paid "Test Packet 2027".
- MIT 2020 Q49: the full intended network is only in the key's drawing (an image). I checked my split-table reading in Python. I did not check the drawing.
- UMSO 2019-20: the division is not printed. I assumed Div C.
- Typical V_f values (red 2.0 V, green 2.2 V, blue 3.0 V) are typical, not from the kits.

Sources

- 2027 Div B Rules Manual, Circuit Lab B, Part II.c.vii and II.d; rule 3.c.vi and 3.c.viii.
- Northview 2025 lab (`downloads/dlG/nv26C_lab.txt`) and test (`dlG/nv26C_test.txt`, Q16, Q26); MIT 2020 test and key (`dlG/mit20_test.txt`, `mit20_key.txt`, Q49); Princeton 2020 (`dlG/pz/p20/Circuit Lab/`, Q40); Berkeley 2026 (`dlB/tests/berk26_key.txt`, Q24-25); Pembroke Hill 2026 key (`dlB/tests/phi26_key.txt`, Q25); BullSO 2026 B and C (`dlB/tests/bullso26_test.txt`, `dlG/bull26C_test.txt`, Q24); ATX 2026 Regional (`dlG/atx26reg_*.txt`).
- Test extractions: `research/tests/ut-a.md` (ATX 2026), `ut-b.md` (UT 2019-20), `ut-c.md` (UT 2018), `misc-a.md` (F-M 2019, North Pocono 2020, UMSO), `misc-b.md` (MIT 2019, NY 2018, UF Northern 2020, Shock re-tryout), `atx-reg.md` (ATX 2021), `nat26.md` §3 (USC 2026 lab), `state.md` §4; `prior/G_tests.md`, `prior/B_topics.md` §6a, `prior/I_bank.md` D5, `prior/H_logic_diodes.md`, `prior/F_slides.md` CTRL-3WAY.
- Kuphaldt, *Lessons in Electric Circuits*, Vol. 4 ch. 4 "Switches" (switch types, poles and throws).
- All numbers and switch tables checked in Python (`threeway.py`, `mit49.py`, 2026-10-16).

Task: Build a Target Equivalent Resistance

Note: Task - Target Resistance

Official task (2027 rules, Part II.c.viii): "Create a specified equivalent resistance from given individual resistors." ✅ **IN at every level.** It is **new for 2027**.

The **equivalent resistance** (R_eq) is the one resistance that could replace the whole network. It draws the same current from the same voltage. The **target** is the R_eq the supervisor asks for.

Key facts

- **Series** (end to end, one path): R_eq = R1 + R2 + … It is always larger than the largest part.
- **Parallel** (side by side, same two nodes): 1/R_eq = 1/R1 + 1/R2 + … It is always smaller than the smallest part.
- Two in parallel: R_eq = R1·R2 / (R1 + R2) ("product over sum").
- n equal resistors R: series nR; parallel R/n.
- **Read the rules of the station first.** Typical rules: "use all resistors exactly once", "use any combination", "at least two in parallel and at least one in series".
- **Search method (top-down):**
  - Target bigger than any one resistor → you need a series part. Subtract a resistor and solve for the rest.
  - Target smaller than every resistor → you need a parallel part. Use 1/R_rest = 1/T − 1/R.
- Each split leaves a smaller target. Repeat until the rest is one resistor or a simple pair.
- If no exact answer exists, report the **closest** value and its **percent error**.
- **Tolerance** is the allowed spread of a real resistor's value. Gold band = ±5%, brown band = ±1%.
- If every resistor has the same tolerance, R_eq has the same percent tolerance. (Every series-parallel R_eq rises when any part rises.)
- **Measure** R_eq with the ohmmeter: power off, network out of the circuit, hands off the leads.
- Many stations end with a **supervisor sign-off**. BullSO 2026 gave 9 of 29 points (B) for the signed, working network.
- Resistor values come in **E-series** (preferred values). E12 per decade: 10, 12, 15, 18, 22, 27, 33, 39, 47, 56, 68, 82.
- E24 adds 11, 13, 16, 20, 24, 30, 36, 43, 51, 62, 75, 91. (E12 is the ±10% series, E24 the ±5% series.)
- A target that is not a standard value (for example 133.33 Ω) often comes from two standard values in parallel.

Formulas

What	Formula	Notes
Series	R_eq = R1 + R2 + …	
Parallel	1/R_eq = 1/R1 + 1/R2 + …	
Two in parallel	R1·R2 / (R1 + R2)	Product over sum
Needed series rest	R_rest = T − R	When T > R
Needed parallel partner	1/R_rest = 1/T − 1/R	When T < R
From a current target	T = V / I	UT 2019 style
Percent error	|built − target| / target × 100%	
Tolerance range	T × (1 − t) to T × (1 + t)	Same t for all parts
Max and min of a kit	all in series; all in parallel	

Number of different values with **all** resistors used once (Python, series-parallel only):

Resistors	Identical	All different
2	2	2
3	4	8
4	9	48

Question types

Easy

1. Target is a sum: plain series

- Scope: ✅ IN.
- Where seen: Shock Value re-tryout (about 2012), Div B, practice, Part V-1 (a near answer is 243 + 223 = 466 Ω). The general skill is in every R_eq question.
- How to solve:
  1. Look for two or three kit values that add up to the target.
  2. Wire them end to end in one row chain on the breadboard.
- Worked example: kit 1.0 kΩ, 2.2 kΩ, 3.3 kΩ. Target 3.2 kΩ. 1.0 + 2.2 = **3.2 kΩ**.
- Traps: two resistor legs in the same breadboard row are joined. Series needs each joint in its own row (see Breadboards A9).
- Calculator: calculate — type: list of R.

2. Target is half (or a third): equal resistors in parallel

- Scope: ✅ IN.
- Where seen: BullSO Invitational 2026, Div B, INV, Lab (two 50 Ω parts; 50 || 50 is a common step). I_bank (identical-resistor networks).
- How to solve:
  1. n equal resistors in parallel give R/n.
  2. Two equal in parallel give R/2.
- Worked example: two 100 Ω in parallel = 100·100/200 = **50 Ω**. Three 1 kΩ: 1 kΩ + (1 kΩ || 1 kΩ) = **1.5 kΩ**.
- Traps: the parallel result is always below the smallest resistor. If your answer is bigger, you added instead.
- Calculator: calculate — type: R, n.

3. Every value you can make from equal resistors

- Scope: ✅ IN.
- Where seen: BullSO Invitational 2026, Div C, INV, SA4 (four equal resistors in a network give a stated R_eq; find R → 27 Ω). Common warm-up in prior sets (I_bank).
- How to solve:
  1. List the networks: all series, all parallel, and the mixes.
  2. For 3 equal R using all three: 3R, R/3, R + R/2 = 1.5R, and 2R || R = 0.667R.
- Worked example: three 100 Ω, all used: **300, 33.3, 150, 66.7 Ω** (4 values). With any subset: also 100, 200, 50 Ω (7 values in all). Python brute force agrees.
- Traps: R || 2R is 2R/3, not R/3.
- Calculator: calculate — type: R.

Medium

4. Use all resistors exactly once, with layout rules (BullSO B)

- Scope: ✅ IN.
- Where seen: BullSO Invitational 2026, Div B, INV, Lab: R1 = 100 Ω, R2 = 200 Ω, R3 = R4 = 50 Ω. "Using all four resistors exactly once, construct a circuit with an equivalent resistance of 100 Ω." At least two in parallel and at least one in series. Points: labeled sketch 8, step-by-step R_eq 8, explanation 4, supervisor sign-off 9.
- How to solve:
  1. Target 100 Ω is below 200 Ω, so 200 Ω sits in parallel with something.
  2. 1/R_rest = 1/100 − 1/200 = 1/200 → the rest must be **200 Ω**.
  3. 100 + 50 + 50 = 200 Ω. Done.
- Worked example: 200 || (100 + 50 + 50) = 200 || 200 = **100 Ω** exactly. Python brute force: the four parts give **32** different values. Only this one network (in its two orders) gives 100 Ω.
- Traps: write each step: "100 + 50 + 50 = 200 Ω; 200 || 200 = 100 Ω". The step-by-step part was worth 8 points.

- Calculator: calculate — type: list of R, target.

5. Any subset, odd target (BullSO C, 133.33 Ω)

- Scope: ✅ IN.
- Where seen: BullSO Invitational 2026, Div C, INV, Lab (24 points): 2 × 100 Ω, 1 × 200 Ω, 4 × 50 Ω. "Using any combination of resistors, construct … 133.33 Ω." At least two in parallel, at least one in series. Points: sketch 8, steps 8, explanation 2, sign-off 6.
- How to solve:
  1. 133.33 = 100 + 33.33. So use 100 Ω in series with a 33.33 Ω part.
  2. 1/R_rest = 1/33.33 − 1/100 = 0.02 → R_rest = **50 Ω**.
- Worked example: 100 + (100 || 50) = 100 + 33.33 = **133.33 Ω** with 3 resistors. The key's example uses all 7: 100 + (300 || 50 || 150) = 133.33 Ω, where 300 = 200 + 100 and 150 = 50 + 50 + 50. Python: both are exact (400/3 Ω).
- Traps: "133.33" means 400/3. Keep fractions or 5 significant figures in your steps.
- Calculator: calculate — type: list of R, target.

6. No exact answer: closest value and percent error

- Scope: ✅ IN.
- Where seen: Shock Value student test (scioly.org wiki, about 2011), Div B, practice, Part VI: 16, 4, 56, 45 Ω; make "about 16.7 Ω", "about 3.9 Ω", "approximately 3.5 Ω". Shock Value re-tryout (about 2012), Div B, practice, Part V-1: 243, 223, 556, 919 Ω; make about 466.4, 483.9, 354.1 Ω.
- How to solve:
  1. Try the top-down method for the closest structure.
  2. Compute the result. Report it with its percent error.
- Worked example (Python brute force of all networks):
  - 16.7 Ω: (16 + 4) || (56 + 45) = 20 || 101 = **16.69 Ω** (0.04% off).
  - 3.9 Ω: (16 + 56 + 45) || 4 = **3.87 Ω**.
  - 3.5 Ω: key 56 || 4 || (45 + 16) = **3.52 Ω**. Closer: 45 || 4 || (16 + 56) = **3.50 Ω** (3.495).
  - 466.4 Ω: (243 + 919) || (223 + 556) = **466.36 Ω**. 483.9 Ω: (243 + 556 + 223) || 919 = **483.88 Ω**. 354.1 Ω: (243 + 223) || (556 + 919) = **354.12 Ω**.
- Traps: "about" targets usually have one intended network. If you find a closer one, show both.
- Calculator: calculate — type: list of R, target.

7. Largest and smallest R_eq from a kit

- Scope: ✅ IN.
- Where seen: Shock Value student test (scioly.org wiki, about 2011), Div B, practice, Part VI-4 (add the largest and smallest; key ≈ 123.84 Ω). Shock Value re-tryout, Part V-4.
- How to solve:
  1. Largest = all in series.
  2. Smallest = all in parallel.
- Worked example: 16, 4, 56, 45 Ω. Series = **121 Ω**. Parallel = 1/(1/16 + 1/4 + 1/56 + 1/45) = **2.84 Ω**. Sum = **123.84 Ω** (key ✓). For 243, 223, 556, 919 Ω the sum is **2028.06 Ω**.
- Traps: the smallest is below 4 Ω, the smallest single part. Check that.
- Calculator: calculate — type: list of R.

8. Target given as a current (or power)

- Scope: ✅ IN.
- Where seen: UT Invitational 2018, Div B/C, INV, Q25a (1 V source; 2, 4, 5, 10 Ω; draw a circuit with 125 mA total). The same test asks for a power target too (ut-c extraction).
- How to solve:
  1. Turn the target into a resistance: T = V / I (or T = V^2 / P).
  2. Then search as usual.
- Worked example: T = 1 V / 0.125 A = **8 Ω**. ((2 + 10) || 4) + 5 = 3 + 5 = **8 Ω**. Python: this is the only network that uses all four.

- Traps: convert mA to A first. 1/125 = 0.008 Ω is wrong.
- Calculator: calculate — type: V, I.

9. Measure the built network with the ohmmeter

- Scope: ✅ IN (meter use, Part II.b).
- Where seen: B_topics §9e ("check with an ohmmeter"). The sign-off at BullSO 2026 needs a working network. How the supervisor checks it is not stated.
- How to solve:
  1. Disconnect the battery. Take the network out of any other circuit.
  2. Choose the lowest Ω range above the target (for 133 Ω: the 200 Ω range).
  3. Touch the probes together first. Subtract the lead resistance (often 0.2-0.5 Ω).
  4. Do not hold both probe tips with your fingers.
- Worked example: built 100 + (100 || 50) = 133.33 Ω. The meter reads 131.8 Ω. Percent error = 1.53/133.33 = **1.15%**. That is inside the ±5% range of gold-band parts.
- Traps: for small targets, lead resistance matters. 0.3 Ω is 3% of a 10 Ω target.
- Calculator: calculate — type: reading, lead R, target.

10. Tolerance range of the network

- Scope: ✅ IN.
- Where seen: not seen in a test. It explains why the sign-off reading is not exact (B_topics §9e).
- How to solve:
  1. Same tolerance on every part → R_eq has the same percent range.
  2. Mixed tolerances → set every part to its low end (min) and its high end (max).
- Worked example: 100 + (100 || 50), all ±5% → **126.67 to 140.00 Ω**. With the series 100 Ω at ±1% and the others at ±5% → **130.67 to 136.00 Ω** (±2.0%). Python checked all 8 corner cases.
- Traps: the network tolerance is never better than the worst part in the main path.
- Calculator: calculate — type: list of R, tolerances.

Hard

11. Top-down search by hand (no brute force)

- Scope: ✅ IN.
- Where seen: not seen as a named method. It solves BullSO B and C quickly.
- How to solve:
  1. Compare T with each resistor R in the kit.
  2. T > R: try R in series. New target T − R.
  3. T < R: try R in parallel. New target 1/(1/T − 1/R).
  4. Stop when the new target is a kit value or a simple sum.
- Worked example: four 100 Ω, target 75 Ω. 75 < 100 → parallel. 1/75 − 1/100 = 1/300 → need **300 Ω** = 100 + 100 + 100. So 100 || 300 = **75 Ω**.
- Traps: a negative new target means that branch fails. Try another resistor.
- Calculator: calculate — type: T, R.

12. Non-standard target from E-series parts

- Scope: ✅ IN (resistor values; E-series names are ⚠️ borderline vocabulary).
- Where seen: not seen in a test. I_bank C39 lists E12/E24.
- How to solve:
  1. Find two standard values whose parallel or series result hits the target.
  2. Compare with the nearest single E24 value.
- Worked example: target 133.33 Ω. Nearest E24 value: **130 Ω** (2.5% off). From E12 parts: 150 || 1200 = **133.33 Ω** exactly. Target 150 Ω from a kit of 100, 220, 470 Ω: 220 || 470 = **149.86 Ω** (2 parts), or 100 + (100 || 100) = **150 Ω**.
- Traps: 150 Ω is itself an E12 value. Ask whether a single resistor is allowed.
- Calculator: calculate — type: target, list of E values.

13. Check the power before you test it on a battery

- Scope: ✅ IN (power, rule 3.c.iv; see Task - Power Supplied [A9]).
- Where seen: not seen in a test. The BullSO key asks for a "battery/complete circuit" in the sketch (1 point).
- How to solve:
  1. I = V / R_eq.
  2. Power in the series part: P = I^2 R.
  3. Compare with the rating (usually 1/4 W).
- Worked example: 100 + (100 || 50) on 9 V: I = 9/133.33 = **67.5 mA**. P in the series 100 Ω = $0.0675^2 \times 100$ = **0.46 W**. That is too much for 1/4 W. On 5 V: I = **37.5 mA**, P = **0.14 W** (safe).
- Traps: low targets on 9 V draw large currents. Measure with the ohmmeter, not with the battery.
- Calculator: calculate — type: V, R_eq, R_series.

Nationals

14. Find the missing resistor so the network hits the target

- Scope: ✅ IN (algebra, one unknown; rule 3.e.i).
- Where seen: BullSO Invitational 2026, Div C, INV, SA4 (find R from a stated R_eq → 27 Ω). I_bank (45 Ω version).
- How to solve:
  1. Write R_eq as an expression with the unknown R.
  2. Set it equal to the target. Solve.
- Worked example: (R || 300) + 100 = 250 Ω. So R || 300 = 150. 1/R = 1/150 − 1/300 = 1/300 → R = **300 Ω**.
- Traps: if the answer is negative, no resistor can do it. Change the structure.
- Calculator: calculate — type: target, known R.

15. Fewest resistors, and how many networks exist

- Scope: ✅ IN.
- Where seen: BullSO 2026 C (the key used 7 parts; 3 are enough). Not asked as a count in a test.
- How to solve:
  1. Try 1 part, then 2, then 3, … with the top-down method.
  2. For a search by brute force (Python), combine every split of the set in series and in parallel.
- Worked example: BullSO C kit, target 133.33 Ω. Fewest = **3** parts: 100 + (100 || 50). Python finds one network with 3 parts and two with 4 parts. Kit 100, 220, 470 Ω (all used): **8** values: 60.0, 87.3, 158.7, 190.4, 249.9, 302.5, 538.8, 790.0 Ω.
- Traps: a bridge network (not series-parallel) is not in this search. Stations so far ask for series-parallel only.
- Calculator: calculate — type: list of R.

Out of scope (skip)

- Thevenin equivalent resistance of a network with sources. ❌ Norton/Thevenin (rule 3.d).
- Impedance targets (R with C or L). ❌ AC circuit theory, capacitance, inductance (rule 3.d).
- Y-Δ (wye-delta) transforms for bridge networks. ⚠️ Borderline: it is algebra, but not in the rule list. KCL handles a bridge at States/Nationals (rule 3.c.ix).
- Resistor temperature coefficients. ⚠️ Borderline trivia.

Mistakes in tests and keys

- **Shock Value student test (scioly.org wiki, about 2011), Div B, practice, Part VI-3:** the key's 3.52 Ω network is not the closest to 3.5 Ω. 45 || 4 || (16 + 56) = 3.495 Ω is closer. Both are "approximately 3.5".
- **BullSO Invitational 2026, Div C, INV, Lab:** the test says the explanation is worth 2 points. The key says 8 points (13 for B) and folds the "total is 133.333" check into it. The point split is unclear.
- **BullSO 2026 C key:** "3 way parallel" here means three branches in parallel. It is not a 3-way switch.

Not confirmed

- No State or Nationals target-resistance station was found. Where I looked: `research/tests/state.md`, `nat26.md` (USC Invitational 2026, not Nationals), and every `*.txt` in `downloads/` for "equivalent resistance of", "target", "exactly once".
- How supervisors check the network at sign-off (ohmmeter or V and I) is not written in any test I found.
- The E12 = ±10% and E24 = ±5% link is the IEC 60063 convention, from memory. I did not open the standard.
- Shock Value re-tryout has no key. The 466.36, 483.88 and 354.12 Ω networks are mine (Python best matches).

Sources

- 2027 Div B Rules Manual, Circuit Lab B, Part II.c.viii; rules 3.c.iv-v, 3.e.i.
- BullSO 2026 B test and key (`downloads/dlB/tests/bullso26_test.txt`, `bullso26_key.txt`); BullSO 2026 C test and key (`dlG/bull26C_test.txt`, `bull26C_key.txt`); Shock Value test and key (`dlG/wiki/Shock_value_test_answer_key.txt`); Shock re-tryout (`dlG/wiki/Shock_Retryout_test_no_key.txt`).
- Test extractions: `research/tests/ut-c.md` (UT 2018 Q25a), `ut-b.md` (UT 2019 Q47), `misc-b.md` (Shock re-tryout); `prior/G_tests.md` (target R_eq lines), `prior/B_topics.md` §9e, `prior/I_bank.md` C39.
- OpenStax, *College Physics 2e*, §21.1 "Resistors in Series and Parallel".
- All numbers checked in Python; every network found by brute force over all series-parallel splits (`target.py`, `target2.py`, 2026-10-16).

TAB T

Topics (appendix)

Circuit Lab - Topics 675
CL 1 - History 676
CL 2 - Static Electricity and Coulomb's Law 678
CL 3 - DC and AC 680
CL 4 - Current, Voltage, Resistance, Power, Energy 682
CL 5 - DC Circuits 684
CL 6 - Switches, Relays, Fuses, Breakers, GFCI 687
CL 7 - PN Junctions, Diodes and LEDs 689
CL 8 - Digital Logic 692
CL 9 - Meters, Breadboards and Hands-On Tasks 696
CL - Out of Scope for 2027 699
CL - Past Tests 700
Question Types 701

The topic notes: what each rule area covers, what is out of scope, and the past tests.

Circuit Lab - Topics

Everything that can be on the 2027 Circuit Lab B test, area by area. Back to Circuit Lab. The official list is in Circuit Lab - Rules 2027 [R].

How to read the tags

- ✅ **IN**: the 2027 rules name it. Study it.
- ⚠️ **BORDERLINE**: the rules don't settle it. A supervisor could ask it. Study it lightly.
- ❌ **OUT**: excluded for 2027. Skip it, even if old tests have it.

How often each area came up is based on three full 2025-26 Div B tests: **BullSO 2026, Berkeley 2026 and Pembroke Hill 2025**. "New" means it is on the 2027 list, but no test has used it yet.

The areas

#	Note	Tag	How often
1	CL 1 - History [T]	✅	Common (BullSO had 6 questions)
2	CL 2 - Static Electricity and Coulomb's Law [T]	✅	Common (Berkeley: 32 of 160 points)
3	CL 3 - DC and AC [T]	✅ (AC circuit theory ❌)	Common
4	CL 4 - Current, Voltage, Resistance, Power, Energy [T]	✅	Common, in every test
5	CL 5 - DC Circuits [T]	✅ (Kirchhoff only at States/Nationals)	**The biggest point block in every test**
6	CL 6 - Switches, Relays, Fuses, Breakers, GFCI [T]	✅	Common (2–4 questions per test)
7	CL 7 - PN Junctions, Diodes and LEDs [T]	✅ now at every level	Sometimes, and it should grow in 2027
8	CL 8 - Digital Logic [T]	✅ **brand new**	New (only 1 question seen in 2026)
9	CL 9 - Meters, Breadboards and Hands-On Tasks [T]	✅	Every hands-on part
—	Question Types [T]	—	Index of every question-type note, with counts
—	CL - Out of Scope for 2027 [T]	❌	30–40% of each old 2026 test
—	CL - Past Tests [T]	—	What the tests look like and where to find them

What the tests look like

- Many one-fact multiple-choice questions: history, safety, devices and units.
- One or two long circuit problems built on one diagram.
- Coulomb's law calculations.
- A breadboard task. You draw a diagram, explain it, and the supervisor signs off.
- **Always write units.** BullSO took off 1 point for a missing unit.

> **WARNING: Mistakes found in the scioly.org wiki**
>
> - The wiki says a multiplexer needs $\sqrt{N}$ select lines. **Wrong:** it needs $\log_2 N$ (4 inputs → 2 lines, 8 → 3, 16 → 4).
> - The wiki lists the gold and silver resistor bands as digits. **Wrong:** they are multipliers (gold ×0.1, silver ×0.01).

> **INFO: Not checked against Science Olympiad sources** These numbers come from general electrical knowledge, not from SO documents. Check them in a textbook before you rely on them: the GFCI trip level, the body-current table, the USB power levels, the LED voltages by color and the logic-chip pin numbers. Each note marks them.

CL 1 - History

Rule area 1: discoveries by **Ampère, Coulomb, Kirchhoff, Volta, Ohm, Hertz and Boole**. Back to Circuit Lab - Topics Ⓣ.

Question style: one-line multiple choice, either "Who did X?" or "Which unit is named after whom?". There are no calculations. **How often:** common. BullSO had 6 history questions, Berkeley 3 and Pembroke 1.

The seven people (✅ IN)

Person	Know this	Unit
Charles-Augustin de Coulomb (1736–1806, French)	**Coulomb's law** (force falls with distance squared), 1785. He invented the **torsion balance** to measure it. He also studied magnetic poles and friction.	coulomb (C)
Alessandro Volta (1745–1827, Italian)	**Voltaic pile**, 1800: the **first battery** (zinc and copper/silver discs with salt-water cloth). He disagreed with Galvani's "animal electricity". Improved the electrophorus (1775) and discovered methane (1776). Napoleon made him a count.	volt (V)
André-Marie Ampère (1775–1836, French)	Built on Ørsted's 1820 discovery that current moves a compass needle. He showed that parallel currents attract and opposite currents repel. He coined the words "solenoid" and "electrodynamics", and he separated the ideas of current and voltage. Maxwell called him **"the Newton of electricity"** (asked in BullSO).	ampere (A)
Georg Simon Ohm (1789–1854, German)	**Ohm's law**, published in 1827 in *Die galvanische Kette, mathematisch bearbeitet*. Critics rejected it at first. He used thermocouples as steady sources, a galvanometer to measure, and wires of different length and thickness.	ohm (Ω); conductance: siemens (S), also called "mho"
Gustav Kirchhoff (1824–1887, German)	**Current law and voltage law**, 1845, written while he was a student. With Bunsen he did spectroscopy and discovered **cesium and rubidium**. He explained the dark Fraunhofer lines in sunlight and coined "black body".	—
Heinrich Hertz (1857–1894, German)	**First to prove that radio (electromagnetic) waves exist**, 1886–1889. He used a spark-gap transmitter and a loop receiver, which confirmed Maxwell's theory. He showed the waves travel at light speed and reflect, refract and polarize. He also saw the photoelectric effect (1887). He thought his discovery had no practical use. **New for 2027; not in any test yet.**	hertz (Hz) = 1 cycle per second
George Boole (1815–1864, English; professor at Queen's College Cork, Ireland)	Invented **Boolean algebra**: logic as algebra on 0 and 1. Books: *The Mathematical Analysis of Logic* (1847) and *An Investigation of the Laws of Thought* (1854). Mostly self-taught. **New for 2027; not in any test yet.**	—

Likely new questions: "Who first made and detected radio waves?" (Hertz). "Who wrote *The Laws of Thought*?" (Boole).

⚠ BORDERLINE extra people

- **Claude Shannon:** his 1937 MIT master's thesis used Boolean algebra for relay and switch circuits. This is the base of all digital electronics.
- **Augustus De Morgan** (1806–1871): De Morgan's Law; Boole's friend.
- **Thomas Edison:** backed DC in the "War of the Currents". Tesla and Westinghouse backed AC. Pembroke asked: "Who championed DC?" (Edison).
- **Benjamin Franklin:** named positive and negative charge, and set the direction of conventional current. Kite experiment (1752) and the lightning rod.
- **Ørsted, Galvani, Gilbert, Maxwell, Joule, Watt, Siemens:** mostly as unit names. Joule: heating $P = I^2R$.

❌ OUT for 2027

- **Michael Faraday** and **Nikola Tesla** were removed from the list. BullSO and Berkeley 2026 asked about both. Skip those questions in old tests.

Sample past questions

- BullSO 2026: "The electric force is proportional to q_1q_2/r^2. Who is the law named after?" → Coulomb
- BullSO 2026: "Who invented the battery?" → Volta
- Berkeley 2026: "Who wrote V = IR?" → Ohm

CL 2 - Static Electricity and Coulomb's Law

Rule area 2: properties of electric charges and fields, sources and hazards of static electricity, and Coulomb's Law. Back to Circuit Lab - Topics T.

How often: common. Berkeley 2026 gave this area 32 of its 160 points. **Question styles:** concept multiple choice, Coulomb's law calculations, "where is the field zero" on a line, and reading field-line drawings.

Basic ideas

- Charge comes in whole units of the electron's charge.
- Like charges repel. Opposite charges attract.
- Ways to charge something: **friction** (triboelectric effect), **conduction** (touching) and **induction** (bringing a charge near without touching).
- A neutral object is attracted by **polarization**, for example a balloon sticking to a wall.
- Conductors versus insulators. Grounding. The electroscope.

Formulas and constants

Item	Formula or value
Coulomb's law	$F = k \cdot q_1 \cdot q_2 / r^2$
k	8.99×10^9 N·m²/C² ($\approx 9 \times 10^9$) = $1/(4\pi\varepsilon_0)$
ε_0	8.85×10^{-12}
Electric field	$E = F/q = k \cdot q/r^2$ (units N/C, which equals V/m)
Potential	$V = k \cdot q/r$
Potential energy	$U = k \cdot q_1 \cdot q_2/r$
Work	$W = q \cdot \Delta V$
Electron charge e	1.602×10^{-19} C
1 coulomb	6.24×10^{18} electrons
Electron mass	9.11×10^{-31} kg
Proton mass	1.67×10^{-27} kg
1 eV	1.602×10^{-19} J

- Forces and fields add (superposition). Trig is banned, so charges will be on one line or at right angles.

Field lines

- Lines start on + charges and end on − charges.
- Lines never cross. Closer lines mean a stronger field.
- Field lines are perpendicular to equipotential lines (asked in BullSO) and to the surface of a conductor.
- The field inside a conductor is zero. Extra charge sits on the outside surface.
- Charge gathers at sharp points. This explains corona discharge and lightning rods.
- Know how to find the point where E = 0, or where V = 0, between or outside two charges.

Triboelectric series

A common order, from becomes + to becomes −: **rabbit fur, glass, human hair, wool, silk, paper, cotton, rubber, PVC, Teflon**

- Glass rubbed with silk becomes **+**: it **loses electrons**. Protons never move.
- Rubber or plastic rubbed with wool or fur becomes **−**.

Static hazards and sources

- Lightning.

- **ESD** (electrostatic discharge) damages electronics. Prevent it with wrist straps, antistatic bags and grounding.
- Sparks can ignite fuel vapor or dust, for example at gas pumps or in grain silos. Grounding and bonding prevent this.
- A Van de Graaff shock is not dangerous: the voltage is high, but the current and total charge are tiny (Berkeley).
- Static is worse in dry air (winter) than in humid air.

⚠ BORDERLINE

- The **Faraday cage** (zero field inside a conductor). The idea counts as electric fields, so it is IN. The person Faraday is OUT.
- Electric potential energy has appeared in tests, and it fits "properties of electric fields".
- Gauss's law: not seen, probably too advanced for Div B.

Sample past questions

- BullSO 2026: +6 μC and −7 μC are 0.5 m apart. Force? → **1.512 N, attractive**. ($9\times10^{9} \times 6\times10^{-6} \times 7\times10^{-6} / 0.5^{2} = 1.512$)
- Berkeley 2026: −3 C at x = −2 and +1 C at x = +2. Where on the x-axis is E = 0?
- Pembroke 2025: Glass rubbed with silk turns positive. Did it gain protons or lose electrons? → **Lost electrons**
- Berkeley 2026: What is charging by rubbing called? → **Triboelectric effect**
- BullSO 2026: How are field lines and equipotential lines related? → **Perpendicular**

CL 3 - DC and AC

Rule area 3: characteristics, sources, uses and hazards of **direct current (DC)** and **alternating current (AC)**. **AC circuit theory is ❌ OUT.** Back to Circuit Lab - Topics [T].

How often: common.

DC

- Current flows in one direction, and the polarity stays the same.
- **Conventional current** flows from + to −. **Electrons** flow from − to +. This is Franklin's convention; Pembroke asked it as true/false.
- Electrons drift slowly (μm/s to mm/s), but the signal moves at nearly light speed.
- **Sources:** batteries and cells, solar cells, DC generators, thermocouples, fuel cells, power adapters (AC → DC), USB ports.
- **Uses:** electronics, phones, cars (12 V), flashlights, electric-car batteries. High-voltage DC power lines are ⚠ borderline.
- **Hazards:** very large short-circuit currents, battery fires or explosions, lithium-ion "thermal runaway", swallowed button cells, hydrogen gas and sparks at car batteries. High-voltage DC causes one strong muscle jerk (it often throws the person away) and burns.

Batteries

- A **cell** is one unit. A **battery** is several cells.
- **Single-use (primary) cells:**

Type	Voltage
Alkaline (Zn/MnO_2)	1.5 V
Carbon-zinc	1.5 V
Lithium coin (CR2032)	3 V
Silver oxide	1.55 V
Zinc-air (hearing aids)	1.4 V

- **Rechargeable (secondary) cells:**

Type	Voltage
Lead-acid	2.0 V per cell (a 12 V car battery = 6 cells)
NiCd, NiMH	1.2 V
Lithium-ion	3.6–3.7 V normal (4.2 V full)

- A 9 V battery is six 1.5 V cells inside. AA, AAA, C and D are all 1.5 V: **size changes capacity, not voltage.**
- **Capacity** is in mAh or Ah. **Energy** in Wh = Ah × V.
- Cells **in series add voltage**. Cells **in parallel add capacity** at the same voltage.
- **Internal resistance r:** V_terminal = ε − I·r. Short-circuit current = ε / r.
- **Electrochemistry:** the **anode** is where oxidation happens; it is the **negative** terminal when the cell is discharging. The **cathode** is where reduction happens (positive terminal). The electrolyte sits between them. A dry cell has a paste; a wet cell has a liquid.
- A lemon battery uses zinc (a galvanized nail) and copper (a penny). It gives about 0.9 V per cell, with very little current.

AC

- The current reverses direction again and again, usually as a sine wave.
- Frequency f is in hertz. Period T = 1/f. There are peak, peak-to-peak and RMS values.
- **US outlets: 120 V RMS, 60 Hz** (asked in Berkeley). Dryers and stoves use 240 V. The peak of 120 V RMS is about 170 V.

- Europe and most of the world: 230 V, 50 Hz.
- ⚠ **RMS:** V_rms = V_peak / √2 ≈ 0.707 × V_peak. BullSO asked: "What DC voltage gives the same power as a 100 V-peak AC source?" → 100/√2 ≈ 70.7 V. This is borderline, because it touches AC theory.
- **Sources:** generators, alternators and power plants (the magnetism inside them is ❌ OUT), and **inverters** (DC → AC; asked in BullSO).
- **Why the power grid uses AC** (⚠ borderline, but likely): AC is easy to step up to high voltage. High voltage means low current, and low current means low I^2R loss in the wires (asked in Berkeley). How a transformer works is ❌ OUT.
- AC → DC uses a **rectifier** made of diodes (see CL 7 - PN Junctions, Diodes and LEDs 🅃).
- **Hazards:** at 50–60 Hz, AC is more dangerous than DC at the same voltage. It can cause **ventricular fibrillation** (the heart quivers instead of pumping) and constant muscle clamping, so you **can't let go** (asked in Berkeley).

⚠ Household wiring (likely, but no longer named in the rules)

- **US wire colors:** black (or red) = **hot**, white = **neutral**, green or bare = **ground** (asked in Berkeley).
- **Outlet:** narrow slot = hot, wide slot = neutral, round hole = ground.
- Polarized plugs, 3-prong grounded plugs, and double-insulated tools (symbol: a square inside a square).
- BullSO asked which shock-prevention practice is the least effective.

Effects of current on the body

General safety-table values for 60 Hz current from hand to hand. They are not from an SO source, and sources differ a little.

Current	Effect
~1 mA	Felt as a tingle
~5 mA	Highest "harmless" level; the GFCI trip level
~10–20 mA	"Let-go" limit: muscles clamp
~50–100 mA	Heart may fibrillate
1–4 A and more	Heart stops; severe burns

- Body resistance: about **100 kΩ with dry skin** and about **1 kΩ with wet or broken skin**.
- "It's the current that kills, but the voltage pushes it": I = V/R.
- Safety habits: keep one hand in your pocket, keep your hands dry, never remove the ground pin, unplug before working, and use GFCIs near water.

❌ OUT

Phase angle, impedance, reactance, power factor, 3-phase power, frequency response, oscilloscopes and the skin effect (BullSO 2026 asked about it).

Sample past questions

- Pembroke 2025: A battery's EMF is 9.0 V. Its terminal voltage is 8.5 V across a 10 Ω load. Find r. → **0.59 Ω**. (I = 8.5/10 = 0.85 A; r = 0.5/0.85)
- Pembroke 2025: Where does oxidation happen in a battery? → **The anode**
- BullSO 2026: Can a battery output AC? → **No**

CL 4 - Current, Voltage, Resistance, Power, Energy

Rule area 4: concepts, calculations and units of current, voltage, resistance, power and energy. Back to Circuit Lab - Topics Ⓣ.

How often: common, with many points in every test. **Question styles:** plug-in calculations, "what happens if" scaling, naming the unit, and energy cost.

Definitions and units

Quantity	Meaning	Unit
Current I	I = Q/t	ampere (A = C/s)
Voltage V	energy per charge (also called potential difference or EMF)	volt (V = J/C = W/A)
Resistance R	R = V/I	ohm (Ω = V/A)
Conductance G	G = 1/R	siemens (S), also called "mho"
Power P	energy per second	watt (W = J/s)
Energy E	—	joule (J) or kilowatt-hour (kWh)
Charge Q	—	coulomb (C) or mAh

- 1 kWh = 3.6×10^6 J
- 1 mAh = 3.6 C

Formulas

- **V = I·R** (Ohm's law)
- **P = I·V = I²·R = V²/R**
- **E = P·t = V·I·t**
- **Q = I·t**
- **Cost = kWh × price per kWh**
- **Efficiency = P_out / P_in**
- **Resistivity: R = ρ·L/A.** R goes up with length and down with cross-section area.

Resistivity values (Ω·m, from the national supervisor practice slides)

Material	ρ
Silver	1.59×10^{-8}
Copper	1.68×10^{-8}
Gold	2.44×10^{-8}
Aluminum	2.65×10^{-8}
Nichrome	$\approx 1.1 \times 10^{-6}$
Silicon	$\approx 6.4 \times 10^{2}$
Glass	$\sim 10^{10}$ to 10^{14}

Temperature

- Metals: resistance **goes up** as temperature goes up.
- Semiconductors and NTC thermistors: resistance **goes down** as temperature goes up.
- Superconductors: R = 0 below a critical temperature.

SI prefixes

Prefix	Value
p (pico)	10^{-12}
n (nano)	10^{-9}

Prefix	Value
μ (micro)	10^{-6}
m (milli)	10^{-3}
k (kilo)	10^{3}
M (mega)	10^{6}
G (giga)	10^{9}
T (tera)	10^{12}

- Answers must be in **metric units with correct significant figures** (rule 3.b).
- SI base units: A, m, kg, s. Derived units: N, J, W, C, V, Ω, S. Hertz is ✅ IN (history and AC). The farad is ❌ OUT, except as trivia.

⚠ BORDERLINE

- **Bulb brightness:** in series, the bulb with the **higher wattage rating is dimmer**. Rare.
- **AWG wire gauge:** a bigger number means a thinner wire. Rare.

Sample past questions

- Pembroke 2025: A 100 W bulb runs for 24 h at $0.15/kWh. Cost? → **$0.36** (2.4 kWh × 0.15)
- Berkeley 2026: A 60 W bulb at 120 V. Find R. → **240 Ω** (120^2 / 60)
- Berkeley 2026: 30 mA flows for 2 minutes. Charge? → **3.6 C** (0.030 × 120)
- BullSO 2026: Wire length ×3 and area ÷2. New R? → **6R**
- Pembroke 2025: Which unit equals a volt? → **J/C**
- Pembroke 2025: V doubles and R halves. Power? → **×8** ($P = V^2/R$)
- Pembroke 2025: A 600 W device at 120 V on a 5 A fuse. Does the fuse blow? → It draws exactly 5 A. **The key says no.**
- Berkeley 2026: Find the energy a circuit uses in 2 hours.

CL 5 - DC Circuits

Rule area 5: calculations, building and arranging DC circuits and parts. This includes series and parallel, open and short circuits, Ohm's Law and schematics. Rule area 9, **Kirchhoff's laws, is for States and Nationals only.** Back to Circuit Lab - Topics T.

How often: common. **This is the biggest point block in every test.**

Series and parallel

- **Series:** the **same current** flows through every part. R_eq = R_1 + R_2 + … The voltages add up.
- **Parallel:** every branch has the **same voltage**. 1/R_eq = 1/R_1 + 1/R_2 + … The currents add up.
  - Two resistors: R_eq = $R_1 \cdot R_2$ / (R_1 + R_2)
  - n equal resistors: R_eq = R / n
  - R_eq is always **smaller than the smallest** branch.
- **Mixed circuits:** simplify one step at a time, and redraw the circuit after each step.
- Ladder networks: sometimes. Symmetry and bridge shortcuts: rare.
- **Voltage divider:** V_x = V_source × R_x / R_total
- **Current divider (2 branches):** I_1 = I_total × R_2 / (R_1 + R_2)

Open and short circuits (named in the 2027 rules)

- **Open circuit:** I = 0. The **full source voltage appears across the gap**. A meter reads "OL".
- **Short circuit:** R ≈ 0, so V ≈ 0 across it, and the current can be huge. This is dangerous, and it is why fuses exist.
- Short one bulb in a series string, and the other bulbs get **brighter**.
- In a series string, one dead bulb turns them **all** off. In parallel, only the dead one goes out.
- A meter reads 0.1 Ω across a part. It is **shorted** (Pembroke).

Sources and ground

- Ideal voltage source versus ideal current source.
- A real battery = an ideal source in series with its internal resistance r.
- Voltage sources in series add up; watch the + and – direction. Don't connect unequal voltage sources in parallel.
- **Ground** is the 0 V reference point. Know the difference between a node's voltage and the voltage across a part.

Schematic symbols (common in tests)

Know both the US (ANSI) and the international (IEC) drawings.

- **Wires:** a dot means connected. No dot, or a small hop, means the wires cross without touching (Pembroke).
- **Resistors:** zigzag (US) or rectangle (IEC). A zigzag with a diagonal arrow = **variable resistor** (Pembroke). Potentiometer (3 terminals), rheostat. ⚠ Thermistor and photoresistor.
- **Sources:** cell (the **long line is +**), battery (several cells), DC source, AC source (a sine wave in a circle), current source (an arrow in a circle).
- **Ground:** earth, chassis and signal ground.
- **Switches:** SPST, SPDT, DPST, DPDT, push-buttons (normally open and normally closed). Relay (a coil plus contacts), fuse, circuit breaker.
- **Loads:** lamp, buzzer, speaker. ⚠ The motor symbol (motor physics is OUT).
- **Diodes:** diode (a triangle and a bar; the **bar is the cathode**), LED (arrows pointing out), Zener (bent bar), Schottky (S-shaped bar). ⚠ Photodiode (arrows pointing in).
- **Meters:** voltmeter (V in a circle), ammeter (A in a circle), ohmmeter (Ω), galvanometer.
- **Logic gates:** see CL 8 - Digital Logic T.
- ✗ Capacitor, inductor, transformer and transistor are OUT as topics. They could still appear as wrong answer choices in a symbol question. BullSO 2026 asked to name a capacitor symbol.

Kirchhoff's laws (✅ States and Nationals only)

Even invitationals used these in 2026.

- **KCL (current law, junction rule):** the current going into a node = the current coming out. It comes from **conservation of charge** (asked in Berkeley and Pembroke).
- **KVL (voltage law, loop rule):** the voltage changes around any closed loop add up to 0. It comes from **conservation of energy** (asked in Berkeley).
- **Sign rules:** going from − to + through a battery is a rise. Going through a resistor in the direction of the current is a drop of I·R.
- **Nodal analysis:** pick a ground node, then write KCL at each unknown node.
- **Mesh analysis:** give each loop its own current, then write KVL for each loop. A resistor shared by two loops carries ($I_1 - I_2$).
- **Math limit:** at most **two equations**. So expect at most 2 unknown nodes or 2 loops.
- ⚠️ **Wheatstone bridge:** balanced when $R_1/R_2 = R_3/R_x$, and then no current flows through the middle meter.
- ⚠️ Superposition, dependent sources, maximum power transfer (R_load = r). Rare.
- Delta-Y: rare (one 2014 forum question).
- ❌ **Norton and Thévenin equivalents: explicitly OUT.**

Resistor color code (sometimes in written tests, likely in hands-on)

Color	Digit	Multiplier	Tolerance
Black	0	×1	—
Brown	1	×10	1%
Red	2	×100	2%
Orange	3	×1 k	—
Yellow	4	×10 k	—
Green	5	×100 k	0.5%
Blue	6	×1 M	0.25%
Violet	7	×10 M	0.1%
Gray	8	×100 M	0.05%
White	9	×1 G	—
Gold	—	**×0.1**	**5%**
Silver	—	**×0.01**	**10%**
No band	—	—	20%

- **4 bands:** digit, digit, multiplier, tolerance.
- **5 bands:** digit, digit, digit, multiplier, tolerance.
- **6 bands:** a temperature-coefficient band is added.
- Example: red-violet-green-gold = 27 × 10^5 = **2.7 MΩ ±5%**.
- ⚠️ The scioly.org wiki wrongly lists gold and silver as digits. They are **multipliers**.
- Standard (E12) values: 10, 12, 15, 18, 22, 27, 33, 39, 47, 56, 68, 82, times a power of 10.
- Resistors have power ratings (1/8, 1/4, 1/2 W). Keep P = I^2R below the rating.
- ⚠️ Surface-mount codes (103 = 10 kΩ, 4R7 = 4.7 Ω): rare.

Which components are in scope

Component	Tag
Resistor, potentiometer, rheostat	✅
Battery, cell	✅
Switch, relay, fuse, breaker, GFCI	✅ (CL 6 - Switches, Relays, Fuses, Breakers, GFCI T)
Diode, LED, Zener	✅ (CL 7 - PN Junctions, Diodes and LEDs T)
Lamp, bulb, buzzer (as loads)	✅

Component	Tag
Thermistor, photoresistor, varistor	⚠ (Berkeley asked to name a thermistor)
Solar cell, photodiode	⚠
"Passive" versus "active" parts	⚠ (BullSO: a transistor and a diode are not passive)
Capacitor	✗ (2026 tests had 4–5 capacitor questions each)
Inductor, transformer, motor, generator, electromagnet	✗
Transistor, thyristor, op-amp	✗

Sample past questions

- BullSO 2026: Parallel parts share ___; series parts share ___. → **Voltage; current**
- Pembroke 2025: 15 V across 1 kΩ + 4 kΩ in series. Voltage across the 4 kΩ? → **12 V**
- Berkeley 2026: Using 3 of the resistors 1–5 Ω, find the smallest possible R_eq.
- BullSO 2026: Four equal resistors in a given network give 45 Ω. Find each R.
- Berkeley 2026: Draw the direction of conventional current. Find the voltage at points 1 and 2.
- Berkeley 2026: A node has 4 A coming in and 7 A going out. Find the other incoming current. → **3 A**
- BullSO 2026: nodal analysis → $V_1 = 28$ V, $V_2 = 30$ V. Mesh analysis → $I_1 = 5$ A, $I_2 = 2.5$ A.
- BullSO 2026: one loop with two sources: $-20 + V_1 - 12 + V_2 = 0$, $I = 0.8$ A.

CL 6 - Switches, Relays, Fuses, Breakers, GFCI

Rule area 6: electrical control devices, including switches, relays, fuses, ground fault circuit interrupters (GFCIs) and breakers. Back to Circuit Lab - Topics [T].

How often: common. Each 2026 test had 2–4 questions.

Switches

- **Pole** = how many separate circuits the switch controls.
- **Throw** = how many positions each pole can connect to.

Type	What it is
SPST	Simple on/off (single pole, single throw)
SPDT	Changeover with 3 terminals: common + 2
DPST	Two SPSTs that move together
DPDT	Two SPDTs that move together. Used to reverse a motor or swap polarity.

- **Momentary** (push-button) versus **latching** (toggle, rocker, slide).
- **Normally open (NO)** versus **normally closed (NC)**.
- ⚠️ Make-before-break versus break-before-make.
- Other types: rotary, DIP, limit, micro switch. ⚠️ Reed switch (it is magnetic).

3-way switch (a 2027 hands-on task)

- Two SPDT ("3-way") switches are joined by **two traveler wires**. Either switch can turn the light on or off.
- The light is on when both switches pick the **same traveler**.
- In logic terms the light = A XNOR B, or A XOR B, depending on how it starts. This links to CL 8 - Digital Logic [T].
- For 3 or more switch locations, put **4-way switches** (DPDT wired as a crossover) between the two 3-way switches.
- **Practice build:** battery + LED + resistor + 2 SPDT slide switches on a breadboard. Connect each switch's common to the supply side or the load side. Connect the two outer terminals of one switch to the two outer terminals of the other.

Relays

- A **switch worked by electricity**. A small current in the coil controls a separate, bigger circuit through the contacts. This keeps the two circuits apart.
- Parts: coil, armature, spring, contacts. Terminals: **COM, NO, NC**. A relay can be SPST, SPDT or DPDT.
- **Flyback diode** across the coil (cathode to +). It absorbs the voltage spike when the coil turns off.
- The coil has a voltage rating. The contacts have current and voltage ratings.
- **Relay logic:** contacts in series = AND, contacts in parallel = OR, an NC contact = NOT. Early computers worked this way.
- ⚠️ Solid-state relays (they are semiconductors). A Hovercraft clarification says SO relays are electromechanical; no Circuit Lab clarification was found.
- The coil works by magnetism, and magnetism physics is ❌ OUT. But how a relay works is ✅ IN, because relays are named in the rules.

Fuses

- A thin metal link placed **in series** on the **hot** wire. It melts when the current is higher than its rating.
- Single use: replace it with the **same rating**, after fixing the cause.
- Ratings: current (A), voltage, and breaking capacity.
- **Fast-blow** versus **slow-blow** (time-delay, for the start-up surge of motors).
- Types: glass cartridge, ceramic, car blade fuses (the color shows the amps). ⚠️ Resettable PTC "polyfuse".
- A fuse is usually faster than a breaker (national supervisor slides).

- A multimeter's current jacks have fuses, often about 200–400 mA for the mA jack and 10 A for the 10 A jack.
- **Does the fuse blow?** Work out I = V / R_eq and compare it with the rating.
  - 2014 forum: three 3 Ω resistors in parallel on 5 V = 1 Ω, so 5 A. **A 4 A fuse blows.**
  - Pembroke 2025: 600 W at 120 V = 5 A on a 5 A fuse.

Circuit breakers

- A **resettable** switch that opens (trips) on too much current.
- **Thermal trip:** a bimetal strip bends as it heats from I^2R. This catches slow overloads.
- ⚠ **Magnetic trip:** a coil pulls the latch during a big short-circuit current. This catches sudden faults.
- Rated in amps. Home circuits: 15 A uses 14 AWG wire, and 20 A uses 12 AWG wire.
- **Fuse versus breaker** (asked in BullSO and Berkeley):
  - A breaker can be reset. A fuse must be replaced.
  - Both protect against overcurrent.
  - A fuse is usually faster and cheaper. A breaker can be used again.
  - **Neither one protects people from shock.** They protect the wires from fire.
- ⚠ AFCI (arc-fault breaker) detects sparking. It is not named in the rules.

GFCI (Ground Fault Circuit Interrupter, also called GFI)

- **What it does:** compares the current in the **hot** wire with the current in the **neutral** wire. If they differ, some current is leaking another way, maybe through a person to ground. Then it cuts the power.
- **Trip level and speed:** about **4–6 mA** (nominal 5 mA in the US) in about **1/40 s (≈ 25 ms)**. *These values are from US electrical standards (UL 943 and the electrical code), not from an SO source. Check them before relying on exact numbers.*
- **Where the US code requires them:** bathrooms, kitchen counters, outdoors, garages, unfinished basements, laundry rooms, near pools, sinks and tubs.
- **Buttons:** TEST (makes a small leak on purpose; it should trip) and RESET. Test once a month.
- **Limits:**
  - It does **not** trip on an overload; that is the breaker's job.
  - It does **not** trip if a person touches hot and neutral together, because the currents still match.
  - It works even with **no ground wire**.
  - One GFCI outlet also protects the outlets wired after it ("load" side).
- **Inside:** a ring-shaped current sensor and a trip coil. Know the idea, not the magnetism.
- BullSO 2026 spelled it "Interruptor". The standard spelling is **"Interrupter"**.

⚠ Other protection

- Surge protectors (they use MOVs, or varistors).
- The ground pin, equipment grounding and bonding.
- Polarized plugs.
- Double insulation (symbol: a square inside a square).

Sample past questions

- Pembroke 2025: Which switches does a hallway light controlled from 2 places need? → **SPDT**
- Berkeley 2026: Why does flipping either SPDT switch toggle the light? → **It is on only when both switches connect to the same traveler.**
- BullSO 2026: What does a 3-way switch setup do? → **Two switches, and either one can change the bulb.**
- Berkeley 2026: One push turns off a red LED and turns on a green LED. Which switch? → **SPDT**
- Pembroke 2025: Which device protects people in wet areas by sensing a hot/neutral mismatch? → **GFCI**

CL 7 - PN Junctions, Diodes and LEDs

Rule area 7: basic electrical behavior of **silicon PN junctions**, and uses of single-junction parts (**diodes, LEDs**). New for 2027: this is now at **every level**; last season LEDs were States/Nationals only. Back to Circuit Lab - Topics T.

How often: sometimes in 2026 (Berkeley: 3 questions, Pembroke: 2, and an LED hands-on task in all three tests). Expect more in 2027.

Semiconductor basics

- Pure silicon (Si) and germanium (Ge) are in group 14: **4 valence electrons**.
- **n-type:** add group 15 atoms (P, As, Sb). They give extra free **electrons**, which are the main carriers.
- **p-type:** add group 13 atoms (B, Ga, In, Al). They make **holes**, which are the main carriers.
- Doped material is still **neutral** overall.
- GaAs and similar compounds are used in LEDs.
- **Band gap:** Si ≈ 1.1 eV, Ge ≈ 0.67 eV, GaAs ≈ 1.42 eV. Insulators have big gaps. Conductors have no gap.
- A semiconductor's resistance goes **down** as it gets hotter.
- ❌ Transistors (BJT, FET) and chip internals are OUT.

The PN junction

- **Depletion region:** electrons move into the p side and holes into the n side. Fixed charged atoms stay behind. They make a built-in field and a **built-in voltage of about 0.7 V for Si and 0.3 V for Ge**.
- **Forward bias:** p side (anode) to +, n side (cathode) to −. The depletion region gets thinner. Current flows once V is above about **0.6–0.7 V** for silicon.
  - Pembroke: current flows when the **anode is at a higher voltage** than the cathode.
- **Reverse bias:** the depletion region gets wider. Only a tiny leakage current flows (nA to µA), until **reverse breakdown**.
  - Breakdown can be avalanche or Zener. Normal rectifier diodes take 50 to 1000+ V in reverse. **LEDs break down at only about 5 V.**
- **Reading the I-V graph:** forward side: almost no current, then a sharp "knee" and fast rise. Reverse side: flat near zero, then a sharp drop at the breakdown voltage.
- ⚠ Extra details: reverse recovery time (µs for PN, ns for Schottky). V_f drops about 2 mV per °C.

Diode models (how to solve circuits)

- **Ideal model:** a closed switch (short) when forward-biased, an open switch when reverse-biased.
- **Constant-drop model:** ideal diode + a 0.7 V drop for Si. In KVL: **I = (V_source – 0.7) / R**.
- ⚠ Piecewise-linear model: 0.7 V plus a small resistance.
- **Method for several diodes:** guess "on" or "off" for each, solve, then check. An "on" diode must carry forward current. An "off" diode must have less than 0.7 V across it. If a check fails, change the guess.
- A diode does **not** follow Ohm's law; it is non-linear.

Diode types

- **Rectifier diode** (1N4001–1N4007) and **signal diode** (1N4148). The **band** marks the cathode.
- **LED:** see below.
- **Zener:** designed to break down in reverse at an exact voltage V_Z (for example 5.1 V). It is used reverse-biased as a voltage regulator or limiter. Series resistor: **R = (V_in – V_Z) / I**. ⚠ How deep questions go is unclear.
- ⚠ **Schottky:** a metal-semiconductor junction (not P-N). V_f ≈ 0.2–0.4 V. Very fast. Low breakdown voltage.
- **Germanium diode:** V_f ≈ 0.3 V.
- ⚠ **Photodiode and solar cell:** a PN junction where light makes current. A silicon solar cell gives about 0.5–0.6 V with no load.

LEDs (light-emitting diodes)

- **How it works:** electrons fall into holes at the junction and give off **photons** (light) (asked in Berkeley). The photon energy is about equal to the band gap, so **the color depends on the material**.
- **Which leg is which: long leg = anode (+)**, **short leg = cathode (–)** (asked in Pembroke). The **flat edge** on the rim is the cathode side. The bigger metal piece inside is *usually* the cathode, but don't trust it.

Forward voltage by color

Typical values from datasheets. They are not from an SO source, and real parts vary.

Color	V_f (typical)	Material
Infrared (~850–940 nm)	1.2–1.5 V	GaAs
Red (~620–700 nm)	**1.8–2.1 V**	AlGaInP, GaAsP
Orange / yellow	2.0–2.2 V	
Green	2.0–2.2 V (old type); ~3.0 V (InGaN)	
Blue (~450–490 nm)	**2.7–3.3 V**	InGaN
White	3.0–3.5 V	blue LED + yellow phosphor
UV	3.1–4.4 V	

- **Rule:** shorter wavelength → more photon energy → bigger band gap → **higher V_f**.
- ⚠ Photon energy: E = h·f = h·c/λ. Shortcut: **E (eV) ≈ 1240 / λ (nm)**. h = 6.626 × 10^{-34} J·s. Not named in the rules.

The current-limiting resistor

- An LED **always needs a series resistor**: **R = (V_source – V_f) / I_LED**
- Normal LED current: **10–20 mA** (the max is about 20–30 mA for a 5 mm LED).
- Example: 9 V supply, red LED (V_f = 2.0 V), 20 mA → R = 7 / 0.020 = **350 Ω**. Use **390 Ω** (the next standard value up).
- Resistor power = I^2R. LED power = V_f × I.
- **Brightness depends on current**, not voltage.
- With no resistor, the LED burns out. More than about 5 V in reverse can also destroy it.

Two LEDs equally bright (a 2027 hands-on task)

- **In series:** both carry the **same current**, so identical LEDs are equally bright. The supply must be more than the two V_f values plus the resistor's drop.
- **In parallel:** each LED needs its **own resistor**. With one shared resistor, the LED with the lower V_f takes most of the current. Red and blue in parallel: the red lights, and the blue may stay dark.
- Two **different colors:** give each LED its own resistor sized for its V_f. Equal current gives *roughly* equal brightness, but eyes see some colors as brighter. Explain this on your answer sheet.
- **Past hands-on tasks:**
  - Berkeley 2026 (40 points): a 9 V battery, resistors, 3 LEDs and 1 blue LED. Make two LEDs dim but equal, one LED bright, and keep the blue LED connected but **not lit**. Trick: put the blue LED in parallel with a lower-V_f LED, or connect it backwards.
  - Pembroke 2025: build 2 resistors in parallel in series with 2 LEDs in parallel. Then build 2 resistors and 2 LEDs all in series. Measure V and I and write the units.
- ⚠ History: Nick Holonyak made the first visible (red) LED in 1962. Nakamura, Akasaki and Amano made the efficient blue LED and won the Nobel Prize in 2014.

Uses of diodes (the rules say "uses")

- **Rectifier:** half-wave (1 diode) or full-wave **bridge** (4 diodes). A bridge loses about 2 × 0.7 V. The smoothing-capacitor part is ❌ OUT (capacitance).
- **Reverse-polarity protection:** a diode in series, or a diode across the supply plus a fuse.
- **Flyback diode** across a relay coil.
- **Diode OR-ing** of two power sources (battery backup).

- **Diode logic gates:**
  - Diode OR: the inputs go to the anodes; the joined cathodes go to the output with a pull-down resistor.
  - Diode AND: the inputs go to the cathodes, with a pull-up resistor.
  - **You cannot make a NOT gate from diodes and resistors alone** (from the supervisor practice slides).
- Dropping voltage: about 0.7 V per diode.
- Indicator lights, LED displays, light sensors.
- Zener voltage reference.

Testing diodes with a multimeter (hands-on)

- **Diode-test mode** shows the forward drop in volts: Si ≈ 0.5–0.7 V, LEDs ≈ 1.6–3 V (the LED may glow faintly). The reverse direction shows "OL".
- An ohmmeter gives confusing readings on a diode.
- Diode mode finds the anode and the cathode, and finds bad diodes: "OL" both ways = **open**; ≈0 both ways = **shorted**. It can also tell LED colors apart.

Sample past questions

- Berkeley 2026: Which LED color has the lowest voltage drop? → **Red**
- Berkeley 2026: Order these from smallest to largest band gap: → **infrared, red, blue**
- Berkeley 2026: How does an LED make light? → **Electrons recombine with holes and release photons.**
- Pembroke 2025: When does current flow through a diode? → **When the anode is at a higher voltage than the cathode.**
- Pembroke 2025: Which LED leg is the anode? → **The long leg**

CL 8 - Digital Logic

Rule area 8: **digital logic gates, truth tables, universal gate sets, Boolean expressions, De Morgan's Law, and combinational logic circuits (for example multiplexers and adders).** Plus binary whole numbers and binary addition (math rule 3.e). Back to Circuit Lab - Topics 🅃.

This is brand new for Div B. soinc.org's event page says "Not a topic in 2026". Only one 2026 question touched it (Pembroke: "Which gate gives 1 only if both inputs are 1?"). No 2027 test exists yet, so we can't see how much will be asked. But every test must have at least one logic question.

❌ **Excluded:** sequential logic (latches, flip-flops, counters, shift registers, memory), clock signals and timing diagrams, negative binary numbers (two's complement), binary subtraction or multiplication, and decimal points in binary.

Logic levels and binary

- 1 = TRUE = HIGH (for example 5 V or 3.3 V). 0 = FALSE = LOW (0 V).
- ⚠ TTL thresholds: low < 0.8 V, high > 2.0 V.
- **Binary place values:** 1, 2, 4, 8, 16, 32, 64, 128, …
- **Decimal → binary:** divide by 2 again and again, or subtract powers of 2.
- n bits can hold 0 to $2^n - 1$. A truth table with n inputs has **2^n rows**.
- Bit, nibble (4 bits), byte (8 bits). LSB = least significant bit, MSB = most significant bit.
- **Binary addition:**

Add	Sum bit	Carry
0 + 0	0	0
0 + 1	1	0
1 + 1	0	1 (= 10)
1 + 1 + 1	1	1 (= 11)

- Example: **1011 (11) + 0110 (6) = 10001 (17)**. A carry out of the top bit is **overflow**.
- ⚠/❌ Hexadecimal and octal: the rules say "whole numbers in binary" only.

The gates

Gate	Output is 1 when…	Expression	US symbol	IEC box
NOT (inverter)	the input is 0	$\bar{A}$, A′, ¬A, ~A	triangle + bubble	"1" + bubble
AND	all inputs are 1	A·B, AB, A∧B, A&B	D shape	"&"
OR	any input is 1	A+B, A∨B	curved shield	"≥1"
NAND	not all inputs are 1	(AB)′, A↑B	AND + bubble	"&" + bubble
NOR	all inputs are 0	(A+B)′, A↓B	OR + bubble	"≥1" + bubble
XOR	the inputs are different (with 3+ inputs: an odd number of 1s)	A⊕B = AB′ + A′B	OR with an extra curve at the back	"=1"
XNOR	the inputs are the same	(A⊕B)′ = AB + A′B′	XOR + bubble	"=1" + bubble

- A bubble always means NOT. ⚠ A buffer (a triangle without a bubble) just passes the input through.
- **Notation trap:** supervisors write NOT as a bar over the letter, a prime (A′), ¬ or ~. The national practice slides say to "have them all in your binder".
- **Switch versions:** switches in series = AND, in parallel = OR, a normally-closed push-button = NOT.

Truth tables (2 inputs)

A	B	AND	OR	NAND	NOR	XOR	XNOR
0	0	0	0	1	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	1	0	0	0	1

Boolean algebra rules

Name	OR form	AND form
Identity	A + 0 = A	A · 1 = A
Null (dominance)	A + 1 = 1	A · 0 = 0
Idempotent	A + A = A	A · A = A
Complement	A + A′ = 1	A · A′ = 0
Double NOT	(A′)′ = A	
Commutative	A + B = B + A	AB = BA
Associative	(A+B)+C = A+(B+C)	(AB)C = A(BC)
Distributive	**A + BC = (A+B)(A+C)** ← not like normal algebra	A(B+C) = AB + AC
Absorption	A + AB = A	A(A+B) = A
"Disappearing opposite"	A + A′B = A + B	
⚠ Consensus	AB + A′C + BC = AB + A′C	

- An older version of the practice slides marked these as "not allowed". **The 2027 rules allow Boolean expressions and De Morgan.** How much simplifying is expected is ⚠ unclear.

De Morgan's Laws (named in the rules; expect a question)

- **(A·B)′ = A′ + B′**
- **(A + B)′ = A′ · B′**
- Memory trick: **"Break the bar, change the sign."**
- Works for more inputs: (ABC)′ = A′ + B′ + C′.
- **Bubble pushing:**
  - NAND = OR with both inputs inverted.
  - NOR = AND with both inputs inverted.
  - Moving a bubble through a gate swaps AND ↔ OR and puts bubbles on the other side.

Universal gate sets (named in the rules)

- A set is **universal** if you can build **any** logic function from it.
- **Universal:** {NAND}, {NOR}, {AND, NOT}, {OR, NOT}.
- **Not universal:** {AND, OR}, because nothing can invert. {XOR} alone is not universal either.

Gate counts from NAND only

Build	NAND gates	How
NOT	1	NAND(A, A)
AND	2	NAND, then a NAND used as NOT
OR	3	NAND(NOT A, NOT B)
NOR	4	OR + NOT
XOR	4	N1 = NAND(A,B); out = NAND(NAND(A,N1), NAND(B,N1))
XNOR	5	XOR + NOT

Gate counts from NOR only

Build	NOR gates
NOT	1 (NOR(A, A))
OR	2
AND	3 (NOR(NOT A, NOT B))
NAND	4
XNOR	4
XOR	5

- A two-level AND-OR circuit turns straight into NAND-NAND. OR-AND turns into NOR-NOR.
- Likely question: "What is the fewest 2-input NAND gates that make an XOR?" → **4**

Truth table ↔ expression ↔ circuit (a core skill and a 2027 hands-on task)

- **Circuit → expression:** label each gate's output and work forward from the inputs.
- **Expression → truth table:** list all 2^n input rows, add a column for each middle step, then the output column.
- **Truth table → expression:**
  - **Sum of products (SOP):** for each row where the output is 1, AND the inputs together (put a NOT on any input that is 0). Then OR all these terms.
  - ⚠ **Product of sums (POS):** for each row where the output is 0, OR the inputs (put a NOT on any input that is 1). Then AND all these terms.
- **Simplify** with the rules above. ⚠ Karnaugh maps (2–4 inputs) are not named in the rules, but they are the normal tool and make the hands-on circuit smaller.
- **Word problems:**
  - "The alarm sounds if the door is open AND the system is armed, OR the panic button is pressed" → **Y = D·A + P**
  - Majority of 3 voters → **Y = AB + BC + AC**
  - The practice slides have a "mammal detector" from 5 animal traits.

Hands-on: build a logic circuit

Expect switches or jumper wires as inputs and LEDs (with resistors) as outputs. The logic may come from chips, or from switches and diodes.

- ⚠ **Common logic chips** (7400 series). *Pin numbers are general knowledge, not from an SO source; check the datasheet.*

Chip	Contents
7400	4 × 2-input NAND
7402	4 × 2-input NOR (**different pinout**: its outputs are on pins 1, 4, 10, 13)
7404	6 × NOT
7408	4 × 2-input AND
7432	4 × 2-input OR
7486	4 × 2-input XOR

- 14-pin chips: **pin 14 = +5 V (Vcc), pin 7 = ground**. Count the pins counter-clockwise from the notch or dot.
- The chip sits **across the center gap** of the breadboard.
- Unused CMOS inputs must be tied high or low, not left loose.
- You may ask the supervisor how the breadboard is wired (rule II.b), and you can ask for datasheets.
- ⚠ Pull-up and pull-down resistors make a switch input read a clean 0 or 1.

Combinational circuits (named: multiplexers and adders)

- **Half adder:** inputs A, B. **Sum = A ⊕ B. Carry = A·B.** Built from 1 XOR + 1 AND.

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

- **Full adder:** inputs A, B, C_in.
  - **Sum = A ⊕ B ⊕ C_in**
  - **C_out = AB + C_in(A ⊕ B)** = AB + A·C_in + B·C_in (the "majority" function)
  - Built from 2 half adders + an OR gate on the two carries.

A	B	C_in	Sum	C_out
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

- **Ripple-carry adder:** n full adders in a chain; each C_out feeds the next C_in. The last C_out is the extra top bit (overflow). A 4-bit adder needs 4 full adders (or 3 full adders + 1 half adder).
- **Multiplexer (MUX, "data selector"):** 2^n data inputs, **n select lines**, 1 output.
  - 2-to-1 MUX: **$Y = S' \cdot D_0 + S \cdot D_1$**
  - 4-to-1 needs 2 select lines, 8-to-1 needs 3, 16-to-1 needs 4. **Select lines = $\log_2$(number of inputs).**
  - ⚠ The scioly.org wiki says √N. **That is wrong.** It only happens to work for 4 and 16.
  - Built from AND gates (one per input) feeding one OR gate.
  - ⚠ A MUX can make any function of its select inputs, if you put 0s and 1s on the data inputs.
- **Demultiplexer (DEMUX):** 1 input sent to one of 2^n outputs by n select lines.
- ⚠ **Decoder** (n inputs → 2^n outputs, one output on at a time; for example 2-to-4) and **encoder** (the reverse; priority encoder).
- ⚠ **Comparator:** 1-bit "equal" = XNOR; "A > B" = A·B′. **Parity checker** = a chain of XORs. Seven-segment decoder: rare.
- The **3-way light switch** is an XOR/XNOR (see CL 6 - Switches, Relays, Fuses, Breakers, GFCI T).

Where to practice (no Div B logic tests exist yet)

- Older Circuit Lab **Div C** tests from 2019–2021 had basic digital logic.
- Georgia Tech ECE 2020 topic pages (recommended in the scioly.org wiki and the 2027 forum thread): https://ece2020.ece.gatech.edu/new/topics.html
- All About Circuits textbook, Digital volume: https://www.allaboutcircuits.com/textbook/
- Falstad circuit simulator: https://www.falstad.com/circuit/circuitjs.html
- Tinkercad Circuits (breadboard simulator).
- National supervisor practice slides, deck #15 (Digital Logic): https://www.soinc.org/sites/default/files/uploaded_files/15_CircuitLabSlides.Practice15.v1.pptx
- The 2027 forum thread links a "Timing and Sequential Logic" page. **Skip it:** sequential logic is ❌ OUT.

CL 9 - Meters, Breadboards and Hands-On Tasks

The hands-on part is **25–50% of the score**. The rules say you must know **breadboards and multimeters**. Back to Circuit Lab - Topics [T].

How often: every test has a hands-on part. Written tests also ask about meters (Pembroke had 4 meter questions).

Multimeter

- **Black lead in COM.** Red lead in **VΩ** for volts, ohms, diode test and continuity. Red lead in **mA** or **10 A** for current.
- The current jacks have fuses. If that fuse blows, the meter reads 0 current.
- Pick **DC volts (V⎓)** or **AC volts (V~)**. Pick a range: start high, or use auto-range. **"OL"** means over the range, or an open circuit. Check the units on the display (mV or V).
- **Voltmeter:** goes **across** the part (in parallel), with the power **on**. An ideal voltmeter has infinite resistance; a real digital one has about 10 MΩ.
  - **Loading effect:** a voltmeter with low resistance across a large resistor reads **lower** than the true value (Pembroke).
- **Ammeter:** open the circuit and put the meter **in series**. An ideal ammeter has 0 resistance (Pembroke).
  - **Never put an ammeter straight across a battery.** That is a short and blows the meter's fuse.
- **Ohmmeter:** power **off**, and take the part out of the circuit (or lift one leg). Otherwise you also measure the other paths around it.
  - Holding both probe tips with your fingers adds your body's resistance in parallel.
- **Continuity** (beep) and **diode test**: see CL 7 - PN Junctions, Diodes and LEDs [T].
- **Indirect methods:**
  - R = V / I, using a voltmeter and an ammeter.
  - Current from the voltage across a known resistor: I = V_R / R.

Breadboard

- Each row of **5 holes** (a–e, and f–j) is connected inside.
- The **center gap** separates the two halves. Chips sit across it.
- The **power rails** are the long lines along the edges: red = +, blue = −. On some long boards **the rail is split in the middle**. Check with the continuity beep.
- Common mistakes:
  - Both legs of a part in the same 5-hole row. That shorts the part.
  - Forgetting to bridge a split rail.
  - An LED in backwards.
  - A missing resistor.
  - A loose jumper wire.
- You may ask the supervisor how the breadboard is wired inside (rule II.b).
- SO handout: https://www.soinc.org/sites/default/files/uploaded_files/Circuit_Board_Resistors.pdf

The official example tasks (2027)

1. Mystery resistor using only voltage

- Put the unknown R_x in series with a known R_k across a battery. Measure V_x and V_k.
- The same current flows through both, so **R_x = R_k × V_x / V_k**.
- Or measure the battery voltage V_s and V_k: **R_x = R_k × (V_s − V_k) / V_k**.

2. Power supplied to a circuit

- Measure the source voltage and the current (or the voltage across a known series resistor). Then **P = V × I**.

3. Two LEDs equally bright

- See CL 7 - PN Junctions, Diodes and LEDs [T]. Short version: LEDs in series get equal current; LEDs in parallel each need their own resistor.

4. USB charger label → power levels and charge time

- **Read the label.** "Input: 100–240 V ~ 50/60 Hz 1.5 A" is the wall (AC) side; it works on any country's voltage.
- "Output" lists the DC levels, for example "5 V ⎓ 3 A, 9 V ⎓ 3 A, 15 V ⎓ 3 A, 20 V ⎓ 3.25 A (65 W max)".
- Symbols: ⎓ **= DC**, ~ **= AC**.
- **Power at each level = V × I.** For that example: 15 W, 27 W, 45 W and 65 W.

Common USB levels. These come from the USB standards as general knowledge, not from an SO source; check the exact values.

Standard	Output
USB 2.0 port	5 V, 0.5 A (2.5 W)
USB 3.x port	5 V, 0.9 A (4.5 W)
USB Battery Charging 1.2	5 V, 1.5 A (7.5 W)
Old Apple cube	5 V, 1 A (5 W)
iPad charger	5 V, 2.1–2.4 A (≈10–12 W)
USB-C (no PD)	5 V at 1.5 A (7.5 W) or 3 A (15 W)
USB Power Delivery (PD)	5, 9, 15 or 20 V; up to 3 A (60 W) with a normal cable, 5 A (100 W) with a special e-marked cable. PD 3.1 adds 28, 36 and 48 V, up to 240 W.
⚠ Qualcomm Quick Charge	5, 9 or 12 V

- The charger only **offers** levels. The phone **asks** for the level it wants. A charger never forces extra current into a device.

Charge time:

- **Simple method:** time = capacity (mAh) / charging current (mA). Example: 3000 mAh at 1000 mA = **3 h**.
- **Energy method** (use it when the voltages differ):
  1. Battery energy (Wh) = capacity (Ah) × battery voltage. A lithium-ion phone cell is **3.7 V** (some are 3.85 V).
  2. Time = Wh / (charger W × efficiency).
     - Example: 5000 mAh × 3.7 V = **18.5 Wh**. With an 18 W charger at 85% efficiency: 18.5 / (18 × 0.85) ≈ **1.2 h**.
- **Common mistake:** dividing 5000 mAh by the charger's current at 9 V. The charger's current is at 9 V, but the battery is at 3.7 V. **Convert to Wh first.**
- ⚠ Real charging slows down near full: lithium-ion charges with constant current, then constant voltage (CC/CV). A supervisor may want you to include efficiency (≈80–90%), or to say you assumed it.
- ⚠ A power bank labeled "10000 mAh 3.7 V 37 Wh" gives less than 10000 mAh at 5 V. 37 Wh is under the 100 Wh airline limit.
- No USB question was in the three 2026 tests we read, even though the task is in both seasons' rules.

5. Logic circuit from a truth table

- See CL 8 - Digital Logic [T] (sum of products, then build with gates or switches).

6. Find the fault in a broken circuit

From the national supervisor practice slides:

1. Check the obvious things first: power connected, battery good (measure it), switch on, fuse or breaker OK, a burned-out part, a loose wire.
2. Draw and label the circuit. Write the voltage you expect at each point.
3. With the power on, measure the voltage across each part and compare it with what you expected.
4. With the power off, measure resistances. Remember the other paths in parallel.

How to read the clues:

- **Series circuit with one open part:** everything goes out, and the **full source voltage appears across the open part**. (Slide example: 6 V across bulb L5 means L5 is open.)
- **Parallel circuit:** only the broken branch goes out.
- **Short:** V = 0 and R = 0 across it.
- **Open:** R = "OL", unless there is another path.
- No current means no voltage drops: both ends of a series resistor read the same voltage.
- Common faults: an LED in backwards, the wrong resistor (color code misread), a part in a single breadboard row, a dead battery, a blown meter fuse.

7. 3-way switch with SPDT switches

- See CL 6 - Switches, Relays, Fuses, Breakers, GFCI 🅃.

8. Make a given equivalent resistance

- List the possible series and parallel combinations, then check with the ohmmeter.
- Resistors have tolerance: a ±5% resistor can read a little off.
- BullSO 2026 task: use 100, 200, 50 and 50 Ω to make exactly **100 Ω**, with at least one parallel group and one series part. One answer: **200 || (100 + 50 + 50) = 200 || 200 = 100 Ω**, using all four.
  - BullSO grading: labeled sketch 8 points, step-by-step R_eq math 8 points, explanation 4 points, supervisor sign-off 9 points.
  - BullSO's answer key says 133.3 Ω, which does not match the 100 Ω task. It is probably a key mistake.

❌ Removed for 2027

- "Build an electromagnet from wire, a bolt and a battery" was a 2026 task. It is gone.

CL - Out of Scope for 2027

Don't study these for 2027. They were on last season's (2026) tests, so old practice tests still have them. They make up about **30–40% of the questions** in each 2026 test we read. Back to Circuit Lab - Topics [T].

Topic	Where it showed up in 2026 tests
Magnetic poles and fields, Earth's field	Berkeley, BullSO, Pembroke
Right-hand rule; force on a moving charge; "magnetic force does no work"	BullSO (4 questions), Pembroke (2), Berkeley (6)
Electromagnets and solenoids (turns, current, iron core)	BullSO, Berkeley (2), Pembroke (2), and a 2026 hands-on task
Transformers (Vs/Vp = Ns/Np, power conserved, no DC)	BullSO (2), Berkeley (2), Pembroke (2)
Motors, generators, induction, Lenz's law	Berkeley; national practice slides 7–8
Capacitance (C = Q/V, capacitors in series and parallel, stored energy, plate area and gap, dielectric, RC time constant $\tau = RC$)	BullSO (4), Berkeley (4), Pembroke (1)
Inductance	—
Superconductors in magnetism questions	BullSO. "R = 0 below a critical temperature" may still come up as a resistance fact.
Faraday and Tesla history	BullSO (2), Berkeley (2)
Skin effect, Hall effect	BullSO
Op-amps	BullSO answer key
Thévenin and Norton equivalents, delta-Y	2014 forum
3-phase power, power factor, phasors, oscilloscopes	scioly.org wiki (AC section)
Sequential logic: flip-flops, latches, counters, clocks, timing diagrams	— (new exclusion with the logic topic)
Math: trigonometry, complex numbers, matrices, calculus, more than 2 equations	—

Exception: a **symbol question** could still show a capacitor or a transformer as a wrong answer choice. Know what they look like.

CL - Past Tests

Past Circuit Lab Div B tests: what they look like and where to get them. Back to Circuit Lab - Topics [T].

> **WARNING: All of these are 2026-season tests** No 2027 test exists yet (checked 2026-10-16). Skip the magnetism and capacitor questions; see CL - Out of Scope for 2027 [T].

Tests we read in full (test + answer key)

Test	Date	Format	Link
BullSO Invitational (Univ. of South Florida)	Jan 24, 2026	100 points: multiple choice 41, short answer 30 (symbols, Ohm and power, R_eq, Kirchhoff, nodal and mesh analysis), lab 29	https://drive.google.com/drive/folders/1O1duR0PqVUilIS5VwtgkbmHq8Oo0fqv4 → "Circuit Lab B"
Science Olympiad @ Berkeley	Feb 28, 2026	160 points: multiple choice 55, DC analysis 33, electrostatics 32, hands-on 40	https://drive.google.com/drive/folders/1MaQqY2djJaWI1ZJc_B2hxEJ7Dd-6PcYe → "Circuit Lab"
Pembroke Hill Invitational	Dec 6, 2025	35 short questions (1–2 points each) + 4 hands-on tasks (15 points)	https://drive.google.com/drive/folders/18_JSo5INjEBEKjrAcYtpdyQOgxTrPj3E → "CircuitLab"

- BullSO's hands-on answer key says 133.3 Ω, but the task asks for 100 Ω. The key is probably wrong.

More 2026 tests (need a free scioly.org login)

These are on the scioly.org Test Exchange. We could not download them without an account.

- NC State Invitational: https://scioly.org/tests/eAllhQ3VCpg
- Archimedean: https://scioly.org/tests/Ivpf06itqXM
- Texas A&M: https://scioly.org/tests/EpCc3X9y5k4
- Brown: https://scioly.org/tests/dG2xNvItLsI
- Milpitas: https://scioly.org/tests/tXaMRY9Esk4
- Greenbrier (B/C): https://scioly.org/tests/KGVD6o10HOU
- Pembroke Hill and Berkeley are also there (the same tests as above).

Other practice

- **National supervisor practice slides** (15 decks) and a homework-generator spreadsheet: https://www.soinc.org/circuit-lab-b
- 2027 Division B National Test Packet (paid): Science Olympiad store.
- Forum: Circuit Lab B/C 2027 thread (study links): https://scioly.org/forums/viewtopic.php?t=29830
- Forum: Shock Value B / Circuit Lab C question marathon (2014): https://scioly.org/forums/viewtopic.php?t=5022
- NMT supervisor slides, "Circuits Lab Division B + C" (2020): https://www.nmt.edu/stem/olympiad/Circuits%20Lab%202020.pdf
- Kuphaldt, *Lessons in Electric Circuits* (free textbook): http://www.ibiblio.org/kuphaldt/electricCircuits/
- PhET circuit simulations; OpenStax College Physics 2e.

Not found

- Older Div B tests (Circuit Lab B 2019–2021, Shock Value B 2013–2014) are not on the current Test Exchange.

Question Types

Every question-type note for Circuit Lab B (2027 rules), area by area. Back to Circuit Lab and Circuit Lab - Topics [T]. The official list is in Circuit Lab - Rules 2027 [R].

When your answer to a past-test question differs from the key, check Circuit Lab - Answer Key Errata first.

How to read this

- Each note lists question types from Easy to Nationals. Each type has a worked example.
- The count is the number of question types in the note. Out-of-scope types are counted too, because the notes keep them for recognition.
- History notes number their types differently (H-A01, K1, Q-V01 and so on). The count is the same idea.
- **Total: 1498 question types** in the notes that exist now.

Summary by area

#	Area	Topic note	Notes	Question types
1	History	CL 1 - History [T]	9	281
2	Static Electricity and Coulomb	CL 2 - Static Electricity and Coulomb's Law [T]	5	125
3	DC and AC	CL 3 - DC and AC [T]	5	128
4	Current Voltage Resistance Power Energy	CL 4 - Current, Voltage, Resistance, Power, Energy [T]	5	127
5	DC Circuits	CL 5 - DC Circuits [T]	9	218
6	Switches Relays Fuses Breakers GFCI	CL 6 - Switches, Relays, Fuses, Breakers, GFCI [T]	7	130
7	Diodes and LEDs	CL 7 - PN Junctions, Diodes and LEDs [T]	7	122
8	Digital Logic	CL 8 - Digital Logic [T]	10	197
9	Hands-On and Meters	CL 9 - Meters, Breadboards and Hands-On Tasks [T]	11	170

A1 History

Ampère, Coulomb, Kirchhoff, Volta, Ohm, Hertz, Boole (rule 3.c.i).

Note	Question types
Ampere [A1]	33
Boole [A1]	27
Coulomb [A1]	34
Hertz [A1]	27
Kirchhoff [A1]	31
Ohm [A1]	31
Other Historical Figures [A1]	41
Units Named After People [A1]	23
Volta [A1]	34

A2 Static Electricity and Coulomb

Charges, fields, static hazards, Coulomb's law (rule 3.c.ii).

Note	Question types
Charge Basics [A2]	21
Charging Methods and Electroscopes [A2]	19
Coulombs Law [A2]	23
Electric Field and Potential [A2]	36
Static Hazards [A2]	26

A3 DC and AC

DC and AC traits, sources, uses and hazards (rule 3.c.iii).

Note	Question types
AC Characteristics A3	25
Batteries and Cells A3	28
Household Wiring A3	25
Other DC Sources A3	24
Shock and Safety A3	26

A4 Current Voltage Resistance Power Energy

Concepts, sums and units (rule 3.c.iv).

Note	Question types
Current and Charge A4	18
Power Energy and Cost A4	35
Resistance and Resistivity A4	35
Units Prefixes and Sig Figs A4	22
Voltage and EMF A4	17

A5 DC Circuits

Series, parallel, open, short, Ohm's law, schematics; KVL/KCL at States/Nationals (rules 3.c.v and 3.c.ix).

Note	Question types
Dividers and Bulb Brightness A5	37
Hard Resistor Networks A5	19
Internal Resistance and Power Transfer A5	20
Kirchhoff Analysis (States-Nationals) A5	24
Meters and Measurement A5	25
Open and Short Circuits A5	23
Resistor Color Codes A5	26
Schematic Symbols A5	23
Series and Parallel Resistance A5	21

A6 Switches Relays Fuses Breakers GFCI

Control devices (rule 3.c.vi).

Note	Question types
3-Way and 4-Way Switching A6	18
Circuit Breakers A6	19
Fuses A6	20
GFCI A6	18
Relays A6	16
Surge Protection and Grounding A6	18
Switch Types A6	21

A7 Diodes and LEDs

Silicon PN junctions, diodes, LEDs (rule 3.c.vii).

Note	Question types
Diode Applications A7	19

Note	Question types
Diode Circuit Analysis A7	14
LEDs A7	36
PN Junction and I-V Curve A7	15
Semiconductors and Doping A7	12
Special Diodes A7	13
Testing Diodes A7	13

A8 Digital Logic

Gates, truth tables, Boolean algebra, De Morgan, MUX, adders (rule 3.c.viii).

Note	Question types
Adders A8	19
Binary Numbers and Addition A8	17
Boolean Algebra and Simplification A8	21
De Morgan and Bubble Pushing A8	15
Logic Design Word Problems A8	24
Logic Gates and Notation A8	17
Logic Hardware and Chips A8	25
Multiplexers and Decoders A8	18
Truth Tables and Boolean Expressions A8	23
Universal Gates (NAND and NOR) A8	18

A9 Hands-On and Meters

Breadboards, multimeters and the 8 listed hands-on tasks (Part II).

Note	Question types
Breadboards A9	11
Lab Skills A9	15
Multimeter Use A9	22
Task - 3-Way Switch A9	17
Task - Equal LEDs A9	15
Task - Logic from Truth Table A9	16
Task - Mystery Resistor A9	12
Task - Power Supplied A9	14
Task - Target Resistance A9	15
Task - Troubleshoot A9	17
Task - USB Charger A9	16

Still missing or unconfirmed

- **Real 2026 Nationals tests:** not found. The "nat26" file is the USC Invitational (Jan 2026), not Nationals. The Nationals tests are sold in the soinc Test Packet.
- **Continued fraction by self-substitution** (USC 2026 Div C Q38a): no note has it. It is pure algebra and only in the Div C test, so it is ⚠ borderline. The infinite-ladder types use the same method.
- **Boole quote attribution** (Boole A1 T27) is new. No test has a Boole quote yet. The quote itself was checked in the *Laws of Thought* text.
- **Coulomb quote** (Florida State 2021 Q6): the attribution is not confirmed. See Coulomb A1.
- **Florida State 2021 long free-response keys** are not in the folder. Q43-45 could not be checked. Most of those items are out of scope.

- **Numbers not from Science Olympiad sources:** GFCI trip level, body-current table, USB power levels, LED voltages by color and logic-chip pins. Check them in a textbook.
- **Calculator notes:** no rule says if notes stored on the calculator are allowed. See Circuit Lab - Calculator Rules [R].

How this index was checked (2026-10-16)

- Every item in rules 3.c (i-ix), 3.e (math limits) and Part II (8 hands-on tasks) was searched in the notes. Each one has question types.
- Every in-scope question type from the USC 2026 and State tests (Florida 2021, MYSO practice) was searched. All appear in a note, except the continued fraction above.
- 47 numbers from worked examples in all 9 areas were re-computed in Python. None failed.

TAB R

Rules and Calculator Rules (appendix)

Circuit Lab - Rules 2027 707

Circuit Lab - Calculator Rules 710

The 2027 rules summary and the calculator rules. Check the official rules if anything here disagrees.

Circuit Lab - Rules 2027

Official rules for **Circuit Lab, Division B**, this season. Back to Circuit Lab.

> **INFO: Which season is this?** Science Olympiad names a season after the year of its national tournament.
> - **This season (2026-27) is called "2027".** Its rules came out on September 8, 2026.
> - Last season (2025-26) was "2026". Its rules were different (see "What changed" below).
> - Circuit Lab B is an event in both seasons.

Quick facts

Item	Rule
Team	Up to 2 people
Time	About 50 minutes
Build / impound	Nothing to build. The supervisor gives you all parts and meters for the hands-on tasks.
Notes	**Unlimited paper notes**, any size, from any source. Binders, books, notebooks, folders, sheet protectors, lamination, tabs and labels are all OK.
Calculators	**Two Class III calculators** per team (a TI-Nspire is OK). See Circuit Lab - Calculator Rules [R].
Multimeter	You may bring your own "basic" multimeter, if the supervisor says yes.
Scoring	Written test = 50–75% of the score. Hands-on tasks = 25–50%. High score wins.
Ties	Broken by questions picked before the test.
Units	Answers in metric units with correct significant figures, unless the question says otherwise.

What the written test covers

The test has **at least one question from each area below, and only these areas.**

1. **History:** discoveries by Ampère, Coulomb, Kirchhoff, Volta, Ohm, **Hertz** and **Boole**.
2. **Static electricity:** properties of electric charges and fields, sources and hazards of static electricity, Coulomb's Law.
3. **DC and AC:** characteristics, sources, uses and hazards of direct current and alternating current.
4. **Basics with calculations:** current, voltage, resistance, power and energy, and their units.
5. **DC circuits:** calculations, building and arranging circuits and parts. This includes series and parallel circuits, open and short circuits, Ohm's Law and circuit schematics (diagrams).
6. **Control devices:** switches, relays, fuses, ground fault circuit interrupters (GFCIs) and breakers.
7. **Diodes:** basic electrical behavior of silicon PN junctions, and uses of single-junction parts (diodes, LEDs).
8. **Digital logic:** logic gates, truth tables, universal gate sets, Boolean expressions, De Morgan's Law, and combinational logic circuits (for example multiplexers and adders).
9. **Kirchhoff's Voltage and Current Laws.** These are for **States and Nationals only.** A regional test has no Kirchhoff circuit analysis.

Not on the test

- semiconductors other than those listed above
- AC circuit theory and frequency analysis
- **capacitance** and inductance
- non-linear parts other than those listed above
- 3-phase power
- Norton and Thévenin equivalents
- oscilloscopes
- sequential logic (for example flip-flops) and clock signals

Math limits

- Functions, algebra, and systems of **at most two** equations.
- 2D geometry for vector directions: points, parallel and perpendicular lines, angles in degrees.
- Whole numbers in **binary**, and **binary addition** only. No negative numbers, no decimals in binary, and no other operations.
- Logic written as Boolean expressions, including De Morgan's Law and truth tables.
- **No** trigonometry, complex numbers, matrices or calculus.

Hands-on tasks

You must know how to use a **breadboard** and a **multimeter**. You may ask the supervisor how the breadboard is wired inside.

The rules list these example tasks. The real tasks "may include but are not limited to" these:

1. Find the value of a mystery resistor using **only voltage** measurements.
2. Calculate the power supplied to a circuit.
3. With wires, batteries, resistors and 2 LEDs, wire them so the **LEDs are equally bright**.
4. Read a **USB charger label**: list its output power levels, and work out how long it takes to charge a given battery.
5. Build a logic circuit that matches a given truth table.
6. Find the fault in a broken circuit.
7. Wire a **3-way switch** using SPDT switches.
8. Make a given **equivalent resistance** from a set of resistors.

Safety rule: no soldering, no high-power circuits and no AC circuits in the hands-on part.

What changed from last season (2026 → 2027)

1. **All magnetism is gone.** No magnetic fields, electromagnets, transformers, motors or generators, no right-hand rule, and no electromagnet-building task. Tesla and Faraday are out of the history list.
2. **Digital logic is new.** Gates, truth tables, Boolean algebra, De Morgan, multiplexers, adders and binary addition.
3. **Capacitance is now excluded.** Last season it was tested.
4. **Diodes and LEDs are at every level now.** Last season LEDs were States and Nationals only.
5. **Hertz and Boole** were added to the history list.
6. New math limits (no trig and no calculus), new hands-on examples, and the new safety rule.

> **WARNING: Old practice tests** Most past Circuit Lab tests are from older seasons. They often have magnetism and capacitor questions. Skip those for 2027. See Circuit Lab - Topics Ⓣ for what is in and out.

Corrections and clarifications

- soinc.org has posted **no rule corrections** for Circuit Lab this season (checked 2026-10-16).
- The Circuit Lab B clarification page has **no answered questions yet** (checked 2026-10-16). You can ask your own question there: https://www.soinc.org/circuit-lab-div-b

Questions the rules leave open

- What counts as a "basic" multimeter?
- The test covers AC "characteristics, sources, uses and hazards", but AC circuit theory is excluded. Where is the line between them?
- How deep can the diode questions go?
- Each tournament picks its own written/hands-on split inside the 50–75% / 25–50% range.

Sources

- 2027 Division B Rules Manual, pages B11–B12. The official soinc.org file is saved at `~/Daniel/Science_Olympiad_Div_B_Rules_2027_for_Web_Secured.pdf`. It matches the text above word for word (checked 2026-10-16).
- 2026 Division B Rules Manual, pages B10–B11: https://www.soinc.org/sites/default/files/uploaded_files/Science_Olympiad_Div_B_Rules_2026_for_Web_Secured.pdf
- Official event page with practice slides and links: https://www.soinc.org/circuit-lab-b
- Rule corrections: https://www.soinc.org/events/rules-corrections
- Not checked: the scioly.org wiki and forum. Their bot protection blocked every attempt to read them.

Circuit Lab - Calculator Rules

Can you use your **TI-Nspire CX II CAS** in Circuit Lab, and can you keep notes on it? Back to Circuit Lab.

Short answer

- **The calculator: yes.** Circuit Lab B allows two Class III calculators per team. "NSPIRE" is a named Class III example.
- **CAS: yes.** The rules never mention "CAS". But the Class III examples include the TI-89 and Voyage, which are CAS calculators, so a CAS Nspire is fine.
- **Notes stored on the calculator: not decided by any rule.**
  - No rule allows it, and no rule bans it.
  - No rule says you must clear the calculator's memory.
  - **The risk:** the notes rule says notes must be "written/printed on paper". A supervisor could say the calculator notes break that rule. The supervisor's decision is final.

> **TIP: What to do**
>
> 1. **Always bring the full paper binder too.** Paper notes are unlimited and always allowed.
> 2. Before each tournament, ask the tournament director whether a stored reference on the calculator is OK in Circuit Lab.
> 3. To get an official answer for the whole season, submit a question on the Circuit Lab B clarification page (cite rules 2.b and 2.c): https://www.soinc.org/circuit-lab-div-b
> 4. If a supervisor makes you turn on **Press-to-Test**, everything stored on the calculator is hidden during that test. See TI-Nspire CX II CAS.

The calculator classes (official wording)

The class in the event rules is the **highest** class allowed. Lower classes are fine too.

Class	What it is	Examples
I	Basic 4- or 5-function calculator	—
II	Scientific, not programmable, not graphing	Casio FX-260, Sharp EL-501, TI-30X
III	**Programmable and/or graphing**	Casio 9750/9850/9860, HP 40/50/Prime, **TI-83/84/89/NSPIRE/Voyage**
IV	Calculator apps on a phone, laptop, tablet or watch	**Not allowed** in Circuit Lab

The page also says: "The decisions of the event supervisors will be final."

The rules that matter (exact words)

- Circuit Lab B, rule 2.b: "Each team may bring two Class III calculators for use during any part of the event."
- Circuit Lab B, rule 2.c: "Each team may bring a collection of notes and resources, **written/printed on paper**, of any size containing information in any form and from any source."
- General Rule 1: "Actions and items (e.g., tools, notes, resources, supplies, electronics, etc.) are permitted, unless they are explicitly excluded in the rules, are unsafe, or violate the spirit of the problem."
- General Rule 2: devices that can communicate with other devices, and calculator apps on phones, laptops or tablets, are not allowed unless the event rule or the supervisor allows them.
- The rules' introduction also says teams should follow "the spirit of the problem" and must not read the rules in a way that gives them an unfair advantage.

Where we looked

- The 2026 and 2027 Division B rule manuals. We searched the full text for "program", "memory", "clear", "stored", "CAS" and "press-to-test" and found nothing about calculators.
- soinc.org Circuit Lab B clarifications (none posted yet), the general Q&A page, and the 2027 rule corrections (nothing about calculators).
- The tournament pages for Princeton and Michigan (GVSU). Neither says anything about clearing memory.
- A 2025 Michigan regional note says notes "cannot be in a digital format". That was one regional only.
- **Not read:** the soinc.org FAQ "What class of calculator is allowed?" (it needs a login), and the scioly.org forums (blocked).

Sources

- Calculator Class Descriptions: https://www.soinc.org/sites/default/files/uploaded_files/Calculator_Class_Descriptions.pdf (also in the 2027 Div B manual, page B70)
- 2027 Div B rules: see Circuit Lab - Rules 2027 R
- General rules: https://www.soinc.org/ethics_rules

TAB G

TI-Nspire App User Guide (appendix)

TI-Nspire - App User Guide 715

How to use the CircuitRef / Circuit Lab app on the TI-Nspire.

TI-Nspire - App User Guide

How to **use** the Circuit Lab app on your TI-Nspire CX II CAS. Every example below was typed into the app on the Mac. The screen lines are copied exactly. Back to Circuit Lab. Also see TI-Nspire - What It Can Answer and TI-Nspire - How to Build the App.

> **WARNING: Two rules for every answer**
>
> 1. **Always write the work on paper.** Write the formula, the numbers and the units. The app only checks your work.
> 2. **Check that the answer makes sense.** Is it the right size? Is the unit right? Is the sign right?

Cheat card (one key opens a tool)

Key	Tool	Key	Tool
`o` / `1`	Ohm's law + power	`j`	Kirchhoff, 2 equations
`s` / `2`	Series / parallel	`y`	3 or more equations
`d` / `3`	Voltage divider	`w`	AC peak / rms / period
`k`	Current divider	`m`	Mystery resistor
`l` / `4`	LED / diode / zener resistor	`p`	Battery pack (cells)
`c` / `5`	Color code	`z`	USB charge time
`q` / `6`	Coulomb's law	`n`	Nearest E6/E12/E24 value
`b` / `7`	Battery capacity	`h`	Hit a target R
`e` / `8`	Power, energy + cost	`t`	Truth table + simplify
`x` / `9`	Binary	`g`	Gate table + De Morgan
`f`	Field + potential	`v`	Key test
`a`	Charge Q = It + electrons	`0`	**Search the facts**
`r`	Resistivity R = ρL/A	**Shift+N**	**Network solver**
`i`	Internal resistance	Shift+W	Wheatstone bridge
`u`	Fuse sizing	Shift+L	Infinite ladder
Shift+P	Plates, electron speed	Shift+R	R(T) with temperature
Shift+J	Energy V²t/R	Shift+F	Loads per fuse
Shift+B	Rated bulbs	Shift+T	Tolerance window
Shift+Q	Point charges on a line	`esc`	Back to the home list

Keys that work in every formula tool:

- A **letter** picks a box. **Digits** type into it.
- `k m u n` after a digit is a prefix: kilo, milli, micro, nano. **Shift+M** is mega.
- `m` after a digit means **milli**. For minutes type `min`: `5min` in a time box = 300 s.
- `(–)` makes a number negative. A `–` after a value subtracts (`24–16` = 8). A `–` at the end makes the box negative.
- The empty boxes are solved after every key.
- **Enter** copies the picked value. It goes to the clipboard and to the variable `cl`.
- **Tab** or the arrows move between boxes. **Ctrl+Del** clears all. **Esc** goes home.

Fast typing (added 2026-10-17)

Shortcuts that save key presses. The printed key card in the binder has all of them on one page. Practice with TI-Nspire - Typing Drills (real past-test questions with their keys).

You want	Type	Example
Use an answer in the next tool	**Enter** copies it. Then `=` in a box (or **Ctrl+V** in any tool) pastes it. Enter on a box you typed copies the first answer; the message says which ("Copied i = 8.333 A enter: p"). Press Enter again for the next answer.	`s 100/100` Enter, Esc, `o i2 r=` → V = 100 V
Do math inside a box	`+ − × ÷ ^ ( )`	`o v4×1.5 r3` → I = 2 A (four 1.5 V cells); `o v24−16 r4` → 2 A. The box shows the value: `24−16 V = 8 V`
A negative number	`(−)` first, or `−` at the **end**	`q a2u−` = −2 µC
Powers of ten	`1.6×10^−19` works in every box. `1.6x10^−19` also works, except in tools with an **x** box (`k`, `m`, `i`, `j`, Shift+W): there type `×`. The **EE** key always works. `e` after a digit also works, except in tools that have an **e** box (`a`, `f`, `i`, `e`, `j`, `z`)	`t2.48 EE −16`
R from V and P in an ohm box	`^` or `×` makes the box plain math	`r120^2/1000` → 14.4 Ω
Centimeters	`c` after a digit, in a **distance** box	`q a2u b3u r5c` → F = 21.57 N
Minutes or hours	`5min` in a time box, or multiply: `t2×60`. `3h` in a **seconds** box = 3 hours (it moves to the hours box `h`, and a note says so). Only numbers under 60 do this: `t7200` then `h` just opens box `h` (= 2 h)	`J p100 t5min v10` → 30 kJ; `a i2 t3h` → Q = 21.6 kC
Units after the number	allowed, the app reads them: `12V`, `12volts`, `4ohm`, `4ohms`, `2000mAh`, `2.5Ah` (= 2500 mAh), `1.2kWh`, `2hrs`; `4R7` = 4.7 Ω	`b c2000mAh i.5` → 4 h
Square millimeters	type the unit: `2mm^2` (or `2u`, because 1 mm² = 10^{-6} m²). `3cm^2` and `2m^2` work too	`r p1.68e−8 l10 a2mm^2` → R = 84 mΩ (10 m of 2 mm² copper)
Battery pack notation	`4S2P` (or `4s2p`) right after opening `p`	`p 4S2P v3.7 c2500` → 14.8 V, 5000 mAh, 74 Wh
`/` in an ohm box	`/` = **parallel**. `100/2` is 100 ‖ 2 = 1.96 Ω, and the app shows a warning. To divide, add `∗`: `100∗1/2` = 50	`o v10 r100∗1/2` → 200 mA
A letter that is not a box	the app warns "No box c here". After a number the letter stays in the box, so the box shows "Can not read": delete it with del	`o v12c` → fix with del
Mega vs milli	capital `M` = mega (10^6), small `m` = milli. The app warns when `M` is in a box that is not ohms	`o i5M` → warning
Parallel pair in an ohm box	`÷` between the values	`d v12 a3.3k b4.7k÷6.8k`
A space before a prefix	joined: `4.7 k` = 4.7k, `20 m` = 20m	`o v12 r4.7 k` → I = 2.553 mA
Paste, then a digit	the digit replaces the pasted value	after a copy, `o v=12` → v = 12 V (not 3 then 12)

Other tools: bulbs `B60 120, 100 120 =120s` (only series) or `=120p` (only parallel). Charges: `e` = 1.602×10^{-19} C, so three protons at 0, 4 and −6 m are `Q e 0, 4, −6`. Charges in a plane: give `q x y` for each, e.g. a triangle `Q 1u 0 0, 2u .1 0, −3u .05 .0866` → total U = −0.629 J; add `=x y` for E and V at a point. Binary: `x 0101^0011` (XOR), `x 1100×1010` (AND), `x 5 or 3`. Network: `2−0×3 6` = three 6 Ω in parallel.

Truth tables: an output column can have **x** for a don't care: `t x1x10x0100010001` → F = A'D + CD. Series/parallel: with `×` or `^` the line is plain math (`s 12×2^2` = 48).

Search codes. In search (key `0`), 2–5 letters are enough. Type the start of a title word: `kir`, `rm`, `gfc`. Two words work too: `co la` = Coulomb's law. Short forms work: `kvl`, `kcl`, `led`, `emf`, `rms`, `gfci`, `sop`, `pos`, `kmap`. The binder

has a list with the code for every fact.

Put it on the calculator

1. Plug the calculator into the Mac with the USB cable.
2. Open **TI-Nspire CX II Connect** in **Chrome**: https://education.ti.com/en/products/computer-software/ti-nspire-cx-ii-connect
3. Connect to the calculator in the web app.
4. Send the file `dist/CircuitLab.tns` to the calculator. It is in `~/Daniel/circuitlab/app/dist/`.
5. Send `dist/CircuitRef.tns` too. It is a second file: search only, with all 736 facts.
6. On the calculator, open **My Documents** and open **CircuitLab**.

First session on the real calculator

The Mac tests use a fake calculator. These things can only be checked on the real one. Do this list once, before the first tournament.

- ☐ **Does the file open?** Open CircuitLab. You should see the home list "Circuit Lab".
- ☐ **Key test `v`.** Press `v`. The top line shows the api level and the window size. On the Mac it says `api 2.7 318x212`.
- ☐ **Shift + letter.** In the key test, press Shift then `n`. It must send a capital `N`. All the Shift tools need this.
- ☐ **Special keys.** In the key test, press `×`, `÷`, `(−)`, `EE`, `x²`, `=` and `,`. Write down what each one sends.
- ☐ **EE key.** Open `r` and type `p1.68`, then EE, then `−8`. The box must show `1.68E−8`.
- ☐ **Fonts and fit.** Open every tool. Check that no text is cut off. The home list may have one or two pages.
- ☐ **Speed.** Press **Shift+V** (speed test). It times the cube, the two "find x" cases, the Ohm tool and a search on this calculator. Write the numbers down. Under 300 ms feels instant.
- ☐ **Copy.** Press Enter on an answer. Check that `cl` works on a Calculator page.
- ☐ **Paste.** Press Enter on an answer, Esc, open `o`, press `r` then `=`. The box must show the number. Then try Ctrl+V in the network solver.
- ☐ **CircuitLab search right after start.** Open CircuitLab, press `0` at once and type `kirch`. The list must appear (the app builds its search index in the first second). Note any pause.
- ☐ **CircuitRef start.** Open CircuitRef. It shows "Loading facts …/736" while it builds the index. Time it. On the Mac it takes 0.04 s.
- ☐ **Rules.** Ask the tournament director if the app is allowed. Read Circuit Lab - Calculator Rules Ⓡ first. Bring the paper binder anyway: `~/Daniel/circuitlab/binder/CircuitLab_QuickRef.pdf` (print first) and `CircuitLab_Binder.pdf`.

1. History

Use **search** (key `0`). See the Search section.

2. Static electricity

`q` Coulomb's law

Solves: any 3 of F, q_1, q_2, r → the 4th. It also says repel or attract. **Example:** 2 µC and 3 µC, 0.1 m apart. Type `q a2u b3u r.1`:

```
f  Force (+ repel)  = 5.393 N
a  Charge q1  2u C
b  Charge q2  3u C
r  Distance  .1_ m
Same signs: REPEL
```

Answers: Coulombs Law A2.

`f` Field + potential

Solves: E = kq/r², V = kq/r, the force on a test charge and U. **Example:** a 2 µC charge, 0.3 m away. Type `f q2u r.3`:

```
e  Field E  = 199.7 kN/C
q  Charge q  2u C
r  Distance  .3_ m
v  Potential V  = 59.92 kV
```

Answers: Electric Field and Potential A2.

Shift+Q Point charges on a line

Solves: the force on each charge, E and V at a point, and where E = 0. **How to type:** `charge position` for each charge, split by commas. `=x` is the point to check. **Example:** +2 µC at 0 m and −3 µC at 0.1 m. Check x = 0.05 m. Type `Q 2u 0, −3u .1 =.05`:

```
E at x=0.05: 17.98 MN/C right
V at x=0.05: −179.8 kV  (F on q = qE)
F on q1 (2 uC at 0): 5.393 N right
F on q2 (−3 uC at 0.1): −5.393 N left
E = 0 at x = −0.44495 m
```

Answers: Coulombs Law A2, Electric Field and Potential A2. For charges in a plane, type `q x y` for each (see Fast typing). Positions take units: `10cm`, `100mm`. Charges take units: `2uC`.

Shift+P Plates and electron speed

Solves: E = V/d, F = qE, a = F/m, energy qV, and the speed from rest. q and the mass are the electron's unless you type others. **Example:** an electron goes through 100 V. Type `P v100`:

```
j  Energy gained qV  = 1.602E−17 J
Energy = 100 eV, speed 5.931E6 m/s
```

The speed box `s` is below the screen. The last line shows the speed. **Answers:** Electric Field and Potential A2.

`a` Charge and electrons

Solves: Q = It, and the number of electrons. **Example:** 2 A for 60 s. Type `a i2 t60`:

```
q  Charge  = 120 C
e  Electrons  = 7.4898E20 electrons
```

Answers: Charge Basics A2, Current and Charge A4.

3. DC and AC

`w` AC peak / rms

Solves: peak, rms, peak-to-peak, frequency and period. **Example:** US mains is 120 V rms. Type `w r120`:

```
p  Peak (amplitude)  = 169.7 V
r  RMS  120_ V
a  Peak−to−peak  = 339.4 V
```

Answers: AC Characteristics A3, Household Wiring A3.

`i` Internal resistance

Solves: EMF, terminal voltage, current → internal r, load R, load power and max power. **Example:** EMF 12 V, 11 V at the terminals, 2 A. Type `i e12 v11 i2`:

```
r  Internal r  = 0.5 ohm
l  Load R  = 5.5 ohm
p  Load power  = 22 W
x  Max power (load = r)  = 72 W
```

Answers: Internal Resistance and Power Transfer A5, Batteries and Cells A3.

`b` Battery capacity

Solves: mAh, Wh, current, time and charge. **Example:** 2000 mAh at 0.5 A. Type `b c2000 i.5`:

```
t  Time (hours)  = 4 h
q  Charge  = 7.2 kC
```

Answers: Batteries and Cells A3.

`p` Battery pack

Solves: cells in series and parallel → pack voltage, capacity and energy. **Example:** 4 cells in series, 2 strings, 1.5 V and 2000 mAh each. Type `p s4 p2 v1.5 c2000`:

```
a  Pack voltage  = 6 V
b  Pack capacity  = 4000 mAh
w  Pack energy  = 24 Wh
```

Answers: Batteries and Cells A3.

4. Current, voltage, resistance, power, energy

`o` Ohm's law + power

Solves: any 2 of V, I, R, P → the other 2. **Example:** 12 V across 4 Ω. Type `o v12 r4`:

```
v  Voltage  12 V
i  Current  = 3 A
r  Resistance  4_ ohm
p  Power  = 36 W
```

Answers: Current and Charge A4, Voltage and EMF A4, Task - Power Supplied A9.

`e` Power, energy + cost

Solves: power, hours per day, days, price → kWh, joules and cost. **Example:** a 60 W bulb, 5 h a day, 30 days, $0.15 per kWh. Type `e p60 h5 d30 r.15`:

```
e  Energy  = 9 kWh
j  Energy (joules)  = 32.4 MJ
c  Cost  = $1.35
```

Answers: Power Energy and Cost A4.

`r` Resistivity

Solves: R = ρL/A. It takes the wire diameter too. **Example:** copper (ρ = 1.68E-8), 10 m long, 0.5 mm across. Type `r p1.68E-8 l10 d.5m`:

```
r  Resistance  = 0.85562 ohm
a  Area  = 1.9635E-7 m^2
d  Diameter (round wire)  .5m_ m
```

Answers: Resistance and Resistivity A4.

Shift+R R(T) with temperature

Solves: $R = R_0(1 + \alpha\,\Delta T)$. T_0 is 20 °C unless you type it. **Example:** a tungsten bulb, 24 Ω cold, at 2520 °C. Type `R o24 a.0045 t2520`:

```
r  R at temperature T  = 294 ohm
d  dT = T − T0  = 2500 C
R/R0 = 12.25
```

Answers: Resistance and Resistivity A4.

Shift+J Energy from V, R and time

Solves: E = V²t/R = I²Rt = Pt. Time is in seconds. **Example:** 1000 V across 500 Ω for 10 ms. Type `J v1000 r500 t10m`:

```
i  Current  = 2 A
p  Power  = 2 kW
j  Energy (joules)  = 20 J
```

Here `10m` means 10 **milli**seconds, not minutes. **Answers:** Power Energy and Cost A4.

5. DC circuits

`s` Series / parallel

Solves: R_eq from a list. `+` = series. `/` = parallel. Brackets work. **Example:** 100 Ω in series with (220 Ω || 330 Ω). Type `s 100+220/330`:

```
Req = 232 ohm
100 + [220 || 330]
3 resistor(s)
G = 1/R = 4.31 mS
```

`/` comes before `+`, like × before + in math. **Answers:** Series and Parallel Resistance A5.

`d` Voltage divider

Solves: Vin, R1, R2 → Vout and the current. **Example:** 9 V, R1 = 1 kΩ on top, R2 = 2 kΩ below. Type `d v9 a1k b2k`:

```
o  Vout (across R2)  = 6 V
i  Current  = 3 mA
```

An ohm box also takes series/parallel, for example `b4.7k/6.8k`. **Answers:** Dividers and Bulb Brightness A5.

`k` Current divider

Solves: total current and 2 resistors → the current in each. **Example:** 30 mA into 100 Ω || 200 Ω. Type `k t30m a100 b200`:

```
x  I1 (in R1)  = 20 mA
z  I2 (in R2)  = 10 mA
v  Voltage across both  = 2 V
```

Answers: Dividers and Bulb Brightness A5.

`j` Kirchhoff, 2 equations

Solves: ax + by = c and dx + ey = f. Write the loop equations on paper first. **Example:** 2x + 3y = 5 and x − y = 1. Type `j a2 b3 c5 d1 e(-)1 f1`:

```
x  x (I1 or V1)  = 1.6
y  y (I2 or V2)  = 0.6
det = ae − bd = −5
```

Answers: Kirchhoff Analysis (States-Nationals) A5.

`y` 3 or more equations

Solves: n equations at once (3 loops). Type each row as numbers with spaces. The last number is the right side. Split rows with commas. **Example:** $3x_1 - x_2 = 5$, $-x_1 + 4x_2 - 2x_3 = 0$, $-2x_2 + 5x_3 = 2$. Type `y 3 -1 0 5, -1 4 -2 0,`

`0 -2 5 2`:

```
x1 = 1.95349
x2 = 0.860465
x3 = 0.744186
det = 43
Negative current = flows the other way
```

Answers: Kirchhoff Analysis (States-Nationals) A5.

Shift+N Network solver

The biggest tool. It has its own section below: Network solver.

Shift+W Wheatstone bridge

Solves: a balanced bridge, a/b = c/x. **Example:** a = 100 Ω, b = 200 Ω, c = 500 Ω. Type `W a100 b200 c500`:

```
x  Rx (bottom right)  = 1 kohm
```

Answers: Hard Resistor Networks A5, Meters and Measurement A5. For a bridge that is **not** balanced, use Shift+N.

Shift+L Infinite ladder

Solves: an endless ladder with series R `a` and shunt R `b`. `r` = R_eq when `a` comes first. `s` = R_eq when `b` comes first. **Example 1:** all 1 Ω. Type `L a1 b1`:

```
r  Req (a first)  = 1.618 ohm
s  Req (b first)  = 0.61803 ohm
Equal R: a first 1.618 R, b first 0.618 R
```

Example 2: it works backwards too. R_eq (b first) = 4 Ω and b = 12 Ω. Type `L s4 b12`:

```
a  Series R (each step)  = 2 ohm
r  Req (a first)  = 6 ohm
```

Answers: Hard Resistor Networks A5.

`c` Color code

Solves: bands → value, or value → bands. Each color has one letter (below). **Example 1:** yellow violet red gold. Type `c yvrd`:

```
4.7 kohm  +/-5%
yellow violet red gold
range 4.465k to 4.935k ohm
```

Example 2: which bands make 4.7 kΩ? Type `c 4.7k`:

```
4-band: yellow violet red gold
   keys yvrd  (5%)
5-band: yellow violet black brown brown
   keys yvknn  (1%)
Standard value? E12 yes, E24 yes
```

Color letters: k black, n brown, r red, o orange, y yellow, g green, b blue, v violet, a gray, w white, d gold, s silver, x none. **SMD codes (small chip resistors):** type the digits. `c 472` also shows `If SMD code: 472 = 4.7 kohm` (47 × 10^2). `c 1002` = 10 kΩ. `c 4R7` = 4.7 Ω (R is the decimal point; Shift+R). **Answers:** Resistor Color Codes A5.

`n` Nearest standard value

Solves: the nearest E6, E12 and E24 values. **Example:** you need 350 Ω. Type `n 350`:

```
E12 nearest 330  (-5.71%)
     below 330   above 390
E24 nearest 360  (+2.86%)
```

Answers: Resistor Color Codes A5, LEDs A7.

Shift+B Rated bulbs

Solves: bulbs rated in W and V. It finds R = V²/P, then puts them in series and in parallel. **How to type:** `watts volts` for each bulb, split by commas, then `=supply`. **Example:** a 60 W and a 100 W bulb, both rated 120 V, on 120 V. Type `B 60 120, 100 120 =120`:

```
SERIES on 120 V: Rtot 384 ohm, I = 312.5 mA
  bulb 1: 75 V, P = 23.44 W
  bulb 2: 45 V, P = 14.06 W
  total 37.5 W; brightest: bulb 1 (biggest R)
PARALLEL on 120 V: Rtot 90 ohm, I = 1.333 A
```

Answers: Dividers and Bulb Brightness [A5].

Shift+T Tolerance window

Solves: the range of a part with any %, and whether a measured value is in it. **Example:** 1520 Ω ± 8%, you measure 1420 Ω. Type `T 1520 8 1420`:

```
1398.4 to 1641.6
1420: IN range (-6.58%)
```

Answers: Resistor Color Codes [A5], Multimeter Use [A9].

6. Switches, relays, fuses, breakers, GFCI

`u` Fuse sizing

Solves: power and voltage → current and R. Then: does the fuse blow? **Example:** a 1500 W heater on 120 V with a 15 A fuse. Type `u p1500 v120 f15`:

```
i  Current  = 12.5 A
r  Total resistance  = 9.6 ohm
Smallest std fuse above I: 15 A
Continuous (x1.25 = 15.62 A): 20 A
I <= rating: fuse holds
```

The "Continuous" line is the NEC 80 % rule. Use it when the load runs for 3 hours or more. **Answers:** Fuses [A6], Circuit Breakers [A6].

Shift+F Loads per fuse

Solves: how many equal loads you can add before the fuse blows. **Example:** 15 A fuse, 120 V, 100 W bulbs. Type `F f15 v120 p100`:

```
a  Current of each load  = 833.3 mA
c  Loads (f / load I)  = 18
Max loads: 18 (exactly at the rating; 19 blows it)
Continuous (80%): 14 loads
```

The 80 % line is for loads that run for 3 hours or more. **Answers:** Fuses [A6].

For switches, relays and GFCI facts, use **search**.

7. Diodes and LEDs

`l` LED / diode / zener resistor

Solves: supply, LED Vf, number of LEDs, current → the resistor and its power. It also gives the next standard value up. **Example:** 9 V supply, 2 V LED, 20 mA. Type `l s9 f2 i20m`:

```
r  Resistor  = 350 ohm
p  Resistor power  = 140 mW
w  LED power (all)  = 40 mW
Next std up: E12 390, E24 360  -> I=17.95 mA
Resistor rating: 1/2 W or more (2x margin)
```

For a zener, type the zener voltage as Vf. For a silicon diode, type 0.7. **Answers:** LEDs A7, Diode Circuit Analysis A7, Task - Equal LEDs A9.

8. Digital logic

`t` Truth table + simplify

Solves: an expression → its truth table and simplest form. `-` or `~` = NOT. `ab` = A AND B. `+` = OR. **Example 1:** AB + C'. Type `t ab+-c`:

```
F = AB + C'
m(0,2,4,6,7)
M(1,3,5)
A B C | F
0 0 0 | 1
0 0 1 | 0
```

Example 2: check De Morgan with `=`. Type `t ~(ab)=~a+~b`:

```
EQUAL (all 4 rows match)
F = A' + B'
```

Example 3: an output column → the expression. Type `t 0110`:

```
Output column for A,B
F = A'B + AB'
This is an XOR gate
```

Minterms (added 2026-10-18): F(A,B,C) = Σm(1,3,5) is `t abc m1,3,5`. Type the letters first, so the app knows there are 3 variables. Don't cares: add `d6`. Maxterms ΠM(0,2): `t abc M0,2`. Brackets are fine: `m(1,3,5)`. Without letters the app guesses the variables from the biggest number and shows a warning. **Multiplexer:** every table also shows a mux line, e.g. `t abc m1,2,6,7` → `Mux 4:1, select A,B: D0.. = C C' 0 1`. The first letters are the select lines. Each data input is 0, 1, the last letter or its NOT. **Answers:** Truth Tables and Boolean Expressions A8, Boolean Algebra and Simplification A8, De Morgan and Bubble Pushing A8, Task - Logic from Truth Table A9.

`g` Gate table

Shows: every gate, its expression, symbol, NAND/NOR count and chip number. Scroll with up/down. Press `g`:

```
NOT   A' (also ~A or A-bar)
   triangle + bubble;  1 NAND, 1 NOR; 7404
AND   A*B (AB)
   D shape;  2 NAND, 3 NOR; 7408
```

Answers: Logic Gates and Notation A8, Universal Gates (NAND and NOR) A8, Logic Hardware and Chips A8.

`x` Binary

Example 1: add 1011 + 110. Type `x 1011+110`:

```
carry 11100
        01011
    +  00110
    =  10001
dec 11 + 6 = 17
```

Example 2: convert 45. Type `x 45`:

```
bin 101101  (6 bits)
hex 2D
= 32+8+4+1
```

Subtract: `x 101-11` → diff bin 10 (5 – 3 = 2). A negative answer also shows its two's complement. **Answers:** Binary Numbers and Addition A8, Adders A8.

9. Meters and hands-on

`m` Mystery resistor

Solves: a known R in series with the mystery R. Measure the voltage across each. **Example:** Rk = 1 kΩ, 3 V across it, 6 V across Rx. Type `m a1k v3 w6`:

```
x  Mystery R (Rx)  = 2 kohm
i  Current  = 3 mA
s  Supply (Vk + Vx)  = 9 V
```

Answers: Task - Mystery Resistor A9.

`h` Hit a target R

Solves: which series/parallel mix of your parts is closest to a target. Up to 4 parts per mix. **Example:** parts 1k, 2.2k, 4.7k, 10k; target 3k. Type `h 1k 2.2k 4.7k 10k=3k`:

```
Target 3 kohm from 4 parts (100 combos)
3.025k (+0.819%)  2.2k+(4.7k||1k)
2.962k (−1.27%)  2.2k+(4.7k||(10k||1k))
2.914k (−2.88%)  ((10k+4.7k)||2.2k)+1k
```

Answers: Task - Target Resistance A9.

`z` USB charge time

Solves: battery mAh and V, charger label, efficiency → hours (energy method). **Example:** 3000 mAh at 3.7 V, a 5 V 2 A charger, 80% efficient. Type `z c3000 v3.7 s5 a2 e.8`:

```
w  Battery energy  = 11.1 Wh
g  Charger power  = 10 W
t  Charge time  = 1.3875 h
```

The app also shows "e chose box e. A power of ten? Use the EE key". That is only a hint: `e.8` is box e = 0.8.
Answers: Task - USB Charger A9.

Also useful in hands-on: `o` for Task - Power Supplied A9, `c` and Shift+T for reading parts, `l` for Task - Equal LEDs A9.

The network solver (Shift+N)

This tool solves **any** resistor circuit with sources. You type the circuit as a list of parts (a **netlist**). It gives every node voltage, and the V, I and P of every part. Use it for Hard Resistor Networks A5 and Kirchhoff Analysis (States-Nationals) A5.

Step 1: number the nodes on paper

- A **node** is a point where wires join. Give each node a whole number.
- Node **0** is ground (the – side of the battery, usually).
- Draw the numbers on your copy of the circuit. Then type the parts.

Step 2: type the parts

Split the parts with commas or spaces. Commas are **optional**: a new part starts at each `a–b`.

You type	It means
`1–2 100`	a 100 Ω resistor between nodes 1 and 2
`1–2 4.7k/6.8k`	a parallel pair between 1 and 2 (`100/2` shows a warning: `/` is parallel, not divide)
`2–0*3 6`	three 6 Ω resistors in parallel between 2 and 0
`2–3 0`	a wire (0 Ω)
`V1–0 9`	a 9 V source, **+ at node 1**
`I2–0 5m`	a 5 mA current source that pushes into node 2

You type	It means
`=1–3`	R_eq between nodes 1 and 3 (sources turned off)
`=2`	copy V2
`V2–3`	a voltmeter: V2 – V3
`1–2–3–4–1 5`	a **chain**: four 5 Ω resistors, 1-2, 2-3, 3-4 and 4-1
`1–5` (no value)	the **same value as the last resistor**
no values at all	every resistor is 1 Ω, so the answer is **in units of R**

The answer updates after every key.

Example: a bulb is removed (before → after)

"L1, then L3 || L4, then L5, all 1 Ω, on 1 V. L3 is removed. How does the power in L4 and L5 change?" Mark the part that changes with `o` (removed after) or `s` (shorted after). Type `N V1–0 1 1–2 1 2–3 o2–3 3–0`:

```
BEFORE -> AFTER (o removed, s shorted)
1–2 1: 160 mW -> 111.1 mW  x0.6944 dimmer
2–3 1: 40 mW -> 111.1 mW  x2.778 brighter
2–3 1: 40 mW -> 0 W  removed
3–0 1: 160 mW -> 111.1 mW  x0.6944 dimmer
```

So L4 is ×2.78 brighter and L5 is ×0.694 dimmer (UT fall 2018 Q24c). Brightness follows power.

Example: the cube (fast typing)

"Twelve equal resistors R form a cube. Find R between opposite corners." Type `N1–2–3–4–1 5–6–7–8–5 1–5 2–6 3–7 4–8 =1–7` (42 presses; with commas and values it was 78):

```
R(1–7) = 0.83333 ohm
     = 0.8333333 ohm (sources off)
No R value given: 1 ohm (answer in units of R)
```

So R = 5R/6. Use `=1–3` for the face diagonal (3R/4) and `=1–2` for one edge (7R/12). A half-typed last part is skipped.

Example: a simple divider

Type `N V1–0 9, 1–2 100, 2–0 220`:

```
V1 = 9 V            V2 = 6.188 V
V1–0 9: 28.12 mA out of +, gives 253.1 mW
R1–2 100: 2.812 V  28.12 mA 1>2  79.1 mW
R2–0 220: 6.188 V  28.12 mA 2>0  174 mW
Sources give 253.1 mW, resistors use 253.1 mW
```

`1>2` means the current flows from node 1 to node 2.

Example: an unbalanced bridge

A 10 V source. 100 Ω and 200 Ω on top, 300 Ω and 400 Ω below, 50 Ω across the middle. Type `N V1–0 10, 1–2 100, 1–3 200, 2–0 300, 3–0 400, 2–3 50`:

```
V1 = 10 V           V2 = 7.258 V
V3 = 7.097 V
V1–0 10: 41.94 mA out of +, gives 419.4 mW
R2–3 50: 161.3 mV  3.226 mA 2>3  520.3 uW
```

Voltmeter across the middle: add `, V2–3` at the end:

```
V2 – V3 = 161.3 mV
```

R_eq of the bridge: leave out the source and add `=1–0`. Type `N 1–2 100, 1–3 200, 2–0 300, 3–0 400, 2–3 50, =1–0`:

```
R(1–0) = 238.5 ohm
     = 238.4615 ohm (sources off)
```

> **WARNING: Leave the source out for R_eq** If the source sits between the same two nodes, `=1–0` gives 0 Ω. The tool turns the source into a wire. So `N V1–0 10, ... , =1–0` shows `R(1–0) = 0 ohm`.

Example: the resistor cube (5R/6)

Number the corners 0 to 7. Each edge is 1 Ω. Corners 0 and 7 are opposite. Type `N 0–1 1, 0–2 1, 0–4 1, 1–3 1, 1–5 1, 2–3 1, 2–6 1, 3–7 1, 4–5 1, 4–6 1, 5–7 1, 6–7 1, =0–7`:

```
R(0–7) = 0.83333 ohm
     = 0.8333333 ohm (sources off)
```

0.8333 = 5/6, so the answer is 5R/6.

Example: two sources

12 V at node 1 with 4 Ω to node 2. 6 V at node 3 with 3 Ω to node 2. 2 Ω from node 2 to ground. Type `N V1–0 12, 1–2 4, 2–0 2, V3–0 6, 3–2 3`:

```
V1 = 12 V           V2 = 4.615 V
V3 = 6 V
V1–0 12: 1.846 A out of +, gives 22.15 W
R2–0 2: 4.615 V  2.308 A 2>0  10.65 W
V3–0 6: 461.5 mA out of +, gives 2.769 W
R3–2 3: 1.385 V  461.5 mA 3>2  639.1 mW
Sources give 24.92 W, resistors use 24.92 W
```

Find an unknown: `x` + one condition

Write `x` for **one** part value. Then add **one** condition part. The tool finds the x that makes the condition true.

Condition	It means
`I2–0=20m`	20 mA flows from 2 to 0 in the resistors there
`I=3.18`	3.18 A flows out of + of the first source
`V2=5`	node 2 is at 5 V
`V2–3=0`	V2 – V3 = 0 (a balanced bridge)
`P2–0=.1`	the resistors between 2 and 0 use 0.1 W
`P=2`	the sources give 2 W
`=1–0 500`	R_eq between 1 and 0 is 500 Ω

Find a resistor (current condition). Which R4 makes the source give 3.18 A? Type `N V1–0 12, 1–0 8, 1–2 5, 2–0 6, 2–3 1, 3–0 10, 3–0 x, I=3.18`:

```
x = 3.043 ohm  (R3–0)
     = 3.043478 ohm for I(source) = 3.18 A
```

Find a source voltage. Which E makes 20 mA flow in the 220 Ω? Type `N V1–0 12, 1–2 330, 2–0 220, 3–2 100, V3–0 x, I2–0=20m`:

```
x = 4.097 V  (V3–0)
     = 4.09697 V for I2–0 = 20 mA
```

Balance a bridge. Which R makes the bridge balance? Type `N V1–0 10, 1–2 1k, 1–3 2k, 2–0 2.35k, 3–0 x, V2–3=0`:

```
x = 4.7 kohm  (R3–0)
     = 4700 ohm for V2–3 = 0 V
```

Find R for a target R_eq. Which R makes the total 150 Ω? Type `N 1–2 100, 2–0 x, 1–0 300, =1–0 150`:

```
x = 200 ohm  (R2–0)
     = 200 ohm for R1–0 = 150 ohm
```

Power: two answers. Which load gets 0.1 W? Type `N V1–0 10, 1–2 100, 2–0 x, P2–0=.1`:

```
2 answers: x = 12.7 ohm or 787.3 ohm
Below: x = 12.7 ohm
```

Both answers are right. Read the question to pick one. The screen shows the circuit for the smaller one.

Other messages: with no answer, it says the range that x can reach. A wrong sign is tried flipped and marked.

Diodes (added 2026-10-18)

Type `D` + anode node + `-` + cathode node. No value = a silicon diode (0.7 V). A value is the drop: `D2-3 2` = an LED with 2 V, `D2-3 0` = an ideal diode. The app tries every ON/OFF guess and keeps the one that works: an ON diode carries current forward, an OFF diode has less than its drop across it. That is the same check you write on paper. **Example (A7 type 5):** 10 V → 1 kΩ → node 2. From node 2 to ground: a red LED (2 V), and two Si diodes in series. Type `N V1-0 10, 1-2 1k, D2-0 2, D2-3, D3-0`:

```
V1 = 10 V           V2 = 1.4 V
D2-0 2: OFF, 1.4 V across
D2-3 .7: ON  8.6 mA 2>3  6.02 mW
D3-0 .7: ON  8.6 mA 3>0  6.02 mW
```

The LED stays dark. **Real diode with r_f:** add a resistor after it: `D3-4, 4-0 1` = 0.7 V + 1 Ω. Up to 8 diodes. With diodes, `=a-b` (R between nodes) is not defined. **Meters read resistors first:** `P2-0` or `I2-0` reads the resistor(s) between 2 and 0. With a load across the zener, give the zener its own node pair: `Z0-3 5.1, 3-2 0` (a wire) and ask `P0-3`. **Smallest load that still regulates:** make the load `x` and ask for the zener voltage: `N V1-0 12, 1-2 1k, Z0-2 5.1, 2-0 x, V2=5.1` → "Any x from 739.1 ohm up". When a whole range of x works, the app gives the range; its edge is the min (or max) value. **Zener:** `Z2-3 5.1` = anode 2, cathode 3, breakdown 5.1 V (forward it drops 0.7 V). A regulator: `N V1-0 12, 1-2 1k, Z0-2 5.1, 2-0 2k` → V2 = 5.1 V, "ON zener (breakdown) 4.35 mA 2>0". With a 500 Ω load it says OFF and V2 = 3 V: the load is too heavy to regulate. Units after a value are fine in the netlist: `V1-0 9V`, `I2-0 5mA`, `1-2 100ohm`.

Dependent sources (added 2026-10-18)

A source whose value depends on another voltage or current. Type the factor, `*`, then what it copies: | On the test | Type | |---|---| | a voltage source 5·Vx, Vx = V1 − V2 | `V3-0 5*V1-2` | | a voltage source 2·V3 (V3 to ground) | `V4-0 2*V3` | | a current source 0.6·ix, ix flows 2 → 0 in the parts between 2 and 0 | `I3-0 .6*I2-0` | | a current source 2 mS · V1 | `I2-0 2m*V1` | | a voltage source 1 kΩ · ix | `V4-0 1k*I1-2` | Write `ix` with its direction from the picture: `I2-0` = current from node 2 to node 0 in the **resistor(s)** between 2 and 0 (no resistor there: in the other part, e.g. a source). This is the same rule as the `I2-0` meter. Always type the `*`: `I3-0 .6 I2-0` (with a space) is read as a source and a meter, and a warning says "Dependent? 5*V1-2". `=a-b` keeps dependent sources on (only the plain sources go off), which is the right way to find a Thevenin R. **Example (UT 2020 B Q74):** `N V1-0 8, 1-2 3, I0-2 .1*I1-2, 2-3 2, 3-0 2, =3-0` → R = 1.455 Ω, V3 = 2.182 V (key: 2.18 V, 1.45 Ω). Dependent sources are **borderline** for the 2027 Div B rules, but some invitational tests use them.

Network keys

- **Del** deletes one character. **Ctrl+Del** clears all.
- **Up/down** scroll the answer.
- **Enter** copies R, V, x, or the total source power.

Search (key 0)

Search finds the facts: history, rules of thumb, safety, switches, chips. There are two files:

- **CircuitLab** (key `0`): 230 core facts, next to all the tools.
- **CircuitRef**: a second file with search only. It has all 748 facts (the 230 core facts and 518 more). Open it from My Documents when CircuitLab does not have the fact. It starts on the search screen. It may take a few seconds to open, because it builds its word list at the start.

The keys are the same in both files:

1. Press `0` (in CircuitLab).

2. Type a few letters or numbers. The list updates after every key. Digits are part of the query, so `e12` and `7400` work.
3. **Enter** opens result 1 (the highlighted one).
4. To open another result: press **tab** or an arrow to go into the list, then press `1-9`. Or move with the arrows and press Enter. Type a letter again to go back to the query.
5. In a result, **tab** goes to the next result. **Esc** goes back.

Example: type `0 kcl`:

```
Search facts  3 found  [0]
? kcl_
1 KCL current law
2 Nodal analysis
3 Kirchhoff: law aliases
enter=open  tab/arrows=list  esc=home
```

Then press tab, then `1`:

```
KCL current law  1/3
Current into a node = current out of the node. It comes
from conservation of CHARGE. Example: 4 A in and 7 A
out -> the other incoming current is 3 A. States and
Nationals only, but invitationals used it in 2026.
tags: kirchhoff keys: kcl, kirchhoff current law, junction
rule, node, conservation of charge
arrows=scroll  tab=next result  esc=back
```

Tips:

- **Use short words.** `nand` puts "NAND gate" first. `gfci` finds "GFCI trip level" in the top 3.
- **Abbreviations work.** `vf` finds "LED Vf by color". `kcl` finds "KCL current law".
- **A start of a word works.** `tri` finds "Triboelectric series" in the top 3.
- **Small typos work.** `kirchof` still finds "Kirchhoff: life".
- **Two words narrow it down.** `led res` puts "LED resistor" first.
- **Numbers work.** `e12` finds "E12 standard values":

```
Search facts  1 found  [0]
? e12_
1 E12 standard values
enter=open  tab/arrows=list  esc=home
```

- **Years work in CircuitRef.** `1826` finds "Ohm's 1826 formula" and "Ohm: inspired by Fourier".

What it cannot do

- **It cannot see the diagram.** You read the circuit, number the nodes and type the parts.
- **Only one unknown in the network solver.** One `x` and one condition. Two unknowns at once do not work.
- **No capacitors or AC** in the network solver. Diodes (`D2–3`) and zeners (`Z2–3 5.1`) work with the constant-drop model.
- **Charges:** on a line or in a plane (`q x y`), point charges only. No 3D.
- **Network size:** up to 40 unknowns (nodes + voltage sources + wires).
- **Search and digits:** `1-9` open a result only after tab or an arrow. Before that, digits are part of the query.
- **ASCII text only.** It writes "ohm", "u" for micro and "rho".
- **Some answer keys are wrong.** The app found mistakes in a few past tests (for example UMSO 2019 Q31 cube). Trust your work on paper, then ask.
- **Not tested on the real calculator yet.** See the first-session list above.

TIP: The 2 habits

1. **Always write the work on paper.** Graders give points for the work, and paper is your backup if the app is not allowed.
2. **Check that the answer makes sense.** A 9 V battery cannot give 900 A. Parallel R is smaller than the smallest part.

TAB S

Search codes (appendix)

History (147) 733
Static electricity (50) 733
DC and AC (41) 733
Batteries (31) 734
Basics: I, V, R, P (14) 734
Units (18) 734
Resistors (45) 734
DC circuits (28) 734
Kirchhoff (13) 734
Meters (26) 734
Hands-on tasks (41) 734
Switches, relays, fuses (59) 735
Safety (21) 735
Diodes (32) 735
LEDs (19) 735
Logic (94) 735
Binary (5) 736
Traps (65) 736
Index by code 737

Short search codes for the CircuitRef facts on the TI-Nspire. Type a code in search (key 0).

Search codes

Type the code in CircuitRef search (key 0) and the fact is first in the list. CircuitLab (229 core facts) uses gen/codes_core.tsv — codes there may be even shorter.

A dash (—) means no short code puts that fact first: search two words of its title instead.

History

af th After the pile: first uses
len Ampere force law: lengths
am na Ampere named galvanometer
am un Ampere unit history
dis Ampere's discoveries
as Ampere: astatic needle
post Ampere: career posts
ampe Ampere: life
sep Ampere: Sept 1820 timeline
he Ampere: words he coined
co pr Before Coulomb: priority
bl no Blue LED Nobel 2014 detail
de le Boole and De Morgan letters
trea Boole quote: treatise
bo la Boole's law x^2 = x
je Boole's OR vs Jevons
sy Boole's symbols
1847 Boole: 1847 book details
books Boole: books
tho Boole: Laws of Thought title
bo Boole: life
roy Boole: Royal Medal 1844
cav Cavendish found it first
grad Coulomb quote: graduating
posth Coulomb unit: posthumous
178 Coulomb's 1785 data
co hi Coulomb's law history
att Coulomb: attraction method
cou Coulomb: life
prov Coulomb: proved vs assumed
gl Crown of cups: wine glasses
shan De Morgan and Shannon
de De Morgan: life
du Du Fay: two kinds of charge
ed Edison's argument vs AC
swa Edison: bulb 1879 and Swan
el ho Electrophorus: how it works
end End of the war 1892
fra Franklin
fr Franklin: battery and Fr unit
ju Franklin: kite June 1752
ga Galvani
179 Galvani: 1791 book title
gi Gilbert: De Magnete 1600
verso Gilbert: electricus, versorium
gu Gray and von Guericke
qu Hertz quotes
unit Hertz unit history
187 Hertz: 1879 Berlin prize
ray Hertz: cathode rays
her Hertz: life
1887 Hertz: photoelectric 1887
rad Hertz: radio waves
rece Hertz: the receiver
he tr Hertz: the transmitter
show Hertz: wave properties shown
hol Holonyak: first visible LED
jo Joule and Watt
pad Joule: paddle wheel
whil Joule: unit named while alive
1857 Kirchhoff 1857 signal speed
sun Kirchhoff quote: gold sun
ki di Kirchhoff's discoveries
stu Kirchhoff: 1845 as a student
bla Kirchhoff: black body
ces Kirchhoff: cesium and rubidium
cit Kirchhoff: cities
ali Kirchhoff: law aliases
kir Kirchhoff: life
spect Kirchhoff: spectroscope trap
ki th Kirchhoff: the other laws
maxw Maxwell: link to Hertz
mh Mho and siemens history
new Newton of electricity quote
oe Oersted
pap Oersted: paper and Arago
pn Ohl: PN junction details
mem Ohm quote: the Memoir
oh un Ohm unit name history
182 Ohm's 1826 formula
aco Ohm's acoustic law
hi Ohm's law history
die Ohm: Die galvanische Kette
ho Ohm: honours
fou Ohm: inspired by Fourier
li Ohm: life
pog Ohm: Poggendorff thermocouple
1827 Ohm: the 1827 controversy
ley Pile vs Leyden jar
pn le PN junction and LED history
names Seven names mnemonic
det Shannon thesis details
194 Shannon: info theory 1948
195 Solar history 1839, 1954
stead Steady-state laws multi-select
tesl Tesla (out for 2027)
tha Thales and amber c. 600 BC
sci The seven 2027 scientists
timel Timeline of the seven
185 Timeline: 1850 to 2014
bc Timeline: 600 BC to 1850
ba ho Torsion balance: how it works
flu Trap: Ampere and fluorine
fo ma Trap: Ampere found magnetism
tr am Trap: Coulomb or Ampere?
el ke Trap: electrodynamics key
mix Trap: Galvani mix-ups
jo oh Trap: Joule before Ohm
stru Trap: kite struck by lightning
nob Trap: Nobel starts 1901
tr wo Trap: not Boole's work
tr ki Trap: not Kirchhoff's work
tr vo Trap: not Volta's
fara Trap: Oersted law = Faraday
bis Trap: Ohm's copper-bismuth
riv Trap: Ohm's rival not Galvani
mag Trap: On Magnetism key
pred Trap: predicted vs confirmed
sod Trap: sodium in Sun, not Moon
tor Trap: torsion balance vs Ohm
mad Trap: Volta made electrophorus
worke Trap: Volta worked in France
whi Trap: which Siemens
wh el Trap: who coined electricity
inven Trap: who invented AC
cong Units at 1881 Paris congress
aft Units named after people
188 Volt unit named 1881
honor Volta honors timeline
180 Volta letter: 1800 details
exp Volta quote: experiment
vo vs Volta vs Galvani: details
qs Volta vs Galvani: past Qs
vo co Volta's contact series
prop Volta's law: Q prop. to V
volta Volta: life
17 Volta: methane 1776
ot Volta: other work
pil Voltaic pile
vo bu Voltaic pile: build details
war War of the Currents
ado Watt and joule adopted 1889
hp Watt: steam engine, hp
wer Werner von Siemens
wes Westinghouse: AC wins
whea Wheatstone bridge history
pe Zener: person vs diode

Static electricity

kv Air breakdown 3 kV/mm
car Car safe in lightning
cons Charge constants
quan Charge quantization
meth Charging methods
cond Conductors in fields
insu Conductors vs insulators
ratio Coulomb ratio: charges
dist Coulomb ratio: distance
co un Coulomb unit = 1 A*s
co la Coulomb's law
disch Discharge types order
el ch Electric charge basics
el fi Electric field
39 Electric vs gravity 10^39
el Electron-volt
el ind Electroscope by induction
leaf Electroscope leaf rules
equi Equipotential rules
es ESD you cannot feel
cag Faraday cage: charge inside
fi Field lines
whe Field where E = 0
fla Flash-to-bang rule
dir Force direction in a field
ig Fuel ignition risk
flo Grounding electron flow
hu Humidity and static
tou Identical spheres touch
indu Induction steps in order
joi Joined spheres: Q ~ r
un k value and units
nu Lightning numbers
3rd Newton 3rd law: equal force
be Pith ball behavior
po Polarization
po en Potential and energy
qua Quark charges
sta Static hazards
cau Static shock cause
thu Thunderstorm charging
ve Tribo series: long version
mos Tribo: most charge pair
trib Triboelectric series
tw sp Two touching spheres induced
unif Uniform field E = V/d
van Van de Graaff safe to touch
wim Wimshurst machine
moh Wrist strap 1 Mohm resistor
ze po Zero-potential points: two

DC and AC

240 120 V and 240 V split phase
ac sy AC and DC symbols
ac AC basics
da AC vs DC danger
ad la Adapter label symbols
ben Bench supply CV and CC

`buc` Buck, boost, DC-DC
`co` Commutator
`ar` DC arcs harder to stop
`dc` DC basics
`ha` DC hazards
`dc us` DC sources and uses
`grov` Fuel cell: Grove 1839
`hot` Hot, neutral, ground roles
`hou` Household wire colors
`hv` HVDC lines
`60` Japan mains 100 V 50/60 Hz
`soc` Lamp socket: shell is neutral
`los` Line loss and voltage
`mai` Mains in other countries
`ne` NEMA name code
`ou me` Outlet meter readings
`ou` Outlet slots
`rm` RMS
`sola` Solar cell numbers
`ac or` Source: AC or DC?
`seeb` Thermocouple: Seebeck 1821
`34` Trap: 340 V peak-to-peak
`120x` Trap: 60 Hz reverses 120x/s
`les` Trap: AC has less resistance
`tr ad` Trap: adapter vs converter
`all` Trap: all AC is a sine wave
`10 pe` Trap: is 100 V peak or RMS?
`125` Trap: rated 125 V not 120 V
`sq` Trap: RMS of a square wave
`nev` Trap: wave never below zero
`110` US 110, 115, 117, 120 V
`us` US and world outlets
`wal` Wall adapter insides
`gri` Why the grid uses AC
`wi` Wire gauge AWG

Batteries

`al` Alkaline cell
`ano` Anode and cathode
`ra` C-rate
`ca en` Capacity and energy
`zi` Carbon-zinc (Leclanche)
`cc` CC/CV slow finish
`cells` Cells in series and parallel
`adds` Cells: r in series, parallel
`cr` CR2032 lithium coin
`18` Daniell cell 1836
`dr` Dry cell vs wet cell
`eve` Everyday source voltages
`int` Internal resistance
`lea` Lead-acid cell
`lem` Lemon or potato cell
`lip` LiPo and LiFePO4
`io` Lithium-ion cell
`max` Max power transfer
`ni` NiCd cell
`nim` NiMH cell
`3s` Pack names like 3S2P
`pa ce` Parts of a cell
`pr ce` Primary cell voltages
`prim` Primary vs secondary
`rech` Rechargeable cell voltages
`stor` Trap: battery stores charge
`used` Trap: copper is used up
`24` Trap: diesel battery 24 V
`nai` Trap: two copper nails = 0 V
`ty ba` Typical battery capacities
`wea` Weak battery reads full

Basics: I, V, R, P

`co fl` Conventional vs electron flow
`cu` Current
`dri` Drift speed of electrons
`ef` Efficiency
`cos` Electricity cost
`em` EMF is not a force
`en` Energy and kWh
`en co` Energy conversions
`la` Ohm's law
`sc` Out of scope 2027
`pow` Power formulas
`ap` Typical appliance powers
`typi` Typical currents
`vo` Voltage

Units

`19` 19 SI units named for people
`am ba` Ampere is the base unit
`am si` Ampere: only SI base unit
`non` Non-SI conversions
`pai` Prefix shortcut pairs
`qu un` Quantities and units table
`sev` Seven SI base units
`si oh` SI base units of V, ohm, S
`si` SI prefixes
`fig` Sig fig counting rules
`sig` Sig figs in math
`tr el` Trap: 1 A = 1 electron/s
`tr la` Trap: Coulomb law vs unit
`plu` Trap: S vs s, plurals
`tr wa` Trap: watt vs joule
`wr` Unit writing rules
`un eq` Units equal to a volt
`un wa` Units equal to a watt or ohm

Resistors

`zer` 3-band and zero-ohm
`sy li` Anti-symmetry: V/2 line
`cop` AWG copper table
`sca` AWG scaling rules
`bala` Balanced Wheatstone bridge
`fil` Bulb filament and inrush
`dig` Color code digits
`gol` Color code gold silver
`mn` Color code mnemonic
`del` Delta-Y conversion
`e1` E12 standard values
`e6` E6 and E24 series
`eq va` Equal-resistor values
`eq` Equivalent resistance tips
`far` Finite ladder: far end first
`ite` Infinite ladder 0.618R
`61` Infinite ladder 1.618R
`metal` Metal resistivity order
`to` Network tolerance
`ohmi` Ohmic vs non-ohmic
`tap` Pot taper: linear vs log
`pot` Potentiometer
`po rh` Potentiometer vs rheostat
`str` Power limit of a string
`re ba` Reading 4, 5, 6 bands
`resi` Resistivity
`re va` Resistivity values
`meta` Resistivity: non-metals
`fe` Resistivity: W, Fe, Pt, Pb
`cub` Resistor cube values
`re po` Resistor power rating
`re ch` Resistor tolerance check
`rh` Rheostat
`gr` Slope of V-I graph
`bun` Stretch or bundle a wire
`su di` Superconductor discovery
`cut` Symmetry: join or cut
`6t` Tempco band (6th band)
`res` Temperature and resistance
`coe` Temperature coefficient
`ld` Thermistor, varistor, LDR
`tr pa` Trap: 1/8 W parts
`unb` Unbalanced bridge voltmeter
`read` Which end to read first
`nich` Why heaters use nichrome

DC circuits

`ie` ANSI vs IEC symbols
`ba co` Battery symbol: count pairs
`pa dimm` Bulb A then B, C in parallel
`bul` Bulb brightness
`bu ra` Bulb rating to resistance
`cu di` Current divider
`dea` Dead lamp: 0 V or full V
`ea` Earth ground symbol
`fu sy` Fuse and breaker symbols
`pa` How parts fail
`rank` Identical bulbs: rank rules
`lam` Lamp symbol: X or loop
`loade` Loaded divider: output drops
`op` Open circuit
`para` Parallel resistors
`sc ba` Schematic symbols: basics
`sch` Schematic symbols: parts
`pa ra` Series / parallel ratio n^2
`ser` Series resistors
`nodes` Series vs parallel by nodes
`sh` Short circuit
`wha` Shorted bulb: what happens
`bra` Shorted parallel branch
`sh se` Shorting one series bulb
`opens` Switch closes or opens
`th sy` Thermistor and LDR symbols
`div` Voltage divider
`ze cu` Zero-current branch trick

Kirchhoff

`deli` Absorb or deliver power
`fixe` Battery to ground fixes a node
`un pi` Count unknowns, pick method
`kc` KCL current law
`ru` KVL sign rules
`kv la` KVL voltage law
`me` Mesh analysis
`nod` Nodal analysis
`mil` One-node formula (Millman)
`su` Supernode
`steps` Superposition steps
`thev` Thevenin and Norton: OUT
`tr ju` Trap: junction rule = charge

Meters

`accu` 3.5-digit meter, accuracy
`am` Ammeter
`bur` Ammeter burden
`box` Black box: find the diode
`conti` Continuity and diode test
`di te` Diode test readings
`jack` Fuse in the mA jack
`galvano` Galvanometer
`idea` Ideal meters memory trick
`ind` Indirect measurements
`ke` Kelvin bridge
`le ze` Lead resistance zero
`le fa` LED fault from voltages
`loa` Loading effect
`me po` Meter polarity: minus sign
`se` Meter settings
`lett` Meter symbols: letters
`ja` Multimeter jacks
`oh` Ohmmeter
`pic` Pick the best range
`red` Red lead is anode
`shu` Shunt vs multiplier
`am le` Trap: ammeter 0, LED lit
`lef` Trap: lead left in mA jack
`voltm` Voltmeter
`vo se` Voltmeter in series

Hands-on tasks

`wa bu` 3-way build checklist
`wa fa` 3-way fault tables
`lay` Breadboard layout
`lim` Breadboard limits
`br ro` Breadboard rows of 5
`33` BullSO C 133.33 ohm task
`unc` Carry uncertainty
`ch` Charge time

co br Coin bridge switch test
le ch Equal LEDs checklist
fa Fault clues
commo Find the SPDT common pin
poi Hands-on scoring points
shar Hands-on share of score
hug Huge mystery R: loading
mini Minimum safe resistor
my Mystery R checklist
my re Mystery resistor task
pc PCB printed circuit board
clo Pick R_k close to R_x
bank Power bank real output
supp Power supplied task
po ch Power task checklist
varie Rail polarity varies
ont Schematic onto breadboard
so Sources of error
spl Split rail check
tar Target R checklist
top Top-down target method
big Trap: big charger, fast charge
bot Trap: both legs in one row
kn Trap: known R in other branch
tr di Trap: LED on a divider
one Trap: one fault only
se le Trap: one series LED reversed
op ba Trap: open-circuit battery V
tro Troubleshooting steps
rou Uncertainty rounding
usb USB label reading
us le USB standard levels
tas USB task checklist

Switches, relays, fuses

trav 3-way common and travelers
wa 3-way switch
loc 4-way switch
ins 4-way switch inside
af AFCI vs GFCI
au Automotive relay pins
bef Break-before-make use
cap Breaking capacity
bre Circuit breaker
con Contact bounce
bl Does the fuse blow
dpd DPDT switch
dp DPST switch
fas Fast vs slow-blow fuse
fl Flyback diode
fu Fuse
2t Fuse heat and I^2t
ty Fuse types
fu vs Fuse vs breaker
forms GFCI forms and history
works GFCI how it works
line GFCI LINE and LOAD
te GFCI TEST button
tri GFCI trip level
gf ti GFCI trip time rule
gf wh GFCI where and limits
lat Latching switch types
mic Limit and micro switch
mu pa Multiway = parity (XOR)
fri NC push button (fridge)
doo NO push button (doorbell)
overl Overload vs short circuit
up Pick-up and drop-out
ptc PTC polyfuse
nc Push button NO vs NC
rc RCD 30 mA
ree Reed switch
re Relay
com Relay COM NO NC states
rati Relay contact rating
compu Relay logic and computers
rese Resetting a breaker
sin Single and double pole
sol Solid-state relay
spd SPDT switch
sp SPST switch
ki Switch kinds
thr Switch poles and throws
ter Switch terminal count
br su Trap: breakers and surges
bac Trap: flyback diode backwards
gf cu Trap: GFCI limits current
pres Trap: NC DPST pressed
overs Trap: oversized fuse/breaker
tr sp Trap: SPST for 2 places
neu Trap: switch on neutral
goe Why a fuse goes in series
prefe Why prefer a fuse
why Why use a relay

Safety

bo cu Body current table
bod Body resistance
go Dalziel AC let-go numbers
dal Dalziel DC numbers
ex Extra-low voltage 50 V
arci Fuse voltage rating: arcing
bon Grounding vs bonding
int bo Internal body resistance
va MOV varistor
pron Polarized and 3-prong plugs
sa Safety habits
fac Shock factors
ai Shock first aid
keye Shock test values keyed
st Step voltage
sur Surge protector
su ra Surge protector ratings
12 Trap: 12 V car battery shock
tet Trap: AC tetanus or DC tetanus
br pr Trap: breakers protect people
hel Why the ground wire helps

Diodes

1n 1N4001 1N4007 1N4148 parts
ba Band gap
ga wa Band gap to wavelength
ge Band gaps: Ge to GaN
br di Bridge diode faults
br rec Bridge rectifier details
dep Depletion region
cla Diode clamp to a reference
curv Diode I-V curve
gates Diode logic gates
mo Diode models
di Diode uses
acc Donors and acceptors
dop Dopant element list
el di Electron flow in a diode
fin Find the cathode
fo Forward bias
four Four regions of diode curve
mino Majority and minority carriers
type N-type and p-type
ph Photodiode facts
piv Rectifier PIV and peaks
re vs Rectifier vs inverter
rec Rectifiers
re bi Reverse bias and breakdown
scho Schottky, Ge, photodiode
sem Semiconductor basics
no Why no silicon LEDs
bea Why silicon beats germanium
ze Zener diode
1n47 Zener parts 1N4733A 1N4742A
av Zener vs avalanche breakdown

LEDs

ant Anti-parallel LED pair
blu Blue and violet LED Vf
gre Green LED: two kinds
makes How an LED makes light
ran LED current range
le LED legs
mate LED materials by color
lit LED not lit trick
le re LED resistor
typ LED typ vs max Vf gap
vf LED Vf by color
cf LED vs CFL vs incandescent
bands LED wavelength bands
ye Orange and yellow LED Vf
63 Red LED Vf 1.63-2.03 V
dim Shunt resistor dims LED
tw Two equal LEDs task
wav Wavelength and Vf
15 White LED and 4.15 V key

Logic

16 16-pin chip power pins
ga ch 3- and 4-input gate chips
7400 7400 quad NAND chip
740 7402 quad NOR chip
7404 7404 hex inverter chip
7408 7408 quad AND chip
741 74138 active-low decoder
742 74266 quad XNOR chip
7428 74283 4-bit adder chip
74 7432 quad OR chip
748 7486 quad XOR chip
abs Absorption
coun Adder gate counts
or na AND and OR from NAND
or no AND and OR from NOR
an AND gate
ans ANSI gate shapes
exa At least k / exactly k
alg Boolean algebra laws
bo id Boolean identities
bu Bubble pushing
pins Chip pins and wiring
ch ro Chips needed: round up
cm CMOS inverter
cm lo CMOS logic levels
comb Combining law
th Consensus theorem
laws De Morgan's laws
dem Demux, decoder, encoder
di ou Diode gate output voltages
do Don't care rows
dua Dual vs complement
en pr Encoder and priority encoder
engl English words to gates
ful Full adder
mi Full adder minterms
ad Half adder
kar Karnaugh map (K-map)
syn Law name synonyms
lo ch Logic chips 7400 series
han Logic hands-on build
lo Logic levels
nota Logic notation symbols
word Logic word problems
maj Majority function
pi Minterm, maxterm, sigma, pi
mu Multiplexer
dec Mux and decoder chips
mu ga Mux as a gate
tr Mux tree N - 1
bigg NAND counts: bigger circuits
na NAND gate
na onl NAND-only gate counts
no na NOR from 4 NANDs
nor NOR gate
on NOR-only gate counts
not NOT gate
or OR gate
or not OR gate from AND and NOT
ord Order of operations

bit Parity bit facts
pri Priority lights
pul Pull-up vs pull-down
ri Ripple-carry adder
sel Select lines: round up
she Sheffer stroke, Peirce arrow
sink Sourcing vs sinking LED
sum Sum of products
tra Trap: + means OR
tr nor Trap: 3-input NOR build
tr xn Trap: 3-input XNOR
tr 74 Trap: 7402 wired like 7400
bo bi Trap: Boolean 1+1 vs binary
buf Trap: buffer does not flip
tr de Trap: De Morgan vs Boole
ha wr Trap: half adder wrong gates
no ti Trap: NAND(A,0) is not NOT
ass Trap: NAND/NOR not assoc.
tr ri Trap: ripple of half adders
tr sh Trap: Shannon vs Boole
sim Trap: simplify before counting
stai Trap: staircase XOR or XNOR
ta or Trap: truth table row order
ta Truth table rows
tt TTL input levels
mar TTL output levels, margins
univ Universal gate sets
unu Unused gate inputs
nors XNOR from 4 NORs
xn XNOR gate
xo bu XOR bubble rules
nands XOR from 4 NANDs
xo XOR gate
id XOR identities

Binary

bin Binary addition
pl Binary place values
nee Bits needed for N
oc Hex, octal, BCD
tr tw Trap: two's complement OUT

Traps

tr si Trap: 1.1 V is not Si Vf
br se Trap: 100 W brighter in series
tr ba Trap: 5-band multiplier
part Trap: ammeter across part
at Trap: ATX 2026 ground key
ol Trap: blue LED reads OL
unk Trap: bridge unknown arm
burn Trap: burns out = open
tr ca Trap: capacitor vs cell
cho Trap: choose the source R
sli Trap: Coulomb unit slips
acr Trap: current "across"
cu us Trap: current is used up
sid Trap: depletion ion sides
di li Trap: diode is non-linear
tr ano Trap: diode vs battery anode
tr ins Trap: E = 0 inside so V = 0
ow Trap: each battery own R
closi Trap: electroscope closing
tr fi Trap: field units N*C
tr bia Trap: forward bias is not on
fu co Trap: fuse count rounds down
gaa Trap: GaAs alone is infrared
gai Trap: gained protons
tr gap Trap: GaP/GaN LED colors
posi Trap: gold by position
tr go Trap: gold is the best
ha vo Trap: half voltage, half power
he se Trap: heat and semiconductor R
tr ci Trap: ideal vs real circuit
tr kc Trap: KCL conserves energy
rev Trap: LED reverse 5 V
look Trap: look-alike symbols
mah Trap: mAh is not energy
po cu Trap: max power vs current
vf li Trap: max Vf is not limit
me pow Trap: meter as power source
minu Trap: minus sign force
nit Trap: N is not nitrogen
neg Trap: negative current answer
watts Trap: no watts in code
tr oh Trap: ohm/m vs ohm m
liv Trap: ohmmeter on live circuit
mode Trap: ohms mode on a diode
of Trap: open means off
2r Trap: P = I^2R for any part
pa bu Trap: parallel bulb removed
varia Trap: pot vs variable R
ov Trap: product over sum
puls Trap: pulsing DC is DC
se sou Trap: R in series w/ I source
ra se Trap: rated watts in series
rod Trap: rod removed first
sam Trap: same reading, 2 faults
tr sc Trap: Schottky is not PN
wa sou Trap: sine wave = AC source
sm Trap: SMD 100 = 10 ohm
tr sup Trap: superposition power
thi Trap: thick wire has more n
sq ch Trap: V = 0 is not E = 0
ac op Trap: V across an open
scala Trap: V is 1/r not 1/r^2
vs th Trap: varistor vs thermistor
wro Trap: yellow Vf key wrong
wired Trap: Zener wired forward

Search codes: index by code

10 pe Trap: is 100 V peak or RMS?
110 US 110, 115, 117, 120 V
12 Trap: 12 V car battery shock
120x Trap: 60 Hz reverses 120x/s
125 Trap: rated 125 V not 120 V
15 White LED and 4.15 V key
16 16-pin chip power pins
17 Volta: methane 1776
178 Coulomb's 1785 data
179 Galvani: 1791 book title
18 Daniell cell 1836
180 Volta letter: 1800 details
182 Ohm's 1826 formula
1827 Ohm: the 1827 controversy
1847 Boole: 1847 book details
185 Timeline: 1850 to 2014
1857 Kirchhoff 1857 signal speed
187 Hertz: 1879 Berlin prize
188 Volt unit named 1881
1887 Hertz: photoelectric 1887
19 19 SI units named for people
194 Shannon: info theory 1948
195 Solar history 1839, 1954
1n 1N4001 1N4007 1N4148 parts
1n47 Zener parts 1N4733A 1N4742A
24 Trap: diesel battery 24 V
240 120 V and 240 V split phase
2r Trap: P = I^2R for any part
2t Fuse heat and I^2t
33 BullSO C 133.33 ohm task
34 Trap: 340 V peak-to-peak
39 Electric vs gravity 10^39
3rd Newton 3rd law: equal force
3s Pack names like 3S2P
60 Japan mains 100 V 50/60 Hz
61 Infinite ladder 1.618R
63 Red LED Vf 1.63-2.03 V
6t Tempco band (6th band)
74 7432 quad OR chip
740 7402 quad NOR chip
7400 7400 quad NAND chip
7404 7404 hex inverter chip
7408 7408 quad AND chip
741 74138 active-low decoder
742 74266 quad XNOR chip
7428 74283 4-bit adder chip
748 7486 quad XOR chip
abs Absorption
ac AC basics
ac op Trap: V across an open
ac or Source: AC or DC?
ac sy AC and DC symbols
acc Donors and acceptors
accu 3.5-digit meter, accuracy
aco Ohm's acoustic law
acr Trap: current "across"
ad Half adder
ad la Adapter label symbols
adds Cells: r in series, parallel
ado Watt and joule adopted 1889
af AFCI vs GFCI
af th After the pile: first uses
aft Units named after people
ai Shock first aid
al Alkaline cell
alg Boolean algebra laws
ali Kirchhoff: law aliases
all Trap: all AC is a sine wave
am Ammeter
am ba Ampere is the base unit
am le Trap: ammeter 0, LED lit
am na Ampere named galvanometer
am si Ampere: only SI base unit
am un Ampere unit history
ampe Ampere: life
an AND gate
ano Anode and cathode
ans ANSI gate shapes
ant Anti-parallel LED pair
ap Typical appliance powers
ar DC arcs harder to stop
arci Fuse voltage rating: arcing
as Ampere: astatic needle
ass Trap: NAND/NOR not assoc.
at Trap: ATX 2026 ground key
att Coulomb: attraction method
au Automotive relay pins
av Zener vs avalanche breakdown
ba Band gap
ba co Battery symbol: count pairs
ba ho Torsion balance: how it works
bac Trap: flyback diode backwards
bala Balanced Wheatstone bridge
bands LED wavelength bands
bank Power bank real output
bc Timeline: 600 BC to 1850
be Pith ball behavior
bea Why silicon beats germanium
bef Break-before-make use
ben Bench supply CV and CC
big Trap: big charger, fast charge
bigg NAND counts: bigger circuits
bin Binary addition
bis Trap: Ohm's copper-bismuth
bit Parity bit facts
bl Does the fuse blow
bl no Blue LED Nobel 2014 detail
bla Kirchhoff: black body
blu Blue and violet LED Vf
bo Boole: life
bo bi Trap: Boolean 1+1 vs binary
bo cu Body current table
bo id Boolean identities
bo la Boole's law x^2 = x
bod Body resistance
bon Grounding vs bonding
books Boole: books
bot Trap: both legs in one row
box Black box: find the diode
br di Bridge diode faults
br pr Trap: breakers protect people
br rec Bridge rectifier details
br ro Breadboard rows of 5
br se Trap: 100 W brighter in series
br su Trap: breakers and surges
bra Shorted parallel branch
bre Circuit breaker
bu Bubble pushing
bu ra Bulb rating to resistance
buc Buck, boost, DC-DC
buf Trap: buffer does not flip
bul Bulb brightness
bun Stretch or bundle a wire
bur Ammeter burden
burn Trap: burns out = open
ca en Capacity and energy
cag Faraday cage: charge inside
cap Breaking capacity
car Car safe in lightning
cau Static shock cause
cav Cavendish found it first
cc CC/CV slow finish
cells Cells in series and parallel
ces Kirchhoff: cesium and rubidium
cf LED vs CFL vs incandescent
ch Charge time
ch ro Chips needed: round up
cho Trap: choose the source R
cit Kirchhoff: cities
cla Diode clamp to a reference
clo Pick R_k close to R_x
closi Trap: electroscope closing
cm CMOS inverter
cm lo CMOS logic levels
co Commutator
co br Coin bridge switch test
co fl Conventional vs electron flow
co hi Coulomb's law history
co la Coulomb's law
co pr Before Coulomb: priority
co un Coulomb unit = 1 A*s
coe Temperature coefficient
com Relay COM NO NC states
comb Combining law
commo Find the SPDT common pin
compu Relay logic and computers
con Contact bounce
cond Conductors in fields
cong Units at 1881 Paris congress
cons Charge constants
conti Continuity and diode test
cop AWG copper table
cos Electricity cost
cou Coulomb: life
coun Adder gate counts
cr CR2032 lithium coin
cu Current
cu di Current divider
cu us Trap: current is used up
cub Resistor cube values
curv Diode I-V curve
cut Symmetry: join or cut
da AC vs DC danger
dal Dalziel DC numbers
dc DC basics
dc us DC sources and uses
de De Morgan: life
de le Boole and De Morgan letters
dea Dead lamp: 0 V or full V
dec Mux and decoder chips
del Delta-Y conversion
deli Absorb or deliver power
dem Demux, decoder, encoder
dep Depletion region
det Shannon thesis details
di Diode uses
di li Trap: diode is non-linear
di ou Diode gate output voltages
di te Diode test readings
die Ohm: Die galvanische Kette
dig Color code digits
dim Shunt resistor dims LED
dir Force direction in a field
dis Ampere's discoveries
disch Discharge types order
dist Coulomb ratio: distance
div Voltage divider
do Don't care rows
doo NO push button (doorbell)
dop Dopant element list
dp DPST switch
dpd DPDT switch
dr Dry cell vs wet cell
dri Drift speed of electrons
du Du Fay: two kinds of charge
dua Dual vs complement
e1 E12 standard values
e6 E6 and E24 series
ea Earth ground symbol
ed Edison's argument vs AC
ef Efficiency
el Electron-volt
el ch Electric charge basics
el di Electron flow in a diode
el fi Electric field
el ho Electrophorus: how it works
el ind Electroscope by induction
el ke Trap: electrodynamics key
em EMF is not a force
en Energy and kWh
en co Energy conversions
en pr Encoder and priority encoder
end End of the war 1892
engl English words to gates
eq Equivalent resistance tips
eq va Equal-resistor values
equi Equipotential rules
es ESD you cannot feel
eve Everyday source voltages
ex Extra-low voltage 50 V
exa At least k / exactly k
exp Volta quote: experiment
fa Fault clues
fac Shock factors
far Finite ladder: far end first
fara Trap: Oersted law = Faraday
fas Fast vs slow-blow fuse
fe Resistivity: W, Fe, Pt, Pb
fi Field lines
fig Sig fig counting rules
fil Bulb filament and inrush
fin Find the cathode
fixe Battery to ground fixes a node
fl Flyback diode
fla Flash-to-bang rule
flo Grounding electron flow
flu Trap: Ampere and fluorine
fo Forward bias
fo ma Trap: Ampere found magne…
forms GFCI forms and history
fou Ohm: inspired by Fourier
four Four regions of diode curve
fr Franklin: battery and Fr unit
fra Franklin
fri NC push button (fridge)
fu Fuse
fu co Trap: fuse count rounds down
fu sy Fuse and breaker symbols
fu vs Fuse vs breaker
ful Full adder
ga Galvani
ga ch 3- and 4-input gate chips
ga wa Band gap to wavelength
gaa Trap: GaAs alone is infrared
gai Trap: gained protons
galvano Galvanometer
gates Diode logic gates
ge Band gaps: Ge to GaN
gf cu Trap: GFCI limits current
gf ti GFCI trip time rule
gf wh GFCI where and limits
gi Gilbert: De Magnete 1600
gl Crown of cups: wine glasses
go Dalziel AC let-go numbers
goe Why a fuse goes in series
gol Color code gold silver
gr Slope of V-I graph
grad Coulomb quote: graduating
gre Green LED: two kinds
gri Why the grid uses AC
grov Fuel cell: Grove 1839
gu Gray and von Guericke
ha DC hazards
ha vo Trap: half voltage, half power
ha wr Trap: half adder wrong gates
han Logic hands-on build
he Ampere: words he coined
he se Trap: heat and semiconduct…
he tr Hertz: the transmitter
hel Why the ground wire helps
her Hertz: life
hi Ohm's law history
ho Ohm: honours
hol Holonyak: first visible LED

`honor` Volta honors timeline
`hot` Hot, neutral, ground roles
`hou` Household wire colors
`hp` Watt: steam engine, hp
`hu` Humidity and static
`hug` Huge mystery R: loading
`hv` HVDC lines
`id` XOR identities
`idea` Ideal meters memory trick
`ie` ANSI vs IEC symbols
`ig` Fuel ignition risk
`ind` Indirect measurements
`indu` Induction steps in order
`ins` 4-way switch inside
`insu` Conductors vs insulators
`int` Internal resistance
`int bo` Internal body resistance
`inven` Trap: who invented AC
`io` Lithium-ion cell
`ite` Infinite ladder 0.618R
`ja` Multimeter jacks
`jack` Fuse in the mA jack
`je` Boole's OR vs Jevons
`jo` Joule and Watt
`jo oh` Trap: Joule before Ohm
`joi` Joined spheres: Q ~ r
`ju` Franklin: kite June 1752
`kar` Karnaugh map (K-map)
`kc` KCL current law
`ke` Kelvin bridge
`keye` Shock test values keyed
`ki` Switch kinds
`ki di` Kirchhoff's discoveries
`ki th` Kirchhoff: the other laws
`kir` Kirchhoff: life
`kn` Trap: known R in other branch
`kv` Air breakdown 3 kV/mm
`kv la` KVL voltage law
`la` Ohm's law
`lam` Lamp symbol: X or loop
`lat` Latching switch types
`laws` De Morgan's laws
`lay` Breadboard layout
`ld` Thermistor, varistor, LDR
`le` LED legs
`le ch` Equal LEDs checklist
`le fa` LED fault from voltages
`le re` LED resistor
`le ze` Lead resistance zero
`lea` Lead-acid cell
`leaf` Electroscope leaf rules
`lef` Trap: lead left in mA jack
`lem` Lemon or potato cell
`len` Ampere force law: lengths
`les` Trap: AC has less resistance
`lett` Meter symbols: letters
`ley` Pile vs Leyden jar
`li` Ohm: life
`lim` Breadboard limits
`line` GFCI LINE and LOAD
`lip` LiPo and LiFePO4
`lit` LED not lit trick
`liv` Trap: ohmmeter on live circuit
`lo` Logic levels
`lo ch` Logic chips 7400 series
`loa` Loading effect
`loade` Loaded divider: output drops
`loc` 4-way switch
`look` Trap: look-alike symbols
`los` Line loss and voltage
`mad` Trap: Volta made electrophorus
`mag` Trap: On Magnetism key
`mah` Trap: mAh is not energy
`mai` Mains in other countries
`maj` Majority function
`makes` How an LED makes light
`mar` TTL output levels, margins
`mate` LED materials by color
`max` Max power transfer
`maxw` Maxwell: link to Hertz
`me` Mesh analysis
`me po` Meter polarity: minus sign
`me pow` Trap: meter as power source
`mem` Ohm quote: the Memoir
`meta` Resistivity: non-metals
`metal` Metal resistivity order
`meth` Charging methods
`mh` Mho and siemens history
`mi` Full adder minterms
`mic` Limit and micro switch
`mil` One-node formula (Millman)
`mini` Minimum safe resistor
`mino` Majority and minority carriers
`minu` Trap: minus sign force
`mix` Trap: Galvani mix-ups
`mn` Color code mnemonic
`mo` Diode models
`mode` Trap: ohms mode on a diode
`moh` Wrist strap 1 Mohm resistor
`mos` Tribo: most charge pair
`mu` Multiplexer
`mu ga` Mux as a gate
`mu pa` Multiway = parity (XOR)
`my` Mystery R checklist
`my re` Mystery resistor task
`na` NAND gate
`na onl` NAND-only gate counts
`nai` Trap: two copper nails = 0 V
`names` Seven names mnemonic
`nands` XOR from 4 NANDs
`nc` Push button NO vs NC
`ne` NEMA name code
`nee` Bits needed for N
`neg` Trap: negative current answer
`neu` Trap: switch on neutral
`nev` Trap: wave never below zero
`new` Newton of electricity quote
`ni` NiCd cell
`nich` Why heaters use nichrome
`nim` NiMH cell
`nit` Trap: N is not nitrogen
`no` Why no silicon LEDs
`no na` NOR from 4 NANDs
`no ti` Trap: NAND(A,0) is not NOT
`nob` Trap: Nobel starts 1901
`nod` Nodal analysis
`nodes` Series vs parallel by nodes
`non` Non-SI conversions
`nor` NOR gate
`nors` XNOR from 4 NORs
`not` NOT gate
`nota` Logic notation symbols
`nu` Lightning numbers
`oc` Hex, octal, BCD
`oe` Oersted
`of` Trap: open means off
`oh` Ohmmeter
`oh un` Ohm unit name history
`ohmi` Ohmic vs non-ohmic
`ol` Trap: blue LED reads OL
`on` NOR-only gate counts
`one` Trap: one fault only
`ont` Schematic onto breadboard
`op` Open circuit
`op ba` Trap: open-circuit battery V
`opens` Switch closes or opens
`or` OR gate
`or na` AND and OR from NAND
`or no` AND and OR from NOR
`or not` OR gate from AND and NOT
`ord` Order of operations
`ot` Volta: other work
`ou` Outlet slots
`ou me` Outlet meter readings
`ov` Trap: product over sum
`overl` Overload vs short circuit
`overs` Trap: oversized fuse/breaker
`ow` Trap: each battery own R
`pa` How parts fail
`pa bu` Trap: parallel bulb removed
`pa ce` Parts of a cell
`pa dimm` Bulb A then B, C in parallel
`pa ra` Series / parallel ratio n^2
`pad` Joule: paddle wheel
`pai` Prefix shortcut pairs
`pap` Oersted: paper and Arago
`para` Parallel resistors
`part` Trap: ammeter across part
`pc` PCB printed circuit board
`pe` Zener: person vs diode
`ph` Photodiode facts
`pi` Minterm, maxterm, sigma, pi
`pic` Pick the best range
`pil` Voltaic pile
`pins` Chip pins and wiring
`piv` Rectifier PIV and peaks
`pl` Binary place values
`plu` Trap: S vs s, plurals
`pn` Ohl: PN junction details
`pn le` PN junction and LED history
`po` Polarization
`po ch` Power task checklist
`po cu` Trap: max power vs current
`po en` Potential and energy
`po rh` Potentiometer vs rheostat
`pog` Ohm: Poggendorff thermocou…
`poi` Hands-on scoring points
`posi` Trap: gold by position
`post` Ampere: career posts
`posth` Coulomb unit: posthumous
`pot` Potentiometer
`pow` Power formulas
`pr ce` Primary cell voltages
`pred` Trap: predicted vs confirmed
`prefe` Why prefer a fuse
`pres` Trap: NC DPST pressed
`pri` Priority lights
`prim` Primary vs secondary
`pron` Polarized and 3-prong plugs
`prop` Volta's law: Q prop. to V
`prov` Coulomb: proved vs assumed
`ptc` PTC polyfuse
`pul` Pull-up vs pull-down
`puls` Trap: pulsing DC is DC
`qs` Volta vs Galvani: past Qs
`qu` Hertz quotes
`qu un` Quantities and units table
`qua` Quark charges
`quan` Charge quantization
`ra` C-rate
`ra se` Trap: rated watts in series
`rad` Hertz: radio waves
`ran` LED current range
`rank` Identical bulbs: rank rules
`rati` Relay contact rating
`ratio` Coulomb ratio: charges
`ray` Hertz: cathode rays
`rc` RCD 30 mA
`re` Relay
`re ba` Reading 4, 5, 6 bands
`re bi` Reverse bias and breakdown
`re ch` Resistor tolerance check
`re po` Resistor power rating
`re va` Resistivity values
`re vs` Rectifier vs inverter
`read` Which end to read first
`rec` Rectifiers
`rece` Hertz: the receiver
`rech` Rechargeable cell voltages
`red` Red lead is anode
`ree` Reed switch
`res` Temperature and resistance
`rese` Resetting a breaker
`resi` Resistivity
`rev` Trap: LED reverse 5 V
`rh` Rheostat
`ri` Ripple-carry adder
`riv` Trap: Ohm's rival not Galvani
`rm` RMS
`rod` Trap: rod removed first
`rou` Uncertainty rounding
`roy` Boole: Royal Medal 1844
`ru` KVL sign rules
`sa` Safety habits
`sam` Trap: same reading, 2 faults
`sc` Out of scope 2027
`sc ba` Schematic symbols: basics
`sca` AWG scaling rules
`scala` Trap: V is 1/r not 1/r^2
`sch` Schematic symbols: parts
`scho` Schottky, Ge, photodiode
`sci` The seven 2027 scientists
`se` Meter settings
`se le` Trap: one series LED reversed
`se sou` Trap: R in series w/ I source
`seeb` Thermocouple: Seebeck 1821
`sel` Select lines: round up
`sem` Semiconductor basics
`sep` Ampere: Sept 1820 timeline
`ser` Series resistors
`sev` Seven SI base units
`sh` Short circuit
`sh se` Shorting one series bulb
`shan` De Morgan and Shannon
`shar` Hands-on share of score
`she` Sheffer stroke, Peirce arrow
`show` Hertz: wave properties shown
`shu` Shunt vs multiplier
`si` SI prefixes
`si oh` SI base units of V, ohm, S
`sid` Trap: depletion ion sides
`sig` Sig figs in math
`sim` Trap: simplify before counting
`sin` Single and double pole
`sink` Sourcing vs sinking LED
`sli` Trap: Coulomb unit slips
`sm` Trap: SMD 100 = 10 ohm
`so` Sources of error
`soc` Lamp socket: shell is neutral
`sod` Trap: sodium in Sun, not Moon
`sol` Solid-state relay
`sola` Solar cell numbers
`sp` SPST switch
`spd` SPDT switch
`spect` Kirchhoff: spectroscope trap
`spl` Split rail check
`sq` Trap: RMS of a square wave
`sq ch` Trap: V = 0 is not E = 0
`st` Step voltage
`sta` Static hazards
`stai` Trap: staircase XOR or XNOR
`stead` Steady-state laws multi-sel…
`steps` Superposition steps
`stor` Trap: battery stores charge
`str` Power limit of a string
`stru` Trap: kite struck by lightning
`stu` Kirchhoff: 1845 as a student
`su` Supernode
`su di` Superconductor discovery
`su ra` Surge protector ratings
`sum` Sum of products
`sun` Kirchhoff quote: gold sun
`supp` Power supplied task
`sur` Surge protector
`swa` Edison: bulb 1879 and Swan
`sy` Boole's symbols
`sy li` Anti-symmetry: V/2 line
`syn` Law name synonyms
`ta` Truth table rows
`ta or` Trap: truth table row order
`tap` Pot taper: linear vs log
`tar` Target R checklist
`tas` USB task checklist
`te` GFCI TEST button
`ter` Switch terminal count
`tesl` Tesla (out for 2027)

tet Trap: AC tetanus or DC tetanus
th Consensus theorem
th sy Thermistor and LDR symbols
tha Thales and amber c. 600 BC
thev Thevenin and Norton: OUT
thi Trap: thick wire has more n
tho Boole: Laws of Thought title
thr Switch poles and throws
thu Thunderstorm charging
timel Timeline of the seven
to Network tolerance
top Top-down target method
tor Trap: torsion balance vs Ohm
tou Identical spheres touch
tr Mux tree N - 1
tr 74 Trap: 7402 wired like 7400
tr ad Trap: adapter vs converter
tr am Trap: Coulomb or Ampere?
tr ano Trap: diode vs battery anode
tr ba Trap: 5-band multiplier
tr bia Trap: forward bias is not on
tr ca Trap: capacitor vs cell
tr ci Trap: ideal vs real circuit
tr de Trap: De Morgan vs Boole
tr di Trap: LED on a divider
tr el Trap: 1 A = 1 electron/s
tr fi Trap: field units N*C
tr gap Trap: GaP/GaN LED colors
tr go Trap: gold is the best
tr ins Trap: E = 0 inside so V = 0
tr ju Trap: junction rule = charge
tr kc Trap: KCL conserves energy
tr ki Trap: not Kirchhoff's work
tr la Trap: Coulomb law vs unit
tr nor Trap: 3-input NOR build
tr oh Trap: ohm/m vs ohm m
tr pa Trap: 1/8 W parts
tr ri Trap: ripple of half adders
tr sc Trap: Schottky is not PN
tr sh Trap: Shannon vs Boole
tr si Trap: 1.1 V is not Si Vf
tr sp Trap: SPST for 2 places
tr sup Trap: superposition power
tr tw Trap: two's complement OUT
tr vo Trap: not Volta's
tr wa Trap: watt vs joule
tr wo Trap: not Boole's work
tr xn Trap: 3-input XNOR
tra Trap: + means OR
trav 3-way common and travelers
trea Boole quote: treatise
tri GFCI trip level
trib Triboelectric series
tro Troubleshooting steps
tt TTL input levels
tw Two equal LEDs task
tw sp Two touching spheres indu…
ty Fuse types
ty ba Typical battery capacities
typ LED typ vs max Vf gap
type N-type and p-type
typi Typical currents
un k value and units
un eq Units equal to a volt
un pi Count unknowns, pick met…
un wa Units equal to a watt or ohm
unb Unbalanced bridge voltmeter
unc Carry uncertainty
unif Uniform field E = V/d
unit Hertz unit history
univ Universal gate sets
unk Trap: bridge unknown arm
unu Unused gate inputs
up Pick-up and drop-out
us US and world outlets
us le USB standard levels
usb USB label reading
used Trap: copper is used up
va MOV varistor
van Van de Graaff safe to touch
varia Trap: pot vs variable R
varie Rail polarity varies
ve Tribo series: long version
verso Gilbert: electricus, versorium
vf LED Vf by color
vf li Trap: max Vf is not limit
vo Voltage
vo bu Voltaic pile: build details
vo co Volta's contact series
vo se Voltmeter in series
vo vs Volta vs Galvani: details
volta Volta: life
voltm Voltmeter
vs th Trap: varistor vs thermistor
wa 3-way switch
wa bu 3-way build checklist
wa fa 3-way fault tables
wa sou Trap: sine wave = AC source
wal Wall adapter insides
war War of the Currents
watts Trap: no watts in code
wav Wavelength and Vf
wea Weak battery reads full
wer Werner von Siemens
wes Westinghouse: AC wins
wh el Trap: who coined electricity
wha Shorted bulb: what happens
whe Field where E = 0
whea Wheatstone bridge history
whi Trap: which Siemens
whil Joule: unit named while alive
why Why use a relay
wi Wire gauge AWG
wim Wimshurst machine
wired Trap: Zener wired forward
word Logic word problems
worke Trap: Volta worked in France
works GFCI how it works
wr Unit writing rules
wro Trap: yellow Vf key wrong
xn XNOR gate
xo XOR gate
xo bu XOR bubble rules
ye Orange and yellow LED Vf
ze Zener diode
ze cu Zero-current branch trick
ze po Zero-potential points: two
zer 3-band and zero-ohm
zi Carbon-zinc (Leclanche)